

PINNACLE

60 DAYS Self--Preparation Module

2nd
Edition



SSC General Studies Theory Book

Based on Latest TCS Pattern

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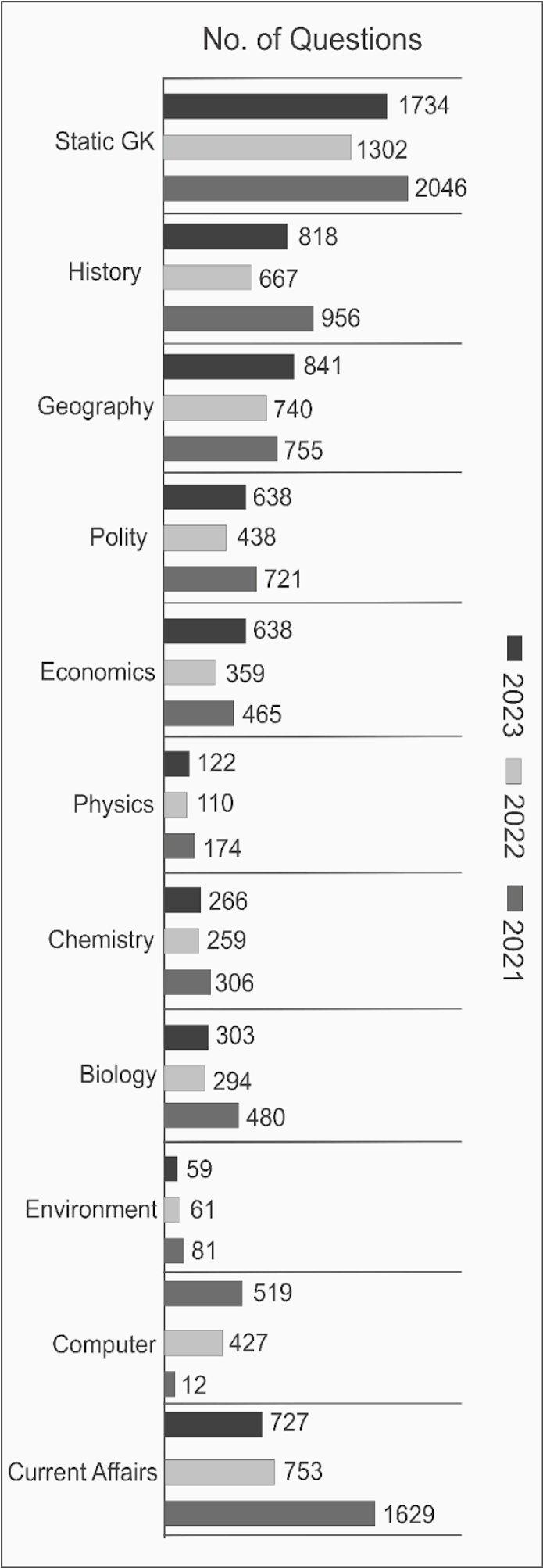
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Last Three Years Trend Analysis (Weightage)

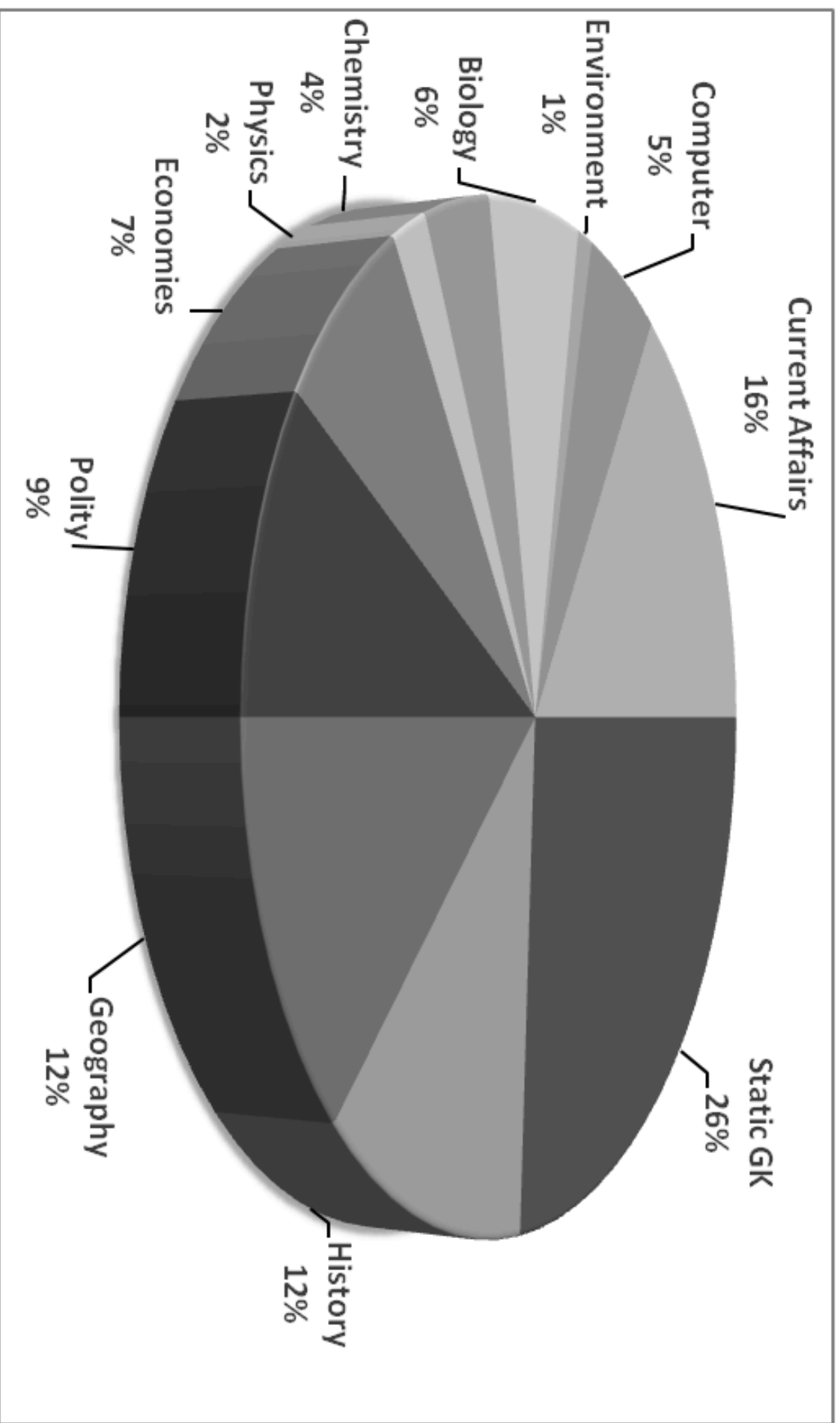
S.N.	Subject		CGL (T1) (No. of Ques)	CGL (T2) (No. of Ques)	CHSL (T1) (No. of Ques)	CHSL (T2) (No. of Ques)	MTS (No. of Ques)	CPO (No. of Ques)	Delhi Police (No. of Ques)	JE (T1) (No. of Ques)	Selection Post (No. of Ques)	Total No. of Ques.
1.	Static Gk	2023	227 (23.3%)	-	216 (21.6%)	1 (1.4%)	274 (40.6%)	117 (26.0%)	694 (25.7%)	117 (26.0%)	88 (29.3%)	1734 (26%)
		2022	243 (24.3%)	5 (2.8%)	226 (25.1%)	1 (2.9%)	438 (30.7%)	118 (26.2%)	96 (13.3%)	110 (24.4 %)	65 (26%)	1302 (24.07%)
		2021	79 (15.0%)	-	497 (47.3%)	-	714 (29.8%)	108 (36%)	294 (15.9%)	29 (9.7%)	325 (27.1%)	2406 (26.8%)
2.	History	2023	155 (15.9%)	3 (6.7%)	93 (9.3%)	4 (5.7%)	65 (9.6%)	62 (13.8%)	350 (13.0%)	36 (8.0%)	50 (16.7%)	818 (12.3%)
		2022	157 (15.7%)	10 (5.6%)	78 (8.7%)	2 (5.7%)	255 (17.9%)	60 (13.3%)	32 (4.4%)	33 (7.3%)	40 (16%)	667 (12.33%)
		2021	68 (13%)	-	20 (1.9%)	-	446 (18.6%)	31 (10.3%)	221 (11.9%)	30 (10%)	140 (11.7%)	956 (12.5%)
3.	Geography	2023	120 (12.3%)	4 (8.9%)	130 (13%)	6 (8.6%)	76 (11.3%)	75 (16.7%)	315 (11.7%)	72 (16.0%)	43 (14.3%)	841 (12.6%)
		2022	119 (11.9%)	14 (7.8%)	132 (14.7%)	3 (8.6%)	237 (16.6%)	76 (16.9%)	62 (8.6%)	67 (14.9%)	30 (12 %)	740 (13.68%)
		2021	61 (11.16)	-	70 (6.7%)	-	260 (10.8%)	31 (10.3%)	164 (8.9%)	42 (14%)	127 (10.6%)	755 (9.9%)
4.	Polity	2023	77 (7.9%)	3 (6.7%)	91 (9.1%)	3 (4.3%)	56 (8.3%)	56 (12.4%)	288 (10.7%)	36 (8.0%)	28 (9.3%)	638 (9.6%)
		2022	81 (8.1%)	16 (8.9%)	58 (6.4%)	2 (5.7%)	125 (8.8%)	56 (12.4%)	48 (6.7%)	32 (7.1%)	20 (8%)	438 (8.10%)
		2021	58 (11%)	-	42 (4%)	-	184 (7.7%)	27 (9%)	258 (13.9%)	29 (9.7%)	123 (10.3%)	721 (9.5%)
5.	Economics	2023	71 (7.3%)	3 (6.7%)	197 (19.7%)	6 (8.6%)	53 (7.9%)	49 (10.9%)	216 (8.0%)	36 (8.0%)	7 (2.3%)	638 (9.6%)
		2022	75 (7.5%)	8 (4.4%)	65 (7.2%)	2 (5.7%)	74 (5.2%)	49 (10.9%)	52 (7.2%)	34 (7.6%)	-	359 (6.64%)
		2021	26 (5%)	-	40 (3.8%)	-	140 (5.8%)	17 (5.7%)	148 (8%)	12 (4%)	82 (6.8%)	465 (6.1%)
6.	Physics	2023	24 (2.5%)	2 (4.4%)	31 (3.1%)	2 (2.9%)	14 (2.1%)	12 (2.7%)	27 (1.0%)	2 (0.4%)	8 (2.7%)	122 (1.8%)
		2022	28 (2.8%)	12 (6.7%)	22 (2.4%)	2 (5.7%)	27 (1.9%)	12 (2.7%)	-	2 (0.4%)	5 (2%)	110 (2.03%)
		2021	16 (3%)	-	24 (2.3%)	-	34 (1.4%)	18 (6%)	34 (1.8%)	23 (7.7%)	25 (2.1%)	174 (2.3%)
7.	Chemistry	2023	63 (6.5%)	3 (6.7%)	59 (5.9%)	6 (8.6%)	13 (1.9%)	35 (7.8%)	45 (1.7%)	34 (7.6%)	8 (2.7%)	266 (4%)
		2022	70 (7%)	4 (2.2%)	28 (3.1%)	2 (5.7%)	49 (3.4%)	35 (7.8%)	22 (3.1%)	34 (7.6%)	15 (6%)	259 (4.79%)
		2021	34 (6.5%)	-	78 (7.4%)	-	104 (4.3%)	18 (6%)	24 (1.3%)	16 (5.3%)	32 (2.7%)	306 (4%)
8.	Biology	2023	56 (5.7%)	2 (4.4%)	31 (3.1%)	3 (4.3%)	23 (3.4%)	28 (6.2%)	99 (3.7%)	45 (10.0%)	16 (5.3%)	303 (4.5%)
		2022	61 (6.1%)	9 (5%)	40 (4.4%)	3 (8.6%)	75 (5.3%)	26 (5.8%)	18 (2.5%)	52 (11.6%)	10 (4%)	294 (5.43%)

	2021	32 (6.1%)	-	83 (7.9%)	-	135 (5.6%)	18 (6%)	116 (6.3%)	43 (14.3%)	53 (4.4%)	480 (6.3%)
	2023	12 (1.2%)	1 (2.2%)	8 (0.8%)	2 (2.9%)	7 (1.0%)	1 (0.2%)	18 (0.7%)	9 (2.0%)	1 (0.3%)	59 (0.9%)
9. Environment	2022	8 (0.8%)	5 (2.8%)	3 (0.3%)	1 (2.9%)	11 (0.8%)	4 (0.9%)	10 (1.4%)	14 (3.1%)	5 (2%)	61 (1.13%)
	2021	9 (1.7%)	-	5 (0.5%)	-	17 (0.7%)	4 (1.3%)	25 (1.4%)	7 (2.3%)	14 (1.2%)	81 (1.1%)
10. Computer	2023	-	20 (44.4%)	-	30 (42.9%)	-	1 (0.2%)	450 (16.7%)	18 (4.0%)	-	519 (7.8%)
	2022	-	80 (44.4%)	70 (7.8%)	15 (42.9%)	-	-	240 (33.3%)	22 (4.9%)	-	427 (7.89%)
	2021	0	-	-	-	-	-	-	12 (4%)	-	12 (0.2%)
11. Current Affairs	2023	170 (17.4%)	4 (8.9%)	144 (14.4%)	7 (10%)	94 (13.9%)	14 (3.1%)	198 (7.3%)	45 (10.0%)	51 (17%)	727 (10.9%)
	2022	158 (15.8%)	17 (9.4%)	178 (19.8%)	2 (5.7%)	134 (9.4%)	14 (3.1%)	140 (19.4%)	50 (11.1%)	60 (24%)	753 (13.92%)
	2021	142 (27%)	-	191 (18.2%)	-	366 (15.3%)	28 (9.3%)	566 (30.6%)	57 (19%)	279 (23.3%)	1629 (21.4%)
Total No. of Questions	2023	975	45	1000	70	675	450	2700	450	300	6665
	2022	1000	180	900	35	1425	450	720	450	250	5410
	2021	525	-	1050	-	2400	300	1850	300	1200	7625

Bar Graph Trend Analysis of Last Three Years SSC Exams Papers



Pie Chart Trend Analysis of Last Three Years SSC Exams Papers



Static GK

Classical and Folk Dances

Sangeet Natak Akademi recognised 8 classical dances of India named as Bharatanatyam (Tamil Nadu), Kathak (Uttar Pradesh), Kathakali (Kerala), Kuchipudi (Andhra Pradesh), Manipuri (Manipur), Mohiniyattam (Kerala), Odissi (Odisha), Sattriya (Assam).

The Ministry of Culture recognised Chhau dance as 9th classical dance of India. The Medieval term of Dancing is "Raqs". The technique of classical dancing was codified by Bharata in about the 5th century B.C. The national award for classical dance is called the Abhinandan Saroja Award.

Bharatanatyam

Bharatanatyam expresses South Indian religious themes and spiritual ideas of Shaivism. It was formerly known as the Sadir Attam. Krishna Iyer first coined the term Bharatanatyam for the Sadir dance. It was banned by the colonial British government in 1910. It is the first of India's traditional dances to be refashioned as a theatre art and to be exhibited widely both at home and abroad.

It evolved from ekaharya, where one dancer takes on many roles in a single performance. It emerged in the temples of South India, performed by Devdassis, thus also known as Dasiattam. Bharatanatyam is traditionally performed to Carnatic music.

It consists of six portions- Alarippu (Invocation), Jathiswaram (Nritta part), Shabdham (Short compositions with word), Varnam (a story, that includes both Nritta and Nritya), Padam (Religious prayer, Bhajan, Keerthanam) and Tillana (Origin in the Tarana of Hindustani Music).

Bharatanatyam poses are depicted on the gopurams of the Chidambaram temple (Tamil Nadu). A Bharatanatyam artist uses Abhinaya (the Sanskrit word and it means 'taking towards') as a tool to convey ideas and evoke emotions among the audience members. Abhinaya can be classified into four types-

Satvika Abhinaya (expressing by evoking the state of mind of the character), Angika Abhinaya (expressing using body movements such as movement of hands,

legs and limbs) Vachika Abhinaya (expressing using medium of speech such as songs, music and dialogues), and Aharya Abhinaya (expressing using decorations such as costumes, jewelry and makeup).

The four Nattuvanars namely Ponaiyah, Vadivelu, Sivanandam and Chinnaiya, who are renowned as Tanjaore Bandhu shaped up the modern day Bharatanatyam.

Different styles of **Bharatanatyam**, called Banis -

Tanjavur Style - Kandappa Pillai one of the famous Nattuvanars (Guru /Teacher) of this style and a direct descendent of the Tanjore Quartette was trained by Kannuswami Pillai.

Pandanallur Style is attributed to the famous Minakshisundaram Pillai who was a direct descendant of the Tanjore Quartette.

Vazhavur style was created by Ramiah Pillai of Vazhuvoor town in Tamil Nadu.

Kalakshetra Style is credited to the disciple of Minakshisundaram Pillai and a renowned Bharatanatyam exponent Rukmini Devi Arundale. Rukmini Devi founded the Kalakshetra Institute in Chennai making it a platform to promote Bharatanatyam as an art form.

Melattur Style is credited to Mangudi Dairairaja Iyer. It is known for its soft footwork and emphasis on the srngara rasa.

Famous Dancers - Kamala Narayan, Bala Saraswati, C. V. Chandrasekhar, Leela Samson, Mrinalini Sarabhai, Padma Subramanyam, Rukmini Devi, Sonal Mansingh, Yamini Krishnamurti, Lakshmana Swamy, Alarmel Valli, Smrithi Krishnamurthy, Hema Malini, Mallika Sarabhai, Vasundhara Doraswamy, etc.

Mohiniyattam

Mohiniyattam (Malayalam) is an Indian classical dance originated in the state of Kerala. Mohiniyattam literally means 'dance of the enchantress'. The dance is performed by women in honor of the Hindu God Vishnu in his incarnation as the enchantress Mohini. Kalyanikutty Amma is known as the 'Mother of Mohiniyattam'.

It is lasya - style based on Natya Shastra. It consists of 'Nritta' and 'Nritya'

mentioned in 'Natya Shastra'. It includes facial expressions and hand gestures.

The main percussion instrument used in the performance of Mohiniyattam dance is Edakka.

It consists of seven portions -

Cholkettu, Jatisvaram, Varnam, Padam (song), Tillana, Shlokam (a hymn in praise), and Saptam (the Expressions (or Abhinaya) .

The techniques of Mohiniyattam consist of Adavus (steps) and Mudras (hand gestures).

Adavus is classified into four- Thaganams, Jaganams, Dhaganams, and Samishrams. These adavus were codified by Kalamandalam Kalyanikutty Amma in 20th century CE.

Mudras are based on the book Hastalakshana deepika. There are a total of 24 mudras. Some are - Pataka, Katakam, Mushti, Karthari, etc.

Famous Dancers - Sunanda Nair, Smitha Rajan, Gopika Varma, Jayaprabha Menon, Pallavi Krishnan and Vinitha Nedungadi, etc.

Kuchipudi

Kuchipudi (short form of 'Kuchelapuram' or 'Kuchilapuri') is India's classical dance originated in a village called Kuchelapuri in Andhra Pradesh.

Kuchipudi is accompanied by Carnatic music. It incorporates all three classical dance elements- Nritta (non-narrative and abstract dancing), Nritya, and Natya.

Daru is the main content of the music in Kuchipudi; it adapts Jaru style of singing in medium and quick tempos.

It is characterized by graceful, swaying body movements with no abrupt jerks or sudden leaps. It belongs to the lasya style ('soft and suitable for female presentation')

Famous dancers - Guru Smt. Vijaya Prasad, Dr. Vempati Chinna Satyam, Kalpalathika, Raja and Radha Reddy, Kaushalya Reddy, Yamini Reddy, Bhavana Reddy, Atisha Pratap Singh, Sreelakshmy Govardhanan, Smt. Vyjayanthi Kashi, Haleem Khan, Prateeksha Kashi, Yamini Krishnamurthy, Arunima Kumar, Achuta Manasa, Shobha Naidu, Uma Rama Rao, Vedantam Satyanarayana Sarma, etc.

Manipuri

Manipuri is known for its Hindu Vaishnavism themes, and Radha - Krishna's love - inspired dance drama Rasleela. Predominant theme of Manipuri dance is Devotion.

Two categories of Manipuri dance are Jagoi (reflects the Lasya item mentioned in Natya Shastra of Bharata) and Cholom (form of classical Tandava dance). The masculine aspect of Manipuri classical dance is known as Choloms.

The ethnic majority of Meitei people call Manipuri dance as 'Jagoi'.

Manipuri dance is Introverted and restrained compared to most dances of India, In this dance the artist never established eye contact with the audience. The Manipuri classical style of singing is called 'Nat'.

Varieties of Manipuri forms of dance-

Manipuri dance has two traditions- one is the sankeertana (which is the devotional aspect) and the other is the raasa.

Manipuri Sankirtana was also inscribed in the Representative List of the UNESCO Intangible Cultural Heritage of Humanity in 2013. It is practiced primarily by the Vaishnava community in Manipur (also in some parts of Tripura and Assam). It is performed to narrate the life and deeds of Lord Krishna. Instruments used are Cymbals (Karataala) and Drums (Choloms).

Pung Cholom (meaning "roar of the drums") is the soul of Manipuri Sankirtana music and classical Manipuri dance.

Dhol Cholom (Drum Dance or Dholak Cholom) is performed in the spring season known as 'Yaosang' meaning Spring festival of Holi.

Kartal Cholam is a Tandava dance of cymbals. It is performed by the male dancers only.

'Ras Leela' refers to the 'Dance of Divine Love' that beautifully portrays the affection between Krishna - the eighth avatar of Lord Vishnu, and Radha - the consort of Lord Krishna. Its three types are Tal Rasak (followed by a clap), Danda Rasak (drums are pounded with two sticks while the dancers' position produces geometric shapes), Mandal Rasak (Lord Krishna in the centre

surrounded by the Gopis).

Famous dancers - Hanjaba Guru Bipin Singha, Jhaveri Sisters (Nayana Jhaveri, Ranjana Jhaveri, Suverna Jhaveri, and Darshana Jhaveri), Nirmala Mehta, Savita Mehta, Yumlembam Gambhini Devi etc.

Odissi

Odissi (Odisha) roots can be traced back to Natya Shastra. Its south-eastern style is known as the Odhra Magadha.

The Pakhawaj syllables are primarily used to conclude Odissi dance.

The performance repertoire includes invocation, nritya (pure dance), nritya (expressive dance), natya (dance drama) and moksha (dance climax connoting freedom of the soul and spiritual release).

Nritya and Abhinaya are the two most interesting attractions of Odissi.

Nritya- Dancers perform delicate physical movements to create ornamental movements.

Abhinaya- Dancers make facial expressions to explain a religious story or legend.

Famous dancers - Sonal Mansingh, Sanyukta Panigrahi, Jhelum Paranjape, Mayadhar Raut, Gangadhar Pradhan, Ileana Citaristi, Leena Mohanty, Chitra Krishnamurti, Bijayini Satpathy, Madhavi Mudgal, Sanchita Bhattacharyaa, Sutapa Talukdar, Deba Prasad Das, Pankaj Charan Das, Raghunath Dutta, etc.

Kathak

In Kathak dance (Uttar Pradesh) The term Kathak comes from the Sanskrit Vedic word Katha which means "story" and Kathakar which means "one who tells a story" or "related to stories". The story of Radha and Krishna is the major theme of Kathak. Hindustani Classical Music associated with this dance. The main Gharanas (houses) of Kathak, are located in Lucknow, Jaipur, Banaras and Chhattisgarh.

It was popularized under the rule of Mughals. The Golden age of this dance comes under the patronage of Wajid Ali Shah (last Nawab of Oudh). Erstwhile Princely state Avadh was primarily associated with Kathak. National institute of Kathak dance is located in New Delhi.

Natwari Nritya and Thumri music is associated with kathak dance.

Three main sections of a Kathak dance are invocation and 'Nritya' and 'Nritya' mentioned in 'Natya Shastra'.

Nritya- Pure dance portrayed by the artist.

Nritya- Here the artist showcases a story or theme through gestures, expressions and slow body movements accompanied by vocal and instrumental music.

Facial expressions and sensuous gestures emote during Kathak performances are mentioned in the book 'The Wrongs of Indian Womanhood' by Marcus B. Fuller published in 1900.

Famous dancers - Pandit Birju Maharaj, Lachu Maharaj, Shambu Maharaj, Shovna Narayan, Kumari Kamala, Sunanya Hazarilal Agarwal, Pandit Durgalal, Prerana Shrimala, Rani Karna, Sitara Devi, Rupa Rani Das Bora, Roshan Kumari, Maya Rao, Kumudini Lakhia, Rohini Bhate, Pandit Munnalal Shukla, Uma Sharma, Kriti Sanon, etc.

Kathakali

In Kathakali (Kerala) the term Kathakali is derived from Katha (Sanskrit) meaning "story or conversation, or traditional story", and Kali (from art) meaning "performance and art".

The dance-drama art form called 'Krishnanattam' and Ramanattam based on Mahabharata and Ramayana are precursors of 'Kathakali'. Traditionally the stage presentation of this dance was immediately preceded by a practice session called sevakali and undertaken in the precincts of a temple. This dance took shape in Southern India in the seventeenth century under the patronage of the Prince of Karnataka.

It is a harmonious combination of 5 forms of fine art - Literature (Sahithyam), Music (Sangeetham), Painting (Chithram), Acting (Natyam) and Dance (Nritham). The dance form combines four aspects of abhinaya (Angika, Aharya, Vachika, Satvika) and the Nritya, Nritya and Natya.

The story Nalacharitham is written by "Unnai Varier". It is the romantic story of unlimited love between Nala and Damayanthi. There are 24 Basic Mudras (hand gestures) and a total of 470 different symbols used in this dance.

Characters in this dance are broadly divided into satvika (noble characters like Krishna and Rama), rajasika (evil characters) and tamasika (beard characters). It follows the traditional sopana sangeet of Kerala.

In Kathakali emphasis is given more on "Aharya Abhinaya", the use of costumes, ornaments and facial make-up.

A Kathakali performance begins with the "Kelikottu", calling the audience to attention followed by the "Thodayam". "Kelikottu" is the formal announcement of the Kathakali dance done in the evening when drums and cymbals are played for a while in the courtyard.

Then the musicians and drummers hold the stage with an exhibition of their skills in "Melappadam". The end of the performance is marked by a piece of pure dance called "Dhanasi". "Tiranokku" is the debut on the stage of all characters other than the "Pacha" or "Minukku".

Kalasams are pure dance sequences where the actor is at great liberty to express himself and display his skills. The leaps, quick turns, jumps and the rhythmic co-ordination make kalasams, a joy to watch.

Famous dancers - Kalamandalam Gopi, Hari Priya Nambodiri, Kalamandalam Krishna Prasad, Kottakal Sivaraman, Kalamandalam Ramankutty Nair, Kalamandalam Vasu Pisharody, Kavungal Chathunni Panicker.

Sattriya dance

Sattriya dance (Assam) originated in Sattri, a monastery, as a part of the neo-Vaishnavite movement started by Srimanta Sankardev in Assam, in the 15th Century. The songs composed by Sankaradeva are known as 'Borgeet'.

The Sattriya dance can be classified into two styles, namely 'Paurashik Bhangi' i.e. Tandava or Masculine style and 'Stri Bhangi' i.e. Lashya or feminine style.

Chali, Jhumura and Nadu Bhangi are associated with Sattriya dance and Ankiya Nat is a subgenre consisting of one-act plays of Sattriya.

Pat Silk saree is the most popular kind of saree used in this dance, which represents the locality through its various colorful motifs and designs.

The basic dance unit and exercise of a Sattriya is called a Mati Akhara. There are 64 Mati Akharas and they may be further sub - divided into eight main types- Ora, Saata, Jhalak, Sitika, Pak, Jap, Lon and Khar.

Sattriya was recognised in 2000 as a Classical Dance by Sangeet Natak Akademi.

Famous dancers - Gunakanta Dutta Barbayan, Manik Barbayan, Jogen Dutta Bayan, Anita Sarma, Sarodi Saikia, Haricharan Bhuyan Borbayan, Ramkrishna Talukdar, Ranjumoni Saikia, etc.

Chhau Dance

Chhau Dance is a folk dance of West Bengal, Jharkhand and Odisha with martial traditions. It is found in three styles- Purulia Chhau (performed during sun festival in West Bengal), Seraikella Chhau (Jharkhand), Mayurbhanj Chhau (Odisha). The Ministry of Culture recognised it as 9th classical dance of India.

The fundamental vocabulary of Chhau dance constitutes mock combat techniques, stylised gaits of birds and animals and movements modelled on the chores of village housewives and mostly performed by male dancers in eastern India.

Dancers enacted stories from the Hindu epics the Ramayana and the Mahabharata, the Puranas and other Indian literature. It is performed with religious themes found in Shaivism, Shaktism and Vaishnavism.

It was inscribed in the UNESCO's Representative List of the Intangible Cultural Heritage of Humanity in 2010.

Famous Dancers - Jagannath Choudhury, Mousumi Choudhury, Biren Kalindi, Binadhar Kumar, Kartik Singh Mura, Baghambar Singh Mura, Upendra Biswal, Banamali Das, Rajendra Pattanayak, Gopal Prasad Dubey etc.

Folk dances

Andhra Pradesh

Dhimsa is a tribal dance performed primarily by Porja caste women in Araku valley, Andhra Pradesh.

Veeranatyam (dance of bravery) is a folk dance of Andhra Pradesh, performed to

honor Lord Shiva. It starts with dancers appearing on stage with a plate called "**Veerabhadra Pallem**" carrying camphor fire.

Kolannalu (stick dance) folk dance is also known as kolkolannalu and commonly known as Kolattam in Telugu (the official language of the state) amongst the people in Andhra Pradesh.

Dappu Nrityam is one of the popular dance forms in Andhra Pradesh and Telangana. This dance form earned its name from the sound produced by a simple percussion instrument, a tambourine-like drum which is called 'Dappu'.

Bhamakalpam is both a dance and a drama. Siddhendra Yogi created this devotional dance form in the 17th century.

Butta Bommalu is popular in the Tanuku region of West Godavari District in Andhra Pradesh.

Tappeta Gullu is a dance that has vigor, rhythm and tempo and is performed to invoke the Rain God.

Bonalu is a special dance wherein the female dancers step to the rhythmic beats and balance pots on their heads.

Vilasini Natyam is the dance form of Devadasis of Telugu hence it is also called Devadasi dance.

Lambadi dance is a folk dance of the Banjara community originated in Andhra Pradesh. It is done to impress gods for a good harvest.

Kalapam is a dance-drama with a moral for the audience. It is a mono-play in which there is a main person and another comparatively less significant person.

Gobbi dance is popular in coastal regions of Andhra Pradesh. It is performed during Sankranti festival when houses are cleaned and court yards decorated with 'rangavalli'.

Dandaria dance is a stick dance which is performed by the Gond tribe in the Northern region of Hyderabad district.

Arunachal Pradesh

Ponung is the harvest dance performed by the Adi tribal community of Arunachal Pradesh.

Aji Lhamo is a folk dance form practiced

by the Monpa Tribe of Tawang district of Arunachal Pradesh. This dance form is performed during the Losar Festival.

Pasi Kongki is a folk dance of Adi tribe of Arunachal Pradesh that depicts the social work of Pasi. It is performed to the tune of a song sung by the locals, called Aabang.

Tapu dance is a war dance performed by the Adi tribe of Arunachal Pradesh. It takes place during the festival of Unying Aran. It is performed in order to ward off evil spirits from the village.

Wancho dance is performed by the Wancho tribe of Arunachal Pradesh. The Wancho tribe's most important festival is known as Oriah.

Buiya is a folk dance of Arunachal Pradesh performed by the Digaru Mishmi tribe. This dance is performed in multiple festivals like Tazampu, Duiya, and Tanuya.

Rikhampada is a dance of Nishi tribe of lower Subansiri District of Arunachal Pradesh.

Popir dance, performed by the Galo tribe of Arunachal Pradesh during Mopi festival.

Shapawng Yawng Manau Poi is an annual festival of the Singpho tribe of Arunachal Pradesh.

Sangphao tribal dance is performed in Arunachal Pradesh and Nagaland.

Assam

Bagurumba is a folk dance of the Bodo community in Assam. It is also called "butterfly dance".

Bhaona is a theatre form. The plays of bhaona are popularly known as Ankiya Nats and their staging is known as bhaona. It is a creation of Mahapurusha Srimanta Sankardeva, written in the early sixteenth century.

Bihu (Rongali bihu, Bohang bihu) dance is an indigenous folk dance from the Indian state of Assam related to the Bihu festival and an important part of Assamese culture.

Deodhani dance, of Boro-kachari origin, is a folk dance of Assam. It is a solo or group performance (comprises 3 or 4 women). It is associated with the worship of the snake goddess Manasa or Marei/Maroi.

Jhumur is a traditional dance of the tea tribal communities (kulis) of Assam and also in some parts of West Bengal. It is usually performed in the harvest season and festivals.

Bhortal dance is performed in a group of six or seven dancers. It is performed to a very fast beat, known as 'Zhiya Nom'.

Bihar

Bidesiya in Bhojpuri, meaning 'from the foreign land', is a folk theatrical performance of western Bihar. It is based on a drama named Bidesiya written by Bhikhari Thakur.

Chaiti folk dance is performed by men with their bodies smeared with 'ramras'.

Jat-Jatin is a folk dance of Bihar, most famous in Mithila and Koshi regions. It is performed on moonlit nights during the monsoon season.

Jhijhiya is a cultural folk dance of Mithila and Bhojpura regions of Bihar and Madhesh province of Nepal. It is usually performed by a group of young women dancers. Portrays offering of prayers to Lord Indra for a good monsoon and a rich harvest. It is performed during the Dussehra festival from the day of Ghatasthapana to Bijaya Dashami in the Hindu month of Ashwin (September /October).

The whole Navaratri nine nights festival is celebrated to worship the three forms of lord Lakshmi, Parvati, and Saraswati. It has its own types of songs either in Maithili or Bhojpuri language. Women balance the lantern made up of clay on their head and dance.

Jadur is the dance form of the oraons tribe of Bihar. It is mainly performed during the occasion of Sarhul festival.

Biraha dance, a folk dance of Bihar depicts the anguish of women whose partners are gone from home. This dance is also performed in Jharkhand and Uttar Pradesh. Bidesiya is a derivative of Biraha dance.

Kajari dance welcomes the season of Monsoon and is usually performed in the months of Shravan and Bhadrapad.

Bhojpuri Jhumar dance is popular in Magadh region. It is performed to welcome the spring season.

Magahi Jhumar dance is usually presented in the form of a duet.

Jharni dance is a ritualistic dance performed by the Julaha community during Muharram.

Sohar Khelwana is a dance performed by women to celebrate the birth of a child.

Kishan Nritya is an expression of pride of the farmers of Bihar.

Natua dance begins with an item called the Natua Kachal.

Chhattisgarh

Suwa dance (Parrot dance) is a tribal dance, mainly performed by the Gonds tribe of Chhattisgarh. It is performed exclusively by women on the occasion of Goura's marriage. Dancers keep a parrot in a pot made of bamboo and form a circular shape around it.

Raut Nacha is a ceremonial dance performed mainly by the tribal community of Chhattisgarh. It is performed during the "dev udhni ekadashi", after the Diwali festival. It is a dance performed by Yadavas, a caste which are descendants of Krishna.

Karma dance or Karma Naach is a traditional dance of Gond tribals of central and Eastern India annually performed during the karma festival (autumnal festival). It is performed in the state of Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha and West Bengal. This folk dance is performed during the worship of the god of fate which is known as Karam Devta.

Thapati dance is a tribal dance performed by the Korku tribes of Chhattisgarh, Madhya Pradesh, Maharashtra. It is performed in the month of Vaishakh. The main instruments of this dance are Dholak and flute.

Saila dance is performed only by boys after the harvest season in the Hindu month of Aghan (November- December).

Panthi dance is a prominent ritual of the Satnami community of Chhattisgarh. It is mainly danced to depict the importance of Durg district to the people. Performed on Maghi Purnima (the birth anniversary of Guru Baba Ghasidas).

Gaur Maria dance is performed in the plateau of Bastar in Chhattisgarh. It is a group dance and both men and women participate enthusiastically in this dance. It is performed on the occasion of marriage.

Pandavani dance ballad portrays the account of Pandavas. Recital of tales of Mahabharata is one of its prominent characteristics. The two main styles of the narrations are Vedamati and Kapalik.

Jhirliti dance is performed in a Halloween like ritual. It is played by the kids in the Bastar region of central India.

Gendi dance- Dancers are mounted on two long bamboo. This is a special dance of the Muria tribe of Bastar. It is completely a dance of balance.

Rahas is a modern folk dance of Chhattisgarh and is mainly performed in Dhamtari district. The theme focuses on the immortal love story of Lord Krishna and Radha.

Chaitra festival dance is a famous dance of the Gonds of Bastar district. It is performed after the harvest to thank goddess Annapurna for the harvested crop.

Goa

Dashavatar is the most developed theatre form of the Konkani and Goa regions. The performers personify the ten incarnations of Lord Vishnu-the god of preservation and creativity. The ten incarnations are Matsya (fish), Kurma (tortoise), Varaha (boar), Narsimha (lion-man), Vaman (dwarf), Parashuram, Rama, Krishna (or Balram), Buddha and Kalki.

Ghode (horse) Modni (dance-like movements) is a warrior dance commemorating the victory of the Ranes, the Maratha rulers of the Sattari taluka in Goa over the Portuguese.

Fugdi dance is mainly the dance of Konkani women. It is performed during festivals like Ganesh Chaturthi, Dhalo festival and Vrata of Goddess Mahalakshmi.

Kunbi dance is a tribal folk dance of the Kunbi community of Goa. Women holding lamps on their heads perform the lamp dance during the Shigmo festival.

Dekhni is a semi-classical dance form of Goa. In Konkani, it means bewitching beauty. It is performed by a group of girls (representing Devadasi) carrying earthen oil lamps in their hands.

Ranmale is a ritualistic and folk theatre form based on mythological stories from the popular Indian epics, the Ramayana,

and the Mahabharata. It is performed during the Holi festival which is celebrated as Shigmo (spring festival) in Goa and the Konkani region. It is also practiced in some parts of Maharashtra and Karnataka.

Zemmado is a folk dance of Goa, in which women folk belonging to the Dhangar community imitate the actions of goats.

Tarangamel is the folk dance performed during Dussehra and Holi in the state of Goa.

Corridinho is a form of Portuguese dance performed in Goa. It is always a couple dance.

Lamp dance is usually performed during the Shigmo Festival (Goa).

Shigmo dance - The objective of performing the dance is to celebrate the new crops in the spring season.

Romat dance is a Goan folk dance and procession that is performed in the month of March (at the Singma festival).

Musal is a pounding stick and the dance is performed during harvest time by the people of Chandor, the ancient capital of Goa.

Dhalo dance is associated with the fertility of the earth and is a women-only dance.

Goff is a celebration of a bountiful harvest. The songs accompanying the dance are dedicated to Hindu God Lord Krishna.

Romta Mel is a way for Goans to express their thanks to their Gods, accompanied with a serpentine procession of people marching their way to a temple.

Gujarat

Garba (Sanskrit term) dance is performed by women around a centrally lit lamp or a picture or statue of the Goddess Shakti. It is performed during the nine-day festival Navratri. It is also performed at the Holi spring festival.

Dandiya Raas is the socio-religious featured dance of Navratri evenings in Gujarat. It is also performed in the Marwar region of Rajasthan. It is believed to be performed in memory of Goddess Durga's victory over Mahishasura.

Siddi Dhamal dance form is performed

by the men of the Siddi communities in Jafrabad and Jambur and came with them from East Africa.

Rathwa ni gher dance is a tribal dance performed by the rathwa tribe of Gujarat on the occasion of Holi festival also known as Kavant festival, named after the place where the Holi carnival takes place. The Gher performances begin on **Dhulendi** ('the day of flying coloured dust').

Vinchhudo is a folk dance of Gujarat. This dance shows strong belief in superstition.

Gheria dance is performed to worship 'Mataji' (Goddess Amba) during Diwali festival. The tribal people of Gujarat perform the 'Gheria' folk dance dressed in brightly colored clothes, garlands of marigold flowers and traditional ornaments.

Hudo dance form is mostly performed by the Bharvad or shepherd community, especially at the Tarnetar Fair in Surendranagar, Gujarat.

Hallisaka, a group dance, The origin of the Hallisaka dance is traced to the Harivamsa Purana.

Tippani dance form came into existence in Chorwad region of Saurashtra in Gujarat. This dance form is performed in festivals and marriages.

Padhar dance is introduced by the people of the Padhar community (fishermen who live along the banks of Nal Sarovar in the Bhal region).

Dangi Nritya is a tribal dance form native to the Dangs District of Gujarat.

Matukadi dance form is mostly performed by the Rabari and Bharvad communities.

Haryana

Phag is performed in the Hindu month of Phalgun (February–March) to celebrate the colorful festival of Holi in the harvesting season.

Jhumar is a folk dance specially performed by young married women of Haryana. In some parts of the state, it is also known as the 'Hariyanvi Gidda'. Types of Jhumar- Satluj Jhumar, Beas Jhumar, Chenab Jhumar, Multani Jhumar, and Jhoomar Taari.

Loor dance is performed around the Holi festival and is very popular in the Bangar and Bagar parts of the region during the month of 'Phalgun' (February/March).

Ratvai dance is a folk dance of Mewati tribes of Mewat region. Mewati (Meo) is an ethnic group from the Mewat region of north-western India which includes the Nuh district in Haryana and parts of adjacent Alwar district and Bharatpur district in Rajasthan.

Dhamal dance is famous in the Gurgaon area, which is inhabited by Ahirs. The origin of the dance dates back during the time of the Mahabharata.

Chaupaiyan, which is a devotional dance and is performed by men and women carrying 'manjiras'.

Deepak dance, men and women carrying earthen lamps, express their devotion through dance, which often lasts the whole night.

Been-bansuri dance - During this dance, the dancers use lute and flute to produce music. This dance is famous in the Bangar region of the state.

Ghoora dance is organized on the occasion of Birth of a baby boy.

Chhathi dance is performed on the birth of a male child. Women perform this dance on the sixth day of the birth of the child.

Khoria dance is a collective form of the variety of the Jhumar dance style & steps, exclusively performed by women.

Gugga dance is performed exclusively by men. It is performed in the procession taken out in the memory of saint Gugga.

Himachal Pradesh

Nati dance is traditionally performed in the states of Himachal Pradesh and Uttarakhand. Several varieties of Nati performed are Kullvi Nati, Mahasuvi Nati, Sirmauri Nati, Kinnauri Nati, Jaunpuri Nati, Seraji Nati, Karsogi Nati, Chuhari Nati, Barada Nati, Bangani Nati.

Dhaman is a folk dance of Himachal Pradesh.

Thoda is a Warrior dance performed by the Rajputs in Himachal Pradesh. It is performed during the Vishu Fair in the month of Chaitra and Vaisakh (April-May).

Dangi Dance or Ghurehi dance is performed exclusively by women of Chamba. During the performance one group asks questions while the other replies.

Cholamba dance is performed in Ropa Valley. This dance is performed while coiling a snake.

Dhure dance is also very popular in Lahaul, dances are performed on epics like Ramayana and Mahabharata.

Shan and Shabu are popular dances of Lahaul Valley. These dances are performed in the Gompa in the memory of Buddha.

Laaldi is a popular female folk dance of Himachal Pradesh.

Swaangtegi dance are performed by wearing lion and garden wooden masks on Diwali.

Namgen dance is performed in Himachal Pradesh in the month of September to celebrate the autumn.

Jharkhand

Paika dance is usually performed by Munda tribe to welcome special honoured guests or during religious processions (shobha yatra). This dance form represents the great war of the Munda community against the British. It is also the folk dance of Odisha which is practiced by the Paikas of Oriya army.

Jenana (Janani) Jhumur is a traditional dance of the women performed mainly during the period of cultivation in the rainy season.

Jhika-Dashain is a form of worship dance practiced by the Santhal tribe to train young men of the community in the art of acquiring spiritual powers to dispel the influence of evil spirits.

Santhal dance is a popular folk dance practiced by Santhal tribes in Jharkhand and West Bengal. It is performed in groups. It is very similar to the bamboo folk dance of Assam and Mizoram.

Borao dance is a celebration of the Oraon community thriving in the state of Jharkhand. It is one of the largest groups dwelling in the hilly region of Hazaribagh Goomla. The Oraon community is also known as Kurukh.

Mundari dance is a famous folk dance of the Munda community of Jharkhand and

Orissa. It is a women centric dance.

Phagua is a dance form popular among the tribes residing in Jharkhand and the adjoining state Bihar. It is performed during Holi – the spring festival.

Birhor dance is a tribal folk dance from Jharkhand. The Birhors are a tribal /Adivasi forest people, traditionally nomadic living in states of Jharkhand, Orissa, Chhattisgarh and West Bengal.

Kadsa dance is a dance style performed carrying a 'Kalasha' (earthen pot). This is a female dominated dance.

Mardani Jhumur is performed by the men of the Nagpuri community and Southern cultures dance after the harvest.

Jhitka and Danga dances are performed by both men and women to celebrate different feudal traditions.

Lahasuya is a Nagpuri folk dance of the Chota Nagpur Plateau region of Jharkhand.

Ghoda Naach dance performed only by men. The wooden puppet is the main prop used.

Sarhul is a traditional dance which is performed by the Oraon tribe in Jharkhand, Chhattisgarh and West Bengal.

Karnataka

Huttari dance (Kodagu dance) is one of the most spirited dances of Karnataka. Different forms of this dance are Bolak-aat, Ummatt-aat, and Komb - aat.

Bolak-aat dance is performed by Kodava men dressed in all black with an oil lamp in the open field.

'Ummatt-aat' is a folk dance form performed in Coorg. This dance is performed by the Kodava women.

Komb-aat, is a devotional dance which is carried out in a temple by the Kodava men. They carry horns of a deer that indicate the horns of the "Krishna murga" (a deer of a Kodava epic) while playing tunes on wind instruments and drums.

Yakshagana is a traditional folk dance form popular in coastal Karnataka districts and some parts of Kerala.

Veeragase / Veerabhadrana kuniitha is one of the dances demonstrated in the Dasara procession held in Mysore, during the hindu months of Shravana (July -

August) and Karthika (October - November).

Kamsale (Beesu Kamsale) is a unique folk art performed by the devotees of God Mahadeshwara. Kamsale also refers to a brass made musical instrument.

Bhootada Kola dance is highly stylized and held in honor of the local deities worshiped by the Tulu speaking population. It has influenced Yakshagana folk theatre.

Bayalata is an open air theatre form. It features stories from Indian epic poetry and the Puranas rendered as dance and drama.

Sadal Kekai is a folk dance of Nagaland performed by the Kuki tribe.

Goravara Kunitha is often a tradition of a dance of Kuruba Gowda's who are the worshippers of Lord Mailana Longa. Bhootha Aradhane represents the worship of the Devil.

Gaarudi Gombe basically means magical puppet in Karnataka. It is especially performed during festivals and processions to ward off evils.

Pata Kunitha is a group dance performed by 10 to 15 persons together.

Dollu Kunitha dance performed primarily by men of the shepherd community called the Kuruba.

Naga mandala dance is usually performed by male dancers, called Vaidyas. Within the dance, they dress up as Nagakannikas or as female snakes.

Suggi is a dance performed at the time of harvesting by the Halakki Vokkaliga tribe.

Aati Kalanja is a ritualistic folk dance performed by the 'Nalke' Community.

Kerala

Velakali is a traditional martial dance performed by the men of the Nair community of Kerala.

Mangalamkali is a dance ritual followed by the Mavilan tribes of North Kerala.

Koodiyattam (Kutiyattam) is a traditional performing art form in the state of Kerala. It is traditionally performed in temple theatres known as Koothambalam. It was officially recognised by UNESCO as a Masterpiece

of the Oral and Intangible Heritage of Humanity.

Margomkali is a ritual folk art dance of the Syrian Christians of Kottayam and Thrissur districts in Kerala.

Mudiyettu, a dance drama is a village ritual performed by members of the Marar and Kuruppu communities in Thrissur, Ernakulam, Kottayam and Idukki districts of Kerala. In 2010 it was inscribed in the UNESCO's Representative List of the Intangible Cultural Heritage of Humanity.

Thullal is a solo satiric dance belonging to the state of Kerala. There are three versions of this art form - Sheethankan Thullal, Parayan Thullal and Ottan Thullal.

Theyyam (Kaliyattom) is a ritualistic folk dance originated in North Kerala. The ritual used to be performed at the sacred groves and shrines. The musical instruments used in this magnanimous art form are Chenda, Kurumkuzhal, Elathalam, and Veek kuchenda. Varieties of Theyyams are Raktha Chamundi, Muchilottu, Kari Chamundi, Bhagavathi, Gulikan, Wayanadu Kulaven, and Pottan.

Pulikali (Leopard Dance) is a recreational folk art performed at the annual harvest festival Onam.

Elelakkaradi is a well-known dance form of the tribal group called Irulas of Attappady in Palakkad District.

Kalaripayattu (Kalari) is an Indian martial art that originated in modern-day Kerala. It is mentioned in the Vadakkan Pattukal, a collection of ballads written about the Chekavar of the Malabar region of Kerala.

Kummattikali (Kummatti) is a colourful mask dance prevalent in Kerala and parts of South Malabar. It is performed during the Onam festival.

Padayani literally means "row of warriors" is a dance form belonging to Kerala. Performed in honour of "Bhadra Kali" (i.e. an avatar of Lord Shiva).

Thiruvathirakali (Kaikottikali) is performed on the auspicious day of Thiruvathira, the birthday of Lord Shiva. It falls in the Malayalam month of Dhanu (December - January).

Chakyar Koothu is a solo performance in which the performer dresses up as a snake.

Chavittu Natakam, is believed that the art form flourished at Kodungalloor in Kerala with the spread of Christianity.

Chavittukali is a folk dance performed by Ullada, Oorali, Cherumar and Kanakkaar communities.

Kakkarissi Natakam is a folk art form popular in Central Travancore.

Kanyarkali is a folk dance ritual performed in Bhagavati temples and Vettakorumakan Temple of Palakkad district.

Arjuna Nritham (Mayilpeeli Thookam), is a ritual art of Kerala performed in the Bhagavati temples of south Kerala, mainly at Kollam, Alappuzha and Kottayam.

Ezhamathukali, a folk art form closely related with the ancient Sanghakali, is also known as Ezhamadukali, Ezhamuthipurappad etc.

Alamikali was a popular regional art form in Kasaragod, performed by 'Alamis', soldiers of Tipu's army.

Vedanpaattu is a traditional folk art form performed in the Malayalam month of Karkidakam (June/July) in remote parts of Kerala.

Onathaar is a traditional folk art form performed during Onam season in Kannur and adjoining areas.

Ivarkali (Aivarnatakam, Thattinmelkali, Kannilkuthikali, Pandavarkali) is played by Ikkudikammalar people belonging to the Vishwakarma community.

Kathaprasangam or the art of storytelling performance is a popular art form which blends the tradition of music and speech together.

Oppana is a song and dance performance popular among the Muslims of Malabar.

Madhya Pradesh

Maach is a form of folk theatre from the Malwa region of Madhya Pradesh. It is performed around the Indian festival of Holi.

Ahirai is a popular dance form of the Bharia tribe.

Bhadam (Bhangam dance) is mainly performed by the Bharia tribe of Madhya Pradesh on the occasion of marriage.

Sahariya dance is a famous dance performed by the Saharia community of Madhya Pradesh and Rajasthan. It is performed during the month of Holi to the beats of the dhol, nagari and matki.

Bhagoriya is a well known dance of the Bhils, a large tribe of the Jhabua district in Madhya Pradesh.

Grida dance is performed to celebrate the "harvesting of crops". It has three phases- Sela, Selalarki and Selabhadoni.

Matki dance is mostly performed in the Malwa region by nomadic tribes.

Phulpati dance is performed in Malwa region of India (western Madhya Pradesh and south eastern Rajasthan) by unmarried girls.

Jawara dance is a popular folk dance of Madhya Pradesh. It is mainly famous in the Bundelkhand region among the peasant community.

Lehangi is a popular folk dance of the Banjara and Kanjar tribe of the Bhopal commissary of Madhya Pradesh and is performed during the blossoming monsoon period.

Baredi is an important folk dance of the Gwalior district. Starting from Diwali, the dance is performed till the day of 'Karthik Purnima'.

Bilma dance is performed by the Gond and Baiga tribes during the festival of Dussehra.

Chatkora dance is done by the Korku tribe. It is in the Chinndwara and Betul districts.

Kangra dance is performed by the Dhobi caste in Bundelkhand.

Gocho dance is performed by the Gond tribals.

Reena dance is performed by the Baiga and Gond tribal women during the festival of Diwali.

Atari dance is the dance of the Bhumiya and Baiga tribes of Baghelkhand region.

Muria dance is performed near the Ghotul.

Maharashtra

Lavani dance is generally performed by Dhangars or shepherds living in the Solapur district of Maharashtra.

Lezim is a folk dance where the dancers carry a small musical instrument with jingling cymbals called the 'Lezim'. The first Lezim performance to an international audience was presented at the 9th Asian Games in the year 1982 at Delhi.

Dhangari Gaja dance (Dhangar dance) is performed by the shepherd community known as Dhangars during the Navratri festival.

Tamasha dance form has been believed to be derived from the ancient form of Sanskrit drama - the 'Prahana' and the 'Bhana'. Kolhati and Mahar groups are associated with the performance of Tamasha.

Waghya Murali dance originated in Maharashtra, associated with Lord Khandoba.

'Povadas' folk dance is presented in the Marathi ballad form. This dance form describes the events in the life of the founder of the Maratha Empire, Shri Chhatrapati Shivaji Maharaj.

Koli dance is a popular folk dance of the Maharashtra and Goa states of India. It was created by the Kolis tribe of Mumbai.

Dindi dance is performed on Ekadashi in Kartik month of the Hindu calendar.

Nakata dance is performed by the fisher community of Maharashtra.

Gafa dance is performed by a group of boys.

Kala dance is a form of folk dance that describes the mood of Lord Krishna. It is a symbol of fertility.

Songi Mukhawate is derived from the two lion masks worn by two dancers who represent Narasimha, an aspect of Lord Vishnu. The dance celebrates the victory of truth over falsehood.

Manipur

Shim Lam dance (Fly dance) is the traditional folk dance of the Rongmei community.

Luivat Pheizak is one of the most popular dances of the Tangkhul Naga community of Manipur.

Nupa dance is a variation of Pung Cholom. It is also known as Cymbal dance or Kartal Cholom. It is accompanied by an ancient classical

drum called "Pung" in Manipuri. It is performed by only male dancers.

Rakhal Nritta is a part of Raas Leela. Manipuri boys are performing a dance at Under the Kadam tree.

Katabenlu Laam (Bangle Dance) is known for its intricate footwork and rhythmic movements.

Mao Naga dance is performed by the Mao Naga community during the annual harvesting and seed-sowing festivals (Chikhuni).

Pao-Sa Jagoi dance is part of the traditional Kanglei Haraoba.

Kabui dance is performed by the Kabui community during the Gang - Ngai festival.

Raas is a highly developed dance drama depicting Lord Krishna's union with his female followers, the gopis, and especially his consort-devotee Radha.

Thabal chongba is traditionally performed during the festival of Yaoshang.

Lai Haraoba dance enacts the 'Creation of the Universe', which was initially a part of the Lai Haraoba festival.

Thang-Ta (martial art) symbolises an old and outstanding Manipur culture. It demonstrates the remarkable combat style of the Thang (sword) and the Ta (spear).

Lhou Sha is a battle dance that is performed at every clash between two villages.

Meghalaya

Laho dance is performed by the Jaintia tribe of Meghalaya. It is performed during the Behdienkhlam festival. It is also known as the Chipiah Dance. When the Harp tribe of Meghalaya performs this dance, it is known as Wangla dance.

Nongkrem dance festival is celebrated during Autumn at Smit, the cultural centre of the Khasi Hills. A five day long religious festival of the Khasis, Ka Pomblang Nongkrem dance is popularly known as Nongkrem dance.

Behdienkhlam is the main dance of the "Jaintias" festival held every July in Jowai, Jaintia Hills.

Chad Sukra, a popular traditional dance

festival of Meghalaya, is celebrated as a Sowing festival. It is celebrated every year by the Pnar people of Jaintia Hills.

Shad Suk Mynsiem is an annual spring dance that celebrates the harvest and planting season.

Dorsekata dance festival is also a dance in which the women try to take the turbans off their male partners during the dance.

Mizoram

Cheraw folk dance performed with bamboo sticks. On the occasion of 'Buza Aih,' or bumper harvest, an individual family performs Cheraw dance.

Khuallam dance is usually performed during the ceremony of 'Khuangchawi'. Literally, the terms 'Khual' and 'lam' mean guest and dance respectively. Khuallam literally means 'Dance of the Guests'.

Chailam dance is mainly practiced during the festival of Chapchar kut and is regarded as one of the most significant festivals of Mizo people.

Solakia dance is prevalent in Mizoram. 'So' means the severed head of an enemy, 'La' means dance and 'Kia' refers to that which has been done with better understanding and knowledge.

Chawnglaizawn is a popular folk dance of one of the Mizo communities known as Pawi.

Tianglam is performed throughout the state using music of Puma Zai.

Zangtalam dance is mainly performed by the 'Paihte' tribe.

Nagaland

Zeliang dance is performed by the Zeliangrong Naga tribe found in the hilly terrain of Mount Barail in Nagaland.

Changai dance is performed by the Chang tribe during the Naknyulum festival, which lasts for three days.

Chang Lo dance (Sua la) is performed by the Chang tribe. It is performed in a three day festival known to be 'Poanglem', marking the beginning of the harvest season in the state.

Aaluyattu is performed by the Konyak tribe.

Udoho dance is a traditional war dance of the Angami Naga tribe.

Rangma folk dance performed by the Naga people. It is usually a part of the Ngada festival. It depicts war culture and men dressed up like warriors.

Leshalaptu is a folk dance performed by the women of Nagaland.

Shankai and Moyashai are the victory dances performed by the Lotha tribe of Nagaland.

Rukhyo Sharu dance is performed by the Lotha Naga tribe.

Langnyu-Khiamtsangshe dance is a traditional folk dance performed by the Khiamniungan Naga tribe of Nagaland during two of their most important festivals i.e. Miu and Tsokum, which are co-related to each other.

Akok-Khi is a dance form performed by the Sangtam tribe during the Mongmong festival.

Changsang dance is a traditional folk dance performed by the Chang Naga Tribe during Naknyulum festival in the month of July.

Monyu Asho dance is performed by the Phom Nagas during the Phom Monyu festival which is the biggest festival of the Phom Naga tribe. It is celebrated in the month of April.

Khupielilie dance is a traditional folk dance performed by womenfolk of the Pochury Naga Tribe of Nagaland.

Kukui Pheto dance is a traditional folk dance performed by Chakesang Nagas.

Kulu-Tsen is the name of a traditional Naga folk dance of the Yimkhiong tribe.

Nokinteker Tsungsang dance is a Naga traditional dance form of the Ao Naga tribe.

Butterfly dance is associated with the Zeliangrong tribe.

Melo Phita dance is performed by the Angami Nagas during the Sekrenyi festival in the month of February. Angushu Kighilhe dance is a traditional war dance performed by the menfolk of the Sumi tribe.

Odisha

Gotipua (Bandha Nritya) is the precursor of Odissi classical dance. In this dance young boys dress as women to praise lord Jagannath and Krishna.

Dalkhai is a popular folk dance of the

Adivasis of Odisha. It originates in the Sambalpur district in Odisha. It is performed in various festivals such as Bhaijiuntia, Phagun Puni and Nuakhai.

Danda Nata (Danda Jatra) is one of the most important traditional dance festivals organized in different parts of South Odisha and particularly in the Ganjam District (ancient Kalinga Empire). The Danda Naata festival is held in the month of Chaitra (March/April) every year.

Dhemsā is a traditional folk dance of tribal people of central India-Southern Odisha and adjacent areas of Chhattisgarh and Andhra Pradesh. Adivasis or Desia people perform this dance form at late nights during occasions such as Pus Puni or Pus Parab and ChaitParab.

Rangabati dance is popular in Odisha and most parts of West Bengal, Bihar, Jharkhand, Andhra Pradesh, and Chhattisgarh.

Ranapa is performed on the stilt and accompanies by drum music, along with songs related to Lord Krishna childhood stories.

Sakhi Kandhei is a string puppetry show popular in the Indian state of Odisha, especially in the Kendrapara district of Odisha.

Baagha Naacha or Tiger Dance is performed in Binka, Sonapur of Subarnapur district and Brahmapur and in some parts of Ganjam district in Odisha.

Chaiti Ghoda dance is mainly performed by Kaibarta Caste during the festivities related to the fishing communities of Odisha.

Medha Nacha is a type of mask dance performed mainly in the coastal districts of Odisha.

Changu dance is performed by almost all the regional tribes of Odisha.

Sambalpuri folk dance is another form of Dalkhai Dance.

Kela Keluni- The Kelas are a wandering group of people who make their living by catching snakes.

Ghumra is a folk dance of the Kalahandi district of Odisha.

Dhanu Jatra (Dhanu Yatra) is an annual

drama - based open-air theatrical performance celebrated in Bargarh, Odisha.

Daskathia dance is the tribal dance performed by two males depicting the historic and Puranic events.

Punjab

Bagga is the martial dance form of Punjab. It is also known as Bhangra. It is performed during the "harvest festival" i.e. Vaishakhi.

Kikli is performed by two girls holding hands and twirling each other in a circle and balancing their positions in circular motions.

Giddha is a folk dance of women in Punjab, known for being high-spirited. It is a celebration of the reaping and harvest period in the spring season.

Viyahula Giddha is a popular folk dance performed during marriages in the Indian state of Punjab and also in Pakistan.

Malwai Giddha is the folk dance of males of Malwa region of Punjab.

Gatka (Dankara dance) is a form of martial art originated in Punjab. It is associated with the Nihang Sikhs of the Punjab and other related ethnic groups, such as Hindkowans.

Julli dance is done by Muslim holy men Juli, called pirs.

Jaago is a beautiful Punjabi cultural celebration by the maternal families of both the Bride and the Groom.

Luddi dance, movement of the body is sinuous and like a snake.

Rajasthan

Teratali/Terataal folk dance is performed by women of Kamar (Kamada) tribes of Rajasthan and Madhya Pradesh.

Ghoomar is a dance performed by girls of border areas of Haryana and Rajasthan at various festivals like Holi, Gangaur Puja and Teej. It was basically developed by the Bhil tribe to worship Goddess Saraswati and was adopted by other Rajasthani communities.

Chari dance is a female group dance. It is prominent in the Gujjar and Saini community of Kishangarh and Ajmer. In the chari dance, women hold chari or pots on their heads, and a lighted lamp is then placed in the pot.

Kalbelia dance is also known by other names like 'Sapera Dance' or 'Snake Charmer Dance'. It was added to UNESCO's representative list of the Intangible Cultural Heritage of Humanity in 2010.

Bhavai is a folk dance in which women dancers balance up to seven or nine brass pots on their heads. Jat, Bhil, Raigar, Meena, Kumhar, and Kalbelia communities are associated with this dance form.

Kachchhi Ghodi dance is an Indian folk dance that originated in the Shekhawati region of Rajasthan. It is performed by men on dummy horses.

Gair Dance is one of the popular folk dances practiced by the Bhil community in Rajasthan.

Chakri is a folk dance of the Kanjar tribe. It is performed exclusively by womenfolk in marriages and festivals. Kanjar tribe inhabited some parts of the Kota and Baran districts of Rajasthan.

Kathputli is a string puppet theatre, native to Rajasthan and is the most popular form of Indian puppetry.

Khayal dance is performed by the Bhawai tribe of Rajasthan. It is full of mockery, humour and irony.

Jhulan folk dance is performed by both men and women.

Fire dance mostly performed by the Banjara tribe who belong to the Churu and Bikaner districts of Rajasthan.

Drum dance has its origin from the Jhalore regions. Gavari is a tribal dance drama performed by the Bhil tribe.

Walar dance is performed by the Garasia Rajasthani tribe.

Sikkim

Bhutia folk dance "Tashi Sabdo" This age-old group dance beautifully and gracefully shows the customs of offering khadas (Scarfs) on auspicious occasions.

Chu Faat Dance - The Lepcha tribe performs this dance during the Pang Lhabso festival. It is performed on the fifteenth day of the seventh month in the Buddhist Calendar. It is performed in honour of Mount Khangchendzonga.

Zo-Mal-Lok is the most popular folk dance of the Lepcha community.

Tendong Lho Rum Faat is celebrated by the Lepcha people in Sikkim. This group dance is performed to save people from the onslaught of mighty, mountainous rivers.

Denong-Neh-Nah is performed by the Bhutia community. This group dance is performed to pay homage to the past saints such as Guru Rimpoche.

Kayged dance is held every year in various Sikkimese Monastery. It is a time of celebration for the Bhutias as they welcome the New Year.

Naumati is a group dance of the Damai community.

Ta-Shi-Yang-Ku is performed by the Bhutia community during the consecration of a new house and to bless a newly married couple.

The Lama dances or Chham is a masked dance performed by Buddhist lamas (monks) during special occasions like the Pang Lhabso festival.

Tamil Nadu

Kolaattam ('Kol' means a small stick and 'Attam' means play) dance also known as stick dance is an ancient village folk art performed only by women in Tamil Nadu. It is also called Kolannalu in Andhra Pradesh. It is performed for ten days starting with the Amavasi or New-moon night after Deepavali.

Paampu attam (snake dance) is a folk dance of Tamil Nadu. It is done by young girls wearing tight-fitted costumes resembling a snake.

Karakattam is an ancient folk dance of Tamil Nadu performed in praise of the rain goddess Mariamman. The performers balance a pot on their head. Its two types are Aatta Karakam, which symbolizes joy and happiness and Sakthi Karakam, performed only in temples as a spiritual offering.

Karagam is performed by balancing a brass pot on the head. This dance is performed on the song called temmanguppau.

Therukoothu is a Tamil street theatre form practiced in Tamil Nadu state of India and Tamil-speaking regions of Sri Lanka.

Kavadi Aattam (Burden Dance) is a dance form performed by men. This

dance involves carrying Kavadi (burdens) through the pilgrimage. This dance form is performed in worship of Lord Murugan (Hindu God of war).

Kummi is a folk dance, popular in Tamil Nadu and Kerala in India, danced mostly by Tamil women in a circle. It is also danced by Tamils of Sri Lanka.

Bamber Dance is performed inside a temple to worship Lord Krishna. It is mainly performed around a lamp during Ram Navami and Gokulashtami.

Devaraattam dance was presented to the Tamil kings and their army after their victorious return from the battle.

Puliyattam (Puli Aattam or Tiger Dance) In this dance, bodies of the dancers are painted by yellow and black paint to create a replica of Tiger.

Shattam Dance is devoted to Lord Vishnu, performed in groups with the instrument named "Urumi".

Kazhai Kothu is similar to a modern-day circus, which includes gymnastics steps.

Mayil Aattam or Peacock Dance is usually performed by women. It is performed in the Hindu temples and offered to Lord Murugan.

Ottan Koothu dance form is presented by tribal people to depict ancient stories during festivals.

Poikkal Kuthirai Aattam or Artificial Horse Dance is a dance form in which dancers wear a dummy horse which is hollow in between so that a person can fit in it.

Koothu is a street party in which music and dance are performed by people during festivals and weddings.

Telangana

Gussadi is a traditional dance form performed by Dandari (Dancing) troupes during the festival of Diwali for a fortnight with participation from Kolam and Raj Gond tribes.

Mathuri dances are special tribal dances by the Mathuri tribes of the Adilabad district of Telangana, performed during the rainy month of Shravana.

Dappu (Tapetta and Palaka) Nrityam is a dance form in Telangana.

Lambadi is performed by the semi -

nomadic tribes called 'Lambadis' or 'Banjaras' or 'Sengalis'.

Perini Sivatandavam (Perini Thandavam) is a typical war dance which owes its origins to the 11th century rulers of Kakatiya dynasty. It is dedicated to Lord Shiva (Lord Rudra).

Mayuri is performed in the tribal areas of Khammam district.

Bathukamma dance is performed during Bathukamma festivities in Telangana and Andhra Pradesh.

Tripura

Wangala Dance is a dance of Tripura performed after the happy harvest 'Wangala' (1-st rice eating ceremony).

Garia dance is performed during Goria puja, the festival to celebrate the sowing of new crops and prayer for a good harvest during the month of Baisakh.

Hoza Giri dance is celebrated to pray for the blessings of Mainuma (Goddess of wealth) by the Bru Reang community of Tripura.

Maimata is a popular dance of the Kalo community.

Lebang Boomani dance is a harvest dance performed by the Tripuri people of Tripura.

Sangrai dance is performed by the Mog community people on the occasion of Sangrai festival falling in the month of Chaitra of Bengali calendar year.

Mamita dance is popular among the Kalo community. It is performed at the Mamita Festival, the harvest festival of the Tripuri people.

Hai Hak dance is connected with Jhum cultivation and performed by the Halam community.

Jhum dance displays the lifestyle, mode of cultivation, culture and tradition of the people.

Bizu dance is performed by the Chakma community. Bizu marks the end of the Bengali calendar.

Welcome dance is performed by Lusai girls whenever any visitor pays visits to their house.

Galamuchamo Dance is celebrated at the end of the harvesting season. It is

performed to express their gratitude to the gods for a good harvest.

Uttarakhand

Mukhota dance is usually performed to worship God Bhunitayal in Uttarakhand.

Bhotiya dance is a ritual of Uttarakhand conducted by the Bhotiya or the Bhot ethnic group.

Pandav Nritya/Pandava Lila dance is dedicated to the Pandavas which is based on the events of Mahabharata.

Hurka Baul dance gets its name from the 'Hurka' or the drum used as an accompaniment to the 'Baul' or song. It is performed during the cultivation of paddy and maize in the Kumaon region.

Chholiya (Hudkeli) is a folk dance form that originated in the Kumaon division in the state of Uttarakhand (India) and in some parts of Sudur Paschim province of Nepal. It is a sword dance.

Langvir Nritya is an acrobatic dance and is performed by the men folk only. It is mainly performed in the Tehri Garhwal region.

Chanchari is the dance style of Danpur region in Kumaon.

Chhapeli, also known as 'Chhabili' is a dance of Kumaon which is considered as the dance of two lovers.

Tandi, in this dance, all the people dance in a series holding each other's hands.

Chaunfula is the dance of joy and gaiety. It holds a special place in the minds of all the people of Uttarakhand.

Jhoda dance by its name is popular for couples.

Jhumelo folk dance is very famous in both Garhwal and Kumaon regions of Uttarakhand.

Uttar Pradesh

Mayur dance is performed by girls dressed as peacocks in the Braj region of Uttar Pradesh. It is carried out as a kind of adoration for Lord Krishna. Peacock dance in other states- Arunachal Pradesh (Monpa tribe), Andhra Pradesh and Tamil Nadu (known as Mayilattam dance).

Rasalila is a popular form of folk theater in the regions of Mathura, Vrindavan in Uttar Pradesh, Nathdwara in Rajasthan

(amongst various followers of pushtimarg or the vallabh sect). It is also seen in gaudiya vaishnavism in Nadia (West Bengal) which is also known for Raas Utsava.

Nautanki is a famous folk theatre of Uttar Pradesh.

Charkula dance is typically performed on the third day of the famous Hindu festival of Holi.

West Bengal

Dhanuchi naach is a devotional dance and tradition in West Bengal, performed during Durga puja.

Gaudiya Nritya is a classical dance form of Bengal as well as of other Eastern parts of the country like Assam, Odisha and Manipur. It has its origins in the Natyashastra. It was widely performed in the Vaishnav communities.

Gomira dance is a rural dance form mainly practiced in the Dinajpur district. It is usually performed by the villagers to please the gods to usher in 'good forces' and ward off 'evil forces'.

Alkap is a Bengali folk dance popular in the districts of Murshidabad, Malda and Birbhum in West Bengal and Chapai Nawabganj, Randajshahi in Bangladesh. It is also popular in Jharkhand. It includes dancing, singing, drama and recital.

Gambhira dance (masked dance) is performed during the festival of Chaitra Sankranti. The masks are made out of neem and fig trees by the local Sutradhar community.

Dhali dances portray the shieldmen in the troops. This dance form is widely popular in the aboriginal community of Bengal.

Jatra is a form of folk drama combining acting, songs, music, dance, characterised by stylised delivery and exaggerated gestures and orations.

Brita dance (Vrita Dance) is especially performed by people after their recovery from certain diseases, often smallpox.

Tusu dance is performed during the month of Pausa, found in the month of December and January.

Lathi dance is performed during the primary ten days of Muharram. It's the

dance to show off the powers of the body.

Baul dance forms part of the religious rites of bauls.

Marisia dance is a tribal dance of West Bengal performed by a group of men.

Dadra and Nagar Haveli and Daman and Diu

Tarpa dance is a tribal dance performed by the Warli, Kokna and Koli tribes of Dadra and Nagar Haveli.

Other famous folk dances of Daman and Diu are Mando Dance, Verdigao and Vira Dance, which are performed on special occasions.

Jammu and Kashmir

Dumhal is a folk dance performed by the Watal tribe.

Rauf Dance - It is primarily performed to celebrate the harvesting season of spring.

Bhand Pather is a form of theatre dance performed in Jammu and Kashmir.

Hafiza dance is widely performed by kashmiris at home weddings or when the wedding processions would be taken out on boats on the Jhelum River.

Hikat dance is usually performed to welcome spring in Kashmir. It is a celebration of the harvesting season.

Bacha Nagma is performed by Young boys, particularly during the harvest season.

Wuegi-Nachun is performed after the marriage ceremony, when the bride has left for her new home.

Kud is one of the art forms which is performed in order to highlight the aficionados of the princely states.

Ladakh

Cha-rtse (Pigeon dance) is a dance form of Ladakh.

Jabro dance is performed by the nomadic people of the Tibetan region.

Koshan dance of Ladakh is presented during horse riding and the person who is riding the horse is known as Landak.

Shondol dance is also known as the 'Royal Dance of Ladakh'. It is performed

by Takshoma or women dancers to praise the King of Ladakh on special occasions.

Spawo dance is an integral part of the Ladakhi culture and is highly appreciated as a folk dance form.

Shone dance is of the Mons and people perform this dance only in Ladakh's rural parts.

Drugpa-Rches dance form is presented by the Dards, people who belong from the Aryan origin. Dards are situated in the Gorkhan and Drass areas of the Ladakh region.

The Bagstanarches dance is performed for several weeks in Ladakhi wedding ceremonies.

Tukhstanmo is a seasonal dance form, which is performed in the Bakhamul and Zanskar area.

Surahi dance is also called as Chang Rches. In Surahi dance, performers carry the Chang's pot over their head and with that pot, they do the performance.

Other Folk Dances

Yak dance (Yak Chham or Tibetan Yak Dance) is performed in the Indian states Arunachal Pradesh, Sikkim, and Assam and in union territory Ladakh.

Mayilattam is performed during the harvest festival of Thai Pongal in the Indian states of Tamil Nadu and Kerala. It is performed in the Hindu temples of Tamil Nadu in reverence to Lord Subrahmanya. Performers wear costumes from head to toe like a peacock with a beak. It is performed in all Murugan (Lord Subrahmanya) temples as a tradition during festivals.

Maruni is a group dance of Sikkim, Darjeeling and Assam, Performed during weddings.

Chham dance is known for the elaborate mask, headgear and costume of the performers and is very popular in Tibetan settlement areas of Himachal Pradesh, Ladakh and Kinnaur.

Domkach is a traditional wedding dance that belongs to Jharkhand and Bihar. Popularly known as Nagpuri dance, Domkach is part of wedding festivities.

Khon is a dance drama genre from Thailand. Khon has been performed

since the Ayutthaya Kingdom. It is based on the tales of the epic **Ramakien** (Thai adaptation of Indian Hindu epic Ramayana).

Lava is a popular folk dance of Minicoy Island of Lakshadweep.

Parichakali is a traditional folk dance form of Lakshadweep.

Kolkali Dance (Stick dance) is a folk dance practiced by the inhabitants of the region of North Malabar, located in Kerala and Lakshadweep.

Saang, also known as Swang (meaning "initiation") or Svang, is a popular folk dance-theatre form in Haryana, Rajasthan, Uttar Pradesh and Malwa region of Madhya Pradesh.

Kaksar dance is a festival dance of Madhya Pradesh and Chhattisgarh.

Waziri dance- It is performed by the Pashtun tribe of Pakistan.

Kandyan dance is a folk dance of Sri Lanka. It encompasses various dance forms native to the area called Kandy of the central hills region in Sri Lanka.

Balti dance of Ladakh is distinct as it has its own costume, music, and songs in the Balti language.

Joenpa Legso is a welcome dance of Bhutan.

Types of Puppets Used in India

Glove Puppets {PavaKoothu (Kerala), and Beni Putul (Bengal)}.

Rod Puppets {Putul Nach (West Bengal), Kathi Kandhe (Orissa), and Yampuri (Bihar)}.

Shadow Puppets {Tholu Bommalata (Andhra Pradesh), Togalu Gombeyata (Karnataka), Tholpavakoothu (Kerala), Chamadyache Bahulya (Maharashtra), Ravana Chhaya (Orissa), and Thol Bommalattam (Tamil Nadu)}.

String Puppets {Putal Nach (Assam), Togalu Gombeyatta (Karnataka), Kalasutri Bahulya (Maharashtra), Gopalila Kundhei (Orissa), and Bommalattam (Tamil Nadu)}.

Arts Personalities

Bharat Ratna Awardees

Lata Mangeshkar- She was also known as 'The Queen of Melody', 'Nightingale of Bollywood', 'also known as Nightingale of India', and 'Swar Kokila'. She recorded her first song "Naachu Ya Gade, Khelu Saari Mani Haus Bhaari" at the age of 13 for a Marathi film (Kiti Hasaal). In her honor the Madhya Pradesh government started Lata Mangeshkar Awards (for music) in 1984. Her teacher was Aman Ali Khan (Bhendibazar gharana). She was born in Indore. **Awards** - Bharat Ratna (2001), Padma Bhushan (1969), Padma Vibhushan (1999), Dadasaheb Phalke Award (1989), Sangeet Natak Akademi Fellowship (1989), National Order of the Legion of Honour - Highest civilian Award of France (2009).

M.S. Subbulakshmi - She was an Indian Carnatic singer from Madurai, Tamil Nadu and known as the 'Queen of Carnatic Music'. She was honored in August 2016, by the United Nations Organization for her contribution to music. She was the first Indian singer to perform before the UN General Assembly (1966).

Awards - Bharat Ratna (1998, first Indian musician to receive), Ramon Magsaysay Award (1974, first Indian musician to receive), Padma Bhushan (1954) and Padma Vibhushan (1975), Sangeet Natak Akademi Award (1956), Sangeetha Kalanidhi (1968), Indira Gandhi Award for National Integration (1990).

Pandit Ravi Shankar- He was an Indian sitar player who belonged to Maihar Gharana. His guru was Allauddin Khan. His birth name was Ravindra Shankar Chowdhury. He is given credit for collaborating with the famous band 'The Beatles'. He was also the first Indian composer to win the Grammy Awards. His autobiography - "My Life, My Music".

Awards - Bharat Ratna (1999), Padma Vibhushan (1981), Padma Bhushan (1967), Sangeet Natak Akademi Award (1962), Sangeet Natak Akademi Fellowship (1975), Kalidas Samman (1987-88) and 4 Grammy (1967, 1973, 2002, 2013, and Lifetime Achievement at the 55th Annual Grammy Awards).

Pandurang Vaman Kane - He was a notable Indologist and Sanskrit scholar. Notable work - History of Dharmasastra. **Awards** - Bharat Ratna (1963), Sahitya Akademi award (1956).

Ustad Bismillah Khan (Qamaruddin Khan)- He was a famous Shehnai player and is known for playing Shehnai at the first Independence day (1947) celebration. A theater is named after him in Teheran (Iran). He also became the first Indian to be invited to perform at the prestigious Lincoln Centre Hall in the United States of America. In his honor, Ustad Bismillah Khan Yuva Puraskar (started in 2006) is given by the Sangeet Natak Akademi for artists under 40 years of age. **Awards** - Bharat Ratna (2001), Padma Vibhushan (1980), Padma Bhushan (1968), Padma Shri (1961), Sangeet Natak Akademi Award (1956), Tahar Mausique from the Republic of Iran (1992).

Pandit Bhimsen Gururaj Joshi- He was a renowned vocalist and belongs to Kirana Gharana known for the khyal form of singing. He made his debut in the live music scene at the young age of 19 in 1941 and, the following year in 1942, released his debut album (featuring a selection of devotional songs in Marathi and Hindi) through HMV (His Master's Voice). His guru was Sawai Gandharva. **Awards** - Bharat Ratna (2009), Padma Shri (1972), Padma Bhushan (1985), Padma Vibhushan (1999), Sangeet Natak Akademi Award (1976), Sangeet Natak Akademi Fellowship (1998), National Film Award for best male playback singer for Ankahee hindi movie in 1985.

Bhupendra Kumar Hazarika - He was an Assamese singer who is known as 'Sudha Kantha'. He was also known as the 'Bard of Brahmaputra'. Dhola - Sadiya Bridge (which connects Assam and Arunachal Pradesh) was named after him. **Awards** - Bharat Ratna (2019 - Posthumous), Padma Vibhushan (2012 - posthumously), Padma Bhushan (2001), Padma Shri (1977), Dadasaheb Phalke Award (1992), Sangeet Natak Akademi Award (1987), Sangeet Natak Akademi Fellowship (2008), Asom Ratna (2008).

Other Art Personalities

Satyajit Ray- He was a film director, writer, and composer of music in India. He is known for the direction of several films such as "Pathar Panchali" (1955), "Aparajito" (1956), "Apur Sansar" (1959), "Devi" (1960), "Two" (1964), "Kapurush" (1965), "Nayak" (1966).

Awards - Bharat Ratna (1992), Padma Vibhushan (1976), Padma Shri (1958), Padma Bhushan (1965), Sangeet Natak Akademi Puraskar (1959), Ramon Magsaysay Award (1967), Sangeet

Natak Akademi Fellowship (1986), Commander of the Legion of Honour By the French government (1987), Lifetime Achievement award by Oscars (1992), Dadasaheb Phalke Award (1984).

Pandit Shiv Kumar Sharma - He was an Indian classical musician and santoor player. He collaborated with Indian flutist Hariprasad Chaurasia under the collaborative name Shiv-Hari and composed music for such hit Indian films as Faasle (1985), Chandni (1989), and Lamhe (1991). **Awards-** Padma Vibhushan (2001), Padma Shri (1991), Sangeet Natak Akademi Award (1986).

Allah Rakha Rahman (A.S Dileep Kumar)- He is an Indian music composer, record producer, singer and songwriter, popular for his works in Indian cinema. He is popular by the nickname 'the Mozart of Madras'. He was born in Madras. **Awards-** Two Oscars (2009 for his work in Slumdog Millionaire), Six National Film Awards, two Academy Awards, two Grammy Awards (Best Compilation Soundtrack Album for visual media of Slumdog Millionaire and Best Song Written for Visual Media Jai Ho, From Slumdog Millionaire), Kalaimamani (Tamil Nadu), and a Padma Shri (2000), Lifetime Achievement Award (2008), Padma Bhushan (2010), Japan's 'Fukuoka prize-2016' for outstanding contribution towards South Asian traditional fusion music.

Shovana Narayan- She is a reputed classical dancer of Kathak dance form. She was a student of 'Pandit Birju Maharaj' and also served as an officer with Indian Audit and Accounts Services. **Awards** - Padma Shri (1992), Sangeet Natak Akademi Award (1999–2000).

Pandit Jasraj Motiram- He was an Indian classical vocalist, belonging to the Mewati Gharana. The International Astronomical Union (IAU) has named minor planet 2006 VP32 (number -300128) between Mars and Jupiter after Pandit Jasraj Motiram in 2019.

Awards- Padma Vibhushan (2000), Padma Shri (1975), Padma Bhushan (1990), Sangeet Natak Akademi Award (1987), Swathi Sangeetha Puraskaram (2008), Sangeet Natak Akademi Fellowship (2010).

Yamini Krishnamurthy- She is an Indian dancer of Bharatanatyam and Kuchipudi styles. She opened Yamini school of Dance in Delhi in the year 1990. She was given the title of Asthana Nartaki

("Resident Dancer") by the Tirumala Tirupathi Devasthanam temple. **Awards-** Padma Vibhushan (2016), Padma Shri (1968), Padma Bhushan (2001), Sangeet Natak Akademi (1977).

Zakir Hussain- He is a famous tabla player and the son of famous Tabla player Ustad Allah Rakkha. He was a part of the band named 'Planet Drum'. He belongs to Punjab Gharana of Tabla. **Awards** - Sangeet Natak Akademi Award (1990), Padma Bhushan (2002), Padma Shri (1988), Padma Vibhushan (2023), Grammy Award for Best Contemporary World Music Album (2009), Grammy (2009), Sangeet Natak Fellowship (2018), Kalidas Samman (2006). In 1999, he was awarded the United States National Endowment for the Arts' National Heritage Fellowship, the highest award given to traditional artists and musicians.

Uday Shankar- He was the first to bring the concept of modern dance to India with his novel approach and works. He came up with his own style of dance called 'Hi-dance', which later he called 'Creative dance'. He is also known for creating a fusion style of dance by adapting European theatrical techniques to Indian classical dance. **Awards-** Sangeet Natak Akademi award (1960), Sangeet Natak Akademi Fellowship (1962), Padma Vibhushan (1971).

Pandit Birju Maharaj - He was an Indian dancer, composer, singer, and exponent of the Lucknow "Kalka-Bindadin" Gharana of Kathak dance. His birth name was Brijmohan Nath Mishra. He is the direct descendant of Ishwari Prasad (the first known Kathak teacher). He was a descendant of the Maharaj family of Kathak dancers (which includes his two uncles, Shambhu Maharaj and Lachhu Maharaj and his father Acchan Maharaj). He established an institution named 'Kalashram' for imparting training in dance and associated disciplines. **Awards-** Padma Vibhushan (1986), Sangeet Natak Akademi Award (1964), Kalidas Samman (1987).

Narthaki Nataraj - She is an Indian transwoman Bharatanatyam dancer. **Awards-** In 2019, she was awarded the Padma Shri, making her the first transgender woman to be awarded India's fourth-highest civilian award.

Alarmel Valli - She popularised the 'Pandananallur Style' in Bharatanatyam. She founded The Dipasikha Dance Foundation in Chennai in 1984. She

performed at Salzburg Festival in 2015. She also became the second youngest dancer to be honoured with the 'Padma Shri' award in 1991. **Awards-** Grande Medaille de la Ville de Paris Award (1997), Sangeet Natak Akademi Award (2001), Padma Bhushan (2004), Chevalier of Arts and Letters award from the French Government (2004).

Mangalampalli Balamuralikrishna- He was an Indian Carnatic vocalist. He is credited with composing Talas such as Trimukhi, Panchamukhi, Saptamukhi and Navamukhi. **Awards-** Padma Vibhushan (1991), Padma Shri (1971), Sangeet Natak Akademi Award for Carnatic Music - Vocal (1975), National Film Awards (1986 and 1987), Chevalier of the Ordre des Arts et des Lettres from the Government of France (2005).

Jagjit Singh (Jagmohan Singh Dhiman) - He was an Indian composer, singer and musician. He was referred to as 'Ghazal King'. His collection of semi-classical music was 'The Unforgettables'. **Awards-** Padma Bhushan (2003), Sahitya Academy Award (1998), Lata Mangeshkar Samman (1998, Madhya Pradesh government).

R. Muthukannammal- She is a seventh generation veteran Sadhir dancer from the Indian State of Tamil Nadu. **Awards-** Padma Shri award (2022).

Usha Uthup Iyer- She is an Indian Pop, filmi, jazz, and playback singer. **Awards-** Padma Shri (2011), Padma Bhushan (2024).

Vempati Chinna Satyam- He was a Kuchipudi dancer. **Awards-** Padma Bhushan (1998), Sangeet Natak Akademi Award (1981), Kalidas Samman (1992).

Tanjore Balasaraswati- She was an Indian Bharatanatyam dancer. Bengali film director Satyajit Ray made a documentary film on Balasaraswati named Bala (1976). **Awards-** Padma Vibhushan - The First female dancer to receive the honor (1977), Padma Bhushan (1957), Sangeet Natak Akademi Award (1955).

Indira P. P. Bora- She is an internationally acclaimed exponent of Bharatanatyam, Kuchipudi and Sattriya. She belongs to Assam. **Awards** - Sangeet Natak Akademi Award (1996), Padma Shri (2020).

Meera Banerjee- She was a renowned

name of the Patiala Gharana. She was trained under Ustad Bade Ghulam Ali Khan at an early age.

Teejan Bai - She is a folk singer (Pandavani of Chhattisgarh). **Awards** - Padma Vibhushan (2019), Padma Shri (1988), Padma Bhushan (2003), Sangeet Natak Akademi Award (1995), Sangeet Natak Akademi Fellowship (2022).

Rukmani Devi Arundale (Rukmani Neelakanta Sastri)- She was a famous Bharatanatyam Dancer. She was the first woman in Indian history to be nominated (1952) as a member of the Rajya Sabha. In January 1936, she established Kalakshetra Dance School at Adyar, at Chennai. **Awards**- Padma Bhushan (1956), Sangeet Natak Akademi Award (1957), Sangeet Natak Akademi Fellowship (1967), Kalidas Samman (1984), Prani Mitra Award (1968).

Purandara Dasa- His pen name was "Purandara Vittala". He was a Haridasa philosopher and a follower of Madhwacharya's Dwaitha philosophy -saint. He is widely referred to as the Father of Carnatic Music in India. He introduced the raga Mayamalavagowla.

Padma Subrahmanyam- She is an Indian classical Bharatanatyam dancer and well known as the developer and founder of the dance form Bharata Nrithyam. **Awards**- Sangeet Natak Akademi Award (1983), Padma Bhushan (2003), Padma Shri (1981), Padma Vibhushan (2024), Kerala Nishagandhi Puraskaram (2015).

Gopika Varma- She is a Kerala born Mohiniyattam dancer and dance teacher who is settled in Chennai, Tamil Nadu, India. **Awards**- Sangeet Natak Akademi Award (2018), Kalaimamani (2004, first dancer to receive Kalaimamani for Mohiniyattam).

Sarodi Saikia is an Indian classical dancer, dance teacher and chetar player from Assam. She is best known for her Sattriya dance. **Awards** - Sangeet Natak Akademi Award (2015).

Haripriya Nambodiri - She is an exponent of Kathakali. She was winner of Stree Ratnam and Devdasi National Award.

Wadali Brothers- Puranchand Wadali and Pyarelal Wadali are Sufi singers and musicians from Guru Ki Wadali in the Amritsar District in Punjab. **Awards**- Sangeet Natak Akademi Award (1991), Punjab Sangeet Natak Akademi Award

(2003), Puranchand Wadali was awarded the Padma Shri (2005).

Guru Bipin Singh was a director, choreographer and teacher of Manipuri dance. He created the Govindji Nartanalaya dance school for women in Imphal and, in 1972, the Manipuri Nartanalaya dance school in Calcutta. **Awards** - Sangeet Natak Akademi Award (1966), Kalidas Samman (1989), The title of Natyacharya and Hanjaba conferred by the Maharaja of Manipur.

Ammannur Madhava Chakyar - He was a master of Kutiyattam, the classical Sanskrit theatrical form indigenous to Kerala. **Award** - Sangeet Natak Akademi Award (1979), Padma Shri (1982), Padma Bhushan (2003).

Kanaka Raju- He is an Indian gussadi dancer. **Award**- Padma Shri (2021).

Aluna Kabuini - She is a Ras exponent and Manipuri dancer. She was the recipient of the Manipur State Kala Academy Award in 2001 for her contribution in the field of Tribal Kabui dance.

Dhananjayans- Vannadil Pudiyaveettill Dhananjayan and Shanta Dhananjayan, are a dancing couple of Bharatnatyam in India. **Awards**- Rashtriya Kalidas Samman (Madhya Pradesh, 2020-21), Padma Bhushan (2009), and Kerala Sangeetha Nataka Akademi Fellowship (1994).

Ustad Asad Ali Khan was famous for Khandarbani schools of Dhrupad as an instrumentalist of rudra veena. **Awards** - Sangeet Natak Akademi Award (1977), and Padma Bhushan (2008).

Ustad Ghulam Mustafa Khan- He was an Indian classical musician in the Hindustani classical music tradition, belonging to the Rampur-Sahaswan Gharana. **Awards**- Padma Vibhushan (2018), Padma Shri (1991), Sangeet Natak Akademi Award (2003), Padma Bhushan (2006).

Ustad Chajju Khan- He along with Nazir Khan and Khadim Hussain Khan founded Bhendi Bazaar gharana in 1890 in Mumbai.

Roshan Kumari Fakir Mohammad (Kathak dancer)- She follows the Jaipur Gharana and is the founder of Kathak Nritya Kala Kendra (Mumbai). **Awards**- Sangeet Natak Akademi Award (1975), Padma Shri (1984).

Shobha Koser- She is an Indian Kathak dancer, writer and teacher. She belongs to the Jaipur Gharana. **Award**- Sangeet Natak Akademi Award (2019).

Maya Rao - She was an Indian classical dancer, choreographer and educator, in Kathak dance. **Awards** - Sangeet Natak Akademi Award (1989), Sangeet Natak Akademi Tagore Ratna (2011).

Kumudini Lakhia- She is an Indian Kathak dancer. She founded Kadamb School of Dance and Music (Ahmedabad) in 1967. **Awards**- Padma Shri (1987), Padma Bhushan (2010), Sangeet Natak Akademi Award (1982).

Rani Machaiah is an Ummathat folk dancer from Karnataka Award- Padma Shri (2023).

Shambhu Maharaj (Shambhunath Mishra)- He was a Guru of the Lucknow Gharana of Kathak. **Awards**- Sangeet Natak Akademi award (1955), Padma Shri (1958).

Bappi Lahiri (Bappida) was an Indian singer and composer from West Bengal. **Awards** - Filmfare Award (1985, 2018), Banga Bhushan Civilian Honour by Government of West Bengal (2017).

Gangubai Hangal- She was an Indian vocalist of Hindustani classical tradition who belonged to the Kirana Gharana. **Awards**- Padma Bhushan (1971), Padma Vibhushan (2002), Sangeet Natak Akademi Award (1973).

Hema Malini (Bharatnatyam dancer)- She is an Indian actress, director, producer, and politician. She is known as the 'dream girl of India'. **Award** - Padma Shri (2000).

Rahul Dev Burman (RD Burman) - He was a music scientist who was known as PanchamDa (the name given by Lata Mangeshkar). He is also known as Shahenshah- E-Music. He was music director of the famous film 'Sholay' released in 1975 and composed the famous song 'Yeh Dosti Hum Nahi Todenge'.

Ustad Bade Ghulam Ali Khan- He was an Indian Hindustani classical vocalist from the Patiala Gharana. **Awards**- Sangeet Natak Akademi Award (1962), Sangeet Natak Akademi fellow (1967), and Padma Bhushan Award (1962).

Uma Dogra- She is an Indian exponent of Kathak, an Indian Classical Dance form.

She is the senior most disciple of Pt. Durga Lal, the Kathak Maestro from Jaipur Gharana. She was awarded Sangeet Natak Akademi Award in 2014.

Pandit Durga Lal- He was a renowned Kathak dancer of the Jaipur Gharana. **Awards-** Sangeet Natak Academy Award (1984).

Mohanrao Kallianpurkar- He was a Kathak dancer and teacher from Karnataka. He belonged to the Jaipur gharana of Kathak. **Awards-** Sangeet Natak Akademi Award (1962).

Nandini Singh- She is considered to be one of the exponents of Jaipur Gharana of Kathak. In 1996, she founded a charitable organization namely, "Vaasuki Natyashala".

Ustad Alladiya Khan- He was an Indian Hindustani classical singer who founded the Jaipur-Atrauli Gharana (Jaipur Gharana). He is recognized for producing disciples like Bhaskarbuwa Bakhale, Kesarbai Kerkar, and Mogubai Kurdikar.

Kesarbai Kerkar- She was an Indian classical vocalist of the Jaipur-Atrauli Gharana. The Maharashtra Government awarded her the title Maharashtra Rajya Gayika. **Awards-** Sangeet Natak Akademi Award (1953), Padma Bhushan (1969).

Mogubai Kurdikar (Gaan Tapasvini)- She was a renowned Indian classical vocalist of the Jaipur-Atrauli gharana. **Awards-** Sangeet Natak Akademi Award (1968), and Padma Bhushan (1974).

Lachhu Maharaj (Pandit Bajinath Prasad)- He was a kathak dancer. He was awarded the 1957 Sangeet Natak Akademi Award. He was the paternal uncle of Pandit Birju Maharaj. He also learnt the Pakhawaj, the Tabla and Hindustani Classical vocal music.

Pandit Ishwari Prasad- He was the founder of the Lucknow Gharana of Kathak dance.

Manjari Chaturvedi- She is an Indian Kathak dancer. She belongs to the Lucknow Gharana.

Bharati Shivaji is an exponent of Mohiniyattam. She is the founder of Center for Mohiniyattam, a dance academy. **Award** - Padma Shri (2004).

Kanak Rele is an exponent of Mohiniyattam. **Awards** - Padma Shri (1990), Padma Bhushan (2013).

Janaki Prasad- He was a Kathak exponent from Bikaner in West Rajasthan. Janaki Prasad Gharana also known as Banaras Gharana named after him.

Manna Dey (Prabodh Chandra Dey)- He was an internationally acclaimed and celebrated Indian playback singer, music director, and a musician. He belonged to the Bhendi Bazaar Gharana and was trained under Ustad Aman Ali Khan. **Awards-** Padma Shri (1971), Padma Bhushan (2005), Ramon Magsaysay Award (1966).

Sitara Devi (Dhanalakshmi)- She was an Indian dancer of the classical Kathak style of dancing, a singer, and an actress. She was born at Cuttack. Rabindranath Tagore described her as Nritya Samragini meaning the empress of dance. **Awards-** Sangeet Natak Akademi Award (1969), Padma Shri (1973), Kalidas Samman Award (1995).

Ustad Sabri Khan (Sarangi player)- He belonged to the Senia Gharana. **Awards-** Sangeet Natak Akademi Award (1986), Padma Shri (1992), Padma Bhushan (2006), Sangeet Natak Akademi Tagore Ratna Award (2012).

Ustad Rehman Khan - He is a famous Pakhawaj player. He belongs to the Patiala Gharana.

Amjad Ali Khan- He is an Indian classical sarod player. He performed 'Raga For Peace' in the 2014 Nobel Peace Prize Concert. **Awards-** Padma Vibhushan (2001), Padma Shri (1975), Padma Bhushan (1991), Sangeet Natak Akademi Award (1989). In his honor, the US state of Massachusetts proclaimed 20 April as Amjad Ali Khan Day in 1984.

Sistla Janaki (S Janaki) - Indian playback singer from Andhra Pradesh. She is referred to respectfully as "Janaki Amma" and Nightingale of South India. **Award -** National Film Award (1977, 1981, 1984, and 1992) for Best Female Playback Singer.

Udit Narayan Jha - Playback singer of India received the Padma Shri in 2009 and Padma Bhushan in 2016.

Gulabo Sapera (Dhanvantri)- She is a folk dancer from Rajasthan who performs Kalbeliya dance. **Award-** Padma Shri (2016) and Bharat Gaurav Award (2021).

Sonal Mansingh- She is a prominent Indian classical dancer who specializes

in Bharatanatyam and Odissi. **Awards -** She was the youngest recipient of Padma Bhushan in 1992. Padma Vibhushan (2003), Sangeet Natak Akademi Award (1987).

Deba Prasad Das connoisseurs as one of the four first generation gurus of the Indian classical dance form of Odissi. He was a recipient of the 1977 Sangeet Natak Akademi Award.

Guru Pankaj Charan Das- He was an Indian classical dancer, choreographer and the Adi Guru of Odissi dance. He is known as the 'Father of Odissi dance'. He was born in Puri, Odisha. He was awarded Padma Shri in 1992. He strictly followed the mahari style in his dance.

Gangadhar Pradhan (Odissi dancer)- He founded Orissa Dance Academy in Bhubaneswar in 1975, and Konark Natya Mandap in Konark in 1986, and also established both the Konark Dance and Music Festival (1986) and the Dhauli dance festival (2001). **Awards-** Sangeet Natak Akademi Award (1998) and the Padma Shri (2008).

Bhanu Ji is associated with Jaipur Gharanas of Kathak.

Rohini Bhate (Kathak dancer)- She established Nritya Bharati Kathak Dance Academy (1947). **Awards-** Sangeet Natak Akademi Award (1979), Sangeet Natak Akademi Fellowship (2006).

Pandit Munnalal Shukla - He is a leading Kathak guru and choreographer of the Lucknow Gharana. He is the recipient of Sangeet Natak Akademi Award in 2006.

Uma Sharma- She is a kathak dancer. **Awards-** Padma Shri (1973), Padma Bhushan (2001), Sangeet Natak Akademi Award (1987).

Akhtari Bai Faizabadi (Begum Akhtar)- She was an Indian singer and actress. She is known as "Mallika-e-Ghazal" (Queen of Ghazals) and is regarded as one of the greatest singers of ghazal, dadra, and thumri genres of Hindustani classical music. **Awards-** Padma Shri (1968), Sangeet Natak Akademi Award (1972), and Padma Bhushan (posthumously, 1975).

C Ramchandra (Ramchandra Narhar Chitalkar or Anna Sahib) - He was an Indian music director and playback singer. He was the music director of the famous song "Aye Mere Watan Ke Logon" sung by Lata Mangeskar in front of the

first Prime Minister of India, Pandit Jawaharlal Nehru.

Vallathol Narayana Menon- He was a Malayalam poet. He founded the Kerala Kalamandalam and is credited with revitalizing the traditional Kerala dance form known as Kathakali. His famous poem is "Ente Gurunathan" ("My Great Teacher"). **Awards-** Padma Bhushan (1954).

Kalamandalam Kallyanikutty Amma- She was a Mohiniyattam dancer from Kerala. **Awards-** Kerala Sangeetha Nataka Akademi Award (1974), Kendra Sangeet Natak Akademi Award.

Waheeda Rehman - Actress-cum-dancer was honoured with Padma Bhushan award in 2011 and Dadasaheb Phalke Award (2021).

Sutapa Talukdar is an odissi dancer. **Awards-** Sangeet Natak Akademi Awards (2019).

Malini Awasthi is folk singer, sings in Bhojpuri, Awadhi and Hindi. **Awards-** Padma Shri (2016), Sangeet Natak Academy Award (2018), Uttar Pradesh Sangeet Natak Academy Award (2017).

Kottakal Nandakumaran Nair - Sangeet Natak Akademi 2019 Awardee for his contribution to Kathakali.

Thokchom Ibemubi Devi is a Manipuri Dancer. **Awards-** Sangeet Natak Akademi Award (2021).

Bhushan Lakandri was awarded the Sangeet Natak academy award 2020 for his contribution to Contemporary Dance.

Shubha Mudgal - She is an Indian singer of Hindustani classical music. Her repertoire includes the genres of khyal, thumri, dadra, and Indian pop. **Awards-** Padma Shri (2000) and Rajiv Gandhi National Sadbhavana Award (2016).

Kamala Narayan (Bharatnatyam dancer) - Her other names include Baby Kamala, Kamala Laxman, Kumari Kamala, and Kamala Lakshmi Narayanan. **Awards** - Central Sangeet Natak Akademi Award (1968), Padma Bhushan (1970).

Mrinalini Sarabhai- She is known for Bharatanatyam and Kuchipudi classical dance forms. She was the co-founder of Darpana Academy of Performing Arts. She was born in Kerala. In 2018, Google Doodle celebrated its 100th birthday. **Awards-** Padma Bhushan (1992), Padma Shri (1965), Sangeet Natak Akademi

Fellowship (1994).

Kaushalya Reddy is a professional Kuchipudi dancer performer. **Awards-** YUVA RATNA AWARD Govt. of Mauritius.

Urmila Sathyanarayana is an Indian classical dancer of bharatanatyam. **Awards-** Kalaimamani (1997) given by Tamilnadu and NrityaChoodamani (2001).

Pandit Vishnu Narayan Bhatkhande was a sitarist and prominent musician who wrote the first modern treatise on Hindustani classical music. He belonged to the Mewati Gharana.

Satyaram Reang (Maitya Ram Reang)- He is an Indian folk performer from Tripura. **Awards-** Padma Shri (2021) and Sangeet Natak Akademi Award (2004).

Ustad Allarakha Khan (Alla Rakha Quresh)- He left home at age 12 to learn a musical instrument under the great master (ustad) Mian Qadir Baksh. **Awards-** Padma Shri (1977) and Sangeet Natak Akademi (1982).

Uppalapu Srinivas was an Indian mandolin player in Carnatic classical music and composer. **Awards-** Padma Shri (2010).

Pandit Kumar Gandharva (Shivaputra Siddaramaiah Komkali Math)- He was a Hindustani classical singer, originally known as Shivaputra Siddharamayya Komkalimath. He was born in Karnataka. **Awards-** Padma Bhushan (1977) and Padma Vibhushan (1990).

Pandit Devabrata Chaudhari (Debu Chaudhuri)- He was an Indian sitarist and teacher. He belongs to Senia Gharana. **Awards-** Sangeet Natak Akademi Award (1996) and Padma Bhushan (1992).

Pandit Kishan Maharaj- He was an Indian tabla player who belonged to the Benaras Gharana of Hindustani classical music. **Awards** - Padma Vibhushan (2002), Padma Shri (1973).

Darshana Jhaveri- She is the youngest of the four Jhaveri sisters (Other three are Nayana Jhaveri, Ranjana Jhaveri, and Suverna Jhaveri), and a leading Indian exponent of Manipuri dance. She is one of the founders of the Manipuri Nartanalaya in 1972. **Awards-** Padma Shri (2002), Sangeet Natak Akademi Award (1996).

Ustad Hassu Khan was one of the

founders of the Gwalior Gharana of music.

Ramkrishna Talukdar received the Sangeet Natak Academy Award 2017 for his contribution to Sattriya Dance.

Akham Lakshmi Devi- She is a pioneer Classical Solo and Manipuri duet dance artist. **Awards-** Sangeet Natak Akademi Award (2018).

Ustad Shahid Parvez Khan- He is an Indian classical sitar maestro from the Imdadkhani Gharana. **Awards-** Kalajyothi Lifetime Achievement Award (2020), Sangeet Natak Akademi Award (2006), and Padma Shri (2012).

Pandit Budhaditya Mukherjee- He is an Indian classical sitar and surbahar maestro of the Imdadkhani gharana. **Award-** Padma Bhushan (2019).

Khemraj (Jammu and Kashmir) received the Sangeet Natak Akademi award in 2015 for his contribution to folk music and dance of 'Kud'.

Dr.P.Rama Devi, is a well known Senior Kuchipudi Dance Exponent and established her institution Sri Sai Nataraja Academy of Kuchipudi Dance in Secunderabad.

Anoushka Shankar- She is a British-American sitar player and music artist. She was the youngest and first woman to receive a British House of Commons Shield.

Pandit Harashankar Bhattacharya is an exponent of Sitar.

Rashid Khan- He belongs to the Rampur-Sahaswan gharana, and is the great-grandson of gharana founder Inayat Hussain Khan. **Awards-** Padma Shri (2006), Sangeet Natak Akademi Award (2006), Global Indian Music Academy Awards (2010), Padma Bhushan (2022).

Vasundhara Doraswamy received the Sangeet Natak Akademi Award in 2019 for Bharatanatyam.

Anokhelal Mishra- He was an Indian Tabla player who belonged to the Banaras Gharana.

Inayat Hussain Khan- He was an Indian classical vocalist, and the founder of Rampur-Sahaswan Gharana (Uttar Pradesh).

Guru Kelucharan Mohapatra- He is a

renowned Odissi dancer. He along with his wife, Laxmipriya Mohapatra (Odissi Dancer), and their son Ratikant Mohapatra built Srjan in 1993. **Awards-** Padma Vibhushan (2000), Padma Bhushan (1988), Padma Shri (1974), Sangeet Natak Akademi Award for Dance - Odissi (1966).

Jyoti Prasad Agarwala was a music composer popularly called 'Rupkonwar' in the state of Assam.

Sampooran Singh Kalra (Gulzar)- He is an Indian Urdu poet, lyricist, author, screenwriter, and film director known for his works in Hindi cinema. **Awards-** Sahitya Akademi Award (2002), Padma Bhushan (2004), Academy Award - Oscars (2009), Grammy Award (2010), Dadasaheb Phalke Award (2013).

Vedantam Satyanarayana Sarma (Kuchipudi). Award- Padma Shri (1970).

Ratikant Mohapatra- He is an Odissi dancer. Award- Sangeet Natak Akademi Award (2016).

Kamalini Asthana and Nalini Asthana- They are a sister duo from Agra in Uttar Pradesh known for their great performances of the Benaras Gharana style of Kathak. **Award-** Padma Shri (2022).

Surupa Sen- She is an Odissi dancer and choreographer who has lived and worked in Nrityagram, a dance village situated in the rural areas of Karnataka. **Award-** Sangeet Natak Akademi Award (2018).

Vidha Lal - She is a disciple of Kathak dancer Geetanjali Lal of Jaipur Gharana. Award - Guinness world record for the most number of spins (103) in one minute (2011) and Ustad Bismillah Khan Yuva Puraskar 2017.

Papanasam Sivan was a composer of Carnatic music and a singer.

Ilyas Khan was born in Lucknow to a family of musicians. He was a Sitarist.

Girija Devi- She is a Thumri Singer who belongs to Banaras Gharana. She is also known as 'Queen of Thumri'. **Awards-** Padma Vibhushan (2016), Padma Shri (1972), Padma Bhushan (1989), Sangeet Natak Akademi (1977), Sangeet Natak Akademi Fellowship (2010).

Pandit Chhannulal Mishra- He is a Hindustani Classical singer belonging to Kirana Gharana. He was a specialist in Khyal and thumri style of singing. **Awards**

- Padma Vibhushan (2020), Padma Bhushan (2010), Sangeet Natak Akademi awards (2010).

Anjanibai Malpekar- She was a noted Indian classical singer, belonging to the Bhandi Bazaar Gharana of Hindustani classical music. **Awards-** Sangeet Natak Akademi Fellowship (1958).

Ustad Allaiddin Khan- He was an Indian sarod player. He was a founder of the Maihar gharana. He was the guru of Pandit Ravi Shankar. He used to play Shehnai, sarod, sitar, and bansuri. **Awards-** Padma Vibhushan (1971), Padma Bhushan (1958), Sangeet Natak Akademi Fellowship (1954).

Nityanand Haldipur- He is a performer and teacher of the Indian bamboo flute (bansuri). He belongs to Maihar Gharana and learned from Maa Annapurna Devi. **Awards-** Sangeet Natak Akademi Award (2010).

Saswati Sen- She is an Indian exponent of Kathak. She was a disciple of Pandit Birju Maharaj. She received a Sangeet Natak Akademi Award in 2004.

Khayyam (Mohammed Zahur Khayyam Hashmi) was a famous music director in Hindi cinema. **Award** - Padma Bhushan (2011).

Talat Aziz is a ghazal singer and actor.

Asghari Bai was an Indian Dhrupad singer from Madhya Pradesh. She was a recipient of Padma Shri in 1990.

Shanti Bardhan, a junior colleague of Uday Shankar, is associated with synthesising classical and modern Indian dance through a theatre group called Ranga Sri Little Ballet Troupe.

Kabichandra Kalicharan Patnaik coined the term 'Odissi' for Odisha's classical dance. Some of his major published works- Kalanka Bhanjana, Milana Madhuri, Daridrya Bhanjana.

Zarin Sharma - She was a famous Sarod player of Khayal (Agra gharana). **Award** - Sangeet Natak Akademi Award (1988).

Asha Bhosle- She belonged to Bhendibazar Gharana of Mumbai where Navrang Nagpurkar was her teacher. She was officially acknowledged as the most recorded artist in music history by the Guinness Book of World Records in 2011. **Awards-** Padma Vibhushan (2008), Dada Saheb Phalke (2000).

Saroja Vaidyanathan- She is a choreographer, guru and a notable exponent of Bharatanatyam. **Awards** - Padma Bhushan (2013), Padma Shri (2002), Kalidas Samman (2009-10).

Ustad Shakoore Khan- He was a Sarangi player associated with Kirana gharana. Padma Shri (1973).

Joravarsinh Jadav - He belongs to Gujarati folk form, Bhavai. Award - Padma Shri (2019).

Bombay Jayashri Ramnath- She is an Indian Carnatic vocalist, singer, and musician. She has sung in multiple languages including Tamil, Telugu, Kannada, Malayalam and Hindi movies. **Award** - Padma Shri (2021).

Ustad Shafat Ahmed Khan was one of the foremost tabla players in the field of Hindustani classical music. **Awards** - Padma Shri Award in 2003.

Jayanthi Kumaresh- She is an Indian Veena musician. **Awards and Honors-** Kalaimamani from Tamil Nadu Government (2006), Veena Nada Mani - Kanchi Kamakoti Peetham (2010 and 2013), Asthana Vidwan at Sringeri Sharadha Peetham (2014).

Ustad Ali Akbar Khan- He was an Indian Hindustani classical musician of the Maihar Gharana. He is known for his virtuosity in playing the sarod. **Awards-** Padma Vibhushan (1989), Padma Bhushan (1967).

Milena Salvini, an Italian born French exponent of Kathakali dance, was awarded Padma Shri in 2019.

Nahid Siddiqui (Kathak dancer) was awarded Milapfest's Lifetime Achievement Award in 2017.

Elam Endira Devi - Exponent of Manipuri dance. Award - Padma Shri (2014).

Ustad Amir Khan (Sur rang)- He was one of the greatest and most influential Indian vocalists in the Hindustani classical tradition. He was the founder of the Indore Gharana (Vocal). He was born in Kalanaur. **Awards-** Sangeet Natak Akademi Award (1967), Padma Bhushan (1971).

Thetakudi Harihara Vinayakram (Vikku Vinayakram)- He is known as the God of Ghatam (Percussion instrument). **Awards-** Padma Shri Award (2002), Sangeet Natak Akademi Fellowship (2012), Padma Bhushan (2014).

Indrani Rahman (Indrani Bajpai) - She was an Indian classical dancer of Bharata Natyam, Kuchipudi, Kathakali and Odissi. She was the first woman to represent India in the introductory edition of the Miss Universe pageant in 1952. **Awards-** Padma Shri (1969) and Sangeet Natak Akademi Award (1981).

Jatin Goswami- He is an Indian dancer and choreographer of the famous classical dance form of Sattriya. He is the founder director of Sattriya Akademi, Guwahati. **Awards-** Padma Shri Award (2008), Sangeet Natak Akademi (2004) and Sangeet Natak Fellowship (2018).

Deepika Reddy- She is a renowned Kuchipudi dancer born at Hyderabad in Telangana. **Awards-** Sangeet Natak Akademi Award (2017), Kala Ratna Award (Government of Andhra Pradesh, 2007), Nritya Choodamani (Krishna Gana Sabha, 2014), Telangana State Government Award (2016) and the Keerthi Puraskar by the Potti Sreeramulu Telugu University (2016).

Kunnakudi Vaidyanathan- He was an Indian classical music violinist and music composer. **Awards-** Padma Shri (2005), Sangeet Natak Akademi Award (1993).

Bimbavati Devi- She is the exponent of Manipuri dance form. She has also undergone training in pung (Manipuri mridang) playing and Thang Ta (Manipuri martial art). **Awards-** Ustad Bismillah Khan Yuva Puraskar (2008).

Vidushi Sharan Rani Backliwal - She was an Indian classical sarod player and music scholar. She was popularly known as 'Sarod Rani'. In 2004, the government of India honoured select artists by conferring upon them the title of 'National Artiste'. **Awards** - Padma Shri (1968), Padma Bhushan (2000) Sangeet Natak Akademi Award (1986).

Shrinivas Joshi- He is an Indian classical vocalist, belonging to Kirana Gharana. He was born to legendary Hindustani vocalist, Bhimsen Joshi and Vatsalabai Joshi.

Brij Bhushan Kabra was an Indian guitarist. **Awards-** Sangeet Natak Akademi Award (2005).

'Bayanacharya' Shri Ghanakanta Bora Muktiyar - He is one of the most eminent exponents and gurus of Sattriya dance, a major classical dance tradition of

Assam. **Awards-** Padma Shri (2013) and Sangeet Natak Akademi Award (2001).

Pandit Pran Nath- He was an Indian classical singer and master of the Kirana gharana singing style. He began performing in the United States in the 1970s, and established the Kirana Center for Indian Classical Music (New York) in 1972.

Rajendra Prasanna- He is an Indian classical flutist and shehnai player from Benares Gharana. He was born in Varanasi, Uttar Pradesh. **Awards-** Grammy Award certificate for his participation in "Concert for George" (2004), Sangeet Natak Akademi Award (2017).

Rasoolan Hussain- She was a leading Indian Hindustani classical music vocal musician. She belonged to the Benaras gharana. She specialized in the romantic Purab Ang of the Thumri musical genre and tappa. **Awards-** Sangeet Natak Akademi Award (1957).

Rajendra Chaturvedi- He is a Kathak Dancer in the Banaras Gharana.

Krishna Ram Chaudhary- He was a shehnai player belonging to Benaras Gharana. **Awards-** Sangeet Natak Akademi Award (2008), Padma Shri (2017).

Ananda Gopal Bandopadhyay- He was an Indian tabla player. He was trained in the style of the Benaras Gharana by his teacher Mahadev Prasad Mishra.

Rajashree Shirke- She is a Kathak dancer. **Awards-** Sangeet Natak Academy Award (2013).

Rani Karnaa Nayak (Kathak dancer)- She founded Samskritika Shreyaskar dance academy in Kolkata in 1955. **Awards-** Padma Shri (2014), Sangeet Natak Akademi Award (1996).

Sundaram Balachander- He was an Indian veena player and filmmaker. **Awards-** Sangeet Natak Akademi Award (1977).

Perumanoor Gopinathan Pillai (Guru Gopinath)- He was a well known Kathakali dancer.

Rajkumar Singhajit Singh- He is a leading Manipuri dancer. **Awards** - Sangeet Natak Akademi Award (1984), Padma Shri (1986).

Kishori Ravindra Amonkar- She was an Indian classical vocalist, belonging to the Jaipur Gharana. She was a performer of the classical genre khayal and the classical genres thumri and bhajan. **Awards-** Padma Vibhushan (2002), Padma Bhushan (1987), Sangeet Natak Akademi Award (1985), Sangeet Natak Akademi Fellowship (2009).

Prerana Shrimali- She is a senior dancer of Jaipur Gharana of Kathak. **Awards-** Rajasthan Sangeet Natak Akademi Award (1993), the Rashtriya Ekta Award (2001), Raza Puraskar Raza Foundation (Delhi, 2004), Keshav Smriti Award Kaladharmi (2008), Sangeet Natak Akademi Award (2009), etc.

Prerana Deshpande is a recognized Indian exponent of Kathak dance. Award - Devadasi National Award (2016).

Pandit Mani Prasad was an Hindustani classical vocalist from the Kirana gharana (singing style). He was awarded Sangeet Natak Akademi Award in 2019.

Deepak Mazumdar is a Bharatanatyam performer. He is the founder and director of the Shiladhish Art and Research Institute, Bangalore. **Award** - Sangeet Natak Akademi Award (2018).

M.S. Gopalakrishnan (Madras Sundaram Iyer Gopalakrishnan) mastered both Carnatic and Hindustani classical music. He was the master of Violin. Award - Padma Shri (1975).

Guru Nileshtar Mukherjee- He is a prominent Manipuri dancer of India. Rabindranath Tagore was fascinated with a Ras performance in Sylhet in 1920 by Guru Nileshtar Mukharjee and he invited him to Shantiniketan.

Dr. M. Lalitha and M. Nandini are popularly known as 'Violin sisters'.

Pandit Hariprasad Chaurasia - He is an Indian music director and classical flutist in the Hindustani classical tradition. He belonged to the Senia gharana. **Awards-** Padma Vibhushan (2000), Sangeet Natak Academy Award (1983), Padma Bhushan (1992).

Prateeksha Kashi- She is an Indian Kuchipudi dancer, a classical dance form of Andhra Pradesh. **Awards-** Ustad Bismillah Khan Yuva Puraskar (2014).

Surekha Punekar- She is an Indian actress and folk artist known for her lavani performances.

Priyadarsini Govind is a Bharatanatyam dancer. She was conferred the Nritya Kalanidhi title by Music Academy in 2019. **Awards** - Kalaimamani (1998).

Shrimati Kumkum Mohanty- She is a world-famous Odissi dancer. **Awards**- Padma Shri (2005), Sangeet Natak Akademi Award (1994), Odisha Sangeet Natak Akademi (1993).

Sharmila Biswas- She is a noted Indian classical dancer and choreographer in Odissi and a disciple of guru Kelucharan Mohapatra. **Awards** - Sangeet Natak Akademi Award (2012).

Ustad Sultan Khan- He was an Indian sarangi player and classical vocalist belonging to Sikar Gharana. **Awards**- Padma Bhushan Award (2010), Sangeet Natak Akademi Award (1992).

Gaddam Padmaja Reddy- She is an Indian Kuchipudi exponent and music teacher. **Awards**- Sangeet Natak Akademi Award (2015), and Padma Shri (2022).

Louise Mary Lightfoot- She was an Australian architect, choreographer and dancer. She was popularly called the Australian Mother of Kathakali by Vallathol Narayana Menon (the founder of Kerala Kalamandalam).

Ustad Abdul Karim Khan and Wahid Khan- They were Indian classical singers who founded Kirana Gharana. Due to frequent visits to Mysore Palace, Karim was conferred the title Sangeet Ratna. Karim founded the Arya Sangeet Vidyalaya in Poona in 1913. Karim was guru of Pandit Balkrishnabuwa Kapileshwari, Ganapat Rao Gurav, Roshan Ara Begum, Sureshbabu Mane, Vishwanathbuwa Jadhav and Sawai Gandharva. Wahid Khan was an Indian surbahar and sitar player.

Sudha Chandran- She is an Indian Bharatanatyam dancer and actress who appears in Indian television and films.

Ustad Bade Fateh Ali Khan- He was among the foremost Khyal vocalists in Pakistan, and a leading exponent of the Patiala Gharana tradition of music. He was born in Hoshiarpur (Punjab). He was a foremost vocalist of Hindustani classical, Dhrupad, Khyal, Ghazal, and Thumri. **Awards**- Pride of Performance Award in 1969 by the President of Pakistan and Tamgha-e-Imtiaz (Medal of Distinction) by the Government of Pakistan.

Ali Baksh Jarnail Khan- He was an Indian classical singer. He and Fateh Ali Khan propounded the Patiala Gharana in the 19th century. Patiala gharana has claimed to combine the musical traditions of Delhi gharana, Gwalior Gharana and Jaipur-Atrauli Gharana. The Patiala Gharana was originally founded by Mian Kallu.

Lakshminarayana Subramaniam- He is an Indian violinist, and music composer. **Awards**- Padma Shri (1988), Sangeet Natak Akademi (1990), Padma Bhushan (2001).

Appa Jalgaonkar (Sakharam Prabhakar Jalgaonkar or Appasaheb Jalgaonkar)- He was an Indian harmonium player from the state of Maharashtra. He was awarded Sangeet Natak Akademi Award in 2000.

Ustad Bundu Khan (Sarangi player)- He was awarded the Pride of Performance Award by the President of Pakistan in 1985.

Aditi Mangaldas is a leading dancer & Choreographer in the classical Indian dance form of Kathak. **Award** - Mahari Award for dance (2022).

Pandit Satya Dev Pawar- He is a violinist trained under Ustad Ali Akbar Khan. He belongs to "the Seni Baba Allaaddin Gharana".

Aziz Warsi - He was associated with Qawwali.

Mallika Sarabhai (Kuchipudi and Bharatanatyam dancer)- She was honoured by the French government with Chevalier des Palmes Academiques (2005), Padma Bhushan (2010).

Sabri Brothers- These were a musical band from Pakistan who were performers of Sufi qawwali music and were closely connected to the Chishti Order. The Sabri Brothers were led by Ghulam Farid Sabri and his brother Maqbool Ahmed Sabri.

Ustad Nusrat Fateh Ali Khan (Pervez Fateh Ali Khan)- He was a Pakistani singer, songwriter, and music director. He was primarily a singer of qawwali – a form of Sufi devotional music. He was called as the "Shahenshah-e-Qawwali" (the King of Kings of Qawwali).

Pandit Nikhil Ranjan Banerjee- He was an Indian classical sitarist of the Maihar Gharana.

Sunayana Hazarilal Agarwal- She is an Indian Kathak dancer, reportedly the only surviving practitioner of the Janaki Prasad Gharana (Banaras Gharana) of Kathak, also known as Benares Gharana. **Awards**- Sangeet Natak Akademi Award (2003) and Padma Shri (2011).

Ileana Citaristi- She is an Italian Odissi and Chhau dancer. **Awards**- Padma Shri (2006).

Meenakshisundaram Pillai- He is a classical dance guru considered as the prominent founder of Pandanallur style of Bharatanatyam.

Meenakshi Sabanayagam Chitharanjan - She is an exponent of the Pandanallur style of Bharatanatyam. Award- Padma shri (2008).

Thanjavur Quartet (Tanjore Bandhu) were four brothers, Chinnayya, Ponnayya, Sivanandam and Vadivelu, who lived during the early 19th century and contributed to the development of Bharatanatyam and Carnatic music.

Tankeswar Hazarika Borbayan- He is associated with Sattriya dance. **Awards**- Sangeet Natak Akademi (2018).

Madhavi Mudgal- She is an Odissi dancer. **Awards** - Padma Shri (1990), Sangeet Natak Akademi Award (1999-2000).

Kadri Gopalnath - He was an Indian alto saxophonist and one of the pioneers of Carnatic music for that instrument. He was an exponent of Indian classical music, film music, and jazz fusion genres. **Awards** - Sangeet Natak Academy Award for Carnatic Music (2004), Padma Shri (2004), Sangeetha Kalasikhamani award (2013), Kamban Pugazh award (2018).

Radha Sridhar- She is a famous Bharatanatyam Dancer who won the Sangeet Natak Akademi Award in 2018.

N. Rajam- He is an Indian violinist who performs Hindustani classical music. **Awards**- Sangeet Natak Akademi Award (1990), Padma Shri (1984), Padma Bhushan (2004).

Rajendra Kumar Gangani - He is a Kathak dancer born in a traditional family of Jaipur Gharana. He has also acquired proficiency in playing the Pakhawaj. **Awards**- Sangeet Natak Akademi Award (2002).

Minati Mishra (Minati Das)- She was an exponent of Odissi. **Awards-** Orissa Sangeet Natak Akademi Award (1975), Sangeet Natak Akademi Award (1999 - 2000), Padma Shri (2012).

Chitti Babu (Veena Chitti Babu)- He was one of the greatest Veena artists in the Carnatic Music genre of South India.

Awards- Sangeet Natak Akademi award (1990).

Yog Sunder Desai, a pioneer who ventured into the world of Indian Dance in the pre-independence era. He was born in Gujarat.

Nooran Sisters - Sultana Nooran and Jyoti Nooran are a devotional Sufi singing duo from Hoshiarpur, Punjab. Born to a family of Sufi musicians, they perform Sham Chaurasia Gharana classical music.

Janki Bai- She was a Hindustani vocalist from Allahabad. She, along with her famed contemporary Gauhar Jaan, had the privilege of performing for Emperor George V at the Delhi Darbar in 1911. She was popularly known as "Chhappanchhurivali". Her guru was Moinuddin Khan. Her fans affectionately called her Bulbul.

Gopal Prasad Dubey- He is an exponent of Chhau. He is a leading exponent of the Seraikella variant of the art form. **Award-** Padma Shri (2012).

Shamshad Begum- She was one of the first Indian playback singers in the Hindi film industry. **Awards-** O.P. Nayyar Award for her contribution to Hindi film music (2009), Padma Bhushan (2009).

Tulsidas Vasant Borkar- He was an Indian composer and harmonium player. **Awards-** Padma Shri (2016), Sangeet Natak Akademi Award (2005).

Guru Makar Dhawja Darogha- He was an exponent of Chhau dance. He was conferred Padma Shri in 2011.

Guru Amubi Singh (Manipuri dancer). **Awards-** Sangeet Natak Akademi (first winner from the state of Manipur, 1956), and Padma Shri (1970).

Annapurna Devi (Roshanara Khan)- She was an Indian surbahar (bass sitar) player of Hindustani classical music. She was given the name 'Annapurna' by former Maharaja Brijnath Singh of the former Maihar state. **Awards** - Padma Bhushan (1977), Sangeet Natak Akademi

Award (1991, only surbahar player awarded), fellow of the Sangeet Natak Akademi (2004).

Kumar Sanu (Kedarnath Bhattacharya)- He is an Indian playback singer. He is known as the King of Melody in Bollywood. He holds the record for winning five consecutive Filmfare Awards for Best Male Playback Singer from 1991 to 1995. He made the Guinness world record in 1993 when he recorded 28 songs in a single day. The Mayor of Dayton, Ohio in the USA declared March 31 as Kumar Sanu Day. **Awards-** Padma Shri (2009), Sangeet Samman Lifetime Achievement Award (2015), Lata Mangeshkar Award (2021).

Vidyagauri Adkar- She is a Kathak dance exponent in India representing the Jaipur Gharana.

Shyama Charan Pati- He was an Indian dancer, known as one of the leading exponents of the Seraikela tradition of Chhau.

Sartaj Khan, Sarwar Khan, and Swaroop Khan are famous for Rajasthani folk music.

Pandit Sunder Prasad- He belongs to Jaipur Gharana of Kathak. He received his training from his father Pandit Chunnilal of Jaipur Gharana. He also trained under Bindadin Maharaj of Lucknow Gharana. He was honored with the Sangeet Natak Akademi Award in 1959.

Laxmibai Kolhapurkar- She won the first National Film Award for Best Choreography in 1992 for her song in the Marathi film 'Ek Hota Vidushak.' She was the first female choreographer in the country to get the award.

Mame Khan- He is an Indian playback and folk singer from Rajasthan. He was born in Satto in Rajasthan. He received the Best Folk Single Award at Global Indian Music Academy Awards (GiMA) 2016.

G. Narayanaswamy Balasubramaniam- He was an Indian Carnatic singer. He along with Semmangudi Srinivasa Iyer and Madurai Mani Iyer, are referred to as the 20th century male trinity of Carnatic music.

Ricky Kej- He is a three-times Grammy Award-winning, Indian music composer and environmentalist. Grammy Awards- 2015 (Best New Age Album for Winds of

Samsara), 2022 (Best New Age Album and Best Immersive Audio Album for Divine Tides), 2023 (New age Album).

Manjamma Jogathi (Manjunatha Shetty)- She is an Indian theatre actress, singer and dancer of Jogathi Nritya. **Awards** - Padma Shri (2021).

Malka Jaan- She belongs to Agra Gharana, and was one of Gauhar Jaan's contemporaries. She was born in Azamgarh. She was a court musician at the durbar of Nawab Wajid Ali Shah in Calcutta.

Maharaj Kalika Prasad- He belonged to the Lucknow Gharana. Kalka Prashad ji had three sons - Achhan Maharaj, Baijnath Prashad (Luchhu Maharaj), and Shambhu Maharaj. He promoted Kathak-style dance and Thumri singing.

Chintaman Raghunath Vyas (C. R. Vyas)- He was an Indian classical singer, known for singing khyal style. He has written a book Raag Sarita which sums up his work in the field of music. **Awards-** Padma Bhushan (1992), Maharashtra Gaurav Puraskar (1990), and Sangeet Natak Akademi Award (1987).

Kalamandalam Kshemavathy Pavithran- She is a Mohiniyattam dancer from Thrissur, Kerala. **Awards-** Padma Shri (2011), Sangeet Natak Akademi Award (1998).

C. V. Chandrasekhar- He is an Indian Bharatanatyam dancer. **Awards-** Sangeet Natak Akademi Award (1993), and Padma Bhushan (2011).

Allah Jilai Bai- She was a folk singer from Rajasthan. She was well versed in Maand, Thumri, Khayal and Dadra. **Awards-** Padma Shri (1982), Sangeet Natak Akademi Award (1988).

Rama Vaidyanathan- She is an Indian Bharatanatyam artist from Delhi. **Awards-** Kerala Sangeetha Nataka Akademi Award (2013), Sangeet Natak Akademi Award (2017), Devadasi National Award (2013).

Sujata Mohapatra (Sujata Mohanty)- She is an Indian classical dancer and teacher of Odissi dancing style. She received the Central Sangeet Natak Akademi Award in 2017.

Ashish Sankrityayan is an exponent of Dhrupad.

Sivamani (Drums Sivamani), is an Indian

percussionist. Awards- Padma Shri (2019).

Keezhpadam Kumaran Nair- He was a Kathakali artist from Kerala. **Awards**- Padma Shri award (2004), Sangeet Natak Akademi award (1988), Kerala Sangeetha Nataka Akademi Fellowship (1996).

Pandit Sandeep Das - He is an Indian tabla player and composer.

Trilichan Mohanta - A Chhau dance exponent. Award - Sangeet Natak Akademi Awards (Akademi Puraskar) for the Year 2020.

Sukhbir Singh is an Indian singer. He is often referred to as the "Prince of Bhangra".

Kishore Kumar (Abhas Kumar Ganguly)- He was an Indian playback singer and actor. He was born in 1929 in Khandwa (Madhya Pradesh). **Awards** - Lata Mangeskar Award (1985). In 1997, Madhya Pradesh Government initiated an award called the "Kishore Kumar Award".

Maguni Charan Das- He was a dancer of Gotipua, a traditional dance form of Odisha. **Awards**- Padma Shri Award (2004).

Lakshmi Shankar (Sastri)- She was an Indian singer and a noted Hindustani classical vocalist of the Patiala Gharana. She was known for her performances of khyal, thumri, and bhajan.

Dr. Mallikarjun Bheemaryappa Mansoor- He was the exponent of Khayal, a type of Hindustani classical music. He belonged to the Jaipur-Atrauli Gharana. **Awards**- Padma Vibhushan (1992), Padma Shri (1970), Padma Bhushan (1976).

Radhe Shyam Barle- He is a Panthi folk dancer and artist. **Awards**- Padma Shri Award (2021).

Tanvi Shah - She is the first Indian woman to win a Grammy for Best Song Written for Visual Media song 'Jai Ho' (Slumdog Millionaire) in 2009. She shared this award with A.R. Rahman and lyricist Gulzar.

Pandit Gopal Mishra- He was a sarangi player in the twentieth century. He was the founder of the Banaras Gharana of the style of khayal singing.

Anup Jalota- He is an Indian singer, musician and actor, best known for his contributions to the Bhajan genre of Indian Music. He is popularly known as

the "Bhajan Samraat" (Emperor of Bhajans).

Lourembam Bino Devi- She is a practitioner and a popularizer of the art of Manipur called Leeba in the Meitei language. The Leeba art is used in creating Monmai which is a decorative circular art piece. **Awards**- Padma Shri (2022).

Raja & Radha Reddy (Kuchipudi). They founded and run the Natya Tarangini Institute of Kuchipudi Dance in New Delhi. **Awards** - Padma Shri (1984), Padma Bhushan (2000), Sangeet Natak Akademi Award (1991).

Ramali Bin Ibrahim- He is a Malaysian choreographer of Bharatanatyam and Odissi dance forms of India. **Awards**- Padma Shri (2018).

Kalamandalam Rajan- He is a Kathakali exponent from Kerala. **Awards**- Sangeet Natak Akademi Award (2009).

Kottakkal Sivaraman- He was a performing artiste who revolutionized the portrayal of female roles in Kathakali. **Awards**- Kerala Sangeetha Nataka Akademi Award (1996), Kerala Sangeetha Nataka Akademi Fellowship (2008) and Central Sangeet Natak Akademi award (1998).

Manju Barggavee was awarded the Sangeet Natak Akademi Award 2019 for her contribution in Kuchipudi.

Dr. Sunil Kothari was awarded Madhabdev Award in 2019 by the Government of Assam for popularising Sattriya dance. **Awards**- Sangeet Natak Akademi Award (1995), Padma Shri (2001).

Kumkum Dhar - Lucknow based Kathak dancer received the Sangeet Natak Akademi Award (2020).

Kailash Kher is an Indian music composer and singer. **Awards**- Padma Shri (2017).

Nirmala Paniker was awarded the Sangeet Natak Akademi in 2019 for her contribution to Mohiniyattam.

Mahendra Kapoor was an Indian playback singer. He received the Padma Shri in 1972. He sang the famous song 'Mere Desh Ki Dharti Sona Ugle' from the film 'Upkar' (1967).

Sang Jangmu is a folk dancer from

Arunachal Pradesh. She received the Sangeet Natak Akademi Award in 2020.

Nargis Dutt was the first Indian actress to receive the Padma Shri Award in 1958.

Utpal Borpujari is a film critic and filmmaker. His documentary, 'Songs of the Blue hills' is based on the folk songs and culture of the Naga tribes.

Shruti Bandopadhyay - An eminent Manipuri dancer. **Awards**- Sangeet Natak Akademi Awards (2020).

Guru Shashadhar Acharya is a Chhau dance exponent from Saraikela (Jharkhand). **Awards**- Padma Shri (2020).

Dr. Prabha Atre was an Indian classical vocalist from the Kirana gharana. **Awards**- Padma Shri (1990), Padma Bhushan (2002) and Padma Vibhushan (2022), and Sangeet Natak Academy Award (1991).

S P Balasubrahmanyam was an Indian playback singer. **Awards**- Padma Shri (2001), Padma Bhushan (2011) and Padma Vibhushan (2021).

Shri Sudarshan Sahoo is an Indian sculpture artist from Puri, Odisha. **Awards**- Padma Shri (1988), Padma Vibhushan (2021).

Ilaiyaraaja is an Indian musician, composer, popular for his works in Tamil films. **Awards**- Padma Bhushan (2010), Padma Vibhushan (2018).

Shri K J Yesudas is an Indian playback singer. **Awards**- Padma Shri (1975), Padma Bhushan (2002), and Padma Vibhushan (2017).

Shri Rajinikanth (Shivaji Rao Gaikwad) is an Indian actor who works mainly in Tamil cinema. **Awards**- Padma Bhushan (2000), Padma Vibhushan (2016), Kalaimamani (1984).

Shri Amitabh Bachchan is an Indian actor who works in Hindi cinema. **Awards**- Padma Shri (1984), Padma Bhushan (2001), Padma Vibhushan (2015), and Legion of Honour (2007).

Shri Mohammed Yusuf Khan alias Dilip Kumar was an Indian actor who worked in Hindi cinema. **Awards**- Padma Bhushan (1991), Padma Vibhushan (2015), and Nishan-E-Imtiaz (1998).

Shri Raghunath Mohapatra was an

Indian architect and sculptor. Awards- Padma Shri (1976), Padma Bhushan (2001), Padma Vibhushan (2013).

Shri Sayed Haider Raza was an Indian painter. Awards- Padma Shri (1981), Padma Vibhushan (2013).

Prof. Kalpathi Ganapathi Subramanyan was an Indian artist. Awards- Padma Shri (1975), Padma Bhushan (2006), Padma Vibhushan (2012).

Shri Mario de Miranda was an Indian cartoonist and painter from Goa. Awards- Padma Shri (1988), Padma Bhushan (2002), Padma Vibhushan (2012).

Dr. (Smt.) Kapila Vatsyayan was a scholar of Indian classical dance, art, architecture. Awards- Padma Shri (1990), Padma Vibhushan (2011).

Dr. Akkineni Nageswara Rao was an Indian actor and producer, who worked majorly in Telugu cinema. Awards- Padma Shri (1968), Padma Bhushan (1988), Padma Vibhushan (2011).

Smt. Homai Vyrawalla (Dalda 13) was India's first woman photojournalist. Awards- Padma Vibhushan (2011).

Dr. Umayalpuram K. Sivaraman is an exponent of the Carnatic percussion instrument, the mridangam. Awards- Padma Shri (1988), Padma Bhushan (2003), Padma Vibhushan (2010).

Shri Ebrahim Alkazi was an Indian theater director and drama teacher. Awards- Padma Shri (1966), Padma Bhushan (1991), Padma Vibhushan (2010).

Smt. Zohra Segal was an Indian actress. Awards- Padma Shri (1998), Padma Vibhushan (2010).

Shri Adoor Gopalakrishnan is an Indian film director. Awards- Padma Shri (1984), Padma Vibhushan (2006).

Pandit Ram Narayan is a sarangi player. Awards- Padma Shri (1976), Padma Bhushan (1991), Padma Vibhushan (2005).

Shri Rasipuram Krishnaswamy Laxman was an Indian cartoonist, illustrator, and humorist. He is best known for his creation "The Common Man". Awards- Padma Bhushan (1971), Padma Vibhushan (2005).

Dr. Shantaram Rajaram Vankudre was an

Indian filmmaker, producer, and actor. Awards- Padma Vibhushan (1992).

Shri Kaloji Narayana Rao was an Indian poet. Awards- Padma Vibhushan (1992). **Shri Maqbool Fida Hussain** was a modern Indian painter. Awards- Padma Bhushan (1973), Padma Vibhushan (1991).

Dr. Sammagudi Radhakrishnier Srinivasier was an Indian Carnatic vocalist. Awards- Padma Bhushan (1969).

Shri Keshav Shankar Pillai was an Indian cartoonist. Awards- Padma Bhushan (1966), Padma Vibhushan (1976).

Shri Binode Behari Mukherjee was an Indian painter. Awards- Padma Vibhushan (1974).

Shri Nandlal Bose was an Indian painter. Awards- Padma Vibhushan (1954),

Padma Vibhushan (2024)

Shri Konidela Chiranjeevi is an Indian actor, film producer and former politician who works predominantly in Telugu cinema.

Padma Bhushan (2024)

Dattatray Ambadas Mayaloo alias Rajdutt is a Marathi Film Director.

Pyarelal Sharma is one of the most successful music composers in Hindi cinema.

Padma Shri (2024)

Badrappan M is an exponent of Tamil Nadu's ancient 'Valli Oylil Kummil' dance.

Kaluram Bamaniya is a Folk singer (Bhajan) from Madhya Pradesh.

Rezwana Choudhury Bannya is an exponent of Rabindra Sangeet, the songs written and composed by Rabindranath Tagore.

Ramlal Bareth is a Kathak Dancer. Awards- Sangeet Natak Akademi Award (1995).

Gita Roy Barman is a singer of the Rajbanshi community's Bhawaiya folk songs.

Som Datt Battu is a Hindustani classical vocalist of the Patiala Gharana.

Mahabir Singh Guddu is a Haryanvi Folk

Artist.

Ms. Anupama Hoskere - She is a master puppeteer and the Founder -Director of the Dhaatu Puppet Theater, Bengaluru, India.

Jankilal is a Bahurupiya ('multi-faceted') artist from Rajasthan.

Ratan Kahar is a folk singer of Bhadu and Jhumur songs of rural Bengal.

Dasari Kondappa is a Burra Veena player.

Uma Maheshwari D is a Harikatha exponent from Andhra Pradesh.

Ram Kumar Mallick is a Hindustani classical vocalist from the Darbhanga Gharana in Bihar.

Surendra Mohan Mishra is a renowned Hindustani classical singer from the Banaras Gharana in Uttar Pradesh

Ali Mohammed and Shri Ghani Mohammed, Maand Folk singing brother-duo from Bikaner.

Narayanan E P is a Theyyam Dance maestro from Kerala.

Bhagabat Padhan is a Sabda Nrutya Folk Dancer from Bargarh.

Binod Kumar Pasayat is a distinguished lyricist, playwright and poet from Sambalpur.

Ms. Silbi Passah is a folk musician from Meghalaya.

Romalo Ram is a Dogri folk artist. He is the inventor of the 'Geetru' musical style, a distinctive and innovative genre within Dogri music.

Ms. Nirmal Rishi is an Indian Punjabi film and television actress.

Pran Sabharwal is a Veteran Theatre Actor from Patiala who is credited for over 5,000 performances over 7 decades.

Gaddam Sammaiah is a Chindu Yakshaganam Artist from Telangana.

Omprakash Sharma is a Mach folk theater artist from Madhya Pradesh.

Seshampatti T Sivalingam is a Nadaswaram maestro from Tamil Nadu.

Ms. Urmila Srivastava is a Kajri singer from Mirzapur.

Nepal Chandra Sutradhar was a Chhau

mask maker from West Bengal.

Gopinath Swain is a Krishna Leela Bhajan player from Ganjam (Odisha).

Laxman Bhatt Tailang was a Hindustani classical music maestro, known for creating a new form of singing called 'Pachrang.'

Jagdish Labhshanker Trivedi is a renowned Gujarati comedian.

Balakrishnan Sadanam Puthiya Veetil is a Kalluvazhi Kathakali Dancer from Kerala.

Godawari Singh is a wood toy maker, dedicated to promoting and preserving Varanasi's Wooden Lacquerware & Toys.

Khalil Ahamad is a carpet Weaver from Mirzapur, Uttar Pradesh.

Naseem Bano is a chikankari artisan from Lucknow, Uttar Pradesh.

Takdira Begum is a Kantha embroidery artist from West Bengal.

Drona Bhuyan is a folk artiste and Ojapali performer from Assam.

Ashok Kumar Biswas is a tikuli folk painter from Bihar.

Smriti Rekha Chakma is a chakma Loinloom Shawl Weaver from Tripura.

Velu Ananda Chari is a master Staphathi Sculptor from Telangana.

Ghulam Nabi Dar is a master wood-carver from Jammu & Kashmir.

Jordan Lepcha is a bamboo craftsperson preserving the traditional Lepcha hats of Sikkim.

Sanatan Rudra Pal is a sculptor known for crafting traditional Sabeki Durga idols. He is from west bengal.

Shanti Devi Paswan & Shri Shivan Paswan duo Godna painters from Bihar.

Machihan Sasa is a master craftsperson who has promoted and preserved the ancient Manipuri tradition of Longpi pottery.

Babu Ram Yadav is a brass Marori craftsperson from Moradabad, Uttar Pradesh.

Padma Bhushan (2023)

Ms. Vani Jairam was an Indian playback singer in Indian cinema.

Ms. Suman Kalyanpur is an Indian playback singer, who is most notable for her work in Hindi cinema.

Padma Shri (2023)

Usha Barle is an Indian Pandavani folk singer, of a traditional performing folk art form, from Chhattisgarh.

Hemant Chauhan is a folk singer and bhajnik (singer of bhajans) from Rajkot (Gujarat).

Ahmed Hussain & Shri Mohd Hussain are a renowned Ghazal singer duo from Jaipur.

Mahipat Kavi from Gujarat is a writer, musician, translator, and educator creating awareness on social issues through puppets or kathputlis.

M M Keeravaani is an Indian music composer, who predominantly works in Telugu cinema. He won the Oscar Award for Best Original Song and the Golden Globe Award for Best Original Song for "Nattu Nattu" for the 2022 Telugu film RRR.

Maguni Charan Kuanr is a Puppeteer from Keonjhar (Odisha), promoting the Kathi Kandhei Nach for 6 decades.

Domar Singh Kunvar is a traditional dance artist from Chhattisgarh.

Risingbor Kurkalang is a craftsman from Meghalaya, he makes traditional musical instruments and promotes Khasi traditional folk music and dance.

Shri Nadoja Pindipapanahalli Munivenkatappa is a veteran Thamate (similar to a drum) player

Ramesh Parmar and Ms. Shanti Parmar are a craftsperson duo who have kept the tradition of tribal doll making alive.

Ms. Krishna Patel is a renowned Sambalpur folk singer.

Guru Thiru K Kalyanasundaram Pillai is a famous Bharatanatyam Dancer, Choreographer and Teacher from Tamil Nadu.

Mangala Kanti Roy is a sarinda player from Jalpaiguri, West Bengal.

K C Runremsangi is a Mizo folk singer from Mizoram.

Ritwik Sanyal is an Indian classical vocal

maestro and senior exponent of the Dagar Gharana of Dhrupad.

Kota Satchidananda Sastry is a veteran Harikatha artist from Andhra Pradesh.

Moa Subong is a Naga musician who invented the musical instrument Bamhum.

Coomi Nariman Wadia is India's 1st Female Choir Conductor'.

Ghulam Muhammad Zaz is a santoor craftsman from Kashmir.

Kapil Dev Prasad is a handloom weaver from Bihar, he is known for keeping the 'Bavan Buti' weaving art alive.

C V Raju is known for preserving and promoting the traditional method of making Etikoppaka toys by using non-toxic paints and natural dyes.

Shah Rasheed Ahmed Quadri is a master craftsman engaged in the field of Bidri Ware Handicrafts.

Pareshbhai Rathwa is a 'Pithora' artist from Chhotaudepur, Gujarat.

Ms. Jodhaiyabai Baiga is a Baiga painter from Madhya Pradesh.

Premjit Baria is painter from Dadra and Nagar Haveli and Daman and Diu.

Bhanubhai Chunilal Chitara is a Kalamkari Artist preserving the 400 years old Mata ni Pachedi craft.

Arts Awards

Sangeet Natak Akademi Award - (Akademi Puraskar)- It is the highest Indian recognition given to people in the field of performing arts. The award earlier in 2003, consisted of Rs. 50,000, a citation, an angavastram (a shawl), and a tamrapatra (a brass plaque). Since 2009 cash prize has been increased to Rs. 1,00,000.

It is an award given by the Sangeet Natak Akademi. First Awarded in 1952 are Mushtaq Hussain Khan (Vocal), Allaudin Khan (Sarod), Ariyakudi Ramanuja Iyengar (Carnatic Music), Karaikudi Sambasiva Iyer (Veena).

Sangeet Natak Akademi Fellowship (Akademi Ratna Sadasyata)- It is an Indian honour for the performing arts presented by Sangeet Natak Akademi. Established - 1954.

First Winners - Ariyakudi Ramanuja Iyengar (Music), Karaikudi Sambasiva Aiyer (Music), and Prithviraj Kapoor (Theatre).

The award recipient is awarded with a cash prize of Rs. 3 lakh an Angavastram (Shawl), and a Tamrapatra (citation plaque) given under the seal of Akademi and signature of its Chairman.

Kalaimamani Award- It is given by the Tamil Nadu Iyal Isai Nataka Mandram for outstanding contribution in the field of art and literature. It was first given in 1954 and it carries a cash purse "Porkizhi" of Rs. 1,00,000 to senior Kalaimamani winners along with a certification and shield.

Kalidas Samman- It is an arts award presented annually by the Government of Madhya Pradesh. It carries a plaque and cash prize of 5 lakh rupees and a citation is presented in the fields of classical music, classical dance, theatre and Plastic art. First recipients (1980) - Semmangudi Srinivasa Iyer (Classical music) and Mallikarjun Mansur (Classical Music).

Bharat Muni Samman- It was started by Nalanda Dance Research Centre in 2011. In 1966, this centre was established by Kanak Rele (exponent of Mohiniyattam) in Mumbai. Hema Malini has been selected for the prestigious Bharat Muni Samman 2012.

Jakanachari Award- It is a state award of Karnataka conferred on talented sculptors and craftsmen from the state to celebrate the contributions of the legendary sculptor Amarashilpi Jakanachari. Established - 1995. It includes plaque, shawl, gardland and cash reward of Rs. 5 lakhs. First recipient - C. Parameshwara Acharya (1995).

Kala Ratna Award- It is a civilian honour of the Andhra Pradesh state conferred annually by the Government of Andhra Pradesh on the occasion of Telugu new year, Ugadi. The award carries an amount of Rs. 30,000, a shawl, a gold plated Hamsa (Swan) memento. The award is conferred to artists from the field of literature, music, dance, painting, sculpture, folk and tribal arts. First awarded - 1999.

Kamal Kumari National Award- It is an Indian Award given to individuals and groups in India for outstanding contribution to Art, Culture and Literature

and Science and Technology. It was instituted by the Kamal Kumari Foundation in 1990 in memory of Kamal Kumari Barooah, the remarkable matriarch of the Khongiya Barooah family of Thengal, Assam. First award - Sobha Brahma (1991). The award carries a cash award of Rs. 2,00,000, a trophy and a citation.

Lalit Kala Akademi Fellowship (Lalit Kala Akademi Ratna)- It is an honour for the fine arts in India. It carries a cash prize of Rs. 25,000, a citation and a plaque. First Award - Jamini Roy (1955).

National Handicrafts Award- It is an Indian Government award conferred to outstanding master craftspersons. The award is presented by the President of India, along with Shilp Guru Awards to Master Craftspersons and Sant Kabir Award to master weavers, introduced in 1965. The award carries a cash prize of Rs. 1 lakh, a shawl, a certificate and a tamrapatra.

Sant Kabir Award- It is an Indian Government award conferred to outstanding weavers who have made valuable contributions in keeping alive the handloom heritage. It is presented by the President of India, along with Shilp Guru Awards and National Awards to Master Craftspersons and master weavers, introduced in 1965. It consists of one mounted gold coin, one shawl and a citation and Cash prize Rs 6 lakh.

Shilp Guru- It is the highest honour in the Indian handicrafts sector awarded by the Development Commissioner, Ministry of Textiles. First introduced - 2002. The award consists of Rs. 2 lakh prize money, a plaque, a gold coin, a shawl and a certificate.

Skoda prize for Indian contemporary art in the Indian art scene to recognize an Indian artist under 45 years of age. First recipient - Mithu Sen (2010).

Tulsi Samman (founded in 1983-84) is an arts award presented annually by the government of Madhya Pradesh. The recipient gets Rs. 5,00,000 along with a citation.

Varnashilpi Venkatappa Award- It was established by the government of Karnataka in 1993, in the name of the great painter late Varnashilpi Venkatappa to reward senior painters for their contributions in the field of art and paints. It includes plaque, shawl, gardland and cash reward of Rs. 5 lakhs.

Poompuhar State Award- It is an award given to outstanding artisans every year by Tamil Nadu Handicrafts Development Corporation Limited.

Hridaynath Award- Established - 2011. First awardee - Lata Mangeshkar in 2011. Other Hridaynath Award Winner- 2014 - Hariprasad Chaurasia, 2015 - AR Rahman, 2016 - Vishwanathan Anand, 2017 - Javed Akhtar, 2018 - Khayyam.

Guru Kelucharan Mohapatra Award instituted in the year 1995 by Sri Ratikant Mohapatra. to confer recognition on an individual for his contribution in extensive areas in the performing arts.

The Tagore Award for Cultural Harmony was instituted by the Government of India. First Tagore Award (2012) - Pt. Ravi Shankar (Sitar Maestro).

National Nritya Shiromani award is an annual award given to eminent dancers and musicians in the world for their excellence and for their immense contribution to the art field.

Guru Kelucharan Mohapatra Yuva Pratibha Samman presented to talented performing artistes under the age of 40 years in the field of classical dance and Odissi music.

Abhinandan Saroja award is a national award given to eminent dancers in promoting and preserving the traditional classical dance forms of India.

Guru Gopinath Natanagaram National Natya Puraskar instituted by Guru Gopinath Natanagaram on behalf of the Department Of Culture, Government Of Kerala.

Devdas Banjare Smriti Panthi Dance Award instituted by the Culture Department in the memory of late Devdas Banjare, a famous Panthi Nartak of Chhattisgarh.

Awards for Films, Music and Television

National Film Awards- It is one of the most prestigious events in the country. The Hon'ble President of India presents the awards culminating in showcasing of award-winning films for the public. Established - 1954.

It is given in three sections – Features, Non-Features and Best Writing on Cinema. Separate awards for artists and technicians were instituted in 1968 for

the films of 1967, Nargis Dutt and Uttam Kumar being the first actress and actor to get the Best Actress and Best Actor Awards respectively. The Marathi film "Shyamchi Aai" received the first National Film Award in 1954 (for best feature film).

Dadasaheb Phalke Award- It is India's highest award in the field of cinema. It was instituted in 1969 to honor Dadasaheb Phalke, the pioneer of Indian cinema who directed India's first full-length feature film, Raja Harishchandra, in 1913.

The award comprises a Swarna Kamal (Golden Lotus) medallion, a shawl, and a cash prize of Rs.10 lakh. First recipient - Devika Rani Roerich (1969).

Academy Awards (Oscars) are awards for artistic and technical merit for the American film industry. The awards are regarded by many as the most prestigious, significant awards in the entertainment industry in the United States.

It is presented annually by the Academy of Motion Picture Arts and Sciences (AMPAS). It is the oldest of the four major annual American entertainment awards - Emmy Awards (Television), Tony Awards (Theatre), and Grammy Awards (Music). First Actor to win an Oscar Award was Emil Jannings and First actress was Janet Gaynor.

BAFTA - It is an annual award show hosted by the British Academy of Film and Television Arts (BAFTA) to honour the best British and international contributions to film. The first BAFTA Awards ceremony was held in 1949, and the ceremony was first broadcast on the BBC in 1956 with Vivien Leigh as the host.

The Palme d'Or is the highest prize awarded at the Cannes Film Festival. First Palme d'Or Award (1955) - Delbert Mann (Movie "Marty").

Order of Arts and Letters is an order of France established in 1957. The Order has three grades- Commandeur (Commander), Officier (Officer) and Chevalier (Knight).

Grammy Awards- It is presented by the Recording Academy of the United States to recognize "outstanding" achievements in the music industry. First awarded - 1959.

Golden Globe Awards- They are accolades bestowed by the Hollywood Foreign Press Association. It was started in January 1944, recognizing excellence in both American and international film and television. The first award was presented to director and producer Cecil B. DeMille.

Lata Mangeskar Award- It is a national-level award in the field of music. First Awardee - Naushad.

Ustad Bismillah Khan Yuva Puraskar- It is an annual Indian award given by the Sangeet Natak Akademi to outstanding artists under 40 who have demonstrated talent in the fields of music, dance and drama. Established - 2006. Award prize money of Rs 25000 and a tamrapatra to be issued under the seal of Akademi and signature of its Chairman.

First Awardee in 2006 are Sheejith Krishna (Bharatnatyam), Kalamandalam Pradeep Kumar (Kathakali), Prashant Shah (Kathak), Vedantam Venkata Nagachalapathi Rao (Kuchipudi), Sijagurumayum Nimita Devi (Manipuri), Bijayini Satpathy, Leena Mohanty (Odissi), Anurupa Roy (Puppetry).

Sri Shanmukhananda National Eminence Award- It was set up with the title 'Sangeetha Kala Vibhushan' in 2002. First recipient - Shri Semmangudi Srinivasaiyer (Carnatic Vocal) in 2002.

Rashtriya Kumar Gandharva Samman- It was established by the government of Madhya Pradesh in 1992 as an annual national honour presented in the field of music. Cash prize - Rs. 1.25 lakh. First recipient - Ajoy Chakrabarty (1992-93).

Nataka Kalasarathy- It is the title awarded annually by Parthasarathy Swami Sabha to a dramatist. It confers a shawl, a citation, a medallion with a cash prize of Rs. 5,000.

Nishagandhi Puraskaram (Nishagandhi award)- It is an annual award instituted by the Tourism Department of Kerala in India to honour personalities in the fields of music and dance. Established - 2013. The award carries a cash prize of 1,50,000 Indian Rupees, a citation, and a statuette. First award - Mrinalini Sarabhai (2013).

Other Arts Awards

Carnegie Art Award (Sweden), Guggenheim International Award (United States), Bucksbaum Award (United

States), Hugo Boss Prize (New York), Marcel Duchamp Prize (France), Praemium Imperiale (Japan), Roswitha Haftmann Prize (Zurich), Turner Prize (Great Britain), Jameel Prize (Saudi Arabia), Malcolm Award (United States), Sobey Art Award (Canada), Kyoto Prize (Japan), Lumen Prize (United Kingdom), Leonardo da Vinci World Award of Arts (Mexico), Taro Okamoto Award (Japanese), Mary Smith Prize (Pennsylvania) etc.

Festivals

Dussehra (Vijayadashami) is a major Hindu festival celebrated at the end of Sharad Navaratri every year. It marks the end of Ramlila and remembers god Rama's victory over Ravana. Ramlila was added to UNESCO's list of Intangible Cultural Heritage in 2008. It is also a famous celebration in Mysore Palace (Karnataka).

Holi (Festival of colours, Phagwa) is a festival for Hindus. It falls on the full moon day in the month of Phalgun (March). It is associated with the events from the lives of Prahlada (devotee of Lord Vishnu) and Holika.

Different names in the different regions - Rang Panchami (Maharashtra), Lathmar Holi (Uttar Pradesh), Holla Mohalla (Punjab), Shigmo (Goa), Royal Holi (Udaipur), Kumaoni Holi and Khadi Holi (Uttarakhand), Manjal Kuli (Kerala), Dol Jatra (West Bengal), Yaosang (Manipur), Phaguwa (Bihar), Phakuwah (Assam).

Makar Sankranti is a major harvest festival celebrated by Hindus in January. The Sun God is worshipped along with Vishnu and the Goddess Lakshmi.

Different names in the different regions - Makar Sankranti (Kerala, Odisha, Karnataka, Maharashtra, Goa, West Bengal), Magh Bihu (Assam), Maghi Saaji (Himachal Pradesh), Maghi Sangrand (Jammu), Sakrat (Haryana, Rajasthan), Sukarat (central India), Thai Pongal (Tamil Nadu), Ghughuti (Uttarakhand), Dahi Chura (Bihar), Khichdi Parv (Uttar Pradesh), Uttarayan (Kite flying is the most popular activity during the Uttarayan festival in Gujarat).

Diwali (Deepawali, Festival of Lights): The festival gets its name from the row (avali) of clay lamps (deepa). It is celebrated on the Amavasya of Krishna Paksha months of Kartika during the Hindu lunisolar Calendar. On this day, Rama returned to his kingdom in

Ayodhya after 14 years of exile defeating the demon Ravana. In Jainism, Diwali marks the nirvana of Lord Mahavira. In Sikhism, it honors the day that Guru Hargobind Ji, the Sixth Sikh Guru, was freed from imprisonment.

Dev Deepavali is predominantly celebrated in Varanasi on a full moon day, 15 days after Diwali, on the ghats of River Ganga.

Ram Navami (Chaite Dashain) is a festival that celebrates the birth of Lord Rama, the son of King Dasharath. It is also a religious festival for the Nepali community inhabiting the Himalayan state of Sikkim.

Navaratri is an annual Hindu festival observed to worship nine forms of Goddess Durga.

Nagula Chavithi (Festival of Snakes) is celebrated on the fourth day after Deepavali. It is a major festival in Andhra Pradesh and some parts of Karnataka and Tamilnadu.

Janmashtami (Gokulashtami) celebrates the birth of Lord Krishna. It is observed on the eighth tithi (Ashtami) of the Krishna Paksha (dark fortnight) of Shraavana Masa or Bhadrapada Masa. The activity of making a human pyramid and putting efforts to break an earthen pot filled with buttermilk is associated with this festival.

Kali Puja (Shyama Puja or Mahanisha Puja), honoring the goddess Kali, is celebrated on the new moon in Kartik (Oct-Nov). Popular in West Bengal, Odisha, and Bangladesh.

Raksha Bandhan is associated with the term "ties of protection". It is a festival celebrated on the Purnima (full moon day) in the month of Shravan according to the Hindu Lunar calendar.

Ganga Dussehra is held every year on the 10th day of Shukla Paksha of Jyeshtha month.

Thiruvathira (Thiruvathirai or Ardra Darshan) is a Hindu festival celebrated in Tamil Nadu and Kerala.

Ugadi is celebrated to mark the first day of Telugu New Year in the states of Andhra Pradesh, Telangana and Karnataka.

Lohri is a harvest festival celebrated in Punjab and some parts of the Jammu

region, Delhi, Haryana, and Himachal Pradesh. It is celebrated every year during the month of Paush, a day before Makar Sankranti.

Baisakhi (Vaisakhi, Vasakhi) commemorates the formation of Khalsa Panth by Guru Gobind Singh in 1699. It is a spring harvest festival usually celebrated on the 13th or 14th of April every year. It is mainly celebrated in the states of Punjab and Haryana.

Sharad Purnima (Kojagari Purnima, and Kaumudi Purnima) is celebrated on the full moon day of the Shukla Paksha of the Hindu lunar month of Ashvin.

Guru Purnima, celebrated by Hindus, Jains and Buddhists in India falls in the month of Ashadha.

Nag Panchami is celebrated in Shravan month of Hindu calendar.

Ashadhi Bij (Kutch New Year) is celebrated in Gujarat, Uttar Pradesh, and some regions. It falls on the second day of Shukla paksha of the Ashadha month, of the Hindu calendar. It is associated with the beginning of rain in the Kutch region.

Gyan Panchami is the 5th day after Diwali. Holy recitation and Pratikraman are performed. It is also known as Labh pancham, Shrut Panchami, and Jnan Panchami.

Maha Shivaratri is a Hindu festival celebrated annually in honour of the God Shiva. It is held on the 13th night and the 14th day of the month Phalguna (February or March).

Gowri Habba is usually observed by married women and is a significant festival in Karnataka and Tamil Nadu. It is known as Hartalika Teej in the North Indian states.

Losar is the Tibetan New Year which falls in the month of February. It is mainly celebrated in Tibet, Bhutan, Nepal (called Lhochhar), and India (Arunachal Pradesh, Sikkim, and Ladakh).

Navanna is a harvest festival celebrating the harvesting of the new crop- Aman paddy. It is usually celebrated with food, dance, and music in Bangladesh.

Nobanno is celebrated in West Bengal, Tripura, and Assam's Barak Valley. It is celebrated with a mela which is called Nabanna Mela.

Shoton Festival is held annually in August in Tibet.

Ongkor festival is one of the most important Tibetan festivals of the Tibetan lunar year. It is held at the end of summer. Ongkar means good wishes for harvest.

Namsong festival, celebrating the Lepcha New Year, takes place in Upper Dzongu (Teesta & Rongyung Chu rivers) around December-January.

Chitragupta puja is held on the second day of Shukla Paksha in Kartik month. It is also known as Yama Dwitiya.

Kalpataru Utsav- On Jan 1st, Ramakrishna Mission marks Sri Ramakrishna's divine awakening with Kalpataru Utsav, named after the wish-fulfilling tree.

Hunter's Moon is called Kartik Purnima in India. Kartika Purnima is a Hindu, Sikh, and Jain festival.

Aadi Perukku is celebrated on the 18th day of the Tamil month of Adi (mid-July). It worships nature in the form of Amman deities organized to shower Nature's bountiful grace on human beings. It is also known as the Aadi monsoon festival.

Ojiale, an agricultural festival in Arunachal Pradesh (March-April), sees Wancho farmers pray for good harvests and share rice beer in bamboo tubes to celebrate.

Mim Kut Festival is a harvest celebration in Nagaland and Mizoram, honoring the maize crop. The Kuki Naga tribes, native to the Mizo Hills, hold this joyous post-harvest festival.

Gajalakshmi puja is observed for 11 days starting from Kumarpurnima or Ashwin Purnima (full moon day in Ashwin month).

Karama is a harvest festival celebrated in Jharkhand, Odisha, West Bengal, Bihar, Madhya Pradesh, Chhattisgarh, Assam, and Bangladesh. It is a tribute to the Karam tree.

Pola is a festival of the state of Maharashtra and Chhattisgarh which is celebrated by the farmers by the worship of the bull.

Dol Jatra (Doul Utsav, Dol Purnima) is a major Holi festival of Braj, Bangladesh,

and the Indian state of West Bengal, Odisha, and Assam. On this day, Chaitanya Mahaprabhu (founder of Gaudiya Vaishnavism) was born.

Gahambars (seasonal festivals) take place six times a year.

Halloween (All Saints' Eve) is a celebration observed on 31 October.

Andhra Pradesh

Tirupati Tirumala Brahmotsavam (The Festival Of Lord Brahma) is celebrated at the Tirumala Venkateswara temple in Tirupati (Andhra Pradesh). It lasts for nine days every year in October.

Atla Tadde is a traditional festival celebrated in Andhra Pradesh. It occurs on the 3rd night after the full moon in the Ashwayuja month of the Telugu calendar. It is the Telugu equivalent of Karva Chauth. It is observed by married Hindu women, for the health and long life of their husbands.

Bathukamma is a flower festival celebrated by the women of Telangana and some parts of Andhra Pradesh. This festival is celebrated for 9 days (mostly before Dussehra) during September - October.

Deccan Festival is an annual celebration in Hyderabad, Andhra Pradesh. It goes on for 5 days in February and March. It is also known as the Hyderabad festival.

Rottela Panduga (Rotiyaan ki Eid) is an annual 3 - 5 day festival held at Bara Shaheed Dargah in Nellore in Andhra Pradesh. It is observed in the month of Muharram (August-September) as a festival of 12 martyrs.

Ratha Saptami is celebrated to mark the birth of the Sun God.

Tirupati Ganga Jatara a week-long event, where devotees offer food and sarees to Gangamma (Goddess who protects Tirupati).

Pyditthallamma utsavam is a cattle trading festival that takes place in Vizianagaram.

Visakha Utsav is a 3 day festival of the arts, crafts and cuisine of Visakhapatnam District. Organized every year by the department of tourism of the Government of Andhra Pradesh.

Chandanotsavam (Chandana Yatra) is

organised during the months of March and April at Sri Varaha Lakshmi Narasimha Temple in Simhachalam.

Vinayaka Chavithi is mainly celebrated in Kanipakam from seven to ten days.

Arunachal Pradesh

Mopin is a five-day-long festival celebrated by the Galo Adi tribe of Arunachal Pradesh. It is celebrated in April.

Chalo Loku, a harvest festival of the Nocte tribe in Arunachal Pradesh, kicks off with animal offerings. It spans three days: Chamkatja (day 2) and Thanglangja (day 3).

Nyokum is celebrated by the Nyishi tribe in Arunachal Pradesh for better productivity, prosperity and happiness of all human beings on earth.

Tamladu is celebrated by the Buddhist community of Arunachal Pradesh. It is celebrated by the Mishmis on 15th February. Main deity - Lord Jebmalu.

Myoko festival in Arunachal Pradesh rotates among three Apatani communities - Diibo-Hija, Hari-Bulla, and Hong.

Murung is the six-day festival of Prosperity, celebrated by the Apatani tribe of the Lower Subansiri district in Arunachal Pradesh.

Dree festival is celebrated on July 5 in Arunachal Pradesh. It is an agricultural festival of the Apatani tribe. Tamu, Harniang, Metii, and Danyi gods are worshipped. It is a 3-day long festival.

Reh is the festival of the Idus tribes. It is celebrated in February-August for six days in Arunachal Pradesh.

Siang river festival is celebrated in different parts of Arunachal Pradesh in December every year. It is held on the bank of the Siang (Brahmaputra) river.

Boori Boot Yullo is a spring harvest festival celebrated by the Nyishi tribe of Arunachal Pradesh.

Solung festival, celebrated by the Adi tribe in Arunachal Pradesh during monsoons (around September), is a three-part harvest festival.

Shapawng Yawng Manau Poi is celebrated by the Singpho tribe. It is

usually celebrated between 12th to 15th February.

Pongtu is an agricultural festival celebrated by the Tutsa tribe on the eve of the rainy season.

Moh-Mol is an agricultural related festival of the Tangsas community.

Pangsau Pass International Festival is a global village event organized every year in Nampong which is situated in Changlang district.

Assam

Bihu is celebrated in Assam. It is celebrated three times in a year as: Bohag Bihu in April (wish for a good harvest), Kaati Bihu in October (mark the cutting and binding of grains), Magh Bihu in January (marks the season of harvesting of grains).

Rongker is an annual spring festival celebrated by the Karbis of Assam. It is also known as Dehal in Dumra Longri (Assam) and Ri-Bhoi district of Meghalaya.

Ambubachi is a festival of Kamakhya temple of Guwahati and is held every year during monsoon (mid-June).

Me-Dum-Me-Phi is the most important ancestor worship communal festival in the Ahom religion in Assam. It is celebrated by the Ahom people on 31 January every year in memory of the departed.

Ali-Aye-Ligang- Assam's Mishing tribe celebrates a spring festival marking the start of sowing seeds. Held on the 1st Wednesday of Phalguna (Feb-March), it's a 5-day agricultural celebration.

Baishagu - Assam's Boro tribe celebrates Baishagu, a springtime new year festival (mid-April) marked by community prayers (Garjasali) and honoring Lord Shiva.

Majuli festival is celebrated in the state of Assam at the bank of River Luit. It starts on the 21st of November each year and continues to the 24th of the same month.

Dehing Patkai Festival is organised at Lekhapani, in the Tinsukia district. This festival was initiated by the Assam Government in 2002.

Kaziranga Elephant Festival has been celebrated at Kaziranga National Park since 2003 for the conservation and

protection of Asiatic elephants.

Assam Tea Festival is held in the district of Jorhat. It is celebrated every year between November and January.

Lachit Diwas (Lachit Day) is a regional public holiday in Assam. It is observed on November 24th each year.

Bare Saharia Bhaona is a 200-year-old drama festival held every 5–6 years at Jamugurihat, a small town in Sonitpur District.

Rajini Gabra and Harni Gabra is the cultural festival.

Bihar

Chhath puja is a four-day long festival in the worship of 'Surya', the sun deity. It is celebrated on the sixth day of the lunar month of Kartika (October - November) in the Hindu calendar. It is celebrated in Bihar, Uttar Pradesh, West Bengal, Jharkhand, and the Nepalese provinces of Madhesh and Lumbini.

Festival of Salhesh is celebrated in the Mithila region of Bihar every year during the auspicious month of Shravan (July-August). Salhesh is the chief deity of the Dusadh Dalit community.

Rajgir dance festival (Rajgir Nritaya Mahotsav) is an annual three-day event in Rajgir, Nalanda (Bihar).

Bihula is celebrated in some portions of Bengal, Bihar, and Jharkhand. It is chiefly famous in the district of Bhagalpur (Bihar). The serpent goddess Mansa is worshipped.

Madhushravani is a 13-day festival celebrated primarily in the Mithila region of Bihar during the monsoon season (Shravan month, roughly July-August).

Sama-Chakeva is a ten-day festival celebrated in November in Mithila. It marks the bond between brothers and sisters.

Chhattisgarh

Bastar Dussehra (Jagdalpur Dussehra) is celebrated in Chhattisgarh. It lasts for 75 days. Main deity - Goddess Danteshwari.

Sirpur Dance and Music Festival is an esteemed cultural carnival celebrated in Sirpur region of Mahasamund district in the Indian state Chhattisgarh.

Goncha festival (Chariot festival) is

celebrated by tribes of Bastar region in Chhattisgarh.

Kajari festival in Chhattisgarh is a monsoon festival that is celebrated by the farmers.

Gaura-Gauri Puja is celebrated on the Amavasya of Krishna Paksha in Kartik month in Chhattisgarh. It celebrates the pastoral life of the people of Chhattisgarh.

Madai is the festival of Sanskritik Chhattisgarh. The Gond Tribe, Kurna, and Charama communities, and tribes of Narayanpur and Bastar celebrate it. It is celebrated from December to March. It is dedicated to Goddess Kesharpal Kesharpalin Devi.

Hareli, a Chhattisgarh harvest festival on Shravan Amavasya (July-Aug), worships Kutki Dai for good crops. It marks the Kharif season and kicks off the 75-day Bastar Dussehra Mahapaur tribal festival.

Bhoramdeo festival is organized near the Bhoramdeo temple located in the Kabirdham district, which is also known as the Khajuraho of Chhattisgarh.

Bastar Lokotsav comes after the end of the rainy season. It is held in the Jagdalpur region of Chhattisgarh.

Maati Ti'ar is celebrated in the Bastar district of Chhattisgarh. Different names in different regions:- Maati Ti'ar in central Bastar, Beeja Pandum in the South Bastar.

Goa

Shigmo festival (Shigmotsav) is the spring festival celebrated by the Konkani diaspora of Goa.

Ladainha is a thanksgiving festival celebrated in Goa.

The Feast of St Francis Xavier (Goinchea Saibache Fest) is celebrated on the third of December every year.

Chikal Kalo is the name of a religious festival, which translates in Konkani as 'playing in the mud'.

Mando festival- Goa Celebrates a unique dance to Mando music, a fusion reflecting its Portuguese colonial past.

Fatorpa Zatra- Goa honors Shantadurga with a five-day December/January festival culminating in a four-day chariot procession.

Sao Joao festival is associated with Goa. It is an annual Catholic festival celebrated on June 24.

Grape Escapade is India's greatest wine festival celebrated in Goa.

Sunburn Festival is Asia's largest three-day Dance and Music festival. It is held in December every year in Goa.

Gujarat

Rann Utsav is celebrated every year in November onwards at Kutch (Gujarat) and marks this celebration till Holi.

Saptak Annual Festival of Music is an annual thirteen-day Indian classical music festival held in Ahmedabad, Gujarat. It is organized by the Saptak School of Music.

Divaso - Dadra and Nagar Haveli's Dhodia and Varli tribes celebrate Divaso, a ritual on Ashada Amavasya (no moon day) to honor Parvati and welcome the holy month of Shravan.

Modhera dance festival is held in the Mehsana district in the third week of January every year.

Other festivals: Mai Bhambhi (Jat Muslim), Abdapir (Rajput/Hindu), Jeshal Toral (Hindu), Pankerser (Ahir/ Rabari), Jogninar (Ahir).

Haryana

Gugga Naumi is the festival related to snake worship in the lunar month of Bhadon and celebrated in Haryana and some parts of Punjab, Rajasthan, Uttar Pradesh and Himachal Pradesh.

International Gita Mahotsav is celebrated mainly in Kurukshetra during November/ December.

Pinjore Heritage festival was started at the Pinjore Gardens in 2006.

Himachal Pradesh

Phulaich festival (festival of flowers) is also called Ookayand Festival or Ukyam Festival. It is celebrated in the Kinnaur district of Himachal Pradesh.

Doongri is Hindu religious festival celebrated at Hidimba Temple in honour of Hidimba Devi (wife of Bhima), situated in the sacred Dhungri forest in Manali, Himachal Pradesh.

Halda is a two-day festival celebrated by

the Lamas in January in Himachal Pradesh every year. It is a festival of lights celebrated in Magha Purnima. It is dedicated to Shiskar Apa, the Goddess of Wealth.

Gochi (Gotsi) festival is celebrated in the Chandra and Bhaga valleys of Himachal Pradesh. It is organized to celebrate the birth of a male child. It is usually celebrated in February every year.

Kullu Dussehra is the International Mega Dussehra festival observed in October in Himachal Pradesh. It was started in the 17th century by the local King Jagat Singh.

Chait is celebrated on the first day of the month of Chaitra in Kangra, Hamirpur and Bilaspur. Dholru, a special genre of songs, are sung on this occasion.

Sair and Nawala is a festival of thanksgiving and is held in September or October.

Bishu (Basoa) festival is celebrated to mark the onset of the summer season.

Sazo Festival is celebrated in the Kinnaur district. It is celebrated in January.

Magha Saaji is celebrated as Makar Sankranti.

Chaitraul is a cultural festival.

Pori Festival is a three-day festival. It is celebrated annually during the 3rd week of August.

Jharkhand

Rohini is a festival of sowing seeds in the field. It is celebrated in Jharkhand. There are some other festivals like Rajsawala Ambavati and Chitgomha also celebrated with Rohini.

Bhagta Parab is celebrated in Jharkhand between the period of spring and summer. It worships Budha Baba.

Tusu is a harvest festival held during the winter on the last day of Pus (Pausha) month. This festival is mostly seen in the area between Bundu, Tamar, and Raidih areas of Jharkhand.

Jawa festival is celebrated on the 11th of the moon in Bhadra month in Jharkhand. It is held for the expectation of good fertility and better households. The girls offer green melons to the Karam deity as a symbol of 'son'.

Sohrai (cattle festival) is a harvest festival of Jharkhand. It is celebrated after harvest and coincides with the festival of Diwali. It is celebrated by Sadan, Oraon, Munda and Santal among other tribes.

Sarhul - Marking the New Year and spring's arrival, Sarhul is a Jharkhand tribal festival (Oraon, Munda, Ho tribes) celebrated on the 3rd day of Chaitra month in the Hindu calendar.

Hal Punhya is a festival which begins with the fall of winter. The first day of Magh month, known as "Akhain Jatra" or "Hal Punhya", is considered as the beginning of ploughing.

Bandana is a festival celebrated in Jharkhand during the black moon of the month of Kartik (Kartik Aamavashya).

Jani-Shikhar festival is held once every 12 years. This festival advocates the right of equality.

Karnataka

Kailpodh (the festival of arms) is celebrated in the first week of September by the Kodava community of the Kodagu district in Karnataka. It is similar to the Ayudha Puja (worship of arms) that is celebrated in South India.

Basavanagudi Kadalekai Parishe (groundnut festival) - Farmers congregate at the Bull Temple each year to seek blessings for a good harvest.

Bengaluru International Film Festival held in February for about a week. The festival is a platform to bring together creative filmmakers and audiences.

Madikeri Dasara festival is celebrated in September- October. The highlight of the festivities in Madikeri is a colorful procession of beautifully lit temple cars.

Pattadakal Dance Festival is a three-day affair celebrated in January in Karnataka. Major Attraction Dances :- (Kathakali, Kuchipudi, and Bharatnatyam), Carnatic Music, and Craft Mela.

Gombe Habba (festival of dolls) is celebrated during Dasara/Navaratri each year in Karnataka. It is celebrated for 10 days and culminates on the day of Vijayadashami or Dasara. It is known as Kolu (Golu) in Tamil Nadu and Bommala Koluvu in Andhra Pradesh.

Kambala festival is celebrated to please the Gods for a good harvest in \

Karnataka. It is dedicated to Lord Kadri Manjunatha (an incarnation of Lord Shiva). It includes the splashing of the mud and racing across the fields a buffalo race.

Kaveri Sankramana is celebrated in October (the first day of Tula month as per the Hindu calendar). It is celebrated at Talakaveri in the Kodagu district in Karnataka.

Huthri is a harvest festival celebrated in the months of November and December in Karnataka.

Hampi festival (vijaya utsav) is Organized annually by the Government of Karnataka, the festival is celebrated over 2 to 3 days.

Vairamudi Utsava is an annual festival that takes place in the Melukote town of the Mandya district.

Karaga festival - 11 days honoring Draupadi (ideal woman, Goddess Shakti) on Chaitra full moon (March/April) at Dharmaraya Swamy Temple.

Kerala

Vettukad is an annual festival at the Madre-de-Deus church (Vettukad church), Kerala. It is observed for 10 days.

Guruvayur Ekadashi is a cultural festival of Kerala.

Nishagandhi Festival is the annual mega-cultural event. It is a one-week festival held at the open-air Nisha Gandhi Theater in Kerala. It is organized by the Kerala Tourism Department.

Chandanakuda Mahotsavam - A 10-day festival (Jama Dul Akbar, Hijri calendar) with flag-hoisting, self-purification rituals (Ratheeb), swordplay art (Dahara Muttu), and Vadithallu.

Ezhara Ponnana festival (Feb-Mar) features a procession of seven - and -a-half golden elephants on the 8th night at Ettumanoor Mahadeva Temple, culminating in the Arattu ceremony on the 10th day.

Thrissur Pooram is an annual Hindu temple festival held in Thrissur (Kerala). It is held at the Vadakkunnathan (Shiva) Temple in Thrissur every year on the Pooram day (the day when the moon rises with the Pooram star) in the Malayalam Calendar month of Medam (April-May).

Onam, a 4-10 day harvest festival in Kerala (Aug-Sept), marks the return of King Mahabali's spirit and celebrates the Vaman avatar of Vishnu. It starts in Chingam, the first Malayalam month.

Kumarakom boat race is conducted annually on Sree Narayana Guru Jayanthi Day in Kerala. It falls in August or September (the asterism of Chathayam in the Malayalam month of Chingam). It took place on the Vembanad Lake in September during Onam.

Vishu (Malayali New Year) is a Hindu festival celebrated in the state of Kerala and some parts of Karnataka. It marks the first day of Medam, the first month of the Malayalam Calendar.

Soorya Classical Dance Festival is hosted in Thiruvananthapuram (Kerala). It is organized by Soorya Stage and Film Society. It is conducted for ten days every year.

Makaravilakku festival is celebrated at the sacred grove of Lord Ayyapa. It is an annual festival held on Makar Sankranti in Kerala.

Puthari (Huttari) is an annual rice harvest festival celebrated by the Kodava tribe in the Kodagu region (North Malabar) in Kerala. It is celebrated on a full moon day in the Kodava month of Birchiar (November - December).

Arattu festival - Kerala's Sree Sundareswara Temple holds an eight-day festival (March-April) culminating in the Arattu ritual (holy bath) with a finale of elephant processions and fireworks.

Arthunkal Perunnal (Perunnal means feast) is the ten-day annual feast at St. Sebastian's church at Arthunkal, a coastal region in Alappuzha District in Kerala.

Sabarimala Shrine festival is celebrated every year in the honour of Lord Ayyappa at Sabarimala on 14 January.

Attukal Pongala festival is celebrated at Attukal Bhagavathy Temple.

Manjinnikara Church festival is celebrated in the Manjinnikara Church to pay respect to Ignatius Elise III, the holy patriarch of Antioch.

Adoor Gajamela is a ten-day annual celebration at the Parthasarathy Temple in Adoor from January to February. In this festival, nine elephants get a chance

to compete with each other for the title of Gajaraja Pattam.

Attuvella Mahotsavam is a 2-day water carnival (March-April) honoring Goddess Bhagavathy.

Chettikulangara Bharani festivals are celebrated at Chettikulangara Temple near Mavelikara in Alappuzha. It is held during the Malayalam month of Kumbham (February - March). Kettukazhcha - primary attraction.

Machattu Mamangam five-day festival celebrated at the Machattu Thiruvannikkavu temple dedicated to Goddess Bhagavathi. It is celebrated in the Malayalam month of Kumbham (Mid February).

Kalpathi Ratholsavam is an annual ten-day chariot festival conducted in November. It is conducted at the Sri Viswanatha Swamy temple.

Thirunakkara Arattu is an annual 8-day festival at the Thirunakkara Mahadeva Temple. It starts every year on the 1st of the Malayalam month of Meenam (March-April).

Kanjiramattom Kodikuthu is an annual festival held in January at the Kanjiramattom Mosque.

Madhya Pradesh

Bhagauria (Bhagoria Bhils) is a harvest festival celebrated by the Bhil tribe in Madhya Pradesh. It is celebrated on the occasion of Holi when the rabi crop is ripened.

Laru Kaj is the festival of Gonds which is associated with the pig sacrifice. It is held in Madhya Pradesh. The ceremony is considered the pig's wedding in honour of Narayan Dev.

Lokrang is a five-day festival that begins every year on the 26th of January. It is organized by the Madhya Pradesh Adivasi Lok Kala Academy. Tribal folk dance and exhibitions of art and crafts.

Khajuraho Dance Festival is a six-day festival organized by the Kala Parishad of the Madhya Pradesh government. It is held at Khajuraho temples. It was started in 1975.

Ghadalya, the main Malwa (Madhya Pradesh) festival, sees girls sing and collect grain in a perforated pot, house-to-house. A similar boys' festival is

Chhala, and in Bundelkhand, it's called Tesoo.

Mandu festival is an annual festival celebrated from December - January in Madhya Pradesh. The first Edition was organized from 28 December 2019 – 3rd January 2020.

Orchha Utsav is celebrated in Madhya Pradesh. The first edition of the Orchha Festival was organized from 6 to 8 March 2020. It aims to promote "Orchha" the Architectural and Heritage city as an International Tourist destination.

Malmaas festival is celebrated in May.

Maharashtra

Gudi Padwa - Celebrates the Marathi New Year and harvest season with a symbolic victory flag (gudi) marking a prosperous year ahead. It falls on the first day of the Hindu lunar calendar.

Banganga Festival is a two-day festival held every year at Malabar Hills in Mumbai, Maharashtra.

Bhaubeej is celebrated on the fifth day of Diwali in Maharashtra. This festival symbolizes the strong relationship between brother and sister. Basundi puri or shrikhand poori is prepared for this occasion.

Elephanta Festival is a festival for dance and music organized by the Maharashtra Tourism Development Corporation. It is organized in February every year on Elephanta islands in Mumbai, Maharashtra.

Sula fest is India's biggest vineyard music festival. It is held in the first week of February in Nashik, Maharashtra.

Marabats and Badgyas- This festival is celebrated on Bhadrapada Shukla Pratipada (approx in August- September) of the Indian lunar calendar.

Vat Poornima is celebrated in the Indian month of Jyestha (May-June). Main God:- Goddess Savitri.

Palkhi Festival : A 22-day pilgrimage in June honoring the god Vithoba. Devotees (warkaris) walk from various locations to Pandharpur carrying palkhis (sacred shrines).

Narali Poornima marks the end of the monsoon season in Maharashtra. It is celebrated on the full moon day of the Indian month of Shravan (July-August).

Ellora Ajanta Dance Festival organized by Maharashtra Tourism Development (MTDC) is celebrated every year for three days.

Kala Ghoda Arts Festival is organized in February every year for nine days in a row.

Ashadhi Ekadashi is celebrated in honour of the God Vishnu. It refers to the 11th tithi of a lunar month. It falls during the Shukla Paksha (Brighter fortnight), during the month of Ashadha. It is also called Devshayani Ekadashi.

Osho Monsoon festival is an international festival of music and meditation held in Pune. It is a 5-day annual event starting from 11 August till 15 August.

Manipur

Kang Chingba is a 10-day festival in July, similar to Puri's Ratha Yatra. The Meitei community celebrates with a procession of deities Jagannatha, Balarama, and Subhadra in chariots (kang) at the Govindjee temple.

Gaang Nigaai (Chakaan Gaan-Ngai) is a post-harvest festival of the Zeliangrong people of Assam, Manipur, and Nagaland.

Lui-Ngai-Ni is celebrated on the 15th day of February every year in Manipur. It is a seed-sowing festival of the Nagas.

Heikru Hidongba is Manipur's boat race festival! Held on the 11th day of their Langban month (around September).

Lai-Haraoba (Merry making of the Gods) is a festival associated with the Meitei people celebrated to please the Umang Lai, the traditional deities of Sanamahism. It is celebrated in May in Manipur.

Sangai Festival (since 2010) celebrates its culture with art, music, dance, sports, and food from Nov 21-30th, named after the state's brow-antlered deer (Sangai).

Yaoshang festival is celebrated in Manipur. It is a five-day long event celebrated on the full moon day of the month of Lamta (February - March).

Cheiraoba (Manipuri New Year) is a festival celebrated by the Meitei tribe of Manipur in April. It marks the beginning of the lunar new year for the Meiteis.

Shirui Lily festival is a 4-day celebration (April-May) honoring the state's unique Shirui Lily, a rare flower that blooms in the Shirui Hills and can't be grown anywhere else!

Chumpha is celebrated for seven days in the month of December after the harvest.

Ningol Chakkouba is observed on the second day of the new moon in the Manipuri month of Hiyangei (October /November). It is a social festival of the Meiteis.

Mera Houchongba is celebrated every year during the Manipuri month of Mera (October) to promote peace and solidarity.

Meghalaya

Wangala Festival (Hundred Drums Festival) is the most popular harvest festival among the Garos of Meghalaya, Assam, and Nagaland.

Khasis celebrate Shad **Suk Mynsiem** ("dance of joyful hearts") in April. This 3-day harvest festival at Weiking Grounds features traditional dance honoring a bountiful year.

Nongkrem is a post-harvest tribal festival celebrated by the Khasi tribe of Meghalaya.

Chad Sukra is celebrated by the Jaintia people every year in April or May in Meghalaya. It marks the beginning of the sowing season.

Strawberry Festival is celebrated to promote the farming of fruit. It is held in February every year in Meghalaya.

PNars celebrate **Behdienkhlam** in July, a harvest festival that chases away evil spirits and brings good fortune.

Laho is a dance-festival of the Jaintias.

Mizoram

Pawl Kut is a harvest festival celebrated either in December or January for two days. The Mara tribes of Mizoram celebrate it.

Khuado Kut is celebrated by the Paite community and observed on full moon nights after the harvest.

Anthurium Festival is a three-day festival celebrated in Mizoram. It is held annually in September. It is held at Reiek Mountain.

Chapchar Kut is a festival of Mizoram. It is a spring festival celebrated during March every year after the completion of their most arduous task of jhum operation.

Lyuva Khutla is celebrated after completion of arduous jhumming tasks (slash and burn).

Hlukhla Kut is celebrated by the Lai people. It is celebrated during March after completion of their most arduous task of jhum operation.

Nagaland

Tuluni (Anni) is a harvest festival celebrated by the Sumi Naga tribe of Nagaland.

Moatsu festival - Ao tribes celebrate Moatsu, a 3-day harvest festival in early May. It features Sangpangtu, where a bonfire unites the community for feasting and celebrating the harvest.

Sekrenyi (Phousanyi) is a major annual festival of the Angami Nagas of Nagaland. It is observed for ten days in February.

Yemshe, a two-part harvest festival, thanks the heavens for a bountiful crop. Celebrated by Nagaland's Pochuri tribe (Khuri, Kupo, Kuchu) in late Sept-early Oct.

Mongmong - 6-day harvest festival (Sept) by Sangtam Nagas. Celebrates unity, good harvest (Lijaba deity) with rituals & meals.

Hornbill Festival (Festival of Festivals) is an annual festival celebrated from 1 to 10 December in Kohima, Nagaland. It was celebrated for the first time in 2000. The festival is named after the hornbill bird.

Tsokum Samai is celebrated by the people of Nagaland to invoke blessings for a rich harvest.

Nazu festival is a very colourful festival celebrated by the Pochury tribal groups just before the annual sowing of seeds.

Suhkruhnye is celebrated by the Chakhesang Tribe.

Monyu is celebrated by Phom Naga.

Ngada is an agricultural festival celebrated by the Rengama Tribes.

Metemneo is a harvest festival celebrated by the Yinchunger tribe.

Aoleang is an annual festival of the Konyaks from northern Nagaland. It is celebrated in the first week of April in the Konyak district.

Bushu Jiba is celebrated by the Dimasa Kachari tribes in Dimapur. It is an annual post-harvest festival celebrated in the last week of January with gusto. People worship Sibrai in this.

Hega is a matrimonial festival celebrated from 10 to 15 February. It is celebrated by the Zeliang tribe of Peren.

Miu is a Naga festival celebrated by the Khiamniungan Nagas, a minority tribe in Nagaland. It is held in the first week of May every year.

Naknyulem Festival is a festival celebrated by the Chang tribe in Tuensang. It is celebrated every 11th month of the Chang calendar which is at the end of July.

Tokhu Emong is a nine-day festival celebrated in the 1st week of November every year by the Lotha tribe. The Lotha Nagas (Kyong) is a major Naga ethnic group native to the Wokha District of Nagaland.

Ahuna is a post-harvest festival celebrated by the Sumi Naga tribe.

Odisha

Sume-Gelirak is an important festival for the Bondas of Koraput (Odisha). It lasts for ten days during which they worship their traditional Gods and demons.

Nuakhai is a harvest festival celebrated primarily in western Odisha by farmers to thank 'Mother Earth'. It is observed to welcome the new rice of the season on the day after Ganesh Chaturthi.

Raja Parba is an agricultural festival. It is held for three days in the month of June.

Bali Jatra - Asia's largest open fair! Celebrated for a week around Kartika Purnima (Oct-Nov) on the Mahanadi River to honor Lord Kartikeswar.

International Sand Art Festival is celebrated every year. It was started in 2011 on Chandrabhaga beach, a short distance from Konark. It was a part of the Konark Dance Festival that was organized during the same time.

Dhanu Yatra - World's largest open-air theater! This 11-day festival (Dec-Jan) in

Bargarh reenacts Lord Krishna's life - from birth to slaying Kansa.

Eco Retreat festival is held every year in Odisha. The objective of this festival is to attract tourists as well as private sectors for investment.

Bali Trutiya (Tritiya) festival falls on the third day of Shukla Paksha of the Bhadra month (usually falls one day before Ganesh Chaturthi). It is dedicated to Lord Shiva and Parvathi. Tritiya fasting is observed by married women in Odisha.

Sakraat festival is celebrated with the 'Budigali' or 'Kalasi' dance in Odisha where women take part in this dance, carrying decorative pots on their heads.

Bija Pandu - Koya tribe's harvest festival (Mar/April) honors Earth Goddess for a prosperous season.

Konark Dance Festival is a five-day dance festival held every year in the month of December mostly from 1 to 5 in the backdrop of the Sun temple in Konark, Odisha. It was started in 1986.

Rath Yatra (Car Festival) is the sacred journey of Lord Jagannath with brother Balabhadra and sister Subhadra from the Jagannath Temple to the divine shrine called the Gundicha Temple.

Nabakalebara is celebrated in Odisha. The year of Nabakalebara is when the full moon occurs twice during the month of Ashadha. It is the ritualistic recreation of the wooden icons of four Hindu deities (Jagannath, Balabhadra, Subhadra, and Sudarshana) at Jagannath Temple, Puri.

Ekamra Utsav is held to promote the rich cultural heritage of Orissa. It is dedicated to Lord Shiva. It lasts for 10 days.

Bhaijiuntia is celebrated on the Maha Ashtami Day of Durga Puja, by undertaking total fasting by girls and women.

Khudurukuni Osa is a ritual meant to be practised by unmarried girls and women every Sunday of the month of Bhadrava (August-September), for the health, happiness and prosperity of their brethren.

Samba Dashami is a unique ritual observed on the 10th day during the Shukla Paksha in Paush month (December - January) as per traditional Odia calendar.

Prathamastami is a rite held in Odisha praying for the life and prosperity of the eldest child of a family.

Chatar Jatra is celebrated by the Khond People from the Kalahandi district. It worships Goddess Manikeswari. It starts on the night of Durga Ashtami.

Sital Sasthi is a sacred Hindu festival. It celebrates the marriage of Lord Shiva with Lord Parvati.

Punjab

Fazilka Heritage festival is an annual art, cultural and food festival celebrated in Punjab.

Shaheedi Sabha (Three day annual festival): Honors the martyrdom of Guru Gobind Singh's sons at Gurdwara Fatehgarh Sahib (December).

Punjabi Teej (Teeyan)-13-day monsoon festival for daughters, sisters, and good harvests (Sawan month).

Hola Mohalla - A 3-day Sikh festival (March) following Holi. Celebrated with mock battles, drills, and religious ceremonies in Anandpur Sahib to honor Guru Gobind Singh's legacy.

Babe-da-viah is a tradition of celebrating the wedding anniversary of Guru Nanak Dev Ji.

Kila Raipur festival (Olympics of Punjab) is a Rural Sports Festival.

Rajasthan

Maru Mahotsav (Jaisalmer desert festival) is an annual event held in the Hindu month of Magh (February), three days before the full moon.

Elephant Festival is a traditional festival held every year at Jaipur Polo Ground (Rajasthan) on the day of Holi.

Gangaur (Rajasthan, Madhya Pradesh, Gujarat, West Bengal): Spring festival where women worship Gauri (Parvati) for marital happiness and good harvest.

Bundi Utsav is celebrated every year in the month of Kartik (November) in Rajasthan.

Bikaner Camel festival is an annual event, celebrated on the second Saturday & Sunday of January month.

Kajli Teej is held every year in the month of Bhadra (July-August). This week-long

celebration filled with gaiety and fanfare pays homage to Goddess Uma.

Momasar Utsav is celebrated by artistes and artisans from all over Rajasthan.

Matsya festival of Alwar is held in November over two days.

Mewar Festival is held every year in the month of March- April. It marks the advent of spring. It coincides with the Gangaur Festival.

Urs festival is an annual festival held at Ajmer, Rajasthan. It commemorates the anniversary of the death of Sufi saint Moinuddin Chishti. It is held over six days. It is celebrated in the seventh month of the Islamic lunar calendar.

Marwar Festival is celebrated for two days during the full moon of Sharad Purnima, which usually falls in September or October. It is celebrated annually in the city of Jodhpur.

Sikkim

Guru Rinpoche's Thrunkar Tshechu is the birth anniversary of saint Lord Padmasambhava.

Indra Jatra is celebrated by the Newar community.

Barahimizong is a festival observed by the Mangar Community in Sikkim. The first Barahimizong Festival was held in 1998. It is celebrated every year on the fifth day of Mangar Purnima (December).

Kagyed dance festival, celebrated annually in Sikkim falls on the 28th and 29th day of the tenth month according to the Tibetan calendar and December month according to the English calendar. It is celebrated by Buddhists in the name of eight tantrik Gods and Goddesses.

Sonam Lhosar is celebrated as the New Year of the Tamang Community.

Bumchu Festival (Sikkim): Unique ceremony at Tashiding Monastery (Feb-Mar). Monks open a sacred vase to predict fortunes for the year using the water level.

Tamu Lachar festival is celebrated as the New Year by the Gurung Community in Sikkim.

Losoong (Sonam Losoong) is the Sikkimese New Year of the Bhutia tribe. It is a harvest festival celebrated every year in December.

Saga Dawa (Sikkim, Bhutan): "Triple Blessed Festival" honoring Buddha's birth, enlightenment, and nirvana (May -June). It's the holiest month for Buddhists in the region.

Maghe Sankranti is celebrated on the first day of the tenth month of the Bikram Sambat calendar. It is observed for three days, usually in mid-January in Sikkim. A bathing festival called Makkar is observed when people take a dip at the confluence of the Tista and Rangit.

Sakewa (Bhoomi puja) is celebrated by the Kirat Khambu Rai community.

Pang Lhabso is held on the 15th day of the 7th month of the Tibetan Lunar Calendar every year in Sikkim. It commemorates the consecration of Mount Khangchendzonga as the guardian deity of Sikkim.

Tendong Lho Rum Faat is one of the oldest festivals of the Lepchas and is usually held in August. The 3-day celebrations begin with the offering of prayers to Mount Tendong in South Sikkim.

Teyongsi Srijunga Sawan Tongnam is observed by the Limboo Community in Sikkim every year. It marks the birth anniversary of the famous 18th-century scholar Teyongsi Sirijunga.

Tamil Nadu

Pongal festival has the meaning 'to boil' in its name. It is a four-day harvest festival celebrated in South India, mainly in Tamil Nadu. It falls in mid-January every year and marks the auspicious beginning of the Uttarayan (the sun's journey northwards). Pongal is celebrated over four days - Bhogi Pongal, Thai Pongal, Mattu Pongal and Kaanum Pongal.

Thaipusam is celebrated by the Hindu Tamil community on the full moon in the Tamil solar month of Thai (January -February). It coincides with the Pushya star, known as Poosam in Tamil. It is also observed as Thaipooyam Mahotsavam in Shree Subrahmanya Temple in Alappuzha, Kerala. It is dedicated to the Hindu God Murugan (Karttikeya), the son of Shiva and Parvati.

Jallikattu is celebrated in the second week of January as a part of Pongal celebrations on Mattu Pongal day in Tamil Nadu. It is also known as Eru Thazhuvuthal and Mancuvirattu.

Mahamaham is a Hindu festival celebrated every 12 years in the Mahamaham tank located in the city of Kumbakonam in Tamil Nadu.

Chithirai festival is celebrated on the full moon day of the first month of the Tamil calendar Chithirai (April and May). It is celebrated in Madurai Temple in Tamil Nadu.

Float Festival (Theppotsavam) is celebrated at the famous Meenakshi Amman Temple in Madurai city of Tamil Nadu. It is celebrated on a full moon night of the Tamil month, Thai (January-February).

Puthandu (Puthuvarudam, Tamil New Year), is the first day of the first Tamil month, Chittirai (14 April) on the Tamil calendar. It is celebrated as a festival by Tamil Hindus of Tamil Nadu and Puducherry.

Valvil Ori Vizha festival of Tamil Nadu is celebrated in the Kolli Hills during August (17th and 18th day of Adi in Tamil Month) every year.

Kanthuri Festival is a 14-day event celebrated in Tamil Nadu. It is also known as the Nagore Dargah Festival. It commemorates the death anniversary of a Muslim saint. It is held at the Shrine of Saint Qadir Wali between July and August.

Natyanjali is an annual dance festival. It commemorates the Hindu deity, Shiva. It originated around 1981. It is organized by collaborative efforts of the Department of Tourism of Tamil Nadu and the Natyanjali Trust.

Modhweth is celebrated at the MuthanaduMund, near Thalaikundha, in the Nilgiris in Tamil Nadu. It is celebrated by the Toda tribe at the Moonpo Temple (the oldest Toda temple) in Muthanadu Mund.

Masi Magam (Maasi Makam) is a Tamil festival that is majorly celebrated by Tamilians in Pondicherry.

Thimithi (firewalking ceremony) is a Hindu festival originating in Tamil Nadu. It is celebrated a week before Deepavali, during the month of Aipasi (or Aippasi) of the Tamil calendar (Gregorian calendar months of October and November).

Arubathu Moovar festival is held every year in the Tamil month of Panguni (March-April).

Karthigai Deepam festival is celebrated in the Tiruvannamalai Temple, where a huge lamp is lit on top of the Tiruvannamalai hillock.

Telangana

Bonalu is a Hindu festival centered on the Goddess Mahakali from Telangana.

Peerla Panduga, also called Muharram, is an important festival in the state of Telangana.

Sammakka Saralamma Jatara (also Sammakka Sarakka Jatara and Medaram Jatara) is a festival to honour the Hindu Tribal goddesses, celebrated in the state of Telangana.

Dunnapothula Panduga (Sadar) occurs on the second day after Diwali. It is a buffalo carnival celebrated annually by the Yadav community of Telangana.

Sri Kurumurthy Swamy Jathara is celebrated in Sri Kurumurthy Swamy Temple, Mahabubnagar district.

Prataparudra Singaraaya Jatara is celebrated at the Lakshmi Narasimha Swami temple located near Karimnagar district.

Nagoba Jaatara is a Hindu tribal festival held in the Adilabad district. It is celebrated by the Mesaram clan of the Gond and Pardhan tribes for 10 days.

Peddagattu or Gollagattu Jathara is the festival celebrated every 2 years. Main deity :- Sri Lingamanthula Swamy (Incarnation of Lord Shiva), and Goddess Choudamma.

Edupayala Jatara is a three-day grand affair celebrated during Shivaratri in February.

Tripura

Lampura Puja is performed by the Noatia community of Tripura.

Kharchi Puja is celebrated at old Agartala (Tripura) in the temple premises of fourteen Gods (the dynasty deity of the Tripuri people). It includes a ritual of cleaning the earth.

Garia Puja is celebrated in the third week of April in Tripura. It is a seven-day festival to honor the deity Baba Garia. It is held on the seventh day of the month of Boishakh every year.

Ker Puja is celebrated in Tripura. It occurs two weeks after the Kharchi Puja to honor Ker, the guardian deity of Vastu Devata. 14 different deities are worshiped in this puja.

Ashokastami Festival (Unakoti) is held every year in February in Tripura. It is associated with the sacred dip in the holy river of the Ashtami Kunda.

Pilak festival is celebrated every year in the month of February/March.

Neermahal festival is celebrated every year in the month of August and December at Melaghr in Rajghat & Rudrasagar Lake.

Uttarakhand

Phool Dei is a significant harvest festival of Uttarakhand. This auspicious festival is celebrated to pay reverence to the Gods for the good crop year.

Harela is a harvest festival celebrated in Uttarakhand and some regions of Himachal Pradesh.

Jagda festival is held every year in August in the honour of Mahasu Devta in the Jaunsar-Bawar region.

Janopunya - People change their janeu (sacred thread worn across the body by Hindu men). The popular Bagwal fair is also held on this day in Champawat.

Kandali festival: This unique festival of Pithoragarh district coincides with the blooming of the kandali flower, which blooms every 12 years. Celebrated between August and October, during this festival, women symbolically destroy the plant, as per a folk legend.

Khatarua festival is celebrated by the pastoral and agricultural community in the Kumaon region.

Tehri Lake festival is a hot pot of culture, adventure, art, and music. This festival is India and Asia's biggest lake festival.

Ramman is a religious festival manifested in the form of a ritual theatre held at the twin villages of Saloor-Dungra (Chamoli). It is recognized by UNESCO in its Representative List of the Intangible Cultural Heritage of Humanity in 2009.

Uttar Pradesh

Lathmar Holi is specifically celebrated in the village of Barsana, located in the outskirts of the Mathura city.

Kicheri is a harvest festival and involves ritual bathing, celebrated majorly in Uttar Pradesh.

Taj Mahotsav in Agra is the most famous fair of art, craft, culture & cuisine organized in the month of March.

West Bengal

Durga Puja (Durgotsava or Sharadotsav) is a ten-day Hindu festival in the month of Ashvin. It is held in September or October in Kolkata (West Bengal) and the rest of India. It was included in the UNESCO List of Intangible Cultural Heritage in 2021. It is observed to celebrate the triumph of the Goddess Durga over Mahishasura.

Jagadhatri Puja is worshiped in the West Bengal region.

Gajan is a Hindu festival celebrated in West Bengal. It is associated with such deities as Shiva-Parvati, Neel, and Dharmaraj.

Bhai Phonta in this festival sisters pray for their brothers, safety, well-being and success.

Ratha Yatra (cart festival) at Mahesh is the second oldest chariot festival in India (after Ratha Yatra at Puri) and the oldest in Bengal. Celebrated in the month of July.

Poila Baishakh is the first day of the Bengali calendar, coinciding with 14 April every year. The traditional Bengali greeting on this occasion is 'Shubho Noboborsho', which literally means 'Happy New Year'.

Jamai Shasti is celebrated on the sixth day of Shukla paksha in Jyestha month (May or June). It personifies the beautiful relationship of a son-in-law with his in-laws.

Union Territories

Andaman and Nicobar Islands

Ossuary festival (Pig Festival) :- It is celebrated in Car Nicobar. It is dedicated to the departed head of the family. Dance:- Nikobarese Dance.

Island Tourism festival begins in January every year in the tropical paradise. It is a 10-day carnival.

Dadra and Nagar Haveli and Daman and Diu

Barash is mostly celebrated by the Varli and Kokna tribes and is similar to Diwali.

Masimagam festival is celebrated at Dadra and Nagar Haveli, on a full moon day of the 'Masi' month (February to March).

Villianur Temple Car festival is celebrated during the middle of the month of May.

Tarpa Festival is celebrated every year in December.

Delhi

Phool Walon Ki Sair or Sair-e-Gul Faroshan (procession of the florists) is an annual three-day festival held in September in the region of Mehrauli, Delhi.

Jammu and Kashmir

Tulip festival is celebrated in Indira Gandhi Memorial Tulip Garden at Srinagar. Large variety of Tulips are displayed on this occasion.

Navreh is the lunar new year that is celebrated in Jammu and Kashmir.

Shikara Festival takes place in July or August to promote tourism. It is celebrated in Dal Lake.

Ladakh

Tak-Thok (Tse-Chu) is a festival in Ladakh. It is celebrated on the 10th day of the Lunar Tibetan Calendar (July and August). It is celebrated by Chham dances at cave Gompa.

Yuru Kabgyat is a two-day festival at Lamayuru that is celebrated in the month of July-August.

Matho Nagrang is celebrated on the 15th day of the first month of the Tibetan calendar in the Matho Monastery of Ladakh.

Dosmoche festival was started by the kings of Ladakh.

Ladakh Harvest festival is celebrated in the month of September. This 2 week long festival features masked and folk dances, theater shows, display of local handicrafts and archery competitions.

Sindhu Darshan is a three-day long

festival held every year in the month of June.

Spituk Gustor Zanskar is celebrated in the special monasteries of Ladakh in October-November.

Religious festivals

Muslim festival

Eid al-Adha (Eid ul-Zuha) is celebrated on the 10th day of the final month Dhu al-Hijjah (June-July) in the Islamic calendar. It is also called Bakr-Id due to the tradition of sacrificing a goat or bakr.

Eid-ul-Fitr (Holiday of Breaking the Fast) is celebrated on the last day of the holy month of Ramadan.

Milad-Un-Nabi/Id-E-Milad (Barawafat) is a precious festival for Muslims, who celebrate the birthday of Prophet Muhammad. It is commemorated in Rabi' al-awwal, the third month in the Islamic calendar.

Muharram commemorates the martyrdom of Hussain Bin Ali, the prophet's grandson, in the war of Karbala in 61 Hijri (680 AD). It takes place in Muharram, the first month of the Islamic calendar.

Shab-e-Miraj (Night of Ascent, Al-Isra Wal-Miraj) is observed in the Holy Month of Rajab by Muslims all over the world.

Rabi' al-Awwal is celebrated in the third month of the Islamic calendar. It means "the first spring" in Arabic but it can fall in any season.

Chehlum (Arbaeen) is a Shia religious observance. It occurs forty days after the Day of Ashura.

Jain festivals

Mahavir Jayanti (Mahavir Janma Kalyanak) is a Jain festival that celebrates the birth of Mahavir, the twenty-fourth and last Tirthankara of present Avasarpiṇi. It is celebrated in the month of Chaitra.

Akshay Tritiya Tapa commemorates Rishabhdev, the first Jain Tirthankara. It is the spring festival.

Maun-Agiyara (Maun Ekadashi) falls on the 11th day of the Magshar month. It is marked by complete silence, fasting and Meditation.

Paryushana (Das Lakshana) is the annual holy event for Jains. It is usually celebrated in August or September in the Hindi calendar. The last day is called Samvatsari. The duration of Paryushana is for 8 days for Svetambara Jains. and 10 days for the Digambara Jains.

Mahamastakabhisheka is a Jain festival held once every twelve years in the Karnataka town of Shravanabelagola.

Navapad Oli is nine days of semi-fasting. During this time, Jains eat only one extremely simple meal every day.

Buddhist festivals

Buddha Purnima (Vesak Poya) is celebrated on the first full moon day of the Vaisakha month. It commemorates the birth, enlightenment (Nibbana), and death (Parinirvana) of Gautama Buddha.

Galdan Namchot festival marks the birthday of a great Buddhist saint Je Tsongkhapa.

Chethiyagiri Vihara festival is a Buddhist festival held at Sanchi (Madhya Pradesh) in late November every year.

Drukpa Tsheshe commemorates the auspicious day on which Lord Buddha delivered his first sermon of four Noble Truths viz Dukha Satya, Samudaya Satya, Nirodha Satya, and Marga Satya to his five disciples.

Sangken (Buddhist festival) is celebrated by Khamtis (one of the oldest tribes of Lohit District, Arunachal Pradesh).

Paro Tshochu is a Buddhist Festival and one of the major annual events of Bhutan.

Hemis festival marks the birth of Guru Padmasambhava. It is held in Hemis Gompa monastery in Ladakh. It is a two-day event and Influence of Buddhism religion. It is celebrated on the 10th day of the fifth month (June-July) of the Tibetan Lunar Calendar.

Lhabab Duchen occurs on the 22nd day of the 9th month on the Tibetan calendar.

Ullambana festival (Ghost Festival) is the festival celebrated by Buddhists in different countries including China, Japan, India, Thailand, and Malaysia.

Lumbini Festival is organised to honour Buddhism. It Starts from the 2nd Friday in the month of December every year and continues for 3 days.

Christian festivals

Good Friday (Friday before Easter) is the day on which Christians annually observe the commemoration of the Crucifixion of Jesus Christ.

Easter is a principal festival of the Christian church, which celebrates the Resurrection of Jesus Christ on the third day after his Crucifixion. Easter is celebrated on Sunday. The custom of decorating eggs is practiced during this festival.

Christmas is an annual festival commemorating the birth of Jesus Christ, observed on December 25.

Sikh festivals

Guru Nanak Jayanti (Gurupurab, Guru Nanak's Prakash Utsav) celebrates the birth of the first Sikh guru, Guru Nanak.

Parsi festivals

Navroz is celebrated by Parsis as Parsi New Year (Iranian New Year). The central feature of this festive day is visiting Fire Temple and offering prayers to deities Khorshed and Meher, who are considered the presiding deities of the sun and moon respectively. It is a festival of Zoroastrian origin. The day is also known as Jamshed-i-Nowruz after the Persian King, Jamshed, who introduced the Parsi calendar.

Pateti (means repentance) festival is followed by the Parsi New year.

Zarthost No Deeso is celebrated on the 11th day (Khorshed) of the 10th month (Dae). It is a commemoration of the death anniversary of the prophet Zoroaster.

Sindhi festivals

Cheti Chand is a spring harvest festival. It marks the beginning of the Lunar Hindu New Year for Sindhi. It falls on the first day of the year, in the Sindhi month of Chet (Chaitra). Main deity - Jhulelal.

Chailo Sahib or Chaliha is a forty-day festival celebrated in the month of July-August. This festival is celebrated for the Sindhi people to please their deity Jhulelal. During this period many Sindhi devotees keep fast for forty days.

Fairs

Kumbh Mela is the largest peaceful congregation of pilgrims on earth, during which participants bathe ('Shahi Snan') or take a dip in a sacred river.

This fair starts on the day of Makar Sankranti. This fair is held every 12 years, with the location of the observance rotating among four pilgrimage sites on the four sacred rivers- Ganga at Haridwar, Triveni Sangam at Prayagraj, the Godavari at Nashik, and Shipra at Ujjain. Ardha (half) Kumbh Mela is held at only two places, Haridwar and Allahabad, every sixth year and a Maha Kumbh is held after every 144 years after 12 Purna Kumbh Mela. In 2017, it was added to the UNESCO'S List of Intangible Cultural Heritage.

Several holy men from different Hindu sects attend the mela, such as the Nagas (who do not wear any clothes), Kalpwasis (who bathe thrice a day) and Urdhawavahurs (who believe in putting the body through severe austerities).

Fairs of Indian States

Andhra Pradesh

Kotappakonda Tirunalla Fair is a two-day fair celebrated at Sri Trikoteswara Swamy temple on the eve of Maha Shivratri.

Assam

Sualkuchi Fair is held at Sualkuchi town of Assam. Sualkuchi is famous for its 'muga' silk.

Ambubachi Mela, is an annual Hindu mela held at Kamakhya Temple in Guwahati, Assam. It is celebrated in the month of June every year.

Jonbeel Mela is a three-day annual indigenous Tiwa Community fair held the weekend of Magh Bihu at a historic place known as Dayang Belguri at Joonbeel in Assam.

Bihar

Vaishali Fair is held in Chitra Trayodashi in Vaishali on the occasion of Mahavir's birthday.

Sitamarhi Fair is held in the Chaitra month on Ram Navami every year. At this fair, Ram-Janki marriage is celebrated.

Bounsi Fair is held in Banks at Mandar

Hill foothills. The fair starts from 14th January (Makar Sankranti day) every year and continues for a month.

Simariya Fair is organised at Simariya village of Begusarai. It is held twice a year for the Ganga bath and later on giving alms.

Sonepur Cattle Fair is one of Asia's largest cattle fairs, organised at the confluence of two rivers, the Ganga and Gandak. It begins on Purnima (full moon night) and Kartik month. It is associated with the Hariharnath temple.

Malmas Fair is organised at Rajgir for taking bath in warm kund (Hot Springs).

Buxar Fair is organised in Chaitra Shukla Paksha Naavami (Ramnavami).

Sahodara Fair is organised by Tharus in the old temple of mother Subhadhra at Sahodara in West Champaran district.

Pitrapaksha Mela is held in Gaya in the month of September.

Saurath Fair is held in Madhubani district of Bihar. This fair is also known as the Marriage Market that is held in the Jayestha Assad month.

Baba Brahmeshwar Nath Fair is organised on Shivratri and Baishakh Krishna Paksh Ekadashi twice a year at Buxar district, Bihar. It is an animal fair in which people first pour water on Brahmeshwar Nath and then throng fair.

Diwali Fair/Pawapuri Fair is organised at Nalanda district, Bihar. It is celebrated during Diwali to represent the Nirvana of Burdhan Mahavira.

Patna Sahib Gurudwara Fair is organised on the birthday of 10th Sikh Guru Govind Singh at Takht Sri Harmandir Sahib Ji Patna.

Urs of Makhdum Sahib in Maner Sharif is located in Danapur Subdivision, Patna. This place bears two well-known Mohammedan tombs, that of Shah Daulat or Makhdum Daulat, known as Chhoti Dargah, and the other that of Sheikh Yahia Maneri or Makhdum Yahia, called the Bari Dargah.

Singheshwar Sthan Fair is organised in Madhepura district, Bihar on Shivratri at a famous Shiv temple in which people pour water over Shivaling.

Shravani Mela - This Mela is observed every year during the month of Shravan

(July-August).

Makar Sankranti Mela is held every year at Rajgir (a town in Bihar) in the month of January.

Rajgir Mahotsav- Rajgir is the ancient capital of the Magadha Empire in Bihar and is held sacred by both Jains and Buddhists because of its long association with Buddha and Mahavira.

Chhattisgarh

Champan Mela is held in the month of 'Magh', from January to February, every year in Raipur, Chhattisgarh. Champan is said to be the birthplace of Saint Vallabhacharya, the founder of the Vedanta philosophy of Shuddha Advaita (Pure Non-dualism). Mainly the devotees of Vaishnav religion visit the Champan Mela.

Sheorinarayan Fair is held every year in the town of Sheorinarayan (Janjgir -Champa) in the month of February during the Magh Purnima.

Narayanpur Mela is held in the Bastar area of Chhattisgarh. It reflects the state's tribal community.

Rajim Maghi Punni Mela (Kumbh Mela of Chhattisgarh) is one of the largest religious congregations in Chhattisgarh. Devotees arrive in Rajim a day ahead to partake in the special Puja performed at midnight and take holy dip in the 'Triveni Sangam' the confluence of the three sacred rivers namely Mahanadi, Pairi and Sondur rivers.

Gujarat

Shamlaji Fair (Kartik Purnima fair) is held in the month of November every year. The Shamlaji Temple is a renowned Vaishnav Shrine and the deity housed here is known by various names including Gadadhar (bearer of the mace) and Shaksi Gopal.

Chitra-Vichitra Mela is the biggest Tribal festival. Celebrated mainly by the 'Gharasia' and 'Bhil' tribes. It is celebrated in the month of March, fortnight after Holi, in Khed Brahma Taluka of Sabarkantha District.

Madhavpur Fair (Held on Ram Navami). is a celebration of the marriage between Lord Krishna and Rukmani, which is said to have taken place at Madhavpur (Porbandar). This mela shares its

connection to the Mishmi Tribe of Arunachal Pradesh.

Bhavnath Mahadev Mela is celebrated near the Bhavnath Mahadev Temple, situated at the foothills of Mount Girnar, in the Junagarh city of Gujarat. It is organized in the month of Mahashivratri and lasts for five days.

Rann Utsav is celebrated every year over a span of four months (October to February) in the Dhordo village of Kutch district in the state of Gujarat. It is famous for Bandhani Sarees.

Tarnetar Mela (Trinetreshwar Mahadev Fair) is organized in a village called Tarnetar, Gujarat. It is an annual three day fair, celebrated during the months of August-September.

Famous Fairs - Kavant Fair, Bhadrpad Ambaji Fair, Vautha Fair (Animal Fair).

Haryana

Kapal Mochan Mela (Gopal Mochan) is held on the eve of Kartik Purnima at Bilaspur (Yamuna Nagar). It is an ancient place of pilgrimage to both Hindus and Sikhs.

Chetar Chaudas Fair is held in the month of March or Spring season in Pehowa in Kurukshetra. All the pilgrims both Hindu and Sikhs, bathe in the holy tank of Saraswati.

Surajkund International Crafts Mela (Faridabad) is unique as it showcases the richness and diversity of the handicrafts, handlooms and cultural fabric of India, and is the largest crafts fair in the world. This crafts mela is held in February every year.

Masani Fair is held in the honour of the goddess of small - pox. Masani Temple is located at Gurugram.

Baldev Chhat Fair is celebrated in honor of the elder brother of Lord Krishna, named Balrama at Palwal. It is believed that Balram has ended the dominion of a demon named 'Palwasur' as he killed him and rescues this place from his evil clutches.

Mansa Devi Fair is held twice a year in March - April and September - October. This fair is held in Bilaspur village near ManiMajra.

Baba Shami Dayal Fair is held in the honor of the famous saint of Bhiwani at

Charkhi Dadri.

Other Fairs - Baba Mungipa Fair at Riwasa (Bhiwani), Shah Chokha Khorai Fair by Meo Muslims at Nuh, Markanda fair at Shahabad (Kurukshetra), Hatkeshwar Fair at Jind, and Shiva Chaudas fair at Ambala.

Himachal Pradesh

Manimahesh Yatra/Fair is a religious fair. Lord Shiva is the presiding deity of this fair.

Ashiwin Navratra Mela celebrated in Himachal Pradesh during the Navaratri.

Lavi Fair (Rampur) is a commercial fair held in Kartika (November).

Doongri Festival also known as Hidimba Devi Fair is celebrated in Kullu, Himachal Pradesh on the occasion of Basant Panchami.

Bawan Dwadshi (Vaman Dwadashi) Fair is celebrated at Sarahan (Pachhad) in district Sirmaur of Himachal Pradesh. It is celebrated to mark the incarnation of Lord Vaman, the Fifth incarnation of Lord Vishnu.

Nalwari Fair (Cattle Fair) is held in Bilaspur town in the month of March or April.

Minjar Fair has been declared as one of the state fairs of Himachal Pradesh. It is held on the second Sunday of the Hindu month of Shravana.

Famous Fairs - Rohru Fair, Bhoj Fair, Bharara Fair, Sipi Fair, Ladaracha Fair in Kaza (in the month of July marking the termination of summers). Dhungri Fair in Manali. Shoolini Fair in Solan (tribute to Shoolini mata). Sair Fair in Arki (known for BullFighting). Piploo Fair in Piploo (in the month of June). Renuka Fair in Renukaji (celebrated annually to commemorate the meeting of Lord Parshuram and his mother Renuka), Magha Saaji (Makar Sankranti).

Jammu and Kashmir

Purmandal Mela (February-March) - Three-day fair is held during the time of Shivratri.

Jhiri Mela (Farmer's fair) is celebrated to worship the holy soul of the farmer named Baba Jitu. It is held in the months of October-November.

Bahu Mela is a major festival held at the Kali Temple in Bahu Fort (Jammu) twice a year during (March-April) and (September-October).

Jharkhand

Bhadli Mela - On the day of Makar Sankranti It is organized in Itkhor in Chatra District. The ancient temple in Itkhor belongs to Goddess Kali and Lord Shiva.

Kolhaiya Mela is celebrated in the Hindu month of Magh, particularly on the day of Basant Panchmi.

Lawalong Mela is the biggest cattle fair in Jharkhand celebrated every year on the occasion of aghan purnima. It is celebrated in the month of November or December.

Famous Fairs - Kunda mela in Pratappur, Chatra mela organised during Durga puja, Tutilawa mela in Simaria, Belgada Cattle Fair in Simaria, Kundri mela in Chatra, Kolhua mela in Hunterganj.

Karnataka

Godachi fair in the month of Kartik, is held in the honor of Shri Veerabhadra.

Banashankari Fair is held to honour Devi Banashankari or Parvathi.

Sri Shidlingappa's Fair is observed on Shivaratri and the deity is taken in a palanquin accompanied by drummers (Dollu and Majalu) from several neighbouring villages.

The Yellamma fair reveres Devi Yellamma or Renuka, wife of sage Jamadagni.

Madhya Pradesh

Heera Bhumiya Mela is held in Gwalior region in the months of August and September. It is organised in memory of Hiranman Baba.

Simhastha Mela (Kumbh Mela) is held on the banks of the Shipra River in Ujjain. Simhastha fair is held at an interval of every twelve years. This fair is held from the full moon of Chaitra month to the Vaishakh Purnima.

Goddess Jageshwari (Jogeshwari) Fair is held at Chanderi in Ashok Nagar district. It is held annually in March-April near Hauz Khas Pond.

Mahamrityunjaya fair is held in Mahamrityunjaya temple located in Rewa district, on Basant Panchami and Shivaratri every year.

Barman's Mela is held on the occasion of Makar Sankranti at the well-known Ret Ghat located on the banks of the Narmada River in Narsinghpur district.

Dhamoni Urs is a fair held at the shrine of Mastan Shah Vali. It is organized in Sagar district located in Dhamoni. It is held in the month of April - May.

Fair of Kana Baba is held at the samadhi of Kana Baba in a village called Sodhalpur in Hoshangabad district.

Baba Shahabuddin Auliya's Urs is held for only four days in the month of February in Neemuch district. The tomb of Baba Shahabuddin is located here.

Mathaghoghara Fair - On the occasion of Shivaratri festival at the place called Bhairavnath in Seoni district, a 15-day monastery fair is organised.

Gotmar Mela is held in Pandhurna of Chhindwara district on the new moon day of Bhadrapada month and on the banks of the Jam river situated in the middle of Savargaon.

Gwalior Trade Fair is the trade fair in India. It was started in 1905 by the King of Gwalior, Maharaj Madhav Rao Scindia.

Fair of Pir Budhan is held every year in the month of August- September in sawnra area of the Shivpuri district, Madhya Pradesh. It is held in memory of Muslim saint Pir Budhan. This fair is 250 years old. The Mausoleum of Pir Budhan is also built here.

Fair of Rampapali- A five-day fair is organized on the occasion of the full moon of Kartik month on the banks of the Chandan river in Balaghat district.

Nagaji's Fair is organized in Porsa village of Morena district in the memory of Nagaji, the saint of Akbar.

The fair of Kaluji Maharaj takes place in Piplya Khurd in Khargone district (Western Nimar) for about a month. In this fair, keema bulls of Nimari breed are bought and sold.

Bhagoria Fair is held every year on the occasion of Holi in Jhabua district.

Fair of Chandi Devi - Chandi Devi is considered to be an incarnation of

Goddess Parvati at a place called Ghoghara in Sidhi district. This fair is held in March-April.

Sonagir Fair- On the occasion of Holi, a five-day fair is organized at the famous Shri Digambar Jain pilgrimage site at Sonagir, located in Datia.

Fair of Garibnath- In the memory of Baba Garibnath, a follower of the Nath sect, this fair is organized in the month of Chaitra in the village of Avantipur Badodiya in Shajapur district.

Sankua Fair is organized every year on the full moon of Kartik month at Sankua salt place situated on the banks of Sindh river in Datia district.

Punjab

Jor Mela - Annual three-day Shaheedi Jor Mela is held at Fatehgarh Sahib Gurdwara in memory of Sahibzada Zorawar Singh and Fateh Singh.

Baba Sodal Mela is held in Jalandhar city. The fair of Baba Sodal is associated with a small boy named Sodal, who is respected as a child-god.

Maghi da Mela is held at the holy city of Sri Muktsar Sahib each year in January or the month of Magh according to the Nanakshahi calendar.

Rauza Sharif Urs is celebrated in the memory of Sufi Saint Sheikh Ahmed Farooqi Sirhindi who was a disciple of Khawaja Baqi Billah.

Rajasthan

Kaila Devi Fair is held during the months of April and March in the Kaila village, it is celebrated for almost a fortnight. It is held on the banks of the river Kalisil in Kaila village.

Pushkar Fair (Pushkar Camel Fair) is an annual multi-day livestock fair and cultural fete held in the town of Pushkar. It is one of India's largest camel, horse and cattle fairs.

Beneshwar Fair is held in the month of January and February during the time of Shivarathri at Dungarpur.

Mahaveerji Fair is celebrated to honour the memory of Shri Mahavir Swami, the 24th tirthankara (saint) of the Jains.

Gogaji Fair which is held at the GogaMedi in Ganganagar district. This is

usually held in the month of August. Ramdevra Fair in Jaisalmer.

Urs Fair Organized on the death anniversary of the great Sufi Saint, Khwaja Moin-ud-din Chishti, it is held at the memorial of the holy saint in the city of Ajmer.

Gomti Sagar Cattle Fair (Jhalawar) takes place at the bank of Gomti Sagar. It is a popular place to buy animals of Malawi breed.

Nagaur Cattle Fair is the second biggest fair in India, held annually for eight-days during the month of January – February.

Veer Tejaji Cattle Fair is held in Parbatar village of Nagaur district. The fair is biggest in terms of income as the famous breed of Nagauri bulls and Bikaneri camels are sold here.

Kolayat Cattle Fair (Kapil Muni Fair) - Bikaner, Month of December.

Jhalawar Cattle Fair (Chandrabhaga Fair) is held every year on the bank of Chandrabhaga River in Jhalawar district. The Fair is celebrated on Kartik Poonima. Famous Malwa breed bulls are sold and purchased here.

Mallinath Cattle Fair is held in Tilwara of Barmer, on the banks of Luni River on the 11th day of Chaitra Krishna Paksh.

Ramdev Cattle Fair is a popular place to trade Nagauri breed bulls.

Gangapur Cattle Fair takes place in Gangapur near Bhilwara in Rajasthan.

Shri Jagannath ji Fair in Alwar city, an important religious fair. The fair starts with a procession of Lord Jagannath passing through the city.

Tamil Nadu

The cattle fair of Thiruvannamalai - It is held during the famous Karthikai Deepam festival is one of the busiest fairs and lasts for four days.

Other fairs- Tourism Fair Chennai, Madurai Book Fair.

Tripura

Poush Sankranti Fair is celebrated in the middle of January. People assemble every year on the occasion of the Uttarayan Sankranti for a holy dip in the river Gomati at its place of origination, known as Tirthamukh.

Uttarakhand

Bagwali Pokhar Fair is celebrated in Kumaon district on Yam Dwitiya (Bhai Dooj). It is celebrated in October /November.

Dudhiyal Devi Fair is celebrated every 12 years at Paunada village located in the Budha Kedar region of Tehri Garhwal district.

Nanda Devi Fair is organised in the Kumaon region in the month of September every year. Almora is the place where the main fair is held.

Baurani Fair is celebrated at Baurani in Pithoragarh town is located in the Kumaon region of Uttarakhand.

Chaiti Mela is held every year during Navratri at Kashipur, in the Nainital district of Uttarakhand.

Gauchar Mela is an annual trade fair held on 14th November in the hilly town of Gauchar, located on the left bank of Alaknanda River in the Chamoli district.

Haatkalika Fair is held on the ashtami of the Chaitra and Bhado, at Kalika temple at Gangolihat.

Jageshwar Fair is held on the fifteenth day of the month of Baishakh at the Shiva temple in Jageshwar.

Pandukeshwar Festival, locally known as Janti Fair is celebrated every 6 years in Pandukeshwar Temple situated on the left bank of Vishnu Ganga River. The fair which lasts for 9-11 days begins during Magh Sankranti.

Gaindi Khagoti Fair is celebrated in the Pauri district of Garhwal region of Uttarakhand. The fair begins on Vasant Panchami and continues till Baisakhi.

Gananath Fair is held at Gananath Temple in Almora.

Chandrabadni fair in Tehri district region of Uttarakhand.

Bagwali Pokhar Fair is celebrated in Kumaon, Uttarakhand on Yam Dwitiya (Bhai Dooj).

Jauljibi Fair is held every year from 14th-21st November at the confluence of Kali and Gori rivers.

Kamleshwar Fair is held every year on the 14th day of Kartik month (early November) which is known as "Vaikuntha

Chaturdashi".

Purnagiri Fair is held on Vishuwat Sankranti every year. Somnath Fair or Masi Fair is held every year in Almora district in the Kumaon region of Uttarakhand. The fair is held at Someshwar Mahadev Temple.

Uttar Pradesh

Gau Charan Fair is one of the famous fairs of Brij region. It is organised in Mathura on the eighth day of Kartik month ('Gopashtami'). In this fair cows are worshipped.

Dewa Sharif Fair is organised every year during the month of Kartik in Deva (a place near the Barabanki district) on the shrine of great Sufi Saint Waris Ali Shah.

Shakumbhari Fair is organised during Navratri, twice a year in Saharanpur.

Nauchandi Fair is organised in Meerut after a few days of Holi festival.

Agra Cattle Fair or Bateshwar Fair takes place near Agra on the banks of holy river Yamuna, every year during the Karthik month November.

Magh Fair is organised in Allahabad, near the confluence of the three rivers-Ganga, Yamuna and Saraswati during the month of Maagh (January/February).

Rath Fair is organised in Vrindavan, Mathura district of Uttar Pradesh in the month Chaitra. This fair is organised in the honour of Lord Ranganath.

Ashwin Amavasya Mela is held in Chitrakoot Dham, Bundelkhand.

West Bengal

Jalpesh Mela is organised on the occasion of Shivratri in the months of February and March at Maynaguri in the district of Jalpaiguri.

Ganga Sagar Mela is conducted at the mouth of the Hooghly river in West Bengal. It is the second biggest fair of India after the Kumbha Mela of India celebrated on Sagardwip (Sagar Island) in West Bengal.

The fair is a six-day fair held in the month of January and on the day of Makar Sankranti, a ritual of pilgrims taking bath in holy water and visiting nearby famous Kapil Muni Temple is performed.

Bhim Mela takes place on the 11th day

of the Bengali month Magha In Purba Medinipur.

Rash Mela is held at Shantipur and Nabadwip (Cooch Behar).

Sotimar Mela at Ghoshpara Kalyani and **Lalan Mela** at Kadamkhali village at Asannagar are also well-known.

Kenduli Mela is held at Kenduli in Birbhum district in January, known for the congregation of 'bauls' or wandering minstrels of Bengal.

Vasanta Utsav (Spring Festival) is held at Shantiniketan in the month of March.

Charak Gajan Melas are held in 17 to 18 places of Kolkata. This mela continues for seven days. Other important fairs of this category are Kandore at Garbeta PS, Basanta Roy of Raskundu, Hajarlinga Shiva of Nayagram, Kedarkunda fair of Chapaleswar Shiva at Debra.

Poush Mela is an annual fair and festival that takes place in Santiniketan, in Birbhum, marking the harvest season.

Musical Instruments

Musical instruments are the tangible and material representation of music which is an auditory art. Natya Shastra by Bharat Muni (composed between 200 BC and 200 AD) clubbed musical instruments into four groups- Avanaddha Vadya (membranophones or percussion instruments), Ghan Vadya (idiophones or solid instruments), Sushir Vadya (aerophones or wind instruments), and Tat Vadya (chordophones or stringed instruments).

Stringed Instruments

Instruments in which sound is produced by striking the strings made of iron, steel, brass or other metals as well as cotton, silk threads etc. The Stringed Instruments are divided into two broad categories - the plucked and the bowed.

Rudra veena / Been - Famous exponents - Zia Mohiuddin Dagar, Mohammed Khan Faridi, Ustad Shamshuddin Faridi Desai, Zahid Faridi Desai, Hindraj Divekar, Jyoti Hegde, R.V. Hegde, Ustad Abid Hussain Khan, Ustad Asad Ali Khan, Bande Ali Khan, Jamaluddin Khan, Murad Khan, Naubat Khan, Rajab Ali Khan, Wazir Khan, Zahid Khan, Krishnarao Kholapure, Sharada Mushti, Madhuvanti Pal, Dattatreya Rama Rao Parvatikar, Bindu

Madhav Pathak, Shrikant Pathak, Peter Row, P. D. Shah, Carsten Wicke, Dr A P J Abdul Kalam.

Sarod - Famous exponents - Ustad Amjad Ali Khan, Ustad Allaaddin Khan, Ustad Ali Akbar Khan, Brij Narayan, Aashish Khan, Bahadur Khan, Jotin Bhattacharya, Buddhadev Das Gupta, Radhika Mohan Maitra, Vasant Rai, Hafiz Ali Khan, Rajeev Taranath, Pandit Vikash Maharaj, Soumik Datta.

Sitar - Famous exponents - Pandit Ravi Shankar, Ustad Vilayat Khan, Pandit Nikhil Banerjee, Annapurna Devi, Ustad Rais Khan, George Harrison, Anoushka Shankar, Prem Joshua, Ustad Shahid Parvez Khan, Ustad Bade Gulam Ali Khan, Nishat Khan, Imrat Khan, Anjan Chattopadhyay, Shujaat Khan, Purbayan Chatterjee, Manju Mehta, Hara Shankar Bhattacharya, Arvind Parikh.

Saraswati veena - Famous exponents - Muthuswami Dikshitar, Veenai Dhanammal, Veena Sheshanna, Veena Venkatagiriappa, Veena Doraiswamy Iyengar, Emani Sankara Sastry, Chitti Babu, Karaikudi Sambasiva Iyer, K. S. Narayanaswamy, Trivandrum R Venkataraman, S. Balachander, Nirmala Rajasekar, Prashanth Iyengar, B.Sivakumar.

Sursringar - Famous exponents - Pyar Khan, Basat Khan, Jaffar Khan, Mohammad Ali Khan (Son of Basat Khan), Bahadur Sen Khan, Allaaddin Khan, Radhika Mohan Maitra, Kumar Birendra Kishore Roy Choudhury, Shaikat Ali Khan, Joydeep Mukherjee.

Sarangi - It is normally played in a vertical position and with an arched bow. Famous exponents - Abdul Latif Khan, Aruna Narayan, Ashique Ali Khan, Bharat Bhushan Goswami, Bundu Khan, Dhruva Ghosh, Ghulam Ali, Harsh Narayan, Kamal Sabri, Manonmani, Ramesh Mishra, Pandit Ram Narayan, Sabir Khan, Siddiqui Ahmed Khan, Suhail Yusuf Khan, Sultan Khan, Ustad Faiyaz Khan, Moinuddin Khan, Sabri Khan.

Santoor - Famous exponents - Ulhas Bapat, Tarun Bhattacharya, Rahul Sharma, Shivkumar Sharma, Abhay Sopori, Bhajan Sopori, R. Visweswaran, Varsha Agrawal, Mohammad Tibet Baqal, Harjinder Pal Singh, Areti Ketime.

Ektara - It is generally played by Nath, Kalbelia saints. Famous exponents - Saieen Zahoor Ahmed, Girish Khyapa.

Mohan Veena - Famous exponents - Vishwa Mohan Bhatt, Harry Manx.

Surbahar - Famous exponents - Annapurna Devi, Imdad Khan, Wahid Khan, Enayat Khan, Imrat Khan, Mushtaq Ali Khan, Pushparaj Koshti, Ashwin M. Dalvi, Irshad Khan, Budhaditya Mukherjee, Manilal Nag, Kushal Das, Rajeev Janardan, Suvir Misra, Deobrat Mishra, Babu Khan.

Violin - Famous exponents - Dwaram Venkataswamy Naidu, N Rajam, Lalgudi Gopala Iyer Jayaraman, Nandini and Ragini Shankar.

Percussion instruments

In this category of Instruments, sound is produced by striking the animal skin which has been stretched across an earthen or metal pot or a wooden barrel or frame.

Ghatam is a large, narrow-mouthed earthenware pot used as a percussion instrument. Famous exponents - TH Vinayakram, EM Subramaniam, Umayalpuram K. Narayanaswamy, Tripunithura N. Radhakrishnan, Trichy Sankaran, V. Selvaganesh, Sukanya Ramgopal, Giridhar Udupa, Vikku Vinayakaram.

Tabla - Famous exponents - Zakir Hussain, Sabir Khan, Allah Rakha, Pt Kishan Maharaj, Sandeep Das, Ustaad Shafaat Ahmad Khan, Pt Jnan Prakash Ghosh, Trilok Gurtu, Anindo Chatterjee, Bickram Ghosh, Alope Dutta, Talvin Singh, Pandit Swapan Chaudhuri, Pandit Nikhil Jyoti Ghosh, Ananda Gopal Bandopadhyay, Faiyaz Khan.

Kanjira - Famous exponents - G. Harishankar, V. Nagarajan, C. P. Vyasa Vittala, Bangalore Amrit, B. Shree Sundarkumar, V. Selvaganesh, Swaminathan Selvaganesh, A. S. N. Swamy, B.S. Purushotham, G. Guru Prasanna, N. Ganesh Kumar, S Sunil Kumar, Nerkunam Sankar, Anirudh Athreya, KV Gopalakrishnan, Sunaad Anoor.

Mridangam - Famous exponents - SV Rajarao, KV Prasad, Palghat Mani Iyer, Palghat Raghu, Karaikudi R Mani, Guruvayur Dorai, Umayalpuram K. Sivaraman, Ramnad V Raghavan.

Pakhawaj - Famous exponents - Totaram Sharma, Gopaldas, Ramshankar Pagaldas, Pt Ayodhya Prasad.

Wind Instruments

A musical instrument that produces sound primarily by causing a body of air to vibrate, without the use of strings or membranes. The wind instruments are roughly divided into two categories on the basis of how sound is produced. They are - The flutes And The reed.

Flute (Sushira Vadya) - Famous exponents - TR Mahalingam, Hariprasad Chaurasiya, N. Ramani, Pannalal Ghosh, Prabhat Sarma.

Harmonium - Famous exponents - Shri Purushottam Walawalkar, Appa Jalgaonkar, Jnan Prakash Ghosh, RK Bijapore.

Shehnai (Mangal Vadya) is a reed instrument with seven holes. Famous exponents - Bismillah Khan, Krishna Ram Chaudhary, Ali Ahmad Hussain, Daya Shankar, Bageshwari Qamar.

Saxophone - Famous exponents - Kadri Gopalnath, Manohari Singh.

Nadaswaram - Famous exponents - Rajaratanam Pillai, Sheikh Chinnamula, Neeruswami Pillai.

Solid Instruments

The earliest instruments invented by man are said to be the Solid Instruments or Ghana Vadya. Once constructed, this variety of instrument does not need special tuning prior to playing.

Example - Manjira, Jaltarang, Kanch-tarang, Kasht-tarang, Jhanj, Khartal, Piccolo, Tic-Toc, Tong, Tasha, Stirdrum, Kabbas, Moracus, Talam, Ago and Chimta.

Folk Music Instruments of India

Andhra Pradesh

Pambai or pamba is a pair of cylindrical drums used in temple festivals and folk music in Tamil Nadu and Andhra Pradesh.

Bobbili Veena is created at Bobbili and Gollapalli of Vizianagaram district and Nuzvid in Krishna District. It is made of Jackfruit wood.

Arunachal Pradesh

Ponu Yoksi is a sword-like instrument used during ceremonial dances.

Assam

Gogona is a type of Jew's harp, made of a piece of bamboo that has a bifurcation on one end.

Pepa is a musical blowing instrument made of buffalo horns. It is an essential part of Bihu Songs.

Khol is also known as Mridanga in Bengali language. It is a terracotta two-sided drum. It is an integral part of Vaishnava music in Assam and Bengal.

Dotara is a two-stringed, plucked musical instrument played in Assam, West Bengal and Bihar.

Chhattisgarh

Dhankul is a musical instrument put together by using a huge pot, a bow and a traditional cane window. It is also played in Madhya Pradesh.

Goa

Ghumat (Ghumot), a membranophone, is a percussion instrument of earthen vessels having both sides open.

Kansallem is a percussion instrument.

Mhadalem, a percussion instrument, is a baked clay cylinder covered with monitor lizard skin at both ends.

Shamel, a percussion instrument, played with two thin cane or bamboo sticks, one of which is straight, and the other is curved at the end.

Other instruments- Surt, Tasso, Nagado, Tambura.

Gujarat

Sundari is a family instrument of the Langa community in Kachchh. It is a double reed wind instrument and has 7 to 9 holes.

Bhorindo is a very antique folk music instrument of Kachchh.

Ghado or Ghaghar is arrhythmic instrument. It is baked earthen Jat of an average size.

Jodia Pawa is a pair of two flutes or double flutes.

Other instruments- Turi, Bungal, Pava, 'Kani' or 'Nar', Nagfani, and Surando.

Himachal Pradesh

Surnai is a wind instrument made of wood and metal.

Sagna is a percussion instrument made of lacquer.

Narsingha is a wind instrument made of copper and metal. It is also found in Bihar.

Karnal is a wind instrument made of brass.

Robana is a stringed instrument made of wood, parchment, gut, steel, and bone.

Other instruments- Nagra, Khal, Dhol, Pohl Bhopal, shamal, and Bugal.

Karnataka

Nagaswaram is a double reed wind instrument, and is among the world's loudest non-brass acoustic instruments.

Karadi Vadya is a percussion instrument, made of wood, iron, cotton, and parchment.

Other instruments- Jagate, Yektar, Tamboori, Chittikatala, Chende, and Kamsale.

Kerala

Chenda, a cylindrical percussion instrument, has two sides, the left side called "Edamthala" and the right side "Valamthala".

Kombu instrument is like a longhorn. Usually played along with Panchavadyam, Pandi Melam, Panchari melam etc.

Mizhavu is a big copper drum played as an accompanying percussion instrument.

Madhya Pradesh

Chinkara is a bowed stringed musical instrument.

Maharashtra

Dholki or Nal is a folk drum, popular in the tamasha or street performance of Maharashtra, bhajans, Lavani, kirtans and dhun.

Ektara is a single string musical instrument made using bamboo and a gourd.

Other instruments- Tuntuni, Daf, Manjeera, Harmonium.

Manipur

Pena is a mono string instrument falling in the lute category.

Meghalaya

Tangmuri is a double-reed conical-bore wind-instrument used by the Hynniew Trep people of Meghalaya.

Duitara is a four-stringed Khasi and Jaintia folk musical instrument.

Other instruments- Sarong, Maryngod, Tangmuri, and Talwak.

Mizoram

Khuang (Drum) is made of hollow tree, wrapped on both sides with animal skin.

Other instruments- Dar (Gong), Bengbung, and Talhkhung.

Nagaland

Asem (drum with animal skin masked upon carved wood) Jemji (Horn made using mithun horn).

Other instruments- Petu, Theku.

Odisha

Mahuri - It is a traditional wind instrument.

Other instruments - Tamak, Gochi Baja, Kendera, Ghanta, Dhenka.

Punjab

Tumbi is a single-string plucking instrument.

Dhadd is an hourglass-shaped traditional musical instrument.

Other instruments- Algoze, Chimta.

Rajasthan

Morchang (morsing) consists of a metal ring in the shape of a horseshoe.

Kamaicha is a bowed flute played by the Manganiars of West Rajasthan.

Other instruments- Chikara, Murli, Sindhi Sarangi, Algoze, Ravanahatha, Ghara, Kartal (Idiophone), Dedh Satara, and Naag Phani.

Sikkim

Tungbank is a stringed instrument made

of softwood and parchment.

Bampanyut is a bird-shaped whistle made of bamboo.

Tamil Nadu

Udukai, is a member of the family of membranophone percussion instruments of India and Nepal used in folk music and prayers in Tamil Nadu.

Thanthi Panai is a stringed instrument made of clay, metal, and parchment.

Other instruments- Thalam, Thavil, Veelai, Venkural.

Tripura

Sarinda is a stringed instrument especially used by most of the tribes and is known as Sarinda Uakharp.

Dangdoo is a small rod idiophone.

Other instruments- Sumui (flute), Kham (Drum), and Chong Preng.

Uttarakhand

Turri and Ransingha are the instruments of war. These are made of brass or copper.

Other instruments- Dhol and Damoun, Daur and Thali, Dholki, Bhankora, and Mushak Been or Bagpipe.

West Bengal

Ghungur is a solid instrument made of metal. This garland of small bells.

Anandalahari is a percussion instrument made of earthenware and wood.

Esraj is a stringed instrument made of wood, goat skin and horse hair.

Dhumsa is a percussion instrument made of wood, metal, iron, and leather. This folk instrument is found in West Bengal and Orissa.

Other instruments- Buang, Banam, Madal.

Jammu & Kashmir

Saz-E-Kashmir is a stringed instrument made of Ivory, softwood, parchment, and steel.

Saranga is a stringed instrument made of wood, steel, horsehair, and gut.

Rabab is a stringed instrument made of wood, parchment, and steel. Majorly used in 'Chakkari', 'Sufiyana Qalam', and other folk forms of Kashmir. Famous exponents- Tansen.

Tasha is a percussion instrument made of wood, metal, brass, leather, cloth, and parchment. It is also used in various parts of North India.

Ladakh

Thun Chen is a wind instrument made of brass. Majorly used by 'Lamas'.

Kangling is a wind instrument made of dried human bone, metal, and black skin.

Damaru is a percussion instrument. It is a local instrument, found in Ladakh, Tamil Nadu, Gujarat, Bihar and other parts of North India.

Other instruments- Daman, Surna and Piwang.

Folk Music

The present Indian classical music has evolved from ancient times when two musical genres prevailed - Gandharva (the formal performances during ceremonies) and Gana (for informal performances).

The first reference to music can be credited to Panini in 500 BCE; and the musical theory credited to 'Rikpratisakhya' in 400 BCE. Natyashastra by Bharat Muni was the first musical work (Natyashastra) in written format that divided music into octaves and 22 keys.

Indian classical music branched out into two different systems of classical music - Hindustani and Carnatic.

Hindustani classical music is practised in North India, while Carnatic music is confined to South India (mainly in Karnataka, Andhra Pradesh, Tamil Nadu and Kerala). Purandaradasa is known as the Father of Carnatic music.

In Hindustani music, there are 10 main forms or styles of singing and compositions- 'Dhrupad', 'Dhamar', 'Hori', 'Khayal', 'Tappa', 'Chaturang', 'Ragasagar', 'Tarana', 'Sargam' and 'Thumri' (Poorab style - Fairly slow and staid, Punjab style - More mercurial).

Six main Ragas in Carnatic Music-

Sampurna has all the seven musical notes (Swaras) present and is the basis of the classification of ragas.

Melakarta (Parent Ragas) are the main point of the musical notes (Swaras) in Carnatic music. There are a total of 72 Melakarta Ragas. In Hindustani music the 'thaat' is equivalent to Melakarta. Janya has different sequential patterns that originate from the parent raga.

Audava are some Janya Ragas that have only 5 musical notes (Swaras).

Shadava Raga are some Janya Ragas that have 6 musical notes or Swaras that ascend (Aroh) and descend (Avaroh) in a steady sequence.

Vakra ragas do not follow a particular order at all and have varied, contrasting musical notes or Swaras in between. Tyagaraja, Muthuswami Dikshitar and Syama Sastri are regarded as the Trinity of Carnatic music.

Folk Music Of India

Andhra Pradesh

Burrakatha is a ballad in which the singers recite mostly mythological stories.

Arunachal Pradesh

Ja-jin-ja (Sung during the marriage ceremony), Baryi, Niyoga, and Roppi.

Assam

Oja-Pali-geet consists of 3 types of songs- biyah-gowa, Sukananni and Manasa and Ramayani.

Other Songs- Bihu-geet (Bihu song), Bon-geet (forest/nature song), Naworiageet (boatmen's song), Nichukoni-geet (lullabies), Biyanam (marriage song) and Dehbisaror-geet (devotional song).

Bihar

Kajri songs are sung by peasant women during the rainy season.

Other Songs- Holi Geet, Jhoola, and Bara-Masa.

Chattisgarh

Pandavani is a folk singing style that involves the narration of tales from the ancient Indian epic Mahabharata.

Other songs- Sohar geet (sung after the birth of a child), bihar geet (sung during marriages), goura geet (sung to propitiate Lord Shiva and Goddess Durga).

Goa

Mando Song is a slow verse dealing with love and tragedy.

Other songs - Fhell, Zagor, Dulpods, and Dekhni.

Gujarat

Gajiyo, Duha, Sorathaa, chaand, nuptial and haveli sangeet (palace songs).

Daryo and Lokvarta (religious and social message), Marasiyas (originates from Marsiya), dandiya (Dandiya Raas), and Fattanna or Lagna-geets (during marriages).

Bhikhudan Govindbhai Gadhvi is an Indian folk singer of Daryo. Gujarati folk music, known as Sugam Sangeet, is a hereditary profession of the Barot, Gadhvi and Charan communities

Haryana

Folk songs in Haryana are called 'Raganis'. Folk singing styles are Gharwa Gayan, Jhoolana, Patka, Rasia.

Himachal Pradesh

Jhoori is a type of song that celebrates extramarital romance. It is popular in Shimla, Solan and Sirmaur.

Other Songs- Laman, Ainchaliyan.

Jharkhand

Jhumar is mainly performed by the Santhal community.

Other Songs- Dohari, Domkach, Janani, Jhumta, Mardana, Daidhara, Pahil sanjha, Adhratiya and Vinsaria.

Karnataka

Janapada Geete, which includes the vachanas of Basavanna.

Kerala

The Bhuta song is based on superstitions.

Pulluvan Paatu, the song dedicated to the serpent deities, is the traditional folk

music of Kerala.

Madhya Pradesh

Kalgiturra, Phag songs, Gavlan songs, Sanjha (sung by young girls in group), Hid songs.

Maharashtra

Abhangas is a form of devotional poetry sung in praise of the Hindu god Vitthal, also known as Vithoba.

Other Songs- Lalita, bharuds, gondhals, Powada (Ballads sung for the heroes), Lavani, Bhalari (sung by farmers), owi (describing women's maternal and marital homes), Kirtan, Bhajan, palane (lullabies), and Tumbadi.

Manipur

The Manipuri folk song is known as 'Khulang Ishei'. Khubak Eshei (devotional folk music sung by a group of people), Khongjom Parva (Ballad genre), and Sana Lamok.

Mizoram

Bawh Hla is the chant or cry raised by the warriors when returning from a successful raid.

Hladu represents the cry of successful hunters.

Other Songs- Lengzem Zai (love songs).

Nagaland

The Hereileu song is known as the war song of Nagaland.

Other songs- Neuleu, Hekaileu (Songs about oneself), Heliameu (Dancing songs).

Odisha

Ganda Baja is a traditional folk music of western Odisha.

Other songs- Daskathia, Geet Kudia, Rasarkeli, Maelajada, Chutkuchuta, Jaiphula, Maleshree, Chhilollai etc.

Punjab

Tappa is a form of semi classical vocal music.

Other songs- Jugni, Mahia, Jindua, Dhola, Kafian, Dohre, Bolian.

Rajasthan

Maand is similar to the thumri and the ghazal.

Lotia music is a regional song usually sung on Holi in Chaitra month.

Pankhida is sung by the peasants of Rajasthan while working in the fields.

Other songs- Panihari (a music form praise for rains), Pabuji Ki Phach.

Sikkim

"Gha To Kito" describes the natural beauty of the sikkim.

Lu Khangthamo, a Bhutia folk-dance, is an age-old dance performed on the thanksgiving.

Other songs- Lhamo, Sherpa Lok Geet, Rumdung (Bhutia community); Tamang Selo, Chyabrun, Mun (Lepcha community); Dohori, Chautari, Lok Geet (Nepali community).

Telangana

Oggukatha, Sarada Kala, suvvi paatalu.

Uttar Pradesh

Kajri is a folk song sung by women of Uttar Pradesh and adjacent regions, during the rainy season.

Birha is an ethnic Bhojpuri folk genre of Ahir communities in Uttar Pradesh, Bihar, and Jharkhand in India.

Rasiya is the form of folk music that is mostly sung in the Braj region on Holi.

Other songs- Barahmasa, Sohar (sung to celebrate birth of a child), Kaharwa (sung by Kahar caste at the time of wedding), Chanayni, Nauka Jhakkad (barber community), Qawwali (brought in India from Persia) etc.

Uttarakhand

Thadya was originally performed in royal courts. It is sung by folk artists, including both men and women.

Other songs- Mangal, Basanti, Khuded and Chhopati. Bedu Pako, Phag, Bair, Baramasa, Saiddhali, Pavada, Hurkiya Baul and Pandav.

West Bengal

Bhawaiya is a popular folk music in

Jalpaiguri and parts of Darjeeling, Kalimpong and North Dinajpur districts.

Padavali Kirtans (literally means "gathering of songs") composed in the praise of Vishnu.

Other songs- Baul, Ramprasadi, Bishnupuri Classical music, Kirtan, Shyama Sangeet, Nazrul Geeti, Dwijendrageeti, Prabhat Samgiita, Agamani-Vijaya, Patua Sangeet, Gambhira, Bhatiali (river song by boatmen), Bengali Rock and Rabindra Sangeet (associated with Rabindranath Tagore).

Jammu and Kashmir

Sufiana Kalam, Chakri, Ladishah, Gwatri, Henzae (Kashmiri Pandits), Wanvun, Bacha Nagma (performed during marriage ceremonies), and Dandaras (performed during the Lohri festival).

Folk Paintings

Andhra Pradesh

Lepakshi Paintings- They were created during the Vijayanagara period and have a religious theme centred on the Ramayana, Mahabharata, and Vishnu's incarnations. They can be found painted on the walls of Veerabhadra temple in Lepakshi.

Srikalahasti Kalamkari- It is a style of Kalamkari work that involves the dyed hand-painting of a fabric. It is produced in Srikalahasti of Chittoor district of Andhra Pradesh.

Other Famous Paintings - Tirupati School of Painting, Leather Puppetry, Adivasi Kolam Painting.

Bihar

Manjusha Painting- It is folk art of Ang region which is based on Folklore of Bihula-Bishari. It is often referred to as Snake Paintings

Madhubani Paintings (Mithila Paintings)- It originated in the Madhubani village of Bihar and was mostly created by women. The art can be found in Nepal's Terai region and its surrounding areas. The paintings all have a common theme and are usually inspired by Hindu religious motifs such as Krishna, Rama, Durga, Lakshmi, and Shiva. It received GI (Geographical Indication) status in 2007. It received official recognition in 1969

when Sita Devi received the State award from the Government of Bihar.

Patna Kalam Painting is a unique style of Indian miniature painting.

Tikuli Art- Tikuli is the term that is locally used for bindis, which are essentially colourful dots that women wear as accessories between their eyebrows.

Chhattisgarh

Bastar arts are a form of tribal art and craft that originated in the Bastar region of Chhattisgarh.

Kosa Silk Art- Kosa silk is known for its fine texture, durability, and natural sheen, and it has been produced in the region for centuries.

Jogimara Cave Paintings- It contains a few paintings and inscriptions in Brahmi script. Paintings include dancing couples as well as creatures such as elephants and fish.

Other Famous Paintings - Godhna, Wrought Iron or Loha Shilp.

Gujarat

Pithora Painting- It is a ritualistic painting done on the walls by the Rathwa, Bhils and Bhilala tribes in Gujarat. It is a wall painting that is mainly characterised by the seven horses. It is also made in Rajasthan, Madhya Pradesh and some parts of Chhattisgarh.

Other Famous Paintings - Mata Ni Pachedi, Rathwa, Rogan.

Haryana

Asthal Bohar paintings are also in the Rajput style, and their influence can be seen even in the Shiva temples in Panchkula and Pinjore, Venumadhava temple in Kaul, the temples in Kaithal and Pabnama, the Kapil temple in Kalayat and the Sarsainth temple in Sirsa.

Bhiti Chitra (Fresco Painting)- Fresco is a technique of mural painting executed on freshly laid or wet lime plaster.

Himachal Pradesh

Kangra Painting- It is the pictorial art of Kangra, named after Kangra, Himachal Pradesh.

Other Famous Paintings - Guler, Chamba.

Jharkhand

Jadopatiya Painting- It is generally practiced by the Santhals in which the artisans make scrolls called Jado or Jadopatia and are drawn with natural inks and colours.

Paitkar Painting- This painting form is popular in West Bengal, Bihar, Orissa and other adjacent states of India. It is considered as the old age tribal painting of India.

Sohrai art - It is practiced by tribal communities. The Sohrai art celebrates fertility in the harvest where the walls are painted with animal motifs.

Other Famous Paintings - Dhokra Art, Kohver Art, Ganju Art, Teli and Prajapati Art, Kurmi Art, Mundas Art, Turi art, Birhor and Bhuiya Art, Ghatwal art.

Karnataka

Badami Cave Paintings - It is a complex of Hindu and Jain cave temples located in Karnataka. They depict various scenes from Hindu mythology, including gods, goddesses, and mythical creatures, as well as scenes of court life and daily activities.

Chittara Painting - It is the art form practised by women of the Devaru community. It is created by the tribal women of Malnad on their red mud-coated houses, as well as rangoli floor designs.

Other Famous Paintings - Chittara, Ganjifa Art, Mysore style, Samavasaran.

Kerala

Mukhathezhuthu - It signifies the painting of the artistes' face in ritual art forms.

Kalamezhuthu- It is practised using natural pigments and powders, usually in five colours.

Padmangal - These are "Kalangal" (drawings made on the floor) to perform Tantrik rituals and sorcery.

Other Famous Paintings - Kerala mural, Kathakali Body painting, Theyyam.

Madhya Pradesh

Bagh Caves Paintings - They are an extension of the Ajanta school. Rang Mahal, Cave No. 4, features exquisite

murals on the walls illustrating Buddhist and Jataka tales, similar to those found in Ajanta.

Nandana- It is a mud resist block-printing craft practised in Tarapur village of Neemuch district of Madhya Pradesh. It is popular among the Bhil tribe, the art includes graceful yet aligned arrangements of motifs on the fabric.

Other Famous Paintings - Gond, Bhil, Mandana.

Maharashtra

Ajanta Caves Paintings- They were sculpted out of volcanic rocks. The paintings depict human values and social fabric, as well as period styles, clothes, and accessories. They are distinguished by the fact that each female figure has a distinct hairstyle. Themes in these paintings span from Jataka stories to Buddha's life to complex flora and fauna decorative patterns. Vegetable and mineral dyes were used as the painting medium.

Warli Painting - It is a form of tribal art mostly created by the tribal people from the North Sahyadri Range in Maharashtra. It can be categorised into four groups - The Gods, The People, The Animals, Rights and Rituals. Jivya Soma Mashe - Modern father of Warli painting.

Ellora Caves Paintings- It is seen in five caverns, with Kailasa temple being the most famous. All three religions (Buddhism, Jainism, and Hinduism) are represented in the artworks.

Other Famous Paintings- Pinguli, Chitrakathi.

Odisha

Saura Painting - It is a style of wall mural paintings associated with the Saura tribes of the state of Odisha. The Saura wall paintings are called Italons or Ikons (or Ekons) and are dedicated to Idital (also Edital) the main deity of the Sauras.

Pattachitra Painting- It is a traditional painting of Odisha and West Bengal. It is scroll painting art usually on cotton depicting Hindu mythological narratives and folktales. It is based on Hindu mythology and specially inspired by Jagannath and Vaishnava sect. Odisha Pattachitra depicting Radha Krishna and West Bengal Patachitra depicting Goddess Durga.

Ravan Chhaya Rock Shelter Paintings- A half-opened umbrella is seen in ancient fresco paintings on a rock shelter.

Other Famous Paintings - Chitra Pothi, Mural paintings, Santhal.

Rajasthan

Phad Painting- It is a style of religious scroll painting and folk painting. Phad is a type of scroll painting that narrates elaborate religious stories of local deities and gods. Traditional examples of this art are Devnarayan Ki Phad (30 Feet) and Pabuji Ki Phad (15 Feet).

Pichwai Painting- It originated over 400 years ago, in the town of Nathdwara near Udaipur.

Mandana Painting- The age-old tribe communities of Meenas are the first painters of Mandana. The core beliefs of drawing Mandana Paintings and art on the walls and floors are to welcome divinity into the house and keep away from evil forces.

Kajali Painting- It is an ancient form of painting originating from the eastern part of Rajasthan. The word 'kajali' is derived from 'kajal' meaning carbon black.

Bagru Print- It is a form of hand block printing done by natural colours followed by the Chippa's in Bagru, Rajasthan.

Sanganeri Printing- It is a method of hand-block printing, and it originated in the village of Sanganeri Rajasthan.

Other Famous Arts- Miniature Art, Meenakari, Jaipur art, Marwar Art, Mewar Art, Bikaner Art, Bundi & Kota, Kishangarh, Dhenu, Kavad, Molela Terracotta, Jogi.

Tamil Nadu

Thanjavur (Tanjore) Painting- It originated under the patronage of the Maratha rulers of Thanjavur in the 17th century CE and flourished under the Maharaja Serfoji II. They are locally known as 'Palagai Padam' as it is mainly done on solid wood planks. Bal Krishna, Lord Rama, and other gods, goddesses, saints, and subjects from Hindu mythology are popular themes in Tanjore paintings.

Sittanavasal Cave (Arivar Koil) Paintings- It is based on the Jain Samavasarana subject. These murals have a striking resemblance to paintings from Bagh and

Ajanta.

Armamalai Cave Paintings - The tales of Astathik Palakas (deities defending eight corners) and Jainism are shown in stunning colourful murals on the walls and roof.

Other Famous Paintings - Tanjore, Mica, Mural Paintings.

Telangana

Cheriyal scroll Paintings (Nakashi art)- It is a dying art form of Telangana practised by the Nakashi family. It received Geographical Indication (GI) status in 2007.

Kalamkari Painting- It is an ancient style of hand painting done on cotton or silk fabric with a tamarind pen, using natural dyes. This art involves 23 steps.

Other Famous Paintings - Nirmal Arts, Deccan Paintings.

Uttar Pradesh

Chikankari Embroidery- It is an art from Lucknow, Uttar Pradesh. It is also popular by the name Lucknow chikan or Lucknow Chikankari or Chikan Embroidery. It entails delicate traditional hand embroidery on fabrics like chiffon, muslin, organza, organdie and silk. It is believed that the wife of Jahangir, Nur Jahan introduced this embroidery.

Other Famous Paintings - Sanjhi, Miniature art.

West Bengal

Kalighat Painting- It is a folk painting of West Bengal. It originated in the vicinity of Kalighat Kali Temple (Kolkata).

Patua Art- These paintings are done on pats or scrolls, and scroll painters, or patuas, have been traveling to different towns to sing their stories in exchange for food or money for generations.

Other Famous Paintings- Chaksudan, Folk paintings from Midnapore, Bengal Scroll, Santhal (Bihar and Jharkhand).

Other Folk Art Crafts & Paintings

Rajput Schools of Paintings- Rajasthani school of paintings is another name of Rajput paintings. It is classified into

Mewar school, Kishangarh school, Bundi school, Amber-Jaipur School, Marwar School, Kota school, Bikaner school, Jodhpur school.

Bani Thani Paintings- Bani Thani is an Indian painting in the Kishangarh school of paintings. It has been labeled as India's Mona Lisa.

Pahari painting, style of miniature painting and book illustration that developed in the independent states of the Himalayan foothills in India. Parts-Basohli, Guler and Kullu Paintings.

Thangka Painting- It is a part of Sikkimese Buddhism and it is painted in the traditional Tibetan style. It is an important tool to depict Buddha and his teachings as well as that of other deities and bodhisattvas.

Bodhisattva Padmapani- It is a painting from cave no. 1 at Ajanta caves. The paintings at Ajanta and Sittannavalas are the earliest Buddhist and Jain murals. The elegant features like a high forehead and downcast eyes represent an impression of dignity and serene meditation.

Margomkali- It is a ritual folk art of the Syrian Christians of Kottayam and Thrissur districts of Kerala.

Meenakari- The process of coating grooves or engravings in ornaments with coloured enamels.

Ajrak- It is a unique form of block printed shawls and tiles found in Sindh, Pakistan; Kutch, Gujarat; and Barmer (Rajasthan).

Other Paintings

Sigiriya Frescoes and Dambulla Cave Paintings- They are found in Sri Lanka.

Arunachal Pradesh - Thangka Painting (Buddhist deity).

Assam - Assamese Scroll paintings.

Manipur - Stone Black Pottery, Wood Carving.

Punjab - Mudwall Painting or Chowk Poorana.

Uttarakhand - Garhwal School of Art.

Jammu and Kashmir - Paper Mache, Basholi.

Leh Ladakh - Tibetan and Thangka Art.

Popular Embroidery styles and Textile Variety of India

The Sujani embroidery (Bihar), Mural embroidery (Udaipur, Rajasthan), Kamal

kadai (Andhra Pradesh), Gota (Rajasthan), Chamba Rumal (Himachal Pradesh), Banni or Heer Bharat (Gujarat), Karchobi (Rajasthan), Khneng (Meghalaya), Kaudi (Karnataka), Phulkari (Punjab), Zardozi (Uttar Pradesh), Pawanpui textile variety (Mizoram).

Popular Painters of India

Raja Ravi Varma (Kerala)- In 1904, the British colonial government awarded Varma with the Kaiser-i-Hind Gold Medal. He is known as the "Father of Indian Modern Art". **Famous Paintings** - Damayanthi Talking to a Swan, Shakuntala looking for Dushyanta, Nair Lady Adorning Her Hair, and Shantanu and Matsyagandha.

Abanindranath Tagore- He was the principal artist and creator of the "Indian Society of Oriental Art". He was also the first major exponent of Swadeshi values in Indian art. **Famous Paintings**- Bharat Mata (1905), Ashoka's Queen (1910), Buddha and Sujata (1901), Tales of Arabian Nights (1928), Journey's End (1913), Krishna Lal series (1901 to 1903).

Maqbool Fida Hussain (M.F. Hussain) predominantly used horses in his painting to depict a lively and free spirit. He was awarded Padma Shri (1966), Padma Bhushan (1973) and Padma Vibhushan (1991). **Famous Paintings**- Mother Teresa, Madhuri Dixit, Hindu Deities, Battle of Ganga and Jamuna, Tribute to Hashmi, Krishna, Lady with Veena, British Raj.

Jamini Roy- He is one of the celebrated artists who was inspired by Kalighat Painting. **Famous Paintings**- Three Pujarins, Rupa Katha, Blue Boy, Krishna with Gopis, Cat and the Lobster, Bal Gopal, Last Supper.

Amrita Sher-Gil- She was a Hungarian-Indian painter. She has been called "one of the greatest avant-garde women artists of the early 20th century" and a pioneer in modern Indian art.

Famous Paintings- Three Girls, Ancient Storyteller, South Indian Villagers Going to a Market, Brahmacharis, Hungarian Market Scene, Haldi Grinders, Mother India.

Satish Gujral- He was an Indian painter, sculptor, muralist, and writer of the post-independent era. Award - Padma Vibhushan (1999). **Famous Paintings**- Days of Glory, Meera Bai, Tree of Life, Mourning en Masse etc.

Tyeb Mehta- He was an Indian painter, sculptor and filmmaker. Awards - Padma Bhushan (2007), Kalidas Samman (1988). **Famous Paintings**- Kali, The Rickshaw Puller.

Famous Painters around the World

Pablo Picasso (Spanish) - His famous paintings are Guernica, The Old Guitarist, The Weeping Woman, Dove.

Leonardo da Vinci (Italian) - His famous paintings are Monalisa, The Last Supper, Salvator Mundi, Vitruvian Man, Annunciation, and Benois Madonna.

Vincent van Gogh (Dutch) - His famous paintings include The Starry Night, Almond Blossoms, The Potato Eaters, Irises.

Claude Monet (French)- His famous paintings include Impression Sunrise, Woman with a Parasol, The Artist Garden, The Water Lily Pond, poppies.

Michelangelo (Italian)- His famous paintings include The creation of Adam, Doni Tondo, Bacchus, Angel.

Salvador Dali (Spanish) - His Famous works- The Persistence of Memory, The Ecumenical Council, The Hallucinogenic Toreador.

Tribes (India/ World)

Andhra Pradesh - Yanadis, Pardhans, Lambadas or Lambadis (Part of Banjara or nomadic tribes), Konda Reddis, Koyas, Bagata, Reddi, Naikpod, Koracha, Sugalis, Bhils etc.

Arunachal Pradesh - Abors, Apatanis, Galos, Monpas, Nyishis, Sherdukpen, Khamba, Khampti, Tagin, Hrusso, Mishmi, Singpho, Momba, Tai-Khampti etc.

Assam - Karbis, Karbis, Bodos, Hajong, Hmar, Lumpheng, Jongbe, Deori, Miri, etc.

Bihar - Kharwars, Gond, Birjia, Asur, Savar, Parhaiya, Chero, Birhor, Agaria, Oraon, Mahli, Kharia, Binjhia, Patar, etc.

Chhattisgarh - Gonds, Agariya, Kol, Oraon, Munda, Sawara, Karku, Bhaina, Nagesia, Mawasi, Bhil Mina, Nagarchi.

Gujarat - Bavchas (Bamcha), Barda, Gamit, Gamta, Rathawa, Dhanka, Dubla, Dang, etc.

Goa - Dhodias, Warlis, Naikda, Gawda.

Himachal Pradesh - Gaddis, Gujjar, Kinnara, Lahaula, Domba, Gara, Zoba.

Jammu & Kashmir - Gujjars, Bakarwals, Balti, Beda, Bot, Changpa, Garra, Gaddi, Sippi.

Jharkhand - Mundas, Oraons, Santhals, Birija, Chero, Parhaiya, Korwa, Ho, Gorait.

Karnataka - Todas, Siddis, Kurumans, Kokna, Hakkipikki, Pardhi.

Kerala - Adiyans, Kani, Kadars, Koragas Eravallan, Kurumba, Karimpalan, Paniyan, Melakudi, Mannan, Kurichian, Maha Malasar, and Kattunaikans

Madhya Pradesh - Baigas, Saharias, Bhils (Belong to the Indo-European Linguistic family), Kols, Madiya, Bhariya, Halba, Gond, Korku, Saur, Baiga.

Maharashtra - Warli or Varli, Koli, Bhaina, Bhunjia, Dhodia, Katkari, Khond, Rathawa, Dhanka, Halba, Kathodi, Kokna, Koli Mahadev, Pardhi, etc.

Manipur - Poumai tribe, Angamis, Aimol, Chiru, Paite, Mao, Lamgang etc.

Meghalaya - Garos, Jaintias, Hajong, Gangte, Hmar, Pawi, Mikir, Boro Kacharis.

Mizoram - Lushai, Lai, Hmar tribe, Dimasa (Kachari), Changsan, Kuki, Rieng, Khawchung, etc.

Nagaland - Rengmas, Naga, Kuki, Kachari, Mikir, Garo, etc.

Odisha - Bondas, Didayi, Banjara, Malhar, Mundari, Nagesia, Kol, etc.

Rajasthan - Damaria, Minas (Meenas), Kathodi, etc.

Sikkim - Bhutia tribes, Lepcha, Limboo, Tamang, Dukpa, Sherpa, Kagatey, Dophthapa, etc.

Tamil Nadu - Todas, Aranadan, Kadar, Kammara, Sholaga, Kanikkar, Malai Vedan, Koraga, etc.

Telangana - Chenchus, Bagata, Bhil, Bodo, Doli Koya, Sugalis, Banjara, Lambadi, etc.

Tripura - Notiya, Bhutia, Chakma, Chaimal, Garo, Uchai, Lepcha, etc.

Uttarakhand - Jaunsari, Tharus, Raji, Buksa, Bhotia.

Uttar Pradesh - Baksa (Buxa / Buksa),

Gond, Dhuria, Saharya, Kharwar, Khairwar, etc.

West Bengal - Rabhas, Kora, Korwa, Lepcha, Gorait, Karmali, etc.

Andaman and Nicobar Islands - Great Andamanese, Jarawas, Onges (Little Andaman), Andamanese, Sentinelese (Sentinel Islands), and Shompens (Great Nicobar).

Dadar and Nagar Haveli - Dhodia tribe, Kathodi, Varli, Naikda or Nayaka, etc.

Daman And Diu - Naikadas, Varli, Siddi (Nayaka), Dubla, (Halpati), etc.

Major Tribes of the World

Amerinds - North and South America

Aleuts - Alaska

Ainus - Japan

Aborigines - Australia

Bantus - Southern Africa

Bedouins - Northern African and Middle Eastern countries

Bushmen - Kalahari desert of Africa.

Bindibu - Australia

Chukchi - North East Asia, Russia, North Siberia.

Eskimos - Greenland, North Canada, Alaska, North Siberia.

Fulani - Western Africa

Gaucho - Uruguay and Argentina

Gobi Mongols - Gobi Desert (Mongolia)

Guicas - Amazon forest area

Hamites - North- West Africa

Hausa - North Nigeria

Hottentots - Hot tropical Africa

Ibanas - Borneo island

Kirghiz - Central Asia

Kalmuk - Central Asia

Kazakhs - Kazakhstan

Koryaks - North Siberia, Eurasian, Northern Kamchatka Peninsula

Maoris - New Zealand.

Masai - East & Central Africa

Meos - Myanmar

Mahsuds - Pakistan

Moors - Morocco.

Moros - Philippines

Orang Asli - Malaysia

Pygmies - Congo basin, Zaire

Red Indian - North America

Swahili - Kenya, Uganda and Tanzania

Semangs - East Sumatra

Tartar - Siberia, Russia
Tapiro - Papua New Guinea
Turregs - Sahara
Tenharim - Brazil
Yakuts - Siberia, Russia
Zulus - South Africa

Languages of India and World

Languages

Language in its literal meaning means a system of communication through speech, a collection of sounds that a group of people understand to have the same meaning.

Most Widely Spoken Languages in the World

English, Mandarin, Hindi, Spanish, French, Arabic, Bengali

Classification of Indian Languages

Indian Languages are broadly divided into following four families - Indo European family, Dravidian family, Austric family, Sino Tibetan family

Official languages

According to the Constitution, India does not have a national language, although **Hindi and English** are the official languages. The first Official Language Commission was established in 1955 under Shri B.G. Kher.

There are **22** scheduled languages which are listed in the 8th schedule of the constitution including Assamese, Bengali, Bodo, Dogri, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Maithili, Marathi, Malayalam, Meitei (Manipuri), Nepali, Oriya, Punjabi, Sanskrit, Santali, Sindhi, Tamil, Telugu, Urdu are included.

There are six languages that enjoy the 'Classical' status in India- Tamil (declared in 2004), Sanskrit (2005), Kannada (2008), Telugu (2008), Malayalam (2013), and Odia (2014).

The **Ministry of Culture** provides guidelines regarding Classical languages. The Classical Languages in India are listed in the Eighth Schedule of the Constitution. The Rajya Sabha's Ministry of Culture released the criteria for designating a language as classical in

February 2014. which are -

1. The high antiquity of its early texts is recorded in history.
2. A body of ancient literature/texts, which is considered a valuable heritage by the generation of speakers.
3. The literary tradition be original and not borrowed from another speech community.
4. The classical language and literature are distinct from modern; there may also be a discontinuity between the classical languages.

States (Official Language) - Other Spoken Languages

Andhra Pradesh (Telugu) - Koya, Urdu, Hindi, Tamil, Marathi, Savara, Kannada, Odia, Konda, Gadaba.

Arunachal Pradesh (English) - Nyishi, Dafla, Miji, Adi, Gallong, Wancho, Tagin, Hill Miri, Mohpa, Bokar, Nocte, Aka, Tangsa, Khamti, Apatani.

Assam (Assamese) - Dimaca, Mishng, Karbi, Rabha, Tiwa, Hindi, Bengali, Bodo.

Bihar (Hindi) - Bhojpuri, Magadhi, Maithili, Urdu, Magahi, Angika, Vajjika and Surjapuri, Santhali.

Chattisgarh (Hindi) - Gondi, Marathi, Urdu, Oriya, Gujarati, and Punjabi.

Goa (Konkani) - Hindi, Kannada, Marathi, Urdu, Portuguese, Gujarati.

Gujarat (Gujarati) - Hindi, Sindhi, English, Marathi, Marwari, Urdu.

Haryana (Hindi) - Haryanvi (Bangru), Punjabi, Ahirwati, Mewati.

Himachal Pradesh (Hindi) - Mandiali, Kulavi, Kehluri, Hinduri, Chameali, Sirmauri, Miahasvi, Pangwali, Churah, and Keonthali.

Karnataka (Kannada) - Urdu, Konkani, Kodava and Tulu, Tamil, Marathi, Telugu, Malayalam.

Kerala (Malayalam) - Tamil, Tulu, Kannada, Gujarati, English.

Madhya Pradesh (Hindi) - Bundeli, Bagheli, Nimari, Marathi, Sindhi, Urdu, and Malwi.

Maharashtra (Marathi) - Hindi, Gujarati, Tamil, Telugu, Konkani, Kannada, Sindhi, Punjabi, Bengali, Tulu.

Manipur (Manipuri) - Meitei, Thadou,

Tangkhu, Poula, Mao, Nepali, Rongmei, Paite, Kabui, Hmar.

Meghalaya (English) - Khasi, Garo, Nepali, Hindi, Bengali, Assamese.

Mizoram (Mizo, English) - Aso, Chho, Halam, Hinar, Lai, Lusei, Mara, Miu - Khumi, Paite and Thado - Kuki.

Nagaland (English) - Angami, Chokri, Kuzhami, Mao, Poula, Pochuri, Ntenyi, Rengma, Sema.

Odisha (Odia) - Bengali, Telugu, Hindi, Santali, Kui, Urdu, Ho, Munda.

Punjab (Punjabi in Gurumukhi Script) - Urdu, Hindi, English, Bagri, Pahari, Doabi, Malwai, Majhi.

Rajasthan (Hindi) - Rajasthani, English, Sindhi, Punjabi, Gujarati, Marwari, Mewari, Bhilodi, and Malvi.

Sikkim (English) - Nepali, Bhutia, Lepcha, Limbu, Newari, Rai, Gurung, Mangai, Sherpa, Tamang, Sunwar.

Tamil Nadu (Tamil) - Telugu, Kannada, Urdu, Malayalam, Eravalla, Kadar, and Muduvur.

Telangana (Telugu and Urdu) - Andh, Chenchu, Gondi, Kolami, Lambadi, Pardhan, Koya.

Tripura (Bengali and Kokborok) - Bengali, Tripuri, Hindi, Mogh, Chakma.

Uttar Pradesh (Hindi)- Urdu, Bhojpuri, Bundelkhandi, Awadhi, Braj Bhasha, English.

Uttarakhand (Hindi) - Garhwali, Kumaoni, Urdu, Sanskrit, Bengali, Jaunsari.

West Bengal (Bengali) - Hindi, Santhali, Urdu, Nepali, Telugu, Kurmal.

Union Territories (Official languages) - Other Languages

Andaman and Nicobar Islands (Hindi, English) - Hindi or Bengali, Tamil, Telugu, and Malayalam.

Chandigarh (English) - Punjabi, Urdu, Hindi

Dadra and Nagar Haveli and Daman and Diu (Hindi, Gujarati and Konkani).

Delhi (Hindi, English) - Punjabi, Bengali, Urdu, Marathi, Sindhi, Gujarati, Nepali, Tamil, Oriya, Malayalam, Konkani.

Jammu and Kashmir (Kashmiri, Dogri, Hindi, Urdu, English).

Lakshadweep (Malayalam)

Ladakh (Hindi, English) - Purkhi, Hindi, Shina, Tibetan, Ladakhi, Purgi.

Puducherry (Tamil and English) - Telugu, Malayalam and French.

Official language of Neighbouring countries of India

Bangladesh	-	Bengali
Afghanistan	-	Dari, Pashto
Bhutan	-	Dzongkha
China	-	Mandarin
Myanmar	-	Burmese
Nepal	-	Nepali
Pakistan	-	Urdu
Sri Lanka	-	Sinhali, Tamil
Maldives	-	Dhivehi

Traditional dishes of Indian states

Andhra Pradesh- Gongura mamsam, gavvalu, ariselu, sunnundalu, Pulihora (rice dish), Pootharekulu, Pesarattu (mung bean dosa), Medu Vada, Chepa Pulusu, Gutti Vankaya koora, Palak Pappu, Gongura Pachadi, Gongura Pickle Ambadi, Bachali Kura, Uppindi.

Arunachal Pradesh- Pika pila (pickle), Thukpa, Apong, Chura Sabji, Lukter, Pehak, Koat pitha, Marua.

Assam- Bhendir sorsori, Paleng xaak or khar, labra, Doi Chira, Assam Laksa Stock, Poora Haah (Roasted duck), Baanhgajor Lagot Kukura, Pani Hamuk, Paro Mangxo, Polu Or Leta, Khaar, Maasor Tenga, Aloo Pitika.

Bihar- Khaja, Khurma, Pua, Thekua, Kebabs, Litti chokha, sattu paratha, Kadhi Badhi, Makhana kheer (specialty of Darbhanga region), Sattu Sarbhat.

Chhattisgarh- Dal pithi, bafauri, kusli, thokva, Bore Basi, Fara, Chila, Muthia, Aamat (The Bamboo Shoot Soup Of Tribal Of Chhattisgarh), Dubki Kadhi, and Bafauri, Khurmi.

Goa- Feni (Alcoholic drink), Balchao, Pork Sorpotel, Khatkatem, Chicken Xacuti, Chicken Cafreal, Pork Vindaloo.

Gujarat- Ganthia nu shaak, Khajli,

Khandvi, Dhokla, Bardoli ki khichdi, Methi ka Thepla, Dal Dhokli, Undhiyu, Khakhra, Ghughra, Khaman and Halvasan, Jalebi Fafda.

Haryana- Bura Roti-Ghee, Kachri Ki Sabzi, and Churma, Bajra Khichdi, Bajra Aloo Roti, Besan ki Masala Roti, Hara Dhania Choliya, Bhatua Raita.

Himachal Pradesh- Guchhi mattar, sepua vadi and kaddu ka khatta, Dhaam, Chana Madra, Babru, Siddu, Aktori, Bhey, Chha Gosht, Aloo Palda and Tudkiya Bhat.

Jharkhand- Handia, Mitha khaja, and Chilka Roti, Dhuska, Koinar Sag, Litti-Chokha, Rugra, and Kudurum Ki Chatni.

Karnataka- Chow Bath, Neer Dosa, Mysore Pak, Ragi Mudde and Soppina Saaru, Bisi Bele Bath, Maddur Vada, Coorg Pandi Curry, and Dharwad Peda, Paramanna, and Tovve.

Kerala- Parippu, Kozhippidi, Puttu and Kadala Curry, Erissery, Sadya, Appam With stew, Nadan Kozhi Varuthathu and Pazham Pori.

Madhya Pradesh- Daal Bafla, Poha, Seekh Kebabs, Chakki Ki Shak, and Mawa Bati, and Bhutte ki Kees.

Maharashtra- Puran Poli, Shrikhand, Pudachi Vadi (Sambar Vadi), Keri Aamti, Modak, Kothimbir Vada, Pandhra Rassa, Pav Bhaji, Thalipeeth, bharli vangi, and Bombil.

Manipur - Hawaii, Nga Aiyaba, Chamthong or Kangshoi, Iromba, Morok Metpa, Singju, Paaknam, Alu Kangmet and Nga-Thongba, Chak -hao kheer (black rice pudding), Ngari (fermented fish).

Meghalaya - Jadoh (Khasi tribe), Doh-Khlieh, Pumaloi, Nakam Bitchi, Tungrymbai, Kyat, Minil Songa, Pudoh, Putharo, and Pukhleim.

Mizoram- Vawksa Rep, Lubrusca (grape wine), Misa Mach Poora, Panch Phoran Tarka, Chhum Han, Bai, and Sanpiau.

Nagaland- Zutho, Fish in Bamboo, Bushmeat or Dog Meat, Akini, Samathu (Sema tribe), Galho, and Naga Pork Curry, Axone (Fermented Soybean).

Odisha- Chakuli Pitha, Chitau Pitha, and Kakara Pitha, Chungdi Malai, Macha Ghanta, Dalma, Pakhala Bhata, Gupchup, Chhena Pada, Rasabali, and Kora Khai.

Punjab- Chole Bhature, Rajma Chawal (rice), Sarso ka Saag and Makki di Roti, Kada Prashad, Punj Ratani Dal, Pork Pickle and Pinni.

Rajasthan- Dal Bati Churma, Laal Maas, Mawa Kachori and Dil Khushal, lahsun ki chutney, mirchi bada, mawa kachori and alwar ka mawa (milk cake).

Sikkim- Thukpa/ Gya-thuk, Sishnu, Saelroti, Chhurpi, Chaang, and Tama, Phagshapa, Sha Phaley, Gundruk, Sinki, Kinema, Dal Bhat and Dhindo.

Tamil Nadu- Paruppu Payasam, Pollachi Nandu Fry, Banana Bonda, Prawn Kuzhambu, Vadai, Dosai, Oothappam and Upma.

Telangana- Chinta chiguru pappu (tender tamarind leaf dal), Gatuka, Sarva Pindi, Chakinalu or Sakinalu, Pachi Pulusu, Sarvapindi.

Tripura- Muya Awandru, Mui Borok, Kosoi Bwtwi, Mosdeng Serma, Bhangui, Chire Doi Aam, and Tohan Mosdeng, Muya bai Wahan.

Uttar Pradesh- Baati Chokha, Petha (Agra), Peda (Mathura), Nimona, and Tehri.

Uttarakhand- Garhwal - Kafili, Phaanu, Gahat (Kulath) Soup, Bhangak Khatai. Kumaon - Bal Mithai, Mandua ki Roti, Singodi, Sisunak Saag, Lesu.

West Bengal- Aloo Posto, Luchi - Alur Dom, Kosha Mangsho, Keemar Doi Bora, Shukto, Pantua, Kadombo, Nikhuti.

Dadra and Nagar Haveli and Daman and Diu- Ubadiyu, Khaman, Potaya, puri, pla, Cozido de peixe (a fish stew), Caldinho de Camarao (shrimp soup).

Delhi- Kebab, Samosa, Chole bhature.

Jammu and Kashmir- Tabaq maz, rogan josh, haleem, shab deg, Kahwa (or saffron tea), Yakhni, Khatta Meat, Chaaman, Gushtaba.

Ladakh- Thukpa, Skyu, Tingmo, Khambhir, Gur-Gur Chai, Chhurpi (or yak cheese).

Traditional Dishes/Foods of Neighbouring Countries

Bhutan- Ema Datshi, Gondomaru, Khabzey, betel leaf (Doma Pani).

China- Peking Roasted Duck (Beijing),

Ling-hu-fen (Hong Kong).

Sri Lanka- Kiribath, Kokis, Kool, Lamprais, Kuzhambu.

Traditional Dresses of Indian States

Andhra Pradesh

Men- Dhoti, kurta and lungi.

Women- Silk saree and langa-voni (like Kalamkari sarees, Dharmavaram Sarees, Uppada Sarees, Venkatagiri Sarees)

Arunachal Pradesh

Men- Adi tribe - Jackets and woolen dresses.

Women- Adi tribe - Jackets and woolen dresses.

'Bauhche' is a traditional bridegroom's sword worn during a tribal wedding ceremony. Monpas tribe - Skull cap, loose sleeveless shirt and a long jacket. Nyishi tribe - Byopa is a traditional headgear. Miji tribe - Long cloaks.

Assam

Men- Dhoti-Gamosa, kurta and tongali (waistcloth), Gamocha (bihuwaan), Sola or Fotua and Eri Chaddar.

Women- Mekhela-chador or Riha-Mekhela.

Bihar

Men- Mirjai (kurta) with dhoti or lungi.

Women- Tussar silk sarees.

Chhattisgarh

Men- Sleeveless kurta, dhoti and cotton headgear.

Women- Lugda (saree) and polkha (blouse).

Goa

Men- Catholic and non-Catholic - Western attire, Fishermen wear colourful shirts and half pants.

Women- Catholic – Gowns and dresses, Non-Catholic– Pano Bhaju, Nauwadi and Kashti.

Gujarat

Men- Chorno-kediyu and phento (turban).

Women- Chaniya-choli (Ghagra Choli), odhani and saree.

The famous 'Tangaliya shawl' is associated with Gujarat. Bandhani is a textile art from Gujarat / Rajasthan.

Haryana

Men- Kurta-dhoti and pagri.

Women- Dimach, Kurti, Ghagra (Daaman) and chunder.

Phargal/Fargal is a cap that children wear during the winter season

Himachal Pradesh

Men- Kurta-churidar (Rajput), kurta-dhoti (Brahmin), embroidered overcoat, turban and soft towel.

Women- Salwar-kurta (Rajput), Ghagra - Choli (Brahmin), rahide (long scarf) and shawl.

Chola and Dora are traditional dresses of Gaddi tribe.

Jharkhand

Men- Dhoti and innerwear with red and white scarves on the waist.

Women- Tussar silk sarees. Tribal-Parhan and Panchi.

Karnataka

Men- Panche Sarong (Dhoti) Kurta and Angavastram, Mysore Peta (turbans during wedding), Kupya (Kodagu Community).

Women- Kanjeevaram silk sarees, Mysore silk sarees and Bangalore silk sarees, Red Kodagu sarees.

Kasuti is a traditional form of folk embroidery practised in the state of Karnataka.

'Mysuru Peta' is a traditional attire worn on the head in Mysuru and Kodagu district.

Madhya Pradesh

Men- Dhoti, saluka (shirt), Mirzai and Bandi (Bundelkhand and malwa region) and safa (turban).

Women- Ghagra-choli and Odhani or

lugra, and bandhej sarees.

Maharashtra

Men- Dhoti-kurta, bandi and pehta.

Women- Nauwadi saadi or lugda, Himroo and Mashru Shawl, Paithani (five yard saree).

Manipur

Men- Dhoti and pagri, Khamen Chatpa.

Women- Phanek, sarong and Innaphi.

Meghalaya

Men- Long cloth on their waist, jacket and turban.

Women- Garo– Eking, wrapped on waist, blouse and Dakmanda (lungi), Paila. Khasi - Jainsem (skirt), blouse and tap-moh Khlieh, (shawl). Jaintia - Thoh Khyrwang, a long sarong and blouse.

Mizoram

Men- Long and wide clothes.

Women- Puanchei (Shawl), Puan, Lal Paun, waist coat, seki, fungki.

Tawlhlohpuan is a Puan shawl of Mizoram.

Nagaland

Men- Alungstu, Rehuke khim or cowrie shawl (Yimchunger tribe).

Women- Azu Jangup Su, neikhro, Hompani, Meyni.

Odisha

Men- Sambalpuri kurta and dhoti.

Women- Shambalpuri and Khandua saree.

Punjab

Men- Punjabi Kurta and Tehmat, Kurta-muksari pyjama and pagri.

Women- Patiala Salwar Suit, Parandi.

Rajasthan

Men- Dhoti, angrakha, kurta-pajama and pagri.

Women- Ghagra-choli and odhani.

Sikkim

Men- Bhutia tribe – Bakhu/Kho.

Women- Bhutia tribe - Bakhu/Kho.
Lepcha tribe - Dumvum.

Tamil Nadu

Men- Lungi or Dhoti, angavastra (shirt) and angavastram.

Women- Kanchipuram silk sarees and Pavada (young girls).

Telangana

Men- Pancha (dhoti), kurta and Hyderabad sherwani.

Women- Salwar kameez and churidar, Pochampally Ikat Saree.

Tripura

Men- Rikutu Gamcha, a loose towel-like cloth and Kubai, a shirt.

Women- Rinai (lower body) and Risa (upper body).

Uttarakhand

Men- Kurta with lungi, dhoti or pyjama, turban (older men), topi woolen jackets and sweaters.

Women- Garhwali – Saree, blouse and shawl. Kumaoni – Ghagra-choli and Odhani (pichhora).

Uttar Pradesh

Men- Kurta-pyjama or dhoti.

Women- Benarasi sarees, ghagra-choli with odhani and salwar kameez.

West Bengal

Men- Panjabi and dhoti.

Women- Cotton or silk sarees.

Jammu and Kashmir

Pashmina shawl woven from the wool is obtained from the under fur of Changthangi Kashmir goats.

Pheran or Phiran is the traditional outfit for both males and females in Kashmir.

Traditional Dresses of Other Countries

Middle East Countries- Keffiyeh,

Shemagh and Ghutra.

France- Coiffe and Brittany.

Japan - Montsuki-haorihakama, Jinbei, Tsumugi, Kimono, Furisode, Tomesode, Homongi.

Bhutan - - Gho (Men), Kira (women), Kabney, Toego, Wonju.

Scotland- Kilt suits.

Germany- Trachts.

Kenya- Massai beadwork.

Vietnam- Conical Hats.

Spain- Flamenco dress.

United States (Texas)- Ten-gallon Stetson.

Italy- Sardinian dress.

Temple architecture in India

Indian Style of Architecture for temples is broadly classified into two categories

1. Nagara or north India style
2. Dravidian or south India style

1. Nagara style Architecture

The temples have four Chambers- Garbhagriha, Jagmohan, Natyamandir, and Bhogamandir. The complex is made up of two buildings- the main shrine, which is taller, and a shorter mandapa adjacent to it. Nagara styles of temple architecture was developed between Himalaya to Vindhya. The Nagara temples belong to Saiva, Vaishnava and Sakta faith.

The following are the basic components of a Hindu Temple-

Garbhagriha- It is a cave-like chamber that holds the temple's major icon. In ancient times, it was a small chamber with only one entrance. In later times, it evolved into a larger chamber. The garbhagriha is always located directly under the tallest tower.

Shikhara- The Garbhagriha (Inner Sanctum) is crowned with a curvilinear-shaped Shikhara. A bell-shaped structure has been erected to the main shrine.

Types of Nagara Temple Style Architecture based on the Shikhara-

Rekha-Prasad or Latina - The most common name for a simple spire which is square at the base and whose walls curve or slope upwards to a point is called 'Latina' or spire of rekha-prasad. The Sri Jagannath Temple of Odisha has been constructed in the Rekha-Prasad Shikhara style and the Sun Temple at Markhera in Madhya Pradesh.

Valabhi- Temples of this style are rectangular in shape and have barrel-vaulted roofs. Wagon vaulted buildings /structures get their name from the vaulted chamber ceiling. Teli Ka Mandir, a Gwalior temple from the 9th century, was built in this manner.

Phamsana- Roofs with multiple slabs that climb upwards in a gradual slope on a straight incline like a pyramid meeting at a single point above the midpoint of the building are shorter but broader structures. The Phamsana method is used to build the Jagmohan of Konark Temple.

Mandapa- In the Hindu temple the mandapa is a porch-like structure through the (gopuram) (ornate gateway) leading to the temple. Ardha Mandapa, Mandapa, and Maha Mandapa are the three forms of Mandapa (main hall).

Amalaka- At the top of the Temple is a stone disc-like structure.

The Kalasha- It is the highest pinnacle of the Shikhara as well as the entire temple, crowns Amalaka.

Antarala- Between the Garbhagriha and the Mandapa, there is a vestibule called Antarala.

Jagati- A raised platform where devotees can sit and worship, which is popular in north Indian temples.

Vahana- It is the mount or vehicle of the temple's main deity along with a standard pillar or Dhvaj which is placed axially before the sanctum.

Famous Temples built in Nagara Style architecture-

Kandariya Mahadeva Temple (Dedicated to shiva) is the largest and most ornate Hindu temple in the mediaeval temple group found at Khajuraho in Madhya Pradesh. It means "the Great God of the Cave" King Dhangadeva from the Chandel Dynasty constructed the temple in 999 CE. In 1986, it was declared a World Heritage Site by UNESCO.

Konark Sun Temple (Puri city, Odisha) is dedicated to the God Sun. It was built in 1250 CE by the Eastern Ganga Dynasty ruler in the 13th century, Narasimhadeva I. It is a classic illustration of the Odisha style of Architecture or Kalinga architecture (sub style of Nagara Architecture). It is called "Black Pagoda".

The chief architect behind this temple was Bisu Maharana.

Khajuraho Group of Monuments is a group of Hindu temples and Jain temples in Chhatarpur district, Madhya Pradesh. They are fine denotations of the Chandela dynasty between 950 and 1050 AD.

Shri Radha Madan Mohan Temple (Vrindavan, Uttar Pradesh) was rebuilt by Sri Sanatana Gosvami in 1580 AD. The presiding deity of the temple is Madan Mohan, a form of the god Krishna.

Markandeshwar temple is situated on the bank of River Wainganga in district Gadchiroli of Maharashtra. It is also known as the "Khajuraho of Vidarbha".

Mukteshwara Temple is a 10th-century Hindu temple dedicated to Shiva located in Bhubaneswar, Odisha. It means "Lord of Freedom". It was built by King Yayati I of the Somavamshi Dynasty.

Kedarnath temple is dedicated to Shiva, located on the Garhwal Himalayan range near the Mandakini river, in the state of Uttarakhand. It was built in the 8th century A.D. by Adi Shankaracharya and is one of the 12 jyotirlingas in India.

2. Dravidian style Architecture

The Dravida temple is enclosed within a compound wall. The front wall has an entrance gateway in its centre, which is known as a gopuram. The shape of the main temple tower known as vimana in Tamil Nadu, is like a stepped pyramid that rises up geometrically rather than the curving shikhara of North India. It is common to find a large water reservoir, or a temple tank, enclosed within the complex. The Pallavas were the first to start the Dravidian style of temple architecture in South India.

Subsidiary shrines are either incorporated within the main temple tower, or located as distinct, separate small shrines beside the main temple. Vimana feature Shikhara/Vimana has been recorded since the 5th century CE. On top is a mountain-like spire. It is known as Shikhara in North India and has a curved shape. It is known as Vimana in south India and looks like a pyramidal tower.

A **Gopuram** or gopura is a monumental entrance tower, usually ornate, at the entrance of a Hindu temple, in the South Indian architecture of the Southern Indian states of Tamil Nadu, Andhra

Pradesh, Kerala, Karnataka, and Telangana. They are a prominent feature of koil, Hindu temples built in the Dravidian style. They are topped by the kalasam, a bulbous stone finial. Nataraja temple in Chidambaram, Tamil Nadu is one of the famous examples of Gopuram. Nataraja temple is also referred to as the Chidambaram Nataraja temple or Thillai Nataraja temple.

Subdivisions in Dravidian architecture temples.

These are basically five different shapes-

1. Square, usually called kuta, and also chaturasra.
2. Rectangular or shala or ayatasra
3. Elliptical, called gaja-prishta or also called vrittayata, deriving from wagon vaulted shapes of apsidal chaityas with a horse-shoe shaped entrance facade usually called a nasi
4. circular or vritta
5. octagonal or ashtasra.

Famous Temples built in Dravidian Style architecture -

Mahabaleshwar Temple is a 4th-century CE Hindu temple located in Gokarna, Uttara Kannada district, (Karnataka), This temple is dedicated to Lord Shiva.

Kapaleeshwarar Temple is dedicated to Lord Shiva located in Mylapore, Tamil Nadu. It was built around the 7th century CE. Shiva is worshipped as Kapaleeswarar and is represented by the lingam.

Brahmapureeswarar Temple is a Hindu Temple located in Tamil Nadu.

Rajarajeshwara temple is a Shiva temple located in Taliparamba in the Kannur district, Kerala. It is built by Bhagwan Parashurama. Rajarajeshwara Temple had been constructed by a 4 km long incline for placing 90 tonnes of stone on top of the tallest shikhara amongst temples of its time.

Shore Temple belongs to the 8th century AD, built with granite blocks. It is located in Mahabalipuram in Tamil Nadu. The site was a busy port during the reign of Narasimhavarmam II of the Pallava dynasty.

Airavatesvara temple is located in Kumbakonam, Thanjavur District in Tamil Nadu. It was built by the Chola emperor Rajaraja II in the 12th century CE. It was declared a UNESCO World Heritage Site in 2004. It is dedicated to Shiva.

Brihadishwara Temple (Periya Kovil or Rajarajeswaram) is a Shaivite Hindu temple built in a Chola architectural style by Chola emperor Rajaraja I between 1003 and 1010 CE. It is located on the south bank of the Cauvery river in Thanjavur, Tamil Nadu. It is also called Dakshina Meru ("Meru of the South"). It became a part of the UNESCO World Heritage Site in 1987. It is the part of "Great Living Chola Temples", along with the chola-era Gangaikonda Cholapuram temple and Airavatesvara temple.

Kailash or Kailasanatha temple is the largest of the rock-cut Hindu temples at the Ellora Caves, Maharashtra. Most of the excavation of the temple is generally attributed to the eighth-century Rashtrakuta king Krishna I (756 – 773). It shows traces of Pallava and Chalukya styles. It is a huge monolithic temple dedicated to lord shiva. The Kailash Temple is the sixteenth cave, and it is one of the 32 cave temples and monasteries forming the magnanimous Ellora Caves.

Virupaksha Temple is located in Hampi in the Vijayanagara district of Karnataka. Virupaksha Temple was built by Queen Lokamahadevi in the 8th century to celebrate her husband's victory (Vikramaditya II) over the Pallavas of Kanchi. This temple was constructed in assistance with Lakkana Dandesha's who was a commander under the ruler Deva Raya II (Prauda Deva Raya) of the Vijayanagara Empire. It is part of the Group of Monuments at Hampi, designated as a UNESCO World Heritage Site in 1986. It is dedicated to Sri Virupaksha (a form of Shiva).

Munneswaram temple (Sri Lanka) is associated with the Indian epic Ramayana and Lord Ram. It was built by Kotte Kingdom King Parakrakra Bahu VI.

The Iravatesvara temple (Sri Jurahareswarar Temple) is dedicated to Lord Shiva and it is located in Kanchipuram in Tamil Nadu. It was built by Pallava King Narasimhavarmam II.

Srikalahasti Temple (Tirupati, Andhra Pradesh) is dedicated to Shiva. This is the only temple in India which remains open during Solar and lunar eclipses.

Other architectural style

Vesara Style of Architecture - Its emphasis is on Vimana and Mandapa. Its

ground plan is basically in a star shape or stellate plan. One of the unique features of India's architecture, Vesara has an open ambulatory passageway. Structures of this style have an unraised platform or Jagati as their base. Pillars, doorways, ceilings are decorated with intricate carvings. Examples of Vesara style temples are Chennakesava temple, Virupaksha temple, Ladkhan temple, Badami cave temples, and Rudreshwar temple (Ramappa temple).

Panchayatana style of architecture

where the main shrine is built on a rectangular plinth with four smaller subsidiary shrines at the four corners (making it a total number of five shrines, hence the name, panchayatana). This style was introduced by Gupta Rulers.

Hoysala Architecture developed under the rule of the Hoysala Empire. Multiple shrines of different deities were built around a central pillared hall. Stellate plan- These shrines are established intricately in the design of a star. Soft soapstone was the main building material. The walls and stairs of the temple followed a zigzag pattern. Examples of the Hoysala architectural style, including the Chennakesava Temple at Belur (Karnataka), the Hoysaleswara Temple at Halebidu (Karnataka), and the Kesava Temple at Somanathapura (Karnataka). These three temples were accorded UNESCO world heritage site status in 2023.

Gothic Architecture is a European style of architecture popular in the 12th-16th century. This architecture has its roots in France and England. Features are Pointed arches, ribbed vaults, and flying buttresses. The Britishers merged some Indian features of the architecture with Gothic architecture, which resulted in the Indo-Gothic style of architecture. Examples of Indo-Gothic style of architecture- The Chhatrapati Shivaji Maharaj Terminus.

Forts of India

Andhra Pradesh

Chandragiri Fort was built in the 11th century by the Yadava rulers located at Chandragiri suburb in Tirupati.

Gooty Fort is located in the Anantapur district.

Gandikota Fort was founded by Kapa

Raja of nearby Bommanapalli village. **Kondaveedu Fort** is located in the Guntur district. It was built in the 13th century by Prolaya Vema Reddy of the Reddy dynasty.

Kondapalli Fort is located in the Krishna district and was built in the 14th century by the Reddy dynasty.

Vizianagaram Fort was built by Vijaya Rama Raju, Raja of Vizianagaram.

Arunachal Pradesh

Bhismaknagar Fort is located in district Dibang valley.

Ita Fort is located in Itanagar. The fort was built in an irregular shape by a king of the Jitari dynasty, Ramachandra.

Assam

Garchuk Lachit Garh is situated in the Guwahati, towards the western part of Ahom Gaon. The fort was built during the reign of Lachit Borphukan.

Garh Douli- A site near Tezpur area of Sonitpur District which has been declared a protected monument by the Archaeological Survey of India or ASI.

Kareng Ghar Fort- It implies 'royal palace' and is existent in Rangpur. This fort was composed of wood.

Talatal Ghar is a seven-storied monument located in Sivasagar and was built in the 18th century by the Ahom king Swargadeo Rajeswar Singha.

Bihar

Jalalgarh Fort was constructed by Syed Muhammed Jalaluddin Khan. Some historians claim that the fort was established by Saif Khan who was the Nawab of Purnia.

Munger Fort is a 14th-century fort that is situated on the top of a hill on the southern bank of the river Ganga.

Rohtasgarh Fort was constructed in the city of Rohtas situated on the banks of Sone River.

Chhattisgarh

Chaiturgarh Fort was constructed by Raja Pratwived First. The fort has three main entrances which are named as Menaka, Humkara, and Simhadwar.

Ratanpur Fort (Bilaspur)- It has four entrances - the Singh, Ganesh, Bhairav and Semar dvars.

Goa

Aguada Fort- It is a seventeenth-century Portuguese fort, located on Sinquerim Beach. The fort was originally constructed in 1612 to guard against the Dutch.

Reis Magos Fort- It is located at the banks of the Mandovi River.

Terekhol Fort - The fort was built by Maharaja Khem Sawant Bhonsle of Sawantwadi in the 17th century. The fort was rebuilt in 1764 when the Portuguese Viceroy Dom Pedro Miguel de Almeida captured it. In 1788, the fort was legally incorporated in Goa.

Gujarat

Aina Mahal was built during the reign of Rao Lakhpatji of the Jadeja Rajput dynasty, in the 18th century. Also called the Hall of Mirrors.

Bhadra Fort is a 15th-century fort that is located in the city of Ahmedabad. It was built by Sultan Ahmed Shah I.

Bhujia Fort- The venerable Bhujia Fort is located outside present-day Bhuj city. It was built by Roa Godji.

Manek Burj (Ahmedabad)- It takes its name from Maneknath, a Hindu saint who lived in the 15th century, who helped Ahmadabad Shah I to build the Bhadra Fort.

Uparkot fort is located in Junagadh. It was established at the foothills of Girnar hill during the reign of the Maurya Empire.

Other Forts- Prag Mahal (Bhuj), Lakshmi Vilas Palace (Vadodara), Kanthkot Fort, Dhoraji Fort.

Haryana

Asigarh Fort is located in the Hisar district at Hansi. It is said to have been built for Prithviraj Chauhan, the king of Ajmer and Delhi, during the 12th century.

Dhosi Hill Fort was built during the reign of the Suri dynasty under the command of Hemu, who was the chief minister of the Suri king, Adil Shah Suri.

Madhogarh Fort is situated in the

Mahendragarh district atop Madhogarh Hill and was built during the rule of a Jaipur king, Madho Singh.

Himachal Pradesh

Gandhola Fort- An 8 storeyed architecture which was built by Raja Man Singh in 1700 AD.

Haripur Fort was built by Raja Hari Chand of Kangra in Guler and is set in a strategic place on the Dhauladhar range.

Kangra Fort was built by the royal Katoch Dynasty. It is also known as Nagarkot fort.

Nurpur Fort belonged to the Pathania dynasty which was earlier called as Dhameri.

Sujanpur Tihra Fort- It was made the capital of the kingdom by Raja Abhaya Chand. He had this fort built in 1758 AD.

Jharkhand

Palamu Fort- It has two forts - the Purana Kila (Old Fort) and the Naya Kila (New Fort). The forts were built during the beginning of the Mughal period by the Chero chiefs Medani Roy and Pratap Roy. The New fort was constructed by Raja Medini Ray.

Karnataka

Bangalore Palace was built in 1887 by the Wodeyar dynasty.

Badami Fort was built by the Chalukya King Pulakeshin.

Bidar Fort was built by Ahmad Shah Wali Bahman. Ahmad Shah I shifted the capital from Gulbarga to Bidar.

Chitradurga Fort was built by Chalukyas or Hoysalas between the 11th to 13th centuries and spread across on the banks of river Vedavati.

Gajendragad Fort was built by famous Maratha ruler Chatrapati Shivaji Maharaj. Gajendra means Elephant God and Gad means fort.

Other Forts- Bellary Fort, Gulbarga Fort (Kalaburagi), Malkhed Fort, Nagara Fort (Shimoga).

Kerala

Anjengo (Anchuthengu Fort), is in Thiruvananthapuram. It was built by the

East India Company and it served as a good location for signalling ships.

Bekal Fort is situated in Kasargod near Karnataka border. It is a marine fort constructed by Shivappa Nayaka. It is the largest fort in Kerala.

Fort Immanuel- The first European fort in India was built by the Portuguese in the year 1503.

Palakkad Fort was constructed by Hyder Ali. It is also called Tipu's (Hyder Ali's son) Fort.

St. Angelo Fort or **Kannur Fort** is located in Kannur on the coast of Arabian Sea. It is the first Portuguese fort which was built by Dom Francisco de Almeida.

Madhya Pradesh

Ahilya Fort is located in Maheshwar, Khargone on the banks of the Narmada River.

Chanderi Fort is located in Chanderi (Ashoknagar) on the hill, Southwest of Betwa River. It has three entrance gates known as 'Hawa Paur', 'Khooni Darwaza', and 'Kati Ghati'.

Gwalior Fort is called the 'Gibraltar of the East' and stands on a steep land of sandstone. It is a hill fort near Gwalior, built by King Suraj Sen during the 6th century AD to honour the saint Gwalipa.

Jai Vilas Palace was constructed in 1874 by Maharaja Jayajirao Scindia.

Mandsaur Fort (Dashpur Fort) was built by Hoshang Shah Ghorī who was one of the first Islamic kings of the Malwa region.

Orchha Fort complex was built by the Bundela Rajputs starting from the early 16th century by King Rudra Pratap Singh of the Orchha State.

Maharashtra

Daulatabad fort, also known as Devagiri Fort, is a fortified citadel located in Aurangabad, Maharashtra. It was the capital of the Yadava dynasty (9th century–14th century CE).

Panhala fort (Panhalgad) is located over a pass in the Sahyadri mountain range which was a major trade route from Bijapur in the interior of Maharashtra to the coastal areas. It is also called the

'Fort of Snakes' as it is zigzagged in shape.

Rajgad (literally meaning Ruling Fort) is a hill fort situated in Pune. The fort was the capital of the Maratha Empire under the rule of Shivaji, after which the capital was moved to the Raigad Fort (first capital of the Maratha Empire).

Sindhudurg Fort was built by Shivaji Maharaj. The fortress lies on the shore of Malvan town of Sindhudurg District in the Konkan region of Maharashtra.

Shivneri Fort is a 17th-century military fortification located near Junnar in Pune. It is the birthplace of Shivaji, the founder of Maratha Empire.

Torna Fort (Prachandagad fort) is a large fort located in Pune. It was the first fort captured by Chhatrapati Shivaji Maharaj in 1646, at the age of 16, forming the nucleus of the Maratha empire.

Other Forts- Malhargarh Fort (Pune), Pratapgargh Fort (Satara), Janjira Fort.

Manipur

Kangla Fort is located in Imphal. The existence of this fort dates back to 33 AD, when the mythical God-King of Manipur, Nongda Lairen Pakhangba first ascended the throne.

Nagaland

Khonoma fort was built by the Angami Naga tribe.

Odisha

Barabati Fort was built by Marakata Keshari of Somavamshi (Keshari) dynasty in Cuttack.

Chudanga Gada is located in Ganjam. It was built during the rule of the Gajapati dynasty.

Raibania Fort- It comprises a group of forts located in Baleswar district. The fort complex houses 161 fort goddesses. It was built by Langula Narasinha Deva, the ruler of Ganga dynasty of Utkala.

Other Forts- Sisupalgarh Fort (Khurda).

Punjab

Bahadurgarh Fort was built in the 17th century by Nawab Saif Khan during the reign of Aurangzeb and was reportedly called Saifabad.

Gobindgarh Fort is a military fort located in Amritsar.

Phillaur Fort was originally a "saran", but was rebuilt as a fort by Maharaja Ranjit Singh.

Qila Mubarak (Bathinda Fort) is a monument in Bathinda. It served as a prison for Empress Razia Sultana after she was defeated by her rebellious general, Malik Ikhtiar-ud-din Altunia.

Rohtas Fort or Rotas Ghur is a 16th-century fortress located near the city of Dina in Jhelum district of the Punjab (In present day Pakistan). An Afghan general, Sher Shah Suri built the fort. The fort was declared a UNESCO World Heritage Site in 1997. UNESCO called it an "exceptional example of the Muslim military architecture of Central and South Asia."

Rajasthan

Amber Fort was originally built by Raja Man Singh. Jai Singh I expanded it.

Chittorgarh Fort is a UNESCO World Heritage Site. The fort was the capital of Mewar and is located in the present-day city of Chittorgarh. Gaumukh is a water reservoir located in this fort.

Junagarh Fort was built by Raja Rai Singh. The fort was originally called 'Chintamani' and was renamed Junagarh or "Old Fort" in the early 20th century. The fort complex was built under the supervision of Karan Chand, the Prime Minister of Raja Rai Singh, the sixth ruler of Bikaner.

Jaigarh Fort is situated on the hill called the "Cheel ka Teela" (Hill of Eagles) of the Aravalli range. The fort was built by Jai Singh II to protect the Amer Fort.

Jaisalmer fort (Sonar Qilla or Golden) was built by Rawal Jaisal.

Kumbhalgarh Fort was built by Rana Kumbha from the Sisodia Rajput dynasty. It has the longest continuous wall in India, the second longest wall in the world.

Lohagarh Fort (or Iron Fort) was built in the 18th century by Raja Suraj Mal.

Mehrangarh Fort was built by Rao Jodha, the ruler of Mandore and the founder of the city of Jodhpur.

Nahargarh Fort is located on the edge of the Aravalli Hills.

Other forts- Jal Mahal or Water Palace (Jaipur), City Palace (Udaipur), City Palace (Jaipur), Ranthambore Fort (Sawai Madhopur), Bhangarh Fort (Alwar).

Tamil Nadu

Gingee Fort was called the 'Troy of the East' by the British. The renowned Maratha King Chhatrapati Shivaji lauded Gingee Fort as the 'most impregnable fortress in India'.

Rock Fort- It features a beautiful temple complex (Ucchi Pillayar Temple, dedicated to Lord Shiva and the Thayumanaswamy Temple, dedicated to Lord Ganesha).

Sadras Fort is situated in Kalpakkam near Mahabalipuram is one of the oldest Dutch forts in the country. This Fort was initially a Hindu temple with Lord Vishnu as the presiding deity.

Other Forts- Padmanabhapuram Palace, Vellore Fort, Udayagiri Fort (Dillanai Kottai or De Lannoy's Fort), Alamparai Fort.

Telangana

Bhongir Fort- It was constructed by the Western Chalukya Dynasty in the 10th century.

Devarakonda Fort is located in Devarakonda town in Nalgonda District.

Golconda Fort (Mankal) is located in the western part of Hyderabad city. It was a mud fort under the reign of Rajah of Warangal and the principal capital of the Qutub Shahi kings. It was built during the Kakatiya dynasty.

Warangal Fort- The fort has four ornamental gates, known as Kakatiya Kala Thoranam. The fort is included in the "tentative list" of UNESCO World Heritage Sites.

Other Forts- Khush Mahal, Chowmahalla Palace.

Tripura

Neer Mahal is a royal mansion located in the middle of the Rudrasagar Lake built by Maharaja Bir Bikram Kishore Debbarman in 1930.

Ujjayanta Palace was built by Tripura king, Maharaja Radha Kishore Manikya of

the Manikya dynasty, between 1899 and 1901.

Uttarakhand

Pithoragarh Fort is also known as London Fort, Baulaki Fort (bowl-shaped fort) and the Gorkha Killa. It was built by the Gorkhas back in the year 1789.

Uttar Pradesh

Agra Fort was the main residence of the emperors of the Mughal dynasty. It is a UNESCO World Heritage Site declared in 1983 and originally built by Akbar from red stone. It comprises Jahangir Mahal, built by Akbar. Sheesh Mahal, Khas Mahal, Diwan-i-Khas, Diwan-i-Am, Moti Masjid (or Pearl Mosque) all built by Shahjahan.

Allahabad Fort is the largest fort built by Emperor Akbar in 1583 AD.

Chunar Fort (also known as Chandrakanta Chunargarh and Charanadri) is located in Mirzapur.

Kalinjar fort is located in the Banda and was built by Chandela ruler Paramaditya Dev.

Ramnagar Fort is situated on the left bank of Kud river in mid Himalayan range. It also served as the capital of the old Bandarla State.

West Bengal

Kurumbera Fort is located on the outskirts of Keshari town in the Gaganeshwar village. It was constructed by Suryavamsi Gajapati, the Odisha king.

Fort William is located on the banks of the Hooghly River in Hastings. It is also known as the 'Black Hole of Calcutta.'

Other forts- Hazarduari Palace (Murshidabad), Buxa Fort (Alipurduar).

Dadra and Nagar Haveli and Daman and Diu

Diu Fort lies at the mouth of the Gulf of Khambhat. A colossal structure built by the Portuguese in 1535.

Delhi

Red Fort was built in the 17th century CE, by the Mughal Emperor Shah Jahan to serve as the palace fort for his capital Shahjahanabad. The Moti Masjid is a

white marble mosque inside the Red Fort complex in Delhi, built by the Mughal emperor Aurangzeb.

Siri Fort was built by Sultan Ala-ud-Din Khilji to the north of Mehrauli.

Other Forts- Purana Qila' (Old Fort)

Ladakh

Leh Palace (Lhachen Palkhar) was built by King Sengge Namgyal during the 1500s and completed by the 17th century.

Puducherry

Fort Louis or Fort Saint Louis is a French fort that stood on the Eastern coast of India. The fort was built around 1701 by Francois Martin.

Caves in India

Andhra Pradesh - Belum Caves, Borra Caves (located in Ananthagiri hills of the Araku Valley), Undavalli Caves, Bojjannakonda and Lingalakonda caves, Guntupalli Buddhist Caves, Moghalrajapuram Caves.

Assam - Jogo Hopa Caves, Umachal rock Caves.

Bihar - Barabar hill caves, Dungeshwari Caves, Nagaragunja Caves (3 caves), Saptarni Caves.

Chhattisgarh - Jogimara and Sitabenga Caves, Kotumsar Cave, Kailash Caves, Singhanpur caves, Bhangra Caves, Kobra Cave etc.

Gujarat - Junagadh Caves, Dhank Caves, Kadia Dungar Caves, Khambhalida Caves, Sana Caves, Siyot Caves, Talaja Caves etc.

Himachal Pradesh - Tabo Caves, Arjun Gufa, Shringi Rishi Caves, Trilokpur Caves etc.

Karnataka - Badami cave, Nellitheertha Cave, Satre's limestone cave (at Karnataka-Goa border), Gavipuram Cave Temple, Narasimha Jharni Cave Temple, Ravan Phadi cave, etc.

Kerala - Edakkal caves, Kaviyoor, Kottukkal, Nedumala, Punarjani Guha etc.

Madhya Pradesh - Adamgarh Caves, Bagh Caves, Bandhavgarh Caves, Bhimbetka Caves, Shankaracharya Cave, Udaygiri Caves, Ambadevi Caves, Lohani

Caves, Pachmarhi Caves, Bharat Neer Cave, Mrigendra Nath Caves, Kajal Rani Caves etc.

Maharashtra - Ajanta caves (recognised as a UNESCO world heritage site in 1983), Bhaja caves, Ellora caves UNESCO world heritage site in 1983), Elephanta caves (Gharapuri Caves UNESCO world heritage site in 1987), Jogeshwari caves, Kanheri caves, Karla caves, Mandapeshwar cave (Mumbai), Ambivali Caves, Aurangabad Caves, Bedse Caves, Bahrot Caves (Zoroastrian or Parsi Cave), Agashiv Caves, Nasik Caves, Pandavleni Caves, Pataleshwar Caves, Mahakali Caves etc.

Meghalaya - Krem Liat Prah-Umim-Labit cave, Krem Puri cave, Mawsmi Caves, Siju Cave, Krem Tyngheng Diengjem, Krem Shriek (tanghub), Krem Lympu Nongjri, Krem lawe.

Odisha - Udayagiri and Khandagiri caves, Lalitgiri Caves, Ratnagiri caves, Gupteswar Cave, Bikramkhola (Vikramkhola) cave etc.

Rajasthan - Binnayaga Buddhist Caves, Hathiagor Buddhist Caves, Kolvi Caves.

Tamil Nadu - Armamalai cave, Sittanavasal Caves, Varaha Cave Temple, Trichy Caves, Trimurti Cave, Kongana Siddhar cave, Kukkal Caves, Pancha Pandava Cave, Mahishasuramardini Cave, Vallimalai Jain caves, and Gudiyam Cave.

Uttarakhand - Koteswar caves, Lakhudiyar cave, Patal Bhuvneshwar cave.

Uttar Pradesh - Muchkund Cave (Lalitpur) - This natural caves are situated in the valley of river Betwa.

West Bengal - Laljal Cave (Jagram district), Mahakal Caves.

Jammu and Kashmir - Amarnath cave, Gufkral, Vaishno Devi Caves, Kalaroo Caves, Bumzuva cave etc.

Samadhi Sthal

Raj Ghat is a memorial dedicated to Mahatma Gandhi in Delhi, India.

Vijay Ghat is the samadhi of Lal Bahadur Shastri, situated in New Delhi.

Chaityabhoomi is the samadhi of B.R.Ambedkar situated in Mumbai.

Mahaprayan Ghat is the samadhi of Dr Rajendra Prasad is situated in Patna.

Hussainiwala National Martyrs Memorial, is the burial place of Bhagat Singh, Rajguru and Sukhdev which is situated at Hussainiwala village, near Firozpur city Punjab, India.

Phool Bagh is the Samadhi of Rani Lakshmi Bai situated at Gwalior, Madhya Pradesh.

Shantivan is the Samadhi of India's first Prime Minister, Jawaharlal Nehru situated in Delhi.

Sardar Vallabhbhai Patel National Memorial is a museum and exhibition centre dedicated to Sardar Vallabhbhai Patel at Moti Shahi Mahal, located in Shahibag, Ahmedabad, Gujarat.

The **Chandrashekhar Azad Park** or Alfred Park in Prayagraj (Allahabad) is the place where Azad got martyred. His body was transported to Rasulabad Ghat for cremation.

Mahabat Maqbara and Bahauddin Maqbra are mausoleums in Junagadh, Gujarat dedicated to Mahabat Khan II, the Nawab of Junagadh State, and his minister Bahauddin Hussain Bhar respectively.

Veer Bhumi in Delhi serves as the Samadhi Sthal of former Prime Minister Rajiv Gandhi.

Shakti Sthal located in Delhi, It is the Samadhi of former Prime Minister Indira Gandhi.

Abhay Ghat in Ahmedabad is the resting place of Morarji Desai.

Karma Bhumi signifies the Samadhi Sthal of Dr. Shankar Dayal Sharma.

Ekta Sthal in Delhi commemorates Giani Zail Singh, a former President of India.

Samta Stal, the resting place of Jagjivan Ram, the former Deputy Prime Minister of India.

Jannayak Stal honors Chandra Shekhar, a former Prime Minister of India.

Smriti Sthal serves as a place of remembrance for IK Gujral.

Tombs and Monuments

Tombs

Andhra Pradesh - Syed Ahmed Sahib's Tomb (Cuddapah district).

Bihar - Tomb of Sher Shah Suri (Sasaram), Tomb of Bakhtiyar Khan, The Tomb of Syed Ibrahim Malik, Tomb of Hasan Shah Suri, Tomb of Mahmud Shah.

Gujarat - Bahauddin Maqbra (Junagadh), Mahabat Maqbara, Tomb of Sikandar Shah, Tomb of Khawaja Safar Sulemani, Hazira or Qutbuddin Mahmud Khan's Tomb, Tomb of Khudawand Khan.

Haryana - Tomb of Sheikh Chilli (Taj Mahal of Haryana), Khwaja Khizr Tomb - Sonapat, Prampir Badshah Tomb (Ancient Gombaj), Tomb of Sheikh Tayyab, Mosque of Pir Turkman and Tomb - Narnaul (Mahendragarh), Tomb of Bu-Ali Shah Kalandar - Panipat, Tomb of Ibrahim Lodhi - Panipat.

Karnataka - Gol Gumbaz (Bijapur), Tomb of Khwaja Bande Nawaz (Gulbarga), Raja's Tomb (Gaddige).

Madhya Pradesh - Tomb of Hoshang Shah (Mandu, considered to be India's first marble tomb), Tombs of Mohammed Ghaus, Tombs of Tansen, Tomb of Sawai Singh, Mahmud Khilji's Tomb, Tomb of Adil Shah Faruki.

Maharashtra - Tomb of Aurangzeb, Bibi-Ka-Maqbara, Asaf Jahan's Tomb, Prince Ajham Shah's Tomb, Shahid Nasir Jung's Tomb, Abdul Hasan Shah's Tomb, Nakkur Hussain's Tomb, Roshan Khan's Tomb, Chand Bibi's Tomb (Ahmednagar).

Punjab - Tomb of Ustad, Tomb of Shagird and Tomb of Amir Ali in Fatehgarh Sahib district.

Rajasthan - Tomb of Fateh Jung (Alwar), Tomb of Allauddin Khan (Ajmer), Tombs of Abdulla Khan (Ajmer).

Tamil Nadu - Kanja Sahib Tomb.

Telangana - The Qutb Shahi Tombs (Hyderabad), Khairat Khan's Tomb, Shams-ul-Umra Tombs (Paigah Tombs). Tomb of Behlul Shah Wali - Nizamabad.

Uttar Pradesh - Tomb of Akbar, Tomb of Itimad-ud-Daulah ('baby Taj'), Tomb of Mariam-uz-Zamani, Tomb of Salim Chishti.

Delhi - Tomb of Humayun (UNESCO World Heritage Site in 1993), Tomb of Muhammad Shah, Sultan Ghari's Tomb, Safdarjung's Tomb.

Monuments

Assam - Ahom Raja's palace, Kareng Ghar (Garhgaon Palace), Powa Mecca, Rang Ghar, Talatal Ghar.

Bihar - Golghar, Pathar Ki Masjid (Saif Khan's Mosque)

Gujarat - Teen Darwaza (Ahmedabad).

Haryana - Vishvamitra ka Tila (Kurukshetra).

Karnataka - Daria Daulat Bagh (Mysore), Mysore Palace (Amba Vilas Palace, Henry Irwin was the architect of this palace), Rangan Mahal.

Kerala - Kanakakunnu Palace (Thiruvananthapuram), Mattancherry Palace (Kochi).

Ladakh - Leh Palace (Lachen Palkar Palace).

Maharashtra - Chand Minar (Tower of the Moon), The Gateway of India (It was designed by George Wittet).

Madhya Pradesh - Jahaz Mahal, Rajwada Palace (Holkar Palace), Sheesh Mahal.

Punjab - Sarai Nurmahal, Sheesh Mahal (Palace of Mirrors).

Rajasthan - Adhai Din Ka Jhonpra (Ajmer), Hawa Mahal (Jaipur, There are 953 windows on the outside walls). Kirti Stambh (75 feet), Lalgah Palace (Bikaner), Monsoon Palace, Sheesh Mahal (Mirror palace), Umaid Bhawan Palace, Amber Fort, City Palace, Jantar Mantar, Mehrangarh Fort, Jaisalmer Fort, Kumbhalgarh Fort, Chittorgarh Fort, Saheliyon-ki-Bari (Udaipur).

Tamil Nadu - Thirumalai Nayakkar palace (Madurai), Vivekananda Rock Memorial (Kanyakumari).

Tripura - Ujjayanta Palace or Nuyungma (Agartala).

Telangana - Charminar, Chowmahalla Palace.

West Bengal - Black Hole, Dakshin Darwaza (Malda), Hazarduari Palace (Bara Kothi), Shaheed Minar (Ochterlony Monument), Victoria Memorial (Taj of the

Raj), Shantiniketan - Rabindranath Tagore. Belur Math - Swami Vivekananda.

Uttar Pradesh - Bara Imambara or Asfi Imambara (Lucknow), Buland Darwaza (The Door of Victory, Fatehpur Sikri), Chota Imambara or Imambara Hussainabad Mubarak (Lucknow), Haveli of Aga Khan and Hathi Khana, Kanch Mahal, Panch Mahal, Rumi Darwaza (60 feet tall), Sheesh Mahal, Taj Mahal, (located on the banks of Yamuna river in Agra), It was built by Shah Jahan (Mughal emperor). It was designated as a UNESCO World Heritage site in 1983. It became one of the seven wonders of the world in 2007).

Chandigarh - The Open Hand Monument (designed by Le Corbusier).

Delhi - Alai Darwaza, India Gate (It was designed by Sir Edwin Lutyens), Jahaz Mahal, Qutb Minar complex, Qutub Minar (It has 5 storeys), Qubbat-ul-Islam Mosque, Qutb-al-Aqtab Khwaja.

Famous Stupas

Stupas were burial mounds prevalent in India from the vedic period. A stupa, Buddhist commemorative usually housing sacred relics associated with the Buddha or other saintly persons.

Structure of Stupa

Anda- Hemispherical mound symbolic of the mound of dirt used to cover Buddha's remains (in many stupas actual relics were used).

Chaityas- It is similar to a prayer room. It featured pillars on each side of a tunnel or walkway that lead to a stupa.

Harmika- Square railing on top of the mound.

Pradakshina path- Stupas have more than one circumambulatory path.

Torana- Indian gateway, usually of stone, marking the entrance to a Buddhist shrine or stupa or to a Hindu temple.

Vedika- Wooden or stone railing that surrounds the stupa.

Viharas (Monasteries) are meditation cells and living quarters for the monks.

Yashti- A Central pillar symbolising the cosmic axis and Supports a triple umbrella structure (Chattra).

Stupas of India

Amaravati Stupa - The great stupa or Mahachaitya at Amaravati was one of the biggest in Andhra Pradesh (Guntur district) with a probable diameter of 50 metres and a height of 27 metres.

Bharhut Stupa (Satna, Madhya Pradesh) was discovered by Alexander Cunningham in 1873. The narrative art of Bharhut, depicts Jatakas of Buddha's previous birth in Sculptures. At times more than one event at one geographical place is clubbed in the picture space. Other Stupas in Madhya Pradesh are Deur Kothar and Sonari.

Dhamek Stupa is located at the archaeological site of Sarnath, in Varanasi, Uttar Pradesh. It was built in 249 B.C. by Ashoka. This solid cylindrical shaped Stupa made of red bricks and stone in its present position stands at a height of 43.6 m with a diameter of 28 m. Another Stupa in Uttar Pradesh is Chaukhandi Stupa.

Kesariya Stupa, a buddhist stupa in East Champaran district of Bihar, was discovered in 1958 during an excavation led by K. K. Muhammed. It is the tallest and the largest Buddhist stupa in the world. Faxian (Fahien) and Xuan Zang (Hsuan Tsang), had visited this place in ancient times. Other Stupas in Bihar-Maha Bodhi (UNESCO World Heritage Sites, 2002).

Nagarjunakonda Stupa is located near Nagarjuna Sagar in Palnadu, Andhra Pradesh. It belongs to the Saka-Satavahana era. Jaggayyapeta, Amaravati, Bhattiprolu, Nagarjunakonda, Goli, etc are found in the principal sites of Vengi. Nagarjunakonda, also known as Vijayapuri, was the capital of the Ikshvaku Kingdom.

Piprahwa Stupa (Siddharthnagar, Uttar Pradesh). Piprahwa-Ganwaria was the site of the ancient city of Kapilavastu, the capital of the Shakya kingdom, where Siddhartha Gautama spent the first 29 years of his life.

Ramabhar Stupa (Kushinagar, Uttar Pradesh) is 49 feet long, and is now a large brick mound. This is where the Buddha was cremated. The Chinese travellers Fa Hien and Hiuen Tsang visited Kushinagar during different centuries.

Sanchi Stupa (Sanchi town, Raisen, Madhya Pradesh) was discovered by Sir

Henry Taylor in 1818. It is a Buddhist complex built by Ashoka (Mauryan Emperor) in the third century BC. Since 1989, Sanchi Stupa has been a UNESCO World Heritage Site. There are 4 gates (torana) around the Stupa in four directions namely East (Buddha's birth), West (First Sermon where he preached his teachings or dharma), North (Nirvana) and South (Enlightenment) which depict the life events of Buddha.

Shanti Stupa is located on a hilltop in Chanspa, Leh district, Ladakh. It was built in 1991 by Japanese Buddhist Bhikshu, Gyomo Nakamura.

Vishwa Shanti Stupa (Peace Pagoda) is located in Rajgir, Bihar on a 400 m high Ratnagiri hill.

Stupas around the world

Boudhanath Stupa (Kathmandu, Nepal) has been a UNESCO World Heritage Site since 1979.

Benalmadena Stupa (Andalusia, Spain) is the tallest stupa in Europe at 108 feet (33 m) high.

Borobudur (Java, Indonesia) is the largest Buddhist temple in the world.

Kyaiktiyo Pagoda (Golden Rock Stupa) Myanmar.

Ruwanwelisaya (Sri Lanka)- The stupa was built in 140 BCE by King Dutugemunu.

Swayambhunath Stupa (Kathmandu, Nepal) was built by King Manadeva, in 460 A.D. It is also known as 'Monkey Temple'. It has been a UNESCO World Heritage Site since 1979.

The Great Stupa of Dharmakaya (USA) was built to inter the ashes of Chogyam Trungpa, who died in 1987.

The Great Stupa of Universal Compassion is a Buddhist monument located near Bendigo in central Victoria, Australia.

The One Hundred and Eight Stupas (China, also known as the 108 Pagodas)- It is a Buddhist monument..

Famous Statues

Statues of India

Statue of Unity (Tallest statue in the world) is located on the banks of the

Narmada river (island named Sadhu Bet) Gujarat. Height - 182 metres (597 ft). It is a statue of Sardar Ballabh Bhai Patel. Sculptor - Ram Vanji Sutar. Built - By Larsen and Toubro (2018).

The Statue of Belief or Vishwas Swaroopam (Nathdwara, Rajasthan) is a statue of the Hindu God Shiva. Height - 112.4m (369 ft). Sculptor - Naresh Kumawat.

Statue of Prosperity (Bengaluru) is a statue of Sri Nadaprabhu Kempegowda. Height - 108 feet (About 33 m). Built - 2022. Sculptor - Ram Vanji Sutar.

Statue of Knowledge (Latur, Maharashtra) is a statue of Dr. Babasaheb Ambedkar.

Statue of Peace (Jainacharya Shree Vijay Vallabh Surishwer Ji Maharaj) - Location - Pali, Rajasthan. The 151 inch tall statue has been made from Ashtadhatu i.e. 8 metals. Built - 2020.

The Statue of Equality (Ranga Reddy, Hyderabad) is a statue of Indian philosopher Ramanuja. Height - 65.8m (216 ft).

The Statue of Ahimsa is the tallest Jain statue in the World that belongs to the first Jain Tirthankara, Rishabhanatha. The 108 feet tall statue has been carved out of the Mangi-Tungi hills near Nashik, Maharashtra.

The Great Buddha statue (Bodh Gaya, Bihar)- The statue is 25 m (80 ft) high representing the Buddha seated in a meditation pose, or dhyana mudra, on a lotus in the open air.

Hanuman Statue (Srikakulam, Andhra Pradesh).

Panchamukhi Hanuman Statue (Tumakuru, Karnataka)- Height - 49m (161 ft).

Muthumalai Murugan Statue (Salem, Tamil Nadu)- Height - 44.5m (146 ft). Built - 2022.

Maa Vaishno Devi Statue (Vrindavan Mathura, Uttar Pradesh). Built - 2010.

Veera Abhaya Anjaneya Hanuman statue (Vijayawada Andhra Pradesh).

Thiruvalluvar Statue (Kanyakumari, Tamil Nadu)- Height - 133 feet (41 metres).

Gommateshwara statue - Location - Shravanbelagola, Karnataka. Height - 17 m (57ft). Statue of Lord Bahubali.

Tathagata Tsal - Location - Ravangla, Sikkim. Statue of Gautam Buddha.

Ganapati Dham Temple - Location - Bahadurgarh, Haryana. Built - 2009.

Navagraha Jain Temple Statue - Location - Hubli, Karnataka. Built - 2010.

Sri Abhaya Anjaneeya Swamy - Location - Hanuman Junction Andhra Pradesh. Built - 2015.

Kuravan and Kurathi Statue - Location - Idukki District, Kerala. Built - 2005

Statue of Mahavira - Location - At Jain temple in Jaisalmer Fort, Rajasthan.

The statue of Padmasambhava (Rewalsar town, Himachal Pradesh) is a statue of Guru Rinpoche. Built - 2012.

Sai Baba statue (Sai Baba of Shirdi)- Location - Repuru, Kakinada, Andhra Pradesh. Built - 2015.

Adiyogi Shiva statue (Coimbatore, Tamil Nadu) is a statue of lord Shiva. Height - 112 ft (34 m).

The reclining nirvana statue of Lord Buddha- Location - Kushinagar, Uttar Pradesh. It represents the 'Dying -Buddha'. It is 6.10 m long.

Statue of Subhas Chandra Bose (Kartavya Path, Delhi)- Height - 8.53 m (28 ft). Sculptor - Shri Arun Yogiraj. Inaugurated on 8th September 2022.

Rampurwa bull, a stone sculpture located in the entrance of the Rashtrapati Bhavan, New Delhi.

Chauri-bearer statue located in Bihar Museum, Bihar.

Sultanganj Buddha, a gupta period statue, is located in Birmingham, England.

Statues of Mahatma Gandhi - World's tallest bronze statue of Mahatma Gandhi- Location - Gandhi Maidan, Patna, Bihar. Height - 72 feet (22 m). Sculptor - Ram Sutar.

His other statues Location- Manchester (London), Sculptor - Ram Vanji Sutar. Built on the 150th anniversary of Gandhi's birth. It is a 9-foot high, 800 kg bronze statue; Johannesburg (South Africa), Sculptor - Ms. Tinka Christopher;

Parliament Square, Westminster (London), Sculptor - Philip Jackson (2015).

Statues in the World

The Spring Temple Buddha (2nd tallest statue in the world)- Location - Henan (China).

Laykyun Sekkya (Khatakan Taung, Myanmar) is a statue of Gautama Buddha. Built on 21 February 2008.

Sendai Daikannon- Location - Japan. Built - 1991.

Garuda Wisnu Kencana statue is located in Bali, Indonesia.

Guishan Guanyin of the Thousand Hands and Eyes (Hunan, China)- Built - 2009.

The Big Buddha of Thailand- Location - Thailand. Built - 2008.

Statue of Lenin is a 16 ft bronze statue of Russian Communist revolutionary Vladimir Lenin in the Fremont neighbourhood of Seattle, Washington, United States.

Buddha of Bamiyan- These were two 6th-century monumental statues carved into the side of a cliff in the Bamiyan valley of central Afghanistan.

Mother of All Asia - Tower of Peace- Location - Batangas City, Philippines.

The Motherland Calls (Russia) was created by sculptor Yevgeny Vuchetich.

Christ the Redeemer (Rio de Janeiro, Brazil)- Sculptor - Paul Landowski.

Famous Monastery

Major Monasteries in India

Andhra Pradesh

Thotlakonda Monastery is a 2000 years old monastery atop Thotlakonda Hill, Andhra Pradesh.

Arunachal Pradesh

Bomdila Monastery is also known as Gentle Gaden Rabgyel Lling monastery.

Tawang monastery is the largest monastery in India and is known as the Tawang Garden Namgyal Lhatse.

Urgelling Monastery - Urgelling Gompa is situated near Tawang.

Himachal Pradesh

Dhankar Monastery is located in the district of Lahaul and Spiti and at an elevation of 3894 meters.

Kye Monastery is located in the Spiti Valley and at an altitude of 4166 meters above sea level.

Kardang Monastery - A famous Drukpa Lineage monastery, situated in Lahaul and is situated on the left bank of River Bhaga. It was renovated in 1912 CE by Lama Norbu Rinpoche.

Namgyal Monastery is the Personal Monastery of the 14th Dalai Lama located in McLeod Ganj, Dharamsala.

Shashur Monastery - A Buddhist monastery of the Drugpa sect in Lahaul and Spiti.

Tsuglagkhang Monastery - This is the home for His Holiness Dalai Lama and is situated in McLeodganj suburb of Dharamshala district.

Tabo Monastery is located in Tabo village of Spiti Valley and built by 'Yeshe O'd'.

Karnataka

Bylakuppe monastery (Namdroling Monastery) is located in Bylakuppe, in the district of Mysore.

Dzogchen Monastery - The original Dzogchen Monastery in Tibet was established by Pema Rigdzin, the first Dzogchen Rinpoche.

Ladakh

Alchi Monastery is situated in Alchi village of Leh District. It was constructed by Guru Rinchen Zangpo.

Hemis monastery is famous for the annual festival (Hemis Gompa) for Guru Padmasambhava which is held in June-July.

Korzok (Karzok) Monastery is located in the Korzok village, on the northwestern bank of Tso Moriri (lake) in Leh District.

Matho Monastery was founded by Lama Tungpa Dorjay.

Phugtal Monastery is located in Lungnak valley in the Zaskar region. It is built around a cave in Ladakh which is believed to have been visited by Sage, Saints and Scholar 2500 years ago.

Shey Monastery and Shey Palace complex are located on a hillock in Shey. It contains a huge Shakyamuni Buddha statue.

Thiksey Monastery is located on top of a hill in Thiksey village in the east of Leh.

Sikkim

Enchey Monastery - The literal meaning of Enchey Monastery is 'the solitary temple'. It was established in 1909 in Gangtok.

Gonjang Monastery was established by Tinkye Gonjang Rimpoche.

Lingdum Monastery is located in Ranka village in Gangtok. This monastery follows the Zurmang Kagyu tradition.

Phodang Monastery - It comes under the Kagyupa Sect, and was founded by the fourth King Gyurmed Namgyal.

Pemayangtse Monastery is located in Pemayangtse near Pelling. The Planned, designed and founded by Lama Lhatsun Chempo.

Rumtek Monastery (Dharmachakra Centre) is located in Gangtok.

Ralang Monastery is a Buddhist monastery of the Kagyu sect.

Uttarakhand

Mindrolling Monastery was established in Clement Town, Dehradun by Khochhen Rinpoche.

West Bengal

Ghoom or Ghum (Darjeeling) Monastery was founded by Lama Sherab Gyatso.

Jammu and Kashmir

Harwan Monastery is located on the outskirts of Srinagar in Harwan.

Major Monasteries of the World

Bagan Monastery (Burma) is located on the banks of the Ayerwaddy river.

Fo Guang Shan is a Chinese Mahayana Buddhist monastery located in Kaohsiung, Taiwan.

Kopan Monastery (Nepal) is located on the outskirts of Kathmandu.

PunaKha Dzong (Bhutan) is standing on an island between the confluence of the Pho Chhu and Mo Chuu rivers.

Santa Catalina Monastery is located in Arequipa, Peru.

Tengboche Monastery (Nepal) is also known as Dawa Choling Gumpa. It is located in the Khumbu valley at an altitude of 3867 metres.

Taung kalat (Burma) is built at the top of an extinct volcano plug, which is also one of the most breathtaking sites in Burma.

Taksang (Tiger's Nest) Monastery (Bhutan) is situated on the edge of a 900 meter cliff and This is the spot where Guru Rinpoche did his meditation.

Wat Rong Khun Monastery (Thailand) is one of the most unique Buddhist monasteries in the world.

XuangKong Monastery (The Hanging Temple) (China) - Built into a rock on Mount Heng.

Sports (Outdoor/Indoor)

There are two types of Sports- Outdoor and Indoor.

Outdoor Sports

Cricket - The first known game in which the teams used county names is in 1709. It was popular and widely documented in England during 1700. The first international cricket game was played between the USA and Canada in 1844.

The first-ever official international Test match was played between Australia and England at Melbourne in 1877. The first Indian Cricket club named the Parsee Oriental Cricket Club was founded in the year 1848 and they played their first match in Mumbai.

India's first ever Test victory came against England at Madras in 1952. India's first series victory was against Pakistan later the same year. In 1967-68, India won their first series outside the subcontinent against New Zealand under Mansur Ali Khan Pataudi.

Golf - It originated in 15th-century Scotland. The 18-hole round was created at St Andrew's in 1764. Golf's (the world's oldest tournament) first major tournament is the Open Championship (British Open) was first played in 1860 at

the Prestwick Golf Club in Ayrshire, Scotland. Golf is a club-and-ball sport in which players use various clubs to hit balls into a series of holes on a course in as few strokes as possible.

Hockey - The International Hockey Federation was established in Paris in 1924, initiated by Paul Leautey. Indian Hockey Federation (IHF) was formed in 1925. The FIH (International Hockey Federation) Women's Indoor Hockey World Cup 2015 held at Leipzig Germany. The FIH Men's Indoor Hockey World Cup 2018 was organized at Berlin, Germany. The FIFA Men's World Cup, held in 2018 in Russia. The 2018 FIH Men's Hockey World Cup in Belgium. India defeated four-time champions South Korea 2-1 and clinched their maiden Women's Junior Hockey Asia Cup in Japan's Kakamigahara.

Football - It was introduced in the nineteenth century. The Football Association Challenge Cup (the world's oldest football championship) was started in 1872. In 1888, the 'Durand Cup' (the second-oldest football tournament in the world) was started in Shimla (India). The first men's football world cup started in 1930.

Polo - It was developed and formalized in ancient Iran as chovgan or Chogan. The modern game of polo is derived from Manipur, where the game was known as 'Sagol Kangjei', 'Kanjai-bazee', or 'Pulu'. It is a ball game played on horseback, a traditional field sport.

Kho-Kho - The modern form of Kho-Kho was shaped by the Deccan Gymkhana of Pune. The Kho-Kho Federation was established in 1956. Each kho-kho team consists of 12 players, but during a contest, only 9 players from each team take the field.

Volleyball - It was originally called Mintonette and was invented in 1895 by William G. Morgan. The Volleyball Federation of India (VFI) was formed in the year 1951.

Judo derives from the martial art of jujutsu and originated in Japan. Jigoro Kano was the founding father of Judo in 1882. The Judo Federation of India (JFI) was formed in India in 1965. The first National Judo Championship was held in 1966 at Hyderabad.

Gymnastics - The Federation of International Gymnastics (FIG) was

formed in Liege in 1881. The Gymnastics Federation of India (GFI) was initiated in 1951. It was affiliated with the Indian Olympic Association (IOC).

Ice hockey - It is a team sport played on ice skates, usually on an ice skating rink with lines and markings specific to the sport. It belongs to hockey.

Cycling - It consists of professional and amateur races. Cycling as a sport officially began on May 31, 1868. Cycle sport is a competitive physical activity using bicycles.

Indoor Sports

Chess - It originated in India during the Gupta dynasty in the 6th century. The recorded history of chess goes back at least to the emergence of a similar game, chaturanga, in seventh-century India. The International Chess Federation or World Chess Federation (FIDE) was founded in Paris, France, on 20 July 1924.

Badminton - It is also known as Poona or Poonah after the garrison town of Poona. India joined the Badminton World Federation (1934) as an affiliate in 1936.

Table Tennis - It originated in England. It is also known as ping-pong and whiff-whaff. It is governed by the International Table Tennis Federation (ITTF), founded in 1926. The first edition of the World Para Table Tennis Championships was held in 1990 Assen, Netherlands.

Kabaddi - It is known as Hu-Tu-Tu in Western India, Ha-Do-Do in Eastern India and Bangladesh, Chedugudu in Southern India. It is the national sport of Bangladesh. The Kabaddi Federation of India (KFI) was founded in 1950, and it compiled a standard set of rules. The Amateur Kabaddi Federation of India (AKFI) was founded in 1973. It made its first appearance as an exhibition sport in 1982 and became an Asian Games event in 1990 in Beijing, China. Until the 2018 edition, India always dominated the event by winning the highest number of gold medals in both men's and women's categories.

Basketball - It was invented in 1891 by Canadian-American gym teacher James Naismith in Springfield, Massachusetts, United States.

Snooker - It originated in the second half of the 19th century in India.

Swimming - It is an aquatic sport that requires the use of one's entire body to move through the water. World Aquatics was formerly known as FINA (Federation Internationale de natation or International Swimming Federation) established on 19 July 1908. The Swimming Federation of India (SFI) was formed in 1948.

Handball - International Amateur Handball Federation formed in 1928. Jagat Singh Chauhan of Haryana, India, was a Founder of 'Handball Federation of India. The First Field Handball World Championships were played in Germany in 1938.

Wrestling - It was developed by ancient Greeks as a way to train soldiers in hand-to-hand combat and military exercises. The first national wrestling tournament was hosted in New York City in 1888. Types of wrestling- Freestyle, Greco-Roman and Sumo Wrestling in Japan.

Squash - It is a racket and ball sport played by two or four players in a four-walled court with a small, hollow, rubber ball. The International Squash Rackets Federation (ISRF) was formed in 1967, its name being changed in 1992 to the World Squash Federation (WSF).

Chaupar - It is a cross and circle board game very similar to pachisi, played in India.

Other Sports

Boxing - Western boxing (Pugilism) is a combat sport in which two people, usually wearing protective gloves and other protective equipment such as hand wraps and mouthguards, throw punches at each other for a predetermined amount of time in a boxing ring. The first governing body for boxing in India, the Bombay Presidency Amateur Boxing Federation was formed in Mumbai in 1925.

Lawn tennis - The Open Era in tennis began in 1968 when the Grand Slam tournaments agreed to allow professional players to compete with amateurs.

Aquatic sports - Swimming, diving, Synchronized swimming, Water polo, open water swimming.

Korfball - It is a ball sport, with similarities to netball and basketball, Its

teams consist of eight players (mixed gender teams) – four male and four female. The size of the court is 20 m × 40 m (22 yd × 44 yd).

Fencing - It is a combat sport that features sword fighting. A contest between the two fencers is known as bout.

Archery - It is the sport, practice, or skill of using a bow to shoot arrows. It was developed in Africa.

Equestrian is most often described as gymnastics and dance on horseback.

Triathlon - It is an endurance multisport race consisting of swimming, cycling, and running over various distances.

Surfing - It is a water sport where an athlete rides a wave standing or any other position on a board, or just with his body (body surfing). It was a central part of ancient Polynesian culture.

Croquet - That involves hitting wooden or plastic balls with a mallet through hoops embedded in a grass playing court.

Roque - It is an American variant of croquet played on a hard, smooth surface.

Lacrosse - It is a team sport played with a lacrosse stick and a lacrosse ball.

Taekwondo - It is a Korean form of martial art.

Basque Pelota - It is the name for a variety of court sports played with a ball using one's hand, a racket, a wooden bat or a basket, against a wall more traditionally with two teams face to face separated by a line on the ground or a net.

Track Event - All running events come under it. Classification of Track Events- Short Distance Races (100 m), Middle Distance Races (800 m), Long Distance Races (3000 m steeplechase).

Field Event - The jumping and throwing events are called field events. Classification of Field Events- Jumping Events and Throwing Events. Jumping Events- Long Jump, High Jump, Triple Jump, Pole Vault. Throwing Events- Shot Put (Glide Technique used in this game), Discus Throw, Javelin Throw, Hammer Throw.

Powerlifting - There are three events-

bench press, squat and deadlift (weightlifting style).

Decathlon - Events are held over two consecutive days. Men's Decathlon- Day 1 - 100 meters, Long jump, Short put, High jump, 400 meters. Day 2 - 110 meters hurdles, Discus throw, Pole vault, Javelin throw, 1500 meters. Women's Decathlon- Day 1 - 100 meters, Discus throw, Pole vault, Javelin throw, 400 meters. Day 2 - 100 meters hurdles, Long jump, Shot put, High jump, 1500 meters.

Traditional Games of India

Uriyadi is played during Onam celebrations and on Shri Krishna Jayanti (Ashtami Rohini).

Kilithattu Kali also known as 'Thattukali', the traditional game of Kilithattu Kali is slowly losing its popularity.

Jallikattu is a traditional sport of Tamil Nadu typically practiced during the Pongal celebrations in January. Pulikulam or Kangayam breed of cattles are used for Jallikattu.

Pachisi was the game of ancient India, which means twenty-five in Hindi and played on a board or circle board.

Silambam is an Indian martial art originating in Tamilnadu. This style is mentioned in Tamil Sangam literature. The World Silambam Association is the Official international body of Silambam.

Mardani khel was a martial art form of Maharashtra. Popularized by Chhatrapati Shivaji.

Dahi Handi is an official sport in the state of Maharashtra, India.

Sagol Kangjei is the other name of polo played in Manipur.

Mallakhamb is the display of aerial yoga and wrestling grips with vertical stationary or hanging wooden pole. It was introduced in the 36th national games. Madhya Pradesh declared Malakhamba as the 'State Sport' in 2013.

Yubi Lakpi is a seven-a-side traditional football game played in Manipur, using a coconut, which has some similarities to rugby.

Kambala is a sport where a pair of buffaloes run one by one. It Played in Karnataka mainly in districts of Dakshina

Kannada and Udupi.

Kalaripayattu is a form of Indian martial arts. Some regional Indian martial arts - lathi khela from Bengal, Huyen langlon from Manipur, Kalari from Kerala, gatka from Panjab and silambam from Tamil Nadu.

Dhopkhel is a traditional ball game played in Assam. The game is played between two teams of 11 on a 125 m x 80 m field bounded by four flags and resembles kabaddi played with a ball.

Vallam kali also known as Snake Boat Race is a traditional boat race in Kerala, India.

Bullock cart race is known as Bailgada Sharyat in Maharashtra.

Pallanguzhi is a traditional indoor board game, played in TamilNadu. It is played by two players on a rectangular board.

Insuknawr game is a rod pushing sport played in Mizoram. In this game two players try to push each other out of a ring by holding a bamboo or wooden staff.

Nadan Panthu Kali is a traditional sport in Kerala. Each team consists of 5 or 7 players.

Thang ta is a form of martial arts in Manipur. The game is played by using a sword or spear against one or more opponents.

Thoda is a form of martial art in Himachal Pradesh.

Mukna is a form of folk wrestling from the north-east Indian state of Manipur.

Gatka is a form of martial art in Punjab. It is a style of fighting with wooden sticks between two players.

Seval Sandai or cockfighting is a game in Tamil Nadu and the coast of Andhra Pradesh.

Maramadi Bull Surfing is a race from Kerala that takes place in Anandapally village where a pair of bulls race with different categories for speed and style in paddy fields.

Ball Badminton is a sport native to India and in early days was also called "Poona". It is played with a yellow ball made of wool. In **Ball Badminton**, each game immediately after the first 9th, 18th and 27th point the teams shall change

the sides and the server shall serve from the right court or left court as he would have done in the normal course if the side was not changed.

Sports Terminologies

Basketball

Free throw - A free throw is an unguarded scoring attempt that gives a basketball team the chance to score extra points after a foul. A free throw is worth one point.

Dribbling - In basketball, dribbling is bouncing the ball on the floor continuously with one hand at a time. It is the only legal way that a player may maintain possession of the ball while walking or running.

Screen Technique - When a player uses their body as a barrier to their teammate's defender, giving their teammate more space to move.

Other Terminologies - Slam dunk, Full-court press, Alley-oop, Jump shot, Playmaker, Point guard, Hoopster, Sixth man, Bonus, Cherry Picking, Overhead Pass, Underhand Pass, Pivoting.

Hockey

Long Corner - A long corner is awarded if a defender accidentally puts the ball over the back line.

Scoop - It is a method of passing or shooting the ball by picking it up into the air and fling it.

Hit - It is to hit the ball with the swinging stick.

Yellow Card- In hockey, It refers to temporary suspension of a player for a minimum of five minutes of playing time.

Other Terminologies - Flick, Striking Circle, Scoop, Back stick, 16 yard hit, Field hockey, Hockey stick interference, Undercutting, Sticks, Sudden death, Quarterback, blue line.

Cricket

Straight Drive- A deliberate shot that aims to hit the ball along the ground to prevent being caught out.

Cover Drive- The act of stroking the ball through the covers with well-timed wristwork and conventional movement of the front foot toward the pitch of a

delivery aimed at or outside the off stump.

Strike Rate- The Average number of runs a batter makes per 100 deliveries in cricket.

Four- A batsman hitting the ball to the boundary along the ground.

Six- A player hitting the ball so that it crosses the boundary at the edge of the playing area before it touches the ground.

Maiden Over - An over in which no run is scored by the batsman.

First slip is the fielding position behind the batsman.

Mid on and Mid off are the two fielding positions located on either side of the pitch within the 30 yard circle.

How's that- An appeal is the act of a player on the fielding team asking an umpire for a decision regarding whether a batter is out or not.

Types of out in cricket -

Caught- To catch a ball hit in the air by a batsman before it falls to the ground.

Bowled- When the bowler hits the ball to the stumps behind the batsman while bowling.

Leg Before Wicket (LBW) - A batsman is considered out LBW if they obstruct a ball which was en route to hit the stumps with any part of their body barring their hands.

Stumped- It is a method of dismissing a batsman in cricket, which involves the wicket-keeper putting down the wicket while the batsman is out of his crease.

Run-out - When the batsman is outside the popping crease and the stumps nearest to him are put down while the ball is in play.

Obstructing the field - When batsman obstructs or distracts the fielding side by words or action.

Hit Wicket - When the batsman dislodges his own stumps with his body or bat, while hitting a shot or taking a run.

Hit the ball twice - When the batsman hits the ball twice – first time lawfully and second time intentionally.

Retired out - When the batsman leaves the field without the Umpire's consent for

reasons other than injury or illness.

Other Terminologies - Off break, Mankading, Follow-on, Fine leg, Mid Wicket, Gully, Third-man, Googly, Short-pitched, Full toss, Square leg, Yorker, Wicket, Pitch, Bye, Beamer, Seamer, leg bye, Chinaman, Pavilion, No ball, Duck.

Table Tennis

Volley- A shot hit, usually in the vicinity of the net, by a player before the ball bounces on their own side of the court.

Spin- It is imparted onto the ball by using a tangential brushing action with your racket.

Rally- The period in which the ball is in play.

Other Terminologies - Twiddle Backhand, Half Volley, Chop, Jamming, Shakehand grip.

Kabaddi

Tackle point- Tackling a raider and stopping him from escaping to the midline amounts to one point.

Lona- In Kabaddi, a team scores a bonus of two points, called a "Lona", if the entire opposing team is declared out.

Other Terminologies - Baulk line, Bonus line, End line, Raid, Live, Cant, Broken Cant, Hu-Tu-Tu, Super Tackle, Super Raid, Do or Die Raid, Raider, Struggle, etc.

Snooker

Break Shot - The first shot of a game.

Cue - It is the stick with which the game is played.

Other Terminologies - Backspin, Cushion, In-Of, Scratch, Jigger, Stroke.

Chess

Stalemate is a position in which the player whose turn it is to move has no legal move and his king is not in check.

Checkmate is a game position in chess and other chess-like games in which a player's king is in check and there is no possible escape. Checkmating the opponent wins the game.

Other Terminologies - Bishop, Gambit, Castling.

Badminton

Smash - A hard, powerful overhead shot sending the shuttlecock downward to the opponent's side.

Other Terminologies - Shuttlecock, Fail, Love, Deuce.

Swimming

Freestyle- The freestyle is a great competitive and exercise stroke, and allows you to swim longer distances without exhaustion.

Butterfly- The butterfly is a swimming stroke swum on the chest, with both arms moving symmetrically, accompanied by the butterfly kick along with movement of the hips.

Backstroke- A way of swimming on your back by moving first one arm then the other backwards while kicking your feet.

Other Terminologies - Tumble turn, Dolphin Kick, Breaststroke.

Wrestling

Greco Roman- The legs may not be used in any way to obtain a fall, and no holds may be taken below the waist.

Other Terminologies - Bridge, Danger Position, Passivity, Technical Fall.

Golf

Drive- The first shot of each hole, made from an area called the tee box, usually done with a driver (a type of golf club).

Putt- A putt is a stroke in golf that you make when the ball has reached the green in an attempt to get the ball in the hole. In golf, the score depends on putting the ball in a hole.

Par- The number of strokes needed to complete a hole in a golf course.

Other Terminologies - Albatross, Ace, Birdie, Bunker, Double eagle, Eagle, Block, Bounce Back, Club-face, Drive, Fairway, Turkey, Address.

Football

Free Kick- A free kick awarded to a player from which a goal may not be scored directly. If a player commits a foul, the opposition team is awarded a free kick.

Long out- A ball that is hit deep into the

outfield (and caught) is a "long out".

Still in play- When the ball strikes the frame of the goal or the referee and remains within the goal and touch lines.

Other Terminologies - Goal, Set, Volley, Service line, Header, Direct Free Kick, Penalty Kick, Hat-trick, Video Assistant Referees, Comebacker, Long out, Sidermer, Stopper, Off-Side, Trapping, Blooper, Bunt, Booking.

Polo

Reins- To control the horses, reins are used. These are positioned in between the rider's hand and the saddle. Another purpose of the reins is to keep the head of the horse down.

Other Terminologies - Bump, Hook, Knock-in, Mallet/Stick, Out-of-bounds.

Volleyball

Wipe - When a hitter pushes the ball off of the opposing block so it lands out of bounds.

Power tip - A ball that is pushed or directed with force by an attacking team.

Other Terminologies - Penetration, Free Throw, Ace, Back-one, Bump (a common term for forearm passing in volleyball), Carry, Cross, Cut, Dump, Dink (a legal push of the ball around or over blockers), Floater, four-two, five-one, Six Pack, Double Block, Jump Serve, Side out, Slide.

Judo

Ukemi - Literally means as in the mat "receiving the person".

Shintai - Advancing and retreating.

Kumi kata - To be linked together as a pair.

Other Terminologies - Time trial, Sprint, Point Race, Swing off.

Lawn Tennis

Backhand - Stroke in which the ball is hit with the back of the racket hand facing the ball at the moment of contact.

Deuce - It is a special score in tennis. When both the opponents have won three points each and the score is 40-40 this is called a deuce.

Other Terminologies - Ace, Service Break, Smash, Slice, Stroke.

Gymnastics

Somersault - A gymnastics maneuver in which a person rotates around the somersault axis, moving the feet over the head.

Aerial - A type of cartwheel where a gymnast's hands do not touch the ground.

Other Terminologies - Pommel horse, still rings/roman rings, vaulting table, Produnova.

Equestrian

Dressage - An equestrian sport where horses and their riders perform a test of movements in a rectangular arena while being judged against set criteria.

Bridle - A harness of leather straps that fits around the horse's head and holds the bit.

Other Terminologies - Balk, Bit, Trot, Vaulting, Show jumping.

Weight Lifting

Snatch - It is the first of two lifts contested in the sport of weightlifting followed by the clean and jerk.

Split - To drop while bending the legs, one forward and one straight back.

Other Sports Terminologies

Squash - Let Ball, stroke ball, No Let, counterdrop, crosscourt, rail, re-drop, the T, the tin, volley.

Kho Kho - Runner, Chaser, Poleby, Out, Foul, Taking Direction, Sudden Change, Tapping, Turning Round the post, Trapping, Diving, Fake Kho, Late Kho, innings.

Bridge - Trump, Dummy, Cuebid, Singleton, Tenace.

Bowling - Arsenal, Tenpins, Tempo, Turkey, Fingertip grip.

Sports Personalities (India/World)

Major Dhyan Chand ('Wizard of Hockey'): Three Olympic gold medals, in 1928,

1932 and 1936, during an era where India dominated field hockey.

Rani Rampal is the first woman hockey player to receive the Major Dhyan Chand Khel Ratna Award (2020).

Prithipal Singh- The first-ever Arjuna Award to a hockey player was conferred to him in 1961.

Mangte Chungneijang Mary Kom (Mary Kom) : She was given the title 'Magnificent Mary' by the International Boxing Association (AIBA). The Government of Manipur has conferred on her the title "Meethoi Leima". First Indian female boxer to win a gold medal in the Asian Games in 2014 and gold at the 2018 Commonwealth Games.

Vijender Singh Beniwal: He became the first Indian boxer to win an Olympic medal (bronze medal in the 2008 Beijing Olympics).

Sachin Tendulkar, renowned as the "Master Blaster," debuted in Test cricket at 16 against Pakistan in 1989. He achieved the milestone of the first batsman to score 100 international centuries, culminating with his 100th against Bangladesh in 2012. He was the youngest Bharat Ratna awardee and the first sportsperson in 2014.

Kapil Dev: He is also known as the "Haryana Hurricane". India won the 1983 Cricket World Cup under his captaincy.

Lala Amarnath Bharadwaj: He became the first player to score a century for the Indian national cricket team in Test cricket. He was the first Test cricket captain of an independent Indian team.

Anil Kumble: He became the second bowler after Jim Laker in Test history to pick up all 10 wickets in an innings against Pakistan at Arun Jaitley Stadium in Delhi.

Sunil Manohar Gavaskar: He was the first Test cricket player to make over 10,000 runs.

World cup victory by Indian Captain in ODI and T20 : 1983 ODI Men's world cup - Kapil Dev, 2007 T20 Men's world cup - MS Dhoni, 2011 Men's ODI world cup - MS Dhoni.

Mithali Dorai Raj: She is the highest run-scorer in women's international cricket.

Milkha Singh (The Flying Sikh) (Track

and field sprinter) :- He is the only athlete to win gold at 400 meters at the Asian Games as well as the Commonwealth Games. He also won gold medals in the 1958 and 1962 Asian Games.

Neeraj Chopra: He is the first Asian athlete to win an Olympic gold medal in Men's Javelin throw (87.58 m) at Tokyo Olympics 2020.

Deepa Malik: She is a Para Athlete. She is the first Indian woman to win a medal at the Paralympics. She won the silver medal in the shot put in 2016 Paralympic Games.

Pilavullakandi Thekkeraparambil Usha (PT Usha): She has won 4 Asian gold medals and 7 Silver medals. She is often associated as the "Queen of Indian track and field". She is the first nominated MP in history to become the Vice Chairperson of the Rajya Sabha.

Hima Das (Dhing Express): She is an Indian sprinter from the state of Assam. She is the first Indian athlete to win a gold medal in a track event at the IAAF World U20 Championships.

Dutee Chand: She is the first Indian to win a gold medal in 100m race in a global competition.

Devendra Jhajharia: He won silver medal in javelin throw in the Paralympic Games in Tokyo. He is the first para-athlete to receive the Rajiv Gandhi Khel Ratna in 2017.

Nilima Ghose (track athlete) became the first Indian woman to compete at an Olympic Games at the 1952 Summer Olympics held in Helsinki, Finland.

Kamaljeet Sandhu is the first female Indian athlete who won a gold medal at the 1970 Bangkok Asian Games in the 400 m race.

Koneru Humpy: She is an Indian chess player. She is the first Indian Grandmaster to win FIDE Women's rapid chess championship in 2019.

Viswanathan Anand: He became the first grandmaster from India in 1988.

Srikanth Kidambi: He became the first Indian male shuttler to win a silver medal in the BWF World Championships (2021).

Prakash Padukone : He is a former Indian badminton player. He became the

first Indian to win the All England Open Badminton Championships in 1980.

PV Sindhu (Pusarla Venkata): She is an Indian badminton player from Hyderabad. She is the first and only Indian to become the badminton world champion in 2019. She is the second individual athlete from India to win two consecutive medals at the Olympic Games.

Pullela Gopichand: He is a former Indian badminton player. He won the All England Open Badminton Championships in 2001. He became the second Indian to achieve this feat after Prakash Padukone.

Saina Nehwal : She won India's first badminton medal in the Olympics. She has represented India three times in the Olympics, winning a bronze medal in her second appearance at London 2012.

Manika Batra - First Indian woman to win a medal (Bronze) at the Asian Cup table tennis tournament in 2022.

Achanta Sharath Kamal: He is the first Indian table tennis player ever to become ten times Senior National Champion.

Vinesh Phogat :- She became the first Indian woman wrestler to win gold in both Commonwealth and Asian Games.

Sakshi Malik: She became the first Indian female wrestler to win a bronze medal in the 58 kg category at the Olympics.

Khashaba Dadasaheb Jadhav: He was an Indian freestyle wrestler. Best known for winning a bronze medal at the 1952 Summer Olympics in Helsinki. He was the first athlete from independent India to win an individual medal in the Olympics.

Joshna Chinappa: She was the first Indian to win the British Junior Squash Championship title in 2005 in the under-19 category and was the youngest Indian women's national champion.

Abhinav Bindra (Shooting): He is the first Indian to win an Individual Olympic Gold Medal (2008).

Manu Bhaker (Shooter): She is the youngest Indian to represent India at the 2018 ISSF World Cup and won two gold medals.

Heena Sidhu: She is the first Indian pistol shooter to reach number one in world rankings.

Saikhom Mirabai Chanu (Weightlifter): She is from Imphal city, Manipur. She won the silver medal at the 2020 Tokyo Olympics in the Women's 49 kg category.

Karnam Malleswari (Weightlifter): She belongs to Andhra Pradesh. She is the first Indian woman to win a medal at the Olympics in 2000.

Leander Paes (Tennis): He holds the record for the most doubles wins in the Davis Cup. He won eight men's doubles and ten mixed doubles Grand Slam titles.

Mahesh Bhupathi (Tennis): The first Indian to win a Grand Slam title. He won at the 2006 Australian Open mixed doubles.

Bhaichung Bhutia: He is often nicknamed the "Sikkimese Sniper" because of his shooting skills in football. He is considered as the torchbearer of Indian football in the international arena. He is the first Indian footballer to play 100 international games.

Sunil Chhetri: He plays as a forward and captains both Indian Super League club Bengaluru and the India national team. He is the second Indian footballer to play 100 international games.

Edson Arantes do Nascimento: He is known by his nickname "Pele". He holds the distinction of having the most wins of the FIFA World Cup by any player. He was a Brazilian professional footballer. He is known as 'The Black Pearl', 'The King of Football'.

Manisha Kalyan: The first Indian footballer to play at the UEFA Women's Champions League.

Wilson Jones, a legendary Indian billiards player, won the amateur World Championship twice (1958, 1964) and was the first Indian World Champion in any sport after independence.

Pankaj Advani: He is the only Indian to have won the Amateur World Title in both snooker and billiards. He won the Asian Billiards title for the 8th time in 2022. He has won the International Billiards and Snooker Federation (IBSF) World Billiards Championship 22 times.

Bhakti Sharma: She is the first Asian woman and the youngest in the world to set a record in open swimming in Antarctic waters.

Mihir Sen: He was the first Indian

swimmer to swim across the English Channel in 1958. He was the only man to swim the oceans of the five continents in one calendar year.

Arati Saha: She was the first Asian woman to swim across the English Channel in 1959.

Michael Phelps (Swimmer): He is the most winning Olympian medal player, having 28 (23 gold, 3 silver, and 2 bronze) medals in both individual and team events.

Deepika Kumari (Archer): She won a silver medal at the 2020 Tokyo Olympics test event. She won a gold medal in the 2010 Commonwealth games in the women's individual recurve event.

Limba Ram: He is an Indian archer who represented India in international competitions including three Olympics. He equalled an archery world record in 1992 at the Asian Archery Championships in Beijing.

Bhavani Devi (Chennai, Tamil Nadu): She is the first Indian fencer to ever qualify for the Olympics after qualifying for the 2020 Summer Olympics.

Fouaad Mirza (equestrian) : He became the first Indian equestrian in the last 20 years to qualify for the Olympics and the first Indian equestrian to reach the finals of the Olympic individual eventing category.

Dipa Karmakar : She is the first gymnast of India who competed in the Olympics. In her debut 2016 Summer Olympics, in the final she reached 4th position. She won a bronze medal at the 2014 Commonwealth Games in Glasgow becoming the first Indian female gymnast to do so in the history of the Games.

Simone Biles: She is an American artistic gymnast.

Kobe Bryant: A Basketball star who won an Oscar award for an animation film. The six-minute-long film is titled "Dear Basketball".

Linthoi Chanambam is a judo player from Manipur, India. She is the first Indian to be crowned world cadet judo champion 2022.

Anju Bobby George - The first Indian athlete ever to win a medal (bronze) in World Championships 2003 (Paris) in

Athletics jumping.

Dipika Pallikal (Squash player) is the first Indian woman to be ranked inside the top 10 in the world squash rankings.

Sahanand Mahadev Shetty was the first kabaddi player to be honored with the Arjuna Award in 1972.

Ms. Shakuntala P Khatavkar was the first women kabaddi player to be honored with the Arjuna Award in 1978.

Roger Federer is a Swiss former professional tennis player.

Shiny Abraham Wilson (Track and field) was the first woman to carry the flag of India at the opening ceremony of any Olympics (1992 Summer Olympics).

Annu Rani Dharayan is an Indian javelin thrower from Uttar Pradesh. She is the first Indian woman to cross a distance of 60 m.

Anjali Bhagwat is an Indian rifle shooter who won the 2002 International Shooting Sport Federation (ISSF) Champion of Champions combined-air-rifle event to become the first Indian to win that competition.

Famous Sportsperson from India

Virat Kohli (Cricket), Mahendra Singh Dhoni (Cricket), Rohit Sharma (Cricket), Sania Mirza (Lawn Tennis), Narain Karthikeyan (Motor Sports), Yuvraj Singh (Cricket), R Ashwin (Cricket), Gautam Gambhir (Cricket), Harmanpreet Kaur (Cricket), Rohan Bopanna (Lawn Tennis), Lakshya Sen (Badminton), Bajrang Punia (Wrestling), Ravi Kumar Dahiya (Wrestling), Lovlina Borgohain (Boxing), Manpreet Singh (Hockey), R Praggnanandhaa (Chess), Amit Panghal (Boxing), Anup Kumar (Kabaddi), Saurav Ganguly (Cricket), Rahul Dravid (Cricket).

Famous Sportsperson from the World

Cristiano Ronaldo (Football), LeBron James (Basketball), Lionel Messi (Football), Neymar Jr. (Football), Roger Federer (Tennis), Kevin Durant (Basketball), Tiger Woods (Golf), Rafael Nadal (Tennis), Kobe Bryant (Basketball), Usain Bolt (Track and Field), Novak Djokovic (Tennis), Maria Sharapova (Tennis), Steffi Graf (Tennis).

Indian Athletes :

Shooting - Bhowneesh Mendiratta,

Rudranksh Patil , Swapnil Kusale, Akhil Sheoran, Mehuli Ghosh, Sift Kaur Samra, Rajeshwari Kumari, Sarabjot Singh, Arjun Babuta, Tilottama Sen, Manu Bhaker, Anish Bhanwala, Shriyanka Sadangi, Varun Tomar, Esha Singh, Rhythm Sangwan, Vijayveer Sidhu, Raiza Dhillon, Anantjeet Singh Naruka, Maheshwari Chauhan.

Athletics - Murali Sreeshankar (long jump), Avinash Sable (3000m steeplechase), Parul Chaudhary (3000m steeplechase), Kishore Jena (Javelin Throw).

20 km Race Walk- Akshdeep Singh, Priyanka Goswami, Vikas Singh, Paramjeet Bisht, Suraj Panwar, Servin Sebastian, Arshpreet Singh.

Wrestling - Antim Panghal, Vinesh Phogat, Reetika Hooda, Anshu Malik.

Boxing - Nikhat Zareen, Preeti Pawar, Parveen Hooda, Lovlina Borgohain.

4 × 100m Relay - (Muhammed Anas Yahiya/ Muhammed Ajmal/ Arokia Rajiv/ Amoj Jacob), (Rupal/ Jyothika Sri Dandi/ MR Poovamma/ Subha Venkatesan).

Badminton - PV Sindhu, HS Prannoy, Lakshya Sen, Satwiksairaj Rankireddy /Chirag Shetty, Ashwini Ponnappa /Tanisha Crasto.

Dhiraj Bommadevara (Archery), Balraj Panwar (Rowing), Nethra Kumanan (Sailing).

Sports Rules and Dimensions

Cricket

The number of players in a cricket team on a field is 11. The cricket pitch is 20.12 meters (22 yards) in length and 3.05 meters (3.33 yards) in width. The height of stumps in cricket is 28 inches.

The cricket ball must weigh between 155.9 and 163 grams. The average circumference of the standard cricket ball in international cricket is between 8.81 inches and 9 inches. The Maximum length of a cricket bat is 38 inches. 1.22 m is the distance between the stumps and the popping crease at the same end of a Cricket pitch.

Maximum 10 overs a bowler can bowl in an ODIs match. In T20 matches a bowler can bowl maximum 4 overs. There are

typically 6 balls in 1 over. Minimum number of overs allowed per day in a test cricket match is 90. The minimum over rate in a Test match is 15 over/h.

Two players can remain outside the circle in the first six overs of a T20 cricket match.

Five penalty runs are given to the batting team if helmet of the wicket keeper is kept on the ground and the ball that the batsman strikes hits the helmet.

The **Duckworth-Lewis-Stern** method is associated with Cricket. It is a mathematical formulation designed to calculate the target score (number of runs needed to win) for the team batting second in a limited overs. This formula was used for the first time in International cricket on 1 January 1997.

The number of on-field umpires in cricket matches is 2. **Umpire Symbols** - Out (umpire raises an index finger above the head), **Not out** (waving the arms in a sweeping motion out in front of the chest), **No ball** (One arm fully extended horizontally), **Free hit** (One hand held above their head and making a circular motion), **Wide Ball** (Both arms extended horizontally), **Six runs** (Both arms held high above their head), **dead ball** (crossing and uncrossing his arms in front of his body) etc.

Table Tennis

The table is 2.74 m (9.0 ft) long, 1.525 m (5.0 ft) wide, and 76 cm (2.5 ft) high. The length of the upper surface of the table is 274 cm. The 40 mm ball was introduced after the end of the 2000 Summer Olympics; previously a 38 mm ball was standard. In table tennis there is no limit on how many lets the server can serve in a row.

Chess

The chess board contains 32 black and 32 white squares. A chess piece is any of the six different types of movable objects used on a chessboard. It consists of 1 king, 1 queen, 2 rooks, 2 bishops, 2 knights, 8 pawns on both sides of the chess board (Total -16).

The **Pawn** moves directly forward in chess, never backward or to the side.

The **queen** is able to move any number of squares vertically, horizontally, or diagonally, combining the power of the

rook and bishop.

The **King** may move to any adjoining square; it may also perform a move known as castling. It can move one square in any direction.

The **Knight** moves in an 'L-shape,' two squares in a straight direction, and then one square perpendicular to that.

The **bishop** moves and captures along diagonals without jumping over intervening pieces.

Badminton

Badminton court - 13.4 m by 6.1 m (singles) and 13.4 m by 5.18 m (doubles). Each match consists of the best of 3 games of 21 points without extra time. The interval between the first and second set is 90 seconds, and between the second and third set 3 minutes. The height of the badminton net is 5.1 feet.

Yellow card - Warning for misconduct, Red Card - Fault for misconduct, Black Card - Disqualified for misconduct.

Kabaddi

Two teams of seven members each in Kabaddi. Each has five supplementary players held in reserve for substitution. Court dimensions of Kabaddi are 13 × 10 meters (43 ft × 33 ft) for men and 12 × 8 meters (39 ft × 26 ft) for women. The duration of a game of kabaddi is 40 minutes. There are two halves of 20 minutes each and both teams switch sides at halftime.

Basketball

Basketball court dimensions are 28 m long and 15 m wide. The shape of a basketball court is a rectangle. Team has 12 players, with five players on the court at any given time. Playing Time - 4 Quarter × 10 minutes (FIBA) and 4 Quarter × 12 minutes (NBA). The maximum number of points one can score is 4 points. In Basketball free throws are worth **one point**. The Basketball center line is part of the **Backcourt**.

Snooker

It includes 22 object balls (15 red and 6 colored balls) and one cue ball. Players have to pot a red ball first, followed by a colored ball, a red ball, a colored ball, and

so on. The red-colored sequence continues until all of the balls have been pocketed.

Handball

Handball is played between two teams of 7 or 11 players who try to throw or hit a ball into a goal at either end of a rectangular playing area while preventing their opponents from doing so. Players that stop dribbling in handball have to pass or shot within 3 seconds. For resin handballs, the circumference ranges from 58 to 60 cm and the weight is between 425 to 475g for men. For women, the ball circumference ranges from 54 to 56 cm, and the weight is between 325 to 375g.

Wrestling

Wrestling Mat- The thickness should be between 5 and 7 centimeters.

Points - Takedown (2 points), Escape (1 point), Reversal (2 points), Near Fall (2 points if lasts for 2 seconds or 3 points if lasts for 5 seconds), Penalty Points (1 or 2 points).

Hockey

The dimension of the Hockey field is with the longer edges (sidelines) 91.4m long (100yd) and the shorter edges (backlines) 55m wide (60yd).

The standard hockey goal size is 12 ft wide and 7 ft tall (approx. 3.66m x 2.13m). The standard weight of a hockey ball is 156 gm - 163 gm.

The maximum number of players that can be on the field during a hockey match in a team is 11 and there is no limit to the number of times a player is allowed to be substituted.

The total duration of a field hockey match is 60 minutes (four quarters of 15 minutes each).

Hockey signals of Umpire - If the umpire points both arms horizontally towards the goal it is penalty Corner. **Free hit** (Indicate the direction with one arm raised horizontally and open hand at face level), **Penalty stroke** (One arm points to the penalty spot, the other points straight in the air), **Goal scored** (point both arms horizontally towards the center of the field).

Football

FIFA recommendations for field dimensions in professional football are 105 metres in length and 68 meters in width.

The maximum number of players that can be on the field during a football match in a team is 11. A football match lasts two equal periods of 45 minutes.

The standard soccer/football goal dimensions are 24ft (7.32 metres) wide and 8 ft (2.44 metres) high which means the goal is 192 square feet. Both goalposts and the crossbar have the same width and depth of 12cm. Penalty mark from the midpoint of the goal post in football is 11m. All free kicks and corners are direct and opposition players must be 3 meters away from kick-in.

Penalty mark from the midpoint of the goal post in football is 11m ($\approx$ 12 yards).

A yellow colour is used to warn a player, a red card results in the player's dismissal from the field of play.

Tennis

Tennis court Dimensions - The court is 23.77m (78 ft) long and for singles matches, 8.23m (27 ft) wide. For doubles matches the court is 10.97m (36 ft) wide.

The court is divided into two equal areas by a net suspended by a cord or metal cable attached to two net posts. The net is 1.07m high.

The first point scored after the deuce is known as an Advantage. 'Advantage in' is the server's point. 'Advantage out' is the score when the receiver wins the next point after the deuce. In tennis 6 - 4 scores is a complete set.

There are four main types of surface for tennis courts- Grass, clay, hard and artificial grass.

Volleyball

Court dimensions are 18 m by 9 m. There are 2 teams of six players each. The antenna is a flexible rod, 1.8m long and 10mm in diameter.

Matches are played best of five sets. The first four sets are played to 25 points, with the final set being played to 15 points. The U.S. came out with the first official compilation of volleyball rules

and was published by the YMCA.

Boxing

International Boxing Association (IBA) Categories - IBA Elite - 19 - 34 years old. IBA Youth - 17 or 18 years old (U19). IBA Junior - 15 or 16 years old (U17).

The International Boxing Association (AIBA) consists of 3 rounds of 3 minutes each in men and four rounds of two minutes in women and break time between each round is 1 minute. In AIBA Boxing Junior Boys and Girls Competitions, the bouts must consist of each round of 2 minutes.

A bout refers to a match, normally meaning an organised fight in a playing area. Area of a bout is called a ring in boxing. In AIBA Boxing Junior Boys and Girls' Competitions, the Bouts must consist of each round of two minutes.

Other sports

Beach volleyball is a team sport played by two teams of two or more players on a sand court divided by a net. Beach Volleyball court dimensions are 16 m long by 8 m wide. There is no substitute player in this game.

Sailing event at Olympic Games categories are Dinghy - (470, 49er, 49erFX, Laser Standard, Laser Radial, Multihull - (Nacra 17), Sailboard - (IQFoil), Kiteboard - (Formula Kite).

Archery - The highest score for a single arrow is 10 for hitting the inner gold ring, while the least (for hitting the outer white ring) is one point. A standard competition archery target has 10 circle regions.

The equipment used in Mountaineering - Bolt, Climbing pack, Rope, Helmet, Harness (with adjustable leg loops), Crampons, Ice axe (with leash), Belay/rappel device, Pulley.

Running events include- Sprints (100m, 200m, 400m), Middle distance (800m, 1500m), Long distance (3000m Steeplechase, 5000m, 10,000m), Hurdles (110/100m, 400m), Relays (4 x100m, 4 x 400m, mixed 4 x 400m).

In running events Lap scorer counts the laps. In case of first false start in a running event, the athlete Can restart.

Javelin throw - The men's javelin must weigh at least 800 grams (1.8 pounds) and be 260 cm (102.4 inches) long while

the women's javelin must weigh 600g and be minimum 220 cm (86.6 inches) long.

110 m hurdle race- Number of Flights is 10. Each flight height is 2 feet 9 inches (83.8 meters).

100 m women's hurdle- The distance between hurdles is 8.50 meters.

In Swimming, the minimum Lane width is 2.13 meters.

Trapshooting (Clay-pigeon shooting), a sport in which participants use shotguns for shooting at saucer-shaped clay targets flung into the air from a spring device called a trap.

The **Triple jump** (the hop, skip and jump) is a track and field event. **White flag** - trail is valid (a fair jump). **Red Flag** - a foul jump. The hop is used in Triple Jump.

A marathon must be greater than or equal to 26 miles 385 yards (42.195 Km). Half marathon – 21.1km (13.1 miles) Ultramarathon- Longer than 42.2km / 26.2 miles.

Number of Players in sports (each team) - Hockey (11), Football (11), Cricket (11), Baseball (9), Basketball (5), Bridge (2), Lacrosse (10), Ice Hockey (6), Handball (7), Kabaddi (7), Kho-kho (9), Rugby Football (15), Volleyball (6), Water Polo (7) etc.

Sports Stadium

Andhra Pradesh

Cricket- Indira Gandhi Stadium (Vijayawada), YS Raja Reddy Stadium (Kadapa), Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium (Visakhapatnam).

Assam

Cricket- Dr. Bhupen Hazarika Cricket Stadium (Guwahati).

Football- Indira Gandhi Athletic Stadium (Guwahati), Satindra Mohan Dev Stadium (Silchar).

Bihar

Cricket- Moin-ul-Haq Stadium (Patna, Bihar).

Football- Patliputra Sports Complex (Patna), Rajendra Stadium (Siwan).

Chhattisgarh

Cricket - Shaheed Veer Narayan Singh International Cricket Stadium (Raipur).

Hockey- International Hockey Stadium (Rajnandgaon).

Goa

Cricket- Railway Stadium, Vasco da Gama, Dr. Rajendra Prasad Stadium (Margao), Arlem Breweries Ground (Margao), Bhausahab Bhandodkar Ground (Panaji).

Football- Fatorda Stadium (Margao), Tilak Maidan Stadium (Vasco Da Gama), Duler Stadium (Mapusa)

Gujarat

Cricket- Narendra Modi Stadium (Ahmedabad), CB Patel International Cricket Stadium (Surat), Moti Bagh Stadium (Baroda), IPCL Sports Complex Ground (Baroda), Madhavrao Scindia Cricket Ground (Rajkot), Saurashtra Cricket Association Stadium (Rajkot).

Haryana

Cricket - Chaudhary Bansi Lal Cricket Stadium (Rohtak), Tau Devi Lal Stadium (Gurgaon).

Football- Nehru Stadium (Gurgaon),

Himachal Pradesh

Cricket- Himachal Pradesh Cricket Association Stadium (Dharamshala).

Maharashtra

Cricket- Vidarbha Cricket Association Stadium (Nagpur), Brabourne Stadium (Mumbai), Maharashtra Cricket Association Stadium (Pune), Wankhede Stadium (Mumbai), Indira Gandhi Stadium (Solapur), Guru Gobind Singh Stadium (Nanded), Jawaharlal Nehru Stadium (Pune).

Jharkhand

Cricket - JSCA International Cricket Stadium (Ranchi), Keenan Stadium (Jamshedpur).

Football - Birsa Munda Football Stadium (Ranchi).

Karnataka

Cricket- M. Chinnaswamy Stadium (Bengaluru), Gangotri Glades Cricket Ground (Mysore).

Football- Sree Kanteerava Stadium (Bengaluru), Mangala Stadium (Mangalore), Visvesvaraya Stadium (Mandy).

Kerala

Cricket- Jawaharlal Nehru Stadium (Kochi), Greenfield International Stadium (Thiruvananthapuram).

Football- Lal Bahadur Shastri Stadium (kollam), Chandrasekharan Nair Stadium (Thiruvananthapuram), Fort Maidan (Palakkad), Maharaja's College Stadium (Kochi).

Madhya Pradesh

Cricket - Captain Roop Singh Stadium (Gwalior), Holkar Cricket Stadium (Indore).

Football- Dr. Rajendra Prasad Football Stadium (Neemuch), Ravi Shankar Shukla Stadium (Jabalpur).

Hockey- Aishbagh Stadium (Bhopal).

Nagaland

Cricket- Nagaland Cricket Association Stadium (Chumukedima).

Football- Indira Gandhi Stadium (Kohima).

Odisha

Cricket- Barabati Stadium (Cuttack).
Veer Surendra Sai Stadium (Sambalpur), East Coast Railway Stadium (Bhubaneswar).

Hockey- Birsa Munda International Hockey Stadium (Rourkela), largest hockey stadium in India.

Punjab

Cricket- Punjab Cricket Association IS Bindra Stadium (Mohali), Dhruv Pandove Cricket Stadium (Patiala).

Football- Guru Nanak Stadium (Ludhiana), Guru Gobind Singh Stadium (Jalandhar).

Hockey- Sector 42 Stadium (Chandigarh), International Hockey Stadium (Ajitgarh), War Heroes Stadium

(Sangrur), Guru Nanak Stadium (Kapurthala).

Rajasthan

Cricket- Sawai Mansingh Stadium (Jaipur), Barkatullah Khan Stadium (Jodhpur).

Sikkim

Cricket- Mining Cricket Stadium (Rangpo).

Football- Baichung Stadium (Namchi), Paljor Stadium (Gangtok).

Tamil Nadu

Cricket- M. A. Chidambaram Stadium (Chennai), Jawaharlal Nehru Stadium (Coimbatore), Mahatma Gandhi stadium (Salem).

Telangana

Cricket- Rajiv Gandhi International Stadium (Hyderabad).

Tripura

Cricket- Maharaja Bir Bikram College Stadium (Agartala).

Jammu & Kashmir

Cricket- Sher-i-Kashmir Stadium (Sri Nagar).

Football- Bakshi Stadium (Srinagar).

Uttarakhand

Cricket- Rajiv Gandhi International Cricket Stadium (Dehradun).

Uttar Pradesh

Cricket - Ekana International Cricket Stadium (New Name - Atal Bihari Vajpayee Stadium Lucknow), K. D. Singh Babu Stadium (Lucknow), Green Park Stadium (Kanpur), Saifai International Cricket Stadium (Saifai), Dr. Akhilesh Das Stadium (Lucknow).

Hockey - Major Dhyanchand Hockey Stadium (Lucknow), Dr Sampurnanda Stadium (Varanasi).

West Bengal

Cricket - Eden Garden (Kolkata).

Football - Salt Lake Stadium (Kolkata), Kanchenjunga Stadium (Siliguri), East

Bengal Ground (Kolkata), Mohun Bagan Ground (Kolkata), Kalyani Stadium (Kalyani), Rabindra Sarobar Stadium (Kolkata), Mohammedan Sporting Ground (Kolkata), Jadavpur Stadium (Kolkata), Mela Ground (Kalmpong).

Famous Sports/Cricket stadium in the World -

Cricket - Melbourne Cricket Ground (Melbourne, Australia), Perth Cricket Stadium (Perth, Australia), Adelaide Oval (Adelaide, Australia), Lords Cricket stadium (London, England), The Oval (London), Sydney Cricket Ground (Australia), Kensington Oval (Barbados), The Wanderers Stadium (Johannesburg).

Other Sports stadium - Vivekananda Yuba Bharati Krirangan (Athletics and Football Stadium in Kolkata), EMS football stadium (Kozhikode, Kerala), Bhimrao Ambedkar International Sports Stadium (Faizabad), JRD Tata Sports Complex Stadium (Jamshedpur).

Sports Institutes- Netaji Subhas National Institute of Sports (Patiala), National Sports University (Manipur), National Institute of WaterSports (Goa), Pullela Gopichand Badminton Academy (Hyderabad).

Sports Cups and Trophies

Table Tennis - Ramanujan Trophy, Electra Gold Cup, Gasper Giest Prize, Kamala Ramanujan Cup, Pithapuram Cup (men), Corbitton Cup (Women), Swaythling Cup (Men), Corbillon Cup, Rajkumari Challenge Cup, Barna-Bellack Cup, Jayalakshmi Cup, Travancore Cup (women).

Chess - Naidu Trophy, Khaitan Trophy, Limca Trophy, Linares City Trophy, FTX Crypto cup.

Badminton - Thomas Cup (Biennial event), Syed Modi, Uber Cup (Women), Aggarwal Cup, Tunku Abdul Rahman Cup, Amrit Diwan Cup, Chadha Cup, Harilela Cup, Ibrahim Rahmatullah Challenger Cup, Konica Cup, Yonex Cup, Sudirman Cup.

Kabaddi - The Pro Kabaddi League, Kabaddi World Cups.

Basketball - Larry O'Brien Championship Trophy, Bangalore Blues Challenge Cup, Basalat Jha Trophy, S.M. Arjuna Raja Trophy, Todd memorial Trophy, William Jones Cup, B.C. Gupta Trophy, Walter A.

Brown Trophy, Bob Cousy Trophy, Oscar Robertson Trophy, Maurice Podoloff Trophy, Nat "Sweetwater" Clifton Trophy, Wayne Embry Trophy, Earl Lloyd Trophy, Willis Reed Trophy, Sam Jones Trophy, Chuck Cooper Trophy.

Golf - FedEx Cup, Canada Cup, Colombo Cup, Prince of Wales Cup, Eisenhower Cup, Ryder Cup, Walker Cup, Topolino Trophy, Nomura Trophy, Solheim Cup, Inter-Continental Cup, Muthiah Gold Cup, President's Cup, Waterford Crystal Trophy.

Hockey - Agha Khan Cup, Beighton Cup (oldest hockey tournament in India which was instituted in 1895), Dhyan Chand Trophy, Sultan of Johor Cup, Essandre Champions Cup, Guru Nanak Championship, Modi Gold Cup, Murugappa Gold Cup, Nehru Trophy, Rangaswami Cup, Ranjit Singh Gold Cup, Sahni Trophy, Scindia Gold Cup, Tommy Eman Gold Cup, Indira Gold Cup, Kuppuswamy Naidu Trophy, Lady Ratan Tata Trophy, MCC Trophy.

Cricket - Syed Mustaq Ali Trophy (T20), C. K. Naidu Trophy (50 overs), Cooch Behar Trophy, Deodhar Trophy, Duleep Trophy, G. D. Birla Trophy, Ghulam Ahmed Trophy, Irani Trophy, Moin Ud Dowla Gold Cup, Rani Jhansi Trophy, Ranji Trophy, Reliance Cup, Rohinton Baria Trophy, Sahara Cup, Benson and Hedges cup, ICC Champions Trophy, ICC World Cup, Charminar Challenge Cup, Gillette Cup, Pataudi Trophy, Frank Worrell Trophy, Southern Cross Trophy, Sir Vivian Richards Trophy, Clive Lloyd Trophy, Warne-Muralitharan Trophy, The Freedom Trophy, Sobers-Tissera Trophy, Benaud-Qadir Trophy.

Football - Durand Cup, Rovers Cup, Santosh Trophy, Subroto Mukherjee Cup, Dr B.C. Roy Trophy, Sri Krishna Gold Cup, Stafford Cup, Todd Memorial Trophy, Begum Hazrat Mahal Trophy, Nixon Gold Cup, Sanjay Gold Cup, Bandodkar Trophy, Chakola Gold Trophy, Colombo Cup, Kalinga Cup, Scissor Cup, Vittal Trophy.

Top Football trophies in the World - FIFA World Cup Trophy, UEFA Champions League, UEFA European Championship Trophy, Copa America, FA Cup, Premier League Trophy, La Liga Trophy, Africa Cup of Nations, Confederations Cup, Copa del Rey, Concacaf Gold Cup, Super bowl championship (USA).

As of April 2024, Most successful teams in the Football cups/ tournaments in

International Competitions.

FIFA (Brazil - 5 tiles) hold the record for the most FIFA World Cup wins, **UEFA** European Football Championship (Germany, Spain - 3 titles), **AFC** Asian Cup (Japan - 4 titles), CAF Africa Cup of Nations (Egypt- 7 titles), **CONMEBOL** Copa America (Argentina, Uruguay - 15 titles), **CONCACAF** Gold Cup (Mexico - 8 titles), OFC Nations Cup (New Zealand - 5 Titles).

Polo - Ezra Cup, Maharaj Prithi Singh Baria Cup, Radha Mohan Cup, Westchester Cup.

Volleyball - Centennial Cup, Federation Cup, Indira Pradhan Trophy, Shivanthi Gold Cup.

Tennis - Edgbaston Cup, Wimbledon Trophy, Grand Prix, Wightman Cup, Thant Cup, Rajendra Prasad Cup, and Davis Cup, Barna Bellack Cup, Jaylaxmi Cup (Women's), Ramanujan Trophy, Billi Jean King Cup Tennis (Women), Australian Open, French Open, US Open, Heineken Cup, Hopman Cup, Tata Open, Ambre Solaire Cup, Evert Cup, Ghafar Cup, Watson's Water Trophy, Wightman cup.

Tennis Grand Slam Tournament

Australian Open (hard courts) - The Australian Open is a tennis tournament held annually at Melbourne Park in Melbourne, Victoria, Australia. The Australian Open starts in the middle of January every year. It was founded in 1905.

French Open (clay courts) - The French Open, also known as Roland-Garros is a major tennis tournament held over two weeks at the Stade Roland Garros in Paris, France, beginning in late May each year. It was founded in 1891.

Wimbledon (grass outdoors) - It is the oldest tennis tournament in the world which was founded in 1877. It is usually held from late June to late July. It is held at the All England Lawn Tennis and Croquet Club.

US Open (hard courts) - The US Open Tennis Championships is a hardcourt tennis tournament held annually (in August) in Queens, New York. It was founded in 1881.

Golden Slam - Player who has won four Grand Slam Tournaments along with the

Olympic Gold Medal is awarded with the Golden Slam.

Auto Racing - Formula One, (also called F1 in short, is an international auto racing sport), A B Crookall Trophy, Australian Tourist Trophy, Borg Warner Trophy, Denny Hulme Memorial Trophy, Douglas Pirie Trophy, Enduro Cup, FIA Pole Trophy, Gordon Bennett Cup, Hawthorn Memorial Trophy, Jimmy Simpson Trophy, John Williams Trophy, Lorenzo Bandini Trophy, Neil Robinson Trophy, Peter Brock Trophy, Torrens Trophy, Wheatcroft Trophy.

Bridge - Basalat Jha Trophy, Holkar Trophy, Ruia Gold Cup, Singhanian Trophy, Hiron Trophy, Bermuda Bowl, Venice Cup, Rosenblum Cup, McConnell Cup, Rand Cup, Camrose Trophy.

Air Racing - Aerial Derby, Bendix Trophy, Dole Air Race, Gordon Bennett Trophy (airplanes), Grosvenor Cup, Halle Trophy Race, Manx Air Derby, Michelin Cup, Thompson Trophy, Jawaharlal Challenge Trophy, King's Cup, Jaswant Singh Trophy.

Rugby Football - Bledisloe Cup, Calcutta Cup, Webb Ellis Trophy, Melrose Cup, Triple Crown, Millennium Trophy, Centenary Quaich, Giuseppe Garibaldi Trophy, Auld Alliance Trophy, Doddie Weir Cup, Cuttitta Cup, Bledisloe Cup, Mandela Challenge Plate, Freedom Cup, Puma European Rugby Champions Cup (Previous Heineken Cup).

Weight Lifting - Burdwan Cup.

Yachting - America Cup.

Shooting - North Wales Cup, Welsh Grand Prix.

Boat race - The Nehru Trophy, Kerala.

Cricket World Cup Facts

India was the winner of the first T20 cricket World Cup held at Johannesburg, South Africa in 2007 under the captainship of M S Dhoni. India hosted the ICC T20 World Cup cricket for the first time in 2016.

India has hosted the ICC Cricket World Cup four times (1987, 1996, 2011 and 2023).

The first ICC Women's Cricket World Cup was held in 1973, England. India never won the title but they were runner-up two times in 2017 and 2005.

The official mascot of the ICC Cricket World Cup 2011 is Stumpy (an elephant). It was played in India, Sri Lanka, and Bangladesh, in which India was the winner.

The ICC Under-19 Cricket World Cup was started in 1988. India has won five times till March 2023.

The first World Deaf T20 Cricket Championship was held by Kerala in January 2023.

ICC Men's ODI Cricket World Cup Year - Winners - Runners-up -

1975 - West Indies - Australia, 1979 - West Indies - England, 1983 (Hosts - England, Wales) - India - West Indies, 1987 - Australia - England, 1992 - Pakistan - England, 1996 - Sri Lanka - Australia, 1999 - Australia - Pakistan, 2003 - Australia - India, 2007 - Australia - Sri Lanka, 2011 - India - Sri Lanka, 2015 - Australia - New Zealand, 2019 - England - New Zealand, 2023 (Host - India) - Australia - India.

(International and National Cup/Trophy, Game, Year Founded)

Davis Cup, Tennis Male, 1900, Thomas Cup, Badminton Men, 1949, Uber Cup, Badminton Women, 1957, Sudirman Cup, Badminton, 1989, Swaythling Cup, Table Tennis Men, 1926, Durand Cup, Football, 1888, Santosh Trophy, Football, 1941, Rovers Cup, Football, 1891, Ranji Trophy, Cricket, 1934, Duleep Trophy, Cricket, 1961-62, Irani Trophy, Cricket, 1959-60, Deodhar Trophy, Cricket, 1973-74, Vijay Hazare Trophy, Cricket, 2002-03, All England Championship, Badminton, 1899, etc.

The first World Cup of various sports (sport, inaugural year, location).

(FIVB Volleyball Men (inaugural), 1965, Poland), (FIFA Football, 1930, Uruguay), (Hockey, 1971, Barcelona, Spain), (Cricket, 1975, England), (Badminton, 1979, Tokyo, Japan), (Table Tennis, 1980, Hong Kong), (Boxing, 1979, New York, USA, Madison Square Garden), (IBSF World Billiards Championship, 1926, London), (FIBA Basketball, 1950, Argentina), (Archery, 2006), (FIDE Chess World Cup, 2000, Shenyang, China), (Kabaddi, 2004, Mumbai).

Sport Association/Federation

The Badminton Association of India is an

association registered under the societies act. It was formed in 1934 and has been holding national-level tournaments in India since 1936. Headquarters- New Delhi.

The Indian Hockey Federation was formed in 1925. Headquarters - New Delhi, India.

The Board of Control for Cricket in India (BCCI) was set up on December 4, 1928. Headquarters - Mumbai. The Maharaja of Vizianagaram was the first Indian cricketer to serve as President of BCCI.

The **ICC** has its headquarters in Dubai, United Arab Emirates. Cricket became an international game with the formation of ICC. It was founded as the Imperial Cricket Conference in 1909 by representatives from Australia, England and South Africa. It was renamed as the International Cricket Conference in 1965, and took up its current name in 1987.

All India Tennis Association was founded in March 1920. Headquarters - Delhi.

The All India Chess Federation was formed in 1951. Headquarters - New Delhi.

The All India Football Federation was formed in 1937. Headquarters - New Delhi. The Indian Football Association was established in Calcutta in 1893.

The Athletics Federation of India was founded in 1946. Headquarters - New Delhi.

The Kabaddi Federation of India (KFI) was founded in 1950, and it compiled a standard set of rules. The Amateur Kabaddi Federation of India (AKFI) was founded in 1973.

The Volleyball Federation of India was founded in 1951. Headquarter- Chennai, Tamil Nadu.

International Table Tennis Federation founded in 1926. Headquarters- Lausanne, Switzerland.

International Hockey Federation established in 1924. Headquarters - Lausanne, Switzerland.

Badminton World Federation (BWF) established in 1934. Headquarters - Kuala Lumpur, Malaysia.

FIFA (Federation internationale de football association) was founded on 21 May 1904 in Paris, France. Headquarters

Zurich, Switzerland.

Sports Authority of India (SAI) is the Apex National Sports Body of India, established in 1984. Headquarters - Delhi, India. The National Anti-Doping Bill, 2021 targets Sports in India.

The International Association of The Athletics Federations (IAAF) was organised at Dusseldorf (Germany) in 1977.

Indian Premier League (IPL) - It was founded in 2008 by the Board of Control for Cricket in India (BCCI) and Lalit Modi. Rajasthan won the inaugural edition of the Indian Premier League.

Indian Premier League teams - Chennai Super Kings (CSK), Delhi Capitals (DC), Gujarat Titans (GT), Kolkata Knight Riders (KKR), Lucknow Super Giants (LSG), Mumbai Indians (MI), Punjab Kings (PBKS), Rajasthan Royals (RR), Royal Challengers Bangalore (RCB), and Sunrisers Hyderabad (SRH).

Indian Super League was founded in 2013. Indian Super League (Football Club) Teams- Kerala Blasters, ATK Mohun Bagan, Bengaluru FC, Chennaiyin, East Bengal, Goa, Hyderabad, Jamshedpur, Mumbai City, North East United, Odisha, Roundglass Punjab. Atletico de Kolkata (ATK) won the first edition of Indian Super League in 2014.

The **Pro Kabaddi League** was founded in 2014. There are 12 teams (as of March 2023). They are Bengal Warriors, Bengaluru Bulls, Dabang Delhi K.C, Gujarat Giants, Haryana Steelers, Jaipur Pink Panthers, Patna Pirates, Puneri Paltan, Tamil Thalaivaa, Telugu Titans, U Mumba, U.P Yoddhas.

Famous Sport Events

Khelo India Youth Games (KIYG), formerly known as Khelo India School Games (KISG), is a pan-Indian multi-disciplinary sports competition for the country's under-18 young talents. KIYG currently (April 2024) has 26 sports under its belt. Games edition - Host States - 1st (2018) - New Delhi, 2nd (2019) - Maharashtra (Pune), 3rd - Assam, 4th - Haryana, 5th - Madhya Pradesh, 6th - Tamil Nadu.

Khelo India University games- 3rd edition (Mascot- Jitu) - Uttar Pradesh (First position - Punjab University). The Khelo India Winter Games started in 2020.

Janjatiya Khel mahotsav - First edition of this event held in Kalinga Institute of Social Sciences, Bhubaneswar, Odisha.

Assam Rifles Public School in Shillong is the first Khelo India Sports school in North East India.

35th National Games - Kerala, 36th National Games - Gujarat, Mascot- Saavaj, 37th National Games - Goa, Mascot- Moga.

The World Badminton Championship event 2026 will be organized by India.

India's first Olympic Values Education Programme (OVEP) was launched in Odisha.

Kabaddi World Cup (Circle style) is an international kabaddi competition administered by the Government of Punjab (India). Pakistan won the circle style Kabaddi World Cup in 2020 after defeating India.

The World Athletics Championships was inaugurated in 1983 at Helsinki, Finland with 41 events.

The first South Asian Federation Games was held in 1984 in Kathmandu.

Sports Awards

Major Dhyan Chand Khel Ratna Award (Rajiv Gandhi Khel Ratna Award)- Established - 1991-1992. It is the highest sporting honour in India. It is awarded annually by the "Ministry of Youth Affairs and Sports". The award comprises a medallion, a certificate, and a cash prize of Rs. 25 lakh. The first recipient - Chess Grandmaster Viswanathan Anand (1991-92).

Games and first recipients of Khel Ratna Award - Chess- Viswanathan Anand (1991-92), Billiards- Geet Sethi (1991-92), Weightlifting- Karnam Malleswari (1994-95), Cricket- Sachin Tendulkar (1997-98), Hockey- Dhanraj Pillay (1999-2000), Badminton- Pullela Gopichand (2000-2001), Shooting- Abhinav Bindra (2001), Gymnastics- Dipa Karmakar (2016), Table tennis- Manika Batra (2020), Football- Sunil Chhetri (2021).

Dronacharya Award- Established - 1985. It is awarded annually for outstanding coaches in Sports and Games by the "Ministry of Youth Affairs and Sports". The award comprises a bronze statuette of Dronacharya, a certificate,

ceremonial dress, and a money prize of Rs 15 lakh (lifetime) and 10 Lakh (regular).

The first recipients of the award were Bhalchandra Bhaskar Bhagwat (Wrestling), Om Prakash Bhardwaj (Boxing), and O. M. Nambiar (Athletics), who were honoured in 1985. First woman (Weightlifting) - Hansa Sharma.

Arjun Award- Established - 1961. It is the second-highest sporting honour of India and awarded annually by the "Ministry of Youth Affairs and Sports". The award comprises a bronze statuette of Arjuna, certificate, ceremonial dress, and a cash prize of Rs. 15 lakh.

Dhyan Chand Award- It is officially known as the Major Dhyan Chand Award for Lifetime Achievement in Sports and Games. It is awarded annually by the "Ministry of Youth Affairs and Sports". The award comprises a statuette, a certificate, ceremonial dress, and a cash prize of Rs. 10 lakh. The first recipients - Shahuraj Birajdar (Boxing), Ashok Diwan (Hockey), and Aparna Ghosh (Basketball), who were honoured in 2002.

Other Sports Awards- Tenzing Norgay National Adventure Award, Gazzetta Sports Awards, Laureus World Sports Award, Honda-Broderick Cup, Polly Umrigar award, Ekalavya Award, Sports awards from haryana (Bhim Award - highest award of Haryana, Rani Laxmi Bai Award, Eklavya Award, Guru Vashisht Award, Maharana Pratap Award, Vikramaditya Award, Dr. Sarvepalli Radhakrishnan Award), Madhya Pradesh (Vikram Award, Vishwamitra Award).

Olympic, Asian, and Commonwealth Games

Olympic Games

The roots of the Olympic Games are to be found in Ancient Greece. The first modern Olympic Games were held in Athens (Greece) in 1896 which was initiated by Pierre-de-Coubertin, he proposed the motto of Olympics. The 245 participants in Athens participated from 14 different countries. Types of Olympic Games - Summer, Winter Olympics, Youth Olympic Games, and Paralympics.

The Olympic Games include the Games of the Olympiad (i.e. the Summer Games) and the Olympic Winter Games. Until

1992, the Summer and Winter Games were held in the same year. The Summer and Winter Games are now organized once every four years. The difference in the number of years between the summer and winter Olympic events is 2 years. The first Winter Olympic Games were held in 1924 in Chamonix (France).

Olympic games were canceled altogether because of World War I (1916) and World War II (1940 and 1944).

The five rings on the olympic flag represent the inhabited continents of the world (the Americas were considered one continent and Europe was treated as distinct from Asia). It was made to contain the colors (blue, black, red, yellow, and green) which are common to almost all flags around the world. The symbol was originally created in 1913 by Coubertin.

Gold medal - It consists of silver with a purity of at least 92.5%, and 6 grams of pure gold. Silver medal - silver 92.5% and remainder as copper. Bronze medal is composed of bronze, an alloy made of mostly copper and another metal (such as tin or zinc). All Olympic medals must be at least 60 mm in diameter and 3mm thick.

Women made their Olympic debut at the 1900 Games in Paris (France), in tennis and golf. Subsequently, (e.g. swimming in 1912, volleyball in 1964, rowing in 1976, cycling in 1984, and football in 1996). Introduction of five track and field events for women in athletics was started during the 1928 Summer Olympics.

The **first Olympic Congress** was held in 1894 in Paris, France. It was this Congress that established the International Olympic Committee (IOC). Baron Pierre de Coubertin founded the International Olympic Committee (IOC) in 1894 in Paris. Paris (France) hosted the 2nd Olympic Games in 1900.

Sports included in the Olympics for the first time -

- 1896** - Artistic Gymnastics, Athletics, Cycling Road, Cycling Track, Fencing, Shooting, Swimming, Tennis, Weightlifting, Wrestling.
- 1900** - Equestrian, Football, Golf, Rowing, Sailing, Water Polo.
- 1904** - Boxing, Diving.
- 1908** - Figure Skating (played in Winter

only), Hockey.

- 1912** - Modern Pentathlon.
- 1920** - Ice Hockey (Winter).
- 1924** - Bobsleigh (Winter), Cross-Country Skiing (Winter), Curling (Winter), Nordic Combined (Winter), Ski Jumping (Winter), Speed Skating (Winter).
- 1928** - Skeleton (Winter).
- 1936** - Alpine Skiing (Winter), Basketball, Canoe Flatwater, Handball.
- 1960** - Biathlon (Winter).
- 1964** - Judo, Luge (Winter), Volleyball.
- 1972** - Canoe Slalom, Archery (from 1900 - 1920 and then reintroduced in 1972).
- 1984** - Artistic Swimming, Rhythmic Gymnastics.
- 1988** - Table Tennis.
- 1992** - Badminton, Baseball Softball, Freestyle Skiing (Winter), Short Track Speed Skating (Winter).
- 1996** - Beach Volleyball, Cycling Mountain Bike.
- 1998** - Snowboard (Winter).
- 2000** - Taekwondo, Trampoline, Triathlon.
- 2008** - Cycling BMX Racing, Marathon Swimming.
- 2016** - Rugby Sevens.
- 2017** - 3 × 3 Basketball.
- 2018** - Beach Handball, Futsal (Yog only).
- 2020** - Cycling BMX Freestyle, Karate, Skateboarding, Sport Climbing, Surfing.
- 2024** - Breakdancing.
- 2026** - Ski Mountaineering (Winter).

Sports not included in Olympics - Cricket, Acrobatic gymnastics, Polo, Darts, Lacrosse, Distance Plunging, Rope Climbing, Croquet, Basque Pelota, Water Motorsports, Tug of War, Squash, and Bowling. Since 1900, Cricket has not been played at the Olympics.

Host City	Year
Athens (Greece)	1896 and 2004 (Summer Olympics)
Paris (France)	1900, 1924 and 2024 (Summer Olympics)
London (England)	1908, 1948 and 2012 (Summer Olympics)
St. Moritz (Switzerland)	1928 and 1948 (Winter Olympics)
Lake Placid (United States of America)	1932 and 1980 (Winter Olympics)

Los Angeles (United States of America)	1932, 1984 and 2028 (Summer Olympics)
Cortina d'Ampezzo (Italy)	1956 and 2026 (Winter Olympics)
Innsbruck (Austria)	1964 and 1976 (Winter Olympics)
Tokyo (Japan)	1964 and 2020 (Summer Olympics)
Beijing (China)	2008 (Summer Olympics) and 2022 (Winter Olympics)
Stockholm (Sweden)	1912 and 1956 (Summer Olympics)

Beijing became the first city in the world to host both the Summer (2008) and Winter Olympic Games (2022).

Lords Cricket Ground hosted Archery in the 2012 olympics game.

India's performance in the Olympics -

Indian Olympic Association (IOA) - Established - 1927. First president - Sir Dorab Tata. It is an autonomous body recognized by the Ministry of Youth Affairs and Sports. It is affiliated to the International Olympic Committee. India first participated in the Olympics in 1900 in Paris. India's rank in Tokyo Olympic 2020 is 48th.

Gold medal - The Indian hockey men's team won 8 Olympic gold medals; Amsterdam 1928, Los Angeles 1932, Berlin 1936, London 1948, Helsinki 1952, Melbourne 1956, Tokyo 1964, Moscow 1980. The Indian Hockey team won the first gold medal after Independence at the 1948 London Summer Olympics.

Abhinav Bindra - He became the first individual gold medalist for India in men's 10m air rifle at the Beijing Olympics 2008. He was the first Indian to receive the International Shooting Federation (ISSF) Blue Cross.

Arpinder Singh won a Gold Medal for India in Men's triple jump event after a gap of 48 years in the 2018 Asian Games in Jakarta.

Gaurika Singh won 4 gold medals in a season at the 2019 South Asian games held in Nepal.

Silver Medal - The Indian men's Hockey team got a silver medal in the 1960 Rome Summer Olympics. Two silver medals won by India in the London Olympics 2012.

Saikhom Mirabai Chanu won silver at the

Tokyo 2020 Olympics by lifting the 202kg (87 kg snatch; 115 kg clean and jerk) weight in the women's 49 kg category.

Bronze Medals - The Indian men's hockey team won three bronze medals, one in the 1968 Summer Olympics and the second in Munich 1972 Summer Olympic Games, and the third in Tokyo 2020 Summer Olympic Games. India won 4 bronze medals in the London Olympics 2012.

Leander Paes (First Indian and Asian Tennis Player) - He won a bronze medal for India in men's singles at the 1996.

India has won three Bronze Olympic medals in **boxing** {Vijender Singh won in Beijing in 2008 (75 Kg), Mary Kom in London in 2012 (51 Kg), and Lovlina Borgohain in Tokyo in 2020 (69 Kg)}.

Khashaba Dadasaheb Jadhav won a bronze medal at the 1952 Summer Olympics in Helsinki.

The Uttar Pradesh government has adopted the wrestling sport and is expected to pump in an investment of Rs 170 crore in infrastructure and support to the wrestlers till the 2032 Olympics. The Odisha government supports hockey.

Youth Olympic Games

The Youth Olympic Games (YOG) are aimed at bringing together talented young athletes aged from 15 to 18 from around the world. The first summer edition was held in Singapore from 14 to 26 August 2010. The 4th Summer Youth Olympic Games 2026 will take place in Dakar, Senegal, the first country on the African continent to be awarded the honor of hosting an Olympic sports event. The latest Winter YOG took place in Lausanne in 2020 and featured 1,872 athletes. The 4th Winter Youth Olympic Games 2024 will be held in Gangwon (Republic of Korea).

Paralympic Games

The Paralympic Games or Paralympics is an international multi-sport event. It is held for specially-abled or differently-abled people. It is also known as the Games of the Paralympiad. The first organized athletic event for disabled athletes that coincided with the Olympic Games took place on the day of the opening of the 1948 Summer Olympics in London, United Kingdom. Jewish-German Dr. Ludwig Guttman hosted a

sports competition for British World War II veteran patients with spinal cord injuries in 1948. They involved 16 injured servicemen and women who took part in archery.

The ninth International Stoke Mandeville Games, considered the first Paralympic Games, took place in 1960 in Rome, six days after the closing ceremony of the Olympic Games. Before 1960, the Paralympics were called the Stoke Mandeville Games. In 1976 the first Winter Games in Paralympics history were held in Sweden, and as with the Summer Games, have taken place every four years, including a Paralympics Opening Ceremony and Closing Ceremony.

The International Paralympic Committee is the global governing body of the Paralympic Movement. It comprises 176 National Paralympic Committees (NPC) and four disability-specific international sports federations.

Summer Paralympics Sports - Para archery, Para athletics, Para badminton, Blind football, Boccia, Para canoe, Para-cycling, Para equestrian, Goalball, Para judo, Para powerlifting, Para rowing, Shooting Para sport, Sitting volleyball, Para swimming, Para table tennis, Para taekwondo, Para triathlon, Wheelchair basketball, Wheelchair fencing, Wheelchair rugby, and Wheelchair tennis.

Winter Paralympics Sports - Para alpine skiing, Para biathlon, Para cross-country skiing, Para ice hockey, Para snowboarding, and Wheelchair curling. The mascots were first created for the Arnhem 1980 Paralympic Games. They were designed by Necky Oprinsen. Name - Noggi and Joggi (a pair of squirrels). Someity was the mascot and tagline "we have wings" for the Tokyo 2020 Paralympic Games. 2024 Summer Paralympics (Paris) Mascot - The Paralympic Phryge - Phrygian Cap.

Three cities have hosted or are scheduled to host Paralympic Games more than once- Innsbruck (Austria) in 1984 and 1988, Beijing (China) in 2008 (summer games) and 2022 (winter games), and Tokyo (Japan) in 1964 and 2020. The 2022 Winter Paralympics was hosted by Beijing, China.

India's Performance -

Gold Medal - Avani Lekhara won a Gold

Medal in the 10m air rifle standing at Tokyo 2020 Paralympics. 1972 Heidelberg (Murlikant Petkar), 2004 Athens (Devendra Jhajharia), 2016 Rio de Janeiro (Devendra Jhajharia, Mariyappan Thangavelu), 2020 Tokyo (Manish Narwal, Pramod Bhagat, Sumit Antil, and Krishna Nagar).

Silver Medal - 1984 Stoke Mandeville (Bhimrao Kesarkar, Joginder Singh Bedi), 2012 London (Girisha Nagarajegowda), 2016 Rio de Janeiro (Deepa Malik), 2020 Tokyo (Bhavina Patel, Nishad Kumar, Yogesh Kathuniya, Devendra Jhajharia, Mariyappan Thangavelu, Praveen Kumar, Singhraj Adhana, and Suhas Lalinakere Yathiraj).

Bronze Medal - Avani Lekhara won a Bronze Medal in the 50m rifle 3 positions at Tokyo 2020 Paralympics. Joginder Singh Bedi won 2 bronze medals at 1984 Stoke Mandeville. 2004 Athens (Rajinder Singh Rahelu), 2016 Rio de Janeiro (Varun Singh Bhati), 2020 Tokyo (Sundar Singh Gurjar, Singhraj Adhana, Sharad Kumar, Avani Lekhara, Harvinder Singh, and Manoj Sarkar).

Asian Games - The Asian Games are held every four years. It was regulated by the Asian Games Federation (AGF) from the first Games in New Delhi, India, until the 1978 Games. Since the 1982 Games, they have been organized by the Olympic Council of Asia (OCA). India organized the Asian Games for the second time in 1982.

The first edition of the Asian Games was hosted in New Delhi in 1951. The motto of the first Asian Games was - "Play the game in the spirit of the game". Second - Manila 1954. Third - Tokyo 1958. Fourth - Jakarta 1962. Fifth - Bangkok 1966. Sixth - Bangkok 1970. Seventh - Tehran 1974. Eight - Bangkok 1978. Ninth - New Delhi 1982. Tenth - Seoul 1986. Eleventh - Beijing 1990. Twelfth - Hiroshima 1994. Thirteenth - Bangkok 1998. Fourteenth - Busan 2002. Fifteenth - Doha 2006. Sixteenth - Guangzhou 2010. Seventeenth - Incheon 2014. Eighteenth - Jakarta and Palembang 2018. Nineteenth - Hangzhou 2022. Twenty - Aichi-Nagoya 2026. Twenty-First - Doha 2030. Twenty-Second - Riyadh 2034.

The **first Asian Games** to introduce a mascot was New Delhi in 1982 when Appu, an elephant, was chosen.

Games with their Mascot - Seoul 1986 (Hodori), Beijing 1990 (Pan Pan),

Hiroshima 1994 (Poppo and Cuccu), Bangkok 1998 (Chai-yo), Busan 2002 (Duria), Doha 2006 (Orry), Guangzhou 2010 (A Xiang, A He, A Ru, A Yi, and Le Yangyang), Incheon 2014 (Baram, Chumuro, and Vichuon), Jakarta-Palembang 2018 (Bhin Bhin, Atung, and Kaka), and Hangzhou 2022 (Congcong, Lianlian, and Chenchen).

India's performance -

India won a total of 70 medals (16 Gold, 23 Silver, 31 Bronze) in the 2018 Asian Games in Jakarta and Palembang. India won 5 gold medals at the 1986 'Asian Games' held in Seoul, South Korea.

India became the first-ever Asian Games football champion, winning the gold medal at the inaugural 1951 and 1962 edition (Indonesia).

Commonwealth Games

The Commonwealth Games is a quadrennial international multi-sport event among athletes from the Commonwealth of Nations. It is often referred to as the Friendly Games or simply the Comm Games. It was first proposed by John Astley Cooper in 1891.

The event was first held in 1930 at Hamilton (Canada) and, with the exception of 1942 and 1946 (canceled due to World War II). It is organized every four years. The Games were called the British Empire Games from 1930 to 1950, the British Empire and Commonwealth Games from 1954 to 1966, and British Commonwealth Games from 1970 to 1974.

The Games are organized by the Commonwealth Games Federation (CGF), which controls the sporting program and selects host cities. The games movement consists of international sports federations (IFs), Commonwealth Games Associations (CGAs), and organizing committees for each specific Commonwealth Games. It was formed in 1932. Its Headquarters is in London.

The opening ceremony starts with the hoisting of the host country's flag and a performance of its national anthem

The game is closed with Flag-bearers from each participating country entering the stadium, followed by the athletes who enter together, without any national distinction. The president of the organizing committee and the CGF

president make their closing speeches and the Games are officially closed.

Host City	Country	Year
Hamilton	Canada	1930
London	England	1934
Sydney	Australia	1938
Auckland	New Zealand	1950, 1990
Vancouver	Canada	1954
Cardiff	Wales	1958
Perth	Australia	1962
Kingston	Jamaica	1966
Edinburg	Scotland	1970, 1986
Christchurch	New Zealand	1974
Edmonton	Canada	1978
Brisbane	Australia	1982
Victoria	Canada	1994
Kuala Lumpur	Malaysia	1998
Manchester	England	2002
Melbourne	Australia	2006
Delhi	India	2010
Glasgow	Scotland	2014
Gold Coast	Australia	2018
Birmingham	England	2022
Victoria	Australia	2026

Only six nations have participated in every Commonwealth Games- Australia, Canada, England, New Zealand, Scotland, and Wales. Of these six, Australia, England, Canada, and New Zealand have each won at least one gold medal in every game.

Sports included - Archery, Athletics, Para Athletics, Badminton, Basketball, Basketball 3x3, Boxing, Cricket, Cycling (Mountain Bike), Cycling (Para Track), Cycling (Road), Cycling (Track), Diving, Hockey, Gymnastics (Artistic), Gymnastics (Rhythmic), Judo, Lawn bowls, Para lawn bowls, Netball (Women), Powerlifting, Rugby sevens, Shooting, Squash, Swimming, Para swimming, Table tennis, Para table tennis, Triathlon, Volleyball (beach), Weightlifting, Wrestling (Freestyle).

India hosted the Commonwealth Games (19th session) for the first time in 2010. It was India's most successful Commonwealth Games to date. India won 101 medals in total, including 38 Gold medals, 27 silver, and 36 bronze medals. **2022-** Host (Birmingham, England England),

Shree Shiv Chhatrapati Sports Complex, Pune was constructed in 1994 for the National Games. The 2008 Commonwealth Youth Games were hosted here.

Southeast Asian Games

The Southeast Asian Games (SEA Games) is a biennial multi-sport event involving participants from the current 11 countries of Southeast Asia. The games are under the regulation of the Southeast Asian Games Federation with supervision by the International Olympic Committee (IOC) and the Olympic Council of Asia (OCA). It occurs every 2 years.

The first event was held at the 1959 Southeast Asian Peninsular Games in Bangkok, Thailand. 2023 Southeast Asian Games - Phnom Penh, Cambodia.

The 2021 Southeast Asian Games is the biennial regional multi-sport event that was held in Hanoi, Vietnam, and its surrounding cities from 12 to 23 May 2022. It was also known as the 31st Southeast Asian Games, and also recognized as Hanoi 2021.

Asian wrestling championship

Asian Wrestling Championship is a Wrestling Asian Championship organized by the Asian Associated Wrestling Committee (AAWC). The men's tournament began in 1979 and the women's tournament was first staged in 1996, and it has been held every year. The first event was hosted by India in 1979 in Jalandhar. Hosted by India - Jalandhar (1979), Mumbai (1987), New Delhi (1991, 2003, 2010, 2017, 2020, 2023).

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Anita Desai- "Clear Light of Day", "Fasting, Feasting", "Fire on the Mountain", "In Custody", "The village by the Sea", "Baumgartner's Bombay", "Diamond dust", "The Zigzag Way", "Voices in the City", "The Peacock Garden", "Sulagata Parvat", "The Artists of disappearance".

Amrita Pritam- "Raseedi Ticket", "Pinjar", "The Skeleton, and Other Writings", "In The Times Of Love And Longing", "Jalte Bujhte Log", "The Other Dimension" (Lotus collection).

Arvind Adiga- "Amnesty", "Last Man in Tower", "Selection Day", "Between The Assassinations", "The White Tiger" (Man Booker Prize in 2008).

Amartya Sen- "The Idea of Justice", "The Argumentative Indian", "The Country of First Boys And Other Essays", "Development and Freedom", "Garibi Aur Akaal", "Identity and Violence", "Inequality Reexamined", "Home in The World".

Anuja Chauhan- "The Zoya Factor", "Battle For Bittora", "Those pricey Thakur Girls", "The House That BJ Built".

Anupam Kher- "Your Best Day Is Today!", "Aap Khud Hi Best Hain", "The Best Thing About You is You".

Amish Tripathi- "The Immortals of Meluha", "The Secret of the Nagas", "Scion of Ikshvaku", "Sita- Warrior of Mithila", "Raavan- Enemy of Aryavarta", "Legend of Suheldev- The King Who Saved In", "The Shiva Trilogy- Omnibus", "The Oath of the Vayuputras", "Immortal India- Articles and Speeches".

Arthur C Clark- "2001- A Space Odyssey", "Rama Series", "Childhood Ends", "City and the Stars", "Fountains of Paradise".

Andy Marino- "Narendra Modi- A Political Biography", "Escape From Chernobyl", "It Rides a Pale Horse", "Escape from East Berlin".

Avni Doshi- "Burnt Sugar", "Girl in White Cotton".

Anuradha Ro- "Sleeping on Jupiter".

Anand Neelakantan- "Ajaya- Roll of the

Dice", "Ajaya- Rise of Kali", "The Rise of Sivagami", "Penn Ramayanam", "Valmiki's Women", "Asura - Tale of the Vanquished".

Adam Smith- "Wealth of Nations".

Bill Gates- "The Road Ahead".

Bipin Chandra- "India's Struggle for Independence", "History of Modern India", "India since Independence", "Freedom Struggle", "Communalism in Modern India", "India After Independence", "In the Name of Democracy", "Indian National Movement".

Barack Obama- "A Promised Land", "The Audacity of Hope", "Of Thee I Sing- A Letter to My Daughters", "Renegades- Born in the USA", "Change We Can Believe In".

Balraj Khanna- "Born in India Made in England".

Bhisham Sahni- "Tamas", "Balraj- My Brother", "Madhavi", "Bhakti Raakh", "Pahala Path", "Patriyan", "Today's Past- A Memoir".

Chetan Bhagat- "One Indian Girl", "Two States", "Five Point Someone" (Society Young Achievers Awards and the Publisher's Recognition Award), "The Girl in Room 105", "One Night at The Call Center", "Half Girlfriend", "The 3 Mistakes of My Life", "Revolution 2020", "Love Corruption Ambition", "400 Days", "One Arranged Murder", "What Young India Wants".

Chitra Banerjee Divakaruni- "The Palace of Illusions", "Black Candle", "Leaving Yuba City", "Before We Visit the Goddess", "The Forest of Enchantments", "Mistress of Spices", "The Last Queen" and "Queen of dreams".

Charles Darwin- "On The Origin of Species", "The Descent of Man, and Selection in Relation to Sex", "The Expression of the Emotions in Man and Animals", "Insectivorous Plants", "On Natural Selection", "The Voyage of the Beagle".

Charles Dickens- "Great Expectations", "The Pickwick Papers", "A Tale of Two Cities", "David Copperfield", "Little Dorrit", "Bleak House", "Oliver Twist".

Dalai Lama- "Freedom in Exile" (Autobiography), "The Art of Happiness", "The Book of Joy", "How to See Yourself

as You Really Are", "Beyond Religion- Ethics for a Whole World", "Destructive emotions", "How to Practice", "An Open Heart", "The Universe in a Single Atom- The Convergence of Science and Spirituality", "The Little Book of Encouragement".

Dr. Raghuram Rajan- "Fault Lines", "Naan Seivathai Seikiren", "The Third Pillar", 'I Do What I Do', 'Monetary Policy and Its Unintended Consequences'.

Dr. Rajendra Prasad- "Satyagraha in Champaran", "Atmakatha", "Enslaved in Paradise", "Champaran mein Mahatma Gandhi". "India Divided".

Devdutt Pattanaik- "Myth = Mithya- Decoding Hindu Mythology", "Jaya", "My Gita", "Krishna's Secret", "7 Secrets of Shiva", "Devlok with Devdutt Pattanaik", "How To Take Decisions", "Sita", "My Hanuman Chalisa", "How to become Rich", "Sita- An Illustrated Retelling of the Ramayana".

E.M Forster- "A Passage to India", "A Room with a View", "Howards End", "Maurice", "E. M. Forster Trilogy".

Etaf Rum- "A Woman Is No Man", "Evil Eye".

Ernest Hemingway- "A Movable Feast", "The Old Man and the Sea", "A Farewell to Arms".

F Scott Fitzgerald- "The Great Gatsby", 'Tender is the Night', 'The Beautiful and Damned', 'This Side of Paradise', 'The Short Stories of F. Scott Fitzgerald', etc.

Fakir Hassen- '150 Years of Celebrating the Mahatma – the South African Legacy'

George Harrison- "I, Me, Mine".

George David Roberts- "Shantaram", 'The Mountain Shadow', 'The Spiritual Path', 'Shantaram Part Two', etc.

George Orwell- 'Animal Farm', "Nineteen Eighty-Four", "Down and out in Paris and London", "Homage to Catalonia", "The Road to Wigan Pier", "Burmese Days" and "Why I Write".

Herbert George Wells- "The War of the Worlds", "The Island of Dr Moreau", 'Seven Famous Novels', "The Invisible Man".

Haruki Murakami- "What I Talk About When I Talk about Running", "Kafka on the Shore", "Norwegian Wood", "1Q84",

"After Dark", "A Wild Sheep Chase", "The Elephant Vanishes".

Harper Lee - "To kill a mockingbird", "Go Set a Watchman", etc.

Indira Gandhi- "My Truth", "Freedom's Daughter", "Remembered Moments".

Indu Sundaresan- 'The Mountain of Light', "The twentieth wife", "The feast of roses".

Jugal Hansraj- "The Coward And The Sword", "Cross Connection- The Big Circus Adventure' (2018)".

Jairam Ramesh- "Indira Gandhi- A life in Nature", "The Light of Asia- The Poem that Defined the Buddha", "Intertwined Lives- P.N. Haksar and Indira Gandhi", "Making Sense of Chindia- Reflections on China and India", "Green Signals- Ecology, Growth, and Democracy in India".

Jeet Thayil- "Narcopolis", "The Book of Chocolate Saints", "Low", "English-Poems", "These errors are correct", "Names of the Women", "Collected Poems".

Jhumpa Lahiri- "In Other Words", "Interpreter of Maladies", "The Namesake" (first novel), "Unaccustomed Earth", "The Lowland", "Dove mi trovo", "Whereabouts", "Translating Myself and Others", "Hell-Heaven", "The Clothing of Books".

Jaswant Singh- "India at Risk- Mistakes, "Misconceptions and Misadventures of Security Policy", "Defending India", "Sauryam Tejo", "Travels in Transoxiana", "A Call to Honour- In Service of Emergent India", "Jinnah India-partition Independence".

JK Rowling- "Fantastic Beasts and Where to Find Them", "The Ink Black Heart", "The Ickabog", "Troubled Blood", "The Cuckoo's Calling", "The Tales of Beedle the Bard", "Harry Potter prequel", "Harry Potter Series", "Harry Potter and Philosopher's stone".

James Mill- "A History of British India", "Analysis of the Phenomena of the Human Mind", "Elements of Political Economy".

James Brown- "The Godfather of Soul", 'I Feel Good- A Memoir of a Life of Soul', 'Soul Classics', 'Roots of Rhythm Series'.

Jaishankar Prasad - 'Kamayani', 'Titli', 'Kankal', 'Inderjaal', 'Ajatshatru', 'Mamta'.

Karan Thapar- 'Devil's Advocate- The untold story', "Face To Face India - Conversations With Karan Thapar", "Sunday Sentiments, Wisdom Tree", "More Salt Than Pepper", etc.

Khushwant Singh - "Train to Pakistan", "Delhi- A Novel", "History of the Sikhs", "Truth, Love and a Little Malice", "India- An Introduction", "The end of India", "Big Book of Malice", "A Bride for the Sahib and Other Stories", "I Shall Never Hear a Nightingale", "Why I Supported The Emergency", "Delhi Through the Seasons", "Godmen and Godwomen of India",

Kiran Bedi- "Fearless Governance", "What Went Wrong and Why", "It's Always Possible", "As I See", "Broom and Groom", "Himmat Hai", "Creating Leadership", "Empowering Women".

Khalid Jawed- "The Paradise of Food", "Tafreeh ki ek dopahar".

Kiran Desai- "The Inheritance of Loss" (Man Booker Prize in 2006), "Hullabaloo in the Guava Orchard".

Lew Wallace- 'Ben-Hur - A Tale of the Christ', 'The Prince of India', 'Fair God or the Last of the 'Tzins'.

Lewis Carroll - 'Alice's Adventures in Wonderland' (1865) and 'Through the Looking-Glass', and 'What Alice Found There' (1872).

Leo Tolstoy- "War and Peace", "My Confession", "Anna Karenina", "The Death of Ivan Ilych", "The Kingdom of God is Within You", "The Kreutzer Sonata", "The Gospel in Brief", "Sevastopol Sketches", "Resurrection", "The Cossacks", "A Confession", "Childhood", "After the Ball", "What Men Live By".

Michael Jackson- "Moon Walk" (Memoir), "Thriller Live", "Dancing the Dream".

Munshi Premchand- "Godan", "Pratigya", "Seva Sadan", "Prema", "Rangbhoomi", "Premashram", "Roothi Rani", "Devasthan Rahasya", "Beti ka Dhan", "Ishwariya Nyaya", "Beton Wali Vidhwa", "Durga ka Mandir".

Maya Angelou- 'I Know Why the Caged Bird Sings'.

Manmohan Singh- "The Quest for Equity in Development", "Collateral and Financial Plumbing", "Changing India",

India's Economic Reforms and Development", "In Ghost's Den".

Margaret Atwood- "The Testaments" is a sequel to 'The Handmaid Tales'. The Testaments was joint winner of the 2019 Booker Prize, alongside Bernardin Evaristo's novel "Girl, Woman, Other".

Mulk Raj Anand - "Untouchable", "Coolie", "Across the Black Waters", "Two Leaves and a Bud", "The Sword and the Sickle", "The Big Heart", "The Private Life of an Indian Prince", "Lajwanti", "The Lost Child", "Apology For Heroism- A Brief Autobiography of Ideas", "The Morning Face" (Sahitya Akademi Award in 1971), "Seven Summers".

Meghna Pant- "Feminist Rani", "One and a Half Wife", "The Terrible, Horrible Very Bad Good News", 'Boys Don't Cry'.

Mahasweta Devi- "Imaginary Maps", "Hazaar Chaurasi ki Maa", "Breast Stories", "Chotti Munda and His Arrow", "Five Plays".

Mukul Kesavan- "Secular Common Sense", "Looking through Glass", "Homeless on Google Earth", "Men in White - A Book Of Cricket".

Nandini Sengupta- "The King Within", "The Poisoned Heart", "The Ocean's Own", "The Blue Horse and Other Amazing Animals from Indian History", "Babies from the Heart- A Complete Guide to Adoption".

Nirad C. Chaudhuri- "A Passage to England", "Hinduism- A Religion to Live By", "Thy Hand, Great Anarch! India, 1921-1952", "To Live or Not to Live", "Why I mourn for England". "The Autobiography of an Unknown Indian".

NR Narayana Murthy- "A, Clear Blue Sky", "Mere Business Mantra", "A Better India- A Better World".

Narendra Modi- "Exam Warriors", "Prem tirth", "Sabka Saath Sabka Vikas", "Abode of Love", "A Journey- Poems", "Jyotipunj", "Social Harmony", "Letters to Mother", "Sakshi Bhaav", "Convenient Action- Continuity for Change", "Nayanam Idam Dhanayam", "India's Singapore Story".

Nelson Mandela- "Conversations with Myself", "Madiba magic", "Prison Letters", "I am prepared to die", "A Prisoner in the Garden", "No Easy Walk to Freedom", "The Struggle Is My Life", "In His Own Words", "Notes to the Future", "Long Walk to Freedom",

Nand Kishore Acharya- "Chheelate Hue Apne Ko", 'Sanskriti ka Vyakarna', 'Sabhyata ka Vikalp', 'Gulam Badshah', etc.

Nayantara Sahgal- "Rich Like Us".

Newton- "Opticks", "Principia".

Pranab Mukherjee- "The Dramatic Decade- The Indira Gandhi Years", "The Turbulent Years", "Thoughts and Reflections", "The Coalition years".

P. Chidambaram - "Fearless in Opposition", "Standing Guard", "Undaunted - Saving the Idea of India", "Speaking Truth to Power- My Alternative View".

Paro Anand- 'Being Gandhi', "Like Smoke- 20 Teens 20 Stories".

Phanishwar Nath 'Renu'- "Maila Anchal" "Parti Parikatha", "Kitne Chauraha", "Paltu Babu Road".

Prem Prakash- "Three Paths of Devotion", "Universal Yoga", "Song of the Guru Gita", "Reporting India- My Seventy-Year Journey as a Journalist".

Rudyard Kipling- "Just So Stories for Little Children", "The Jungle Book", "Kim", "Captains Courageous", "Rikki Tikki Tavi", "Gunga Din", "The Phantom Rickshaw and Others Tales".

R.K. Narayan- "The Guide", "Malgudi Days", "Swami and Friends", "The Dark Room", "The Vendor of Sweets", "The English Teacher", "Waiting for the Mahatma", "The Bachelor of Arts".

Ruskin Bond- "A Book of Simple Living", "The Room on the Roof", "The Blue Umbrella", "Cherry Tree", "Angry River", "The Great Train Journey", "How To Live Your Life", "A Flight of Pigeons", "Delhi is not far", "Wind on the Haunted Hill", "The Tunnel", "Roads to Mussoorie", "A Handful Of Nuts", "The Tree Lover", "When the Tiger was King", "Our Trees Still Grow in Dehra" (Sahitya Akademi Award in 1992), "The Banyan Tree".

Ravi Shankar- "Raga Mala" (Autobiography), "My music my life", "Yours in Music- A Graphic Autobiography".

Rohinton Mistry- "A Fine Balance", "Such a Long Journey", "Family Matters", "Tales from Firozsha Baag", "Black Water 2- More Tales of the Fantastic", "Firozeshah Bagh Ke Kisse".

Ramachandra Guha- "A Corner of a Foreign Field- The Indian History of a British Sport", "India After Gandhi- The History of the World's Largest Democracy", "Gandhi Before India", "Environmentalism- A Global History", "Gandhi- The Years That Changed the World", "Rebels Against the Raj", "Makers of Modern India", "Democrats and Dissenters", "The States of Indian Cricket", "The Commonwealth of Cricket".

Roald Dahl- 'Boy - "Tales of Childhood"

Ramanujan (Mathematician)- "A Life of the Genius" (biography written by Robert Kanigel).

Rishi Singh Gherwal- "Kundalini- The Mother of the Universe", "Patanjali's Raja Yoga", "Yoga Vasistha Or Heaven Found", "A Treatise on Yoga".

Raja Rao- "Kanthapura"

Rajendralal Mitra- "Buddha Gaya- The Hermitage of Sakya Muni", 'The Chandogya Upanishad', 'Antiquities Of Orissa', 'The Sanskrit Buddhist literature of Nepal' etc.

Ravi Subramanian- "If God Was A Banker", "The Incredible Banker", "In the Name of God", "God Is A Gamer", "I Bought the Monk's Ferrari", "Don't Tell The Governor".

Shashi Tharoor- "The Great Indian Novel" (Commonwealth Writers Prize), "India- From Midnight to the Millennium and Beyond", "Nehru- The Invention of India", "The Paradoxical Prime Minister", "Why I Am a Hindu", "The Hindu Way- An Introduction to Hinduism", "Inglorious Empire- What the British did to India", "The Battle of Belonging", "India- The future is Now", "Kerala- God's Own Country", "Bookless in Baghdad".

Salman Rushdie- "The Satanic Verses", "Midnight's Children" (Man Booker Prize in 1981), "Quichotte", "Shame", "The Ground Beneath Her Feet- A Novel", "The Moor's Last Sigh", "Languages of Truth- Essays 2003-2020", "The Golden House", "Fury", "Imaginary Homelands- Essays and Criticism 1981-1991", "Shalimar the Clown".

Sarvepalli Radhakrishnan- "The Hindu View of Life", "The Pursuit of Truth".

Swami Vivekananda- "Raja Yoga", "My Master", "The East and the West", "Jnana Yoga", 'Karma Yoga and Bhakti Yoga',

'Pathways to Joy- The Master", "Meditation and Its Methods According", "Vedanta- Voice of Freedom", "My India- The India Eternal", "Chicago Addresses", "My Idea of Education", "Pearls of Wisdom", "Religion of Love".

Shobha De- "Sultry Days", "Sethji", "Bollywood Nights", "Snapshots", "Speed Post", "Socialite Evenings", "Second Thoughts", "Sisters".

Sudha Murty- "Wise and Otherwise- A Salute to Life", "The Magic Drum and Other Favourite Stories", "Three Thousand Stitches", "Dollar Bahu", "Grandma's Bag of stories", "The Sage with Two Horns- Unusual Tales from Mythology", "The Daughter from a Wishing Tree", "The Mother I Never Knew", "Gently Falls the Bakula", "The Day I Stopped Drinking Milk".

Subramanian Swamy- "Reset- Regaining India's Economic Legacy", "2G Spectrum Scam", "Human Rights and Terrorism in India", "Sri Lanka in Crisis- India's Options".

Siddharth Dhanvant Shanghvi- "The Last Song of Dusk", "AIDS Sutra- Untold Stories From India", "The Lost Flamingoes of Bombay".

Stephen King (American)- 'On Writing- A Memoir of the Craft', "Fairy Tale", "Billy Summers", "It", "The Shining", "Carrie".

Salim Ali - "The books of Indian birds", "Indian hill birds", "A pictorial guide to the birds of the Indian subcontinent", 'The Fall of a Sparrow'.

Shivaji Sawant- "Mrityunjay", "Chhava" (Novel), 'Yugandhar' (Novel), 'Kavadase', 'Kanchan Kan', 'Ladhat', 'Ashi Mane Ase Namune'.

Sitaram Yechury- "Modi Government- New Surge of Communalism", "Socialism in a Changing World", "Left Hand Drive- Concrete Analysis of Concrete Conditions", "Communalism vs. Secularism", "Ghrina Ki Rajniti".

Sri Aurobindo- "The Secret of the Veda", "The Life Divine", "Integral Yoga", "Hidden Forces of Life", "The Future Evolution of Man", "Powers Within", "Rebirth and Karma", "The Human Cycle", "The Mind of Light", "The Yoga on Sleep and Dreams".

Shiv Kumar Sharma- Journey with a Hundred Strings- My Life in Music.

Sanjay Baru - "The Accidental Prime

Minister", "1991- How P.V. Narasimha Rao Made History", "India and the World- Essays on Geoeconomics and Foreign Policy", "India's Power Elite", "A New Cold War- Henry Kissinger and the Rise of China".

Stephen Hawking- "A Brief History of Time" (autobiography). **Other books** - "Brief Answers To The Big Questions", "The Theory of Everything", "Black Holes- The Reith Lectures", "The Universe In A Nutshell".

Taslima Nasrin- "Lajja", "My Girlhood", "French Lover", "Revenge", "Selected Columns".

Thomas Hardy- "Tess of the d'Urbervilles", "Far from the Madding Crowd", "Jude the Obscure", "The Return of the Native".

VS Naipaul- "In a Free State" (Booker Prize in 1971), "A House for Mr. Biswas", "One out of many", "A Bend in the River", "Miguel Street", "The Enigma of Arrival", "An Area of Darkness", "India- A Wounded Civilization", "India- A Million Mutinies Now".

Vikram Chandra- "Red Earth and Pouring Rain", "Sacred Games", "Love and Longing in Bombay", "Mirrored Mind- My Life in Letters and Code", "Shanti- Faber Stories", "Bombay Paradise- Roman".

Vikram Seth- "A Suitable Boy", "The Golden Gate", "From Heaven Lake- Travels Through Sinkiang and Tibet", "Two Lives- A Memoir", "A Suitable Girl", "Riot at Misri Mandi", "The Humble Administrator's Garden", "All You Who Sleep Tonight", "Three Chinese Poets", "The Frog and the Nightingale".

Vrindavan Lal Verma- "Mrignayani", "Gadh Kundar", "Virata ki Padmini", "Musahibju", "Jhansi ki Rani", "Kachnar", "Madavji Sindhia", "Tute Kante", "Bhuvan Vikram", "Ahilya Bai".

Vikas Swarup- "Slumdog Millionaire", "Q & A- Slumdog Millionaire", "Six Suspects", "The Accidental Apprentice".

Viktor E Frankl- 'Man's Search for Meaning', "The Doctor and the Soul", "On the Theory and Therapy of Mental Disorders", "Psychotherapy and Existentialism", "The Will to Meaning", "The Unheard Cry for Meaning", "Man's Search for Ultimate Meaning".

William Dalrymple- "The Anarchy- The

Relentless Rise of the East India Company", 'Kohinoor- The Story of the World's Most Infamous Diamond', 'Return of a King', 'The Last Mughal', 'Nine Lives', 'White Mughals'.

Wendell Rodricks- "Moda Goa- History and Style", "Poskem - Goans in the Shadow on your Kindle in under a minute".

William Shakespeare - "Hamlet", "Romeo and Juliet", "Othello", "The Merchant of Venice", "A Midsummer Night's Dream", "King Lear", "The Tempest", "As You Like It", "The Winter's Tale", "All's well that ends well", "Measure to Measure", "The Tragedy of Antony and Cleopatra", "Twelfth Night".

Yann Martel- "Life of Pi", "Self", "Letters to a Prime Minister", "The Facts Behind the Helsinki Roccamatios".

Indira Nooyi - "My Life in FULL- Work, Family and Our Future" (Memoir).

Paramahansa Yogananda- Autobiography of a Yogi, "Man's Eternal Quest", "Metaphysical Meditations", "Scientific Healing Affirmations", "Journey to Self Realization", "Inner Peace- How to be Calmly Active".

Shrilal Shukla- "Raag Darbari", "Makaan", "Bisrampur Ka Sant and Soonie Ghaati Ka Sooraj".

Dadabhai Naoroji- "Poverty and Un-British Rule in India", "The Wants and Means of India", "The Manners and Customs of the Parsees", "The European and Asiatic Race".

Romesh Chunder Dutt- "The Economic History of India", "The Epic Of The Bharatas Mahabharatas", "The peasantry of Bengal".

Tahira Kashyap Khurrana- "The 12 Commandments of Being a Woman", "Cracking the Code- My Journey in Bollywood", "Be Atmanirbhar", "Souled Out", "I Promise...", "The 7 Sins of Being a Mother".

Shuja Nawaj- "Crossed Swords- Pakistan, Its Army and the War Within", "The Battle for Pakistan", "The Inner World", "FATA—A Most Dangerous Place".

Baby Kamble- "The Prisons We Broke"

Andimuthu Raja- '2G Saga Unfolds',

Kamala Das- "My Story"

Vishnu Sharma- Panchatantra

Meghan Markle- "The Bench",

Amit Shah- "Karmayodha Granth"

Hitler - 'Mein Kampf'

Akshat Gupta- 'The Hidden Hindu'

Nandikeshwara- "Abhinaya Darpan",

Shantabai Krushnaji Kamble- "Majya Jalmachi Chittarkatha"

Johnny Marr- 'Set the Boy Free'

Ishtiaq Ahmed- 'Jinnah- His Successes, Failures and Role in History'.

Satya Nadella- 'Hit Refresh',

Nirupama Yadav- 'Vishesh- Code To Win'

Sarbpreet Singh- 'Night of the Restless Spirits'

N Chandrasekaran- 'Bridgital Nation - Solving Technology's People Problem'

Sunil Gangopadhyay- 'Those Days'

Teju Behun- 'Drawing From the City'.

Venkat Raman Singh Shyam - "Finding My Way".

Indra Nooyi- 'My Life in Full- Work, Family and Our Future'.

RK Laxman- 'The Tunnel of Time',

Banarasi Das Jain- "half story"

Dr. Verghese Kurien - 'The Man Who Made the Elephant Dance by Kurien'.

Fali S Nariman- "Before Memory Fades- An Autobiography".

Autobiography of Sports Person

Yuvraj Singh - "The Test of My Life".

Sachin Tendulkar - "Playing It My Way".

Rahul Dravid - "The Nice Guy Who Finished First".

VVS Laxman - "281 and Beyond".

Sunil Gavaskar - "Sunny Days".

Suresh Raina - "Believe- What Life and Cricket Taught Me".

Saurav Ganguly - "A Century is not Enough- My Roller-coaster Ride to Success".

Anil Kumble - "Wide Angle".

Chris Gayle - "Six Machine".

Shane Warne - "No Spin".

Ravi Shastri - "Stargazing- The player in My Life".

Saina Nehwal - "Playing to Win".

Mitali Raj - "Unguarded".

PT Usha - "Golden girl".

Sania Mirza - "Ace Against Odds".

Mary Kom - "Unbreakable".

Sanjay Manjrekar - "Imperfect".

Major Dhyan Chand - "GOAL".

Milkha Singh - "Race of my Life".

Maria Sharapova - "Unstoppable- My Life So Far".

Abhinav Bindra - "A Shot at History".

Viswanathan Anand - "Mind Master".

Novak Djokovic - "Serve to win".

David Beckham - "My World, My Side".

Mike Tyson - "Undisputed Truth- My Autobiography".

Dilip Vengasakar - "One Day Wonders".

Usain Bolt - "Faster Than Lightning- My Autobiography".

Rickey Ponting - "At the Close of Play".

Rafael Nadal - "RAFA- My Story".

Arunima Sinha - "Born Again on the Mountain- a story of losing everything and finding it back".

Autobiography of Politicians

Mahatma Gandhi - "The Story of My Experiment With Truth".

B.R. Ambedkar - "Waiting for a Visa".

Jawaharlal Nehru - "An Autobiography".

Inder Kumar Gujral - "Matters of Discretion".

J.B. Kripalani - "My Times".

Sheila Dixit - "Citizen Delhi- My Times, My Life".

Mamata Banerjee - "My Unforgettable Memories".

VK Singh - "Courage and Conviction".

Sita Ram Goel - "How I Became a Hindu".

Manmohan Singh - "Changing India".

Bhagat Singh - "Why I Am An Atheist".

L.K. Advani - "My Country My Life".

K. Natwar Singh - "One Life Is Not Enough".

Sharad Pawar - "Life On My Terms- From the Grassroots to the Corridors of Power".

Margaret Alva - "Courage and Commitment".

Sheikh Muhammad Abdullah - "Aatish-i-Chinar".

Hillary Clinton - "Living History".

Barack Obama - "Dreams from my father".

Benazir Bhutto - "Daughter of Destiny".

James Brown - "The Godfather of Soul".

Arnold Schwarzenegger - "Total Recall - My Unbelievably True Life Story".

Bill Clinton - My Life

Arjun Singh - 'A Grain Of Sand In The Hourglass of Time'

Meira Kumar- 'Indian Parliamentary Diplomacy - Speaker's Perspective'

Famous Books of 2023

Salman Rushdie - "Victory City".

J.P. Nadda - "Modi- Shaping a Global order in flux".

Satyjit Ray - "The Best of Satyajit Ray".

P. Sainath - "The Last Heroes".

Shashi Tharoor - "Ambedkar- A Life".

Shubhra Gupta - "Irfan Khan- A Life in Movies".

Shri Rishi Raj Kargil - "Ek Yatri Ki Jubani".

Abhay K - "Monsoon".

PS Sreedharan Pillai - "Heritage Trees of Goa".

Dr. Dinesh Prasad Saklani - "Let's Move Forward".

T.N. Seshan - "Through The Broken Glass".

Ranjit Pratap - "As the Wheel Turns".

Satish Chandra - "A Life Well Spent- Four Decades in the Indian Foreign Service".

A.K. Bhattacharya - "India's Finance Ministers".

Kasturi Ray - "Droupadi Murmu".

Narayanan Vaghul - "Reflections".

Amitabh Kant - "Made In India".

Boria Majumdar - "Sachin @ 50".

Ashok Kumar and OP Manocha - "Cyber Encounters".

Ajai Chowdhry - "Just Aspire".

Indian History Books and Authors

Ancient

Panini - "Ashtadhyayi".

Maharishi Charak - "Charak Samhita".

Shudraka - "Mrichchakatikam".

Kautilya (Chanakya) - "Arthashastra".

Patanjali - "Mahabhashya".

Bharat Muni - "Natyashastra".

Harshavardhana - "Nagananda", "Ratnavali", "Priyadarsika".

Vatsyayana - "Kamasutra".

Harisena - "Brhatkathakosa".

Banabhatta - "Kadambari", "Harsh Charitra".

Kalhana - "Rajatarangini".

Jayadeva - "Gita Govinda".

Chand Bardai - "Prithviraj Raso".

Kalidasa - "Abhigyan Shakuntalam", "Kumarasambhava", "Vikramorvasiyam", "Ritusamhara", "Meghdoot", "Malavikagnimitra", "Raghuvansh".

Vishakhadatta - "Mudrarakshasa", "Devichandraguptam".

Jaimini - "Mimamsa - sutras"

Medieval

Hasan Nizami - "Tajul Masir".

Minhaj Us Siraj - "Tabakate Nasiri".

Ziauddin Barani - "Tarikh-i-Firozshahi", "Fatwa-e-Jahandari".

Babur - "Tuzuk-i-Babri", "Diwan", "Risal-i-Usaj".

Gulbadan Begum - "Humayunnama".

Abul Fazl - "Akbarnama", "Ain-i-Akbari".

Jahangir - "Tuzuk-i-Jahangiri".

Amir Khusrau - "Tughlaqnama", "Kiranusain", "Mifataulfatuh", "Devalrani Khizr Khan", "NuheSipehar", "Khazain Ul Futuh", "Five Diwans", "Khamasa", "Aijaz Khusravi", "Faljalul Vaid".

Alberuni - "Tahqiq-e-Hind".

Ibn Battuta - "Rehla".

Firoz Shah Tughlaq - "Futuh-e-Firoz Shahi".

AL Utbi - "Tarikh-e-Yamini".

Firdausi - "Shahnama".

Mir Hasan Dehlavi - "Fawa'id-ul-Fawad".

Hiuen Tsiang - "Si-yu-ki".

Faizi - "Nal-Damayanti".

Dara Shikoh - "Majma-ul-Bahrain", "Sirr-e-Akbar", "Yoga Vasistha", "Safinat-al-Auliya", "Shakinat-al-Auliya".

Allasani Peddana - "Manucharitra".

Abul Fazl - "Razmnama", "Bahar-e-Damish", "Kalila and Dimna".

Krishnadevaraya - "Amuktamalyada", "Jambati Kalyanam".

Tulsidas - "Vinaya Patrika", "Ramcharitmanas", "Barvai Ramayana", "Parvati Mangal", "Ramlala Nahchu".

Surdas - "SurSagar", "Nal - Damayanti".

Kabir - "Sakhi", "Bijak", "Rekhta", "Gyan sagar".

Malik Muhammad Jayasi - "Akhrwat", "Akhir Kalam", "Padmavat", "Kaharnama".

Sharngadeva - "Sangita Ratnakara".

Modern

Raja Ram Mohan Roy - "Tuhfatul Muwahhidin", "A Gift to the Monotheists".

Bal Gangadhar Tilak - "Geeta Rahasya",

"The Arctic Home of the Vedas".

Dinabandhu Mitra- "Nil Darpan".

Rabindra Nath Tagore- "Home and the World", "Gitanjali", "Gora", "The Homecoming", "The Crescent Moon", "Galpaguchchha" (short Stories).

Lala Lajpat Rai- "Unhappy India", "History of the Arya Samaj", "Young India", "Hindu Muslim Unity", "Message of Bhagavad Gita".

Mahatma Gandhi- "Hind Swaraj", "Peace", "Non-violent Resistance", "The Way to God", "All Men are Brothers", "A Tale of Two Cities and Related", "India of My Dream".

Dayanand Saraswati- "Satyarth Prakash".

Surendra Nath Banerjee- "A Nation in Making", "Thought of India".

Bankim Chandra Chatterjee- "Anandamath", "Dharma Tattva", "Durgesh Nandini".

Annie Besant - "How India Wrought for Freedom", "India a Nation", "India Bond or Free", "The Future of the Indian Politics".

Sarojini Naidu - "Broken Wings", "Songs of India", "World of Time", "The Golden Threshold".

Vinayak Damodar Savarkar - "The War of Indian Independence", "Hindutva".

Jawaharlal Nehru - "A bunch of Old letters", "Discovery of India", "A Glimpse of World History", "Indian Foreign Policy", "Panchsheel", "An Autobiography- Toward Freedom".

Dr B R Ambedkar - "Castes in India", "The Problem of the Rupee", "Bahishkrit Bharat", "The Annihilation of Caste", "Federation Versus Freedom", "Thoughts on Pakistan", "Ranade, Gandhi and Jinnah", "Mr. Gandhi and Emancipation of Untouchables", "Pakistan Or Partition Of India", "Who were the Shudras", "The Untouchables", "Buddha Or Karl Marx", "The Buddha and his Dhamma".

Jyotiba Phule - "Gulamgiri", "The Buddha and His Dhamma".

M N Roy - "Planning in India", "The Communist International".

Maulana Abul Kalam Azad (First Education Minister of India) - "India Wins Freedom" (autobiography), "Ghubar-e-Khatir".

Kishwar Desai - "Jallianwala Bagh, 1919- The Real Story".

Dominique Lapierre and Larry Collins - "Freedom at Midnight", "Kohinoor- The Story of the World's Most Infamous Diamond", "Return of a King", "Nine Lives", "White Mughals", "From the Holy Mountain", "The Anarchy- The Relentless

Rise of the East India Company".

Ramchandra Guha - "Makers of Modern India", "Democrats and dissenters", "India after Gandhi".

Amartya Sen - "The Argumentative Indian", "Development As Freedom", "The Idea of Justice", "Predictions- Thirty Great Minds on the Future", "Inequality Reexamined".

William Dalrymple - "The Last Mughal- The Fall of a Dynasty, Delhi 1857", "The Anarchy", "City of Djinns- A Year in Delhi", "White Mughals- Love and Betrayal in Eighteenth Century India".

A.L. Basham - "The Wonder That Was India", "The Origins and Development of Classical Hinduism", "Hare Krishna, Hare Krishna", "History and Doctrines of the Ajivikas", "The Sacred Cow".

Harivansh Rai Bachhan - "In The Afternoon of Time", "Madhushala", "Nisha Nimantran", "Need ka Nirman Phir", "Milan Yamini".

Satyajit Ray - "Jakhan Choto Chilam", "The Adventures of Feluda", "The Chess Players and Other", "Twenty Stories", "One Dozen Stories", "Childhood Days - A Memoir".

Famous Personalities

Bharat Ratna Awardees

Chakravarti Rajagopalachari - Popularly known as Mootharignar Rajaji. He was First Governor of West Bengal after Independence, 2nd chief minister of Madras state, and also the only Indian-born Governor-General of Dominion of India. Awards - Bharat Ratna (1954), Sahitya Akademi Award (1958).

Sarvepalli Radhakrishnan - He served as the second president of India. He was also the first vice president of India. He was the first Indian to hold a chair at the University of Oxford. Award - Bharat Ratna (1954).

Chandrasekhar Venkataraman - He was an Indian physicist known for his work in the field of light scattering. Awards - Nobel Prize in Physics (1930), Bharat Ratna (1954).

Bhagwan Das - He co-founded Kashi Vidyapith and worked with Madan Mohan Malaviya to establish the Banaras Hindu University. Award - Bharat Ratna (1955).

Sir Mokshagundam Visvesvaraya - He was a civil engineer and statesman. Award - Bharat Ratna (1955).

Jawahar Lal Nehru - He was born on 14th November 1889 in Allahabad, Uttar Pradesh. He was the first and the longest-serving Prime Minister of India from 1947 to 1964. Award - Bharat Ratna (1955).

Govind Ballabh Pant - He successfully abolished the exploitative Zamindari system. He was the First Chief Minister of Uttar Pradesh. In Kashipur, he established an organization called Prem Sabha. Awarded - Bharat Ratna (1957).

Rajendra Prasad - He was an Indian politician, lawyer, Indian independence activist, journalist & scholar who served as the first President of India from 1950 to 1962. Award - Bharat Ratna (1962).

Lal Bahadur Shastri - He was born 2 October 1904 in Mughalsarai in Uttar Pradesh. He served as the second Prime Minister (1964-66) of India. He gave the famous slogan 'Jai Jawan Jai Kisan'. Awards - Bharat Ratna (1966), and a memorial "Vijay Ghat" was built for him in Delhi.

Indira Gandhi - She was Born on November 19, 1917. She was the daughter of Pt. Jawaharlal Nehru. She was Known as the "Iron Lady of India". In January 1966, She became the first female Prime Minister of India. The Southernmost point of India 'Indira Point' is named after her. Awards - Bharat Ratna (1971). Bangladesh Swadhinata Sammanona (2011).

Kumaraswami Kamaraj (Kamarajar) - He was born in Virudhunagar, Tamil Nadu. He served as the Chief Minister of Madras State (Tamil Nadu). Award - Bharat Ratna (1976).

Mother Teresa - She was an Albanian-Indian Catholic nun. She established Nirmal Hriday ("Place for the Pure of Heart"). Awards - Bharat Ratna (1980), Padma Shri (1962), Ramon Magsaysay Peace award (1962), Nobel Peace Prize (1979).

Khan Abdul Gaffar Khan (Bacha Khan) - He was a close friend of M.K. Gandhi. He was known as "Frontier Gandhi". He founded the Khudai Khidmatgar ("Servants of God"), commonly known as the "Red Shirts" (Surkh Posh), during the 1920s. He received Bharat Ratna, the highest Indian honour in 1987.

Bhimrao Ramji Ambedkar - He was born on 14 April 1891 in the town and military

cantonment of Mhow (now officially known as Dr Ambedkar Nagar in Madhya Pradesh). He headed the committee drafting the Constitution of India. He served as the first Law and Justice minister. He is also known as "Father of Constitution of India" or "Chief Architect of Constitution of India".

Nelson Mandela - He was the leader of the Anti-Apartheid Movement in South Africa and later served as the President of South Africa (1994–99). Often called the "Gandhi of South Africa". Award - Nobel Peace Prize (1993), Bharat Ratna (1990).

Rajiv Gandhi - He was an Indian politician who served as the 6th prime minister of India from 1984 to 1989. Award - Bharat Ratna (1991).

Sardar Vallabh Bhai Patel - He was born on 31st October 1875 in Nadiad, Gujarat. He was the first Home Minister and Deputy Prime Minister of India. Women of Bardoli bestowed the title 'Sardar' on Vallabhbhai Patel. He played a key role in the integration of about 565 princely states into the Indian Union. He is also known as the "Iron Man of India" and "Bismarck of India". Award - Bharat Ratna (1991).

Morarji Desai - He was the 4th Prime Minister of India (1977-79). He was the first non-Congress Prime Minister of India. In 1952, he became the Chief Minister of Bombay. He is the oldest person to hold the office of prime minister, at the age of 81. **Awards** - Bharat Ratna (1991), Nishan-e-Pakistan (1990).

Maulana Abul Kalam Azad - His original name is Muhiyuddin Ahmad. In 1912, he started a weekly journal in Urdu called Al-Hilal. In 1923, he was elected as the youngest president of Indian National Congress. In 1947, he became the first education minister of free India. Award - posthumously Bharat Ratna in 1992.

Jehangir Ratanji Dadabhoy Tata - He was born on 29 July 1904. He was the first person in India to own a flying license and known as the father of Indian civil aviation. Award - Bharat Ratna (1992), Padma Vibhushan (1955).

Gulzarilal Nanda - He was the Acting Prime Minister of India for two short periods following the deaths of Jawaharlal Nehru in 1964 and Lal Bahadur Shastri in 1966. Award - Bharat

Ratna (1997).

Aruna Asaf Ali - She was known for hoisting the tricolor flag in Bombay during the Quit India Movement in 1942. Post Independence, Ali was elected as Delhi's first mayor in 1958. Award - Bharat Ratna (1997).

Avul Pakir Jainulabdeen Abdul Kalam (Missile Man) - He was Born on 15th October 1931 at Rameswaram in Tamil Nadu. His birth anniversary is celebrated as the National Innovation Day. He was India's 11th President (2002-2007). Award - Padma Bhushan (1981), Padma Vibhushan (1990) and Bharat Ratna (1997).

Chidambaram Subramaniam - He served as the minister of agriculture from 1964 to 1966. He is known for his contribution towards the Green Revolution in India. Award - Bharat Ratna (1998).

Jayaprakash Narayan - He is popularly referred to as JP or Lok Nayak. He is remembered for leading the mid-1970s opposition against Prime Minister Indira Gandhi, for whose overthrow he had called for a "total revolution". In 1952, he formed the Praja Socialist Party (PSP). Awards - Bharat Ratna (1999), Magsaysay award (1965).

Amartya Sen - He is an Indian economist and philosopher. Awards - Nobel Memorial Prize in Economic Sciences (1998), Bharat Ratna (1999).

Atal Bihari Vajpayee - He was born on December 25, 1924, in Gwalior. He became Prime Minister thrice in 1996, 1998-1999 and 1999-2004. During his tenure as prime minister, India carried out the Pokhran-II nuclear tests in 1998. Awards and Recognitions - Bharat Ratna (2015) and Padma Vibhushan (1992). His birthday, December 25, is observed as the Good Governance Day by the government.

Chintamani Nagesa Ramachandra (C.N.R) Rao - He is a chemist and a scientist specializing in solid state chemistry. Award - Bharat Ratna (2014).

Pranab Mukherjee - He was a politician who served as the 13th President of India. Award - Bharat Ratna (2019).

Nanaji Deshmukh - He was a social activist and politician. He was a leader of the Bharatiya Jana Sangh and also served as a member of the Rajya Sabha. Award - Bharat Ratna (2019).

Karpoori Thakur - He was a politician, who served two terms as the 11th Chief Minister of Bihar. Award - Bharat Ratna (2024).

Lal Krishna Advani - He is a politician who served as the 7th Deputy Prime Minister of India from 2002 to 2004. He is one of the co-founders of Bharatiya Janata Party and is credited with scripting the rise of the BJP as a major political force through the Ram Janmabhoomi Movement. Award - Bharat Ratna (2024).

P. V. Narasimha Rao - He was an Indian lawyer and politician who served as the 9th prime minister from 1991 to 1996. He was the first Prime Minister from South India. Award - Bharat Ratna (2024).

Charan Singh - He was an Indian politician and an independence activist who served as the 5th prime minister from 1979 to 1980. He is known as the "Champion of India's peasants". Award - Bharat Ratna (2024).

M. S. Swaminathan - He was one of the major architects of the green revolution in India known for his leadership and role in introducing and further developing high-yielding varieties of wheat and rice. Award - Bharat Ratna (2024).

Other Famous Personalities

Mohandas Karamchand Gandhi - (Father of the Nation)- He was born in Porbandar (Gujarat) on 2 October 1869. His Father's name is Karamchand Gandhi, Mother - Putlibai Gandhi and Wife - Kasturba Gandhi. He regarded Gopal Krishan Gokhale as his political guru.

He returned to India from South Africa on 9th January 1915, on this day celebrated on Pravasi Bharatiya Divas every year. Albert Einstein was inspired by Mahatma Gandhi's idea of non-violence.

Titles - Gandhiji was awarded the 'Kaiser-i-Hind' in 1915 by Lord Hardinge of Penshurst for his ambulance services in South Africa. Mahatma Gandhi returned his medal of 'Kaiser-i-Hind' in 1920. Rabindranath Tagore gave the title "Mahatma" to Mohandas Karamchand Gandhi. Gandhiji gave CF Andrews the title 'Deen Bandhu' (Friend of the poor) for his contributions to the Indian Independence Movement. He gave the Gurudev title to Rabindranath Tagore. He said that "English education has enslaved us".

Mahatma Gandhi was nominated for the Nobel Peace Prize five times. He was assassinated by Nathu Ram Godse on 30 January 1948 at age 78 in the compound of Birla House (now Gandhi Smriti), a large mansion in central New Delhi.

Rabindranath Tagore - He was also referred to as 'Gurudev', 'Kaviguru' and Vishwakavi. His compositions were chosen by two nations as national anthems- India's "Jana Gana Mana" and Bangladesh's "Amar Shonar Bangla". Award- He was the first non-European to receive the Nobel Prize in Literature in 1913 for his work on "Gitanjali".

Swami Vivekananda - He was born on 12th January 1863 in a Bengali family in Calcutta and was originally named Narendranath Datta. National Youth Day is held every year on his birth anniversary. In 1893, upon the request of Maharaja Ajit Singh of the Khetri State, he took the name 'Vivekananda'. He is Best known for his speech at the World Parliament of Religion in Chicago in 1893.

Sarojini Naidu - She is famous by the name of Nightingale of India. She was the first Indian woman to be president of the Indian National Congress in 1925 (Kanpur Session). After India gained independence, she was appointed as the governor of Uttar Pradesh, becoming the first woman to hold a governor's office in India. In 1917, she founded the Women's India Association with Annie Besant.

Bal Gangadhar Tilak - He was born on 23 July 1856 in Ratnagiri, Maharashtra (then Bombay Presidency). He is also known as Lokmanya Tilak. He was the founder of the Deccan Education Society (1884) along with his associate Gopal Ganesh Agarkar and others. A book 'Indian Unrest' written by Valentine Chirol, an English journalist, stated Tilak the 'father of Indian unrest'. He founded the All India Home Rule League in April 1916 at Belgaum. Newspapers- Weeklies Kesari (Marathi) and Mahratta (English).

Homi Jehangir Bhabha - He was an Indian nuclear physicist. He was the founding director of Tata Institute of Fundamental Research and the Atomic Energy Establishment, Trombay (AEET) which is now named the Bhabha Atomic Research Centre in his honor. He is also known as "Father of the Indian nuclear programme". He was awarded with the Adams Prize (1942) and Padma Bhushan (1954). He was also nominated for the

Nobel Prize for Physics in 1951 and 1953–1956. He was the first Chairman of the Atomic Energy Commission of India.

Swami Dayanand Saraswati - He was born in the town of Tankara, Gujarat. He was an Indian philosopher, social leader and founder of the Arya Samaj, a reform movement of the Vedic dharma. He was the first to give the call for Swaraj as "India for Indians" in 1876, a call later taken up by Lokmanya Tilak.

Kailash Satyarthi - He is an Indian social reformer who campaigned against child labor in India and advocated the universal right to education. He was the co-recipient of the Nobel Peace Prize in 2014, along with Malala Yousafzai, "for their struggle against the suppression of children and young people and for the right of all children to education."

Subramanian Chandrashekhara - Awards- FRS (1944), Adams Prize (1948), Nobel Prize in Physics (1983), Padma Vibhushan (1968).

Sam Manekshaw - Field Marshal Sam Hormusji Framji Jamshedji Manekshaw, widely known as Sam Manekshaw and Sam Bahadur ("Sam the Brave"), was the Chief of the Army Staff of the Indian Army during the Indo-Pakistani War of 1971, and the first Indian Army officer promoted to the rank of Field Marshal. Awards- Padma Vibhushan, Padma Bhushan, Military Cross.

Savitribai Phule - She published Kavya Phule in 1854 and Bavan Kashi Subodh Ratnakar in 1892. In 1852, She started the Mahila Seva Mandal. In 1863, they started a home for the prevention of infanticide in their own house, for the safety of pregnant, exploited Brahman widows and to nurture their children.

Jagjivan Ram - He was born in Bihar. He was popularly known as Babuji. On Malaviya's invitation, he joined the Banaras Hindu University.

Gopal Krishna Gokhale - He was born on 9 May 1866 in Kotluk village in present-day Maharashtra (then part of the Bombay Presidency). He became president of INC in 1905 in Banaras session. His political guru was M.G Ranade.

Sir Syed Ahmed Khan - He was born on 17 October 1817. He was an Indian Muslim reformer, philosopher, and educationist in nineteenth-century

British India. Many orthodox groups at the time labeled him a 'kafir'.

He wrote a profound booklet titled 'Asbab-e-Baghawat-e-Hind' (Reasons for the Indian Revolt of 1857). The British government bestowed upon him the Order of the Star of India in 1869.

Jyotiba Phule - He was born on 11th April, 1827 in present-day Maharashtra. He along with his followers formed Satyashodhak Samaj in 1873 which meant 'Seekers of Truth' in order to attain equal social and economic benefits for the lower castes in Maharashtra.

Jivatram Bhagwandas Kripalani - He is popularly known as Acharya Kripalani. He served in the Interim government of India (1946–1947) and the Constituent Assembly of India. He became one of the founders of the Kisan Mazdoor Praja Party (KMPP) after independence. His wife, **Sucheta Kripalani** was India's first female Chief Minister, serving as the head of the Uttar Pradesh government from 1963 to 1967. He moved the first-ever No confidence motion in Lok Sabha in 1963, immediately after the India-China War (1962).

Venkatraman Ramakrishnan - He is an Indian-born British and American structural biologist who shared the 2009 Nobel Prize in Chemistry with Thomas A. Steitz and Ada Yonath, "for studies of the structure and function of the ribosome". Awards- Knight Bachelor (2012), Padma Vibhushan (2010), Nobel Prize in Chemistry (2009).

Kasturba Mohandas Gandhi - She was an Indian political activist. She was married to Mohandas Gandhi. The Government of India in the year 2003 designated 11 April as National Safe Motherhood Day. The day also marks the birth anniversary of Kasturba Gandhi. Her memorial is in Aga Khan Palace, Pune.

Srinivasa Ramanujan - He was born on 22 December 1887 in Erode, Tamil Nadu. He published the first of his papers in the Journal of the Indian Mathematical Society in 1911. He was elected to the Royal Society of London in 1918. In 2012, former Prime Minister Manmohan Singh declared December 22 as National Mathematics Day.

Elattuvalapil Sreedharan - He is an Indian engineer and politician from the Indian state of Kerala. He is credited with changing the face of public transport in India with his leadership in building the

Konkan Railway and the Delhi Metro while he served as the managing director of Delhi Metro Rail Corporation (DMRC) between 1995 and 2012. He is Known as the Metro Man.

He was awarded the Padma Shri in 2001, the Padma Vibhushan in 2008, the Chevalier de la Légion d'honneur in 2005 by the French government.

Satyendra Nath Bose - He was an Indian mathematician and physicist specializing in theoretical physics. Padma Vibhushan (1954).

Amrita Pritam - She was born as Amrit Kaur in Gujranwala, Punjab, in British India. She was active in writing and editing 'Nagmani' a Punjabi monthly magazine till her last. She is most remembered for her poem titled "Ajj Aakhaan Waris Shah Nu," (Today I Invoke Waris Shah), addressed to Punjabi Sufi poet Waris Shah in which she laments the tragedy of the partition of Pakistan from India. **Awards** - Sahitya Akademi Award (1956), Padma Shri (1969), Padma Vibhushan (2004).

Dadabhai Naoroji - He was born on 4 September 1825. He was a Businessman. He is known as the Grand old man of India and country's Unofficial Ambassador. He was the founder of the East India Association in London in 1866. He is known as "The Father of India Politics and Economics". He is the creator of "Brain Drain Theory". He highlighted the draining of wealth from India by the British and its effect in his book "Poverty and Un-British Rule in India" (1901). He was the first Indian to be elected to the British House of Commons.

Mohammad Ali Jinnah - He was born on 25 December 1876 in Karachi, now in Pakistan. From 1906 to 1920, the role of Jinnah during the freedom struggle of India was Nationalist and Secular. In 1916, he was elected as the president of the Muslim league.

Salim Moizuddin Abdul Ali - He was born on 12 November 1896. He was an Indian ornithologist and naturalist. He is also referred as the "Birdman of India". Awards- Padma Bhushan (1958), Padma Vibhushan (1976).

Bipin Chandra Pal - He was born on 7 November 1858 in the village of Poil, Habiganj, Sylhet District, Bengal Presidency of British India. He was third of the "Lal Bal Pal" trio. He was one of

the main architects of the Swadeshi movement along with Sri Aurobindo.

Annie Besant - She was a British socialist, theosophist, freemason, women's rights and Home Rule activist, educationist, and campaigner for Indian nationalism. She was born on 1 October 1847 in London to an Irish-origin family. She founded the All India Home Rule League in 1916. In December 1917, she was elected president of the Indian National Congress for a one-year term. She was one of the founders of the Banaras Hindu University. She also founded the Central Hindu School to promote education.

Lala Hardayal - He was born on 14th October 1884. He was one of the co-founders of the Ghadar Party. He moved to Paris and associated himself with a newspaper Vande Mataram over there. In 1911 he settled in San Francisco and indulged in Industrial Unionism.

Mario de Miranda - He was an Indian cartoonist and painter based in Loutolim in the Indian state of Goa. Awards- Padma Shri (1988), Padma Bhushan (2002), Padma Vibhushan (2012), All India Cartoonists's Association Lifetime achievement award.

Vallabhacharya - He was also known as Vallabha, Mahaprabhuji, Vishnuswami, and Vallabha Acharya. He founded the philosophy of Shuddha Advaita which believed in absolute union. He prescribed Pushtimarg as the means for salvation (extreme devotion to God).

Shiv Dayal Singh - He was an Indian spiritual guru. He was known by the honorific "Param Purush Puran Dhani Huzur Soami Ji Maharaj" by his disciples and devotees. He founded Radhasoami Satsang at Agra in 1861.

Bhaktivedanta Swami Prabhupada - He established the International Society for Krishna Consciousness (ISKCON). in 1966 in New York City. Its core beliefs are based on Hindu scriptures, particularly the Bhagavad Gita and the Bhagavata Purana, and the Gaudiya Vaishnava tradition. Swami ji also established over a hundred temples and wrote several books, teaching the path of Bhakti Yoga to the world.

Chaitanya Mahaprabhu - He was also known as Sri Gauranga, was a popular Vaishnava saint and reformer from Nadia, West Bengal. He was the chief

proponent of the Vedantic philosophy of Achintya Bheda Abheda Tattva. He composed the Shikshashtakam. He preached Vaishnavism in Bengal and Odisha. Puri became the main center of his activities. He was also the first to introduce Kirtans in Bhakthi movement. The form of Vaishnavism that he preached came to be called as 'Gudik Vaishnavism'. He was a devout Krishna follower who popularized the "Hare Rama, Hare Krishna" chant.

Narayana Guru - He was a philosopher, spiritual leader and social reformer in India. He led a reform movement against the injustice in the caste-ridden society of Kerala in order to promote spiritual enlightenment and social equality. He was of the ideology of "oru jati, oru matam, oru daivam manushyanu" (one caste, one religion, one god for all men).

Pingali Venkayya - He was an Indian freedom fighter and a Gandhian. He was the designer of the flag on which the Indian National Flag was based. At the age of 19, He joined the army during the Boer war. It was during this stint in the British Army that he met Gandhi. He wrote a book titled "National Flag for India," which was published in 1916.

Sheila Dikshit - She is the longest-serving Chief Minister of Delhi, as well as the longest-serving female chief minister of any Indian state. She represented India at the United Nations Commission on Status of Women for five years (1984–1989).

Sunita Williams - She is a renowned astronaut of Indian origin, who spent a total of 322 days in space on two missions.

Arunima Sinha - She became the world's first female Amputee to scale Antarctica's Highest Peak Mount Vinson on 4th January 2019. Apart from this, she was also the world's first woman amputee to climb Mount Everest in 2013.

Vijaya Lakshmi Pandit - She was an Indian diplomat and politician who was the first female elected as 6th Governor of Maharashtra and 8th President of the United Nations General Assembly.

Dr. Verghese Kurien - He was the founder Chairman of National Dairy Development Board from 1965 to 1998. He was the architect of India's White Revolution, which helped India emerge as the largest milk producer in the world. In the late 60s, he drew up a project called

Operation Flood. Awards- World Food Prize (1989), Order of Agricultural Merit (1997), Padma Vibhushan (1999), Padma Bhushan (1966), Padma Shri (1965), Ramon Magsaysay Award (1964).

Satyendra Nath Tagore - He was the first Indian to join the Indian Civil Service. He was the first Indian to qualify for covenanted service. He was assigned to the Bombay ICS, where, from 1864 to 1897, he spent his entire career. He was a member of Brahmo Samaj. He was the brother of Rabindranath Tagore.

Pandit Motilal Nehru - He was a leader of the Indian independence movement. He was the cofounder of the Swaraj ("Self-rule") Party, and the father of India's first prime minister, Jawaharlal Nehru.

Baba Ram Chandra - He was an Indian trade unionist who organized the farmers of Awadh, India into forming a united front to fight against the abuses of landlords in 1920s and 1930s. He had earlier worked in Fiji as an indentured laborer.

Margaret Elizabeth Noble - She was also known as Sister Nivedita. Sister Nivedita was an Irish teacher, author, social activist, school founder and disciple of Swami Vivekananda.

Malavath Purna - She is an Indian mountaineer. On 25 May 2014, Purna scaled the highest peak of Mount Everest and, aged 13 years and 11 months, became the youngest Indian and the youngest female in the world to have reached the summit.

Ban ki Moon - He was born on 13 June 1944. He is a South Korean politician and diplomat who served as the eighth secretary-general of the United Nations between 2007 and 2016. His autobiography is titled "Resolved- Uniting Nations in a Divided World."

Le Corbusier - He was a Swiss-French architect, designer, painter, urban planner, writer. He prepared the master plan for the city of Chandigarh in India. He designed the Open Hand Monument in Chandigarh.

Professor Muhammad Yunus - He is the founder of the Grameen Bank of Bangladesh (1983). Grameen Bank of Bangladesh was founded as a means of providing small loans to poor individuals in 1983. He was awarded the Nobel Prize for Peace (2006).

Kalpna Chawla - She was an Indian-born American astronaut and aerospace engineer who was the first woman of Indian origin to go to space. She first flew on Space Shuttle Columbia in 1997 as a mission specialist and primary robotic arm operator. Her second flight was on STS-107, the final flight of Columbia, in 2003.

Sirimavo Bandaranaike - She was the first woman Prime Minister of Sri Lanka and the world. She also served as the Minister of Defence and External Affairs of Sri Lanka. She created the Ministry of Women and Child Affairs in Sri Lanka in 1975. She joined the Lanka Mahila Samiti (Lankan Women's Association), the country's largest women's voluntary organization in 1941.

Dhanpat Rai Srivastava (Munshi Premchand) - He began his career under the pen name "Nawab Rai". He is an Indian author of novels and short stories in Hindi and Urdu.

Nawab Sir Khwaja Salimullah Bahadur - He was born at the ahsan manzil in Dacca, Bengal Presidency, British India. He was an architect of the Muslim League. He was Nawab of Dhaka. In 1906 He planned to create an all-India political party called 'Muslim All India Confederacy'. On 30 December 1906, the last day of the convention, the 'All India Muslim League' was formed. He was made its vice-president.

Fatima Sheikh - She was born in Pune. She was an Indian educator and social reformer, who was a colleague of the social reformers Jyotirao Phule and Savitribai Phule. She is widely considered to be India's first Muslim woman teacher.

Syed Ameer Ali (1849–1928) - He was an Indian jurist, a prominent political leader. He was a signatory to the 1906 Petition to the Viceroy and was thus a founding-member of the All India Muslim League. He played a key role in securing separate electorates for the Muslims in British India and promoting the cause of the Khilafat Movement.

Balshastri Jambhekar - He is also known as Father of Marathi journalism for his efforts in starting journalism in Marathi language with the first newspaper in the language named 'Darpan'. Digidarshan, the first monthly magazine in Marathi was started by him.

Nawab Abdul Latif - He was a Bengali Muslim aristocrat, educator and social

worker. His title, Nawab, was awarded by the British in 1880. He is known as the 'father of Muslim renaissance' in Bengal.

Begum Rokeya Sakhawat Hossain - She was a Bengali feminist thinker, writer, educator and political activist from British India (present-day Bangladesh).

Bankim Chandra Chattopadhyay - He was born on 27th June 1838 in the village of Kanthapura in the town of North 24 Parganas, Naihati, present day West Bengal. He composed the song Vande Mataram in Sanskrit. His epic Novel Anandamath - set in the background of the Sanyasi Rebellion (1770-1820). India got its national song, Vande Mataram, from Anandamath. He also founded a monthly literary magazine, Bangadarshan, in 1872.

Sachchidananda Hirananda Vatsyayan - He was born on 7 March 1911 in Kasia, a small town near Kushinagar in Uttar Pradesh. He is popularly known by his pen name Agyeya. He took an active part in the Indian freedom struggle and spent several years in prison for his revolutionary activities against British colonial rule. He was awarded the Sahitya Akademi Award (1964), Jnanpith Award (1978) and Golden Wreath Award for poetry.

Maqbool Fida Husain - He was born on 17 September 1915 in Pandharpur, Bombay Province (present - day Maharashtra). He was an Indian artist known for executing bold, vibrantly coloured narrative paintings in a modified Cubist style. As per Howard Gardner's theory of multiple intelligence, M. F. Husain demonstrates a Visual-Spatial type of Intelligence.

Tanguturi Prakasam Pantulu - He was chief minister of the Madras Presidency, and subsequently became the first chief minister of the new Andhra state, created by the partition of Madras. He was also known as "Andhra Kesari".

Madeleine Slade - She is also known as Mirabehn or Meera Behn, was a British follower who in the 1920s left her home in England to live and work with Mahatma Gandhi. She devoted her life to human development and the advancement of Gandhi's principles.

Ibrahim Sutar - He was a social worker from Karnataka, India. He was recipient of India's fourth highest civilian award, the Padma Shri (2018) and Karnataka

State's Rajyotsava Award in the year 1995. He is fondly referred to as the "Kabir of Kannada".

Bhanubhakta Acharya - He is widely regarded as the first poet in the Nepali language. He is best known for translating the great epic Ramayana from Sanskrit to Nepali for the first time. He is conferred with the title Aadikabi. His other notable works include- Amarawati Kantipuri Nagari, Ghansi, Badhu Shiksha, Bhakta Mala, etc.

Matangini Hazra - She was fondly called Gandhi Buri (Old Lady Gandhi) for her dedication towards Gandhian principles. Matangini Hazra was an Indian revolutionary and social activist who participated in the Indian independence movement. She was shot dead by the British Indian police in front of the Tamluk Police Station. The first statue of a woman put up in Kolkata, in independent India, was her in 1977.

Fathima Beevi - She was born on 30 April 1927 at Pathanamthitta in the Kingdom of Travancore (Kerala). She is a former judge of the Supreme Court of India. She became the first female judge to be a part of the Supreme Court of India and the first Muslim woman to be appointed to any of the higher judiciary in the country.

Syed Fazl-ul-Hasan - He was known by his pen-name Hasrat Mohani. He was an Indian activist, freedom fighter in the Indian independence movement. He coined the slogan Inquilab Zindabad in 1921.

Kamla Chaudhary - She had come from an affluent family in Lucknow but had struggled to get an education. She was one of the most active women in the Civil Disobedience Movement in 1930.

Shri Krishna Sinha - He is also known as Sri Babu and Bihar Kesari. He served as the first Chief Minister of Bihar from 1946 to 1961.

Rahmat Ali - He wrote the pamphlet 'Now or Never'. The "Pakistan Declaration" was a pamphlet written and published by him on 28 January 1933. He is credited with creating the name "Pakistan" for a separate Muslim homeland in South Asia and is generally known as the originator of the Pakistan Movement.

Ramnath Kovind - He was born on 1

October 1945. He is an Indian politician and lawyer who served as the 14th president of India from 2017 to 2022. He is a member of the Bharatiya Janata Party.

Boutros Boutros-Ghali - He was an Egyptian politician and diplomat who served as the sixth Secretary-General of the United Nations from 1992 to 1996.

Dineshchandra Sircar - He is known for his work of deciphering inscriptions. He was the Chief Epigraphist of the Archaeological Survey of India.

Helena Petrovna Blavatsky - She is often known as Madame Blavatsky. She was born in the Russian Empire and she co-founded the Theosophical Society in 1875.

Edmund Percival Hillary - He was a New Zealand mountaineer, explorer, and philanthropist. On 29 May 1953, Hillary and Sherpa mountaineer Tenzing Norgay became the first climbers confirmed to have reached the summit of Mount Everest.

Alexander Cunningham - He was the first Director-General of the Archaeological Survey of India and is called the father of Indian Archaeology.

Potti Sriramulu - He wanted the capital of Andhra Pradesh to be built in Rayalaseema, and it was due to his 56 day fast that in 1952 a separate Andhra state was carved out of the erstwhile Madras Presidency with Kurnool as its capital.

RK Shanmukham Chetty - He was India's delegate to the World Monetary Conference at Bretton Woods in 1944. He was an Indian lawyer, economist, and politician who served as independent India's first finance minister from 1947 to 1948.

Rassundari Debi - She was born in 1810 in the village of Potajia, now Pabna, Bangladesh. She is the first Bengali and the first Indian woman to write an autobiography. She is a renowned author of 'Amar Jibon' (My Life), which was published in 1876.

Maurice Linford Gwyer - He was the first Chief Justice of the Federal Court of India. He is credited with having founded the college Miranda House in 1948 in Delhi.

Mylai Chinna Thambi Pillai Rajah - In 1932, He formed the All India Depressed Classes Association in Nagpur and M. C. Rajah was its first elected President.

Colonel Henry Steel Olcott - He was first president of the Theosophical Society.

Edwin Lutyens - He was an English architect. Famous buildings built by him are Rashtrapati Bhavan, India Gate, The Cenotaph Whitehall, (London). The Old Parliament of India was also built by Herbert Baker and Lutyens.

Matthew Nowicki - He was chief architect of the new Indian city of Chandigarh.

Mohan Singh - He established Azad Hind Fauz in Singapore in 1942.

AO Hume - He was a British civil servant who worked in British India. He was the founder of the Indian National Congress.

Homai Vyrawalla - She was India's first female photojournalist. Award - Padma Vibhushan (2011).

Baby Kamble (Babytai Kamble) - She was an Indian activist and writer.

Lakshmi Bai Tilak - She was a writer from Maharashtra.

Basanti Dulal Nagchaudhuri - He was an Indian physicist. Award - Padma Vibhushan (1975).

Dr. Yelavarthy Nayudamma - He was a chemical engineer and a scientist. Award - Padma Shri (1971).

Kocheril Raman Narayanan - He served as the ninth vice president of India from 1992 to 1997 and tenth president of India from 1997 to 2002.

G Parthasarathi - He was the first Vice-Chancellor of Jawaharlal Nehru University in 1969.

Kadambini Bose Ganguly - She was a medical doctor. She was the first woman graduate of Calcutta University.

Herbert Baker - He was an English architect remembered as the dominant force in South African architecture.

Lawrence Wilfred "Laurie" Baker - He was a British-born Indian architect. Award - Padma Shri (1990).

Vikram Sarabhai - He was an Indian physicist and astronomer. Award -

Padma Bhushan (1966), Padma Vibhushan (1972).

Dhirubhai Ambani - He was an Indian businessman. He was the founder of Reliance Industries. Awards - Padma Vibhushan (2016).

Tanguturi Prakasam Pantulu - He was an Indian jurist, known as "Andhra Kesari".

Tenzin Gyatso - He is the 14th Dalai Lama .Dalai Lama is a significant position in the Tibetan administration. Award - Nobel Peace Prize (1989).

Bindeshwar Pathak - He was the founder of Sulabh International. Awards - Padma Bhushan (1991), Padma Vibhushan (2024).

Neela Satyanarayan - She was the first woman Election Commissioner of Maharashtra.

Hemant Karkare was awarded the Ashok Chakra (2009) for his bravery in the 26/11 Mumbai terrorist attacks.

Important Days and Years

Important Days

(National and International)

January

- 1st January - Global Family Day
- 4th January - World Braille Day
- 6th January - World Day of War Orphans
- 9 January - Pravasi Bharatiya Divas
- 10th January - World Hindi Day
- 11th January - National Human Trafficking Awareness Day,
- 12 January - National Youth Day
- 14th January - Indian Armed Forces Veterans Day
- 15th January - Indian Army Day
- 24th January - National Girl Child Day
- 24th January - International Day of Education
- 25th January - Tourism Day, National Voters Day
- 30th January - Martyrs Day

February

- 1st February - Indian Coast Guard Day,
- 2nd February - World Wetlands Day,
- 4th February - International Day of Human Fraternity, World

Cancer Day

- 10th February - World Pulses Day, National Deworming Day
- 11th February - International Day of Women and Girls in Science
- 12th February - International Day for the Prevention of Violent Extremism as and when Conducive to Terrorism
- 13th February - World Radio Day, National Women's Day
- 20th February - World Day of Social Justice
- 21st February - International Mother Language Day
- 23rd February - World Peace and Understanding Day
- 24th February - Central Excise Day
- 27th February - World NGO Day
- 28th February - National Science Day.

March

- 1st March - World Seagrass Day, Zero Discrimination Day, World Civil Defence Day
- 3rd March - World Wildlife Day
- 4th March - National Security Day
- 5th March - International Day for Disarmament and Non - Proliferation Awareness,
- 8th March - International Women's Day
- 15th March - World Consumer Rights Day
- 16th March - National Vaccination Day
- 20th March - World Sparrow Day, International Day of Happiness, World Oral Health Day
- 22nd March - World Water Day
- 23rd March - World Meteorological Day
- 24th March - World Tuberculosis Day,
- 27th March - World Theatre Day

April

- 2nd April - World Autism Awareness Day
- 4th April - International Day of Mine Awareness,
- 5th April - National Maritime Day, International Day of Conscience,
- 6th April - International Day of Sport for Development and Peace,
- 7th April - World Health Day,
- 10th April - World Homeopathy Day,
- 11th April - National Pet Day, National Safe Motherhood Day

- 12th April - International Day of Human Space Flight,
- 14th April - World Chagas Disease Day
- 17th April - World Haemophilia Day
- 18th April - World Heritage Day,
- 21st April - National Civil Services Day, World Creativity and Innovation Day
- 22nd April - International Mother Earth Day
- 23rd April - World Book and Copyright Day
- 24th April - National Panchayati Raj Day,
- 25th April - World Malaria Day
- 28th April - World Day for Safety and Health at Work
- 29th April - International Dance Day,
- 30th April - Ayushman Bharat Diwas, International Jazz Day

May

- 1st May - International Labour Day
- 2nd May - World Tuna Day,
- 3rd May - World Press Freedom Day
- 5th may - International Day of the Midwife, World Hand Hygiene Day
- 7th May - World Athletics Day
- 8th May - World Thalassaemia Day
- 11th May - National Technology Day
- 12th May - International Day of Plant Health, International Nurses Day
- 16th May - International Day of Light
- 17th May - World Hypertension Day
- 17th May - World Telecommunication and Information Society Day
- 18th May - International Museum Day,
- 20th May - World Metrology Day, World Bee Day
- 21st May - International Tea Day, Anti-Terrorism Day
- 22nd May - International Day for Biological Diversity
- 23rd May - International Day to End Obstetric Fistula, World Turtle Day
- 24th May - Commonwealth Day
- 25th May - World Thyroid Day
- 28th May - International Day of Action for Women's Health,
- 29th May - International Day of UN Peacekeepers, International Mount Everest Day),
- 31st May - World No Tobacco Day.

June

- 1st June - World Milk Day
- 3rd June - World Bicycle Day
- 5th June - World Environment Day
- 7th June - World Food Safety Day
- 8th June - World Ocean Day
- 12th June - World Day Against Child Labour
- 13th June - International Albinism Awareness Day
- 14th June - World Blood Donor Day
- 17th June - World Day to Combat Desertification and Drought,
- 19th June - World Sickle Cell Day
- 20th June - World Refugee Day
- 21st June - World Music Day, International Yoga Day, International Day of the Celebration of the Solstice
- 23rd June - United Nations Public Service Day, International Olympic Day
- 24th June - International Day of Women in Diplomacy
- 29th June - National Statistics Day
- 30th June - International Asteroid Day,

July

- 1st July - National Doctor's Day
- 2nd July - World UFO Day, World Sports Journalists Day
- 4th July - Swami Vivekananda Death Anniversary
- 6th July - World Zoonoses Day
- 11th July - World Population Day
- 15th July - World Youth Skills Day
- 17th July - World Day for International Justice
- 18th July - Nelson Mandela International Day
- 20th July - International Chess Day,
- 25th July - World Drowning Prevention Day,
- 26th July - Kargil Memorial Day ,
- 28th July - World Nature Conservation Day,
- 29th July - International Tiger Day,
- 30th July - International Day of Friendship, World Day against Trafficking in Persons

August

- 6th August - Hiroshima Day

- 7th August - National Handloom Day
- 9th August - Nagasaki Day, World Tribal Day
- 10th August - World Biofuel Day, World Lion Day
- 12th August - International Youth Day
- 14th August - Pakistan Independence Day
- 19th August - World Humanitarian Day
- 20th August - World Mosquito Day
- 29th August - National Sports Day, International Day against Nuclear Tests

September

- 2nd September - World Coconut Day
- 5th September - Teachers Day, International Day of Charity
- 7th September - International Day of Clean Air for Blue Skies
- 8th September - International Literacy Day
- 9th September - International Day to Protect Education from Attack
- 14th September - Hindi Day
- 15th September - International Day of Democracy, National Engineer's Day
- 16th September - World Ozone Day, International Day for Interventional Cardiology,
- 21st September - International Day of Peace, World Alzheimer's Day
- 22nd September - World Rhino Day
- 23rd September - International Day of Sign Languages
- 26th September - International Day for the Total Elimination of Nuclear Weapons
- 27th September - World Tourism Day
- 28th September - World Maritime Day
- 29th September - World Heart Day

October

- 1st October - International Day of Older Persons
- 2nd October - International Day of Non-Violence
- 4th October - World Animal Welfare Day
- 5th October - World Teachers' Day
- 7th October - World Cotton Day
- 8th October - Indian Air Force Day
- 9th October - World Post Day

- 10th October - National Post Day
- 11th October - International Day of the Girl Child
- 12th October - World Arthritis Day
- 13th October - World Day for Natural Disaster Reduction
- 15th October - International Day of Rural Women
- 16th October - World Food Day
- 17th October - International Day for the Eradication of Poverty
- 20th October - International Day of the Air Traffic Controller, World Osteoporosis Day
- 24th October - United Nations Day, World Polio day
- 27th October - World Day for Audiovisual Heritage
- 31st October - National Unity Day (Rashtriya Ekta Diwas), World Cities Day

November

- 2nd November - International Day to End Impunity for Crimes against Journalists
- 5th November - World Tsunami Awareness Day
- 9th November - World Legal Services Day
- 10th November - World Immunization Day; World Science Day for Peace and Development
- 11th November - National Education Day
- 12th November - World Pneumonia Day
- 13th November - World Kindness Day
- 14th November - World Diabetes Day
- 15 November - Janjatiya Gaurav Diwas
- 17th November - International Students Day
- 18th November - World Adult Day
- 20th November - Universal Children's Day
- 21st November - (World Television Day, World Fisheries Day)
- 25th November - International Day for the Elimination of Violence against Women
- 26th November - National Law Day, Constitution Day, National Milk Day

December

- 1st December - Border Security Force (BSF) Raising Day
- 2nd December - World Computer Literacy Day

- 3rd December - International Day of Disabled Persons
- 4th December - Indian Navy Day, International Day of Banks
- 5th December - World Soil Day, International Volunteer Day for Economic and Social Development
- 7th December - Indian Armed Force Flag Day, International Civil Aviation Day
- 10th December - Human Rights Day, International Animal Rights Day
- 11th December - International Mountain Day
- 14th December - National Energy Conservation Day,
- 18th December - International Migrants Day
- 20th December - International Human Solidarity Day
- 22nd December - National Mathematics Day
- 23rd December - National Farmers Day
- 24th December - National Consumer Day
- 27th December - International Day of Epidemic Preparedness

Important International Decades

(2018–2028) - International Decade for Action "Water for Sustainable Development", (2019–2028) - United Nations Decade of Family Farming, (2019–2028) - Nelson Mandela Decade of Peace, (2021–2030) - United Nations Decade of Ocean Science for Sustainable Development, (2021–2030) - United Nations Decade on Ecosystem Restoration, (2021–2030) - United Nations Decade of Healthy Ageing, (2022–2032) - International Decade of Indigenous Languages, (2024–2033) - International Decade of Sciences for Sustainable Development.

States G.K.

State - India has states and union territories (UTs) clubbed into zones, while each state is further divided into districts. As per 2011 census, India has 640 districts. Presently, there are 28 states and 8 Union Territories in India.

States

Andhra Pradesh (Amaravati) came into

existence on 1 November 1956. Assembly seat - 175, Council - 58, Lok sabha - 25, Rajya sabha - 11, 1st Governor - Chandulal Madhavlal Trivedi, 1st Chief Minister - Neelam Sanjiva Reddy. State Bird - Rose-ringed Parrot, Animal - Blackbuck, Flower - Jasmine and Tree - Neem.

Arunachal Pradesh (Itanagar) - The North-East Frontier Agency was renamed Arunachal Pradesh in 1972, and it became a union territory. Later on Arunachal Pradesh became a state on 20 February 1987. Assembly seat - 60, Lok sabha - 2, Rajya sabha - 1, 1st Governor - Bhishma Narain Singh, 1st Chief Minister - Prem Khandu Thungan. State Bird - Hornbill, Animal - Mithun, Flower - Foxtail orchid and Tree - Hollong.

Assam (Dispur) got full statehood on 26 January 1950. Assembly seat - 126, Lok sabha - 14, Rajya sabha - 7, 1st Governor - Sir Muhammad Saleh Akbar Hydari, 1st Chief Minister - Gopinath Bordoloi. State Bird - White-winged wood duck, Animal - Indian One-horned Rhino, Flower - Foxtail orchid and Tree - Hollong.

Bihar (Patna) was formed by the Britishers on 22 March 1912 (Bihar Diwas) but it got full statehood on 26 January 1950. Assembly seat - 243, Council - 75, Lok sabha - 40, 1st Governor - Jairamdas Daulatram, 1st Chief Minister - Sri Krishna Singh. Rajya sabha - 16, State Bird - House Sparrow, Animal - Gaur, Flower - Kachnar and Tree - Peepal.

Chhattisgarh (Raipur), the 26th State formed by partitioning ten Chhattisgarhi and six Gondi-speaking southeastern districts of Madhya Pradesh. Formation - 1st November, 2000. Assembly seat - 90, Lok sabha - 11, Rajya sabha - 5, 1st Governor - D. N. Sahay, 1st Chief Minister - Ajit Jogi. State Bird - Hill Myna, Animal - Wild buffalo, and Tree - Sal.

Goa (Panaji), the 25th state was formed on 30th May, 1987. Goa Liberation Day is observed on 19th December. Assembly seat - 40, Lok sabha - 2, Rajya sabha - 1, 1st Governor - Gopal Singh, 1st Chief Minister - Dayanand Bandodkar. State Bird - Black Crested Bulbul, Animal - Gaur, and Tree - Matti.

Gujarat (Gandhinagar), the 15th state, The bilingual state of Bombay was divided into two separate states (Maharashtra for Marathi speaking people and Gujarat for Gujarati speaking people) by the Bombay Reorganisation

Act, 1960 on May 1, 1960. Assembly seat - 182, Lok sabha - 26, Rajya sabha - 11, 1st Governor - Mehdi Nawaz Jung, 1st Chief Minister - Jivraj Narayan Mehta. State Bird - Greater flamingo, Animal - Asiatic lion, Flower - Marigold and Tree - Banyan.

Haryana (Chandigarh) was carved out of Punjab on 1st November 1966, as the 17th Indian state. It was formed on the recommendation of Sardar Hukum Singh Parliamentary Committee. Assembly seat - 90, Lok sabha - 10, Rajya sabha - 5, 1st Governor - Dharma Vira, 1st Chief Minister - Bhagwat Dayal Sharma. State Bird - Black francolin, Animal - Blackbuck, Flower - Lotus and Tree - Peepal.

Himachal Pradesh (Shimla is the summer capital and Dharamshala is the winter capital), on 18th December, 1970 the State of Himachal Pradesh Act was passed by Parliament and the new state came into being on 25th January, 1971. Assembly seat - 68, Lok sabha - 4, Rajya sabha - 3, 1st Governor - S. Chakravarti, 1st Chief Minister - Yashwant Singh Parmar. State Bird - Western tragopan, Animal - Snow leopard, Flower - Rhododendron and Tree - Deodar.

Jharkhand (Ranchi) was created on 15 November 2000 by bifurcating the southern part of Bihar. Assembly seat - 81, Lok sabha - 14, Rajya sabha - 6, 1st Governor - Prabhat Kumar, 1st Chief Minister - Babulal Marandi. State Bird - Koel, Animal - Indian elephant, Flower - Palash and Tree - Sal.

Karnataka (Bengaluru) came into existence as the Mysore state in 1953, It was formed as Mysore State on 1 November 1956, with the passage of the States Reorganisation Act, and renamed Karnataka in 1973. Assembly seat - 224, Legislative Council - 75, Lok sabha - 28, Rajya sabha - 12, 1st Governor - Jayachamarajendra Wadiyar, 1st Chief Minister - K. Chengalaraya Reddy. State Bird - Indian roller, Animal - Indian elephant, Flower - Lotus and Tree - Sandalwood.

Kerala (Thiruvananthapuram) - Former states of Travancore, Cochin, and Malabar were merged together to form the state of Kerala in 1956. It got full statehood on November 1, 1956. Assembly seat - 140, Lok sabha - 20, Rajya sabha - 9, 1st Governor - Burgula Ramakrishna Rao, 1st Chief Minister - E. M. S. Namboodiripad. State Bird - Great hornbill, Animal - Indian elephant, Flower

- Golden shower tree and Tree - Coconut.

Madhya Pradesh (Bhopal) was formed in 1956. Assembly seat - 230, Lok sabha - 29, Rajya sabha - 11, 1st Governor - B. P. Sitaramayya, 1st Chief Minister - Ravishankar Shukla. State Bird - Indian paradise flycatcher, Animal - Swamp Deer, Flower - White Lily and Tree - Banyan.

Maharashtra (Mumbai) - Maharashtra and Gujarat were the Part of the state of Bombay Province. On 1 May 1960, the states of Maharashtra and Gujarat were formed as independent states (1 May - Maharashtra and Gujarat Day). Assembly seat - 288, Legislative Council - 78, Lok sabha - 48, Rajya sabha - 19, 1st Governor - Raja Maharaj Singh, 1st Chief Minister - Yashwantrao Chavan. State Bird - Green Imperial Pigeon, Animal - Indian giant squirrel, Flower - Jarul and Tree - Mango.

Manipur (Imphal) - Maharaja Budhachandra of princely state Manipur signed the Instrument of Accession in 1947 to join India. In 1956, it became a Union Territory and got full statehood on 21 January 1972. Assembly seat - 60, Lok sabha - 2, Rajya sabha - 1, 1st Governor - B.K. Nehru, 1st Chief Minister - Mairembam Koireng Singh. State Bird - Mrs. Hume's Pheasant, Animal - Sangai, Flower - Siroi Lily and Tree - Toon.

Meghalaya (Shillong) was formed on 21 January 1972 by carving out two districts from the state of Assam- United Khasi Hills and Jaintia Hills, and the Garo Hills. Assembly seat - 60, Lok sabha - 2, Rajya sabha - 1, 1st Governor - Braj Kumar Nehru, 1st Chief Minister - Williamson A. Sangma. State Bird - Hill myna, Animal - Clouded leopard, Flower - Lady's Slipper Orchid and Tree - Gamhar. **"Living root bridges"** is an iconic structure found in Meghalaya.

Mizoram (Aizawl) was formed on 20 February 1987. It became UT in 1972 by the 27th Amendment before that it was part of Assam. By the 53rd amendment, It became a full-fledged state. Assembly seat - 40, Lok sabha - 1, Rajya sabha - 1, 1st Governor - S.P. Mukherjee, 1st Chief Minister - C. Chhunga. State Bird - Mrs. Hume's pheasant, Animal - Himalayan serow, Flower - Red Vanda and Tree - Iron Wood.

Nagaland (Kohima) became the 16th state of India on 1 December 1963. Assembly seat - 60, Lok sabha - 1, Rajya

sabha - 1, 1st Governor- Vishnu Sahay, 1st Chief Minister - P. Shilu Ao. State Bird - Blyth's tragopan, Animal - Mithun, Flower - Rhododendron and Tree - Alder.

Odisha (Bhubaneswar) was established on 1 April 1936. Assembly seat - 147, Lok sabha - 21, Rajya sabha - 10, 1st Governor - Kailash Nath Katju, 1st Chief Minister - Harekrushna Mahatab. State Bird - Indian Roller, Animal - Sambar deer, Flower - Asoka and Tree - Sacred fig (Ashwattha).

Punjab (Chandigarh) was established on 1st November 1966. Chandigarh is the joint capital of Punjab and Haryana. Assembly seat - 117, Lok sabha - 13, Rajya sabha - 7, 1st Governor - Chandulal Madhavlal Trivedi, 1st Chief Minister - Gopi Chand Bhargava. State Bird - Northern goshawk, Animal - Blackbuck, Flower - Gladiolus and Tree - Sheesham. The state is rich in fertile soil and natural resources.

Rajasthan (Jaipur) - On March 30, 1949 the four states like Jodhpur, Jaipur, Bikaner and Jaisalmer joined the integration and the region came to be known as Greater Rajasthan. But 'Rajasthan' finally formed on November 01, 1956. March 30th is now celebrated as Rajasthan day. Assembly seat - 200, Lok sabha - 25, Rajya sabha - 10, 1st Governor - Gurmukh Nihal Singh, 1st Chief Minister - Heera Lal Shastri. State Bird - Godawan or Great Indian Bustard, Animal - Camel and Chinkara, Flower - Rohida and Tree - Khejri.

Sikkim (Gangtok) became the 22nd state of India on 16th May 1975. Assembly seat - 32, Lok sabha - 1, Rajya sabha - 1, 1st Governor - B. B. Lal, 1st Chief Minister- Kazi Lhendup Dorjee. State Bird - Blood pheasant, Animal - Red panda, Flower - Noble Orchid and Tree - Rhododendron.

Tamil Nadu (Chennai) was reorganised as a state in 1950 and the Madras state officially renamed as Tamil Nadu on 14 January 1969. Assembly seat - 234, Lok sabha - 39, Rajya sabha - 18, 1st Governor- Sardar Ujjal Singh, 1st Chief Minister - C. N. Annadurai. State Bird - Emerald dove, Animal - Nilgiri tahr, Flower - Kandhal and Tree - Palm.

Telangana (Hyderabad) - On 2nd June, 2014 the northwestern part of Andhra Pradesh was separated and 29th state Telangana was created. Assembly seat - 119, Legislative Council - 40, Lok sabha -

17, Rajya sabha - 7, 1st Governor - Ekkadu Srinivasan Lakshmi Narasimhan, 1st Chief Minister - K. Chandrasekhar Rao. State Bird - Indian roller, Animal - Spotted deer, Flower - Thangedu and Tree - Shami (Jammi chettu).

Tripura (Agartala) was an erstwhile princely State, merged with the Indian Union on 15th October 1949 and became a Union Territory without a legislature with effect from November 1, 1956. It became a full-fledged State on 21st January, 1972. Assembly seat - 60, Lok sabha - 2, Rajya sabha - 1, 1st Governor - B. K. Nehru, First Chief Minister - Sachindra Lal Singh.

State Bird - Green imperial pigeon, Animal - Phayre's Langur, Flower - Nageswar and Tree - Agar.

Uttarakhand (Dehradun) - On 9 November 2000, It became India's 27th state, carved out of the hilly regions of undivided Uttar Pradesh. Assembly seat - 70, Lok sabha - 5, Rajya sabha - 3, 1st Governor - Surjit Singh Barnala, 1st Chief Minister - Nityanand Swami. State Bird - Himalayan Monal, Animal - Musk deer, Flower - Brahm Kamal and Tree - Burans.

Uttar Pradesh (Lucknow) was created on 1 April 1937 by the British regime under the name of United Provinces and post independence it was renamed as Uttar pradesh in 1950. Assembly seat - 403, Legislative Council - 100, Lok sabha - 80, Rajya sabha - 31, 1st Governor - Sarojini Naidu, 1st Chief Minister - Govind Ballabh Pant. State Bird - Sarus crane, Animal - Swamp Deer, Flower - Palash and Tree - Ashoka.

West Bengal (Kolkata) - Bengal was divided on October 16, 1905, by Lord Curzon. It divided the Bengal province into two separate entities, Bengal and Eastern Bengal. The Western Bengal, Bihar and Orissa were retained under Bengal while the remaining portion of Bengal and Assam formed the Eastern Bengal. Assembly seat - 294, Lok sabha - 42, Rajya sabha - 16, 1st Governor - Chakravarthi Rajagopalachari, 1st Chief Minister - Prafulla Chandra Ghosh. State Bird - White throated kingfisher, Animal - Fishing cat, Flower - Sefali and Tree - Devil's Tree.

Union Territories

Andaman and Nicobar Islands (Port Blair)- Year of establishment - 1956. They are a group of islands at the southeastern edge of the Bay of Bengal,

consisting of 572 islands. State Bird - Andaman wood pigeon, Flower - Pyinma, Animal - Dugong and Tree - Andaman Padauk.

Chandigarh was made a Union Territory in 1966, It is the capital of both Punjab and Haryana. Lok sabha - 1, State Bird - Indian grey hornbill, Flower - Dhak, Animal - Indian grey mongoose and Tree - Mangifera indica.

Dadra and Nagar Haveli and Daman and Diu (Daman)- Year of establishment - 2020. It was constituted through the merger of the former territories of Dadra and Nagar Haveli and Daman and Diu. Lok Sabha - 2.

Jammu and Kashmir (Srinagar)- Year of establishment - 2019. State Bird - Black necked Crane, Flower - Lotus, Animal - Hangul and Tree - Chinar.

Ladakh (Leh)- Year of establishment - 2019. State Bird - Black-necked crane and Animal - Snow leopard.

Lakshadweep (Kavaratti)- Year of establishment - 1956. India's smallest Union Territory Lakshadweep is an archipelago consisting of 36 islands but 10 inhabited. State Bird - Sooty Tern, Flower - Neelakurinji, Animal - Butterfly Fish and Tree - Breadfruit.

Delhi was made a Union Territory in 1956. Assembly seat - 70, Lok sabha - 7, Rajya sabha - 3, State Bird - House sparrow, Flower - Alfalfa, Animal - Nilgai and Tree - Gulmohar.

Puducherry- The Union Territory of Puducherry comprises the former French establishments of Puducherry (Tamil Nadu), Karaikal (Tamil Nadu), Mahe (Kerala) and Yanam (Andhra Pradesh), which lie scattered in South India. State Bird - Asian Koel, Flower - Nagalingam, Animal - Squirrel and Tree - Bael Fruit.

Famous Cities and their Nicknames

Ahmedabad - (Manchester of India, Commercial Capital of Gujarat, Financial Capital of Western India, Boston of India, Heritage City),

Anand - Gujarat (Milk City),

Asansol - West Bengal (City of Brotherhood, Land of Black Diamond),

Bengaluru - Karnataka (Silicon Valley of India, Science city of India, Garden City of India),

Chennai - Tamil Nadu (Detroit of Asia, Healthcare Capital of India, Gate Way of South India, City of Flyovers, Electronic Manufacturing Hub in India, Banking Capital of India),

Coimbatore - Tamil Nadu (Manchester of South India).

Coorg (Kodagu), Karnataka (Scotland of India),

Dhanbad - Jharkhand (Coal Capital of India)

Dibrugarh - Assam (Tea City of India),

Erode - Tamil Nadu (Turmeric City of India),

Gaya, Bihar (The City of Enlightenment),

Guntur, Andhra Pradesh (Chilli Capital of India),

Jamshedpur, Jharkhand (Steel City of India, Pittsburg of India),

Jodhpur, Rajasthan (Blue City, Sun City),

Jaisalmer (Golden City).

Kanpur, Uttar Pradesh (Leather City),

Kolkata, West Bengal (City of Joy, Cultural Capital of India, Heaven of the Aged, City of Buildings, City of Palaces, City of Castles).

Kollam, Kerala (Cashew Capital of the World),

Kurnool, Andhra Pradesh (The Gateway of Rayalaseema),

Kuttanad, Alappuzha district, Kerala (Rice Bowl of Kerala).

Lucknow, Uttar Pradesh (City of Nawabs),

Madurai, Tamil Nadu (City of Festivals, City of Jasmine, Athens of the East, Capital of Pandya Nadu),

Mangaluru, Karnataka (Ice cream capital of India, Gateway of Karnataka, Rome of The East),

Mumbai, Maharashtra (City of Seven Islands, Gateway of India, Hollywood of India),

Mussoorie, Uttarakhand (Queen of the Mountains),

Mysuru, Karnataka (Sandalwood City, Cultural Capital of Karnataka)

Nagpur, Maharashtra (Orange City),

Nasik, Maharashtra (Wine Capital of India),

Pune, Maharashtra (Oxford of the East, Queen of Deccan),

Rishikesh, Uttarakhand (Adventure Capital of India, Rafting Capital of India, Yoga Capital of the World),

Surat, Gujarat (City of Diamonds, Silk City of India),

Tenali, Andhra Pradesh (Andhra Paris),

New names of Cities of India

Peshawar (Purushapura), Patna (Pataliputra), Hyderabad (Bhagyanagar) Guwahati (Pragjyotishpura), Bidar (Muhammadabad), Warangal (Orugallu), Prayagraj (Allahabad), Kolkata (Calcutta), Kozhikode (Calicut), Chennai (Madras), Mumbai (Bombay), Vadodara (Baroda), Kochi (Cochin), Varanasi (Benares), Thoothukudi (Tuticorin), Kanya Kumari (Cape Comorin), Kalaburagi (Gulbarga), Belagavi (Belgaum), Bengaluru (Bangalore), Mysuru (Mysore), Tumakuru (Tumkur), Ujjain (Avantika).

Famous cities on rivers

Surat (Tapi), Vijayawada (Krishna), Badrinath (Alaknanda), Jabalpur (Narmada), Durgapur (Damodar), Gwalior (Chambal), Kota (Chambal), Dholpur (Chambal), Jhansi (Betwa), Jamshedpur (Subarnarekha), Nasik (Godavari), Ujjain (Shipra), Ahmedabad (Sabarmati), Kolkata (Hugli), Aurangabad (Kauna), Agra (Yamuna), Delhi (Yamuna), Allahabad (Sangam of Ganga, Yamuna, Saraswati),

Haridwar (Ganga), Kanpur (Ganga), Patna (Ganga), Srirangapatna (Kaveri), Tiruchirapalli (Kaveri), Lucknow (Gomti), Jaunpur (Gomti), Dibrugarh (Brahmaputra), Guwahati (Brahmaputra), Cuttack (Mahanadi), Sambalpur (Mahanadi), Srinagar (Jhelum), Madurai (Vaigai), Panji (Mandovi), Pune (Mutha and Mula) Ayodhya (Saryu (Ghagra)), Hyderabad (Musi), Quepem and Curchorem (Zuari).

Famous Indian Handicrafts

Andhra Pradesh (Kondapalli Toys, Kalamkari Art), Arunachal Pradesh (Thangka Painting), Assam (Bamboo and Cane Handicraft, Water Hyacinth Handicrafts, Xorai and Bota), Bihar (Madhubani Painting, Sikki Grass Craft), Chhattisgarh (Bell Metal Craft or Dhokra Craft, Terracotta craft), Goa (Seashell craft, Azulejos or Hand painted tiles), Gujarat (Bead Craft, Khavda Pottery), Haryana (Sarkanda Craft),

Himachal Pradesh (Wood Carving, Chamba Rimal), Jharkhand (Jadupatia Painting, Sohrai Painting), Karnataka (Sandalwood craft, Mysore Painting), Kerala (Kathakali Masks, Coir Products), Madhya Pradesh (Leather Crafts,

Chanderi Craft), Maharashtra (Warli Painting, Kolhapuri Chappals), Manipur (Kaun Grass Craft, Stone Carving), Meghalaya (Bamboo and Cane Craft, Weaving),

Nagaland (Cane and bamboo work, wood carving, blacksmithing and pottery), Odisha (Silver Filigree of Cuttack, Pattachitra), Punjab (Phulkari Art, Paranda), Rajasthan (Blue Pottery- Jaipur is famous for blue pottery), Sikkim (Choktse Table, Carpet Weaving), Tamil Nadu (Toda Embroidery, Tanjore Painting), Telangana (Bidri Craft, Banjara, Needlecraft), Tripura (Cane and Bamboo Handicrafts, Handlooms), Uttar Pradesh (Chikankari, Banarasi Saree),

Uttarakhand {(Woolen handicrafts– Kaangsuk (socks), gloves, shawls, Dumkar (blanket), Paagad (belt) and Saai (cap))}, West Bengal (Madur, Shola craft), Jammu and Kashmir (Kashmir shawl, Wagoo (Kashmiri mat).

Breeds of Buffaloes

Bhadawari (Uttar Pradesh and Madhya Pradesh), Jaffarabadi, Mehsana, Surti, Banni (Gujrat), Marathwadi, Nagpuri, Pandharpuri (Maharashtra), Murrah (Haryana), Nili Ravi (Punjab), Toda, Bargur (Tamil Nadu), Chilika, Manda (Odisha), Gojri (Punjab and Himachal Pradesh).

World G.K.

Major Countries in world

Australia - Capital - Canberra, Currency - Australian dollar. James Cook discovered Australia in 1770. The Ayers Rock of Australia is the world's largest rock. The Great Barrier Reef is the biggest reef system in the world. It is a continent and a country (the only country in the world that covers a whole continent). Australia is the only place in the world where Kangaroos exist in the wild.

Argentina - Capital - Buenos Aires, Currency - Peso, Official language - Spanish. Argentina was the first country in South America to legalize same-sex marriage, in 2010. Argentina was the first country to use fingerprinting as a method of identification.

Bolivia - Capital - La Paz (executive capital) while the constitutional capital is Sucre. Currency - Bolivian Boliviano (BOB).

Belarus - Capital - Minsk, Currency - Belarusian ruble. The term Belarus means 'White Russia.' The city of Minsk has also been given the title of a "hero city".

Brazil - Capital - Brasília, Official language - Portuguese, Currency - The Brazilian real, National Animal - Jaguar, National Bird - Rufous-bellied thrush. Brazil is the largest country in South America and the fifth largest nation in the world. The Amazon river flows through Brazil and is the 2nd longest river in the world (the Nile is the longest). Voting in Brazil is compulsory.

Canada - Capital - Ottawa, Currency - The Canadian dollar, National animal - Beaver, Official languages - English and French. Maple Leaf is the Symbol of Canada. Canada has the longest coastline in the world. Lacrosse (summer) and Ice Hockey (winter) are the national sports of Canada.

Chile - Capital - Santiago, Currency - Chilean peso. Chile is the world's narrowest country. The national dance of Chile is Cueca.

Cambodia - Capital - Phnom Penh, Currency - Cambodian Riel, Official language - Khmer. Buddhism is the state religion of Cambodia. Angkor Wat temple, the world's largest religious building, is in Cambodia.

Denmark - Copenhagen, Currency - Danish Krone, Official language - Danish. The Oresund Bridge connects Denmark and Sweden. Handball was invented in Denmark in 1897. The Parliament of Denmark is called 'Folketinget'. The swan is Denmark's national bird.

Egypt - Capital - Cairo, Official language - Arabic, Currency - Egyptian Pound, National sport - Soccer, National language - Egyptian Arabic, National animal - Steppe Eagle.

Ethiopia - Capital - Addis Ababa, Currency - Ethiopian Birr, Official language - Amharic, National animal - lion. Africa's first Olympic gold medal was won by an Ethiopian.

France - Capital - Paris, Official language - French, Currency - Euro, National Day - 14 July. France is the largest country in the EU and is called the hexagon. French gastronomy was awarded UNESCO World Heritage Status in 2010. France, including its overseas territories, has the most time zones with 12.

Finland - Capital - Helsinki, Currency - Euro. Finland was the first country in Europe to give all women the right to vote. National animal - Brown Bear. Finland is known as the "Land of Thousand Lakes". Angry Birds originated in Finland by Rovio in 2009.

Germany - Capital - Berlin, Official language - German, Currency - Euro. Germany is called the "Land of Poets and Thinkers." Germany was the first country in the world to adopt Daylight saving time - DST, also known as summer time. The city of Dresden is known as the Jewelry Box.

Iraq - Capital - Baghdad, Official language - Arabic and Kurdish, Currency - Iraqi Dinar.

Iceland - Capital - Reykjavík, Currency - Icelandic Krona. Official language - (Icelandic). Iceland is also known as "The Land of Fire and Ice". Iceland is the only NATO member country without an army, navy or air force.

Indonesia - Capital - Jakarta, Official language - Indonesian, Currency - Indonesian Rupiah. Rafflesia arnoldii, or corpse flower, the world's largest flower, is found in Indonesian rainforests, especially on Sumatra Island. Indonesia is the hottest spot on the Pacific Ring of Fire.

Italy - Capital - Rome, Currency - Euro, Official language - Italian. The longest river in Italy is the Po river. Christopher Columbus (Christopher Columbus is credited with discovering the Americas in 1492) was Italian.

Japan - Capital - Tokyo, Official language - Japanese, Currency - Yen, National sport - Sumo Wrestling. Japan is the 11th most populous country in the world. There are four main islands in Japan - Hokkaido, Honshu (Largest of Japan), Shikoku and Kyushu.

Kazakhstan - Capital - Astana, Official language - Russian and Kazakh, Currency - Tenge, National animal - Steppe eagle. Kazakhstan is the largest landlocked country in the world. It is also the largest uranium producing country in the world.

Kenya - Capital - Nairobi, Currency - Kenyan Shilling, National Animal - East African Lion. Lake Turkana is Kenya's largest lake and also the world's largest desert lake, is known as the largest alkaline lake in the world. National Day -

Jamhuri Day 12 December (Independence Day). The first woman (Baroness Bertha von Suttner) to win a Nobel Peace Prize was from Kenya.

Kyrgyzstan - Capital - Bishkek, Official language - Russian and Kyrgyz, Currency - Som. Independence day - 31 August 1991 {from Russia (USSR)}.

Laos - Capital - Vientiane, Currency - Lao kip, Official language - Lao. Laos is the only landlocked country in Southeast Asia. Dok Champa is Laos' national flower.

Libya - Capital - Tripoli, Currency - Libyan Dinar, Official language - Arabic. Official religion - Islam. Ghadames is known as the "Pearl of the Desert".

Madagascar - Capital - Antananarivo, Currency - Ariary. Madagascar has two official languages (Malagasy and French). Madagascar is also known as the Red Island. Madagascar is located in the Indian Ocean. Wild lemurs are only found in Madagascar.

Malaysia - Capital - Kuala Lumpur, Currency - The Malaysian Ringgit, National animal - Malayan tiger, National bird - Rhinoceros Hornbill (buceros hornbill), National sport - Sepak Takraw. The highest point in Malaysia is Mount Kinabalu.

Mongolia - Capital - Ulaanbaatar, Currency - Mongolian Tugrik, Official language - Mongolian. Mongolia is a landlocked country in East Asia. "Naadam" is an annual traditional festival in Mongolia which is celebrated between 11th and 13th July.

Mexico - Capital - Mexico City. Spoken Language - Spanish. Currency - Mexican Peso, National animal - Golden Eagle, National Flower - Dahlia. National sport - Charrería. The National Dish of Mexico is Mole Poblano. Mexico City has the largest taxi fleet in the world.

Morocco - Capital - Rabat, Currency - Moroccan Dirham, National sport - Soccer. National Animal - Barbary Lion. National bird - Moussier's redstart. Mint tea is the national drink of Morocco.

New Zealand - Capital - Wellington, Official languages - English, Maori and New Zealand Sign Language. National bird - Kiwi.

Netherlands - Capital - Amsterdam, Currency - Euro, Official language - Dutch.

The Netherlands (Hague) is the Legal Capital of the World. The Netherlands is the world's biggest flower exporter.

Norway - Capital - Oslo, Currency - Norwegian Krone, Norway has two official languages- Norwegian and Sami. Norwegian is called the "Land of the Midnight Sun". The Nobel Peace Prize is awarded in Oslo every year. The country is popular for fjords. It is one of the three Scandinavian countries.

Oman - Capital - Muscat, Official language - Arabic, Currency - Omani Rial, National animal - Arabian Oryx. The National Symbol of Oman is Khanjar. The National Beverage of Oman is Coffee.

Philippines - Capital - Manila, Currency - Peso, National animal - Carabao, National bird - Philippine Eagle. The two official languages of the Philippines are Filipino and English. Halo-halo is a popular dessert in the Philippines.

Peru - Capital - Lima, Currency - Nuevo sol, National animal - Vicuna, National bird - Andean cock-of-the-rock. There are three official languages in Peru - Quechua, Aymara, and Spanish. Lake Titicaca, the world's highest navigable lake located in Peru. National dance - Coquettish marinara. The Largest Flying Bird on Earth can be found in Peru, The Giant Andean Condor. The source of the Amazon river (longest river) lies in Peru.

Portugal - Capital - Lisbon, Currency - Euro, Official language - Portuguese. The Iberian wolf is the official national animal of Portugal. Vasco D Gama was a renowned Portuguese explorer. Portuguese is the official language of 9 countries - Portugal, Brazil, Cape Verde, Angola, Guinea Bissau, Mozambique, Principe, Sao Tome, and Equatorial Guinea.

Poland - Capital - Warsaw, Currency - Zloty, Official language - Polish. The most famous rivers in Poland are the Oder and the Vistula.

Qatar - Capital - Doha, Official language - Arabic, Currency - Qatari Rial, National sport - Football. National animal - Arabian Oryx.

Romania - Capital - Bucharest, Currency - Romanian leu, Official language - Romanian. National animal - Lynx. National bird - Golden eagle.

Russia - Capital - Moscow, Official language - Russian, Currency - Russian Ruble, National sport - Bandy (Ice Hockey), National animal - Eurasian brown bear, National bird - Eagle. Major Mountain ranges - Ural, Altay. Major Rivers - Amur, Irtysh, Lena, Ob, Volga, Yenise.

As the world's largest country, Russia occupies one-tenth of all the land on Earth. It spans 11 time zones across two continents (Europe and Asia).

Saudi Arabia - Capital - Riyadh, Official language - Arabic, Currency - Riyal. Saudi Arabia is the largest country in the world without a river. Petroleum (crude oil) was discovered in 1938 in Saudi Arabia. 'Kabsa' is an important cultural dish. Saudi Arabia is the birthplace of Islam. Saudi women wear specific clothes - abayas. Saudi Arabia is sometimes called the "country of two mosques" (Makkah the Blessed and Madinah the Radiant).

South Korea - Capital - Seoul, Official language - Korean, Currency - South Korean won, National sport - Taekwondo. The Korean alphabet is called 'Hangul'. South Korea is known as The Plastic Surgery Capital Of The World. The national anthem of the Republic of Korea is "Aegukga," also known as "The Patriotic Song."

Sweden - Capital - Stockholm, Currency - Swedish Krona. The multinational group IKEA hails from Sweden. The main official language of Sweden is Swedish (Svenska), but the country also has five other official languages - Finnish, Yiddish, Sami, Meankieli and Romani. It is one of the three Scandinavian countries. Sweden is the biggest Scandinavian nation.

Switzerland - Capital - Bern, Official languages - German, French, Italian, and Romansh. Currency - Swiss Franc, National sport - Skiing.

South Africa - Capitals - Pretoria (administrative), Cape Town (legislative), Bloemfontein (judicial). South Africa's Constitution recognises 11 official languages. Currency - Rand.

Spain - Capital - Madrid, Currency - Euro, National animal - Bull. Spanish is the 2nd most widely spoken language in the world. According to The Guinness Book of World Records, the world's oldest restaurant is in Spain.

Sudan - Capital - Khartoum, Currency - Sudanese pound, Official language - Arabic. Sudan was the first Muslim and Arab country to appoint a female judge.

Tanzania - Capital - Dodoma, Currency - Tanzanian shilling. Tanzania is home to the famed Mt. Kilimanjaro, the tallest mountain in Africa, with its summit at 19,340 feet (5,895 metres). The people of Tanzania speak Swahili.

Thailand - Capital - Bangkok, Official language - Thai, Currency - Thai Baht, National animal - Elephant, National sport - Muay Thai boxing (the art of eight limbs). Thailand was originally known as Siam until 1939. Thailand is known as the "Land of Smiles".

Turkey - Capital - Ankara, Official language (Turkish). Currency (Turkish Lira). National sport - (oil wrestling). Turkey is the largest producer of hazelnuts. National Flower (Tulips). Santa Claus was born in Turkey.

Turkmenistan - Capital - Ashgabat, Official language - Turkmen, Currency - Manat. Turkmenistan has 70% desert. The Darvaza Gas Crater or Gates to Hell Is Man-Made is situated in this country.

United States of America (USA) - Capital - Washington D.C., Currency - United States dollar, National sport - Baseball, National animal - Bald Eagle, National mammal - The American Bison. The United States has no official language or religion. The lowest point in the country - Death Valley {-282 feet (-86 meters)} and the highest peak - Denali (Mt. McKinley) at 20,320 feet (6,198 meters). The US is the only country to have all five major climate zones- tropical, dry, temperate, continental, and polar. The Navajo Nation is the largest tribe in the USA. Silicon Valley (California) is the biggest hub of the IT industry in the world.

United Kingdom (UK) - Capital - London, Currency - Pound Sterling, National sport - Cricket, The UK consists of four countries- England, Scotland, Wales and Northern Ireland. Great Britain is made up of England, Wales, and Scotland. The first postage stamp in the world was produced in England.

United Arab Emirates - Capital - Abu Dhabi, Official religion- Islam, Official language - Arabic, Currency - U.A.E. Dirham, National Bird - Falcon, National animal - Arabian Oryx. UAE's seven

emirates – Abu Dhabi, Dubai, Sharjah, Ajman, Umm Al Quwain, Ras Al Khaimah and Fujairah. Ajman is the smallest emirate in the UAE. Abu Dhabi, is the largest emirate.

Uzbekistan - Capital - Tashkent, Official language - Uzbek, Currency - Uzbekistan Som. It's a double land-locked country. The only two double landlocked countries of the world are Uzbekistan and Liechtenstein. Al-Biruni from Uzbekistan, travelled to India in the 11th century.

Ukraine - Capital - Kyiv, Currency - Hryvnia, Official Language - Ukrainian. Vyshyvanka is the national costume. The tallest peak in Ukraine is Mount Hoverla. Ukraine is known as The Breadbasket Of Europe.

Vietnam - Capital - Hanoi, Official language - Vietnamese, Currency - Dong, National flower - Lotus. The Son Doong Cave, located in Vietnam, is the largest cave in the world.

Venezuela - Capital - Caracas, Currency - Bolivar. The cuatro is their national instrument. The Angel Falls, also called Salto Angel, is the world's tallest waterfall with 979 m (3,211 ft) and is located in eastern Venezuela. The world's biggest rodent is found in Venezuela – Capybara. Venezuela has the largest oil reserves.

Yemen - Capital - Sanaa, Official language - Arabic, Currency - Rial. Yemen is the only state in the Arabian Peninsula with a republican government. The national symbol of Yemen is the golden eagle.

Important Boundary Lines of the World

1. Radcliffe Line (Between India-Pakistan-Bangladesh)
2. Hindenburg Line (Germany-Poland)
3. Durand Line (Pakistan-Afghanistan)
4. McMahon Line (India-China)
5. Mannerheim Line (Russia-Finland)
6. Medicine Line (USA - Canada)
7. Oder-Neisse Line (Germany-Poland)
8. 17th Parallel North (North Vietnam - South Vietnam)
9. 20th Parallel North (Sudan-Libya)
10. 31st Parallel (Iraq-Iran)
11. 38th Parallel (North Korea-South Korea)
12. 42nd Parallel (New York - Pennsylvania)

Continents

There are seven continents in the world. Their descending order on the basis of area - Asia, Africa, North America, South America, Antarctica, Europe, and Australia (oceania).

Asia

Asia is the largest continent and the most populated continent. It covers about one third of the total land area of the earth. It lies in the Eastern Hemisphere. Asia is separated from Europe by the Ural mountains on the west. The combined landmass of Europe and Asia is called Eurasia. It is called the 'continent of contrasts'.

Some major countries in Asia - India, China, Japan, Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam.

Important Facts -

Highest mountain peak - Mount Everest (Nepal)
 Longest river system - Yangtze (China)
 Largest country (area-wise) - China
 Largest desert - Gobi Desert (China)
 Wettest place - Mawsynram, Meghalaya (India)
 Longest railway network - Trans-Siberian railway (Russia)
 Largest city (population-wise) - Tokyo
 Biggest lake - Caspian Sea
 Smallest country - Maldives
 Busiest waterways - Suez Canal
 Highest plateau - Tibet

Africa

It is the second largest and second most populous continent in the world. It is called "the dark continent". "The Great Wildebeest Migration" in Africa is the largest wildlife migration on Earth. The only continent through which the Tropic of Cancer, the Equator and the Tropic of Capricorn pass

Africa has the highest number of countries among continents. Some of these Countries- Algeria, Botswana, Burkina Faso, Djibouti, Egypt, Ethiopia, Gabon, Ghana, Kenya, Libya, South Africa, South Sudan, Sudan, Tanzania, Togo.

Important Facts -

Highest mountain - Mount Kilimanjaro
 River that crosses the Equator twice - River Congo
 Largest lake - Lake Victoria

Largest desert - The Sahara Desert
 Longest river - River Nile (Longest in world also)
 Largest dam -Grand Renaissance Dam
 Largest island - Madagascar
 Lake, which has dried and turned into a desert - Lake Faguibine.

North America

It is a continent entirely within the Northern Hemisphere and almost all within the Western Hemisphere. It can also be considered a northern subcontinent of the Americas. North America is the third largest continent after Asia and Africa.

There are 23 countries in North America- Some of these Countries - USA, El Salvador, Bahamas, Canada, Cuba, Jamaica, Haiti, Mexico, Barbados, Puerto Rico.

Important Facts -

Highest mountain - Mount McKinley (Denali)
 Longest river - River Missouri
 Largest desert - Chihuahuan Desert
 Largest canyon - Grand Canyon (Colorado River)
 Great Lakes of North America - Lake Superior, Lake Huron, Lake Ontario, Lake Michigan, Lake Erie
 River which forms a boundary between the USA and Canada - St. Lawrence River.

South America

It is a long triangular shaped continent. South America lies to the west of the Prime Meridian. Fjords are deep inlets of the sea into mountainous land. Continent that receives the largest amount of rainfall - South America. Continent with the largest river system (basin area) - South America (Amazon basin).

South America consists of 12 countries - Chile, Argentina, Uruguay, Paraguay, Bolivia, Peru, Ecuador, Brazil, Colombia, Venezuela, Guyana, and Suriname—plus an overseas department of France called French Guiana.

Important Facts -

Highest mountain - Mount Aconcagua
 Longest river - River Amazon
 Highest waterfall - Angel Fall (On Caroni river)
 Highest navigable lake - Lake Titicaca
 Only pass in the Andes mountain ranges - Uspallata pass
 Most mineral-rich province - Minas Gerais (Brazil).

Europe

Europe is considered the most developed continent in the world. It is bounded by the Arctic Ocean in the north, the Mediterranean Sea in the south, the Atlantic Ocean in the west and the Ural Mountains in the east. Europe is the first continent to be industrialised. Europe has the world's smallest country, the Vatican City. Europe is separated from Asia by the Caspian Sea and Ural mountains.

Norway, Sweden, Iceland and Denmark are together called 'Scandinavia'. Yugoslavia, Greece, Romania and Albania, situated on the coast of Black Sea and Mediterranean Sea are called 'Balkan states'. Lithuania, Latvia and Estonia are together called 'Baltic states'

Some European Countries - France, the United Kingdom, Finland, Portugal, Russia, Austria, Belgium, Denmark, Czech Republic, Finland, Germany, Croatia, Belarus, Bulgaria, Greece, Hungary, Iceland, etc.

Important Facts -

Highest Peak - Mt. Elbrus
 Longest River - River Volga
 Port of five seas - Moscow
 Textile hub - Manchester (England)
 Largest country (area wise) - Russia
 Important cities located along the Danube River - Vienna, Budapest, Belgrade, Bucharest.

Australia/Oceania

It is the smallest continent that lies entirely in the Southern Hemisphere. It is called an island continent. River Murray is the longest river in Australia. There are two types of grasslands found in the continent-tropical and temperate. These are called 'Savana And 'Downs Respectively.

Important Facts -

Highest mountain - Mount Kosciuszko
 Largest desert - Great Victoria Desert
 Longest river - River Murray
 Largest lake - Lake Eyre
 Man who discovered Australia - Captain James Cook.

Antarctica

It is completely in the Southern Hemisphere, and is a huge continent. It is permanently covered with thick ice sheets. There are no permanent human settlements.

All countries of the world under the United Nations have agreed to use Antarctica only for research and scientific purposes and not for resource exploitation.

It is divided into two parts by Trans-Antarctic ranges. Mount Vinson is the highest point in Antarctica. Prominent active volcanoes of Australia include Mt. Erebus. Antarctica is also famous for the phenomenon of shining light in the night sky called Aurora Australis.

Landlocked country

A landlocked country is a nation with borders that do not touch an ocean or any waterway that directly connects to an ocean. There are approximately 44 landlocked countries in the world.

Kazakhstan is the largest landlocked country while Vatican city is the smallest landlocked country.

Landlocked countries in Asia -

Afghanistan, Nepal, Bhutan, Armenia, Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Uzbekistan, Laos, Mongolia.

Landlocked countries in Africa -

Chad, Ethiopia, Malawi, Mali, Niger, Rwanda, South Sudan, Zambia, Zimbabwe, Botswana, Burkina Faso, Burundi, Central African Republic, Swaziland, Lesotho.

Landlocked countries in Europe -

Austria, Belarus, Hungary, Kosovo, Luxembourg, Macedonia, Moldova, Serbia, Slovakia, Switzerland, San Marino, Vatican City, Andorra, Liechtenstein,, Czech Republic.

Landlocked countries in South America -

Bolivia and Paraguay.

Organisations

Association of South-East Asian Nations

(ASEAN)- Headquarters - Jakarta, Indonesia. Established - 1967. Members- 10 - Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam.

Asian Infrastructure Investment Bank

(AIIB)- Headquarters- Beijing, China. Established in - 2016.

Asian Development Bank (ADB)-

Established in - 1966. Headquartered - Mandaluyong (Manila), Philippines.

Takeshi Watanabe was its first President. There are 68 member countries of the Asian Development Bank. India is a founding member of the Asian Development Bank.

The South Asian Association for Regional Cooperation (SAARC) - Headquarters - Kathmandu (Nepal). Established - 8 December, 1985. Members - 8 countries - Afghanistan, Bangladesh, Bhutan, India, the Maldives, Nepal, Pakistan and Sri Lanka.

South Asian Free Trade Area (SAFTA) is the free trade arrangement of the SAARC formed in 2004 and came into force in 2006.

Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC)- Headquarters - Dhaka (Bangladesh). Established - 1997. Members- 7 countries - Bangladesh, Bhutan, India, Myanmar, Nepal, Sri Lanka, and Thailand. Myanmar joined BIMSTEC in 1997. Nepal and Bhutan joined in 2004.

Shanghai Cooperation Organisation (SCO)- Headquarters - Beijing, China. Established - 2001. Members- 8 (China, Kazakhstan, Kyrgyzstan, Russia, Tajikistan, and Uzbekistan, India, Pakistan); India and Pakistan joined SCO as full members in 2017.

BRICS- Established - 2009. Members - Brazil, Russia, India, China, South Africa (South Africa joined BRIC in December 2010). After that the group adopted the acronym BRICS.

European Union (EU)- Established - 1993. Members - 27. European Union - Brexit- On January 31, 2020, the United Kingdom (UK) formally left the European Union. The U.K is the first country to leave the E.U. Headquarters - Brussels, Belgium.

United Nations (UN) is an International organisation founded in 1945. It is currently made up of 193 Member States. India joined the United Nations after ratifying the UN Charter on 30 October 1945. The principal organs of the United Nations are- General Assembly (GA), Security Council (SC), Economic and Social Council (ECOSOC), Trusteeship Council, International Court of Justice (ICJ), and UN Secretariat.

New Development Bank (NDB)- Headquarters- Shanghai, China. Established in - 2014. Members - Brazil, Russia, India, China and South Africa.

World Health Organization (WHO)- A specialised agency of the United Nations responsible for international public health. Established on - 7 April 1948. Headquartered in - Geneva, Switzerland.

Organisation for Economic Co-operation and Development (OECD)- Headquarter - Paris (France), Founded - 1961.

World Food Programme (WFP)- Founded in - 1961. Headquarters - Rome, Italy.

International Atomic Energy Agency (IAEA)- Headquarters - Vienna, Austria. Founded - 1957.

Intergovernmental Panel on Climate Change (IPCC)- Founded in - 1988. Headquarters - Geneva, Switzerland.

Climate Action Network-International (CAN) Founded in - 1989 by Michael Oppenheimer. Headquarters - Bonn, Germany.

Australian Youth Climate Coalition (AYCC)- Formed in 2006. Headquarters - Melbourne, Australia.

World Economic Forum (WEF)- Founded in - 1971 by Klaus Schwab. It publishes - Global Competitiveness Report, Global Gender Gap Index, Travel and Tourism Competitiveness Report, Enabling Trade Report, Global Energy Architecture Performance Index Report, Global Environment Performance Index, World Power Language Index, Inclusive Development Index, Human Capital Index.

International Solar Alliance (ISA) Founded - 30 November 2015. Headquarters - Gurugram, Haryana.

International Social Security Association (ISSA)- Founded in 1927, Headquarters - Geneva, Switzerland.

International Court of Justice (ICJ)- Headquarters - The Hague, Netherlands. Established - 1945.

International Monetary Fund (IMF)- Formed in - 1944 Headquarters - Washington, D.C. It publishes the World Economic Outlook, Global Financial Stability Report and Fiscal Monitor. Gita Gopinath is the first Indian-American woman to become Chief Economist at IMF.

The World Bank- Founded - July 1944. Headquarters - Washington, D.C., United States. Reports published by the World Bank are Commodity Markets Outlook,

Doing Business, Global Economic Prospects, Global Financial Development Report, etc.

The International Bank for Reconstruction and Development (IBRD)- Established - 1944. Headquarters - Washington D.C., United States.

The General Agreement on Tariffs and Trade (GATT) was signed by 23 nations at the Palais des Nations in Geneva on 30 October 1947.

World Trade Organisation (WTO) - Established on - January 1, 1995 by replacing GATT. Headquarter - Geneva, Switzerland.

Group of Seven (G7) - Established in- 1975. Members- Canada, USA, UK, France, Germany, Japan and Italy.

Group of 20 (G-20)- Established - 1999. Members - Argentina, Australia, Brazil, Canada, China, France, Germany, India, Indonesia, Italy, Japan, Republic of Korea, Mexico, Russia, Saudi Arabia, South Africa, Türkiye, United Kingdom, United States and European Union. In September 2023 The African Union was made a permanent member of the G-20.

India, Israel, UAE, USA (I2U2) - Established - 2021. The countries share a number of global challenges, including the food security crisis and defence, which will be a focus of the four-nation summit. 'I2U2' initiative is also referred to as the West Asian Quad.

Organization of the Petroleum Exporting Countries (OPEC) is a cartel of 13 countries. It was founded in 1960 in Baghdad by Iran, Iraq, Kuwait, Saudi Arabia and Venezuela. It is headquartered in Vienna, Austria.

North American Free Trade Agreement (NAFTA) was an agreement signed by Canada, Mexico and the United States that created a trilateral trade bloc in North America. The agreement came into force in 1994.

International Maritime Organisation (IMO)- Headquarters - London, United Kingdom, Founded In - 1958.

World Meteorological Organisation (WMO)- Founded - 1950. Headquarters - Geneva, Switzerland.

World Intellectual Property Organization (WIPO)- Headquarter - Geneva, Switzerland. Established - 1967.

International Organization for Standardization (ISO) is a non-governmental organisation. Headquarters - Geneva, Switzerland. Founded - 1947.

The Food and Agriculture Organization (FAO)- Founded - 1945. Headquarters at - Rome, Italy. It has 195 members.

North Atlantic Treaty Organisation (NATO)- Headquarter - Brussels (Belgium). Established- 1949.

International Union for Conservation of Nature (IUCN)- An international organisation founded on 5 October 1948, and headquartered at Gland, Switzerland. It works in the field of natural conservation and sustainable use of natural resources.

Amnesty International- Founded - 1961 by the lawyers Peter Benenson and Philip James. Headquarter - London, United Kingdom. It is an International non-governmental organisation focused on human rights. An Amnesty International office - First set up in Bihar, India in 1966.

International Seabed Authority- Established in - 1994. Headquartered in - Kingston, Jamaica. Government of India signed a 15 year contract with International Seabed Authority (ISA) for exploration of polymetallic nodules from the Central Indian Ocean Basin (CIOB) in 2002.

Quadrilateral Security Dialogue (QUAD)- Established- 2007. Members- USA, Australia, India and China.

The International Financial Services Centres Authority (IFSCA) - Established on 27 April 2020 under the International Financial Services Centres Authority Act, 2019. Headquarters in GIFT City, Gujarat.

The National Association of Software and Service Companies (NASSCOM) is the apex body for the Indian technology industry. Established in - 1988. Headquarters - Noida, Uttar Pradesh.

Defence Research and Development Organisation (DRDO) works under the Ministry of Defence of the Government of India, charged with the military's research and development. The Post of Secretary DD and R & D is in the Level I (top level) in DRDO. Headquarters - New Delhi. Established - 1958.

Bombay Textile Labour Union (BTLU)

was the first trade union registered under India Trade Union Act of 1926. It was formed by N M Joshi in 1926 and was one of the first city wide textile worker's unions.

Marine Products Export Development Authority (MPEDA)- Established in - 1972. It is a statutory body, Which is responsible for development of the marine products industry with special reference to exports. Its headquarters is at Kochi, Kerala.

The Indian National Trust for Art and Cultural Heritage (INTACH) is a non-profit charitable organisation registered under the Societies Registration Act, 1860. INTACH was founded in 1984 in New Delhi.

Financial Stability and Development Council (FSDC) is a non-statutory apex council under the Ministry of Finance constituted in 2010. Raghuram Rajan committee (2008) proposed its creation. It is chaired by the Finance Minister and its sub-committee is headed by the Governor of RBI.

National Bank for Agriculture and Rural Development (NABARD) was established on the recommendations of B. Sivaraman Committee in 1982. Headquartered at- Mumbai, Maharashtra. Functions of NABARD include planning, organising and controlling agricultural activities.

Organisation for the Prohibition of Chemical Weapons- Headquarters - The Hague, The Netherlands.

International Council on Monuments and Sites (ICOMOS)- Founded - 1965, Headquarter - Charenton-le-Pont, France. **United Nations Educational, Scientific and Cultural Organization (UNESCO)**- Formation - 16 November 1945 Headquarters- Paris, France.

The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) entered into force on 1 July 1975. India became a signatory to the CITES in 1976.

Global Innovation Index was started in 2007 by INSEAD and World Business.

Bank of Bhutan, is the oldest bank in Bhutan, was established in 1968.

Food Safety and Standards Authority of India (FSSAI) was established in 2008 by the former Union Minister (Dr. Anbumani

Ramadoss) under the Food Safety and Standards Act, 2007. It is headquartered at New Delhi.

National Green Tribunal (NGT) was formed as a statutory body on 18 October, 2010 under the National Green Tribunal (NGT) Act, 2010 to deal with environmental cases and speedy implementation of decisions relating to it.

Madras Labour Union is the first organised Trade Union in India. It was formed in 1918 by B.P Wadia with V. Kalyanasundaram Muladiar.

National Culture Fund (NCF) was set up as a Trust on 28th November, 1996 by Government of India under the Charitable Endowment Act, 1890.

Indira Gandhi Institute of Development Research (IGIDR) was inaugurated in 1987. It is an advanced research and educational institution located in Mumbai.

National Institution for Transforming India (NITI Aayog) was formed on 1 January 2015. It is a think-tank of the government of India. The Prime Minister is the Chairperson and vice chairman is nominated by the Prime Minister.

Employees Provident Fund Organization (EPFO) administers the mandatory provident fund, a basic pension scheme and a disability/death insurance scheme. It was formed on 4 March 1952.

Central Statistical Organization (CSO) working under Ministry of Statistics and Programme Implementation is responsible for preparation of National Income estimates in India. Established - 2 May 1951. Headquarters - New Delhi.

National Sample Survey Office (NSSO)- Established in 1950 under the Ministry of Statistics, Planning and Programme Implementation. It conducts periodic socio-economic surveys. It was merged with the Central Statistical Office (CSO) to form the National Statistical Office (NSO) in 2019. The National Sample Survey organisation collects data related to Unemployment in India.

Food Corporation of India (FCI)- Founded on 14 January 1965 which is headquartered in New Delhi. It is a statutory body setup under the Food Corporation Act 1964.

Geological Survey of India (GSI) - Established - 1851. Headquarters - Kolkata, West Bengal.

National Statistical Commission (NSC) was set up in 2005 on the recommendation of the Rangarajan Commission.

Cotton Corporation of India (CCI) was established in 1970. It is headquartered in Mumbai.

Industrial Finance Corporation of India (IFCI) is a development finance institution under the ownership of the Ministry of Finance, Government of India. It was established in 1948 in New Delhi.

Industrial Development Bank of India (IDBI)- A development finance institution. Established in 1964 and headquartered at Mumbai.

National Committee on Environmental Planning and Coordination (NCEPC) formed in 1972. It gradually evolved and became a full-fledged Ministry of Environment and Forests in 1985.

News Broadcasters Association (NBA)- A private association of different current affairs and news television broadcasters in India, has decided to change its name to News Broadcasters and Digital Association (NBDA). Established on 3 July 2007.

Centre for Cultural Resources and Training (CCRT)- Set up in 1979 as an autonomous organisation by the Government of India.

Border Roads Organisation (BRO)- Formed on 7 May 1960 and headquartered in New Delhi. It develops and maintains road networks in India's border areas and friendly neighbouring countries.

The Board for Industrial and Financial Reconstruction (BIFR)- Set up in 1987. It was a development finance institution under the ownership of the Ministry of Finance and it dissolved on 1 December 2016.

India Brand Equity Foundation (IBEF) is a trust established by the Ministry of Commerce and Industry in 2003.

Telecom Regulatory Authority of India (TRAI) is a regulatory body set up by the Government of India on 20 February 1997. It is the regulator of the telecommunications sector in India.

The Associated chambers of commerce and industry of India (ASSOCHAM) is a 100 years old trade association of India which was established in 1920.

Indian Association for the Cultivation of Science (IACS) was founded by Mahendralal Sircar in Kolkata, West Bengal. It was founded on 29 July 1876.

Federation of Indian Chambers of Commerce and Industry (FICCI) is a non-governmental trade association and advocacy group based in India. It was founded in 1927 and headquartered in New Delhi. It was established on the advice of Mahatma Gandhi by Indian businessman G.D. Birla and Purshottamdas Thakurdas.

Press Council of India (PCI) is a regulatory body established in 1966 by the parliament under the Press Council of India Act of 1978.

Credit Rating Information Services of India Limited (CRISIL) was founded in 1987 and headquartered at Mumbai.

Credit Analysis and Research (CARE) was incorporated as Credit Analysis and Research Limited in 1993 at Mumbai, Maharashtra.

National Disaster Management Authority (NDMA) is the apex body for Disaster Management in India. The Prime Minister of India is the ex-officio chairperson of the NDMA. Established through the Disaster Management Act enacted by the Government of India on 23 December 2005.

Securities and Exchange Board of India (SEBI)- Established - 1988. It was given statutory powers in 1992 through the SEBI Act, 1992. Headquarter - Mumbai, Maharashtra.

HelpAge India- An Indian organisation focused on the concerns of elders and supports geriatric initiatives. It was founded by Cecil Jackson-Cole in 1978.

Wildlife Crime Control Bureau (WCCB) is a statutory multi-disciplinary body under the Ministry of Environment, Forest and Climate Change to combat organised wildlife crime in the country. Its headquarters is in New Delhi and was established in 2006.

Sports Authority of India (SAI) is the apex national sports body of India, was set up in 1984. Headquarters - New Delhi.

The National Sports Development Fund (NSDF) was established in November, 1998. The Indian Olympic Association (IOA) was established in 1927 by Sir Dorabji Tata.

National Anti-Doping Agency (NADA) is the national organisation founded in 2005 which works toward the vision of dope free sports in India. Headquarters - New Delhi.

National Dope Testing Laboratory located in New Delhi. It functions under the Ministry of Youth Affairs and Sports.

World Anti-Doping Agency (WADA) is a foundation initiated by the International Olympic Committee in 1999 to promote, coordinate, and monitor the fight against drugs in sports. Headquarters - Montreal, Canada.

National Rifle Association of India (NRAI) was founded in 1951. The first speaker of Lok Sabha, G.V. Mavlankar was the founder and the first president of NRAI.

The Serum Institute of India launched the oral polio vaccine in 2013. It was founded in 1966 by Dr. Cyrus Poonawalla with the aim of manufacturing life-saving immuno-biologicals, which were in shortage in the country and imported at high prices.

National Stock Exchange (NSE) of India was incorporated in 1992. It started operations in April 1993. Headquarters - Mumbai, Maharashtra.

Employees State Insurance Corporation (ESIC)- Established in - 1952. Headquarters - New Delhi.

Pension Fund Regulatory and Development Authority (PFRDA)- Founded in 2003 and headquartered at New Delhi.

World Customs Organisation (WCO)- Founded - 26 January 1952. Headquarters - Brussels, Belgium.

Insurance Regulatory and Development Authority of India (IRDAI)- Established - 1999. Headquarters - Hyderabad, Telangana.

ICRA Limited (formerly Investment Information and Credit Rating Agency of India Limited). Established - 1991.

Indian council of Historical Research (ICHR) was established on 27th March 1972.

Indian Academy of Sciences, founded in 1934 by Sir C V Raman. Location - Bangalore.

Indian National Science Academy established in 1935. Location - New Delhi.

Sangeet Natak Academy was established on 31 May 1952 by a resolution of the Ministry of Human Resource development. The Akademi was officially inaugurated on 28 January 1953 by the first President of India Rajendra Prasad and PV Rajmanner was appointed as its first Chairman.

Forest Research Institute (FRI)- Established - 1878. It was established as "Imperial Forest Research Institute" in 1906. Location - Dehradun.

Spices Board of India established in 1987. Headquarters - Kochi, Kerala.

Hindustan Aeronautics Limited (HAL)- Founded - 23 December 1940. Headquarters - Bengaluru, Karnataka.

Popular Front of India (PFI) is an Islamic political organisation founded in 2006. It was banned in 2022 for five years.

National Institute of Oceanography CSIR-NIO established on 1 January 1966. Headquarters - Dona Paula, Goa.

Indian Institute of Remote Sensing (IIRS) established in 1966. Headquarters - Dehradun, Uttarakhand.

Council of Scientific and Industrial Research (CSIR) was established on 26 September 1942. It comes under the Ministry of Science and Technology. First woman director general - Nallathambi Kalaiselvi.

Chennai Petroleum Corporation Limited (CPCL), formerly known as Madras Refineries Limited (MRL) formed in 1965. Headquarters- Chennai.

Indian Institute Of Skiing and Mountaineering (IISM) is located in Gulmarg, Jammu and Kashmir.

Lalit Kala Akademi (National Academy of Art) was inaugurated on 5th August 1954. Headquarters - New Delhi.

National Archives of India was established on 11 March 1891 at Calcutta (Kolkata).

Tagore Centre for the Study of Culture and Civilization was established in 2013 in Shimla, Himachal Pradesh.

The Panchsheel Treaty or Five Principles of Peaceful Coexistence was signed between India and China in 1954.

Indian Citizens' Association (ICA) of Nepal was formed on 14 September 1990. Branches at Pokhara, Damak and Bhairahawa.

Agriculture Organisation and Headquarters

Indian Agricultural Research Institute (IARI), commonly known as the Pusa Institute. Formation - 1905. Location - New Delhi. ICAR-Directorate of Groundnut Research - Junagadh, Gujarat. ICAR-Directorate of Rapeseed & Mustard Research - Bharatpur, Rajasthan. ICAR-Directorate of Soybean Research - Indore, Madhya Pradesh. ICAR-Indian Institute of Soil Science - Bhopal, Madhya Pradesh.

First in India and World

First Man

Yuri Gagarin - First man in the world to go into space. He was from Russia.

Acharya Vinoba Bhave - First Indian to receive the Ramon Magsaysay award in 1958.

Pandit Ravi Shankar - First Indian to win a Grammy for 'West Meets East', the 1968 album with American violinist Yehudi Menuhin.

J.R.D Tata - First Indian citizen to get a commercial pilot's license on February 10, 1929.

Abul Ahsan - First Secretary General of SAARC in 1987.

Yuan Shikai - First president of the Republic of China in 1912.

Zhou Enlai - First Prime Minister of China.

Dadabhai Naoroji - First Indian to win a seat in the House of Commons.

Womesh Chandra Banerjee - First president of the Indian National Congress.

Alexander Cunningham - First Director-General of the Archaeological Survey of India (ASI).

Chintaman Dwarkanath Deshmukh - First

Indian Governor of the Reserve Bank of India.

Sir Osborn Smith - First Governor of the Reserve Bank of India.

Ramanathan Krishnan - First Asian tennis player to win the Wimbledon's Boys singles title and the first Indian to reach the semi-final in Wimbledon Tennis Championship.

Baba Amte- He became the only Indian to have won the United Nations Award in the field of human rights in 1988.

Arthur Travers Crawford - First Municipal Commissioner and collector of Bombay (now Mumbai).

Satyarup Siddhanta - He became the youngest mountaineer in the world and the first from India to climb the Seven Summits as well as Volcanic Summits.

First Woman

Vijaya Lakshmi Pandit - The first woman and first Indian to serve as the President of the United Nations General Assembly from 1953 to 1954.

V. S. Ramadevi - First lady to become the Chief Election Commissioner of India in 1990.

Surekha Yadav - India's first female loco pilot in 1988.

Anna Chandy - First Indian woman to serve as a judge at a High Court in 1959.

Nirja Bhanot - First Indian woman to get the Ashok Chakra in 1987.

Chonira Belliappa Muthamma - First woman to join the Indian Foreign Service in 1949. She became the first Indian woman IFS Officer and Ambassador.

M. Fathima Beevi - First female judge of the Supreme Court of India. She also became the first Muslim woman to be appointed to any of the higher judiciary in the country.

Arati Saha - First Indian and Asian woman to swim across the English Channel in 1959.

Fathimath Dhiyana Saeed - First woman and youngest Secretary General of SAARC in 2011.

Sucheta Kriplani - First female Chief Minister, serving as the head of the Uttar Pradesh government (1963 to 1967).

Rajkumari Amrit Kaur - First Indian woman to join the Cabinet of Independent India as its health minister in 1947.

Dr. Seema Rao - First woman commando trainer in India. She is popularly known as the 'Wonder Woman of India'.

Savitribai Phule - First female teacher in India (1848).

Meera Sahib Fatima Bibi - First female judge who was appointed to the Supreme Court of India in 1989.

Mother Teresa - First Indian woman to win a Nobel Peace Prize in 1979.

Indira Gandhi - First woman Prime Minister of India and served from 1966 to 1977.

Pratibha Patil - First woman President of India and held office from July 2007 to July 2012.

Nargis Dutt - First Indian actress to receive the Padma Shri (1958).

Kalpna Chawla - The first woman of Indian origin to go to space.

Bachendri Pal - First Indian woman to reach the summit of Mount Everest in 1984.

Usha Padhee - First woman to be given the charge of Director General of the Bureau of Civil Aviation Security.

Kamala Harris - America's first female Vice President.

Madhabi Puri Buch - First woman Chairperson of the Securities and Exchange Board of India (SEBI) in 2022.

Karnam Malleswari - First Indian woman to win a medal at the Olympics in 2000 and also the first Indian weightlifter to win an Olympic medal.

Avani Chaturvedi - First Indian woman Air Force Officer to fly a MiG-21 Bison.

Bhawna Kanth - First-ever woman fighter pilot to have taken part in a Republic Day Parade.

Shivangi Singh - Indian Air Force's first woman pilot to fly the Rafale fighter aircraft.

Arunima Sinha - World's first female amputee to scale Mount Everest, Mount Kilimanjaro, Mount Elbrus, Mount

Kosciuszko, Mount Aconcagua, Carstensz Pyramid, and Mount Vinson.

Santosh Yadav - First woman in the world to climb Mount Everest twice.

Premlata Agrawal - First Indian woman to scale the Seven Summits, the seven highest continental peaks of the world.

Priyanka Mohite - First Indian woman to scale five peaks above 8,000 meters Mount Annapurna, Mt. Everest, Mt. Lhotse, Mt. Makalu, and Mt. Kanchenjunga.

Kanchan Chaudhary Bhattacharya - First woman to become Director General of Police of Uttarakhand.

Archana Ramasundaram - First woman officer in India to be inducted as an additional director in the Central Bureau of Investigation in 2014.

P V Sindhu - First Indian woman to claim two Olympic medals after she won a badminton bronze medal in Tokyo.

Geraldine Jerrie Mock - First female pilot to circumnavigate the world alone in 1964.

Prem Mathur - First Indian woman commercial pilot and started flying for Deccan Airways.

Bhanu Athaiya - First Indian personality to win the Oscar award in the 'Best Costume Design' category for the film 'Gandhi' in 1983.

Justice Gita Mittal - First woman Chairperson of the Broadcasting Content Complaints Council (BCCC).

Santishree Pandit was appointed as the first woman Vice-Chancellor of Jawaharlal Nehru University (JNU), New Delhi.

Zeenat Aman - Won the Miss Asia Pacific 1970 title, becoming the first Indian and South Asian woman to win the title.

First in Indian State/UT

South Maubang (Mizoram) - First Open-Defecation Free village.

Mysore (Karnataka) - India's first open defecation-free city.

Radha Krishna Mathur - First Lieutenant Governor of Ladakh.

Aditya Nath Jha - First Lieutenant

Governor of Delhi.

Praful Khoda Patel - First Lieutenant Governor of Dadra and Nagar Haveli and Daman and Diu.

Kewal Singh - First chief commissioner of Puducherry.

Girish Chandra Murmu - First Lieutenant Governor of Jammu and Kashmir.

M L Kampani - First Lieutenant Governor of Andaman and Nicobar Islands

Bhairab Dutt Pandey - First Administrator of Chandigarh.

Mridula Sinha - First woman Governor of Goa.

Padmaja Naidu - First woman Governor of West Bengal.

Mamata Banerjee - First female Chief Minister of West Bengal

Sundari Nanda - First woman to head the Director General of the Police force in the Puducherry Union Territory.

Leila Seth - First woman judge in Delhi High Court. She was also the first woman to become Chief Justice in the High Court of Himachal Pradesh.

Dong Village - In India, the sun rises first in a place called Dong village in the state of Arunachal Pradesh.

The Tamil Nadu government has introduced India's first elephant death audit framework.

Rajasthan has become the first state in the country to get the L-root server.

Sikkim became the first fully organic farming state in India in 2016.

First in India (Governance)

First President of India - Dr. Rajendra Prasad.

First Vice-President of India - Sarvepalli Radhakrishnan.

First Prime Minister of India - Jawaharlal Nehru.

First Chief Election Commissioner of India - Sukumar Sen.

First Viceroy of India - Lord Canning.
First Governor-General of India - Lord William Bentinck.

First Chief Justice of India - Hiralal Jekisundas Kania.

First Speaker of Lok Sabha - G. V. Mavlankar.

First Home Minister of India - Sardar Vallabhai Patel.

Meira Kumar - First Woman Speaker of Lok Sabha in 2009.

Sarojini Naidu - First Indian woman to be president of the Indian National Congress (1925) and First woman Governor of a State in free India.

Pinaki Chandra Ghose - The first Lokpal of India who was appointed on 19 March 2019.

The First Law Commission was set up in independent India in 1955.

The Calcutta High Court is the oldest high court in the country, established on 1 July 1862.

The first Backward Classes Commission of India was headed by Kaka Kalelkar.

The budget was first introduced on 7th April 1860. The first Finance Member, who presented the Budget, was James Wilson. The first Union Budget of post-Independence India was presented on November 26, 1947, by RK Shanmukham Chetty.

First in India (Defence)

First Defence Minister of India - Baldev Singh Chhokar.

First Commander-in-Chief of Free India - Kodandera Madappa Cariappa.

First Commander-in-Chief of the Indian Air Force - Subroto Mukherjee.

First Param Vir Chakra Winner - Major Somnath Sharma.

First female Jawan in the Army - Shanti Tigga.

First cosmonaut - Sqn. Ldr. Rakesh Sharma.

First woman in the Indian Army awarded the coveted 'Sword of Honour' prize - Divya Ajith Kumar.

First Ashoka Chakra - Havildar Bachittar Singh and Naik Narbahadur Thapa.

Sam Manekshaw - First Indian Army officer to be promoted to the rank of field marshal.

General Bipin Rawat - India's first Chief of Defence Staff.

Kalvari-class submarines - First submarines inducted into the Indian Navy.

The first Indigenous Aircraft Indian Carrier Ship is INS Vikrant.

Vijayalakshmi Ramanan - First woman officer in the Indian Air Force.

Punita Arora - First woman in the Indian Armed Forces to be promoted to a Three-star rank.

Neerja Bhanot- She was the first woman and youngest recipient of the Ashok Chakra Award in 1987.

Abhilash Tomy (Indian Naval Officer) - First Indian to circumnavigate the globe, solo and non-stop, over a 150-day voyage in 2013.

Anjali Gupta - First female officer in India and in Air Force to be court-martialed.

Abhilasha Barak - First woman officer to join the Army Aviation Corps.

Aparna Kumar - First woman Indian Police Service DIG and Indo-Tibetan Border Police (ITBP) officer to successfully complete the South Pole expedition.

Kiran Bedi - First Woman Indian Police Service Officer in 1972.

Mawya Sudan - First woman fighter pilot in the Indian Air Force (IAF) from Jammu and Kashmir.

Flight Lieutenant Bhawana Kanth - First woman to qualify for combat missions on a fighter jet in May 2019. She also became the first-ever woman fighter pilot to have taken part in a Republic Day Parade and became the second Indian Woman fighter pilot to fly a MiG-21 Bison Solo.

Famous personality

First Indian to win the Booker Prize was Arundhati Roy for her book 'God of Small Things' in 1997.

First Indian to receive the Nobel Prize in Economics is Amartya Sen in 1998 for his work in welfare economics.

Gobind Behari Lal - First Indian origin to win a Pulitzer Prize in the year 1937.

Reita Faria - First Indian and the first woman of Asian origin to win the Miss World 1966 pageant.

Saina Nehwal - First-ever badminton player from India to clinch an Olympic medal.

Jwala Gutta - First badminton player from India to qualify for two events in the Olympics-women's doubles with Ponnappa and mixed doubles with V. Diju in London.

Neeraj Chopra - First Indian javelin thrower to win an Asian Games gold medal in 2018. He is the first Asian athlete to win an Olympic gold medal in Men's Javelin throw.

Civilian Awards

First Bharat Ratna - Sarvapalli Radhakrishnan, Sir C.V. Raman, and Chakravarti Rajagopalachari in 1954.

Sachin Tendulkar - First sportsperson and the youngest to receive Bharat Ratna in 2014.

Khan Abdul Ghaffar Khan was the First Non-Indian to Receive the Bharat Ratna in 1987.

Indira Gandhi became the first woman to be honored with Bharat Ratna in 1971.

Padma Vibhushan- First recipients - Satyendra Nath Bose, Nandalal Bose, Zakir Husain, Balasaheb Gangadhar Kher, V. K. Krishna Menon, and Jigme Dorji Wangchuck, who were honored in 1954.

Padma Bhushan- Sisir Kumar Bhaduri - First person to refuse the Padma Bhushan Award in 1959.

Padma Shri- Arati Saha, first female sportsperson to be awarded Padma Shri in 1960.

Balbair Singh Dosanjh - First sportsperson to receive the Padma Shri in 1957.

Syeda Tahira Safdar - First female Chief Justice of any court in the history of Pakistan.

Film Awards

National Film Award - Nargis Dutt and Uttam Kumar being the first actress and actor to get the Best Actress and Best Actor Awards respectively.

Dada Saheb Phalke Award - The first recipient of this award was Devika Rani who received it in 1969 on the occasion of the 17th National Films Awards.

The Filmfare Awards - The first filmfare award ceremony was held on 21st March

1954. The first winners - Bimal Roy for his direction of Do Bigha Zameen, Dilip Kumar for his performance in Daag, Meena Kumari for her performance in Baiju Bawra.

Tanvi Shah - First female from India to win a Grammy award for her work in the song 'Jai Ho' in Slumdog Millionaire.

Literature Awards

Jnanpith Award- First recipient - G. Sankara Kurup (Malayalam writer) in 1965. The first Hindi author to get this award was Sumitranandan Pant in 1968. First woman - Ashapurna Devi in 1976.

Sahitya Akademi Award- First recipient - Mahkhanlal Chaturvedi (First Hindi writer) in 1955. Amrita Pritam was the first female recipient of the Sahitya Akademi Award in 1956.

First in World

Valentina Tereshkova - First woman in the world to travel in space on 16 June 1963.

Bula Chaudhury (India) - First woman in the world to cross seven important seas of the world by swimming.

Mark Inglis - First double amputee in the world to reach the summit of Mount Everest.

The Netherlands was the first country in the world to legalize same-sex marriage in 2001. Second - Belgium (2003). Taiwan becomes the first in Asia to legalise same-sex marriage in 2019.

First Hospital Train - Lifeline Express (Jeevan Rekha Express), started running from July 1991. This is the world's first hospital train that covers the country's remote areas, providing free treatment to people in need.

First CNG Terminal - The World's first Compressed Natural Gas (CNG) terminal is located at Bhavnagar in Gujarat.

First Sailing Boat - Flipflop is the world's first sailing boat made entirely from plastic waste.

Roald Amundsen's polar party - First in the world to reach the South Pole in December 1911.

Robert Edwin Peary - First person in the World to reach the North Pole after eight trials in 1909.

Neil Armstrong - First human to step on the Moon in the world.

Junko Tabei - First woman in the world to reach the summit of Mount Everest and the first woman to ascend the Seven Summits, climbing the highest peak on every continent.

Alexei Leonov - First person in the world to conduct a spacewalk, exiting the capsule during the Voskhod 2 mission for 12 minutes 9 seconds on 18 March 1965.

Chicago Pile-1 - World's first nuclear reactor and was created in 1942.

Idaho Gem - The world's first cloned mule.

Copycat - The world's first cloned kitten.

ANDi - The world's first genetically modified rhesus monkey.

Sirimavo Bandaranaike - In 1960, she was the first female Prime Minister of Sri Lanka as well as the first female Prime Minister of the world.

Elisabeth Domitien - First and only woman Prime Minister of the Central African Republic from 1975 to 1976.

Golda Meir - First woman to become the Prime Minister of Israel.

Akademik Lomonosov - World's first floating nuclear reactor launched by Russia in 2018.

Kami Rita Sherpa - The first person in the world to climb Everest 27 times till 2023.

Duncan White - First athlete to win an Olympic medal for Sri Lanka.

United Kingdom - First nation in the world to declare a climate emergency to eliminate net zero greenhouse gas emissions completely by 2050.

Sports

First Formula of Motor Race - The first F1 Indian Grand Prix took place in 2011 at the renowned Buddh International Circuit, a 5.14 km (3.2 miles) long racetrack designed for high-speed racing. The first-ever ABB FIA Formula-E World Championship race in India is to be held in Hyderabad.

First Asian Games - It was a multi-sport event celebrated in New Delhi, India from 4 to 11 March 1951.

First Hockey Association - The first hockey association in India was formed in Kolkata in 1908 - the Bengal Hockey Association.

First Gold Medal in Hockey - India won its first Olympic hockey gold in 1928 at the Amsterdam Games.

First Wimbledon tournament - It was held in 1877 and has been contested every year.

First Paralympic Games - It was organized in Rome (Italy) in 1960.

Mahesh Shrinivas Bhupathi - First Indian to win a major Grand Slam tennis tournament in 1997.

Vishwanath Anand - India's first chess grandmaster in 1988.

Achanta Sharath Kamal - First Indian table tennis player ever to become ten times Senior National Champion.

Sakshi Malik - First Indian woman wrestler to win a medal at the 2016 Rio Olympics.

Abhinav Bindra - First Indian to receive the International Shooting Federation Blue Cross's highest honor.

Anjali Bhagwat - First Indian to win the 2002 International Shooting Sport Federation (ISSF) Champion of Champions in the 10m Air Rifle event.

Purma Banerjee - India's first olympian to be the flag-bearer for the Indian contingent at the 1920 Olympics held in Antwerp, Belgium.

Yuki Bhambri - First Indian to win the under-18 title in the Orange Bowl tennis tournament in 2008.

Heena Sidhu - First female pistol shooter from India to reach number 1 in the world ranking by the International Shooting Sport Federation and first Indian pistol shooter to win a gold medal in the 10-meter air pistol event in an ISSF World Cup finals.

Balbir Singh Dosanjh - First sports person to receive the Padma Shri in 1957. Demetrius Vikelas was the first president of the International Olympic Committee (IOC).

Mohammed Salim was the first player from the Indian subcontinent to play for a European Football club (Celtic FC).

The First Women's Cricket World Cup was hosted by England.

Sir Dorabji Tata was the first president of Indian Olympic Association (IOA).

The first hockey association in the world was formed in the United Kingdom in 1876.

First World Weightlifting Championship was organized by the World Weightlifting Federation in London, UK in March 1891.

Space

First Mission to Moon- Chandrayaan-1 is India's first mission to the Moon. It was launched on 22 October 2008 using the Polar Satellite Launch Vehicle (PSLV-C11) from the Satish Dhawan Space Center in Sriharikota, India.

First Mission to Mars- Mars Orbiter Mission or Mangalyaan was India's first interplanetary mission.

First Rocket - First Rocket in India was launched in 1963. The launchpad for the first rocket 'RH200' was Thumba Balasore and took off from Thumba near Thiruvananthapuram.

RH-75 - The first rocket developed in India was launched on 20 November 1967, from the Thumba Equatorial Rocket Launching Station (TERLS).

Vikram-S - India's first privately developed rocket, was launched by the Indian Space Research Organisation (ISRO) from its spaceport in Sriharikota in November 2022.

EOS-03 - India's first Earth-imaging satellite in a geostationary orbit. It is also called GISAT-1.

APPLE - First Communication Satellite of India.

IRNSS-1A - First regional satellite in the Indian Navigation Satellite System.

IRS-1A - India's first operational remote sensing satellite. It was launched in 1988.

Kalpna-1- First meteorological satellite of India. It was launched in 2002 and built by ISRO.

GSAT-3 - India's first exclusive educational satellite. It was launched in 2004.

STUDSAT - It is India's first pico-satellite (weighing less than 1 kg). It was launched in 2010.

Sputnik 1 - First artificial Earth satellite in the World. It was launched into an

elliptical low Earth orbit by the USSR on 4 October 1957 as part of the Soviet space program. Launch Site- Gagarin's Start (Baikonur Site). Rocket- Sputnik 8K71PS.

Aryabhata - First artificial satellite launched by India in 1975.

Vikram Sarabhai - First chairman of the Indian Space Research Organisation (ISRO).

SpiceJet - First Indian budget carrier to join the International Air Transport Association (IATA).

RISAT-1 - India's first indigenous all-weather Radar Imaging Satellite. Launch year - 2012.

Bhaskara-I - First experimental remote-sensing satellite of India launched in 1979.

Astrosat - India's first dedicated multi-wavelength space observatory. Launch year - 2015.

Rakesh Sharma - First Indian to go to space.

Gaganyaan is India's first Human spaceflight programme.

Other Facts (India)

First Cloned Animal - Assamese buffalo male calf named Sach-Gaurav was born in Hisar, Haryana.

First Jute Mill - It was set up at Rishra in Kolkata in 1855.

First Cotton Mill - The first cotton mill Bombay Spinning and Weaving Company was set up as a spinning mill in Bombay in 1854.

First Anti-Nuclear Medical Kit - Institute of Nuclear Medicine and Allied Sciences (INMAS) has developed India's first indigenous anti-nuclear medical kit.

First Refinery of India - Digboi Refinery (Assam).

First 3D planetarium - Swami Vivekananda Planetarium at Pilikula in Mangalore is the first 3D planetarium in India.

First Nuclear Research Reactor - APSARA was the first nuclear research reactor in India and also Asia.

First Workable Prototype - ARPANET (Advanced Research Projects Agency Network) is the first workable prototype

of the Internet which came in the late 1960s.

First Crude Oil Pipeline (India) - The Naharkatia-Nunmati-Barauni pipeline.

Radhanath Sikdar - First person to calculate the height of Mount Everest, in 1852.

Brajesh Mishra - First National Security Advisor of India in 1998 by the Atal Bihari Vajpayee Government.

Nallathamby Kalaiselvi - First women director general of the Council of Scientific and Industrial Research (CSIR).

Bachi Karkaria - First Indian board member of the World Editors Forum.

Prime Minister Narendra Modi - Received the 'first-ever' Philip Kotler Presidential award in 2019.

Neecha Nagar - First Indian film to win the Palme d'Or at the Cannes Film Festival in 1946.

Sushmita Sen - First Indian to win the Miss Universe pageant in 1994.

The first radio show was broadcast in India in 1923.

India's first floating post office was established in Jammu and Kashmir. It is located in Dal Lake, Srinagar.

The President of Indonesia (Sukarno) attended the first Republic Day celebrations of India as its chief guest.

Cochin International Airport Limited (CIAL), Kerala is India's first to be completely powered by solar energy.

Humayun's Tomb (Maqbara-i Humayun) is the first garden - tomb in the Indian subcontinent.

India's first underwater metro project has been launched in Kolkata.

James Augustus Hickey was behind publishing the first newspaper in India.

National Organic Chemical Industry Limited was the first petrochemical company in India.

Gandhinagar Railway Station, Jaipur is India's first all-women-managed railway station.

The first playback song in Hindi Movies was recorded by music directors RC Boral and Pankaj Mullick in 1935.

Awards (National / International)

National Awards

Bharat Ratna - Highest civilian award of India. Established - On 2 January 1954. The recommendations for the Bharat Ratna - By the Prime Minister to the President (a maximum of three in a particular year but in 1999 four awards were given). The first recipients of the Bharat Ratna - Sarvepalli Radhakrishnan, CV Raman, and C. Rajagopalachari in 1954.

Padma Awards - Established - 1954. The Padma Awards are announced annually on the eve of Republic Day. These awards are presented by the President of India. The awardees are presented a Sanad (certificate) signed by the President and a medallion. The award is given in three categories are Padma Vibhushan, Padma Bhushan and Padma Shri.

Padma Vibhushan is awarded for exceptional and distinguished service. The first recipients of the Padma Vibhushan - Satyendra Nath Bose, Nandalal Bose, Zakir Husain, Balasaheb Gangadhar Kher, V.K. Krishna Menon, and Jigme Dorji Wangchuck, who were honored in 1954.

Padma Bhushan is awarded for distinguished service of high order. The first recipients of the Padma bhushan - Amarnath Jha along with other 22 people who were honoured in 1954.

Padma Shri is awarded for distinguished service in any field. The first recipients of the Padma Shri - Asha Devi Aryanayakam along with other 18 people who were honoured in 1954.

Gallantry Awards

Paramvir Chakra, Ashok Chakra and Vir Chakra are presented during the celebration of Republic day.

Param Vir Chakra : India's highest Gallantry award. Established - 26 January 1950. Param Vir Chakra translates as the 'Wheel of the Ultimate Brave'. The first award recipient - Major Somnath Sharma (1950). Yogendra Singh Yadav - The Youngest person to have received the Param Vir Chakra award.

Mahavir Chakra : Second-highest Gallantry award. Established - 26 January 1950.

Vir Chakra : Established - 26 January 1950.

Ashoka Chakra: Established - 4 January 1952. First recipient - Naik Narbahadur Thapa.

Kirti Chakra : Established - 1952 . It is given to civilians as well as military personnel, including posthumous awards. It is the second highest peacetime award and is equivalent to the Maha Vir Chakra.

Shaurya Chakra : Established - 1952 as Ashoka Chakra Class-III and renamed in 1967 as Shaurya Chakra.

Sena Medal : Established - 26 January 1960. Five other medals are the Vishisht Seva Medal series (Class I, Class II, Class III), the Sainya Seva Medal, Videsh Seva Medal, Nao Sena Medal and the Vayu Sena Medal.

Yudh Seva Medal is one of India's military awards for distinguished service during wartime. The award is equivalent to the Vishisht Seva Medal, which is a peacetime distinguished service decoration. Established - 1980. First woman awardee (2019) - Minty Agarwal.

Literary Awards

Vyas Samman (literary award): It is awarded annually by the K.K.Birla Foundation and includes a cash of Rs 4,00,000. The first recipient - Ram Vilas Sharma for 'Bharat ke Pracheen bhasha Parivar aur Hindi' (1991). Second recipient - Shiv Prasad Singh for 'Neela Chand' (1992).

Saraswati Samman: Annual award for outstanding prose or poetry literary works in any of the 22 languages of India listed in Schedule VIII of the Constitution of India. Established - 1991 by the K. K. Birla Foundation. First recipient of Saraswati Samman - Harivanshrai Bachchan for his four-volume autobiography, "Kya Bhoolon Kya Yaad Karoon", "Needh Ka Nirman Phir", "Basere Se Door", and "Dasdwar se Sopan Tak". Ramakant Rath (1992) for his work "Shri Radha".

Jnanpith Award (Highest literary award): Established - 1961. First recipient - Malayalam writer G. Sankara Kurup (1965) for his collection of poems,

Odakkuzhal (The Bamboo Flute). The second recipient - Tarasankar Bandyopadhyay (1966). First woman awardee - Ashapoorna Devi for her novel "Pratham Pratishruti".

The Sahitya Academy Award: Established - 1954. The award comprises a copper plaque and a cash prize of Rs.1,00,000. The Sahitya Akademi Fellowship is a literary honor in India bestowed by the Sahitya Academy.

Sahitya Akademi Yuva Puraskar: Established - 2011. It recognizes young writers under the age of 35, with the aim of encouraging and promoting young writers. It comprises a cash prize of Rs. 50,000 and an engraved copper plaque.

Bihari Puraskar: It is one of the three literary awards instituted by the K K Birla Foundation in 1991. It is named after the famous poet Bihari. First recipient - Jaisingh Neeraj for his poetry compilation "Dhani Ka Admi" in 1991.

The Gitanjali Literary Prize: Established - 2012 as a Franco-Indian literary award. It was founded by the Mohsen Hachtroudi Foundation and CHG Earth Group (a hotel company in India).

Science Awards

The Breakthrough Prize: Established - 2012. The first recipient of the Fundamental Physics Prize - Ashoke Sen for his work on 'String Theory'. Each prize is 3 million dollars and presented in the fields of Life Sciences, Fundamental Physics and Mathematics and founded by Sergey Brin, Priscilla Chan and Mark Zuckerberg, Yuri and Julia Milner, and Anne Wojcicki.

The Shanti Swarup Bhatnagar Prize (SSB) for Science and Technology : It was first awarded in 1958.

The Borlaug Award (for research and contributions in the field of agriculture and the environment): Started - 1972. First awardee - Dr. Atmaram Bhairav Joshi (1972).

The Bidhan Chandra Roy Award: Established - 1962 in memory of Dr. B. C. Roy. First awarded - Sandip Mukerjee (1973) in the field of medicine.

SASTRA Ramanujan Prize: Established - 2005. The first recipients - Manjul Bhargava and Kannan Soundararajan in 2005.

The **ICTP Ramanujan Prize** is awarded annually for excellence in Mathematics by the International Centre for Theoretical Physics in Italy. It was first awarded in 2005 to Marcelo Viana.

International Awards

Nobel Prize

The Nobel Prize is a set of annual international awards bestowed in a number of categories by Swedish and Norwegian institutions in recognition of academic, cultural, and scientific advances for outstanding work in Physics, Chemistry, Physiology or Medicine, Literature, Economics, and Peace.

The foundations for the prize were laid in 1895 when Alfred Nobel wrote his last will, leaving much of his wealth to the establishment of the Nobel Prize. The Nobel Peace Prize is awarded in Norway while the other Nobel prizes are awarded in Sweden. Each recipient (known as a "laureate") receives a gold medal, a diploma, and a monetary award of 11,000,000 SEK (Swedish Krona). It was first awarded in 1901.

The laureates in the Memorial Prize in Economics are selected by the Royal Swedish Academy of Sciences. It was first awarded in 1969.

The youngest Nobel Prize winner is Malala Yousafzai (Peace Prize, 2014). The oldest Nobel Prize winner is John B. Goodenough (Chemistry, 2019). Jean-Paul Sartre (1964 Nobel literature Prize) and Le Duc Tho (1973 Nobel Peace Prize) declined the prize.

The first Nobel Prize in Literature was awarded to Sully Prudhomme in 1901.

Selma Lagerlof (Swedish author) - The first woman to be awarded the Nobel Prize in Literature in 1909.

Linus Pauling is the only person to have been awarded two individual Nobel Prizes - The 1954 Nobel Prize in Chemistry and the 1962 Nobel Peace Prize.

The Nobel Prize has only been awarded posthumously twice to Dag Hammarskjöld (Nobel Peace Prize 1961) and Erik Axel Karlfeldt (Nobel Prize in Literature 1931).

The scientist who won the Nobel Prize twice or more - J. Bardeen (Physics 1956,

Physics 1972), Madam Curie (Physics 1903, Chemistry 1911), L. Pauling (Chemistry 1954, Peace 1962), F. Sanger (Chemistry 1958, Chemistry 1980), K. Barry Sharpless (Chemistry 2001, Chemistry 2022), and International Committee of the Red Cross (Peace 1917, Peace 1944, Peace 1963), and United Nations High Commissioner for Refugees (Peace 1954, Peace 1981).

Important Nobel Prize in Chemistry -

1901 - Jacobus Henricus Van't Hoff "in recognition of the extraordinary services he has rendered by the discovery of the laws of chemical dynamics and osmotic pressure in solutions".

1902 - Hermann Emil Fischer "in recognition of the extraordinary services he has rendered by his work on sugar and purine synthesis".

1903 - Svante August Arrhenius "in recognition of the extraordinary services he has rendered to the advancement of chemistry by his electrolytic theory of dissociation".

1908 - Ernest Rutherford "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances".

1911 - Marie Curie "in recognition of her services to the advancement of chemistry by the discovery of the elements radium and polonium, by the isolation of radium and the study of the nature and compounds of this remarkable element".

1921 - Frederick Soddy "for his contributions to the chemistry of radioactive substances, and his investigations into the origin and nature of isotopes".

1930 - Hans Fischer "for his research into the constitution of haemin and chlorophyll and especially for his synthesis of haemin".

1944 - Otto Hahn "for his discovery of the fission of heavy nuclei".

1954 - Linus Carl Pauling "for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances".

1958 - Frederick Sanger "for his work on the structure of proteins, especially that of insulin".

1961 - Melvin Calvin "for his research on the carbon dioxide assimilation in Plants".

1975 - Vladimir Prelog received the Nobel Prize for his research into the stereochemistry of organic molecules and reactions.

2009 - Venkatraman Ramakrishnan, Thomas Steitz, and Ada E. Yonath "for studies of the structure and function of the ribosome".

2020 - Emmanuelle Charpentier and Jennifer A. Doudna "for the development of a method for genome editing".

2021 - Benjamin List and David MacMillan "for the development of asymmetric organocatalysis".

2022 - Carolyn R. Bertozzi, Morten Meldal, and Karl Barry Sharpless "for the development of click chemistry and bioorthogonal chemistry".

2023 - The Prize was awarded to Moungi G. Bawendi, Louis E. Brus and Alexei I. Ekimov "for the discovery and synthesis of quantum dots."

Important Nobel Prize in Physics -

1901 - Wilhelm Conrad Röntgen "in recognition of his extraordinary services in the discovery of Röntgen rays (X rays), named after him".

1903 - Antoine Henri Becquerel "in recognition of the extraordinary services he has rendered by his discovery of spontaneous radioactivity". Pierre Curie and Marie Curie "in recognition of the extraordinary services they have rendered by their joint research on the radiation phenomena discovered by Professor Henri Becquerel".

1909 - Guglielmo Marconi and Karl Ferdinand Braun "in recognition of their contributions to the development of wireless telegraphy".

1910 - Johannes Diderik van der Waals "for his work on the equation of state for gasses and liquids".

1911 - Wilhelm Wien "for his discoveries regarding the laws governing the radiation of heat".

1915 - Sir William Henry Bragg and William Lawrence Bragg "for their services in the analysis of crystal structure by means of X-rays".

1918 - Max Karl Ernst Ludwig Planck "in recognition of the services he rendered to the advancement of Physics by his discovery of energy quantum".

1919 - Johannes Stark "for his discovery of the Doppler effect in anode rays and the splitting of spectral lines in electric fields".

1921 - Albert Einstein "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect".

1923 - Robert Andrews Millikan "for his work on the elementary charge of electricity and on the photoelectric effect".

1926 - Jean Baptiste Perrin "for his work on the discontinuous structure of matter, and especially for his discovery of sedimentation equilibrium".

1927 - Arthur Holly Compton "for his discovery of the Compton effect named after him". Charles Thomson Rees Wilson "for his method of making the paths of electrically charged particles visible by condensation of vapor".

1928 - Owen Willans Richardson "for his work on the thermionic phenomenon and especially for the discovery of the Richardson law named after him".

1930 - Sir Chandrasekhara Venkata Raman "for his work on the scattering of light and for the discovery of the Raman effect named after him".

1932 - Werner Karl Heisenberg "for the creation of quantum mechanics, the application of which has, inter alia, led to the discovery of the allotropic of hydrogen".

1935 - James Chadwick "for the discovery of the neutron".

1936 - Victor Franz Hess "for his discovery of cosmic radiation". Carl David Anderson "for his discovery of the positron".

1969 - Murray Gell-Mann "for his contributions and discoveries concerning the classification of elementary particles and their interactions".

1979 - Sheldon Lee Glashow, Abdus Salam, and Steven Weinberg "For their contributions to the theory of the electromagnetic weak interaction between elementary particles, and

predicting the weak neutral current".

1983 - Subramanyan Chandrasekhar "for his theoretical studies of the physical processes of importance to the structure and evolution of the stars". William Alfred Fowler "for his theoretical and experimental studies of the nuclear reactions of importance in the formation of the chemical elements in the universe".

2020 - Roger Penrose "for the discovery that black hole formation is a robust prediction of the general theory of relativity". Reinhard Genzel and Andrea Ghez "for the discovery of a supermassive compact object at the center of our galaxy".

2021 - Syukuro Manabe and Klaus Hasselmann "for the physical modeling of Earth's climate, quantifying variability and reliably predicting global warming". Giorgio Parisi "for the discovery of the interplay of disorder and fluctuations in physical systems from atomic to planetary scales".

2022 - Alain Aspect, John F. Clauser, and Anton Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science". The Nobel Prize in Physics 2022 was awarded in the field of quantum mechanics.

2023 - Pierre Agostini, Ferenc Krausz and Anne L'Huillier for their experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter.

Important Nobel Prize in Medicine -

1902 - Ronald Ross "for his work on malaria, by which he has shown how it enters the organism and thereby has laid the foundation for successful research on this disease and methods of combating it".

1905 - Robert Koch "for his investigations and discoveries in relation to tuberculosis".

1906 - Camillo Golgi and Santiago Ramón y Cajal "in recognition of their work on the structure of the nervous system".

1909 - Emil Theodor Kocher "for his work on the physiology, pathology, and surgery of the thyroid gland".

1924 - Willem Einthoven "for his discovery of the mechanism of the electrocardiogram".

1930 - Karl Landsteiner "for his discovery of human blood groups".

1945 - Sir Alexander Fleming, Ernst Boris Chain and Sir Howard Walter Florey "for the discovery of penicillin and its curative effect in various infectious diseases".

1968 - Robert W. Holley, Har Gobind Khorana, and Marshall W. Nirenberg "for their interpretation of the genetic code and its function in protein synthesis".

2019 - William G. Kaelin Jr, Sir Peter J. Ratcliffe, and Gregg L. Semenza "for their discoveries of how cells sense and adapt to oxygen availability".

2020 - Harvey J. Alter, Michael Houghton, and Charles M. Rice "for the discovery of Hepatitis C virus".

2021 - David Julius and Ardem Patapoutian "for their discoveries of receptors for temperature and touch".

2022 - Svante Paabo "for his discoveries concerning the genomes of extinct hominins and human evolution".

2023 - The Nobel Prize in Physiology or Medicine to Katalin Karikó and Drew Weissman for their work in the realm of mRNA vaccines that helped in tackling the COVID-19 pandemic.

Nobel Peace Prize

1931 - Jane Addams and Nicholas Murray Butler "for their assiduous effort to revive the ideal of peace and to rekindle the spirit of peace in their own nation and in the whole of mankind".

1962 - Linus Carl Pauling "for his fight against the nuclear arms race between East and West".

1964 - Martin Luther King Jr. "for his non-violent struggle for civil rights for the Afro-American population".

1970 - Norman E. Borlaug "for having given a well-founded hope – the green revolution".

1979 - Mother Teresa "for her work for bringing help to suffering humanity".

1982 - Alva Myrdal and Alfonso García Robles "for their work for disarmament and nuclear and weapon-free zones".

1993 - Nelson Mandela and Frederik Willem de Klerk "for their work for the peaceful termination of the apartheid regime, and for laying the foundations for a new democratic South Africa".

1997 - International Campaign to Ban Landmines (ICBL) and Jody Williams "for their work for the banning and clearing of anti-personnel mines".

2009 - Barack H. Obama 'for his extraordinary efforts to strengthen international diplomacy and cooperation between peoples".

2014 - Kailash Satyarthi and Malala Yousafzai "for their struggle against the suppression of children and young people and for the right of all children to education".

2021 - Maria Ressa and Dmitry Andreychuk Muratov "for their efforts to safeguard freedom of expression, which is a precondition for democracy and lasting peace".

2023 - Narges Mohammadi won the Nobel Prize in Peace 2023 for "for her fight against the oppression of women in Iran and her fight to promote human rights and freedom for all."

Amartya Sen received the Nobel Prize for Economics in 1998 for his contributions to welfare economics and social choice theory and for his interest in the problems of society's poorest members.

The Abel Prize (presented by the King of Norway to mathematicians): It is named after the Norwegian mathematician Niels Henrik Abel. First awardee - Jean-Pierre Serre (2003)

Alexander Dalrymple Award (presented by the United Kingdom (UK) Ministry of Defence): Established - 2006. It is given for efforts in raising the standards of hydrography, cartography and navigation around the world.

Von Karman Award (in Engineering Mechanics): Established - 1960 in honor of Theodore von Karman by the Engineering Mechanics Division of the American Society of Civil Engineers (ASCE). First recipient - William Prager (1960).

The Crystal Award (by the World Economic Forum): The Indian recipients of the award are A.R. Rahman (2011), and Deepika Padukone (2020).

The New York Dance and Performance Awards or Bessie Awards: The broad categories of the awards are choreography, performance, music composition, and visual design. Established - 1983 by Dance Theater Workshop and was named in honor of Bessie Schonberg.

The Golden Reel Award for Outstanding Achievement (by Motion Picture Sound Editors): The first Asian to win the award for best sound - Resul Pookutty (for the documentary "India's Daughter").

The "Officer de la Legion d'Honneur" (highest French order of merit, both military and civil): Established - 1802 by Napoleon Bonaparte.

The Pritzker Architecture Prize: Established - 1979 by Jay A. Pritzker and Cindy of Chicago. First awardee - Philip Johnson in 1979.

Ramon Magsaysay Award (Nobel Prize of Asia): Established - 1957 by the Rockefeller Brothers. It is an annual award established to perpetuate former Philippines President Ramon Magsaysay's example of integrity in governance, courageous service to the people, and pragmatic idealism within a democratic society.

The Pulitzer Prize: Established - 1917. Gobind Behari Lal won the Pulitzer in 1937 in the journalism category.

Booker Prize (The Man Booker Prize): It is a literary prize awarded each year for the best novel written in English and published in the United Kingdom or Ireland. It was formerly known as the Booker Prize for Fiction (1969–2001). It was first awarded in 1969. Arundhati Roy was the first Indian to win the Prestigious Booker prize for her work 'The God Of Small Things' in 1997.

Fukuoka Prize to honor the outstanding work of individuals or organizations in preserving or creating Asian culture. It was established in 1990.

Other Awards

Om Prakash Bhasin Award for Science and Technology: Established - 1985. It has been instituted by Shri Om Prakash Bhasin Foundation. First recipient - B. P. Pal in 1985.

G.D. Birla Award for Scientific Research: Established - 1991 by the K. K. Birla Foundation in honor of the Indian

philanthropist Ghanshyam Das Birla. First recipient - Asis Datta for his work in Molecular Biology in 1991.

The Infosys Prize: It was first awarded in 2008 to Manindra Agrawal for his outstanding contribution to the field of complexity theory. Upinder Singh was the recipient of the inaugural Infosys Prize for Social Sciences – History, 2009.

Gomant Vibhushan Award (highest civilian honour of the State of Goa): The first recipient - Anil Kakodkar in 2010.

Gauri Lankesh Memorial Award for Journalism: It is conferred by the Gauri Lankesh Memorial Trust. It was announced in September 2019 on the 2nd death anniversary of Gauri Lankesh. First recipient - Ravish Kumar.

Sansad Ratna Awards - The Sansad Ratna Awards were instituted in 2010, inspired by the teachings of former President APJ Abdul Kalam, to honour "top performing Members of Parliament". TK Rangarajan received the Lifetime Achievement Award during the 13th Sansad Ratna.

The **Amrita Devi Bishnoi Wildlife Protection Award** is a national award instituted by the government of India for wildlife conservation in remembrance of Amrita Devi Bishnoi. It was started at the State level by Rajasthan in 2001.

Golden Peacock Awards are regarded as a benchmark of Corporate Excellence worldwide. It was instituted annually by the Institute of Directors (IOD), India in 1991.

Geeta Chopra Award was introduced for the first time in 1978, in memory of a child named Geeta Chopra. It is a prestigious civilian award given to children under the age of 16 years, for their acts of bravery. It is given under the National Bravery Awards category.

Swachh Vidyalaya Puraskar - It was instituted by the Ministry of Education in 2016-17. Its aim is to honor schools that have undertaken significant steps towards fulfilling the mandate of the Swachh Vidyalaya Campaign.

Indira Gandhi Award for National Integration is a prestigious award accorded by the Indian National Congress in memory of Indira Gandhi. It is given annually from 1985.

Sardar Patel National Unity Award is given in the field of contribution to the unity and integrity of India on the occasion of national unity day (31 october).

The Lokmanya Tilak National Award was started in 1983. The award is annually given on August 1, the death anniversary of Lokmanya Tilak. It is given by Lokmanya Tilak Smarak Trust for strengthening democratic principles.

Religious Places

Famous Temples of India

Andhra Pradesh - Lord Venkateshwara Swamy Temple (Tirupati), Varaha Lakshmi Narasimha Temple (Visakhapatnam), Sri Mallikarjuna Temple (Nandyala), Suryanarayana Swamy Temple (Srikakulam), Ranganatha Temple (Nellore), Veer Bhadra Temple (Anantapur), Srikalahasti Temple (Nellore), Sri Bhramaramba Mallikarjuna Temple (Nandyala), Yaganti Temple (Nandyal), Ahobilam Temple (Nandyal), Kanipakam Vinayaka Temple (Chittoor), Simhachalam Temple (Visakhapatnam), Sri Padmavathi Ammavari Temple (Tirupati).

Arunachal Pradesh - Parshuram Kund Temple (Lohit), Akashganga or Malinithan temple (West Siang).

Assam - Kamakhya Temple (Kamrup), Negheriting Shiva Doul (Golaghat), Siva Dol (Sivasagar), Shri Kedar Temple (Kamrup), Asvakraanta Temple (Kamrup), Basistha Temple (Guwahati), Dhekiakhwa Bornamghar (Jorhat), Bhairabi Temple (Sonitpur), Umananda Temple (Guwahati), Ugra Tara Temple (Kamrup metropolitan), Maha Bhairav Temple (Sonitpur), Tamresveri Temple (Tinsukia), Navagraha Temple (Guwahati), Da Parbatia Temple (Sonitpur), Tilinga Mandir (Tinsukia), Mahamaya Temple (Kokrajhar).

Bihar - Mahabodhi Temple (Bodh Gaya), Vishnupad Mandir (Gaya), Mangala Gowri Temple (Gaya), Mithila Shakti Peeth (Darbhanga), Sundarnath Temple (Araria), Budhanath Temple (Bhagalpur), Shri Adinath Akhara (Buxar), Mittheswarnath Shiv Temple (Darbhanga), Chaumukhi Mahadev (Vaishali), Mundeshwari Temple (Kaimur), Sita Kund (Sitamarhi), Vishnu Dham temple (Siwan).

Chhattisgarh - Maa Bamleshwari Devi Temple (Rajnandgaon), Mahamaya Temple (Bilaspur), Chandahasini Devi temple (Janjgir -Champa), Boramdeo Temple (Kabirdham), Danteshwari Temple (Bastar), Banjari Mata Mandir (Raipur), Jatmai Ghatarani (Gariaband), Shivani Maa Temple (Kanker), Patala Bhairavi Temple (Rajnandgaon), Hatkeshwar Temple (Raipur), Ganga Maiya Temple (Balod).

Goa - Shanta Durga Temple (Ponda), Mangeshi Temple (North Goa), Mahadev Temple (Tambdi Surla), Sri Nagesh Maharudra Temple (Bandora), Shri Saptakoteshwar Temple (North Goa), Maruti Temple (Fontainhas), Shri Damodar Temple (South Goa), Mahalasa Temple (Ponda), Shri Nagesh Temple (Ponda).

Gujarat - Somnath Temple (Gir Somnath), Swaminarayan Akshardham Temple (Gandhinagar), Dwarkadhish Temple (Dwarka), Sun Temple (Modhera), Rukmini Temple (Dwarka), Shri Shatrunjaya Temples (Palitana), Nilkanth Dham Swami Narayan Temple (Poicha), Sandipani Temple (Poicha).

Haryana - Sheetla Mata Mandir (Gurugram), Sita Mai Temple (Karnal), Bhuteshwar Temple (Jind), Geeta Mandir (Kaithal), Chitta Temple (Yamunanagar), Mata Mansa Devi Mandir (Panchkula), Chandi Mandir (Panchkula), Bhadrakali Temple (Kurukshetra), Shiv Mandir (Baghot), Sthaneshwar Mahadev Temple (Kurukshetra), Birla Mandir (Thanesar).

Himachal Pradesh - Bajreshwari Temple (Kangra), Baijnath Temple (Kangra), Jwalamukhi Devi Temple (Kangra), Shri Chamunda Devi Mandir (Kangra), Naina Devi Temple (Bilaspur), Jwala Devi Temple (Kangra), Mata Chintpurni Devi Temple (Una), Bijli Mahadev Temple (Kullu), Jakhu Temple (Shimla).

Jharkhand - Baidyanath Temple (Deoghar), Maluti Temples (Dumka), Jagannath Temple (Ranchi), Chinnamasta Temple (Ramgarh), Sri Sri Kalika Maharani Temple (Bokaro), Harihar Dham (Giridih), Jharkhand Dham (Dhanwarh), Dewri Temple (Ranchi), Naulakha Temple (Deoghar).

Karnataka - Udupi Sri Krishna Temple (Udupi), Vidyashankara Temple (Chikkamagaluru), Chennakeshava Temple (Hassan), Kotilingeshwara Temple (Kolar), Murudeshwar Shiva Temple (Bhatkal), Mahabaleshwar

Temple (Gokarna), Mallikarjuna Temple (Pattadakal), Kedareshwara Temple (Balligavi), Virupaksha Temple (Hampi) Vitthala Temple (Hampi), Meguti Jain Temple (Bagalkot).

Kerala - Sabarimala Temple (Pathanamthitta), Attukal Bhagavathy Temple (Attukal), Sree Krishna Temple (Ambalapuzha), Chottanikkara Temple (Ernakulam), Guruvayur Temple (Thrissur), Shri Padmanabhaswamy Temple (Thiruvananthapuram), Attukal Bhagavathy Temple (Thiruvananthapuram), Ettumanoor Mahadeva Temple (Kottayam), Thirunelli Temple (Wayanad Valley), Tali Temple (Kozhikode).

Madhya Pradesh - Virateshwar temple (Shahdol), Mahakaleshwar Jyotirlinga (Ujjain), Omkareshwar Temple (Khandwa), Rama-Janardana Mandir (Ujjain), Kandariya Mahadev Temple (Khajuraho), Adinath Temple (Khajuraho), Kal Bhairav Temple (Ujjain), Matangeshwar Temple (Khajuraho), Gauri Somnath Mandir (Omkareshwar).

Maharashtra - Grishneshwar Jyotirlinga Temple (Sambhaji Nagar), Yamai Devi Temple (Satara), Moreshwar Temple (Pune), Trimbakeshwar Jyotirlinga (Nashik), Chintamani Temple (Pune), Shirdi Sai Temple, Shani Shingnapur (Ahmadnagar), Bhimashankar Temple (Pune), Siddhivinayak Temple (Mumbai), Harihareshwar Temple (Raigad), Walkeshwar Temple (Mumbai).

Manipur - Mahabali Temple (Imphal), Shri Govindaji Temple (Imphal), Sanamahi Temple (Imphal West), Jorebangla Temple (Bishnupur), Shree Govindajee Temple (Imphal), Gopinath Mandir (Ningthoukhong).

Meghalaya - Mawsynram Temple (Mawsynram), Nartiang Durga Temple (West Jaintia Hills), Mahadev Khola Dham (Shillong), Matri Mandir (Shillong).

Mizoram - Solomon's Temple (Aizawl).

Nagaland - Kalibari Temple (Dimapur).

Odisha - Jagannath Temple (Puri), Sun Temple (Konark), Lingaraja Temple (Bhubaneswar), Rajarani Temple (Bhubaneswar), Mukteshvara Temple (Khurda), Chausath Yogini Temple (Hirapur), Ananta Vasudeva Temple (Bhubaneswar), Gundicha Temple (Puri), Tara Tarini Temple (Ganjam), Mausi Maa Temple (Puri).

Punjab - The Golden Temple (Amritsar), Durgiana Temple (Amritsar), Shri Krishna Mandir (Ludhiana), Mukteshwar Mahadev Temple (Doong), Kali Mata Mandir (Patiala).

Rajasthan - Brahma Temple (Pushkar), Dilwara Temple (Mount Abu), Shrinathji Temple (Nathdwara), Khatu Shyam Temple (Sikar), Eklingji temple (Udaipur), Ambika Mata temple (Jagat), Karni Mata Mandir (Deshnoka), Salasar Balaji Temple (Churu), Mehandipur Balaji Temple (Dausa).

Sikkim - Kirateshwar Mahadev Temple (Gyalshing), Solophok Chardham (South Sikkim), Hanuman Tok (Gangtok), Viswa Vinayak Temple (Reshi), Baba Harbhajan Singh Temple (East Sikkim).

Tamil Nadu - Brahmapureeswarar Temple (Trichy), Kailasanathar Temple (Kanchipuram), Meenakshi Temple (Madurai), Brihadisvara Temple (Thanjavur), Ramanathaswamy Temple (Ramanathapuram), Airavatesvara Temple (Kumbakonam), Suchindram Temple (Kanyakumari), Kapaleeshwarar Temple (Chennai), Nagaraja Temple (Nagercoil).

Telangana - Ramappa Temple (Mulugu), Thousand Pillar Temple (Hanamkonda), Padmakshi Temple (Hanamkonda), Sri Lakshmi Narasimha Swamy Temple (Yadadri), Chilkur Balaji Temple (Hyderabad), Gnana Saraswathi Temple (Adilabad), Sanghi Temple (Hyderabad), Karmanghat Hanuman Temple (Champapet), Bhadrakali Temple (Warangal).

Tripura - Tripurasundari Temple (Matabari), Bhubaneswari Temple (Rajnagar), Kasba kali Temple (Sipahijala), Chaturdash Devata Temple (West Tripura)

Uttar Pradesh - Kashi Vishwanath Temple (Varanasi), Ram Janmabhoomi (Ayodhya), Hanuman Garhi (Ayodhya), Shri Krishna Janambhoomi (Mathura), Dwarkadheesh Temple (Mathura), Sarnath Temple (Varanasi), Maa Vindhyavasini Temple (Vindhyachal), Shri Banke Bihari Temple (Vrindavan).

Uttarakhand - Kedarnath Temple (Rudraprayag), Gangotri Temple (Uttarkashi), Har ki Pauri (Haridwar), Yamunotri Temple (Uttarkashi), Neelkanth Mahadev Temple (Rishikesh), Maa Naina Devi Temple (Nainital), Koteswar Mahadev Temple

(Rudraprayag), Kainchi Dham Temple (Nainital), Madhyamaheshwar Temple (Garhwal), Rudranath Temple (Chamoli), Kartik Swami Temple (Rudraprayag).

West Bengal - Kalighat Kali Temple (Kolkata), Dakshineswar Kali Mandir (Kolkata), Taraknath Temple (Hooghly), Chinese Kali Mandir (Tangra), Belur Math (Belur), Taraknath Temple (Tarakeswar).

Jammu & Kashmir - Amarnath Temple (Anantnag), Vaishno Devi Temple (Reasi), Shankaracharya Temple (Srinagar), Raghunath Temple (Jammu), Martand Sun Temple (Anantnag).

Ladakh - Manjushree Temple (Alchi), Vajra Bhairav Shrine (Spituk), Serzang Temple (Leh), Lamokhang Temple (Thiksey), Matho Temple (Leh).

Puducherry - Arulmigu Manakula Vinayagar, Sri Gokilambal Thirukameswar Temple, Sree Krishna Temple.

Delhi - Lotus Temple (New Delhi), Chhatarpur Temple, Kalkaji Temple (New Delhi), Swaminarayan Akshardham, Gauri Shankar Temple (Chandni Chowk).

Dadra and Nagar Haveli and Daman & Diu - The Om Temple (Silvassa), Bindrabin Temple (Khadoli).

Andaman and Nicobar Islands - Sri Vetrimalai Murugan Temple (Port Blair), Shiva Temple (Port Blair).

Famous Temples of World

Swaminarayan Akshardham - Largest Hindu temple in the USA (New Jersey).

Angkor wat, Cambodia - One of the world's largest religious monuments. It was built by Khmer King Suryavarman II in the 12th century.

Sri Subramanya Swamy Temple, Malaysia - statue of Bhagwan Murugan standing at the entrance of the Batu Caves is considered to be his highest figure in the world.

Arulmigu Sri Rajakaliamman Glass Temple - It is the world's only Hindu temple built of glass situated in Malaysia.

Sri Siva Subramaniya Temple : A Dravidian style Hindu temple in Nadi, Fiji. It is the largest Hindu temple in the Pacific.

Tanah Lot Temple : one of the seven seaside temples, each temple lining up with the next and forming a lengthy chain on Bali's coast, Indonesia. It houses ancient Balinese mythology.

Sri Venkateswara Balaji Temple : The first temple of Lord Venkateswara in Europe. This temple was inaugurated on August 23, 2006. The statue of Lord Vishnu here is about 12 feet high.

Shri Swaminarayan Mandir, London : It was inaugurated in 1995. The stones used in the construction of the temple were not used before in the construction of any Hindu temple.

Ganesh Temple, Paris : It was established in 1985. All priests of this temple come from Madras. The Ganesh statue, made in Sri Lanka with five special metals, is the main attraction of this temple.

Pashupatinath Temple : Dedicated to Lord Pashupati (Lord Shiva), and is located in Kathmandu, Nepal near the Bagmati River. The temple was classified as a World Heritage Site in 1979.

Prambanan Temple : A 9th-century Hindu temple compound in Java, Indonesia. It is dedicated to the Trimurti (Brahma, Vishnu, Mahesh). It is the largest Hindu temple site in Indonesia and the second-largest in Southeast Asia after Angkor Wat. It was designated UNESCO World Heritage Site in 1991.

Shri Hinglaj Mata Temple is the largest Hindu pilgrimage center in Pakistan. It is situated in Balochistan province about 250 km west-northwest of Karachi.

Dhakeshwari Temple, the National Temple (state owned) of Bangladesh, is the largest Hindu temple in Bangladesh. The temple is one of the 51 Shakti Peethas.

Dubai Hindu Temple : The second Hindu temple in the UAE (the other being the BAPS Hindu Mandir in Abu Dhabi). The architecture of the temple has a unique Arabian look.

Shri Kali Temple : A Hindu temple located in Yangon, Burma. It was built by Tamil migrants in 1871, whilst Burma Province was part of British India.

Famous Churches

Basilica of Bom Jesus church: Located - Goa regions. This heritage site is said to

have the earthly remains of St. Francis Xavier, one of the founders of Jesuits.

St. Thomas Church: Located - Thrissur district in Kerala is the oldest church in India. It was established in 52 AD by St Thomas.

St. Francis Church: Located - Kochi city. It is India's oldest European church. "Vasco da Gama" was laid to rest in this Church.

Padri Ki Haveli: Located - Patna. It is the oldest church in Bihar.

St. Peter's Basilica church: Located - Vatican City, Rome. It is the largest church in the world. 'Church' is related to 'Christians' in the same way 'Synagogue' is related to 'Jews'.

Some other Churches in India- Santhome Cathedral Basilica (Chennai Tamil Nadu), Santa Cruz Basilica (Fort Kochi, Kerala), St. Philomena's Church (Mysore Karnataka), Christ Church (Shimla, Himachal Pradesh), St. Paul's Cathedral (Kolkata, West Bengal), Immaculate Conception Cathedral (Gurugram, Haryana), St. Mary's Basilica (Bangalore, Karnataka), Velankanni Church (Tamil Nadu), Medak Cathedral (Telangana), Cathedral Church of The Redemption (New Delhi, Delhi), Basilica of Our Lady of the Mount (Mumbai, Maharashtra), Sacred Heart Cathedral (New Delhi, Delhi), All Saints Cathedral (Allahabad, Uttar Pradesh), St. Thomas Major Archdiocesan Shrine, Palayoor (Thrissur district in Kerala), St. Thomas Cathedral (Mumbai), St. James Church (Delhi), All Saints Church (Pune, Maharashtra).

Famous Gurudwaras

Anandpur Sahib: Located - Rupnagar district (Roper) Punjab, India. It is considered as the holiest place of Sikh religion where Shri Guru Gobind Singh Ji spent 25 years of his life teaching and guiding his disciplines.

Goindwal Sahib: Located - On the bank of the river Beas and is one of the main centres of small scale industries in Taran Taran district. Guru Amar Das, the third sikh Guru, stayed in Goindwal for 33 years.

Hemkund Sahib (Gurudwara Shri Hemkund Sahib Ji): Located - Chamoli district, Uttarakhand. It is devoted to Guru Gobind Singh, the tenth Sikh Guru,

and finds mention in Dasam Granth.

Nankana Sahib: Located - Nankana Sahib District in the Punjab province of Pakistan. Nankana Sahib had previously been known as Rai Bhoi Ki Talvandi but was eventually renamed in honour of Guru Nanak.

Patna Sahib (Takht Sri Harimandir Ji): Located - On the banks of Ganga River Patna. It is the place where Guru Gobind Singh Ji was born in the year of 1666. It was also constructed by Maharaja Ranjit Singh.

The Fatehgarh Sahib (Sirhind): Located - Fatehgarh Sahib district of Punjab. The two young sons of Sikh Guru Gobind (Tenth Guru of Sikhs) were killed at this Sirhind, during the rule of Wazir Khan.

Taran Taran Sahib: Located - Tarn Taran district Punjab, India. It was founded by Guru Arjan Dev, who was the fifth Guru of Sikh community.

Baba Bakala: Located - Amritsar District of Punjab, India. It is closely associated with the 9th Sikh Guru, Guru Tegh Bahadur.

Gurdwara PatalPuri: Located - Rupnagar district in Punjab. It is built on the banks of the River Sutlej. Guru Hargobind in 1644 as well as Guru Har Rai in 1661 were cremated in Gurdwara PatalPuri.

The Golden Temple (Sri Harmandir Sahib Amritsar): Located - Amritsar, Punjab, India. Harmandir Sahib also known as Golden Temple was built by Guru Arjan Dev (1581-1606).

Kartarpur Sahib corridor: It connects Dera Baba Nanak Shrine in Gurdaspur, India to Kartarpur Gurudwara in Pakistan, which is located near the banks of the river Ravi.

Gurdwara Sis Ganj Sahib: Located in Delhi. Guru Tegh Bahadur was beheaded here on 11 November 1675 on the order of Aurangzeb.

Takht Sri Damdama Sahib: Located in Talwandi Sabo, Punjab. It was constructed in the memory of Guru Gobind Singh. It is also known as Guru Ki Kashi.

Famous Masjids

Jama Masjid, Delhi: It was built by Mughal emperor Shah Jahan. Shah Jahan was the 5th Mughal Emperor.

Moti Masjid (In Red Fort of Delhi): It was built by Emperor Aurangzeb in 1659-60.

Moti Masjid (in Red Fort of Agra): It was built by Shah Jahan.

Jama Masjid (Agra): Located - Opposite to the Agra fort.

Nagina Masjid: Located - Agra fort. It was built by Mughal emperor Shah Jahan in Bengal style.

Taj-UI-Masjid: Located - Bhopal and is considered as one of the largest mosques in India, and also one of the largest mosques in Asia.

Ajmer Sharif Dargah: Located - Ajmer, Rajasthan. It is a dargah of Sufi saint, Moinuddin Chishti. It was built by the Mughal Emperor Humayun.

The Haji Ali Dargah: Located - On an islet off the coast of Worli in the southern Mumbai. It is a mosque and dargah or the monument of Pir Haji Ali Shah Bukhari.

Kabuli Bagh Mosque: Located - Panipat, Haryana. It was built by the emperor Babur to mark his victory over Sultan Ibrahim Lodhi at the first Battle of Panipat in 1526.

Moorish Mosque: Located - Kapurthala Punjab. It is patterned on the lines of the Grand Mosque of Marrakesh, Morocco. It was commissioned by Maharaja Jagatjit Singh, the last ruler of Kapurthala.

Katra Masjid: Located - Murshidabad, West Bengal. It was built during the 18th century.

Al-Haram Mosque: Located - Mecca. It is the largest mosque in the world. It was first built under the leadership of the Caliph Omar Ibn al-Khattab.

Al-Aqsa Mosque (Qibli Mosque or Qibli Chapel): It is the main congregational mosque or prayer hall in the Al-Aqsa mosque compound in the Old City of Jerusalem.

Famous Places in India

Mawsynram is a town in the East Khasi Hills district of Meghalaya. Mawsynram receives the highest rainfall in India. It is reportedly the wettest place on Earth.

Salar Jung Museum is an art museum located at Dar-ul-Shifa, on the southern bank of the Musi River in the city of

Hyderabad, Telangana.

Varanasi is also known as Banaras and Kashi, is a city on the Ganges river. It is one of the world's oldest cities.

Aurangabad is officially known as Chhatrapati Sambhaji Nagar. It is situated on the Kham river in Maharashtra.

Nathdwara is a city of Shreenathji (Swarup of Krishna) in the Rajsamand district of the state of Rajasthan.

Kushinagar is a town in the Kushinagar district in Uttar Pradesh. It is an important and popular Buddhist pilgrimage site, where Lord Buddha attained Mahaparinirvana.

Panipat is a historic city in Haryana.

Kurukshetra is a city in the Indian state of Haryana. It is also known as Dharmakshetra ("Realm of duty") and as the "Land of the Bhagavad Gita".

Surajkund is an ancient reservoir of the 10th century located on Southern Delhi Ridge of Aravalli range in Faridabad city of Haryana.

Silchar is a city and the headquarters of the Cachar district of the state of Assam. It is located on the banks of the Barak River.

Dhubri is an old town and headquarter of Dhubri district in the Indian state of Assam.

Dibrugarh serves as the headquarters of Dibrugarh district in the state of Assam in India. It serves as the headquarters of the Sonowal Kachari Autonomous Council.

Tezpur is a city and urban agglomeration in Sonitpur district, Assam. It is located on the banks of the river Brahmaputra.

Capitol Complex, located in Chandigarh city in India, is a government compound designed by the architect Le Corbusier. It was added to the UNESCO World Heritage Site List in 2016 along with sixteen other works by Le Corbusier for its contribution to the development of modernist architecture.

Pachmarhi is a hill station in Hoshangabad district of Madhya Pradesh state of central India. It is widely known as Satpura ki Rani ("Queen of Satpura"), situated at a height of 1067 m

in a valley of the Satpura Range in Narmadapuram district.

Amarkantak is a pilgrim town in Anuppur, Madhya Pradesh. This is where the Narmada River, the Son River and Johilla River (tributary of Son) originate.

Jagdalpur is a city in Bastar district in the state of Chhattisgarh. It is the fourth largest city of Chhattisgarh. The city is one of the most important commercial, financial and political hubs in Chhattisgarh.

Kodaikanal is a hill station which is located in Dindigul district in the state of Tamil Nadu. Its name in the Tamil language means "The Gift of the Forest". Kodaikanal is referred to as the "Princess of Hill stations".

Mount Abu is a hill station in the Aravalli Range in Sirohi district of the state of Rajasthan. The highest peak on the mountain is Guru Shikhar.

Badrinath is a town in Chamoli district in the state of Uttarakhand. It is a Hindu holy place and it is one of the four sites in India's Char Dham pilgrimage and is also part of India's Chota Char Dham pilgrimage circuit.

Panhala is a city and a Hill station in Kolhapur district in Maharashtra. The main historical attraction here is the Panhala fort.

Kalinga is a historical region of Odisha, India. The Kalingas have been mentioned as a major tribe in the legendary text Mahabharata.

Nankana Sahib, earlier known as Rai Bhoi di Talvandi is birthplace of the founder of Sikhism, Guru Nanak. It is a city in Pakistan's Punjab province.

Netaji Birth Place Museum is located at Cuttack district of Odisha.

Kanyakumari, earlier known as Cape Comorin, is a city in Kanyakumari district in Tamil Nadu. It is the southern tip of the Indian subcontinent and the southernmost city in mainland India, and thus referred to as "The Land's End".

The **Rampurva** capitals are the capitals of a pair of Ashoka Pillars discovered in 1876 by A. C. L. Carlleyle. The archaeological site is called Rampurva, and is located in the West Champaran district of the Indian state of Bihar.

Jai Prakash Narayan Museum of Socialism is situated in Gomti Nagar, Lucknow. He joined the Indian National Congress in 1929 when he returned from the United States.

Sher Mandal is a 16th-century historic building within the Purana Qila located in Delhi. Designed in a blend of Timurid and Safavid architecture.

Kolhapur is a city on the banks of the Panchganga River in the southern part of the Indian state of Maharashtra.

The **Nehru Memorial Museum & Library** (NMML) is a museum and library in New Delhi, which aims to preserve and reconstruct the history of the Indian independence movement. Housed within the Teen Murti House complex, it is an autonomous institution under the Indian Ministry of Culture.

The **Nehru Memorial Museum and Library Society** was formed on 1 April 1966. It was designed by Robert Tor Russell who also designed Connaught Place and a few parts of Janpath. It aims to foster academic research on modern and contemporary history.

Bharat Bhavan is an autonomous multi-arts complex and museum in Bhopal, Madhya Pradesh. The architect of the Bharat Bhavan is Charles Correa.

The **Mattancherry Palace** is a palace popularly known as the Dutch Palace, in Mattancherry, Kochi, in the Indian state of Kerala.

Hill Palace is an archaeological museum and palace located in the Tripunithura neighbourhood of Kochi, Kerala.

The **Indian Museum** in central Kolkata, West Bengal, is the ninth oldest museum in the world, the oldest in India as well as the largest in Asia.

Superlatives (India/World)

India

Largest bird sanctuary - Bharatpur Bird Sanctuary (Rajasthan)

Largest Fort - Chittorgarh Fort (Rajasthan)

Longest glacier - Siachen Glacier

Longest road tunnel - Syama Prasad Mookerjee Road Tunnel (Jammu and Kashmir)

Longest canal - Indira Gandhi canal	on Brahmaputra river (Assam)	Largest sea-bird - Wandering Albatross
Longest Beach - Marina beach (Chennai)	Largest river in South India - Godavari	Largest country (area) - Russia
Longest corridor - Ramanathaswamy Temple (Tamil Nadu)	Largest river without delta - Narmada	Largest diamond - Cullinan
Longest bridge - Dhola-Sadiya Bridge (Bhupen Hazarika Setu) Assam	Longest river - Ganga River	Largest volcano - Mauna Loa (Hawaii Islands)
Longest railway platform - Sri Siddharudha Swamiji Railway Station (Hubli, Karnataka)	Longest tributary river - Yamuna River	Largest church - St. Peter's Basilica (Italy)
Longest Train Route - Dibrugarh (Assam) to Kanyakumari (Tamil Nadu)	Shortest river - Arvari River (Rajasthan)	Largest mosque - Al-Haram Mosque (Mecca, Saudi Arabia)
Longest coastline - Gujarat	Deepest River Valley - Bhagirathi and Alaknanda	Largest temple - AngkorVat (Cambodia)
Highest peak - Kanchenjunga (8586m)	Oldest Church - St. Thomas Church (Palayar, Kerala)	Largest freshwater lake (By Surface Area) - Lake Superior (North America)
Highest waterfall - Kunchikal Falls (Karnataka)	Largest church - St. Francis Church (Kerala)	Largest freshwater lake (by volume) - Lake Baikal (Russia)
Highest rainfall - Mawsynram (Meghalaya)	Largest state (Area) - Rajasthan	Largest artificial lake - Lake Kariba (Africa)
Highest gateway - Buland Darwaza (Fatehpur Sikri)	Largest museum - Indian Museum (Kolkata)	Largest saltwater lake - Caspian Sea
Highest literacy state - Kerala	Largest zoo - Arignar Anna Zoological Park (Chennai)	
Highest Civilian Award - Bharat Ratna	Largest dome - Gol Gumbaz (Karnataka)	
Smallest state - Goa	Largest mosque - Taj-ul-Masajid (Bhopal)	
Largest state (by forest area) - Madhya Pradesh	Largest desert - Thar (Rajasthan)	
Smallest state (by forest area) - Haryana	Largest delta - Sundarbans delta	
Busiest Bridge - Howrah (Kolkata)	Largest cave temple - The Kailash temple (Ellora caves, Maharashtra)	
Oldest oil Refinery - Digboi Refinery (Assam)	Largest animal fair - Sonepur fair (Bihar)	
Largest freshwater lake - Wular lake (Jammu and Kashmir)	Largest library - National Library (Kolkata)	
Largest saltwater lake - Chilka Lake (Odisha)	Largest gurudwara - Golden Temple (Amritsar)	
Largest Inland Saline Water Lake - Sambhar lake (Rajasthan)		
Largest artificial lake - Gobind Ballabh Pant Sagar Lake (Uttar Pradesh)		
Oldest Dam - Kallanai Dam (Tamil Nadu)		
Largest dam - Bhakra Nangal Dam (Himachal Pradesh)		
Longest dam - Hirakud (Odisha)		
Largest arch dam - Idukki dam (Kerala)		
Largest plateau - Deccan Plateau		
Largest river island - Majuli river island		

Roadways

Classification of Roads in India

In India, roads are classified in the six classes.

1. Golden Quadrilateral Highways
2. National highway
3. State highway
4. District Roads
5. Rural / Village Roads
6. Border Roads

National Highway Authority of India - It was set up under NHA act, 1988. NHA was operationalised in 1995. It comes under the ministry of Road Transport and Highways. It is responsible for the development, maintenance and management of National Highways.

Important national Highways in India

National Highway 1 (old) / Sher Shah Suri Marg - Its old name was the Grand Trunk road. It was built by Sher Shah Suri. It linked the national capital New Delhi to the town of Attari in Punjab near the India–Pakistan border.

National Highway 1 (NH 1) - It runs between the union territories of Jammu & Kashmir and Ladakh.

National Highway 3 - It starts from Atari and terminates at Leh in Ladakh.

World

Largest desert (Cold) - Antarctic Desert

Largest Flower - Rafflesia arnoldii
Biggest library - Library of the Congress (United State)

Smallest Bird - Hummingbird

Smallest country - Vatican City

Lowest water body - Dead Sea

Longest river - Nile River, Largest River - Amazon

Highest Waterfall - Angel Falls (Salto Angel) Venezuela

Deepest Lake - Lake Baikal (Russia).

National Highway 4 - It runs through Port-Blair to Diglipur. It is also known as the great Andaman Trunk Road.

National Highway 5 - It runs through Ferozepur in Punjab to the Sino-Indian border at Shipki La in Himachal Pradesh traversing Haryana and Chandigarh.

National Highway 7 - It connects Hindu pilgrim centers of Rishikesh, Devprayag, Rudraprayag, Karnaprayag, Chamoli, Joshimath and Badrinath. It passes through the Indian states of Punjab, Haryana, Himachal Pradesh, and Uttarakhand.

National Highway 10 - It connects the Indo/Bangladesh border via Siliguri to Gangtok. It passes through the Indian states of West Bengal and Sikkim. Atal Setu Bridge constructed by NHIDCL, the longest roadway bridge of Sikkim is also a part of National Highway 10.

National Highway 11 - It connects Jaisalmer in Rajasthan to Rewari in Haryana.

National Highway 16 - It runs along the east coast of India which lies around Bay of Bengal and connects West Bengal, Odisha, Andhra Pradesh, and Tamil Nadu.

National Highway 19 - It connects Agra to Kolkata. It was previously referred to as Delhi-Kolkata Road or NH 2.

National Highway 23 - It runs entirely in the state of Rajasthan.

National Highway 27 - East - West highway that connects Porbandar in Gujarat to Silchar in Assam. It passes through the 7 states: Gujarat, Rajasthan, Madhya Pradesh, Uttar Pradesh, Bihar, West Bengal, Assam.

National Highway 44 - It is the longest national highway in India running over 4,112 km from Srinagar to Kanyakumari. It passes through 13 State / UTs. It covers the North-South Corridor of NHDP.

National Highway 48 - It connects Delhi to Chennai and passes through 7 states Delhi, Haryana, Rajasthan, Gujarat, Maharashtra, Karnataka and Tamil Nadu. Its stretch from Delhi to Mumbai was earlier designated as NH.

National Highway 548 - It is the shortest (5 Km) highway of India. It is a spur road of National Highway 48 and is situated in Maharashtra.

Expressways of India

Ahmedabad Vadodara Expressway / National Expressway 1 (NE 1) - It is the 93.1 km long expressway.

Eastern Peripheral Expressway (KGP expressway) - It connects Kundli in Haryana to Palwal and Ghaziabad. It is named National Expressway - 2 (NE 2).

Delhi Meerut Expressway / National Expressway 3 (NE 3) - It is the widest expressway in India.

Delhi Mumbai Expressway / National Expressway 4 (NE 4) - It is the longest Expressway in India. It passes through states of Delhi, Haryana, Rajasthan, Madhya Pradesh, Gujarat and Maharashtra.

Yamuna Expressway - Its former name was Taj expressway. It starts from Pari Chowk in Greater Noida and ends at Kuberpur, Agra. The project was implemented by Jaypee Infratech Limited.

Western Peripheral Expressway - It is known as Kundli - Manesar - Palwal expressway. It is in Haryana.

Mumbai-Nagpur Expressway - Its official name is Hindu Hrudaysamrat Balasaheb Thackeray Samruddhi Mahamarg. It is a 701 km long access - controlled expressway in Maharashtra.

Char Dham project : The objective of this project is to improve connectivity between four pilgrimages centers (Badrinath, Kedarnath, Gangotri, Yamunotri).

National Common Mobility Card (NCMC) : This card is launched to pay multiple kinds of transport charges, including metro services and toll tax, across the country.

Tunnels

Shyama Prasad Mukherjee Tunnel : It is India's 2nd longest highway tunnel (9 km long). It is located in Jammu & Kashmir.

Banihal Qazigund Tunnel: It is in the Pir Panjal range of Jammu and Kashmir connecting Banihal and Qazigund.

Jawahar Tunnel: It is also called Banihal Tunnel or Banihal Pass. The tunnel facilitates connectivity between Srinagar and Jammu.

Zojila Tunnel: It is located on the Srinagar - Leh section of the NH 1. The tunnel is between Baltal and Minamarg.

Atal Tunnel: It is located in Himachal Pradesh. It is the world's longest highway tunnel. Its length is 9.02 km.

Trivandrum Port Tunnel: It is located in Kerala.

Karwar Tunnel: It is located in Karnataka.

Railways

The first railway proposals for India were made in Madras in 1832. India's first railway line was named Red Hill Railroad and was built by Arthur Cotton to transport laterite stone for road-building.

India's first passenger train, operated by the Great Indian Peninsula Railway and hauled by three steam locomotives (Sahib, Sindh and Sultan), ran for 34 kilometers (21 miles) with 400 people in 14 carriages on 5 ft 6 inch broad gauge track between Bori Bunder (Mumbai) and Thane on 16 April 1853. Lord Dalhousie was known as the "Father of Indian Railways".

Indian Railways is the fourth-largest network in the world (United States, China, Russia, India, Canada).

In 1864, the north got its first station - The Delhi Junction. The oldest one in the city, it was a major station and junction and remains so till date. It was first established near Chandni Chowk in 1864 when trains from Howrah/Calcutta started operating up to Delhi. The current building was made operational in 1903.

Important Facts about Railway

The Railway Board was made subordinate to the Government. In 1905, its powers were formalized by Lord Curzon's government.

Indian Railways was nationalized in the year 1951.

The Stockton and Darlington Railway (S&DR) was the world's first public railway to use steam locomotives in 1825.

The Deccan Queen (Introduced in June 1930) was India's "first superfast train, first long-distance electric-hauled train, first vestibuled train, the first train to have

a 'women-only' car, and the first train to feature a dining car".

Dibrugarh – Kanyakumari Vivek Express is the longest train of Indian Railways which runs from Dibrugarh (Assam) to Kanyakumari (Tamil Nadu). The train covers a distance of 4189 kilometers and traverses through nine states in India.

'Bholu the Train Manager' (Train Guard) is the mascot of Indian Railways, represented as a cartoon of an elephant holding a signal lamp with a green ring in one hand.

The Trans-Siberian Railway is the longest railway system connecting St. Petersburg in Western Russia to Vladivostok on the Pacific coast.

The London Underground first opened as an underground railway in 1863 and its first electrified underground line opened in 1890. It is the world's oldest metro system. The Shanghai Metro is the world's longest metro network.

Steam Locomotives: The manufacture of steam locomotives in the country was stopped in 1972.

Electric Trains: First Electric coach ran on the 3rd February 1925 between Bombay VT and Kurla stations.

Computerized Reservations: First Reservation through the computer System commenced in New Delhi in the year 1986.

Railway Zones (Headquarters)

Central Railway (Mumbai), Eastern Railway (Kolkata), East Central Railway (Hajipur), East Coast Railway (Bhubaneswar), Northern Railway (New Delhi), North Central Railway (Allahabad), North Eastern Railway (Gorakhpur), Northeast Frontier Railway (Maligaon, Guwahati), North Western Railway (Jaipur), Southern Railway (Chennai), South Central Railway (Secunderabad), South Eastern Railway (Garden Reach, Kolkata), South East Central Railway (Bilaspur), South Western Railway (Hubli), Western Railway (Church Gate, Mumbai), West Central Railway (Jabalpur), Metro Railway (Kolkata), Konkan Railway (Navi Mumbai), South Coast Railway (Visakhapatnam)

The city of Kolkata has the maximum number of Railway Headquarters in the country and some of the oldest. Moreover, the biggest Railway station in

the country is the Howrah Junction, located in the state of West Bengal.

Metro Rail in India

The introduction of metro rail in India began in Kolkata (formerly Calcutta) with the Kolkata Metro. It was the first underground metro system in the country and started its operations in 1984.

The project was executed by the Kolkata Metro Rail Corporation (KMRC) and the planning and design assistance was provided by the Soviet Union (now Russia).

The second oldest after the Kolkata Metro is the Delhi Metro (2002), which is the largest and busiest network of all.

Railway gauges used in India

There are four types of railway gauges used in India: Broad Gauge- wide gauge or large line - 5 ft 6 inch (1676 mm).

Standard gauge (Delhi Metro) - 4 ft 8 $\frac{1}{2}$

inches (1435 mm). Meter Gauge- 3 ft 3 $\frac{3}{8}$

inches or 1,000 mm. Narrow Gauge (Small Line) - 2 ft 6 in (762 mm).

Famous Railways Training Institutes

National Academy of Indian Railways (Vadodara) ; Indian Railways Institute of Civil Engineering (Pune) ; Indian Railways Institute of Mechanical and Electrical Engineering (Jamalpur) ; Indian Railways Institute of Signal and Telecommunications Engineering (Secunderabad) ; Indian Railways Institute of Electrical Engineering (Nasik) Indian Railways Institute of Transportation Management (Lucknow).

Toy Trains of India

Toy Trains are running to the hill station on the narrow gauge. The bogies are also very small like toys, so they are called toy trains.

Kalka-Shimla Railway - It was inaugurated by the then British Viceroy, Lord Curzon in November 1903, covering a distance of 96 km.

Darjeeling Himalayan Railway - This 88.48 kilometer long section connecting Darjeeling with the railhead at Siliguri and

has been given the status of a World Heritage Site.

Nilgiri Mountain Railway - It is 46 kilometers long and runs on a 1000 millimeter gauge railway connecting Mettupalayam to Ooty. It is the only rack railway in India that was declared a UNESCO World Heritage Site in 2005.

Neral-Matheran Railway - Matheran Railway links Neral, which lies on the Mumbai-Pune route. It is a heritage railway in Maharashtra, built between 1901 and 1907 by Abdul Hussein Adamjee Peerbhoy.

Pathankot - Jogindernagar Railway (Kangra Valley Railway) - The work on this line started in 1926. Three years later, this 163 kilometer route was opened to traffic.

Railway Production Units In India

Integral Coach Factory (Perambur, Chennai), Chittaranjan Locomotive Works (Chittaranjan, West Bengal), Diesel Locomotive Works (Varanasi), Rail Wheel Factory (Yelahanka, Bangalore), Diesel Loco Modernisation Works (Patiala), Rail Coach Factory (Kapurthala), Modern Coach Factory (Raebareli), Rail Wheel Plant (Bela, Bihar).

Famous Trains in India

Rajdhani Express - It runs from New Delhi, the country's capital, to several capitals of states or the major cities in multiple states. It was started on 1 March 1969.

Shatabdi Express - A series of fast passenger trains operated by Indian Railways to connect important metro cities. The trains were first introduced in 1988 on the birth centenary of India's first Prime Minister Jawaharlal Nehru.

Duronto Express - These deluxe, superfast, long-distance trains link various Indian state capitals and metropolises. These trains were introduced in 2009 and were named 'Duronto', meaning 'speedster' in Bengali.

Jan Shatabdi Express - It is a less expensive variant of the Shatabdi Express or a more affordable and economical version of the Shatabdi Express, It Started in 2002.

Humsafar Express - The first service was inaugurated on 16 December 2016 between Gorakhpur and Anand Vihar Terminal (New Delhi).

Garib Rath - Garib Rath Premium trains are low-cost, fully air-conditioned trains that offer subsidized high-speed connections. It is a 3-tiered long-distance train. It started in October 2006.

Kavi Guru Express - The train is named after Rabindranath Tagore, the Nobel laureate, also known as Kavi Guru in Bengali. It was introduced by Indian Railways in 2011.

Vande Bharata Express : This express is the Second Indian semi-high-speed under the Indian government's Make in India initiative. This Express is also known as 'Train 18'. The train was launched on 15 February 2019 between New Delhi and Varanasi Junction. It was designed and manufactured by Integral Coach Factory (ICF) Chennai.

Sampark Kranti Express - They are high-speed premium trains that travel from New Delhi, the national capital, to each state's capital or major cities.

Gatimaan Express : This express is the first semi high-speed train service started between Delhi and Agra.

Bandhan Express : This train is an international express rail service which runs between the Indian city Kolkata and the Bangladeshi city Khulna every week.

Thar Express : The Thar Express was an international passenger train that ran between the Bhagat Ki Kothi, a suburban area of Jodhpur in India and Karachi Cantonment of Karachi in the Pakistani.

Train runs between India and Pakistan (Samjhauta Express), Train runs between India and Bangladesh (Maitree Express, Mitali Express).

World's Longest Railway Platform Stations

Shri Siddharoodha Swami ji station, Hubli Junction (Karnataka) - 1507 meters. Gorakhpur Junction (Uttar Pradesh) - 1366 meters. Kollam Junction (Kerala) - 1180.5 meters.

Tunnels

Natuwadi Tunnel : It is located in Maharashtra.

Tike Tunnel : It is located in the Western Ghats region of the state of Maharashtra.

Maliguda Tunnel : It is located in Odisha.

Berdewadi Tunnel : It is located in Maharashtra.

Pir Panjal Railway Tunnel : It is India's longest transportation railway tunnel. The tunnel stretches through the Pir Panjal mountain range between Qazigund and Baramulla.

Sangaldan Railway Tunnel : It is Situated in the middle Himalayan region of Jammu and Kashmir.

Karbude Tunnel : It is located on the Konkan railway. It is the longest railway tunnel in the Konkan railway.

Patalpani Rail Tunnel : It is located in Madhya Pradesh.

Tunnel T-49 : It is located in Jammu & Kashmir.

Rapur Tunnel : It is located in Andhra Pradesh. It is India's longest electrified rail tunnel in India.

Savarde Railway Tunnel : It is located in Maharashtra.

Barcem Tunnel : It is located in Goa.

Borali BG Tunnel : It is located in Assam.

Airports

Airports Authority of India (AAI) was constituted by an Act of Parliament and came into being on 1st April 1995 by merging erstwhile National Airports Authority and International Airports Authority of India.

On the occasion of the Kumbh Mela, the world's first airmail service was launched in India at Allahabad on February 18, 1911. In 1912, the first domestic air route between Karachi and Delhi was opened by the Indian State Air Services in collaboration with Imperial Airways, United Kingdom.

Airports in India

Andhra Pradesh - Tirupati Airport (Renigunta, Tirupati), Vijayawada International Airport (Gannavaram), Rajahmundry Airport (Madhurapudi).

Arunachal Pradesh - Donyi Polo Airport (Itanagar), Daparizo Airport, Tezu Airport,

Pasighat Airport, Along Airport.

Assam - Lokpriya Gopinath Bordoloi International Airport (Guwahati), Jorhat Airport, Tezpur Airport, Dibrugarh Airport (Mohanbari).

Andaman and Nicobar Islands - Veer Savarkar International Airport (Port Blair), INS Baaz (Campbell Bay), Car Nicobar Air Force Base (Car Nicobar).

Bihar - Gaya International Airport (Gaya), Jayaprakash Narayan International Airport (Patna), Darbhanga Airport.

Chandigarh - Shaheed Bhagat Singh International Airport (Jhiurheri in Punjab, It is a customs airport serving the union territory of Chandigarh).

Chhattisgarh - Swami Vivekanand International Airport (Raipur), Bilaspur Airport, Jagdalpur Airport.

Delhi - Indira Gandhi International Airport (New Delhi), Safdarjung Airport.

Goa - Goa International Airport (Dabolim), Manohar International Airport (Mopa, North Goa).

Gujarat - Sardar Vallabhai Patel International Airport (Ahmedabad), Surat International Airport (Magdalla), Rajkot International Airport (Hirasar), Jamnagar Airport, Bhavnagar Airport, Vadodara Airport, Porbandar Airport.

Haryana - Maharaja Agrasen International Airport (Hisar), Bhiwani Airport, Pinjore Airport (Panchkula).

Himachal Pradesh - Kangra Airport (Gaggal), Shimla Airport, Kullu Manali Airport.

Jharkhand - Birsa Munda Airport (Ranchi), Chakulia Airport (Near Singhbhum District), Sonari Airport (Jamshedpur), Bokaro Airport.

Jammu and Kashmir - Sheikh ul-Alam International Airport (Budgam), Jammu Airport.

Karnataka - Kempegowda International Airport (Bangalore), Mangaluru International Airport (Bajpe).

Kerala - Cochin International Airport (Kochi), Calicut International Airport (Karipur, Malappuram), Thiruvananthapuram International Airport (Chackai), Kannur International Airport (Mattanur).

Lakshadweep - Agatti Airport (Agatti Island).

Madhya Pradesh - Devi Ahilyabai Holkar International Airport (Indore), Raja Bhoj International Airport (Bhopal), Jabalpur Airport (Dumna), Rajmata Vijaya Raje Scindia Airport (Gwalior), Khajuraho Airport (Chhatarpur).

Maharashtra - Chhatrapati Shivaji Maharaj International Airport (Mumbai), Bharat Ratna Dr. Baba Saheb Ambedkar International Airport (Nagpur), Akola Airport, Jalgaon Airport, Sholapur Airport, Shri Guru Gobind Singh Ji Airport (Nanded), Chhatrapati Rajaram Maharaj Airport (Kolhapur).

Manipur - Bir Tikendrajit International Airport (Imphal)

Meghalaya - Shillong Airport (Umroi)

Mizoram - Lengpui Airport (Aizawl)

Nagaland - Dimapur Airport (Chumukedima)

Odisha - Biju Patnaik International Airport (Bhubaneswar), Jeypore Airport, Veer Surendra Sai Airport (Jharsuguda).

Punjab - Sri Guru Ram Das Jee International Airport (Amritsar), Bathinda Airport, Adampur Airport (Jalandhar).

Puducherry - Puducherry Airport (located at Lawspet).

Rajasthan - Jaipur International Airport (Sanganer), Jodhpur Airport, Jaisalmer Airport, Kishangarh Airport (Ajmer), Kota Airport, Maharana Pratap Airport (Udaipur), Bikaner Airport, Kolana Airport (Jhalawar), Lal Garh Airport (Sri Ganganagar).

Sikkim - Pakyong Airport (Gangtok)

Telangana - Rajiv Gandhi International Airport (Shamshabad), Begumpet Airport (Hyderabad), Warangal Airport, Ramagundam Airport.

Tamil Nadu - Chennai International Airport (formerly known as Madras Airport), Coimbatore International Airport, Tiruchirappalli International Airport.

Tripura - Maharaja Bir Bikram Kishore Manikya Airport (Agartala), Kamalpur Airport, Khowai Airport, Kailashahar Airport.

Uttar Pradesh - Chaudhary Charan Singh International Airport (Lucknow), Kushinagar International Airport, Lal Bahadur Shastri International Airport (Varanasi), Maharishi Valmiki International Airport (Ayodhya), Hindon Airport (Ghaziabad), Noida International Airport (Gautam Buddh Nagar).

Uttarakhand - Jolly Grant Airport (Dehradun), Pantnagar Airport, Chinyalisaur Airport (Uttarkashi).

West Bengal - Netaji Subhash Chandra Bose International Airport (Kolkata), Cooch Behar Airport, Kazi Nazrul Islam Airport (Paschim Bardhaman), Burnpur Airport (Asansol).

Famous Airports in the World

Chengdu Shuangliu International Airport (China), Frankfurt Airport (Germany), John F Kennedy International Airport (New York, USA), Heathrow Airport (London, UK), King Khalid International Airport (Riyadh, Saudi Arabia), Zurich International Airport (Switzerland).

Waterways

Waterways are bodies of water, such as rivers, channels, or canals, that are navigable. The approximate length of the navigable inland waterways of India is 14,500 km. As per the National Waterways Act, 2016 a total of 111 have been declared as National Waterways.

Inland Waterways Authority of India (IWAI)

IWAI is the statutory body in charge of the waterways in India. It came into existence in 1986 by IWAI, Act 1985 for development and regulation of inland waterways for shipping and navigation. It is headquartered at Noida.

Important Waterways in India

National Waterway -1 (NW-1) or Ganga - Bhagirathi-Hooghly river system is located in India and runs from Prayagraj in Uttar Pradesh to Haldia in West Bengal via Patna and Bhagalpur in Bihar across the Ganges river. Its length is 1620 km and is the longest National waterway in India. States covered - Uttar Pradesh, Bihar, Jharkhand, West Bengal. It was declared a National Waterway in 1986.

National Waterway -2: It is a section of the Brahmaputra River having a length of

891 km between the Bangladesh border near Dhubri and Sadiya in Assam. It was declared a National Waterway No. 2 on 1 September 1988. States covered - Assam.

National Waterway - 3 : The West Coast Canal or National Waterway - 3 is a 205 km long island navigational route located in Kerala which runs from Kollam to Kottapuram. It was declared a National Waterway in 1993. It has been extended to Kozhikode by the National Waterways Act, 2016.

National Waterway - 4 : It connects Krishna and Godavari Rivers. The National Waterways Act, 2016 has extended the length of NW-4 from 1,078 km to 2,890 km. The NW-4 connects four Indian states which are Telangana, Andhra Pradesh, Tamil Nadu and Puducherry (Union Territory). It was declared a national waterway on 24 November 2008. The NW-4 has 15 terminals between Kakinada-Pondicherry and runs along the Coromandal Coast.

National Waterway - 5 : It runs a total length of 623 km of which 91 km is within West Bengal and the remaining 532 km is in Odisha. It was declared a National Waterway in November 2008.

National Waterway - 16 : The navigable portion of Barak River in India is the 121 km stretch between Laxhipur and Bhanga which has been declared as NW-16 in the year 2016.

Other Important National waterways

NW-7 (West Bengal, Ajay River), NW-8 (Alappuzha to Changanassery Canal in Kerala), NW-9 (Alappuzha - Kottayam - Athirampuzha Canal in Kerala), NW-27 (Cumberjua River in Goa), NW-37 (Gandak River, Bihar and Uttar Pradesh), NW-40 (Ghagra River, Bihar and Uttar Pradesh), NW-44 (Ichamati River, West Bengal), NW-52 (Kali River, Karnataka), NW-57 (Kopili River, Assam), NW-68 (Mandovi River, Goa), NW-73 (Narmada River in Maharashtra and Gujarat), NW-86 (Rupnarayan River in West Bengal), NW-97 (Sunderbans Waterways, West Bengal). NW - 99 (Thamirabarani River, Tamil Nadu).

Canal

Grand Canal: The Grand Canal of China is the world's longest man-made waterway.

Panama Canal was created in 1914 and is 77 km long that connects the Atlantic Ocean with the Pacific Ocean and divides North and South America.

Corinth Canal is an artificial canal in Greece that connects the Gulf of Corinth in the Ionian Sea with the Saronic Gulf in the Aegean Sea.

The Suez Canal is an artificial sea-level waterway in Egypt, connecting the Mediterranean Sea to the Red Sea.

Twante Canal is a canal that connects the Irrawaddy River and the Yangon river in Myanmar.

Canals in India

Indira Gandhi Canal: It is the longest canal in India.

Sharda Canal (Uttar Pradesh), Shipping Canal Project (Tamil Nadu, Kerala), Indira Gandhi Canal (Sultanpur, Punjab and Rajasthan), Buckingham Canal (Andhra Pradesh, Tamil Nadu), Upper Ganges Canal (Uttarakhand, Uttar Pradesh), Lower Ganges Canal (Uttarakhand, Uttar Pradesh), Agra Canal (Uttar Pradesh, Haryana, Rajasthan), Telugu Ganga project (Andhra Pradesh), Handri-Neeva (Andhra Pradesh), K. C. Canal (Andhra Pradesh), Tajewala Canal (Haryana), Western Yamuna Canal (Haryana & Delhi), Buddha Nullah Canal (Punjab), Sutlej Yamuna link Canal (Punjab), Narmada Canal (Gujarat, Rajasthan), Conolly Canal (Kerala), Ainsley canal (Tamil Nadu), Kalingarayan Canal (Tamil Nadu), Basavarajeshwari Camp (Karnataka), Lower Bhavani Project Canal (Tamil Nadu), Kaveri-Vaigai link canal (Kerala, Karnataka, Tamil Nadu), Kakatiya Canal (Telangana), Zamania Canal (Uttar Pradesh), Deokali Canal (Uttar Pradesh), Karmanasa Canal (Bihar), Kutte Kol (Jammu and Kashmir), Nallah Mar or Mar Canal (Jammu and Kashmir), Tsoont Kol (Jammu and Kashmir), Cumbarjua Canal (Goa), Soundane Cut (Maharashtra).

Other Facts

Maritime India Vision 2030 - It is a ten-year blueprint for the maritime sector which was released by the Prime Minister at the Maritime India Summit in November 2020. It will supersede the Sagarmala initiative and aims to boost waterways, give a fillip to the shipbuilding industry and encourage cruise tourism in India.

Sagarmala Project - Launched in 2015, the project aims to expand inland waterways in addition to developing coastal trade routes, to drive industrial development. It intends to save logistical costs by tripling the number of domestic waterways in the modal mix from the existing 6%.

Jal Marg Vikas Project - II (Project Arth Ganga) : To energize economic activity along river banks in Uttar Pradesh, Bihar, Jharkhand and West Bengal. It envisages to re-engineer the JMVP by involving the local community with a focus on economic activities in and around the Ganga river.

Wildlife Projects

The Zoological Survey of India (ZSI) was established in **1916**. It is a national centre for faunistic survey and exploration of the resources leading to the advancement of knowledge on the exceptionally rich faunal diversity of the country. It is headquartered at **Kolkata** and **16 regional stations** located in different geographic locations of the country.

Project Elephant is a Centrally funded programme that began in February 1992 to safeguard elephants, their habitats, and corridors. Scientific name of Elephant is *Elephas maximus*.

Project Tiger was launched in 1973 with 9 tiger reserves for conserving our national animal, the tiger. Tiger became the national animal of India in 1973. It was Classified as Endangered as per IUCN Red Data Book Schedule 1st animal as per Wildlife Protection Act, 1972.

Project Snow Leopard was launched in 2009 to promote an inclusive and participatory approach to conserve snow leopards and their habitat. Snow Leopard is a globally endangered species as well as the most important flagship species of the mountain region. Scientific Name → *Panthera uncia*.

Project Red Panda (Cat - Bear) - The project was started in 1994 at Padmaja Naidu Himalayan Wildlife Park. Padmaja Naidu Himalayan Zoological Park is located in Darjeeling, West Bengal. Red Panda is found in Arunachal Pradesh, Sikkim, and the Himalayas around Darjeeling.

Project Hangul (Kashmiri-stag) - Hangul or Kashmiri stag is a critically

endangered species found mainly in the Dachigam National Park (Srinagar) and adjoining areas in the state of Jammu & Kashmir.

Project Secure Himalaya, launched in 2017, By India in association with UNDP, the GEF -Global Environment Facility. Project duration: 6 years. Objective: to secure livelihoods, conserve, restore & sustainably use Himalayan ecosystems.

Project Sea Turtle: It was launched in 1999, by the Ministry of Environment Forest and Climate Change in association with the United Nation Development Program. It was implemented by Wildlife Institute of India, Dehradun.

Project Crocodile was launched in 1975, by Gol in association with the United Nation Development Program, Food and Agriculture Organisation. There are three crocodilian species found in India.

Operation Kachhapa was launched in 1998 by Wildlife Protection Society of India (WPSI). Aim - To reduce turtle mortality and try to safeguard the future of the Olive Ridley Sea Turtle.

Convention of Migratory Species (CMS) of Wild Animals, also known as the Bonn Convention. It was signed in 1979 in Bonn, Germany and entered into force in 1983. Its headquarters are in Bonn, Germany. It is the only convention that deals with the removal of species from the wild.

Tiger Reserve in India

There are 55 Tiger Reserve in India (As of April 2024)

Andhra Pradesh - (Nagarjunsagar Srisailem)

Arunachal Pradesh - (Pakke, Kamlang, Namdapha)

Assam - (Kaziranga, Manas, Nameri, Orang)

Bihar - (Valmiki)

Chhattisgarh - (Achanakmar, Indravati, Udanti-Sitanadi)

Jharkhand - (Palamau)

Karnataka - (Bandipur, Bhadra, Biligiri Ranganatha Temple, Dandeli-Anshi Nagarhole Tiger)

Kerala - (Parambikulam, Periyar)

Madhya Pradesh - (Bandhavgarh, Kanha, Panna, Pench, Sanjay-Dubri, Satpura)

Maharashtra - (Nawegaon-Nagzira, Pench, Sahyadri, Tadoba-Andhari, Bor,

Melghat)

Mizoram - (Dampa)

Odisha - (Satkosia, Similipal)

Rajasthan - (Dholpur – Karauli Tiger Reserve, Ranthambore, Sariska, Mukundra, Ramgarh Vishdhari,)

Tamil Nadu - (Anamalai, Kalakad - Mundanthurai, Srivilliputhur Megamalai, Mudumalai, Sathyamangalam)

Telangana - (Amrabad, Kawal)

Uttar Pradesh - (Dudhwa, Pilibhit, Ranipur, Amanagarh)

Uttarakhand - (Corbett, Rajaji)

West Bengal - (Buxa, Sunderbans)

Elephant reserves in India

There are 33 Elephant reserves in India (As of April 2024)

Andhra Pradesh - (Rayala),

Arunachal Pradesh - (Kameng, South Arunachal),

Assam - (Sonitpur, Dihing-Patkai, Kaziranga Karbi Anglong, Dhansiri - Lungding, Chirang-Ripu),

Chhattisgarh - (Badalkhol-Tamorpingla, Lemru)

Jharkhand - (Singhbhum)

Karnataka - (Mysore, Dandeli)

Kerala - (Wayanad ,Nilambur, Anamudi, Periyar)

Meghalaya - (Garo Hills)

Nagaland - (Intanki, Singphan)

Odisha - (Mayurbhanj, Mahanadi, Sambalpur)

Tamil Nadu - (Nilgiri, Coimbatore, Anamalai, Srivilliputtur, Agsthyamalai)

Uttar Pradesh - (Terai Elephant Reserve)

Uttarakhand - (Shivalik)

West Bengal - (Mayurjharna, Eastern Doars)

UNESCO World Heritage Sites

A World Heritage Site is a place that is listed by UNESCO for its special cultural or physical significance. As of April 2024 There are 42 UNESCO World Heritage Sites in India.

Cultural World Heritage Sites (34)

Taj Mahal (Uttar Pradesh, 1983),

Ellora Caves (Maharashtra, 1983),

Ajanta Caves (Maharashtra, 1983),

Agra Fort (Uttar Pradesh, 1983),

Sun Temple, Konarak (Odisha, 1984),

Group of Monuments at Mahabalipuram (Tamil Nadu, 1984),

Khajuraho Group of Monuments (Madhya Pradesh, 1986),

Group of Monuments at Hampi (Karnataka, 1986),

Fatehpur Sikri (Uttar Pradesh, 1986),

Churches and Convents of Goa (Goa, 1986),

Group of Monuments at Pattadakal (Karnataka, 1987),

Great Living Chola Temples (Tamil Nadu, 1987),

Elephanta Caves (Maharashtra, 1987),

Buddhist Monuments at Sanchi (Madhya Pradesh, 1989),

Qutub Minar and its Monuments (Delhi, 1993),

Humayun's Tomb (Delhi, 1993),

Mountain Railways of India (Tamil Nadu, 1999),

Mahabodhi Temple Complex at Bodh Gaya (Bihar, 2002),

Rock Shelters of Bhimbetka (Madhya Pradesh, 2003),

Chhatrapati Shivaji Terminus (Maharashtra, 2004),

Champaner-Pavagadh Archaeological Park (Gujarat, 2004),

Red Fort Complex (Delhi, 2007),

The Jantar Mantar (Rajasthan, 2010),

Hill Forts of Rajasthan (2013),

Rani-Ki-Vav (Gujarat, 2014),

Archaeological Site of Nalanda Mahavihara (Nalanda University) (Bihar, 2016),

The Architectural Work of Le Corbusier, an Outstanding Contribution to the Modern Movement (Chandigarh, 2016),
Historic City of Ahmedabad (Gujarat, 2017),

Victorian and Art Deco Ensemble of Mumbai (Maharashtra, 2018),

Jaipur City (Rajasthan, 2020),

Dholavira (Gujarat, 2021),

Kakatiya Rudreshwara (Ramappa) Temple (Telangana, 2021),

Sacred Ensembles of the Hoysalas (Karnataka, 2023),

Santiniketan (West Bengal, 2023)

Natural World Heritage sites (7)

Kaziranga National Park (Assam, 1985).

Keoladeo Ghana National Park (Rajasthan, 1985).

Manas Wildlife Sanctuary (Assam, 1985).

Nanda Devi National Park and Valley of Flowers (Uttarakhand, 1988).

Sundarbans National Park (West Bengal, 1987).

Western Ghats (2012)

Great Himalayan National Park (2014)

Mixed World Heritage Site (1)

Khangchendzonga National Park (Sikkim, 2016)

Intangible Cultural Heritage (15)

Kutiyattam, Sanskrit theater (2008)

Tradition of Vedic chanting (2008)

Ramlila (the traditional performance of the Ramayana) (2008)

Ramman, religious festival and ritual theater of the Garhwal Himalayas (2009)

Chhau dance (2010)

Kalbelia folk songs and dances of Rajasthan (2010)

Mudiyettu (ritual theater and dance drama of Kerala (2010)

Buddhist chanting of Ladakh: Recitation of sacred Buddhist texts in the trans-Himalayan Ladakh region, Jammu and Kashmir (2012)

Sankirtana, ritual singing, drumming and dancing of Manipur (2013)

Traditional brass and copper craft of utensil making among the Thatheras of Jandiala Guru, Punjab (2014)

Nawrouz, Novruz (2016)

Yoga (2016)

Kumbh Mela (2017)

Durga Puja in Kolkata (2021)

Garba of Gujarat (2023)

UNESCO released a list of 50 iconic Indian heritage Textiles Crafts

Some of the Important Textiles Crafts: Toda embroidery and Sungudi (Tamil Nadu), Himroo weaves (Hyderabad), Bandha tie and dye weaving from Sambalpur (Odisha), Kunbi weaves (Goa), Mashru weaves and Patola (Gujarat), Himroo (Maharashtra), Garad-Korial (West Bengal), Ilkal and Lambadi or Banjara embroidery (Karnataka), Sikalnayakanpet Kalamkari

(Tamil Nadu), Khes (Haryana), Chamba rumals (Himachal Pradesh), Thigma or wool tie and dye (Ladakh), Awadh Jamdani (Varanasi).

Ramsar Wetland Sites

Wetland is a land area where a large part of the ecosystem is saturated with water or remains submerged in it, either permanently or annually. Based on the types, wetlands are classified (8) as marsh, swamps, bogs, fens, estuaries, lakes and ponds, river floodplains and oxbow lakes.

Ramsar Wetland Sites - Ramsar Wetland Sites is an international treaty that was created to ensure the conservation of wetlands and their sustainable use. This treaty recognizes the ecological functions of wetlands, their economic, cultural, scientific and recreational importance.

It is named after the Iranian city of Ramsar, on the Caspian Sea, where the treaty was signed on 2 February 1971 by UNESCO and came into force in 1975. World Wetlands Day, observed on February 2. India Ratified the Convention on 1 February 1982.

As of March 2024, there are 80 Ramsar Sites in India.

Indian Ramsar Wetland Sites (Designated Year)

Andhra Pradesh - Kolleru Lake (2002)

Assam - Deepor Beel (2002)

Bihar - Kabartal Wetland (2020)

Goa - Nanda Lake (2022)

Gujarat - Khijadiya Wildlife Sanctuary (2021), Nalsarovar Bird Sanctuary (2012), Thol Lake Wildlife Sanctuary (2021), Wadhvana Wetland (2021).

Haryana - Bhindawas Wildlife Sanctuary (2021), Sultanpur National Park (2021)

Himachal Pradesh - Pong Dam Lake (2002), Chandertal Wetland (2005), Renuka Wetland (2005).

Jammu and Kashmir - Wular Lake (1990), Surinsar Mansar Lakes (2005), Hokera Wetland (2005), Hygam Wetland Conservation Reserve (2022), Shallbugh Wetland Conservation Reserve (2022).

Karnataka - Ranganathittu Bird Sanctuary (2022), Ankasamudra Bird Conservation Reserve (2024), Aghanashini Estuary (2024), Magadi Kere Conservation Reserve (2024).

Kerala - Ashtamudi Wetland (2002), Sasthamkotta Lake (2002), Vembanad Kol Wetland (2002).

Ladakh - Tso Kar Wetland Complex (2020), Tsomoriri Lake (2002).

Madhya Pradesh - Bhoj Wetlands (2002), Sakhya Sagar (2022), Sirpur Wetland (2022), Yashwant Sagar (2022).

Maharashtra - Lonar Lake (2020), Nandur Madhameshwar (2019), Thane Creek (2022).

Manipur - Loktak Lake (1990)

Mizoram - Pala Wetland (2021)

Odisha - Chilka Lake (1981), Ansupa Lake (2022), Bhitarkanika Mangroves (2002), Hirakud Reservoir (2022), Satkosia Gorge (2021), Tampara Lake (2022).

Punjab - Ropar Lake (2002), Beas Conservation Reserve (2019), Harike Lake (1990), Kanjli Wetland (2002), Keshopur-Miani Community Reserve (2019), Nangal Wildlife Sanctuary (2019).

Rajasthan - Keoladeo Ghana NP (1981), Sambhar Lake (1990).

Tamil Nadu - Chitrangudi Bird Sanctuary (2022), Gulf of Mannar Marine Biosphere Reserve (2022), Kanjirankulam Bird Sanctuary (2022), Karikili Bird Sanctuary (2022), Koonthankulam Bird Sanctuary (2021), Pallikarainai Marsh Reserve Forest (2022), Pichavaram Mangrove (2022), Point Calimere Wildlife and Bird Sanctuary (2002), Suchindram Theroor Wetland Complex (2022), Udhayamarthandapuram Bird Sanctuary (2022), Vaduvur Bird Sanctuary (2022), Vedanthangal Bird Sanctuary (2022), Vellode Bird Sanctuary (2022), Vembanur Wetland Complex (2022), Karaivetti Bird sanctuary (2024), Longwood Shola Reserve Forest (2024).

Tripura - Rudrasagar Lake (2005)

Uttar Pradesh - Bakhira Wildlife Sanctuary (2021), Haiderpur Wetland (2021), Nawabganj Bird Sanctuary (2019), Parvati Agra Bird Sanctuary (2019), Saman Bird Sanctuary (2019), Samaspur Bird Sanctuary (2019), Sandi Bird Sanctuary (2019), Sarsai Nawar Jheel (2019), Sur Sarovar (2020), Upper Ganga River (2005).

Uttarakhand - Asan Conservation Reserve (2020).

West Bengal - East Kolkata Wetlands (2002), Sundarbans Wetland (2019).

Gardens

India is a home to many beautiful luxurious gardens. Bangalore is known as the 'Garden city of India'.

Famous Gardens of India

Andhra Pradesh - TTD (Tirumala Tirupati Devasthanam) Gardens, Padmapuram Botanical garden, Kumbhkaran Garden, Oravakallu Rock Garden, Vuda Park, Shilotharnam Rock Garden.

Arunachal Pradesh - Tezu Botanical Garden, Polo Park, Bomdila and Tawang are important and famous places for their orchards and gardens.

Assam - Monabarie Tea Estate, Ghograjan Tea Garden, Barduar Tea Garden, Kanaklata Botanical Park, Apurugiya Brikhya Udyan.

Bihar - Sanjay Gandhi Botanical Garden, Anandbag Botanical Garden, Yamuna Singh Botanical Garden, Litchi Garden.

Chhattisgarh - Atal rock garden, Collectorate garden.

Chandigarh - Rock Garden of Chandigarh, Rose Garden (rosarium), Zakir Hussain Rose Garden, Bougainvillea Garden, Garden of Silence, Terraced Garden, Japanese Garden, Hibiscus Garden, Butterfly Park.

Delhi - Lodhi Gardens (Lady Willingdon Park), Roshanara Garden, Garden of Five Senses, Talkatora Garden, Mughal Garden, Amrit Udyan, Buddha Jayanti Park, National Bonsai Park, Deer Park, Shalimar Garden, Indraprastha Park.

Goa - Campal Gardens, JohnAdel Garden, Panjim Municipal Garden, Margao Municipal Park.

Gujarat - Sayaji Baug, Waghai Botanical Garden, Jubilee Garden, Law Garden, Dutch Garden, Sarita Udyan.

Haryana - Yadavindra Gardens, Pinjore Gardens, Sarkaria Cactus Garden, Maharana Pratap Swarn Jayanti Park, NeverEnuf Garden Railway.

Himachal Pradesh - Viceregal Lodge and Botanical Gardens, Him lehri garden, Katoch garden.

Jharkhand - Rungta Garden, Royal Garden Park, Raj Bhavan Garden, Birsamunda Museum Garden.

Karnataka - Brindavan Gardens, Lalbagh Botanical Garden.

Kerala - Malampuzha gardens, Heritage garden (Thrissur), Abraham's Spice Garden, Thumboormuzhi Garden, Periyar Spice and Ayurvedic Garden, Banasura Spices Garden, Kuzhipallam Botanical Garden, Carmelgiri Botanical Garden, Evergreen Spice Garden.

Jammu and Kashmir - Indira Gandhi Memorial Tulip garden, Mughal Gardens, Nishat Bagh, Shalimar Bagh, Achabal Bagh, Chashma Shahi, Pari Mahal, Verinag, Naseem Bagh, Achabal Gardens.

Madhya Pradesh - Meghdoot Garden (Indore), Upper Lake Garden (Bhopal), Gandhi Bal Bhawan Garden (Indore).

Maharashtra - Hanging Gardens, Mango Garden Belapur, Osho Teerth Park, Rajmachi Garden, Five Gardens, Bamboo Garden, Empress Botanical Gardens.

Manipur - Kombirei Garden, Matai Garden, East Garden.

Meghalaya - Garden Of Caves, Lady Hydari Park.

Mizoram - Rokhawmthanga Garden, Damdai Botanical Garden, Biblical Garden.

Nagaland - Niathu Garden, Duolhou garden.

Odisha - Smurtivana Botanical Garden, Nandankanan Botanical Garden, Kolab Botanical Garden, Kelucharan Park, Ekamra Kanan Botanical Gardens.

Punjab - Baradari Garden, Bharthala Garden, Jallianwala Bagh, Nehru Rose Garden.

Rajasthan - Ram Niwas Garden, Gulab Bagh (Sajjan Niwas Garden), Vidyadhar Bagh, Mandore Garden, Kesar Kyari Garden, Chambal Garden.

Sikkim - Saramsa Garden, Jawaharlal Nehru Botanical Garden, Temi Tea Garden, Namchi Rock Garden, Sewaro Rock Garden.

Tamil Nadu - Government Botanical Garden, Vasanthamani Garden, Ahnaf Garden, Sim's Park, Bryant Park, Thread Garden, Semmozhi Poonga Botanical Garden, Butterfly Park (Srirangam).

Telangana - NTR Gardens, Public Garden (Bagh -e-Aam), Mughal Garden (Ramoji Film City), Kakatiya Musical Garden,

Priyadarshini Park, Landscape Garden.

Tripura - Kanu Tripura Garden, Manu Valley Tea Garden.

Uttarakhand - Foothills garden, Company Garden (Mussoorie), Moss Garden (Nainital), Chaubatia Garden.

Uttar Pradesh - National Botanical Garden (Lucknow), Dilkusha Garden, Company Garden (Prayagraj), Jhansi Botanical Garden, Mehtab Bagh, Saharanpur Botanical Garden.

West Bengal - Botanical Garden (Kolkata), Prakriti Tirtha, Lloyd's Botanical Garden, Chatterjee's Garden, Acharya Jagadish Chandra Bose Indian Botanic Garden.

Gardens in the World

Shalimar Gardens - Lahore, Pakistan. Construction of the gardens began in 1641 during the reign of Emperor Shah Jahan, and was completed in 1642. In 1981 the Shalimar Gardens were inscribed as a UNESCO World Heritage Site.

Kenrokuen (Kanajawa, Japan), Keukenhof (Lisse, Netherlands), Gardens of Versailles (France), Longwood Gardens (Pennsylvania, US), Claude Monet's Garden (Giverny, France), Gardens By the Bay (Singapore), Shinjuku Gyoen National Garden (Tokyo, Japan), Bodnant Gardens (Colwyn Bay, UK).

Hill Stations of India

A hill station is a town located at a higher elevation than the nearby plain or valley, popular as a holiday resort during the hot season.

Araku Valley - A hill station in the Visakhapatnam district of Andhra Pradesh. This tourist spot is present in the Ananthagiri Hills of Eastern Ghats.

Darjeeling - A hilly town in India's West Bengal state, in the Himalayan foothills. Darjeeling Tea is world renowned for its flavour.

Dharamshala - Winter capital of Himachal Pradesh. It is situated in Dhauladhar Range which has dense coniferous forests. Tibetan spiritual leader, the Dalai Lama resides here. Himachal Pradesh Cricket Association Stadium, situated in Dharamshala is the world's highest cricket stadium.

Leh - The Joint capital of Ladakh along with Kargil. Leh is the highest hill station in India. It is situated at an altitude of 3,505 meters, amidst the Himalayan and Karakoram mountain ranges.

Manali - A hill station in Kullu District of Himachal Pradesh. In Manali, tourists visiting in winter experience living in igloos. Igloos are traditional winter homes made up of snow. A gift of the Himalayas to the world, Manali is a beautiful township nestled in the picturesque Beas River valley.

McLeod Ganj - A suburb of Dharamshala in the Kangra district of Himachal Pradesh. It is known as "Little Lhasa" or "Dhasa" because of its large population of Tibetans.

Mussoorie - The scenic location is at the foothills of the Garhwal Himalayan Ranges and is also described as "The Queen Of Hills. Famous Kempty Waterfall is situated near mussoorie. The Dalai Lama established the independent Tibetan government in exile in the north Indian hill station of Mussoorie when India gave shelter to him in 1959.

Nainital - A hill station located at the foothills of the Kumaon region of shivalik ranges in Uttarakhand. It is famous for Naina Devi Temple.

Pachmarhi - It is widely known as Satpura ki Rani (Queen of Satpura), situated at a height of 1100 m in a valley of the Satpura Range in Hoshangabad district, Madhya Pradesh.

Shimla - The capital of Himachal Pradesh and was also the summer capital in pre-Independence India. The hill station, Shimla is built on seven major hills namely – Prospect Hill, Observatory Hill, Summer Hill, Jakhoo Hill, Inveram Hill, Elysium Hill and Bantony Hill

North India

Haryana - Morni Hills is the only hill station in Haryana. It is a hill station situated on the outskirts of Panchkula in Haryana close to Chandigarh.

Himachal Pradesh - Chail, Kasol, Kullu, Kufri, Khajjiar, Mandi, Shoja, Kalpa, Narkanda, Tabo, Dalhousie.

Jammu and Kashmir - Srinagar, Sonmarg, Gulmarg, Pahalgam, Patnitop etc.

Uttarakhand - Nainital, Binsar, Mussoorie,

Dhanaulti, Dehradun, Auli, Bhimtal, NaukuchiaTal, Mukteshwar, Ranikhet, Kausani, Lansdowne, Almora, Chamba etc.

South India

Andhra Pradesh - Araku Valley, Horsley Hills, Nagari Hills, Maredumilli, Tirumala, Nallamala Hills, Lambasingi, Papikondalu, Ananthagiri.

Kerala - Munnar, Thekkady, Athirapally, Wayanad, Ponmudi, Malampuzha, Vagamon, Ramakkalmedu.

Karnataka - Coorg (Madikeri), Chikmagalur, Agumbe, Sakleshpur.

Tamil Nadu - Ooty (Udhagamandalam), Kodaikanal, Yercaud, Yelagiri, Coonoor, Kotagiri, Anaimalai Hills, Valparai.

East and North East India

Arunachal Pradesh - Tawang, Daporijo, Khonsa, Roing.

Assam - Haflong.

Jharkhand - Netarhat "Queen of Chotanagpur Plateau".

Manipur - Chandel, Kaina, Langthabal, Phangrei, Kangchup.

Meghalaya - Cherrapunji, Tura, Shillong is also popularly known as 'Scotland of East'.

Mizoram - Mamit, Lunglei, Champhai.

Nagaland - Phek, Mokokchung, Tuensang, Pfutsero.

Sikkim - Gangtok, Pelling, Geyzing, Mangan.

West Bengal - Siliguri, Kurseong ("The Land of White Orchids").

West India

Gujarat - Saputara ("The abode of serpents"), Wilson Hills, Dediapada, Subir.

Madhya Pradesh - Amarkantak, Tamiya Hill, Kailaras Hill Station, Chhogala Kanjwani Hill.

Maharashtra - Mahabaleshwar ("The land of Strawberries"), Lonavala, Khandala, Malshej Ghat, Lavasa, Bhandardara, Chikhaldara, Karjat, Dapoli, Igatpuri, Panchgani, Amboli, Panhala, Toranmal, Rajmachi, Durshet, Bhimashankar, Tamhini Ghat, Mhaismal, Suryamal.

Rajasthan - Mount Abu, Sajjanganrh.

Ports of India / World

A port or harbor is an area that can accommodate a large number of boats and vessels. India has a coastline of 7516.6 Km. The Indian coastline touches nine states - Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal and two union territories - Daman and Diu and Dadra and Nagar Haveli, and Puducherry. The two island territories of India are - Andaman and Nicobar Islands in the Bay of Bengal and Lakshadweep Islands in the Arabian Sea.

India has 13 major seaports (12 Government-owned and one private) and 205 notified minor and intermediate ports. About 95 percent by volume and 65 percent by value of India's total international trade are carried on through maritime transportation. Major ports are managed by the Union Ministry of Shipping Ports and Waterways while minor ports are managed by the state government.

Major ports in India

Eastern coast: Kolkata port (West Bengal), Haldia port (West Bengal), Paradip port (Orissa), Visakhapatnam port (Andhra Pradesh), Kamarajar port (Tamil Nadu), Chennai port (Tamil Nadu), V. O. Chidambaram or Tuticorin port (Tamil Nadu).

Western coast: Cochin port (Kerala), New Mangalore port (Karnataka), Mormugao port (Goa), Mumbai port (Maharashtra), Jawaharlal nehru port (Maharashtra), Deendayal port (Gujarat).

Chennai Port is an artificial harbour built in 1859. Formerly known as Madras Port, is the second-largest container port of India. It is due to the existence of the port that the city of Chennai known as the 'Gateway of South India'.

Cochin or Kochi Port is situated at the head of Vembanad Kayal, popularly known as the 'Queen of the Arabian Sea', is also a natural harbour. It is also the first transshipment terminal in India. The port lies on two islands in the Lake of Kochi: Willingdon Island and Vallarpadam, towards the Fort Kochi river mouth opening onto the Laccadive Sea.

Deendayal Port is situated at the head of Gulf of Kutch in Gujarat. The Ministry of Shipping has renamed Kandla Port Trust as Deendayal Port Trust with effect

from 25th of September, 2017. It is a tidal port.

Haldia port has been built at the meeting place of the Haldi River and Hooghly river. Kolkata Port Trust has been created in this port as the port's partner. So it is not a port. It is an official dock complex.

Jawaharlal Nehru Port (Nhava Sheva) is the largest artificial port and the largest container port in India. It is located in east Mumbai, Maharashtra. Trade: Textiles, sporting goods, carpets, pharmaceuticals, chemicals etc.

Kandla port is located in Kutch District, Gujarat. Kandla port was developed after independence to reduce the volume of trade at Mumbai port.

Kamarajar Port was formerly known as Ennore port, located on the east coast of India, Coromandel Coast, Chennai, Tamil Nadu. It is India's First corporatised port. Trades: Iron Ore, Coal, petroleum products and chemicals.

Kolkata Port (officially known as Syama Prasad Mookerjee Port Trust) is the only riverine major port of India. It is a freshwater port with no variation in salinity. It is known as Diamond Harbour and the 'Gateway of Eastern India'.

Mormugao Port is situated on the estuary of the river Zuari, is a natural harbour in Goa. It is known as the youngest major port of India.

Mumbai Port is a natural harbour and the biggest port of the country. This port has 3 enclosed wet docks: Prince's Dock, Victoria Dock, Indira Dock. It is the busiest Port in India.

New Mangalore Port is located in the state of Karnataka. It handles fertilizers, petroleum products, edible oils, coffee, tea, wood pulp, yarn, granite stone, molasses, etc.

Paradip Port is a natural, deep-water and the First Major Port commissioned after Independence. It is located at the confluence of the Mahanadi river and the Bay of Bengal in Odisha. It deals with the export of iron and aluminum.

Tuticorin Port: It has been renamed as V O Chidambaram Ports. It is an artificial port located in the Gulf of Mannar, Tamil Nadu. It is famous for pearl fishery in the Bay of Bengal and is thus also known as the pearl city. It deals with a variety of cargo, including coal, salt, food grains,

edible oils, sugar, chemicals and petroleum products.

Visakhapatnam Port is a land-locked harbour, located in Andhra Pradesh. It is a natural harbor and also the second largest port by volume of cargo handled. Andhra Pradesh and Telangana are the main hinterlands for this port.

Minor ports in India: Azhikkal Port (Mouth of Valapattanam River Kerala), Kannur (Kerala), Kundapur (Udupi District Karnataka), Dahej (Gujarat), Jafrabad (Gulf of Cambay Gujarat), Kasargod (Kerala), Karaikal (Puducherry), Gopalpur (Odisha) etc.

Important ports around the World

Shanghai Port (China), Ningbo-Zhoushan, Guangzhou Harbor, Tianjin (China), Port Klang (Malaysia), Antwerp (Belgium), Piraeus (Greece), Kaohsiung (China), Xiamen (China), Port of Busan (South Korea), Tanjung Pelepas (Malaysia), Port of Mergui, Port of Moulmein, Port of Rangoon (Myanmar), Hamburg (Germany), Laem Chabang (Thailand), Keihin Ports (Japan), Tanjung Priok (Indonesia), Colombo (Sri Lanka), Hanshin Port (Japan), Manila (Philippines), Algeciras (Spain), Hodeidah Port (Yemen), Khasab Port (Oman), Bandar Abbas, Bandar Anzali, Bandar Khomeini (Iran), Ketu Bunder Sea Port, Gwadar Port (Pakistan), Lamu port (Kenya), Tanger med (Morocco), Port of Rotterdam (Netherlands), Chabahar Port (Iran).

Some Important Ports: India: Deepest port - Krishnapatnam Port (Andhra Pradesh), Largest port - Mumbai (Maharashtra), Smallest port - Port Blair (Andaman and Nicobar Islands), Newest Port - Wadhawan (Maharashtra), Oldest Port - Syama Prasad Mookerjee Port Trust (Kolkata), Private Major Port - Mundra (run by Adani Ports and SEZ Limited - APSEZ) (Gujarat). **World:** Largest port - Port of Shanghai (China), Busiest port - Port of Shanghai (China), Biggest man-made harbor - Port of Jebel Ali (UAE).

National Parks

National parks are a protected area under Wildlife Protection Act 1972, to conserve the Flora, fauna and other Biological diversity. No human activity is permitted. There are 106 national parks

in India covering an area of 44,402.95 km², which is 1.35% of the geographical area of the country (National Wildlife Database, 2024).

Andhra Pradesh

Sri Venkateswara National Park (1989) is a part of the Seshachalam Hills Biosphere Reserve.

Other National Parks: Papikonda National Park (2008) and Rajiv Gandhi (Rameswaram, 2005) National Park.

Arunachal Pradesh

Namdapha National Park (1983) was designated as Tiger reserve and is recognised under UNESCO Tentative list.

Mouling National Park (1986) is located on the right bank of Siang river.

Assam

Dibru-Saikhowa National Park was established as a National Park in 1999. It is bounded by the Brahmaputra and Lohit Rivers in the north and the Dibru river in the south. It was also designated as a Biosphere Reserve in 1997.

Dehing Patkai National Park was declared a National Park in June 2021 and became 7th National park of Assam. It is located in the Dibrugarh, Charaideo and Tinsukia districts of Assam.

Kaziranga National Park was established as a National Park in 1974 and Tiger Reserve in 2007. It was declared a UNESCO World Heritage Site in 1985. It is inhabited by the world's largest population of one-horned rhinoceroses.

Manas National Park was established as a National Park in 1990 and Tiger reserve in 2008. It has been recognised under UNESCO World Heritage Sites (1985).

Nameri National Park was established as a National Park in 1998. It is a part of the Eastern Himalayan Biodiversity Hotspot. It shares its northern boundary with the Pakhui Wildlife Sanctuary of Arunachal Pradesh.

Orang National Park was established as a National Park in 1999. It was designated as the 49th Tiger Reserve in 2016 (Newest in Assam). It is located in the Darrang and Sonitpur districts.

Raimona National Park was declared a National Park on 5 June 2021. It is

famous for its Golden Langur, an endemic species which is known as the mascot of BTR.

Bihar

Valmiki National Park (1989) - It was declared as a Tiger Reserve. It is located in the West Champaran District of Bihar. It is the only national park in Bihar.

Chhattisgarh

Guru Ghasidas National Park (1981) - It was the last known habitat of the Asiatic cheetah in the country.

Other National Parks: Indravati National Park (1982), Kanger Ghati National Park (1982).

Goa

Mollem National Park (1992) is situated in the Western Ghats of West India, in Dharbandora taluka.

Gujarat

Blackbuck National Park (Velavadar) was established as a National Park in 1976 in Bhavnagar district of Gujarat.

Gir National Park (Sasan Gir) was established as a National Park in 1975. It is the only known habitat of the Asiatic Lion.

Marine National Park (1982) is located in the Gulf of Kutch and is the largest Marine National Park in India.

Vansda (Bansda) National Park was established as a National Park in 1979. It is located on the banks of Ambika River.

Haryana

Kalesar National Park was established as a National Park in 2003. It is located in Yamunanagar district of Haryana. It is probably the only forest in Haryana with a natural Sal tree belt.

Sultanpur National Park was established as a National Park in 1989. It is situated in Gurgaon district of Haryana.

Himachal Pradesh

Great Himalayan National Park (1984) is located in Kullu region in the state of Himachal Pradesh. In June 2014, It was added to the UNESCO list of World Heritage Sites.

Other National Parks: Inderkilla (2010), Khirganga (2010), Col. Sherjung Simbalbara (2010), and Pin Valley (1987).

Jharkhand

Betla National Park was established as a National park in 1986. It is part of the core area of Palamau Tiger Reserve. North Koel river passes through the park.

Karnataka

Nagarahole National Park (1988) was declared as the 37th Tiger reserve under Project Tiger in 2007. It lies in the Western Ghats and is a part of the Nilgiri Biosphere Reserve. Nagarahole River flows through the park.

Other National Parks: Anshi (1987), Bandipur (1974), Bannerghatta (1974) and Kudremukh (1987).

Kerala

Silent Valley National Park was established as a National Park in 1984. River Kunthipuzha is passing through the Park. Lion Tailed Macaques are the flagship species of the Park. The Kadalundi River has its origin in Silent Valley.

Other National Parks: Anamudi Shola (2003), Pambadum Shola (2003), Mathikettan Shola (2003), Eravikulam (1978), Periyar National Park (1982).

Madhya Pradesh

Bandhavgarh National Park is located in the Vindhya Hills of Madhya Pradesh's Umaria district. It was designated as a National Park in 1968 and Tiger reserve in 2007.

Kuno National Park is situated in the center of Kuno Wildlife Division of Sheopur district. The cheetahs brought from Namibia were kept in this park.

Other National Parks: Fossil (1983), Pench (1975), Kanha (1955), Madhav (1959), Panna (1981), Sanjay (1981), Satpura (1981) and Van Vihar (1979).

Maharashtra

Sanjay Gandhi (Borivali) National Park (1983) is located in Mumbai. Kanheri caves is located within the park.

Other National Parks: Chandoli (2004), Gugamal (1975), Nawegaon (1975), Pench (Jawaharlal Nehru, 1975) and

Tadoba (1955).

Manipur

Keibul-Lamjao National Park (1977) is the only floating National Park in the world. It is located on the Loktak Lake. It is the natural habitat of the 'Sangai', the dancing deer of Manipur. It is famous for phumdis.

Shiroi National Park (1982) is famous for shiroi lily.

Meghalaya

Balpakram National Park (1986) is located near the Garo Hills in Meghalaya and is famous for the golden cat and the barking deer.

Nokrek Ridge National Park (1997) is also recognised under UNESCO List of biosphere reserve in 2009.

Mizoram

Phawngpui (Blue Mountain) National Park (1992) is located in the Mizo hills and the highest peak of Mizoram (Blue Mountain) is located in the park.

Other National Parks: Murlen National Park.

Nagaland

Intanki (Ntangki) National Park (1993) is famous for Hoolock gibbon (Endangered species under IUCN Red data book).

Odisha

Bhitarkanika National Park (1988) is the breeding place for the Salt Water Crocodiles. Gahirmatha Beach is located on the eastern boundary of the park, famous for Olive Ridley Turtles.

Simlipal National Park (1980) was declared Tiger reserve in 2007 and part of the UNESCO World Network of Biosphere Reserve since 2009.

Rajasthan

Ranthambore National Park was established as a National Park in 1980 and Tiger Reserve in 2007.

Mukundara Hills National Park was established as a National Park in 2006 and Tiger Reserve in 2013. It is located on the eastern bank of the Chambal River. It is located near Kota, Rajasthan.

Desert National Park (1992) is famous for Critically Endangered (IUCN Red List) Great Indian Bustards. It has been recognised under UNESCO Sites on the Tentative List (2009).

Sariska National Park was established as a National Park in 1992 and Tiger Reserve in 2007.

Keoladeo Ghana National Park (formerly known as the Bharatpur Bird Sanctuary) is located in Bharatpur.

Sikkim

Kanchenjunga National Park was established as a national park in 1977. It is recognized under the UNESCO list of biosphere reserves. It is the only recognized World Heritage Site under UNESCO's "Mixed Heritage Site".

Tamil Nadu

Gulf of Mannar Marine National Park (1980) is recognised under UNESCO list of Biosphere reserve. It is famous for Critically Endangered Dugong (Sea Cow).

Indira Gandhi (Annamalai) National Park (1989) was also designated as Tiger Reserve in 2007. Annamalai Hills is located in the park.

Other National parks: Guindy (1976), Mukurthi (1990) and Mudumalai (1990).

Telangana

Mrugavani National Park (1994) is located near Hyderabad.

Other National Parks: Kasu Brahmananda Reddy National Park (1994) and Mahavir Harina Vanasthali National Park (1994).

Tripura

Clouded Leopard National Park (2007) is India's first dedicated wildlife park for Clouded Leopards.

Bison (Rajbari) National Park (2007) was established to protect the Indian gaur, a type of bison.

Uttarakhand

Nanda Devi National Park (1982) is recognised under the UNESCO Natural list of world heritage sites. It is a Biosphere Reserve Recognised under UNESCO's Man and Biosphere Programme. River Rishi Ganga flows

through it.

Valley of Flowers was established as a National Park in 1982. It is recognised under the UNESCO Natural list of world heritage sites. It is located in the Chamoli district of Uttarakhand.

Rajaji National Park was established as a National Park in 1983 and Tiger Reserve in 2015.

Gangotri National Park was established as a National Park in 1989. It is situated on the catchment of the Bhagirathi River. It forms an international boundary with Tibet (China).

Govind Pashu Vihar National Park was established as a National Park in 1990. River Tons (Tributary of Yamuna) flow through this park.

Jim Corbett National Park is the oldest national park in India, established in the year 1936 as Hailey National Park. It was declared a Tiger Reserve in 2010. Project Tiger was first launched in 1973 in this park. River Ramganga flows through this park.

Uttar Pradesh

Dudhwa National Park (1977) is located on the Indo-Nepali border in the Lakhimpur Kheri District. Mohana and Suheli river flow through the park. It was declared a Tiger Reserve in 2007.

West Bengal

Gorumara National Park was established as a National Park in 1992. It is located in Jalpaiguri district, West Bengal.

Buxa National Park was established as a National Park in 1992 and Tiger Reserve in 2009.

Other National Parks: Jaldapara (2014), Neora Valley (1986), Sundarban (1984) and Singalila (1986).

Andaman and Nicobar

Campbell Bay National Park was established as a National Park in 1992. It is located on Great Nicobar Island.

Other National Parks: Galathea Bay (1992), Mount Harriet (1987), Rani Jhansi Marine (1996), Saddle Peak (1987), Mahatma Gandhi Marine (Wandoor, 1983).

Jammu and Kashmir

Dachigam National Park (1981) is located in the Zabarwan Range of the western Himalayas.

Kazinag National Park (2000) is situated on the north bank of Jhelum river.

Kishtwar National Park (1981) is surrounded by the Rinnay River in the north, Kibar Nala catchments in the southern part, the Great Himalaya in the east and Marwa River in the west.

City Forest (Salim Ali) National Park was established as a National park in 1992.

Ladakh

Hemis National Park (1981) is the largest national park of India. Globally famous for its snow leopards. It is the only national park in India that is north of the Himalayas.

Wildlife Sanctuaries

Andhra Pradesh

Kolleru Bird Sanctuary, Nelapattu Bird Sanctuary, Pulicat Lake Bird Sanctuary, Coringa Wildlife Sanctuary, Nagarjunsagar-Srisailem Tiger Reserve, Sri Venkateswara National Park, Rollapadu Wildlife Sanctuary, Sri Lankamalleswara Wildlife Sanctuary, Krishna Wildlife Sanctuary, Koundinya Wildlife Sanctuary, Gundla Brahmeswaram Wildlife Sanctuary, Sri Penusila Narasimha Wildlife Sanctuary, Kambalakonda Wildlife Sanctuary.

Arunachal Pradesh

D'Ering Memorial, Itanagar, Mahao, Eagle Nest, Sessa Orchid Sanctuary, Kamlang, Dibang, Kane, Yordi Rabe Supse.

Assam

Pobitora Wildlife Sanctuary: It has the highest density of one-horned rhinos in the world and It has the second highest number of one-horned rhinos in India after Kaziranga National Park.

Manas Wildlife Sanctuary: It spans the Manas river and is bounded to the north by the forests of Bhutan. Manas was originally a game reserve since 1928 and became a Tiger Reserve in 1974, a World Heritage Site in 1985, and a Biosphere Reserve in 1989 then was declared as a National Park in 1990.

Wildlife Sanctuaries: Garampani Wildlife Sanctuary, Laokhowa Wildlife Sanctuary, Bura Chapori Wildlife Sanctuary, Bornadi Wildlife Sanctuary, Chakrashila Wildlife Sanctuary, Pani Dihing Wildlife Sanctuary, Hoollongapar Gibbon Sanctuary, Sonai Rupai Wildlife Sanctuary.

Bihar

Vikramshila Gangetic Dolphin Sanctuary is in Bhagalpur District of Bihar.

Wildlife Sanctuaries: Bhimbandh, Gautam Buddha sanctuary, Pant (Rajgir), Udaypur, Kaimur, Nagi Dam, Nakti Dam, Kanwar Lake Bird Sanctuary, Kusheshwar Asthan Bird Sanctuary, Barela Jheel Salim Ali-Jubba Sahni Bird Sanctuary.

Chhattisgarh

Sitanadi Wildlife Sanctuary, Achanakmar Wildlife Sanctuary, Badalkhol Wildlife Sanctuary, Barnawapara Wildlife Sanctuary, Tamor Pingla Wildlife Sanctuary, Semarsot Wildlife Sanctuary, Bhairamgarh Wildlife Sanctuary, Bhoramdeo Wildlife Sanctuary, Sarangarh - Gomarda Wildlife Sanctuary, Udanti Wildlife Sanctuary.

Goa

Bhagwan Mahavir Wildlife Sanctuary: It was declared a wildlife sanctuary in 1969. It was Originally known as the Mollem Game Sanctuary. The sanctuary is particularly known for its Leopards, Elephants, Deers & Gaur also known as Indian Bison.

Wildlife Sanctuaries: Cotigao, Bondla, Salim Ali Bird Sanctuary, Netravali, Bhagwan Mahavir, Molem.

Gujarat

Little Rann of Kachchh, Gujarat – The major population of Indian Wild Ass (Indian onager or Ghudkhur and Khur) is in Little Rann of Kachchh and the area is declared as Wild Ass Sanctuary in 1972 and came under the Wildlife Protection Act of 1972.

Barda Wildlife Sanctuary is located in Gujarat. It is situated west of Gir Forest National Park. It was established as a Wildlife Sanctuary in 1979.

Wildlife Sanctuaries: Nal Sarovar Bird Sanctuary, Hingolghadh Nature Reserve, Khijadiya Bird Sanctuary, Ratanmahal Sloth Bear Sanctuary, Shoolpaneshwar,

Kutch Desert, Gaga, Thol Bird Sanctuary, Rampara Vidi, Porbandar Bird Sanctuary, Balaram Ambaji, Paniya, Purna, Jambughoda, Narayan Sarovar Sanctuary, Mitiyala, Girnar.

Haryana

Bhindawas, Chhilchhila, Bir Shikargah, Nahar, Khaparwas, Khol Hi-Raitan.

Himachal Pradesh

Bandli, Chail, Chandra Taal, Churdhar Sanctuary, Daranghati Sanctuary, Dhauladhar Wildlife Sanctuary etc.

Jharkhand

Dalma, Hazaribagh, Koderma, Lawalong, Mahuadanr, Palkot, Parasnath, Topchanchi, Udhwa Lake.

Karnataka

Dandeli Wildlife Sanctuary is located in Uttara Kannada District of Karnataka. It is most famous for the great hornbill (great Indian hornbill or great pied hornbill) and the Malabar pied hornbill.

Ranganathittu Bird Sanctuary also known as the 'Pakshi Kashi' of Karnataka.

Wildlife Sanctuaries: Adichunchanagiri Hills Wildlife Sanctuary, Arabithittu Wildlife Sanctuary, Attiveri Bird Sanctuary, Bhadra Wildlife Sanctuary, Bhimgad Wildlife Sanctuary, Brahmagiri Wildlife Sanctuary, Cauvery Wildlife Sanctuary, Chincholi Wildlife Sanctuary, Daroji Sloth Bear Sanctuary, Gudekote Sloth Bear Sanctuary, Malai Mahadeshwara Wildlife Sanctuary etc.

Kerala

Chinnar Wildlife Sanctuary is located in the Idukki district of Kerala. It was established in 1984. It is surrounded by Kodaikanal Wildlife Sanctuary to its East, Indira Gandhi Wildlife Sanctuary to its North, and Eravikulam National Park to its South.

Periyar National Park and Wildlife Sanctuary is a protected area located in the districts of Idukki and Pathanamthitta in Kerala. Declared a Sanctuary during 1950 and declared as Tiger Reserve during 1978.

Wildlife Sanctuaries: Aralam Wildlife Sanctuary, Chimmony Wildlife Sanctuary,

Idukki Wildlife Sanctuary, Karimpuzha Wildlife Sanctuary, Kottiyoor Wildlife Sanctuary, Kurinjimala Sanctuary, Malabar Wildlife Sanctuary, Mangalavanam Bird Sanctuary, Neyyar Wildlife Sanctuary, Parambikulam Wildlife Sanctuary, Pathiramanal Bird Sanctuary, Peechi-Vazhani Wildlife Sanctuary, Peppara Wildlife Sanctuary etc.

Madhya Pradesh

Ratapani, located in the Raisen district of Madhya Pradesh. The Betwa river originates from this sanctuary.

Wildlife Sanctuaries: Bori, Gandhi Sagar Sanctuary, Nauradehi, National Chambal Sanctuary, Ghatigaon, Karera, Ken Gharial Sanctuary, Ralamandal, Kheoni Sanctuary, Narsinghar, Bagdara, Orchha, Panpatha, Phen, Sailana, Sardarpur, Singhori, Son Gharial Sanctuary.

Maharashtra

Amba Barwa, Andhari, AnerDam, Bhamragarh, Bhimashankar, Chaprala, Gautala Autramghat Sanctuary, Ghodazari, Great Indian Bustard Sanctuary, Kalsubai Harishchandragad, Karanja Sohal, Katepurna, Koka, Koyna, Lonar, Malvan Marine, Mansingdeo.

Manipur

Khongjaingamba Ching, Yangoupokpi Lokchao, Zeilad, Thinungei Bird Sanctuary.

Meghalaya

Baghmara Pitcher Plant Wildlife Sanctuary, Nongkhylllem Wildlife Sanctuary, Siju Wildlife Sanctuary.

Mizoram

Khawnglung, Lengteng, Ngengpui.

Nagaland

Fakim, Puliebadze, Rangapahar, Singphan.

Odisha

Badrama, Bhitarkanika, Satkosia Gorge, Hadagarh, Nandankanan, Baisipalli, Kotagarh, Chandaka Elephant Sanctuary, Khalasuni, Balukhand-Konark, Kuldiha, Debrigarh, Lakhari Valley, Nalbana Bird Sanctuary etc.

Punjab

Abohar, Takhni-Rehmapur, Nangal, Harike lake, Bir Dosanjh, Bir Mehaswala, Bir Moti Bagh, Bir Bhunerheri, Bir Bhadson, Bir Gurdialpura.

Rajasthan

Mount Abu, Kumbhalgarh, Tal Chhapar Sanctuary, National Chambal Sanctuary, Sita Mata, Phulwari ki Nal, Todagarh Rawali Sanctuary, Sajjangarh, Bassi, Bhainsrorgarh, Jaisamand, Jamwa Ramgarh, Jawahar Sagar Wildlife sanctuary, Kailadevi, Nahargarh, Ramgarh Vishdhari, Ramsagar, Sawai Madhopur, Sawai Man Singh, Shergarh, Van Vihar.

Sikkim

Kyongnosla Alpine Sanctuary, Fambong Lho, Shingba Rhododendron Sanctuary, Maenam, Barsey Rhododendron Sanctuary, Pangolakha, Kitam Bird Sanctuary.

Tamil Nadu

Vedanthangal Bird Sanctuary, Mudumalai National Park, Point Calimere Wildlife and Bird Sanctuary, Indira Gandhi, Kalakkad Mundanthurai Tiger Reserve, Vettangudi Bird Sanctuary, Pulicat Lake Bird Sanctuary, Vallanadu Black Buck Sanctuary, Karikili Bird Sanctuary, Chitrangudi Bird Sanctuary, Kanjirankulam Bird Sanctuary, Vaduvor Bird Sanctuary, Udayamarthandapuram Bird Sanctuary, Koonthankulam Bird Sanctuary, Vellode Birds Sanctuary, Karaivetti Bird Sanctuary.

Tripura

Sepahijala Wildlife Sanctuary is a wildlife sanctuary in Tripura, located in Bishalgarh. It has an artificial lake and natural botanical and zoological gardens. It is famous for its clouded leopard enclosures.

Wildlife Sanctuaries: Gumti Wildlife Sanctuary, Rowa Wildlife Sanctuary, Trishna Wildlife Sanctuary.

Telangana

Pocharam, Pakhal, Eturnagaram, Kawal, Kinnerasani, Manjira, Nagarjunsagar - Srisailem Tiger Reserve, Pranahita.

Uttar Pradesh

Chandra Prabha, Kishanpur, Katarniaghat, Ranipur Sanctuary, Bakhira Sanctuary, Kaimoor, Hastinapur, Samaspur Bird Sanctuary, Sohagi Barwa, Kachhua Sanctuary, Okhla Bird Sanctuary, Saman Bird Sanctuary, Sandi Bird Sanctuary, Vijai Sagar, Patna Bird Sanctuary, Suhelwa Sanctuary, Mahavir Swami, Nawabganj Bird Sanctuary, Parvati Aranga Bird Sanctuary.

Uttarakhand

Govind Pashu Vihar, Kedarnath, Askot Musk Deer Sanctuary, Sonanadi, Binsar, Mussoorie, Nandhaur.

West Bengal

Kulik Bird Sanctuary or Raiganj Wildlife Sanctuary is situated near Raiganj in Uttar Dinajpur district in the Indian state of West Bengal.

Lothian Island Wildlife Sanctuary is situated in South 24 Parganas district, West Bengal.

Jaldapara Wildlife Sanctuary - The preserve was established in 1941 mainly for the protection of the great Indian rhinoceros (*Rhinoceros unicornis*).

Wildlife Sanctuaries: Chapramari, Haliday Island, Mahananda, Senchal, Ballabhpur, Bethuadahari, Bibhutibhushan, Ramnabagan, Chintamani Kar Bird Sanctuary, JorePokhri, Raiganj, Buxa Tiger Reserve (It is threatened by the ongoing Dolomite mining).

Andaman and Nicobar Islands

Interview Island Wildlife Sanctuary: Main attraction of the sanctuary is elephants.

Wildlife Sanctuaries: Arial Island, Bamboo Island, Barren Island, Batti Malv Island, Belle Island, Bennett Island, Bingham Island, Blister Island, Bluff Island, Bondoville Island, Brush Island, Buchanan Island, Channel Island, Cinque Island, Clyde Island, Cone Island.

Delhi

Asola Bhatti, Najafgarh Drain Bird Sanctuary.

Jammu and Kashmir

Trikuta, Surinsar Mansar, Gulmarg, Hirpora, Limber, Overa-Aru.

Biosphere Reserves

Biosphere Reserve (BR) is an international designation by the United Nations Educational, Scientific and Cultural Organization (UNESCO) for representative parts of natural and cultural landscapes extending over large areas of terrestrial or coastal/marine ecosystems or a combination of both. The first biosphere reserve of the world was established in 1979.

There are 18 biosphere reserves in India

Achanakmar-Amarkantak - It Covers parts of Anuppur and Dindori districts of Madhya Pradesh and parts of Bilaspur districts of Chhattisgarh. It was declared a biosphere reserve in 2005 and has been part of the UNESCO World Network of Biosphere Reserve since 2012. Key fauna - Indian wild dog (*Cuon alpinus*), Saras crane (*Grus antigone*), Asian white-backed vulture (*Gyps bengalensis*).

Agasthyamala is located near Thiruvananthapuram or Trivandrum, the capital city of Kerala. It is contiguous to Neyyar Wildlife Sanctuary and Peppara Wildlife Sanctuary. It was declared a biosphere reserve in 2001 and has been part of the UNESCO World Network of Biosphere Reserve since 2016.

Cold Desert (2009) – Himachal Pradesh. Key Fauna - Snow leopard. It was declared as India's 16th Biosphere Reserve in 2009, noted for the conservation of Red Fox, Blue Rock Pigeon and Lammergeier.

Dehang-Debang (1998) - Arunachal Pradesh, Key fauna - Mishmi takin, musk deer.

Dibru-Saikhowa (1997) - Assam, Key fauna - Tiger, Elephant, Leopard, Jungle Cat, Bears.

Great Nicobar is the southernmost island of the Nicobar Islands Archipelago. It was declared a biosphere reserve in 1989 and has been part of the UNESCO World Network of Biosphere Reserve since 2013. Key fauna - Saltwater crocodile.

Gulf of Mannar (1989) is located in the Gulf of Mannar (between India and Sri Lanka), east of Tamil Nadu. It consists of 21 small islands (islets) and adjacent coral reefs located in the Indian Ocean.

The Gulf of Mannar Biosphere Reserve is also home to the largest endangered marine mammal, the dugong, and sea turtles. It has been part of UNESCO since 2001.

Kachchh (2008) - Gujarat (Largest Area), Key fauna - Indian wild ass.

Khangchendzonga is located in Sikkim and is one of the highest ecosystems in the world, reaching elevations of 1220 meters above sea-level. World's third highest peak, Mt. Khangchendzonga, is situated within the Park. It was declared a biosphere reserve in 2000 and has been part of the UNESCO World Network of Biosphere Reserve since 2018. Key fauna - Snow leopard, red panda

Manas (1989) - Assam, Key fauna - Asiatic elephant, tiger, Assam roofed turtle, hispid hare, golden langur, pygmy hog.

Nanda Devi (1988) is situated around the peak of Nanda Devi (7816 m) in the state of Uttarakhand. It was inscribed a World Heritage Site by UNESCO in 1988. Nanda Devi Park was designated as a World Network of Biosphere Reserve by UNESCO in 2004. Key fauna - Snow leopard, Himalayan black bear.

Nilgiri (1986) - It is located in the Western Ghats. The Nilgiri Biosphere Reserve encompasses parts of Tamilnadu, Kerala and Karnataka. UNESCO declared it as a World Heritage Site in 2012. It is one of the world's biodiversity hotspots and provides habitat for the probably largest South Indian populations of tiger (*Panthera tigris*), elephant (*Elephas maximus*) and other large mammals. Bandipur Tiger Reserve and national park forms part of the Nilgiri Biosphere Reserve. Key fauna - Nilgiri tahr, tiger, lion-tailed macaque.

Nokrek (1988) - This Biosphere Reserve extends to three districts namely West Garo Hills, East Garo Hills and South Garo Hills District of Meghalaya. Nokrek is the highest peak of the Garo hills, rising up 1,412 meters. UNESCO added the Nokrek Biosphere Reserve to its list of Biosphere Reserves in 2009. Key fauna - Red panda.

Pachmarhi (1999) is Located in Hoshangabad and Chhindwara districts of Madhya Pradesh. The area has been identified on the World Network of Biosphere Reserves by UNESCO on 26th May 2009. Key fauna - Giant squirrel, flying squirrel.

Panna (2011) is located in Panna and Chhatarpur districts in Madhya Pradesh. Ken river (a tributary of the Yamuna River) flows through the reserve. Madhya Pradesh has the highest number of tigers in the country followed by Karnataka and Uttarakhand. It was declared a biosphere reserve in 2011 and has been part of the UNESCO World Network of Biosphere Reserve since 2020. Key fauna - Tiger, Leopard and Chinkara.

Seshachalam (2010) - Andhra Pradesh, Key fauna - Tigers, leopard, Elephants, sloth bear.

Similipal is located in the Mayurbhanj district of Orissa. The highest peak in the Similipal hill range is Khairiburu (1178 meters). It was declared a biosphere reserve in June, 1994 and has been part of the UNESCO World Network of Biosphere Reserve since 2009. Key fauna - Gaur, royal Bengal tiger, Asian elephant.

Sundarban (1989) - It is Part of delta of Ganges and Brahmaputra river system located in west bengal, famous for its well known predator - The Bengal Tiger. It contains the world's largest mangrove forests. It was designated a UNESCO World Heritage site in 1989. Key fauna - Royal Bengal tiger.

Man and Biosphere Programme: Launched in 1971, UNESCO's Man and the Biosphere Programme (MAB) is an intergovernmental scientific programme that aims to establish a scientific basis for the improvement of relationships between people and their environments. There are a total **12 biosphere** reserves of India which have been recognized internationally under the Man and Biosphere Reserve program: Nilgiri Biosphere, Gulf of Mannar, Sunderban, Nanda Devi, Nokrek, Pachmarhi, Similipal, Achanakmar - Amarkantak, Great Nicobar, Agasthyamala, Khangchendzonga, Panna.

Beaches, Stepwells and Waterfalls

A beach is a narrow, gently sloping strip of land that lies on the sea. Beaches are formed when waves deposit sand and gravel along the shoreline. Most beach material is the product of weathering and erosion.

Blue Flag Certification of Beaches

The Blue Flag programme is a world-renowned eco-label certification for beaches, marinas and boats. It was started in France in 1985 and in areas outside of Europe in 2001. It is an internationally recognized eco-label that is accorded based on 33 criterias. These criterias are divided into 4 major heads : Environmental education and information, Bathing water quality, Environmental management, and Conservation and Safety services in the beaches.

It is accorded by the international jury composed of eminent members - United Nations Environment Programme (UNEP), United Nations World Tourism Organisation (UNWTO), Denmark-based NGO Foundation for Environmental Education (FEE) and International Union for Conservation of Nature (IUCN).

List of Blue Flag Beaches of India

Shivrajpur (Gujarat), Ghoghla (Daman & Diu), Kasarkod (Karnataka), Padubidri (Karnataka), Kappad (Kerala), Rushikonda (Andhra Pradesh), Golden beach (Odisha), Radhanagar (Andaman and Nicobar), Kovalam (Tamil Nadu), Eden (Puducherry), Thundi (Lakshadweep), Kadmat (Lakshadweep).

Important Beaches in India

Marina beach - India's longest and world's second longest beach situated in Chennai along the Bay of Bengal.

Colva beach - Longest beach in South Goa.

Calangute - largest beach in North Goa. Calangute is a census town in North Goa in the State of Goa. It is known as the Queen of beaches.

Radhanagar Beach - Located in Havelock Island (Swaraj Dweep) of Andaman and Nicobar Islands.

Muzhappilangad beach - A beach on the Malabar Coast in the district of Kannur, Kerala. It is considered as the longest drive-in beach in Asia.

Ramakrishna beach is situated on the east coast of Bay of Bengal in Visakhapatnam, Andhra Pradesh. It gets its name from the Ramakrishna Mission ashram situated near the beach.

Rabindranath Tagore beach - It is located

in Karwar city of Karnataka, is formed at the confluence point of River Kali and Arabian sea. It is named after Nobel Laureate Rabindranath Tagore, he was so impressed by the beauty of Karwar that he called it the Kashmir of Karnataka.

Puri Beach - It is also known as Golden beach. It is located in the city of Puri in the State of Odisha on the shore of the Bay of Bengal. A Hindu sacred place, this beach is the site of the annual Puri Beach Festival & sand displays.

Chandrabhaga - A beach on the coast of Konark, Odisha becoming Asia's first to get the Blue Flag certification.

Gahirmatha - A beach in Kendrapara district of Odisha. It is the world's most important beach for olive ridley sea turtles.

Beyt Dwarka Beach (Beyt Sankhodhar) - It is located at Dwarka in Gujarat and is believed to be the residence of Lord Krishna. Before the development of Okha, this was the beach that was responsible for the trading and shipping for Gujarat.

West Coast Beaches in India

Gujarat - Dumas, Suvali, Ubharat, Dandi, Diu, Tithal, Okha Madhi etc.

Maharashtra - Aksa, Aare Ware, Anjarle, Ganpatipule, Alibaug, Kashid, Karde (Murud), Ganeshgule, Devgad, Kondura, Nivati, Tarkarli (Malvan), Guhagar etc.

Goa - Morjim, Baga, Candolim, Sinquerim, Arossim, Majorda, Varca, Mobor, Cola, Palolem, Agonda, Arambol, Benaulim, Cavelossim, Chapora, Mandrem, Mariposa beach Grove, Colva, Miramar, Bambolim, Cabo de rama, Anjuna, Utorda, Betul, Querim, Vagator, Ashvem, Betalbatim, Caranzalem, Dona Paula etc.

Karnataka - Kasarkod, Padubidri, Gokarna, Kudle, Malpe, Murudeshwara, Panambur, Maravanthe, Tannirbhavi, Devbagh, Sasihithlu, Apsarakonda, Kaup, Someshwar, Half Moon etc.

Kerala - Alappuzha (Alleppey), Hawa, Samudra, Azheekal, Bekal, Vizhinjam, Chavakkad, Cherai, Chootad, Dharmadam, Fort Kochi, Kollam, Kanhangad, Haripad, Marari, Meenkunnu, Muzhappilangad, Mundakkal, Payyambalam, Saddam, Shangumugham, Snehatheeram,

Thirumullavaram, Thangassery, Kannur, Kappil etc.

East Coast Beaches in India

Tamil Nadu – Marina, Dhanushkodi, Mahabalipuram, Edward Elliot, Tharangambadi, Covelong, Olaikuda, Ariyaman/kushi, Pamban, Velankanni, Sothavilai, Kanyakumari, Vattakottai, Sanguthurai, Sengumal etc.

Andhra Pradesh – Manginapudi, Baruva, Bheemunipatnam (Bheemili), Kalingapatnam, Kakinada, Koduru, Mypadu, Perupalem, RamaPuram, Ramakrishna, Rushikonda, Sagarnagar, Suryalanka, Tenneti Park, Tupilipalem, etc.

Odisha – Sonapur, Talsari, Dagara, Gahirmatha, Satabhaya, Pentha Sea, Paradeep sea, Astaranga, Beleswar, Ramachandi, Satapada, Ganjam etc.

West Bengal – Bakkhali sea beach, Digha Sea beach, Junput, Frasergunj Sea beach, Gangasagar Sea beach, Tajpur beach, Bankiput Sea beach, Mandarmani, Shankarpur, Falta etc.

Puducherry – Paradise, Serenity, Promenade, Karaikal, Yanam, Auroville, Veerampattinam etc.

Beaches in Island Territories

Lakshadweep – Bangaram, Agatti, Kalpeni, Kavaratti, Minicoy etc.

Andaman and Nicobar Islands – Bharatpur, Laxmanpur, Kala Patthar, Elephant, Baratang Island, Wandoor etc.

Dadra and Nagar Haveli and Daman and Diu: Devka Beach

Important Beaches of the World

Praia do Cassino (Casino beach) is the longest beach in the world. It is located in the Southernmost part of Brazil (South America).

Monolithi beach - The longest beach of Europe, situated in Greece.

Coorong beach - The longest beach in Australia, situated in South Australia.

McDonald beach - The longest beach in Antarctica.

Padre Island National Seashore is located in Texas, United States of

America.

Cox Bazar Beach is located at Cox's Bazar, Bangladesh, is the longest unbroken sandy beach in the world.

Stepwells

A **stepwell** is a well or a pond that is designed elaborately with a staircase and is used for storing water. The main purpose of building a stepwell is to ensure all-year-round availability of the water. Stepwells are known as baori, baoli or 'vavs' and are found spread across India, mostly in the states of Gujarat and Rajasthan.

Rani ki Vav is a stepwell located in Patan, Gujarat. Its picture has been featured on ₹100 note by RBI (Reserve Bank of India) in July 2018 and it was included in UNESCO World Heritage Site in 2014. Built by his wife Rani Udayamati in the loving memory of King Bhimdev I of Solanki rule.

Chand Baori is a stepwell situated in the village of Abhaneri in Rajasthan. It is one of the deepest and largest stepwells in India. It was built by King Chanda of the Nikumbh Dynasty in 9th century AD.

Agrasen Ki Baoli is a historical stepwell in New Delhi. It was designated a protected monument by the Archaeological Survey of India (ASI) under the Ancient Monuments and Archaeological Sites and Remains Act of 1958.

Pushkarani stepwells are situated in Hampi, Karnataka, and traces its origins back to the Vijayanagara Empire. These stepwells were used to host the annual boat festival during the reign of the Vijayanagara Empire.

Surya Kund is a stepwell that is situated in the vicinity of the Surya Temple, which is located in the Modhera Village, Mehsana, Gujarat. This stepwell was constructed in the 11th Century under the rule of Bhima Deva I of the Chalukya Dynasty.

Adalaj Vav was built in the 15th Century by Mahmud Begada to commemorate Queen Rudabai, who was the wife of Veer Singh the chieftain of Vaghelas. It is located in Adalaj, Gujarat.

Raniji Ki Baoli is a stepwell that was constructed in the 17th Century by Rani Nathavati Ji. It is a 46 meters deep stepwell. It is located in Bundi, Rajasthan.

Panna Meena Ka Kund is located in Jaipur, Rajasthan.

Rajon Ki Baoli was built in the 16th Century by Daulat Khan and was used as a bathing place for the masons. It is located in Mehrauli, Delhi. There is a mosque adjoining the stepwell which was used by Daulat Khan to offer prayers.

Baoli Ghaus Ali Shah is a stepwell that was constructed by Ghaus Ali Shah, a local chief during the reign of Mughal emperor Farrukh Siyar. It was built in the 18th Century.

Shahjahan ki baoli (Choron ki baoli) is a baoli in Maham, Haryana. It was built during the reign of Emperor Shah Jahan.

Dada Harir Stepwell is located in Ahmedabad, Gujarat. It is also known as Bai Harir Sultani.

Birkha Bawari is a stepwell located in Jodhpur, Rajasthan.

Shahi Baoli is located in Lucknow. It was built by Nawab Asaf-ud-Daulah between 1784 and 1794.

Charthana Stepwell, Pingli Stepwell, Arvi Stepwell, all lie in Maharashtra.

Waterfalls

Waterfalls In India

Andhra Pradesh: Kailasakona, Katiki, Thatiguda, Penchalakona, Kalyan Revu, Kaigal, Ganganna Sirrassu.

Arunachal Pradesh: Sirki, Nuranang, Bap Teng Kang.

Assam: Akashiganga, Champawati, Kakochang, Sivakunda, Dikrut, Kathilangso, Panimur, Bendao Baglai, Tegheria.

Bihar: Kakolat, Karkat, Telhar, Tutla Bhawani, Manjhar, and Dhua Kund,

Chhattisgarh: Chitrakote falls (on Indravati River, located in Bastar district), Amritdhara Waterfall, Tirathgarh, Rajpuri, Chitradhara, Rakasganda, Ghatarani, MalanjKudum, Kendai, Devdhara, Kailash gufa, Tamra Ghoomar, Mandawa.

Goa: Dudhsagar falls or "Sea of milk", Tambdi Surla, Netravali, Kesarval, Bamanbudo, Charavane.

Gujarat: Juna Ghata, Gira Dhodh, Hathni, Jamjir, Ninai, Zanzari.

Himachal Pradesh: Rahla, Sissu, Chadwick.

Jharkhand: Hundru Falls (Ranchi District), Rajrappa waterfalls, Dassam, Hirni, Jonha, Lodh, Lower Ghaghri, Panchghagh, Sadni, Usri.

Karnataka: Gokak falls (Niagara of Karnataka) is a waterfall located on the Ghataprabha River in Belagavi district. Kunchikal falls is formed on Varahi river in the Shimoga district. The height of this waterfall is 455 mt (1,493 ft) making it the highest waterfall in India. Other Waterfalls: Gaganachukki and Bharachukki, Apsarakonda, Lalguli, Nisargadhama, Burude, Dondole, Shanti, Barkana, Chunchanakatte, Godchinamalaki, Hebbe, Koosalli, Muthyala Maduvu, Varapoha, Chunchi, Sathodi, Shivasamudram.

Kerala: Adyanpara, Aruvikkuzhi Athirappilly, Charpa falls Cheeyappara, Kumbhavurutty, Lakkom Waterfall, Marmala, Meenmutty, Palaruvi, Pattathippara, Perunthenaruvu, Soochipara, Thommankuthu, Thusharagiri, Valanjanganam, Valara, Vazhachal.

Madhya Pradesh: Dhuandhar falls (Jabalpur district), Bahuti (highest waterfall in Madhya Pradesh), Patalpani, Raneh, Pandav, Tinch, Rajat Prapat, Keoti, Bee, Pawa, Purwa, Gatha, Sultangarh, Bhoora Khon, Kapildhara, Chachai, Dhanora, Shambhu Dhara.

Maharashtra: Bhambavli Vajrai Waterfall (Satara district), Amboli Ghat, Chinchoti, Dabhosa, Dhobi, Dugarwadi, Garambi, Gavlideo Hill, Kune, Lingmala, Malshej Ghat, Mandakini, Napne, Pandavkada, Randha, Sahastrakund, Tapalwadi.

Manipur: Barak, Khayang, Sadu Chiru, Taphou, Khoupum, Ngalo.

Meghalaya: Kynrem falls, Elephant, Nohsngithiang (Seven Sisters falls), Dainthlen, Langshiang, Bishop And Beadon – The Twin Brother, Sweet (Kshaid Weitden), Bophill, Mawphlang, Imil-Chang-Dare, Weinia, Tyrshi, Pelga.

Odisha: Khandadhar waterfall (Sundargarh district), Badaghagara, Bagra, Beniadhush, Duduma, Handi Bhanga, Joranda, Koilighugar, Pradhanpat, Sanaghagara.

Rajasthan: Chulia waterfall (on Chambal River), Gaipernath, Dhruddiya, Padajhar

Mahadev, Menal, Bhimlat, Alewa.

Tamil Nadu: Hogenakkal falls (Tamil Nadu and Karnataka), Agasthiyar, Agaya Gangai, Aintharuvu, Catherine, Gaur Vellaian, Halasana, Kiliyur, Kolakambai, Kuthiraiyar, Kutladampatti, Poombarai, Pykara Lower, Pykara upper, Thalaiyar, Thirparappu, Monkey, Kumbakarai, Siruvani, Thirumoorthy, Kodiveri.

Telangana: Kuntala, Pochera, Mallela Theertham.

Uttar Pradesh: Lakhaniya Dari, Sidhnath Dari, Vindham.

Uttarakhand: Birthi, Corbett, Gauri Kund, Kempty, Tiger, Vasudhara.

Jammu and Kashmir: Aharbal, Siar Baba Temple, Kokernag, Noori Chamb, Verinag, Baba Dhanisar.

Five Highest Waterfalls in India: Kunchikal falls (Karnataka), Barehipani falls (Odisha), Nohkalikai falls and Nohsngithiang falls (Meghalaya), Dudhsagar Waterfalls (Goa).

Highest Waterfalls in the World

Angel falls is located in Canaima National Park, Bolivar, Venezuela (highest waterfall in the World).

Tugela falls is located in KwaZulu-Natal, South Africa.

Tres Hermanas falls is located in Otishi National Park, Junin, Peru.

Olo'upena falls lies in Molokai, Hawaii, United States.

Yumbilla falls is located in Amazonas, Peru.

Lakes of the India/World

Lake: A lake is a body of water that is surrounded by land.

Types of Lakes in India

Freshwater lakes: These have unsalted water that is found in low-lying areas and these are fed from rivers, streams, and runoffs. Wular Lake in Jammu and Kashmir is the largest freshwater lake in India.

Saltwater Lake (brackish water lake): These have a high concentration of salts. Sambhar Lake in Rajasthan is an example of Saltwater Lake.

Artificial Lake: These lakes are man-made lakes that are used to store water for public use. Gobind Ballabh Pant Sagar located in Sonbhadra district, Uttar Pradesh, is the biggest man-made lake of India.

Oxbow Lake: They are the remains of the bend in the river. They are stillwater lakes. This means that water does not flow into or out of them. There is no stream or spring feeding the lake, and it doesn't have a natural outlet. Chandubi in Assam is an example of Oxbow lake.

Crater Lake: The crater lakes are formed when there is a collapse or explosive activity of volcanic eruptions. World's largest basaltic meteor impact crater located in the Buldhana district of Maharashtra. The lake formed on this crater is Lonar Crater Lake.

Lakes of India

Andhra Pradesh

Pulicat Lake is the second largest brackish water lake. It is famous for its wonderful flora and fauna. The State Government organized Falmingo Festival for tourists and visitors as huge number of migratory birds arrive at Pulicat Lake for breeding.

Other Lakes: Kolleru Lake (Freshwater lake), Kondakarla Ava (Natural lake).

Assam

Haflong Lake (Freshwater lake), Son Beel (Freshwater Lake), Chandubi Lake (Oxbow).

Bihar

Kanwar Lake is Asia's largest freshwater oxbow lake. It draws water from the confluence of the Gandak, the Bia and the Kareh river.

Other Lakes: Ghora Katora (Natural Lake).

Chandigarh

Sukhna Lake (Artificial lake).

Gujarat

Sudarshan lake is located in mid-Girnar Hill in Saurashtra. It was built by Pushyagupta, the provincial ruler of Chandragupta Maurya. This artificial reservoir was repaired in 150 AD by Rudradaman, which is mentioned in

Junagarh inscription.

Other Lakes: Hamirsar Lake (Artificial Lake), Kankaria Lake (Artificial Lakes), Thol Lake (Lentic Lake), Shakoor Lake, Lakhota Lake (Artificial lake).

Haryana

Surajkund Lake : A man-made lake in Faridabad that was built in the 10th century by Raja Suraj Pal of Tomar dynasty.

Damdama Lake is the largest natural lake of Haryana located in Gurugram.

Other Lakes: Badkhal, Blue Bird Lake, Brahma Sarovar, Karna, Tilyar.

Himachal Pradesh

Gobind Sagar Lake (Artificial lake): It is a reservoir situated in Una and Bilaspur districts. It is formed by the Bhakra Dam. The reservoir is on the river Sutlej and is named in honour of Guru Gobind Singh, the tenth Sikh guru.

Renuka lake: It is the largest Natural lake of Himachal Pradesh. This lake was named after the goddess Renuka and has been designated as Ramsar site since November 2005.

Other Lakes: Suraj Taal, Dehnasari Lake, Kamrunag Lake, Kareri Lake, Prashar Lake, Chandra Taal.

Karnataka

Agara, Kukkarahalli, Honnamana, Pampa Sarovar, Ulsoor lake.

Kerala

Ashtamudi lake is also called the gateway to the backwaters of Kerala and is well known for its houseboat and backwater resorts.

Vembanad Lake is the longest lake in India, as well as the largest lake in the state of Kerala. Kuttanad lake (The Rice Bowl of Kerala) lies on the southern portion of Vembanad lake.

Other Lakes : Karingalichal Lake, Kuttanad (Brackish Water Lake). Freshwater Lake - Sasthamkotta Lake, Pookode Lake.

Madhya Pradesh

Bhojtal (Upper Lake), Lower Lake, Sharangpani Lake, Sagar Lake (Lakha

Banjara Lake).

Maharashtra

Lonar Lake was created due to a meteorite. It is the only major hoverback in basalt rock. Its water is alkaline. Lonar Lake has been declared as Geological Heritage Sites.

Other Lakes: Artificial Lakes (Gorewada Lake, Powai Lake), Freshwater lakes (Salim Ali, Shivsagar), Mehrun Lake (Natural Lake).

Manipur

Loktak Lake is the largest freshwater lake in Northeast India. It is also home to unique floating islands called "phumdi". Keibul Lamjao National Park is built over the lake, and today it is famous for being the world's only floating national park.

Meghalaya

Umiyam Lake (Barapani Lake) was created by damming the Umiyam River in the early 1960s.

Odisha

Chilika Lake is Asia's largest brackish water lake. It spread across the districts of Puri, Khurda and Ganjam in eastern India. It was designated as the first Ramsar site of India (1981).

Ansupa Lake is a horseshoe-shaped freshwater oxbow lake on the left bank of the Mahanadi river, in Cuttack district.

Rajasthan

Pichola Lake is an artificial freshwater lake. It is located in the Udaipur city. Two islands, Jag Niwas and Jag Mandir are located within Pichola Lake. It was built in 1362 AD by "Banjara" tribesmen. Later, Maharana Udai Singh enlarged the lake by constructing a stone masonry dam in the Badipol region on the shore of the lake.

Pachpadra Lake is a salt lake near Pachpadra in Barmer District. Its sodium chloride level is marked at 98%.

Sambhar Salt Lake is India's largest inland saline water lake. The lake receives water from six rivers: Mantha, Rupangarh, Khari, Khandela, Medtha and Samod.

Dhebar Lake (Jaisamand Lake) is the second largest artificial lake in Asia. It is located in the Udaipur District. It was built by Maharana Jai Singh in 1685.

Other Lakes: Balsamand lake, Kaylana Lake, Nakki Lake, Rajsamand Lake, Ramgarh Lake, Talwara Lake, Siliserh Lake, Pushkar lake (Ajmer).

Sikkim

Gurudongmar Lake is one of the highest lakes in the world. It has religious significance for the Buddhists, Sikhs, and Hindus. It is situated in the Mangan District in the Great Himalayas.

Other Lakes: Khecheopalri Lake, Lake Cholamu, Tsomgo Lake (Changu Lake), Samiti Lake, South Lhonak Lake.

Tamil Nadu

Chembarambakkam (Artificial Lake), Ooty Lake (Freshwater), Berijam Lake, Kaliveli Lake, Kodaikanal Lake, Perumal Eri (Fresh water lake), Red Hills Lake, Sholavaram Lake, Veeranam Lake, Stanley Reservoir.

Telangana

Hussain Sagar (Connects the twin cities of Hyderabad and Secunderabad), Himayat Sagar, Bhadrakali Lake (Freshwater), Fox Sagar Lake (Man made lake), Laknavaram Lake, Osman Sagar, Pakhal Lake, Saroornagar Lake, Waddeppally Lake.

Uttar Pradesh

Govind Ballabh Pant Sagar is the reservoir of Rihand Dam and is India's largest artificial lake.

Other Lakes: Belasagar (Freshwater), Surha Tal (Natural lake), Keetham Lake (Ramsar Sites in 2020), Moti Jheel.

Uttarakhand

Tehri Lake, Bhimtal Lake, Nainital Lake, Roopkund Lake, Deoria Tal, Kedar Tal, Suryadhar lake.

West Bengal

Rabindra Sarobar, Rasikbil Lake, Santragachi Lake, Senchal Lake.

Jammu and Kashmir

Dal lake is located in Srinagar. It is named the 'Jewel in the crown of

Kashmir' or 'Srinagar's Jewel'.

Wular Lake is the largest fresh-water lake of India, situated on the foothills of Harmukh Mountain. Main source of water for Wular Lake is River Jhelum. This lake also has a small island in its center called the 'Zaina Lank'. It was shaped as a result of "tectonic activity".

Other Lakes: Anchar Lake, Nigeen Lake, Krishansar Lake (Oligotrophic Lake), Manasbal Lake (Freshwater Lake), Nundkol Lake.

Ladakh

Pangong Lake is the world's highest saltwater lake. It is a long boomerang-shaped endorheic (landlocked) water body. It is formed from Tethys geosyncline.

Puducherry

Bahour Lake, Ousteri Lake (Ossudu Lake), Velrampet Lake, Nallambal Lake.

Lakes of World

Australia: Lake Eyre (Salt lake) is Australia's largest salt lake.

Africa: Lake Victoria is the largest freshwater lake in Africa.

Antarctica: Lake Vostok (Freshwater lake).

Asia: Lake Baikal (Deepest Lake) is a rift lake located in southern Siberia, Russia.

Europe: Lake Ladoga is one of the largest freshwater bodies in Europe.

Finland is also known as the "Land of Thousand lakes".

North America: Lake Superior is the largest freshwater lake in the world by surface area and the third-largest by volume.

South America: Lake Titicaca is the highest, commercially navigable lake in the world. It is situated between Peru to the west and Bolivia to the east.

Dead Sea (Salt sea): It is a landlocked salt lake between Israel and Jordan in southwestern Asia.

Red Lake is a lake in Beltrami County in northern Minnesota, United States.

The Caspian Sea is the world's largest saltwater lake by volume.

Lake Van: It is the largest lake in Turkey and second Largest in the Middle east.

Great Lakes of North America are a series of interconnected freshwater lakes which connect to the Atlantic Ocean through the Saint Lawrence Seaway. It consists of Lakes Superior, Michigan, Huron, Erie, and Ontario (in the order of west to east). Superior, Huron, Michigan, Erie, and Ontario (In the order of largest to smallest).

African Great Lakes is Series of lakes constituting the part of the Rift Valley lakes in and around the East African Rift. They include Lake Victoria (the Second -largest freshwater lake in the world by area), Lake Tanganyika (the world's second-largest freshwater lake by volume and depth), and Lake Malawi.

Loch Lomond is a freshwater Scottish loch (Lake) which crosses the Highland Boundary Fault, often considered the boundary between the lowlands of Central Scotland and the Highlands.

Valleys of the India / World

Valleys of the India

Chumbi Valley (Sikkim), The valley is on the south side of the Himalayan drainage divide, near the Chinese border with Sikkim, India and with Bhutan.

Dang valley is located in the Terai belt region of Nepal. It lies between the Mahabharat Range in the north and the Churia Range in the south. It is the largest valley in Asia.

Dibang Valley : Dibang is named after the Dibang River or the Talon as the Mishmis. It has an area of 9,129 square Kilometers (3,525 sq mi). Dibang Valley is India's least densely populated district of Arunachal Pradesh.

The Doon Valley is an abnormally wide, long valley inside the Shivalik Hills and the Lesser Himalayas, in the Indian provinces of Uttarakhand, Himachal Pradesh, and Haryana.Examples: Kotli Dun and Patli Dun, dehradun are some of the well-known Duns.

Kangra Valley is located in the lap of Western Himalayas, extends from the foot of the Dhauladhar range of mountains to the south of the Beas River in Himachal Pradesh.

Kullu Valley - The valley mainly spreads

on both banks of Beas river and also on the lower parts of its major tributaries like Parvati and Tirthan river in Himachal Pradesh.

Kashmir valley is situated between the Pirpanjal and Himadri ranges. It has been called the "Paradise on Earth" and "Switzerland of Asia". In valleys grow the rarest of trees and herbs, including the most precious of all flowers - The Zaafran (Saffron). The Kashmir valley is distinctly basin shaped and it has a length of about 140 km (84 miles) and a width varying from 53-55 km (20 to 25 miles).

Lahaul Valley and Spiti Valley are located in the Lahaul and Spiti districts of Himachal Pradesh. The valley is inhabited by snow leopards, foxes ibex, Himalayan brown bear, musk deer and Himalayan blue sheep.

Laitlum Canyons is located about 24 kms from Shillong (Meghalaya) literally translates to the 'End of the Hills.

Letein is a beautiful valley rolling down the southern slopes of Jaintia Hills District, in Meghalaya.

Markha Valley is located between the Zaskar and Ladakh, in the Himalaya region, Ladakh.

Mawdok Dympep is located in East Khasi Hills District -Cherrapunji, Meghalaya.

Nubra (Orchard of Ladakh) and Shyok Valley, situated in the North of Ladakh, between Karakoram and Ladakh ranges of Himalayas.

Puga Valley is located in Changthang Valley in Ladakh. It lies close to Salt Lake Valley.

Suru Valley is located about 140 km from Kargil up to the base of the Penzi-la pass into Zaskar, the Suru Valley is inaccessible in winters because of heavy snowfall. Rich crops of wheat and barley are produced in a major part of this valley. The lower portion of the valley offers spectacular views of the Nun-Kun peaks.

Silent Valley is located in the South-western of Nilgiris in Kerala.

The Stot River Valley begins from Pensi La pass, opening the Zaskar region towards Kargil in the north. The descent into the main Stot valley opens out into a

wide U- shaped valley.

The Valley of Flowers is situated in the Chamoli-Pithoragarh district of Uttarakhand. It is also home to a large number of wild animals such as Asiatic black bear, red fox.

Yumthang Valley or Sikkim Valley of Flowers sanctuary, is a nature sanctuary with river, hot springs, yaks and grazing pasture on rolling meadows surrounded by the Himalayan mountains in the North Sikkim district of Sikkim state in India. It is also called Zero Point Sikkim because this is where civilian roads end and civilians are not allowed to go beyond this point.

Zaskar Valley lies between the Great Himalayan and Zaskar Ranges. It is a semi-arid region, Located around 105 km away from Leh (Ladakh).

Valleys in the world

Alai (Alay) valley is one of the largest highland valleys in Kyrgyzstan through which passes the Pamir tract.

Chuy valley is a large valley located in northern Kyrgyzstan and Southern Kazakhstan, in the northern part of the Tian shan. There are deposits of zinc ore, lead, gold, and construction materials.

Ferghana valley in Central Asia lies mainly in eastern Uzbekistan, but also extends into southern Kyrgyzstan and northern Tajikistan.

Other valleys of world: Kaghan Valley (Pakistan), Lauterbrunnen (Switzerland), Great Rift Valley (Africa), Santa Maria Valley (California), Danum Valley (Malaysian), Valley of Geysers (Russia), Waipio Valley (Hawaii), Barun Valley (Nepal), Jiuzhaigou Valley (China), Yosemite Valley (USA).

Plateaus of the India / World

It is a flat, elevated landform that rises sharply above the surrounding area on at least one side. It takes up a third of the Earth's land.

Types of plateaus

Dissected plateaus - It forms as a result of upward movement in the Earth's crust. The uplift is caused by the slow collision of tectonic plates. Example - Colorado Plateau (United States), Tibetan plateau

(China).

Volcanic plateaus - It is formed by numerous small volcanic eruptions that slowly build up over time, forming a plateau from the resulting lava flows. The North Island Volcanic Plateau covers most of the central part of the North Island of New Zealand. This volcanic plateau still has three active volcanoes: Mount Tongariro, Mount Ngauruhoe, and Mount Ruapehu. Example: Columbia Plateau (United States) and the Deccan Plateau (India).

Indian Peninsular Plateau

The Peninsular plateau is a tableland composed of the old crystalline, igneous and metamorphic rocks. It is part of an ancient land mass called Gondwana level. It is spread over the states of Gujarat, Maharashtra, Bihar, Karnataka and Andhra Pradesh. Its black soil area is known as Deccan Trap. It is divided into two parts: Central highlands, Deccan plateau.

Central highlands - The part of the Peninsular plateau lying to the north of the Narmada river, covering a major area of the Malwa plateau. The Central Highlands are wider in the west but narrower in the east. The eastward extensions of this plateau are locally known as the Bundelkhand and Baghelkhand. The Chotanagpur plateau marks the further eastward extension, drained by the Damodar river.

Bundelkhand plateau - It lies between the Indo-Gangetic Plain to the north and the Vindhya Range to the south. It lies in the east of the plateau of Malwa and the Aravalli hills lie in its North-West. It is made of hard crystalline igneous rock. Ken, Betwa, and Dasan rivers flow through it.

Baghelkhand plateau - Baghelkhand lies in the Eastern part of Madhya Pradesh. Various types of soils are found in this area and they are: black, red, yellow etc. To the west lie elevated plains, and to the east are ridges of the Vindhya Range. The area is divided into two natural regions by the Kaimur Range. It is made up of limestones and sandstone rocks.

Chotanagpur Plateau - It is the Indian Peninsula's north-eastern extension, with an average elevation of about 700 metres above sea level. Mostly in Jharkhand, northern part of Chhattisgarh and Purulia district of West Bengal. The

Son river travels through the plateau and enters the Ganga. It is famous for minerals like Mica, iron ore, bauxite, copper and limestone. This is mainly composed of Gondwana rocks.

Marwar Plateau or Mewar Plateau - It is made up of sandstone, shales and limestones of the Vindhyan period. The Banas river, along with its tributaries (Berach river, Khari rivers) originate in the Aravali Range and flow northwest into Chambal river.

Malwa Plateau roughly forms a triangle based on the Vindhyan Hills, bounded by the Aravali Range in the west and Madhya Bharat Pathar to the north and Bundelkhand to the east. The average elevation of the plateau is 500 meters above the mean sea level. This plateau has two systems of drainage; one towards the Arabian sea (The Narmada, the Tapi and the Mahi), and the other towards the Bay of Bengal (Chambal and Betwa, joining the Yamuna).

The Deccan plateau is one of the oldest plateaus in India. It is a triangular landmass that lies to the south of the river Narmada.

Meghalaya Plateau - It extends east beyond the Rajmahal hills to Meghalaya or the Shillong plateau. The western, central and the eastern parts of the plateau are known as the Garo Hills (900 m), the Khasi-Jaintia Hills (1,500 m) and the Mikir Hills (700 m). Shillong (1,961 m) is the highest point of the plateau.

Karnataka Plateau (Mysore plateau) - Its highest peak is Mullayanagiri in Baba Budan Hills in Chikmagalur district. The plateau is divided into two parts called Malnad and Maidan.

Telangana Plateau - This plateau consists of Archaean gneisses. Its average elevation is 500-600m. The region is drained by three river systems, Godavari, Krishna and Penneru. The entire plateau is divided into Ghats and Peneplains (a vast featureless, undulating plain which is the last stage of the deposition process).

Dandakaranya Plateau - It is located in the Bastar region of the present-day state of Chhattisgarh in the central part of India. Extending over a neighborhood of about 35600 square miles, it includes the Abujmad Hills within the west and borders the Eastern Ghats within the east. The Dandakaranya includes parts of the Chhattisgarh, Odisha, Telangana,

and Andhra Pradesh states.

Plateaus of the world

Tibetan Plateau - Highest and largest plateau in the world with a height of 4,000 to 6,000 meters above the mean sea level and called the 'roof of the world'. It stretches through the countries of Tibet, China, and India. It is surrounded by imposing mountain ranges that harbor the world's two highest mountain peaks, Mount Everest and K2.

Colorado Plateau - It lies in the western part of the U.S.A. It is the largest plateau in America. It is divided by the Colorado River and the Grand Canyon.

Columbia – Snake Plateau (Columbia Basin) is located in Washington, Oregon, and Idaho (USA). It is formed due to the volcanic eruptions of basalt lava. Hence, it is called "flood basalt plateau".

Kimberley Plateau - It is caused by a volcanic eruption, Lies in the northern part of Australia.

Mascarene Plateau - It is a submarine plateau in the Indian Ocean, north and east of Madagascar.

Katanga Plateau - It is located in Congo and famous for copper production.

Mexican Plateau - It is also known as the Mexican Altiplano. World's biggest silver mine is situated in the plateau.

Anatolian Plateau - It is also known as Asia Minor, most of Turkey lies on this plateau. Tigris and Euphrates Rivers flow through this plateau.

Laurentian Plateau - It is known as the "canadian shield".

Patagonian Plateau - It is located in Argentina. It is a semi-arid scrub and rain shadow desert plateau.

Altiplano Plateau - It is located in the South-eastern region of Peru and western region Bolivia. It is famous for Tin reserves.

Loess Plateau - It is located in china. It is known as "Huangtu Plateau". Famous for its highly erodible nature.

Pothohar Plateau - Pakistan.

Bavarian Plateau - Germany.

Ahaggar Plateau (Hoggar) is located in Algeria.

Important fact

Oceanic Plateaus - Plateaus in the ocean are divided into two groups. One group is made of continental crust. The other is made of igneous rock. Igneous oceanic plateaus represent an age between the older, less-dense continental crust and the younger, more dense oceanic crust.

Passes of the India / World

Himachal Pradesh

Baralacha La Pass is situated in the Zaskar range. It links the Lahaul district of Himachal Pradesh to the Leh district of Ladakh.

Debsa Pass - Kullu and Spiti are connected by this road. This pass was discovered in 1995 by a group of mountaineers led by Joydeep Sircar.

Rohtang Pass is a dividing point with one side leading to Kullu Valley and the other to Lahaul-Spiti.

Shipki La connects the districts of Kinnaur and Tibet. It is mostly used for short-distance trade between India and Tibet.

Lepcha La Pass connects the southeastern end of the Spiti valley with Shaikhar and Chango of the Hangrang Valley in Kinnaur District, Himachal Pradesh.

Uttarakhand

Lipu Lekh Pass - Between Uttarakhand, Tibet and Nepal, it acts as a tri-junction. It was India's first border post to allow trade with China. The Manasarovar Yatra passes across this mountain pass.

Mana Pass - The Dungri la pass or Mana Pass connects the Uttarakhand district of Mana with Tibet encompassing Nanda Devi Biosphere Reserve.

Mangsha Dhura (Pithoragarh district) connects of Uttarakhand and Tibet.

Muling La is located in the Great Himalayas, north of Gangotri. During the winter, this seasonal pass connecting Uttarakhand and Tibet is snow-covered.

Niti Pass connects of Uttarakhand and Tibet. It was closed in 1962.

Traill's Pass connects the Pindari and Milam valleys and is located at the end of the Pindari glacier.

Ladakh

Aghil Pass is located to the north of the K2 peak in the Karakoram Range (part of Trans-Himalayas). The pass joins Ladakh with the Shaksgam Valley (Xinjiang Province) of China.

Chang La connects the Ladakh region with Tibet. It provides access to Pangong Lake from Leh. It lies in between Shyok river valley and Leh district.

Fotu La (Photu La) is a mountain pass on the Srinagar-Leh highway in the Zaskar Range of the Himalayas.

Imis La links Ladakh and Tibet.

Khardung La is located in the Ladakh range of mountains. It has an elevation of 5,359 meters. It connects Leh and Nubra Valley.

Karakoram Pass is located in Ladakh between Ladakh and Xinjiang province of China.

Tanglang La Pass is a high altitude mountain pass in Ladakh. It is traversed by the Leh Manali Highway.

Umling La pass is located in Eastern Ladakh and has recently become the world's highest motorable pass (Project Himank). It connects Chisumle and Demchok villages.

Zoji La Pass connects Srinagar and Leh. This pass is located in Dras. It connects three regions of Kashmir Valley in the West, Indus Valley in East, Suru Valley in North-East.

Jammu and Kashmir

Banihal Pass connects Jammu and Srinagar. Jawahar Tunnel was located beneath it. This pass is no longer in use.

Burzil Pass connects Kashmir's Astore Valley and Ladakh's Deosai Plains.

Digar La is a mountain pass situated in Kashmir. Its trekking starts from Sabu to Digar.

Pensi La connects Kashmir with Kargil.

Pir Panjal Pass (Peer ki Gali) connects the Kashmir valley with Rajouri and Poonch. It serves as a vital connection between India and Pakistan. Pir-Panjal Range at an elevation of 2835 metres.

Passes in North Eastern States

Bom Di La connects Arunachal Pradesh with the city of Lhasa in Tibet. It is located at an elevation of 4331 meters.

Dihang La pass connects the states of Arunachal Pradesh and Myanmar in the north-east.

Diphu Pass is located on the MacMohan Line, which serves as a dividing line between Tibet and India's north-east.

Jelep-La is a high mountain pass between India and Tibet in Sikkim Himalayas.

Nathu La connects Sikkim and Tibet.

Pangsau Pass - Arunachal Pradesh and Myanmar are connected by this pass. It is located on the crest of the Patkai Hills on the India-Burma (Myanmar) border.

Bum La Pass is a mountain pass located at the Indo-China border about 15,200 ft above sea level, 37 km away from Tawang in Arunachal Pradesh.

Tulung La is a border pass between the Tsona County in the Tibet region of China and India's Tawang district in Arunachal Pradesh.

Passes in Western Ghats

Bhor Ghat (Maharashtra) - This pass is used to connect three different locations: Railway route between Palasdari and Khandala and Road route between Khopoli and Khandala.

Palakkad Ghat connects Tamil Nadu and Kerala. It is a narrow chasm. The Pal Ghat passes through the Nilgiri Hills and the Anaimalai Hills.

Shencottah Gap connects the Tamil Nadu city of Madurai with the Kerala district of Kottayam. It connects Shencottah and Punalur. Road-rail lines pass through this gap which connects Shencottah and Punalur.

Thal Ghat (Kasara ghat) - It can be found on Mumbai-Nashik Road. The railway line that runs through this pass is one of India's steepest. In Maharashtra, it connects Kalyan and Kasara.

Other important Passes

Bolan Pass is a valley and natural gateway. It is located in the Toba Kakar Range in the Balochistan province of

Pakistan. It connects Quetta to Sibi by road and railway.

Khyber Pass is a mountain pass in the Khyber Pakhtunkhwa province of Pakistan on the border with Afghanistan (Nangarhar Province). It had been an integral part of the ancient Silk Road.

Taungup Pass is a mountain corridor connecting India with Myanmar.

Straits of the India / World

It is a naturally formed, narrow navigable waterway that connects two larger bodies of water. It is a channel of water that lies between two landmasses. Some straits are not navigable, because they are too shallow, or because of an unnavigable reef or archipelago.

Major Straits of India

Nine Degree Channel: It separates the island of Minicoy from the main Lakshadweep archipelago.

Ten Degree Channel: It separates the Little Andaman and Car Nicobar in the Bay of Bengal.

The Grand Channel is between the Great Nicobar islands and the Sumatra islands of Indonesia. It is known as the six degree channel.

Duncan Passage is a strait in the Bay of Bengal. It separates Rutland Island (part of Great Andaman and South Andaman administrative division) to the north, and Little Andaman to the south.

Palk strait: It is located between Tamil Nadu (India) and Jaffna (Sri Lanka).

Mannar strait: It separates India and Sri Lanka. The Gulf of Mannar is a shallow bay, part of the Laccadive Sea in the Indian Ocean.

Gulf of Khambhat (Gulf of Cambay): It divides Kathiawar Peninsula of Gujarat.

The Coco Strait is between the North Andaman islands and the Coco Islands of Myanmar.

Eight Degree Channel (8 degrees north latitude) : It separates the islands of Minicoy (India) and Maldives. Its traditional name is Maliku Kandu.

Major Straits of the World

Gibraltar Strait: It is a waterway that runs

between the southernmost point of Spain and the westernmost point of Africa. It connects the Mediterranean Sea and the Atlantic Ocean.

Bab al-Mandab: It separates Yemen on the Arabian Peninsula and Djibouti and Eritrea in the Horn of Africa. It connects the Red Sea to the Gulf of Aden.

Sunda Strait: It is located between the east Java and Sumatra islands. It connects the Indian Ocean to the Java Sea in the Pacific Ocean (south).

Bosphorus Strait: It separates the European part from the Asian part of Istanbul. It connects the Black sea with the sea of Marmara.

Hudson Strait: It lies between Baffin Island (Nunavut) and northern Quebec, Canada. Links the Atlantic Ocean and the Labrador Sea to Hudson Bay in Canada.

Bass Strait: It separates Tasmania from the Australian mainland.

Dardanelles Strait: It connects the Sea of Marmara with the Aegean Sea in Turkey.

Mozambique Strait: It is situated between the eastern island country of Madagascar and Mozambique on the continent of Africa (west).

Strait of Dover: It separates the United Kingdom from the rest of Europe. It is located in the English Channel's narrowest section, connecting to the North Sea.

Torres Strait: It is situated in the Pacific Ocean, between Australia's Cape York Peninsula and Papua New Guinea.

Strait of Magellan : It is located near southern Chile along the southern edges of the South American continent. It links the Atlantic and Pacific oceans.

Davis Strait: It is the broadest Strait in the world. It is situated north of the Labrador Sea (Greenland) and Baffin Island (Nunavut, Canada). It links Baffin Bay with the Atlantic Ocean.

Bering Strait: It separates Russia and Alaska. It connects the Arctic Ocean's East Siberian Sea to the Pacific Ocean's Bering Sea.

North Channel: It separates Northern Ireland and Scotland and connects the Irish Sea to the Atlantic Ocean.

Yucatan Strait: It connects the Gulf of Mexico to the Caribbean Sea and is located between Mexico and Cuba.

Straits of Florida: It connects the Gulf of Mexico with the Atlantic Ocean. It lies between Florida (United States) and Cuba.

Strait of Otranto: It separates Italy from Albania. It connects the Adriatic Sea and the Ionian Sea.

Taiwan Strait (Formosa Strait): It is located between Taiwan and Mainland China. It links the South China Sea with the East China Sea to the north.

The Denmark Strait (Greenland Strait): It is an oceanic strait between Greenland to its northwest and Iceland to its southeast.

Malacca strait: It separates the Malay Peninsula and Indonesian Island of Sumatra. It connects the Pacific and Indian oceans. It allows for a shorter route from the Andaman Sea to the South China Sea and is now the world's busiest waterway.

Cook strait: It connects the Tasman Sea to the South Pacific Ocean. It is located between New Zealand's north and south islands.

Power Projects in India

India is home to several important hydroelectric projects, such as the Tapovan Vishnugad and Vishnuprayag projects in Uttarakhand, as well as the joint venture Chambal Project between Rajasthan and Madhya Pradesh.

Major Hydroelectric Power Plants

Andhra Pradesh - Nagarjuna Sagar Dam hydroelectric plant (816 MW), Srisailem Right Bank Power Station Project (770 MW).

Arunachal Pradesh - Kameng (600 MW), Ranganadi Hydro power station (405).

Assam - Kopili (275 MW), Karbi Langpi (100 MW).

Chhattisgarh - Hasdeo Bango Dam (120 MW).

Gujarat - Sardar Sarovar Project (1200 MW - River Bed Power House, 250 MW - Canal Head Power House).

Himachal Pradesh - Bhakra Hydroelectric Project (1325 MW), Pong Hydroelectric Project (396 MW), Dehar Hydroelectric Project (990 MW), Nathpa Jhakri Hydroelectric Project (1500 MW), Koldam Hydroelectric Plant (800 MW).

Jammu and Kashmir - Baglihar Hydroelectric Project (450 MW), Salal Hydro Electric Power Plant (720 MW).

Jharkhand - Subarnarekha Hydel Power Project (130 MW)

Karnataka - Almatti Hydroelectric Project (290 MW), Sharavathi Hydro Power Plant (1035 MW), Shivanasamudra Hydro Power Project (42 MW).

Kerala - Idukki Hydro Electric Project (780 MW),

Madhya Pradesh - Indira Sagar Hydroelectric Project (1000 MW), Bansagar Hydroelectric Project (435 MW).

Maharashtra - Koyna Hydroelectric Power Plant (1960 MW)

Odisha - Hirakud HydroElectric Project (276 MW), Balimela Hydro Electric Project (510 MW).

Punjab - Anandpur Sahib Hydroelectric Project (134 MW), Ranjit Sagar Dam Hydroelectric Project (600 MW).

Tamil Nadu - Kadamparai Hydroelectric Project (400 MW), Kundah Hydroelectric Project (585 MW).

Telangana - Srisailem Left Bank hydroelectric Project (900 MW).

Uttarakhand - Tehri Hydroelectric Project (2400 MW).

Uttar Pradesh - Rihand Hydroelectric Project (300 MW).

West Bengal - Sidrapong Hydroelectric Power Plant.

Major Solar Power Plant

Andhra Pradesh - Kurnool Ultra Mega Solar Park (1000 MW, 2017), NP Kunta (978 MW, 2021), Galiveedu Solar Park (400 MW, 2020), Kadapa Ultra Mega Solar Park (250 MW, 2020).

Madhya Pradesh - Rewa Ultra Mega Solar (750 MW, 2018), Mandsaur Solar Farm (250 MW, 2017), Tata Power Solar System, Rajgarh (50 MW, 2014),

Welspun Solar MP Project (151 MW, 2014).

Karnataka - Pavagada Solar Park (2050 MW, 2019), Karnataka I Solar Power Plants (40 MW, 2018).

Gujarat - Charanka Solar Park (690 MW, 2012), Bitta Solar Power Plant (40 MW, 2012)

Tamil Nadu - Kamuthi Solar Power Project (648 MW, 2017)

Maharashtra - Sakri Solar Plant (125 MW, 2013), Maharashtra I Solar Power Plants (67 MW, 2017)

Odisha - Green Energy Development Corporation (50 MW, 2014)

Uttar Pradesh - Lalitpur Solar Power Project (30 MW, 2015).

Rajasthan - Bhadla Solar Park (2245 MW)

Thermal power plants

Madhya Pradesh - Amarkantak Thermal Power Plant, Satpura Thermal Power Plant, Sanjay Gandhi, Birsinghpur Thermal Power Plant, Shri Singaji Thermal Power Station Dongalia, Vindhyachal Thermal Power Station.

Maharashtra - Amravati Thermal Power plant, Khaperkheda Thermal Power plant, Tiroda Thermal Power plant, Chandrapur Thermal Power plant, Solapur Super Thermal Power Station, Mauda Super Thermal Power Plant.

Uttar Pradesh - Anpara Thermal Power plant, Dadri Thermal Power plant, Feroz Gandhi Unchahar Thermal Power plant, National Capital Thermal Power plant, Obra Thermal Power plant, Rihand Super Thermal Power plant, Rosa Thermal Power plant.

Bihar - Barauni Thermal Power Station, Kahalgau Super Thermal Power Project.

Karnataka - Raichur Thermal Power station, Bellary Thermal Power station, Yermarus Thermal Power Station, Udupi Thermal Power plant.

Chhattisgarh - Sipat Thermal Power Plant, Lara Super Thermal Power Plant, Korba Thermal Power Plant, Bhilai Thermal Power Plant.

Jharkhand - Bokaro Thermal Power plant, Patratu Thermal Power plant.

Rajasthan - Chhabra Thermal Power

plant, Kalisindh Thermal Power plant, Kota Thermal Power plant, Suratgarh Super Thermal Power Plant, Barsingsar Thermal Power Station, Anta Thermal Power Station, Ramgarh Gas Thermal Power Station.

West Bengal - Durgapur Thermal Power plant, Farakka Thermal Power plant, Mejia Thermal Power Station, Kolaghat Thermal Power Station, Bakreshwar Thermal Power Station, Durgapur Steel Thermal Power Station, Budge Budge Thermal Power Plant, Sagardighi Thermal Power Station.

Tamil Nadu - Ennore Thermal Power plant, Mettur Thermal Power plant, Neyveli Thermal Power Station, Tuticorin Thermal Power Station, IND Barath Thermal Power Plant.

Gujarat - Gandhinagar Thermal Power plant, Mundra Thermal Power plant, Sikka Thermal Power plant, Ukai Thermal Power plant, Wanakbori Thermal Power plant, Akrimota Thermal Power Station, Kutch Lignite Thermal Power Station, Sabarmati Thermal Power Station.

Odisha - Hirakud Captive Thermal Power Plant, Jharsuguda Thermal Power Plant, Talcher Thermal Power Plant.

Andhra Pradesh - Simhadri Thermal Power Plant.

Telangana - Ramagundam Thermal Power Plant.

Assam - Namrup Thermal Power Plant.

Nuclear Power Plant

Kaiga Atomic Power Station: It is located on Kadra Reservoir on Kali river in Uttar Kannada district in Karnataka. It provides power to five states (Karnataka, Andhra Pradesh, Tamil Nadu, Kerala, and Pondicherry). It has been in operation since 2000 and has a capacity of 880 MW.

Jaitapur Nuclear Power Project: This Project is a key component of the strategic partnership between India and France. It would be the largest nuclear power generating station in the world by net generation capacity, at 9,900 MW. It would be built at Madban village in Maharashtra on the Arabian Sea. Its construction started in 2018.

Chutka Nuclear Power Plant: It is a proposed nuclear power plant to be built in Chutka Village, Madhya Pradesh.

Kakrapar Atomic Power Station: It lies in the proximity of the Mandvi, Surat, and Tapi rivers in Gujarat. Its commercial electricity production started in 1993.

Kalpakkam (Madras) Atomic Power Station: It is located at Kalpakkam in Chennai. It started its operation in 1983-85.

Narora Atomic Power Station: It is located in Narora, Bulandshahar District in Uttar Pradesh. The commercial operation began in 1991.

Rajasthan Atomic Power Station: It is located at Rawatbhata in Rajasthan. It was commissioned in 1973.

Tarapur Atomic Power Station: It is located in Tarapur, Maharashtra. It was the first commercial nuclear power station built in India. It started commercial operation in 1969.

Under Construction: Gorakhpur (Haryana), Bhimpur (Madhya Pradesh), Mahi Banswara (Rajasthan), and Haripur (West Bengal).

Geographical Indication (GI) Tags

GI Tag is a name or sign given to a certain product that has a specific relation with geographical locations. It came into effect on 15 September 2003. GI tags in India are valid for a period of 10 years. It can be applied to agricultural, handicraft, industrial, and food products.

State/UTs Wise GI Tags

Assam: Gamocha, Muga Silk, Assam (Orthodox), Karbi Anglong Ginger, Tezpur Litchi, Joha Rice, Chokuwa Rice, Boka Chaul, Kaji Nemu.

Arunachal pradesh: Arunachal Orange, Idu Mishmi Textiles, Wancho Wooden Craft, Adi Kekir.

Andhra Pradesh: Bangenapalli mango, Araku Valley Arabica Coffee, Guntur Sannam Chilli, Bandar Laddu, Tirupati laddu, Allagadda Stone Carving, Etikoppaka Toys, Durgi Stone, Carvings, Udayagiri Wooden Carving, Dharmavaram Handloom Pattu Sarees and Paavadas, Bobbili Veena, Mangalagiri Sarees and Fabrics, Venkatagiri Sarees, Uppada Jamdani Sarees, Andhra Pradesh Leather Puppetry, Budithi Bell & Brass Metal

Craft, Machilipatnam Kalamkari, Kondapalli Bommallu, Srikalahasti kalamkari.

Bihar - Madhubani Paintings, Applique (Khatwa), Sujini Embroidery, Sikki Grass, Bhagalpur Silk, Bhagalpuri Zardalu, Katarni Rice, Maghai Paan, Shahi Litchi, Silao Khaja, Mithila Makhana (2022), Marcha rice (2023).

Chhattisgarh - Bastar Dhokra, Bastar Iron Craft, Bastar Wooden Craft.

Goa: Fenni, Khej, Harmal Chilli, Mandoli, Banana (Moirá Banana), Khola Red Chilli.

Gujarat: Sankheda Furniture, Bhalia Wheat, Agates of Cambay, Gir Kesar Mango, Kutch Embroidery, Pethapur Printing Blocks, Tangaliya Shawl, Rajkot Patola, Surat Zari Craft, Jamnagar Bandhani, Kachchh Shawls, Kachchhi Kharek.

Himachal Pradesh - Kangra Tea, Himachali Chulli Oil, Kala Zeera, Chamba Rumal, Kinnauri Shawl, Kullu Shawl (Logo), Kangra Paintings.

Jammu & Kashmir - Kani Shawl, Kashmir Pashmina, Kashmir Paper Mache, Kashmir Walnut Wood Carving, Khatamband, Kashmiri Hand Knotted Carpet, Saffron (Mongra, Lachha, Guchhi), Gucci Mushroom, Ramban Anardana.

Jharkhand - Sohrai- Khovar Painting.

Karnataka: Gulbarga Tur Dal, Mysore Silk, Bidriware, Channapatna Toys & Dolls, Kasuti Embroidery, Ilkal Sarees, Mysore Rosewood Inlay, Navalgund Durries.

Kerala: Tirur Betel Leaf (Tirur Vettilla), Aranmula Kannadi, Alleppey Coir, Navara Rice, Palakkadan Matta Rice, Malabar Pepper, Monsooned Malabar Arabica Coffee, Maddalam of Palakkad, Shell Crafts of Kerala, Pokkali Rice, Vazhakulam Pineapple.

Ladakh (Union Territory): Raktsey Karpo Apricot, Pashmina Wool, Ladakh Shingskos.

Madhya Pradesh: Ujjain Batik Print, Chanderi fabric, Leather Toys of Indore, Bagh Prints, Dhalapathar Parda & Fabrics, Bell Metal Ware of Datia and Tikamgarh (Logo), Nagpur Orange, Ratlami Sev, Jhabua Kadaknath Black Chicken Meat, Balaghat Chinnor, Morena Gajak.

Maharashtra - Alibag white onion, Alphonso mango, Ajara Ghansal Rice, Mangalwedha Jowar, Sindhudurg & Ratnagiri Kokum, Solapur Chaddar, Solapur Terry Towel, Puneri Pagadi, Nashik Valley Wine, Paithani Sarees and fabrics, Mahabaleshwar Strawberry.

Manipur - Shaphee Lanphee, Chak-Hao, Wangkhei Phee, Kachai Lemon, Moirang Phee.

Meghalaya - Khasi Mandarin, Memong Narang.

Mizoram - Pawndum, Mizo Chilli, Ngotekherh, Mizo Puanchei, Tawla Lohan, Hmaram.

Nagaland - Chak-Hao, Chakhesang Shawl, Naga Tree Tomato, Naga Mircha, Naga Cucumber.

Odisha: Gopalpur Tussar Fabrics, Ganjam Kewda Rooh, Ganjam Kewda Flower, Dhalapathar Parda & Fabrics, Sambalpuri Bandha Saree & Fabrics, Bomkai Saree & Fabrics, Habaspuri Saree & Fabrics, Berhampur Patta (Phoda Kumbha) Saree & Joda, Pattachitra, Araku Valley Arabica Coffee, Kandhamal Haladi, Odisha Rasgulla, Similipal Kai Chutney, Kapdaganda Shawl, Lanjia Saura Painting, Koraput Kala Jeera Rice, Odisha Khajuri Guda, Dhenkanal Magji, Nayagarh Kantei Mundi Brinjal.

Punjab and Haryana: Phulkari, Basmati.

Rajasthan: Bikaneri Bhujia, Kota Doria, Phulkari, Bagru Hand Block Print, Thewa Art Work, Makrana Marble, Molela Clay, Blue Pottery of Jaipur, Pokaran Pottery, Sojat Mehndi.

Tamil Nadu: Marthandam Honey, Sholavandan Vetrilai, Negamam Cotton Saree, Cumbum Paneer Thratchai, Authoor Vetrilai, Salem Sago (Javvarisi), Thaikkal Rattan Craft, Ooty Varkey, Manamadurai Pottery, Manapparai Murukku, Ramanathapuram Mundu Chilli, Vellore Spiny Brinjal, Myladi Stone Carvings.

Telangana - Tandur Redgram, Warangal Durries, Adilabad Dokra, Telia Rumal, Pochampally Ikat, Silver Filigree of Karimnagar, Nirmal Toys and Crafts.

Tripura: Tripura Queen Pineapple.

Uttar Pradesh: Sambhal Horn Craft, Mahoba Gaura Patthar Hastshilp, Mainpuri Tarkashi, Banaras Pan (Betel Leaf), Muzaffarnagar Gur (Jaggery),

Ramnagar Bhanta (Brinjal), Banaras Langda Aam (Mango), Hathras Hing.

Uttarakhand: Uttarakhand Thulma, Kumaon Chura Oil, Tamta Product, Ringal Craft, Munsyari Razma, Uttarakhand Aipan.

West Bengal: Dalle Khursani, Madur kathi, Darjeeling Tea, Nakshi Kantha, Shantiniketan Leather Goods, Malda Laxman Bhog Mango, Malda Fazli Mango, Santipur Saree, Dhaniakhali Saree, Joynagar Moa, Tangail Saree, Garad Saree, Korial Saree, Kalo Nunia Rice, Sundarban Honey.

Famous GI tag Products

Darjeeling Tea - It became the first geographical indication tagged product in India in 2004-05. The tea leaves are processed as black tea. Tea plants were first planted in the Darjeeling region in the mid-1800s.

Shahi Litchi - It is mainly cultivated in Muzaffarpur (Litchi Garden and Lychee Kingdom), Samastipur, Vaishali, East Champaran and Begusarai and some parts of adjoining areas in agro-climatic areas of the Bihar.

Warli Painting - (Maharashtra, Gujarat, Dadra & Nagar Haveli, Daman Diu): It is a traditional art form of Maharashtra. Its roots can be traced to as early as the 10th century A.D. These paintings are mainly dominated by basic geometric shapes like circles, triangles and squares.

Kanjivaram Silk - Traditionally, the Kanjeevaram is a sari that is usually handwoven in mulberry silk and has pure gold or silver zari that renders it a festive quality. It received a GI Tag in 2005-06.

Chanderi Saree (Madhya Pradesh) - It is light-weight, has glossy transparency and has a sheer texture. The Chanderi sari tradition began in the 13th century. Chanderi sarees are produced from three kinds of fabric: Pure silk, Chanderi cotton and Silk cotton.

Kashmir Saffron - There are three types of saffron available in Kashmir - Lachha Saffron, Mongra Saffron and Guchhi Saffron. It also helps in revitalizing health. It is used in cosmetics and for medicinal purposes.

Indian States Area and Boundaries

Rajasthan (342,239 km²) - It is the largest Indian state in terms of area and the seventh largest in terms of population. It forms the inter-state borders with Madhya Pradesh (longest, Southeast), Haryana (Northeast), Gujarat (South), Punjab (North) and Uttar Pradesh (Shortest, East).

Madhya Pradesh (308,245 km²) - Its inter-state boundaries are Uttar Pradesh (longest, Northeast), Rajasthan (Northwest), Maharashtra (South), Chhattisgarh (East), and Gujarat (Shortest, West).

Maharashtra (307,713 km²) - It forms the inter-state borders with 7 states or Union Territories. Its inter-state boundaries are Madhya Pradesh (longest, North), Karnataka (South), Telangana (Southeast) and Chhattisgarh (East), Dadra and Nagar Haveli and Daman and Diu (Northwest), and Goa (Shortest, South).

Uttar Pradesh (240,928 km²): It is the most populous and fourth largest state in terms of area. It forms the inter-state borders with 9 states or Union Territories. Its inter-state boundaries are Madhya Pradesh (longest, South), Bihar (East), Jharkhand and Chhattisgarh (Southeast), Haryana, Rajasthan (West), Uttarakhand (North), Delhi (Northwest), and Himachal Pradesh (Shortest, Northwest). It shares borders with Nepal.

Gujarat (196,024 km²): It forms the inter-state borders with 4 states or Union Territories, Its inter-state boundaries are Maharashtra (longest, Southeast), Rajasthan (Northeast), Madhya Pradesh (East), and Shortest with Dadra and Nagar Haveli and Daman and Diu (South). Its coastline is about 1214 kms the longest in the country.

Karnataka (191,791 km²): It forms the inter-state borders with 5 states, Its inter-state boundaries are Andhra Pradesh (longest, East). Maharashtra (North), Telangana (Northeast), Tamil Nadu (Southeast) and Kerala (Southwest) and Goa (Shortest, northwest).

Andhra Pradesh (162,968 km²) - It forms the inter-state borders with 5 states. Its inter-state boundaries are Karnataka (longest, West). Telangana (Northwest),

Odisha (North-East), Tamil Nadu (South) and Chhattisgarh (Shortest, North). It has the Second longest Coastline in India after Gujarat, of about 974 km.

Odisha (155,707 km²) - It forms the inter-state borders with 4 states. Its inter-state boundaries are Chhattisgarh (longest, West), Andhra Pradesh (South), Jharkhand (North) and Shortest - West Bengal (North).

Chhattisgarh (135,191 km²) - It forms the inter-state borders with 7 states. Its inter-state boundaries are Odisha (longest, East), Madhya Pradesh (Northwest), Maharashtra (Southwest), Jharkhand (Northeast), Telangana (North), Uttar Pradesh (North), and Andhra Pradesh (Shortest, South).

Tamil Nadu (130,058 km²) - It forms the inter-state borders with 5 states or Union Territories, Its inter-state boundaries are Kerala (longest), Karnataka, Andhra Pradesh, and with Puducherry (Shortest).

Telangana (112,077 km²) - It forms the inter-state borders with 4 states, Maharashtra (North), Andhra Pradesh (East and South), Karnataka (West) and Chhattisgarh (Northeast).

Bihar (94,163 km²) - It forms the inter-state borders with 3 states. Its inter-state boundaries are Uttar Pradesh (longest, West), Jharkhand (South) and West Bengal (Shortest, East). It shares borders with Nepal.

West Bengal (88,752 km²) - It forms the inter-state borders with 5 states. Its inter-state boundaries are Jharkhand (west), Bihar (North-west), Orissa (South), Sikkim (Shortest, North), Assam (east). It has the longest International border with Bangladesh. It shares borders with Nepal.

Arunachal Pradesh (83,743 km²) - It forms the inter-state borders with 2 states. Its inter-state boundaries are Assam (south), Nagaland (south).

Jharkhand (79,714 km²) - It forms the inter-state borders with 5 states. Its inter-state boundaries are Bihar (longest, North), West Bengal (East), Chhattisgarh (West), Odisha (South) and Uttar Pradesh (Shortest, Northwest).

Kerala (38,863 km²) - It forms the inter-state borders with 2 states. Its inter-state boundaries are Tamil Nadu

(longest, East and South), Karnataka (North and Northeast).

Assam (78,438 km²) - It forms the inter-state borders with 7 states. Its inter-state boundaries are Meghalaya (longest, South), Arunachal Pradesh (North), Nagaland (East), Manipur (East), Mizoram (South), West Bengal (West) and Tripura (Shortest, South). It shares the Shortest international border with Bangladesh.

Himachal Pradesh (55,673 km²) - It forms the inter-state borders with 6 states or Union Territories. Its inter-state boundaries are Punjab (longest, West), Uttarakhand (southeast), Jammu and Kashmir (North), Ladakh (North), Haryana (Southwest) and Uttar Pradesh (Shortest, South).

Uttarakhand (53,483 km²) - It forms the inter-state borders with 2 states. Its inter-state boundaries are Uttar Pradesh (South), Himachal Pradesh (West and North-West). It shares borders with Nepal.

Punjab (50,362 km²) - It forms the inter-state borders with 5 states or Union Territories. Its interstate boundaries are Himachal Pradesh (North and Northeast), Rajasthan and Haryana (Southwest), Jammu and Kashmir (North) and Chandigarh (Shortest, East).

Haryana (44,212 km²) - It forms the inter-state borders with 6 states or Union Territories. Its inter-state boundaries are Punjab (North), Rajasthan (West and South), Uttar Pradesh (East), Himachal Pradesh (North), Delhi and Chandigarh (Shortest, Northeast).

Meghalaya (22,429 km²) - It forms the inter-state border with only Assam.

Manipur (22,327 km²) - It forms the inter-state borders with 3 states, Nagaland (North), Mizoram (South) and Assam (West).

Nagaland (16,579 km²) - It forms the inter-state borders with 3 states. Its inter-state boundaries are Assam (longest, West), Manipur (South) and Arunachal Pradesh (North).

Mizoram (21,081 km²) - It forms the inter-state borders with 3 states. Its inter-state boundaries are Tripura (Northwest), Assam (North) and Manipur (Northeast).

Tripura (10,486 km²) - It forms the inter-state borders with 2 states. Its inter-state boundaries are Assam and Mizoram (East). It is surrounded on three sides by Bangladesh (North, South and West).

Sikkim (7,096 km²) - It forms the inter-state borders with 1 state - West Bengal. It shares international borders with three nations – China, Bhutan and Nepal.

Goa (3,702 km²) - It forms the inter-state borders with 2 states - Maharashtra (north), and Karnataka (east and south), It is India's smallest state in terms of area.

Andaman and Nicobar Islands (8,249 km²) - It has no interstate boundaries.

Chandigarh (114 km²) - It shares borders with 2 states, Punjab (north, west and the south), Haryana (east).

Dadra and Nagar Haveli and Daman and Diu (603 km²) - It shares borders with Maharashtra and Gujarat.

Delhi (1,484 km²) - It shares borders with 2 states, Uttar Pradesh (east) and with the state of Haryana in the remaining directions.

Jammu and Kashmir (125,535 km²) - It forms the inter-state borders with 3 states or Union Territories, Himachal Pradesh (North), Punjab (North), and Ladakh (West).

Ladakh (96,701 km²) - It forms the inter-state borders with 2 states or Union Territories, Himachal Pradesh (South) and Jammu and Kashmir (West).

Lakshadweep (32.62 km²) - It has no interstate boundaries.

Puducherry (479 km²) - It shares boundaries with 3 states, Andhra Pradesh, Kerala and Tamil Nadu.

Indian Coastline

India has a coastline of about 7,516.6 km.

Coastline of states : Gujarat (1214.7 km) has the longest coastline followed by Andhra Pradesh, Tamil Nadu, Maharashtra, Kerala, Odisha, Karnataka, West Bengal and shortest coastline with Goa (101 km).

Coastline of Union Territories : Andaman and Nicobar Islands (1962 km) have the longest coastline among the Union Territories.

The transit point between India and

Pakistan : Sikar district is situated in the eastern part of Rajasthan. Ganda Singh Wala, Vagha and Atari are transit points on the border of India and Pakistan.

Neighbouring Countries of India

India shares land Boundaries with 7 countries: Bangladesh (4,096.7 Kms, Radcliffe line), China (3,488 Kms, McMahon line), Pakistan (3,323 Kms, Radcliffe line), Nepal (1,751 Kms), Myanmar (1,643 Kms), Bhutan (699 Kms), Afghanistan (106 Kms).

Afghanistan - Capital - Kabul. Official language - Pashto and Dari (Afghan Persian). First Prime Minister - Shir Ahmad. First President - Hamid Karzai (elected President). Currency - Afghani. National sport - Buzkashi. Highest Peak - Noshaq Peak. Longest River - Helmand River.

National bird - Golden eagle. National flower - Tulip. Independence Day - 19 August 1919. It shares its borders with India (P.O.K), Pakistan, Iran, China, Turkmenistan, Uzbekistan, and Tajikistan.

Bangladesh - Capital - Dhaka. Independence Day - 26 March 1971. Victory Day (Vijay Diwas) - 16 December 1971. Bangladesh was a part of Pakistan from 1947 - 1971. National language - Bengali. Currency - Bangladeshi Taka. National sport - Kabaddi. National animal - Royal Bengal Tiger (*Panthera tigris*). National Bird - Magpie robin, National fruit - Jackfruit.

First President and Father of Bangladesh - Sheikh Mujibur Rahman. First Prime Minister - Tajuddin Ahmed. National Flower and National Emblem - White Water Lily or Sada Shapla. Parliament Name - Jatiya Sangsad Bhaban. Land Boundaries - India and Myanmar.

Longest River - Padma. Highest Peak - Keokradong. Its national flag was designed by Quamrul Hassan. Chakma (Changma) is the largest tribal community of Bangladesh.

India shares the largest land boundary with this nation (4096 km). Indian states that share borders with Bangladesh are

Assam, West Bengal, Mizoram, Meghalaya, and Tripura.

Bangladesh is one of the biggest exporters of Garments including knitwear and hosiery.

Bhutan - Capital and largest city - Thimphu. Official language - Dzongkha. Currency - Ngultrum. First Prime Minister - Jigme Yozor Thinley (first democratically elected). National sport - Archery. National Animal - Takin. National Bird - Raven. First monarch (1907) - Gongsar Ugyen Wangchuck. Parliament Name - Tshogdu. Bhutan has four major rivers, named Drangme Chhu, Mo Chhu, Wang Chhu and Torsa Chhu. Manas river system is the largest river system in Bhutan.

Highest Peak - Gangkhar Puensum. The only Carbon Negative Country in the World. It has never been conquered. It is also called "Land of the Thunder Dragon". Paro town is a famous tourist destination.

On 18 July 2008 Bhutan adopted its constitution. Buddhism is the major religion in Bhutan. It touches the boundary with four Indian states - Assam, Arunachal Pradesh, Sikkim, and West Bengal. International Boundaries - India and China.

China - Capital - Beijing. Official language - Mandarin. Currency - Chinese Yuan, Renminbi. National animal - Giant panda. National bird - red-crowned crane. National sport - Table Tennis. Largest city by population - Shanghai.

National/Independence Day - October 1, 1949. Parliament Name - Quanguo Renmin Daibiao Dahui (National People's Congress). Longest River - Yangtze River. Its Parliament is the biggest parliament in the world. China shares borders with 14 countries.

The most important of these are Afghanistan, Bhutan, India, Kazakhstan, North Korea, Kyrgyzstan, Laos, Mongolia (longest border), Myanmar (Burma), Nepal, Pakistan, Russia, Tajikistan, and Vietnam.

The boundary between Arunachal Pradesh (India) and China is known as 'McMahon Line'. The sino-indian Border is divided into 3 sectors.

China is the biggest producer of cereals, cotton, fruit, vegetables, meat, poultry, eggs, fishery products, coal, silicon, etc. Techniques of making silk were first

invented in China around 7000 years ago.

Maldives - Capital - Male. Official language - Dhivehi. Currency - Maldivian rufiyaa. National Bird - White-breasted Waterhen. First President - Mohamed Amin Didi. Parliament Name - People's Majlis. Independence Day - 26 July 1965. World's First Underwater Cabinet Meeting Was Held in Maldives (2009).

Myanmar - Capital - Naypyidaw (Nay Pyi Taw). Official language - Burmese. Currency - Kyat. First Prime Minister - Thakin Nu. First President - Sao Shwe Thaik. National sport - Chinlone (caneball). National Bird - Green peafowl (Daung). Largest ethnic group - Bamar.

National Flower - Padauk. Independence Day - 4 January 1948. Renamed from "Burma" to "Myanmar" - 18 June 1989. Longest river - Irrawaddy. Highest Peak - Mount Hkakabo. The country shares borders with Thailand, Laos, China, India and Bangladesh.

Nepal - Capital and largest city - Kathmandu. Official language - Nepali. First Prime Minister - Bishweshwar Prasad Koirala (the first democratically elected). First President - Ram Baran Yadav. Currency - Nepalese Rupee. National sport - Volleyball. National Animal - Cow. National flower - Rhododendron arboreum. Highest Peak - Mt. Everest.

Longest river - Karnali River. Independence day - 21 December 1923. Parliament Name - National Assembly or Rastriya Sabha. National Bird - Himalayan monal. International Boundary - India and China. Nepali flag is the only triangular flag in the world.

Pakistan - Capital - Islamabad. Official Language - Urdu. Currency - Pakistani Rupee. National sport - Hockey. National Animal - Markhor. National Bird - Chukar Partridge (*Alectoris chukar*). National anthem - "Qaumi Tarana".

National Flower - Jasmine. Independence /Formation Day - 14 August 1947. Longest River - The Indus River. First Prime Minister - Liyaqat Ali Khan. First President - Mohammad Ali Jinnah. Parliament Name - Majlis-e-Shoora.

Sri Lanka - Capital - Sri Jayawardenepura Kotte (legislative) and Colombo (executive and judicial). Official language - Sinhala. Currency - Sri Lankan Rupee. National sport - Volleyball. National animal - Elephant. National Flower -

water lily. National symbol - A gold lion passenger holding a sword in its right forepaw. Longest River - Mahaweli Ganga.

Independence Day - 4 February 1948. Highest Peak - Pidurutalagala (Mount Pedro). First Prime Minister - D. S. Senanayake. First President - William Gopallawa. Its conventional name is Democratic Socialist Republic of Sri Lanka. Sri Lanka, also known as the 'Pearl of Indian Ocean' and 'teardrop of India'.

Sri Lanka is famous for producing finest tea, rubber, coconut and spices.

Inventions

Invention in Physics

Adding Machine (Pascaline, Arithmetic Machine, and the first calculator) - Blaise Pascal (France) in 1642.

Centigrade Scale - Anders Celsius (Sweden) in 1742.

Cyclotron - Ernest Orlando Lawrence and MS Livingston (USA).

Electric Bulb and phonograph - Thomas Alva Edison (USA).

Electromagnet - William Sturgeon (England).

Mercury Barometer - Evangelista Torricelli (Italy) in 1643.

Photoelectric effect - Heinrich Rudolf Hertz.

Transistor - William Shockley, John Bardeen and Walter Brattain (USA).

Mercury Thermometer - Daniel Gabriel Fahrenheit in 1714.

Laws of planetary motion - Johannes Kepler.

Solar System - Nicolaus Copernicus.

Cosmic radiation - Victor Francis Hess.

First electromagnetic telegraph - By Carl Friedrich Gauss and Wilhelm Eduard Weber in 1833.

Electromagnetism - Hans Christian Oersted.

Radar - Heinrich Hertz (Germany).

X-ray - Wilhelm Conrad Roentgen (Germany) in 1895.

Invention in Chemistry

Oxygen - Joseph Priestley (England).

Phosphorus - Hennig Brand (German alchemist).

Chlorine - Carl Wilhem Scheele (Sweden).

Benzene - Michael Faraday in 1825 (England).

Periodic Table - Dmitri Mendeleev (Russian).

Radioactivity - Henri Becquerel (France).

Fullerenes - Robert Curl, Harold Kroto, and Richard Smalley.

Pasteurization - Louis Pasteur (France).

Sodium - Humphry Davy (England).

Iodine - Bernard Courtois (France).

Radium - Marie Curie and Pierre Curie in 1898.

Radon - Friedrich Ernst Dorn (German physicist).

Structure of an atom - Ernest Rutherford and Niels Bohr.

Electron - Joseph John Thomson in 1897 (England).

Plum pudding model - Joseph John Thomson (England).

Photon and Theory of relativity - Albert Einstein.

Energy Quanta - Max Karl Ernst Ludwig Planck.

Synthesis of urea - Friedrich Wohler (German chemist).

Left-step periodic table - Charles Janet.

Hydrogen Isotope - Harold C Urey.

Antiparticles - Paul Dirac (Introduced the concept in 1930).

Law of Octaves - John Newlands (England) in 1865.

First Bose-Einstein condensate (Fifth state of matter) - Carl E. Wieman (USA).

Plutonium - Glenn T Seaborg in 1940.

Law of Definite proportions - Joseph Proust.

Invention in Biology

Theory of Evolution - Charles Robert Darwin (England).

Vaccination and Smallpox vaccine - Edward Jenner (England).

Cholera vaccine - Jaime Ferran.

TB vaccine - Leon Calmette and Camille Guerin.

Penicillin (First antibiotic) - Alexander Fleming (Scotland).

Classification of five kingdoms - R. H. Whittaker (USA).

First human heart transplant - Christiaan Barnard (South Africa).

Insulin - Frederick Banting.

Bacteria - Antonie Van Leeuwenhoek (Netherlands).

Mitochondria - Albert von Kolliker (Switzerland).

Nucleus (In Cell) - Robert Brown (Scotland).

Polio vaccine - Jonas Salk (USA).

Rabies Vaccine - Louis Pasteur (France).

DDT (Dichloro-diphenyltrichloroethane) - Paul Hermann Müller (Switzerland).

Vitamin - Casimir Funk.

Vitamin A - Frederick Gowland Hopkins (England).

Vitamin C - Albert Szent-Györgyi.

Double helical structure of DNA - James Watson and Francis Crick.

HIV - Luc Montagnier (France).

Invention in Computer

@ sign for Email - Ray Tomlinson.

C Programming Language - Dennis M. Ritchie.

C++ Programming Language - Bjarne Stroustrup.

JAVA Computing Language - James Gosling.

Python programming Language - Guido van Rossum.

World Wide Web (WWW) - Tim

Berners-Lee in 1989.

Computer - Charles Babbage (Britain).

Other Inventions

Aeroplane - Wright brothers (Wilbur and Orville, USA) in 1903.

Balloon (Rubber) - Michael Faraday (England) in 1824.

Dynamo - Michael Faraday in 1831.

Balloon (Hot air) - Montgolfier brothers (Joseph-Michel and Jacques-Étienne).

Ball-Point pen - John Jacob Loud (USA) in 1888.

Bicycle - Baron Karl von Drais (Germany).

Compound Microscope - Hans Janssen and Zacharis Janssen (Netherlands) in 1590.

Capillary feed fountain pen - Lewis Edson Waterman (USA).

Cinema (First to present projected moving pictures) - Lumiere brothers (Auguste and Louis Lumière, France).

Clock (Pendulum) - Christian Huygens (Netherlands) in 1656.

Diesel engine - Rudolf diesel (Germany).

Dynamite - Alfred Nobel (Sweden).

Electric Iron - Henry Seely White (USA) in 1882.

Fountain Pen - Petrache Poenaru (Romanian).

Gatling Gun - Dr. Richard Jordan Gatling (USA).

Jet Engine - Frank Whittle (England).

Railway steam engine - George Stephenson (England).

Revolver - Samuel Colt (U.S.A).

Safety match - Gustaf Eric Pasch (Sweden) in 1844.

Motor Car (Petrol) - Karl Benz (Germany).

Morse code - Samuel Finley Breese Morse (USA).

Neon Lamp - Georges Claude (France).

Nylon - Wallace Hume Carothers (USA).

Photography (Paper) - Invented by

William Henry Fox Talbot (England).

Digital Camera - Steven Sasson (USA).

Pneumatic Tyres - John Boyd Dunlop (Scotland) in 1888.

Printing Press - Johannes Gutenberg (Germany).

Wireless Telegraph - Guglielmo Marconi (Italy).

Razor (Safety) - King Camp Gillette (USA).

Razor (electric) - Jacob Schick (USA).

Rubber (Vulcanized) - Charles Goodyear (USA).

Waterproof raincoat - Charles Macintosh (Scotland).

Safety Lamp - Sir Humphry Davy (England).

Safety pin - Walter Hunt (USA).

Sewing Machine - John Greenough patented the first sewing machine in the United States in 1842. Elias Howe created a sewing machine in 1845.

Steam Engine - Thomas Newcomen (England) in 1712.

Spinning jenny - James Hargreaves (England).

Telephone - Alexander Graham Bell and Antonio Meucci.

Telescope - Hans Lippershey in 1608.

Television - John logie baird (Scotland).

Tractor (first successful petrol engine that could go back and forth) - John Froelich (USA).

Portland cement - Joseph Aspdin.

Paper - Cai Lun (China).

Kevlar - Stephanie Louise Kwolek.

Cotton Gin - Eli Whitney.

Superconductivity - Heike Kamerlingh Onnes.

Vacuum Pump - Otto von Guericke.

Films

Movies based on Sportsperson Biopic

Sachin A Billion Dreams - It is based on the life of Sachin Tendulkar. It is directed by James Erskine.

M. S. Dhoni: The Untold Story - This is the biopic of MS Dhoni. Director - Neeraj Pandey.

Mary Kom - Director - Omung Kumar. It won the Best Popular Film Providing Wholesome Entertainment at the 62nd National Film Awards.

Bhag Milkha Bhag - Based on the life of Indian runner Milkha Singh. Director - Rakeysh Omprakash Mehra. Award : Best Entertainer Award at the 61st National Film Awards.

Shabash Mithu - The biopic of Mithali Raj (Former Captain of Indian Women's Cricket Team). Director - Srijit Mukherji.

Soorma - It is a biographical sports drama film based on the life of Indian hockey player Sandeep Singh. Director - Shaad Ali.

Azhar (2016) - It is a sport-related movie based on the life of Former national team cricketer Mohammad Azharuddin. Director - Tony D'Souza.

Paan Singh Tomar - The story of an Indian athlete and seven-time national steeplechase champion who becomes one of the most feared dacoits in Chambal Valley. Director - Tigmanshu Dhulia.

Indian movies based on sports

Dangal - This movie is based on the true story of wrestlers Mahavir Singh Phogat and his daughters Geeta Phogat and Babita Phogat. Director - Nitesh Tiwari.

Gold - Based on the Indian hockey team who got India its first Gold medal at the 1948 Olympic Games. Director - Alphonse Puthren.

83 - Based on India's incredible cricket World Cup victory in 1983, under the captaincy of Kapil Dev. Director - Kabir Khan.

Saala Khadoos - The film is inspired by a real-life coach named Mohammed Ali Qamar (boxer) from Kolkata. Director - Sudha Kongara.

Love All (2023) - A film inspired by the true story of the Indian Blind Tennis Team, who defied all odds to win the gold medal at the 2018 World Blind Tennis Championship. Director - Sudhanshu Sharma.

Ghoomer (2023) - Inspired by a true story and revolves around a paraplegic woman who becomes a successful cricketer with the guidance of her coach. Director - R Balki.

Jhund (2022) - Based on the life of Vijay Barse, a retired sports teacher who founded an NGO called Slum Soccer. Director - Nagraj Manju.

Kacchey Limbu (2023) - A Indian Hindi -language sports drama film. Director - Shubham Yogi.

Oscar Nominated Indian Movies

Mother India (1957) - It was nominated for the Best Foreign Language Film category. This film starred Nargis, Sunil Dutt, Raaj Kumar, and Rajendra Kumar. Director - Mehboob Khan.

The House That Ananda Built (1979) - It was nominated for an Academy Award for Best Documentary Short. Director - Fali Bilimoria.

An Encounter with faces (1978) - The film was nominated for an Academy Award under the category of Documentary Short Subject. Director - Vidhu Vinod Chopra.

Salaam Bombay (1989) - Director - Mira Nair. It was nominated for Best Foreign Language Film category.

Lagaan (2001) - It was nominated under the Best Foreign Language Film Category. Director - Ashutosh Gowariker. Raju Khan won the National Film Award in 2002 for his choreography.

Little Terrorist (2004) - It was nominated in the Best Live Action Short Film category at the Academy Awards. Director - Aswin Kumar.

The White Tiger (2021) - A black comedy-drama film nominated in the Best Adapted Screenplay category. Director - Ramin Bahrani.

Writing with Fire (2021) - It was nominated in the Best Documentary Feature category. Directors - Rintu Thomas, Sushmit Ghosh.

All That Breathes (2022) - It was nominated in the Best Documentary Feature category. Director - Shaunak Sen.

The Elephant Whisperers (2022) - It was nominated for Best Documentary Short film. Director: Kartiki Gonsalves.

RRR (2022) - It was nominated for Best Foreign Language Film. Director - S. S. Rajamouli. Movie's song 'Naatu Naatu' was nominated for 95th academy awards 2023, and won it.

Famous Indian Directors

Satyajit Ray - His famous Movies - The Apu Trilogy (1955–1959), The Music Room (1958), The Big City (1963) and Charulata (1964) and the Goopy-Bagha trilogy.

Prithviraj Kapoor - Movies - Mughal-E-Azam, Maharathi Karna and Draupadi.

Raj Kapoor - Famous Movies by him - Neel Kamal, Barsaat, Awaara, Shree 420, Sangam, Mera Naam Joker, Sapno Ka Saudagar, Boot Polish, etc.

Mrinal Sen - Bhuvan Shome, Calcutta 71, Ek Din Pratidin, Interview, Akash Kusum, Chorus, Kharij, Baishey Shravana, Aamar Bhuvan.

Bimal Roy - Madhumati, Do Bigha Zamin, Devdas, Biraj Bahu, Sujata, Bandini, Parakh, Naukari, Yahudi, Parineeta (1953).

Shekhar Kapur - Masoom, Bandit Queen, Elizabeth, What's Love Got to Do with It?, The Four Feathers, Mr. India, Elizabeth : The Golden Age.

Films based on Books and Novels

3 Idiots - Based on Novel "Five Point Someone". Director - Rajkumar Hirani.

2 States - Based on the novel 2 States. Director - Abhishek Verma.

Kai Po Che - Based on the novel "The 3 Mistakes Of My Life". Director - Abhishek Kapoor.

Half Girlfriend - Based on the novel "Half Girlfriend". Director - Mohit Suri.

Devdas - Released in 1955, based on the Novel "Devdas" by Sarat Chandra Chattopadhyay. Director - Bimal Roy.

Aisha - Based on the novel "Emma".

Director - Rajshree Ojha.

Omkara - Based on the Novel "Othello". Director - Vishal Bhardwaj.

Haider - Based on the Novel "Hamlet". Director - Vishal Bhardwaj.

Maqbool - Based on the Novel "Macbeth". Director - Vishal Bhardwaj.

The Blue Umbrella - Based on the novel "The Blue Umbrella". Director - Vishal Bhardwaj.

Lootera - Based on the Novel "The Last Leaf". Director - Vikramaditya Motwane.

Junoon - Based on the Novel "A Flight Of Pigeons". Director - Shyam Benegal.

Pinjar - Based on the Novel "Pinjar". Director - Chandraprakash Dwivedi.

Shatranj Ke Khiladi - Based on the Novel "Shatranj Ke Khiladi". Director - Satyajit Ray.

Noor - Based on the Novel "Karachi You're Killing Me". Director - Sunhil Sippy.

Tamas - Directed by Govind Nihalani in 1988.

Guide - Movie directed by Vijay Anand and based on the novel "The guide".

Movies on Indian Freedom Fighters

Gandhi (1982) - Based on Mahatma Gandhi. Director - Richard Attenborough. The Music Director - Pandit Ravi Sankar.

Mangal Pandey: The Rising (2005) - Based on the life of Mangal Pandey. Director - Ketan Mehta.

Shaheed Udham Singh (2000) - Based on Udham Singh. Director - Chitraarth.

Netaji Subhash Chandra Bose: The Forgotten Hero (2004) : Director - Shyam Benegal.

Chittagong (2012) - Based upon events of British India's Chittagong Uprising. Director - Bedabrata Pain.

Kaalapani (1996) - Director - Priyadarshan.

Kerala Varma Pazhassi Raja (2009) - Based on the life of Pazhassi Raja, who fought against the East India Company in the 18th century. Director - Hariharan.

Sardar (1994) - Director - Ketan Mehta. Based on Sardar Vallabhbhai Patel.

Gandhi, My Father (2007) - It depicts Mahatma Gandhi's tumultuous relationship with his eldest son, Harilal Gandhi. Director - Feroz Abbas Khan.

Movies on Politicians

Thackeray (2019) - Based on the life of political leader Balasaheb Thackeray. Director - Abhijit Panse.

An Insignificant Man (2016) - The socio-political documentary is centered on Delhi Chief Minister Arvind Kejriwal.

Thalaivii (2021) - Actress Kangana Ranaut plays late actor-politician Jayalalithaa in the Indian biographical drama. Director - AL Vijay.

The Accidental Prime Minister (2019) - It starred actor Anupam Kher as then-Prime Minister Manmohan Singh.

PM Narendra Modi (2019) - Director - Omung Kumar. Lead role - Vivek Oberoi as Narendra Modi.

Movies Based On Wars

Haqeeqat (1964) - Movie is set in the backdrop of the India-China War of 1962. Director - Chetan anand.

Border (1997) - Based on the Battle of Longewala, it covers the Indo-Pakistan War of 1971. Director - JP Dutta.

LOC Kargil (2003) - Based on the 1999 "Operation Vijay". Director - JP Dutta.

1971 (2007) - This movie talks about the Indo-Pakistan War of 1971 and tells the story of 6 prisoners of war.

Uri: The Surgical Strike (2019) - A fictional dramatisation of true events of the retaliation of 2016 Uri attack.

Kesari (2019) - Based on the battle of Saragarhi that was fought between just 21 soldiers of the 36 Sikh Regiment of the British Army and 10,000 Afridi and Orakzai Pashtun tribesmen in 1897.

Shershah (2021) - The story of Captain Vikram Batra, who was responsible for India's victory during the 1999 Operation Vijay.

Important Film Cities

Ramoji Film City (Hyderabad) - It is the largest film studio complex in the world and as such has been certified by the Guinness World Records. Established by

Ramoji Rao in 1996. Maa Naannaki Pelli (1997) was the first film shot in the studio.

Mumbai Film City - Situated near Sanjay Gandhi National Park and is also known as Dadasaheb Phalke Chitra Nagari.

Innovative Film City, Bengaluru - It is located at bidadi.

Noida Film City, Noida - It has AAFT and Marwah Films & Video Studios along with India's prominent news channels.

MGR Film City, Chennai - A popular tourist hotspot.

Important Facts about Indian Cinema

Dada Saheb Phalke - He is known as the father of Indian cinema.

Raja Harishchandra (1913) - First silent film in India.

Alam Ara (1931) - First talking film of India. Director - Ardeshir Irani.

Kisan Kanya (1937) - First indigenously made color film of India.

Elphinstone Picture Palace (Calcutta) - First cinema theater in India, established in the year 1907.

Bhanu Athaiya - First Indian to receive an Oscar award for the costume Design of the Film Gandhi (1982).

Devika Rani - First ever recipient of the Dadasaheb Phalke Award (1969).

Government Schemes

Pradhan Mantri MUDRA (Micro Units Development and Refinance Agency Limited) Yojana - It was launched on April 8, 2015. The scheme has been classified under three categories as 'SHISHU (loans upto Rs.50,000)', 'KISHORE (loans above Rs.50,000/- and upto Rs. 5 lakhs)' and 'TARUN (loans above Rs. 5 lakhs and upto Rs. 10 lakhs)'.

Pradhan Mantri Gram Sadak Yojana (PMGSY) - It was launched on 25th December 2000 as a Centrally Sponsored Scheme. Nodal ministry - Ministry of Rural Development.

Pradhan Mantri Jan Dhan Yojana (PMJDY) - This scheme was announced on 15th August 2014 and launched on 28 August 2014. It is implemented by the

Ministry of Finance. It has facility Basic savings bank accounts with an overdraft facility (OD) of Rs. 10,000 for every household.

Pradhan Mantri Kaushal Vikas Yojana - It was launched in 2015 to encourage skill development in the country by providing free short-duration skill training. Ministry - Ministry of Skill Development and Entrepreneurship (MSDE).

Pradhan Mantri Awas Yojana (Urban) Mission - It was launched on 25th June 2015 and intends to provide housing for all in urban areas by the year 2022. It was launched by the Ministry of Housing and Urban Affairs. Pradhan Mantri Awas Yojana Gramin (PMAY-G) was launched in 2016.

Pradhan Mantri Garib Kalyan Rojgar Abhiyaan (PMGKRA) - It was launched in 2020 for a period of 125 days with a resource envelope of Rs.50,000 crore, by the Ministry of Rural Development.

Pradhan Mantri Rojgar Protsahan Yojana (PMRPY) was launched on 9th August, 2016. Objective - To incentivise employers for creation of employment. Nodal Ministry - Ministry of Labour and Employment.

Pradhan Mantri Jeevan Jyoti Bima Yojana (PMJJBY) - Launched - 9th May, 2015 under the Ministry of Finance. It is a one year life insurance scheme renewable from year to year offering coverage for death. It is available to people in the age group of 18 to 50 years having a bank account.

Pradhan Mantri Suraksha Bima Yojana (PMSBY) is a one-year accidental insurance scheme renewable from year to year. It was launched on 9th May, 2015. Ministry - Ministry of Finance.

Pradhan Mantri Annadata Aay Sanrakshan Abhiyan (PM AASHA) was launched in 2018. It is a scheme that guarantees a Minimum support price (MSP) to the farmers.

PM WANI Scheme stands for PM Wi-Fi Access Network Interface. It was approved in December 2020. It aims to enable easily accessible public Wi-Fi hotspots spread across the country and aims at helping accelerate the uptake of broadband internet services.

PM Formalisation of Micro food processing Enterprises Scheme:

Objective - To provide financial, technical and business support for setting up / upgradation of micro food processing enterprises. The scheme is operational for a period of five years from 2020-21 to 2024-25. The PMFME Scheme primarily adopts One District One Product (ODOP).

Pradhan Mantri Krishi Sinchayee Yojana - Launched on 1st July, 2015 with the motto of "Har Khet Ko Paani". It is being implemented to expand cultivated areas with assured irrigation, reduce wastage of water and improve water use efficiency. Nodal Ministry - Ministry of Jal Shakti.

PM-ABHIM (Prime Minister Ayushman Bharat Health Infrastructure Mission) was launched in October 2021. Aim - to strengthen health care infrastructure across the country with focus on primary, secondary and tertiary care services.

The PM Street Vendor's Atma Nirbhar Nidhi (PM SVANidhi) was launched on June 01, 2020 by the Ministry of Housing and Urban Affairs.

The Prime Minister's Employment Generation Programme (PMEGP) is related to the Ministry of Micro, Small and Medium Enterprises. It was launched in 2008.

Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN) - Shri Narendra Modi has approved the Pradhan Mantri Swasthya Suraksha Nidhi (PMSSN) as a single non-lapsable reserve fund for a share of Health from the proceeds of the Health and Education Cess levied under Section 136-b of the Finance Act, 2007.

PM-YUVA (Yuva Udyamita Vikas Abhiyan) - Launched by the Ministry of Education on 29 May 2021 for young writers up to the age of 30 years. YUVA 2.0 Scheme: It is a mentoring programme for young and budding authors below the age of 30.

Pradhan Mantri Vaya Vandana Yojana - It was launched in 2017 by the Ministry of Finance to offer a guaranteed payout of pension to senior citizens every month. Eligibility - Minimum Entry Age: 60 years (completed) and Maximum Entry Age: No limit.

The PM CARES for Children Scheme was launched on 29th May 2021. It aims to support children who have lost both the Parents or legal Guardian or Adoptive Parents or Surviving Parent to COVID-19 pandemic.

The Pradhan Mantri Fasal Bima Yojana (PMFBY) was launched in 2016, to provide financial support to the farmers and to provide insurance coverage to farmers in case of crop loss due to natural calamities, pests, or diseases.

Pradhan Mantri Ujjwala Yojana (PMUY) was launched on 1 May 2016 to release deposit free LPG connections in the name of adult female members. PMUY phase-2 (Ujjwala 2.0) was launched in August 2021 with a target to release 1 crore additional PMUY connections.

Pradhan Mantri Matsya Sampada Yojana (PMMSY) was launched on 10th September 2020. It is being implemented in all States and UTs for a period of 5 years from FY 2020-21 to FY 2024-25.

Pradhan Mantri Suryodaya Yojana launched in 2024. Aim - To install rooftop solar power systems in one crore households across the nation.

PM Shram Yogi Mandhan - Social welfare scheme launched by the Ministry of Labour and Employment in February 2019 for poor labourers in the unorganised sector from minimum 18 years of age to maximum 40 years.

PM Kisan Urja Suraksha Evam Utthan Mahabhiyan (PM KUSUM Scheme) - Launched in March 2019 by the Ministry of New and Renewable Energy (MNRE), to subsidize farmers to install solar irrigation pumps for cultivation.

Stand Up India Scheme - It was launched on 5th April, 2016 by the Ministry of Finance. The Scheme facilitates bank loans between Rs.10 lakh and Rs.1 crore to at least one Scheduled Caste/ Scheduled Tribe borrower and at least one Woman borrower per bank branch for setting up greenfield enterprises.

Soil Health Card Scheme - It is a Centrally Sponsored scheme, launched on 19 February 2015. Its objective is to issue soil health cards every 2 years, to all farmers. It is implemented by the Ministry of Agriculture and Farmers' Welfare.

Start-Up India Scheme - It was first announced by Indian Prime Minister, Narendra Modi during his speech on 15 August 2015. The event was inaugurated on 16 January 2016. It intends to build a strong ecosystem for nurturing innovation and Startups in the country that would drive sustainable economic growth and generate large-scale

employment opportunities.

Saansad Adarsh Gram Yojana (SAGY) - It was launched by the Prime Minister of India on 11th October, 2014 on the birth anniversary of Jai Prakash Narayan. Reviewed periodically by the Ministry of Rural Development.

Swachh Bharat mission - It was launched in 2014 by Shri Narendra Modi to fulfill the vision of a cleaner India by 2 October 2019. It was launched on the occasion of Mahatma Gandhi's 145th birth anniversary. It aims to achieve neat and clean tourist places which attract more Foreign and Domestic Tourists.

Swadesh Darshan Scheme is a Central Sector scheme launched in 2014 - 15 by the Ministry of Tourism and Culture for the integrated development of theme - based tourist circuits.

Swadesh Darshan 2.0 (SD2.0) - It aims to develop sustainable and responsible destinations with a tourist and destination-centric approach.

Smart cities and Academia Towards Action and Research (SAAR) Scheme - It was launched in 2022 by the Ministry of Housing and Urban Affairs (MoHUA) as a part of the Azadi Ka Amrut Mahotsav (AKAM) celebrations under the Smart Cities Mission. It is a joint initiative of MoHUA, the National Institute of Urban Affairs (NIUA), and some of the leading academic institutions in India.

Atal Mission for Rejuvenation and Urban Transformation (AMRUT) - It was launched on 25th June 2015 in selected 500 cities and towns across the country. Ministry - Ministry of Housing and Urban Affairs. Jawaharlal Nehru National Urban Renewal Mission was renamed to Atal Mission for Rejuvenation and Urban Transformation and then relaunched by Prime Minister of India Narendra Modi in June 2015. It has been subsumed under AMRUT 2.0, which was launched on 1 October 2021. AMRUT 2.0 would promote the circular economy of water through the development of a City Water Balance Plan (CWBP) for each city focusing on recycling or reuse of treated sewage, rejuvenation of water bodies, and water conservation.

Ayushman Bharat Yojana (AB PM-JAY, Ayushman Bharat Pradhan Mantri Jan Arogya Yojana, People's Health Scheme) - Launched by the Prime Minister, Shri Narendra Modi on 23rd September, 2018 at Ranchi, Jharkhand.

Jal Jeevan Mission - It is envisioned to provide safe and adequate drinking water through individual household tap connections by 2024 to all households in rural India. It was launched on 15 August 2019. Nodal ministry - Ministry of Jal Shakti.

Mid-Day Meal Scheme - It was started in India on 15th August 1995 under the name of the 'National Programme of Nutritional Support to Primary Education (NP-NSPE)'. In 2021, It was renamed as the '**Pradhan Mantri Poshan Shakti Nirman**' scheme or PM Poshan Scheme and it also covers students of balvatikas (children in the 3–5-year age group) from pre-primary classes. It would be implemented for a five-year period (2021-22 to 2025-26). It aims to improve the nutritional outcomes of children, pregnant women, and breastfeeding mothers. It has been shaped in part by the NITI Aayog.

Mahatma Gandhi National Rural Employment Guarantee Scheme - It was launched on September 7, 2005. It came into force on February 2, 2006, in 200 backward districts. It provides a legal right for 100 days of employment in every financial year to at least one member of every rural household. It is a social security scheme and poverty alleviation program which provides the legal Right to Work (Article 41) in exchange for money to the citizens of the country.

Rural Employment Generation Programme (REGP) - The Khadi and Village Industries Commission (KVIC) launched REGP on 1st April 1995 for the generation of two million jobs under the Khadi and Village Industry sector in the rural areas of the country.

New India Literacy Programme Scheme - It is a Centrally Sponsored Scheme namely, the 'New India Literacy Programme' (NILP) for implementation during five years from 2022-23 to 2026-27. The program has **5 objectives**: (i) Foundational Literacy and Numeracy, (ii) Critical Life Skills, (iii) Vocational Skills Development, (iv) Basic Education, and (v) Continuing Education.

Mission Antyodaya - It was adopted in the Union Budget 2017-18. It is a convergence and accountability framework aiming to bring optimum use and management of resources allocated by 27 Ministries or Departments of the Government of India under various

programs for the development of rural areas. Nodal Ministry - Ministry of Rural Development.

Target Olympic Podium Scheme (TOPS) - It was formulated by the Ministry of Youth Affairs and Sports in 2014. It facilitates selected athletes with support staff and training personnel, sports psychologists, physiotherapists, etc including direct financial support of 50,000 rupees per month.

Universal Immunization Programmes - The Expanded Programme on Immunization (EPI) was started in India in 1978. Its objective is to reduce morbidity and mortality from tetanus, poliomyelitis, diphtheria, pertussis, and childhood tuberculosis by providing immunization services to all eligible children and pregnant women by 1990.

Nikshay Poshan Yojana - It was launched in 2018 by the Ministry of Health and Family Welfare. It aims to help every Tuberculosis (TB) Patient by providing a Direct Benefit Transfer (DBT) of Rs 500 per month for nutritional needs.

UDAN (Ude Desh ka Aam Nagrik) - It is aimed at enhancing connectivity to remote and regional areas of the country and making air travel affordable. Launched: 2016. Nodal ministry: Ministry of Civil Aviation.

Make In India - Launched on 25th September 2014, aims to transform the country into a leading global manufacturing and investment destination.

PRASAD Scheme (Pilgrimage Rejuvenation And Spirituality Augmentation Drive) - It was launched in the year 2014 - 2015 under the Ministry of Tourism.

Fit India Movement - It was launched by the Prime Minister on 29th August 2019 on National Sports Day to encourage citizens to take up physical activity and sports. The Fit India Movement is a 'multi-ministry effort', including the Sports, Education Ministry, Panchayati Raj, and Rural Development Ministries.

Swasthya Sathi Scheme - It is a health insurance scheme launched by the Chief Minister of West Bengal on 30th December 2016. Under this scheme, basic health insurance cover up to Rs 5 lakh is provided for both tertiary and secondary medical care for each family.

Training of Rural Youth for Self Employment (TRYSEM) scheme was launched to provide technical skills to the rural youth of families below poverty line by the Government of India in 1979.

Chief Minister Learn and Earn Scheme (2023) of Madhya Pradesh, the government will provide Financial assistance, Training and Employable skills.

Power Sector's Revamped Distribution Sector Scheme was launched by the Prime minister in 2022. This scheme proposes to provide 25 crore Smart Prepaid meters to consumers all over the country.

Nand Baba Milk Mission scheme was launched by the Uttar Pradesh government to boost milk production
Swarna Jayanti Shahari Rojgar Yojana (SJSRY) was launched in the year 1997 by the Government of India.

Mukhyamantri Ladli Behna Yojana was implemented on 28 January 2023 by Madhya Pradesh Government. Rs 1000 per month will be given to women under this scheme.

Scheme for Safeguarding the Intangible Heritage and Diverse Cultural Traditions of India: Objective - Reinvigorating and revitalizing various institutions, groups, individuals, identified non-MOC institutions, non-government organizations. Ministry - Ministry of Culture.

Mission Amrit Sarovar was launched on 24th April 2022. The Mission is aimed at developing and rejuvenating 75 water bodies in each district of the country as a part of celebration of Azadi ka Amrit Mahotsav.

PANKH (Protection, Awareness, Nutrition, Knowledge, and Health) Abhiyan was launched for a year (2021-22) to aid in the holistic development of the girl child.

Smart Cities Mission was launched by Narendra Modi in 2015. 100 Cities were proposed to be covered under this scheme.

National Urban Health Mission (NUHM) was Launched by the Ministry of Health and Family Welfare in 2013.

Jal Kranti Abhiyan was inaugurated by the Union Minister of Water Resources, River Development and Ganga Rejuvenation in Rajasthan in June 2015.

PRASHAST (Pre - Assessment Holistic Screening Tool) was Launched in 2022. It is a Disability Screening Mobile App developed by NCERT launched in Delhi schools to screen students with suspected disabilities.

Annapurna Yojana provides Food Security to Senior Citizens above 60 years of age and to those eligible old-aged persons who remained uncovered under Indira Gandhi NOAPS (National Old Age Pension Scheme).

The Pudhumai Penn scheme: Tamil Nadu Chief minister M K Stalin launched this Scheme on 5 September 2022.

Swarnajayanti Gram Swarozgar Yojana was launched in 1999-2000. Ministry - Ministry of Rural Development. The Direct Benefit Transfer of LPG or **PAHAL** (Pratyaksh Hanstantrit Labh) scheme was earlier launched on June 1, 2013. This scheme is related to LPG Subsidy.

The National Literacy Mission (NLM) was set up in 1998 with an aim to impart functional literacy to all non-literate persons in the 15 - 35 age group.

Saksham Anganwadi and Poshan 2.0 scheme was launched by the Ministry of Women and Child Development

Surakshit Hum Surakshit Tum Abhiyan was launched by NITI Aayog and Piramal Foundation.

Startup India Seed Fund Scheme - It was launched in February 2021 and was implemented with effect from 1 April 2021. Ministry - Ministry of Commerce and Industry.

Rural Backyard Piggery Scheme was launched by Meghalaya.

The National Family Benefit Scheme was started in August 1995.

Mission Karmayogi - It was launched in September 2020 by the Ministry of Personnel, Public Grievances, and Pensions. It is a National Programme for Civil Services Capacity Building (**NPCSCB**) scheme for its 46 lakh civil servants including the officers and employees at all levels. Karmayogi Bharat owns and operates the digital assets and platforms for the online training of Civil servants.

The National Rural Employment Programme was merged with Jawahar

Rozgar Yojana in 1989.

Panchamrut Yojana was launched by the Uttar Pradesh Government to increase the production of sugarcane and Double the Farmers' Income.

Weekly Iron and Folic Acid Supplementation (WIFS) program was launched by the Ministry of Health and Family Welfare of India in 2012.

The Twenty Point Programme (TPP) was launched in 1975. This was a package of Social Sector Schemes/Programmes having a bearing on the lives of the poor and the underprivileged.

SVAMITVA (Survey of Villages Abadi and Mapping with Improved Technology in Village Areas) was Launched on April 24, 2020. It aims to provide an integrated property validation solution for rural India.

Mukhyamantri Chaa Srami Kalyan Prakalpa Scheme: It was launched by the Tripura Government for tea garden workers.

Gujarat Semiconductor Policy 2022 to 2027 launched with a view to generate at least 2,00,000 new employment opportunities.

Kudumbashree is the poverty eradication project of the State government of Kerala.

District Investment Promotion Agencies (DIPA) implemented by the Government of Odisha.

Sanjeevani scheme launched by the Himachal Pradesh government for dairy farmers.

The Village Defence Guards scheme: It was launched by the Jammu and Kashmir.

Sukanya Samridhi Yojana - Launched on 22 January 2015 under the Ministry of Finance. A small deposit scheme of the Government of India meant exclusively for a girl child and is launched as a part of Beti Bachao Beti Padhao Campaign.

Atal Pension Yojana (APY) - It was launched on 9th May, 2015 under the Ministry of Finance.

Mission Indradhanush - A health mission. Launched: 2014. Nodal ministry: Ministry of Health and Family Welfare.

Unnat Jyoti by Affordable LEDs for All (UJALA) - Launched on January 05, 2015, to provide energy-efficient LED bulbs to domestic consumers at an affordable price. Nodal Ministry - Ministry of Power.

Members of Parliament Local Area Development Scheme (MPLADS) - Under the scheme, each MP has the choice to suggest to the District Collector for works to the tune of Rs.5 Crores per annum to be taken up in his/her constituency. Launched: 1993.

Agnipath Scheme - Scheme approved by the Government of India on 14 June 2022, for recruitment of soldiers below the rank of commissioned officers into the three services of the armed forces. All recruits will be hired only for a four year period.

Heritage City Development and Augmentation Yojana (HRIDAY) - Launched in January, 2015 under the Ministry of Urban Development. It aims to preserve and rejuvenate the rich cultural heritage of the country.

UMANG (Unified Mobile Application for New-age Governance) is developed by the Ministry of Electronics and Information Technology (MeitY) launched in 2017.

Nai Roshni Scheme - It aims to empower and enhance confidence among minority women by generating awareness about women rights and interventions for 'Leadership Development'. Nodal ministry: Ministry of Minority Affairs. Launched: 2012-13.

Eklavya model residential school (EMRS) - Scheme for model residential school for Indian tribals (ST, Scheduled Tribes) across India. It was started in 1997-98 under the Ministry of Tribal Affairs.

Faster Adoption and Manufacturing of Electric (and Hybrid) Vehicles (FAME India Scheme) - It aims to promote the adoption of electric vehicles in India. The scheme was launched in 2015. Nodal ministry - Ministry of Heavy Industries.

Strengthening Teaching - Learning and Results for States (STARS) - They are aligned with the objectives of National Education Policy (NEP) 2020 of Quality Based Learning Outcomes. It was launched by the Government of India and the World Bank in 2020.

The Airport as Venue for Skilled

Artisans Of The Region (AVSAR) Scheme was launched in 2022.

One Stop Crisis Centre (Sakhi) - Centrally sponsored scheme of the Ministry of Women and Child Development (MWCD), for addressing the problem of violence against women. Launched: 2015.

Mahila Samman Savings Certificate (MSSC) - A new small savings scheme launched by the Ministry of Finance, aimed at encouraging women investors. The minimum investment amount is Rs. 1,000 and the maximum amount is Rs. 2 lakhs. Interest rate for MSSC is 7.5% per annum, compounded quarterly. The scheme is operational during the period 01.04.2023 to 31.03.2025.

eSanjeevani Programme (Online OPD) - It is the first-ever online OPD (outpatient) consultation service offered by the government of India to citizens. Launched on: 13 April 2020. Nodal ministry: Ministry of Health and Family Welfare.

National Career Service (NCS) - A career service in India operated by the Ministry of Labour and Employment. Launched: 20 July 2015.

World Class Institutions Scheme - Launched in the year 2017 under which University Grants Commission (UGC) has declared 10 Public and 10 Private institutions as Institution of Eminence (IoEs).

Deen Dayal Upadhyaya Gram Jyoti Yojana (DDUGJY) - It was launched under leadership of Prime Minister Shri Narendra Modi in November 2014.

Namami Gange Programme - A Flagship Programme launched by the Union Government in June 2014 under the Department of Water Resources, River Development and Ganga Rejuvenation, Ministry of Jal Shakti.

National Beekeeping and Honey Mission (NBHM): Central Sector Scheme for the overall promotion and development of scientific beekeeping. Launched in the year 2020 for three years till 2023. Nodal ministry: Ministry of Agriculture and Farmers Welfare.

Remission of Duties or Taxes on Export Products (RoDTEP) - It has emerged as a critical tool in supporting India's exporters. It became operational on 1st January 2021 replacing the existing

export incentive scheme, Merchandise Exports from India (MEIS). ministry: Department of Revenue.

Digital India - A campaign launched by the Government of India to make its services available to citizens electronically through improved online infrastructure and by increasing Internet connectivity. Launched on 1 July 2015.

Military Exercises

Maitree Exercise - It is a bilateral annual exercise between the Indian Army and the Royal Thai Army, started off in 2006 at Sattahip, Thailand. It is also called the Indo-Thailand Joint Exercise.

MALABAR - It is a multilateral naval exercise that was started in 1992. It began as a bilateral exercise between the navies of India and the United States. Japan and Australia first participated in 2007, and since 2014, India, the US and Japan have participated in the exercise every year. In 2020 Australia rejoined the Malabar Exercise. It is aimed to support free, open and inclusive Indo-Pacific and remain committed to a rules based international order.

RIMPAC (Rim of the Pacific Exercise, 26 Countries) - It is the world's largest international maritime warfare exercise. The first RIMPAC, held in 1971, involved forces from Australia, Canada, New Zealand, the United Kingdom (UK), and the United States (US). Australia, Canada, and the US have participated in every RIMPAC since then.

India first participated in RIMPAC in 2014 when the indigenously built Shivalik class stealth frigate INS Sahyadri took part in the exercise.

Mitra Shakti - It is a bilateral joint annual military exercise between armies of India and Sri Lanka. Its first edition was held in 2012 at Pune.

Indradhanush - It is a joint air force exercise conducted by the British Royal Air Force (United Kingdom) and the Indian Air Force. The exercise started in 2006.

Milan Exercise - It is a biennial multilateral naval exercise incepted by Indian Navy in 1995 at Andaman and Nicobar Command. Its starting participants were Indonesia, Singapore, Sri Lanka and Thailand.

Garuda Shakti - It is an exercise between India and Indonesia that started off in 2012 in Indonesia.

Varuna - It is a bilateral maritime exercise between India and France that was initiated in 1993. The exercise was later christened as 'Varuna' in 2001.

Surya Kiran - It is India-Nepal military exercise Surya Kiran started in 2011 in Mizoram.

AL NAGAH - It is a Joint Military Exercise between armies of India and Oman. The first edition of the exercise was held at Muscat, Oman in 2015.

Table Top Ex - It is an Air force exercise held between India and Bangladesh. Its one and only edition was held in 2017.

Austra-Hind Exercise - It is a bilateral army exercise between Australia and India. Its first edition was started in December 2022.

Garuda Exercise - It is a bilateral exercise between Air Forces of India and France. Its first edition was held in 2003 in Gwalior.

Vajra Prahar - Vajra Prahar is the joint military exercise between the special forces of India and the United States (US). It started in 2010.

AUSINDEX - It is conducted in India and Australia alternatively. It commenced in 2015 as a bilateral IN-RAN (Indian navy and Royal Australian navy) maritime exercise.

IMBEX - It is a joint military exercise between armies of India and Myanmar.

Sampriti - It is a joint military exercise between the armies of India and Bangladesh. Its first edition was held in 2009 at Assam.

Yudh Abhyas - Yudh Abhyas is a joint military exercise, bilaterally occurring between India and the US. It started in 2002 at Agra.

Indra - It is a bilateral exercise between the Armies, Navies and Air forces of India and Russia. The first Exercise was held in 2003.

Khanjar - 'Khanjar' is a joint military exercise between India and Kyrgyzstan. It was started in 2011.

SLINEX (Sri Lanka India Naval Exercise) - It is a series of naval exercises between

the Indian Navy and the Sri Lanka Navy. The first SLINEX exercise took place in 2005.

Hand in Hand Exercise - This exercise is India-China joint military exercise started in 2007, but was put on hold for five years before being restarted in 2013.

Shakti Exercise - It is a bilateral exercise between the armies of India and France. It was started in 2011 in Uttarakhand.

Ekuberin - It is a joint military exercise held between India and Maldives. It has been in place since 2009.

Harimau Shakti - Indian and Malaysian Armies.

Domestic exercise - Gandiv Vijay, Paschim Lehar, Vayu Shakti and Vijay Prahar.

Other Exercises

Shaheen - Pakistan Air Force and the People's Liberation Army Air Force (China).

Northern/Interaction-2023 - It was held between Russia and China.

Eagle Dash-I - It is a joint military exercise between Pakistan, Sri Lanka, and Maldives.

Steadfast Defender - It is the largest military exercise held by NATO (North Atlantic Treaty Organization).

Indian Forces

The Supreme Command of the Indian Armed Forces vests in the President. The responsibility for national defence rests with the Cabinet. This is discharged through the Ministry of Defence. The Defence Minister is the head of the Ministry of Defence.

Indian Armed Forces

Indian Army- Its professional head is the Chief of Army Staff (COAS), who is responsible for the command, control, and administration as a whole.
Headquarters - New Delhi
Founded - 26 January 1950
Motto - Seva Paramo Dharmah
Army Day - 15 January

Indian Navy - It is the maritime branch of the Indian Armed Forces. The President of India is the Supreme Commander of the Indian Navy. The Chief of Naval Staff,

a four-star admiral, commands the navy.

Three Commands of Navy - The Western Naval Command (Mumbai), The Eastern Naval Command (Visakhapatnam), The Southern Naval Command (Kochi).
Headquarters - Nausena Bhawan, New Delhi, Founded - 26 January 1950
Motto - Sam no Varunah
Navy Day - 4 December

Indian Air Force - It is the air arm of the Indian Armed Forces. The Chief of the Air Staff is the only serving Four Star Air Officer in the Indian Air Force.
Headquarters - New Delhi
Founded - 26 January 1950
Motto - Nabbah Sprsam Diptam
Air Force Day - 8 October

Paramilitary forces of India

The Indian Coast Guard (ICG) is a maritime law enforcement and search and rescue agency of India with jurisdiction over its territorial waters including its contiguous zone. It operates under the Ministry of Defence. Formed - 18 August 1978. Headquarters - New Delhi.

Central Armed Police Forces - The collective name of central police organizations in India under the Ministry of Home Affairs. These are technically paramilitary forces formerly known as Central Paramilitary Forces. These forces are responsible for internal security and guarding the borders.

Seven central armed police forces -

Border Security Force (BSF)- It is currently the world's largest border security force. It is also called 'India's First Line of Defence'.
Formed - 1 December 1965
Headquarters - New Delhi
Motto - Jeevan Parayant Kartavya
Constituting instrument - Border Security Force Act (1968)

Assam Rifles (AR)- It is considered the oldest of all paramilitary forces. It is a central paramilitary force responsible for border security, counter - insurgency, and maintaining law and order in Northeast India.
Formed - 1835
Headquarters - Shillong (Meghalaya)
Motto - Friends of the Hill People.

Constituting instruments- Assam Rifles Act (2006),
Assam Rifles Rules (2010),

Assam Rifles Regulations (2016).

Central Industrial Security Force (CISF) - It provides security cover to Public sector establishments, airports, seaports, power plants, sensitive government buildings, heritage monuments and large special economic zones.
Formed - 10 March 1969
Headquarters - New Delhi
Motto - Protection and Security
Constituting instrument - Central Industrial Security Force Act (1968).

Central Reserve Police Force (CRPF)- The primary mission is counter -insurgency operations. It also assists the State and Union Territories in police operations to maintain law and order.
Formed - 28 December 1949 (On 27 July 1939, as Crown Representative's Police)
Headquarters - New Delhi
Motto - Service and Loyalty
Constituting instrument - Central Reserve Police Force Act (1949)

Indo Tibetan Border Police (ITBP)- It was established after the 1962 Sino-Indian conflict. It was created under the CRPF Act. It guards the Indo-Tibetan border and the mountainous regions of the India-China border and monitors the northern borders.
Formed - 24th October 1962
Headquarters - New Delhi
Motto: Shaurya - Dridhata - Karm Nishtha
Constituting instrument: Indo-Tibetan Border Police Force Act (1992).

National Security Guard (NSG)- It is a major national security force deployed by the Government of India specifically to perform high security tasks. It is commonly known as Black Cats. They played a crucial role in countering the 26/11 Mumbai terrorist attacks.
Headquarters - New Delhi
Formed - 22 September 1986
Motto - Sarvatra Sarvotta Suraksha

Sashastra Seema Bal (SSB)- The force was originally set up under the name Special Service Bureau in 1963 in the aftermath of the Sino-Indian War to strengthen India's border areas against enemy operations.
Headquarters - New Delhi
Motto - Service, Security and Brotherhood.
Constituting instrument - Sashastra Seema Bal Act (2007).

Space Missions

Famous Mission of World

Sputnik I - It was the world's first artificial satellite. It was launched by the Soviet Union on October 4, 1957, marking a historic milestone in the Space Age and launched from Baikonur Cosmodrome in Kazakhstan.

Vostok 1 - It was the first human spaceflight, carrying Soviet cosmonaut Yuri Gagarin into orbit on April 12, 1961. The spacecraft was launched from the Baikonur Cosmodrome in Kazakhstan, and Gagarin completed one orbit of the Earth before landing safely back on Earth.

Apollo 11 - It was the first human spaceflight to land on the Moon. The mission was launched on July 16, 1969, and landed on the Moon on July 20, 1969. The astronauts on board were Neil Armstrong, Buzz Aldrin, and Michael Collins.

Voyager 1 - It is a space probe launched by NASA on September 5, 1977, as part of the Voyager program to study the outer Solar System and interstellar space beyond the Sun's heliosphere.

The Hubble Space Telescope (HST) - It was launched in 1990. It has revolutionized our understanding of the universe and has produced some of the most iconic images in astronomy.

Ariane-5 - European space agency has launched the Jupiter Icy Moons Explorer (JUICE), via the Ariane-5 rocket from Kourou, French Guiana.

Taifa-1 - Kenya has launched its 1st operational 3U Earth Observation Satellite (EOS) 'Taifa-1' using a SpaceX Falcon 9 rocket from California, United States of America (USA).

Fengyun 3G - China launched Fengyun 3G, its first satellite dedicated to monitor heavy rainfalls during catastrophic weather conditions in global regions of low- and mid-latitude.

Starlink - It is the name of a satellite network developed by the private spaceflight company SpaceX to provide low-cost internet to remote locations.

International Space Station

It is a spacecraft supporting human crew

in low Earth orbit. It is a joint project of 5 countries: Roscosmos (Russia), JAXA (Japan), ESA (Europe), CSA (Canada), NASA (USA). Launch date: 20 November 1998.

Launch Vehicles of India

India has three active operational launch vehicles:

Polar Satellite Launch Vehicle (PSLV) - It is a light- to medium-lift launch vehicle that is capable of launching payloads up to 1425 kilograms to sub-GTO (Geostationary transfer orbit).

Geosynchronous Satellite Launch Vehicle (GSLV) Mk II - It is a medium- to heavy-lift launch vehicle that is capable of launching payloads up to 2250 kilograms to geostationary orbit (GTO).

Geosynchronous Satellite Launch Vehicle Mk III (LVM3) - It is a heavy-lift launch vehicle that is capable of launching payloads up to 8000 kilograms to low Earth orbit (LEO).

Famous Missions of India

Aryabhata - It was India's first satellite and launched by a Soviet Kosmos-3M rocket from Kapustin Yar on April 19, 1975. Aryabhata was a 360-kilogram (794-pound) satellite with a 1.4-meter (4.6-foot) diameter. It was equipped with instruments to study the Earth's ionosphere, solar X-rays, and X-ray sources.

Bhaskara-I - India's first experimental remote sensing satellite carried TV and microwave cameras.

Rohini Technology Payload (RTP) - It was a 35 kg experimental spin stabilized satellite designed with a power handling capability of 3W. It was launched onboard SLV-3 on its maiden flight on August 10, 1979.

Rohini Satellite RS-1 - India's first indigenous satellite launch, used to measure the in-flight performance of the second experimental launch of SLV-3E2.

INSAT-1A - It is First operational multipurpose communication and meteorology satellite (On Apr 10, 1982).

IRS-1A - It is India's first operational remote sensing satellite (Mar 17, 1988).

Kalpana-1 - First meteorological satellite built by ISRO (On Sep 12, 2002). It was

launched by PSLV-C4.

GSAT-3 - It is also known as EDUSAT. India's first exclusive educational satellite (Sep 20, 2004). It was launched by GSLV-F01.

Chandrayaan-1 - It was India's first unmanned lunar mission (On October 22, 2008). It was launched by PSLV - C11.

YouthSat - It is a joint Indo-Russian stellar and atmospheric satellite mission with the participation of students from Universities at graduate, post graduate and research scholar level and launched by PSLV- C16.

Megha-Tropiques (On October 12, 2011) - It is an Indo-French Joint Satellite Mission for studying the water cycle and energy exchanges in the tropics and launched by PSLV- C18.

RISAT-1 - It is India's first indigenous all-weather Radar Imaging Satellite. It was launched by PSLV-C19 on April 26, 2012.

Mars Orbiter Mission (MOM) or Mangalyaan-1- India's first interplanetary mission to planet Mars was launched onboard PSLV-C25 on November 05, 2013.

Astrosat - It is India's first dedicated astronomy mission aimed at studying celestial sources in X-ray, optical and UV spectral bands simultaneously. It was launched on September 28, 2015 using PSLV-C30.

CartoSat-2D - It is an Earth observation satellite in a Sun-synchronous orbit (SSO) and the fifth of the Cartosat series of satellites. It was launched by PSLV-C37, on 15 February 2017.

Chandrayaan-2 - It was India's second lunar mission, and the first to attempt a soft landing on the Moon's south pole. It was launched on July 22, 2019 using GSLV Mk III. The mission consisted of an orbiter, a lander named Vikram, and a rover named Pragyan.

EOS-03 (Earth observation satellite) - India's first real-time Earth observation satellite and the first satellite of the GISAT constellation. It was launched by GSLV-F10, on Aug 12, 2021.

PSLV-C55 - Indian Space Research Organisation (ISRO) has launched PSLV-C55 with two Singaporean satellites - TeLEOS-2 as the primary

satellite and Lumelite-4 as a co-passenger satellite from Satish Dhawan Space Center (SDSC)-Sriharikota Range (SHAR), Andhra Pradesh.

Chandrayaan-3 - It is a follow-on mission to Chandrayaan-2 to demonstrate end-to-end capability in safe landing and roving on the lunar surface. It was launched by LVM3 M4, on July 14, 2023 and its soft landing took place on August 23, 2023.

Aditya-L1 - It is a satellite dedicated to the comprehensive study of the Sun. Aditya in Sanskrit means the Sun. L1 here refers to Lagrange Point 1 of the Sun-Earth system. L1 is a location in space where the gravitational forces of two celestial bodies, such as the Sun and Earth, are in equilibrium. It was launched by PSLV-C57, On Sep 02, 2023.

World Space Agencies

National Aeronautics and Space Administration (NASA) - It is a US-based organization that was founded in 1958. Which has visited completely undiscovered planets like Jupiter and Saturn in 1972. Its most famous space research programs include the Hubble Space Telescope, Viking 1, Chandra X-ray Observatory, Voyager 1 and Voyager 2, Skylab, Space Shuttle, Apollo, etc.

Indian Space Research Organisation (ISRO) - It is an Indian organization which was established in 1969.

China National Space Administration (CNSA) - It is a Chinese government organization established in 1993. They performed the first soft landing on the Moon in 2013, using Chang'e 3 as their first robotic lunar lander and rover.

Russian Federal Space Agency (RFSA) - It is a Russian organization which was established in 1992 and also known as Roscomos. This was the first country to send animals and the first female astronaut, Valentina Tereshkova into space.

European Space Agency (ESA) - It is a multi-government (22 European countries) organization which was established in 1975 and headquartered in Paris. Its most famous space missions include the Venus Express, Rosetta mission, Mars Express, Corot space telescope mission, etc.

Japan Aerospace Exploration Agency

(JAXA) - It is a Japanese government's organization which was established in 2003 and headquartered in Tokyo.

Space Exploration Technologies Corporation (SpaceX) - It is the world's first private space research company. It was founded by Elon Musk in 2002 and headquartered in Hawthorne, California.

National Centre for Space Studies (NCSS) - It is an organization of France which was established in 1961 and headquartered in Paris. This space agency is also known as Centre national d'études spatiales (CNES).

German Aerospace Center (DLR) - This space agency is one of the oldest agencies in the world. It was founded in 1969, headquartered in Cologne.

Italian Space Agency (ASI) - It was established in 1988 and headquartered in Rome. Its most famous space research programs include the Mars Reconnaissance Orbiter, Juno, Cassini-Huygens, and XMM-Newton.

Research Institutes

Space Research Centres in India

Thiruvananthapuram - Vikram Sarabhai Space Centre.

Thiruvananthapuram - Liquid Propulsion Systems Centre.

Sriharikota- Satish Dhawan Space Centre.

Bengaluru - ISRO Satellite Centre.
Ahmedabad - Space Applications Centre.

Hyderabad - National Remote Sensing Center.

Thiruvananthapuram - ISRO Inertial Systems Unit.

Ahmedabad - Development and Educational Communication Unit.

Hassan - Master Control Facility.

Bengaluru - ISRO Telemetry, Tracking and Command Network.

Bengaluru - Laboratory for Electro-Optics Systems.

Dehradun- Indian Institute of Remote Sensing.

Ahmedabad - Physical Research Laboratory.

Gadanki (Andhra Pradesh) - National Atmospheric Research Laboratory.

Umiam (Meghalaya) - North-Eastern Space Applications Centre.

Thiruvananthapuram - Indian Institute of Space Science and Technology.

Bengaluru - NewSpace India Limited.

Antrix Corporation is the marketing arm of ISRO.

Government Research Institutes in India

Kolkata - Variable Energy Cyclotron Center (Natural Resource).

Noida (Uttar Pradesh) - V. V. Giri National Labour Institute (Civil Service).

New Delhi - VallabhBhai Patel Chest Institute.

Mumbai, Hyderabad - Tata Institute of Fundamental Research (Civil Service).

Agartala - State Institute of Public Administration and Rural Development (Civil Service).

Vijayawada, Bhopal, Delhi - School of Planning and Architecture.

Srinagar - Sher-i-Kashmir Institute of Medical Sciences (Medical).

Kolkata - Satyajit Ray Film and Television Institute (Entertainment).

Kolkata - Saha Institute of Nuclear Physics (Civil Service).

Indian Council of Medical Research (ICMR)

The ICMR, New Delhi, the apex body in India for the formulation, coordination and promotion of biomedical research.

Agra (Uttar Pradesh) - National JALMA Institute for Leprosy and Other Mycobacterial Diseases.

Belagavi, Karnataka - National Institute of Traditional Medicine.

Bengaluru, Karnataka - National Centre for Diseases Informatics and Research.

Bhopal, Madhya Pradesh - National

Institute for Research in Environmental Health.

Chandrapur, Maharashtra - Centre for Research, Management and Control of Haemoglobinopathies.

Ayapakkam, Chennai - National Institute of Epidemiology.

New Delhi - National Institute of Medical Statistics.

Dibrugarh (Assam), Bhubaneswar (Odisha), Gorakhpur (Uttar Pradesh), Port Blair (Andaman and Nicobar Islands)- Regional Medical Research Centre.

Hyderabad - National Animal Resource Facility for Biomedical Research.

Jabalpur, Madhya Pradesh - National Institute of Research in Tribal Health.

Kolkata - Centre for Ageing & Mental Health.

Madurai, Tamil Nadu - Centre for Research in Medical Entomology.

Mumbai, Maharashtra - National Institute for Research in Reproductive and Child Health.

Patna, Bihar - Rajendra Memorial Research Institute of Medical Sciences.

Puducherry - Vector Control Research Centre.

Pune, Maharashtra - National Institute of Virology.

Ahmedabad, Gujarat - National Institute of Occupational Health.

Bengaluru - Regional Occupational Health Centre.

Chennai, Tamil Nadu - National Institute for Research in Tuberculosis.

New Delhi - National institute of Pathology.

New Delhi - National Institute of Malaria Research.

Hyderabad - National Institute of Nutrition.

Jodhpur, Rajasthan - National Institute for Implementation Research on Non-Communicable Diseases.

Kolkata - National Institute of Cholera and Enteric Diseases.

Mumbai, Maharashtra - National Institute

of Immunohaematology.

Noida, Uttar Pradesh - National Institute of Cancer Prevention and Research.

Pune, Maharashtra - National AIDS Research Institute.

Kolkata - All India Institute of Hygiene & Public Health.

Thiruvananthapuram, Kerala - International Institute of Advanced Virology.

National Institutes - Headquarters

National Rice Research Institute - Cuttack, Odisha. NDDB (National Dairy Development Board) - Anand, Gujarat. National Dairy Research Institute - Karnal, Haryana. Central Coffee Research Institution - Chikmagalur. Jute Technology and Research Institute - Barrackpore, West Bengal. Central Fertilizer Quality Control and Training Institute Agriculture - Faridabad. National Institute of Plant Health Management - Hyderabad. Chaudhary Charan Singh National Institute of Agricultural Marketing - Jaipur. Central Island Agricultural Research Institute - Port Blair. The Indian Institute of Natural Resins and Gums - Ranchi.

Census

The First Non-synchronous Census was conducted in India in 1872 during the reign of Governor-General Lord Mayo. The First synchronous census was taken under British rule on February 17, 1881, by W.C. Plowden (Census Commissioner of India), under the British viceroy Lord Ripon (1880-1884).

The year 1921 is known as the year demographic divide because it is the only census year when there was a decrease in the growth of the population.

The first Census of Independent India began on 9th to 28th, February 1951. It is conducted by the Office of the Registrar General and Census Commissioner, Ministry of Home Affairs. All censuses are conducted under the Census of India Act 1948. It is under the 7th schedule of the Indian Constitution, it is listed as a Union Subject (Article 246).

2011 Census

15th Census and 7th Census after independence (2011) - India was the

second most populous country with a population of 1,210 million after China. India occupies 2.4 percent of the world's land surface area and is home to about 17.5 percent (between 2001 and 2011) of the world's population. The growth rate of females was 18.3%, which was higher than the growth rate of males, which was 17.1%. Bihar has experienced the largest decadal population growth among the main states (25.4%).

A Separate code for 'Other' (The first recorded data on transgenders) was introduced under the sex column in this census.

Slogan - Our Census, Our Future.

Literacy

It is the ability to read, write and comprehend information in order to communicate effectively. It is the Percentage of the literate population of the age group 7 years and above.

The top 5 highest literacy rate states are Kerala (94.00%) followed by Mizoram (91.33%), Goa, Tripura, and Himachal Pradesh.

The top 5 lowest literacy rate states are Bihar (61.80%) followed by Arunachal Pradesh, Rajasthan, Jharkhand and Andhra Pradesh.

The literacy rates from census 1951 and 2011 are mentioned as: 18.33% (1951), 74.04% (2011).

The highest male literacy rate state is Kerala (96.11%), The highest female literacy rate state is Kerala (92.07%), The lowest male literacy rate state is Bihar (71.20%), The lowest female literacy rate state is Bihar (51.5%).

The union territories with Highest literacy rates are Lakshadweep (91.85%) and Daman and Diu (87.10%). The Serchhip district of Mizoram with 97.91% is the most literate district in India. Alirajpur district of Madhya Pradesh with 36.1% is the least literate district in India.

The literacy rate at all India levels is 74.04% and the literacy rate for females and males is 65.46% and 82.14% respectively.

Rural and Urban Population

The urban population increased from 17.3% in 1951 to 31.2% in 2011. In NCT Delhi, the proportion of urban residents is

the highest (97.5 percent).

The top five states in terms of the percentage of people living in cities are Goa (62.2%), Mizoram (52.1%), Tamil Nadu (48.4%), Kerala (47.7%), and Maharashtra (45.2 percent). Highest Populated State - Uttar Pradesh (19.98 crore), Least Populated State - Sikkim. Highest Populated (Union territory) - National Capital Territory of Delhi, Least Populated - Lakshadweep.

Population Density

The number of people living in a unit area of the earth's surface is called population density. It is expressed as a number per square meter, per hectare, or per square kilometer.

To be Qualified as an urban settlement the density of the population must be at least 400 people per sq km and at least 75% of the population must be involved in non-agricultural activities.

The density of population in India in 2011 was 382 and the lowest is Arunachal Pradesh (17) persons per sq km.

The states with highest population density are Bihar (1106) followed by West Bengal, Kerala, Uttar Pradesh, and Haryana.

The top union territories in terms of population density are Delhi (11320) and least is Andaman and Nicobar (46).

Areas of Extremely Low Density: Areas having 100 persons per sq km and less than that are included in this class. Arunachal Pradesh (17), Mizoram (52), Andaman and Nicobar Islands (46) Sikkim (86) and Jammu and Kashmir (56) are included in this category.

Areas of Low Density: Areas having population density of 101 to 250 persons per sq km are included in this class. Meghalaya (132), Manipur (128), Himachal Pradesh (123), Nagaland (119), Chhattisgarh (189), Uttarakhand (189), Rajasthan (200), Madhya Pradesh (236) are included in this category.

Areas of Moderate Density: This class includes those areas which are having 251 to 500 persons per sq km. The average for the whole of India (382 persons per sq km) also falls in this class. Gujarat (308), Karnataka (319), Andhra Pradesh (308), Tripura (350), Maharashtra (365), Jharkhand (414),

Assam (398), Goa (394), and Orissa (270) are included in this category.

Areas of High Density: These are areas having population density of 501 to 1000 per sq km. Uttar Pradesh (829), Kerala (860), Dadra and Nagar Haveli (700), Haryana (573), Tamil Nadu (555), Punjab (551) are included in this category.

Areas of Very High Density: Areas having more than 1000 persons per sq km are termed as areas of very high population density. Daman and Diu (2191), Lakshadweep (2149), Pondicherry (2547), Chandigarh (9258), and Delhi (11320) and Bihar (1106), West Bengal (1028) are areas of very high population.

Child Population

There has been a 0.4% increase (from 2001 to 2011 census) in the number of children aged 0 to 6 years.

The states that do the poorest in terms of the sex ratio for children aged 0 to 6 are Gujarat, Haryana, Punjab, and Jammu and Kashmir.

Schedule Castes

States having the highest population of scheduled caste - Uttar Pradesh, West Bengal, Bihar, Tamil Nadu, Andhra Pradesh, and Maharashtra. Punjab has the highest percentage (31.9%) of scheduled castes in relation to the overall population.

Union territories such as Andaman and Nicobar, Lakshadweep, and the state of Nagaland have no scheduled caste population.

Arunachal Pradesh, Goa, Manipur, Meghalaya, Mizoram, Dadra Nagar Haveli, Daman, and Diu have less than 5% scheduled caste population.

Schedule Tribes

Punjab, Haryana, Chandigarh, Delhi and Puducherry have no population of Scheduled tribes.

The states with the highest tribal population are Madhya Pradesh, Orissa, Maharashtra, Rajasthan, Chhattisgarh, Gujarat, Jharkhand, Andhra Pradesh, West Bengal, and Karnataka.

Mizoram has the highest proportion of Scheduled Tribes, 94.4% of the total population. Uttar Pradesh has the lowest proportion of scheduled tribes, 0.6% of

the total population.

Below Poverty Line (BPL)

It is a benchmark used by the government of India to identify individuals and households in need of government assistance and aid.

According to the Planning Commission, the poverty estimates for 2011-12, Chhattisgarh (39.93%) and Jharkhand (36.96%) has the highest percentage of population below the poverty line. Kerala (7.05%) and Himachal Pradesh (8.06%) have the lowest percentage of population below the poverty line.

The percentage of the population below the poverty line in Chandigarh, Delhi, Lakshadweep, and Andaman & Nicobar Islands are 21.81%, 9.91%, 2.77%, and 1.00% respectively.

According to Multidimensional Poverty Index (MPI) 2019-21:

Bihar, the state with the highest MPI value in NFHS-4 (2015-16), saw the fastest reduction in MPI value in absolute terms with the proportion of multidimensional poor reducing from 51.89% to 33.76% in 2019-21.

The next fastest reduction in the MPI value was seen in Madhya Pradesh and Uttar Pradesh. The proportion of multidimensional poor in Madhya Pradesh and Uttar Pradesh in NFHS-5 (2019-21) are 20.63% and 22.93% respectively.

In terms of number of MPI poor, Uttar Pradesh topped the list with 3.43 crore people escaping multidimensional poverty in the last five years, followed by Bihar (2.25 crore) and Madhya Pradesh (1.36 crore).

State with lowest MPI value: Kerala (0.55%), Goa (0.84%), Tamil Nadu (2.20%).

Union Territories with highest MPI value in NFHS (2019-21): Dadra and Nagar Haveli and Daman and Diu (9.21%), Jammu and Kashmir (4.8%), Ladakh (3.53%), Chandigarh (3.52%), Delhi (3.43%), Andaman and Nicobar Islands (2.30%), Lakshadweep (1.11%), Puducherry (0.85%).

Sex Ratio

It is defined as the number of females per 1000 males in the population. India

has a sex ratio of 943. Child sex ratio is 919.

The top 5 states with the highest sex ratio are Kerala (1084) followed by Tamil Nadu, Andhra Pradesh, Chhattisgarh, and Meghalaya. Haryana (879) has the lowest sex ratio.

The top five union territories with the highest sex ratio are Puducherry (1037), Lakshadweep, Andaman and Nicobar, Delhi and Chandigarh.

The top five union territories with the lowest sex ratio are Daman and Diu (618), Dadra and Nagar Haveli, Chandigarh, Delhi, Andaman and Nicobar Islands.

Important facts

The top ten most populated metro cities are Mumbai (1.83 crore) followed by Delhi, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Pune, Surat and Kanpur.

The state with negative growth of population between 2001 and 2011 is Nagaland (-0.6%).

According to the Census-2011, the average annual growth rate of India is 1.64%.

Among 121.09 crores population (religions) are: Hindu 96.63 crores (79.8%), Muslim (14.2%), Christian (2.3%), Sikh (1.7%), Buddhist (0.7%), Jain (0.4%), others (0.9%). Muslims (14.2%) form the largest religious minority in India.

As per the census of 2001, the population of India was 1027 million and in 2011 the population of India was 1210.19 million (17.64% which is more than 181 million population).

Hindi has retained its position as a predominant language spoken by about 43.63%, followed by Bengali (8.03%), Marathi, Telugu, Tamil, Gujarati, Urdu, Kannada, Odia, Gujarati, Malayalam, Punjabi, Assamese, Maithili.

The Sanskrit language is spoken by the least number of people among the 22 languages mentioned in the Eighth schedule.

Population of state and union territories as per 2011 census descending order

Uttar Pradesh, Maharashtra, Bihar, West Bengal, Andhra Pradesh (including Telangana), Madhya Pradesh, Tamil Nadu, Rajasthan, Karnataka, Gujarat, Odisha, Kerala, Jharkhand, Assam, Punjab, Chhattisgarh, Haryana, National Capital Territory of Delhi, Jammu and Kashmir, Uttarakhand, Himachal Pradesh, Tripura, Meghalaya, Manipur, Nagaland, Goa, Arunachal Pradesh, Puducherry, Mizoram, Chandigarh, Sikkim, Andaman and Nicobar Islands, Dadra and Nagar Haveli and Daman and Diu, Lakshadweep.

Population order of some countries -

The highest population in the world in decreasing order: China, India, United states, Indonesia, Pakistan, Brazil, Nigeria, Bangladesh, Russia, Mexico.

History

Ancient History

Prehistoric Age

The prehistoric era lacks written records and is known as the Stone Age due to early stone tools. It's divided into the Palaeolithic, Mesolithic, and Neolithic based on different types of tools and techniques.

Palaeolithic

It derives its name from two Greek words, 'paleo' meaning old and 'lithic' meaning stone. It refers to the old stone age. It covered a long period ranging broadly from 20,00,000 to 12,000 B.C.

Ostriches were found in India during this period. On the basis of tool types and techniques, it is divided into three phases:

Lower Palaeolithic Age : Quartzite tools of the lower Palaeolithic period were excavated in that region extending from Pinjore in Haryana to Nalagarh (Himachal Pradesh).

Several sites of stone age were discovered in riverbeds of Sabarmati, Mahi river, and lower Narmada rivers of Gujarat.

In that age, people learned to control fire for roasting meat and warding off animals. Hunting and food gathering. They lived on trees and in caves.

Important Sites : Pahalgam in Kashmir, Belan valley in Uttar Pradesh, Nevasa in Maharashtra, Hunasagi in Karnataka and Attirampakkam in Tamil Nadu (the evidence of a set of animal foot-prints along with Acheulian tools). Shivalik range of Kashmir, Himachal Pradesh and Punjab, Berach basin and the hilly area of Rajasthan, Narmada and Some valleys in Madhya Pradesh, Malaprabha and Ghataprabha basins in Karnataka, Chhota Nagpur plateau and several areas of Maharashtra.

Middle Palaeolithic Age : It was characterized by flake tools and the widespread use of fire. The language was invented in this period. Refined and lighter tools made of harder stone material like flint. In the Indian subcontinent, it is dated to around 10,000 BC.

Important Sites : Luni valley in Rajasthan; Valleys of the Belan, Son river, Narmada river, and their tributaries in central India (including Bhimbetka), and some sparse sites in Chota Nagpur plateau, Deccan plateau, and Eastern Ghats.

Upper Palaeolithic Age : It was characterized by the emergence of regional stone tool industries. Other hominin species were eliminated by this time. The most famous examples of art were cave paintings. It took animals and human figures as their subject material. It coincided with the phase of the Ice Age.

In this period **pressure flaking** was a method of forming points, grooves, and notches on stone tools in which a tool was pressed up with a sharp instrument.

Important Sites : Sohan valley (Present day Pakistan). Belan valley (Uttar Pradesh). Narmada valley, Kashmir, Punjab, Rajasthan (Thar Desert). World site Western Europe, Africa.

Mesolithic Period

It derived its name from the Greek words 'meso' and 'lithic'. Meso means middle, and lithic means stone. It saw some new chipped stone tools like microliths and very small stone tools. In the Indian subcontinent, it was dated back to around 8000 BC. The Mesolithic and Neolithic phases are from the Holocene era. Dog was the first animal to be domesticated.

Important Sites: Adamgarh in Madhya Pradesh. Bagaur and Tilwara in Rajasthan. Langhnaj, Akhaj, Valasana, Hirpura, Amrapur, Devnimori, Dhekvadlo, Tarsang in Gujarat. Patne, Pachad, Hatkhamba in Maharashtra. Morkhana, Lekhahia, Baghai Khor, Sarai Nahar Rai, Mahadaha, Damdama, Chopani Mando, Baidha Putpurihwa in Uttar Pradesh. Pachmarhi, Adamgarh, Putli Karar, Bhimbetka, Baghor II, Baghor III, Ghagharia in Madhya Pradesh. Paisra in Bihar. Kuchai in Odisha. Birbhanpur in West Bengal. Muchatla Chintamanu Gavi, Gauri Gundam in Andhra Pradesh. Sanganakallu in Karnataka. Tenmalai in Kerala.

Neolithic Period

The term 'Neolithic' was coined by John Lubbock. The word 'neo' refers to new, while 'lithic' refers to stone. It is also called the new stone age.

The chief characteristic of that age was the new type of ground and polished stone tools. It marked the beginning of the cultivation of plants and the domestication of animals. Pottery was first traceable to this period. They used Microlithic blades along with tools made of polished stones.

Tools and weapons made from bones were used in spear points and arrowheads. These tools were used to cut, pound, crush, and access new foods—including meat from large animals. A celt is a tool from the Neolithic period.

Megalith is a large prehistoric stone that has been used to construct a structure or monument and in a circular shape around the grave. The practice of erecting it began about 3000 years ago in India.

Cairn-circles are the prehistoric stone row which is a linear arrangement of parallel megalithic standing stones.

Dolmen is a type of single-chamber megalithic tomb, usually consisting of two or more vertical megaliths supporting a large flat horizontal capstone or "table".

Menhir is the name used in Western Europe for a single upright stone erected in prehistoric times; sometimes called a "standing stone".

Important Sites : Mehrgarh, Burzahom, and Gufkral in Kashmir Valley. Belan Valley, Koldihwa in Uttar Pradesh. Chirand in Bihar. Sanganakallu, Brahmagiri, Maski, Piklihal, Hallur in Karnataka. Utnur, Nagarjunakonda, Budihal in Andhra Pradesh. Paiyampalli in Tamil Nadu.

Important Sites of Prehistoric Age

Bhimbetka : It is a paleolithic site. It was declared a UNESCO World Heritage site in 2003. It was discovered by Dr. Vishnu Wakankar in 1957-58. It is known for its prehistoric rock shelter. It is located in the Raisen District of Madhya Pradesh at the southern edge of the Vindhya Range.

Chirand : It is an archaeological site in the Saran district of Bihar. It is located on the northern bank of the Ganga River. It has a large prehistoric mound which is known for its continuous archaeological record from the Neolithic age.

Gufkral : It is a site inhabited by potters who utilize caves. It is located in Banmir village in Jammu and Kashmir.

Important findings: Copper bangle, copper pin, stone celts, stone points, ring stone, pounders, querns, bone tools, awls, scape, piercer, polished bone needle, beads, terracotta marble.

Mehrgarh : It is the earliest (First) Neolithic site. It is located in Pakistan. The people lived in houses built of sun-dried bricks and cultivated crops like cotton and wheat. It is the first place where people started growing barley and wheat. People lived in square or rectangular houses. Cotton was probably grown about 7000 years ago.

Lingusuguru : It is the first Paleolithic site discovered in Karnataka.

Mahagara: It provides the pre-historic sites with archaeological evidence of cattle rearing in the form of hoof-marks on clay surfaces in Uttar Pradesh.

Burzahom : It is an archaeological site located in the Kashmir Valley of Jammu and Kashmir. It was discovered by Yale-Cambridge in the late 1930s. Pit-houses dug into the ground, with steps leading into them. The domestic dogs were buried along with their masters in their graves and people lived in pits and used tools made of polished stones as well as bones.

Hunasagi : It is a paleolithic site located in Karnataka.

Daojali Hading : It is a neolithic site in Dima Hasao District of Assam. It was excavated in 1961-63 by a team led by M C Goswami and T C Sharma.

Artifacts - Jadeite stone (must have been transported from China). 4 types of ceramic - cord-marked, incised, stamped, and plain fine red ware.

Paiyampalli : It is a neolithic and megalithic site in Tirupattur taluk of Tirupattur district in Tamil Nadu.

Koldihwa : It is a neolithic site in Uttar Pradesh. It is situated in the valleys of Belan River near village Devghat. It has evidence for the domestication of rice.

Gilund : It was first excavated over half of a century ago by B.B. Lal. It is a small village in Rajasthan. It is also known as "Modiya Mangari". It is also famous for its celebration of "Khekhara Utsav".

Chichali: It is an archaeological site located on the left bank of the Narmada River in Madhya Pradesh. It was discovered in the 1990s by Shri S.B. Ota.

Inamgaon : It is situated near the Ghod river, (a tributary of the river Bhima) in Maharashtra. It is the post-Harappan archaeological site in Maharashtra. Usually adults were buried at this site. It is considered to be the 'regional centre' of the Bhima Valley.

Atranjikhera : It is an archaeological site located in the east of Uttar Pradesh. It is a prehistoric and historic excavated archaeological site on the bank of the Kali Nadi, a tributary of Ganga.

Hastinapur : It is located in Uttar Pradesh. It is described in Hindu texts Mahabharata and Puranas as the capital of the Kuru Kingdom. The iron implement site of Hastinapur was excavated in the early 1950s by B.B. Lal. It is famous for many historical temples like Karna temple, Pandeshwar temple, Kailash Parvat.

Hathnora : It was located near the banks of the Narmada river in Madhya Pradesh. The skull of a 'Homo erectus' was found at this site. The fossil found is also known as Narmada Fossil. Homo erectus was the species of humans from the Pleistocene period.

Gola Dhoro: It is located in the village Bagasra in the Gulf of Kutch in Gujarat. It is famous for the shell bangle workshop excavated within the walls at Gola Dhoro.

Catal Huyuk : It is an important neolithic site located near the modern city of Konya in south central Turkey.

Features - There were no roads or streets in the city. People used to come and go through the flat roofs only. These terraces were connected to each other by wooden stairs.

All the houses were rectangular and made of rectangular mud bricks. These bricks were joined to each other with mortar and set on wooden structures. According to the archaeologists, these houses used to have a main room and some other rooms.

Hanging Gardens of Babylon : It is associated with Mesopotamian ancient culture. It was among Seven Wonders of the Ancient World. It was described as an interesting feat of engineering with an ascending series of tiered gardens. It

was said to have been built in the ancient city of Babylon in Iraq.

Chalcolithic Period

It has more of a transitional period between the Neolithic and the Bronze age. The main tools were made of both stones and copper. The unique feature of this period was painted black pottery as well as Red Ware. They were unaware of Iron, horse and script. It marks the beginning of use metal in place of stone yet burnt brick was generally absent.

Major Chalcolithic Culture: Banas (2600 BC–1900 BC) in south-east Rajasthan, with Ahar near Udaipur and Gilund as its key-sites; Kayatha (2100 BC–2000 BC) with Kayatha in Chambal as its chief site in Madhya Pradesh; Malwa (1700 BC–1400 BC) with Navdatoli in Western Madhya Pradesh as an important site; Jorwe (1400 BC– 700BC) with Inamgaon and Chandoli near Pune in Maharashtra as its chief centers.

The evidence of the chalcolithic cultures also comes from eastern Uttar Pradesh, Bihar and Bengal.

Balathal : It is one of the ninety Ahar - Banas culture sites located in the basins of the Banas River and its tributaries. The site was discovered by V.N Mishra during a survey in 1962 - 63.

Ahar : It is a site of the Banas culture. It is a Chalcolithic archaeological settlement or culture that was located on the banks of the Ahar River in the Indian state of Rajasthan.

Adichanallur : It is an archaeological site located in Thoothukudi district on the banks of the river Thamirabarani which flows in the Thiruvaiyandam circle. It is one of the oldest civilized sites in Tamil Nadu.

It was first excavated by Alexander Rea in 1903-04. It is associated with a pre-1600 BC civilization. The excavations revealed that it was abundant in iron and bronze.

Indus Valley Civilisation

The civilization is often referred to as the Harappan Civilization because of one of its major sites, Harappa. It was first discovered during an excavation campaign under Sir John Hubert Marshall in 1921–22 at Harappa following the discovery of seals by J

Fleet. Mesopotamian people of Tigris-Euphrates valley called the Indus valley civilization Meluhha.

The Harappan civilization is divided into three phases: Early Harappan phase (3500 BC–2600 BC), Mature Harappan phase (2600 BC–1900 BC), and Late Harappan phase (1900 BC–1400 BC).

Features: The Indus Valley towns featured a rectangular grid layout with roads at right angles. The Citadel housed the ruling class, while commoners lived in the Lower Town. Houses had courtyards. An advanced drainage system with burnt brick drains, stone lids, and an underground sewage network existed. The presence of widespread seals, standardized writing, and systems of weights and measures underscored the importance of trade. Additionally, the Indus script, though pictographic and undeciphered, along with the well-planned urban centers and extensive trade networks, reflected the sophistication of this ancient civilization.

Major Indus Valley Civilisation Sites

Harappa : It was discovered in 1921-22 by Dayaram Sahni. It is situated on the bank of river Ravi in Sahiwal (Montgomery) district of Punjab (Pakistan). It is divided into **three parts** - the Citadel, the Middletown, and the Lower town. Harappans got Copper from the present-day Rajasthan.

Important findings : Two rows of 6 granaries, workmen quarters, 2 sandstone statues depicting human anatomy, dog attacking deer, little bullock carts (ekkas).

Mohenjo-daro (Mound of the Dead): It was discovered in 1922 by R. D. Banerji. It is a group of mounds and ruins on the right bank of the Indus River, southern Pakistan.

The town was divided into two parts. It was not a perfect grid-patterned town. It had the world's best-channeled drainage system. There were multi-storey buildings in Mohenjo-Daro. It had a total of 700 wells.

The Great Bath is part of a larger complex discovered in the 1920s. It has a rectangular shape and was lined with bricks, coated with plaster, and made watertight with a layer of natural tar. Built

with fine brickwork, it spans 897 square feet and is positioned 8 feet lower than the surrounding pavement. Likely utilized for religious purposes, it also served as the water tank for the city.

Important findings: a large granary, a piece of woven cotton, bronze dancing girl, seal of Pashupati Mahadeva, Steatite statue of a bearded man supposed to a priest, bronze buffalo, carved seals, balance-scales and weights, gold and jasper jewelry, and children's toys, standing figures, copper.

Khetri is situated at the foothills of the Aravalli Range and is famous for copper mines.

Lothal : It was discovered by R.Rao in 1953. It is located on Bhogva river near Gulf of Cambay (Khambhat) in Gujarat. It is called mini - Harappa. In 1948, Ernest Mackay mentioned that the drainage system in Lothal was made of burnt bricks.

Important Findings : First manmade port, Dockyard, Rice husk, Fire altars, Chess playing, Port town, Evidence of Rice, Graveyard, scale of ivory, Copper dog.

Chanhudaro : It was discovered by N. G. Majumdar in 1931. It is situated in Sindh on the Indus river. It was the only Harappan city without Citadel. It is known as the Lancashire of India.

Important findings : Bead makers' shop, inkpot, footprint of a dog chasing a cat, shell cutting, metal-working, seal-making and weight making, copper knives, spears, razors, tools, axes.

Dholavira: It was discovered by Jagat Pati Joshi in 1968 in Rann of Kachchh in Gujarat. It is situated on the banks of river Luni. It is located on the Tropic of Cancer. It has a fortified citadel, a middle town, and a lower town with walls made of sandstone or limestone. The first astronomical observatory of the Harappan Civilization is located in Dholavira.

Important findings : Water harvesting system, Water reservoir, only site to be divided into three parts, Dams, Embankments, a stadium, Rock – cut architecture, large open area with terraced stands, An inscription comprising ten large-sized signs of Harappan script.

Banawali : It was discovered by R.S Bisht in 1974. It is situated on the ancient river

Sarasvati in Hisar district of Haryana.

Important findings : Beads, Barley, Evidence of both pre-Harappan and Harappan culture, Beads and bangles of semi-precious stones, terracotta and shell, chert blades, ivory, beads in gold, terracotta animal figurines, inscribed steatite seals and terracotta seals, copper fish-hooks, charred barley grains.

Surkotada : It was discovered by J.P Joshi in 1964, in Gujarat. Important finding: Remains of horse bones..

Rakhigarhi : The site was excavated by Amarendra Nath. It is an archaeological site belonging to the Indus Valley civilisation in Hisar District of Haryana. It was a port city and largest Harappan site during the period of the mature phase of the Indus Valley Civilisation, dating to 2600-1900 BCE. The world's largest museum of Harappan culture is coming up in Rakhigarhi.

Rangpur : It was discovered by M. S. Vats in 1935. It is an ancient archaeological site near Vanala in Gujarat. It lies on the tip between the Gulf of Khambhat and Gulf of Kutch.

Important Findings: plant fragments, Rice, bajra, and millet.

Sanauli : It is an archaeological site in western Uttar Pradesh. It gained attention for its Bronze Age solid-disk wheel carts.

Important findings : copper helmets, copper antenna swords, copper swords, a ladle made of copper, grey-ware pottery, large terracotta pots, red vases with flaring rims, copper nails and beads.

Ropar : It is an Indus Valley settlement located on the banks of the Sutlej River. It is the location of the earliest Indus Valley excavations in independent India.

Daimabad : It was discovered by B. P. Bopardikar in 1958. It is an abandoned settlement on the left bank of the Pravara River in Maharashtra.

Important Findings : Copper Rhinoceros, Copper Elephant, and Copper Chariot, microlithic blades, bone tools, a solitary piece of worked elephant tusk, beads, and a few remnants of graded terracotta rings.

Kalibangan : It is located in Rajasthan. The site contains both pre-Harappan and Harappan remains.

Important findings : Ploughed field, artifacts, built structures, human impact on the environment, Fire altars, garbage, stratigraphy, mortuary practices, plant remains, or animal remains, furrow marks.

Kot Diji : It was discovered by Fazal Ahmad Khan at Sindh in Pakistan. It is situated on the bank river Indus.

Important findings : pots, pans, storage jars, toy carts, balls, bangles, beads, terracotta figurines of mother goddesses and animals, bronze arrowheads, and well-fashioned stone implements.

Amri : It was discovered by N.G. Majumdar. It is an ancient settlement in modern-day Sindh, Pakistan.

Balakot : It is one of four known ancient coastal sites in Pakistan. It is famous for the Bangle factory.

Sotka Koh : It is a site on the Makran coast, Pakistan.

Shortughai : It is located in Darkad district of Afghanistan.

Bhirrana : It is considered to be the oldest discovery of the Indus Valley Civilization. Its history stretches back to pre-Harappan times, as revealed by archaeological discoveries.

Allahdino was the smallest site. Rakhigarhi (Hisar district, Haryana) was declared the largest site of the Harappan civilization in 2014 (earlier Mohenjo-daro).

Northernmost sites - Manda (Jammu-Kashmir). Southern-most site - Daimabad (Maharashtra).

Easternmost site - Alamgirpur (Uttar Pradesh). Western-most site - Sutkagendor (Pakistan - Iran border).

Other Important Sites :

Gujarat - Bet Dwarka, Bhagatruv, Desalpur, Kaj, Kanjetar, Kotada Bhadli, Kuntasi, Loteswar, Babar Kot, Balu.

Haryana - Balu, Bhirrana, Hisar mound, Jogna Khera, Kunal, Farmana.

Jammu and Kashmir - Manda.

Uttar Pradesh - Alamgirpur, Bargaon, Hulas, Mandi.

Rajasthan - Baror, Karanpura.

Punjab - Chapuwala, Ganweriwala.

Pakistan - Allah Dino, Kot Bala.

Agriculture

They grew wheat and barley on a large

scale. Other crops that they grew included pulses, cereals, cotton, dates, melons, pea, sesamum, and mustard. Finds of rice are relatively rare. It was an agro-commercial civilization. They were the earliest people to grow cotton.

Economy

They traded goods such as shells, metal, and stone, often using the barter system. Evidence suggests trade extended to Central Asia and the Tigris-Euphrates region. Lapis lazuli trade particularly enhanced the social status of rulers. Seals, primarily square and made of steatite, commonly depicted bulls. Crafts like bangle and shell ornament making were prevalent.

Indus Valley Civilization thrived on both land and sea trade, connecting with regions referenced in Sumerian texts like Meluha (Indus region), with intermediary stations at Dilmun (Bahrain) and Makan (Makran coast). Transportation involved boats and bullock-carts.

Import : Gold, Silver, Copper, Tin, Jade, Steatite.

Export : Agricultural products, cotton goods, terracotta figurines, beads from Chanhudaro, conch-shell from Lothal, ivory products, copper.

Administration

Central authority may have contributed to uniform culture. No clear idea of an organized force or standing army.

Seals And Sealings

Most of the seals are square plaques made mostly from Steatite. Seals had an animal (except Cow) or human figure on one side and an inscription on other/both sides. Seals were primarily used for commercial purposes, as amulets, as a form of identification. Seals with symbols similar to 'Swastika' design have also been found.

Art

Bronze Casting was practiced on a wide scale using the 'lost wax' technique or Cire Perdue. They mainly consist of human and animal figures.

Example: 'Dancing Girl'. Stone Statues - Bearded man (made of Steatite, interpreted as a priest). Red sandstone - figure of a male torso. Terracotta Figures

- Mother Goddess, mask of horned deity, toys.

Religion

The Pashupati seal found in Mohenjo-daro depicts a yogi posture surrounded by animals like buffalo, tiger, elephant, rhinoceros, and deer. This imagery suggests similarities with the god Shiva. Although Harappans worshiped a mother goddess, there's no evidence of temples. Instead, many amulets were found.

Decline

After 2000 BC IVC declined and gradually faded away by 1800 BCE.

Possible reasons - There are believed to be many reasons like, declined soil fertility, depression in land, Aryans invasion, decline of trade, Floods, Earthquake but Most acceptable reason is ecological imbalance.

Vedic Age

Vedic Civilization (1500 - 600 BCE) started with the migration of Aryan people in the north - western part of India. Aryans originally lived in the Steppes region (Indo-Iranian area). Vedic civilization flourished along with the river Saraswati.

Later they moved to central Asia and then came to the Punjab region of India around 1500 BC. They spoke the Indo-Aryan language (Sanskrit).

They first settled in small groups in the northwestern lowlands and the Punjab plains. They later spread into the Indo-Gangetic plains.

They were primarily a cattle-keeping tribe. By the sixth century B.C. They had conquered all of North India, which was known as **Aryavarta**.

Theories of Aryan Origination : Max Muller - Aryans migrated from Central Asia and the Eurasian Steppe. Sir William Jones - Europe was the homeland of Aryans. Bal Gangadhar Tilak - Northern Arctic was the homeland of Aryans. Swami Dayananda Saraswati - Tibet was the original home of Aryans.

Battle of the Ten Kings or Dasrajan Yudha: It was first alluded to in the 7th Mandala of the Rigveda. The battle was fought between the king of Bharatas

(tribe), led by king Sudas and the confederacy of the other ten tribes – Puru, Yadu, Turvasa, Anu, Druhyu, Alina, Paktha, Bhalanas, Shiva and Vishanin. The Bharatas emerged victorious in the battle which was fought on the banks of **Parushni River** (Ravi).

Vedic Literature

Vedic Literature refers to the four Vedas in their Samhita. It is classified into 2 categories: Smriti and Shruti.

Smriti :- It is a Sanskrit word that means "to remember". **Smriti Literature**: Vedangas, Shatdarsana, Puranas, Smriti Dharmashastras, Kalpa - sutras (significant religious manuals) and the Ramayana and Mahabharata. The term "Smriti" came to be used specifically in reference to works dealing with social behavior and the law, such as the renowned **Manu-smriti** (Laws of Manu).

Shruti :- It is a Sanskrit word that means "what is heard". Shruti's works are heard and transmitted by earthly sages.

Shruti Literature :- Rig Veda, Yajur Veda, Samveda, Atharva Veda, Brahmanas, Aranyakas, Upanishads.

Vedas: The word 'Veda' is derived from the root 'vid', which means 'to know'. In other words, the term 'Veda' signifies wisdom, knowledge or vision. There are four Vedas: Rig Veda, Sama Veda, Yajur Veda and Atharva Veda.

Rigveda :- It is the oldest veda. The meaning of the word 'Rigveda' is to praise knowledge. It is also called "the first testament of mankind".

It has 1028 hymns called 'Suktas' and is a collection of 10 books called 'Mandalas.' It has 10600 verses.

It consists of a hymn in the form of a dialogue between Sage Vishvamitra and two rivers (Beas and Sutlej). The dialogue is known as "Vishvamitra Nadi Samvad". The chief author of Mandala 7 of the Rigveda was Vasishtha.

The meters used to form hymns are Gayatri, Anushtubh, Trishtubh, and Jagati. Trishtubh and Gayatri are most important. Brahmana - Aitareya, Kaushitaki. Upaveda - Ayurveda (medicine).

Mandalas of Rig-Veda :

Mandala I and Mandala X - The longest.

Mandala II - It is primarily concerned with the worship of Indra (rain and war god).
Mandala III - Contains Gayatri mantra.
Mandala VII - Battle of the Ten Kings.
Mandala IX - Devoted to Soma.
Mandala X - Contains Purush Sukta, Nasadiya Sukta.

Important Hymns (Sukta) :- Purusha, Nasadiya, Vishwakarma, Hiranyagarbha, Aksha, Yama-Yami Samvad.

Sindhu (Indus) is the most mentioned and Saraswati is the most regarded (holy river) in Rigveda. There are seven Rivers mentioned in the Rig-Veda are Sindhu, Vitasta, Asikini, Purushni, Vipasha, Shutudri, Saraswati. Saraswati Valley was called Brhmavarta. Sindhu is known as the River Goddess. The name 'Bharat' for India originated from the Bharat tribe, mentioned in Rigveda.

In 2007, UNESCO added 30 Rig Veda manuscripts to the Memory of the World Register at the Bhandarkar Oriental Research Institute in Pune.

Samaveda :- It is known as the veda of melodies and chants. It is related to public worship. There are 1549 verses (except 75 verses, all have been taken from Rigveda). There are two Upanishads embedded in Samaveda – Chandogya Upanishad and Kena Upanishad. It is considered as the root of Indian classical music and dance. Brahmana - Tandya, Sadavimsha. Upaveda - Gandharva Veda (Music).

Yajurveda : It stands to mean 'Worship Knowledge'. It compiles ritual-offering mantras or chants. It has two types – Krishna (Black or Dark) and Shukla (White or Bright). Krishna has an un-arranged, unclear, motley collection of verses. Shukla has arranged and clear verses.

Brahmana - Taittiriya, Shatapatha.

Upaveda - Dhanurveda (archery).

Upanishad : Brihadaranyaka, Isha, Taittiriya, Katha, Shvetashvatara, Maitri. Jabala Upanishad is part of the Shukla Yajur Veda. Four ashrams (Brahmacharya, Grihastha, Vanaprastha, and Sannyasa) are taken from Jabala Upanishad.

Atharvaveda : It is the tatpurusha compound of Atharvan. It is divided into 20 Kandas (Volume), which are divided into 36 Prapathakas. It has 730 hymns,

6000 mantras, and 20 books. Paippalada and Saunakiya are two surviving recensions of Atharvaveda. It is known as the veda of magical formulas. It contains a collection of magic spells and charms to fend off evil spirits and diseases.

It includes three primary **Upanishads** – Mundaka, Mandukya, and Prashna. **Brahmana** - Gopatha. **Upaveda** - Arthashastra (economics). Satyamev Jayate is taken from Mundaka Upanishad.

Four styles of texts of Vedas : -

Samhitas (mantras, benedictions),

Brahmanas (text on rituals, ceremonies, sacrifices, symbolic-sacrifices),

Aranyakas (commentaries on rituals, ceremonies and sacrifices), and

Upanishads (text discussing meditation, philosophy and spiritual knowledge).

Samhitas (Karma-khanda) : It is a collection of mantra texts of each of the Vedas consisting of mantras, hymns, and prayers.

Brahmana (Jnana-khanda): They explain the hymns of the Vedas. They act as the appendices to the Vedas. They are written in prose and they elaborately describe the various sacrifices and rituals, along with their mystical meanings.

Aranyakas : It is a part of the ancient Indian Vedas concerned with the meaning of ritual sacrifice.

Upanishads: It is commonly referred to as Vedanta (End of Veda). These are 108 Upanishad. Brihadaranyaka and the Chandogya are the two earliest Upanishads. The latest Upanishad is Muktika Upanishad. Katha Upanishad tells the story of Nachiketa and Yama. Aitareya Upanishad talks about the creation of Atman (Soul) and Consciousness.

Vedanga: They are six auxiliary disciplines of Hinduism that developed in ancient times and have been connected with the study of the Vedas. It is referred to as "limbs of the Vedas". Six Vedangas: Shiksha (Pronunciation of the words and education), Chhanda (Metrics used in Sanskrit verses), Vyakarana (Sanskrit grammar), Nirukta (Origin of the words), Kalpa (Knowledge of rituals), Jyotisha (Understanding of astronomy).

Puranas: It is very vast and has 18 main Puranas and 18 subsidiary Puranas. It was compiled by Sage Vyas. It mentions four ages: Krita, Treta, Dvapara and Kali. It makes the five fundamental pillars of Puranic texts or 'Itihasa' - 'Sarga' (Evolution of the Universe), 'Pratsarga' (Involution of Universe), 'Manvantara' (Recurring nature of time), 'Vamsa' (List of Kings and Sages) and 'Vamsanucharita' (Selected character based life Stories).

18 Main Puranas are as follows: Vishnu, Naradiya, Padma, Garuda, Varaha, Bhagavata, Matsya, Kurma, Linga, Shiva, Skanda, Agni, Brahmanda, Brahma Vaivarta, Markandeya, Bhavishya, Vamana, Brahma Purana.

Ramayana: It is the oldest Epic. It was written by Maharshi Valmiki (It was known as Ratnakar). It started with 6000 verses and 24000 verses after many additions in different times.

Seven kanda of Ramayana: Bala, Ayodhya, Aranya, Kishkindha, Sundara, Yuddha, Uttara. The entire Aranya Kanda of Ramayana was set in Panchavati (Maharashtra). The first printed edition of the Ramacharitmanas by Tulsidas was published from Calcutta in 1810.

Mahabharata: It is the Longest Epics. It was written by Maharshi Veda Vyas. It began with 8800 verses and the final compilation has 100,000 verses.

The Adiparva is the first of eighteen books of the Mahabharata. "Adi" in Sanskrit means "first". It has 19 parts and 236 adhyaya.

The Shanti Parva is the twelfth of eighteen books of the Indian Epic Mahabharata. The Shanti parva recites the duties of the ruler, dharma and good governance, as counseled by the dying Bhishma and various Rishis.

Parts of time period of Vedic age: Early or Rig Vedic Period (1500 B.C - 1000 B.C) and Later Vedic Period (1000 B.C - 600 B.C).

Early Vedic Age Or Rig Vedic Age (1500 BC - 1000 BC)

The early Aryans were located in and around the areas surrounding Indus and its tributaries. The area was referred to as the Sapta-Sindhu region (the land of

seven rivers).

Religion

Indra is the God of Lightning, Thunder, Storms and Rain. He was known as Purandhar or Destroyer of forts. Rig Veda contains 250 hymns in praise of Indra. He is also known as "Lord of Plough".

Rudra is a God of Destruction. He is also worshiped for healing from diseases. He is considered to be the caretaker of the cattle.

Varun is a God of Water and Morals. He maintained cosmic order and laws.

Other Gods: Vayu - God of Air, Surya - God of Life Source, Aditi - Mother of Gods, Agni - God of Fire, Pushan - God of the Shudras, Usha - Goddess of Dawn, Vishnu - An aspect of Surya, Marut - God of Wind, Prithvi - Goddess of Fertility, Aranyani (Aryani) - Goddess of Forest, Parjanya - God of Rain, Prajapati or Adipurush - Supreme God.

According to the Puranas, Lord Vishnu took the shape of a boar in order to rescue the earth, which had sunk into the water. Varaha, ("Boar") third of the 10 incarnations (avatars) of the Hindu god Vishnu. According to Hindu iconography, the boar is a symbol of all aspects of sacrifice.

10 Avatars of Lord Vishnu: First Avatar - Matsya (Fish), Second Avatar - Kurma (Tortoise), Third Avatar - Varaha (Boar), Fourth Avatar - Narasimha (Man-Lion), Fifth Avatar - Vamana (Dwarf), Sixth Avatar - Parasurama (Angry Man), Seventh Avatar - Lord Rama (Perfect Man), Eighth Avatar - Lord Krishna (Divine Statesman), Ninth Avatar - Gautam Budha (Light of Asia), Tenth Avatar - Kalki (Mighty Warrior).

Rivers with Ancient Names: Indus - Sindhu, Jhelum - Vitasta, Chenab - Asikini, Ravi - Purushni, Beas - Vipasha, Sutlej - Shutudri, Ghaghar- Saraswati, Krishna- Krsnaveni, Narmada - Namade, Luni- Sagarmati, Betwa - Vetravati, Chambal - Charmanvati, Gandak - Sadanira, Kosi - Kaushiki, Ghaghara - Karnali, Kabul - Kubha, Swat - Suvastu.

Important Terms

Rig Vedic Terms and their meaning: **Avesta** - oldest text in the Iranian Language. **Naditarna** - another name for River Saraswati (called the best of the

rivers in the Rig Veda). **Dasa or Dasyus** - indigenous inhabitants of India during Aryan migration. **Panchjana** - 5 tribes of Aryans. **Gavisthi** - Term for war and (search for cows in Rig Veda). **Samudra** - collection of water. **Purohita** - head priest. **Bali** - voluntary offerings by people to the tribal chief. **Vrajapati** - when the units settled after wars, the Gramini became head of the village, with course of time he came to be identified as Vrajapati. **Gomat** - A wealthy person who owned many cattle. **Duhitri** - a daughter. **Vrata** (Gana or Grama or Sardha) - Different tribal groups. **Purcharishnu** - moving fort made for fighting a war. **Kshira pakamodanam** - grain cooked with milk. **Sura and Soma** - alcoholic drinks. **Godhuli** - Time. **Samsara** - transmigration of Soul. **Gavyuti** - Distance. **Gauri** - Buffalo. **Gojit** - Winner of Cow or a Hero. **Goghna** - Guests fed on cattle. **Gau** - cow. **Orohippus** - Horse. **Ayas** (Metal), **Hiranya** (Gold), **Shyama** (Iron), **sisa** (Lead), **Trapu** (Tin), **Lohita** (Copper).

Later Vedic Age (1000 BC - 600 BC)

In the later Vedic period, the Aryans moved to the Eastern areas with the help of fire and iron tools. The area inhabited by them included the Guru Panchala region (the Indo-Gangetic divided and the upper Ganga valley).

Religion

Vedic Religion is dominated by the Brahmins due to which it is also known as Brahmanism. Sacrifices remained important. Few instances of idol worship are reported. Prajapati or Adipurush (the creator) became the Supreme God in the later Vedic phase. Vishnu - the protector. Rudra - the destroyer.

Varna system: It was originally used to distinguish between Vedic and non-Vedic people. They were birth-based rather than profession-based in this period.

According to the **10th mandala** of the Rigveda, known as 'purush sukta'. They are formed from different portions of the God's body, such as Brahmin from the mouth, Kshatriya from the arms, Vaishya from the thighs, and Shudra from the feet. Brahmins - priestly, academic class. Kshatriyas - Rulers, administrators, warriors. Vaishyas - Artisans, tradesmen, farmers, merchants. Shudras - Manual laborers. Kshatriya was responsible for

protecting people and administering justice as per rules laid down by the Dharmasutras and Dharmashastras.

Types of Yajna :

Vajapeya was a Soma Yajna, which involves minor animal sacrifice. Soma yajnas are a type of yajna associated with the lunar cycle, and regarded to be performed for the maintenance of the cosmic order.

Rajasuya (Imperial Sacrifice or the king's inauguration sacrifice) is a Sruta ritual of the Vedic religion. It is a consecration of a king.

Ashvamedha is a horse sacrifice ritual followed by the Śrauta tradition of Vedic religion.

Terms of Later Vedic Age

Satapatha Brahmana - text talks about ploughing ritual. Vrihi - rice in Vedic texts. Vis - common people. Vis or Vaishya - refractory people or common people. Yajamana - one who performed Yajna. Purohita - the priest. The guest was known as Ghoghana or the eater of cattle.

Political Terms : Rajanyas - close kinsmen of the King. Rathakar - chariot maker. Gotra - cowpen or place where cattle belong to the whole clan. Suta or Sarathi - charioteer. Senani - General. Kshata - gateman or chamberlain. Sangrahitri - treasurer. Govikarta - head of forest department. Ganas - corporations of merchants.

Political and Social Structure in Vedic Age

The societies were mostly patriarchal, and the eldest male was the head of a family. A family was a basic unit of political organization and was termed as a 'kula'. The head of the family was called a 'Kulapa'. A 'grama' was formed when many such families came together. A village headman called a 'Gramini' was responsible for leading a 'grama'.

A 'visu' was formed by a group of villages. A 'Vishayapati' headed the 'visu'. A group of visu formed the 'Jana' (community). The head of a 'Jana' was known as the 'Rajan'. The people regarded the Rajan as their protector rather than a king. Rajan was also known as 'Gopati Janasya' or 'Janasya Gopa' which means "lord of cattle". The wars in

this period were fought mostly for the control of cattle wealth and not for territory. These wars were fought by Rajan on behalf of his 'Jana'.

There were two assemblies called the 'Sabha' and the 'Samiti' which helped the Rajan make important decisions. The Rajan was also assisted by a 'Purohita' in day-to-day administration.

Sabha referred to a council in the Rig Vedic period. The elder members of the 'Jana' took part in the 'Sabha'. This assembly was also attended by women known as 'Sabhavati'. In the later Vedic period, however, women stopped attending the 'Sabha'. The participation of women in the 'Sabha' is indicated only once in the Rig Veda.

Samiti referred to an assembly in which political business and discussions were performed. It was responsible for electing the 'Rajan'. It also discussed philosophical issues. Prayers and religious ceremonies were a major concern for this assembly. This assembly, however, gained importance only towards the end of the Early Vedic period.

Vidhata was the earliest tribal assembly. The purpose of this assembly was to make decisions for secular, military, economic, social, and religious purposes. It was the earliest folk assembly of the Rig Vedic period. The different clans and tribes used the 'Vidhata' as a common ground for the worship of their Gods.

Major Concepts

Chaturashrama or **Four Ashramas** is part of the ancient Indian concept which identified four stages of human life, with distinct differences based on natural human needs. Four Ashramas: Brahmacharya, Grihastha, Vanaprastha, and Sannyasa.

Brahmacharya Ashram (Student): It is a divine world. It is the sum and substance of Yoga. It is the conduct by which one attains or reaches god. It is a movement towards God or the Atman (self).

Grihastha Ashram (Householder): It embodies a married life with the duties of maintaining a home, family, raising children, their education, and leading a social life. It was the feeder of all other Ashrams.

Vanaprastha Ashram (forest dweller): It

refers to handing over all the household duties and responsibilities to the next generation, taking an advisory role, gradually withdrawing from the world and going to the forest.

Sannyasa (Renunciate): It means to put down everything. He breaks all attachments with the world by giving away all property and world and devotes his time to meditation to reorganize the subtle nature of the supreme soul and seek Moksha.

Purushartha refers to a goal, end or aim of human existence. Four Purusharthas: Dharma - Righteousness, Artha - Acquisition of wealth by proper means and its right use, Kama - Fulfillment of noble desires, Moksha- Liberation or the Final Goal.

Mahajanapadas

Janapadas were key kingdoms in Vedic India, dominated by the powerful Aryan tribes known as 'Janas'. 'Vish' refers to the common people. Around 1500 BCE, these monarchies thrived, with around 22 distinct Janapadas by the 6th century BCE.

Mahajanapadas (big country) existed between 600 BCE and 345 BCE. They were mainly Oligarchical. The Indian Subcontinent had sixteen Mahajanapadas between 600 B.C. and 325 B.C.

Monarchical states: Magadha, Kashi, Vatsa, Chedi, Matsya, Shurasen, Avanti, Gandhara, Kuru, Kosala, Panchala, Anga, Assaka, and Kamboja.

Non-monarchical states or Republican states: Malla, Vajji.

16 Mahajanapadas

They were either monarchical or republican. Republican states had a Ganparishad (assembly of seniors) as the supreme authority.

Anga (Capital - Champa)

The Mahabharata and Atharva Veda both mention Anga. The Anga country was very ancient as was evident from the reference made to it in the ancient Brahmanical works like the Aitareya Brahmana, the Mahabharata, the Puranas, the Ramayana.

It is located in modern-day Bihar and

West Bengal. It was located in the east of Magadha and west of RajMahal hills. Its capital was situated at the confluence of the Ganga and Champa rivers. It was seized by the Magadha Empire under the reign of Bimbisara.

Magadha **(Capital- Rajagriha / Girivraja)**

The name Magadha first appears in the Atharvaveda. It was located between Anga and Vatsa. It covers Patna and Gaya districts of Bihar and eastern parts of Bengal separated by the Champa River. Later, Magadha became a Jain centre, and Rajagriha hosted the first Buddhist Council.

Rivers such as the Ganga and Son flowed through it. Rajagaha or Girivraja, protected by five hills, was the early Magadhan capital. Later, the capital was shifted to Patliputra.

Kashi **(Capital - Varanasi)**

According to the Matsya Purana, this city was named after the rivers Varuna and Asi. Kashi was in constant conflict with the kingdoms of Kosala, Magadha, and Anga. The 'Kashaya' (orange brown) robes of the Buddhist monks manufactured in this city. It was famous for cotton textiles and market for horses.

Vatsa **(Capital - Kaushambi)**

The Vatsa appears to be one from the Rig Vedic period. It was said to be an offshoot of the Kurus, who shifted from Hastinapur and settled in Kaushambi. It was often spelled Vamsa and located on the Yamuna's banks. Udayana (the ruler of this country in the sixth century BCE) had to struggle against king Ajatasatru of Magadha and king Pradyota of Avanti. Later, during the reign of Palaka, Vatsa was annexed to the Avanti kingdom.

Kosala **{Capital - Shravasti (northern), Saketa (southern)}**

The Shatapatha Brahman gives a description of Kosala Mahajanapadas. It was in Awadh, Uttar Pradesh. On the west, Kosala was bound by the river Gomati, on the southern side by the river Sarpika or Syandika, on the east by the Sadanira (Gandak) which separated it

from Videha and on the north by the Nepal hills. Ayodhya, Saketa and Shravasti were three important Kosalan cities.

The kings of Kosala favoured Brahmanism and Buddhism. Prasenjit, the Kosalan king was the contemporary of king Bimbisara and Ajatasatru of Magadha. Kosala also included the tribal republican territory of the Sakyas of Kapilvastu.

Shurasena **(Capital - Mathura)**

During Megasthenes' time, the location was a centre of Krishna devotion. Its capital was Mathura on the bank of the river Yamuna. It was located at the junction of two famous ancient Indian trade routes, the Uttarapath and the Dakshinapatha.

Panchala **(Capital - Ahichchhatra and Kampilya)**

Its capital for northern Panchala was Ahichchhatra (modern Bareilly), and its capital for southern Panchala was Kampilya (modern Farrukhabad). The famous city of Kannauj was located in the kingdom of Panchala.

Kuru **(Capital - Indraprastha)**

It was born in the vicinity of Kurukshetra. It adopted a republican form of government. The Mahabharata (an epic poem) describes a war between two branches of the ruling Kuru clan. It was the most important kingdom of the later Vedic period, but during the sixth century BCE, the Kurus did not occupy the same position.

Matsya **{Capital - Viratanagara (modern Bairat)}**

A hymn in the Rigveda refers to the Matsyas who were attacked by Turvasa (a famous king). It lay to the west of the Panchalas and to the south of the Kurus. It is located in the Rajasthan region of Jaipur, Alwar, and Bharatpur. The Matsyas appear to have been one of the prominent Ksatriya tribes that made up the Vedic Aryan people in the earliest period of their residence in India.

Chedi **(Capital - Sothivati)**

It was mentioned in the Rigveda. Its Sanskrit name in the Mahabharata was Sothivati. It was located in the Bundelkhand area (Central India) in Banda district of Uttar Pradesh. The king there was Shishupala. It formed one of the most ancient tribes among the Ksatriyas in early Vedic times. It was closely connected with the Matsya and Kashi.

Avanti **{Capital - Ujjaini (northern part) or Mahishmati (southern part)}**

It had an important role in the emergence of Buddhism. It is located in Malwa and Madhya Pradesh. The river Betravati divided Avanti into north and south. The ruler of Avanti during the time of the Buddha was Chanda Pradyota. The kingdom of Avanti was finally annexed to the Magadha Empire by Sishunaga.

Gandhara **{Capital - Taxila (Takshashila)}**

The Atharva Veda mentions Gandhara. Ambhi was king of Gandhara near the modern city of Attock (Pakistan), at the time of Alexander the Great's expedition to India.

Pushkarasarin was a powerful king. They were captured by the Persians at the end of the 6th century BC.

Kamboja **{Pooncha (present- Rawalakot)}**

Asoka's edicts refer to Kamboja along with Yona as frontier provinces. It was located in modern-day Kashmir and the Hindukush. Around little before 530 BCE, it fell prey to the attack of Cyrus, the Achaemenid emperor of Persia. Cyrus I was said to have destroyed the Kamboja city of Kapisi (modern Begram).

During Kautilya's time, it transformed from a monarchy to a republic. Sumangalavilasini describes Kamboja as the home for horses. The horses of Kamboja were famous throughout all periods of Indian history.

Asmaka **(Capital - Pratisthan)**

It is situated on the banks of the

Godavari. It was the only Mahajanapada in Dakshinapatha and located south of the Vindhya Range. It includes the Pratisthan or Paithan area.

Vajji (Capital - Vaishali)

The Vajjis ruled the state north of the Ganga in the Tirhut division. The most powerful clans were the Lichchhavis (Capital – Vaishali), Videhans (Capital – Mithila), and Jnatikas (based in Kundapura). Mahavira was a member of the Jnatikas clan. Ajatashatru triumphed over the Vajjis.

Malla (Capital - Kusinara)

It was mentioned in Buddhist and Jain scriptures, as well as in the Mahabharata. Its territory extended to the Vajji state's northern boundary. Kusinara and Pava are the capitals. Malla was a democratic republic. The Buddha ate his final supper at Pava before entering Mahaparinirvana in Kusinara.

Important Aspects of Mahajanapada Period

Economy

Most urban settlements were inhabited by merchants and artisans (organised into a 'sartha' guild). Practice of trade and crafts was hereditary. Trade was facilitated through use of money called 'nishka' and 'satamana' (mentioned in Vedic texts; no archaeological evidence). Vessa meant Merchants Street.

Agriculture was made easier with the use of iron tools like axes, adzes, knives, razors, nails, and sickles. Paddy transplantation was practised. Besides, barley, cotton, pulses, millets and sugarcane were also produced.

Bhaga Tax - 1/6th of the farm produce was to be paid to the royal agent. There were no intermediate landlords. 'Balisadhakas' collected the compulsory tax called 'bali' from peasants and 'vaishyas' only. Rich peasants were called 'grihapatis'.

Punch-marked coins: the earliest coins belong to the 5th century BCE. They circulated for the first time in eastern Uttar Pradesh and Bihar. They were made

of silver and copper. They are classified into two groups: local and universal. Local punch-marked coins have one to four symbols depending on their location and size. Universal marked coins have five symbols and were mainly issued by the Magadha dynasty.

Society

People lived in three types of villages: groups of various castes and communities living together headed by 'Bhojaka', suburban and dominated by craftsmen and linked the rural to urban, villages in the outskirts of the countryside and consisted of fowlers, hunters. The term 'Kammakaras' was related to landless agricultural labourers.

Writing had started and was used for bookkeeping in trade, taxation, and the large size of the army. Lower varnas were subject to many discriminations. Large-scale use of iron tools, spread of agriculture and North Black Polished **Pottery** was also associated with this age. Position of women degraded further except in Buddhist and Jain orders.

Administration and Law

The political centre shifted from the west to the east of the Indo-Gangetic plains during this period. The king was primarily a warlord. He was supported by other officials called Mahamatras who performed the functions of Mantrin (minister) and Senanayaka (Commander), judge and chief accountant. Aayuktas - Another class of officers performing similar functions. Administration of the village was under the village headman called as gramabhojaka, gramini or gramika.

Decline

The epic Mahabharata has one of the first allusions to Magadha, in which the whole Yadava clan abandons their homes on the Gangetic plains in order to avoid continual clashes with their eastern neighbour, Magadha.

Despite Magadha's tremendous riches and military prowess, its ruler was extremely hated due to his harshness and the enormous rates of taxes he imposed on the public. The formidable Magadha monarchy had given birth to an even more powerful state, the Mauryan Empire.

Magadh Empire

Magadha Empire was the strongest of the sixteen Mahajanapadas that were present in the Indian subcontinent in the pre Mauryan times. It is known as the second period of urbanization in India.

The kingdom's heart was in Bihar, south of the Ganges River. It was the first empire in ancient India to use elephants in war on a large scale.

Dynasties of Magadha Empire

Haryanka Dynasty (544 - 412 BCE)

Bimbisara (544 - 492 BCE) is considered to be the most powerful ruler of Magadha due to his conquest of the Anga kingdom to the east. He was the founder of the Haryanka dynasty. He was the son of Bhattiya. His first capital was at Girivraja (identified with Rajagriha) and was famous in Buddhist writings.

He was the contemporary of Buddha. Kosala Devi (princess of Kosala) married Bimbisara and brought Kashi as dowry. Chellana (Bimbisara's second wife) was a Lichchhavi princess from Vaishali. His third wife was the daughter of the Punjabi Madra clan head. The famous physician Jeevaka was appointed in the court of Bimbisara.

Courtiers: Sona Kolivisa, Sumana (flower gatherer), Koliya (minister), Kumbhaghosaka (treasurer) and Jeevaka (physician). The royal physician Jeevaka was sent by Bimbisara to treat Avanti King Pradyota who was ailing with Jaundice. He was killed by his son Ajatashatru.

Ajatashatru (492 BC - 460 BC) was the second important king of the Haryanka Dynasty of Magadha in East India. He killed his father Bimbisara and snatched the kingdom. He was a contemporary of both Mahavira and Gautam Buddha. He was known as Ajatasattu in Buddhist tradition and Kunika or Kuniya in the Jain histories.

He invented two weapons used in war, the Rathamusala (scythed chariot) and the Mahashila Kantak. He conquered Varanasi and Vaishali to his kingdom. His meeting with Buddha was depicted in sculptures of Bharhut.

He wanted to attack the Vajjis, so he sent his minister named Vassakara to the Buddha to get his advice on the matter. He sent his chief minister Vassakara to infiltrate the Vajji confederacy and within three years he had managed to split the Vajjis and also demolished the altars in Vaishali.

Udayin (460 BC - 444 BC) succeeded Ajatashatru. He shifted the capital from Rajgriha to Pataliputra.

Naga-Dasak (444 BC - 412 BC) was the last ruler of the Haryanka dynasty. He was found unworthy to rule by the people and was forced to abdicate his throne in favor of his minister Shishunaga.

Shishunaga Dynasty (412 -344 /345 BCE)

Shishunaga was an Amatya (minister) during the Magadha Period. He became king and founded the Shishunaga dynasty. Most important achievement of Shishunaga was the destruction of the Pradyota dynasty of Avanti. He shifted the capital Rajagriha to Vaishali. He was succeeded by Kalashoka.

Kalashoka was the son and successor of Shishunaga. He divided his kingdom between his ten sons and crowned his ninth son, Nandivardhana as the king of Magadha. He shifted the capital Vaishali to Patliputra. Mahanandin, son of Nandivardhana, was the last Shishunaga king. Mahanandin was killed by Mahapadma (illegitimate son from a Shudra wife).

Nanda Dynasty (344 - 322 BCE)

Mahapadma Nanda : according to the Puranas, was the first Emperor of the Nanda Empire of ancient India. He founded the 1st non-Kshatriya dynasty. He was known as the "Ekarat" (Empire builder). He acquired Kalinga and Kosala to Magadha. He was titled the "second Parshuram". He was succeeded by his eight sons. Collectively, they were called Navanandas (nine Nandas).

He founded the first great North Indian empire, with Magadha as its capital. He shifted the capital to Pataliputra (Patna, Bihar). He defeated all the Kshatriya ruling dynasties of Northern India, like the Kurus, Panchalas, Maithili, Ikshvakus, Asmaka, Kalinga, and the Surasenans. He was also known as "Sarva Kashtriyantaka" (Kshatriya Destroyer).

Dhana Nanda was the last ruler of the Nanda dynasty. Alexander invaded northwestern India during his reign.

Features of Magadh Empire

Iron mines in Jharkhand and the large scale use of iron tools were among the reasons for the rise of the Magadha Kingdom. The spread of agriculture and North Black Polished Pottery is also associated with this age.

The Sramana culture evolved in "Greater Magadha". Rice (paddy transplanting was common), Barley, Pulses, Millets, Cotton, and Sugarcane are the main crops. Customs officials in the Magadha empire were known as Kammika. Toll officials were known as Shulk - adhyaksha.

Control of Trade and Transportation over a vast section of the Ganges River. Strategic marriage connections prevailed. A standing army commanded by professionals. Centralised System of Governance.

Decline of Magadha Empire

Vishnugupta (Chanakya or Kautilya) in order to avenge his humiliation at the hands of the monarch. There was much intrigue, counter-intrigues, scheming and counter-plotting that Kautilya resorted to in order to weaken Dhanananda by weaning away his major allies, adherents, and supporters.

Position of women degraded. A number of castes appeared and the condition of untouchability further worsened.

Somewhat before the time of the Mauryan empire, about 2400 years ago, emperors in China began building the Great Wall.

Jainism

Jainism is the religion which teaches that the path to enlightenment is through nonviolence and reducing harm to living things (including plants and animals) as much as possible.

The Barli Inscription belonging perhaps to 5th - 4th century B.C. is one of the earliest Jaina inscriptions reported from Rajasthan.

Tirthankaras

There is a succession of great teachers

or Tirthankaras in the Jain tradition. There are a total of 24 Tirthankaras associated with Jainism.

Rishabhanath (Rishabhadeva, Ikshvaku): He was the first Tirthankara of Jainism and the founder of Ikshvaku dynasty. He is known by many names including Adinatha, Adishwara, Yugadeva and Nabheya. He was born in Ayodhya. He was called a "ford maker" because his teachings helped one across the sea of interminable rebirths and deaths.

Parshvanatha: He was the 23rd Tirthankaras of Jainism. He is the only Tirthankara who gained the title of Kalikalkalpataru (Kalpavriksha in this "Kali Yuga"). He is popularly seen as a propagator and reviver of Jainism. He was born in Varanasi, India. He belonged to the Ikshvaku dynasty.

Mahavir (Vardhaman)

He was the 24th tirthankara of Jainism. He was born in 540 BC into a royal Kshatriya Jain family in Kundagrama near Vaishali district, Bihar. He belonged to the Jnatika clan. His birth is celebrated as Mahavir Jayanti. His mother's name was Trishala and his father's name was Siddhartha (the head of Jnatika clan).

He was married to Yasoda and had a daughter Anojja or Priyadarshana. At the age of 30, he became an ascetic and left his home in the search for truth. In Pali literature, he was known as Nigantha Nataputta.

Kaivalya: He attained Kaivalya (highest spiritual knowledge) at the age of 42 under Sal tree. He is called Mahavir, Jina (Conqueror), Jitendriya (one who conquered his senses), Nigrantha (free from all bonds), Kevalin. He delivered his first sermon at Pava. He attained salvation at the age of 72 at the Pawapuri in Bihar.

Tenet

Jainism recognised the existence of god but placed them lower than Jina (Mahavira). It did not condemn the varna system but attempted to mitigate the evils of the varna order and the ritualistic Vedic religion.

According to Mahavira, a person is born in higher and lower varna as the consequence of the sins or the virtues in the previous birth. It believes in the

“transmigration of the soul” and the “theory of Karma.” It mainly aims at the attainment of liberation, for which no ritual is required.

Three Jewels (Triratna): Right Faith - Samyak Darshan. Right Knowledge - Samyak Gyana. Right Conduct - Samyak Charitra.

Five Doctrines of Jainism: Ahimsa - Non-injury to living beings. Satya - Do not lie. Asteya - Do not steal. Aparigraha - Do not acquire property and Brahmacharya - Observe continence.

Council

First Jain Council: Time - 300BC, Place - Patliputra, Chairperson - Sthulbhadra, Output - Compilation of 12 Angas.

Second Jain Council: Time - 512 AD, Place - Vallabhi and Mathura (simultaneously), Chairperson - Nagarjuna (Vallabhi), Skandila (Mathura), Output - Final compilation of 12 Angas & 12 upanga.

Doctrine

Anekantavada: It emphasizes that the ultimate truth and reality is complex, and has multiple-aspects that is “theory of plurality”. It refers to the simultaneous acceptance of multiple, diverse, even contradictory viewpoints.

Syadvada: All judgments are conditional, holding good only in certain conditions, circumstances, or senses. It literally means the method of examining different probabilities.

Important Term Related to Jainism :

Asrav - Inflow of karmas to the soul that occurs at every second in life.

Samvara - Stoppage of the influx of the material karmas into the soul consciousness.

Nirjara - Shedding or removal of accumulated karmas from the soul, essential for breaking free from samsara, the cycle of birth-death and rebirth, by achieving moksha, liberation.

Sallekhana (Santhara) - The religious practice of voluntarily fasting to death by gradually reducing the intake of food and liquids.

Tirthankaras and Symbol

No.	Tirthankaras	Symbol
1	Rishabhanatha	Bull
2	Ajitanatha	Elephant
3	Sambhavanatha	Horse
4	Abhinandananaatha	Monkey
5	Sumatinatha	Heron
6	Padmaprabha	Red Lotus
7	Suparshvanatha	Swastika
8	Chandraprabha	Crescent Moon
9	Pushpadanta	Crocodile
10	Shitalanatha	Kalpavriksha
11	Shreyansanatha	Rhinoceros
12	Vasupujya	Buffalo
13	Vimalanatha	Boar
14	Anantanatha	Porcupine
15	Dharmanatha	Vajra
16	Shantinatha	Deer
17	Kunthunatha	Goat
18	Aranatha	Fish
19	Mallinatha	Kalasha
20	Munisuvrata	Tortoise
21	Naminatha	Blue Lotus
22	Neminatha/ Arishtanemi	Shankha
23	Parshvanatha	Snake
24	Mahavira	Lion

Jain literature

Jain literature is classified into two major categories. Agam Literature and Non-agam Literature.

Agam Literature: The texts are collectively known as Agams, the sacred books of the Jain religion. Agam literature is also divided into two groups - Ang-agama and Ang-bhaya-agams.

Ang-agama: These texts contain the direct preaching of Lord Mahavir. They were compiled by Ganadharas. All Ganadharas possessed perfect knowledge (keval-gyan).

Ang-bahya-agams: These texts are expansions of Ang-agams. They were compiled by Shrutakevalin. Monks who had knowledge of a minimum of ten Purvas were known as Shrutakevalin. The twelfth Ang-agam is called Drastivad. The Drastivad consists of fourteen Purva texts, also known as Purvas or Purva-agams. Among Ang-agams, Purvas were the oldest sacred texts. They are written in the Prakrit language.

Non-agam Literature: It consists of explanations of Agam literature and independent works, compiled by elder

monks, nuns, and scholars. They were written in many languages such as Prakrit, Sanskrit, Old Marathi, Gujarati, Hindi, Kannad, Tamil, German, and English.

Prabandha - Chintamani is an Indian Sanskrit language collection of prabandhas (semi-historical biographical narratives). It was compiled in 1304 CE by Jain scholar Merutunga.

Kalpa Sutra: It is a Jain text containing the biographies of the Jain Tirthankaras, notably Parshvanatha and Mahavira.

Sects of Jainism

Jain order has been divided into two major sects: Digambara and Svetambara. The division occurred mainly due to famine in Magadha which compelled a group led by Bhadrabahu to move South India.

Digambara: Monks of this sect believe in complete nudity. Male monks do not wear clothes while female monks wear unstitched plain white sarees. Follow all five vows (Satya, Ahimsa, Asteya, Aparigraha and Brahmacharya). Digambara believes that women cannot achieve nirvana or liberation directly in their current lifetime but need to be reborn as a man to attain it. Bhadrabahu was an exponent of this sect. Major Sub-Sects: Mula Sangh, Bisapantha, Terapanth, Taranpantha or Samaiyapantha. Minor Sub-Sets: Gumanapantha, Totapantha.

Svetambaras: During the 12 years famine, the group in South India stick to the strict practices while the group in Magadha adopted a more lax attitude and started wearing white clothes.

Follow only 4 vows (except brahmacharya). Believe women can achieve liberation. Sthulabhadra was an exponent of this sect. Major Sub-Sects: Murtipujaka, Sthanakvasi, Terapanthi.

Contribution of Jainism

Jainism paved the right path for mankind to create a healthy society, which worked alongside their philosophy of non-violence and peace. This religion was the first to take important steps to stop the caste system and Vedic rituals.

Buddhism

Buddhism originated in South Asia

around the 6th century B.C.E. with Siddhartha Gautama.

Gautam Buddha

He was born into the royal family of the **Sakya** clan which ruled in Kapilvastu, Lumbini (Nepal) in 563 BC. Vesakh is known as Buddha Jayanti. He was a Kshatriya. It is testified by the inscription on the pillar erected by the Mauryan Emperor Ashoka in 249 BC. He was the son of Maha Maya and Raja Shuddhodana.

Mahapajapati Gotami was the step-mother and maternal aunt (mother's sister) of Buddha. She was known to be the first woman to seek ordination for women and became the first female bhikkhuni.

Mahabhiniraskramana: At the age of 29, Gautama left home and rejected his life of riches and embraced a lifestyle of asceticism or extreme self-discipline (Mahabhiniraskramana).

Enlightenment: After 49 consecutive days of meditation, Gautama attained Bodhi (enlightenment) under a pipal tree at Bodhgaya (situated west of the Phalgu River (Niranjana River), a village in Bihar.

Sermon: He gave his first sermon in the Deer Park at the village of Sarnath, near the city of Varanasi in Uttar Pradesh. The event was known as Dharma Chakra Pravartana (turning of the wheel of law). He also delivered his last sermon at Kushinagar.

The maximum number of disciples of Gautam Buddha was from the kingdom of Kosala. He delivered his maximum number of sermons at Shravasti, the capital of the kingdom of Kosala.

Mahaparinirvana: He died at the age of 80 in 483 BCE at a place called Kushinagar (a town in Uttar Pradesh) and was cremated at Ramabhar Stupa (Mukutbandhan Chaitya). The event is known as Mahaparinirvana (salvation).

Teacher: The first teacher of Gautama Buddha was Alara Kalama. After reaching the highest point Buddha left Alara and went to another teacher named Uddalaka Ramaputra, who taught immaterial attainments to Buddha.

Symbols and Great Events of Buddha's Life: White Elephant - Avakranti (conception or descent). Lotus

and Bull - Jati (birth). Horse - Mahabhinishkramana (Great Renunciation). Bodhi tree - Nirvana (enlightenment or Sambodhi). Wheel - Dharmachakra Parivartana (first Sermon). Stupa - Mahaparinirvana (Death).

Teachings

Buddhists believe that human life is a cycle of suffering and rebirth. The state of Enlightenment (nirvana) is a possible way to escape this cycle forever.

Madhyam Marg: He taught a Middle Way between sensual indulgence and severe asceticism leading to Nirvana. It leads to freedom from ignorance, craving, rebirth, and suffering.

Five precepts (Panchshila): No killing, Respect for life, Abstinence from theft, Abstinence from sexual misconduct, Abstinence from falsehood, and Abstinence from intoxication.

Three Jewels of Buddhism (Triratna): The three main aspects of Buddhism are the Buddha, the Dharma and the Sangha.

Four Noble Truths: Sabbam Dukkham - World is full of misery. Pratitya Dukkha Samutpada - Desire is the root cause of sorrow. Dukkha Nirodha - Desire can be conquered. Dukkha Nirodha Gamini Pratipada - there is a path leading to the extinction of dukkha.

Eightfold Noble Path: It consists of eight practices: Right view, Right resolve, Right speech, Right action, Right livelihood, Right mindfulness, Right effort, Right concentration.

Literature

The earliest Buddhist texts were passed down orally in Middle Indo - Aryan languages (Prakrits), Gandhari language, and Pali through the use of repetition, communal recitation and mnemonic devices. These texts were later compiled into canons and written down in manuscripts. The Pali Canon was preserved in Sri Lanka where it was first written down in the first century BCE.

Three Pitakas: Tripitaka (Triple Basket) is the traditional term for ancient collections of Buddhist sacred scriptures. It was compiled in around 5th to 4th century BCE. **Tripitaka:** Vinaya Pitaka, Sutta Pitaka, and Abhidhamma Pitaka.

Vinaya Pitak : consists of rules of conduct and discipline applicable to the monastic life of monks and nuns.

Sutta Pitaka: consists of the main teaching or Dhamma of Buddha. It is divided into five Nikayas or collections: Digha, Majjhima, Samyutta, Anguttara, and Khuddaka.

Abhidhamma Pitaka is a philosophical analysis and systematization of the teaching and scholarly activity of the monks.

Buddhacharita : Written by Ashvaghosha is an epic on the life of Buddha, composed in the early second century CE.

Other literature related to Buddhism - Divyavadana, Milinda Panha (written by Nagasena), Dipavamsa and Mahavamsa.

Milinda Panha: It claims to be a discussion between the Indian Buddhist scholar Nagasena and the Indo-Greek monarch Menander I of Bactria.

Jatakas: They refer to stories of Buddha's Birth. Jataka means Birth in Sanskrit. They were written in the Pali language. Some of the Jataka stories are Chhadanta Jataka, Vidurpundita Jataka, Ruru Jataka, Sibi Jataka, Vessantara Jataka, and Shama Jataka.

Scholars

Moggaliputta Tissa: He was a Buddhist monk and scholar who was born in Pataliputra, Magadha. He launched the Ashoka Dhamma campaign.

Nagarjuna: He was a Mahayana Buddhist thinker, scholar-saint, and philosopher. He is considered to be the founder of the Madhyamaka school of Buddhist philosophy and a defender of the Mahayana movement.

Buddhaghosa: The Visuddhimagga is the 'great treatise' on Buddhist practice and Theravada Abhidhamma written by Buddhaghosa in the 5th century in Sri Lanka.

Dharmakirti: He was an influential Indian Buddhist philosopher who worked at Nalanda.

The Shailendra dynasty who founded their empire in South-east Asia in the 8th Century AD were the followers of Buddhism. Their reign is known for its significant contributions to Buddhist art

and architecture.

Council

First Buddhist Council: Time - 483 BC. Place - Rajagriha. Patronized by - Ajatashatru. President of the council - Mahakasyapa. Features - the teachings of Buddha were compiled into three Pitakas.

Second Buddhist Council: Time - 383 BC. Place - Vaishali. Patronized by - Kalasoka. President of the council - Sabakami. Features - Division into Sthaviravadins and Mahasanghika.

Third Buddhist Council: Time - 250 BC, Place - Pataliputra. Patronized by - Ashoka. President of the council - Mogaliputta Tissa. Features - Buddhist missionaries were sent to other countries, Compilation of Abhidhamapittaka.

Fourth Buddhist Council: Time - 72 AD. Place - Kashmir. Patronized by - Kanishka. President of the council - Vasumitra. Vice president- Ashvaghosa. Features - Divided into Mahayana and Hinayana.

Different Schools of Buddhism

Mahayana: The term Mahayana is a Sanskrit word that literally means "Great Vehicle". It believes in the heavenliness of Buddha and Idol worship of Buddha and Bodhisattvas embodying Buddha Nature.

Hinayana: It literally means a Lesser vehicle. It believes in the original teaching of Buddha or the Doctrine of elders. It does not believe in Idol worship. It tries to attain individual salvation through self-discipline and meditation.

Theravada: It is the most ancient branch of extant Buddhism today. It remains closest to the original teachings of the Buddha. It developed in Sri Lanka.

Vajrayana: It means "The Vehicle of the Thunderbolt", also known as tantric Buddhism. It developed in India around 900 CE.

Zen: It is a school of Mahayana Buddhism that originated in China during the Tang dynasty as the Chan school of Chinese Buddhism and later developed into various schools. It spread to Japan in the 7th century C.E.

Important Terms

Arhat : Liberated beings. Pabbajja - Refers to when a layperson leaves home to live the life of a Buddhist renunciate among a community of bhikkhus.

Chaityas - Buddhist prayer halls that house a stupa.

Vihara - A monastery for Buddhist renunciates.

Sramana : One who labors, toils, or exerts themselves (for some higher or religious purpose) or "seeker, one who performs acts of austerity, ascetic.

Upasampada - Refers to the rite and ritual of ascetic vetting ascetic life.

Vassa : Three to four month (July to October) annual retreat observed by Theravada practitioners. Pavarana - It marks the end of the 3 lunar months of Vassa. Uposatha - It is the day for "the cleansing of the defiled mind," resulting in inner calm and joy.

Bodhisattvas

In Mahayana, Bodhisattva is a person who can reach nirvana but delays doing so through compassion for suffering beings.

Bodhisattvas and Traits: Maitreya - Future Buddha and Earliest bodhisattva. Samantabhadra - Universal Bodhisattva, associated with Meditation. Vajrapani - Depicted as one of the 3 protective deities around Buddha. Avalokiteshvara - Manifests Buddha's compassion, and is kind-hearted. Kshitigarbha - Guardian of purgatory and children, Bodhisattva of hell - beings or earth immortal beings.

Amitabha - Buddha of Heaven.

Sadaparibhuta - Manifests Never disparaging spirit.

Manjushri - Stimulator of understanding and he holds a book describing 10 paramitas. He depicts male bodhisattva wielding a flaming sword in his left hand.

Akasagarbha - Manifestation of wisdom.

Decline of Buddhism

Causes of Decline of Buddhism - Corruption in Buddhist Sanghas, Reform in Hinduism, Division among the Buddhists, Use of Sanskrit Language, Patronage of Brahmanism, Role of Hindu Preachers, Rifts in Buddhist Order, Worship of the Buddha, Lose of Royal Patronage, Emergence of Rajputs, Hun, and Muslim Invasion.

Contribution

The concept of Ahimsa was its chief contribution. Its contribution to the art and architecture of India was notable. The stupas at Sanchi, Bharhut, and Gaya are wonderful pieces of architecture.

It promoted education through residential universities like those at Taxila, Nalanda, and Vikramasila. The language of Pali and other local languages developed through the teachings of Buddhism. It also promoted the spread of Indian culture to other parts of Asia.

Differences between Jainism and Buddhism

Goal of Philosophy: Buddhism tends to eliminate mental anguish. Jainism is a way of life in which devotees follow the teachings of the Tirthankaras, respect all living things, and strive for Nirvana.

Clergy: Bhikkhus (male monks) and bhikkhunis (female monks) make up the Buddhist Sangha. Nuns and Monks are there in Jainism.

Means to Liberation: Buddhism teaches the Great Eightfold Path as the way to reach Enlightenment or Nirvana, while Jainism aims to free the soul from the chains of karma.

Path: Buddhism follows the middle path. Jainism practices rigorous asceticism and self-mortification.

Language of Literature: Pali was used by Theravada tradition while Sanskrit was used by Mahayana and Vajrayana traditions of Buddhism. The most common language of Jainism was Magadhi.

Views on Sins: Buddhists do not believe in sin. Sin is defined as causing harm to another person in Jainism.

Varna System: Buddhism condemns varna system. Jainism does not condemn it.

Mode of Liberation: Jainism's goals are the liberation of the soul through non-harm and non-action - to stop karma directly by stopping actions. Buddhism's goals are the liberation of the mind through the three higher trainings leading to an experientially verified engagement in the four noble truths.

Geographical Distribution and Predominance: Buddhism is primarily practiced in Thailand, Cambodia, Sri Lanka, India, Nepal, Bhutan, Tibet, Japan, Myanmar (Burma), Laos, Vietnam, China, Mongolia, Korea, Singapore, Hong Kong, and Taiwan. Jainists are mostly found in India, the lower Asian subcontinent, and America. Most nations have small groupings.

Similarities Between Buddhism and Jainism

Buddhism and Jainism both promoted the notion of giving up the world and living an austere life.

Both Buddhism and Jainism recognise that we are bound to the cycle of suffering by karma (actions) and unwholesome mental states, primarily, ignorance. Both Buddhism and Jainism are Sramanic traditions.

Mauryan Empire

The Maurya Empire (322-185 BCE) was the first pan-Indian empire, founded by Chandragupta Maurya after he overthrew the Nanda Dynasty. At its peak, it encompassed most of the Indian subcontinent (central and northern) and parts of modern-day Iran.

The Mauryan Empire was centered around the Indo-Gangetic Plain, with Pataliputra (modern Patna) as its capital. Monarchy flourished under the Mauryas. Ashoka's Lion Capital, India's national emblem, originated during this era. The Mauryan Empire ended by 185 BCE, with Brihadratha as the last ruler.

The **Mauryan Empire** divided its entire kingdom into various provinces.

Name of the provinces and Capital:

Uttarapatha (Northern) - Taxila, Dakshinapatha (Southern) - Suvarnagiri, Prachyapatha (Eastern) - Tosali, Avanti, Rashtra (Western) - Ujjaini, Central Province: Magadha - Pataliputra
Mauryan sites are located in Topara in the northern part of India. Sanchi Site (Bhopal) is associated with the Ashoka Pillar.

Rulers

Chandragupta Maurya: (321 - 297 BCE): At age 25, he captured Pataliputra from the last Nanda ruler, with the help of Kautilya (Chanakya or Vishnugupta).

Chanakya is often known as the "**Indian Machiavelli**". Chanakya was the prime minister under the reign of Chandragupta Maurya. He also served as the chief advisor to his son Bindusara.

After the Seleucid-Mauryan war, Chandragupta made a treaty with Seleucus Nicator I. Through this, Chandragupta gained territories west of the Indus, including Balochistan and Afghanistan. In return, Chandragupta gave Seleucus 500 war elephants, aiding his victory at the Battle of Ipsus. Additionally, Chandragupta married Seleucus's daughter. The Hindu Kush became the boundary between their empires.

The Mauryan Empire aimed to expand their rule across India. They conquered most of the subcontinent, except Kalinga and the far south. Chandragupta later converted to Jainism and is said to have died by fasting at Shravanabelagola.

In the third century B.C. famous Jain saint **Srutakevali Bhadrabahu** predicted a long and severe famine in the kingdom of Magadha (Bihar). Bhadrabahu along with 12,000 monks, migrated from Pataliputra to Shravanabelagola (Karnataka) in South India. To avoid the terrible effects of famine, Chandragupta Maurya also went with Acharya Bhadrabahu.

Megasthenes, a Greek historian, visited Chandragupta Maurya's court in Pataliputra. His book "Indica" earned him the title "Father of Indian History". Megasthenes described Pataliputra as a large, beautiful city surrounded by a massive wall, deep water, 570 towers, and 64 gates.

According to Megasthenes, Indian society was divided into seven classes: philosophers, cultivators, herdsmen, artisans or traders, soldiers, overseers, councilors.

Bindusara (297 - 269 BCE): He also known as "Amitrochates" or "Amitragata" by the Greeks, was the son of Chandragupta Maurya. Chanakya served as his prime minister. Bindusara expanded the Mauryan Empire to Mysore, uniting sixteen states into the empire and conquering nearly the entire Indian peninsula.

Bindusara had cordial diplomatic ties with Greece. Antiochus-I sent his envoy Deimachus to Bindusara's court. Bindusara patronized the Ajivikas and was taught by the Ajivika Brahmin

Pingalavatsa. Bindusara died around 273 BCE.

Ashoka (269 - 232 BC): The son of Bindusara, was one of India's greatest emperors. He was known as "Ashoka the Great", was the first ruler to communicate with people through inscriptions. After becoming king, the Kalinga War was the only conflict he fought.

He had sent his son Mahendra and daughter Sanghamitra to Sri Lanka to spread Buddhism. Ashoka brought further unification in the Mauryan empire by one dharma, one language and one script.

Kalinga War

The conquest of Kalinga in 261 BCE was a pivotal event in Ashoka's reign. Horrified by the carnage, described in the XIII Rock Edict, Ashoka abandoned military conquest (Bherigoshha) and embraced the path of Dhamma, spreading Buddhism. Kalinga refers to the coastal region between the Godavari and Mahanadi rivers, modern-day Odisha.

Dhamma Policy

Initially Ashoka followed Brahmin religion. But after the Kalinga war he converted himself to Buddhism.

He entered into Buddhist sangha and became a **Bhikshu Gatik** (those who lived for sometime in viharas are known as Bhikshu Gatik).

The definition of Dhamma: Ashoka defined Dhamma in his 2nd and 7th pillar edicts, mentioning principles like non-violence, respecting elders/teachers, good conduct towards servants, minimalism.

In the 13th Major Rock Edict, Ashoka considered Dhammavijaya to be the greatest victory. He was the first ruler to pursue a policy of non-violent imperial expansion, known as Dhamma imperialism. Ashoka appointed Dhammamahamatras (ministers) for propagating Dhamma policy. Other officials like Yukta, Rajukka, Pradeshika were also appointed for this purpose.

Inscriptions

He built pillars and edicts all over the empire. These edicts are also present in today's Nepal, Pakistan, Afghanistan, and

Bangladesh preaching about Lord Buddha's teachings.

There are four types of Ashoka edicts :
Major rock edicts, Minor rock edicts, Major pillar edicts, Minor pillar edicts.

Most of Ashoka's pillar Inscriptions were in Brahmi script.

James Princep deciphered Ashoka's inscriptions in Brahmi and Kharosthi scripts in 1837. Ashoka's decrees were erected at public places. The Kandahar Edicts, written in Greek and Prakrit, were discovered in 1958.

Alexander Cunningham published a set of Ashokan inscriptions in the year 1877.

Allahabad pillar inscription: It is an Ashoka Stambha, constructed by Ashoka.

Mughal emperor Akbar moved the pillar from its original location and installed it within Akbar's Allahabad Fort in Uttar Pradesh.

Major Rock Edicts

There are 14 separate major rock edicts of Indian Emperor Ashoka.

Major Rock Edict I: Prohibition of animal sacrifice, especially during festive seasons.

Major Rock Edict II: It refers to southern kingdoms like Cholas, Pandyas etc.

Major Rock Edict III: Generosity towards Brahmins, appointment of officers to spread Dhamma.

Major Rock Edict V: Concern for welfare of slaves, appointing Dhamma Mahamatras.

Major Rock Edict VI: Desire to be informed about people's condition, welfare measures.

Major Rock Edict VII: Requests tolerance for all religions.

Major Rock Edict VIII: Describes Asoka's first Dhamma Yatra to Bodhgaya and Bodhi Tree.

Major Rock Edict IX: Condemns popular ceremonies.

Major Rock Edict X: Condemns the desire for fame and glory. Stresses the popularity of Dhamma.

Major Rock Edict XI: Elaborates Dhamma.

Major Rock Edict XII: Directed and determined requests for tolerance among different religious sects.

Major Rock Edict XIII: Victory over Kalinga through Dhamma, outreach to Greek kings and southern regions.

Major Rock Edict XIV: Inscribing edicts across the country.

The **Jaugada** inscription in **Dhauli hills**, Odisha contains Major Edicts 1-10, 14 and two Separate Edicts, located near Kalinga War site.

Major Rock Edict Sites: Dhauli (Odisha), Erragudi (Andhra Pradesh), Kalsi (Uttarakhand), Kandahar (Afghanistan), Mansehra (Pakistan), Girnar (Gujarat), Sopara (Maharashtra), Jaugada (Odisha), Sannati (Karnataka).

Minor Rock Edicts: Kandahar (Afghanistan), Lampaka (Afghanistan), Panguraria (Madhya Pradesh), Sohagaura (Uttar Pradesh), Sasaram (Bihar), Barabar Caves (Bihar), Mahasthan (Bangladesh), Rajula-Mandagiri, (Andhra Pradesh), Suvarna Giri (Andhra Pradesh), Brahmagiri (Karnataka), Maski (Karnataka), Nittur (Karnataka), Udegolam (Karnataka), and Palkigundu and Gavimath (Karnataka).

Brahmagiri is situated at the southern boundary of Maurya Empire.

Maski minor rock edict confirms Ashoka's titles Devanampiya and Piyadasi.

Minor Rock Edict 3 lists Buddhist scriptures for clergy to study.

Pillar

Ashoka's First pillar was found from Vaishali known as Koluha pillar.

Sarnath pillar (Eastern India) of Ashoka is the best example of art and architecture. Mauryan pillars are different from the pillars in the other parts of the world as they are rock cut pillars out of a single monolith.

They were made of sandstone. Stone pillars were erected all over the Mauryan Empire.

Motifs associated with pillars:

One Lion - It appears on the pillars of **Vaishali**, Lauriya Nandangarh and pillars.

Elephant - It was found at Sankisha.

Bull - It was found on the Rampurva pillars in Bihar.

Rampurva Bull Capital is located in Rashtrapati Bhavan.

Four Lions - On Sarnath and Sanchi pillars.

Major Pillar Edicts refers to those exclusively inscribed on the Pillars of Ashoka. Except for two, all the other major pillar edicts are found in Central India. There are seven Major pillar edicts.

Major Pillar Edict I: Ashoka's principles relating to protection of his people.

Major Pillar Edict IV: The duties of Rajukas are mentioned.

Major Pillar Edict VI - Policy of Dhamma is explained.

Major Pillar Edict VII - Tolerance towards all religious sects.

Minor pillar edicts are inscribed on the **pillars** of Ashoka. They are preceded by minor rock edicts. These edicts belong to the initial period of Ashoka's reign.

Minor pillar edicts:

The Schism edicts - Warning of punishment for dissent in Samgha.

The Queen's edict - Ashoka's queen Karuvaki mentions the donation.

Nigali Sagar pillar inscriptions - Mentions about Ashoka's dedication to increase the height of Kanakamuni Buddha.

Rummindei pillar inscriptions - Mentions about Ashok's visit to Lumbini.

Other Kings of Maurya Empire: Dasaratha, Samprati, Salisuka, Devavarman, Satadhanvan, Brihadratha. Brihadratha was the last Ruler of the Maurya Empire.

Literary Sources

Arthashastra was written by Kautilya, Chandragupta Maurya's contemporary, covers the legal and administrative framework for governing the Mauryan kingdom.

Mudrarakshasa was written during the Gupta era by Vishakhadatta, and provides insights into the socioeconomic conditions under the Mauryas. The work elaborates on Chandragupta Maurya's victory over the Nanda dynasty, guided by Kautilya.

Mahavamsa, the Ceylonese chronicle, acted as an important literary source of the Mauryan empire.

Indica was written by Megasthenes, a Greek ambassador to Chandragupta Maurya. who was sent by Seleucus to Chandragupta Maurya's court.

Stupa were burial mounds with a cylindrical drum, circular dome, and crowned by a Harmika and Chhatra. The structure is enclosed by a wall with gateways, as seen in Sanchi, Piprahwa, Bharhut, Dharmarajika, and Bodhgaya.

Caves

The Mauryan period witnessed the emergence of rock-cut cave architecture. The Barabar and Nagarjuni hills contain simple yet highly polished caves commissioned by Ashoka and Dasharatha, dedicated to the Ajivikas. The only sculptural element was a relief carving on the Lomas Rishi Cave doorway.

Society

Varna system was fully developed, work determined by varna. Entertainment: Vihara-yatra, Samaj, Pravahana (social gatherings). Senabhakta - villages supplied provisions to the royal army. Citizens' occupations based on varna.

Economy

The primary source of revenue for the state was land revenue. It generally amounted to one-sixth Part. Samaharta was in-charge of the land revenue department.

The royal share of the produce of the soil was known as Bhaga.

There were two types of lands:

First - State land or Rajkiya Bhoomi - the income derived from this land was known as Sita.

Second - Niji Bhoomi - the income derived from this land was known as Bhaga. This land belonged to the farmers. They had to pay one-fourth or one-sixth as a tax.

Rajukka did the measurement of land. Tax free villages were known as Pariharaka. Tax free land was known as Udwalik or Parihar.

There was also a concept of emergency tax known as Pranay tax.

Other Taxes:

Pindakara - It was paid in kind for a group of villages.

Prakriya - Royalty.

Hiranya - It was paid in cash.
Prasavam - Surcharge.
Parigha - Monopoly tax.
Pravesya - Import tax.
Vartani - Road cess.
Pratikara - land tax.
Sulka - Customs Duty.
Yatratvetana - Tax levied on pilgrims.

Agriculture

Kautilya mentioned three crops per year, while **Megasthenes** noted two crop types. The Mauryan economy was based on agriculture, animal husbandry, and trade, collectively known as Varta.

The state monopolized mining and metallurgy, a major source of revenue. Akradhyakshas managed mining, while Sitadhyakshas oversaw agriculture.

The lake and canal system of irrigation was known to be modern during the Maurya period. Pushyagupta during the reign of Chandragupta Maurya built **Sudarshan lake** in Junagarh (Gujarat).

Coins

In **Arthashastra** coins were called Roop.

Other types of coins were:

Nishaka or Suvarna - Gold coins.

Karshapan (Dharan or Pann) - Silver coin.

Mashak (kakini) - Copper coins.

It is mentioned that there were state minting factories, headed by Lakshanadhyaksha. Rupdarshaka was the examiner of coins. Mauryan economic year started in July.

Administration

The king was the **supreme commander** of the military, assisted by a council of ministers headed by the Mantriparishadadhyaksha, who oversaw the civil service **Adhyakshas/Amatyas**. The Ashwadhyaksha is the commanding officer of cavalry.

Kautilya in his Saptanga Theory stated that a state or a well-governed kingdom is composed of seven essential elements or limbs that contribute to its stability and prosperity. These include: Swami (The Ruler), Amatyas (The Minister), Janapada (The People and the Territory), Durga (Fortification), Kosha (The Treasury), Danda (The Army), Mitra (Alliance).

In the Arthashastra, Kautilya outlined the principles of administration and the central administrative machinery, which consisted of 18 key departments called

Tirthas.

18 Tirthas:

Mantrin - Chief Minister.

Purohita - Priest.

Senapati - Commander of the Army.

Yuvaraja - Prince.

Dauvarika - Chief of Palace Attendants.

Antarvamsika - Chief of the King's Guards.

Prasasta - Magistrate.

Samahata - Collector General.

Samnidhata - Chief Treasurer.

Pradesta - Commissioner.

Nayak - Town Guard.

Paur - Chief of the Town.

Karmanta - Superintendent of Mines.

Dandpala - Officer of the Army Department.

Durgapala - Guardian of the Forts.

Antapala - Officer-in-charge, Boundaries.

Ativanika - Officer-in-charge, Forests.

Other ministers:

Panyaadhyaksha - commerce superintendent.

Lavanadhyaksha - Salt department.

Sitadhyaksha - Agriculture minister.

Suvarnaadhyaksha - gold superintendent.

Bandhanagaradhyaksha - looked after the Jail.

Paura Janapada was the assembly of the people of the town and the country.

District Administration (Aahar or Vishay): Each was administered by Pradeshika, Yukta and Rajukka officers.

Pradeshika was a senior.

Rajukka was subordinate.

Yukta was subordinate to both of them. It was the duty of **Pradeshika** to tour the kingdom every five years.

Judicial Administration

The king was the fountainhead of justice. There were two classes of courts: Dharmasthiya (diwani) were civil courts which decided cases relating to contract, gift, agreement, sales, marriages. Kantakshodhan (faujdari) courts were criminal courts which dealt with the cases of theft, robbery, murder, sex offenses.

Decline

The last Mauryan king Brihadratha was

assassinated by his commander-in-chief Pushyamitra Shunga in 185 BC. Pushyamitra Shunga established the Shunga dynasty.

Sunga Dynasty

The Sunga Dynasty ruled eastern India from 185 to 73 BCE, succeeding the Mauryas in Magadha. They were a Brahmin clan from the Bharadvaja clan.

The Sunga Empire extended from the Ganges to the Narmada Valley and parts of northern India. The capital shifted from Pataliputra to Vidisha.

The founder was Pushyamitra Sunga, who had been a Mauryan viceroy in Ujjain. Major Sunga cities included Ujjain, Saket, Sanchi, Mathura, and Kapilavastu. Information about the Sungas is found in texts like the Gargi Samhita, Mahabhasya, Divyavadana, and works by Kalidasa and Banabhatta.

Rulers

Pushyamitra Sunga: He was the founder of the Sunga Dynasty. A Brahmin and army chief under the last Mauryan king, he overthrew Brihadratha and established the Sunga Empire in 180 BCE with the capital at Pataliputra.

Pushyamitra fought off attacks from Greek kings Menander and Demetrius, and defeated the Kalinga king Kharavela. He also conquered the Vidarbha region.

Divyavandana's book, Ashokavadana, describes him as the destroyer of Buddhist monasteries and their places of worship. He was a follower of Buddhism. While some sources portray him as anti-Buddhist, he renovated Buddhist sites like Sanchi and Bharhut, and performed Vedic rituals. The Sanskrit scholar Patanjali was patronized by Pushyamitra, whose reign lasted around 36 years.

Agnimitra: He was the son of Pushyamitra Sunga. He ruled the Sunga Empire from 149 BC to 141 BC. By the time he succeeded to the throne, the Vidarbha region broke away from the Sunga empire and became independent.

Kalidasa's play 'Malvikagnimitra' is a love story of the King Agnimitra and a handmaiden Malvika. After him, his son Sujyesta and his son Vasumitra succeeded to the throne.

Later Rulers

The details of Vasumitra successors are not known. Historians came up with different names such as Andhraka, Vajramitra, Ghosha, and Pulindaka. Heliodorus was the Greek ambassador and stayed at the court of King Bhagavat. Bhagavat was identified with Kasipurra Bhagabhadra of Vidisha pillar inscription.

Society

The Sunga dynasty revived Brahmanism, Bhagavatism, and the caste system as evident from Manusmriti reaffirming the position of Brahmins in the fourfold society. A major development was the emergence of mixed castes and integration of foreigners into Indian society.

Sanskrit gained prominence as the court language during this period. The Sungas issued gold and silver coins, and continued with the Mauryan administrative structure.

Decline

Devabhuti was the last known Sunga King. As per Bhanabhatta, he was killed by his minister (Vasudeva Kanva). Vasudva established the Kanva dynasty which ruled in eastern and central India.

Kanva dynasty

The Kanva dynasty ruled from 73 BC to 28 BC, with its capital at **Pataliputra** (former Mauryan capital). It was founded by Vasudeva Kanva, who murdered **Devabhuti** of the Sunga dynasty in **73 BCE**. The Kanvas were descendants of Rishi Kanva and described as 'servants of the Sungas'.

Under the Kanvas, Magadha lost territories in the west to Greeks and in the east to the Kosala Kingdom. The Kanva rulers allowed the remnants of the Sunga dynasty to continue ruling obscurely. They paved the way for the Mitra dynasty, which was eventually overthrown by the Shakas in Magadha.

Prominent Rulers

Vasudeva Kanva (73 - 64 BCE): He was the founder of the Kanva dynasty. He was originally an Amatya (minister) of the last Shunga ruler Devabhuti. Bana's Harshacharita informs us that he came to power after the death of Devabhuti. He was succeeded by his son **Bhumimitra**.

Bhumimitra (64 - 50 BCE): He ruled for fourteen years and was later succeeded by his son Narayana. Coins bearing the legend Bhumimitra have been discovered from Panchala realm. Copper coins with the legend "Kanvasya" have also been found from Vidisha, as well as Kaushambi in the Vatsa realm.

Narayana (50 - 38 BCE): He ruled for twelve years. He was succeeded by his son Susharman.

Susarman (38 - 28 BCE): He was the last king of the Kanva dynasty. He was overthrown by a successor of Simuka of the Satavahana dynasty.

Decline

According to the **Puranas**, the Kanva dynasty was brought to an end by the **Satavahanas** in **28 BCE**.

Chedi Dynasty

After the Mauryan Period, the Cheti or Chedi Dynasty emerged in Kalinga in the 1st century. The other name of the dynasty was Mahameghavahana. Chedis are referred to as Cheta or Chetavamsa, and Mahameghavahana. Suktimati was the Capital of Chedi.

The founder of the Chedi Dynasty was **Mahameghavahana**. The most notable ruler of the dynasty was King Kharavela (third Cheti king). Kharavela was a Jain follower who donated caves to Jain Monks in Udaigiri Hills.

Art and Architecture

Udayagiri and **Khandagiri** caves were built by King Kharavela. It is located in Bhubaneswar in Odisha. Udayagiri (Sunrise Hill) has 18 caves while Khandagiri (Broken Hill) has 15 caves.

The **Hathigumpha cave** has the Hathigumpha Inscription written by Raja Kharavela (king of Kalinga). It is a stone inscription and the text is in Brahmi script and has 7-8 lines. It gives us information about the dynasty and its rulers. It refers to the conquest of Kalinga with Nandas.

Decline

Samudragupta defeated Mahendra of **Kosala** (Cheti King). As a result During the 4th century AD, it became part of the Gupta empire.

Satavahana Dynasty

The Satavahanas or Andharas founded their kingdom in the 1st century BCE after the Mauryan decline. Ruling for around 450 years, it spanned modern Andhra Pradesh, Telangana and Maharashtra. Their capitals were Pratishthana and Amaravati. The **Nasik and Naneghat** inscriptions shed light on the reign of Gautamiputra Satakarni. Simuka was the dynasty's founder.

Important Rulers

Simuka founded the Satavahana dynasty and destroyed the Shunga power with the help of Rathikas and Bhojakas. He was beheaded by his brother Kanha who succeeded him.

Kanha was the Satavahana Dynasty's second ruler. He expanded the empire southward. Satakarni-I (Simuka's son), succeeded him.

Satakarni I: He was also known as Sri Satakarni. He was a great ruler among the Early Satavahanas and the son of Simuka. His queen's name was Naganika, and he was known as the Lord of Dakshinpatha. He performed the Ashvamedha and Rajsuya Yajnas.

Hala: He was the 17th King of the Satavahana line and another great Satavahana king. He had compiled the "Gatha saptasati" or Gaha Sattasai, which was primarily a love text. He is also mentioned in the Lilavati text.

Gautamiputra Satakarni: He was the greatest king of the dynasty in 2nd century CE. He claimed to be Brahmana, who had dethroned Shakas and annihilated many rulers of Kshatriya.

He was known to destroy the **Kshaharata heritage** of Nahapana. He is known as the destroyer of Yavanas, Pahlavas, and Shakas. The restorer of the Satavahanas' glory in Nasik inscription of his mother Gautami Balashri. He was known as Ekabrahmana, and Khatiya Dapa Manamada. He was given the effluent titles of Maharaja and Rajaraja. He donated his land to Buddhism.

The Karle inscription mentions Karajika village near Pune. Junagadh inscription suggests Gautamiputra lost some Kshaharata provinces to the Kardamaka against the Shaka Kshatrapas. His son Vasisthputra Sri Pulumayi succeeded

him.

Gautamiputra's empire spanned Eastern and Western Malwa, Narmada Valley, Vidarbha, Rajputana, Saurashtra, and Kalinga, as described in the **Nasik Prasasti**. His reign extended to Kanchi in the south.

Vashishthiputra Pulumayi succeeded Gautamiputra Satakarni and expanded the Satavahana Empire. Yagnashri Satakarni, another ruler, issued trade-focused ship coins. **Inscriptions in Junagadh** show Vashishthiputra Pulumayi married Rudradaman I's daughter.

Yajna Sri Satakarni, brother of Vashishtiputra Satakarni, reclaimed Satavahana territories lost to the Western Satraps, defeating them and retaking their southern lands. But after his reign, the Satavahana dynasty started declining, while the Western Satraps prospered for two more centuries.

Sources

The Jain sources mention Satavahana as the first king of the dynasty. The Kathasaritsagara also contains a story about him. The Kondapur coins bear the legend 'Satavahana', likely referring to the first ruler Simuka, as evident from the **Kotilingala coins** and a Nanaghat inscription mentioning "Simuka Satavahana".

Political and Social structure

The Satavahana polity was decentralized, with trade as a critical component of their economy. They were the first to grant royal lands to Buddhism and Brahmanism, and saw the writing of a **Yajnavalkya Smriti** commentary.

The **Satavahanas** were the earliest Indian rulers to issue coins with portraits, starting with Gautamiputra Satakarni. Their architectural legacy includes the Amaravati Stupa and numerous other stupas, while their **Ajanta paintings** are the earliest surviving Indian art.

Decline of Satavahanas

Pulamavi IV was the last Satavahana king, reigning until 225 AD. After his death, the empire fragmented into five smaller kingdoms, with the Ikshvakus replacing the Satavahanas in the east. The west was ruled by the Abhiras, the southwest by the Chutus, the southeast by the Pallavas, and the northern

provinces by a Satavahana collateral branch.

Indo Greeks

Sakas

The Saka Empire (Indo-Scythians) was probably established by King Maeus around the 2nd century A.D. They were the group of Iranian nomadic people who invaded India by destroying the Indo-Greeks in North-West India. They invaded Parthia and Bactria and they defeated the Parthian King.

They were divided into 5 distinctive branches: Afghanistan, Punjab (Taxila as capital), Mathura, Western India, Upper Deccan (Ujjain as capital).

Maeus (King Moga) established the Indo-Scythian rule in India, ruling over Gandhara with his capital at Sirkap. His son Azes I later conquered the remaining Indo-Greek territories by defeating Hippostratos.

Azilises and Azes II: He was the last great ruler of the Saka dynasty.

Nahapana: He was defeated by Gautamiputra Satakarni, ruler of Satavahana Dynasty.

Chashtana was a Saka leader of the Western Kshatrapas (Satraps) line who controlled Ujjain. Ptolemy mentions him as "Tiasthenes" or "Testenes".

Rudradaman I, grandson of Shaka king Nahapana, became the 'Mahakshatrapa' and issued the first long Sanskrit inscription. He converted to Hinduism and oversaw the translation of the Yavanajataka from Greek to Sanskrit. He repaired Sudarshan Lake.

Decline of Shaka

The Saka Empire started declining after their defeat at the hands of the Satavahana Emperor Gautamiputra Satakarni. Their rule in northwest India and Pakistan came to an end after the death of Azes II when the region came under the Kushanas.

Parthians

The Parthian (Arsacid) Empire was founded by Arsaces I, rebelling against the Seleucids. It extended from Turkey to eastern Iran. The Parthians were later succeeded by the Gondophares, a Scythian (Saka) king, in the late 1st

century BC.

The Parthians' **first capital was Taxila**, later shifting between Kabul and Peshawar. After Gondophares I, the empire fragmented. The Parthian Empire was a major power in **Iran** from 247 BC to 224 AD. Parthia was a region captured by the Medes in the 7th century BC, stretching from the Mediterranean to China.

Kushanas

The **Kushanas** were a branch of the **Yuezhi tribe** in Central Asia. Their movement into India dates to the 1st century CE. The first Kushan king was Kujula Kadphises.

Menander: A Greek king who ruled over the Indo-Greek kingdom. He made Sialkot as the capital of his kingdom. He has been identified with King Milinda mentioned in the famous Buddhist text Milindapanha.

Kushana Dynasty

The Kushan Empire flourished from the 1st to 3rd centuries CE, spanning from Tajikistan to northern India. Mathura became their second capital around 2000 years ago.

The Kushans were a branch of the Yuezhi confederation, unifying much of Central Asia. Many rulers took the title "devaputra" or "son of god".

The first Kushan ruler was Kujula Kadphises, who issued India's first gold coins.

Rulers

Kujula Kadphises: He laid the basis for the Kushan Empire which was rapidly expanded by his **descendants**. He adopted the title 'Great King, King of Kings'.

Vima Taktu: He expanded the Kushan Empire into the northwest of the Indian subcontinent.

Vima Kadphises: The third Kushan ruler added to their territory through conquests in Afghanistan and northwest India. He introduced the first gold coinage in India.

Kanishka I (78-144 CE) was the successor of Vima Kadphises. He expanded the Kushan Empire from Uzbekistan and Tajikistan to Mathura and

Kashmir, with his capital at Purushapura (Peshawar). Kanishka founded the Shaka Era in 78 AD and was a patron of Buddhism, convening the 4th Buddhist council in Kundalvana (Kashmir) in 78 AD.

Scholars like Asvaghosa, Nagarjuna, and Charaka resided at his court. Kanishka defeated the Chinese Han Emperor Ho-ti and is known as the "Second Ashoka."

Vasishka: He had been a Kushan emperor, who had a short reign following Kanishka. His rule extended as far south as Sanchi.

Huvishka: His rule was a period of retrenchment and consolidation for the Empire. Huvishka was the successor of Kanishka.

Vasudeva I: He ruled as the last of the "Great Kushans." The end of his rule coincided with the invasion of the Sassanids as far as northwestern India.

Administration

The Empire extended from Central Asia to Bihar and from Kashmir to Sindh. It was an organized administrative system, probably in three tiers, at central, provincial and local levels.

They seem to have followed the earlier existing pattern of the Indo-Greeks and Parthians by appointing kshatrapas and mahakshatrapas for different units of the empire.

Other inscriptions mention other officials performing both civil and military functions, called '**dandanayaka**' and '**mahadandanayaka**', indicating prevalent feudal elements.

The inscriptions mention two terms – '**gramika**' and '**padrapala**', both signifying '**village headman**'. He collected the king's dues and took cognizance of crimes in his area. The **Kushan rulers** accepted the prevalent Indian and Chinese concept of the **divinity of kingship**.

They borrowed the Achaemenid and subsequently Indo - Greek and Indo-Parthian system of appointing satraps as provincial governors, while the feudal lord (dandanayaka) was their own creation.

Inscriptions

The Kushan inscriptions were in Bactrian (Greek script) and Prakrit

(Brahmi/Kharosthi). The **Rabatak Inscription** established Kanishka's lineage, naming his ancestors Kujula Kadphises, Vima Taktu, and Vima Kadphises.

Coins

Vima Kadphises introduced India's first gold coins, which followed the Greco-Bactrian style of a deity on one side and the king on the other.

Religion

A fusion of **Hellenistic** and Buddhist cultural elements expanded into central and northern Asia as Mahayana Buddhism. He also had the original Gandhari vernacular (Prakrit) Buddhist texts translated into the language of Sanskrit.

Decline

After **Vasudeva I**, the Kushan Empire split into western and eastern halves. The Sassanid Persians conquered the Western Kushans in Afghanistan, deposing them and installing Persian vassals known as Kushanshas or Indo-Sassanids. The White Huns in the 5th century and the spread of Islam later eliminated the remnants of the Kushan Empire.

Gupta Dynasty

The Gupta Rule (320 - 550 CE) for witnessing all round progress in spheres of art, architecture, literature, science and technology, metallurgy and philosophy, regarded as the **Golden Age of Ancient India** or Golden Period of Indian History. The Gupta dynasty was founded by King Shrigupta.

Kalidasa credited the Guptas with conquering around 21 kingdoms within and outside India, including the Parasikas, Hunas, Kambojas, Oxus tribes, Kinnaras, Kiratas.

Rulers

Sri Gupta I: He was the founder of the Gupta dynasty of northern India in late 3rd century CE and was succeeded by his son, Ghatotkacha.

Chandragupta I (319 CE - 334 CE): The first famous ruler was Chandragupta-I who is considered the actual founder of the dynasty. He assumed the title of

Maharajadhiraj (King of Kings).

Chandragupta-I married the **Licchavi princess Kumardevi** and increased his influence in the North Bihar region (Nepal). He got Pataliputra in dowry from the Licchavis. Gold Coins bearing the figures of Kumardevi and Chandragupta - I known as Kumardevi Coins were issued.

SamudraGupta (335 - 375 CE): He was the successor to Chandragupta I. Harisena also wrote the book 'Vrihat Katha Kosha'.

He issued **Ashwamedh coins**, Tiger slayer coin, Battle axe coin and Veena Coin in which he was displayed playing the Veena. He was not only a conqueror but also a great poet, musician and patron of learning. Vincent Arthur Smith gave the title as Napoleon of India to him. He assumed imperial titles like **Maharajadhiraja** and **Kaviraja**.

Ramagupta: He was the elder son of Samudragupta. He succeeded Samudragupta but Chandragupta II killed him and married his queen Dhruvadevi.

ChandraGupta II (375 CE - 414 CE): The peak of the territorial expansion of the Gupta empire reached its heights during the reign of Chandragupta-II, the son of Samudragupta. Prabhavati Gupta was the daughter of Chandragupta II, the ruler of the Gupta empire, and queen Kuberanaga. She was the Vakataka queen who was the consort of Maharaja Rudrasena II. She ruled the Vakataka kingdom.

The most important military achievement of Chandragupta-II (Vikramaditya) was his war against the **Sakas** of western India (Gujarat, Kathiawar, and west Malwa).

He appointed nine jewels (Navratna) named Amarasimha (Amar Singh), Dhanvantari, Ghatakarapara, Kalidasa, Kshapanaka, Shanku, Varahamihira, Varuchi, Vetala- Bhatta. Vetala Bhatta wrote 'Mantrashastra'.

He was the first Gupta ruler to issue silver coins bearing lion figure similar in pattern with Saka coins.

Vikramaditya began the **Vikrama Samvat era** in 57 BCE after defeating the Shakas, and those who believe that he is based on a historical figure place him around the first century BCE. However, this era is

identified as Vikrama Samvat after the ninth century CE.

The Chinese pilgrim Fa Hien visited India during the reign of Chandragupta II, suggesting that he ruled over a peaceful and prosperous kingdom. He began his journey back home from Bengal by boarding a ship belonging to some merchants. He traveled from China to India on foot to acquire Buddhist texts. He recorded his observations in a travelogue titled Fo-Kwo-Ki (a Record of Buddhistic Kingdoms).

Kumar Gupta I (415 - 455 CE): He was the son of Chandragupta II and queen Dhruvadevi. He maintained control over inherited territory spanning from Gujarat to the Bengal region. Titles: Maharajadhiraja, Parama-Bhattaraka, Paramadvaita, and Mahendraditya. Coins bear variations of Mahendra, such as Shri-Mahendra and Mahendra-simha. His garuda coins are found in western India, peacock coins in the Ganges valley. Founded Nalanda Mahavihara (Nalanda University) in the 5th century CE.

Skanda Gupta (455 - 467 CE): He was the son of the Gupta emperor Kumaragupta I. He issued five types of gold coins: Archer, King and queen, Chhatra type, Lion - slayer, and Horseman type. His silver coins are of four types: Garuda, Bull, Altar, and Madhyadesha type.

Rulers Succeeded Skandagupta : Budhagupta, Vainyagupta, Bhanagupta, Narasimhagupta, Baladitya, Kumaragupta-II and Vishnugupta. They ruled over northern India from 467-550 CE.

Vishnugupta (540 - 550 CE): He was one of the lesser-known kings of the Gupta Dynasty. He was generally considered to be the last recognized king of the Gupta Empire. From the fragment of his clay sealing discovered at Nalanda during the excavations of 1927-28, it was revealed that he was the son of Kumaragupta III.

Inscription Related to Gupta Dynasty

Allahabad Pillar Inscription (Prayaga Prashasti inscription) : It was originally engraved on the Asokan pillar at Kaushambi near Allahabad, later moved to Allahabad Fort. Composed by poet and minister Harisena in the court of King Samudragupta, it describes the king

as equal to gods like Dhanada, Varuna, and Indra. Samudragupta conquered eight northern Indian kings, followed by performing the Ashwamedh Yajna.

Mehrauli iron pillar inscription: Near the Qutub Minar in Delhi, is associated with Chandragupta II. It stands in the Quwwat ul Islam mosque. The pillar, over 1600 years old, remains uncorroded. Chandragupta II fought against Bengal's enemies and defeated the Valhikas in Bactria after crossing the Sindhu region.

Junagadh Rock or Girnar Rock inscription: Near Girnar hill in Junagadh, Gujarat, was inscribed by Western Satraps ruler Rudradaman I. It features inscriptions of Ashoka (I-XIV Major Rock Edicts), Rudradaman I, and Skandagupta. The first eight lines record water management at Sudarshana Lake from Chandragupta Maurya's time to around 150 CE, with the last twelve lines praising Rudradaman I.

The Bhitari Pillar Inscription of Skandagupta, found in Village Bhitari, Ghazipur, Uttar Pradesh, is written in Sanskrit and stands 15 feet tall. It's crucial for understanding Gupta rulers' chronology and details conflicts with the Pushyamitras and Hunas.

Administration

The kingdom was divided into administrative units. Bhukti (province) and Vishaya (province division) were administrative units headed by Uparika and Vishayapati respectively. Vishaya was further divided into Vithi and village was the smallest unit.

Important Offices: Mahanandanayaka - Justice Delivery, Mahapratihara - Chief of the Guards, Dutakas - Associated with Gifts and Grants, Sandhi - Vighraha - Minister of Peace and War, Pilupati - Headed Elephants, Asvapati - Headed Horses, Narapati - Headed by Foot Soldier, Ranabhandagrika - In charge of the store, Akshapataladhiakrita - Superintendent of Records and Accounts.

Economy

Agriculture: During the Gupta period, agraharas, tax-free villages granted to Brahmanas through copper plate charters, proliferated. Rural society included mahattaras (village elders), kutumbins (wealthy peasants), and karshakas (ordinary peasants).

Types of Land: The Amarakosha refers to at least 12 types of land: urvara, ushara, maru, aprahata (field which has not been ploughed), sadvala (grassy land), pankila, jalaprayanupam (land situated near a waterbody), kachchha (marshy land), sarkara, sarkati, nadimatrika (river - fed) and devamatrika (rain - fed). The land that did not yield revenue was known as Apratikara (Utpatikara or Sunyapratikara). Atavika Rajya referred to the Forest Kingdom. Aranya (forest land) has been mentioned in the Damodarpur copper plate of Gupta era.

Trade and Commerce: The transactions related to business have been mentioned in the Amarakosha as krayavikraya. Kalidasa refers to various types of vipani (shops) and also to apanamarga (a path with shops on both sides). The Amarakosha contains abundant references to sreshthi (wealthy merchant) and sarthavaha (caravan trader).

Coinage: The Gupta gold coins were known as Dinara introduced by Skandagupta. The silver coins were introduced by Chandragupta II who conquered western India by ousting the Saka Kshatrapas. They also issued copper coins.

Taxation : - Bhaga (one-sixth of produce), Uparikara (additional tax), Halvakara (plough tax), Hiranya (cash tax), Kara (irregular tax), Shulka (custom or toll tax), Udinanga (social security tax), and Klipta (land transaction tax).

Society and Religion

During this period, society became feudalistic with Brahmins and feudal chiefs receiving land grants. Women and Shudras experienced improved status. Crime was scarce, and death penalty rare, according to Fahien. Vishtis served in the royal army. Vaishnavite or Shaivite kings constructed temples. Religious texts like Puranas, Mahabharata, and Ramayana expanded. The first inscriptional example of Sati emerged in this period.

Art and Architecture

The period's finest painting, sculpture, and architecture are showcased in **Ajanta, Ellora, Sarnath, Mathura, Anuradhapura, and Sigiriya**. Additionally, vocal music, dance, and musical

instruments like the veena, flute, and mridangam were practiced.

Literature

Literature flourished in the Gupta period, encompassing poetry, drama, philosophy, religion, science, mathematics, physiology, and astronomy. Key texts included Dharmashastras, Narad Smriti, Vishnu Smriti, portions of Ramayana and Mahabharata, as well as Bhasa's 12 plays.

Decline

Important Causes for the Fall of the Gupta Dynasty: The Pushyamitras dealt a significant blow during Skandagupta's rule. Later, the Huns, led by Toramana and Mihiragula, further weakened the empire. Successors lacked strength. Trade decline with Rome due to Hun attacks on the Roman Empire also played a role.

Contribution

It was considered as one of the most glorious chapters in the history of ancient India. It witnessed changes in the social structure, religion, learning and education, architecture, art, painting, and also developments in pure sciences. The Sanskrit language became prominent during the Gupta period. Dev nagari script had evolved from the Brahmi script. Numerous works in classical Sanskrit came to be written in the forms of epic, lyrics, drama, prose and poetry. The **Nalanda University** founded by Kumaragupta I became the most celebrated Buddhist educational centre in North India.

Successors

Maitrakas: They were tributary chiefs of the Guptas, who later established an independent kingdom in western India, ruling over Saurashtra in Gujarat. Dhruvasena II was their most prominent ruler, married to Harshavardhana's daughter. Valabhi, their capital, thrived as a port town for trade. Their rule lasted until the eighth century, weakened by Arab attacks.

Maukharis: They ruled Kannauj, gradually becoming a political center in North India. They were Gupta subordinates, titled Samantas. Grihavarman, married to Harshavardhana's sister, faced attacks

from Shashanka of Bengal and Devgupta, leading to Kannauj's merger with the Pushyabhutis.

Pushyabhutis rose to prominence post-Gupta, with Thanesar as their capital in Kurukshetra. Prabhakarvardhana's accession marked their influential era.

Vakatakas: The founder of the dynasty was Vindhyashakti. They ruled over Central - Western India.

Chalukyas: They set up their kingdom in western Deccan with the capital at **Vatapi**.

The kingdom rapidly rose to prominence during the reign of **Pulakesin II (AD 610 - 642)**.

Pallavas: The Pallavas gained power under Mahendravarman and Narasimhavarman I, later replaced by the Cholas in South India. Their era was culturally significant for the growth of Tamil bhakti literature and Dravidian art and architecture.

Vakataka Dynasty

It ruled parts of Central and South India between the 3rd and 5th centuries. He succeeded the Satavahana Dynasty in the Deccan region. They were believed to be the Brahmins belonging to Vishnu Vriddha Gotra. The dynasty was founded by the ruler Vindhyashakti around 250 CE. They were the contemporaries of the Guptas in northern India. In the Puranas, the Vakatakas are referred to as the Vindhyakas.

Rulers

Vindhyashakti : He was the founder of the Vakataka Dynasty. He was a Brahmin. He was succeeded by his son Pravarasena.

Pravarasena I : He was the real founder of the Vakataka dynasty.

His titles: Samrat, Dharmamaharaja, Haritiputra. His empire extended in the parts of northern India and the Deccan. He conducted battles with the Nagas. He had four children according to the Puranas.

After his demise, there were two divisions of the Vakatakas: Pravarapura - Nandivardhana branch and the Vatsagulma branch.

He was succeeded by Rudrasena I.

Pravarapura-Nandivardhana Branch : It ruled from various sites like Pravarapura (Maharashtra), Mansar, and Nandivardhan in the present Nagpur locale.

Sarvasena : He was the child of Pravarasena I. He created Harivijaya in Prakrit. He adopted the title "Dharma-Maharaja".

Rudrasena I : He was Grandson of Pravarasena I. He was preceded by his child Prithvishena. He was the founder of the Nandivardhana branch of the Vakataka kingdom. He was a worshiper of Lord Mahabhairava, the fierce form of Shiva.

Rudrasena II : He was a child of Prithvisena I. He got married to Prabhavatigupta, the daughter of Chandragupta II.

Pravarasena II : He was named Damodarasena. He was son of Rudrasena II. He established the city of Pravarapura. He had a matrimonial alliance with the contemporary Kadambas (near Mysore).

Vatsagulma Branch : It controlled the area between the Sahyadri Range and the Godavari River with its capital at Vatsagulma.

Harishena : He was the Fifth-generation descendant of Sarvasena. He disparaged Buddhist craftsmanship and engineering.

Vardhana Dynasty or Pushyabhuti Dynasty

The Gupta Empire's downfall led to the emergence of several warrior kingdoms in North India, except in areas subdued by the Huns. The Pushyabhuti dynasty was founded at Sthaneshwar (modern Thanesar) in the early 6th century, having been feudatories of the Guptas but gaining independence after the Hun invasions.

The first important Pushyabhuti ruler was **Prabhakara Vardhana**, who held the titles of Maharajadhiraja and Parama Bhataraka. He was succeeded by his son Rajyavardhana, who was killed by Shashanka in 606 AD. Other notable Pushyabhuti rulers included Adityavardhan and Harsha. The Pushyabhuti state rose to prominence under the reign of Harsha.

Harshavardhana (606 CE – 647 CE)

Harshavardhana was an influential 7th century Indian ruler. He was born in 590 AD, belonged to the Vardhana dynasty of Sthaneshwar (Thanesar).

He formed an alliance with the Kamarupa king Bhaskaravarman to confront the **Gauda king Shashanka**, who eventually fled to Bengal. This allowed Harshavardhana to become the ruler of Kannauj.

He established a vast empire stretching from northern and northwestern India to the Narmada River in the south, with Kannauj as his capital. He was regarded as one of the most important Indian monarchs of the 7th century.

Two primary sources of Pushyabhuti Kingdom: The Harshacharita (The Deeds of Harsha) - a prose biography written on Harshavardhana in Sanskrit by Banabhatta. The chronicle of the Chinese traveller Hsuan Tsang.

Reign

Under Harshavardhana, there were two types of territories - Those directly under his control, and feudatory states that he commanded. This marked the rise of feudalism in India during his reign.

In 641 AD, Harshavardhana sent an envoy to the Tang Emperor of China, and received Chinese embassies in return. The famous Chinese pilgrim Xuanzang visited India during this period and praised Harshavardhana's generosity and justice.

His court poet was Banabhatta who wrote the Harshacharita, which describes his life and actions. He reserved one-fourth of the income to the development of Nalanda University. He had an excellent tax structure. A quarter of the taxes collected were spent for charitable and cultural causes. He was a capable military conqueror and administrator. Before the Muslim invasions, he was the last ruler to preside over a wide realm in India.

Administration

There was no slavery in his empire and people were free to lead their life according to their wish.

Important officers of Harsha's

administration: Sreshti - Chief Banker or Merchant. Sarthavaha - Leader of Merchant Caravans. Prathama kulika - Chief Craftsman. Kayasthas - head of the scribes.

According to Hsuan Tsang, people were lightly taxed, and the monarch received one-sixth of the farmer's yield as his grain portion. Inscriptions describe dues such as bhaga, bhoga, kara, and hiranya. He describes the army as composed of infantry, cavalry, chariots, and elephants.

Banskhera and Madhuban inscriptions mention the king's victorious camp, which includes boats, elephants, and horses. During his reign, trade and commerce had decreased. It is evidenced by the loss of commercial centres, a decrease in the quantity of coinage, and the sluggish activity of merchant guilds.

Military Conquests

His most significant military expedition was against the Western Chalukya ruler Pulakesin II in 618 CE. Pulakesin defeated Harsha on the banks of the Narmada River, earning the title of Paramesvara. He conquered Bengal, Bihar, and Odisha after the death of Sasanka. He also vanquished the Vallabhi king in Gujarat, leading to a marriage alliance between Harsha's daughter and the Vallabhi king Dhruvabhatta.

Kanauj was eventually taken over by the Pushyabhutis, while Harshavardhana maintained control over Punjab and central India.

Nepal had accepted his rule. He took control of Kashmir, and the ruler paid tribute to him. He also maintained friendly connections with Assam's monarch, Bhaskaravarman.

Harsha cemented his dominance over all of north India. He had direct influence over contemporary Rajasthan, Punjab, Uttar Pradesh, Bihar, and Orissa, as well as outlying states such as Kashmir, Sind, Valabhi, and Kamarupa.

Society

The society was divided into four classes: Brahmin, Kshatriya, Vysya, and Sudra. Women's positions were unsatisfactory. Swyamvara (the institution of choosing her spouse) has decreased. Widow remarriage was

forbidden, especially among the upper classes. Sati was another popular ritual.

Religion

According to inscriptions, the early Pushyabhuti monarchs were Surya worshippers. Harshavardhana appears to have been a Shiva devotee who was also interested in Buddhism. He called a large gathering at Kannauj, where Xuanzang and others delivered lectures on Mahayana principles. Rajyavardhana was a Buddhist devotee.

Art

King Harshavardhana is credited with authoring three dramas: Ratnavali, Priyadarshika and Nagananda.

The Nagananda tells the story of the bodhisattva Jimutavahana, while Ratnavali and Priyadarshika are love comedies.

Harshavardhan himself wrote the inscriptions of Madhuban and Banskhedha.

Decline of Pushyabhuti dynasty

After Harshavardhana's death in 647 AD, the Pushyabhuti dynasty declined, ending their rule and ushering in Muslim influence in North India. Political instability persisted until the 8th century, when the Gurjara Pratiharas and Rajput rulers became major powers. Around 75 years after Harshavardhana's death, Yashovarman rose to power in Kannauj.

Sangam Age

The Sangam Period spanned the 3rd century BC to 3rd century AD in South India, south of the Krishna and Tungabhadra rivers. Named after the Sangam poet assemblies in Madurai under Pandya kings' patronage, it saw the flourishing of early Dravidian literature. Tamil legends speak of three Sangams/Muchchangams: The First at Madurai attended by gods and sages; the Second at Kapadapuram from which only Tolkappiyam (Tamil grammar) survives; the Third again in Madurai producing a few surviving Tamil literary works - the earliest Dravidian specimens.

Sangam Literature

It includes Tolkappiyam, Ettutogai, Pattuppattu, Pathinenkilkanakku. He had

two epics - Silappathikaram, Manimegalai.

Tolkappiyam by Tolkappiyar is considered the earliest Tamil literary work, providing insights into the political and socio-economic conditions despite being a grammar text.

Ettutogai (Eight Anthologies) consist of **eight works** - Aingurunooru, Narrinai, Aganaooru, Purananooru, Kuruntogai, Kalittogai, Paripadal and Padirrupattu.

Pattuppattu (Ten Idylls) consists of **ten works** - Thirumurugarrupada, Porunararrupada, Sirupanarrupada, Perumpanarrupada, Mullaipattu, Nedunalvada, Maduraikkanji, Kurinjippattu, Pattinappalai and Malaipadukadam.

Pathinenkilkanakku contains 18 ethical works, most notably Tirukkural by the renowned poet-philosopher Thiruvalluvar. **The two epic works** Silappathikaram (around 1800 years ago) by Elango Adigal and Manimekalai by Sittalai Sattanar provide valuable details about Sangam society and polity.

Other Sources

Greek authors like Megasthenes, Strabo, Pliny and Ptolemy mention about commercial trade contacts between the West and South India. Ashokan inscriptions mentioned the Chera, Chola and Pandya rulers to the south of the Mauryan empire. Hathigumpha inscription of Kharavela of Kalinga also has mention of Tamil kingdoms.

Political History

South India during the Sangam Age was ruled by three dynasties - Cheras, Cholas, and Pandyas.

Cheras: It controlled the central and northern parts of Kerala and the Kongu region of Tamil Nadu. Vanji was their capital. The ports of the west coast (Musiri and Tondi) were under their control. The emblem of Cheras was "bow and arrow".

The greatest ruler of Cheras was Senguttuvan. His military achievements have been chronicled in epic Silappathikaram, with details about his expedition to the Himalayas where he defeated many north Indian rulers.

Senguttuvan introduced the Pattini cult.

He was the first to send an embassy to China from South India.

The Cholas ruled central and northern Tamil Nadu from their capital Uraiyur, with the Kaveri delta (later Chola-mandalam) as their core region. The renowned Sangam Chola king Karikala, whose life is depicted in Pattinappalai, defeated a confederacy of Cheras, Pandyas and 11 chieftains at the famous Battle of Venni mentioned in many Sangam poems.

Pandyas: It ruled from their capital Madurai. Nedunjeliyan II was a famous early ruler. Korkai was their main port known for pearl fishery and chank diving. Their emblem was the fish. Sati, caste, idol worship were prevalent practices.

After a decline due to the Kalabhra invasion, King Kadungon took the title 'Pandyadhiraja' and ended Kalabhra rule, ushering a new era.

Maravarman Rajasimha III was the last Pandya king who fought the Cholas and Cheras. Post Sangam age, the Pandyas lost significance for over a century before rising again in the late 6th century under a revived dynasty.

Polity and Administration

During the Sangam period hereditary monarchy was the form of government. The king was assisted by a wide body of officials who were categorized into five councils: ministers (amaichar), priests (anthanar), envoys (thuthar), military commanders (senapathi), and spies (orral).

The chief source of state's income was land revenue while a custom duty was also imposed on foreign trade. Major source of fulfilling the royal treasury was the booty captured in wars. The roads and highways were maintained and guarded to prevent robbery and smuggling.

Society

Tolkappiyam refers to the Five-fold division of lands - Kurinji (hilly tracks), Mullai (pastoral), Marudam (agricultural), Neydal (coastal), and Palai (desert). It also refers to four castes namely arasar (Ruling Class), anthanar, vanigar (carried on trade and commerce), and vellalar (Agriculturists). Ancient primitive tribes like Thodas, Irulas, Nagas and Vedars lived in this period. In Sangam poems,

Muvendar means three chiefs. The social classification of Varna was known to Sangam poets.

Decline

It declined towards the end of the 3rd century A.D. The Kalabhras occupied the Tamil country post-sangam period between 300 AD to 600 AD, whose period was called 'dark age' by earlier historians.

Chola Dynasty

The longest-ruling dynasties in the southern regions of India. The Cholas rose to prominence in the 9th century CE when Vijayalaya seized Thanjavur from Pallava feudatories. Their rule lasted over five centuries until the 13th century. Thanjavur was the Chola capital, and at their peak, their kingdom extended from the Tungabhadra to Kanyakumari.

The greatest Chola kings were Rajaraja and his son Rajendra I. The Cholas defeated the kings of **Malaya**, **Java**, and **Sumatra**. They gave titles like Muvendavelan and Araiyan to rich landowners.

Rulers

Vijayalaya Chola: He was a South Indian king, established the Chola Empire. Initially a Pallava feudatory, he conquered Tanjore in AD 850. By the end of the ninth century, he defeated both the Pallavas and the Pandyas, uniting the southern Tamil country. He captured the Kaveri delta from the Muttaraiyar, building Thanjavur town there. He was succeeded by Aditya I.

Rajaraja I (985 - 1014 A.D.): He was a formidable Chola ruler, expanded his empire through naval conquests, claiming victories from the West Coast to Sri Lanka, the Maldives, and the Lakshadweep Islands. He was a devout Saivite, defeated the Cheras and Chalukyas, and established local self-government before abdicating in favor of his son, Rajendra Chola I.

Rajendra I (1014-1044 A.D.): He was the son of Rajaraja I, defeated Mahipala I of Bengal, founding Gangaikonda Cholaपुरam to commemorate the victory. He earned the title "Gangai-konda" and was known as Pandita-Chola for his patronage of learning. Rajendra built the Rajesvaram temple and a Shiva temple with a Sun

pedestal seized from the Chalukyas. He led a naval expedition to Kadaram (Indonesia), earning the title Kadaramgonda, and annexed Sri Lanka. He referred to the Bay of Bengal as "The Chola Lake."

Kulathunga Chola 1: He was the grandson of Rajendra Chola and preserved the legacy of Imperial Cholas. He abolished many taxes and got the title "Sungam Tavirtta". He sent trade embassies to China. He united the Vengi kingdom with the Chola Empire. He was contemporary to Ramanujacharya.

Rajendra III : He was the last Chola king who was defeated by Jatavarman Sundarapandya II.

Rulers of Chola Empire: Vijayalaya, Aditya I, Parantaka I, Gandaraditya, Arinjaya, Sundara, Uttama, Rajaraja I, Rajendra I, Rajadhiraja, Rajendra II, Virarajendra, Athirajendra, Kulothunga Chola I, Vikrama, Kulothunga II, Rajaraja II, Rajadhiraja II, Kulothunga III, Rajaraja III, Rajendra III.

Administration

King was at the top of the administration. Chola rulers appointed Brahmins as spiritual preceptors or rajagurus (the kingdom's guide).

They granted huge estates of land to Brahmins as Brahmadeyams and Caturvedimangalams.

Rajaraja I and Rajendra I mention the names of rajagurus and Sarva-sivas in their inscriptions.

Provinces: The Chola Empire had provinces (Mandalams) divided into sub-divisions (Valanadus and Nadus). Each Nadu contained autonomous villages. Royal officials (Mudaligal and Dandanayakam) governed Nadus. Princes/officers administered Mandalams. Towns (Nagarams) were governed by councils (Nagarattar).

Military Administration: They maintained a large army consisting of infantry, cavalry, and elephants which were called the three limbs of the army.

Revenue Administration: For tax assessment, the Cholas conducted extensive land surveys and revenue settlements.

Rajaraja I, Kulotunga I, and Kulotunga III appointed officials to survey and classify land for taxation purposes. The

Department of Land Revenue was called Puravuvuri-tinaikkalam.

Various units of the **land measurement** were kuli, ma, veli, patti, padagam. The tax rates were fixed depending on the fertility of the soil and the status of the land. Devadana, palliccanda, kavimurruttu and vettaperu have been identified as tax-free villages.

Kadmai - tax on land.

Nagara Viniyogam - a tax for sustenance of nagaram.

Vetti - forced labour tax.

Local Administration: There were two types of villages at the local area, in the Chola empire. Ur - Settlements of the peasants. Agrahara - tax-free land given to Brahmins. Sabha - assembly of agrahara type of village. Uttarameruru inscription issued by Prantaka Chola gives details of election to these committees.

Important terms and meaning

Kaikkolaperumpadai - Royal troops.

Velaikkarar - Personal troops to defend kings.

Kadagram - Military cantonment.

Puravuvanithinaikkalam - Revenue department.

Peruvazhis - Trunk roads.

Perundanam and serundanam were important officials.

Nagarattars - The trading community Chola inscriptions mention several categories of land:

Vellanvagai - The land of non-Brahmana peasant proprietors.

Brahmadeya - Lands given to the brahmanas.

Shalabhoga - Land for the maintenance of a school.

Pallichchhandam - Land donated to Jaina institutions.

Devadana (Tirunamattukkani) - Land gifted to temples.

Different types of lands rights

Karanmai which referred to the right to cultivated during the Chola period.

It was further divided into two types—

Kudi Nikki which means a situation in which people previously settled in the village were either removed or deprived of their rights.

Kudi Ninga which refers to a situation in which such people were not to be disturbed.

Mitatchi was a superior possessive right.

Society and Economy

Divided on caste lines: Periyar (Untouchables), Vellallas (Farming groups).

Emergence of supra caste dichotomy: Idangai (left hand caste groupings) - Consisted mainly of artisanal and trading groups.

Valangai (right hand groupings) - Consisted mainly of agricultural groups.

Women were heads of some villages, as per some inscriptions. Sabha was the assembly in the villages which were inhabited predominantly by the Brahmanas. Expansions in agrarian economy, use of Agrahtta (Persian wheel) for smooth irrigation.

Emergence of various crafts centres: Kanchipuram - weaving industry centre. Kudamukku - centre for betel nut and areca nut production.

Emergence of trading caste:

Garveras - Northern merchants migrated southwards.

Gaudas or Gavundas - originally cultivators. Chola had trade relations with Java, Sumatra, Arabia and China.

Inscriptions: Ennayiram, Thirumukkudal, and Thirubuvanai inscriptions give details of the education system and colleges.

A new social evil Devadasi system came into practice.

The practice of sati became sacred.

Kaveripattinam (chief Chola port) was located at Kaveri river mouth, identified as present-day Puhar, Nagapattinam district of Tamil Nadu. Also known as Kaveripattinam/Kaveripumpattinam. Founded by Chola King Karikala.

Literature

Tamil and Sanskrit languages flourished during the Chola period. Tamil literature witnessed a golden age.

Kamban Ramayan was written by Kamban (Tamil Ramayan).

Periyapuramam (Tiruttondarpuramam) was written by Sekkilar, and Kalladam by Kalladanar.

Nalavenba was written by Pugalandi.

Nannaranava Samkshevam was a

Sanskrit work composed by Kesava swami.

Sivakasintamani was written by Thiruthakkadevar and Kundalakesi belonged to the 10th century.

The Telugu version of **Mahabharata** was started by Nanniah and completed by Tikkana.

Kalingattupparani written by Jayankondar describes the Kalinga war fought by Kulotunga I.

Moovarula was written by Ottakuthar and depicts the life of three Chola kings.

Pampa, Ponna, and Ranna were considered 3 gems of Kannada literature in the 10th century.

Tamil grammar like Kalladam by Kalladanar, Yapperungalamby Amirthasagarar, a Jain, Nannul by Pavanandhi, and Virasoliyam by Buddhmitra were written during Chola rule.

Art and Architecture

Dravidian-type of temple architecture reached its climax under the Cholas. Chola kings built lofty temples and inscribed long inscriptions on the wall narrating their achievements.

Vijayalaya Chola built the Nishumbha sudini Goddess temple in Thanjavur. Brihadeshwara temple at Tanjore was built by Rajaraja I in 1078 and was dedicated to Shiva. Bhillasvamin town was grown as a temple town during the Chola dynasty. Other famous temples built by Cholas: Kailasanath Temple of Kanchipuram, Airavatesvara temple, and temple at Gangaikondacholapuram, Nataraja Temple at Chidambaram.

Sculptures and Bronzes: Dancing figure of Nataraja.

Religion

Cholas were ardent Shaivites. Shiva was preeminent, worshipped as Lingodbhava and Nataraja idols. Shaiva Siddhanta philosophy evolved during this period. **Kulothunga Chola** persecuted Vaishnavites, forcing Ramanujacharya to leave Srirangam for Melkote, Karnataka.

Reason for Decline of Cholas

One of the reasons for the decline of the Cholas lack resources. Dishonesty and

infiltration of armed forces were the causes of inefficiency that led to the empire's demise. The Cholas themselves were plagued by succession rivalries and palace disputes, which weakened the empire even further.

Maravarman Sundara Pandyan dealt the Chola Empire its final blow, defeating Kulothunga Chola III, sacking Thanjavur, and driving Rajaraja Chola III into exile.

Pallava Dynasty

It spanned from 275 to 897 CE, reaching its peak under Mahendravarman I and Narasimhavarman I. They conflicted with the Chalukyas and Tamil kingdoms. The Pallavas are known for their Shore Temple architecture.

Historians suggest the Pallavas may have Parthian origins, with the first Pallava kings ruling from the early 4th century AD. By the 7th century, the Pallavas, Chalukyas, and Pandyas were the major powers competing for dominance in southern India.

Rulers of Pallavas

Early Pallavas: Simhavarman I, Shivskandavarman, Vijayskandavarman, Skandavarman, Vishnugopa I, Kumaravishnu I, Skandavarman II, Viravarman, Skandavarman III, Simhavarman II, Skandavarman IV, Nandivarman I, Kumaravishnu II, Buddhavarman, Kumaravishnu III, Simhavarman III.

Later Pallavas: Simhavishnu, Mahendravarman I, Narasimhavarman I, Mahendravarman II, Paramesvaravarman I, Narasimhavarman II, Paramesvaravarman II, Nandivarman II, Dantivarman, Nandivarman III, Nirupathungan, Aparajitavarman.

Sivaskanda Varman: He was the most powerful of the early rulers. He ruled at the start of the fourth century AD.

Simhavarman or Simhavishnu (575-600 AD): He defeated the Kalabhras and other southern kingdoms, expanding Pallava influence from Krishna to the Kaveri rivers. Kanchipuram became the Pallava capital, earning the moniker "The City of Thousand Temples." Simhavishnu was a devout Vishnu worshipper, bearing the title Avanshimha. Renowned poet Bharavi is said to have visited his court.

Mahendravarman (600-630 AD): He was the son of Simhavishnu, who succeeded him. He was a multi-talented genius. His titles: Chitrakarapuli, Vichitrachitta, Gunabhara, and Lalitankura. Pulakesin II defeated Mahendravarman I near Kanchi in a Battle of Pullalur.

Narasimhavarman I, also known as Mahamalla or Mamalla, was the most eminent Pallava ruler, significantly enhancing the dynasty's power. He continued his father's conflict with the Chalukyas, defeating Pulakesin II in three battles, including one near Kanchi in 642 CE. Narasimhavarman claimed the title Vatapikonda after Pulakesin II's demise. He allied with the Pandyas and Chalukyas to suppress the Kalabhra uprising against Brahmadeya land grants. Narasimhavarman captured Vatapi (Badami) following his victory over the Chalukyas.

Major Literary Works

The Pallavas of Kanchi prioritized education, making it a hub of learning. Notable figures like Dinganaga and Bharavi flourished during their reign. Dandin was esteemed in Narasimhavarman II's court, and Mahendravarman I authored the Sanskrit play "Mattavilasa Prahasana." Tamil devotional saints used music and dance to bring the "concept of compassionate God" to life.

Art and Architecture

The Pallavas pioneered Dravidian architecture, evolving from cave temples to structural marvels like the Pancha Rathas at Mahabalipuram. The sculptures, including the "Descent of Ganges," are classical masterpieces. Narsimhavarman I constructed the Pancha Rathas, representing characters from the Mahabharata, including the Pandava brothers.

They are the architectural edifices of the nine monolithic temples of Mahabalipuram situated on the Coromandel Coast of the Bay of Bengal, near Chennai. The five Rathas are named 'Dharmaraja Ratha', 'Bhima Ratha', 'Arjuna Ratha', 'Nakula Sahadeva Ratha', and 'Draupadi Ratha'.

Society and Culture

There were three types of places: Ur -

Peasants lived here and it was headed by a headman who collected and paid the taxes to the government. Sabha - These were the land granted to the Brahmins, also called as Agrahara villages. These were tax-free. Nagaram - Merchants and traders resided at this place.

During the Pallava period, the Brahmins replaced the Jains and the Buddhists in formulating policies. Brahmadeya - land granted to Brahmins. They worshiped Shiva and Vishnu. They were liberal of Buddhism and Jainism too but these religions eventually lost their relevance and popularity.

Decline

The attack and temporary occupation of Kanchi by **Vikramaditya II** may be regarded as the beginning of the end of Pallava supremacy over South India. The Pallava kingdom was attacked by the Pandyas, the western Gangas, and the Rashtrakutas. Pallava rule lasted until the end of the ninth century CE.

Chalukya Dynasty

The Chalukyas ruled in the Western Deccan region of southern India from 543 to 755 CE. They were a powerful dynasty during the 6th to 12th centuries. Their rule marked a golden age in the history of Karnataka. Initially, their capital was at **Aihole**, later shifting to Badami. Pattadakal, known for its eclectic 7th-8th century architecture under the Chalukyas, was declared a UNESCO World Heritage Site in 1987.

Chalukya of Badami

Pulakeshin I (543 - 566 AD): He was the founder of the Chalukya dynasty, established a kingdom with Vatapi (Badami) as its capital in present-day Karnataka. He conducted Ashwamedha sacrifices and was known by titles like Satyashraya, Vallabha, and Dharmamaharaja in inscriptions.

Kirtivarman I (566 - 597 AD): He consolidated the newly founded Chalukya Kingdom through the subjugation of the Kadambas, the Nalas of Nalavadi (Bellari), the Alupas of South Kanara and the Maurya chiefs of Konkan.

Mangalesha (597 - 609 AD): He ruled as regent as the heir to the throne. He assumed the titles of Ururanaparakrama, Ranavikrama and Paramabhogavata, and

excavated the Vaishnava temple at Badami.

Pulakeshin II (609 - 642 AD): He was the first ruler in South India to issue gold coinage. His childhood name was Erya and ascended the Chalukya throne as Pulakeshin II. He assumed the title Chalukya Parameshwara.

The Battle of Pullalur (618-619 CE) was fought between Chalukya king Pulakesin II and Pallava king Mahendravarman I, resulting in a Chalukya victory and confinement of Pallava power around Kanchipuram.

Pulakesin II defeated Harshavardhana, became Dakshinapatheshvara (lord of south), but was killed by Pallava king.

Aihole Inscription by Ravikirti praised Pulakesin II (Satyasraya).

Vikramaditya I (655 - 680 AD): He restored order in the fractured kingdom and made Vatapi as capital.

Vinayaditya (680 - 696 AD): His reign was marked by general peace and harmony. He assumed the titles Yuddhamalla, Sahasarasika, Satyashraya.

Vijayaditya (696 - 733 AD): His reign was marked by general peace and harmony. He built a number of temples.

Vikramaditya II (733-746 AD): The famous Virupaksha Temple (Lokeshwar temple) and Mallikarjuna Temple was built during his reign.

Kirtivarman II (746 - 753 AD): He was also known as Rahappa. He was the last king of the Badami dynasty.

Chalukya of Vengi (Eastern Chalukya)

Kubja Vishnuvardhana (624 - 641 AD) was the brother of Pulakesin II, became independent Vengi king after the latter's death, titled Vishmasiddhi (conqueror of difficulties).

Other Ruler: Jayasimha I (641- 673 AD), Indra Bhattaraka (673 AD), Vishnu Vardhana II (673-682 AD), Mangi Yuvaraja (682-706 AD), Jayasimha II (706-718 AD), Kokkili (719 AD), Vishnuvardhana III (719-755 AD).

Vijayaditya I (755 - 772 AD): He succeeded his father to the throne. The Satara plates of his son claim that Vijayaditya I reinstated the fallen lords of

Sthanaka and Goa.

Rajaraja Narendra (1018-1061 AD): Renowned for cultural patronage, including Kannada works Vikramarjuna Vijayam, Gadayuddam popularizing Mahabharata, and Tamil Mahabharata translations.

Vijayaditya VII: He was the last king of this dynasty.

Chalukya of Kalyani (Western Chalukya)

Tailapa II (957 - 997 AD): He assumed the titles Trailokyamalla, Nurmadi Taliapa and Satyashraya Kulatilaka.

Satyashraya (997 - 1008 AD), also known as Sattiga or Irivabedanga, assumed titles like Akalavarsha, Akalankacharita, and Sahasabhima. He patronized the renowned Kannada poet Ranna, one of the 'three gems' of classical Kannada literature. Ranna's epic Sahasabhimavijaya favorably compared Satyashraya to the mighty Bhima of Mahabharata for his strength and valor.

Vikramaditya V (1008 - 1015 AD): He was nephew of Satyashraya and had a very uneventful short reign.

Jayasimha II (1015 -1042 AD): He was also known as Jagadekhamalla II and Mallikamoda.

Someshvara I (1042 - 1068 AD): He was known as "Ahavamalla" or "Trilokamalla". He patronized the Kannada language scholar Shridharacharya who wrote Jatakatilaka (1049 AD), the earliest available work on astrology in the language.

Someshvara II (1068 - 1076 AD): His reign was full of civil war and conflict.

Vikramaditya VI (1076-1126 AD) abolished the Saka era, initiated the Chalukya-Vikrama era. Held titles Permadideva, Tribhuvanamalla. Jurist Vijaneshwara wrote Mitakshara. Court poet Bilhana from Kashmir composed king's biography Vikramankadevacharita as Vidyapathi.

Someshvara III (1126 - 1138 AD): He was a noted historian, scholar and poet. The Sanskrit encyclopedic text Manasollas was written by him. He assumed the titles as Sarvadnya-bhupa (the king who knows everything) and Bhulokamala (the king who was lord of all living beings)

due to his scholarly pursuit.

Jagadekamalla II (1138 - 1151 AD): His rule saw the slow decline of the Chalukya empire with the loss of Vengi. He patronised Kannada grammarian Nagavarma II, who wrote Kavyavalokana and Karnataka Bhashabhushana. Jagadekamalla II himself was a merited scholar and wrote in Sanskrit Sangithachudamani a work on music.

Tailapa III (1151 - 1164 AD): His rule saw the beginning of the end of the Chalukya empire. He was killed by Hoysala Vira Narasimha in 1162.

Jagadekamalla III (1163-1183 AD): His rule was completely overshadowed by the emergence of the Southern Kalachuris under Bijjala II who took control of Basavakalyan.

Someshvara IV (1184 -1200 AD): He was the last king of the Western Chalukya empire. He managed to capture Basavakalyan briefly but failed to prevent feudatories like the Seuna, Hoysala and the Kakatiya dynasty from completely overwhelming the Chalukya empire.

Administration

The administrative system was not centralized. The divisions mentioned in the epigraphical records are: Rashtra, Vishaya and Nadu. The copper-plate grants refer to Vishyapatis, Samantas, Gramabhogikas, Maharattaras. Village was the smallest part of the administrative system. The Gamunda was the royal representative at the village level. The Karanas were the village accountants. Mahajans constituted the village elders.

The **Lakslunesvar inscription** gives us details regarding the links between the royal machinery and local administration. **Achara Vyasthe** - charter of rights and duties.

Nagaras - commercial interests.

18 Prakritis (classes) - refers to royal personnel.

Desadhipatis - officers who collected taxes.

The Hyderabad grant of Pulkesin II refers to the village being granted together with the Nidhi (treasure), Upanidhi, Klipta and Uparikara (dues). Members of the royal family and trade associations also made gifts to the temples in kind (millet, betel leaves).

Economy

Rocky and infertile land limited the income from land for them. Overall decline of trade and commerce in India during this period, therefore they conducted invasions and plunder of the neighbouring regions. Arab traders were given patronage at the west coast under Chalukyas.

Religion

The Chalukyas of Badami were Brahmanical, but respected other faiths. They built temples to Vishnu, Shiva and other gods, reflecting Brahmanical expansion. Jainism also grew, while Buddhism declined. Religion became more ritualistic.

Literature, Art and Architecture

The Chalukyas made significant contributions to the growth of **Prakrit language** and literature. They also patronized the development of Ajanta art. The Chalukyas are known for the fusion of Nagara and Dravidian architectural styles, resulting in the Vesara style of temples.

Important Chalukya centers include Aihole, **Badami**, and **Pattadakal**, where they built intricate and gigantic structural temples. Chalukya kings like Pulakeshin I and Vikramaditya VI were prolific temple builders, constructing major Hindu temples in these regions.

The Mallikarjuna and Virupaksha temples at Pattadakal were built by the two queens of Vikramaditya II to commemorate the Chalukya victory over the Pallavas.

Decline

The last ruler of the Badami Chalukyas was overthrown in the 8th century CE by Dantidurga, founder of the Rashtrakuta Empire.

Pala Dynasty

The Pala Empire (750 - 1161 CE) was an imperial power originated in the region of **Bengal**. It was founded by Gopala, probably in 750 AD. The Pala period is considered one of the golden eras of Bengali history. "Pala" is a Sanskrit word which means "protector".

Rulers

Gopala (750 - 770 AD): He unified Bengal and brought Magadha (Bihar) under his control. Gopala is considered the first Buddhist monarch of Bengal, and he founded the monastery at Odantapuri in Bihar.

Gopala's reign was marked by a tripartite struggle between the Palas, Pratiharas, and Rashtrakutas for control of Kanauj and North India.

Dharmapala (770 - 810 AD): He was the second ruler of the Pala Empire. He was the son and successor of Gopala. He defeated Indraraja, the Pratihara vassal ruler of Kannauj, and installed his own nominee Charkayudha on the throne.

This Kannauj dispute led to a struggle between Dharmapala and the Pratihara king Vatsaraja. Vatsaraja ultimately defeated Dharmapala in a battle near Prayag and declared himself the "Lord of Northern India".

Devapala (810–850 AD): He was the son of Dharmapala and Rannadevi (princess of the Rashtrakuta). He had expanded the empire to eastern India, including the kingdoms of Assam, Odisha, and Kamarupa. Buddhist scholar Vajradatta was the court poet of Devapala. Vajradatta wrote *Lokesvarashataka*.

Mahipala I: He revived the Pala empire in the late 10th century.

Other Rulers of Pala Dynasty

Mahendrapala, Shurapala I, Gopala II, Vighrapala I, Narayanapala, Rajyapala, Gopala III, Vighrapala II, Nayapala, Vighrapala III, Mahipala II, Shurapala II, Ramapala, Kumarapala, Gopala IV, Madanapala, Govindapala.

Art and Architecture

Dharmapala was a great patron of Buddhism. He granted 200 villages to Nalanda university and revived it. He founded the Vikramashila monastery in the ninth century which later evolved into a great learning centre of Buddhism.

Vikramshila University - It is located now in the site of Antichak village, Bhagalpur district (Bihar). He built the great Somapura Mahavihara (largest Buddhist vihara) in Paharpur (Bangladesh).

Miniature Painting flourished in eastern

India, particularly Bengal, Bihar and Orissa, during the 9th-10th century Pala period, with Buddhist themes illustrated in palm-leaf manuscripts.

Administration

The Pala Empire had a monarchical system, with the King as the head of state and commander-in-chief. The kingship was hereditary within the Pala dynasty.

The empire had both directly administered territories and vassal states under autonomous chiefs. The vassal chiefs paid tributes and provided troops to the Pala King.

The territories were divided into provinces called *bhukti*, governed by officials called *Uparika*. These provinces were further divided into districts called *mandalas* or *vishayas*, headed by *Visayapatis*. Smaller chieftains known as *Samanthas* or *Bhogapatis* dominated the villages.

Culture and Religion

Palas were devout Buddhists and they practised **Mahayana Buddhism**. They sent many Buddhist scholars to other parts of the world which helped to spread Buddhism in Tibet, Nepal, Bhutan, Myanmar, and Indonesia.

Decline

Due to constant wars and defeats from Rashtrakutas and Pratiharas, Pala Empire disintegrated. The Hindu Sena dynasty ruler '**Vijayasena**' overthrew the Pala Empire in the 12th century.

Pratihara Dynasty

The **Pratiharas** were also known as the Gurjara-Pratiharas (8th century CE - 10th century CE). They ruled over western and northern India. The most well-known king of this dynasty was Bhoja or Mihira Bhoja (836 CE - 885 CE).

Rulers

Nagabhata I (730 - 756 A.D.): He was the founder of the Pratihara dynasty. He resisted the Arab rulers of Sindh, who were trying to encroach on Rajasthan, Gujarat, Punjab.

He fought against **Rashtrakutas**, but he was defeated. His nephew Kakustha

succeeded him and was followed by his brother Devaraja. These two ruled from around 760 to 775 CE.

Vatsaraja (775-800 CE): He conquered most of central Rajasthan after defeating the Bhandi or Bhatti clan. He then tried his hand at Kanyakubja politics and defeated Palas and secured the throne for his Ayudha nominee. Dhruva Rashtrakuta dealt him a severe blow, capturing the royal insignia that Vatsaraja had captured from the Palas and driving him to seek refuge in the Rajasthan deserts.

Nagabhata II (800-833 CE): Nagabhata II, son of Vatsaraja, conquered Sindh, eastern India, and parts of Gurjaradesha, defeating the Palas and Arabs. However, he was defeated by the Rashtrakuta ruler Govinda III in the fight for control of Gujarat, losing south Gujarat and Malwa.

He captured Kanyakubja, making it the Pratihara capital and ending Ayudha rule. He was succeeded by his son Ramabhadra (833-836 CE), whose reign saw some territorial recovery.

Mihir Bhoja (836 - 885 CE): Grandson of Nagabhata II, ruled for nearly 50 years. With Chedi and Guhila support, he defeated the Palas and Rashtrakutas, expanding his empire into eastern India, Gujarat, Rajasthan, and Madhya Pradesh. He reclaimed Kanauj as the Pratihara capital for a century and founded Bhopal. A Vishnu devotee, he adopted the title Adivaraha.

Mahendrapala I (885-910 CE): He expanded the empire and conquered new territory in the east. He was defeated by the king of Kashmir and ceded some territories in Punjab to him. Following his death, a civil war erupted between his son Mahipala and his half-brother Bhoja II.

Rajshekhar was his court poet and his works include *Karpoormanjari* (written in Sauraseni Prakrit), *Kavya Mimamsa*, *Balabharata*, *Bhrinjika*, *Vidhasalbhanjika*, *Prapancha Pandava*.

Mahipala I (912-944 CE): He was able to secure the throne but was defeated by the Rashtrakutas. Al-Masudi came to India (Gujarat) in 915 AD in his reign.

Mahendrapala II (944-948 CE) and his successors made no significant contributions during their reigns.

Rajyapala: He was defeated by the

Rashtrakuta king Krishna III. Mahmud Ghazni raided Kannauj by forcing Rajyapala to flee the battlefield. Vindhyadhar Chandela assassinated him.

Yashpala (1024-1036 CE): He was the last ruler of the dynasty. Kannauj was conquered by the Gandhavalla by 1090 CE.

Political and Social Structure

Uparika (governor) - Collect land revenue.

Vishayapati (District head) - Maintain law and order.

Viswaroopa form of Vishnu and the Marriage of Shiva and Parvati from Kannauj are two notable sculptures from this period. Mihir Bhoja built Teli Ka Mandir (Hindu Temple) in Gwalior, Madhya Pradesh.

Decline

Following Bhoja's reign, the Pratihara dynasty faced defeats, notably under Indra III, who devastated Kanyakubja in the early 10th century. Weak central power led to assertive feudatories and governors declaring independence, causing the empire's disintegration to only the Kanyakubja region. It fell to the Ghaznavid Turks in the early 11th century.

Rashtrakuta Dynasty

The Rashtrakuta Dynasty ruled parts of South India from the 8th to 10th centuries. At its peak, the kingdom spanned modern Karnataka, Tamil Nadu, Andhra Pradesh, Telangana, Maharashtra and Gujarat.

Their capital was Malkhed (Manyakheta). While early rulers were Hindu, later ones were Jain. The Elichpur clan of Rashtrakutas were feudatories of the Chalukyas of Badami.

Rulers

Dantidurga founded the main Rashtrakuta dynasty, establishing Manyakheta as the capital in 753 CE. He performed Hiranya-Garbha (Golden Womb) ritual, attacked Kanchi, and allied with Nandivarman Pallavamalla. **His titles** included Prithivivallabha, Maharajadhiraja, Parameshvara, and Paramabhattacharaka.

Krishna I (756-774 AD): Dantidurga's uncle, succeeded him after his childless

death. Krishna expanded the kingdom, and the Kailasanatha Temple at Ellora was built during his reign. He commissioned 18 Shiva temples. Krishna's son Govinda II became the next ruler. **His titles** included Shubhatunga, Akaalavarsha, Rajadhiraja, and Parameshvara.

Govinda II (774-780): Govinda II, Krishna I's eldest son, succeeded him. He traveled to the Eastern Chalukyas, assisted a Ganga king, and intervened in Vengi, leading to the defeat of the Eastern Chalukya ruler Vishnuvardhana IV. Govinda II was later deposed by his younger brother Dhruva Dharavarsha. His titles included Prabhatavarsha and Vikramavaloka.

Dhruva Dharavarsha (780-793) : His accession marked the golden age of the Rashtrakutas. He was one of the Tripartite rulers who defeated kings friendly to his elder brother, and then conquered Kannauj, the Gurjara-Pratiharas, and the Palas. This began the Tripartite struggle for control of the Indian heartland.

His other victories included subjugating the Vengi king, who could only ensure peace by marrying his own daughter to Dhruva Dharavarsha. He added the emblem of Ganga and Yamuna to his imperial emblem after his successful campaigns in north India. He earned titles: Kalivallabha, Srivallabha, Dharavarsha, Maharajadhiraja, and Parameshvara.

Govinda III (793-814) : He succeeded his father Dhruva Dharavarsha, and proved to be the most powerful Rashtrakuta ruler. He expanded the kingdom from Kannauj to Cape Comorin, and from Banaras and Bengal to Gujarat. Govinda III defeated the Gurjara-Pratiharas, Palas, Pallavas, Cholas, and Pandyas. **His titles** included Jagattunga, Kirti-Narayana, Janavallabha, Tribhuvanadhavala, and Prabhatavarsha.

Amoghavarsha I (814-878): He was the son of Govinda III, was the greatest Rashtrakuta king. He established a new capital at Manyakheta and made Broach a key part of his kingdom. Amoghavarsha was a patron of education and literature, and converted to Jainism. His titles included Nripatunga, Atishayadhavala, Maharaja-shanda, and Vira-Narayana. The Arab merchant Suleman described

Amoghavarsha as one of the four greatest kings in the world, along with the Caliph of Baghdad. He ruled for 64 years.

Krishna II (878-915): He succeeded his father (Amoghavarsha I) as Rashtrakuta ruler. His queen was Mahadevi, a Haihaya princess. Amoghavarsha renounced state affairs for religion in his later years. Krishna II cultivated a Chedi alliance for military advantage. **His titles** included Akalavarsha and Shubhatunga.

Indra III (914-929): Indraraja was the son of Jagattunga and Lakshmi (a Kalachuri dynasty princess). His coronation took place in Kurundaka (a village near the confluence of the Krishna and Panchganga rivers). According to a copper plate, he destroyed Meru, which was ruled by Pratihara Mahipala.

During his reign, the author Trivikrama Bhatta published Damayanti Katha and Madalasa Champu.

He assumed the titles: Nityavarsha, Rattakandarapa, Kirti-Narayana, and Rajamarathanda.

Krishna III (939-967): He was the last great Rashtrakuta Ruler. His empire extended from the Narmada to Kaveri, including northern Tamil regions and Ceylon. He defeated the Paramars of Malwa, Eastern Chalukyas of Vengi, and Chola king Parantaka I, annexing northern Chola lands. Krishna III set up a victory pillar at Rameshwaram and built a temple there. He held the title 'Akalavarsha'.

Karka II became Rashtrakuta king after Khottiga was killed by Paramaras around 972 CE. The Malwa invasion had weakened the Rashtrakutas. Chalukya king Tailapa II then attacked in 973 CE, completing the Paramara destruction. A Kalachuri inscription confirms Tailapa killed Karka II during this Chalukya invasion.

Rulers of Rashtrakuta Empire: Danti-varman, Indra I, Govindaraja, Kakka I, Indra II, Dantidurga, Krishna I, Govinda II, Dhruva, Govinda III, Amoghavarsha I, Krishna II, Indra III, Amoghavarsha II, Govinda IV, Amoghavarsha III, Krishna III, Khottiga, Karka II.

Administration

The inscriptions show that the succession of rulers was done on a hereditary basis. The kingdom was

divided into: Rashtra (province), Visaya and Bhukti. The head of a Rashtra was called Rashtrapati (governor) who sometimes enjoyed the status and title of a vassal king. The Visaya was a district under Visayapati, and the Bhukti was a smaller unit than it. A body of assistants called the Rashtramahattaras and Vishayamahattaras respectively assisted provincial governors and district level governors in the Rashtrakuta administration.

It was a powerful monarchy with many feudatories, based on the Gupta, Harsha and Chalukya models.

The Rashtrakutas had a large and well-equipped military, with infantry, cavalry, and war elephants, using horses imported from Arabia, West Asia, and Central Asia, as mentioned by Arab travelers.

Literature

Kannada literature rose to prominence during its reign. The Prakrit and Sanskrit eras came to an end during this time period. Court poets produced literary works in Kannada and Sanskrit. 'Kavirajamarga' (written by King Amoghavarsha), was the first Kannada book available. In the court of King Amoghavarsha I, bilingual writers such as Asaga rose to prominence, and noted scholars such as Mahaviracharya wrote on pure mathematics.

Adipurana was written by the Jain writer Adikavi Pampa. It is widely regarded as one of the most influential Kannada writers.

Sri Ponna was another notable Jain writer in Kannada, patronized by King Krishna III. He was best known for Shantipurana (an account of the life of Shantinatha, the 16th Jain Tirthankara).

Art and Architecture

The Rashtrakuta contributions to art and architecture are reflected in the magnificent rock-cut cave temples at Ellora and Elephanta in Maharashtra. They renovated Buddhist caves and also created Jain cave temples, with the walls adorned by spectacular sculptures of Hindu mythological figures.

The Kailash Temple at Ellora is an architectural marvel, featuring sculptures depicting the Buffalo monster being slain by Goddess Durga. The Navalinga temple complex in Karnataka was also built

during the Rashtrakuta reign.

The **Antroli-Chharoli inscription** dated 757 CE mentions four generations of a collateral Rashtrakuta branch ruling over the Lata region in Gujarat. Bharuch was a major city and former capital.

The **Samangadh inscription** of 754 CE records that Dantidurga overthrew the last Chalukya ruler, Kirtivarman II, and assumed imperial titles like Prithivivallabha, Maharajadhiraja, Parameshvara, and Paramabhattaraka. His territory encompassed around 400,000 villages, over half the Chalukyan empire. Dantidurga died childless, leading to a dispute between his uncle Krishnaraja I and other family members.

Decline of Rashtrakutas

Its decline began with the reign of Khottiga and Karka II. Khottiga was defeated and killed by a Paramara dynasty ruler in 972 CE. Karka II succeeded Khottiga. The Bhandana grant of Aparajita (997 CE) and the Kharepatan plates dated 1008 CE state that Tailapa (Chalukya ruler) defeated Karka in battle and the kingdom ceased to exist. It marked the end of the Rashtrakuta Empire.

Other Dynasties of Ancient India

Varman dynasty: It was the first historical dynasty of the Kamarupa kingdom ruled over between the 4th century to 7 century.

It was established by Pushyavarman, a contemporary of Samudragupta. Kumar Bhaskaravarman was the last and the greatest ruler of the Varman dynasty. He was one of the well-known kings of mediaeval Kamarupa.

Kadamba dynasty: The Kadambas ruling from 345 to 540 CE in Karnataka and the Konkan, was founded by Mayura Sharma in Banavasi. Their legacy endures in Halasi's 5th-century monuments, notably the Kadamba Shikara. However, around 540 CE, the Chalukyas of Badami conquered their realm.

Maukhari dynasty: It ruled the Ganges-Yamuna plains from their capital in Kannauj for over six generations. Isanavarman was the first independent Maukhari ruler, adopting the title of Maharajadhiraja. Their political

supremacy in northeast India is attested by numerous coins of rulers like Ishanavarman, Sarvavarman, and Avantivarman.

The **Ningthouja (Mangang) dynasty** was the ruling dynasty of the erstwhile kingdom of Manipur. It was founded by King Nongda Lairen Pakhangba in 33 CE and Kangla was their seat of power. The term "Ningthou" refers to the king, while "Leima" is the term for the queen in Manipuri.

Kalachuris of Ratnapura: The Kalachuris of Ratnapura were a central Indian dynasty during the 11th and 12th centuries.

They ruled parts of present-day Chhattisgarh from their capital at Ratnapura (modern Ratanpur in Bilaspur district). Kalingaraja (founder) conquered the Dakshina Kosala region and made Tummana is his capital.

Namgyal Dynasty : The Namgyals (Chogyal Monarchy) were the ruling dynasty of the former Kingdom of Sikkim, belonging to the Bhutia community. The dynasty was founded by Phuntsog Namgyal, a fifth-generation descendant of Guru Tashi, and they ruled as absolute monarchs from 1642 to 1973, and as constitutional monarchs until 1975.

Sharabhapuriya Dynasty: It ruled parts of Chhattisgarh and Odisha during the 5th-6th centuries. Initially Gupta vassals, they became independent as the Gupta empire declined. They are known through inscriptions and coins, with Sharabha as the earliest known king. Historian D.C. Sircar suggests they may have also been called the Amararyakula.

Utpala Dynasty: It was a Kashmiri dynasty that ruled the Kashmir region from the 8th to 10th centuries CE. It was founded by Avanti Varman in 855 CE, ending the Karkota dynasty. The Utpala rulers built Hindu temples dedicated to Vishnu and Shiva, including the notable Avantiswara and Avantiswami temples.

Chalukya dynasty (Solanki dynasty): The Chalukya dynasty had three branches. The oldest ruled from Mattamayura in central India, the second was founded by Mularaja in Anhilwara (Patan, Gujarat) in the 9th century, and the third was founded by Barappa in Lata, with its capital at Broach in southern Gujarat.

Karkota Dynasty: It was founded by Durlabh Vardhana in 7th century Kashmir. They emerged as a powerful dynasty in central Asia and northern India, and were known for building impressive Hindu temples in their capital Parihaspur.

Philosophies & Philosophers of Ancient India

Six Schools of Philosophy

The Vedic period saw the development of ideas and philosophies about the Atman (soul) and the Brahman (cosmic principle). These concepts later led to the emergence of the six orthodox schools of Indian philosophy, known as the Shada Darshana, which all recognize the Vedas as the ultimate authority.

There are six schools of philosophy: Vaisheshika, Nyaya, Samkhya, Yoga, Purva Mimamsa and Vedanta (Uttara Mimamsa).

Samkhya: It was founded by Kapila, who wrote the Samkhya sutra. It holds that reality is constituted of two principles, one female (Prakriti) and the other male (Purusha).

Prakriti and Purusha are completely independent and absolute. Purusha is mere consciousness, hence it cannot be modified or changed.

Prakriti is constituted of three attributes - thought, movement and the change (transformation of these attributes brings about the change in all objects).

Yoga: The origin of yoga is found in the Yogasutra of Patanjali believed to have been written in the second century BC.

The achievement of spiritual liberation (moksha) occurs when the spirit (purusha) is freed from the bondage of matter (prakriti).

It is also considered as a means of achieving **freedom** (mukti). Freedom can be achieved by various modes: practising self control (yama), observation of rules (niyama), fixed postures (asana), breath control (pranayama), choosing an object (pratyahara) and fixing the mind (dharna), concentrating on the chosen object (dhyana) and complete dissolution of self, merging the mind and the object (Samadhi). Yoga admits the existence of

God as a teacher and guide.

Nyaya: It is a logical philosophical system that defines valid knowledge as understanding reality. It considers God as the creator, sustainer, and destroyer. The Nyaya Sutras are attributed to Gautama.

Vaisheshika: It is a realistic and objective philosophy that views the universe as composed of five elements: earth, water, air, fire, and ether (sky). The foundational text of Vaisheshika was written by Kanada.

The seven categories are: substance (dravya); quality (guna); action (karma); generality (samanya); particularity (vishesha); inherence (samavaya); and non-being (abhava).

The living beings were rewarded or punished according to the law of karma, based on actions of merit and demerit.

Vaisheshika School of philosophy explained the phenomena of the universe by the atomic theory, the combination of atoms and molecules into matter and explained the mechanical process of formation of the Universe.

Mimamsa (Purva Mimamsa): It is a system that analyzes the interpretation and application of the Vedic texts. It considers the Vedas as eternal and the source of all knowledge, with religion being the fulfillment of Vedic duties. The main text is the Sutras of Jaimini, with key figures including Sabar Swami and Kumarila Bhatta. Mimamsa focuses on Dharma, the law of righteousness and the fruits of one's actions. Mimamsa is known as Purva-Mimamsa or Karma-Mimamsa, as it focuses on the earlier parts of the Vedas and the study of actions.

Vedanta (Uttara Mimamsa): It is the philosophy of the Upanishads, the concluding part of the Vedas. Shankaracharya's commentaries on the Upanishads, Brahmasutras, and Bhagavad Gita gave rise to Advaita Vedanta, which propounds non-dualism and the unity of Brahman and the self. Ramanuja was another prominent Advaita scholar.

Charvaka School

Brihaspati is considered the founder of Charvaka philosophy, which is mentioned in the Vedas and Brihadaranyaka Upanishad. Charvaka, also known as

Lokayata, is an ancient Indian materialist philosophy that recognizes only the material world, rejecting the existence of God, soul, and heaven. It holds that knowledge arises from the combination of the four elements of earth, water, fire, and air, which cease to exist after death. Charvaka does not recognize the fifth element of ether.

Purana Kassapa

He was a 6th century BC Indian ascetic who taught a theory of Akriya or non-action. Contemporary with Mahavira and Buddha, his philosophy separated the body and soul as distinct entities, rejecting any notion of reward, punishment or the aftereffects of one's actions.

School of Art and Architecture

Gandhara School of Art

The Gandhara region was situated in the northwestern part of the Indian Subcontinent. This region was successively ruled by the Greeks (Greece), Mauryas, Sungas, Shakas, and Kushanas for many centuries.

The Gandhara school of art flourished in the 1st century BCE to 4th century CE, primarily in the areas of Afghanistan, including Taxila, Peshawar, Begram, and Bamiyan. This school, influenced by Graeco-Roman, Indo-Greek, and Graeco-Buddhist styles, was developed by the Kushanas in ancient India, with the Shakas and Kushanas as its chief patrons.

The school is characterized by its portrayal of the Buddha in Graeco-Roman fashion, with curly hair and draped garments, using the blue-grey schist stone.

Features: The fusion of Greco-Roman and Indian ideas. Its features include the halo around the Buddha's wavy hair, brow furrows, ornaments, and the drape of the garments, which show influences from foreign traditions like China and Iran. The use of **black stone** and **stucco structure**, including chaityas, viharas, and stupas, were also distinctive of this art style.

Mathura School

It originated in the **1st century BC** and flourished until the 12th century AD. It

utilized local red stone with black spots to create sculptures, producing large numbers of Jain deities and ayagapatas (stone slabs for worship). During the Kushana period, it also yielded sculptures of Hindu deities like Kartikeya, Vishnu, and Kubera, reflecting the diverse religious influences of the time.

Amaravati School of Art

The Amaravati school of art flourished in the **Krishna-Godavari** region of Andhra Pradesh from 150 BC to 350 AD, patronized by the Satavahanas and Ikshvakus.

Important Feature: Its sculptures, mainly found on stupas, depict narratives from the Buddha's life using white marble and naturalistic human figures. The medallions carved on the stupas presented entire incidents, such as the Buddha taming an elephant, in a fluid, natural manner.

Other Art and school

Tagore Centre for the Study of Culture and Civilization is located in Shimla, Himachal Pradesh. It also addressed the first Rabindranath Tagore Memorial Lecture.

Sanjhi is a ritualistic practice of unveiling rangoli at dusk with chanting, associated with Mathura and Vrindavan in Uttar Pradesh.

Calico was the cotton textile traded between India and Europe, originating in Calicut in the 11th century.

Kalamezhuthu is an ancient art woven into the ritualistic mores of Bhagavati, Naga, and Ayyappa temples in Kerala. It refers to the pictorial representation of deities on the floor using colored powders during ceremonies.

Khurja pottery is a traditional Indian pottery from Uttar Pradesh, tagged with geographical indication.

Paika Akhada is an Odia term meaning warrior gymnasium. In former times, they served as the training schools of the peasant-militias in Odisha.

Silambam is a weapon-based Indian martial art originating in modern-day Tamil Nadu.

Janapada Geete, which includes the vachanas of Basavanna, is associated with the state of Karnataka. In Janapada

Geete, Jana means people or tribe and pada means verse or rhyme.

Tholu Bommalata is the shadow puppet theater tradition of the state of Andhra Pradesh in India. Tholu means leather in Telugu and bommalu means dolls.

Margomkali is a ritual folk art form of Kerala. It is a dance form of the Syrian Christians of Kottayam and Thrissur districts in which dancers sing and dance around a lighted wick lamp (nilavilakku).

Foreign Invasion in Ancient India

Iranian Invasion

Darius I (550-486 BC) also known as Darius the Great, was the third Persian king of the Achaemenid Empire. He led military campaigns in Europe, Greece, and the Indus Valley, invading and occupying parts of northwestern India in 516 BC, which remained under Persian rule until Alexander's invasion.

Impact of the Invasion on India

Darius I's conquest of northwestern India exposed the region's military weakness, paving the way for Alexander's later invasion. The Persian Satrap system of administration in these provinces served as a model for later Indian dynasties. The increased trade and contacts between India and Persia led to the settlement of foreign populations in northwest India, and the circulation of refined Persian silver coins.

Alexander's Invasion

Alexander the Great (356-323 BCE), born in Pella, **Macedonia**, created one of the largest empires of the ancient world stretching from Greece to northwestern India. In 326 BCE, he invaded India, crossing the Indus and advancing towards Taxila.

After 19 months of fighting in India, Alexander's army returned westward in July 325 BCE. He founded the cities of Alexandria Boukephala and Alexandria Nikaia during his invasion of the Indian subcontinent. Alexander died in Babylon on June 13, 323 BCE.

Battle of the Hydaspes

It was fought between Alexander the Great and Porus in May of 326 BC. Alexander arrived with his troops at the

banks of Hydaspes (Jhelum) river in Punjab. Porus, who ruled over the territory between the Jhelum and Ravi, resisted his advance. Finally he crossed the river in the dark.

A fierce battle was fought and Porus was defeated and captured. Alexander was impressed by the courage and heroism of this Indian prince, reinstated Porus and made him his ally.

Impact of the Invasion on India

Alexander's invasion of India in 326 BCE weakened the various states and warlike tribes of Punjab. It opened four new trade routes between India and Europe, allowing direct relations. Indians learned the art of making beautiful idols and coins from the Greeks. The Gandhara School of Art emerged as a consequence of Greek artistic influences.

Seleucid Invasion

The Seleucid-Mauryan War, fought from 305 to 303 BC, began when Seleucus I Nicator aimed to reclaim Indian territories from Emperor Chandragupta Maurya. It concluded with a settlement granting the Mauryan Empire the Indus Valley and part of Afghanistan, alongside a marriage alliance. This strengthened the Mauryan Empire as the preeminent power in the Indian subcontinent, while the Seleucid Empire shifted its focus westward.

Huna Invasions

From Kumaragupta-I's reign, the Gupta Empire faced threats from the Hunas, notably under chiefs like **Tormana** and **Mihirakula**. These attacks severely weakened Gupta control in northern and western territories by the end of the fifth century AD.

Ancient Scripts and Sculptures in India

Hathigumpha Inscription

The inscription was first discovered by **A. Sterling** in 1825 and was copied by Colonel Mackenzie. The Hathigumpha inscription was deciphered by James Prinsep in 1837. Engraved in Prakrit using Brahmi script, it's found in Udayagiri caves near Bhubaneswar, significant in Jainism. Udayagiri and Khandagiri are also called locally as

Kumari and Kumara Parvata (hill) respectively.

It reveals **Kharavela's** rule over Kalinga and mentions rulers like Satakarni, Nandaraja, Yavanaraja, and Brihaspatimitra, placing Kharavela in the Chedi-Mahameghavahana dynasty's third generation.

Kandahar Bilingual Rock Inscriptions

It is known as the Kandahar Edict of Ashoka and less commonly as the Chehel Zina Edict, is an inscription in the Greek and Aramaic languages that dates back to 260 BCE and was carved by the Mauryan emperor Ashoka at Chehel Zina, a mountainous outcrop near Kandahar, Afghanistan.

Clay Tablets

Clay tablets were the primary writing medium in the Ancient Near East, used for cuneiform writing from the Bronze Age into the Iron Age. Cuneiform characters were pressed into wet clay with a stylus, typically made of reed. In **Mesopotamia**, writing initially started as simple counting marks on clay tokens or occasionally on wood, stone, or pottery.

Manuscripts

A manuscript is a handwritten composition on various materials like paper, bark, cloth, metal, or palm leaf. In India, the distinctive long rectangular palm leaf manuscript was used from ancient times until the 19th century. Globally, manuscripts were produced on vellum, parchment, papyrus, and paper.

Poets, Books and Authors of Ancient India

Charaka Samhita is a Sanskrit text on Ayurveda (Indian traditional medicine). Along with the Sushruta Samhita, it is one of the three works that constitute the Bhṛat Trayī. It was written by Charaka.

Susruta Samhita is an ancient Sanskrit text book on surgery. It is one of the two foundational Hindu texts on the medical profession that have survived from ancient India. The second is Charak Samhita. It was written by Sushruta.

Bṛhat Samhita is a 6th century Sanskrit encyclopedic text written by Varaha Mihira.

Manusmṛiti or Laws of Manu, is an ancient Hindu legal text and Dharmashastra attributed to the legendary lawgiver Manu, dated around 100 CE.

Ashtadhyayi is a Sanskrit grammar text written by Panini in the 6th century BCE. Panini is regarded as the founder of Sanskrit language and literature.

Arthashastra, written by Kautilya in Sanskrit, is an ancient Indian text outlining administration and policies for the Maurya Empire, including statecraft, economics, and military strategy.

Mudrarakshasa is a Sanskrit play by Vishakhadatta from the Gupta period, depicting Chandragupta Maurya's victory over the Nandas with Kautilya's guidance, providing socio-economic glimpses of Mauryan times.

Indica was written by Megasthenes. It documented observations of Pataliputra's administration and the Mauryan Empire under Chandragupta Maurya.

Ramacharitam is a Sanskrit epic poem. It was written by Sandhyakar Nandi. He is considered to be the greatest poet of the Pala dynasty.

Malatimadhava, is a coronation of Rama. A romantic social play in ten acts was written by Bhavabhuti.

Mṛichchhakatika (The little clay cart): It is a ten-act Sanskrit drama. It was written by Shudraka. Three Sanskrit plays are ascribed to him - Mṛichchhakatika, Vinavasavadatta and Padmaprabhritaka.

Kadambari is a romantic novel in Sanskrit. It was written by Banabhatta.

Harsha Charita is a biography of King Harshavardhana. It is authored by Sanskrit poet and scholar Banabhatta.

Mahaviracharita is a play written by Bhavabhuti in the 8th century. It is based on the early life of Rama. His two known later works: Malatimadhava, and Uttararamacharita.

Buddhacharita is an epic poem in the Sanskrit mahakavya style on the life of Gautama Buddha. It was written by Ashvaghosha.

Panchatantra is an ancient Indian collection of interrelated animal fables in Sanskrit verse and prose. The surviving

work is dated to about 200 BCE. It is written by Vishnu Sharma.

Hitopadesha is an Indian text in the Sanskrit language consisting of fables with both animal and human characters. The surviving text is believed to be from the 12th-century. It is written by Narayana Pandit.

Kathasaritsagara is a famous 11th-century collection of Indian legends, histories and folk tales as retold in Sanskrit. It is written by Shaivite Somadeva.

Si-Yu-Ki was written by Chinese pilgrim, monk, and scholar Hiuen Tsang. He resided in India for 17 years in search of Buddhist texts.

The Travels of Marco Polo: It is part of the knowledge base of civilization as we know it. It is written by Chae Hee-Soon, Rustichello da Pisa.

Fo-kwo-ki: It is written by Fa-Hien. This book recounts his documents of his visit to India and neighboring countries, and reflects the condition of those countries during his time, including temples, culture, traditions and fest.

Matla-us-Sadain wa Majma-ul-Bahrain (The Rise of the Two auspicious constellations and the Confluence of the Two Oceans): It was written by Abdur Razzak.

Manu Charitra by the early sixteenth-century poet Allasani Peddana, is the definitive literary monument of Telugu civilization.

Prithviraj Raso is a Braj language epic poem about the life of Prithviraj Chauhan. It was written by Chand Bardai.

Mimamsa Sutra or **Purva Mimamsa Sutras:** The work is divided into twelve adhyayas (chapters), which are further divided into sixty padas. It is written by Rishi Jaimini.

Brahma Sutras: It is a Sanskrit text. It is written by Badarayana.

Pancha-siddhantika: It is a 6th-century Sanskrit language text. It is written by Varahamihira. It basically summarizes the five other books of astronomy: Surya, Romaka, Paulisa, Vasistha and Paitamaha Siddhanta.

Natya Sastra is a Sanskrit treatise on the performing arts. It is written by Bharata

Muni. I

Digha Nikaya is a Buddhist scriptures collection. It is a part of Sutta Pitaka. It is written by Buddhaghosa.

Siddhanta Shiromani is credited with the discovery of differential calculus. It is used in astronomical problems and computations. It was written by Bhaskaracharya (Bhaskara II).

Aryabhatiya (Aryabhatiyam): It is a Sanskrit astronomical treatise. It is the magnum opus and only known surviving work of the 5th century. It was written by Indian mathematician Aryabhata.

Silappatikaram is the first Tamil epic. It is written by Ilango Adigal. The epic is a tragic love story of an ordinary couple, Kannaki and her husband Kovalan.

Valaiyapadhi (Valayapathi): It is one of the five great Tamil epics. It was written by a Tamil Jain monk.

Manimekalai is a Tamil-Buddhist epic. It was written by Kulavanigan Sithalai Sattanar.

Kundalakesi is a Tamil Buddhist epic. It was written around the 10th-century. It was written by Nathakuthanaar.

Nitisara: It is an ancient Indian treatise on politics and statecraft. It is written by Kamandaka.

Nagananda: It is a Sanskrit play attributed to emperor Harsha. It is written by Harsha.

Abhijnanashakuntalam: It is a Sanskrit play. It is written by Kalidasa.

Kumarasambhava, Shakuntala and Ritusamhara: It was written by Kalidasa.

Malavikagnimitram: It is a Sanskrit play written by Kalidasa. It is the love story of King Agnimitra and the court dancer Malavika.

Vikramorvasiyam: It is a five - act drama written by Kalidasa. It depicts the love of Pururavas (the king of Pratishthana) and Urvashi (apsara).

Meghdoot: It is a lyric poem written by Kalidasa (4th - 5th century CE).

Raghuvansha: It is a Sanskrit epic poem written by Kalidasa. The poem speaks in 19 sargas, about the Raghu dynasty, which included Raghu, Dasaratha, Rama.

Tolkappiyam is written by Tolkappiyar (epithet). It is the oldest extant Tamil grammar. He is believed by various traditions to be one of the twelve disciples of Agasthiyar.

Brahmasphutasiddhanta and **Khandakhadyaka** are written by Brahmagupta. He was an Indian mathematician and astronomer. He used terms Kuttaka and Kuttakaganita to refer to algebra.

Rajatarangini is a metrical legendary and historical chronicle of the north-western part of India, particularly the kings of Kashmir. It is written by Kalhana.

Vivekadeepini is a compilation of 36 verses drawn from 67 verses of Prashnottara Ratnamalika. It is written by Adi Shankaracharya.

Priyadarsika: It is a Sanskrit play attributed to king Harshavardhan.

Bhṛigu Samhita: It is a Sanskrit astrological (Jyotisha) treatise attributed to Bhṛigu, one of the Saptarishis (seven sages) of the Vedic period.

Brihat katha: It is an ancient Indian epic said to have been written by Gunadhya.

Kirtarjuniya: It was written by Bharavi (Sanskrit poet) in the sixth century A.D.

Tanakh (the Hebrew Bible): It is the main sacred text of the Jews. It consists of three parts: Torah (the law), Nevi'im (the prophets), and Ketuvim (the writings).

Arab Religious, Social, and Political leader

Islam was founded in the 7th century by Prophet Muhammad, an Arabian merchant from Mecca. The Quran is considered the direct word of God revealed to Muhammad. Islam means "submission to God", and Mecca is the religion's holiest city.

Prophet Muhammad was born in Mecca in 570 AD. He received divine revelations that became the Quran. In 622 CE, he migrated with followers to Medina, marking the start of the Muslim calendar. At Medina, he gained more converts and formed the first umma. By his death in 632 CE, Muslim forces had conquered the Arabian peninsula.

Muhammad-bin-Qasim was an Arab military commander who led the Muslim

conquest of Sindh for the Umayyad Caliphate in 711 AD. Born in Taif, Saudi Arabia, he was the first Muslim to capture Hindu territories in India, defeating the Sindhi Brahman ruler Raja Dahir. He annexed the Sindh region and dubbed Multan the "City of Gold " after his invasion.

The Arab conquest of Sindh is briefly mentioned in Al-Baladhuri's work, while details are scarce in other early accounts. The key source on this conquest is the Persian text Chach Nama, written in 1226 CE.

The Ahmadiyya sect of Islam originated in India, founded by Mirza Ghulam Ahmad in 1889. It promoted liberal principles and the spread of Western education among Indian Muslims.

Zoroastrianism

Zoroaster (Zarathustra, Zarathushtra Spitama, or Ashu Zarathushtra) was an ancient Iranian spiritual leader who founded Zoroastrianism. The Zend Avesta, written in Old Avestan, is the sacred book of Zoroastrianism, containing the teachings of Zoroaster.

Ahura Mazda, the "Wise Lord," is the supreme creator deity in Zoroastrianism. Zoroaster's teachings emphasize "Good thoughts, Good Words, and Good Deeds." Zoroastrianism was a major religion in ancient Iran, and some Zoroastrians later migrated to India, becoming the ancestors of today's Parsis. Fire is an important symbol, with fire temples known as Atash Bahram, and the deceased are given "sky burials" in Dakhmas or Towers of Silence.

Famous Towns of Ancient India

Ujjain: It was an ancient city situated on the eastern bank of the Shipra River that served as the capital of the Avanti kingdom and later the western province of the Maurya Empire. Founded around 600 BCE, it is a major Hindu pilgrimage center known for hosting the Kumbh Mela every 12 years and the Mahakaleshwar Jyotirlinga temple.

Kodungallur: It is formerly known as Mahodayapuram, Shingly, Vanchi, and Muziris, is a historically significant town on the Malabar Coast of Kerala. It is believed that St. Thomas the Apostle landed here in 52 AD to spread

Christianity in India.

Kodungallur was known as the "Rome of India" until the 15th century and was the site of the first Christian church and Muslim mosque in India, built in 629 AD. It was also the capital of the Cherman Kings in the 7th century.

Thanjavur: It means city surrounded by cold farmlands. It gained prominence under the Cholas (9th-12th centuries), when Rajaraja I built the Brihadeeswarar Temple there. It was also the site of the Battle of Venni between Chola king Karikala and the Cheras-Pandyas.

Tamralipta (Tamralipti): It was a port city and capital of the Suhma Kingdom. It is located on the coast of the Bay of Bengal. The Tamluk town in West Bengal is generally identified as the site of Tamralipti. It was a major trade emporium during the Gupta period.

Arikamedu: It is an archaeological site in Pondicherry, identified as the ancient port of Podouke. It was known as an emporium in historical texts and located on the Ariyankuppam River.

Sringaverapura: It had an advanced water harvesting system channeling the Ganges river floods in the 1st century BC. It is mentioned in the Ramayana as the capital of the Nishadraj kingdom, and excavations found a temple of Shringi Rishi.

Pragjyotisha was the ancient name for Assam, meaning "City of Eastern Astrology", mentioned in epics (Mahabharata and Ramayana) and Puranas.

Hallur: It is an early Iron Age archaeological site in Karnataka, located on the Tungabhadra river and containing Neolithic remains of millet, mung, and other crops.

Kingdom of Kolathunadu (Kola Swarupam) : It was ruled by the Kolathiri royal family, was a powerful kingdom on the Malabar Coast during the Portuguese era. Its capital was Ezhimala, and it covered North Malabar, spanning from the Chandragiri river in the north to the Korapuzha river in the south.

Kingdom of Valluvanad : It was an independent chiefdom in central Kerala, thrived from the 12th to the 18th century. Initially part of the Chera Perumal kingdom, it later became autonomous. Situated between Pandalur Hills and

Ponnani seashore, its capital was Angadipuram, home to the historic Thirumandhamkunnu Temple.

Thekkumkur (Thekkumkooor) : It was A southern Kerala kingdom, was ruled by the Edathil Family. It spanned from the Meenachil River to the Pamba River, emerging after changes in princely states post-Chera Kulasekhara dynasty.

Medieval History

Foreign Invasions

The foreign invasions of India during the medieval period began with the attacks led by the Arabs and the Turks.

Muslim Invasions in India

Muhammad Bin Qasim- He led the first Muslim invasion of India in 711 AD, resulting in the conquest of Sindh. He was an Arab military commander in the Umayyad Caliphate. He defeated the Brahmin ruler Raja Dahir (last Hindu ruler of Sindh) in the Battle of Rewar (Battle of Aror) in 712 AD. However, he was later defeated by Bappa Rawal (Guhila dynasty), Nagabhata (Gurjara-Pratihara dynasty), and Vikramaditya II (Chalukya dynasty).

Mahmud Ghaznavi - He was the son of Sabuktigin and was renowned as the first independent ruler of Ghazni. He was a Turkish invader and the first ruler to be given the title of "Sultan." Mahmud of Ghazni attacked India seventeen times between 1000 and 1027 AD, primarily for the sake of money, under the guise of Jihad. He fought several major battles, including the Battle of Peshawar (1001 AD), the Battle of Chach (1008 AD), and the conquest of Thanesar (1014 AD) and Kannauj (1021 AD).

Mahmud's most notable invasion was the attack on the Somnath temple in 1026 AD, which was the last of his invasions outside of Punjab. He sacked the city of Somnath and destroyed the temple during the reign of Bhimdev Solanki I of the Chalukya dynasty. Mahmud of Ghazni died in 1030 AD, during his last invasion, due to malaria. He was also known as the "Idol Breaker" for his campaigns against Hindu temples.

Muhammad Ghori - He was the Sultan of the Ghurid Empire from 1173 to 1202. He was of Persian origin and established Muslim rule in India. His first invasion in India was against the Shia ruler of Multan in 1175 AD, whom he defeated. However, he was later defeated by Queen Naikidevi and Bhima II in 1178 AD.

Ghori fought two significant battles against Prithviraj Chauhan. The **First Battle of Tarain** in 1191 AD resulted in Ghori's defeat, but he returned the following year and won the **Second**

Battle of Tarain in 1192 AD, with the support of Jayachandra, the ruler of Kannauj. Ghori also defeated Jai Chandra in the **Battle of Chandawar** in 1194 AD. Ghori was eventually killed by the Khokhar tribe of the Salt Range in 1206 AD.

Malik Kafur - He invaded the Chola Empire in 1310. He was a slave and military commander of Alauddin Khalji. He led a series of expeditions in the southern part of India, against the Yadavas, the Kakatiyas, the Hoysalas, and the Pandyas.

Nadir Shah - He was an Iranian ruler and the founder of the Afsharid dynasty of Persia. He raided India during the reign of Emperor Muhammad Shah (1719-48), taking away the Peacock throne and the Kohinoor diamond. In 1739, Nadir Shah defeated the Mughal army in the **Battle of Karnal**, in which the Mughal Emperor Muhammad Shah was defeated.

Mongol Invasions in India

Genghis Khan - He was the founder of the Mongol Empire, invaded India in 1221 during the reign of Iltutmish, but his attack was prevented by the Delhi Sultan. The Mongol Empire launched several invasions into the Indian subcontinent from 1221 to 1327, and Genghis Khan also invaded the north-eastern part of Iran, Transoxiana, in 1219.

The Chagatai Khanate - It is a Mongol successor state, invaded Punjab in 1292, but Alauddin Khalji defeated and imprisoned the Mongol leader in the Battle of Kili in 1299.

Timur Lang - He was the founder of the Timurid Empire, invaded India in 1398 during the reign of Nasir-ud-Din Mahmud Shah (Tughluq), conquering several regions and appointing Khizr Khan as the governor of Lahore, Multan, and Dipalpur before leaving India in 1399.

Rajput Dynasty

The Rajputs (647 AD - 1200 AD) are divided into 36 clans, each belonging to one of three basic lineages (vanshas)- Suryavanshi, Chandravanshi, and Agnivanshi. The Suryavanshi Rajput clan is related to Surya, the Hindu God Sun, and is also known as the Solar Dynasty. The Chandravanshi Rajput Gotra are descendants of Chandra (Moon), and this lineage is known as the Lunar

Dynasty. The Agnivanshi Rajputs are descendants of Agni, the Hindu god of fire.

The Rajputs were followers of Hinduism. They also patronized Buddhism and Jainism. The Bhakti Cult was started During their period.

Chauhan Dynasty

The Chauhans of Delhi and Ajmer also known as the Chahamanas of Shakambhari, ruled between the 6th to 12th century and were members of the Agnivanshi Clan. They controlled Delhi, Ajmer, and Ranthambore, along with the Pratiharas, Paramaras, and Chalukyas being other important Rajput dynasties.

Important Rulers

Vasudeva, the founder of the Chauhan dynasty, was gifted the Sambhar Salt Lake by Vidyadhara.

Ajayaraja II (1105-1130 AD) established Ajaymeru (Ajmer) and issued silver coins in his name, while his queen Somalekha also issued coins.

Arnoraja (1130-1150 AD) defeated Turkish invaders and built the Anasagar lake in Ajmer. Arnoraja was defeated twice by Chalukya Kumarapala.

Prithviraj Chauhan III was the most famous Chahamanas ruler, also known as Rai Pithora. His famous Bundelkhand campaign against the Chandela king resulted in the death of the notable Chandela warrior, Udal, who, along with Alha, were part of the Chandela Rajput States.

In the first fight at Tarain, Ghori took over Bathinda in 1191, but Prithviraj's army, led by his prince Govind Rai, won against Ghori. But in the second battle, Prithviraj's defeat led to the end of his dynasty's power in India.

Chandellas of Bundelkhand

They originally served as feudatories to the Gurjara Pratiharas and claimed descent from sage Chandratreya. The Chandella dynasty was founded by Nannuka (831 - 845 AD). The capital city of Chandelas was Khajuraho which was later changed to Mahoba.

Important Rulers

Yashovarman (925-950 AD) - He conquered Kalinjar Hill. He also

commissioned the Lakshmana Temple at Khajuraho.

Vidyadhara (1003 - 1035 AD) - He protected his kingdom against the invasion of Mahmud Gazini.

Tomars of Delhi (8th - 12th Century)- They ruled Haryana with their capital at Dhillika (Delhi) and were the feudatory of the Pratiharas.

Important Rulers

The most important king was **Anangapala Tomara**. He established Delhi and he was described in the 11th - century inscription of the iron pillar at Mehrauli.

Anangapala II was the founder of the citadel of Lal Kot in the Mehrauli area and also built a tank known as the Anang Tal.

The famous Suraj Kund reservoir (Haryana) was commissioned by the tomar king **Surajpala** in the 10th century.

Man Singh Tomar was Tomar Rajput ruler of Gwalior. He ascended the throne in 1486 CE.

Gahadavalas of Kannauj (11th-12th century AD)

They were Suryavanshi Kshatriyas who ruled over Banaras, Ayodhya, and Kanauj.

Important Rulers

Chandradeva - He founded this dynasty and successfully wrested Delhi from the Pratiharas and the Rashtrakutas.

Vijayachandra - He successfully faced the aggressions of the Ghaznavids. During his reign, Delhi was lost, and the Tomar rulers stopped recognising Vijayachandra as their sovereign.

Jaichandra - During his reign, the Chauhans from Ajmer, under Prithviraj Chauhan, annexed Delhi. Jaichandra's grandson, Siyaji, founded the Rathore clan, which ruled the princely state of Marwar from Jodhpur.

The glory of Kannauj ended with the conquest of Iltutmish.

Paramaras of Malwa (9th - 14th century AD)

They were successors of the

Rashtrakutas, who controlled Malwa from the early ninth century.

Upendra or Krishnaraja founded the Paramara dynasty, with branches ruling in Rajputana, Mount Abu, Bhinmal, and other regions. Initially centered in Ujjain, their capital shifted to Dhara later. The last Paramara king, Mahalakadeva, met his demise at the hands of Alauddin Khalji's forces from Delhi.

Senas Dynasty

The Senas dynasty ruled Bengal during the 11th and 12th centuries, safeguarding the region from anarchy after the fall of the Pala Dynasty. The dynasty originated from the southern Indian region of Karnataka, with Vijayasena deposing Madanapala, the last Pala ruler, to establish the Sena dynasty.

Ballala Sena, Vijayasena's son, instituted social reforms known as Kulinism in Bengal. Lakshmana Sena, the greatest ruler of the dynasty, succeeded Ballala Sena.

Maharajas of Jaipur

Maharaja Sawai Madho Singh II was the Maharajadhiraj of Jaipur from 1880 to 1922, being the adopted son of Raja Ram Singh II. Sawai Raja Jai Singh founded the new capital of Jaipur and was given the subedari of Agra in 1722. He established five Jantar Mantar observatories between 1724 and 1735.

The Hawa Mahal was built by **Maharaja Sawai Pratap Singh**, the grandson of Maharaja Sawai Jai Singh, in 1799. Maharaja Man Singh was the last ruling Maharaja of Jaipur State, ruling from 1922 to 1949.

Literature during the Rajput Dynasty

Chand Bardai, the court poet of Prithviraj Chauhan, wrote the Prithviraj Raso, detailing the military exploits of the Chauhan ruler. The Prithviraj Vijaya was authored by Jayanak.

Bhaskara Charya wrote the Siddhanta Shiromani, a treatise on astronomy. Other notable literary works from this period include the Gita Govinda by Jayadeva, Keerthanarjuneeya by Bharavi, Ravanavada by Bharthruhari, and Kavya Meemamse by Rajashekhara.

Kalhana's Rajatarangini and Hemachandra's Kumarapla charitha are among the other significant works associated with the Rajputs. Bilhana, the court poet of Vikramaditya IV of the Chalukya dynasty, wrote the biography 'Vikramankadeva Charitha'.

Kingdom of Mewar

The Sisodia dynasty, also known as the "House of Mewar," governed the Kingdom of Mewar (Udaipur State). They belonged to the Suryavanshi lineage.

Famous Rulers of Mewar

Maharana Hamir Singh (1326-1364)- He was the founder of this dynasty. He was the first ruler who took the title of Rana. He built the Annapurna Mata temple at Chittorgarh Fort. In the Kirti Stambh inscriptions he was titled as Visham Ghati Panchanan.

Maharana Kumbha (1433-1468) - He won the Battle of Sarangpur against Mahmud Khilji. He was an accomplished veena player who patronized musicians and artists in his court. Some of his music texts - "Sangeet raj," "Sangeet mimansa," "Sangeet ratnakar," and "Shudrabandha".

Maharana Sanga (1509-1528) - He was the son of Maharana Raimal and grandson of Rana Kumbha. He defeated Ibrahim Lodhi at the Battle of Khatoli in 1517.

Maharana Pratap Singh (1540- 1597)- He was the son of Uday Singh. The Battle of Haldighati was fought between Maharana Pratap Singh and the Mughal Army under the leadership of Man Singh on 18 June 1576. James Tod gave him the title of 'Leonidas of Rajasthan'.

Amar Singh I (1597-1620) - He was the eldest son of Maharana Pratap. Amar Singh defeated the Mughal commander Sultan Khan in the Battle of Dewair. James Tod called this war the marathon history of Marwar. Later He accepted Mughal services during the reign of Jahangir.

Rao Jodha - He founded the city Jodhpur in 1459. The city is named after him. It was previously known as Marwar.

Bhagwat Singh - He was the last ruler of the marwar dynasty. He ruled after the independence of India.

Slave dynasty (Mamluk Dynasty)

The Slave Dynasty (Mamluk Dynasty) was the first dynasty of the Delhi Sultanate. It was also the first Muslim dynasty to rule India. It governed the Delhi Sultanate from 1206 to 1290.

Qutb-ud-Din Aibak (1206-1210 AD) - He was the founder of the slave dynasty, establishing Turkish rule in India. He earned the title Lakh Baksh and Quran Khan. Muhammad Ghorī appointed him commander of Indian territories after the Second Battle of Tarain in 1192. He fought Taj al-Din Yildiz for control of India. Aibak passed away due to injuries from falling off his horse during a Chaugan game. He appointed Iltutmish as the first Muslim governor of Badaun.

Aram Shah succeeded him briefly. His notable military feat was Bakhtiyar Khilji's conquest of Bengal and Bihar in 1197. He built two mosques, one in Delhi (Quwwat ul Islam) and another in Ajmer (Adhai Din Ka Jhonpra). He laid the foundation of Qutub Minar after the name of famous Sufi saint Khwaja Qutubuddin Bakhtiyar Kaki.

Iltutmish (1210 - 1236) - He was a slave and son-in-law of Qutubuddin Aibak, considered the real founder of the Delhi Sultanate. He moved the capital from Lahore to Delhi and was from the Ilbari tribe, leading to his dynasty being named the Ilbari dynasty.

He was given the titles 'Sultan-i-Azam' and 'Nasir-Amir-al-Mommin' by the Caliph of Baghdad in 1229. He completed the Qutub Minar, the tallest stone tower in India, and built a magnificent mosque in Ajmer, making Delhi the "second Baghdad."

Iltutmish organized his nobles into a group of forty known as the Turkan-i-Chahalgani or Dal Chalisa. He introduced the Iqtadari system, dividing the kingdom into Iqtas assigned to nobles in exchange for a salary, which influenced later kingdoms.

Iltutmish captured Bihar in 1210, invaded and annexed Bengal in 1230, and reoccupied Kanauj and Benaras in 1231. He also resisted an invasion by Genghis Khan in 1221 and annexed Malwa and Ujjain to his empire in 1239.

Rukn-ud-Din Firoz was a ruler of the

Delhi sultanate for less than seven months in 1236.

Razia Sultana (1236 - 1240) - She was the daughter of Iltutmish and the first woman ruler of India. She was named Raziya-ud-din and was married to Malik Altunia, the governor of Bathinda. Razia Sultana was killed by her brother's forces, and her brother Muizuddin Bahram Shah succeeded her. She was buried in Kaithal, Haryana, and her tomb is made of baked bricks.

She was imprisoned by Malik Ikhtiyar-ud-din Altunia in the Qila Mubarak Fort (Bathinda). The governors of Multan, Badaun, Hansi, and Lahore openly revolted against her, and Wazir Muhammad Junaidi and many other nobles refused to accept a woman as the ruler. The book "Razia Queen of India" was written by Rafiq Zakaria about her life.

Minhaj-i-Siraj (Persian historian) wrote about Raziya that "she was more able and qualified than all her brothers".

Bahram Shah (1240 - 1242) - He ruled briefly after Razia's time. In 1241, Mongols invaded India, reaching Lahore. Bahram Shah was killed in 1242. Following him, Masud Shah (1242 - 1246) became Sultan but was later overthrown by Balban with the help of Nasiruddin Mahmud's mother, Malik-i-Jahan.

Nasiruddin Mahmud (1246 - 1266), another grandson of Iltutmish, came next. Balban, appointed as Prime Minister, wielded significant power during Nasiruddin Mahmud's reign. Mahmud died in 1266, and Balban took over as Sultan.

Ghiyas ud din Balban (1266 - 1287), a former slave of Iltutmish, rose to power as the ninth sultan. He was given the position of Amir-i-Shikar (Lord of the Hunt) under the reign of Raziya Sultana. He was given the titles of "Ulugh khan", "Naib-i-mamlikat", "Niyabat-i-Khudai" and "Zil-i-Ilahi".

He brought the practices of Sijda (kneel down and touch the ground with their heads) and Paibosi (kissing Sultan's feet) to India. He defeated the Chahalgani and brought the Persian festival Navroz to India. Tughril Beg revolted against him and declared himself ruler of Bengal in 1279.

Balban proclaimed his grandson Kaiqubad as his successor and died in 1287. Kaiqubad was struck with paralysis and was finally killed in a pitiable manner by the Khilji Maliks. The rule of the Ilbari dynasty came to an end.

Khilji Dynasty (1290 to 1320)

Khajis were of mixed Turkish-Afghan origin. They came to India at the time of the Ghurid invasion and moved to Bengal and Bihar, searching for an opportunity for advancement.

Jalaluddin Khilji (1290-1296 AD) - He was the founder of Khilji Dynasty. He was also called as "Clemency Jalal-uddin" as he followed peace and wanted to rule without violence. He suppressed Malik Chhaju's revolt at Kara. He defeated the Mongols. He was assassinated by Ala-ud-din Khilji.

Alauddin Khilji (1296-1316 AD) - He succeeded Jalal-ud-din Firoz Khilji and was appointed governor of Kara in 1291. He defeated the Mongols near the Ravi river in 1306, with Zafar Khan, Ulugh Khan, and Malik Kafur as his military commanders. His court poet was Amir Khusro.

He implemented several reforms during his reign. He made nobles pay land tax for their holdings, curbing their excess wealth, and imposed detailed regulations on markets, appointing intelligence officers and spies. In the agrarian sector, he brought villages under Khalisa, fixed land revenue at half the produced grain, and imposed taxes on residences and grazing.

To prevent revolts, Ala-ud-din abolished the Iqta system, imposed a 50% kharaj tax, restructured the spy system, and introduced a permanent standing army.

He started the system of branding horses (dagh) and a descriptive roster of individual soldiers (Chahra) to inhibit corruption. He also banned the sale and consumption of alcoholic beverages and seized jagirs, estates, and religious endowments, appointing the Amir-i-Tuzuk to oversee administrative matters.

Agrarian Reforms- The villages were brought under Khalisa (land not assigned to any of the nobles as iqta). The three types of taxes were Ghari (tax on

residence), Jizya (tax imposed on the non-muslims) and Charai (grazing tax).

Expeditions of Alauddin Khilji

Gujarat Expedition (1299 AD)- Alauddin's forces, led by Ulugh Khan and Nusrat Khan, defeated Karan Baghela of Gujarat, acquiring Malik Kafur during the campaign.

Ranthambore Expedition (1301 AD)- Alauddin's chief-in - commander Ulugh Khan and Nusrat Khan conquered Ranthambore, ruled by Hindu King Hamir Deva, resulting in a jauhar by Rajput ladies.

Chittor Expedition (1303 AD) - Following Ranthambore, Alauddin attacked Chittor, led by Rana Ratan Singh, renaming it Khizrabad and assigning power to his son Khizra Khan.

Devagiri Expedition (1308 AD) - Under Malik Kafur, Alauddin's forces attacked Devagiri, ruled by Ramchandra Deva Yadava, obtaining the title Rai Raiyana for Ramchandra.

Warangal Expedition - Malik Kafur confronted Kakatiya King Pratap Rudra Deva, promising recognition and taxes to the Sultanate upon defeat.

Successors after Ala-ud-din-Khilji- Qutb-ud-din Mubarak Shah (1316-1320 A.D.), Nasir-ud-din Khusrav Shah (1320A.D.).

Tomb - The tomb of Alauddin Khilji is located inside the madrasa in the Qutb Minar complex, Mehrauli, Delhi.

Decline of the Dynasty

Ala-ud-din Khilji died in 1316 A.D. The Governor of Punjab Ghazi Malik led a group of nobles, conquered Delhi and captured the throne in 1320 A.D.

Tughlaq Dynasty (1320 - 1414 AD)

It was the third dynasty of the Delhi sultanate. The dynasty reached its pinnacle between 1330 and 1335 AD during the period of Muhammad Bin Tughlaq.

Famous Rulers

Ghiyas-ud-din Tughluq, also known as Ghazi Malik, founded the Tughlaq

dynasty. Ghiyas-ud-din, once the governor of Dipalpur, also founded Tughlaqabad city and changed the land revenue system, reducing taxes to one-third of the produce.

He prioritized infrastructure projects like irrigation and built the Tughlaqabad fort. Additionally, he improved postal services, agriculture, and law enforcement.

He rebelled against the last Khalji Sultan Nasiruddin Khusrau and expanded his rule to Bengal, Orissa, and Warangal. Sadly, he died due to a pavilion collapse while returning from Bengal. This event, orchestrated by his son Jauna Khan, led to suspicions of foul play.

Mir Imard Ahmad, who built the pavilion, was believed to be involved in Ghiyas-ud-din's demise.

His son Jauna Khan led expeditions against Prataparudra II of Warangal, capturing and imprisoning him. However, Prataparudra tragically committed suicide while being taken to Delhi, leading to the annexation of Kakatiya territories by the Tughlaq empire.

Muhammad-bin-Tughlaq (Jauna Khan) - He was proficient in Arabic, Persian, and various sciences, including philosophy, astronomy, and mathematics. Under his rule, a new department for agricultural development, Diwan-i Kohi, was established. He even appointed the Moroccan traveler Ibn Batutah as the chief Qazi of Delhi and dispatched him on a diplomatic mission to China in 1342.

During his reign, Muhammad-bin-Tughlaq embarked on five significant endeavors. Notably, he relocated the capital from Delhi to Daultabad (Devagiri) in 1327 to shield it from Mongol invasions. However, his attempt to introduce copper currency failed when its value plummeted due to easy replication. His initiative to issue leather coins also met with limited success in India.

In 1332-1333 A.D. He planned an ill-fated expedition to Khurasan, which ended in failure. Similarly, the Qarachil expedition in 1333 A.D. targeting the Kullu-Kangra region was unsuccessful due to the detachment of commanders from the main defending forces. Additionally, he imposed higher taxes in the Doab region and appointed individuals from unconventional backgrounds, like Aziz Khummar, Firuz Hajjam, Mank a Tabbakh,

and Ladha and Pira, to positions in his court. Muhammad-bin-Tughlaq passed away in March 1351.

Firoz Shah Tughlaq, a Muslim ruler of Turkic descent, governed the Sultanate of Delhi from 1351 to 1388. He bestowed the title Khan-i-jahan (true ruler) upon his commander Malik Maqbul. Firoz Shah took measures to alleviate the burdens on the populace by reinstating rent-free lands to theologians, scholars, and the less privileged. He also abolished twenty-three taxes, including unpopular levies like grazing and house taxes, retaining only four in accordance with Islamic law- Kharaj, Zakat, Jizya, and Khums.

Introducing an irrigation tax amounting to ten percent of the produce from lands irrigated by state-built canals, Firoz Shah sought to enhance state revenue. Additionally, he rescinded all agricultural loans granted by Mohammed - bin - Tughlaq and increased the salaries of revenue officers. Despite his efforts to streamline taxation, he imposed Jaziya on Brahmins. Moreover, he established social welfare institutions such as hospitals (Dar-ul-shafa), to care for orphans and widows (Diwan-i-khairat), and an Employment Bureau, alongside Diwan-i-Ishtiaq to provide financial aid to the impoverished.

In terms of military campaigns, Firoz Shah led expeditions to Bengal in 1353 A.D. and 1359 A.D., capturing Jainagar and ravaging the Jagannath Temple at Puri. He also undertook urban development projects, founding towns like Firozabad, Hissar, Jaunpur, and Fatehabad. Under his rule, Khawaja Hisamuddin conducted a comprehensive revenue survey.

Futuh-i-Firoz Shahi is an autobiography of Firoz Tughlaq. He patronized scholar Zia-ud-din Barani.

Later Successors- Ghiyas-ud-din Tughlaq Shah II, Abu Bakr Shah, Alauddin Sikander Shah, Nasir-ud-din Muhammad Tughlaq.

Decline

By the end of the 14th century, most of the territories became independent. Only Punjab and Delhi remained under the Tughlaqs. Timur's (1398 A.D.) invasion further devastated the dynasty.

Sayyid Dynasty (1414 to 1451)

The Sayyid dynasty, ruling from 1414 to 1451, marked the fourth dynasty of the Delhi Sultanate. Descendants of the Prophet Muhammad, they came to power during a period when the central authority of the Delhi sultanate was weakened by the invasion of Timur in 1398 AD.

Timur appointed Khizr Khan as the ruler of Multan, who later defeated Sultan Daulat Khan and established the Sayyid dynasty after occupying Delhi.

Khizr Khan ruled independently for seven years without taking the title of sultan. His conquests included Gujarat, Bayana, and Gwalior. He was succeeded by his son Mubarak Shah, who ruled from 1421 to 1434.

Mubarak Shah, as he came to be known, proved to be a competent military leader, suppressing revolts and expanding the kingdom's territories. However, he was assassinated in 1434 by his own nobles.

Following Mubarak Shah's death, Muhammad Shah ascended the throne, but real power lay with Sarwarul Mulk, the Wazir.

Ala-ud-din Shah, Muhammad Shah's son, also known as Alam Shah, became the last ruler of the Sayyid dynasty. However, his reign was marked by incompetence, leading to the dynasty's decline.

Decline

Economically, trade suffered during the Sayyid rule, worsened by the corn taxes imposed and later removed by the Lodhi dynasty. Internal revolts in various provinces further weakened the Sayyids, eventually leading to Bahlul Lodhi's invasion of Delhi. With the help of Sayyid wazir Hamid Khan, Lodhi took over, marking the end of the Sayyid dynasty and the rise of the Lodhis in Delhi.

Lodi Dynasty (1451 - 1526)

It was the fifth and last dynasty of the Delhi Sultanate. It was founded by Bahlul Khan Lodi in 1451.

Rulers

Bahlul Lodi (1451-1489 A.D.) - He

ascended the throne in 1451 and assumed the title "Bahlul Shah Ghazi". He conquered Gwalior, Samthal, Sakit and Mewat. He was awarded the title of Islam Khan and appointed governor of Sirhind in 1419. He annexed Jaunpur to the Delhi Sultanate.

Sikandar Khan Lodi (1489-1517) - Bahlul's third son, took the title Sikandar Shah. He founded Agra in 1504. He shifted capital from Delhi to Agra. He developed agriculture and industry. Sikandar Shah enjoyed "Shehnai" Music. A music work "Lahjati-Sikandar Shahi" was created during his reign. He was a renowned poet who wrote under the pen name Gulrukhi.

He annexed Bihar. He destroyed Hindu Temples in Mathura and Naga Port. He imposed Jizya on Hindus. He attempted to conquer Gwalior Fort five times but was defeated each time by Raja Man Singh. He introduced the land measuring system Gaz-i-Sikandari. He died in 1517 and was succeeded by his son, Ibrahim Khan Lodi.

Ibrahim Lodi (1517-1526 A.D.) - The last sultan of the Lodi dynasty, he spent most of his reign at war with both the Afghans and the Mughal Empire, ultimately perishing in an attempt to prevent the dynasty's extinction.

The Battle of Khatoli, fought in 1517 between the Lodi dynasty (under Ibrahim Lodi) and the Kingdom of Mewar (under Rana Sanga), saw Rana Sanga of Mewar defeating Ibrahim Lodi and imprisoning him. He was later released by Rana Sanga after collecting ransom.

Decline

Daulat Khan Lodi, the governor of Punjab, backed by Afghan nobles, led a movement against Sultan Ibrahim Lodhi. He, along with Alak Khan Lodi, invited Babur to overthrow Ibrahim Lodhi. Babur accepted the offer and defeated Ibrahim Lodhi in the **First Battle of Panipat (1526)**. Thus ending the Delhi Sultanate

Administration and Architecture of Delhi Sultanate

Central Administration

The Sultan was aided in administration by a council of ministers (Majlis-i-Khalwat), personally selected by

him and serving at his discretion

Diwan-i-Wizarat - Supervised by the wazir, overseeing all departments.

Diwan-i-Risalat - Managed justice administration, led by the Sadr-us-Sadr, who also served as Qazi-i-mumalik.

Diwan-i-Ariz- Established by Balban, responsible for military organization, led by Ariz-i-Mumalik.

Diwan-i-Mazalim- Headed by Amir-i-Dad, supervised qadis, kotwal, and muhtasib in the Sultan's absence.

Diwan-i-Bandagan- Managed the Department of Slaves.

Diwan-i-Ishtiaq- Handled pension affairs.

Diwan-i-Khairat- Created by Firuz Shah Tughluq for charity purposes.

Diwan-i-Mustakhraj- Formed by Alauddin Khilji to investigate revenue arrears.

Diwan-i-Insha- Led by the Dabir-i-Khas, responsible for royal correspondence.

Diwan-i-Amir-Kohi- Introduced by Muhammad bin Tughlaq, focused on agricultural improvement.

Qazi- Legal officer responsible for dispensing Sharia law.

Mustaufi-i-Mumalik- Auditor General overseeing expenditure.

Mushrif-i-Mumalik- Accountant General managing income.

Majmudar- Keeper of loans and balances from the treasury.

Dahir-i-mumalik - Head of Royal correspondence.

Qazi-ul-kazat - Head of chief judicial department. **Amir-i-Akhur** - Commanding Officer of Royal Horses. **Shahna-i-pil** - Superintendent of Royal Elephants.

Amir-i-bahr - Officer in charge of police and naval port.

Officers at Provincial Level

Muqti or Wali - Governor of provinces Responsible for law and order and collection of taxes in their iqtas. **Nazir or Wakuf** - Officials responsible for collecting revenue in the provinces.

Sahib-i-diwan - He maintained the financial accounts of the provincial income and expenditure

Territorial Administration

The empire was divided into Sarkar. A group of shiq formed Sarkar. Shiqs were administered by Shiqdars. Shiq was divided into Praganas (sub-district). It was further divided into Gram (villages). The village was the basic unit of administration. Gram was headed by Muqaddam.

The Architecture of the Delhi Sultanate

Adhai Din Ka Jhonpra- Constructed by Qutub ud-din Aibek in Ajmer in 1192 CE, designed by Abu Bakr of Herat.

Moth Ki Masjid- Built by Miya Bhoiya in 1505 during Sikandar Lodi's reign.

Jammat Khana Masjid- The oldest mosque in Delhi, erected by Khizr Khan in 1315-1325 A.D.

Quwwat-ul-Islam Mosque- Located within the Qutub Minar complex, built by Qutb-ud-din Aibak.

Tomb of Ghiyasuddin Tughlaq- Constructed in 1321-25 AD, featuring red sandstone and white marble.

Alauddin Khilji's Tomb- Located at the Qutb Minar complex, built by Alauddin Khilji in 1315.

Tughlaqabad Fort- Erected by Ghiyasuddin Tughlaq in 1321.

Khirki Mosque- Built by Khan-i-Jahan, the prime minister of Feroz Shah Tughlaq.

Lodhi Garden- Originally built by Ibrahim in 1517, redesigned by J.A. Stein in 1968.

Jama Masjid- Constructed by Shah Jahan in the 17th century, also known as Masjid-i-Jahan Numa.

Hauz Khas Complex- Historic site built during Alauddin Khilji's reign.

Begumpuri Masjid- Constructed by Muhammad Tughluq for the city of Jahanpanah.

Tomb of Safdarjung- Built in New Delhi in 1754 by Nawab Shujaud Daula.

Feroz Shah Kotla- Constructed by Feroz Shah Tughlaq due to water shortage in Tughlakabad.

Shish Gumbad- Built during the Lodhi dynasty's reign, featuring a square dome.

Hauz-i-Shamsi- Water reservoir constructed by Iltutmish in 1230 CE.

Jahaz Mahal- Constructed by Ghiyas-ud-din Khilji in Mandu, Madhya Pradesh.

Fort of Siri- Built during Alauddin Khilji's reign, situated between Hauz Khas and Mehrauli.

Poets

Amir Khusro - He served mainly in the court of Alauddin Khilji. He wrote the famous poem 'Hasht - Bihisht'.

Fakhr-i-Mudabbir - He was a contemporary of Iltutmish. His famous works are the Shajara-yi-ansab ("The tree of genealogies") and the Adab al-harb wa-l-shaja'a ("The etiquette of war and valour").

Other Facts of Delhi Sultanate

The sultan used titles like "Sikandar-i-Sani" (Second Alexander) and "Zil-e-Ilahi" (Shadow of God) to emphasize their authority and divine right to rule.

Persian was the language of the Delhi Sultanate, with coins minted known as dehlwial.

'Ulema' during this period referred to learned theologians and jurists.

Zakat Tax, a religious levy, targeted affluent Muslims.

The Double Dome, characteristic of Indo-Islamic architecture, was introduced by the Lodhi Dynasty.

Bhakti Movement

Bhakti, originating in 7th - 12th Century Kerala and Tamil Nadu, later spread to Karnataka, Maharashtra, and reached North India by the 15th Century. It's rooted in the Sanskrit word "bhaj," meaning participation or sharing, and is a path to salvation. Mentioned in ancient texts like the Shvetashvatara Upanishad, Katha Upanishad, and Bhagavad Gita, as well as the Narada Sutra and Shandilya Sutra, it's discussed alongside 'GyanMarg' and 'Karma Marga'.

Bhakti tradition has been classified into two main categories-

Nirguna School - Worship of an abstract form (nirankar) of God. It was introduced by Adi Shankara. Some philosophers of this thought are Kabir, Guru Nanak, and Dadu Dayal.

Saguna Bhakti - The emphasis is on worshipping human-like figures of deities and their avatars such as Rama and Krishna. Some philosophers are Ramanuja, Nimbaraka, Madhva, Vallabha, Meera Bhai, Chaitanya Mahaprabhu, Tulsidas, and Surdas.

Bhakti Movement in South India

During the Bhakti Movement in South India, two main groups emerged- the Nayanars, devoted to Shiva, and the Alvars, devoted to Vishnu. They opposed Jainism, Buddhism, and Brahmanical dominance, advocating personal devotion to God as the path to salvation.

The Alvars - Its meaning "those steeped in God," were Vaishnava poet-saints who praised Vishnu and Krishna. Comprising 12 saints, they wrote hymns collected in the Divya Prabandha, celebrating Vishnu's avatars and the "Divya Deshams," or Vaishnavite deities' abodes. Andal, the sole female Alwar, is revered as the 'Meera of the South' and considered the incarnation of Bhudevi.

Nayanars - Originating as a group of 63 Tamil saints devoted to Shiva, the Nayanars' lives are chronicled in the "Tevaram" or Dravida Veda, emphasizing their devotion to Lord Shiva.

Famous Bhakti Saints

Ramanuja (1060-1118 AD) - A Tamil Brahmin and prominent Vaishnavist influenced by the Alvars, he developed the philosophy of Vishishtadvaita, emphasizing salvation through Karma, Gyan, and Bhakti.

Madhvacharya (1238-1317 AD) - Founder of Dvaita Vada school of Vedanta, emphasizing the dualism of Jivatma and Paramatma, and the worship of Vishnu.

Vallabhacharya (1479-1531 AD) - Known as Achinto Bhadra, he founded Shuddhadvaita and the Rudra Sampradaya school, with notable works including Patravallamban and

Madhurashtakam.

Dnyaneshwar (Jnanadev) - An early Maharashtra Bhakti saint, known for his commentary Dnyaneshwari on the Bhagavad Gita and Amrutanubhav on Yoga.

Eknath - A Maharashtrian saint, known for writing Eknathi Bhagavatam and promoting the Varkari Panth.

Shankaracharya - Credited with Advaita Vedanta doctrine, he founded mathas across India.

Nimbaraka - A disciple of Ramanujacharya, he founded the Sanak Sampradaya and introduced the worship of Radha Madhav.

Basavanna - A 12th-century Lingayat saint known for promoting Visishtadvaita and founding Anubhava Mantapa.

Ramananda - Considered the first Bhakti saint in North India, with notable disciples including Kabir and Ravidas.

Kabir - A 15th-century poet-saint reconciling Hinduism and Islam, advocating devotion to a formless God. He was born in Kashi (Uttar Pradesh). His verses are found in Guru Granth Sahib.

Chaitanya Mahaprabhu - A Bhakti reformer from Bengal known for introducing Kirtans and preaching Vaishnavism.

Tulsidas - He was a contemporary of Akbar and wrote 'RamcharitManas' in the language of Avadi. His other writings are the Dohavali, Sahitya ratna or Ratna Ramayan, Gitavali, Krishna Gitavali or Krishnavali and Vinaya Patrika.

Dadu Dayal - A poet-saint from Gujarat, preaching Nirgunabhakthi and non-sectarian devotion.

Shankar Dev - Introduced Vaishnavism in Assam and contributed to various art forms.

Purandaradasa - Known as the Father of Carnatic music, he emphasized devotion to Lord Krishna.

Gnaneshwar - Early Maratha reformer known for commentary on Bhagavad Gita.

Manikkavachakar - A Tamil devotee of Shiva, known for composing devotional songs.

Namdev - Promoted cultural unity among Marathas and emphasized equality.

Tukaram - A Marathi Bhakti reformer preaching universal brotherhood.

Samard Ramdas - Guru of Shivaji and devotee of Lord Rama, author of Dasabodha.

Baba Dayal Das - Founder of the Nirankari Movement in Punjab.

Sunderdas - A disciple of Dadu Dayal known for his compositions.

Appar - A Tamil devotee of Shiva, also known as Tirunavukkarasar.

Surdas - A Vallabhacharya devotee and composer of devotional songs in Braj and Awadhi.

Jaidev - Court poet of Bengal known for composing Geet Govind.

Mukund Das - Disciple of Ramanand Swami, renowned for his Vedas and music knowledge.

Swami Haridas - A spiritual poet and musician devoted to Radha-Krishna.

Meerabai - Rajasthan Bhakti saint known for her devotion to Lord Krishna.

Karaikal Ammaiyar - A devotee of Shiva, one of the 63 Nayanmars.

Akkamahadevi - A passionate Shiva devotee from Karnataka.

Janabai - A lower caste devotee and poetess who worked for Namdev.

Veerashaivas - The 12th-century Veerashaiva movement, led by Basavanna, advocated worship of the Linga, symbolizing Lord Shiva. Also known as Lingayats.

Sufi Movement

Sufism, known as "tasawwuf" in Islamic texts, began in Persia and entered India in the 11th Century. The term 'Sufi' originates from the Arabic word 'suf,' meaning 'one who wears wool.'

Abd-Allah Ibn Muhammad ibn al-Hanafiyyah is credited as the first person referred to as a "Sufi."

Sufi philosophy progresses through four stages- Shariat (outward religion or law), Tariqa (the path), Haqeeqat (the Inner truth), and Maarifat (divine knowledge).

Silsila

The Silsila, a golden chain of knowledge and religious orders passed down from Prophet Muhammad, typically led by a notable mystic residing in a Khanqah or hospice. In the 12th century, Sufis organized into 12 orders, with the Chisti, Suhrawardi, Naqshbandi, and Qadri Silsilas being the most prominent.

Chisti Silsila - Founded by Khwaja Moin-Ud-din Chishti (Gharib Nawaz) in India, emphasizing the unity of being with God (Wahdat Al-wujud) and renouncing materialism.

Suhrawardi Silsila - Established by Shihabuddin Suhrawardi in Baghdad, it entered India through Bahauddin Zakariya.

Naqshbandi Silsila - Introduced to India by Khwaja Bahauddin Naqshbandi, propagated by Sheikh Baqi Billah and Sheikh Ahmad Sirhindi. Focused on adherence to Shariat and practiced silent meditation, earning the moniker "silent Sufis."

Qadri Silsila - Popular in Punjab, established by Sheikh Abdul Qadir of Badaun. Supported by Akbar, notable figures include Shah Badakhshani, with Mughal princess Jahanara and her brother Dara among its disciples.

Important Sufi Terms

Sufi, Murshid - Saint. **Murid** - Followers.

Khalifa - Disciples. **Zikr** - Recitation of God's name. **Tauba** - Repentance. **Fanaa** - Spiritual merging with the Almighty.

Sama - Musical gathering. **Urs** - Death.

Raqs - It refers to Dancing.

Places of Significance

Idgah - A designated area for public prayers, especially on Muslim festivals.

Pir - Spiritual guides and philosophers of Sufism, respected for their wisdom and guidance.

Dargah - Shrine or tomb built over the grave of a revered religious figure, often a Sufi saint.

Khanqah - A gathering place for Sufi brotherhood and seekers, where spiritual discussions and practices take place. Also known as Ribat.

Famous Sufi Saints

Khwaja Muinuddin Chishti - Founder of the Chishti Order, emphasizing unity with God. Popularized Sama (music recitations). Notable disciples include Khwaja Qutbuddin Bakhtiyar Kaki, Fareeduddin Ganj-e-Shakar, Nizamuddin Auliya, and Naseeruddin Charagh. Shrine located at Ajmer Sharif.

Amir Khusrau - Born in 1253 in Delhi, known as the "Parrot of India". Pioneered Qawwali and Urdu Literature. A favorite disciple of Sufi saint Nizamuddin Auliya.

Bahauddin Zakariya Suhrawardi - Leading Sufi saint in Northwestern India, part of the Suhrawardi silsila. Mausoleum in Multan, died in 1267.

Sheikh Salim Chisti - Renowned Sufi saint of the Chishti Order during Emperor Akbar's era. Akbar named his first son 'Salim' in his honor. Tomb constructed at Fatehpur Sikri by Akbar.

Sufi Saint Abdul-Karim Abd-ur-Razak - Known as Sheikh Chilli, a Qadiriyya Sufi and master of Mughal Prince Dara Shikoh. His tomb, known as the 'Taj Mahal of Haryana', is located in Kurukshetra.

Khwaja Fariduddin Masud - He was known as Ganjshakar (Baba Farid). He built his Khanqah at Ajodan, Punjab, and was the Punjabi poet of Sufism. His Dargah is situated in Ajodhan.

Ali al Hujwiri - First great Sufi saint to visit India, known as Data Ganj Bakhsh. Disciple of Mohammed al Hasan al Khuttali. Spread Islam in South Asia.

Vijaynagar Empire

Founded in 1336 by Harihara and Bukka, blessed by Guru Vidyaranya, on the south bank of the Tungabhadra River. Hampi served as its capital, known as the Karnata Kingdom. Constant conflicts with the Bahmani kingdom over fertile regions like Raichur doab and Tungabhadra doab. Rivalry with the Sultans of Madurai in the south. Contemporary accounts describe it as Karnataka Samrajyamu.

Capitals of Vijayanagara Empire- Vijayanagara (1336–1565), Penukonda (1565–1592), Chandragiri (1592–1604), Vellore (1604–1646).

Vijayanagar Empire was ruled by **four**

dynasties- Sangama, Saluva, Tuluva, and Aravidu.

Sangama Dynasty

It was the first dynasty of the Vijayanagara Empire, with its capital at Vijayanagara.

Notable Rulers (Reign) -

Harihara I (1336–1356) founded the Vijayanagara Empire and the Sangama Dynasty in 1336. He was from the Kuruba clan and built a fort at Barkuru. He ruled over the northern parts of the Hoysala Empire.

Bukka I (1356-1377) served in the Warangal army earlier. He fought with the Bahmani Sultan and signed a treaty. His son Kumara Kampana led the Madurai campaign in 1371.

Harihara II (1377-1406) had titles like 'Vedamarga Pravartak'. The poet Madhura was in his court.

Deva Raya I (1406-1422) was known for military campaigns and irrigation projects.

Deva Raya II (1423-1446), titled 'Gajabetakara' or 'Elephant Hunter', was an able administrator, warrior and scholar. He wrote works like Sobagina Sone and appointed Lakkanna to regulate trade. He expanded the empire greatly. His commander was Lakkanna Dandesha. He recruited a large number of Muslim archers in the army.

Decline of Sangama dynasty

After Devaraya's prosperous rule ended in 1446, succession issues arose, leading to Vijayalaya II briefly taking the throne before Mallikarjuna's accession around 1447. Mallikarjuna's reign weakened the dynasty, exploited by Alauddin II of the Bahmani Sultanate and Kapileswara Gajapati of Orissa. Eventually, Virupaksha II became the final ruler of the Sangama Dynasty.

Saluva Dynasty

It was founded by Narasimha Saluva. It was the second dynasty of the empire. They ruled over the whole of South India. Narasimha Raya II (Immedi Narasimha) succeeded eldest brother Thimma Bhupala as the last ruler of the Saluva Dynasty. Rulers- Narasimha Deva Raya

(1485–1491 CE), Thimma Bhupala (1491 CE), Narasimha Raya II (1491 - 1503).

Tuluva Dynasty

Founded by Tuluva Narasa Nayaka, the Tuluva Dynasty marked the third era of the Vijayanagar Empire. Its most renowned ruler, Krishna Deva Raya (1509-1529 AD), known as Andhra Bhoja, expanded his empire from the river Krishna to the Cauvery in the south, and from the Arabian Sea to the Bay of Bengal. Krishna Deva Raya was a prolific writer, producing works such as "Amuktamalyada" and "Jambavati Kalyanam."

His reign witnessed the construction of significant landmarks like the Hazara Rama temple and the Vittala temple. He established Nagalapuram and patronized scholars in Telugu, Kannada, Sanskrit, and Tamil. Notable among his court scholars were the "Ashtadiggajas." Under Krishna Deva Raya's leadership, the Battle of Raichur in 1520 saw the defeat of the Sultanate of Bijapur.

His successor, Achyuta Deva Raya (1529-1542 CE), continued his legacy, while Sada Siva Raya (1542-1570 CE) ruled in the later years of the 16th century.

Aravidu Dynasty

It was the fourth and last Hindu dynasty of Vijayanagara Empire in South India. Its founder was Tirumala Deva Raya, whose brother Rama Raya had been the masterful regent of the last ruler of the previous dynasty.

Decline

The Battle of Talikota in 1565 marked the downfall of the Vijayanagara Empire. Aliya Rama Raya, its king, was defeated and killed by an alliance of Deccan sultanates, including Ahmednagar, Bijapur, Golconda, and Bidar. This defeat ultimately led to the political collapse of the empire.

Foreign Travelers - During the Reign of

Ibn Battuta (Morocco) - Harihara I

Nicolò de Conti (Italy) - Deva Raya I

Abdur Razzak (Persia) - Deva Raya II

Duarte Barbosa Domingo Paes (Portugal) - Krishnadevaraya

Fernao Nuniz (Portugal) - Achyuta Deva Raya

Administration

Administration in the Vijayanagara Empire was structured with Naiks serving as provincial governors and Mahanayakacharyas acting as intermediaries between villages and the central administration. The Nayakas maintained a standing army, and succession to the throne was hereditary. Administrative units were divided into Mandalams, Nadus, Sthalas, and Gramas.

The empire's income primarily came from sources like land revenue, tributes, customs at ports, and taxes on various professions. Military chiefs were granted land in the Nayankar System, with Amara Nayakas serving as military commanders. Kudirai chettis were crucial in supplying horses, while Pattanulkars migrated from Gujarat as silk weavers.

Currency

The chief gold coin was the varaha or pagoda. The Perta was half a Varahaand. Fanam was one tenth a pertha. Tar was a silver coin. Jittal was a copper coin.

Religion

Early Vijayanagar rulers were followers of **Saivism**. **Virupaksha** was their family God. Later they came under the influence of Vaishnavism.

Cultural Contribution

The chief characteristics of the Vijayanagara architecture were the construction of **tall Raya Gopurams** or **gateways** and the Kalyanamandapam with carved pillars in the **temple premises**. The **horse** was the most common animal found on the pillars. The Provida style is also known as the Vijayanagara style. It was a unique blend of architectural influences, incorporating elements from Dravidian, Islamic, and local South Indian styles.

Decline of Vijayanagara's Empire

The last ruler, Sri Ranga III, witnessed the empire's decline. Conflict with the Sultans began in 1564, culminating in the decisive Battle of Talikota in January, 1565. Ramaraya's defeat led to the city's plunder by Muslim forces, marking the end of Vijayanagara's glory days.

Bahmani sultanate

The Bahmani Kingdom (1347-1526) was founded by Alauddin Bahman Shah in 1347 AD. With its capital at Gulbarga and later Bidar a total of eighteen Sultans ruled over this kingdom. The first Independent Islamic Kingdom in South India was the Bahmani Sultanate or the Bahmani Kingdom.

Rulers of the Bahmani Kingdom (Reign)

Alauddin Hasan Bahman Shah (1347 – 1358) - He was a nobleman in the court of Muhammad bin Tughluq. He was also known as Hasan Gangu. He divided his realm into four provinces- Gulbarg, Daulatabad, Berar and Bidar. He established an independent state in the Deccan in 1347 with the capital at Ahsanabad (now Gulbarga) in Karnataka.

Muhammad Shah-I (1358-1377) - He succeeded Bahman Shah. He defeated Kapaya Nayaks of Warangal and the Bukka-I of Vijayanagara.

Muhammad Shah-II - He ascended the throne in 1378 A.D.

Feroz Shah Bahmani (1397-1422) - He defeated the Vijayanagar ruler Deva Raya I.

Ahmad Shah (1422-1436) - Ahmad Shah succeeded Feroz Shah Bahmani, conquered Warangal, shifted the capital to Bidar, and died in 1436. His shrine is in Ashtoor, Karnataka.

Muhammad Shah-III (1463-1482) - Muhammad Shah-III became Sultan at nine with Muhammad Gawan as regent. Gawan defeated rulers from Konkan, Orissa, Sangameshwar, and Vijayanagar. Muhammad Shah-III died in 1482. Shams Al-Din Daud Shah II ruled Bahamani Sultanate. The kingdom split into Bijapur, Ahmednagar, Berar, Golconda, and Bidar.

Bijapur Dynasty

Yusuf Adil Shah was the founder of Bijapur Dynasty. He was governor of Bijapur province, before creating a de facto independent Bijapur state. This dynasty Annexed by Aurangzeb in 1686.

Ibrahim Adil Shah II - He ruled Bijapur Sultanate from 1580 to 1626, known for his patronage of culture and Hindu-Muslim harmony. He had the title

of 'Abla Baba' of the Deccan, Jagadguru Badshah. He authored 'Nauras-nama', renovated temples, and built 'Ibrahim Rauza'.

Ali Adil Shah I - He was the fifth Sultan of Bijapur Sultanate. He was adopted by Ibrahim Adil Shah II.

Ahmednagar Sultanate

It is also known as the Nizam Shahi dynasty, established in 1490 by Malik Ahmed Shah Bahri after defeating the Bahmani Army. Junnar was the initial capital, later shifted to Ahmednagar in 1494. Aurangzeb, the Mughal viceroy of the Deccan, annexed the sultanate in 1632.

Berar Sultanate

It was known by the name Imad Shahi dynasty. It was founded by Fath-Ullah Imad- ul-Mulk in 1490. He established the capital of the Berar Sultanate at Achalpur and also fortified Gavilgad and Narnala. This dynasty was annexed by Ahmednagar in 1574.

Golconda Sultanate

It was founded by Sultan Quli Qutub-ul-Mulk in 1518, ruled until Aurangzeb's conquest in 1687. Char Minar, built in 1591 by Sultan Mohammed Quli Qutb Shah, honors his wife, Bhagmati.

Bidar Sultanate

It was founded by Qasim Barid. The Barid Shahi dynasty ruled until annexed by the Bijapur Sultanate in 1619 AD. Rangin Mahal was built during the reign of King Ali Barid Shah (1542 - 1580).

Art and Architecture

The Jama Masjid at Gulbarga is a mosque located in Karnataka, India. It was built by Bahamani Sultan Mohammed Shah after he defeated Kapaya Nayaka of Warangal.

Bijapur - Gol gumbaz (built by Mohammad Adil shah in 1656 A.D.), Ibrahim roza, Bara Kaman, Anand mahal, Chand Bawdi etc.

Monuments at Gulbarga - Shah bazaar mosque, Hafta Gumbaz, Jama masjid.

Bidar, Karnataka - Solah khamba mosque, Janata mahal. The Madrasas of

Muhammad Gawan - It was built by Khwaja Muhammad Gilani (Mahmud Gawan) at Bidar in 1472 AD. Rangeen Mahal was built in the 16th century at Bidar, during the reign of King Ali Barid Shah of Barid Shahi dynasty.

The craftspersons of Bidar were so famed for their inlay work in copper and silver that it came to be called Bidri.

Chand Minar - It was built by King Ala-ud-din Bahmani in Daulatabad, Maharashtra.

Babur

(1526 - 1530 A.D.)

Babur (1526 - 1530 A.D.) was the founder of the Mughal empire, born Zahir-ud-din Muhammad, with ancestry tracing back to Timur and Genghis Khan. He ascended to the throne of Ferghana in 1494 at the age of 12 but was ousted by Uzbeks. Invited by Daulat Khan, he defeated Ibrahim Lodhi, annexing Kabul and Qandhar into his empire, proclaiming himself Emperor of Hindustan. He assumed the title of Padshah.

His autobiography, Baburnama or Tuzuk-i-Baburi, written in Chagatai Turkic, details his life and conquests. Babur seized Kabul in 1504 and undertook successful campaigns into India from 1519 to 1523, capturing territories like Bhira and Sialkot. After initial victories and retreats, he decisively defeated Ibrahim Lodhi in the First Battle of Panipat in 1526, establishing Mughal rule.

Babur's military innovations included popularizing gunpowder, artillery, and cavalry in northern India. He introduced Tulughma warfare, employing cannons and flanking tactics.

Notable battles

Battle of Hisar Firoza (1526)- This battle was held between Babar and Hamid Khan.

Battle of Khanwa- Babar defeated Rana Sangha (ruler of Mewar) on 16 March 1527 AD. and took the title of Ghazi.

Battle of Chanderi- Babar defeated Medini Rai of Malwa in 1528.

Battle of Ghagra- He defeated Mahmud Lodi and Nusrat Shah's combined forces in 1529.

He died in Agra On December 26, 1530 at the age of 47. He was first buried at Arambagh in Agra in 1530 but was later taken to Kabul.

Humayun and Sher Shah Suri

Nasiruddin Humayun - He was the eldest son of Babur, who inherited the Mughal throne in December 1530. He ruled over Afghanistan, Pakistan, and northeastern India from 1530 to 1540 and again from 1555 to 1556. Initially the Governor of Badakhshan, he divided the empire among his four brothers as per his father's will- Mirza Kamran (Kabul and Kandhar), Hindal (Alwar and Mewar), Askari (Sambhal), and cousin Suleman (Badakhshan).

In the Battle of Daurah or Dadrah (1532), Humayun triumphed over Mahmud Lodhi's Afghan forces and laid siege to Chunar fort, held by Sher Khan.

The pivotal Battle of Chausa (1539) and the subsequent Battle of Kannauj (1540), also known as the Battle of Bilgram, saw Humayun facing defeat against Sher Shah Suri. Subsequently, Humayun spent fifteen years in exile, seeking refuge under Shah Tahmasp of Persia, where he wedded 'Hamida Bano Begum'.

After a decade and a half, Humayun reclaimed his kingdom, securing victory over Sher Shah Suri in the Battle of Sirhind in 1555, with assistance from the Safavids.

Death - Humayun died in 1556 due to his fall from the staircase of the library of Din Panah Palace.

Tomb - Humayun's Tomb, located in New Delhi, was commissioned by his widow, Biga Begum, in 1569-70 and constructed by architect Mirak Mirza Ghiyath. Built in red sandstone, it served as an inspiration for the design of the iconic Taj Mahal. Akbar later undertook its renovation. This architectural marvel has been a UNESCO World Heritage site since 1993.

Sur Empire (1540 - 55)

Sher Shah Suri (1540 - 1545) - His real name was Farid-Ud-Din Abul Muzaffar. He was also known as Farid Khan. He was born in 1486 in Sasaram district of Bihar. He was an ethnic Pashtun. He was the son of an Afghan Jagirdar of Jaunpur

'Hasan Khan'. He was given the title 'Sher Khan' by Afghan King Muhammad Bahar Khan Lohani for killing a tiger.

He was the founder of the Suri Empire, with its capital in Sasaram in modern-day Bihar. He began his military career as a Jagirdar and rose to become a commander in Babur's Mughal army and later the governor of Bihar.

After defeating Humayun, he completely destroyed Din Panah and built his new capital, the Sher Shahi (now known as Purana Qila) in southeastern Delhi.

Administration - He followed Khilji's Chehra and Daggh system. He used Gaz-i-Sikandari (introduced by Sikander Lodhi) of 32 digits and applied 'Zarib' for land measurement. He fixed the area of a bigha 360 square yards.

Economy - He standardized the silver coin (178 grains) and named the currency as rupee from the Sanskrit term of 'Rupyakam' (means Silver coin). He also introduced the 380 grain copper coins called Dam. His agricultural tax was one-third of the yield.

Sher Shah Suri introduced Kabuliyaat and Patta for agricultural settlement. Patta was a government document recording land given to farmers, while Kabuliyaat was the agreement detailing the rent the farmer agreed to pay. Land was categorized into three based on yield-good, middling, and low.

Architecture - Sher Shah Suri reconstructed and extended the Mauryan roadway, renaming it Sadak-e-Azam or Grand Trunk Road, linking Kabul to Sonargaon. He also built Qila-i-Qunah (Old Fort Mosque, Delhi), Rohtas Fort in Pakistan, and the Sher Shah Suri Masjid in Patna. His roads and sarais served as vital travel arteries and stages for Dak-chowki news services. Additionally, he constructed his mausoleum in Sasaram.

Expeditions

Battle of Surajgarh- He defeated the Lohani chiefs of Bihar and Mohammad Shah of Bengal at Surajgarh in 1533.

Conquest of Punjab- He swiftly conquered Punjab from Kamran, Humayun's brother, after ascending the throne.

Suppression of Khokhars- He defeated the Khokhars of the northern region in 1542.

Conquest of Malwa- He attacked and defeated Maldeo Rathore, the ruler of Malwa, in 1542.

Conquest of Raisen- Purnamal, a Rajput ruler, surrendered to Sher Shah.

Conquest of Multan and Sind- He annexed Multan and Sind in 1543.

Conquest of Marwar- He brought Marwar under his control by defeating Maldev, the ruler of Mewar, between 1543 and 1545.

Conquest of Kalinjar- He won Kalinjar from Kirat Singh in 1545, but tragically lost his life due to injuries sustained in a blast.

Death and Tomb - He died in an accidental gunpowder explosion in the fort of Kalinjar in 1545 AD. After his death, his son Islam Shah ascended to the throne. His tomb was built in Sasaram in Bihar. It was built of red sandstone.

Akbar (1556-1605 A.D)

The real name of Akbar was Abul-Fath Jalal-ud-din Muhammad Akbar. He was born in 1542 in Umarkot (Pakistan). He was crowned at Kalanaur (Punjab) in 1556.

He appointed Bairam Khan as his regent. Bairam Khan was appointed as the commander-in-chief of the Mughal army in 1556-1560. Bairam Khan was assassinated in 1561 near Sahastralinga Tank (a shrine near Anhilwad Patan) by a group of Afghans during his visit to Gujarat. At the age of 18 Akbar ended Bairam Khan's regency and assumed authority.

He shifted his capital from Lahore to Fatehpur Sikri and it remained its capital from 1571 to 1585. He shifted his capital again from Fatehpur Sikri to Lahore in 1585. He again shifted his capital from Lahore to Agra in 1599.

He married the Rajput princess, Harka Bai (daughter of Raja Bharmal). She was given the title of "Mariam-uz-Zamani" (Jodha Bai).

Major Expeditions of Akbar

Second Battle of Panipat (1556) - Hemu and the Afghan forces were defeated in 1556 by the Mughal forces under Akbar.

Malwa (1562) - Akbar sent his army under the leadership of Adham Khan and Pir Mohammad who defeated Baz Bahadur.

Gondwana (1564 A.D) - The battle was fought between Asaf Khan of the Mughal Empire and Bir Narayan and Durgavati of Gondwana on 24 June 1564.

Mewar (1568) - Captured Chittor after the retreat of Rana Udai Singh. With the fall of Chittor, several Rajput states, such as Ranthambore and Jodhpur, also surrendered.

Jaimal and Patta were the warriors who were given charge of defending the fort of Chittor.

Gujarat (1572-73 A.D) - Akbar defeated Muzaffar Shah III. During the conquest of Gujarat, Akbar met Portuguese and saw the sea for the first time.

Battle of Haldighati (1576) - Akbar secured a decisive win in the Battle of Haldighati where Maharana Pratap suffered severe defeat at the hands of the Mughal army under Man Singh.

Bihar and Bengal (1574-1576) - Akbar attacked and occupied Bihar in 1574 A.D. He then left Munim Khan (general) to complete the conquest of Bengal and Orissa.

Kabul (1581-82) - Mirza Hakim (third son of Humayun) revolted in Kabul with his nobles but it was suppressed successfully by Akbar with the help of Man Singh.

Kashmir (1586) - This expedition was led by Bhagwan Das and Kasim Khan. Yusuf Yakub Khan (the last sultan of Kashmir) was defeated in this battle.

Sindh (1591) - This expedition was led by Abdul Rahim Khan-i-Khana. Jaani Beg of the Tarkhan Dynasty was defeated in the battle by Akbar in 1591.

Deccan region- In 1591, occupied the Khandesh region. By 1596, Berar was acquired from Chand Bibi, the regent of Ahmednagar. By 1600, parts of Ahmednagar were under Mughal control.

Gems of Akbar

Abul Fazl- Historian and author of Akbarnama and Ain-i-Akbari, Abul Fazl led the Mughal army in Deccan. He was assassinated by Vir Singh Bundela under Prince Salim's orders. He also translated the Bible into Persian.

Faizi- Persian poet and brother of Abul Fazl, Faizi oversaw the translation of the Mahabharata and Lilavati into Persian.

Tansen- Born Ram Tanu Pandey, Tansen was a renowned musician in Akbar's court. He was a courtier in the court of Rewa before coming to Akbar. Akbar gave him the title of "Mian" (learned man). He created 'Raga Miya Ki Malhar'.

Raja Birbal- Also known as Mahesh Das, Birbal was a poet proficient in Sanskrit, Persian, and Hindi. He was titled "Kavipriya" and was the only Hindu to embrace Din-i Ilahi.

Raja Todar Mal- Revenue minister under Akbar, Todar Mal standardized measurements, implemented land survey systems, and rebuilt the Kashi Vishwanath Temple in 1585.

Raja Man Singh- Chief of the Mughal Army and Raja of Amber, Man Singh defeated the Cheros tribe in 1591.

Fakir Aziao Din- Religious minister of Akbar.

Abdul Rahim Khan-i-Khana- Son of Bairam Khan, he was a poet, defense minister, and translator of Baburnama into Persian.

Mirza Aziz Koka - Also called Khan-i-Azam, he was Akbar's foster brother and Subedar of Gujarat.

Mullah Do Piazza - An advisor to Akbar, some believe he may be a fictional character.

Literary Works

Akbar established a translation department where Mahabharata and Ramayana were translated into Persian. Abul Fazal translated Panchtantra into Persian, and Razmnama is a Persian version of Mahabharata.

In 1583, Akbar initiated the Illahi Era and honored calligrapher Muhammad Hussein with the title Zarin Qalam. Abdur Samad, Basawan, and Miskina were renowned painters in his court.

Among the notable singers were Tansen, Baj Bahadur, Baba Ramdas, and Baiju Bawara. Akbar himself played the Nakkara, while Utbi and Naziri served as his court poets.

Religious Policy

Akbar was liberal in his religious policies. He abolished jizya in 1562. He abolished the Pilgrimage tax in 1563. Akbar built Ibadat Khana (House of worship) in Fatehpur Sikri in 1575.

He issued the famous "Infallibility Decree" or Mahzarnama in 1579, which made him the religious head. In 1582, he created a new religion called Din-i-Ilahi or Tawhid-i-Ilahi (Divine Monotheism). His aim was to unite different religions. This new religion had only 15 followers including Birbal.

Death - Akbar died of dysentery in 1605 CE and was buried at Sikandra (near Agra). His tomb is located at a mausoleum at Sikandra in Agra. He was succeeded on the throne by his son Jahangir.

Jahangir (1605-1627 A.D)

Jahangir was the fourth Mughal Emperor. He reigned from 1605 to 1627.

He was born on 31st August 1569 at Fatehpur Sikri in Uttar Pradesh. His real name was Nur-ud-din Muhammad Salim. He was born to Akbar and Jodha Bai (Mariam-uz-Zamani Begum). He revolted against his father Akbar in 1599.

He married Mehr-un-Nissa (Nur Jahan) in 1611. The meaning of her name is "Light of the World". She was the widow of Sher Afghan.

She was the only Mughal Queen whose name was inscribed on the coins and royal farmans.

Jahangir gave the title of Itmad-ud-daula ("Pillar of the Empire") to Mirza Ghiyath Beg (his father-in-law). Nur Jahan was the daughter of Ghiyath Beg. Her brother Asaf Khan was appointed to a post of 'Khan-i-Saman'.

During his reign, he installed the 'Zanjir-i-Adl' (Golden Chain of Justice) at Agra and banned tobacco around 1617 after its introduction by the Portuguese

in the late 16th century during Akbar's reign.

His autobiography, known as Tuzuk-e-Jahangiri, Karnama-i-Jahangiri, or Jahangir Nama, was written in Persian in 1609 and mentioned the tower, tank, and pavilion twice.

Jahangir captured and imprisoned Guru Arjan Dev, the fifth Sikh Guru, in Lahore Fort in 1606 as the growing influence of Sikhism threatened the orthodox Muslim clergy and Hindu elite. This was after Guru Arjan Dev had blessed Jahangir's rebellious son Khusrau.

Foreign visitors during Jahangir's reign - Captain William Hawkins, who visited his court in 1608 and was given the title "English Khan", and Sir Thomas Roe, who visited in 1615 and resided at Agra for three years until 1619. Jahangir allowed the British East India Company to establish factories in Surat in 1613.

Expeditions - Jahangir fought the Battle of Dewar against Amar Singh I of Mewar and led campaigns against the Ahom kingdoms in Burma, Kangra and most of the Deccan regions. He sent his son Khurram (later Shah Jahan) to fight against Golconda, Bijapur, and Ahmednagar. In 1613, he seized a Portuguese ship in Surat.

Death - Jahangir died in 1627 while traveling from Kashmir to Lahore. His body was temporarily buried in Baghsar Fort and later moved to Lahore, where it was buried in Shahdara Bagh. He was succeeded by his son Khurram, who took the title Shah Jahan. Jahangir's 17th-century mausoleum is located in Shahdara Bagh along the banks of the Ravi River.

Shah Jahan (1627-1658 A.D)

Shah Jahan, originally Shahab-ud-din Muhammad Khurram, ascended to the Mughal throne in 1627, succeeding his father Jahangir. His mother, Jagat Gosain, hailed from Jodhpur. In 1612, he married Arjumand Banu Begum, famously known as Mumtaz Mahal. His reign (1627-1658 AD) is called the "Golden Age of Architecture". He earned the title of the "architect king."

His accomplishment was the re-imposition of the pilgrimage tax, although exceptions were made in

response to requests from figures like Kavindracharya Saraswati. He also halted Christian conversions. The four notable sons of Shah Jahan were Dara Shikoh, Murad Bakhsh, Shah Shuja, and Aurangzeb.

Literature - Inayat Khan authored 'Shah Jahan Nama.' (an unillustrated chronicle of the reign of Shah Jahan). His court historian Abdul Hameed Lahori authored 'Badshah Nama.' (an illustrated group of works written as the official history of the reign of Shah Jahan).

Major events/battles During his time

Revolt of Khan-i-Jahan Lodi- Khan-i-Jahan Lodi was appointed as the Subahdar (Governor) of Bengal. He was assigned to reclaim the Deccan territories, but he lost to the Nizam Shahi ruler. He later joined Nizam Shah.

Revolt of Bundelas- Jujhar Singh (ruler of Orchha)- Shah Jahan sent Mahavat Khan and Khan-i-Jahan Lodhi against Jhujhar Singh. He surrendered before the Mughal army.

Annexation- He annexed Ahmednagar in 1632. He captured the Daulatabad fortress of Maharashtra and imprisoned Husain Shah of Ahmednagar 'Nizam Shahi Kingdom'. (Founder of Nizam Shahi Dynasty - Malik Ahmed). Aurangzeb was appointed as governor of the south. He was able to take control of Bijapur and Golconda in 1636.

Suppression of the Portuguese - Shah Jahan ordered Qasim Khan (governor of Bengal) to suppress the Portuguese in Hooghly in 1632.

Mughal Safavid War (1649–1653)- It was fought between the Mughal and Safavid in modern Afghanistan. The Safavid army captured the fortress city of Kandahar and other strategic cities that controlled the region. The Mughals attempted to regain the city, but their efforts were proven unsuccessful.

Famous painters of his time - Muhammad Fakir Ulla, Mir Hasim, Murar, Hunar Muhammad Nadir, Muhammad Nadir Samarqandi and Chitra were prominent.

War of succession - In 1657-1658, there was conflict over succession amongst Shah Jahan's sons. Aurangzeb was victorious and his three brothers,

including Dara Shukoh were killed. Shah Jahan was imprisoned for the rest of his life in Agra. Shah Jahan died in 1666 at the age of 74 years.

Dara Shikoh

He was the eldest son of Shah Jahan and Mumtaz Mahal. He was favored as a successor by his father and his elder sister (Jahanara Begum). He was declared heir and got the title of 'Sultan Buland Iqbal' in September 1642 by Shah Jahan. He was a disciple of the Sufi saint Mullahshah Bakshi.

He translated Bhagavad Gita, and Yoga Vashishtha from Sanskrit to Persian. He translated the 52 Upanishads into Persian under the name Sirr-i-Akbar. He composed the Persian book named Safinat-ul-Auliya and Majm-ul-Bahrain.

Death- Aurangzeb and Dara Shikoh fought the Battle of Samugarh in 1658. He lost the battle and was killed by Aurangzeb in 1659.

Aurangzeb and later Mughals

He was the sixth Mughal Emperor. His real name was Abdul Muzaffar Muhi-ud-Din Mohammad Aurangzeb. He was born at Dohad near Malwa in 1618. He was the son of Shah Jahan. He was also known as Zinda Pir (living saint), Fakir, Darbesh, and Alamgir (The World Conqueror).

He received a sword called Alamgir as a present from his father Shah Jahan. He was married to Dilras Banu Begum. He used to play Veena. He is considered as the last powerful Mughal ruler of India. Under the reign of Aurangzeb, the Mughal Empire reached its peak. The maximum number of books in Persian on classical Indian music was written in his reign.

He gave the French permission to build a factory in Surat in 1667. Mir Sayyid Muhammad Qanauji was a Sufi scholar who served as tutor to Aurangzeb. He got his coronation done twice, in 1658 in Agra and in 1659 in Delhi. He accorded the title "Mirza Raja" to the two highest-ranking nobles Jai Singh (King of the Amber) and Jaswant Singh (King of Marwar) by the Mughal emperor.

Major events/battles

The Battle of Dharmatpur- It was fought on the bank of river **Narmada** on 15 April 1658. In this battle Mughal emperor Aurangzeb defeated Maharaja Jaswant Singh.

Annexation of Bijapur - Aurangzeb invaded Bijapur from Sikandar Adil Shah and annexed it in 1686.

Conflict with Marathas- Chhatrapati Shivaji Maharaj fought continuously with the Mughals and established an independent state of the Marathas in Maharashtra. Aurangzeb captured Shivaji when he came to Diwan-e-Aam of Agra in 1666. After Shivaji, his son Sambhaji fought continuously with the Mughals. In 1689, Aurangzeb captured Sambhaji and killed him and took over the Maratha state.

Revolt under Aurangzeb's reign

Jat revolt - They revolted in the areas around Mathura under the leadership of Gokula and Rajaram in 1669. Abdul Nabi destroyed a Keshavraj temple of Mathura and built a mosque on its ruins in 1661 AD. which ignited the revolt.

Revolt of Satnamis - The Satnamis revolted in the Mewat and Narnaul regions in 1672.

Sikh revolt - Guru Tegh Bahadur (9th Guru of the Sikhs) opposed the religious policies of Aurangzeb for which he was executed brutally in 1675.

Mirza Muhammad Akbar Revolt - He was the son of Aurangzeb. He rebelled and took refuge under Sambhaji Maharaj.

Religious Policies

Aurangzeb commissioned the compilation of Fatawa-i-Alamgiri, which served as a comprehensive guide covering various aspects of statecraft, ethics, military strategy, economic policy, and justice, firmly establishing Sharia as the governing body of his empire.

Aurangzeb ruled the Mughal Empire according to Sharia law, which he enforced rigorously. He reintroduced the religious tax Jizya on non-Muslims in 1679. His reign saw the destruction of Hindu temples, including the famous Somnath Temple in Gujarat, Kashi Vishwanath Temple in Banaras, and Keshav Rai Temple in Mathura.

He also banned cultural practices like singing, dancing, and playing musical instruments, as well as inscriptions of Kalma on coins. Moreover, he attempted to destroy the Kailash Temple at Ellora. French physician Francois Bernier visited his court during this time.

Death- Aurangzeb died in 1707 in Ahmednagar (Maharashtra) while campaigning against Marathas. He was buried at Aurangabad.

Notable Rulers after Aurangzeb

Bahadur Shah I (Shah Alam or Muazzam) (1707 – 1712) - He was the nominal ruler, with real power held by his wazir, Zulfiqar Khan, known as "Shah-i-Bekhabar".

Jahandar Shah (1712 – 1713) - He granted titles like "Maharaja" to Ajit Singh and "Mirza Raja Sawai" to Jai Singh of Amber. He introduced revenue farming, called Ijarah.

Farrukh Siyar (1713 – 1719) - He ascended the throne with the support of the Saiyyad brothers (the kingmakers) - Saiyyad Abdullah Khan (Wazir), and Hussain Ali Khan (Mir Bakshi). The Saiyyad brothers completely did away with jizya and also abolished pilgrimage tax at a number of places. Indira Kanwar was the second wife of Emperor Farrukhsiyar.

Muhammad Shah or Roshan Akhtar (1719 – 1748) - He was also known as Rangeela. Nadir Shah invaded during his reign. He was defeated by Nadir Shah in 1739 in the Battle of Karnal.

Nadir Shah carried the Peacock throne erected by Shah Jahan with him. He also stole the fabled "Koh-i-noor" diamond.

Ahmad Shah Durrani or Abdali (1748 – 1754)- The ruler of Afghanistan, he invaded India multiple times, recognizing Alamgir II as Mughal emperor.

Alamgir II (1754 – 1759) - The Battle of Plassey (1757) was fought during his tenure. He had to face repeated invasions of Ahmad Shah Abdali.

Ali Gauhar or Shah Alam II (1759 – 1806)- He was the first Mughal ruler who became a pensioner of the East India Company. He appointed the East India Company as the Diwan of Bengal on 12 August 1765. His reign saw significant battles of Buxar (1764) and Panipat (1761).

Akbar II (1806 – 1837)- He conferred the title of "**Raja**" on Ram Mohan Roy.

Bahadur Shah II or (Bahadur Shah Zafar) (1837 – 1857)- He was the last Mughal Emperor. In 1856, Governor-General Canning decided that Bahadur Shah Zafar would be the last Mughal king. He was arrested by British officer Hudson due to involvement in the 1857 revolt and exiled to Rangoon, Burma where he died in 1862.

Decline of Mughal Empire

The Mughal Empire declined due to instability after Aurangzeb's rule. Powerful nobles controlled weak rulers, leading to poor administration. Invasions by Nadir Shah and Ahmad Abdali exposed military weakness. Autonomous states emerged, weakening central power. Aurangzeb's harsh treatment of Hindus and other groups fueled resentment and rebellion.

Mughal Administration

Under the Mughals, the empire was divided into provinces known as Subas, each governed by a Subadar who oversaw political and military affairs. These Subas also had a Diwan responsible for financial matters.

Subas were further divided into Sarkar (district), Pargana (group of villages), and Gram (village). Jagirs were granted to nobles and royals, Khalisa was reserved for the royal treasury, and Inam was bestowed upon religious leaders.

Akbar modeled his administration after Sher Shah Suri's, with Ajmer serving as a prominent Suba headquarters.

Fatehpur Sikri (City of Victory) was the first planned city built by Akbar.

Key Elements of Mughal Governance

In the Mughal administration, Faujdar maintained law and order, while Amalguzar handled land assessment and revenue collection. Muqaddam acted as the headman of the panchayats.

The land revenue system, known as Zabti, Bandobast, or Dahsala, was introduced by Raja Todar Mal, fixed based on a decade's average yield. Akbar standardized measurement units, introducing Gaz-i-ilahi and defining

measurements for land units.

Land was categorized into Polaj (best quality), Parauti/Parati (temporarily uncultivated), Chachar (fallow for 3-4 years), and Banjar (uncultivated for over 5 years). Abwabs comprised additional government taxes.

The Mansabdari System graded officers from 10 to 5000 ranks, including Mansabdars (up to 500), Amirs (500-2500), and Amir Umda/Amir-i-azam (2500 and above). Ranks were Zat (numerical) and Sawar (cavalry), with Do-Aspa and Si-Aspa similar to Sawar, introduced by Jahangir.

The Jagirdari System, derived from the Delhi Sultanate's Iqta, assigned revenue collection areas (Jagir) to officials (Jagirdars). Jagirs included Tankha (in lieu of salaries), Mashrut (based on conditions), Watan (for local rulers), and Altamgha (for Muslim nobles).

Important Terminology

Farman - A royal edict or order. It used to bear the seal of the Mughal Emperor.

Department -

Diwan-i-Wizarat - Department of revenue and finances.

Diwan-i-Arz - Military department.
Diwan-i-Rasalatmuhtasib- Foreign affairs department.

Diwan-i-insha - Custodian of government papers.

Diwan-i-quza - Justice department.
Diwan-i-Barid - Intelligence department.
Diwan-i-Saman - Department in charge of royal household.

Officers (Center) -

Wazir - Head of revenue department.
Diwan - Responsible for all income and expenditure; control over Khalisa and Jagir.

Mir Bakshi - Headed military department, nobility, information and intelligence agencies.

Mir Saman - Incharge of imperial households (Karkhanas).

Diwan-i-Bayutat - Maintained roads, government buildings.

Mir Manshi - Royal correspondence.

Sadr-us-Sadr - Incharge of charitable and religious endowments.

Qazi-ul-Quzat - Head of judicial department.

Muhtasib - Censor of public morals.

Mushrif-i-Mumalik - Accountant general.

Mustauf-i-Mumalik - Auditor general.

Daroga-i-dak-chauki - Officer in charge of imperial post.

Mir-i-arz - Officer in charge of petition.

Waqia Navis - News reporter.

Officers (Province) -

Sipahsalar - The Head executive.

Diwan - Incharge of the revenue department.

Bakshi - Incharge of the military department.

Officers (District) -

Fauzdar - Administrative Head.

Amal or Amalguzar - Revenue collection.

Kotwal - Maintenance of law and order, trial of criminal cases, price regulation.

Officers (Pargana and Village) -

Shiqdar - Administrative head, combined fauzdar and kotwal.

Amin, Quanungo - Revenue officials.

Muqaddam - Headman.

Bhoga - It refers to the revenue assignments during the medieval period.

Mughal Architecture

It is a distinct Indo-Islamic architectural style, prevalent in northern and central India from the 16th to 18th centuries, nurtured by Mughal rulers and known as the Mehrab style, it reflects Islamic architectural principles, with Mehrab indicating the Qibla direction towards Mecca, significant for Muslim prayer orientation.

Features

Mughal architecture, blending Indian, Persian, and Turkish styles, featured grand structures like gates, forts, mausoleums, palaces, mosques, and rest houses (sarais). They often used red sandstone and white marble.

Characterized by bulbous domes, tall minarets with cupolas, large halls, massive arched doorways, and intricate decorations. Mughals also introduced Charbagh gardens from Persia, with "char" meaning four and "bagh" referring to gardens.

In Mughal architecture, the formal gateway to the Iwan (a vaulted rectangular hall) is called Pishtaq. It's a distinctive feature along with the open end of the Iwan.

Major Structures Across Dynasties

Babur

Aram Bagh - It is the oldest Mughal garden in India. It was constructed near the Taj Mahal in Agra by Babur in 1528.

He also built Jama Masjid in Sambhal in 1526 AD, and Kabuli Bagh Mosque at Panipat in 1527 AD.

Humayun

Sher Mandal is an octagonal building inside the Purana Qila complex which was built by Sher Shah Suri. It was used as a library by Emperor Humayun, who died after falling stairs accidentally.

Purana Qila - It is located in the South Eastern part of the present city of New Delhi. The construction started in the reign of Humayun and was completed by Sher Shah Suri.

He also started Dinpanah but it was destroyed by Sher Shah Suri.

Akbar

He Coined the term "Tudor arch" (four centered arch) and constructed Agra Fort. Jahangir Mahal is the largest part of Agra Fort.

Buland Darwaza (gate of magnificence) - It was built to celebrate Akbar's victory over Gujarat. It is a gateway to Jama Masjid.

Ibadat Khana (House of Worship)- It was built in 1575 CE at Fatehpur Sikri.

Tomb of Salim Chisti - It was built in 1581.

Jodha Bai's Palace - It is also known as Mariam-uz-Zamani's Palace.

Pachisi Court - Akbar used to play chess in the courtyard.

Murshidabad (Makhsudabad) was reputedly founded by Akbar in the 16th century.

Panch Mahal - It is known as a "dream in stone".

Jahangir

There were several buildings constructed during his reign such as Akbar's Tomb in Sikandra, and Shalimar Bagh in Kashmir.

His son Parwez Shah built Pathar Ki Masjid on the banks of River Ganga. It is located in Patna, Bihar.

The Moti Masjid (Lahore Fort)- "Pearl Mosque" inside the Lahore Fort was built by Mughal emperor Jahangir earlier in the 17th century.

Shah Jahan

Taj Mahal is made up of white marble with symmetrical architecture including an Iwan (an arch-shaped gateway) capped by a huge dome. It was built in memory of Arjumand Banu Begum (Mumtaz Mahal).

Red Fort (Delhi)- It was commissioned in 1638, when Shah Jahan decided to shift his capital from Agra to Delhi. It was the residence of the Mughal emperor for nearly 200 years, until 1857.

Musamman Burj - It was built between 1631-40. It is a beautiful octagonal tower standing close to the Shah Jahan's private hall Diwan-e-Khas in Agra Fort.

Shahjahanabad was built between 1639 and 1648.

Pari Mahal or **Peer Mahal** is a seven-terraced Mughal garden located at the top of the Zabarwan mountain range, Srinagar. It was built by Mughal prince **Dara Shikoh**, (eldest son of Shah Jahan's).

Delhi's Jama masjid - It was built by the Mughal emperor, Shah Jahan in the honor of his eldest daughter, Jahanara Begum Sahiba in 1648 C.E.

Jama Masjid of Agra was built by his daughter Jahanara.

The Moti Masjid (Agra Fort)- "Pearl Mosque" in Agra was built by the Mughal emperor Shah Jahan in the 17th century.

Important places- Diwan-i-aam (Hall of Public Audience), Diwan-i-Khas (Hall of Private Audience), and Sheesh Mahal (Turkish Bath).

Aurangzeb

Moti Masjid in Red fort, Delhi - "Pearl Mosque" built inside the Red Fort complex in Delhi from 1659-1660.

The **Badshahi Mosque** or "Emperor's Mosque" was built in 1673 in Lahore, Pakistan.

Bibi ka Maqbara monument is also known as the 'Taj of the Deccan'. It is located in Aurangabad (now Sambhaji Nagar), Maharashtra. It was built by Aurangzeb in memory of his wife Dilras Banu Begum.

Zeenat-ul-Masjid (Ghata Masjid meaning cloud mosque) is an 18th-century Mughal mosque located in Delhi. It was commissioned by Zeenat-un-Nissa, the second daughter of Aurangzeb.

Sikh Gurus

The term "Sikh" is derived from the Sanskrit word "Shishya," meaning disciple or learner. Sikhism, also known as "Sikhi," emerged during the early Medieval period, influenced by the Bhakti movement and Vaishnavism. Sikhs follow Gurmat, which translates to "the Way of the Guru," and worship the Nirgun form of God.

The Sikh Gurus are revered figures in Sikhism, a religion that originated in the Punjab region of the Indian subcontinent during the late 15th century.

Bangla Sahib (Delhi) is one of India's structures, and it is inextricably related to Sikhism's history.

The Akali movement was started in the 1920s by the sikh reformers to purify their religious places by removing the evil social practices that had slowly crept into them.

Sikh Gurus (with Birth and Passing Years)

Guru Nanak Dev (1469-1539) - Guru Nanak Dev is regarded as the founder of Sikhism and the first of the Sikh Gurus. Born on April 15, 1469, in Talwandi village (now Nankana Sahib) in Lahore, Guru Nanak emphasized the concept of one God and preached equality among all. He established the institution of Guru Ka Langar, promoting communal dining irrespective of caste or creed. Guru Nanak's teachings were compiled in the Janamsakhis, with the most significant being the Mool Mantar, a foundational scripture in Sikhism.

Guru Angad Dev (1539 to 1552) - He is the second Sikh Guru. He invented and introduced the Gurmukhi (written form of Punjabi) script.

Guru Amardas Sahib (1552 to 1574) - He is the third Sikh Guru. He started Manjis

and Piris to strengthen the Sikh movement. He was the contemporary of Mughal emperor Akbar. He opposed "Parda Pratha" and "Sati Pratha". He composed the poem 'Anand'.

Guru Ram Das (1574 to 1581) - He is the fourth Sikh Guru. He founded Amritsar in 1574 on the land granted by Akbar and started the construction of the Golden Temple. He made the post of Guru hereditary. Mian Mir laid the foundation of the Golden temple.

Guru Arjun Dev (1581 to 1606) - He is the fifth Sikh Guru. He composed the Adi Granth and also founded Tarntaran Sahib (1596). He led to the completion of construction of the Golden temple. He introduced a compulsory spiritual tax. He was assassinated by Jahangir on 16 June 1606 for helping prince Khusrau. He became the first great martyr in Sikh history.

Guru Har Gobind Sahib (1606 to 1644) - He is the sixth Sikh Guru. He built the Akal Takhat. 'Akal Takht' (time-less throne) is a building in the premises of Golden Temple in Amritsar.

Guru Har Rai Sahib (1644 to 1661) - He is the seventh Sikh Guru. He devoted most of his life in maintaining peace with Aurangzeb and doing missionary work.

Guru Har Krishan Sahib (1661 to 1664) - He is the eighth Sikh Guru and the youngest guru of all. He was summoned by Aurangzeb against anti-islamic blasphemy.

Guru Tegh Bahadur Sahib (1665 to 1675) - He is the ninth Sikh Guru, founded **Anandpur Sahib**. He earned the devoted title of "Hind-di-Chadar" or the Shield of India. He was assassinated by Aurangzeb in 1675.

Guru Gobind Singh Sahib (1675-1708) - Guru Gobind Singh Sahib, the tenth and final Sikh Guru, established the Khalsa Panth in 1699, transforming Sikhism into a martial community dedicated to righteousness and social justice. He played a significant role in shaping Sikh identity and principles.

A commemorative coin of Rs. 350 was released by Prime Minister Narendra Modi to mark his 350th birth anniversary

He founded Patna Sahib city (Bihar) and wrote Dasam granth. He established the five symbols of Sikhism- Kangha (comb),

Kesh (long hair), Kada (iron bracelet), Kachcha (underwear) and Kirpan (sword or dagger).

The Zafarnama (Epistle of Victory) was a spiritual victory letter sent by Guru Gobind Singh Ji in 1705 to Aurangzeb after the Battle of Chamkaur. The letter is written in Persian verse.

Sikh Literature

The Adi Granth, the holy book of Sikhs was compiled by Guru Arjun Dev in 1604 AD. It is also known as 'Guru Granth Sahib'. It is a collection of 6000 hymns of Sikh Guru.

Duties of sikhism

Nam japna - To maintain a constant remembrance of God. **Kirt Karna** - To earn a livelihood through honest means. **Vand Chhakna** - To practice charity and care for the well-being of others.

Meanwhile, Sikhs are encouraged to avoid five vices - lust, covetousness and greed, attachment to worldly possessions, anger, and pride.

Takhts of Sikhism

Takhts ('the seat of divine authority') are the Sikhs' sites of prayer.

Five Takhts - Takht Damdama Sahib (Talwandi Sabo), Akal Takht (Amritsar), Takht Keshgarh Sahib (Anandpur Sahib), Takht Hazur Sahib (Nanded, Maharashtra) and Takht Patna Sahib (Bihar).

Sikh Empire

Maharaja Ranjit Singh

Maharaja Ranjit Singh, also known as Buddh Singh, was the founder of the Sikh Empire. Born in Gujranwala, he established Lahore as the empire's capital in 1799. His military prowess led to the formation of the Khalsa army, recruiting soldiers from diverse backgrounds. Under his leadership, the Sikh Empire flourished, extending its reach from Lahore and Multan to Peshawar and Ladakh. His legacy includes the golden beautification of the Harmandir Sahib Gurdwara and the funding of Hazoor Sahib Gurudwara. Ranjit Singh's reign marked a significant chapter in Sikh history, ending with his death in 1839.

The Treaty of Amritsar and Legacy

The Treaty of Amritsar, signed in 1809, delineated the boundary between the Sikh Empire and British territories, ensuring non-interference in each other's affairs. Additionally, it facilitated the sale of Kashmir to the Dogra dynasty. Ranjit Singh's legacy is further punctuated by the Tripartite Treaty of 1838, aligning the Sikh Empire with Shah Shuja against the British, albeit unsuccessfully.

Kharak Singh (Reign 1839) - He was the second Maharaja of the Sikh Empire. Maharani Chand Kaur was the wife of Maharaja Kharak Singh.

Nau Nihal Singh (Reign - 1839 - 1840) - He was the Third Maharaja of the Sikh Empire.

Sher Singh (Reign - 1840 - 1843 - He was the fourth Maharaja of the Sikh Empire.

Maharaja Sir Duleep Singh (Reign - 1843 - 1849) was known as the "Black Prince of Perthshire". He was the last Maharaja of the Sikh Empire. He ascended the throne at the age of 10 years in 1849.

Battle/ Treaties of Sikh Empire

Battle of Attock- It was also known as the Battle of Chuch or Haidru. It took place between the Sikh Empire and the Durrani Empire in 1813. Sikh defeated Durani empire and it was sikh's first victory over Durani.

First Anglo-Sikh War (1845-46)- It Was fought between Daleep Singh and the East India Company in the Ferozpur region of Punjab. It consisted of five important battles- Battle of Mudki (1845, British victory), Battle of Ferozeshah (1845, British victory), Battle of Budhewal (21 January 1846, Sikh victory), Battle of Aliwal (28 January 1846, British victory), Battle at Sobraon (1846, British victory). After the Battle of Sobraon, the **Treaty of Lahore** was signed between Sikh and British in 1846. According to the treaty, the area of Kashmir Hazara and Jalandhar Doab was to be handed over to the british.

Second Anglo-Sikh War (1848-49) - It was fought between the British East India company and sikh empire. Sir Hugh Gough was the commander in chief of the British East India company. Sikh Regent and Queen Jindan Kaur was not

treated well by the British. Before the final annexation of Punjab, three important battles were fought- Battle of Ramnagar (1848), Battle of Chillianwala (1849), Battle of Gujarat (1849). After the war, Sir John Lawrence was appointed as the first Chief Commissioner of Punjab.

Misals of sikh

Misl or sikh confederacy literal meaning "fighting clan or fighting band" which ruled over Punjab region after decline of Mughal Empire. Among them, notable ones included the Ahluwalia Misal led by Jassa Singh Ahluwalia, the Bhangi Misal founded by Chajja Singh, and the Kanaihya Misal, which flourished under Jai Singh and later his daughter-in-law, Sada Kaur.

Jat Kingdom

The Jats, an agrarian community in northern India and Pakistan, rose to prominence in the 17th century through their rebellion against the Mughal Emperor Aurangzeb, eventually establishing the formidable Kingdom of Bharatpur.

In 1669, the Jats initiated their first significant revolt against Mughal imperial forces in the Mathura region, marking the beginning of their struggle for autonomy.

Chudaman and Badan Singh were instrumental in laying the foundations of the Jat Kingdom of Bharatpur in the 18th century. However, it was under the visionary leadership of Suraj Mal that the Jat community reached its zenith of power.

Maharaja Suraj Mal (Reign 1755 - 1763)

He, also known as Sujan Singh, emerged as the paramount figure in Jat history. He was revered as "the Plato of the Jat people," he established Bharatpur city and transformed Ballabgarh into a thriving trading hub.

Under Suraj Mal's rule, the renowned Lohagarh Fort was erected in Bharatpur, standing as a testament to Jat resilience and military prowess. After the 3rd Battle of Panipat, Suraj Mal's compassionate treatment of sick and wounded guests exemplified his magnanimity and hospitality.

Battle of Tilpat (1669)- It was fought between Jats and Aurangzeb under the leadership of Gokul Singh at Tilpat in Faridabad district of Haryana.

Decline

Maharaja Surajmal died in battle in December of 1763 beside the Hindon river, after being ambushed by the army of Mughal chief Najib-ad-Dawlah and this led to the continuous decline of jat rule in India.

Mysore Kingdom

The Mysore Kingdom, ruled by the Wadiyar dynasty from 1399 to 1950 A.D., experienced a brief interruption in the late 1700s. Poor domestic governance and entanglement in conflicts on the plains led to Hyder Ali seizing power in 1761.

Haider Ali - He ruled Mysore from 1761 to 1782. Born in 1722, he initially served as a horseman in the Mysore army under King Chikka Krishnaraja Wodeyar's ministers. In 1761, with the assistance of French troops, he established himself as the de facto ruler of Mysore, introducing Western training methods to his army. He conducted multiple invasions against the Marathas, reclaiming territories between 1774 and 1776. Haider Ali passed away on December 6, 1782.

Tipu Sultan - He was also known as the Tiger of Mysore, ruled from 1782 to 1799 as the son of Haider Ali. He introduced innovative administrative measures such as his coinage and a new land revenue system that spurred the growth of the Mysore silk industry. Tipu was a pioneer in rocket artillery and implemented a new lunisolar calendar. He supported the French and established a Jacobin Club in Seringapatam, planting the Tree of Liberty there. Tipu planned for a significant naval fleet and built dockyards in Mangalore, Wajedabad, and Molidabad. He met his end in 1799 during the Battle of Srirangapatna against English forces.

Anglo-Mysore Wars

First Anglo Mysore War (1767-1769) - Haider Ali defeated a coalition of the Nizam of Hyderabad, the Marathas, and the British. The Treaty of Madras (1769) resulted, returning conquered territories and establishing mutual defense agreements.

Second Anglo Mysore War (1780-1784) - A coalition of Mysore, Marathas, and Nizam fought against the British. Despite the death of Haider Ali, Tipu Sultan emerged victorious. The Treaty of Mangalore (1784) was signed, leading to the return of acquired territories by both parties. The Battle of Pollilur (1780) was part of this war.

Third Anglo Mysore War (1790-1792) - Tipu Sultan initiated hostilities by attacking Travancore, an ally of the English. Tipu Sultan defeated the British army in 1790 under General Meadows. However, in 1791, Lord Cornwallis led a large army to Bangalore and then to Seringapatam, ultimately leading to the British victory in the war.

The Treaty of Seringapatam, signed in 1792, marked the end of the conflict. As per the treaty, Tipu Sultan was obligated to pay a war indemnity of Rs. 3 Crore.

Fourth Anglo-Mysore War (1799) - Tipu Sultan aligned with the French against the British. However, the British emerged victorious at the Battle of Seringapatam in 1799, resulting in the death of Tipu Sultan. Following Tipu Sultan's defeat, the Wodeyars ruled Mysore under the subsidiary alliance with the British.

Maratha Empire

Founded by Shivaji, who established Hindavi-Swarajya after leading Marathas against the Sultan of Bijapur. Five major chiefs- Peshwa at Poona, Bhonsle at Nagpur, Scindia at Gwalior, Holkar at Indore, and Gaekwad at Baroda.

Chatrapati Shivaji Maharaj (Reign 1674 – 1680)

He was born in 1630 at Shivneri fort in Pune. He was son of Shahaji Bhosle (father) and Jijabai (mother). Shivaji had a profound influence of his mother in his life. He inherited the jagir of Poona from his father in 1637.

He captured Torna and Kondana forts from Adil Shah. In 1674, he was formally crowned the Chhatrapati of his realm at Raigad Fort.

Battle

Battle of Pratapgad (1659)- This took place in 1659 at the fort of Pratapgad in Satara (Maharashtra) between the forces of Shivaji and Afzal Khan, the general of

Adil Shah. Afzal Khan was killed by Chhatrapati Shivaji during this battle.

Battle of Pavan Khind (1660)- Fought at the fort of Vishalgad in Maharashtra and occurred in 1660 between the Maratha Sardar Baji Prabhu Deshpande and Siddi Masud of Adilshahi. The Marathas were defeated in this confrontation.

Battle of Surat (1664)- In 1664, the Battle of Surat transpired between Shivaji, representing the Marathas, and Inayat Khan of the Mughal Empire.

Battle of Purandar (1665)- The Battle of Purandar took place in 1665 between Jai Singh I and Shivaji, resulting in the signing of the Treaty of Purandar.

Treaty of Purandar (1665)- The Treaty of Purandar was signed between Raja Jai Singh, commander of Aurangzeb's forces, and Shivaji. According to the treaty, Shivaji was granted the right to hold twelve forts and a net income area of up to 1 lakh Huns. The Marathas were obligated to assist Shivaji in the war against the Mughals, while Sambhaji was bound to fight on the side of the Mughal Empire.

Battle of Sinhagad (1670)- Also known as the Battle of Kondana, the Battle of Sinhagad occurred in 1670 in Pune between Tanaji Malusare, a Maratha commander, and Udaybhan Rathore, the fort keeper under Jai Singh. The Marathas emerged victorious in this battle, recapturing the annexed ports.

Battle of Kalyan (1682-83)- Bahadur Khan of the Mughal Empire defeated the Maratha army and took over Kalyan in the years 1682-83.

Battle of Sangamner (1679)- The Battle of Sangamner, fought in 1679, was the last battle of Shivaji's reign, occurring between the Mughal Empire and the Maratha Empire.

Administration

Shivaji Maharaj organized his territory into three provinces, each further divided into Prants and then Parganas or Tarafs. His army comprised a regular force known as Paga, alongside auxiliary troops called silahdars overseen by havildars. Soldiers received cash salaries, while chiefs sometimes received revenue grants.

Assisted by a council of ministers known as the Ashtapradhan Mandal, Shivaji

Maharaj's administration included key positions such as the Peshwa (prime minister), Senapati (military commander), Majumdar (Accountant General), Waqenavis (Intelligence), Sachiv (royal correspondence), Dabir (advising king in foreign affairs), Nyayadhish (dispensing justice), and Panditrao (High Priest). Each minister was directly accountable to Shivaji.

Revenue

Shivaji's revenue system was modeled after Malik Amber of Ahmednagar. Land measurement utilized a rod known as Kathi, where twenty Kathis constituted a bigha, and 120 bighas made a chavhr.

To streamline administration, Shivaji diminished the authority of existing Deshmukhs and Kulkarnis, installing his own revenue officers known as Karkuns.

Taxes

Chauth and Sardeshmukhi were two types of taxes collected in the Maratha Empire. Chauth, constituting 25 percent or 1/4th of the land revenue, was claimed by zamindars and collected by the Marathas. Sardeshmukhi, an additional levy of ten percent or 1/10th of the standard land revenue, was imposed on lands where the Marathas held hereditary rights.

Judicial System

Hazar Majils was the highest court. Panchayats handled disputes between various parties in the communities, and the criminal cases were decided by the village 'Patel'.

Military Administration

The military comprised infantry, cavalry, and the Navy. Maratha cavalry had two divisions- bargirs, the royal army, and Silahdars, soldiers who provided their own equipment. Ordinary soldiers received cash payments, while chiefs and commanders were granted jagirs. The regular army consisted of 30,000 to 40,000 cavalymen overseen by havildars.

Marathas after Shivaji

Sambhaji, born in 1657 to Shivaji and his first wife Saibai, became the second Chhatrapati of the Maratha empire. He was married to Yesubai and had a son

named Chhatrapati Sahu and a daughter named Bhavani Bai. In the Battle of Wai in 1687, Sambhaji emerged victorious against the Mughal forces.

Tragically, Sambhaji was captured, subjected to torture, and ultimately executed by the Mughals in 1689. Following his demise, his brother Rajaram I succeeded him as the next Chhatrapati.

Rajaram Bhosle I, Shivaji's second son from Soyarabai, ascended to the position of the third Chhatrapati of the Maratha Empire. He passed away in 1700 CE in Satara. After his demise, his minor son, Shivaji II, assumed the throne with his mother Tara Bai serving as regent.

Shahu, the son of Sambhaji Maharaj and Yesubai, ruled from 1707 to 1749. During his reign, he appointed Balaji Vishwanath as his first Peshwa. Shahu played a significant role in supporting the Saiyyad brothers in the execution of Farrukh Siyar.

Rajaram II, also known as Ramraja, reigned from 1749 to 1777. He was adopted by Shahu, and Tarabai presented him as the grandson of Rajaram and herself to assert control over the state.

Shivaji II, who ruled from 1710 to 1714 CE, was the son of Tarabai and Rajaram. He is notable for authoring a Marathi drama titled "Natesa Vilasa."

Sambhaji II, who reigned from 1714 to 1760 CE, was the son of Rajaram and his second wife Rajabai. He signed the Treaty of Wana with his cousin Shahu to formalize the separation of the Bhonsle family into two distinct seats.

The Peshwas

The word "Peshwa" in Persian means "foremost." In the Maratha Empire, the Peshwas served as prime ministers to the Chhatrapatis, assisting in administrative and political matters.

Balaji Vishwanath (Reign - 1713-1720) was bestowed with the title of Sena Karte in 1708 by Shahu and later became the first Peshwa in 1713, establishing the position as hereditary.

Baji Rao I (Reign - 1720-1740), Balaji Vishwanath's eldest son, was renowned for expanding Maratha power, employing effective guerrilla tactics, and capturing territories from the Portuguese. He signed the Treaty of Durai Sarai with

Nizam-ul-Mulk and famously remarked about the Mughals, "Let us strike at the trunk of the withering tree and the branches will fall off themselves." Baji Rao I shifted the administrative capital from Satara to Pune in 1728 CE.

Balaji Baji Rao I (Reign - 1740-1761) succeeded his father as Peshwa and significantly expanded Maratha influence, defeating Alivardi Khan and signing agreements with Ahmad Shah, the Mughal emperor. He met his end in the disastrous Third Battle of Panipat in 1761.

Madhav Rao (Reign - 1761-1772) restored lost territories, defeated rivals, and established semi-independent states like the Holkars, Scindias, and Gaekwads. His reign witnessed struggles for power between Raghunath Rao and Narayan Rao.

Narayan Rao (Reign - 1772-1773), also known as Nana Sahib, was assassinated on the orders of Raghunath Rao.

Madhav Rao (Reign - 1774-1795), the son of Narayan Rao, became Peshwa at just 40 days old, with Nana Phadnavis and the Barbhai Council managing the empire. Nana Phadnavis was instrumental in signing the Treaty of Purandar (1776) with the British and defeating them at the Battle of Wadgaon.

Ahilyabai Holkar (Reign - 1767-1795) was a prominent queen of the Maratha Empire, moving her capital to Maheshwar and contributing significantly to its governance.

Baji Rao II (Reign - 1796-1818 CE), the son of Raghunath Rao, served as the last Peshwa of the Maratha Empire.

The Treaty of Mandsaur, signed in 1818 between Sir Thomas Hislop and Malhar Rao Holkar II, followed the Battle of Mahidpur and resulted in Holkar ceding the southern part of the Satpura hills to the British.

Anglo-Maratha wars

First Anglo-Maratha War (1775-1782) - The power struggle between Sawai Madhav Rao and Raghunath Rao saw the latter receiving support from the British. In 1779, the Marathas emerged victorious against the British in the Battle of Wadgaon. This battle formed part of the First Anglo-Maratha War (1775-1782), initiated by the Treaty of Surat in

1775 and concluded by the Treaty of Salbai in 1782, signed between Raghunath Rao and the British.

The Treaty of Salbai (17 May 1782) - It was signed between the Maratha Empire and the British East India Company. According to it, control of Salsette and Bassein were taken by the company.

Second Anglo - Maratha War (1803-05) The conflict was instigated by Peshwa Baji Rao II, the final Peshwa, and the Holkar, representing various Maratha clans. It arose due to Baji Rao's agreement to the subsidiary alliance in 1802. The Marathas suffered defeat at the hands of the British, leading to the signing of the Treaty of Bassein. Which restored Baji Rao II as Peshwa and accepted all the terms of the Subsidiary Alliance system.

Third Anglo-Maratha War (1817-1818) - It is also known as Pindari war. It was fought between Lord Hasting (Governor General of British) and Marathas.

This formally ended the Peshwa rule. The Peshwa surrendered in 1818. He was dethroned and sent away to Bithur (near Kanpur).

The Battle of Palkhed- It was fought between the Maratha Empire and Asaf Jah I (Nizam-ul-Mulk of Hyderabad) in 1728 at the village of Palkhed in Maharashtra. The Marathas defeated the Nizam.

The battle of Kharda- It was fought between the Nizam of Hyderabad and the Maratha Empire in 1795.

Decline of Maratha Kingdom

The Third Anglo-Maratha War led to the end of the Maratha Empire. The Maharaja of Satara was restored as the ruler of his princely state, but most of his territory was annexed and became part of the Bombay Presidency. In 1848, this territory was also annexed under the "doctrine of lapse policy" introduced by Lord Dalhousie.

Ahom Dynasty

It was established in 1228 by Chaolung Sukapha in the Brahmaputra Valley of Assam. Treaty of Yandaboo in 1826 finished the Ahoms rule in Assam.

Chaolung Sukapha or (Siu-Ka-Pha) - He was the founder of the Ahom kingdom

and the architect of Assam. He is believed to be a descendant of the god Khunlung. He established the capital at Charaideo in 1253 and developed large farms for rice cultivation. Widely known as the architect of 'Bor Asom' or 'Greater Assam', Sukapha died in 1268. Assam celebrates 'Asom Divas' on 2nd December annually to honor Sukapha and his legacy.

The last Ahom King was Purandar Singha.

Other Kings of the Dynasty - Sukapha, Swargadeo Rudra Singha, Pratap Singha (1744-1751), Gaurinath Singha (1751-1769) and Rajeswar Singha (1769-1780).

Ahom Kingdom re-organised in 1497 by Suhungmung.

Battle of Saraighat - It was fought between the Mughal Empire and the Ahom kingdom in 1671. The Ahom army led by general Lachit Barphukan, defeated the Mughal forces.

Important Cities

Charaideo - It was the first Capital of the Ahom kingdom established by **Sukapha** in 1253 AD.

Charagua - It was made capital after Charaideo. It was founded by Sudangpha (Bamuni konwar).

Rangpur - It was established by Rudra Singha.

Jorhat - It was established by Gaurinath Singha, was the final capital until British occupation in 1826.

Pragjyotishpur - It was ancient Guwahati, served as the capital under the Kamarupa kingdom, with Durjaya being the ancient name for North Guwahati.

Administration

Ahom society was divided into clans or khels. The Ahom state depended on forced labour (Paik system).

Military Organization

The Paiks were the main army of the state. Types of Paiks- serving and non-serving. Barbarua was the Commander in Chief of upper Assam and the Barphukan of lower Assam.

Architecture

King Swargadeo Pramatta Singha built Rang Ghar (amphitheater).

Shiva Doul - It is one of the highest altitude Shiva temples in India. It was built in 1734.

Other temples of the Ahoms - Surya Mandir, the Ranganath Dol, the Fakua Dol, the Ganesa Mandir, and the Visnu Dol.

Joysagar tank was excavated by Rudra Singha in 1707. Phuleswari Devi had excavated Gaurisagar tank in 1723.

Queen Ambika devi had excavated the Sivasagar Tank in 1733. She was the wife of Swargadeo Siva Singha.

Literature

Buranjis was a manuscript that was originally written in Ahom and later in Assamese.

Culture and Occupation

Agriculture became the primary source of income. The new rice cultivation techniques were implemented. Mohur, Rupee, and Pana were their coins.

Religion

The religion of the Ahom Kingdom was based on the worship of Deities known as Phi and Dam (Ancestor Spirit).

Vaghela Dynasty

The Vaghela Dynasty, ruling from 1244 A.D to 1304 A.D, marked the final Hindu rule in Gujarat with its capital at Dhavalakka (Dholka). They expanded dominion across Gujarat, parts of Rajasthan, Madhya Pradesh, and Maharashtra. Someshvara, a prominent figure in the court, was known for his role as a priest and Sanskrit poet.

Lawan Prasad, a general under Bhimdev-II Solanki, rebelled and seized control of Dhawalgadh, Dhandhuka, and parts of southern Gujarat.

Notable Rulers

Virdhaval - He was the son of Lavanaprasada and Madanarajni. He had assumed the title Maharaja ("great king").

Vishal Dev - He won military campaigns against Malwa (Parmara).

Karandev (1297-1304) - He was the last Vaghela king and also the last Hindu ruler of Gujarat. His reign saw Gujarat's annexation by Alauddin Khilji's forces, resulting in the dynasty's downfall in 1304.

Suryavamsi Gajapati Dynasty

The dynasty ruled parts of Orissa and Andhra Pradesh in the 15th century. Kapilendra Deva was the founder of the dynasty.

Rulers

Kapilendra Deva (Reign - 1434 – 1467 A.D.) - The ruler of the Gajapati Empire, reached its zenith in 1464. He expanded the empire, promoted cultural and religious development, and abolished taxes for Brahmins. He divided the empire into administrative divisions and claimed descent from the Mahabharata's Surya Vamsha. He died around 1466-67 due to conflict between his sons.

Purushottama Deva - He reigned from 1467 to 1497 A.D.

Prataparudra Deva - He reigned from 1497 to 1540 A.D.

Kakharua Deva - Last ruler of this dynasty.

Pancha Sakhas were the collective name for the famous five Saints (Balarama Dasa, Jagannatha Dasa, Ananta Dasa, Achyutananda Dasa, and Jasobanta Dasa) of medieval Odisha.

Military Arrangement

The Odia poet Sarala Das has given descriptions about the military divisions in his Odia Mahabharata. Hantakaru Dala is the first division of the army. It was the marching army and was responsible for forward scouting, clearing jungles and marking roads for the army.

Thata is the second advance unit to march or charge in the battle formations. Main army - Denkiya, Banua or Dhanuki, Cavalry.

Gajapati Infantry Units

These are major Infantry Units- Dala, Bhuiyan, Vahini, Chamu.

Decline

The Gajapatis had lost much of their southern dominion to Vijayanagar and Golconda by the early 16th century.

Kakatiya Dynasty

It was a South Indian dynasty that ruled most of the eastern Deccan region comprising present day Telangana and Andhra Pradesh, and parts of eastern Karnataka and southern Odisha between the 12th and 14th centuries. Its capital was established in 1195 at Orugallu, now known as Warangal (Telangana).

Notable Rulers

Prataparudra I - The dynasty assumed sovereignty under Prataparudra I in 1163. He ruled between around 1158–1195. He was the son of Prola II.

Rudrama Devi (Reign - 1262–1289) - She succeeded the Ganapati Deva. She was a significant ruler of the Kakatiya dynasty in South India. Marco Polo made note of Rudramma Devi's rule. She also captured important forts like Renadu, Eruva Mulikinadu, and Satti.

Prataparudra II - His reign began in 1289 and ended with the demise of the dynasty in 1323. Alauddin Khalji invaded Kakatiya kingdom in 1303 AD.

Administration

The administration was military-based, with territory divided among Nayankara chiefs who maintained troops for the king. Villages were grouped into Sthala (10-60 villages) and Nadu (multiple Sthalas), managed by Ayagar officials, making the village the basic administrative unit.

Art and Architecture

The Kakatiya Dynasty owned the **Koh-i-Noor Diamond**, mined from the Kollur mines. They built the iconic Kakatiya Thoranam in the 12th century, now emblematic of Telangana.

The **Thousand Pillar Temple**, built by Kakatiya King Rudra Deva, is a star-shaped triple shrine dedicated to Vishnu, Shiva, and Surya. Ramappa Temple and Golconda Fort in Telangana were also constructed during the Kakatiyan period.

Decline

Prince Jaunakhan defeated and annexed the Kakatiya kingdom in 1323, renaming the capital Oragallu as Sultanpur. Prataprudra Deva committed suicide while being taken captive to Delhi.

Nawabs and Begums of Bhopal

The Nawabs of Bhopal were the Muslim rulers of Bhopal, now part of Madhya Pradesh, India.

The Nawabs first ruled under the Mughal Empire from 1707 to 1737, under the Maratha Empire from 1737 to 1818, then under British rule from 1818 to 1947, and independently thereafter until it was acceded to the Union of India in 1949.

Nawabs of Bhopal

The Nawabs of Bhopal were Nawab Dost Mohammad Khan (1723-1728), Sultan Mohammad Khan (1728-1742), Faiz Mohammad (1742-1777), Hayat Mohammad (1777-1807), Ghaus Mohammad Khan (1807-1826), Muiz Mohammad (1826-1837), Jhangir Mohammad (1837-1844), Begum Sultan Shah Jehan (1844-1860) and (1868-1901).

Al-Haj Nawab Sir Hafiz Muhammad Hamidullah Khan Bahadur ruled from 1926-1947. He was the Last of the sovereign Nawabs of Bhopal.

Begum Rule

The Begums of Bhopal played a vital role, from the early 20th century, in promoting education among Muslim women. They founded a primary school for girls at **Aligarh**.

Qasida Begum - She was the first woman in South Asia to successfully assert the right of Muslim women to legally be the ruler of a state.

Sikander Begum - She was the only daughter of Qadsia Begum, became the Nawab of Bhopal in 1844.

Shahjehan Begum - She became rightful heir after the death of her mother, enigmatic Sikandar Begum.

Sultan Jahan Begum - She was the oldest Begum of Bhopal. She was invited

to attend the Coronation of King George V in London.

She founded educational institutions in Bhopal, establishing free and compulsory primary education in 1918. From 1920 until her death, she was the founding Chancellor of Aligarh Muslim University. In 1914, she was the President of the All-India Muslim Ladies' Association.

Nizams of Hyderabad

The first nizam of Hyderabad was Nizam al-Mulk, who was intermittently viceroy of the Deccan under the Mughal emperors from 1713 to 1721 and who resumed the post again under the title Asaf Jah in 1724.

Battle of Shakar Kheda - It took place on 11 October 1724 at Berar near Aurangabad. It was held between Nizam-ul-Mulk and Mubariz Khan, Subedar of Deccan. Nizam ul-Mulk defeated and killed Mubariz Khan in the battle. Nizams established power in Hyderabad state in this battle.

Nizam-ul-Mulk Asaf Jah I (1724 - 1748) - He was also known as Chin Qilich Qamaruddin Khan. He was also one of the most influential members of the court of Mughal Emperor Farrukh Siyar.

He was elevated to the rank of 2000 Zaat and 500 Sawar in 1688 for attacking the fort of Adoni. Aurangzeb dispatched him to Bijapur, to put down a mutiny in 1698.

Governor of Bijapur - In 1702, Aurangzeb appointed Mir Qamaruddin as the subahdar (governor) of Bijapur and presented him with a horse.

Nizam Ali Khan (1762–1803) - He was the second Nizam of Hyderabad state. He shifted the state capital from Aurangabad to Hyderabad in 1763.

Mir Akbar Ali Khan (1803–29) - He was the third nizam of Hyderabad. He was a contemporary of Mughal emperor Shah Alam II.

Mir Tahniyat Khan (Afzal-ud-Daula or Asaf Jah V) - He ruled from 1857 to 1869. He also built the Afzalgunj mosque.

Mir Mahbub Ali Khan or Bahadur Asaf Jah VI (1869–1911) - He was 17 years old when he assumed sovereign rights in 1885.

Mir Osman Ali Khan or Bahadur Asaf Jah VII (1911–67) - He was the last nizam of Hyderabad. He abolished the death penalty from the criminal code for civilians in 1911. He separated the judiciary from the executive in 1921.

Nawabs of Awadh

The ancient name of Awadh was Kosala. Avadh is located in Uttar Pradesh on the bank of the Gomti river. The capital of Oudh/Awadh was Ayodhya in ancient times.

Notable Rulers (Reign)

Saadat Khan founded Awadh as an autonomous state in 1722 AD and made Faizabad his capital.

Saadat Ali Khan or Burhan ul Mul Sa'adat Khan was the first Nawab of Awadh. He was appointed the Subadar Nawab in 1722 and ruled till 1739.

He was appointed as the governor of Awadh by Muhammad Shah, the Mughal Emperor.

Safdar Jung or Adbdul Mansur- He was Saadat Khan's son-in-law and fought alongside him in the Battle of Manpur against Ahmed Shah Abdali.

Suja-ud-daula (1754-1775) was the third nawab of awadh. He supported Shah Alam II and Mir Qasim in the battle of Buxar. His burial place is known as Gulab Bari, Faizabad.

Asaf-ud-daula, Awadh's 4th Nawab (1775-1797), sheltered Shah Alam II, shifted capital to Lucknow in 1755, built Bara Imambara/Asafi Imambara in 1784.

Asif Jah Mirza (1797-1798) was also known as Wazir Ali Khan. He was the 5th Nawab of Awadh. He ascended to the throne with the support of the British.

Ghazi-ud-din-Haider (1814-1819) - He ascended the throne in 1814. He built two houses, the Mubarak Manzil and Shah Manzil in the Moti Mahal complex. He introduced the animal fighting sport for the first time in Lucknow. He died in 1827.

Nasir-ud-din Haider (1827-1873) - He ascended the throne after Ghazi-ud -din haider. He set up an observatory called the Tarawali kothi in lucknow. He established Darshan vilas (a european style kothi) at farhat Buksh

complex in 1832.

Muhammad Ali Shah (1837-1842) - He was 63 years of age when he ascended the throne. He started building the Chhota Imambara.

Amjad Ali Shah (1842-1847) - After Muhammad Ali Shah, his son Amjad Ali Shah ascended the throne.

Wajid Ali Shah (1847-1856) was the last nawab of Awadh. Lord Dalhousie annexed the state on the basis of misgovernance in 1856.

He was popularly known as Jaan-i-Alam and Akhtarpiya. In 1851 Governor General Lord Dalhousie described the kingdom of Awadh as "a cherry that will drop into our mouth one day".

The Nawab of Awadh gifted the illustrated Badshah Nama to King George III in 1799.

Nawabs of Bengal

Murshid Quli Khan (real name Surya Narayan Mishra) was the first Nawab of Bengal from 1717-1727, under whom Bengal became independent from Mughal rule. He transferred his capital to Mukhusabad (later Murshidabad) in 1704. Murshidabad on the Bhagirathi river banks emerged as a silk center.

Sarfaraz Khan - His maternal grandfather, Nawab Murshid Quli Khan (Bengal, Bihar and Orissa) nominated him as the direct heir to his empire as there was no direct heir. After Murshid Quli's death in 1727, Sarfaraz ascended to the throne of the Nawab.

Shuja-ud-Din Muhammad Khan, 2nd Nawab of Bengal 1727-1739, appointed Subedar of Odisha and Bengal, married Murshid Quli Khan's daughters.

Alivardi Khan was the Nawab of Bengal from 1740 to 1756. He defeated Sarfaraz Khan in 1740, toppling the Nasiri dynasty of Nawabs. He encountered Maratha raids under Raghuji Bhonsle, surrendering Orissa in 1751. He founded the Afshar Dynasty after overthrowing the Nasiri Nawabs.

Siraj-ud-Daula was the last independent Nawab of Bengal. At age of 23, he succeeded his maternal grandfather Alivardi Khan in 1756. He established Nizamat Imambara in Murshidabad in 1740. Betrayed by Mir Jafar, he lost the

Battle of Plassey in 1757 to the British. He was the son-in-law of Murshid Quli Khan.

The Battle of Plassey on June 23, 1757 was between Siraj-ud-Daula and British East India Company led by Robert Clive. Siraj was betrayed by Mir Jafar, leading to British victory - their first major and decisive victory for the East India Company in India. It occurred during the reign of Mughal Emperor Aziz-ud-din Alamgir II after the Black Hole Tragedy of 1756.

Mir Jafar betrayed Nawab Siraj-ud-Daulah and helped the British conquer Bengal in the Battle of Plassey in 1757. He was made Nawab by the British but was replaced by Mir Qasim in 1760. Mir Qasim revolted against the British. In 1762 - 63, Mir Jafar was reinstated as Nawab by the British until his death in 1765.

The Battle of Buxar in 1764 pitted the British East India Company led by Hector Munro against the combined Mughal-Bengal-Awadh forces, resulting from a trade dispute. The British victory established their dominance in Bengal, Bihar, Orissa, making them a major power in North India.

The Treaty of Allahabad (1765) was signed by Robert Clive with the Nawab of Awadh and Mughal Emperor Shah Alam II. The Nawab ceded territories, paid Rs 50 lakh to the Company, and gave his estate away. The Emperor granted the Company the Diwani of Bengal, Bihar, Orissa for Rs 26 lakh yearly and Nizamat functions for Rs 53 lakh. This gave the Company control over revenue collection and military affairs in those provinces.

Dual Government in Bengal (1765 - 72) - Robert Clive established a dual system of government in Bengal where the East India Company controlled revenue collection (diwani) and military/judicial functions (nizamat), while appointing deputy diwans. In 1772, Warren Hastings abolished this system, consolidating Company power.

The Treaty of Alinagar in 1757 between Robert Clive and Nawab Siraj-ud-Daulah recognized the trade concessions granted to the British East India Company by the Mughal emperor Farrukhsiyar's farman in 1717.

Manikya Dynasty

The Kingdom of Tripura was founded by Maharaja Maha Manikya (Chang Thung Pha) in 1400 AD in northeast India.

His successors conquered Bengal, Assam, and Burma, and the Manikya dynasty ruled Tripura as a princely state under British influence from 1761 until 1949.

Bir Chandra Manikya (ruled 1870s) initiated reforms and befriended Rabindranath Tagore.

Bir Bikram Kishore Debbarman (1923-1947) is considered the father of modern Tripura architecture.

Maharani Kanchan Prava Devi signed the Tripura Merger Agreement in 1949, integrating the princely state into India on October 15, 1949.

Other Dynasties

Patiala dynasty

The founder of the Patiala dynasty was Ala Singh. This dynasty ruled from 1761 A.D to 1967 A.D. In 1763, Baba Ala Singh laid the foundation of the Patiala Fort, known as Qila Mubarak, around which the present-day Patiala city is built.

Bahadurgarh Fort - It is located in Patiala, Punjab. This grand fortress was built by Nawab Saif Khan during the reign of Aurangzeb and was originally called Saifabad. The fort was later renamed to commemorate the stay of the ninth Sikh Guru, Guru Tegh Bahadur.

Notable Rulers of Patiala Dynasty -

Maharaja Karam Singh - He was the fourth ruler of Patiala. He ascended the throne of Patiala on 30 June 1813.

Maharaja Narinder Singh - He was Maharaja of the princely state of Patiala. During his reign, the Moti Bagh Palace was constructed at Patiala. He was a Member of the Indian Legislative Council during Lord Canning's Viceroyalty.

Maharaja Bhupinder Singh - He was the son of Maharaja Rajinder Singh I of Patiala State. He was the most popular king. They were the first Indian to own a car and aircraft (Wright brothers model-B craft). Moti Bagh Palace was completed at the time of Maharaja Bhupinder Singh.

Yadavindra Singh - He was the ninth and last ruling Maharaja of Patiala. He was also an Indian cricketer who played in one Test in 1934.

Sultanate of Jaunpur (Sharqi dynasty)

It was a Persian Muslim kingdom in northern India between 1394 and 1479. It was ruled by the Sharqi dynasty. It was founded by Khwajah-i-Jahan Malik Sarwar in 1394.

In 1479, Sultan Hussain Shah was defeated by the forces of Afghan chieftain Bahlul Lodhi and this dynasty came to an end.

Safavids Dynasty

It ruled modern day Iran. It lasted from 1501 to 1736. The founder of this dynasty was Shah Ismail. Humayun took refuge in the court of the Safavid ruler of Iran. During the reign of Akbar, Kandahar was wrested from the Safavid rulers.

Notable Rulers of Kashmir

Queen Didda - She is also known as "The Witch Queen" and "The Catherine of Kashmir". She was the ruler of Kashmir from 980 AD to 1003 AD. Rajatarangini written by Kalhana mentions her.

King Lalitaditya Muktapida - He ruled over Kashmir. He was from the Nagvanshi Karkota Kayastha Dynasty of Kashmir. His birth name was Muktapida and his older brothers were Chandrapida and Tarapida. In the year 760 AD, the Lalitaditya era came to an end by his sudden death.

Notable Dynasty in China

Ming Dynasty - It was an imperial dynasty of China that ruled from 1368 to 1644 after the fall of the Mongol-led Yuan Dynasty.

Qing dynasty - It was officially the Great Qing. It was the last dynasty in the imperial history of China. It was established in 1636 and ruled China from 1644 to 1912.

Yuan dynasty - It is also called the Mongol dynasty. It was established by Mongol nomads that ruled portions and eventually all of China in the 13th century.

Sui dynasty - It was a short-lived imperial dynasty in China that lasted from 581 to 618. It was founded by Yang Jian.

Books and Authors of Medieval India

Books

Humayun Nama - It was written by Gul-Badan-Begum (Humayun's sister) in the Persian Language. It provides a detailed account of the life of Mughal Emperor Humayun.

Akbarnama - It is written by Abul Fazl in Persian Language. It was translated into English by Henry Beveridge in the early twentieth century. It is divided into three books - The first book dealt with Akbar's ancestors. The second recorded the events of Akbar's reign. The third is the Ain-i Akbari. It deals with Akbar's administration, household, army, the revenues and geography of his empire.

Mazma-ul-Bahrain ("The Confluence of the Two Seas") - It is written by **Dara Shukoh**. It is written in Persian.

Shah Jahan Nama - It is written by Inayat Khan in the Persian language. It was written in Hindi by Muhammad Sadiq.

Badshah Nama - It was written by Abdul Hameed Lahori. It is also known as "Padshahnama".

Prabandha Chintamani - It was compiled in 1304 CE, in the Vaghela kingdom, by Jain scholar Merutunga.

Muntakhab-ut-Tawarikh - It was written by Abdul Qadir Badauni in Persian language.

Munis al Arwah - It was written by Jahanara. It is the biography of Shaikh Muinuddin Chishti and Mullah Shah.

Tarikh-i-Mubarak Shahi - It was written by Yahya bin Ahmad Sirhindi in Persian language.

Futuh-us-Salatin - It was written by Abdul Malik Isami. It is known as Shahnama of Medieval India.

Kitab-ul-Hind - It was written by Al-Biruni in Arabic language. It is also known as Tarikh-ul-Hind. It is divided into 80 chapters.

Man Kautuhal - It was written by Raja Mansingh Tomar.

Indica - It is written by Megasthenes. In this book it is defined that the royal palace of Magadha is made of wood.

Amuktamalyada - It was written by Krishnadevaraya of the Vijayanagara dynasty. It was written in Telugu language.

Tarikh-i Dilkusha - It was written by Bhimsen in 1707 in the Persian language. It is the memoir of Aurangzeb.

Padmavat - It was written by Malik Muhammad Jayasi. It tells the story of a Sri Lankan princess Padmavati. She was also the queen of Chittor. Other books of Jayasi are Akharwat and Aakhiri Kalam.

Taj-ul-Masir - It was written by Sadruddin Hasan Nizami.

Siyar-e-Mutakherin - It was written by Ghulam Hussain. It describes the Battle of Plassey.

Mirat-i-Ahmadi - It was written by Diwan Ali Muhammad Khan in 1750.

Shahnameh (Book of Kings) - It is an epic composed by the Iranian poet Hakim Abul-Qasim Mansur (Ferdowsi Tusi).

Tarikh-i Firoz Shahi - It is written by Zia ud Din Barani. It was completed in 1357 CE and was dedicated to the reigning monarch.

Tantraloka - It was written by Abhinavagupta.

Tabaqat-i-Akbari - It is written by Khwaja Nizamuddin Ahmad.

Vikramanka Charita - It was written by Kavi Bilhana. He was a Kashmiri poet.

Amarakosha - It is also known as Namalinganushasan. It is a Sanskrit Thesaurus written by Amarasimha.

Tolkappiyam - It is the oldest Tamil grammar book written by Tolkappiyar.

De Gradibus - It was written by Al-Kindi. It describes the theory and nature of compound medicine.

Geet Govinda - It was composed by Jaidev in the 12th century. This poem is a depiction of the love of Radha and Lord Krishna.

Ramcharitmanas - It is an epic poem in the Awadhi dialect of Hindi, composed by Goswami Tulsidas.

Aparajitparcha - It is written by Bhuvanadeva. It is written in Sanskrit language.

Karpuramanjari - It was composed by Rajashekhra. It was written in prakrit Language.

Lilatilakam - It was written in Manipravalam language of the fourteenth century.

Sangita-Ratnakara - It was composed by sarangadeva. It is written in Sanskrit which describes music and drama.

Raag Tarangini - It was written in the 15th century. It is ascribed popularly to Lochan Kavi (Lochan Pandit). It initiated an alternative system of the division of rags, 'the Thaata system'.

Natural History - It is often known as Naturalis historia. It was written by the Roman writer Pliny the elder.

Authors

Nannaya Bhattarak wrote Mahabharatamu (11th century), which is in Telugu literature.

Al-Masudi - He wrote **Akhbar-al-zaman** ("The History of Time"). His major works were created in 30 volumes. His famous works are - Kitab-al-awsaf, Muruj-al-dhahab wa ma'adin-al-jawahir, Kitab at-Tanbih wa-i-Ishraf, Kitab Muruj al-Dhahab wa-Ma'adin al-Jawhar, Kitab al-tanbih wa al-ishraf, among others.

Ziauddin Barani - He wrote his chronicle first in 1356 and another version two years later. His writings - Tarikh-i-Firoz shahi, Fatwa-i-Jahandari, Sana-i-Muhammadi, Tarikh-i-Barmaki.

Minhaj-i-Siraj - He was a Persian historian of the 13th century. He wrote Tabaqat-e-Nasiri.

Other Authors and Books

Bhavabhuti - "Malatimadhava," "Mahaviracharya," and "Uttararamacharita." **Ibn Battuta** - "Rihla - My Travels" in Arabic. **Mutamid Khan** - "Iqbal Naam-e Jahangiri". **Al-Khwarizmi** - "Kitab al-Jaber."

Poets

Khawaja Moinuddin Chishti - He was born in 1141 CE in Chishti, Herat, Afghanistan, known as 'Garib Nawaz'. He established

the Chishtiya Sunni mystical order in the Indian subcontinent in the early 13th century, with Qutubuddin Bhaktiyar Kaki as his immediate student.

Fariduddin GanjShakar (Baba Farid) - He was born in Multan, Pakistan in the 12th century, was a renowned Sufi preacher and a leading figure in the Chishti Sufi order. He spread Sufi teachings through poetry in Persian, Arabic, and the local Hindawi dialect.

Syed Muhammad Nizamuddin Auliya - He, also known as Hazrat Nizamuddin, was a Sunni Muslim scholar and Sufi saint born in 1238 in Badaun (Uttar Pradesh). In those days, Badaun was called Qubut-ul-Islam or Madinat-ul-Aulia. Among his famous disciples were Nasiruddin Chiragh Dehlavi, Amir Khusrow, and Burhanuddin Gharib.

Amir Khusrau - He was a prolific Sufi poet born in Patiyali (Kasganj, U.P.), known as "Voice/Parrot of India" (Tuti-e-Hind). A disciple of Nizamuddin Auliya, he pioneered Qawwali music, served as Alauddin Khilji's court poet, and was the first to record Jauhar practice in Persian. His major works include Nuh-i-Sipahr, Ashiq, Tughlaqnama, and Khamsa.

Abul Fazl - He was born in Agra and He was one of Akbar's prominent Navratnas. He authored the influential works Akbarnama, Ain-i-Akbari, Ruqayat and Insha-i-Abul Fazl. Abul Fazl composed the 'infallibility decree' and played a key role as Akbar's advisor before being assassinated by Vir Singh Bundela in 1602.

Malik Muhammad Jayasi - He was an influential Sufi poet from Awadh. He wrote the famous Awadhi epic Padmavat (1540) and other works like Akhiri Kalam, pioneering Bhakti literature in medieval India. He belonged to the Nirguna Premashrayi stream of the Bhakti movement.

Shaikh Salim Chishti - He was a revered Sufi saint, descended from Khawaja Moinuddin Chishti. The Mughal Emperor Akbar founded the city of Fatehpur Sikri near Agra in his honor. Salim Chishti's tomb is located within the mosque complex at Fatehpur Sikri.

Mulla Daud - He was a 14th century Sufi poet from Dalmu (now Raebareli) who pioneered Sufi literature with his Awadhi work "Laur Chanda". He served as a

patron to the noble Jaunashah under the Tughlaqs and wrote Tarikh-i-alfi.

Hafiz (Hafez) - He was a 14th-century Persian (Iranian) lyric poet. His collection of poems is known as 'Diwan' or 'Diwan-e-Hafiz' (Persian). Hafiz became a poet at the court of Abu Ishak.

Lal Ded - She is also known as Lalleshwari, Lalla Yogini and Lalla Arifa. She was a Kashmiri saint of the 14th century. She was the creator of Vakhs, a kind of poetry.

Goswami Tulsidas - He was a renowned Hindu poet and a contemporary of Akbar. He wrote the epic Ramcharitmanas in Awadhi. A devotee of Lord Rama, he authored works like Hanuman Chalisa and founded the Sankatmochan Temple in Varanasi. His other major works include Vinaya Patrika, Geetawali, and Janaki Mangal.

Chaitanya Mahaprabhu - He was a Bengali spiritual leader who founded Gaudiya Vaishnavism. Considered an avatar of Krishna, he revived Vaishnavite traditions through his six principal disciples known as the Goswamis of Vrindavan.

Nimbarka - He is also called Nimbadiya. He was a Telugu philosopher who founded the Nimbarka Vaishnava sect that worshipped Krishna-Radha. He propounded the 'Dvait advaita' philosophy and prescribed five paths to salvation, including his commentary Vedanta-Parijata-Saurabha.

Vallabhacharya - He was a 16th-century saint considered the founder of the Vaishnavite school of Hinduism. He was a philosopher who propounded the Shuddhadvaita Vedanta philosophy.

Major composition of Vallabhacharya - Patravallamban, Madhurashtakam, Gayatribhashya, Purushottam Sahastranaam, Girirajdharyashtakam and Nandkumarashtakam.

Agha Hasan Amanat - He was a 19th-century Urdu poet, writer, and playwright from Lucknow, affiliated with the court of Wajid Ali Shah. He is known for writing the Urdu play 'Inder Sabha'.

Mirza Ghalib (Asadullah Khan) - He was born in Agra (1797), and was the greatest Urdu poet and a contemporary of the last Mughal emperor Bahadur Shah Zafar II. He mastered ghazals, masnavis and

qasidahs, and received the titles Dabir-ul-Mulk and Najm ud-Daulah.

Mirza Wajid Ali Shah, - The last King of Awadh, he was also a prolific poet (pen-name Qaisar/Akhtarpiya) and patron of arts., who authored over 40 works, including ghazal collections and an autobiography "Huzn-e-Akhtar".

Foreign Travelers (Timeline)

The Indian Continent was known as Swarn Bhoomi by ancient foreign travelers.

Deimachos (3rd Century BCE) - He was a Greek of the Seleucid Empire. He was the ambassador in the court of Bindusara. Syrian King Antiochus-I sent him to the court of Bindusara.

Megasthenes (302 to 298 B.C.)- He was ambassador of Seleucus Nikator, who arrived in India in 305 B.C. during Chandragupta Maurya's reign. He resided in Pataliputra and authored "Indica," referring to Chandragupta as 'Sandrocottus.'

Heliodorus - He was an Indo-Greek ambassador from Antialcidas, was sent to King Bhagabhadra's court in 113 BCE. He erected the "Heliodorus Pillar," also known as "Khamb Baba," near Bhopal, which still stands. Bhagabhadra ruled during the Sunga dynasty.

Sung Hyun and his companions visited India in 518 CE during the reign of Empress Hu of the Northern Wei Dynasty. Their travels, detailed in Yang Xianzhi's Loyang Jielanji, included a trip to the Swat region, where Sung Hyun originated the story of the Gandhara dynasty.

Fa-Hien (405 A.D. to 411 A.D.)- He was a Chinese Buddhist monk. He travelled to India during the rule of Vikramaditya (Chandragupta II) and he visited the Lumbini (birth place of Buddha). He wrote about India in his book Fo-kwo-ki (Travels of Fa-hien).

Hiuen Tsang (630A.D. - 645 A.D.)- He was a Chinese traveller. He was popularly known as Xuanzang, travelled from China to India to collect Buddhist texts. He was also known as Tsang hee. He wrote a detailed description of India during the reign of Harsha Vardhana in his book 'Si-yu-ki' ('The Record of the Western World').

I-tsing (671A.D. - 695 A.D.)- He was a Chinese traveller. He visited India during the reign of Harshavardhana. He stayed at Tamralipti for three years and learned Sanskrit. He resided at the Nalanda monastery for ten years (676-685) and devoted himself to the study of the **Vinya**.

Sulaiman- He is an Arab traveler who visited India in the 9th century. He described Amoghavarsha as one of the four great kings of the world who respected the Muslims and allowed the construction of mosques.

Al-Masudi (957 A.D.)- He was an Arab traveller. He has given an account of India in his book 'Muruj- al-Zahab'. He was known as the "Herodotus of the Arabs." He was a prolific author who is thought to have produced 34 works in total.

Alberuni (1024–1030 A.D.)- He came from Uzbekistan. He was also known as Abu Rehan Mahmud. He came along with Mahmud Ghazni during one of his Indian raids. He wrote Kitab-ul-hind or Tahqiq-i-Hind. He translated Sanskrit works on astronomy, mathematics and medicine into Arabic. His notable works are Patanjali's grammar work. He translated the works of Euclid (a Greek mathematician) into Sanskrit.

Marco Polo- He was a traveler from Venice (Italy). He came to India during Pandya kingdom. He visited Southern India during the reign of Rudramma Devi of the Kakatiyas. He wrote about Indian geography and economic history in his book "The Book of Ser Marco Polo."

Ibn Battuta (1333 to 1347)- He was a Moroccan traveler. He came to the court of Muhammad Bin Tughlaq. He was appointed Qazi or judge by Muhammad bin Tughlaq. He also went to the court of zamorin. He Wrote Kitab-ul-rahla (Rihla).

Shihabuddin al-Umari (1348 A.D.)- He came from Damascus. He gives a vivid account of India in his book "Masalik albsar fi-mamalik al-amsar".

Afanasy Nikitin (1470- 1474 A.D.)- He was a Russian merchant. He came to the Bahmani kingdom during the reign of Muhammad III. He describes the condition of the Bahmani kingdom under Muhammad III in his books "The journey beyond Three seas".

Abdur Razzak (1443 – 1444)- He was a Persian traveller and Ambassador to the

Shahrukh of Timurid dynasty, visited India during Deva Raya II's reign. He wrote "Matla-us-Sadain wa Majma-ul-Bahrain," offering a famous description of Vijayanagara.

Nicolo Conti (1420 - 1421)- He was an Italian merchant. He visited India during the reign of Deva Raya I (Sangam dynasty ruler) of Vijayanagar. He gave a comprehensive account of the Hindu kingdom of Vijayanagar.

Duarte Barbosa (1500-1516 A.D.)- He was a Portuguese traveller. He visited Vijayanagar Empire.

Ralph Fisch- He was the first English traveler to visit India during the reign of Akbar in 1585 AD.

Domingo Paes (1520 A.D.-1522 A.D.)- He was a portuguese traveler who visited Vijayanagara in 1510 in the reign of Krishnadevaraya. He wrote "Chronicle of the Vijayanagar kings" (Chronica dos Reis de Bisnaga).

Fernao Nunes (1535 -1537 A.D.)- He was a Portuguese traveller. He visited India during the reign of Achyutaraya of Vijayanagar empire.

Abdus Samad and Sayyid Ali- They were two famous Iranian painters and calligraphers who came to India on the invitation of Humayun.

Roberto de Nobili - He was known as the White Brahman. He came to India with Jesuit missions in the court of the Mughal emperor Akbar.

William Hawkins (1608-1611 A.D.) - He was a representative of King James I of England. He arrived in India during the rule of the Mughal Emperor Jahangir along with William Finch. He came to seek permission to open a factory at Surat.

Thomas Roe (1615 A.D. – 1619 A.D.) - He was an English diplomat. He visited India during the reign of Jahangir in 1615. He got permission for trade and established a factory at Agra, Ahmedabad and Baroch. He wrote a book "Journal of the Mission to the Mughal Empire".

Francois Bernier (1656 A.D. – 1668 A.D.)- He was a French traveller and doctor. He visited India during the reign of Shah Jahan. He served Prince Dara Shikoh as a doctor before joining Aurangzeb's court. 'Travels in the Mughal Empire' was

written by him. He compared India's situation with Europe. He called kashi as the Athens of India.

Jean Baptiste Tavernier (1638-1663 A.D.)- He was a French traveller who visited India 6 times during the reign of Shahjahan and Aurangzeb. He wrote "Travels in India".

Ibn Habal - He was an Arab physician and scientist born in Baghdad. He was known primarily for his medical compendium titled "Kitab al-Mukhtarat fi al-tibb" (The Book of Selections in Medicine).

Shaikh Ali Hazin - He visited India during the reign of Mughal Ruler Muhammed Shah.

Antonio Monserrate - He was a Portuguese priest who came to Akbar's court in 1580 with the first Jesuit mission. He became a tutor of Akbar's second son Murad. He accompanied Akbar in the expedition to Kabul in 1581.

Marco Ricci was an Italian painter of the Baroque period.

Peter Mundy (1630 AD - 1634 AD)- He was an Italian traveler who came to India during the reign of Shahjahan.

Niccolo Manucci- He was a versatile traveler from Venice, Italy. He visited India during the reign of Shahjahan. "Storia do Mogor" was written by him which gives information about the mughal empire.

Famous towns of Medieval Period

Thanjavur/Tanjore - The Rice bowl of Tamil Nadu and former Chola capital, is built around Brihadeshwara temple.

Calicut/Kozhikode - The medieval 'City of Spices' in Kerala, is where Vasco da Gama arrived in 1498.

Agra - It was known as Akbarabad, was the Mughal capital on Yamuna river under Akbar to Aurangzeb.

Kochi/Cochin - Major port city in Kerala known as 'Queen of the Arabian Sea'. Vasco Da Gama was first buried at St. Francis Church here. It's the industrial capital of Kerala.

Bijapur/Vijayapura - Founded by Kalyani Chalukyas in Karnataka. It has the

17th-century Gol Gumbaz mausoleum. Yusuf Adil Shah of Adil Shahi dynasty developed Bijapur into a significant town.

Fatehpur Sikri - It is a town in the Agra district of Uttar Pradesh. It was the capital of Mughal Emperor Akbar. The place had several beautiful monuments like Buland Darwaza, Panch Mahal, Jodha Bai's Palace and Salim Chisti's tomb.

Lahore - It is mentioned in the ancient 'Hudud al-'Alam' text, known as the "City of Gardens" due to its rich Mughal heritage. It served as the capital of the Mamluk Dynasty (1206) and under Emperor Akbar.

Patna - Its ancient name was Patliputra. In the book of 'Ain-i-Akbari' it is referred to as the paper, stone and glass industries.

It was the capital of the Magadha Empire during the Haryanka, Nanda, Mauryan, Shunga, Gupta and Pala dynasties.

Shahjahanabad - Founded by Mughal Emperor Shah Jahan on the western bank of Yamuna river, it served as his capital after shifting from Agra.

Madurai - It is a major city in Tamil Nadu. Its ancient name is Kadambavanam. It is located on the banks of river Vaigai. It is also known as Thoonga Nagaram (the city that never sleeps).

Hampi - It is situated on the Tungabhadra river in Karnataka, was the 14th century fortified capital of Vijayanagara Empire, also known as Pampa Kshetra, Kishkindha Kshetra and Bhaskara Kshetra. It's a UNESCO World Heritage Site and was the second largest medieval city after Beijing.

Surat - Once known as Suryapur, lies on the banks of the Tapi River. It flourished as a hub of western trade during the Mughal era, alongside Cambay. Referred to as the "gate to Mecca" and "the diamond city."

Kanchipuram - It is also called Conjeevaram, served as the capital of the Pallavas in Tamil Nadu. Mahendravarman, a Pallava king, ruled over parts of present-day Andhra and northern Tamil Nadu.

Ajmer - Ajmer was originally known as Ajaymeru. It was annexed to the Delhi sultanate's Slave dynasty in 1193. It was taken in 1556 by the Mughal emperor

Akbar. It was ceded to the British in 1818.

Masulipatam (Andhra Pradesh)- It was the first British trading settlement (1611) on the Bay of Bengal.

The British captured the city and fort from the French in 1759. Masulipatnam literally means "fish port town."

Krishnadeva Raya is credited with building some fine temples and adding impressive gopurams to many important south Indian temples.

He also founded a suburban township called Nagalapuram after his mother.

Hyderabad - This city was founded in the year 1591 by Mohammed Quli Qutub Shah, the fifth sultan of Qutb Shahi dynasty.

He also built the architectural centerpiece the Charminar in Hyderabad.

Tughlaqabad - The city of Tughlaqabad (near Delhi) was built by Ghiyasuddin Tughlaq (Ghazi Malik).

Aurangabad - It is officially known as Chhatrapati Sambhaji Nagar. Malik Ambar is regarded as the founder of Aurangabad (originally known as Khadki) city in 1610. It's known for the 17th-century marble Bibi ka Maqbara shrine.

Salem - It is situated in the state of Tamil Nadu. It is a city best known for steel.

Tirupati - It is situated in the state of Andhra Pradesh. It is famous for the Venkateswara temple in the sacred Tirumala hills.

Kurnool - It is a city in the state of Andhra Pradesh. It is famous for cultivation of cotton. It formerly served as the capital of Andhra State (1953-1956).

King Achyuta Raya, successor of Sri Krishnadeva Raya constructed the Kurnool Fort during the 16th century.

Udaipur - It was founded by Rana Uday Singh after the fall of Chittor to Akbar in 1568.

Modern History

Advent of Europeans

The 15th century in Europe marked a period of significant geographical discoveries. Christopher Columbus found America in 1492, while Vasco da Gama discovered a new sea route to India in 1498. Portuguese traders were the first to arrive in India, followed by the British, Dutch, Danes, and French.

Portuguese

Portuguese Arrival and Expansion - Led by Vasco da Gama, the first Portuguese mission reached Calicut in May 1498 via the Cape of Good Hope. They established trade posts at Cochin, Goa, Daman, Diu, Salsette, Bassein, and Bombay. Pedro Alvares Cabral arrived in Calicut in 1500 and set up the first European factory in India, followed by another in Cochin in 1501. The Portuguese also established their first factory in Bengal at Chittagong, and they gifted Bombay to King Charles as part of a dowry.

Portuguese Governance and Expansion - Francisco de Almeida became the first Portuguese Viceroy in 1505, implementing the 'Blue water policy' and establishing four ports on the southwestern coast. Alfonso de Albuquerque, the second governor, captured Goa from Bijapur in 1510 and expanded Portuguese influence by acquiring Malacca in Southeast Asia in 1511. Lopo Soares de Albergaria, the third governor, arrived in India in 1515.

Portuguese Colonies and Contributions - Portuguese colonies in India included Goa, Diu, Daman, Dadara, Nagar Haveli, Mangalore, Kannur, Kollam, Nagapattinam, Hooghly, Kannonore, and Surat. Sri Lanka was discovered by the Portuguese in 1505. Duarte Barbosa, a Portuguese writer, visited India from 1500 to 1516 and documented trade and society in South India in his manuscript, 'The Book of Duarte Barbosa'. The Portuguese also introduced several crops to India, including tobacco, pineapple, papaya, cashew nuts, and chili.

British

Establishment and Trade - In 1599, merchants formed the 'Governor of Company of Merchants of London

Trading to the East Indies', receiving a royal charter from Queen Elizabeth I on 31st December 1600 for exclusive Eastern trade for 15 years. Calcutta became British India's capital in 1772. The company traded in cotton, indigo dye, silk, salt, saltpeter (used in gunpowder), opium, and tea. Sawar referred to mounted soldiers in the East India Company Army, while muskets were heavy guns used by infantry soldiers.

Early Engagements - In 1608, British Captain William Hawkins sought permission from Jahangir to establish trading posts in India, with a factory permitted in Surat in 1613. Sir Thomas Roe was sent in 1615 for further concessions. The first southern factory was established in Masulipatnam in 1611, followed by Madras in 1639 and the first Bengal factory in Hugli in 1651. Fort St George, constructed in 1644 in Madras, was the first British fort in India.

Expansion and Agreements - In 1765, the Mughal Emperor Shah Alam II granted the Diwani of Bengal, Bihar, and Orissa to the East India Company. Calcutta became British India's capital in 1772. The Treaty of Seringapatam in 1792 was signed between Tipu Sultan of Mysore and Lord Cornwallis. Fort William was established in 1698 by the British East India Company in Kolkata.

Battle of Goa (9–10 December 1510): The first part of India to fall to European colonial rule was Goa on the west coast. The Battle of Goa refers to a series of naval engagements between the Portuguese Armada and the Dutch East India Company. Goa became a Portuguese colony in 1510, when Admiral Afonso de Albuquerque defeated the forces of the sultan of Bijapur, Yusuf Adil Shah.

The Battle of Swally, also known as the Battle of Suvali, occurred on 29-30 November 1612, near Surat, where the British East India Company forces, under Captain Thomas Best, emerged victorious against the Portuguese. This battle marked a turning point, signaling the decline of Portugal's commercial dominance in India and the rise of the English East India Company's influence.

Treaties signed by the British

The Treaty of Titalia - It was signed between the Chogyal (monarch) of the Kingdom of Sikkim and the British East

India Company (EIC). It was negotiated by Captain Barre Latter in February 1817. The treaty guaranteed the security of Sikkim by the British and returned Sikkimese land annexed by the Nepalese.

Treaty of Sugauli - It was signed on 4 March 1816 between the East India Company and Guru Gajraj Mishra (Gurkha chiefs of Nepal) following the Anglo-Nepalese War of 1814–16. The treaty that established the boundary line of Nepal.

The Treaty of Lhasa - It was signed between Great Britain and Tibet on 2 September 1904. It was signed by Younghusband and Thubten Gyatso, the 13th Dalai Lama of Tibet.

The Treaty of Yandabo - It was the peace treaty that resulted at the end of the First Anglo-Burmese War. It was signed on 24 February 1826. It led to the annexation of Assam by the Britishers.

Dutch

The United East India Company, founded in 1602 in the Netherlands, established trade centers in India, with Cornelis de Houtman being the first Dutch citizen to arrive in the country. The Dutch East India Company, set up its first factory at Masulipatnam in 1605, with significant factories later established at Pulikat, Surat, and other locations. Notably, the Dutch traders played a pivotal role in making India a hub for textile exports.

In 1741, the Dutch suffered a significant defeat by the King of Travancore, Marthanda Verma, at the Battle of Colachel. Additionally, they were decisively defeated by the British, led by Robert Clive, at the Battle of Bedara in 1759. The Mattancherry Palace, also known as the Dutch Palace, in Kochi, serves as a testament to Dutch influence in India.

The Battle of Bedara, fought on 25 November 1759 between the British East India Company and the Dutch Army, with assistance from Mir Jafar, the Nawab of Bengal, resulted in the overthrow of Mir Jafar and the installation of Mir Kasim as Nawab of Bengal by the victorious British.

Danes

The Danes, from Denmark, had colonies in India like Tranquebar (Tamil Nadu), Serampore (Bengal), and the Nicobar

Islands, with Serampore as their main base.

Dutch adventurer Marcellis de Boshouwer sparked Danish involvement in India. They set up their first factory in Tranquebar in 1620 and later established Serampore in Bengal in 1676.

The Serampore Mission Press, founded by Danish missionaries in 1799, was important in the region. However, Danish efforts in India eventually failed, leading them to sell all their settlements to the British in 1845.

French

The French were the last European company to come to India, entering during the reign of Louis XIV in 1664. The French East India Company was established on September 1, 1664. Francois Martin served as the first French governor of Pondicherry, while the last was Georges Escaragueil in 1954. The first French factory in India was set up in Surat in 1667 by Francois Caron, followed by one in Masulipattam by Marcara in 1669, after gaining permission from the ruler of Golconda.

Chandranagar's trading post was established with land granted by the Nawab of Bengal, Shaista Khan. Under Count de Lally, the French Army captured English forts in Vizianagaram. Renewed conflict between France and England erupted in 1756 in Europe. The Battle of Wandiwash in 1760, fought in Tamil Nadu, confirmed British supremacy in India, crushing the French ambition to establish a kingdom in the region.

French Colonies in India

Pondicherry - It served as the capital of the French Colony in India. The Basilica Church (Sacred Heart of Jesus) is located there. The foundation of Pondicherry was laid in 1673 by Martin after obtaining permission from the Qiladar of Valikandapuram under the Sultan of Bijapur.

Mahe - The French established their presence on the west coast of Mahe in 1721. Thalassery was their first site in Kerala. They built their initial fort named St. George at Cherukkalai in 1739 and completed another large fort, "Fort Mahe," in 1769.

Karaikal - Raja Pratap Singh of Tanjore ruled over Karaikal before 1739. It was

sold to the French by the Rajah of Thanjavur in 1739 and remained under French rule until 1954.

Yanam - A French colony until 1954, Yanam is still known as "French Yanam" and is a blend of French and Telugu culture. Situated on the Coromandel coast.

Chandernagar - Established as a French colony in 1673 with permission from Ibrahim Khan (Nawab of Bengal), Chandernagar became a permanent French colony in 1688. Joseph Francois Dupleix served as governor in 1730. Captured by the British Navy on 23 March 1757.

Anglo - French war

First Carnatic War (Battle of Adyar) - Fought between English and French forces from 1740 to 1748, it was an extension of the Anglo-French War in Europe sparked by the Austrian War of Succession. Joseph Francois Dupleix (French Governor-General) and Anwaruddin Khan (Nawab of the Carnatic) allied against Stringer Lawrence (British). The war concluded with the Treaty of Aix-la-Chapelle.

Second Carnatic War (Battle of Ambur) - It was fought between the Nizam of Hyderabad, and the Nawab of the Carnatic in 1749 - 1754. The war ended with the Treaty of Pondicherry in 1754. Muzaffar Jung became Nizam of Hyderabad. Muhammad Ali became the Nawab of Karnataka. The "Siege of Arcot" was the part of the Second Carnatic War between Chanda sahib, the Nawab of the Karnataka and British East India company.

Third Carnatic War (Battle of Wandiwash) - It was a conflict between the French and the British during 1758-63. The French forces, commanded by Count Thomas Arthur de Lally, captured the English forts at St. David and Vizianagaram in 1758. However, under the leadership of Eyre Coote, the English emerged victorious at the battle on January 22, 1760, in Wandiwash, Tamil Nadu. The war concluded with the Treaty of Paris in 1763, which saw the return of Pondicherry and Chandannagar to French control.

The Seven Years' War - It took place 1756 to 1763, spanning five continents and known in America as the 'French and

Indian War'. It solidified Britain's dominance as the foremost European power worldwide. The conflict stemmed from territorial ambitions in North America, with both England and France vying for control.

The Revolt of 1857

The Revolt of 1857, often referred to as the Sepoy Mutiny, was a violent uprising against British rule in India. Though ultimately unsuccessful, it marked a significant turning point in the history of the British Raj and the Indian independence movement.

Start of the Uprising - The revolt ignited on May 10th, 1857, with sepoys (Indian soldiers in the British East India Company's army) stationed in Meerut rebelling against perceived religious and cultural violations.

On May 11th, 1857, the rebellion rapidly spread to other parts of northern and central India, reaching Delhi within a day. Seeking his support, the sepoys arrived at the gates of the Red Fort to meet with the Mughal Emperor Bahadur Shah Zafar on 11 May 1857. The revolt finally ended in Gwalior on 20 June 1858.

The revolt is known by several names- The Sepoy Mutiny (by the British Historians), the Indian Mutiny, the Great Rebellion (by the Indian Historians) and the First War of Independence (by Vinayak Damodar Savarkar).

Causes of The Revolt

The Immediate cause of the revolt was the use of new Enfield rifles with cartridges made of cow and pig skin, offending Hindu and Muslim soldiers. Other causes of the 1857 revolt include grievances among Indian soldiers (sepoys) over British policies, including cultural insensitivity, religious concerns, economic grievances of people, and resentment of Indian rulers towards British authority over Annexation policies like doctrine of Lapse.

The **Doctrine of Lapse policy** devised by **Lord Dalhousie** was a policy through which the British East India Company could annex any princely state if it lacked a natural heir. This led to the annexation of several states, including Satara, Jhansi, and Nagpur.

Major centers of the Revolt

Barrackpore - Mangal Pandey, a Hindu sepoy in the 34th Bengal Native Infantry regiment of the British East India Company, led the revolt by attacking two British officers (Hughes and Baugh). Arrested and sentenced to death on March 29, 1857, in Barrackpore, he is regarded as one of the first freedom fighters of the revolt.

Delhi - The leadership at Delhi was nominally in the hands of Bahadur Shah II (Bahadur Shah Zafar) but the real control was exercised by General Bakht Khan. Bahadur Shah II was the last Mughal emperor and a symbol of Muslim resistance to British rule. After the revolt of 1857, Bahadur Shah Zafar was overthrown by the British and sent to Rangoon jail in 1858. He died in Burma in 1862 at the age of 87. His revolt was suppressed by John Nicholson.

Lucknow - It was the capital of Awadh. Begum Hazrat Mahal, one of the begums of the ex-king of Awadh, took up the leadership of the revolt. Her tomb is located in Kathmandu, Nepal and her revolt was suppressed by Colin Campbell.

Jhansi - Rani Lakshmi Bai, daughter of Moropant Tambe, led rebels after the British rejected her adopted son's claim to the throne of Jhansi. Her childhood name was 'Manikarnika', she fought the British but died on the battlefield in 1858. Her son, Damodar Rao, died four months after birth. General Hugh Rose suppressed her revolt.

Bihar (Jagdishpur) - Kunwar Singh, Zamindar of Jagdishpur, led the revolt in Bihar during 1857. His uprising was suppressed by William Taylor. Veer Kunwar Singh Jayanti celebrates his achievements in the Indian rebellion movement of 1857.

Allahabad and Banaras: Maulvi Liyakat Ali, who was a Muslim religious leader from Allahabad (Prayagraj), in the state of Uttar Pradesh led this revolt. His revolt was suppressed by Colonel O'Neill.

Faizabad: Maulvi Ahmadullah led the revolt in the Awadh region, declaring it a Jihad against the English. Known as the 'Lighthouse of Rebellion,' in Awadh region. His nickname was Danka Shah - The maulvi with the drum. The mutinous 22nd Native Infantry elected him as their leader.

Bareilly: Khan Bahadur Khan was the leader of this region. His revolt was suppressed by Sir Colin Campbell.

Kanpur - The revolt in Kanpur was led by Nana Saheb, the adopted son of Peshwa Baji Rao II and proclaimed as Peshwa. He joined the revolt primarily because he was deprived of his pension by the British. Tantia Tope led his troops. The revolt was suppressed by Sir Colin Campbell.

Baraut Shah Mal led the Jats of Baraut (Uttar Pradesh) in rebellion against the East India Company. He was killed in battle in July 1857.

Gwalior - Laxmibai, accompanied by Tantia Tope, led the Gwalior revolt. Tantia Tope, originally named Ramchandra Pandurang, was a trusted general of Nana Saheb. He died on April 18, 1859 at Shivpuri. His revolt was suppressed by General Hugh Rose.

Causes of failure - The Revolt was failed due to localization and poorly organized, lacked mass support, lack of common ideal among sepoys, lack of leadership, and did not have any support from the ruling prince.

Aftermath of 1857 Revolt

Queen Victoria's Proclamation (1858) was declared openly by Lord Canning (first viceroy of India) on November 1, 1858, at Allahabad (Prayagraj) through Durbar. The Act of 1858 was known as the Act for the better Government of India.

Main provisions: Indian territories of Britain were to be governed in the name of the British Queen, The act ended the controversial 'Doctrine of Lapse'. The designation of Governor-General of India was changed to Viceroy of India, ended the double government rule, and introduced the Secretary of State for India (Lord Stanely) who was directly responsible and accountable to the British Parliament.

Number of Europeans in the army was increased so that they have better control over administration.

Note - The skeletal remains of the Indian soldiers, who were killed by the British troops during the 1857 revolt, were exhumed at Ajnala well in Amritsar district, Punjab in 2014.

Sepoy mutiny before 1857

Bengal Mutiny - Sepoy mutiny occurred in Bengal in 1764.

Vellore mutiny (10 July 1806) - The first scene of a large mutiny in South India occurred in Madras (presently Tamil Nadu) by Indian sepoys against the East India Company. The Vellore mutiny was suppressed by General Gillesby.

47th Native Infantry Unit mutiny (1824)- A parade was held on 1 November, during which the soldiers of the 47th Native Infantry appeared without their knapsacks and refused to bring them. They again demanded their carriage bullocks or payment of double batta and refused to march without redress of grievances.

Solapur Mutiny - Dissatisfaction about the terms of payments led to the disgrace of a native regiment at Solapur when it refused to turn out for parade on 24 Nov 1838.

Socio religious reforms

In the first half of the 19th century, Indian society was entrenched in caste divisions and rigidity. British colonial rule exposed these flaws, introducing English language and modern ideals of liberty, equality, and justice. This led to a social Renaissance, with movements aiming to reform Indian society which laid down the foundation of modern India.

Brahmo Samaj : Raja Ram Mohan Roy founded the Atmiya Sabha (Forerunner of Brahmo Samaj) in 1814, the Calcutta Unitarian Association in 1821, and the Brahmo Sabha in 1828 in Kolkata which later became the Brahmo Samaj.

It was the first intellectual reform movement in modern India, aiming for the worship of the eternal God while opposing priesthood, rituals and sacrifices. It was the forerunner of all social, religious and political movements of modern India.

It split into two in 1866, namely Brahmo Samaj of India led by Keshub Chandra Sen and Adi Brahmo Samaj led by Debendranath Tagore. Raja Ram Mohan Roy is popularly known as 'Harbinger of Modern India' and 'father of Indian Renaissance'. He abolished Sati with the help of Lord William Bantynck in 1829.

Dharam Sabha - It was founded in 1830 in Calcutta by Radhakanta Deb. It was a conservative Hindu society. It campaigned against the Hindu Widow Remarriage Act, 1856.

Arya Samaj - The first Arya Samaj unit was formally set up by Dayananda Saraswati (born in Gujarat) at Bombay in 1875 later establishing its headquarters in Lahore. The aim was to re-establish the Vedas, the earliest Hindu scriptures, as revealed truth. Its slogan was Krinvanto Vishvam Aryam, which means Make the World Noble.

Mool Shankar was the paternal name of Swami Dayanand Saraswati. He translated the vedas and wrote three books Veda Bhasya, Satyarth Prakash and Bhasya Bhumika. He gave the slogan Go Back to Vedas.

In his book Satyarth Prakash, Swami Dayanand Saraswati propounded that the original home of the Aryans was Tibet. The Arya Samaj initiated the Shuddhi Movement, conducting purificatory ceremonies for Hindus converted to other religions like Islam and Christianity.

Arya Mahila Samaj - Founded in 1882 by Pandita Ramabai, Arya Mahila Samaj aimed to empower and educate women for dignified lives. Ramabai, a women's rights activist and social reformer, also established Sharda Sadan and Mukti Mission, advocating for women's education, particularly for widows and children. She received the title Pandita from the faculty of Calcutta University for her knowledge of Sanskrit.

Ramakrishna Mission - Swami Vivekananda, born as Narendranath Dutta, established the Ramakrishna Mission in 1897 in Belur, West Bengal, stressing salvation through social service and selfless action. It promoted Advaita Vedanta and four yogic ideals: Jnana, Bhakti, Karma, and Raja yoga. The motto was "Atmano Mokshartham Jagaddhitaya Cha." He authored 'The East and the West'. Vivekananda's death anniversary is observed on July 4th.

Ramakrishna Paramahansa was Swami Vivekananda's guru. In his 1893 Chicago speech, Vivekananda emphasized Hinduism's values of tolerance and universal acceptance, stating that it accepts all religions as true.

Satyashodhak Samaj - Jyotiba Phule founded the Satyashodhak Samaj in Pune, Maharashtra, on September 24,

1873. It aimed to reform society, promote education, and advocate for social rights, justice, and political access for the marginalized.

His wife, Savitribai, managed women's activities. It was an anti-caste movement. Phule wrote "Gulamgiri," dedicated to those fighting slavery in the US.

Veda Samaj - It was founded in Madras in 1864 by Keshab Chandra Sen and K. Sridharalu Naidu, aimed to abolish caste distinctions, promote widow remarriage, and advocate for women's education. It later became the Brahmo Samaj of Southern India in 1871 under Naidu's leadership. Chembeti Sridharalu Naidu was the most popular leader of the Veda Samaj.

The **Widow Remarriage Association** was founded by Vishnu Shastri Pandit in the 1850s, followed by Mahadev Govind Ranade in 1861. D.K. Karve established another Widow Remarriage Association in 1893.

Dev Samaj - Founded on February 16, 1887 in Lahore, by Shiv Narayan Agnihotri, the Dev Samaj aims to propagate beauty, truth, and goodness in all aspects of life, irrespective of their color, creed, caste, or country.

Prarthana Samaj - Founded in 1867 by Dr. Atmaram Pandurang, influenced by Keshab Chandra Sen, the Prarthana Samaj advocated monotheism, opposed caste prejudice, and critiqued priestly authority. It preceded the Paramahansa Sabha, advocating for liberal principles and caste/community barrier removal. Mahadeo Govind Ranade, joining in 1870, played a pivotal role in the Prarthana Samaj's success, earning it a nationwide reputation. R.G. Bhandarkar and N.G. Chandavarkar also served as a prominent leader.

The Young Bengal movement, led by Henry Louis Vivian Derozio in Calcutta in the 1820s, promoted radical thinking and criticized orthodox Hindu religious practices. Derozio encouraged free-thinking and advocated for liberty, equality, and freedom.

Paramahansa Mandali - It was started by Durgaram Mehtaji, Dadoba Pandurang in Bombay in 1849, aimed to reform Hinduism by opposing Brahmanical orthodoxy and advocating belief in One God. It was associated with the Manav Dharam Sabha's ideology and focused on

breaking caste rules.

Dadoba Pandurang composed Dharma Vivechan under the name of Jagadvasi Arya. He was also known as the panini of the Marathi language. Dharma Vivechan emphasized seven principles: worshiping only God, religion based on love and moral conduct, unity of spiritual religion, freedom of thought, actions consistent with reason, humanity as one caste, and universal access to the right knowledge.

Tattwabodhini Sabha - It was founded by Debendranath Tagore in Calcutta on 6 October 1839 as a splinter group of the Brahmo Samaj. It aimed to promote a rational and humanist form of Hinduism based on the Vedanta and the Upanishads.

The Theosophical Society, founded in 1875 by Madame Blavatsky and Colonel Olcott in New York (later moved to Madras), emphasized the study of ancient Hindu, Buddhist, and Zoroastrian philosophies. It advocated for universal brotherhood based on principles from the Upanishads and Vedas. Annie Besant joined the society in 1889.

The Aligarh Movement, initiated by Syed Ahmad Khan in 1875, focused on the educational and social reform of Muslim society, marking a significant revival and resurgence.

Sarvodaya Samaj - This movement was started by Vinoba Bhave in the 1950s. Its first conference was held in Madhya Pradesh.

All India Depressed Class Association was founded by Dr.B.R.Ambedkar in Nagpur on August 8, 1930. He fought for the rights of the Dalit community, workers and peasants. He belonged to the Mahar community. He initiated temple entry movements between 1927 and 1935 to combat caste discrimination. Additionally, he established the Bahishkrit Hitkarini Sabha in Bombay on July 20, 1924, to foster socio-political awareness among the untouchables.

The Harijan Sevak Sangh, founded by Mahatma Gandhi in 1932, aims to eradicate untouchability and uplift Dalit communities. Its headquarter is in Delhi, with branches across 26 states. Gandhi published the weekly journal "Young India" from 1919 to 1931.

The Servants of India Society, founded by Gopal Krishna Gokhale in Pune on

June 12, 1905, aimed at nation-building through education, social welfare, and upliftment of the underprivileged. Gokhale also established the **Deccan Sabha**.

The Depressed Class Mission, founded by V.R. Shinde in Mumbai in 1906, aimed to combat untouchability nationwide and restore respect and equality for all castes and classes.

Deccan Education Society - It was established in 1884 by Gopal Ganesh Agarkar and Lokmanya Tilak and later built by Gopal Krishan Gokhle and Dhondo Keshav Karve.

The Hindu Mahasabha, founded by Madan Mohan Malviya in 1915, aimed to protect Hindu community rights in British India. It held its first All India Conference in Haridwar in the same year. From 1929 onwards, it became more aggressive, advocating for the concept of Hindu Rashtra.

The Anushilan Samiti, a revolutionary organization in Bengal established in 1902, had over 500 branches. Barindrakumar Ghosh, brother of Aurobindo Ghosh, served as its chief. Other members included Rash Behari Bose and Sachindra Nath Sanyal.

The Asiatic Society of Bengal, founded on January 15, 1784, by Sir William Jones, aimed to study ancient languages and customs and promote indological studies during the British era.

The Akali Movement also known as the Gurdwara Reform Movement was initiated in the 1920s by Sikh reformers, aimed to purify religious places and remove social evils.

Rahnumai Mazdayasnan Sabha : It was founded in 1851 by a group of Parsis including Naoroji Furdonji, Dadabhai Naoroji, K.R. Cama, and S.S. Bengalee. It aimed to regenerate social conditions and restore the Zoroastrian religion.

Mohammedan Literary Society : It was founded in Calcutta in 1863 by Nawab Abdul Latif. It aimed to spread modern education and remove social practices like polygamy among upper and middle-class Muslims.

Faraizi movement : It was started and led by Haji Shariatullah in 1819. The movement extolled Muslims of Bengal to give up un-Islamic practices and act

upon their duties as Muslims. It became revolutionary under the leadership of Dudu Mian in 1840.

The **Pagal Panthis** sect founded by Karam Shah was a religious movement that arose in the northern Mymensingh and Sherpur District areas of Bengal's province.

The Child Marriage Restraint Act, 1929, passed on September 28, fixed the marriage age for girls at 14 and boys at 18. Subsequent amendments raised the age to 15 for girls and 21 for boys in 1949, and then to 18 for girls and 21 for boys in 1978. It's commonly known as the Sharda Act.

The Hindu Widows Remarriage Act, 1856, drafted by Lord Dalhousie and passed by Lord Canning, was enacted in the Bombay Presidency due to the efforts of social reformer **Ishwar Chand Vidyasagar**. He advocated for the abolition of child marriage and polygamy and supported the establishment of the first permanent girls' school in India, the Bethune School. Vidyasagar also served as Principal of a Sanskrit College promoting a blend of Indian and Western culture.

The **Satnami group**, founded in 1820 in Chhattisgarh by Guru Ghasidas, aimed to improve the social status of Dalits, primarily leather workers.

Some Important Reformers

Sree Narayana Guru, born on August 22, 1856, in Chempazhanthy, Kerala, was a social reformer from the Ezhava caste. He advocated the anti-caste movement and famously promoted the slogan "One Caste, One Religion, One God for All." He belonged to the Sree Narayana Dharma Paripalana Yogam organization and built a temple dedicated to Lord Shiva at Aruvippuram in 1888. He became a prominent advocate of Advaita Vedanta, the principle of non-duality. Books written by him are Advaita Deepika, Asrama, Thevarappathinkangal etc.

Shiv Dayal Saheb, a banker from Agra, founded Radha Soami in 1861. **Radha Soami Satsang** teaches the significance of meditation and inner spiritual growth for achieving God-realization.

Dhondo Keshav Karve (Maharshi Karve), was a social reformer in India in the field of women's welfare. On the line of promoting women's education, he

founded the first women's university in India - SNTD Women's University.

The Government of India awarded him the Bharat Ratna (1958) and Padma Vibhushan (1955). His two autobiographies were Atmawrutta (1928) in Marathi, and Looking Back (1936) in English.

Debendranath Tagore wrote the Brahmo Covenant, which was a statement of the creed of the Brahmo Samaj and made a list of the duties and obligations of its members. He founded the Brahmo religion, in 1848 which today is synonymous with Brahmoism. In 1863 he founded Shantiniketan (Abode of Peace).

Dadabhai Naoroji, known as the Grand Old Man of India and the Unofficial Ambassador of India, founded the East India Association in 1867. He authored the influential book "Poverty and Un-British Rule in India" and co-founded the Anglo-Gujarati paper Rast Goftar in 1854, championing social reform among Parsis in Western India.

Nawab Abdul Latif Khan, titled by the British in 1880, is hailed as the Father of Muslim Renaissance in Bengal. He was among the first Muslims in 19th-century India to advocate for modernization.

Sri Aurobindo, born on August 15, 1872, in Kolkata, was a philosopher, yogi, guru, poet, and nationalist. He authored "The Life Divine" and developed Integral Yoga in Pondicherry. Following his acquittal in the Alipore Bomb case, he launched "Karmayogin" in 1909, a weekly English journal covering national religion, literature, science, and philosophy.

Durgabai Deshmukh founded the Andhra Mahila Sabha in 1937 in Andhra Pradesh, aiming to empower women through education and training to overcome socio-economic and political backwardness and participate in national development.

Narayan Malhar Joshi, an Indian trade union leader, founded the Social Service League and started the All India Trade Union Congress (1920). He was the general secretary of AITUC.

Lala Hansraj was a famous Indian educationist and a follower of Arya Samaj. He founded the Dayanand Anglo-Vedic Schools (D.A.V.) in 1886 at Lahore.

Kandukuri Veeresalingam was a social reformer and writer from Andhra

Pradesh. He is considered as the father of the Telugu Renaissance movement. He also fought against child marriage and the dowry system.

Syed Ahmed Khan was the founder of Aligarh Muslim University and a forward looking Islamist reformer. He is widely regarded as the "father of the two-nation theory".

Gopal Hari Deshmukh is known as Lokhitwadi. He emphasized the importance of education for all, irrespective of caste or gender. He wrote a weekly Prabhakar under the pen name "Lokhitawadi" on social reform issues.

Gadadhar Chattopadhyay, later known as Ramakrishna Paramahansa, was a Brahmin priest whose profound spiritual experiences and universal perspective transcend religious barriers, guiding people in their search for truth.

Annie Besant, a British socialist, educator, and women's rights activist, arrived in India in 1893. She established the All-India Home Rule League in 1916 and became the first woman president of the Indian National Congress in 1917. Besant laid the foundation of Central Hindu College in Banaras in 1898, integrating Hindu religion with Western scientific subjects.

Mother Teresa, born on August 26, 1910, founded the Missionaries of Charity, dedicating her life to serving the poor and destitute worldwide. She was awarded the Nobel Peace Prize in 1979 for her charitable and selfless work.

Organisations before Indian National Congress

Landholders' Society - It is also known as the Zamindari Association and is considered to be the first political association of modern India. It was formally launched in Calcutta in March 1838. Raja Radhakant Dev, Dwarkanath Tagore, Prasanna Kumar Tagore, Rajkamal Sen, and Bhabani Charan Mitra were its founding members.

The East India Association was established in 1866 by Dadabhai Naoroji in London, replacing the London Indian Society. It served as a platform for discussing Indian affairs and providing representation to Indians. Lord Lyveden was its first President. Meetings were held in Caxton Hall, Westminster. In 1949, it merged with the National Indian

Association, becoming the Britain, India, and Pakistan Association.

The Bombay Presidency Association was established in 1885 by Badruddin Tyabji, Pherozshah Mehta, and K. T. Telang. It aimed to oppose Lord Lytton's policies and the controversial Ilbert Bill.

The Poona Sarvajanik Sabha : It was founded by Mahadeva Govind Ranade and GV Joshi in 1870 in Pune. They fought for legal rights of farmers, Connected common people with the British Government. Bal Gangadhar Tilak was also a member of this Sabha.

The Bombay Association, formed in 1852 by Jagannath Shankarshet, included notable members like Sir Jamshedji Jejeebhoy, Naoroji Fursungi, Dadabhai Naoroji, and Vinayak Shankarshet. Sir Jamshedji Jejeebhoy served as its first president. It was the first political organization in the Bombay Presidency.

The Bangabhasaha Prakasika Sabha, founded in 1836 by associates of Raja Ram Mohan Roy, focused on administrative reforms, Indian participation in administration, and spread of education. It played a crucial role in fostering modern nationalism among the masses.

The Indian Association of Calcutta : Established in 1876 by Surendranath Banerjee and Ananda Mohan Bose, the Indian Association of Calcutta aimed for the advancement of people in political, intellectual, and material aspects. It laid the foundations of the Indian National Movement and merged with the Indian National Congress in 1886.

Surendranath Banerjee later joined Indian National Congress, he supported the Montagu-Chelmsford reforms in contrast to congress. Later he left congress and formed **Indian National Liberation Federation** in 1919.

The India League, initiated by Sisir Kumar Ghosh in 1875, aimed to foster nationalism among the people. Ghosh, a prominent Indian journalist and freedom fighter from Bengal, founded the Amrita Bazar Patrika in 1868.

The Bengal British India Society, formed in 1843 by William Adam, a friend of Raja Ram Mohan Roy, aimed to collect and disseminate information about people's health and well-being.

Madras Mahajan Sabha, founded in May 1884 by M. Veeraraghavachariar, G. Subramania Iyer, and P. Ananda Charlu, was a prominent Indian nationalist organization based in the Madras Presidency.

Madras Native Association - This organization was formed by Gazulu Lakshminarasu Chetty in 1852 in the Madras Presidency. It was the first political organization in Madras.

Indian National Congress and Muslim League

On 28 December 1885, the Indian National Congress was founded at Gokuldas Tejpal Sanskrit College in Bombay, with 72 delegates in attendance.

A.O. Hume became General Secretary and Womesh Chandra Bonnerjee was elected First President of the Indian National Congress. Lord Dufferin, was the viceroy at that time. He criticized the Congress in 1887, calling them a "microscopic minority".

Some Important Session of INC

Bombay Session (1885)- It was the First Session which was held in Bombay in December, 1885. W.C.Banerjee was the president of this session. Indian National Congress formed in this session.

Calcutta Session (1886) - The second session of INC led by Dadabhai Naoroji, saw 434 delegates. At the session's end, the Congress resolved to establish Provincial Congress Committees nationwide.

Madras Session (1887)- It was the third Session of INC. Syed Badruddin Tyabji (first muslim president) was the President of this session.

Allahabad Session (1888) - It was the fourth Session. George Yule (first English president) was its President.

Calcutta session (1901)- Dinshaw E.Wacha was the President of this session. First time Gandhiji appeared on the Congress platform.

Madras Session (1903)- It was the 19th session of the INC. It was headed by Lal Mohan Ghosh.

Banaras session (1905)- Gopal Krishan Gokhale was the President. Formal proclamation of Swadeshi movement

against the government was passed and the famous Boycott Resolution was passed.

Calcutta Session (1906)- Dadabhai Naoroji was the President of this session. Four resolutions on Swaraj (Self Government) were passed. 'Swaraj' was declared the ultimate goal of the Indian National Congress. The Boycott Movement, Swadeshi, and National Education were adopted.

Surat Session (1907)- Rash Bihari Ghosh was the President of this session. Congress split into Moderates & Extremists.

The moderate leaders were Madan Mohan Malviya, M.G.Ranade, Gopal Krishna Gokhale, Dadabhai Naoroji, Surendranath Banerjee and the extremists were led by Lala Lajpat Rai, Bal Gangadhar Tilak, Bipin Chandra Pal, Aurobindo Ghosh, Rajnarayan Bose, V O C Pillai.

Calcutta Session (1911)- B.N. Dhar was the President of this session. The National Anthem of India "Jana Gana Mana" was first sung in this session.

Bombay Session (1915)- Sir S.P. Sinha was the president. The Constitution of the Congress was altered to admit the delegates from the extremist section.

Lucknow Session (1916)- A.C. Majumdar was the President of this session. Unity between two factions of Moderates and Extremists of Congress took place. Lucknow Pact signed between Congress and Muslim League to build political consensus.

Calcutta Session (1917) - Annie Besant presided over this session. She was the first woman President of the INC.

Special Session (1920) was held at Calcutta in September 1920. Lala Lajpat Rai was the President of this session. In this session, Mahatma Gandhi moved the Non cooperation resolution.

Nagpur Session (1920) - C. Vijayaraghavachariar was the President of this session. Reconstitution of Working committees of Congress on Linguistic basis. MA Jinnah left the Indian National Congress in this session.

Belgaum Session (1924)- M.K. Gandhi was the President of this session. This is the only Session which was presided over by Mahatma Gandhi.

Kanpur Session (1925)- This session was presided over by Sarojini Naidu (the first Indian woman president).

Calcutta Session (1928)- Motilal Nehru was the President of this session. Formation of All India Youth Congress took place.

Lahore Session (1929)- The resolution for Poorna Swaraj was passed on December 19, 1929, declaring January 26 as Independence Day. The first Independence Day in Colonial India was celebrated on January 26, 1930. The demand of INC was changed from the dominion status to Complete Independence or 'Purna Swaraj' of India.

Karachi Session (1931)- Vallabhbhai Patel was the President. It was memorable for a Resolutions on Fundamental Rights and National Economic Programme. Gandhi was nominated to represent INC in the Second Round Table Conference to be held in London and Endorsement of Gandhi-Irwin pact was done. In this session, Mahatma Gandhi said, "Gandhi may die but Gandhism will remain forever".

Faizpur Session (1937)- Jawahar Lal Nehru was the President of this session. It was the First Session to be held in a village.

Haripura Session (1938)- Subhas Chandra Bose was the President of this session. The National Planning Committee was set up under Jawahar Lal Nehru.

Tripuri Session (1939)- Rajendra Prasad was the President of this session. Subhas Chandra Bose was re-elected but had to resign. He defeated Pattabhi Sitaramayya. He became the only President who resigned. Rajendra Prasad was appointed in his place.

Meerut Session (1946)- J.B Kripalani was the President of this session. It was the last session before independence.

Other Important Sessions of Congress

Calcutta Session (1896)- Rahimtullah Sayani was the President of this session. National Song Vande Mataram sung for the first time by Rabindranath Tagore.

Lucknow Session (1899)- Romesh Chandra Dutt was the President of this

session. There was a demand for permanent fixation of Land revenue in this session.

Allahabad Session (1910)- Sir William Wedderburn was the President of this session. M.A Jinnah publicly rejected the separate electorate system introduced by act of 1909.

Special session (1918) was held at Bombay. Syed Hasan Imam was the President of this session. The session was convened to deliberate the contentious Montagu - Chelmsford Reforms Scheme.

Amritsar Session (1919)- Motilal Nehru was the President of this session. Congress extended support to Khilafat Movement during this session.

Gaya Session (1922)- C.R. Das was the President of this session. CR Das and other leaders left INC. The Swaraj Party was formed.

Madras Session (1927)- Dr. M.A. Ansari was the President of this session. A resolution against the use of Indian troops in China, Iran and Mesopotamia was Passed and also a resolution against boycott of Simon Commission was passed. The INC set up an All Parties Conference to draft a Constitution for India.

Bombay Session (1934) - Rajendra Prasad was the President of this session. Amendment in the Constitution of Congress took place.

Lucknow Session (1936)- Jawahar Lal Nehru was the President of this session. He pushed towards socialist ideas. In this session the Government of India Act of 1935 was rejected.

Ramgarh Session (1940)- Abul Kalam Azad was the President of this session. Civil Disobedience movement to be launched at appropriate time and circumstances was decided during this session.

Other Important Facts

In 1923, Abul Kalam Azad became the youngest person at the age of 35 to serve as the President of the INC. He was also a noted writer, poet and journalist. He was a prominent political leader and was elected as Congress President in 1923 and 1940.

Sarojini Naidu was the first Indian woman to chair the Indian National Congress and the first woman to hold the post of the Governor of Uttar Pradesh. She was given the title of Bharat Kokila or the Nightingale of India by Mahatma Gandhi. She was an Indian political activist and poet.

Jivatram Bhagwandas Kripalani also known as Acharya Kriplani was the president of the Indian National Congress during the transfer of power. He was the husband of Sucheta Kriplani, who was the first woman chief minister of Uttar Pradesh.

Gopal Krishna Gokhale claimed for Swaraj or self government within the British Empire, on the model of self-governing colonies like Australia and Canada, was put forward from the Congress platform in 1905.

The Constitution drafted in 1928 (Nehru Report) was drafted by a committee of 9 members. They were Motilal Nehru (Chairman), Jawaharlal Nehru (Secretary), Ali Imam, Tej Bahadur Sapru, Madhav Shrihari Aney, Mangal Singh, Shuaib Qureshi, Subhas Chandra Bose, and G. R. Pradhan.

The Indian National Congress adopted the tricolour flag with a Charkha at the centre in 1931. The first national flag was hoisted in Kolkata in 1906, followed by Madame Bhikaji Cama raising the second flag in Paris in 1907. Lokmanya Tilak and Dr. Annie Besant raised the third flag during the Home Rule movement in 1917.

Jawaharlal Nehru proposed the design of the Indian National Flag with a horizontal tricolour of saffron, white, and dark green in equal proportions, with a navy blue wheel at the centre. The design, adopted by the Constituent Assembly on July 22, 1947, was created by Pingali Venkayya. The flag's width to length ratio is 2:3, with the navy blue wheel representing the chakra placed in the white band.

The National Anthem of India Jana-gana-mana, composed originally in Bengali by Rabindranath Tagore, was adopted in its Hindi version by the Constituent Assembly as the National Anthem of India on 24 January 1950.

Muslim League

The Muslim League, founded by Nawab Khwaja Salimullah in 1906, aimed to

protect Muslim interests in British India as an alternative to the Indian National Congress.

The first Honorary President of the League was Sir Sultan Muhammad Shah (Aga Khan III). Muhammad Ali Jinnah joined the league in 1913.

The Muslim League observed the '**Day of Deliverance**' on 22 December 1939, when Congress members resigned from government positions in protest against India's involvement in World War II without proper consultation.

The Muslim League declared 16th August 1946 as '**Direct Action Day**' to press for a separate Muslim nation after British departure. It led to widespread communal riots, notably in Calcutta, known as the Great Calcutta Killings.

The first annual session of the Muslim League was held in Karachi on 29 December 1907. The Muslim League's Constitution was framed in 1907 at Karachi.

In 1927 the League was divided into two factions regarding the issue of a joint electorate. Those who supported the joint electorates were led by Muhammad Ali Jinnah and those who opposed were led by Sir Muhammad Shafi.

In 1931 the party again split into two when Muhammad Ali Jinnah moved to London abandoning politics. The two factions were led by **Abdul Aziz and Hafiz Hidayat**. The two factions merged again when Jinnah returned to India in 1934.

The last session was held in Karachi in 1943 and was presided by Muhammad Ali Jinnah.

The name Pakistan was first coined by **Choudhry Rahmat Ali** in 1933.

The idea of Pakistan was first conceived by **Muhammad Iqbal**, a Philosopher and poet. He was elected president of the Muslim League in 1930 at its session in Allahabad. He is called the 'spiritual father of Pakistan'.

Partition of Bengal and Swadeshi movements

In 1905, Viceroy Lord Curzon divided Bengal into two provinces - first (Western Bengal, Bihar and Orissa) and second (Eastern Bengal and Assam), sparking protests. At that time Bengal was the biggest province of British India. The real

motive was to weaken Bengali unity and implement divide and rule. This partition was announced in July 1905 and came into effect on 16 October 1905. On that day, Rabindranath Tagore symbolised unity by urging Hindus and Muslims to tie rakhi. The partition lasted until King George V reversed it in 1911 at the Royal Darbar in Delhi.

The Partition of Bengal spurred the adoption of Swadeshi and Boycott as protest methods. It also led to the emergence of India's first mass movement, the Swadeshi movement, from 1905 to 1911.

Its chief architects were Aurobindo Ghosh, Lokmanya Bal Gangadhar Tilak, Bipin Chandra Pal and Lala Lajpat Rai, V. O. Chidambaram Pillai, Babu Genu.

Swadeshi Movement: The Swadeshi movement began on August 7, 1905, at Calcutta's Town Hall, aiming to promote domestic production over foreign goods. While nationwide, it was dubbed the Swadeshi movement, in regions like Andhra Pradesh, it was known as the Vande Mataram movement.

It spread to a few different parts of India: In Poona and Mumbai under Bal Gangadhar Tilak, In Punjab under Lala Lajpat Rai and Ajit Singh, In Delhi under Syed Haider Raza, and In Madras under Chidambaram Pillai. Mahatma Gandhi described it as the soul of swaraj (self-rule).

The Swadeshi Movement promoted indigenous goods, boycotted foreign, mainly British, goods, and aimed to block colonial trade. After the partition, people of Bengal expressed widespread opposition by singing "Vande Mataram". School and college students were most active participants in the Swadeshi Movement, notably in Bengal, Poona, Guntur, Madras, and Salem. The police adopted a repressive attitude towards students. Traditionally home-centered women also actively participated in the movement.

During the Swadeshi Movement in Bengal, the tricolour flag with red, green and yellow colours was designed. It had 8 lotuses which represented 8 provinces, a crescent moon representing Hindus and Muslims.

The **Extremists** (or the Garam Dal) gained a dominant influence over the Swadeshi Movement in Bengal after

1905 till 1908. It is also known as the Era of Passionate Nationalists. Lala Lajpat Rai, Bal Gangadhar Tilak and Bipin Chandra Pal (Lal-Bal-Pal) were important leaders of this Radical group. Tilak Started Shivaji festival to generate Nationalism among common People in 1894.

The **Indian National Congress (INC)** in a meeting in 1905 resolved to condemn the partition of Bengal and support the anti-partition and Swadeshi Movement. The radical nationalists wanted the movement to be taken outside Bengal and go beyond just the boycott of foreign goods.

All India Muslim League supported the move for the partition of Bengal.

Moved by the Swadeshi Movement, **Abanindranath Tagore** painted his famous image of Bharat Mata portraying her as an ascetic figure in 1905. The painting is depicted as a saffron-clad divine woman, holding a book, sheaves of paddy, a piece of white cloth and a rosary in her four hands.

The impacts of this movement were decline in imports, the growth of extremism, Morley- Minto reforms, the establishment of Swadeshi institutions, and Growth in Swadeshi industries.

The Gradual Suppression of the Swadeshi Movement

Government Repression: By 1908, the Swadeshi Movement was almost over in an open phase due to government's violent repression.

Absence of Leaders and Organisation: The movement failed to create an effective organisation. It was rendered leaderless as most of the leaders were either arrested or deported by the time. Maintaining the high intensity of such a mass movement was a difficult task in absence of effective leaders.

Internal Conflicts - The internal conflicts and differences in ideologies among the leaders did more harm to the movement.

Limited Extent - The movement failed to reach the peasantry and was confined to the upper and middle classes only. Annulment of Participation of Bengal took place in 1911 by **Lord Hardinge** primarily to curb revolutionary terrorism. Bihar and Orissa were taken out of Bengal and Assam was made a separate

province. The capital was also transferred to Delhi in 1911.

World Wars

World War I (1914 - 1918) - It was fought between the Allied Powers and the Central Powers. The main members of the Allied Powers were France, Russia, Italy, Japan, Britain. The United States joined after 1917.

The main members of the Central Powers were Germany, Austria-Hungary, the Ottoman Empire, and Bulgaria. It ended with the defeat of the Central Powers.

Treaty of Versailles - It was a peace treaty signed on 28 June 1919. World War I officially ended with the signing of this treaty.

Other Treaty - The Treaty of Neuilly (1919) signed with Bulgaria. The Treaty of Sevres (1920) signed with Turkey.

World War II (1939 - 1945) - It was fought as part of two opposing military alliances: the Allies and the Axis power. The main members of the Allies' power were France, Great Britain, the United States, and the Soviet Union.

The main members of the Axis power were Germany, Italy, and Japan.

It began in Europe on September 1, 1939, when Germany (ruled by Hitler) invaded Poland.

It came to an end with the defeat of Germany in May 1945. The **Paris Peace Treaties** (French) were signed on 10 February 1947 following the end of World War II in 1945

Battle of Waterloo

It was fought on 18 June 1815 between Napoleon's French Army and a coalition led by the Duke of Wellington and Marshal Blücher. Napoleon Bonaparte was the Emperor of the French and was defeated in the Battle of Waterloo.

Gandhian Era

At the age of 46, Mahatma Gandhi came to India from South Africa on the request of Gokhale, conveyed by CF Andrews (Deenbandhu) in January 1915. He established the Satyagraha Ashram in Ahmedabad, later moving it to Sabarmati River's banks.

Gandhiji dominated the Indian freedom struggle from 1919 to 1948. That is why this period is known as the Gandhian era.

Important Events in Gandhian Era

Champaran Satyagraha (1917)

In 1917, on invitation of Rajkumar Shukla, Mahatma Gandhi led the Champaran Satyagraha, India's first civil disobedience movement, addressing British-imposed indigo cultivation and farmer exploitation. Movement led to the abolition of the Tinkathia system and compensation for peasants via the Champaran Agrarian Act of 1918.

Ahmedabad Mill Movement (1918)

The workers in Ahmedabad's textile mills fought for economic justice after the mill owners discontinued their plague bonuses. The hunger strike was started on March 15, 1918. This was the first hunger strike led by Gandhi Ji. The strike was successful and the workers got a 35% wage increase.

Kheda Satyagraha (1918)

In 1918, Mahatma Gandhi led the Kheda Satyagraha in Ahmedabad, Gujarat, the first major non-cooperation movement against British land revenue demands. Peasants, affected by crop failures and a plague epidemic, demanded relaxation in revenue collection. Vallabhbhai Patel and Indulal Yajnik supported Gandhi, organizing village tours and encouraging peasants to resist the government.

Rowlatt Act (1919)

The Rowlatt Act, or the Anarchical and Revolutionary Crimes Act of 1919, was termed the Black Act by Gandhi due to its unjust nature. The Act was described as No Dalil, No Vakil, No Appeal. It allowed for the detention of political prisoners without trial for two years. Gandhi organized the Rowlatt Satyagraha and set up a Satyagraha Sabha on 24th February 1919 at Bombay.

Jallianwala Bagh Massacre (1919)

The Jallianwala Bagh massacre occurred on April 13, 1919, during a peaceful protest against the Rowlatt Act in Amritsar, Punjab. British troops, led by General Dyer, fired on unarmed

protesters, resulting in numerous casualties. In response, Gandhi returned his title Kaiser-i-Hind in 1920, and Rabindranath Tagore renounced his Knighthood awarded by King George V in 1915 for Services to Literature. The Hunter Commission was appointed to investigate the incident.

Khilafat and Non-Cooperation Movement (1920)

The Non-cooperation Movement (1920-1922) opposed British rule in response to the Jallianwala Bagh Massacre, Rowlatt Act, and Montagu-Chelmsford reforms. It was withdrawn in February 1922 after the Chauri Chaura incident. In Chauri Chaura, Uttar Pradesh, a mob set fire to a police station during a clash with protesters, killing 22 policemen. During the movement, C.R. Das and Motilal Nehru abandoned their legal practices.

The Khilafat Movement, led by the Ali brothers (Shaukat Ali and Muhammad Ali) and Abul Kalam Azad, aimed to retain the Ottoman Sultan's rule as Caliph of Islam after WWI. The All-India Khilafat Conference in November 1919 in Delhi threatened to withdraw cooperation from the British government if their demands were not met.

Swaraj Party (1923)

The Swaraj Party, established as the 'Congress-Khilafat Swaraj Party', was a political party formed in India on 1 January 1923 by Chittaranjan Das and Motilal Nehru after the 'Gaya annual conference' in December 1922 of the National Congress. Chittaranjan Das was its President and Motilal Nehru was its Secretary. The aims of this party were attaining dominion status, establishing control over the bureaucracy and engaging in the constructive programmes of the Congress.

Simon Commission (1927)

The Simon Commission (7 members), led by Sir John Simon, arrived in India in 1928 to study constitutional reform. Its composition of solely British members sparked strong opposition among Indian leaders. Lala Lajpat Rai died from injuries sustained during a peaceful protest against the Commission in Lahore.

Nehru Report - The Nehru Report, drafted by Motilal Nehru and eight other Congress leaders in 1928, proposed a

constitution for India advocating for dominion status and a federal system of government.

Bardoli Satyagraha (1928)

The Bardoli Satyagraha of 1928 was a farmers' agitation against increased taxation by the colonial government in the Bombay Presidency. Led by Sardar Vallabhbhai Patel, it demanded the cancellation of a 22% tax hike. The movement's success elevated Patel's leadership role, and he was honored with the title "Sardar" by the women of Bardoli.

Dandi March (1930)

The Salt Satyagraha, or Dandi March, began on March 12, 1930, from Sabarmati Ashram, reaching Dandi on April 6, 1930. With the dandi march, Gandhi started the Civil disobedience movement.

C. Rajagopalachari led a salt march from Trichinopoly to Vedaranyam on Tanjore coast in Tamil Nadu, while Kamaladevi Chattopadhyay advocated for women's participation. Gandhi's breaking of the Salt Laws led to massive success. The civil disobedience movement ended with the Gandhi-Irwin pact on March 5, 1931. The National Salt Satyagraha Memorial in Dandi honors its participants.

Gandhi Irwin Pact (1931)

The Gandhi-Irwin pact, signed on March 5, 1931, resulted in the release of imprisoned activists and allowed Indians to produce salt for domestic use. Gandhi called off the civil disobedience movement and attended the second round table conference in London as Congress's representative. However, upon his return, Gandhi relaunched the movement, which eventually lost momentum by 1934.

Round Table Conferences (1930 - 1932)

These were peace conferences initiated by the British Government and Indian leaders to discuss constitutional reforms in India. They ran from November 1930 to December 1932.

First round table conference was held from 12 Nov 1930 to 19 Jan 1931, inaugurated by King George V. Ramsay MacDonald was the British PM at that time. The key participants from India were B.R. Ambedkar, Jinnah, Sir Tej

Bahadur Sapru, V. S. Srinivasa Sastri, Sir Muhammad Zafrulla Khan, K. T. Paul and Mirabehn.

Second round table conference was held from 7 Sep 1931 to 1 Dec 1931 in London. The key participants from India were Mahatma Gandhi, Pyarelal Nayyar, Sarojini Naidu, Annie Besant, Madan Mohan Malviya, Mahadev Desai, Sir Tej Bahadur Sapru, B.R. Ambedkar.

Third round table conference was held from 17 Nov 1932 to 24 Dec 1932. The key participants from India were Aga Khan III, Sir Tej Bahadur Sapru, B. R. Ambedkar.

Gandhi attended only the second round table conference. Dr. B.R. Ambedkar and Sir Tej Bahadur Sapru attended all three round table conferences.

Poona Pact (1932)

The Poona Pact was signed between Madan Mohan Malviya on behalf of Mahatma Gandhi and B.R. Ambedkar in 1932 at Yerwada Central Jail in Poona. The background to this Pact was the Communal Award of August 1932 made by British prime minister Ramsay MacDonald which provided a separate electorate for depressed classes. The pact established reserved seats for depressed classes in legislative councils, to be elected by the general electorate, not a separate one.

August Offer (1940)

To seek Indian support in World War II, an offer was made by Viceroy Linlithgow in 1940 promising the expansion of the Executive Council of the Viceroy of India to include more Indians, the establishment of an advisory war council, giving full weight to minority opinion, and the recognition of Indians' right to frame their own constitution. Further, Dominion status was explicitly offered. While rejecting the August Offer Nehru said, "Dominion status concept is dead as a doornail".

Individual Satyagraha - In response to the failure of the August Offer, Gandhi initiated a small-scale Individual Satyagraha in each locality, led by a few carefully chosen individuals. Vinoba Bhave, Jawaharlal Nehru, and Brahmdukt were the first three satyagrahis, respectively.

Cripps Mission (1942)

The Cripps Mission, led by Sir Stafford Cripps, aimed to secure Indian cooperation in World War II. It arrived in India in March 1942 during Winston Churchill's tenure as Prime Minister of England.

The main proposal of the Cripps Mission was to offer India Dominion status after World War II, granting full self-government and the right to determine its own constitution. However, its failure to promise immediate independence and lack of agreement on the future Indian government's composition led to its rejection by Indian leaders.

Quit India Movement (1942)

The Quit India Movement began on August 8, 1942, in Bombay by Mahatma Gandhi at the August Kranti Maidan (formerly Gowalia Tank Maidan) with a "Do or Die" call. He was detained the next day at the Aga Khan Palace in Pune, Maharashtra. The remaining leaders were imprisoned in Ahmednagar Fort.

Usha Mehta operated a secret Congress radio station during the movement. Aruna Asaf Ali, popularly known as the 'Grand Old Lady', hoisted the Indian flag at Gowalia Tank Maidan. Lakshman Naik, a tribal leader from Orissa, played a significant role in the movement. Tara Rani Srivastava was also a part of the Quit India Movement. Matangini Hazra is known as old lady Gandhi due to her non-violent protest in 1942. The chief areas of the movement were UP, Bihar, Maharashtra, Midnapore (West Bengal), and Karnataka.

Though the government quelled the movement, Viceroy Linlithgow deemed it the most serious rebellion since 1857. **Kushal Konwar** was the sole martyr hanged during the movement.

Important Facts related to Gandhi

Natal Indian Congress was an organization that aimed to fight discrimination against Indians in South Africa. It was formed in 1895 by Mahatma Gandhi, along with other Indians. Gandhi was the Honorary Secretary and Abdoola Hajee Adam Jhaveri was elected president.

The **Indian Opinion** was an important newspaper published in 1903 in Durban, South Africa, by Mahatma Gandhi and M. H. Nazar. It was the primary

communication organ of the Natal Indian Congress.

Gandhiji set sail for India from Cape Town on 19 July 1914 and reached Mumbai on 9 January 1915 at Apollo Port.

Gopal Krishna Gokhale was a mentor to both Mahatma Gandhi and Mohammed Ali Jinnah and political Guru to Mahatma Gandhi. In 1905, he founded the Servant of Indian Society to train Indians to dedicate their lives to the cause of the country. He belongs to the moderate section of the Indian Congress. He advised Gandhi to spend a year traveling around British India to know the people of India.

Gandhiji in 1942 referred to Subhash Chandra Bose as the Patriot of Patriots and Prince among the Patriots. Subhash Chandra Bose called Mahatma Gandhi, the Father of the Nation in a radio address from **Singapore** on 6 July 1944. The title "Mahatma" was awarded to Gandhiji by Rabindranath Tagore.

Gandhiji called Dr Rajendra Prasad Ajatashatru, which means who has no enemies. Rajendra Prasad was the first President of India.

In October 1937, the **All-India National Educational Conference** in Wardha adopted Mahatma Gandhi's proposal for free and compulsory education for seven years in the mother tongue, integrating manual and productive work into the learning process nationwide.

Mahatma Gandhi was born on 2nd October 1869. He was assassinated at the age of 78 on January 30, 1948, at Birla House, now Gandhi Smriti.

Nathuram Vinayak Godse, a member of the Hindu Mahasabha, was his assassin. He expressed his desire for keeping his ashes intact till India is reunited and throwing them into the Indus after the reunification has been achieved.

Gandhi inspired Albert Einstein on the idea of non-violence.

Revolutionaries

Birsa Munda, born November 15, 1875, in Ranchi, was an Indian tribal freedom fighter and religious leader of the Munda tribe in Jharkhand. The Janjatiya Gaurav Divas honors him on his birth anniversary. He protested unfair land grabbing by colonial and local authorities, leading to his arrest in 1895

and a two-year imprisonment. Ranchi Airport (Birsa Munda Airport) is named after him.

Rani Laxmibai was born as Manikarnika Tambe (Nickname Manu) on November 19, 1828, in Varanasi, was a Marathi Karhade Brahmin. When the Maharaja of Jhansi died in 1853, Lord Dalhousie applied the Doctrine of Lapse, annexing the state. Laxmibai died in battle on June 17, 1858.

Alluri Sitarama Raju, born in Andhra Pradesh in 1897, was a revolutionary known as Manyam Veerudu (Hero of the Jungle). He led the Rampa Rebellion of 1922 against the British in response to the 1882 Madras Forest Act. Raju became a sanyasi at 18 and gained mystical influence among hill and tribal people with his knowledge of astrology, medicine, and ability to tame wild animals.

Bhagat Singh, born September 28, 1907, in Banga, Punjab, was a revolutionary leader who joined the National College, Lahore, in 1923 which was founded and managed by Lala Lajpat Rai and Bhai Parmanand.

He, along with Rajguru and Sukhdev, was involved in the Lahore conspiracy case (murder of a British police officer, JP Saunders) and hanged on March 23, 1931, at the age of 23.

He was a member of the Hindustan Socialist Republican Association and threw a harmless bomb in the Central Legislative Assembly in 1929 to protest against oppressive laws. He founded the Naujawan Bharat Sabha in 1926 in Lahore.

He gave the Slogan "He did not wish to glorify the cult of the bomb and pistol but wanted a revolution in society".

Subhash Chandra Bose was born on 23 January 1897 in Cuttack. His Jayanti is celebrated as 'Parakram Diwas'. In 1919, he had cleared the Indian Civil Services (ICS) examination. He was the founder of 'Tiger Legion' and 'Free India Legion'.

He was placed under house arrest in his Elgin Road house in Calcutta. In the year 1941 (16th January), he escaped from there and reached Russia via Afghanistan. His nephew Sisir Kumar Bose helped him during this escape. Rabindranath Tagore honored him with the title 'Desh Nayak' for his service to the nation.

Chandrashekhar Azad, born July 23, 1906, in Alirajpur, Madhya Pradesh, joined the Non-Cooperation Movement in 1921 and later the Hindustan Republican Association (HRA). He shot himself dead on 27 February 1931 at Alfred Park in Allahabad, Uttar Pradesh, when he was surrounded by police.

Batukeshwar Dutt, born on November 18, 1910, in Oari, West Bengal, was a revolutionary. Alongside Bhagat Singh, he threw bombs in the Central Legislative Assembly in 1929. Sentenced to life imprisonment, he was sent to Cellular Jail in Port Blair. After release, he participated in the Quit India Movement and was jailed for four more years.

Kartar Singh Sarabha, a communist and member of the Gadar Party, inspired Shaheed Bhagat Singh. He was hanged by the British Government in 1915 at the age of 19 for his involvement in the Ghadar Conspiracy.

Veer Savarkar, born May 28, 1883, in Bhagur, Maharashtra, founded Mitra Mela in 1899, later renamed Abhinav Bharat Society in 1904, advocating armed revolution and socialism. He was associated with India House and the Free India Society. Serving as president of Hindu Mahasabha from 1937 to 1943, he was arrested in 1909 and 1910 for his involvement in revolutionary activities. Convicted for abetment to murder of collector Jackson, he was sentenced to 50 years in the Cellular Jail in 1911.

Khudiram Bose, born in Midnapore District in 1889. He and Prafulla Chaki threw a bomb at Chief Presidency Magistrate of Muzaffarpur DH Kingsford's carriage in 1908. Known as the Muzaffarpur Conspiracy, it led to Bose's imprisonment and his sacrifice for the freedom struggle at the age of 18.

Ashfaqulla Khan, born October 22, 1900, in Shahjahanpur, Uttar Pradesh, was involved in the Kakori train action on August 9, 1925. He was subsequently hanged in Faizabad jail for his role in the conspiracy.

The Chapekar brothers, Damodar Hari Chapekar and Balkrishna Hari Chapekar, assassinated British official W. C. Rand and Lieutenant Ayerst on June 22, 1897, in Pune, Maharashtra. They were hanged in 1899, marking the first case of militant nationalism in India after the 1857 Revolt.

Manabendra Nath Roy, a prominent twentieth-century Indian philosopher, founded the Communist Party in Mexico in 1917 and was known as the Father of Indian Communism. He initially joined the Indian National Congress but later left due to their stance on World War II. Roy founded the Indian Renaissance Institute in Dehradun in 1946 to advance the Indian Renaissance Movement.

Aurobindo Ghosh, born August 15, 1872, in Calcutta, was involved in the struggle for Indian independence from 1902 to 1910 and was imprisoned in the Alipore Bomb case. He later fled British India, and found refuge in the French colony of Pondicherry where he dedicated his life to developing integral yoga for spiritual transformation.

Barindra Kumar Ghosh, also known as Barin Ghosh was the younger brother of Sri Aurobindo Ghosh. He was a founding member of the revolutionary Bengali weekly Yugantar Bengali.

Matangini Hazra was an Indian revolutionary killed by British Indian police on September 29, 1942, in front of Tamluk Police Station. In 1977, the Government of Bengal erected its first statue honoring a female freedom fighter, Matangini Hazra, in Kolkata Maidan.

Udham Singh was born on 26 December 1899. He belonged to the Ghadar Party and Hindustan Socialist Republican Association. He killed Michael O'Dier who was responsible for the Jallianwala Bagh massacre. He identified himself as 'Ram Mohammad Singh Azad' while in custody. He was executed on July 31, 1940.

Kanaklata Barua, also known as Birbala and Shaheed, was a teenage freedom fighter from Assam. She was shot dead by British police while leading a procession with the National Flag during the Quit India Movement of 1942. She died at the age of 17 on September 20, 1942, in Barangabari, Gohpur, Assam.

Veerapandiya Kattabomman, an 18th-century Palayakarrar from Panchalankurichi, Tamil Nadu, was also known as Kattabomman Nayak. A dispute with the British over rent escalated into violence, resulting in the death of a British officer by Kattabomman. He was pursued by the British and ultimately captured after betrayal by Ettappan, the Pudukottai Raja. Kattabomman was sentenced to

death and publicly hanged on October 16, 1799, at Kaiyattar.

Ashwini Kumar Dutta founded the Swadesh Bandhab Samiti in 1905 in the Barisal district of Bangladesh. He also played a key role in establishing the Cooperative Hindustan Bank in 1908.

Shyamji Krishna Verma, born October 4, 1857, in Mandvi, Gujarat, founded the Indian Home Rule Society, Indian House (in London, 1905), and the Indian Sociologist in London. He served as the first President of Bombay Arya Samaj. Verma passed away on March 30, 1930.

Chittaranjan Das (C.R. Das) was the political guru of Subhash Chandra Bose. He was the first president of Swaraj Party (1923), Acting president of Indian National Congress in the Ahmedabad session (1921). In 1909, he defended Aurobindo Ghosh on charges of involvement in the Alipore bomb case.

Ramprasad Bismil, born June 11, 1897, in Shahjahanpur, Uttar Pradesh, was a key founder of the Hindustan Republican Association (HRA) along with Sachindra Nath Sanyal and Jadugopal Mukherjee. He participated in the Mainpuri Conspiracy of 1918. While in Lucknow Central Jail, Bismil wrote his autobiography and the iconic song "Mera Rang De Basanti Chola". He was hanged on December 19, 1927, at Gorakhpur Jail and cremated at the site later known as Raj Ghat, on the banks of river Rapti.

Surya Sen, also known as Master Da, born March 22, 1894, in Chittagong (Bangladesh), was a leader of the Anushilan Samiti. He led the 1930 Chittagong armory raid against the British government. After brutal torture by the police, he was hanged on January 12, 1934. Ardhendu Dastidar was also associated with the Chittagong Armoury Raid.

Pandit Madan Mohan Malviya was born on 25th December 1861, in Allahabad (now Prayagraj) in Uttar Pradesh. Dr. S. Radhakrishnan gave him the status of a Karmayogi. He popularized the slogan "Satyameva Jayate".

Madan Lal Dhillon, born February 18, 1883, in Amritsar, assassinated Curzon Wylie on July 1, 1909, in London, as a patriotic act against British atrocities in India. He was found guilty, sentenced to death, and executed at Pentonville Prison on August 17, 1909.

Sukhdev Thapar, born May 15, 1907, was an active member of the Hindustan Socialist Republican Association and joined the Naujawan Bharat Seva. Alongside Shivram Rajguru and Bhagat Singh, he was involved in the killing of John Saunders. He was arrested by the British police and hanged on March 23, 1931.

Shivaram Rajguru, born August 24, 1908, in Khed, Maharashtra, joined the Hindustan Socialist Republican Association (HSRA) and was known as the Gunman of HSRA. He was arrested in a bomb factory in Lahore in March 1929 and was hanged on March 23, 1931.

Veer Surendra Sai, a freedom fighter from Odisha, initiated his revolt against British rule before 1857 and continued guerrilla warfare until 1864. He started protesting at the age of 18 and surrendered in 1862. In 2005, the Government of India decided to erect a statue of him at the Parliament premises.

Bhikaiji Cama, born on 24 September 1861, was the first Indian to unfurl the tricolor on foreign land at the International Socialist Congress in Stuttgart, Germany in 1907. She is regarded as the 'Mother of Indian Revolution'.

NG Ranga, a freedom fighter from Andhra Pradesh, was the first general secretary of the All India Farmer's Congress. He received the Padma Vibhushan in 1991 for his contributions to the peasants' movement.

Rani Gaidinliu, born on January 26, 1915, was a Naga spiritual and political leader who led a revolt against British rule in India. Joining the Heraka religious movement at 13, she was dubbed the "Daughter of the Hills" by Jawaharlal Nehru and given the title Rani. The Rani Gaidinliu Tribal Freedom Fighters Museum is under construction in Manipur's Tamenglong district.

Lala Lajpat Rai, born on January 28, 1865, in Ferozepur district of Punjab, was involved in the activities of Punjab National Bank and Laxmi Insurance Company in their early stages in 1894. He joined INC and was deported to Burma without trial in 1907, he opposed the partition of Bengal and founded the Home Rule League of America in 1917 in New York. He was also elected President of the All India Trade Union Congress.

He supported the non-cooperation movement of Gandhi at the **Nagpur session** of the Congress in 1920. He protested against the Rowlatt Act and the Jallianwala Bagh massacre that followed. He was elected deputy leader of the Central Legislative Assembly in 1926.

He founded Hindu Relief movement in 1897, the Servants of People Society in 1921. In 1928, he was leading a silent protest against **the Simon Commission** in Lahore when he was brutally lathi-charged by Superintendent of Police, James Scott. He died of injuries sustained a few weeks later.

Kanailal Dutta was arrested by the British in the Alipore Bomb Conspiracy Case, a historic trial where the British government attempted to implicate Sri Aurobindo in revolutionary activities.

Bagha Jatin conceived an armed insurrection against the British in cooperation with Germany. The epithet Bagha Jatin was earned by young Jatindranath Mukherjee in 1906 when he single-handedly killed a Bengal tiger in 1906. He was deeply involved in revolutionary activities with groups like Anushilan Samiti and Yugantar. His motto was "Amra morbo, jagat jagbe," (We shall die to awaken the nation).

Phirozshah Mehta, along with K.T. Telang and Badruddin Tyabji, founded the Bombay Presidency Association. He was knighted by the British Government in India for his legal service and was known as 'Bombay's Lion'. From 1885 to 1915, he was regarded as the uncoronated king of Bombay.

Mahadev Govind Ranade, born on January 18th, 1842, in Nashik, was a key figure in founding the Deccan Educational Society and the Indian National Congress. He established the Poona Sarvajanic Sabha in 1870, and the Widow Remarriage Association in 1861.

Palavai Rangaiah Naidu along with Raja Savalai Ramaswami Mudaliar and Panapakkam Anandacharlu founded Madras Mahajana Sabha in 1884.

Swami Shraddhanand, also known as Mahatma Munshi Ram Vij, was an Indian nationalist and Arya Samaj sannyasi who promoted Dayananda Saraswati's teachings. He initiated the Shuddhi program to reintegrate Christians and Muslims into Hindu society, leading to increased communal tensions. Abdul

Rashid assassinated him on December 23, 1926. Shraddhanand authored "Hindu Sangathan" and established the Hindu Mahasabha with Madan Mohan Malaviya's support.

Struggle for Independence

Ghadar Party initially named the Pacific Coast Hindustan Association, was formed on 15 July 1913 in San Francisco (USA) under the leadership of Lala Har Dayal, Sant Baba Wasakha Singh Dadehar, Baba Jawala Singh, Santokh Singh and Sohan Singh Bhakna.

It aimed to overthrow British rule in India through armed struggle.

The **Komagata Maru incident** in 1914 involved a Japanese steamship carrying mainly Sikh immigrants, denied entry into Canada due to racial discrimination. Passengers were forced to stay onboard in harsh conditions for two months before being sent back to India. This incident underscored racial prejudices in Canadian immigration policies and was used by the Ghadar Party to galvanize support for Indian independence.

The **Surat Split of 1907** divided the Indian National Congress into Moderates and Extremists due to ideological differences. Moderates favored constitutional agitation, while Extremists advocated for radical methods like boycott and swadeshi.

Lucknow Pact : In 1916, a joint session of the Indian National Congress and the All India Muslim League was held in Lucknow, with Ambica Charan Mazumdar as president. The agreement aimed to demand greater autonomy from British rule. Bal Gangadhar Tilak represented Congress, while Muhammad Ali Jinnah (joined Muslim League in 1913) represented the Muslim League. Sarojini Naidu gave Jinnah the title 'the Ambassador of Hindu-Muslim unity' for his efforts. The pact also unified Moderate and Extremist factions within Congress.

The **Indian Home Rule Movement** was a campaign for self-governance and independence from British rule in India along the lines of the Irish Home Rule movement, led primarily by Bal Gangadhar Tilak and Annie Besant in the early 20th century. It advocated for Indian self-rule through peaceful and constitutional means, as well as through civil disobedience.

Tilak launched the Indian Home Rule League in April 1916 at Belgaum. Annie Besant launched the Home Rule League in September 1916 at Madras.

In 1920 All India Home Rule League changed its name to Swarajya Sabha.

Flag Satyagraha is a part of civil disobedience during the Indian independence movement. Its main aim was to exercise the right and freedom to hoist the nationalist flag. It was held in Nagpur in 1923.

Hindustan Republican Association was founded in 1924 by Ram Prasad Bismil, Sachindra Nath Bakshi, Sachindranath Sanyal and Jogesh Chandra Chatterjee. Its objective was to establish a "Federated Republic of the United States of India" through an organized and armed revolution.

The Hindustan Socialist Republican Association (HSRA) was founded at a meeting in Feroz shah kotla ground in Delhi with an aim to overthrow the British. It was founded by Chandrasekhar Azad, Bhagat Singh, Sukhdev Thapar in 1928.

Kakori Train Action - The Kakori Train robbery occurred on August 9, 1925, near Lucknow, as part of the Indian independence movement. Ram Prasad Bismil, Ashfaqullah Khan, Rajendra Lahiri, Keshav Chakraborty, Mukundi Lal, Banwari Lal from the Hindustan Republican Association (later, Hindustan Socialist Republican Association) were accused. Bismil and Khan were hanged for their involvement.

The **Nehru Report** of August 15, 1928, drafted by a committee led by Motilal Nehru, with Jawaharlal Nehru as Chief Secretary, advocated for dominion status and a federal government for India. It proposed Joint Electorates with reserved seats for minorities, recommended nineteen Fundamental Rights, and endorsed equal rights for men and women. In response, Muhammad Ali Jinnah formulated the **Fourteen Points** in 1929, representing Muslim community demands.

Declaration of Purna Swaraj - The Indian National Congress, on 19 December 1929, passed the historic 'Purna Swaraj' – (total independence) resolution – at its Lahore session. On December 31, 1929, Nehru hoisted the tricolor on the banks of the Ravi river and demanded "Purna Swaraj" or complete self-rule, and the

date set for independence was January 26, 1930. The day was then celebrated as Purna Swaraj day for the next 17 years.

The **Khudai Khidmatgar**, or Red Shirt movement, led by Abdul Ghaffar Khan, was a non-violent resistance against British rule in the North-West Frontier Province. Ghaffar Khan, also called Frontier Gandhi, was known by various names such as Bachcha Khan, Badshah Khan, or Sarhadi Gandhi.

Qissa Khwani Bazar was the site of a massacre perpetrated by British soldiers against non-violent protesters of the Khudai Khidmatgar movement on April 23, 1930, because of the arrest of Abdul Ghaffar Khan and other leaders of this movement.

Koraput Uprising in the Koraput district of Odisha during the Quit India Movement in 1942 presents an excellent example of the convergence of existing (uprising) traditions of tribal rebellions with the Pan-India freedom struggle.

The **Indian National Army (INA)** was initially formed by Captain General Mohan Singh in Singapore in February 1942 but was disbanded due to differences with the Japanese. It was revived by Subhas Chandra Bose on October 21, 1943, with the motto "Ittehad, Itmad aur Qurbani" (Unity, Faith, and Sacrifice). In 1942, a conference of Indians was held in Tokyo to form the Indian Independence League. A subsequent conference in Bangkok, with Rash Behari Bose elected president, decided to establish the Indian National Army.

The women's regiment of the INA was called the **Rani of Jhansi Regiment**.

The **Bombay Plan**, issued in 1944 by industrialists like JRD Tata, GD Birla, and Lal Shri Ram, advocated for state intervention and planning to drive economic development in India.

The Wavell Plan And Simla Conference (1945) was first presented at the Simla in 1945. It was named after the Viceroy, Lord Wavell.

The **Shimla** Conference was convened in order to agree on the Wavell Plan for Indian self-government, which provided for separate representations on communal lines. Both the plan and the conference failed on account of the Muslim League and the Indian National Congress not coming to an agreement.

Cabinet Mission (1946) was a high-powered mission sent in February 1946 to India by the Attlee Government (British Prime Minister.) The mission had three British cabinet members – Pethick Lawrence, Stafford Cripps, and A.V. Alexander. The Cabinet Mission's aim was to discuss the transfer of power from British to Indian leadership.

Rajagopalachari's formula (C. R. formula or Rajaji formula) was a proposal formulated by Chakravarti Rajagopalachari to solve the political deadlock between the All India Muslim League and the Indian National Congress on the independence of British India. The first Indian Governor General of independent India was C. Rajagopalachari.

The **Tebhaga movement** (1946-47) was a significant peasant agitation, initiated in Bengal by the All India Kisan Sabha of the peasant front of the Communist Party of India. All India Kisan Sabha is the peasant or farmers' wing of the Communist Party of India, an important peasant movement formed by Sahajanand Saraswati in 1936.

The **Royal Indian Navy Mutiny** (RIN mutiny) began at Bombay harbour on 18 February 1946 by Ratings (non-commissioned officers and sailors) against the British. The mutiny was a protest against various grievances, including poor pay, racial discrimination, and oppressive treatment by British officers. Despite being ultimately suppressed by British authorities, it hastened the process of British withdrawal from India.

The Partition of India in 1947 was the change of political borders and the division of other assets that accompanied the dissolution of the British Raj in South Asia and the creation of two independent dominions : India and Pakistan.

Princely States

The Indian Independence Act of 1947 gave princely states an option to accede to the newly born dominions India or Pakistan or continue as an independent sovereign state.

At that time more than 500 princely states covered 48 percent of the area of pre Independent India and constituted 28% of its population.

These kingdoms were not legally part of British India, but in reality they were completely subordinate to the British Crown. Sardar Vallabhbhai Patel (India's first deputy prime minister and the home minister) with the assistance of V.P. Menon (the secretary of the Ministry of the States) was given the formidable task of integrating the princely states.

Travancore- A prominent princely state in southern India, possessed strategic maritime trade routes and abundant resources. Initially hesitant to join the Indian Union, its Dewan, Sir C. P. Ramaswami Aiyar, considered forming an independent state but later changed course after surviving an assassination attempt in July 1947. Consequently, Travancore acceded to India on July 30, 1947 and became the first to sign "Instrument of Accession".

Jodhpur- The Kingdom of Marwar, later known as Jodhpur state, was under British rule from 1818 to 1947. Despite its Hindu king and population, Jodhpur showed a leaning towards Pakistan. Prince Hanwant Singh sought better terms from Pakistan due to its geographic proximity. Muhammad Ali Jinnah offered favorable deals including access to Karachi port and military support. However, on August 11, 1947, Maharaja Hanwant Singh signed the Instrument of Accession, integrating Jodhpur into the Indian Dominion.

Bhopal- The state was ruled by Muslim Nawab Hamidullah Khan, despite a Hindu majority. Khan was a delegate at the Round Table Conference and a key figure in the All-India Muslim League, closely associated with Muhammad Ali Jinnah. By July 1947, observing numerous princes acceding to India, he chose to join India.

Hyderabad- The princely state, under Nizam Mir Usman Ali, was the largest and wealthiest princely state, with a Hindu majority. Despite Nizam's insistence on independence, Indian troops were deployed on September 13, 1948, in Operation Polo. After a four-day armed encounter, Hyderabad was integrated into India.

Junagarh- Situated in southwestern Gujarat, the princely state, led by Nawab Muhammad Mahabat Khanji III, contained a significant Hindu population. Despite initial reluctance to accede to India, a plebiscite in February 1948

overwhelmingly favored accession, with 80% Hindu participation.

In 1960, Bombay state was split into the linguistic states of Maharashtra and Gujarat, in which Junagadh was located and since then Junagadh is part of Gujarat.

Kashmir- Princely state led by Hindu ruler Maharaja Hari Singh, Kashmir had a predominantly Muslim population. Despite proposing a standstill agreement to India and Pakistan, Pakistan invaded Kashmir. Seeking assistance, Maharaja Hari Singh signed the Instrument of Accession to India, formalizing Kashmir's integration. On March 5, 1948, an interim government was formed with Sheikh Mohammed Abdullah as Prime Minister. Article 370, added to the Indian Constitution on October 17, 1949, granted special status to Jammu and Kashmir.

Bengal Province - Bengal was the most fertile and the richest province of the Mughal Empire and included present day Bangladesh and states of Bihar and Odisha.

The Bengal famine of 1943 occurred in the Bengal province of British India during World War II. The social group which mostly suffered were Fishermen, Agricultural labourers and Transport workers.

The Shaheed Minar formerly known as the Ochterlony Monument, is a monument in Kolkata that was erected in 1828 in memory of Major-general Sir David Ochterlony, commander of the British East India Company.

Wars and tribal Revolts before Independence

Sanyasi revolt (1763-1800) - It is also known as the monk rebellion. It was a revolt by sanyasis and sadhus in Bengal. The revolt took place under the leadership of Pandit Bhabhani Charan Pathak and Majanum Shah Fakir. It was made famous by Bankim Chandra Chattopadhyay through his novel Anandamath.

Polygar War - It was started in September 1799 in Tirunelveli district in modern Tamil Nadu. The Polygars were led by Kattabomma Nayak (also called Veerapandi Kattabomman) who was in charge of Panchalankurichi Palayam.

Velu Thampi Revolt (1808- 1809) - Velu Thampi was the Dalawa or Prime Minister of the Indian Kingdom of Travancore, best known for being one of the earliest individuals to rebel against the British East India Company. He sought to assault the Cochin fort and kill the British residents there to weaken their influence in the region.

The **Bheel Revolt** of 1818, one of the first uprisings undertaken by a tribal group in the country.

Paika Revolt (Khurda revolt) 1817 - It was the first ever armed rebellion against Company rule in Odisha, India in 1817 against the British. The standing army of various independent kings were known as Paika. In 1803, the English conquered Odisha. Its leader was Bakshi Jagabandhu Bidyadhar.

Chero revolt - Bhukhan Singh led the Chero revolt in 1800 in Palamu in Jharkhand which was the estate of Raja Shiv Prasad Singh.

Ramosi Revolt - It was a series of uprisings in 19th-century Maharashtra, India, against British rule and local landlords. The first phase occurred in 1822 under Chittur Singh near Satara. The second phase, led by Umaji Naik in 1825-26, was sparked by famine in Pune. The third phase in 1879, led by Vasudev Balwant Phadke, protested British failure to address famine in western Maharashtra. The Ramosi were the hill tribes of western Ghats.

Satara Revolt (1840) - Dhar Rao Pawar and Narsing Dattatreya Pettkar were the important leaders of it. It was the Follow-up of the Ramosi revolt in the Satara region of Maharashtra.

The Kharwar movement, led by Bhagirath Manjhi of the Santhal Tribe, began in 1868 in Bihar. It protested injustices like extensive rack renting and the ejection of village headmen.

The Santhal Rebellion of 1855-56 was a peasant revolt in India led by Sido Murmu and Kanhu Murmu. It was against the East India Company and the zamindari system. A large area of land demarcated as Damin-i-Koh in 1832 was declared as the land of the Santals. The revolt led to the creation of the Santhal Pargana, encompassing 5,500 square miles from Bhagalpur and Birbhum districts. This was the first peasant revolt that occurred in British India.

The Sambalpur rebellion - It was a tribal rebellion led by Surendra Sai who was a direct descendant of Madhukar Sai, former King of Sambalpur. The uprising occurred in two phases from 1827-1840 and 1857-1862.

The Indigo Rebellion (Blue Mutiny) -

It took place in Bengal in 1859-60 and was a revolt by the farmers against British planters who had forced them to grow indigo under terms that were greatly unfavorable to the farmers. Digambar Biswas and Bishnu Biswas of Nadia district of Bengal led the revolt.

Deccan Riots - Landowners in parts of Pune and Ahmednagar districts of Maharashtra revolted against the increase in land revenue in 1875, leading to the Deccan Riots. The root cause was the implementation of the Ryotwari system, a new revenue system introduced by the British. Its purpose was to obtain and destroy the bonds, decrees, and other documents in possession of the moneylenders.

The Kuka Movement - It was started by Satguru Ram Singh. It marked the first major reaction of the people in the Punjab to the new political order initiated by the British after 1849.

The Ahom Revolt of 1828 was sparked by British attempts to incorporate Assam into their dominions after the First Anglo-Burmese War. Led by Gomidhar Konwar, it was a reaction to broken promises of British withdrawal from Ahom territory after the Anglo-Burmese War.

The Khasi Rebellion - It took place in the North-Eastern Hills (between Khasi Hills and Jaintia Hills) in 1829 - 33 in Assam. It arose from the forced labor of local tribes by the British in road construction. Led by Tirath Singh, the rebellion was eventually suppressed by the British army in 1833.

Kittur Revolt - Rani Chennamma of Kittur led an anti-British movement when the British tried to annex Kittur under Doctrine of Lapse. Married to ruler Mallasarja Desai, she became a folk hero of Karnataka.

The Pabna revolt (1873-1876), initially named Pabna agrarian league formed by the peasants (Ryots), was a peaceful movement in Bengal against landlords who forcefully collected rents and land taxes from peasants. It Originated in

Yusuf Shahi Pargana of Pabna (now in Bangladesh).

The Koya Revolts of 1880 occurred in the Godavari tract of Andhra Pradesh and parts of the Malkangiri region in Odisha. Led by Tama Dora, the Koya leader, the grievances included new regulations, denial of traditional forest rights, and oppression by police and moneylenders.

The Munda Rebellion (Munda Ulgulan) in the late 19th century was one of the most significant tribal revolts in Indian Independence history. The Mundas, based in Chhota Nagpur of Jharkhand, practiced the Khuntkatti System of joint land ownership. British policies seized their land, forcing them into landless labor. **Birsa Munda** led the rebellion.

The Battle of Chinhat - It was fought on 30 June 1857 between British forces and Indian rebels at Ismailganj near Chinhat (or Chinhat). It was led by Sir Henry Lawrence, the Chief Commissioner of Awadh (present - day Uttar Pradesh). Maulvi Ahmadullah Shah fought, defeating the forces under Henry Lawrence.

The Kol Rebellion (Kol mutiny) of 1832 was staged by the adivasi Kol people of Chhota Nagpur. Led by Budhu Bhagat, Joa Bhagat, Jhindrai Manki, Madara Mahato, Buli Mahato, and others, it involved various tribes like the Kols, Bhils, Ho, Mundas, and Oraons.

The Patharughat uprising was associated with the Peasant Uprising against Britishers taxation policy in Assam in 1894. Patharughat is a place in Assam and is currently known as Patharighat.

Baba Ramchandra mainly led the peasant struggle during colonial rule in Awadh. He led a peasant movement in Awadh against Talukdars and Landlords.

The Bastar Rebellion of 1910 began in the Kanger forest of Chhattisgarh and quickly spread across the region. Initiated by the Dhurwas, led by rebel leader Gunda Dhur of Nethanar village, it involved widespread destruction, including looting, burning of bazaars, schools, police stations, and the houses of officials and traders throughout Bastar.

Guerrilla movement - In the early 1920s, a militant guerrilla movement spread in the Gudam Hills of Andhra Pradesh.

Guerrilla warfare tactics were first applied in India by Sher Shah Suri in 1545.

The Self-Respect Movement, initiated by E.V. Ramaswamy Naicker in Tamil Nadu in 1925, advocated for breaking down Brahminical hegemony, promoting equal rights for backward classes and women, and revitalizing Dravidian languages like Telugu, Tamil, Kannada, and Malayalam.

Revolt of Songram Sangma - One of the most important rebellions in Assam in 1906, led by Songram Sangma, arose due to British exploitation of tribal forest rights.

Eka Movement or Unity Movement is a peasant movement which surfaced in Hardoi, Bahraich and Sitapur during the end of 1921. Madari Pasi was the leader of the Eka Movement.

Vaikom Satyagraha - It was initiated by K. Kelappan (Kerala Gandhi), Madhavan and K. P. Kesav Menon. It was a movement in Travancore (modern-day Kerala) for temple entry of the depressed classes. It took place near the Shiva Temple at Vaikom, Kottayam district, Kerala during 1924-25. Vaikom was at that time a part of the princely state of Travancore.

Moplah Revolt (1921)- It is also known as the Malabar Rebellion. It was communal violence between Hindus and Muslims centered in Tirurangadi, Kerala. Sparked by the Pookottur incident, it was an armed revolt led by Variyamkunnath Kunjahammed Haji and Ali Musliyar.

The Tebhaga movement (1946-1947) : It was a significant peasant agitation, initiated in Bengal by the All India Kisan Sabha of the peasant front of the Communist Party of India. The leaders of this movement were Kampram Singh and Bhavan Singh.

The Battle of Imphal, fought between British Indian forces and the Japanese, along with the INA, lasted from March 8 to July 18, 1944. It was one of the most grueling campaigns of World War II, as the Japanese attempted to invade British India but were repelled with heavy losses.

The battle of Kohima - It started on 4 April 1944, in which British and Indian troops defeated the Japanese offensive in the northeast of India.

The Battle of Saragarhi was fought on 12 September, 1897. The battle took place in Tirah which is in present-day Pakistan. It was fought between The British Indian Empire and Afghan tribesmen of the area. Only 21 Sikh soldiers fought for the British against about 10,000 tribesmen.

Opium wars were fought between the British and China. The Opium Wars consisted of two wars, spanning the years of 1839 to 1842 and 1856 to 1860.

British acts and Policies

The Regulating Act of 1773 - This act permitted the company to retain its territorial possessions in India. It changed the post of Governor of Bengal to Governor-General of Bengal. The act created a Supreme Court at Calcutta. For the first time, the British cabinet was given the right to exercise control over Indian affairs. The Presidencies of Madras and Bombay became subordinate to the Bengal Presidency.

Pitt's India Act 1784 - It initiated the Dual government scheme, distinguished between the commercial and political functions of the Company. It allowed the court of directors to manage commercial affairs but created a new body called board of control to manage political affairs.

Charter Act of 1793 - This act was passed for extension of Power to Governor-General, Appointment of senior officials and Payments of officials to be paid out of Indian revenues. It was also known as the East India Company Act 1793.

Charter Act of 1813 - It ended the monopoly of the East India Company in India. The company's monopoly in trade with China and trade in tea was kept intact. There was also a provision that the Company should invest Rs. 1 Lakh every year on the education of Indians. The company's dividend was fixed at 10.5% per annum. The company's rule was extended to another 20 years.

The Charter Act of 1833 - It renewed the East India Company's charter for another 20 years. The Governor - General of Bengal was re-designated as the Governor - General of India.

The Governors of Bombay and Madras lost their legislative powers. The civil and military affairs of the company were controlled by the Governor-General in council. It provided for the establishment of a Law Commission for the consolidation and codification of Indian Laws.

The Charter Act of 1853 - For the first time, the legislative and executive functions of the Governor - General's council were separated. It extended the company's rule for an indefinite period, unlike the previous charter acts.

Local representation was introduced into the legislative council in the form of four members from the local governments of Bengal, Bombay, Madras and North Western Provinces.

Government of India Act 1858 - The act replaced the East India Company's rule with direct British control over India. It ended the Dual government system initiated by Pitt's act and ended Doctrine of Lapse, establishing Queen Victoria as Empress of India. The Governor-general became the Viceroy, and provincial autonomy was introduced.

Indian Councils Act 1861 - This Act transformed the viceroy's executive council to function as a cabinet run on the "portfolio" system. It provided that the viceroy should nominate some Indians as non official members of his expanded council. Thus Lord Canning nominated three Indians to his legislative council - Raja of Benaras, Maharaja of Patiala and Sir Dinkar Rao. It empowered the Viceroy to issue ordinances in case of emergency. It restored legislative powers to Bombay and Madras presidencies.

Indian Councils Act, 1892 - Indian representatives, for the first time, allocated seats for election in Indian Legislative Councils. It increased the number of non official members in the central and provincial legislative councils, but maintained the official majority in them. Central council to have not less than 10 members and not more than 16 members.

The members were also given the right to ask questions on the budget but without power to vote.

Indian Councils Act of 1909 - It is also known as **Morley Minto Reforms**, named after the then Secretary of State for India, Lord Morley, and the then Viceroy of

India, Lord Minto. The Central legislative councils increased from 16 to 60 members, while the provincial councils varied in size across provinces. Members were allowed to ask supplementary questions and move resolutions on budget. Satyendra Prasad Sinha became the first Indian to join the viceroy's executive council. He was appointed as law member. This act introduced a 'separate electorate for Muslims', thus Lord Minto came to be known as 'Father of Communal Electorate'.

Government of India Act, 1919 - It is also known as the **Montagu-Chelmsford Reforms**. Edwin Montagu was the Secretary of State and Lord Chelmsford was the Viceroy. The Act introduced **dyarchy** (rule of two individuals/parties) for the executive at the level of the provincial government. Provincial subjects were divided into two parts - Reserved subjects and Transferred subjects. It also introduced Bicameralism (upper house and lower house) in the central legislative assembly. It provided for the establishment of a central Public service commission.

Government of India Act, 1935 - The office of the Secretary of the Council of States for India or Dyarchy in the province was abolished by this Act. Dyarchy was adopted at the centre. It provided for the establishment of all Indian federations consisting of provinces and princely states as units. The act divided the powers between centre and units in terms of three lists - The Federal list, the Provincial list and the Concurrent list. Residuary powers were given to the viceroy.

Bicameralism was introduced in **six provinces** - Bengal, Bombay, Madras, Bihar, Assam and the United Provinces. This was the longest act passed by the British government with 321 sections and 10 schedules. It provided for the establishment of Provincial public service commission and joint public service commission for two or more states. It extended the separate electorates for Women and Labourers.

Under the Government of India Act, 1935, Provincial elections held for the first time in 1937, granted Indians the right to manage provincial governments. Indian National Congress won in seven provinces: Bombay, Madras, Central Provinces,

United Provinces, North-West Frontier Province, Bihar, and Orissa.

Communal Award - It was created by the British prime minister James Ramsay MacDonald, based on the findings of the Indian Franchise Committee (also called the Lothian Committee).

It established separate electorates and reserved seats for minorities, including the depressed classes which were granted seventy-eight reserved seats.

The Indian Independence Act, 1947 - It was passed on July 5th and receiving royal assent on July 18th, facilitated the transfer of power from the British Crown to India. It led to the partition of India into two dominions, India and Pakistan, and set the end of British rule for June 30, 1948. Drafted by the Labour government under Clement Attlee, it was based on the Mountbatten Plan (3rd June plan), agreed upon by leaders of the Indian National Congress and the Muslim League.

Revenue Settlements

The Permanent Settlement - It was introduced by Lord Cornwallis, established fixed land revenue agreements between the East India Company and Zamindars in Bengal, Bihar, Odisha, northern Madras Presidency, and Varanasi district. It recognized Zamindars as landowners, with 10/11th of revenue going to the government and the remainder to the Zamindars. It is also known as the Zamindari System.

Mahalwari system - The Mahalwari system, introduced by Holt Mackenzie in 1822 and reviewed under Lord William Bentinck in 1833, aimed to safeguard village-level autonomy. Implemented in regions like the North-West Frontier, Agra, Central Province, Gangetic Valley, and Punjab, it established average rents based on soil classes. The state's revenue share was 66% of the rental value, settled for 30 years and subject to periodic revision.

Ryotwari system - This system is also known as the Munro system. It is prevalent in southern India, particularly in the Madras Presidency. Devised by Capt. Alexander Read and Sir Thomas Munro, it treated peasants as landowners with rights to sell, mortgage, or gift land. Taxes, set at 50% in dryland and 60% in wetland, were collected directly from

peasants by the government. Failure to pay resulted in eviction. This system covered areas in Madras, Bombay, Assam, and Coorg provinces, encompassing 51% of the region.

Some Other Important Acts and Policies

Calico Act - It was passed by the British government in 1720 to ban the use of Printed cotton textiles - Chintz - in England.

Policy of Paramountcy - A new policy of paramountcy was instituted under Lord Hastings. The company mentioned that because its power was preeminent or superior, it could annex or threaten to annex any Indian state. Sindh was conquered and Punjab was annexed after two prolonged wars in 1849 ruled by Maharaja Ranjit Singh.

The Policy of Ring Fence, implemented by Warren Hastings, aimed to create buffer zones by bolstering the defenses of Company allies against Marathas and Mysore. The local ruler's defense was supported by East India Company troops, with costs borne by the ruler, making them dependent on the Company.

The Cornwallis Code of 1793 implemented a separation of revenue and justice administration in India. It also brought European subjects under the jurisdiction of Indian courts and made government officials answerable to civil courts for actions taken in their official capacity.

The Subsidiary Alliance System, implemented by Lord Wellesley in 1798, required Indian rulers to pay for British forces stationed in their territories. They couldn't maintain independent armed forces. Failure to pay resulted in territorial loss. States like Hyderabad, Mysore, Tanjore, Awadh, Peshwa, and Berar accepted this alliance.

The Doctrine of Lapse, initiated by Lord Dalhousie, allowed the East India Company to annex Indian kingdoms if their rulers died without male heirs. Lord Dalhousie described the Kingdom of Awadh "as a cherry that will drop into our mouth one day" and annexed Awadh in 1856, citing misrule. Other annexations under this doctrine included Satara (1848), Jaitpur and Sambalpur (1850), Udaipur (1852), Jhansi (1854), Nagpur (1853).

The Caste Disabilities Removal Act, 1850 - It was passed during British rule in India, aimed to ease conversion to Christianity. It abolished laws impacting the rights of individuals converting to another religion or caste. The Act granted equal rights, particularly in inheritance, to Hindus converting to other religions.

The Royal Titles Act 1876, passed by the British Parliament, bestowed the title of Empress of India (Kaisar-i-Hind) upon Queen Victoria during the Delhi Durbar of 1877. The title was assumed in 1876 under Prime Minister Benjamin Disraeli's encouragement, marking the transfer of control of British India from the East India Company to the Crown. Formerly known as the Empress of India Act.

The Ilbert Bill, introduced by Lord Ripon in 1883, allowed senior Indian magistrates to preside over cases involving British subjects in India. CP Ilbert, a member of Lord Ripon's council, was involved in its drafting. The bill was enacted as the Criminal Procedure Code Amendment Act 1884 on 25th January 1884.

The Opium Act of 1857 regulated the cultivation, manufacture, sale, and export of opium in India, establishing a government monopoly on opium production and trade. It aimed to control the opium trade and prevent its abuse.

The Vernacular Press Act of 1878, proposed by Lord Lytton, allowed the government to seize assets of newspapers for publishing "objectionable" content. It targeted Indian-language publications to control seditious writing against British policies in India.

The Arms Act of 1878, enacted by Lord Lytton, prohibited Indians from carrying weapons without a licence, while Europeans could do so freely. The law was repealed in 1881 by Lord Ripon, Lytton's successor as Viceroy.

The Bengal Sati Regulation Act of 1829 made the practice of Sati illegal in all areas under British East India Company rule, with practitioners liable for prosecution.

The English Education Act of 1835, introduced by Lord William Bentinck, made English the medium of instruction for higher education in India. Lord Macaulay's Minute on Indian Education

emphasised teaching Western sciences and literature in English, outlining the government's educational policy. Macaulay served as India's first Law member of the Governor-General's Council.

The Indian Official Secrets Act of 1904, enacted during Lord Curzon's tenure as Viceroy, restricted press freedom. It was later replaced by the Official Secrets Act of 1923.

The Defence of India Act, enacted in 1915 under Governor General Lord Hardinge, granted the British government in India broad powers to quell dissent and uphold public order during World War I.

The Commonwealth of India Bill, drafted by Mrs. Annie Besant and Indian associates, was approved at a convention in Kanpur in April 1925. Introduced in the House of Commons by Mr. George Lansbury in December 1925, it advocated for seven fundamental rights.

Governor Generals and Viceroy

When the East India Company arrived in India, it governed Bengal through the position of Governor of Bengal, first held by Robert Clive. After the Regulating Act of 1773, Governor of Bengal became the Governor-General of Bengal, with Warren Hastings as the first appointee.

Subsequently, with the Charter Act of 1833, it evolved into the Governor-General of India, with William Bentinck as the first Governor-General of India.

The Supreme Court was set up in Calcutta by the East India Company in 1773, where Sir Eliza Impey was first chief justice.

Government of India **Act 1858** passed which changed the name of post Governor General of India to Viceroy of India.

Warren Hastings (1773-85) : He was the first Governor-General of Bengal. He abolished the Dual System established by Robert Clive and faced accusations regarding the judicial execution of Raja Nand Kumar. During his tenure, significant events included the Regulating Act of 1773, the Rohilla War

of 1774, the First Maratha War (1775-82), the Treaty of Salbai in 1782, and the Second Mysore War (1780-84). He abolished the Dastak system.

Sir William Jones established the Asiatic Society of Bengal in 1784 during his time. Other notable events include the Pitts India Act of 1784, the introduction of the Policy of Paramountcy, and the establishment of the first post office. Additionally, Hastings ordered the arrest of Maharaja of Benares, Chet Singh, at Shivala Bhavan.

Lord Cornwallis (1786-93) - He was the first to bring into existence and organise the civil services. He professionalized, bureaucratized and Europeanised the administration of East India Company. He abolished the District Faujdari Courts and established circuit courts at Calcutta, Dacca, Murshidabad and Patna. Third Mysore War (1790-92), Permanent settlement (Zamindari System-1793) and Treaty of Seringapatam (1792) were held in his regime. He also introduced a code in 1793, separating revenue administration from judicial administration.

Lord Wellesley (1798-1805) - The fourth Anglo-Mysore war (1799) and the Second Anglo-Maratha War (1803-05) occurred during his regime. Wellesley followed the "subsidiary alliance" policy, first alliance with Nizam of Hyderabad. Took over the administration of Tanjore (1799), Surat (1800), and Carnatic (1801). The Censorship of the Press Act of 1799 was also passed during his tenure, and Fort William College was founded in 1800 to train civil servants.

Lord Minto I (1807-13) - He signed the Treaty of Amritsar in 1809 with Ranjit Singh, Charter Act of 1813 passed in his tenure.

Lord Hastings (1813-23) - Treaty of Sagauli (1816), Anglo-Nepal War (1814-16), Colin Mackenzie became the first Surveyor General of India (1815), Establishment of Ryotwari System (1820), Third Anglo-Maratha War (1817-19) and dissolution of Maratha Confederacy, Initiation of Policy of Paramountcy were held.

Lord Amherst (1823-28) - The principal events of his government were annexation of Assam leading to the first Burmese war of 1824, resulting in the cession of Arakan and Tenasserim to the British Empire.

Lord William Bentinck (1828-35) - Abolition of sati and other cruel rites (1829), Suppression of thuggee (1830), Charter Act of 1833, Macaulay Minute on Education (1835) was passed, Annexation of Mysore (1831), Coorg (1834), and Central Cachar (1834), abolished Provincial Courts of Appeal and Circuit in 1831 which were established by Lord Cornwallis.

Lord Metacafe (1835-36) - He is called the 'Liberator of the Indian press'.

Lord Auckland (1836-1842) - First Anglo-Afghan war (1838-42), Death of Ranjit Singh (1839). He signed the Tripartite Treaty with the Maharaja Ranjit Singh (Sikhs), and Shah Shuja (who had been deposed from the Afghan throne in 1809 and had been living since then as a British pensioner at Ludhiana).

Lord Hardinge I (1844-48) - First Anglo-Sikh War (1845-46), Treaty of Lahore (1846) took place in his regime. He was Commander-in-Chief of the Forces during the Crimean War.

Lord Dalhousie (1848-56) - Second Anglo-Sikh War (1848-49), annexation of Lower Burma (1852), Awadh (1856), Punjab (1849), Introduction of doctrine of lapse, laying down of railway line connecting between Bombay to Thane in 1853, establishment of Public Work Department and Indian Telegraph system.

The first state to be annexed into the British empire under the Doctrine of Lapse was Satara (1848).

Wood's Despatch was issued to Lord Dalhousie in 1854 by Sir Charles Wood. Wood's Despatch recommended using vernacular languages in primary schools, anglo-vernacular medium in high schools, and English as the medium for college-level education.

Lord Canning (1856-62) - Revolt of 1857, establishment of three universities at Calcutta, Madras and Bombay in 1857, Abolition of East India Company and transfer of control to the crown by the Government of India Act, 1858, Hindu Widows Remarriage Act, 1856. He took the decision that Bahadur Shah Zafar would be the last Mughal emperor and no descendants would be recognized as king after his death.

Lord John Lawrence (1864-69) - High Courts were established at Calcutta, Bombay and Madras in 1865. The Bhutan

War, 1865 and Treaty of Sinchula, the Punjab and Oudh Tenancy Act, 1868, was enacted. He is famously called "the Saviour of Punjab" for his developmental activities there. He officially declared Shimla as the summer capital of the British empire in 1864.

Lord Mayo (1869-72) - First Census in India (1872), Establishment of the Statistical Survey of India were held. He took the first step to separate provincial finance from central finance and introduced financial decentralization in India in 1870. He was assassinated in Port Blair.

Lord Northbrook (1872-76) - Arya Samaj was formed by Swami Dayananda Saraswati in 1875 in Bombay. Kuka Movement in Punjab.

Lord Lytton (1876-80) - He passed the Indian Arms Act in 1878 and Vernacular Press Act (1878). The Second Anglo-Afghan War (1878-80) was fought in his tenure. He decreased the Maximum age of appearing in civil services from 21 to 19. Royal Titles Act (1876), Queen Victoria assumed the title of 'Kaiser-i-Hind' or Queen Empress of India.

Lord Ripon (1880-84) - Repeals of the Vernacular Press Act (1882), The first Factory Act (1881), Government resolution on local self-government (1882), The Ilbert Bill controversy (1883 - 84), formation of Hunter Commission on education (1882) were held in his regime.

Lord Dufferin (1884-88) - Third Burmese War (1885-86), Establishment of the Indian National Congress (1885).

Lord Lansdowne (1888-94) - Major Events of his time period were Factory Act (1891), Indian Councils Act (1892), Setting up of Durand Commission (1893).

Lord Curzon (1899-1905) - Lord Curzon succeeded Lord Elgin. He became the youngest Viceroy of India at the age of 39. He was one of the most controversial and consequential holders of that post. He established a Police commission in 1902 under Sir Andrew Frazer. Munda rebellion (1899-1900), Partition of Bengal (1905), Banaras Hindu Girls School (by Annie Besant in 1904), Indian Universities Act (1904), Curzon-Kitchener controversy. He took keen interest in the Restoration and Protection of Historical Monuments. The construction of the

Victoria Memorial Hall at Kolkata was envisaged by him. He established the Archeological Department under the Ancient Monuments Preservation Act, 1904.

Lord Minto II (1905-1910) - Swadeshi Movements (1905-11), Surat Split of Congress (1907), Establishment of Muslim League (1906), Morley-Minto Reforms (Indian Council Act), 1909.

Lord Hardinge II (1910-16) - Annulment of Partition of Bengal (1911), Transfer of capital from Calcutta to Delhi (1911), King George V assume the Crown of India (1911), World war I (1914-18), Establishment of the Hindu Mahasabha (1915) by Madan Mohan Malaviya, Ghadar Mutiny (1915) and Foundation of Banaras.

Lord Chelmsford (1916-21) - Lucknow Pact (1916), Khilafat Movement, Emergence of Gandhi as national leader, passing of Rowlatt Act and Jallianwala Bagh Tragedy (1919), Non-Cooperation Movement, Third Afghan War and Treaty of Rawalpindi, August Declaration (1917), Montague-Chelmsford Reforms (1919).

Lord Reading (1921-1926) - Chauri Chaura incident (1922), Withdrawal of Non-Cooperation Movement (1922), establishment of Swaraj Party (1922), Kakori train robbery (1925).

Lord Irwin (1926-31) - Simon Commission (1927), Butler Indian States Commission (1927), Nehru Report (1928), Deepavali Declaration (1929), Lahore session of the Congress (Purna Swaraj Resolution) 1929, Dandi March and the Civil Disobedience Movement (1930), First Round Table Conference (1930), Gandhi-Irwin Pact (1931) were held in the reign of Lord Irwin.

Lord Willingdon (1931-1936) - Communal Award (1932), Second Round Table Conference (1931) and Third Round Table Conference (1932), Poona Pact (1932), Government of India Act of 1935.

Lord Linlithgow (1936-44) - Resignation of the Congress ministries after the outbreak of the Second World War (1939), Tripuri Crisis & formation of Forward Bloc (1939), Lahore Resolution of the Muslim League (demand for a separate state for Muslims) 1940, August Offer (1940), Formation of the Indian National Army (1941), Cripps Mission (1942), Quit India Movement (1942), Bengal Famine (1943).

Lord Wavell (1944-1947) - C. Rajagopalachari's CR Formula (1944), Wavell Plan and the Simla Conference (1942), Cabinet Mission (1946), Direct Action Day (1946), Announcement of end of British rule in India by Clement Attlee (1947) were held.

Lord Mountbatten (1947-1948) - He was the last Viceroy of India. He was the viceroy of India at the time of India's independence from colonial rule on 15 August 1947. He announced the bifurcation of the British Indian Empire into a secular India and Islamic Pakistan. The 3rd June 1947 Plan was also known as the Mountbatten Plan.

Chakravarti Rajgopalachari (1948-50) - He was the last governor general of India and the only Indian to hold that post.

Elphinstone - He was the Governor of Bombay province of India during 1819-27. He was known for the compilation of the Elphinstone code (a uniform scheme of criminal law). In 1808 he was sent to negotiate an alliance with the Afghan ruler Shah Shuja to prevent a Napoleonic advance upon India.

Kisan Sabha

The Kisan Sabha movement started in Bihar under the leadership of **Sahajanand Saraswati**. He had formed the Bihar Provincial Kisan Sabha (BPKS) in 1929 in order to mobilize peasant grievances against the zamindari attacks on their occupancy rights, and thus sparking the farmers movements in India.

Gradually the peasant movement intensified and spread across the rest of India. All these radical developments on the peasant front culminated in the formation of the All India Kisan Sabha at the **Lucknow session** of the Indian National Congress in April 1936, with Swami Sahajanand Saraswati elected as its first president. The other prominent members of this Sabha were N.G. Ranga, Ram Manohar Lohia.

The **motives** of the All India Kisan Sabha were to abolish the Zamindari system, to reduce land revenue and to institutionalize credit.

Various Kisan Sabha

Bihar Provincial Kisan Sabha was formed in 1929 by Sahajanand Saraswati.

UP Kisan Sabha was formed in 1918 by Gauri Shankar Mishra and Indra Narayan Dwivedi and was supported by Madan Mohan Malaviya.

Oudh Kisan Sabha (Awadh Kisan Sabha), was formed by Jawaharlal Nehru, Baba Ramachandra, and others on 17 October 1920. It was formed to fight against the landlords and talukdars, who demanded exorbitant taxes and rents.

Committees and Commissions before Independence

The Sadler Commission, or Calcutta University Commission, was established in 1917, chaired by Michael Sadler. Its purpose was to investigate the conditions and prospects of the University of Calcutta.

The Indian Education Commission, or the Hunter Commission, was initiated by Lord Ripon in 1882, chaired by Sir William Hunter who was a member of the Viceroy's Executive Council. Its objective was to assess and suggest improvements for elementary education in India.

Wood's Despatch - Charles Wood prepared a Despatch on an educational system for India in 1854. It was considered the Magna Carta of English Education in India.

National Education Commission (1964 - 1966), popularly known as Kothari Commission, was an ad hoc commission set up by the Government of India to examine all aspects of the educational sector in India under the chairmanship of Daulat singh Kothari.

The Indian Universities Commission, or Raleigh commission, formed in 1902 by Viceroy Lord Curzon, aimed to propose reforms for university education in India. Led by Thomas Raleigh, its members included Syed Hussain Belgrami and Justice Gurdas Banerjee.

The Sapru Committee, headed by Sir Tej Bahadur Sapru, released its Constitutional Proposals in 1945. Appointed to address communal issues after the Gandhi-Jinnah talks, the committee made 21 recommendations on constitutional matters and safeguarding minority rights in India.

The National Planning Committee, formed in 1938 by Netaji Subhash Chandra Bose and endorsed by Jawaharlal Nehru, submitted reports on various topics including economic policy, central planning, industrialization, labor laws, rural development, cottage industries, and public finance in India.

The First Famine Commission, initiated in 1878 under Sir Richard Strachey, advocated state intervention in food trade during famines.

The Second Famine Commission in 1897, led by Sir James Lyall, emphasized irrigation development.

The Third Famine Commission in 1900 under Sir Anthony McDonnell recommended year-round famine prevention efforts by official machinery.

The Aitchison Commission, established in 1886 under Sir Charles Aitchison, aimed to enhance Indian representation in public service. It suggested setting minimum and maximum age limits for Indians in civil service exams at 19 and 23 years, respectively.

The Hartog Committee, formed in 1929 by Lord Irwin and chaired by Sir Philip Joseph Hartog, focused on expanding education in India. It highlighted issues of waste and stagnation in primary education.

The Frazer Commission, established by Lord Curzon in 1902 and led by Sir Andrew Fraser, examined police administration. It proposed creating two service branches: the Imperial Service Branch with recruitment from England and the Provincial Service Branch with recruitment solely in India.

Educational Institutions

The Asiatic Society of Bengal was established in 1784 by Sir William Jones in Calcutta with the aim of advancing Oriental knowledge, including history, scriptures, and regional texts of India and the Indian subcontinent.

The Hindu College was founded in 1791 to promote the study of ancient Sanskrit texts. Sri Jonathan Duncan and Governor General Lord Cornwallis established it with the aim of developing and preserving Sanskrit literature, showcasing British support for Indian education.

Fort William College, established in 1800 by Lord Richard Wellesley, aimed to instruct East India Company officials in the vernacular languages of India.

Vedanta College was founded by Raja Ram Mohan Roy in Calcutta in 1825, where he introduced Mechanics and Voltaire's Philosophy.

The Universities of Calcutta, Bombay, and Madras were established as the outcome of Sir Charles Wood's dispatch in 1857, promoting English and Western education in India.

The Muhammadan Anglo - Oriental College in Aligarh was founded in 1875 by Sir Syed Ahmad Khan. It began as a primary school, later evolving into a college-level institution. It started operations on Queen Victoria's 56th birthday, 24 May 1875.

Shantiniketan was established by Rabindranath Tagore near Bolpur on December 22, 1901, with five students (including his eldest son) and an equal number of teachers. He originally named it Brahmacharya Ashram in the tradition of ancient forest hermitages.

Visva-Bharati, founded by Rabindranath Tagore in 1921, is a public central university located in Shantiniketan, West Bengal, India. Initially a college, it attained Central University status in 1951 through an act of Parliament. The Prime Minister of India serves as its Chancellor.

Calcutta Mathematical Society was founded by Ashutosh Mukherjee in 1908.

Central Hindu College, established by Annie Besant in 1898, taught Hindu religion alongside Western scientific knowledge.

Banaras Hindu University was jointly established in 1916 by Madan Mohan Malaviya, Annie Besant, Maharaja Rameshwar Singh of Darbhanga, and Prabhu Narayan Singh and Aditya Narayan Singh of the Narayan dynasty.

Bharatiya Vidya Bhavan, founded by Dr. Kanhaiyalal Maneklal Munshi on November 7, 1938, aims to promote education and culture as a charitable public trust.

The **Dayanand Anglo Vedic (DAV)** schools were established in 1886 to fulfill Swami Dayanand Saraswati's vision. The first school, with Mahatma

Hansraj as headmaster, was founded in Lahore.

Books, Newspapers and Magazines before Independence

Mahatma Gandhi's books - Hind Swaraj or Indian Home Rule (1909), Sermon on the Sea, Dakshina Africana Satyagraha Itihasa, The Story of My Experiments With Truth, Mangal Prabhat, India's Case for Swaraj (1931), Songs from Prison (1934), The Indian States' Problem (1941), Gandhi and the Good life (1943), Gandhi Against Fascism (1944), From Yeravda Mandir: Ashram Observances (1945), Conquest of Self (1946), Women and Social Injustice (1947), Self-restraint versus Self - Indulgence (1947), Gandhigrams (1947), India of My Dreams (1947), The Bhagavad Gita According to Gandhi, Gandhi on Non-Violence, The Power of Non-violent Resistance, The Words of Gandhi, Third Class in Indian Railways.

Newspapers started by Gandhiji : Young India, Harijan, Navjivan, Indian Opinion.

Rajendra Prasad - India Divided, Atmakatha, Satyagraha in Champaran, At the feet of Mahatma Gandhi, The Unity of India.

Jawaharlal Nehru - The Discovery of India, A Bunch of Old Letters, Glimpses of World History, Letters from a Father to his Daughter, An Autobiography: Towards Freedom.

Dayanand Saraswati - Satyarth Prakash (1875).

Sarojini Naidu: The Golden Threshold, Palanquin Bearers, The Village Song, The Bird of Time, The Broken Wing, The Indian weavers, The Feather of the Dawn, The Sceptred Flute, In the Bazaars of Hyderabad, Essential Reader.

Dadabhai Naoroji- Poverty and Un-British Rule in India, Poverty of India, The Wants and Means of India, The European and Asiatic Races.

Rast Goftar was an Anglo-Gujarati paper operating in Bombay that was started in 1854 by Dadabhai Naoroji and Kharshedji Cama and championed social reform among Parsis in Western India.

Lala Lajpat Rai - Unhappy India, Arya Samaj, The Story of My Deportation.

He published an Urdu daily newspaper named Vande Mataram and an English daily newspaper called People. Lala Lajpat Rai was an active member of Arya Samaj.

Bankim Chandra Cheterjee - Rajmohan's Wife (1864) - first Indian novel in English, Anandamath (1882) - a Bengali historical novel which provides information about the Sannyasi rebellion. Durgesh Nandini, Kapalkundala, Devi Chaudhurani, Bishabriksha are some notable books by him.

Rabindranath Tagore - Visva Parichay (1937), dedicated to scientist Satyendranath Bose.

Other literary works : Manasi (1890), The Ideal One, Sonar Tari (1894), The Golden Boat, Gitanjali (1910), Song Offerings, Gitimalya, Dakghar (1912), Chitra : A Play in one Act (1912), Raktakarabi (1926), Red Oleanders, Chokher Bali (1903), The Home and the World, The Postmaster, Fireflies, Gora, Waiting, Friend, On the Nature of Love, Paper Boats.

Bal Gangadhar Tilak - Gita Rahasya (1915) while he was in prison at Mandalay, Burma. The Arctic Home in the Vedas, Orion.

Newspapers : Kesari (In Marathi), Mahratta (in English).

Tarabai Shinde - Stri Purush Tulana, originally published in Marathi in 1882.

Jyotiba Phule: Gulamgiri (Written in Marathi in 1873) - dedicated to the people in the US who were working to end slavery. Other books are Trutiya Ratna (drama) (1855), Powada (1869), Brahmnanche Kasab (1869), Shetkaryacha Asud (1883), Satsaar, Ishara and Sarvajaneek Satyadharma Pustak (1891).

Raja Ram Mohan Roy's books - A Defence of Hindu Theism, The Only True God: Works on Religion, The Precepts of Jesus : the Guide to Peace and Happiness.

Newspaper - Sambad Kaumudi, published in English and Bengali in November 1821. Mirat-ul-Akhbar (Mirror of News), a Persian journal in 1822. In 1809, he wrote Tuhfat-ul-Muwahhidin (Gift to Monotheists - originally written in Persian language) in which he put forward the idea that people must worship a single God.

Books by Aurobindo Ghosh - Savitri: A Legend and a Symbol, The Mother, The

Life Divine, Synthesis of Yoga, The Hour of God, Karmayogini, The Secret of the Veda.

Ishwar Chandra Vidyasagar - "Barnaparichay" is a renowned Bengali primer, published in 1855, and considered highly influential in Bengal. He was regarded as the 'Father of Bengali Prose'. He was the Principal of a Sanskrit College which represented a blend of Indian and Western culture.

Maulana Abul Kalam Azad - India Wins Freedom (autobiography), Ghubar -e-Khatir (Sallies of Mind), The Dawn of Hope, Tarjumanul Quran, Tazkirah.

Bhimrao Ambedkar - Waiting for a Visa (autobiography), The Annihilation of Caste, The Buddha and His Dhamma, Mook Nayak, Janta, The Untouchables, Buddha or Karl Marx, Who Were the Shudras, Pakistan or the Partition of India, Finance of the East India Company, Thoughts on Pakistan.

Annie Besant Books - The Ancient Wisdom, My Path to Atheism. Newspapers - New India, Commonweal

Other Important Author and Books

Bhagat Singh - Why I am an Atheist.

Lala Har Dayal - Hints for Self Culture, Glimpses of World Religions

Romesh Chunder Dutt - Economic History of India, The Civilization of India

MG Ranade - Religious and Social Reform: A collection of Essays and Speeches, Essays on Indian Economics

Surendranath Banerjee - A Nation in the Making (autobiography)

On 29 January 1780, India's and Asia's first English newspaper "Hicky's Bengal Gazette" started its publication. It was a weekly English newspaper started by an Irishman named James Augustus Hickey.

"**Young India**" was a weekly journal founded by Lala Lajpat Rai in 1916 and later published by Mahatma Gandhi. In 1931, Gandhi outlined his vision for the Indian Constitution in "Young India".

'**My Experiments with Truth**' was originally written in Gujarati Language by Mahatma Gandhi. Mahadev Desai translated it into English Language.

"**The History of British India**" is a three-volume work by James Mill, a Scottish historian, economist, and

philosopher. Published in 1817, it examines the history of Company rule in India and achieved immediate success, bringing prosperity to Mill.

Mill categorized Indian history into the Hindu, Muslim and British periods on the basis of dominant political powers and their religious affiliations.

In 1785, **Sir Charles Wilkins** translated the Bhagavad Gita into English. He did this in the special interest of Warren Hastings, the Governor General of India.

Tehzeeb-ul-Ikhlaq is a journal established by the Muslim reformer Sir Syed Ahmad Khan in 1871. The journal published alternative Muslim perspectives, written in plain language.

Nibandhmala was a monthly periodical started by Vishnushastri Chiplunkar.

The diary of manu gandhi was edited and translated by Tripid Suhrud.

The East Indian was a daily English newspaper started by Henry Vivian Derozio in 1831. He was a professor at Hindu College and inspired the Young Bengal Movement.

Periyar EV Ramasamy started **Kudi Arasu**. It was a Tamil weekly magazine in the Madras Presidency in Tamil Nadu.

Ananda Bazar Patrika was started by Sisir Kumar Ghosh and his son Tushar Kanti Ghosh in 1876. Later on, in 1922, Anandabazar Patrika was revived and re-launched by Suresh Chandra Majumdar and editor Prafulla Kumar Sarkar.

Amrita Bazar Patrika was started by Sisir Kumar Ghose and Moti Lal Ghosh in 1868. It became a daily newspaper (initially it was published in Bengali and then in English).

The Hindu was founded in the year 1878 by G. Subramania Iyer in Madras. This was a newspaper that was published in British India.

Dinabandhu Mitra, the playwright of 19th century Bengal's most powerful play, Neel Darpan (1860), wrote on the oppressive practices of the indigo planters in Bengal in the 1850s.

"Vande Mataram" is a poem by Bankim Chandra Chatterjee, included in his 1882 novel "Anandamath." Rabindranath Tagore sang it first at the 1896 Indian National Congress session.

Rabindranath Tagore wrote his first substantial poems titled Bhanusimha Thakurer Padabali in Brajabuli under the pseudonym Bhānusiṃha (Sun Lion) when he was just sixteen.

Indian Industries during British Rule

Before the rise of the modern Industrial system, India was an exporter of Cotton and Silk textile. In the 18th century, the development of the cotton industry in Britain resulted in decline of textile production in India.

The **first English factory** was established on the banks of the Hugli River in Bengal in 1651. Merchants soon settled nearby, and by 1696, the East India Company began constructing a fort around the Hugli settlement.

First cotton textile factory opened in Ghusuri near Kolkata in 1818 but closed soon due to raw material shortages. The Bombay Spinning and Weaving Company, founded on July 7, 1854, in Tardeo, was the first successful cotton mill established by Cowasjee Nanabhai Davar and his associates. The company was designed by Sir William Fairbairn.

Bombay became an Important port for exporting raw cotton to England and China. By 1900, over 84 mills were operating in Bombay, with many established by Parsi and Gujarati businessmen.

Maharashtra has the largest number of cotton textile mills in India followed by Tamil Nadu and Gujarat.

Jute textile industries : Jute was grown mainly in East Bengal (now Bangladesh), while jute mills were situated in West Bengal (now part of India), near the banks of the River Ganga.

The Acland Mill, India's first jute mill, was established in 1855 in Rishra, West Bengal, by British entrepreneur George Acland and Bengali financier Babu Bysumber Sen.

Iron and steel industries : Tata Iron and Steel Company (TISCO), founded by Jamsetji Tata and Dorabji Tata on August 26, 1907, in Jamshedpur, Jharkhand, was India's first iron and steel manufacturing plant. It began production of pig iron in 1911 and steel in 1912, becoming one of Asia's largest manufacturing plants. By

1939, it operated the largest steel plant in the British Empire.

Sugar Industries - The **first sugar mill** in British India was established in Bettiah, Bihar, in 1840. First sugar mill in Uttar Pradesh was founded in Pratappur in 1903. The Sugar Cane Act of 1934 empowered the State Government to set minimum prices for sugarcane. The Sugarcane Development Department was established in Uttar Pradesh in 1935. The first sugar plant in Bihar was set up in Saran in 1904. Ahmednagar district had six joint-stock sugar factories, three in Kopergaon Taluka. Karnataka's first modern sugar industry, Mysore Sugar Company (MySugar), was established in Mandya in 1933.

Cement industries - The common cement or Portland cement was prepared and Patented by Joseph Aspdin in 1824. India's cement era began in 1914 when the Indian Cement Company Ltd. started manufacturing cement in Porbandar, Gujarat. A small cement factory was established in Madras in 1904 by South India Industrial Ltd.

Railway Network - In 1832, the first railway plans in India were presented in Madras. In 1837, the Red Hill Railways, the nation's first train, operated from Red Hills to Madras' Chintadripet Bridge. It was pulled by a William Avery locomotive with a rotary steam engine. The railway, constructed by Arthur Cotton, was primarily used to deliver laterite rock for road construction in Madras.

Private businesses were allowed to start a railway system under Lord Hardinge, the Governor-General of India, in 1844.

The railway era in India commenced on April 16, 1853, under Dalhousie, with the first Passenger train journey from Bori Bunder in Bombay to Thane, covering 34km, marked by a 21-gun salute. The train, pulled by three steam locomotives named Sahib, Sindh, and Sultan, consisted of 14 carriages. Sikkim is the only northeastern state not connected to the rail network.

Other Important Organisations

The Deccan Sabha, founded in 1896 in Pune, India, advocated for social and political reforms in the Deccan region. It was pivotal in the Indian independence movement.

The Communist Party of India - The Indian Communist Conference in Kanpur in 1925 led to the formal establishment of the Communist Party of India (CPI), with M.N. Roy playing a significant role. However, the CPI was initially formed in 1920 in Tashkent by M.N. Roy, Abani Mukherjee, and others. It was officially founded in India in 1925 by the same founders. CPI is one of the oldest political parties in India.

Swatantra Party - The Swatantra Party was an Indian classical liberal political party that existed from 1959 to 1974. It was founded by C Rajagopalachari, running against the Indian National Congress Party. Maharani Gayatri Devi was a member of the swatantra party from 1967-71.

The Congress Socialist Party (CSP) was a socialist faction within the Indian National Congress, established in 1934 by Jai Prakash Narayan, Ram Manohar Lohia, and Acharya Narendra Deva to address ideological disagreements with Mahatma Gandhi.

Revolutionary Socialist Party (RSP) is a communist party in India and was founded on 19 March 1940 by Tridib Chaudhuri. It has its roots in the Bengali liberation movement Anushilan Samiti and the Hindustan Socialist Republican Army. The first general secretary of the party was Jogesh Chandra Chatterjee.

The Forward Bloc, a political party, was formed on May 3, 1939, by Netaji Subhas Chandra Bose in Makur Unnao, Uttar Pradesh, after resigning from the presidency of the Indian National Congress on April 29.

The Paris Indian Society was an Indian nationalist organization founded in 1905 at Paris under the patronage of Madam Bhikaji Cama, Munchersah Burjorji Godrej and S. R. Rana. It was a branch of the Indian Home Rule Society which was established in London in 1905.

Madras Labour Union was founded by Wadia, along with V. Kalyanasundaram Mudaliar, on April 13, 1918, marking one of India's earliest organized labor unions.

Bal Charkha Sangh - Indira Gandhi founded the Bal Charkha Sangh, where children were taught to spin and weave. In her early childhood she founded the Bal Charkha Sangh and in 1930, the Vanar Sena of children to help the

Congress party during the Non-Cooperation Movement.

The Satara Parallel government in Maharashtra from August 1943 to May 1946 against British rule was legendary in the glorious freedom struggle of India. The leader of the Satara Parallel government was **Kranti Singh Nana Patil**. He later joined the Communist Party and was elected to the Lok Sabha on the CPI ticket from Satara in 1957 and again from Beed in 1967.

The **All India Trade Union Congress (AITUC)** is the oldest trade union federation in India. It was founded in 1920. The first President of the AITUC was Lala Lajpat Rai.

Nicknames and Slogans of Freedom Fighters

Subhash Chandra Bose - He was popularly called as Netaji. He gave the slogan 'Dilli Chalo'. He used this slogan to motivate the Indian National Army (INA) in 1944, when the INA commenced its military campaign against Burma. Other slogans given by him were 'Tum Mujhe Khoon Do, Me Tumhe Azadi Dunga' (Give me blood and I shall give you freedom) and 'Jai Hind'.

Bal Gangadhar Tilak, known as Lokmanya Tilak, was called the "Maker of Modern India" by Mahatma Gandhi and the "Father of the Indian Revolution" by Jawaharlal Nehru. Valentine Chirol labeled him the "Father of Indian Unrest." His famous slogan was "Swaraj is my birthright and I shall have it."

Dadabhai Naoroji - He is known as the Grand Old Man of India and Unofficial Ambassador of India. 'Our country is India, our nationality is Indian' was the slogan of Dadabhai Naoroji.

Bhagat Singh - Saheed-e-Azam is the nickname of Bhagat Singh. His slogans were Inqalab Zindabad (which was first phrased by Maulana Hazrat Mohani) and Samrajyavad Ka Nash Ho.

Lala Lajpat Rai - He is also called Punjab Kesari and Lion of Punjab. Simon go back, Defeat and failure are sometimes necessary steps of victory, the shots that hit me are the last nails to the coffin of British rule in India, were popular slogans given by him.

Sardar Vallabhbhai Patel - India's first home minister and also the first Deputy Prime Minister. He is known as the Iron Man of India and Bismark of India. The title Sardar was conferred to him by Mahatma Gandhi.

"Manpower without unity is not a strength unless it is harmonized and united properly, then it becomes a spiritual power" was the slogan of Sardar Vallabhbhai Patel.

Chittaranjan Das - He was a nationalist leader popularly called Deshbandhu (Friend of the Nation), was an Indian freedom fighter, political activist and lawyer during the Indian independence movement. He was the Political Guru of Subhash Chandra Bose.

Jawaharlal Nehru - He is popularly known as Chacha Nehru. His famous slogan was "Aaram haraam hai". On behalf of the INC, he made a statement that 'the Constitution of free India must be framed, without outside interference, by a Constituent Assembly elected on the basis of the adult franchise'.

Mohammad Ali Jinnah - The title of Quaid-e-Azam was given to him by Maulana Mazharuddin Shaheed in 1938.

Mahatma Gandhi - Mahatma Gandhi, known as 'Bapu' and hailed as 'The father of the Nation', was mentored by Gopal Krishna Gokhale. His famous slogans include "Be the change you wish to see in the world", "A flag is a necessity for all nations; millions have died for it", "No one can hurt me without my permission", "Non-violence is the strongest weapon", "Do or die", and "British Quit India". He was awarded the title Kaiser-e-Hind by the British government for his humanitarian services during the Boer War (1899-1902). It was awarded by the Governor-General of India Lord Hardinge II (1910-1916) in 1915.

Jayaprakash Narayan - He is popularly referred to as JP or Lok Nayak (People's Leader).

Sri Krishna Sinha - He is popularly known as Bihar Kesari. "**Amar Sonar Bangla**" also known as "My Golden Bengal" is the national anthem of Bangladesh. An Ode to Mother Bengal, the lyrics were written by Bengali polymath Rabindranath Tagore in 1905.

Bipin Chandra Pal also known as the 'Father of Revolutionary Thoughts in

India'. He was given the title of Bengal Tiger. The famous slogan was 'You Wanted Magic, I Tried to Give You Logic'.

Some Important slogans

The National Anthem of India "Jana Gana Mana" was originally composed in Bengali by Rabindranath Tagore on 11 December 1911.

The National Song of India 'Vande Mataram' was written by Bankim Chandra Chatterjee.

Muhammad Iqbal- Sare Jahan Se Achha Hindustan Hamara.

Lal Bahadur Shastri-Jai Jawan Jai Kisan
Atal Bihari Vajpayee - Jai Jawan, Jai Kisan, Jai Vigyan.

Mangal Pandey- Maro Firangi Ko.

Madan Mohan Malviya- Satyameva Jayate.

Sri Narayana Guru- One caste, One Religion, One God for Humankind.

Ramprasad Bismil- Sarfaroshi ki tamanna ab hamare dil me hai, dekhna hai zor kitna baazu-e-qaatil mein hai.

Events after Independence

On 17 september 1948 The Princely state of Hyderabad was brought into the Indian Union. After the State Reorganisation Act 1956 it was merged with Andhra state and later renamed as Andhra Pradesh.

The Bhoodan or Land Gift movement, started by Vinoba Bhave in 1951, aimed at voluntary land reform in India, urging wealthy landowners to donate land to the landless.

On November 1, 1954 the French possessions in India were transferred to the Indian Union and Puducherry became a Union Territory. About 280 years of French rule finally came to an end.

The '**Instrument of Surrender**' which ended Portuguese rule in India was signed on 19th December 1961, ending 450 years of Portuguese rule in India.

The first meeting of the National Integration Council was held in 1962. The **National Integration Council (NIC)** is a group of senior politicians and public figures in India that looks for ways to address the problems of communalism, casteism, and regionalism.

The Nehru-Liaquat Pact was signed in New Delhi by the Prime Minister (India)

Jawahar Lal Nehru and the Prime Minister (Pakistan) Liaquat Ali Khan on April 8, 1950. It aimed to safeguard minority rights after the Partition. The Nehru-Liaquat pact made it possible that the two countries signed multiple agreements after that - including Indus Water Treaty, 1960 and Shimla Pact, 1972.

India and Pakistan had signed the **Shimla Pact**, under which the two countries agreed to sort out differences and disputes bilaterally. The treaty was signed in Shimla in 1972 by Zulfikar Ali Bhutto, the President of Pakistan, and Indira Gandhi, the Prime Minister of India. The **Bangladesh Liberation War** ended on 16 December 1971.

The Silent Valley Movement began in 1973 in Kerala, led by teachers and the Kerala Sastra Sahitya Parishad (KSSP) to prevent the flooding of the Silent Valley due to a hydroelectric project. In 1985, it was declared a national park.

India's National Flag, in its current form, was adopted by the Constituent Assembly on July 22, 1947, just before India gained independence from British rule on August 15, 1947.

Some Important Facts

Annexation of Goa - In 1961, the Indian Army liberated Goa from Portuguese rule, declaring it a Union Territory. The Portuguese had colonized many western parts of India like Goa, Daman, Diu, Dadar and Nagar Haveli, and Anjediva island in 1510, but on 18-19 December 1961, Indian forces captured the territory, leading to the resignation of the governor-general of Portuguese India.

Merger of Manipur - A few days before Independence, the Maharaja of Manipur, **Bodhachandra Singh**, signed the Instrument of Accession and agreed to become a part of the Union of India.

The Nyaya Yudh (Struggle for Justice) movement was led by Chaudhary Devi Lal in 1987. He founded the Lok Dal party and served twice as Chief Minister of Haryana and also served as the Deputy Prime Minister of India.

Radha Krishna Commission (1948) - It was established to review the current state of university education in India and make recommendations for improvement.

Polity

Basics of Constitution

Constitution : The basic principles and fundamental laws of a nation, state, or social group that determine the powers and duties of the government and guarantee certain rights to the people. It is also described as the 'Fundamental law of the land'.

Types of Constitution

Written Constitution - provisions are written in a book. Example - USA, Canada, India.

Unwritten Constitution - provisions are found in conventions, customs, and practices. Example - UK.

Federal Constitution - Divides power between national and regional governments. Example - USA.

Unitary Constitution - Concentrates power in the national government. Example - United Kingdom.

Rigid Constitution - Requires a special procedure for amendment. Example - Australia.

Flexible Constitution - Amended like ordinary laws. Example - New Zealand.

Indian Constitution

The Indian Constitution is the supreme law of India and the longest of any sovereign nation in the world.

It took precisely 2 years, 11 months, 18 days to create the Constitution of India. The original text of the Constitution contained 395 articles in 22 parts and 8 schedules. Presently, the Indian Constitution is made up of 448 Articles in 25 Parts and 12 Schedules.

The constitution was handwritten in English by Prem Behari Narayan Raizada and in Hindi by Vasant Krishan Vaidya.

The original version of the constitution was beautified and decorated by artists from Shantiniketan including Nand Lal Bose and Beohar Rammanohar Sinha.

The Constitution of India is kept in special helium - filled cases in the Library of the Parliament of India.

On 19 November 2015, the government formally notified 26 November as Constitution Day. Earlier, this day was

celebrated as National Law Day.

Political philosophy : The political philosophy of the Indian Constitution encompasses sensitivity towards historically disadvantaged groups, openness to community values, consideration for the needs of religious and linguistic minorities, promotion of individual freedom and social justice, commitment to secularism, universal franchise, federalism, and fostering a cohesive national identity.

Major Features of Indian Constitution:

The Indian Constitution is the world's lengthiest written constitution. It is a blend of rigidity and flexibility. It is drawn from various sources. It embodies a federal system with a hint of unitary bias. It establishes the parliamentary form of government and harmonizes parliamentary sovereignty with judicial supremacy.

It ensures the rule of law and establishes an integrated and independent judiciary. It provides provisions for fundamental rights, directive principles of state policy, fundamental duties, a secular state and universal adult franchise. It promotes inclusive development with a three-tier government structure and support for cooperative societies.

Federal features : The Indian Constitution has federal features such as dual government, division of powers, supremacy of the constitution, a written constitution, a supreme judiciary, and bicameralism.

Unitary features : On the other hand it has certain unitary features like Strong Centre, Emergency provision, Single Citizenship, All India Service, Governor Appointment by Center, One Election Commission, Judicial Review etc.

Terminologies

Democracy : It is a form of government in which rulers are elected by the people in a free and fair election, a universal adult franchise and governed on the basis of certain fundamental rules like a constitution. In terms of population, India is the largest democracy in the world whereas America is the second largest democracy.

Types of democracy : Direct and Indirect (Representative).

Direct Democracy - In this, people hold sovereign power and make decisions. Example - Switzerland. Four devices of direct democracy are Referendum, Initiative, Recall, and Plebiscite.

Referendum : A public vote on a specific law or policy proposed by the government.

Initiative : A process where citizens can propose a new law or amendment by gathering enough signatures on a petition.

Recall : A procedure allowing citizens to remove an elected official from the office through a public vote.

Plebiscite : A direct vote by eligible citizens to express their opinion on a significant public question, often related to territorial changes.

Indirect Democracy : In an indirect democracy, also known as a representative republic, citizens elect representatives who then make decisions and govern on their behalf.

Republic Day : It is celebrated on 26th January every year to mark the day on which the constitution of India came into effect in 1950. 26th January held significance as it was the day the Indian National Congress urged Indians to celebrate Independence Day in 1930.

The 1st Republic Day parade was conducted at Irwin Stadium (now known as Major Dhyani Chand National Stadium) in Delhi. Indonesian President Sukarno was the chief guest of India's first Republic Day parade on 26 January 1950.

The Republic Day celebrations are majorly divided into three parts (The Republic Day Parade, The Award Distributions, and The Beating Retreat). The opening Ceremony starts with laying a wreath at the Amar Jawan Jyoti at India Gate. The Beating Retreat Ceremony is the concluding event of the Republic Day.

Federalism : It is a system of government in which powers have been divided between the centre and its constituent parts such as states or provinces. It accommodates two sets of polity, one at the centre or national level and second at the regional or provincial level.

Quasi-federal : It refers to a system of government where the distribution of

powers between the Center and the state are not equal. India is referred to as a quasi-federal state because of its strong central government.

The Indian Constitution has been variously described as federal in form but unitary in spirit, quasi-federal by K.C. Wheare, bargaining federalism by Morris Jones, Co-operative federalism by Granville Austin, federation with a centralizing tendency by Ivor Jennings, etc.

Presidential System : It is one in which the head of state is also the head of government, and the executive branch is independent of the legislative branch. It is based on individual leadership.

Semi-Presidential System : It is a republican system of governance that combines elements of presidential democracy with parliamentary democracy. In this system, the President has more power than the Prime Minister.

Parliamentary System : It is a form of government where executives hold the power with the majority support of the legislature. For example - India. India follows parliamentary democracy where real power is vested in the council of ministers headed by the Prime Minister.

Semi - Parliamentary System : It is a form of government that combines elements of both parliamentary and presidential systems. In this system, the head of government (Prime Minister) has more powers than the head of state (President).

Coalition Government : It is a form of government in which political parties cooperate to form a government.

Liberalism : It is a political and philosophical ideology to establish individual freedom, consent and equality.

Radicalization : It is a process in which an individual or group adopts extreme political, social, or religious ideals and aspirations.

Coup : A sudden, violent, and unlawful seizure of power from a government.

Monarchy : It is a form of government in which a person, the monarch, is head of state for lifetime or until abdication.

Oligarchy : It is a type of government in which power is controlled by few people. These people may or may not be

distinguished by one or several characteristics, such as nobility, fame, wealth, education, or corporate, religious, political, or military control.

Autocracy : It means a system of government by one person with absolute power.

Dictatorship : It is a form of government characterized by a leader or group of leaders that hold government power with few to no limitations.

Communist : It is a classless society with common ownership. Karl Marx is known as the father of communism.

Totalitarian Government : In this, the government has absolute power with no opposition (single-party dictatorship).

Order of Precedence : It is the protocol list in which the officials and functionaries are listed as per their rank and office in the Government of India.

Order of Important officials : 1. President
2. Vice-President, 3. Prime Minister, 4. Governors of States, 5. Former Presidents, 5A. Deputy Prime Minister, 6. Chief Justice of India and Speaker of Lok Sabha, 7. Cabinet Ministers of the Union, Chief Ministers of States within their respective States, NITI Aayog, Former Prime Ministers, and Leaders of Opposition in Rajya Sabha and Lok Sabha, 7A. Holders of Bharat Ratna etc.

Judicial Doctrines

It is a principle or a position that is generally applied and upheld by the courts.

Doctrine of Basic Structure : The origins of the basic structure doctrine are found in the German Constitution. Parliamentary democracy, fundamental rights, secularism, federalism, judicial review etc. are all held by courts as the basic structure of Indian Constitution. This principle of basic structure was propounded by the Supreme Court in the landmark Kesavananda Bharati case of 1973.

Related Judgements : Kesavananda Bharati case (1973), Indira Nehru Gandhi v Raj Narain case (1975), Minerva Mills case (1980) and Waman Rao case (1981).

Doctrine of Separation of Powers : Signifies the division of powers between various organs of the state: executive,

legislature and judiciary.

Related Judgements - Ram Jawaya vs State of Punjab case (1955), Indira Nehru Gandhi vs Raj Narain (1975).

Doctrine of Pith and Substance : Pith means true nature and Substance means the most important or essential part of something. The doctrine was first acknowledged in the Canadian Constitution. It is applied where the question arises of determining whether a particular law relates to a particular subject (mentioned in Seventh Schedule).

Related Judgements - Prafulla vs Bank of Commerce (1946).

Doctrine of Incidental or Ancillary Powers : The evolution of this Doctrine can be traced back to R. V. Waterfield (1963), a decision of the English Court of Appeal. It has been developed as an addition to the Doctrine of Pith and Substance and invoked when there is a need to aid the principal legislation in question. Parliament can make provision for ancillary power of the supreme court.

Related Judgements - State of Rajasthan vs G Chawla (1958).

Doctrine of Severability : It is also known as the doctrine of separability and protects the Fundamental Rights of the citizens. It originated from England.

Related Judgements - A.K. Gopalan vs State of Madras (1950), State of Bombay vs F.N. Balsara (1951).

Doctrine of Eclipse : It is applied when any law/act violates the Fundamental Rights (FR). FR overshadows the law/act and makes it unenforceable but not void ab initio (Having no legal effect from inception). It is only against the citizens that these laws/acts remain in a dormant condition but remain in operation as against non-citizens who are not entitled to fundamental rights.

It is contained in Article 13(1) of the Indian Constitution and does not apply to post-constitutional laws.

Related Judgements : Bhikaji Narain Dhakras vs State of Madhya Pradesh (1955).

Doctrine of Territorial Nexus : It says that laws made by a State Legislature are not applicable outside the state, except when there is a sufficient nexus between

the state and the object. It derives its power from Article 245 of the Indian Constitution.

Related Judgements - A.H. Wadia v. Income Tax Commissioner (1948), State of Bombay vs RMDC (1952).

Doctrine of Colourable Legislation : This Doctrine is also called Fraud on the Constitution, traces its origin to a Latin Maxim which, in this context, implies Whatever legislature cannot do directly, it cannot do indirectly.

The Doctrine of Colourable Legislation comes into play when a Legislature does not possess the power to make law upon a particular subject but nonetheless indirectly makes one. The doctrine is usually applied to Article 246.

Related Judgements - R.S Joshi vs Ajit Mills (1977).

Doctrine of Pleasure : The doctrine of pleasure has its origins in English law as per which, a civil servant holds office during the pleasure of the Crown. Under Article 155, the Governor of a State and Under Article 310, the civil servants hold office at the pleasure of the President.

Related Judgements - State of Bihar vs Abdul Majid (1954), and Union of India vs Tulsiram Patel (1965).

Doctrine of Harmonious Construction : The term harmonious construction refers to such construction by which harmony or oneness amongst various provisions of an enactment is arrived at. It originated through interpretations given by courts in a number of cases.

Related Judgements - CIT vs Hindustan Bulk Carriers (2003), Re-Kerala education bill 1951 case, East India hotels Ltd vs Union of India (2001).

Constituent Assembly

The idea of a constituent assembly was first proposed in 1934 by M.N. Roy. The actual constituent assembly was formed in 1946 on the basis of the Cabinet mission plan. The Constituent Assembly was recognised by Section 8(1) of the Indian Independence Act, 1947.

The Constituent Assembly was established with the goal of drafting a constitution for independent India, with S.N. Mukherjee served as its chief draftsman. Initially, Dr. Sachchidananda

Sinha acted as the temporary chairman, succeeded by Dr. Rajendra Prasad who was elected as President, and H.C. Mookerjee as Vice President.

BN Rau was appointed as the legal advisor to the Constituent Assembly.

The Elections to the Constituent Assembly were held in July 1946. The Constituent Assembly was to be a partly elected and partly nominated body. The members were to be indirectly elected by the members of the provincial assemblies, who themselves were elected on a limited franchise.

The Constitutional assembly had 389 members which included 93 from Princely States and 292 from British India and 4 were the chief commissioner of the provinces of Delhi, Ajmer - Merwara, Coorg, and British - Baluchistan. The seats allotted to the British Indian provinces and princely states were allotted in proportion to their respective population.

The first meeting of the Constituent Assembly took place on 9th December, 1946 at New Delhi in the Constitution Hall which is now known as the Central Hall of Parliament House. The Constituent Assembly existed for approximately three years, and the last session was held on 24th January 1950.

The Constituent Assembly formally began its task of framing the Constitution of India on the 13th of December, 1946 with Jawaharlal Nehru moving the Objectives Resolution which established the concept and guiding principles for building the Constitution and eventually took the shape of the Preamble to the Indian Constitution.

Important Sessions of constituent assembly : The Constituent Assembly held eleven sessions covering a total of 165 days. First Session : 9 - 23 December, 1946 , Second Session : 20 - 25 January, 1947 , Seventh Session : 4 Nov, 1948 - 8 Jan, 1949, Tenth Session : 6-17 October, 1949, Eleventh Session : 14-26 November, 1949.

Committees of the Constituent Assembly

The Committees of the Constituent Assembly were divided into 8 major and 14 minor committees.

Major Committees and their Chairman :

Union Power Committee (Jawaharlal Nehru), Rule of Procedure Committee (Dr Rajendra Prasad), Provincial Constitution Committee (Sardar Patel), Steering Committee (Dr Rajendra Prasad), Union Constituent Committee (Jawaharlal Nehru), State Committee (Jawaharlal Nehru), Drafting Committee (Dr. B. R. Ambedkar), Advisory Committee on Fundamental Rights, Minorities, Tribal and Excluded area (Sardar Patel was the head of this committee).

The Advisory committee has the following five committees : Fundamental Rights Sub - Committee (J.B. Kripalani), Minorities Sub - Committee (H.C. Mukherjee), North - East Frontier Tribal Areas & Assam Excluded & Partially Excluded Areas Sub-Committee (Gopinath Bardoloi), Excluded and Partially Excluded Areas (other than those in Assam) Sub-Committee - (A.V. Thakkar) and North-West Frontier Tribal Areas Sub-Committee.

Minor Committees and their Chairman :

Finance and Staff Committee (Dr. Rajendra Prasad), Credentials Committee (Alladi Krishnaswami Ayyar), House Committee (B. Pattabhi Sitaramayya), Order of Business Committee (Dr. K.M. Munshi), Ad-hoc Committee on the National Flag (Dr. Rajendra Prasad), Committee on the Functions of the Constituent Assembly (G.V. Mavalankar), Committee on Chief Commissioners' Provinces(B. Pattabhi Sitaramayya), Special Committee to Examine the Draft Constitution (Jawaharlal Nehru), Press Gallery Committee (Usha Nath Sen), Expert Committee on the Financial Provisions of the Union Constitution [Nalini Ranjan Sarkar (Not an Assembly Member)], Linguistic Provinces Commission [S.K. Dhar (Not an Assembly Member)], Ad-hoc Committee on the Supreme Court [S. Varadachari (Not an Assembly Member)], Ad-hoc Committee on Citizenship [S. Varadachari (Not an Assembly Member)], and Committee to Examine the Effect of the Indian Independence Act of 1947.

Drafting committee - The drafting committee was set up on **29th August 1947**, by the constituent assembly to prepare the draft of the text of the Constitution of India.

It consisted of **seven members** that included Alladi Krishnaswami Ayyar, N. Gopalaswami, B.R. Ambedkar (Chairman), K.M. Munshi, Mohammad

Saadulla, B.L. Mitter and D.P. Khaitan. N. Madhava Rau replaced B.L. Mitter who resigned due to ill-health and T. T. Krishnamachari replaced D.P. Khaitan who died in 1948.

Female in constituent assembly -The Constituent Assembly that framed the Indian Constitution included 15 women: Durgabai Deshmukh (Madras), the only female chairperson; Hansa Mehta (Mumbai); and Begum Aizaz Rasul (United Provinces), only Muslim woman in the assembly. Joining them were Ammu Swaminathan, Annie Mascarene, Dakshayani Velayudhan, Kamla Chaudhary, Leela Roy, Malati Choudhury, Purnima Banerjee, Rajkumari Amrit Kaur, Renuka Ray, Sarojini Naidu, Sucheta Kriplani, and Vijalakshami Pandit.

Interim Government

Interim Government - A government that is meant to last only a limited period before a permanent government is elected. The Interim Government of India, also known as the Provisional Government, was created on 2 September 1946, by the newly elected Constituent Assembly, which lasted until 15 August 1947.

The Viceroy's Executive Council transitioned into the Interim Government's Council of Ministers, with Jawaharlal Nehru as its de facto Prime Minister. The Muslim League also participated, with Liaquat Ali Khan leading the Finance Department.

Members of the Interim Government of 1946 and their Portfolios : Viscount Wavell (till February 1947), and Lord Mountbatten (from February 1947) - President of the Executive Council (Viceroy and Governor-General of India), Sir Claude Auchinleck - Commander-in-Chief, Jawaharlal Nehru (Indian National Congress, INC) - Vice President and in charge of External Affairs and Commonwealth Relations, Sardar Vallabhbhai Patel (INC) - Home Affairs and Information and Broadcasting, Rajendra Prasad (INC) - Agriculture and Food, Ibrahim Ismail Chundrigar (Muslim League, ML) - Commerce, Baldev Singh (INC) - Defence, Liaquat Ali Khan (ML) - Finance, C. Rajagopalachari (INC) - Education and Arts, Ghazanfar Ali Khan (ML) - Health, Jagjivan Ram (INC) - Labour, Jogendra Nath Mandal (ML) - Law, Asaf Ali - Railways and Transport, Abdur Rab Nishtar (Muslim League, ML) -

Post and Air, C. H. Bhabha (INC) - Works, Mines and Power, John Mathai - Industries and Supplies.

Independent India's first cabinet ministers

India became independent in 1947, the country's first Prime Minister Jawaharlal Nehru formed a 15-member cabinet.

First cabinet ministers and their Portfolios : Jawaharlal Nehru (first Prime Minister, Minister of Commonwealth Relations, Scientific Research and External Affairs), Sardar Vallabhai Patel (Home affairs and Information and Broadcasting, and States), Sardar Baldev Singh (Defence), R. Shanmugam Chetty (Finance), Maulana Abul Kalam Azad (Education), B.R. Ambedkar (Law), Dr. Rajendra Prasad (Food and Agriculture), Rajkumari Amrit Kaur (first female cabinet minister, Health), Dr. Shyama Prasad Mukherjee (Industry and Supply), John Mathai (Railway), Rafi Ahmad Kidwai (Communications and Information), V.N. Gadgil (Works, Mines and Energy Ministry), Homi Jahangir bhabha (Commerce). N. Gopalaswami Ayyangar and Mohanlal Saxena were the Ministers without portfolio.

Important Facts

The Constituent Assembly ratified India's membership of the Commonwealth in May 1949.

National Anthem : The first stanza of the song Bharoto Bhagyo Bidhata was adopted in its Hindi version by the Constituent Assembly of India as the National Anthem on 24 January 1950 which was composed originally in Bangla by Rabindranath Tagore. Captain Abid Ali translated and Captain Ram Singh composed it in Hindi.

It was first publicly sung on 27 December 1911 at the Calcutta Session of the Indian National Congress. The playing time of the national anthem is approximately **52 seconds**.

National Song: On 24 Jan 1950, the Indian Constituent Assembly adopted Vande Mataram as the National Song.

National Flag : The design of the National Flag was adopted by the Constituent Assembly of India on 22 July 1947.

National Emblem : It was adopted on 26th January 1950.

Preamble and Objective Resolution

The Constitution of India begins with a Preamble which contains the ideals, objectives and basic principles of the Constitution. The Preamble is called the Introduction Letter of the Indian Constitution and widely recognised as the Soul of the Constitution. It clearly states that the source of power of the constitution is the people of India.

The ideals behind the Preamble to India's Constitution were laid down by Jawaharlal Nehru's Objectives Resolution, on 13th December 1946 indicating the objectives of the constitution. The objective resolution proclaimed India to be an Independent Sovereign Republic, guaranteeing its citizens justice, equality and freedom. It was adopted by the Constituent Assembly on January 22, 1947.

The preamble was amended once by the 42nd Constitutional Amendment Act in 1976, in which three new words Socialist, Secular, and Integrity were added. Although not enforceable in court, the Preamble states the objectives of the Constitution, and acts as an aid during the interpretation of Articles when language is found ambiguous.

Preamble of the Indian Constitution

We, the People of India, having solemnly resolved to constitute India into a Sovereign Socialist Secular Democratic Republic and to secure to all its citizens : JUSTICE, social, economic and political; LIBERTY of thought, expression, belief, faith and worship; EQUALITY of status and of opportunity and to promote among them all FRATERNITY assuring the dignity of the individual and the unity and integrity of the Nation; In Our Constituent Assembly this **26th day of November, 1949**, do hereby Adopt, Enact and Give to Ourselves this Constitution.

Keywords in the Preamble

Sovereign : The Preamble proclaims that India is a Sovereign State. Sovereign means that India has its own independent authority and it is not a dominion or dependent state of any other external power.

Socialist : India has adopted Democratic Socialism, which holds the faith in a

mixed economy where both private and public sectors co-exist side by side.

Secular : India embodies the positive concept of secularism, that is all the religions in India get equal respect, protection and support from the state.

Democratic : The term Democratic indicates that the Constitution has established a form of government that gets its authority from the will of the people expressed in an election. The Indian Constitution provides for a representative parliamentary democracy under which the executive is responsible to the legislature for all its policies and actions.

Republic : In a Republic, the head of the state is elected by the people directly or indirectly. In India, the President is the head of the state. Moreover, in a republic country, political sovereignty is vested in the people rather than a monarch.

Justice : The term Justice in the Preamble embraces three distinct forms : Social, economic and political, secured through various provisions of the Fundamental and Directive Principles. Social Justice in the Preamble means equal treatment of all citizens without any social distinction. Economic Justice means non-discrimination between people on the basis of economic factors. Political Justice means that all citizens have equal rights in political participation.

Liberty : Liberty implies the absence of restraints or domination on the activities of an individual such as freedom from slavery, serfdom, imprisonment, despotism, etc. The liberty conceived by the Preamble or Fundamental Rights is not absolute but qualified.

Equality : Equality means the absence of privileges or discrimination against any section of the society.

Fraternity : Fraternity means the feeling of brotherhood. The Preamble seeks to promote fraternity among the people assuring the dignity of the individual and the unity and integrity of the nation.

Sobriquets given to the Preamble

Kanhaiyalal Maniklal Munshi described the preamble of Indian Constitution as the **Political horoscope** of the Indian Constitution because it defines the

establishment of governance in the country.

Ernest Barker said that the preamble is "The **key note** of the constitution".

Thakurdas Bhargav called Preamble as the **Soul** of the Constitution.

N.A. Palkhiwala called it the **Identity card** of the constitution.

Sources of Indian Constitution

The Constituent Assembly of India drafted the constitution considering features of constitutions of the other countries as well as from the Government of India Act 1935.

Government of India Act 1935 - Federal Court, Emergency Provisions, Public Service Commissions, Office of Governor, Division of Powers, Judiciary, Administrative Details.

United Kingdom - Parliamentary government, Rule of Law, Legislative procedure, Single Citizenship, Cabinet system, Prerogative writs, Parliamentary privileges, Bicameralism, Law making procedure, Institution of Speaker and his role, Political Part of the Indian constitution, and Equality before the law.

Australia - Concurrent list, Freedom of trade, commerce and intercourse, Joint-sitting of the two houses of Parliament, Language of Preamble.

Ireland - Directive Principles of State Policy, Nomination of members to Rajya Sabha, Method of election of the President.

United States - Preamble, Fundamental Rights, Independence of Judiciary, Judicial review, Impeachment of the President, Removal of Supreme Court and High Court judges, Post of Vice-President.

France - Republic, Ideals of liberty, equality and fraternity in the Preamble.

Germany - Suspension of Fundamental Rights during emergency.

Soviet Union (USSR) - Fundamental Duties, Ideals of justice (social, economic and political).

Canada - Federation with a strong Centre, Vesting of residuary powers in the Centre, Appointment of state governors

by the Centre, Advisory jurisdiction of the Supreme Court.

South Africa - Amendment in the Indian Constitution, Election of members of Rajya Sabha.

Japan - Procedure Established by law.

Formation of States and Union Territories

Part 1 of the Constitution deals with The Union and its Territory (Article 1 - 4).

Article 1 - It states that India, that is Bharat, shall be a Union of States. It provides for a governance structure primarily federal in nature.

The term '**Union of India**' includes only the States that are members of the federal system and share the powers with the Union. According to Dr. B.R. Ambedkar, the phrase 'Union of States' has been preferred to 'Federation of States' for two reasons: first, the Indian Federation is not the result of an agreement among the states like the American Federation; and second, the states have no right to secede from the federation. The word federalism is not mentioned in the Indian Constitution.

The Parliament of India holds the sole authority for creating new states and altering existing ones.

Article 1(2) - The States and the territories thereof shall be as specified in the First Schedule.

Article 1(3) - States that the territory of India will comprise the following -

- (i) The territories of the States;
- (ii) The Union territories mentioned in the First Schedule;
- (iii) Such other territories that may be acquired.

Article 2 - It deals with admission or establishment of new States.

Article 3 - It deals with formation of new States, Alteration of areas, boundaries or names of existing States.

Article 4 - It deals with laws made under Articles 2 and 3 to provide for the amendment of the First and the Fourth Schedules and supplemental, incidental and consequential matters.

The process involves passing a law through the Parliament, requiring a simple majority vote.

Reorganization of States

In 1950, the Constitution contained a four-fold classification of the states and territories of the Indian Union - Part A, Part B, Part C, and Part D.

Part A - All British Provinces (Governor province of British India) - Assam, Bihar, Odisha, Madras, Bombay, Madhya Pradesh, Punjab, United Provinces, West Bengal.

Part B - These were former princely states or groups of princely states. Hyderabad, Jammu and Kashmir, Madhya Bharat, Mysore, Patiala and East Punjab States Union (PEPSU), Rajasthan, Saurashtra and Travancore-Cochin.

Part C - This part contains provinces of Chief commissioners and some princely states. These states are listed as follows: Ajmer, Bhopal, Vindhya Pradesh, Tripura, Himachal Pradesh, Bilaspur, Coorg state, Delhi, Kutch, Manipur.

Part D - This state was administered by a lieutenant governor appointed by the central government. It consists of Andaman and Nicobar Islands only.

State Reorganisation Commissions (SRC) : There are some important commissions constituted for the Reorganisation of the State such as the Dhar Commission, JVP Committee and Fazal Ali Commission.

Dhar Commission - It was constituted in June 1948 and headed by S.K. Dhar. It recommended that states should be organized on the basis of administrative convenience.

JVP Committee - It was set up in December 1948 and submitted its report in April 1949. It formally rejected language as the basis for the reorganization of states.

Fazal Ali Commission (State Reorganisation Commission) - It was set up in December 1953. Its members were Fazal Ali (chairman), K.M. Panikkar and H.N. Kunzru. It submitted its report in September 1955. It recommended that factors like language, finances, administration, the welfare of people, and unity of the nation should be considered in state reorganization.

States Reorganisation Act 1956

The Parliament passed the State Reorganisation Act in November 1956 on the recommendation of the Fazl Ali Commission. The 7th Constitutional Amendment Act, 1956 was passed to replace the four types of states, known as Part A, B, C, and D.

It made extensive changes in the boundaries of various states for the purpose of meeting the linguistic, regional and local demands. It created 14 states and 6 union territories. The States were: Andhra Pradesh, Assam, Bihar, Bombay, Jammu and Kashmir, Kerala, Madhya Pradesh, Madras, Mysore, Orissa, Punjab, Rajasthan, Uttar Pradesh and West Bengal. The Union Territories were: Andaman and Nicobar Islands, Delhi, Himachal Pradesh, Laccadive, Minicoy and Amindivi Islands, Manipur and Tripura.

States Created before 1956

Jammu and Kashmir (1948) - The king of this state signed the "Instrument of Accession" to India and became part of the Indian Union. Jammu and Kashmir completed the process of merging with the Indian Union earlier than 1956, with significant steps taken in 1947 and subsequent years, culminating in the abrogation of Article 370 in 2019.

Rajasthan (30 March 1949) - Princely states of the former Rajputana Agency were merged to create the state of Rajasthan.

Madhya Pradesh (1 November 1950) - It achieved statehood in 1950. It was part of the Central Provinces and Berar, Princely State of Makrai and the princely states of the former Eastern States Agency.

Assam - It achieved statehood in 1950.

West Bengal (1950) - It achieved statehood in 1950. It was part of Bengal Province, British India.

Uttar Pradesh (24 January 1950) - It achieved statehood as UP in 1950.

Odisha (1950) - It achieved statehood in 1950, renamed Odisha in November, 2011.

Andhra Pradesh (1953) - It came into existence on October 1, 1953, by separating from the Telugu-speaking northern districts of Madras State. Potti Sreeramulu was a leader who demanded Andhra Pradesh on Linguistic basis.

States Created After 1956

Karnataka (1 November 1956) - It was formed by the States Reorganisation Act, 1956 as Mysore State, renamed as Karnataka in 1973.

Kerala (1 November 1956) - Formed by the States Reorganisation Act, 1956. It was part of Madras State and Travancore-Cochin.

Tamil Nadu (1 November 1956) - It was formed by the States Reorganisation Act, 1956 as Madras State, renamed Tamil Nadu in 1969. It was part of Madras State and Travancore-Cochin.

Gujarat and Maharashtra (1 May 1960) - The State of Bombay was divided into two States i.e. Maharashtra and Gujarat by the Bombay (Re-organisation) Act, 1960.

Nagaland (1 December 1963) - It was carved out from Assam by the State of Nagaland Act, 1962.

Punjab and Haryana - The Punjab Reorganisation Act, 1966, led to the establishment of Punjab and Haryana as separate states on November 1, 1966.

Himachal Pradesh (1971) - The union Territory of Himachal Pradesh achieved statehood in 1971 by the State of Himachal Pradesh Act 1970.

Manipur (21 January 1972) - It was made a Union Territory in 1956, and a fully-fledged State in 1972 by the North-Eastern Areas (Reorganization) Act, 1971.

Meghalaya (21 January 1972) - It received the status of full-fledged state by the North - Eastern Areas (Reorganization) Act, 1971. It was carved out from Assam.

Tripura (21 January 1972) - It achieved statehood by the North-Eastern Areas (Reorganization) Act, 1971.

Sikkim (16 May 1975) - It got the status of full state by the 36th Constitutional Amendment Act, 1975. 16 May is celebrated as Sikkim Day.

Mizoram (20 February 1987) - It was created as a Union Territory by the North-Eastern Areas (Reorganization) Act, 1971. Converted to a state by the State of Mizoram Act, 1986.

Goa (30 May 1987) - Goa was separated

from the Union territory of Goa, Daman and Diu and made a full-fledged State by Goa, Daman and Diu Reorganisation Act, 1987. But Daman and Diu remained as Union territory.

Arunachal Pradesh (20 February 1987) - Until 1972, it was known as the North-East Frontier Agency (NEFA). It gained the Union Territory status on January 20, 1972 and was renamed as Arunachal Pradesh. It became a full-fledged State on February 20, 1987.

Chhattisgarh (1 November 2000) - It was formed by the Madhya Pradesh Reorganisation Act, 2000. It was Part of Madhya Pradesh. 1st November is celebrated in Chhattisgarh formation day.

Jharkhand (15 November 2000) - It was formed by the Bihar Reorganisation Act, 2000. It was part of Bihar. Jaipal Singh Munda (captain of Indian Hockey Team) demanded separate state Jharkhand from Bihar.

Uttarakhand (9 November 2000) - It was formed by the Uttar Pradesh Reorganisation Act, 2000 as Uttaranchal, renamed Uttarakhand in 2007. Earlier it was Part of Uttar Pradesh. 9 November is celebrated as Uttarakhand Diwas.

Telangana was established as a separate state on June 2, 2014, following the Andhra Pradesh Reorganisation Act, 2014. Previously, the region was part of Andhra Pradesh.

Union Territory

Article 239 deals with the administration of Union territories. They are also known as centrally administered territories.

The President of India appoints Lieutenant Governors (LGs) to administer the Union Territories (UTs) under the provisions of the Indian Constitution.

There are two types of Union Territories in India, **with Legislature** - Delhi, Puducherry, and Jammu and Kashmir, and **without Legislature** - Andaman and Nicobar, Chandigarh, Dadra and Nagar Haveli, and Daman and Diu, Ladakh and Lakshadweep.

There is no separate representation in the Rajya Sabha (Upper House) except for Delhi, Puducherry, and Jammu and Kashmir.

Union Territories, their Capitals and their date of establishment

Lakshadweep (Kavaratti) - Formed on 1st November 1956. Lakshadweep is India's smallest Union Territory.

Delhi (New Delhi) - formed on November 1, 1956, under the States Reorganisation Act.

Andaman and Nicobar Islands (Port Blair) - formed on 1 November 1956.

Puducherry (Puducherry) - formed on 16 August 1962.

Chandigarh (Chandigarh) - formed on 1 November 1966

Ladakh {Leh (Summer) Kargil (Winter)} - 31 October 2019. It was established following the passage of the Jammu and Kashmir Reorganization Act, 2019. It is the largest (area wise) and the second least populous union territory of India.

Jammu and Kashmir {Sri Nagar (Summer), Jammu (Winter)} - 31 October 2019

Dadra and Nagar Haveli, and Daman and Diu (Daman) - 26 January 2020.

National Capital Territory

The National Capital Territory of Delhi (NCT). It is a special union territory of India jointly administered by the Central government, the NCT elected government and three municipal corporations.

Government of National Capital of Delhi (GNCTD) Act, 1991 - It is an Act to supplement the provisions of the Constitution relating to the Legislative Assembly and a Council of Ministers for the National Capital Territory of Delhi. The first election in the union territory was held in 1993 in which the first elected Chief Minister of the national capital territory of Delhi was Madan Lal Khurana.

Citizenship

Citizenship is the status of a person recognized under the custom or law of a sovereign state. Citizenship recognized by the state confers civil, political and social rights on a person.

There are two types of Citizenship :

Single Citizenship: In this, the people carry citizenship of the country as whole rather than the states separately.

Dual Citizenship: In this, the people carry citizenship in one as a citizen of a country and other as citizen of states of that country. In India, there is no concept of dual state citizenship.

Part II (Articles 5 to 11) of the Indian Constitution outlined citizenship rules. The Indian constitution provides for single citizenship.

According to the Constitution, the four categories of persons became the citizens of India at its commencement i.e., on January 26, 1950. The Citizenship Act of 1955 prescribes five ways of acquiring citizenship, via, birth, descent, registration, naturalization and incorporation of territory.

Conversely, citizenship can be lost through renunciation, whereby individuals voluntarily relinquish their Indian citizenship. Additionally, citizenship may be deprived if obtained through fraudulent means or false representation. Termination of citizenship is also possible under certain circumstances, such as acquiring citizenship of another country.

The Citizenship Act of 1955 along with its amendments provides specifications for citizenship by registration and citizenship by naturalisation.

Related Articles

Article 5 - Citizenship initially (November 26, 1949), for those who are born in India or whose parents are born in India.

Article 6 - Rights of Citizenship of Certain Persons who have Migrated to Indian Territory (before July 19th, 1948) from Territory of Pakistan.

Article 7 - Rights of people who had migrated to Pakistan after March 1, 1947, but subsequently returned to India.

Article 8 - Rights of Citizenship of Persons of Indian origin residing outside India.

Article 9 - Persons voluntarily acquiring citizenship of a foreign State not to be citizens.

Article 10 - Continuance of the rights of citizenship.

Article 11 - Parliament to regulate the right of citizenship by law.

Normal Residents of the country: They are individuals or institutions who typically reside in the country and have their center of economic interest there. They encompass citizens residing in the country, citizens of other nations residing in the country for over one year with economic ties, and citizens of the country employed in international organizations or embassies abroad.

Ordinary Residents : Ordinary residents of India typically include Indians working in foreign embassies, India's ambassadors abroad, and foreign nationals residing in India for over one year, excluding those here for studies or medical treatment.

Person of Indian Origin (PIO): A person who or whose ancestors was an Indian national and who is presently holding another country's citizenship/ nationality. It was merged with OCI in 2015.

Overseas citizenship of India (OCI) : It is a form of permanent residency available to people of Indian origin and their spouses which allows them to live and work in India indefinitely. OCI status is not citizenship and does not grant the right to vote in Indian elections or hold constitutional office.

Non Resident of India (NRI) : A person who stayed away from India for more than 182 days in a calendar year is known as NRI.

National Register of Citizens: It is an official record of those who are legal Indian citizens. It includes demographic information about all those individuals who qualify as citizens of India as per the Citizenship Act, 1955.

The 2003 Amendment in the NRC mandated the Central Government to create and maintain a National Register of Citizens and to issue national identity cards to all the registered citizens.

National Population Register: It is a database containing a list of all usual residents of the country. A usual resident for the purposes of NPR is a person who has resided in a place for six months or more and intends to reside there for another six months or more.

The Citizenship (Amendment) Act, 2019

The act was introduced to amend the Citizenship Act, 1955. By providing a

pathway to Indian citizenship for religious minorities from Afghanistan, Bangladesh and Pakistan who arrived in India before the end of December 2014. The act sought to make illegal migrants belonging to **six religions** (Hindus, Sikhs, Buddhists, Jains, Parsis and Christians) eligible for citizenship.

Fundamental Rights

Fundamental Rights (FR) are enshrined in Part III of the Constitution (Article 12-35). This Part is popularly described as the Magna Carta of India. Magna Carta is the 'Charter of Rights' issued by King John of England in 1215. It was the first written document related to the fundamental rights of citizens.

FR provides necessary conditions for moral and material protection. They lay down the foundation for social justice.

The Constitution of India provides for 6 Fundamental Rights. These are, (1) Right to equality (Articles 14-18), (2) Right to freedom (Articles 19-22), (3) Right against exploitation (Articles 23-24), (4) Right to freedom of religion (Article 25-28), (5) Cultural and educational rights (Articles 29-30) and (6) Right to constitutional remedies (Article 32). B.R. Ambedkar termed Article 32 as the '**heart and soul**' of the Constitution of India'.

Originally, the constitution also included Right to Property (Article 31) making a total of 7 FR, however, it was deleted from the list of FRs by the 44th Amendment Act, 1978 and has been made a legal right under **Article 300-A** (Right to Property) in Part XII of the Constitution.

Related Articles

Article 12 - Defines the State (Legislative and Executive Organs of the Union Government and State government, All local authorities and Statutory and Non-Statutory Authorities).

Article 13 (Judicial Review) - It declares that all laws that are inconsistent with or in derogation of any of the fundamental rights shall be void.

Right to Equality (Art 14-18)

Article 14 - Equality Before Law and equal protection of law. The concept of 'equality before law' is an element of the concept of 'Rule of Law', propounded by A.V. Dicey,

Article 15 - Prohibition of discrimination on grounds of religion, race, caste, sex or place of birth.

Article 15(1) - The State shall not discriminate against any citizen on grounds only of religion, race, caste, sex, place of birth or any of them.

Article 15(2) - No citizen, on grounds only of religion, race, caste, sex, place of birth or any of them, be subject to any restriction to public services.

Article 15(3) - Nothing in Article 15 shall prevent the State from making any special provision for women and children.

Article 15(4) - Nothing in Article 15 or in clause (2) of Article 29 shall prevent the State from making any special provision for the advancement of any socially and educationally backward classes of citizens or for the Scheduled Castes and the Scheduled Tribes.

Article 16 - Equality of Opportunity in matters of Public Employment.

Article 16(1) - There shall be equality of opportunity for all citizens in matters relating to employment or appointment to any office under the State.

Article 16(2) - No citizen shall, on grounds only of religion, race, caste, sex, descent, place of birth, residence or any of them, be ineligible for, or discriminated against in respect or, any employment or office under the State.

Article 16(3) - It empowers the union parliament to make any law in the matter of public employment for a Particular class in their respective jurisdictions.

Article 16(4) - It states that the state can make a provision of reservation for backward castes if they are not adequately represented.

Article 17 - Abolition of Untouchability.

Article 18 - Abolition of Titles, except for military and academic.

Right to Freedom (Art 19-22)

Article 19 - Protection of certain rights regarding freedom of speech, etc. It gives the protection of 6 Rights.

Article 19(a) - Right to freedom of speech and expression [In 2020, The Supreme Court of India stated that access to the internet is a fundamental right].

Article 19(b) - Right to assemble peaceably and without arms.

Article 19(c) - Right to form associations or unions or co-operative societies.

Article 19(d) - Right to move freely throughout the territory of India.

Article 19(e) - Right to reside and settle in any part of the territory of India.

Article 19(f) - Right to property (**Omitted**)

Article 19(g) - Right to practise any profession or to carry on any occupation, trade, or business.

Article 20 - Right to Protection in Respect of Conviction for Offences.

Article 20 (1) - No person shall be (i) convicted of any offence except for violation of a law in force at the time of the commission of the act, nor (ii) subjected to a penalty greater than that prescribed by the law in force at the time of the commission of the act.

Article 20 (2) - no person shall be prosecuted or punished for the same offence more than once (Double jeopardy).

Article 20 (3) - no person accused of any offence shall be compelled to be a witness against himself.

Article 21 - Protection of life and personal liberty. No person shall be deprived of his life or personal liberty except according to procedure established by law.

Some Rights declared as FR under Article 21 are - Right to privacy, Right to choose one's life partner, Right to Health, Right to sleep, and Right to freedom from noise pollution.

Article 21A - Provide free and compulsory education to all children of the age of six to fourteen years. It was added by the 86th Amendment Act.

Article 22 - Protection Against Arrest and Detention i.e. to consult a legal practitioner of a person's choice.

Right Against Exploitation (Art 23-24)

Article 23 - Prohibition of Human Trafficking and Forced Labour.

Article 24 - Prohibition of Child Labour i.e. no child below 14 years of age shall be employed to work in any factory or hazardous work.

Right to Freedom of Religion (Hallmark of Democracy) (Art 25-28)

Article 25 - Freedom of conscience and

free profession, practice, and propagation of religion.

Article 26 - Freedom to Manage Religious Affairs.

Article 27 - Freedom as to payment of taxes for promotion of any particular religion.

Article 28 - Freedom as to attendance at religious instruction or religious worship in certain educational institutions.

Educational and Cultural Rights (Art 29-30)

Article 29 - Protection of Interests of Minorities.

Article 30 - Right of linguistic and religious Minorities to Establish, Conserve and Administer their Educational Institutions.

Right to Constitutional Remedies

Article 32 - Right to remedies for the enforcement of the fundamental rights using five writs. A Writ is a formal written order issued by the executive or judicial body that directs the individual or authority to do or refrain from doing a particular act.

5 writs issued by the Supreme court under Article 32 and by High courts under Article 226. If the fundamental right of a citizen of India is violated, he/she has the choice of either approaching the Supreme court directly or the High courts. These writs are :

Habeas Corpus - It is a Latin term which means "**to have the body of**". It is issued against forcible detention of a person and to release him if such detention is found illegal.

Mandamus - It means '**we command**', issued by the court to a public official asking him to perform the official duties that he has failed or refused to perform. It can also be issued against any public body, a corporation, an inferior court, a tribunal, or a government for the same purpose.

Prohibition - It means '**to forbid**'. A higher court issues a writ of prohibition against the lower court to prevent it from exceeding its jurisdiction or usurping a jurisdiction that it does not possess.

Certiorari - It means 'to be certified' or 'to be informed', issued by a higher court to a lower court or tribunal either to transfer

a case pending with the latter to itself or to squash the order of the latter in a case.

Quo-Warranto - It means 'by what authority or warrants'. Issued by the court to enquire into the legality of the claim of a person to a public office.

Article 33 - Empowers the Parliament to restrict or abrogate the fundamental rights of the 'Members of the Armed Forces, paramilitary forces, police forces, intelligence agencies, and analogous forces'.

Article 34 - Provides for the restrictions on fundamental rights while martial law (military rule) is in force. Martial rule refers to a situation where civil administration is run by the military authorities according to their own rules and regulations framed outside the ordinary law.

Article 35 - Empowers the Parliament to make laws on specified Fundamental Rights.

The Fundamental Rights guaranteed by Articles 15, 16, 19, 29, and 30 are available **only to citizens** of India and the rest of the articles are available to all persons (except enemy aliens).

Suspension of Fundamental Rights

FRs can be suspended during emergencies declared by the President. Six freedoms guaranteed under Article 19 are automatically suspended. Other rights can be suspended by presidential order, but require parliamentary approval. Articles 20 & 21 can never be suspended.

Article 359 - This article authorises the President of India to suspend the right to move any court for the enforcement of Fundamental Rights during National Emergency except Article 20 and Article 21.

The 44th Amendment restricted suspensions by ensuring judicial review of laws not directly related to the emergency, and permanently protecting Articles 20 and 21.

Exceptions to Fundamental Rights :

1. Article 31A saves five categories of laws from being challenged and

invalidated on the ground of contravention of the fundamental rights conferred by Article 14 and Article 19. Article 31A does not immunise a State law from judicial review unless it has been reserved for the consideration of the President and has received his assent.

2. Article 31B immunises any law included in the Ninth Schedule. However, in I.R. Coelho case (2007), the Supreme Court ruled that there could not be any blanket immunity from judicial review of laws included in the Ninth Schedule.

3. Article 31C, as inserted by the 25th Amendment Act of 1971, states that No law that seeks to implement the socialistic directive principles specified in Article 39(b) or (c) shall be void on the ground of contravention of the fundamental rights conferred by Article 14 or Article 19.

Directive Principles of State Policy (DPSP)

Part IV of the Constitution of India (Article 36-51) contains the DPSP. The origin of DPSP can be traced to the Karachi resolution. The DPSP resembles the 'Instrument of Instructions' enumerated in the Government of India Act of 1935. They are non Justiciable (not enforceable by law) in nature.

Article 36 - Defines the "state".

Article 37 - The provisions contained in this Part shall not be enforceable by any court, but it shall be the duty of the State to apply these principles in making laws.

Classification of Principles : The Directive Principles are classified on the basis of their ideological source and objectives. These are Directives based on : Socialist Principles, Gandhian Principles, Liberal and Intellectual Principles.

Directives based on Socialist Principles

Article 38 - The State should ensure social, economic and political justice and by minimising inequalities in income, status, facilities and opportunities.

Articles 39 - To secure (a) the right to adequate means of livelihood (b) the equitable distribution of material resources (c) prevention of concentration of wealth (d) equal pay for equal work (e) preservation of the health of workers and

children against abuse (f) opportunities for healthy development of children.

Article 39A - To promote equal justice and to provide free legal aid to the poor.

Article 41 - To secure the Right to work, to education and to public assistance in cases of unemployment, old age, sickness and disability.

Article 42 - To make provisions for securing just and humane conditions of work and for maternity relief.

Article 43 - Secure to all workers a living wage and a decent standard of life.

Article 43A - Participation of workers in the management of industries.

Article 47 - To raise the level of nutrition and the standard of living of people and to improve public health.

Directives based on Gandhian Principles:

Article 40 - Organisation of village panchayats.

Article 43 - Promote cottage industries.

Article 43B - Promotion of cooperative societies. Added by the 97th Amendment Act 2011.

Article 46 - Promotes educational and economic interests of the weaker sections like Scheduled Castes (SCs), Scheduled Tribes (STs) and other weaker sections.

Article 47 - To prohibit the consumption of intoxicating drinks and drugs.

Directives based on Liberal Intellectual Principles

Article 44 - Uniform Civil Code.

Article 45 - Early childhood care and education for children until the age of 6 years.

Article 48 - To organise agriculture and animal husbandry on modern and scientific lines.

Article 48A - To protect and improve the environment and to safeguard forests and wildlife.

Article 49 - Protection of monuments and places and objects which have national importance.

Article 50 - Judiciary should be separate from the Executive.

Article 51 - Establishment of international peace and security.

Amendments in DPSP

42nd Constitutional Amendment, 1976:

It changed the subject matter of Article 39 [added 'opportunities for healthy development of children' to Article 39] and added three new DPSP - Article 39A, Article 43A, Article 48A to the Indian Constitution.

44th Constitutional Amendment, 1978 :

It inserted Section-2 to Article 38 (minimise economic inequalities in income and eliminate inequalities in status, facilities and opportunities not amongst individuals but also amongst groups).

86th Amendment Act of 2002 :

It changed the subject-matter of Article 45 and made elementary education a fundamental right under Article 21A.

Fundamental Rights v/s DPSP

FR	DPSP
Negative as they pose limitation on State	positive as they require the State to do certain things
Promote political democracy	Promote social and economic democracy
justiciable	Non-justiciable
Have legal sanction	Have moral and political sanction
Promote welfare of Individual	Promote welfare of community

Conflicts Between Fundamental Rights and DPSP

The justiciability of FR on one hand, and the moral obligation of the state to implement DPSP on the other hand led to the conflict between the two. The Supreme Court (SC) in *Champakam Dorairajan v the State of Madras* (1951) established the supremacy of FR over DPSP. In *Golaknath v the State of Punjab* (1967), the SC held that FR could not be amended by the Parliament even for the implementation of DPSP.

The SC overruled its judgement of the *Golaknath* case in *Kesavananda Bharati v the State of Kerala* (1973). It held that Parliament can amend any part of the Constitution except "Basic Structure" of the Constitution. The SC reiterated its stand in *Minerva Mills v the Union of India* (1980).

Fundamental Duties

Fundamental Duties are eleven legal obligations prescribed for all Indian citizens, set out in Part IV-A (Article 51A) of the Indian Constitution. They were added by the 42nd Constitutional Amendment Act, 1976. This amendment came into effect on January 3rd, 1977.

They are statutory and not enforceable by law but are taken into account by the courts while adjudicating any matter. It applies only to citizens and does not extend to foreigners.

Sardar Swaran Singh committee: The fundamental duties have been adopted on its recommendation. It had 12 members. It was set up by the Congress Party (Indira Gandhi) in 1976. It proposed 8 duties that had to be included in a separate section (Part- IVA) in the Indian Constitution.

Some of its recommendations were not included in the duties like the duty to pay taxes and punishment for violation of duties prescribed by the Legislature.

The 11th Fundamental Duty was added to the Constitution by the 86th Constitutional Amendment Act.

List of Fundamental Duties

- (a) To abide by the Constitution and respect its ideals and institutions, the National Flag and the National Anthem.
- (b) To cherish and follow the noble ideals which inspired our national struggle for freedom.
- (c) To uphold and protect the sovereignty, unity and integrity of India.
- (d) To defend the country and render national service when called upon to do so.
- (e) To promote harmony and the spirit of common brotherhood amongst all people of India transcending religious, linguistic and regional or sectional diversities and to renounce practices derogatory to the dignity of women.
- (f) To value and preserve the rich heritage of our composite culture.
- (g) To protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures.

(h) To develop the scientific temper, humanism and the spirit of inquiry and reform.

(i) To safeguard public property and to abjure violence.

(j) To strive towards excellence in all spheres of individual and collective activity, so that the nation constantly rises to higher levels of endeavor and achievement.

(k) To provide opportunities for education to children between the age of 6 and 14.

President

The President is the Executive head of India. All executive actions of the Government of India shall be expressed to be taken in the name of the President. He is the Supreme Commander of the army forces. He acts as the symbol of unity, integrity, and solidarity of the nation. He is the first citizen of India. The President of India is the Visitor to all central universities of India.

In the Constitution, Part V (The Union), Chapter I (The Executive), Articles 52 to 78 of the Indian Constitution deals with the Union Executive. The President is not a member of any house but an integral part of the Parliament because without the President's assent, a bill that has been passed by both houses of Parliament cannot be enacted into law.

Articles related to President

Article 52 - There shall be a President of India.

Article 53 - The executive power of the Union shall be vested in the President.

Article 54 - Election of President.

Article 55 - Manner of election of President.

Article 56 - Term of office of President

Article 57 - Eligibility for re-election

Article 58 - Qualifications for election of President.

Article 59 - Conditions of the President's office.

Article 59 (1) - The President shall not be a member either of Parliament or of the Legislature of any State.

Article 60 - Oath or affirmation by the President.

Article 61 - Procedure for impeachment

of the President

Article 62 - Time of holding election to fill vacancy in the office of President and the term of office of person elected to fill casual vacancy.

Article 70 - Discharge of President's functions in other contingencies.

Article 71 - Matters relating to or connected with the election of a president or Vice President.

Article 72 - Power of President to grant pardons, etc and to suspend, remit or commute sentences in certain cases.

Article 73 - Extent of executive power of the Union.

Article 86 - Right of President to address and send messages to the House.

Article 87 - Special address by the President.

Article 123 - Power of President to promulgate Ordinances during recess of Parliament.

Article 143 - Power of the President to consult the Supreme Court.

Election of President

The President is chosen by an electoral college consisting of the elected members of both the houses of Parliament (M.P.s), the elected members of the State Legislative Assemblies (Vidhan Sabha) of all States and the elected members of the legislative assemblies (MLAs) of two union territories, i.e National Capital Territory (NCT) of Delhi and Union Territory of Puducherry.

The President election is held on the principle of a Single Transferable Vote by way of Proportional Representation. It is a completely secret ballot and no whip can be issued.

The value of the vote of 1 MLA

$$= \frac{\text{Total population of state}}{\text{total no. of elected MLAs}} \times \frac{1}{1000}$$

The value of the vote of 1 MP =

$$\frac{\text{Total value of votes of all MLAs of all states}}{\text{total no. of elected MPs}}$$

Formula -
$$\frac{\text{Total no. of valid votes}}{\text{No. of seat} + 1} + 1$$

The members that are not involved in the election of the President are the 12 **nominated** members of the Rajya Sabha, nominated members of the State Legislative Assemblies and in the bicameral legislature both elected and

nominated members of the Legislative Councils, and nominated members of the Union Territories of Delhi and Puducherry.

Any dispute related to the election of the President is taken up by the Supreme Court, the Supreme Court's decision is final.

Oaths, Qualifications and tenure of President

He should be a citizen of India and have completed 35 years of age. He should be qualified for election as a member of Lok Sabha. He should not hold any office of profit under the Union government or any State government or any Local authority.

The nomination of a candidate for election to the office of President must be subscribed by at least 50 electors as proposers and 50 electors as seconders. Every candidate has to make a security deposit of Rs 15,000 in the Reserve Bank of India.

The oath to the President is administered by the Chief Justice of India and in his absence, by the senior most judge of the Supreme Court. The tenure of the President of India is for five years.

Impeachment of President

The Parliament can impeach the President of India. The President can be removed from office by a process of impeachment for 'violation of the Constitution'. It is a quasi-judicial procedure.

The impeachment charges can be initiated by either House of Parliament. A Resolution containing the proposal is moved after 14 days' notice to the President signed by not less than 1/4th of the total members of that House. The resolution is then passed by a majority of not less than 2/3rd of the total members of the House. No President has so far been impeached.

Vacancy in President's Office

The Chief Justice of India acts as the Acting President of India, in the absence of both the President and the Vice President. The Vice President acts as President if the office becomes vacant due to resignation, removal, and death. A new President is then elected within six months. In case of expiration of term, the outgoing President continues to hold

office (beyond his term of five years) until his successor assumes charge.

The President may resign from his office by submitting his resignation to the Vice - President.

Salary and Perks of the President

Salary of Rs.5 lacs per month (Non Taxable), Free medical facilities, Free accommodation, Free treatment (life long), etc.

Post Retirement Perks- The President receives ₹2.5 lakh per month as pension after retirement. He is entitled to a secretarial staff comprising a private secretary, a personal assistant, an additional private secretary, two peons, and office expenses up to ₹1 lakh per annum. He is also entitled to free highest-class travel anywhere in India, accompanied by one person, by rail, air, or steamer.

Powers and Functions of President

Executive Powers - All the International Treaties are signed in his name subject to ratification by Parliament. Allocates ministries among ministers on the advice of the Prime Minister. The President of India is to be informed on affairs of the Union and proposed legislative power. The Prime Minister and Council of Ministers are appointed by the President.

Legislative Powers - He can summon and summon or prorogue the Parliament and dissolve the Lok Sabha. He can also summon joint sitting of Parliament. He nominates 12 members of the Rajya Sabha. He can promulgate ordinances when the Parliament is not in session or when either of the two Houses of Parliament is not in session. The President of India decides on the question as to the disqualification of Members of Parliament in consultation with the Election Commission of India.

Reports presented to Parliament by President - The reports of the Union Public Service Commission, Annual Financial Statement, Comptroller and Auditor General of India, Finance Commission and other constitutional and statutory bodies for introduction of any bill.

Veto Power - The President of India enjoys the three types of Veto power : A bill passed by the Parliament can

become an act only if it receives the assent of the President. When such a bill is presented to the President for his assent, he has three alternatives (Article 111): 1. He may give his assent to the bill, or 2. He may withhold his assent to the bill, or 3. He may return the bill (if it is not a Money bill) for reconsideration of the Parliament.

Absolute Veto - It refers to the power of the President to withhold his assent to a bill passed by the Parliament. The bill then ends and does not become an act.

The President uses his absolute veto in the following two cases: When the bill passed by the Parliament is a Private Member Bill, and When the newly appointed cabinet advises the President to not give his assent to the bill passed by the old cabinet.

In India, the President has exercised his absolute veto before. In 1954, it was exercised by Dr. Rajendra Prasad as a President and later in 1991, it was used by the President R. Venkataraman.

Suspensive Veto - The President exercises this veto when he returns a bill for reconsideration of the Parliament. However, if the bill is passed again by the Parliament with or without amendments, it is obligatory for the President to give his assent to the bill. The President cannot exercise his suspensive veto in relation to the **Money Bill**.

Pocket Veto - The President may delay the bill for an unlimited period. Example - Giani Zail Singh used the pocket veto for the Indian Post Office (Amendment) Bill, 1986.

Financial Powers - Only with the prior recommendation of the President, a money bill can be introduced in Parliament. The Annual Financial Statement which is the Union budget is laid before the Parliament by the President. He constitutes a Finance Commission after every 5 years.

Judicial Powers - He can seek advice from the Supreme Court on any question of law or fact (Presidential reference). The President holds the authority to transfer judges from one high court to another high court. He has the power to appoint judges of the Supreme Court and High Court.

The President can grant pardons to persons who have been tried and convicted of any offense in all cases

where : 1. Punishment or sentence is for an offense against a Union Law; 2. Punishment or sentence is by a court martial (military court); and 3. Sentence is a sentence of death.

The pardoning power of the President includes **pardon** (forgive), **commute** (reduce sentence of death to less severe one), **remission** (reduce the period of the sentence but the character of the sentence remains the same), **respite** (to award a lesser sentence in place of one originally awarded to the convict) and **reprieve** (stay on execution of sentence).

Diplomatic Powers - The President represents India in International Forums. Negotiations and conclusions of the International Treaties and Agreements are done on the behalf of the president.

Military Powers - The President appoints the chiefs of the Army, the Navy, and the Air Force. He is the supreme commander in chief of the defense forces of India. He can declare war or conclude peace, subject to the approval of the parliament.

Emergency Powers - The President has extraordinary power with the following **three** types of emergency: National Emergency under Article 352. President Rule under Articles 356 and 365 on the basis of the report of the Governor or otherwise. Financial Emergency under Article 360.

Appointment Powers of President - Prime Minister and council of ministers, and the attorney general. He appoints Chief Election Commissioner and Other Election Commissioner, Comptroller and Auditor General, Chairman and members of Union Public Service Commission, finance commission, governors of the state, interstate councils, administrators of Union territories.

Removal Powers of President - Attorney -General of India, The Governor of a State, The Chairman or a Member of the PSC of the Union or of a State, on the report of the Supreme Court, A judge of the Supreme Court or of a High Court or the Chief Election Commissioner or the Comptroller-General of India on an address of Parliament.

List of Presidents (1947 - Present)

Dr. Rajendra Prasad (26 Jan 1950 - 13 May 1962) - He was the first President of

India, who had worked as president for two terms. He was awarded Bharat Ratna in 1962.

Dr. Sarvepalli Radhakrishnan (13 May 1962 - 13 May 1967) - He was the first vice President and later on became the president. He also held the position of vice-chancellor of the Andhra University and Banaras Hindu University.

Dr. Zakir Husain (13 May 1967 - 3 May 1969) - Dr. Zakir Husain became the first Muslim President of India and died at his post.

V. V. Giri (3 May 1969 - 20 July 1969 and 24 August 1969 - 24 August 1974) - Varahagiri Venkata Giri became the only person to be elected President as an independent candidate. In 1975, he was awarded Bharat Ratna. He was elected Vice President of India in 1967. Between his term, Chief Justice of the Supreme Court Mohammad Hidayatullah became acting President from 20 July 1969 to 24 August 1969.

Fakhruddin Ali Ahmed (24 August 1974 - 11 February 1977) - Fakhruddin Ali Ahmed was the fifth President of India. He was the second President who died in the office. He was the president during the Emergency in 1975. B.D. Jatti was made Acting President.

Neelam Sanjeeva Reddy (25 July 1977 - 25 July 1982) - He was the first Chief Minister of Andhra Pradesh post reorganization. He was directly elected to the post of Lok Sabha speaker and became the youngest President.

Giani Zail Singh (25 July 1982 - 25 July 1987) - He was the Chief Minister of Punjab. During his presidency, many incidents took place, such as Operation Blue Star, the assassination of Indira Gandhi, and the 1984 anti-Sikh riots. He was the president when Rajiv Gandhi was sworn in as Prime Minister of India.

R. Venkataraman (25 July 1987 - 25 July 1992) - He was Vice-President of India from 1984 to 1987. He is a recipient of Tamra Patra for his contributions to India's freedom struggle.

Dr. Shankar Dayal Sharma (25 July 1992 - 25 July 1997) - From 1952 to 1956 he was the Chief Minister of Madhya Pradesh.

K.R. Narayanan (25 July 1997 - 25 July 2002) - He was the first person from Kerala to hold the post and the first Dalit to ascend to this high office. He was the

first President to vote in the Lok Sabha elections and addressed the state assembly. He was the president at the golden jubilee celebrations of Indian Independence.

Dr. A.P.J Abdul Kalam (25 July 2002 - 25 July 2007) - He is known as the Missile Man of India. He was the first scientist who took over the post of President and the first President of India who won the most votes.

Shrimati Pratibha Singh Patil (25 July 2007 - 25 July 2012) - She was the first female president. She was the Governor of Rajasthan before becoming the President. From 1962 to 1985, she was a member of the Maharashtra Legislative Assembly five times and was elected from Amravati to the Lok Sabha in 1991.

Pranab Mukherjee (25 July 2012 - 25 July 2017) - He was the finance minister in the central government before contesting the presidential election. He was also the deputy chairman of the Planning commission (1991 - 1996). He was awarded the best Parliamentary Award in 1997 and Padma Vibhushan in 2008.

Ram Nath Kovind (25 July 2017 - 25 July 2022) - He is a member of the Bharatiya Janata Party. He is the former Governor of Bihar.

Droupadi Murmu (25 July 2022 - Incumbent) - Droupadi Murmu became the 15th President of India. She is the first tribal president of India and belongs to the Santhal tribal community. She is a former Governor of Jharkhand. She is also the first Indian president to be born in independent India. Droupadi Murmu, in 2007, received the Nilkanth Award for the best MLA by Odisha Legislative Assembly.

Vice President

The Vice President of India is the second highest constitutional position after the President in India.

Related Articles

Article 63 - The Vice President of India.

Article 64 - The Vice President to be ex officio Chairman of the Council of States.

Article 65 - The Vice President to act as President or to discharge his functions during casual vacancies in the office, or during the absence of the President.

Article 66 - Election of Vice President.

Article 67 - Term of office of Vice President.

Article 67 (a) - A Vice President may, by writing under his hand addressed to the President, resign his office.

Article 68 - Time of holding election to fill vacancy in the office of Vice-President and term of office of person elected to fill casual vacancy.

Article 68 (1) - The election to fill the vacancy caused by the expiration of the term of office of the Vice-President shall be completed before the expiration of the term of office.

Article 69 - Oath or affirmation by the Vice President.

Qualifications, Tenure, and Re - election

To be qualified as a Vice President, a person must - be a citizen of India, be at least **35 years** of age, not holding any office of profit, and must be qualified for election as a member of the Rajya Sabha.

The constitution of India lays down that the Vice President should not be a member of either the House of Parliament or a House of the state legislature. If any such person is elected Vice President, he is deemed to have vacated his seat in that House before the date on which he enters upon his office as Vice President.

The Vice President holds office for the term of five years but he can be re-elected any number of times.

Election

The Vice President is elected indirectly by the system of **proportional representation** using single transferable votes and the voting is done by secret ballot. The Vice President is elected by the members of the Electoral College.

The Electoral College consists of both elected and nominated members of the Parliament. The members of **state legislatures** are not part of the electoral college. The election must be held no later than 60 days after the end of the term of office of the Vice President.

Removal

There is no formal impeachment process

for the Vice President in the Indian constitution. He can be removed from the office by first giving 14 days advance notice and then a resolution passed by an **effective majority** of all the members of the Rajya Sabha and agreed to by the Lok Sabha with a simple majority. introduced only in the Rajya Sabha and not in the Lok Sabha.

The **Supreme Court** can also remove the Vice President for committing electoral malpractices and not fulfilling the eligibility criteria for the Rajya Sabha member while in office as per Article 71(1) of the constitution.

Functions of Vice President

He acts as the **ex-officio Chairman** of the Rajya Sabha. In that capacity, The Vice President receives a salary of ₹400,000 per month.

He acts as President when a vacancy occurs in the office of the President. He can act as President only for a maximum period of six months.

Vice President who became President of India

Dr S Radhakrishnan, Dr Zakir Hussain, V. V. Giri, R Venkataraman, Dr Shankar Dayal Sharma, and K R Narayanan.

List of All The Vice Presidents

Vice - President	Term of Office
Sarvepalli Radhakrishnan	13 May 1952 - 12 May 1962
Zakir Hussain	13 May 1962 - 12 May 1967
V. V. Giri	13 May 1967 - 3 May 1969
Gopal Swarup Pathak	31 August 1969 - 30 August 1974
B. D. Jatti	31 August 1974 - 30 August 1979
Mohammad Hidayatullah	31 August 1979 - 30 August 1984
R. Venkataraman	31 August 1984 - 24 July 1987
Shankar Dayal Sharma	3 September 1987 - 24 July 1992
K. R. Narayanan	21 August 1992 - 24 July 1997
Krishan Kant	21 August 1997 - 27 July 2002
Bhairon Singh Shekhawat	19 August 2002 - 21 July 2007

Mohammad Hamid Ansari	11 August 2007 - 11 August 2017
Venkaiah Naidu	11 August 2017 - 11 August 2022
Jagdeep Dhankhar	11 August 2022 - Incumbent

Governors

The governors of the states have similar powers and functions at the state level as those of the President of India at the central level. He is the nominal head of the state. Governors exist in the states, while lieutenant governors exist in union territories including the National Capital Territory (NCT) of Delhi.

Related Articles

Article 153 - There shall be a Governor for each State.

Article 154 - Executive power of State vested in the Governor.

Article 155 - Appointment of Governor.

Article 156 - Term of office of Governor.

Article 157 - Qualifications for appointment as Governor.

Article 158 - Conditions of Governor's office.

Article 159 - Oath or affirmation by the Governor.

Article 160 - Discharge of the functions of the Governor in certain contingencies.

Article 161 - Power of Governor to grant pardons, etc., and to suspend, remit or commute sentences in certain cases.

Article 162 - Extent of executive power of State.

Article 200 - Assent to Bills.

Article 213 - Power of Governor to promulgate Ordinances during recess of Legislature.

Appointment and Tenure

The Governors and Lieutenant Governors are appointed by the **President** for a term of 5 years.

The 7th Constitutional Amendment Act of 1956 facilitated the appointment of the same person as a governor for two or more states. When one person serves as governor for multiple states, their pay and benefits are divided among the states according to the President's decision. These payments cannot be reduced during their term.

Qualifications of Governor

A governor must be at least **35 years** of age, should not be a member of either house of the parliament or house of the state legislature, and should not hold any office of profit.

Powers of Governor

The Governor of the state shall possess executive, legislative, financial and judicial powers. But he does not possess diplomatic, military or emergency powers which the President of India has.

Executive Powers : The executive power is exercised by the council of ministers in the name of the Governor. He can recommend the imposition of constitutional emergency in a state to the President. He acts as the chancellor of universities in the state.

The Governor appoints the Chief Minister, other state ministers (based on the Chief Minister's advice), and the Advocate General, and they hold office at the pleasure of the Governor.

Legislative Powers : He has the power to summon, prorogue the state legislature and can also dissolve the legislative assembly (Article 176). He nominates one-sixth of the members of the state legislative council. He decides on the question of disqualification of members of the state legislature in consultation with the Election Commission.

When a bill other than money bill is presented before Governor for his assent, he either gives assent to the bill, withhold his assent to the bill, return the bill for reconsideration of houses, but if the bill is passed again by state legislature with or without amendments, he has to give his assent or **reserve** the bill for consideration of President.

The Governor cannot return money bills for reconsideration since they are introduced with his prior consent.

He can promulgate an ordinance only when the legislative assembly (in case of a unicameral legislature) is not in session or (in case of a bicameral legislature) when both the Houses of the state legislature are not in session or when either of the two Houses of the state legislature is not in session. He can promulgate or withdraw any ordinance only on the advice of the Chief Minister and the **Council of Ministers**.

The ordinance will cease to operate on the expiry of **six weeks** from the reassembly of the State Legislature if an ordinance is neither accepted nor disapproved /withdrawn by the state legislature.

Financial Powers : Money bill can only be introduced in the state legislature on his prior recommendation. No demand for grant can be made except on his recommendation. Money from the contingency fund can be withdrawn after his recommendation. He constitutes a **finance commission**.

Judicial Powers : The President consults the Governor of the concerned state while making appointments to the judges of the High Court. He makes appointments, postings and promotions of the district judges in consultation with the state high court.

He can grant pardons, reprieves, respites and remissions of punishment or suspend, remit and commute the sentence of any person convicted of any offence against any law relating to a matter to which the executive power of the state extends.

Governor pardoning power does not extend in case of sentence given by court martial and in case of death penalty. But, the governor can suspend, remit or commute a death sentence.

Constitutional Discretion : The governor has the constitutional discretion in case of reserving the bill for President's consideration, recommending for imposition of President's Rule, etc.

Removal of Governor

Governors can be removed by the President any time. This power cannot be exercised in an arbitrary manner. It is to be exercised in rare and exceptional circumstances for valid and compelling reasons. The decision to remove a Governor can be challenged in any court of law.

Important Fact : Sarojini Naidu was the first woman Governor of a State in free India. She governed Uttar Pradesh from 15 August 1947 to 2 March 1949. **M M Jacob** has served the longest number of days as governor of an Indian state. He was the governor of Meghalaya.

Parliament

Parliament is the "temple of democracy" which provides the highest forum to discuss, debate and decide issues for the well-being of people. Parliament is the supreme legislative body of India. Article 79 - 122 in Part V of the Constitution deals with the provisions of Parliament.

The Indian Parliament consists of Lok Sabha, Rajya Sabha and the President of India. India has a bicameral system i.e. two Houses of Parliament. At the state level, the equivalent of the Lok Sabha is the Vidhan Sabha or Legislative Assembly, that of the Rajya Sabha is the Vidhan Parishad or Legislative Council.

Lok Sabha

It is known as the House of the People. It is the lower house of India's bicameral Parliament. On the basis of universal adult suffrage, MPs (Member of Parliament) are chosen by direct election to the Lok Sabha. The First Session of this Lok Sabha commenced on 13 May 1952.

Composition - The maximum strength of the house is 550 members. Originally, Article 331 reserved two seats for the Anglo-Indian community. However, the 104th Constitutional Amendment Act in 2019 nullified Article 331, removing the reserved seats.

The composition consists of 530 members representing states across India and 20 members representing Union Territories (UTs). The current strength of Lok Sabha is 543, out of which 530 members represent the states and 13 represent the UTs.

Duration : The term of Lok Sabha is five years from the date of its first meeting after the general elections. But the President, on the advice of the Council of Ministers, may dissolve it before the expiry of five years.

In the case of a national emergency, its term can be extended for one year at a time. But it will not exceed six months after the emergency is over.

Eligibility Criteria for Membership : He must be an Indian citizen. He must be at least 25 years old. Must be a voter for any parliamentary constituency in India.

Disqualification of Member : Hold office of profit. Must not be convicted by the

court with imprisonment of two or more years. Members can also be disqualified based on the grounds of Defection mentioned in the 10th Schedule.

Speaker of Lok Sabha - The Speaker serves as both the ceremonial and legal head of the Lok Sabha and serves as its chief spokesperson. He is elected by the Lok Sabha from amongst its members.

He can allow for a 'secret' sitting of the House at the request of the Leader of the House. He also decides whether the bill is a money bill or not, and his/her decision on this matter is final.

He can adjourn the proceedings of the House or suspend the meeting in absence of a quorum. He decides the questions of disqualification of a member of the Lok Sabha. He acts as the ex-officio chairman of the Indian Parliamentary Group.

Deputy Speaker - When the Speaker is not present, the Deputy Speaker serves as the House's presiding officer and has all of the Speaker's authority.

Pro-tem Speaker - The Pro-tem Speaker presides over the first sitting of the Lok Sabha, administers the oath of office to the newly elected MPs. The President appoints the pro-tem speaker to preside over the sittings of the newly elected house.

He conducts the vote for the speaker and deputy speaker and also administers the floor test.

Panel of Chairpersons - At the commencement of the House or as the case may be, the speaker shall nominate from amongst the members a panel of not more than ten Chairpersons. Anyone from this panel may preside over the House in the absence of the Speaker and the Deputy Speaker. But he will not preside when both these posts are vacant.

Seats of States and UTs in Lok Sabha :

States/UT - Seats	States/UT - Seats
Uttar Pradesh - 80	Chhattisgarh - 11
Maharashtra - 48	Haryana - 10
West Bengal - 42	NCT Delhi - 7
Bihar - 40	Uttarakhand - 5
Tamil Nadu - 38	Himachal Pradesh - 4
Madhya Pradesh - 29	Arunachal Pradesh - 2
Karnataka - 28	Goa - 2

Gujarat - 26	Manipur - 2
Rajasthan - 25	Meghalaya - 2
Andhra Pradesh - 24	Tripura - 2
Odisha - 21	Mizoram - 1
Kerala - 20	Nagaland - 1
Telangana - 17	Sikkim - 1
Assam - 14	Punjab - 13
Jharkhand - 14	Puducherry - 1
Lakshadweep - 1	

Lok Sabha and their Speakers :

Lok Sabha	Speaker	Term of office of Speaker
1st	Ganesh Vasudev Mavalankar	15 May 1952 - 27 February 1956
2nd	M. A. Ayyangar	8 March 1956 - 16 April 1962
3rd	Sardar Hukam Singh	17 April 1962 - 16 March 1967
4th	Neelam Sanjiva Reddy	26 March 1977 - 13 July 1977
5th	Gurdial Singh Dhillon	8 August 1969 - 1 December 1975
6th	Bali Ram Bhagat	15 January 1976 - 25 March 1977
7th	Neelam Sanjiva Reddy	26 March 1977 - 13 July 1977
8th	K. S. Hegde	21 July 1977 - 21 January 1980
9th	Balram Jakhar	22 January 1980 - 18 December 1989
10th	Rabi Ray	19 December 1989 - 9 July 1991
11th	Shivraj Patil	10 July 1991 - 22 May 1996
12th	P. A. Sangma And G. M. C. Balayogi	23 May 1996 - 23 March 1998 And 24 March 1998 - 3 March 2002
13th	Manohar Joshi	10 May 2002 - 2 June 2004
14th	Somnath Chatterjee	4 June 2004 - 31 May 2009
15th	Meira Kumar	30 May 2009 - 4 June 2014
16th	Sumitra Mahajan	6 June 2014 - 17 June 2019
17th	Om Birla	19 June, 2019 - till present

Note : The Lok Sabha secretariat released a booklet of more than 50 pages of words considered unfit for being used in Parliament. The latest list of unparliamentary words - Jumlaajeevi, Baal buddhi, Chakkar, Covid spreader, abused, ashamed, betrayal, chamchagiri, cheated, shakuni, nautanki, girgit, etc.

Rajya Sabha (The Council of States)

It is the **Upper House** (Second Chamber) and it represents the states and union

territories of the Indian Union. The Rajya Sabha or the Council of States held its first sitting on 13 May 1952.

Composition - The maximum strength of the Rajya Sabha is 250. Out of 250, 238 members are representatives of the states & UTs (elected by the elected members of the State Legislative Assembly) and 12 are nominated by the President who has 'special knowledge or practical experience in literature, science, art, and social service'.

The representatives of each UT in the Rajya Sabha are indirectly elected by members of an electoral college specially constituted for the purpose.

The current strength of the house is 245. 229 members represent the states, 4 members represent the UTs and 12 are nominated by the president.

Tenure - The Rajya Sabha is called the permanent House of the Parliament as it is never fully dissolved. Every Rajya Sabha MP has a tenure of six years and elections to one-third seats are held every two years.

Qualifications - He should be a citizen of India and at least 30 years of age. Any Indian citizen can contest the Rajya Sabha elections, irrespective of the state in which he/she resides.

Power - It reviews and alters the laws initiated by the Lok Sabha. The Rajya Sabha provides representation to the States. Therefore, any matter that affects the States must be referred to it for its consent and approval.

If the Union Parliament wishes to remove/transfer a matter from the State list, the approval of the Rajya Sabha is necessary.

Chairman of Rajya Sabha - The Vice-President is the ex-officio chairman of the Rajya Sabha. In his absence, the Deputy Chairman (elected by its members from amongst themselves) presides over the meeting of the House. S. Radhakrishnan was the first chairman of Rajya Sabha.

The Chairman of the Rajya Sabha is empowered to adjourn the House or to suspend its sitting in the event of the absence of quorum. The Chairman's consent is needed to raise a question of breach of privilege in the House.

Deputy Chairman : The Deputy Chairman

is elected by the Rajya Sabha itself from amongst its members. He is directly responsible to the Rajya Sabha.

Panel of Vice-Chairpersons : Under the Rules of Rajya Sabha, the Chairman nominates from amongst the members a panel of vice-chairpersons.

Seats of States and UTs in Rajya Sabha:

States/UT - Seats	States/UT - Seats
Uttar Pradesh - 31	Chhattisgarh - 5
Maharashtra -19	J & K - 4
Tamil Nadu -18	Himachal Pradesh -3
West Bengal -16	Uttarakhand - 3
Bihar -16	NCT Delhi - 3
Karnataka -12	Assam - 7
Madhya Pradesh -11	Goa - 1
Andhra Pradesh -11	Manipur - 1
Gujarat - 11	Meghalaya - 1
Odisha - 10	Mizoram - 1
Rajasthan -10	Nagaland - 1
Kerala - 9	Puducherry - 1
Punjab - 7	Sikkim -1
Telangana - 7	Tripura -1
Jharkhand - 6	Arunachal Pradesh -1
Haryana - 5	

The first general elections under the new Constitution were held during the year 1951-52 and in the year 1952, the first Elected Parliament with two houses came into effect in India. Total Lok Sabha seats were 489 at that time.

Vacant of Seats in Both Houses

A member of Parliament vacates his seat in the following cases :

Double Membership : According to the Representation of People Act (1951), a person cannot simultaneously hold membership in both Houses of Parliament. If elected to both, they must choose within 10 days which House they wish to serve in; otherwise, their seat in the Rajya Sabha becomes vacant. Likewise, one cannot be a member of both Parliament and a state legislature concurrently. If a member fails to resign from the state legislature within 14 days of election then their seat in Parliament becomes vacant.

Disqualification : If a member of Parliament falls under any of the disqualifications outlined in the Constitution, their seat automatically

becomes vacant. This includes disqualifications such as defection as per the provisions of the Tenth Schedule of the Constitution.

Resignation : A member may resign his seat by writing to the Chairman of Rajya Sabha or Speaker of Lok Sabha, as the case may be.

Absence : A House can declare the seat of a member vacant if he is absent from all its meetings for a period of sixty days without its permission. In computing the period of sixty days, no account shall be taken of any period during which the House is prorogued or adjourned for more than four consecutive days.

Joint Sitting of Two Houses

The president can summon both the Houses to meet in a joint sitting for the purpose of deliberating and voting on the bill. It is held to resolve the deadlock between the two houses. Joint sittings apply to ordinary or financial bills, not money or Constitutional amendment bills. The Lok Sabha holds overriding powers for money bills, while Constitutional amendment bills must pass each House separately. The Speaker of Lok Sabha presides over the Joint Sitting.

Leaders in Parliament

Leader of the House - According to the Rules of Lok Sabha, the 'Leader of the House' means the Prime Minister (or another minister who is a member of Lok Sabha and is nominated by the PM to function as the Leader of the House).

Leader of the House in the Rajya Sabha is a minister and a member of the Rajya Sabha who is nominated by the PM to function as such.

Leader of the Opposition : The leader of the largest Opposition party having not less than one-tenth seats of the total strength of the House is recognised as the leader of the Opposition in that House. The leader of Opposition in the Lok Sabha and the Rajya Sabha were accorded statutory recognition in 1977.

Whip

The office of the whip is based on conventions of the parliamentary government. Whips issue a written order to members of the political members for being present for an important vote, or

that they vote only in a particular way.

The Ministry of Parliamentary Affairs organizes the 'All India Whips Conference'. This conference brings together whips from various political parties across India. The first All India Whips Conference was organized at Indore in 1952.

Sessions in Parliament

Summoning - The President calls Parliament to meet, ensuring a gap of no more than 6 months between sessions. It is mandatory for Parliament to meet at least twice a year as per the Constitution of India.

By convention, Parliament meets for three sessions in a year.

Budget Session - Longest session, starts towards the end of January/February, and concludes by the end of April/May.

Monsoon Session - usually begins in June/July and finishes in August/September.

Winter Session - Third session, held from November to December.

Adjournment - It suspends the work in a sitting for a specified time, which may be hours, days or weeks. An adjournment only terminates a sitting and not a session of the House. The power of adjournment lies with the presiding officer of the House.

Adjournment Sine Die - It terminates a sitting of Parliament for an indefinite period.

Prorogation - It terminates both the sitting and session of the House. The power to prorogue or to dissolve the Lok Sabha vests with the President.

Dissolution - It is the formal term for the end of Lok Sabha. It occurs ahead of a general election for a new Lok Sabha.

Quorum - The minimum number of members of a House of Parliament required to be present for proceedings in the House to take place.

The Constitution has fixed one-tenth strength as quorum for both Lok Sabha and Rajya Sabha. It is the duty of the speaker to adjourn the House or suspend the meeting if the quorum is absent.

Lame Duck Session - It refers to the last session of the existing Lok Sabha after a new Lok Sabha has been elected. The members of the existing Lok Sabha who

could not get re-elected to the new Lok Sabha are called lame-ducks. The term lame duck is widely used in the USA congress.

Recess - A break during the parliamentary session (year) in which neither of the House meets to conduct business.

Devices of Parliamentary Proceedings

Zero Hour : It is an Indian parliamentary innovation. It is not mentioned in the parliamentary rules book. Under this, MPs can raise matters without any prior notice.

Question Hour : The first hour of every parliamentary sitting is slotted for the Question Hour. There are three types of questions :

Starred Questions - Those questions for which an oral answer is expected and hence supplementary questions can follow. These questions are printed in green color.

Unstarred questions - Those questions that require a written answer and hence, supplementary questions cannot follow. These questions are printed in white color.

Short notice questions - Questions asked by giving a notice of less than ten days. It is answered orally. These questions are printed in light pink color.

In addition to these questions, the questions can also be asked to the private members. These questions are related to private member's bills, parliamentary committees, and private member resolutions. These questions are printed in Yellow color.

Motions

A Motion refers to a formal proposal made by a member of the legislature to initiate discussion on a matter of general public importance. A motion is also a parliamentary device of the parliamentary proceedings. The various kinds of motions are -

Cut Motion - It is a power given to the members of the Lok Sabha that enables its members to oppose any demand in a Finance Bill proposed by a government.

Adjournment Motion - It is used to call the attention of the Lower House in

matters of urgent public importance and needs the support of 50 members to be admitted. It can be moved only in Lok Sabha. Rajya Sabha is not permitted to make use of this device.

Call-Attention Motion - It is introduced in the Parliament by a member to call the attention of a minister to a matter of urgent public importance. This motion can be presented in both houses of Parliament.

Short Duration Discussion : It is a procedural device to enable members to raise discussion on matters of urgent public importance without a formal motion on a matter of urgent public importance.

Censure Motion - It can be moved against one minister or the entire Council of Ministers. If it is passed in the Lok Sabha, the council of ministers need not resign from the office.

No-Confidence Motion - It is a formal proposal in Lok Sabha to evoke a decision of the house. It can only be introduced in the Lok Sabha. For the motion to be granted, at least 50 members of the Lok Sabha should support the motion. If it is passed, the government has to resign.

The first no-confidence motion moved against the cabinet in independent India during the third Lok Sabha in 1963 when Jawaharlal Nehru was the Prime Minister. It was moved by minister Acharya J.B Kriplani and was backed by 40 MPs. The motion, however, was rejected.

No-Day-Yet-Named Motion - It is a motion that has been admitted by the Speaker but no date has been fixed for its discussion.

Motion of Thanks - The President addresses the opening session of each fiscal year and the first session following each general election. If this motion cannot be passed, the government is deemed to have lost its majority.

Closure Motion - It is a proposal that the Lok Sabha or Rajya Sabha should stop debating and vote on the matter being discussed. There are four kinds of closure motions :

Simple Closure - It is one when a member moves that the 'matter having been sufficiently discussed be now put to vote'.

Closure by Compartments - The clauses of a bill or a lengthy resolution are grouped into parts before the commencement of the debate. The debate covers the part as a whole and thus the entire part is put to vote.

Kangaroo Closure - Only important clauses are taken up for debate and voting. The other clauses are skipped over and then they are considered as passed after the bill is passed.

Guillotine Closure - It is one when the undiscussed clauses of a bill or a resolution are also put to vote along with the discussed ones due to lack of time.

Bills in Parliament

A bill is a proposal for legislation that becomes an act or law when duly enacted.

The Legislation starts with proposals of bills. After passing through the various stages in both houses, the bill becomes an act. Public bills, introduced by the government, aim to affect everyone and cover broad topics like healthcare or taxes. Private bills, proposed by individual MPs, target specific individuals, groups, or local areas. The bills introduced in the Parliament can also be classified into four categories:

Ordinary bills - They are concerned with any matter other than financial subjects.

Financial bills - They are concerned with financial matters (but are different from money bills). Types of Financial Bills : Financial bills (I) - Article 117 (1), and Financial bills (II) - Article 117 (3).

Money bills - They are considered government bills and can only be introduced by a minister. They are concerned with financial matters like taxation, public expenditure, etc. The Speaker of the Lok Sabha has the power to certify whether a Bill is a Money Bill or not.

Money Bill can only be introduced in the Lok Sabha and only with the recommendation of the President. The Rajya Sabha can only make recommendations and cannot reject or amend the bill. The Rajya Sabha must return the bill to the Lok Sabha within 14 days, with or without recommendations. The Lok Sabha can accept or reject any recommendations made by the Rajya Sabha.

Constitution Amendment Bills - They are concerned with the amendment of the provisions of the Constitution.

Powers/Functions of Parliament

Legislative Functions - Parliament can make laws on the subjects of the Union List and Concurrent List. If a subject is not mentioned in any list, the residuary powers are vested with the Parliament.

Financial Functions - It is the custodian of the public money. The Government can neither impose any tax on the public nor spend the money without the approval of the Parliament. The Parliament also scrutinizes government spending and financial performance with the help of its financial committees.

Electoral Functions - It participates in the election of the President of India and also elects the Vice-President. The Lok Sabha elects its Speaker and Deputy Speaker and the Rajya Sabha elects its Deputy Chairman.

Judicial Powers - Certain high functionaries may be removed from office on the initiative of the Parliament. It can remove the President, Judges of the Supreme Court and High Courts through impeachment for violation of the Constitution.

Amendment of the Constitution - Parliament can amend most of the part of the constitution. The Parliament cannot change the basic structure of the Constitution.

Power over Executive - Parliament exercises control over the Executive through question-hour, zero hour, calling attention notice, adjournment motion etc.

Parliamentary Privileges

Parliamentary privileges are special rights, immunities and exemptions enjoyed by the two Houses of Parliament, their committees and their members.

Funds of India

The Constitution of India provides for the following three kinds of funds for the Central Government : Consolidated Fund of India (Article 266), Contingency Fund of India (Article 267), and Public Accounts of India [Article 266(2)]. All three funds can be created by each state.

Consolidated Fund of India : It is a fund to which all receipts are credited and all payments are debited. This fund is filled by direct and indirect taxes and loans taken by the Indian government and by returning of loans/interests of loans to the government by anyone /agency that has taken it.

All the legally authorized payments on behalf of the Government of India are made out of this fund. No money out of this fund can be appropriated (issued or drawn) except in accordance with a parliamentary law.

Contingency Fund of India : Finance Bill 2021 increased its corpus from Rs. 500 crore to Rs. 30,000 crores. It is in the nature of an imprest (money maintained for a specific purpose). It is also operated by executive action. The Secretary of the Finance holds this fund on behalf of the President of India. The president is the custodian of this fund. This fund is used to meet unexpected or unforeseen expenditure. Each state can have its own contingency fund established under **Article 267(2)**.

Public Accounts fund of India : All other public money (other than those covered under the Consolidated Fund of India) received by or on behalf of the Indian Government are credited to this account/fund. This account is operated by executive action. It consists of Bank savings accounts of the various ministries/departments, National small savings fund, defense fund, National Investment Fund (money earned from disinvestment), National Calamity Contingency Fund (NCCF), Provident fund, Postal insurance, etc.

The government does not need permission to take advances from this account.

Indian Parliamentary Group

The Indian Parliamentary Group is an autonomous body formed in the year 1949 in pursuance of a motion adopted by the Constituent Assembly (Legislative) on 16th August, 1948.

Membership of the Indian Parliamentary Group is open to all Members of Parliament and ex-Members of Parliament. The Speaker of the Lok Sabha is the ex-officio President of the Group.

The Group aims to promote personal

contacts between Members of Parliament and disseminate information to the Members of Parliament by conducting seminars etc. on the issues that are likely to come up before Parliament.

Parliamentary Committees

Parliamentary committees are of two types - Standing Committees (permanent) and Ad Hoc Committees (temporary). Some Parliamentary Committees are :

Committee on Public Accounts - Its main purpose is auditing the revenue and the expenditure of the Government of India. It presently comprises 22 members (15 members elected by the Lok Sabha Speaker, and 7 members elected by the Rajya Sabha Chairman) with a term of one year only.

Committee on Estimates - Its main purpose is to scrutinize the functioning of government ministries and departments in terms of expenditure and utilization of funds. It does not include any member from the Rajya Sabha. This Committee has 30 members all from Lok Sabha. For the first time in India, the Estimates Committee was formed in 1950.

Committee on Public Undertakings - Its main purpose is to examine the reports and accounts of the public sector undertakings. It consists of 22 members, fifteen elected by Lok Sabha, and not more than seven members of Rajya Sabha.

Business Advisory Committee of Rajya Sabha - It was constituted for the first time in 1952. It consists of 11 members including the Chairman of the Rajya Sabha. The function of the Committee is to recommend to the government to bring forward specific subjects for discussion in the House and recommend allocation of time for such discussions.

Business Advisory Committee of Lok Sabha - It came into existence for the first time on 14th July 1952. It consists of 15 members including the Speaker of Lok Sabha. The Speaker of Lok Sabha is the presiding officer and highest authority of the house of people.

Articles Related to Parliament

Article 33 - Power of Parliament to modify the rights conferred by this part

in their application to Forces.

Article 34 - Restriction on rights conferred by this part while martial law is in force in any area.

Article 79 - There shall be a Parliament for the Union which shall consist of the President and two Houses to be known respectively as the Council of States and the House of the People.

Article 80 - Composition of the Council of States (Rajya Sabha)

Article 81 - Composition of the House of the People (Lok Sabha).

Article 82 - Readjustment of seats after each census.

Article 83 - Duration of Houses of Parliament.

Article 84 - Qualification for membership of Parliament.

Article 85 - Sessions of Parliament, prorogation and dissolution.

Article 86 - Right of President to address and send messages to Houses

Article 86 (1) - The President may address either House of Parliament or both Houses assembled together.

Article 86 (2) - The President may send messages to either House of Parliament.

Article 87 - Special address by the President at the commencement of the house after the general election and the first session of each year.

Article 89 - The Chairman and Deputy Chairman of the Council of States.

Article 90 - Vacation, resignation, and removal of the Deputy Chairman of the Rajya Sabha (Council of States).

Article 91 - Power of the Deputy Chairman or other person to perform the duties of the office of, or to act as, Chairman.

Article 92 - The Chairman or the Deputy Chairman not to preside while a resolution for his removal from office is under consideration.

Article 93 - The Speaker and Deputy Speaker of the House of the People.

Article 94 - Vacation and resignation of, and removal from, the offices of Speaker and Deputy Speaker.

Article 95 - Power of the Deputy Speaker or other person to perform the duties of the office of, or to act as, Speaker.

Article 98 - Secretariat of Parliament.

Article 99 - Oath or affirmation by members

Article 100 - Voting in Houses, power of Houses to act notwithstanding vacancies and quorum

Article 101 - Vacation of seats of Parliament.

Article 102 - Disqualifications for membership of parliament.

Article 104 - Penalty for sitting and voting before making oath or affirmation under article 99 or when not qualified or when disqualified.

Article 105 - Powers, Privileges and Immunities of Parliament and its Members.

Article 106 - Parliament determines the salaries of the members of either house of parliament.

Article 107 - Provisions as to the introduction and passing of Bills.

Article 108 - Joint sitting of both Houses in certain cases.

Article 110 - It defines Money Bill.

Article 112 - Annual financial statement (Budget).

Article 114 - Introduction of a Bill (Appropriation Bills) to provide for the appropriation out of the Consolidated Fund of India.

Article 115 - Supplementary, additional or excess grants.

Article 119 - Regulation by law of procedure in Parliament in relation to financial business.

Article 120 - Language to be used in Parliament.

Article 121 - Restriction on discussion in Parliament.

Article 122 - Courts not to inquire into proceedings of Parliament.

Article 123 - Power of President to promulgate Ordinances during recess of Parliament.

Article 247 - Power of Parliament to provide for the establishment of certain additional courts.

Article 249 : Rajya Sabha can authorize the Parliament to make a law on a subject enumerated in the State List.

Article 302 - Power of Parliament to impose restrictions on trade, commerce and intercourse.

Article 312 : Rajya Sabha can authorize the Parliament to create new All-India Services common to both the Centre and states.

Article 327 - Power of Parliament to make provisions with respect to elections to Legislatures.

Article 330 - Reservation of seats for Scheduled Castes and Scheduled Tribes in the House of the People.

Article 331 - Representation of the Anglo-Indian community in the House of the People. (Repealed)

State legislature

In India, each state has its own legislature, which is responsible for making laws and governing the state. Articles 168 to 212 in Part VI of the Constitution deal with the organisation, composition, duration, officers, procedures, privileges, powers and so on of the state legislature.

The state legislature is bicameral in some states and unicameral in others, depending on the population and historical factors. At present, **6 states** of India have bicameral legislature viz. Andhra Pradesh (58), Maharashtra (78), Karnataka (75), Telangana (40), Bihar (75), Uttar Pradesh (100). The Legislative Council of Jammu and Kashmir was formally abolished on 16 October 2019.

Legislative Assembly (Vidhan Sabha)

The State Legislative Assembly is a legislative body in the states and union territories of India.

It consists of members elected directly by the people for five-year term. Its maximum and minimum strength is fixed at 500 and 60 respectively. However, in the case of Sikkim and Mizoram the minimum number is fixed at 30 and 40 respectively.

Uttar Pradesh Legislative Assembly has the highest number of MLAs (403) whereas **Sikkim** Legislative Assembly has the lowest number of MLAs (32). The National Capital Territory of **Delhi** and Union Territory of **Puducherry** each have a legislative assembly and council of ministers.

The total number of ministers, including the Chief Minister, in the Council of Ministers in a state shall not exceed 15% of the total strength of the Legislative Assembly of that state.

But, the number of ministers, including the Chief Minister, in a state shall not be less than **12**. It was inserted into Article 75 of the Indian Constitution through the 91st Constitutional Amendment Act of 2003.

The Speaker and Deputy Speaker are elected by the assembly itself from amongst its members.

Legislative Council (Vidhan Parishad)

The State Legislative Council is the upper house in those states of India that have a bicameral state legislature.

Parliament can abolish or create a Legislative Council if the concerned legislative assembly passes a resolution to that effect with simple majority.

The members are indirectly elected. The maximum strength is fixed at one-third of the total strength of the assembly and the minimum strength is fixed at 40.

Uttar Pradesh Legislative Council has the highest number of MLC of 100 whereas **Telangana** has the lowest number of MLC of 40.

Of the total number of Members of the Legislative Council, 1/3 of Members are elected by electorates consisting of the Members of Local Authorities. 1/12 are elected by electorates consisting of graduates residing in the State and 1/12 are elected by electorates consisting of persons engaged in teaching. Other 1/3 are elected by the Members of Legislative Assembly and the 1/6 are nominated by the Governor.

Each Member of the Legislative Council (MLC) is directly elected to serve 6-year terms. One-third of the members of the State Legislative Council retire after every two years.

The **Chairman** and **Deputy Chairman** are elected by the members of the Legislative Council among themselves.

Differences between State Legislature and Parliament

The duration, membership, disqualification, functioning, powers,

procedures, privileges, and so on of the State Legislature are similar to Parliament at the center. But there is a key difference: State Legislative Assemblies can be unicameral or bicameral while the Parliament of India consists of two houses, Lok Sabha and Rajya Sabha.

Qualifications of members of State Legislature : He must be a citizen of India. He must be not less than 30 years of age in the case of the legislative council and not less than 25 years of age in the case of the legislative assembly.

Ordinary Bills : The legislative assembly can override the legislative council by passing the bill for the second time and not vice versa. If the legislative council does not pass the bill within 1 month then it is deemed to have been passed by both the houses..

Assent to Bills : Every bill, after it is passed by the assembly or by both the Houses in case of a bicameral legislature, is presented to the governor for his assent. There are four alternatives before the governor -

1. he may give his assent to the bill;
2. he may withhold his assent to the bill;
3. he may return the bill for reconsideration of the House or Houses;
4. he may reserve the bill for the consideration of the President.

When a bill is reserved by the governor for the consideration of the President, the President may either give his assent to the bill or withhold his assent to the bill or return the bill for reconsideration of the House or Houses of the state legislature.

Money Bills : A Money Bill cannot be introduced in the legislative council. It can be introduced in the legislative assembly only and that too on the recommendation of the Governor.

Important Articles

Article 168 - Constitution of Legislatures in States.

Article 169 - Abolition or creation of Legislative Councils in States.

Article 170 - Composition of the Legislative Assemblies.

Article 171 - Composition of the Legislative Councils.

Article 172 - Duration of State Legislatures.

Article 172(1) - every state assembly shall continue for five years unless dissolved.

Article 172(2) - Legislative Council is not subject to dissolution.

Article 173 - Qualification for membership of the State Legislature.

Article 174 - Sessions of the State Legislature, prorogation and dissolution.

Article 175 - Right of Governor to address and send messages to the House or Houses.

Article 176 - Special address by the Governor. The Governor addresses the Legislative Assembly at the start of both the first session after each election and the first session of each year.

Article 177 - Rights of the Ministers and the Advocate General with respect to the Houses.

Article 178 - The Speaker and Deputy Speaker of the Legislative Assembly.

Article 179 - Vacation and resignation of, and removal from, the offices of Speaker and Deputy Speaker.

Article 180 - Power of the Deputy Speaker or other person to perform the duties of the office of, or to act as, Speaker.

Article 181 - The Speaker or the Deputy Speaker not to preside while a resolution for his removal from office is under consideration.

Article 182 - The Chairman and Deputy Chairman of the Legislative Council.

Article 183 - Vacation and resignation of, and removal from, the offices of Chairman and Deputy Chairman.

Article 184 - Power of the Deputy Chairman or other person, to perform the duties of the office of, or to act as, Chairman.

Article 185 - The Chairman or the Deputy Chairman not to preside while a resolution for his removal from office is under consideration.

Article 186 - Salaries and allowances of the Speaker and Deputy Speaker and the Chairman and Deputy Chairman.

Article 187 - Secretariat of State Legislature.

Article 188 - Oath or affirmation by members.

Article 189 - Voting in Houses, power of Houses to act notwithstanding vacancies and quorum.

Article 190 - Vacation of seats.

Article 191 - Disqualifications for membership.

Article 192 - Decision on questions as to disqualifications of members.

Article 193 - Penalty for sitting and voting before making oath or affirmation under Article 188 or when not qualified or when disqualified.

Article 194 - Powers, privileges, etc., of the Houses of Legislatures and of the members and committees thereof.

Article 195 - Salaries and allowances of members.

Article 196 - Provisions as to introduction and passing of Bills.

Article 197 - Restriction on powers of Legislative Council as to Bills other than Money Bills.

Article 198 - Special procedure in respect of Money Bills.

Article 199 - Definition of "Money Bills".

Article 200 - Assent to Bills.

Article 201 - Bills reserved for consideration.

Article 202 - Annual financial statement.

Article 203 - Procedure in Legislature with respect to estimates.

Article 204 - Appropriation Bills.

Article 205 - Supplementary, additional or excess grants.

Article 206 - Votes on account, votes of credit and exceptional grants.

Article 207 - Special provisions as to financial Bills.

Article 208 - Rules of procedure.

Article 209 - Regulation by law of procedure in the Legislature of the State in relation to financial business.

Article 210 - Language to be used in the Legislature.

Article 211 - Restriction on discussion in the Legislature.

Article 212 - Courts not to inquire into proceedings of the Legislature.

Prime Minister and Chief Minister

The Constitution of India provides for the Prime Minister (PM) and Chief Minister (CM) to be the real executive authority (de facto executive) at the center and state respectively.

Prime Minister

He is the head of the Government. He acts as the communication channel between the Council of Ministers and the President. The Prime Minister plays these functions, that's why Pandit Jawaharlal Nehru called him the **linchpin** of Government.

The PM serves as the **Cabinet's chairman** and leads meetings of the Cabinet. The real executive power is vested in the **Council of Ministers** with the Prime Minister as its head. In case of disagreement, the decision of the PM is final.

The Prime Minister is the head of many departments and services, such as the Nuclear Command Authority, NITI Aayog, the Cabinet Appointments Committee, the Atomic Energy Department, the Space Department, and the Ministry of Personnel, Public Grievances and Pensions. He is Chairperson of National Disaster Management Authority.

Every year on **Independence Day** (15 August) the **Prime Minister hoists the national flag at Red Fort**.

Appointment : The PM is appointed by the President. In 1997, the Supreme Court held that a person who is not a member of either House of Parliament can be appointed as Prime Minister for six months, within which, he should become a member of either House of Parliament; otherwise, he ceases to be the Prime Minister. Constitutionally, the Prime Minister may be a member of any of the two Houses of parliament. For example, three Prime Ministers, Indira Gandhi (1966), Deve Gowda (1996) and Manmohan Singh (2004), were members of the Rajya Sabha.

Qualification: He/she must be a citizen of India. He/she must be a member of either Rajya Sabha or Lok Sabha.

Tenure : The Prime Minister's tenure is not fixed, and he serves at the President's pleasure. The Prime Minister cannot, however, be removed by the President so long as he has the support of the majority of lawmakers in the **Lok Sabha**. The Prime Minister must resign if the Lok Sabha loses confidence in him (by a majority), or the President may remove him.

Powers and Functions : He advises the President with regard to summoning and

proroguing of the sessions of the Parliament. He can recommend dissolution of the Lok Sabha to the President at any time. He announces government policies on the floor of the House. He is the crisis manager-in-chief at the political level during emergencies.

Prime Ministers of India

Jawaharlal Nehru (1947-1964) - Jawaharlal Nehru was the first and till date the longest-serving PM of India. He was a leading figure in the Indian independence movement and served as the PM until his death in 1964. He was the first PM to die in office.

Gulzarilal Nanda (1964) - His earlier 13-day stint as the acting PM of India followed the death of prime minister Jawaharlal Nehru in 1964.

Lal Bahadur Shastri (1964-1966) - Lal Bahadur Shastri was the second PM of India. He came up with the slogan of 'Jai Jawan Jai Kisan' that became popular during the 1965 Indo-Pakistan war. He died the day following the Tashkent Agreement.

Gulzarilal Nanda (1966) - Gulzarilal Nanda took office in 1966 following the death of Lal Bahadur Shastri for 13 days as the acting PM of India.

Indira Gandhi (1966-1977, 1980-1984) - Indira Priyadarshini Gandhi was the first and so far the only woman PM of India who served the second-longest term as a PM. Indira Gandhi also served as the minister of external affairs (1984), minister of defense (1980 - 82), minister of home affairs (1970 - 73), and minister of information and broadcasting (1964 - 66).

She was the first Prime Minister of India who was a member of the Rajya Sabha and was conferred with Bharat Ratna (the only female prime minister). During her tenure, Indo-Pak (1971) war happened, emergency was imposed in 1975 Following Operation Blue Star, she was assassinated in 1984 by her own bodyguard.

Morarji Desai (1977-79) - He belonged to the Janata Party. He was the first non-congress PM. He is the only Indian to receive the highest civilian awards from both India and Pakistan, the Bharat Ratna and Nishan-e-Pakistan (The highest civilian honor in Pakistan). He

was the oldest to become PM and first to resign.

Charan Singh (1979-1980) - Charan Singh was the champion of peasants' rights. Just before he was to prove his majority in Lok Sabha, Indira Gandhi withdrew support to his government, and after just 23 days in power, he resigned. He was the only Prime Minister who had never addressed the Parliament.

Rajiv Gandhi (1984-89) - He took after the assassination of Indira Gandhi in 1984 after the Sikh riots. He served as a pilot for Indian Airlines.

VP Singh (1989-1990) - He is the 12th CM of Uttar Pradesh. From 1984 to 1987, he was the minister of finance and from 1989-90, the minister of defense under PM Rajiv Gandhi when the Bofors scandal surfaced. The Mandal Commission Report for reservation was implemented during his tenure.

Chandra Shekhar (1990-1991) - He headed a minority government of a Janata Dal breakaway faction with the support of Congress to delay the election process.

PV Narasimha Rao (1991-1996) - He was the first PM to come from southern India. Narasimha Rao served as the defense minister from 1993-96 and the minister of external affairs from 1992 to 1994. He was also the home minister under Rajiv Gandhi in 1986.

He is also referred to as Chanakya of Modern India for his ability to get legislation passed even while heading a minority government. The duo of PV Narasimha Rao (PM) and Manmohan Singh (Finance Minister) led the Economic reforms of 1991 and saved India from an Economic crisis.

H. D. Deve Gowda (1996-1997) - Haradanahalli Doddegowda Deve Gowda held the office of the Karnataka chief minister from 1994 to 1996. Deve Gowda was chosen as the PM when no party had won enough seats to form the government and the United Front formed the government with Congress support.

Inder Kumar Gujral (1997-1998) - The 12th PM of India, Inder Kumar Gujral was a participant in the Quit India Movement under Gandhi's leadership. As the minister of external affairs, he is remembered for the Gujral Doctrine (a set of five principles to guide the foreign

policy of India with its immediate neighbors, particularly Pakistan).

Atal Bihari Vajpayee (1996, 1998-99, 1999-2004) - Atal Bihar Vajpayee has served three terms as the PM of India. He was first elected in 1996 as the Prime Minister of India and served for a period of 13 days only. After that for 13 months in 1998 and for five years from 1999 to 2004.

He gave the slogan of Jai Jawaan, Jai Kisaan, Jai Vigyan. Born on Christmas Day, December 25, his birthday is marked in India as Good Governance Day.

Dr. Manmohan Singh (2004-2014) - He has served two complete terms as the PM and headed United Progressive Alliance (UPA) governments two times. A member of the Rajya Sabha, Manmohan Singh was the leader of the upper house from 1998 to 2004. Manmohan Singh was also the 15th governor of the Reserve Bank of India.

Narendra Damodardas Modi (2014 - Present) - The first ever PM to be born after Independence. He also has the distinction of being the longest serving Chief Minister of Gujarat.

Chief Minister

The CM is the **head of the state government**.

Functions of Chief Minister : He acts as vice chairman of the zonal council by rotation for a period of 1 year, chairman of state planning board, member of inter - state and governing council of NITI Aayog, chief spokesman of state government.

He advises the Governor with regard to summoning and proroguing of the sessions of the State Legislature. He can recommend dissolution of the Vidhan Sabha to the Governor at any time. He announces government policies on the floor of the House. He is the crisis manager-in-chief at the political level during emergencies. He is Chairperson of State Disaster Management Authority.

Qualifications : He/she must be an Indian citizen, over 25 years of age, must be a member of his state legislature.

Tenure : The Chief Minister's term is not fixed and he serves at the governor's pleasure. The State Legislative Assembly can remove him by passing a vote of no-confidence against him.

Facts about Chief Ministers of India

Pawan Kumar Chamling (Sikkim) has been India's longest serving chief minister followed by **Jyoti Basu (West Bengal)**, **Naveen Patnaik (Odisha)**.

Of the current chief ministers in India (as of April 2024) Nitish Kumar has taken oath as Chief Minister of Bihar for the highest number of times.

Pema Khandu, the chief minister of Arunachal Pradesh, was born on August 21, 1979, making him the **youngest chief minister in India** at the moment.

Short serving Chief Ministers - Harish Rawat (Uttarakhand), Devendra Fadnavis (Maharashtra), Jagdambika Pal (Uttar Pradesh), all served for one day.

List of Chief Ministers who became Prime Ministers

Morarji Desai - CM of Bombay State.

Charan Singh - CM of the undivided Uttar Pradesh.

V. P. Singh - CM of the Uttar Pradesh

Narasimha Rao - First Prime Minister from South India, CM of Andhra Pradesh.

H.D. Deve Gowda - CM of Karnataka.

Narendra Modi - four-time CM of Gujarat.

Articles related to Chief Minister

Article 163 - The governor is advised by the council of ministers which is headed by the Chief Minister.

Article 164 - The Governor appoints the Chief Minister, who then recommends the appointment of other ministers. The Governor appoints the remaining ministers based on the Chief Minister's advice.

Article 167 - Chief Minister has to communicate all administrative decisions that are taken up by him and the council of ministers to the Governor.

Article 167(a) - To communicate to the Governor of the State all decisions of the council of Ministers relating to the administration of the affairs of the State and proposals for legislation.

Article 167(b) - It mentions the duty of the Chief minister to furnish information relating to the administration of the affairs of the State and proposals for

legislation whenever asked by the Governor.

Article 167(c) - If the Governor so requires, to submit for the consideration of the Council of Ministers any matter on which a decision has been taken by a Minister but which has not been considered by the Council.

Council of Ministers

The Prime Minister and other Ministers are collectively known as the Council of Ministers (CoM). Three categories of the Ministers are Cabinet Ministers, Minister of State and Deputy Ministers. At the top of all these ministers stands the Prime Minister.

Cabinet Ministers - These head the important ministries of the Central government like home, defence, finance, external affairs and so forth. Cabinet is the chief policy formulating body of the Central government.

Ministers of State - These can either be given independent charge of ministries/departments or can be attached to cabinet ministers.

Deputy Ministers - They are attached to the cabinet ministers or ministers of state and assist them in their administrative, political, and parliamentary duties.

At times, the Council Of Ministers may also include a **Deputy Prime Minister**.

91st Constitutional Amendment Act, 2003 - It limits the size of the Council of Ministers (including PM) to **15%** of the total strength of the Lok Sabha. The same is applicable to the Council of Ministers in the State legislature. But, the number of ministers, including the Chief Minister, in a state shall not be less than 12.

Related Articles

Article 74 - There shall be a Council of Ministers with the Prime Minister to aid and advise the President.

Article 74(1) - There shall be a Council of Ministers with the Prime Minister at the head to aid and advise the President who shall, in the exercise of his functions, act in accordance with such advice.

Article 74(2) - The question shall not be inquired into in any court whether the Ministers gave any advice to the

President, and if so what.

Article 75 - Other provisions as to Ministers.

Article 75(1) - The Prime Minister shall be appointed by the President and the other Ministers shall be appointed by the President on the advice of the Prime Minister.

Article 75(2) - The Minister shall hold office during the pleasure of the President.

Article 75(3) - The Council of Ministers shall be collectively responsible to the House of the People.

Article 75(4) - Before a Minister enters upon his office, the President shall administer to him the oaths of office and of secrecy according to the forms set out for the purpose in the Third Schedule.

Article 75(5) - A Minister who for any period of six consecutive months is not a member of either House of Parliament shall at the expiration of that period cease to be a Minister.

Article 75(6) - The salaries and allowances of Ministers shall be such as Parliament may from time to time by law determine and, until Parliament so determines, shall be as specified in the Second Schedule.

Article 77 - Conduct of business of the Government of India.

Article 78 - It shall be the duty of the Prime Minister with regard to the giving of information, etc., to the President.

Article 88 - Every minister shall have the right to speak and take part in the proceedings of either House, any joint sitting of the Houses and any Committee of Parliament of which he may be named a member. But he shall not be entitled to vote.

Responsibility of Ministers

Collective Responsibility : Article 75 clearly states that the Council of Minister is collectively responsible to the Lok Sabha. This means that all the ministers own joint responsibility to the Lok Sabha for all their acts of omission and commission.

Individual Responsibility : It states that the ministers hold office during the pleasure of the President, which means that the President can remove a minister even at a time when the council of

members enjoys the confidence of the Lok Sabha.

Council of Ministers in States

The Council of Ministers in the states is constituted and functions in the same way as the Council of Ministers at the Centre.

Cabinet Committees

Cabinet committees are extra - constitutional in nature. They are not mentioned in the Constitution. However, These are established under the Rules of Business.

There are eight cabinet committees:

Appointments Committee of the Cabinet,
Cabinet Committee on Economic Affairs,
Cabinet Committee on Political Affairs,
Cabinet Committee on Investment and Growth,
Cabinet Committee on Security,
Cabinet Committee on Parliamentary Affairs,
Cabinet Committee on Employment & Skill Development,
Cabinet Committee on Accommodation,

All committees except Cabinet Committee on Accommodation and Cabinet Committee on Parliamentary Affairs are usually headed by the Prime Minister. The Cabinet Committee on Accommodation is headed by the Home Minister and Cabinet Committee on Parliamentary Affairs is headed by the Defence Minister.

Emergency Provisions

The emergency provisions are contained in Part XVIII of the Constitution of India, from (Article 352-360). The Constitution stipulates three types of emergencies : National Emergency, State Emergency, Financial Emergency.

National Emergency

As per Article 352, National emergency can be declared on the basis of war, external aggression or armed rebellion. The Constitution employs the expression proclamation of emergency to denote an emergency of this type.

The 44th Amendment Act changed the cause of National Emergency from internal disturbance to armed rebellion.

Proclamation of emergency : It must be approved by both the houses of

parliament within one month from the date of its issue. If approved by both the houses, the Emergency continues for 6 months and can be extended to an indefinite period with an approval of the Parliament for every six months.

The President can revoke it through a subsequent proclamation without needing parliamentary approval. The emergency must be revoked if the Lok Sabha passes a resolution by a simple majority disapproving its continuation.

During an emergency, the Lok Sabha's tenure can be extended beyond the normal term for up to one year at a time, but not beyond six months after the emergency ends.

Similarly, the state Legislative Assembly's tenure can be extended by the Parliament during a national emergency, with a maximum extension of six months after the emergency ends.

According to Article 358, when a proclamation of National Emergency is made, the six fundamental rights under Article 19 are automatically suspended.

The 44th Amendment Act 1978 laid out that Article 19 can only be suspended when the National Emergency is laid on the grounds of war or external aggression and not in the case of armed rebellion.

The 44th Amendment 1978 Act mandates that the President cannot suspend the right to move the court for the enforcement of Fundamental Rights guaranteed by Article 20 and 21.

This type of emergency has been proclaimed three times so far : 1962 (India-China war), 1971 (India-Pakistan war) and 1975 (On the basis of Internal Disturbance declared by Fakhruddin Ali Ahmed).

The third national emergency of India started at the mid night of 25th June 1975 and was extended till 21 March 1977.

State Emergency

It is the duty of the centre to take over the government of a state in case of failure of constitutional machinery in a state. This is popularly known as President's Rule.

Article 356 empowers the President to issue a proclamation if he is satisfied that a situation has arisen in which the

government of a state cannot be carried on in accordance with the provisions of the constitution.

A proclamation imposing the president's rule must be approved by both the houses of parliament within two months from the date of its issue.

The President acquires extraordinary powers when the President's rule is imposed in a state. He can take up the functions of the state government and powers vested in the governor or any other executive authority in the state. He can declare that the powers of the state legislature are to be exercised by the parliament. He can take all other necessary steps including the suspension of the constitutional provisions relating to any body or authority in the state.

In India, The President's Rule continues for six months, but it can be extended for a maximum period of three years with the approval of the Parliament, every six months.

President's rule was used for the first time in Punjab on 20th June 1951. Manipur has undergone the maximum number (10) of President rule from 1951 to 2019 followed by Uttar Pradesh (09), Punjab (8) and Bihar (8).

Financial Emergency

Article 360 empowers the president to proclaim a Financial Emergency if he is satisfied that a situation has arisen due to which the financial stability or credit of India or any part of its territory is threatened.

A proclamation declaring financial emergency must be approved by both the Houses of Parliament within two months from the date of its issue. Once approved by both the houses of Parliament, the Financial Emergency continues indefinitely till it is revoked. Till now Financial emergency has not been proclaimed in India.

Articles related to Emergency

Article 353 - Effect of Proclamation of Emergency.

Article 354 - Application of the revenue-distribution provisions when an emergency proclamation is in effect.

Article 355 - Every State must be protected by the Union against external

invasion and internal unrest, and every State's government must be run in line with the requirements of this Constitution.

Article 357 - Exercise of legislative powers under Proclamation issued under Article 356.

Article 358 - Suspension of provisions of Article 19 during emergencies.

Article 359 - Suspension of the enforcement of the rights conferred by Part III (except articles 20 and 21) during emergencies.

Supreme Court

The Supreme Court (SC) is the Apex Court, and the highest Court of appeal in India. It is known as the Guardian Of The Constitution. The Federal Court of India was established on **1 October 1937**. After Independence, the Federal Court was replaced by the Supreme Court of India.

The Supreme Court came into existence on 26 January 1950 with the coming into force of the Constitution. On 28 January 1950, the Supreme Court was inaugurated. The inauguration took place in the Chamber of Princes in the old Parliament building where the Federal Court of India sat for 12 years from 1937 to 1950.

The Union Judiciary is mentioned in Chapter IV under Part V (Articles 124-147) of the Indian Constitution.

Strength - The sanctioned strength of the court is 34 judges. Parliament increased the number of judges (including the Chief Justice) from the original 8 in 1950 to 34 in 2019.

Qualification - He must be a citizen of India. He has been for at least five years a Judge of a High Court or of two or more such Courts in succession. He has been for at least ten years an advocate of a High Court or of two or more such Courts in succession; or is in the opinion of the President, a distinguished jurist.

Appointment - The CJI is appointed by the President after consultation with judges of the Supreme Court and high courts. In the Second Judges Case (1993), the Supreme Court ruled that the seniormost judge of the Supreme Court should alone be appointed to the office of the chief justice of India.

The other judges are appointed by the President after consultation with the chief justice and such other judges of the Supreme Court and the high courts as he deems necessary. In the Third Judges case (1998), the Court opined that the consultation process to be adopted by the Chief Justice of India requires 'consultation of plurality judges'. CJI should consult a collegium of four senior most judges of the Supreme Court.

The judge of the Supreme Court shall hold office until he/she attains the age of **65 years**.

Removal and Resignation - A Judge of the Supreme Court shall not be removed from his office except by an order of the President passed after Impeachment by Parliament. A Judge may also resign his office, by submitting his resignation letter to the President.

Related Articles

Article 124 - Establishment and Constitution of Supreme Court.

Article 125 - The Salaries, Pension and Allowances of the Supreme Court Judges are charged upon the Consolidated Fund of India. Its salary is determined by Parliament by law.

Article 126 - Appointment of Acting Chief Justice.

Article 127 - Appointment of Ad Hoc Judges. In the absence of a Quorum; CJI with the previous consent of the President and after consultation of the Chief Justice of the High Court concerned; appoint any Judge of High Court as Ad Hoc Judges of the Supreme Court.

Article 128 - Attendance of retired Judges at sittings of the Supreme Court.

Article 129 - Supreme Court to be a Court Of Record and shall have all the powers of such a court including the power to punish for contempt of itself.

Article 130 - seat of the Supreme Court.

Article 131 - Original Jurisdiction of the Supreme Court.

Article 132 - Appellate Jurisdiction of Supreme Court in appeals from High Court in certain cases.

Article 133 - Appellate jurisdiction of Supreme Court in appeals from High Courts in regard to civil matters.

Article 134 - Appellate jurisdiction of

Supreme Court in regard to Criminal Matters.

Article 136 - Special leave to appeal by the Supreme Court.

Article 137 - Review of Judgements or Orders by the Supreme Court.

Article 138 - Enlargement of the Jurisdiction of the Supreme Court.

Article 139 - Conferment on the Supreme Court of powers to issue writs.

Article 140 - Ancillary powers of the Supreme Court.

Article 141 - Law declared by the Supreme Court to be binding on all courts.

Article 142 - It provides discretionary power to the Supreme Court as it states that the SC in the exercise of its jurisdiction may pass such decree or make such order as is necessary for doing complete justice in any cause or matter pending before it.

Article 143 - Advisory Jurisdiction.

Article 144 - Civil and judicial authorities to act in aid of the Supreme Court.

Article 145 - Rules of Court, etc.

Article 146 - Officers and servants and the expenses of the Supreme Court.

Powers of Supreme Court

It decides the disputes regarding the election of the President and the vice-president.

Original Jurisdiction : It means the cases or matters that are referred to the respective courts at the first instance. For example, disputes between the Government of India and one or more States; or between the Government of India and any State or States on one side and one or more other States on the other; or between two or more States, etc.

Appellate Jurisdiction : The Supreme Court is a court of appeal and hears appeals against the judgements of the lower courts. The Supreme Court is the final court of authority in all cases including Articles 133 and 134. It enjoys a wide appellate jurisdiction which can be classified under four heads: Appeals in constitutional matters, Appeals in civil matters, Appeals in criminal matters, Appeals by special leave.

Advisory Jurisdiction : The President of India to seek an advisory opinion from the Supreme Court in matters of public importance and of any question arising out of pre-constitution, treaty, agreement, engagement, Sanad, or other similar instruments.

Judicial Review : It examines the constitutionality of legislative enactments and executive orders. The grounds of review are limited by Parliamentary legislation or rules made by the Supreme Court.

Power to punish for contempt (civil or criminal) of court with simple imprisonment for 6 months or fine up to Rs. 2000/- Civil contempt means wilful disobedience to any judgment. Criminal contempt means doing any act that lowers the authority of the court or causes interference in judicial proceedings.

Judicial Terms

Judicial Activism - It is the practice of using the courts' authority to examine state actions. The higher judiciary has the power to consider any legislative, executive, or administrative action as unconstitutional and void.

Public Interest Litigation (PIL) - The concept of PIL originated and developed in the USA in the 1960s. Kapila Hingorani is called the 'Mother of PILs'.

It was introduced in India by Justice PN Bhagwati. It means litigation filed in a court of law, for the protection of "Public Interest". Any matter where the interest of the public at large is affected can be redressed by filing a Public Interest Litigation.

High Courts and Other Courts

The high courts of India are the highest courts of appellate jurisdiction in each state and union territory of India.

The Indian High Courts Act 1861 was an act of the Parliament of the United Kingdom to authorize the Crown to create High Courts in the Indian colony. Queen Victoria created the High Courts in Calcutta, Madras, and Bombay through Letters Patent in 1862.

The **Calcutta High Court** is the oldest High Court in India. It has jurisdiction

over the State of West Bengal and the Union Territory of the Andaman and Nicobar Islands.

There are currently **25** high courts in India as of April 2024. Telangana and Andhra Pradesh is the newly formed high court in India.

Judicial Appointment : The Chief Justice of a High Court shall be appointed by the President in consultation with the Chief Justice of India (CJI) and the Governor of the State. In the case of appointment of a Judge other than the Chief Justice, the Chief Justice of the High Court is also consulted.

Collegium System of High Court : A High Court collegium is led by its Chief Justice and four other senior-most judges of that court.

Qualifications : He/she should be an Indian citizen. He/she must have held a judicial office in India for ten years.

Salaries and allowances : The salary, allowances, privileges, leave, and pensions of Supreme Court justices are set by Parliament on a regular basis.

Term of office - He holds office until he attains the age of 62 years. He can resign his office by writing to the president. The 15th Constitutional Amendment Act amended the retirement age of a high court judge from the earlier 60 years to 62 years.

Impeachment : A judge may be removed from office through a motion adopted by Parliament on grounds of 'proved misbehavior or incapacity'. The impeachment process is the same as that of a Supreme Court Judge.

Powers of High Courts

All High Courts have the power to punish all cases of contempt by any person or institution.

High court superintends and controls all the subordinate courts. It can appoint its administration staff and determine their salaries and allowances, and conditions of service.

High Courts have the power of Judicial Review. They have the power to declare any law or ordinance unconstitutional if it is found to be against the Indian Constitution.

Original Jurisdiction - It entails the power of HC to hear disputes in the first instance (applicant can directly go to the High Court) not by the way of appeal.

Matters related to the election of the MLAs and MPs also come under the original jurisdiction of the High Court.

Writ Jurisdiction : Under Article 226, the constitution empowers a High Court to issue writs for the enforcement of the Fundamental Rights and for enforcement of an ordinary legal right.

Appellate Jurisdiction : The High Court hears appeals against the judgments of subordinate courts functioning in its territorial jurisdiction. The High Court has appellate jurisdiction in both civil and criminal matters.

Supervisory Jurisdiction : The High Court has the power of superintendence over all courts and tribunals functioning in its territorial jurisdiction (except military courts or tribunals).

Related Articles

Article 214 : High Courts for States.

Article 215 : High Courts to be courts of record.

Article 216 : Constitution of High Courts.

Article 217 : Appointment and conditions of the office of a Judge of a High Court.

Article 218 : Application of certain provisions relating to the Supreme Court to High Courts.

Article 219 : Oath or affirmation by Judges of High Courts.

Article 220 : Restriction on practice after being a permanent Judge.

Article 221 : Salaries, etc., of Judges.

Article 222 : Transfer of a Judge from one High Court to another.

Article 223 : Appointment of acting Chief Justice.

Article 224 : Appointment of additional and acting Judges.

Articles 224A : Appointment of retired Judges at sittings of High Courts.

Article 225 : Jurisdiction of existing High Courts.

Article 226 : Power of High Courts to issue certain writs.

Article 226A : [Repealed.]

Article 227 : Power of superintendence over all courts by the High Court.

Article 228 : Transfer of certain cases to the High Court.

Article 228A : [Repealed.]

Article 229 : Officers and servants and the expenses of High Courts.

Article 230 : Extension of jurisdiction of High Courts to Union territories.

Article 231 : Establishment of a common High Court for two or more States.

Article 232 : Omitted by the Constitution (7th Amendment Act, 1956).

Article 241 - High Courts for Union territories.

List of High Courts

Bombay (Maharashtra, Dadra and Nagar Haveli, Daman Diu Goa) - It is situated in Mumbai, Its benches are located in Panaji, Aurangabad, and Nagpur.

Jammu and Kashmir & Ladakh High court is situated in Srinagar.

Kolkata (West Bengal, Andaman and Nicobar islands) - It is situated in Kolkata, Its bench located in Port Blair.

Madras (Tamil Nadu, Pondicherry) - It is situated in Chennai, Its bench is located in Madurai.

Allahabad (Uttar Pradesh) - It is situated in Allahabad, Its bench located in Lucknow.

Karnataka - It is situated in Bengaluru. Its benches are located in Dharwad and Gulbarga.

Patna (Bihar), It is situated in Patna.

Guwahati (Assam, Nagaland, Mizoram, Arunachal Pradesh), It is situated in Guwahati.

Odisha - It is situated in Cuttack.

Rajasthan - It is situated in Jodhpur. Its bench is located in Jaipur.

Madhya Pradesh It is in Jabalpur. Its benches are located in Gwalior and Indore.

Kerala (Kerala & Lakshadweep) - It is situated in Ernakulam.

Gujarat - It is situated in Ahmedabad.

Delhi - Delhi is the only union territory which has its own High Court. The High Court of Delhi was established in 1966. The High Court of Delhi continued to exercise jurisdiction over Himachal Pradesh until the State of Himachal Pradesh Act, 1970 was enforced on 25th January, 1971.

Himachal Pradesh - It is situated in Shimla.

Punjab & Haryana (Punjab, Haryana and Chandigarh), It is situated in Chandigarh.

Sikkim (Sikkim), It is situated in Gangtok.

Chattisgarh - It is located in Bilaspur.

Uttarakhand - It is situated in Nainital.

Jharkhand - located in Ranchi.

Tripura - It is situated in Agartala.

Manipur - situated in Imphal.

Meghalaya - located in Shillong.

Telangana - It is situated in Hyderabad.

Andhra Pradesh - situated in Amravati.

Subordinate Court

Articles 233 to 237 in **Part VI** of the constitution make provisions to regulate the organization of subordinate courts and to ensure their independence from the executive

Appointment of District judges: The appointments are made by the governor in consultation with the high court. Article 233 deals with it.

Article 234 - Recruitment of persons other than district judges to the judicial service.

Qualification of District Judges : He should not already be in the service of the central or the state government. He should have been an advocate or a pleader for seven years. He should be recommended by the High Court for appointment

Control over subordinate courts : It is vested with the concerned state high court.

Structure and Jurisdiction : There are three-tiers of civil and criminal courts below the high court.

Criminal Courts - Sessions judge's court (district level), Chief Judicial Magistrate court, and Judicial magistrate court.

Civil Court - District Judge, Subordinate Judge's court and Munsiff's Court.

The **district judge** is the highest judicial authority in the District. He possesses original and appellate jurisdiction in both Civil as well as Criminal Matters.

When he deals with Civil cases, he is known as the District Judge and when he

hears the Criminal cases, he is called the Sessions Judge.

National Legal Services Authority

It has been constituted under the Legal Services Authorities Act, 1987 to monitor and evaluate implementation of legal aid programmes and to lay down policies and principles for making legal services available under the Act.

Family Court Act

The Family Courts Act, 1984 was enacted to provide for the establishment of Family Courts with a view to promote conciliation and secure speedy settlement of disputes relating to marriage and family affairs.

It states that a family court shall have and can exercise the jurisdiction exercisable by any district court or any subordinate civil court under any law.

The jurisdiction is exercisable by the magistrate of the First class relating to the order for maintenance of wife, children, and parents.

To deal with the property matter of the parties of the marriage, the family court exercises its jurisdiction by satisfying two conditions : The property dispute must be between the parties to the marriage only and the dispute should be with respect to the property of either of the parties.

The family court has no jurisdiction to deal with the property matters between the members of the joint family.

Family Courts (Amendment) Bill 2022 seeks to amend the Family Courts Act, 1984 to establish family courts in Himachal Pradesh and Nagaland.

Tribunals

Tribunal is a quasi-judicial institution to solve administrative or tax-related disputes. Part XIV-A is entitled as Tribunals and consists of two Articles namely Article 323A and 323B.

Article 323 A - It deals with Administrative Tribunals.

Article 323-B - It deals with tribunals for other matters.

Central Administrative Tribunal (CAT) : It was set up in 1985. It holds original

jurisdiction over recruitment and service matters concerning public servants within its scope, including all-India services, central civil services, central civil posts, and civilian employees of defense services.

National Green Tribunal (NGT) : It was established in 2010 under the National Green Tribunal Act 2010 as a statutory body. It was set up for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources.

New Delhi is the Principal Place of Sitting of the Tribunal and Bhopal, Pune, Kolkata and Chennai shall be the other four places of sitting of the Tribunal.

Lokpal & Lokayuktas

Lokpal

A Lokpal is an **anti-corruption authority** or body of **ombudsman** who represents the public interest in the Republic of India. In **1809**, the institution of the ombudsman was inaugurated officially in **Sweden**.

In India, the concept of a constitutional ombudsman was first proposed by the then law minister Ashok Kumar Sen in parliament in the early 1960s. In 2005, the 2nd Administrative Reform Commission (ARC) recommended the appointment of the Lokpal and Lokayuktas. The term Lokpal and Lokayukta were coined by **Dr. L. M. Singhvi**.

India Against Corruption (IAC) movement, led by Anna Hazare, resulted in the introduction of the Lokpal and Lokayukta Bill, 2013 in Parliament. The Bill received assent from the President on 1 January 2014 and came into force on 16 January 2014.

Lokpal is a multi-member body that consists of one chairperson and a maximum of **8 members (1+8)**. The Chairperson of the Lokpal should be either the former Chief Justice of India or the former Judge of the Supreme Court or an eminent person.

The members are appointed by the **President** on the recommendation of a Selection Committee.

The term of office for Lokpal Chairman and Members is 5 years or till the age of

70 years. They are not eligible for reappointment, and cannot hold any constitutional or governmental office.

The Jurisdiction of Lokpal includes the Prime Minister, Ministers, members of Parliament, Groups A, B, C, and D officers, and officials of the Central Government.

Former Supreme Court judge Justice Pinaki Chandra Ghose was appointed as the country's first Lokpal.

Lokayukta

Maharashtra was the first state that established its own Lokayukta in **1971**.

The Lokayukta is appointed by the Governor of state or Lieutenant Governor of the UT.

In most of the states, the term of office fixed for lokayukta is of 5 years duration or 65 years of age, whichever is earlier. He is not eligible for reappointment for a second term.

Lokpal and Lokayukta (Amendment) Act 2016

This act amended the Lokpal and Lokayukta Act 2013. It eliminates the 30-day time frame in section 44, and public officials will report their assets and liabilities as outlined by the government.

Panchayats and Municipalities

The Indian Constitution initially established a two-tier federal system, consisting of the Union and State governments. However, the 73rd and 74th Constitutional Amendment Acts of 1992 further democratized the structure by introducing a third tier of local self-government through Panchayats and Municipalities. Lord Rippon is called the father of the local self-government.

Panchayats

It is the system of local self-government of villages in **rural India**. **Part IX** of the Indian Constitution is related to the Panchayats.

Rajasthan, was the first state to implement Panchayati Raj in India (on October 2nd, 1959, in Nagaur district), followed by Andhra Pradesh later in 1959. India celebrates 24th April every

year as National Panchayati Raj Day.

The 11th Schedule contains the provisions that specify the powers, authority and responsibilities of Panchayats. This schedule was added by the 73rd Amendment Act of 1992. It has 29 matters.

Currently, the Panchayati Raj system exists in all states except Nagaland, Meghalaya, and Mizoram, and in all Union Territories except Delhi.

Evolution of Panchayati Raj

The following committees have been formed for the efficient functioning of the Panchayati Raj :

Balwant Rai Mehta committee (1957) : The committee suggested three-tier PRIs, namely, Grama Panchayats (GPs) at the village level, Panchayat Samiti (PSs) at the block level, and Zilla Parishad (ZPs) at the district level. District Collector to be made the chairman of the Zila Parishad.

Ashok Mehta Committee (1977) : It suggested a two-tier system, that is, zila parishad at the district level, and below it, the mandal panchayat. Zila Parishad should be the executive body and responsible for planning at the district level.

G.V.K Rao Committee (1985) : The Zila Parishad should be the principal body in India's Panchayati Raj System, managing developmental programs at the district level. Post of District Development Commissioner to be created. He will be the chief executive officer of the Zila Parishad.

L.M Singhvi Committee (1986) : The Panchayati Raj Institutions (PRIs) should be constitutionally recognised. Judicial tribunals should be established to adjudicate judicial matters related to PRIs.

It also recommended that one-third of the total seats at all levels of rural and urban bodies have to be reserved for female candidates. Bihar is the first State to offer a 50% reservation for women in Panchayati Raj.

P.K.Thungon Committee (1988) : It recommended constitutional recognition for the local government bodies. A constitutional amendment to provide for periodic elections to local government

institutions, and enlistment of appropriate functions to them, along with funds, was recommended.

Gadgil Committee (1988) : The committee on Policy and Programmes was constituted in 1988 by the Congress Party under the chairmanship of V.N. Gadgil. It led to the establishment of a State Election Commission to oversee the conduct of Panchayat elections.

Features of Panchayati Raj

Gram Sabha - It is a village assembly consisting of all the registered voters in the area of a Panchayat. It performs the functions assigned by the State Legislature.

Election - All the members shall be elected directly by the people. Election is conducted by the State Election Commission.

An election to constitute a Panchayat shall be completed before the expiration of a period of **6 months** from the date of its dissolution.

Three-tier system - Panchayati raj institutions exist at the village, intermediate and district levels.

Duration of Panchayats - Every level of panchayat is given a five-year term of office under the statute.

Powers and functions- It plays a vital role in the village local government to develop the village in special areas, such as health, child development, women participation in local government, agricultural development, etc.

Finances - The state legislature may authorize a panchayat to levy, collect and appropriate taxes.

Finance Commission - The governor has the authority to establish an annual finance commission to review the panchayats' financial standing.

Articles Related to Panchayati Raj System

243 - Definitions

243A - Gram Sabha

243B - Constitution of Panchayats

243C - Composition of Panchayats

243D - Reservation of seats

243E - Duration of Panchayats

243F - Disqualifications for membership

243G - Powers, authority and responsibilities of Panchayats.

243H - Powers to impose taxes by, and Funds of, the Panchayats

243I - Constitution of Finance Commission to review financial position

243J - Audit of accounts of Panchayats

243K - Elections to the Panchayats

243L - Application to Union territories

243M - Part not to apply to certain areas.

243N - Continuance of existing laws and Panchayats.

243O - Bar to interference by courts in electoral matters.

Municipalities

It is the system of local self-government of town or city in urban India. **Part IX-A** of the Indian Constitution is related to the Municipalities.

The 74th Constitutional Amendment Act of 1992 elevated Municipalities, now known as Urban Local Governments (ULBs), to the status of constitutional entities. This act empowered ULBs to handle the 18 functions listed in the 12th Schedule of the Indian Constitution.

With a term of five years and led by a **Mayor** who presides over council meetings, Municipal Corporations are the largest category of urban local government in India. The minimum age to contest elections to any municipality is **21 years**.

Article 243Q - It provides for the establishment of 3 kinds of Municipalities in every state depending upon the size and area.

Nagar Panchayat : It is for those areas which are transitional areas i.e. transiting from Rural Areas to Urban areas.

Municipal Council : It is for smaller urban areas.

Municipal Corporation : It is for larger urban areas.

Structure of Urban Local Government in India :

It consists of eight types of Urban local bodies.

Municipal Corporations : They are usually found in big cities such as Bangalore, Delhi, Mumbai, Kolkata, etc.

Municipality : The smaller cities tend to have the provision of municipalities. The

Municipalities are often called upon by other names such as the municipal council, municipal committee, municipal board.

Notified Area Committees : They are set up for the fast-developing towns and the towns lacking the basic amenities. All the members of the notified area committee are nominated by the state government.

Town Area Committee : It is found in small towns. It has minimal authority such as street lighting, drainage roads and conservancy.

Cantonment Board : It is usually set up for a civilian population living in the cantonment area. It is created and run by the central government.

Township : It is another form of urban government to provide basic facilities to the staff and workers living in the colonies established near the plant. It has no elected members and is merely an extension of the bureaucratic structure.

Port Trust : Port trusts are established in the port areas such as Mumbai, Chennai, Kolkata, etc. It manages and takes care of the port. It also provides basic civic amenities to the people living in that area.

Special Purpose Agency : These agencies undertake the designated activities or specific functions belonging to the municipal corporations or municipalities.

Articles related to the Municipalities

243P - Definitions.

243Q - Constitution of Municipalities.

243R - Composition of Municipalities.

243S - Constitution and composition of Wards Committees.

243T - Reservation of seats.

243U - Duration of Municipalities

243V - Disqualifications for membership.

243W - Powers, authority and responsibilities of Municipalities.

243X - Power to impose taxes by, and Funds of, the Municipalities.

243Y - Finance Commission.

243Z - Audit of accounts of Municipalities.

243 ZA - Elections to the Municipalities.

243ZB - Application to Union territories.

243ZC - Part not to apply to certain areas.

243ZD - Committee for district planning.

243ZE - Committee for Metropolitan planning.

243ZF - Continuance of existing laws and Municipalities.

243ZG - Bar to interference by courts in electoral matters.

Committees and Commissions

Rajamannar Commission : It was set up by the DMK Government of Tamil Nadu in 1969 under the Chairmanship of Dr P.V. Rajamannar to look at the overall issue of Centre-state ties.

Sarkaria Commission : It was set up in 1983 and submitted its report in 1988. It was created to explore the relationship between the states and the central government. It recommended the formation of the Inter State Council (Article 263).

Punchhi Commission : It was constituted by the Union Government in April 2007 under the chairmanship of former Chief Justice of India (CJI) Madan Mohan Punchhi. The Commission presented its report to the government in March 2010.

Key Recommendations of Punchhi Commission - Setting up of National Integration Council, State's consultation on subjects in the Concurrent List, regarding appointment and Removal of Governors, regarding appointment of Chief Ministers, and about Union's Power of Making Treaties.

Tarapore Committee : It was constituted in 1997 and headed by S. S Tarapore . It defined Customer Acquisition Cost (CAC) as the freedom to convert local financial assets with foreign financial assets and vice-versa at market determined rates of exchange.

L.C.Gupta Committee : It was formed in November 1996 for the regulation of derivatives trading.

YV Reddy Commission : It was set up in 2001 and had strongly recommended slashing administered rates on small savings.

Kelkar Committee : It was set up by the central government and headed by Vijay Kelkar. It was set up to study and evaluate the extant public-private

partnership (PPP) model in India. The committee submitted its report to the Finance Ministry in November 2015.

National Population Commission : It is chaired by the Prime Minister. It was constituted on 11 May, 2000 to review, monitor and give directions for the implementation of the National Population Policy (NPP), 2000.

Indrajit Gupta Committee : The 8-member committee that was set up by the all-party conference in May 1998, submitted its report in January 1999. It was related to electoral reforms.

Some of the key classic committees towards electoral reforms in the country include Tarkunde Committee Report (1975), Goswami Committee Report (1990), Election Commission's recommendations (1998), Indrajit Gupta Committee Report (1998).

Dinesh Goswami Committee (1990) : It was concerned with electoral reforms. It recommended government funding of parliamentary elections.

Kothari Committee : The Government of India established an ad hoc commission to investigate all aspects of the Indian educational system and create a general framework for education. Its head was Daulat Singh Kothari.

T.S.R Subramanian committee : The Committee for Evolution of the New Education Policy (NEP) (Chaired by Mr. T. S. R Subramanian) submitted its report on May 7, 2016.

The Committee was constituted under the Ministry of Human Resource Development (MHRD) in October, 2015. The report proposes an Education Policy, which seeks to address challenges faced by the current education system.

Justice B. N. Srikrishna Commission : It was constituted by the Government of Maharashtra for investigating the causes of the Bombay riots.

MN Venkatachaliah Committee : This National Commission (NCRWC) was created to review the working of the constitution. It was established by a resolution of the Atal Bihari Vajpayee-led NDA (National Democratic Alliance) government on 22 February 2000 to suggest possible amendments to the Constitution of India.

UC Banerjee Committee : It was constituted to probe 2002 Godhra incident.

Advisory Planning Board: It was set up in 1946 by the Interim Government of India under the chairmanship of K.C. Niyogi. It gave recommendations for the establishment of the Planning Commission.

Malimath Committee (2000) : It was headed by Justice V.S. Malimath. Its report was related to the Criminal Justice System.

The Committee of Administrators (COA) : It was appointed by the Supreme Court of India in 2017, to oversee the implementation of the Lodha reforms in the Board of Control for Cricket in India (BCCI).

Tapas Majumdar Committee (1999) was established, in which the insertion of Article 21-A was included. In 2002, by the 86th Constitutional Amendment Act, the Right to Education was made a fundamental right in Part III of the Constitution.

J S Verma committee : It was appointed to operationalise the suggestions to teach Fundamental Duties to the citizens of the country.

Election Commission

The Election Commission of India (ECI) is an independent, constitutional body. It was established by the Constitution of India to conduct and regulate elections in the country.

Article 324 of the Constitution provides that the power of superintendence, direction, and control of elections to parliament, state legislatures, the office of the president of India, and the office of vice-president of India shall be vested in the election commission. The Election Commission is an all-India body.

The Election Commission was a single-member body when it was set up in 1950 till October 1989, with only the Chief Election Commissioner as its only member. After the Election Commissioner Amendment Act 1989, it has been made a multi-member body. It is a three-member commission with one Chief Election Commissioner and two election commissioners.

The headquarters of the Election

Commission is located in New Delhi which is operated by the Secretariat.

Process of Appointment : The President appoints the Chief Election Commissioner and two other Election Commissioners.

Sukumar Sen was the first election commissioner of India. **V. S. Ramadevi** was the first woman to become Chief Election Commissioner of India.

Term of Office : Chief Election Commissioner shall hold office for a term of six years, or up to the age of 65 years, whichever is earlier, from the date on which he/she assumes his/her office.

Process of Impeachment : The Chief Election Commissioner of India can be removed from office in a manner similar to the removal of a judge of the Supreme Court of India on the grounds of proven misbehavior or incapacity. Other Election Commissioners can be removed by the President of India on the recommendation of the Chief Election Commissioner.

NOTA (None of the above) : This was an option introduced in India following a 2013 Supreme Court directive in the People's Union for Civil Liberties v Union of India. Its symbol was introduced in 2015.

State Election Commission

The State Election Commission in each state is responsible for conducting elections to Panchayati Raj institutions and Municipalities. Appointed by the governor, this commission ensures the smooth functioning of the electoral process. Article 243K(1) mandates the establishment of a State Election Commission.

At the state level, the chief electoral officer, appointed by the chief election commissioner in consultation with the state government, assists the Election Commission. Additionally, at the district level, the district collector serves as the district returning officer, overseeing the electoral process within their jurisdiction. This hierarchical setup ensures the smooth conduct of elections to local bodies across the state.

Related Articles

Articles 324 - Superintendence, direction and control of elections to be vested in an Election Commission.

Articles 325 - No person to be ineligible for inclusion in, or to claim to be included in a special, electoral roll on grounds of religion, race, caste or sex.

Articles 326 - Elections to the House of the People and to the Legislative Assemblies of States to be on the basis of adult suffrage.

Articles 327 - Power of Parliament to make provision with respect to elections to Legislatures.

Articles 328 - Power of Legislature of a State to make provision with respect to elections to such Legislature.

Articles 329 - Bar to interference by courts in electoral matters.

Electoral Bonds

Electoral bonds are an instrument through which anyone can donate money to political parties. Only political parties registered under Section 29A of the Representation of the People Act, 1951 and which secured not less than 1% of votes polled in the last general election to the House of the People or the Legislative Assembly of the State, were eligible to receive electoral bonds.

They were struck down as unconstitutional by the Supreme Court on 15 February 2024.

Model Code of Conduct

The Model Code of Conduct, issued by the Election Commission of India, provides guidelines for political parties and candidates during elections. It covers various aspects such as speeches, polling day, booths, portfolios, manifestos, processions, and general conduct.

Universal Adult Franchise

Article 326 defines a universal adult franchise as the basis for elections to all levels of the elected government.

The **universal Adult Franchise** refers to all citizens who are 18 years and above irrespective of their caste or education, religion, color, race, and economic conditions are free to vote. It is based on the idea of equality.

Manipur was the first part of India to hold an election based on universal adult franchise.

EVM and VVPAT

India has been the first country to adopt Electronic Voting at such a large scale. Electronic Voting Machines (EVM) were introduced by the Election Commission in order to reduce malpractices and improve efficiency.

They were tried for the first time on an experimental basis for the 1982 Kerala State Legislative Assembly Elections.

Voter verifiable paper audit trail (VVPAT) is an independent verification system for voting machines as it allows voters to verify whether their vote was cast correctly or not. It was first used in Noksen (Assembly Constituency) of Nagaland in September 2013.

Political Parties Registration Tracking Management System (PPRTMS)

It was implemented by the Election Commission. In this system, the applicant, who is applying for party registration from 1st January 2020 onwards can track the progress of his/her application and get status updates.

Comptroller and Auditor General (CAG)

The Constitution of India provides for an independent office of the CAG in Chapter V under Part V (The Union), specifically in Articles 148-151.

He is the head of the Indian Audit and Accounts Department. He is the guardian of the public purse and controls the entire financial system of the country at both levels - the centre and state.

The first Auditor General (Sir Edward Drummond) was appointed in 1860 and had both accounting and auditing functions. The Government of India Act, of 1919 established the statutory position of the 'Auditor General in India', responsible for the audit of expenditure in India from the revenues of India. The Government of India Act, 1935, outlined appointment and service conditions for the Auditor General. The first CAG after independence was **V. Narhari Rao**.

He is appointed by the President. He can vacate the office on attaining the age of 65 years or 6-year term, whichever is

earlier. He can resign by addressing a letter to the President. After retiring or resigning from the position of CAG, they are ineligible to hold any government office at the central or state level.

Constitutional Provisions

Article 148 (1) - There shall be a Comptroller and Auditor-General of India who shall be appointed by the President by warrant under his hand and seal and shall only be removed from office in like manner and on like grounds as a Judge of the Supreme Court.

Article 148 (2) - Oath of office before President to CAG.

Article 148 (3) - Salary and other conditions of service of CAG.

Article 149 - Duties and Powers of the Comptroller and Auditor-General of India.

Article 150 - The accounts of the Union and of the States shall be kept in such form as the President may, on the advice of the CAG, prescribe.

Article 151 (1) - The reports of the Comptroller and Auditor-General of India relating to the accounts of the Union shall be submitted to the President, who shall cause them to be laid before each House of Parliament.

Article 151 (2) - The reports of the CAG of India relating to the accounts of a State shall be submitted to the Governor of the State.

Article 279 - Calculation of **Net proceeds** is ascertained and certified by the Comptroller and Auditor-General of India, whose certificate is final.

Roles and Functions

He audits expenses from the Consolidated Fund of India, each state's Consolidated Fund, and Union Territories with a legislative assembly. He also audits spending from the Contingency Fund of India, Public Account of India, and equivalent funds in states.

He advises the President with regard to the prescription of the form in which the accounts of the Centre and States shall be kept.

CAG also acts as a guide, friend and philosopher of the Public Accounts Committee of the Parliament.

Attorney General and Advocate General

The Attorney General and Advocate General play crucial roles in India's legal system at the Center and State. While the Attorney General champions legal matters for the nation, the Advocate General safeguards the legal interests of their respective state.

Attorney General

The Attorney General of India is the highest law officer in the country. It is a constitutional post. The First Attorney General of independent India was **Motilal Chimanlal Setalvad**.

There are other 2 law officers, the Solicitor General of India and an Additional Solicitor General of India (both of them are non-constitutional posts), considered as second law officers of the country. They assist the Attorney General. There is no mention of solicitor general in the constitution.

The Attorney General is appointed by the President. The term of office is not fixed by the Constitution.

Qualification : The person appointed must be a citizen of India. He must be qualified to be a Supreme Court judge, i.e. he must either be a judge of a High Court for **five** years or an **advocate** in a High Court for **10 years** or an eminent jurist in the President's opinion.

Removal : The Constitution does not contain the procedure and grounds for his removal.

Power and Functions : He has the right of audience in all courts of the country. He/she enjoys all the immunities and special privileges available to an MP. He/she has the **right to participate** in the proceedings of both the houses without the right to vote and can be designated as a member. He/she is not a Government servant & not debarred from private legal practice.

Related Articles

Article 76 (1) - Appointment of Attorney-General for India.

Article 76(2) - It shall be the duty of the Attorney General to give advice to the Government of India upon such legal matters, and to perform such other duties of a legal character, as may from

time to time be referred or assigned to him by the President and to discharge the functions conferred on him by or under this Constitution or any other law for the time being in force.

Article 76 (3) - Attorney-General shall have right of audience in all courts in the territory of India.

Article 76 (4) - Attorney-General shall hold office during the pleasure of the President, and shall receive such remuneration as the President may determine.

Article 105 (1) - Guarantees freedom of speech in Parliament. This extends to the Attorney General if they are addressing the House.

Advocate General

The Advocate General of State is the highest law enforcement official in the state. He is appointed by the Governor.

Qualification : He must be an Indian citizen who has served as a judge for ten years or as a high court advocate for ten years.

Removal : He Holds office during the Governor's pleasure.

Power and Functions : They are in charge of helping the state government with legal matters, and they guard and protect the state government's appeals.

Related Articles

Article 165 - Office of the Advocate General of the State

Article 177 - Advocate General's Rights concerning the State Legislature and Its Proceedings.

Article 194 - Advocate General's power privileges and immunities within the State Legislature.

National Commission for Backward Classes (NCBC)

NCBC is an Indian constitutional body under the Ministry of Social Justice and Empowerment. It was established on August 14, 1993, under the National Commission for Backward Classes Act, 1993. It was granted constitutional status through the **102nd Amendment Act** of the Indian Constitution in 2018.

The Government of India established two Backward Class Commissions in the mid-20th century. The first commission, led by Kaka Kalelkar, was set up in the 1950s. The second commission, headed by B.P. Mandal (Babu Bindheshwari Prasad Mandal), came in the 1970s.

First National Commission of Backward Classes (Kalelkar Commission) : The First Commission of Backward Classes was set up by a presidential order on **29th Jan 1953** under the Chairmanship of Shri Kaka Kalelkar. It submitted its report on **30th March 1955**. In its report, it prepared a list of 2339 backward castes or communities for the entire country and 837 of these were classified as Most backward. The report was not implemented by the Government.

Second National Commission of Backward Classes (Mandal Commission) : Janta party announced the formation of a second backward class commission in **1978**. At that time, Morarji Desai was the Prime Minister of India. The commission was formed in **1979** to investigate the conditions of socially and educationally backward classes within the territory of India. It submitted its report in **December 1980**. In its report, the commission identified **3743** castes as backward.

Building on these efforts, the landmark **Indra Sawhney case of 1992** came before the Supreme Court. The court's direction led the parliament to pass the National Commission for Backward Classes Act in 1993. This act established the National Commission for Backward Classes (NCBC) in 1993.

Members : The Commission consists of five members which includes a Chairperson, a Vice-Chairperson and three Members. All five members are appointed by the President of India. The conditions of service and tenure of office of the Members are determined by the President.

Powers and Functions : Commission investigates and monitors all matters relating to the constitutional and other legal safeguards for the socially and educationally backward classes. It has all the powers of a civil court.

Related Articles

Article 340 - Appointment of a Commission to investigate the conditions of backward classes

102nd Constitution Amendment Act inserted new Articles 338 B and 342 A.

Article 338B - There shall be a Commission for the socially and educationally backward classes to be known as the National Commission for Backward Classes.

Article 342 A empowers the President, after consultation with the Governor, to specify socially and educationally backward classes in various states and union territories.

Union Public Service Commission

It is the central recruiting agency in India. It is an independent constitutional body.

Article 315 to 323 in Part XIV of the Indian Constitution lays the foundation for both the Union Public Service Commission (UPSC) and State Public Service Commissions (SPSCs).

It was established on **1 October 1926** as Public Service Commission, it was later reconstituted as Federal Public Service Commission by the Government of India Act 1935 and renamed as Union Public Service Commission after the independence by virtue of Clause (1) of Article 378 of the Constitution in 1950. The first chairman of UPSC in independent India was H. K. Kripalani.

Related Articles

Article 315 (1) - Public Service Commissions for the Union and for the States.

Article 315 (2) - Two or more States may agree that there shall be one Public Service Commission for that group of States.

Article 315 (3) - Any such law as aforesaid may contain such incidental and consequential provisions as may be necessary or desirable for giving effect to the purposes of the law.

Article 315 (4) - UPSC can, if requested by a state government, conduct examinations and interviews to select candidates for state government jobs.

Article 316 - The Chairman and other members of the Union Public Service Commission shall be appointed by the President.

Article 317 - The Chairman or any other member of a Public Service Commission

shall only be removed from their office by order of the President on the ground of "misbehavior" after the Supreme Court, on reference being made to it by the President, has, on inquiry reported that the Chairman or such other member ought to be removed.

Article 318 - The President is empowered to determine the number of members of the Commission and their conditions of service.

Article 319 - A person who holds office as Chairman shall, on the expiration of his term of office, be ineligible for re-appointment to that office.

Article 320 - It shall be the duty of the Union Public Service Commissions to conduct examinations for appointments to the services of the Union. It shall also assist two or more States, if requested so.

Article 321 - An Act made by Parliament may provide for the exercise of additional functions by the Union Public Service Commission services of the Union.

Article 322 - The expenses of the Union Public Service Commission, including any salaries, allowances and pensions payable to or in respect of the members or staff of the Commission, shall be charged on the Consolidated Fund of India.

Article 323 - It will be the duty of the Union Commission to annually present a report to the President of the work done by the Commission. On receipt of such a report, the President shall present a copy before each House of Parliament.

Term of Office - Any member of the UPSC shall hold office for a term of **six years** or till the age of **65 years**, whichever is earlier.

Reappointment - The chairman and the members of UPSC are ineligible for reappointment.

Resignation - A member of the Union Public Service Commission may resign from his/her office by submitting a written resignation to the President of India.

State Public Service Commission

A State Public Service Commission (SPSC) comprises a chairman and other members appointed by the **Governor** of the state.

Term of Office - A member of the SPSC shall hold office for a term of **six years** or till the age of **62 years**, whichever is earlier.

Reappointment - The chairman and the members of UPSC are ineligible for reappointment. They can be appointed to other state PSC and UPSC.

Resignation - The members may resign from his/her office by submitting a written resignation to the Governor of the State.

Removal/Suspension of Members - The Chairman or any other member of SPSC although appointed by the Governor but shall only be removed from his/her office by order of the President of India.

The SPSC shall present the annual report of its work to the Governor of the State.

Joint State Public Service Commission (JSPSC)

It can be created by an act of Parliament at the request of the state legislatures concerned. It is a statutory body.

The number of members of a JSPSC and their conditions of service are determined by the President. A JSPSC presents its annual performance report to each of the concerned State Governors. Each governor places the report before the State Legislature.

Appointment of Members - The Chairman of JSPSC is appointed by the President.

Term of Office - A member of a Public Service Commission shall hold office for a term of six years or until they reach the age of 62, whichever comes first.

Resignation - They can also resign from their offices at any time by submitting their resignation letters to the President.

Removal/Suspension of Members - They can be suspended or removed by the President.

NOTE - Mission Karmayogi or National Programme for Civil Services Capacity Building (NPCSCB) was launched by the Union Cabinet and was approved in September 2020. The programme aims at building a future-ready civil service with the right attitude, skills, and knowledge, aligned to the vision of New India.

National Human Rights Commission (NHRC)

The National Human Rights Commission is an independent statutory body established to promote and protect human rights in India. It is a watchdog of human rights in the country.

It was established on **12th October 1993** under the 'Protection of Human Rights Act' (PHRA), 1993. The act was amended by the Protection of Human Rights (Amendment) Act, 2006 and Human Rights (Amendment) Act, 2019.

Composition of NHRC : It is a multi-member body consisting of a chairman, five full-time members, and seven deemed members.

The chairman should be a retired chief justice of India or a judge of the Supreme Court. The members should be a serving or retired judge of the Supreme Court, a serving or retired chief justice of a high court.

Appointment : The chairman and members are appointed by the **President** on the recommendations of a **six-member committee** consisting of the Prime Minister as its head, the Speaker of the Lok Sabha, the Deputy Chairman of the Rajya Sabha, leaders of the Opposition in both the Houses of Parliament and the Union Home Minister.

Term of office : The chairman and members hold office for a term of three years or until they attain the age of 70 years, whichever is earlier. After their tenure, the chairperson and members are not eligible for further employment under the Central or a state government.

Removal : The President can remove the chairman or any member from the office.

Role and Function : It has all the powers of a **civil court** and its proceedings have a judicial character. The commission is not empowered to inquire into any matter after the expiry of one year from the date on which the act constituting violation of human rights is alleged to have been committed.

The commission can recommend action but cannot punish or grant compensation. It does not cover private parties. Its jurisdiction is limited in the case of armed forces.

The commission submits its annual or special reports to the Central government and to the state government concerned.

State Human Rights Commission

The State Human Rights Commission (SHRC) in India is a statutory body. It serves as a watchdog to ensure the protection and promotion of human rights within each state. It was established as per the provisions of the Protection of Human Rights Act (PHRA), 1993.

A State Human Rights Commission can inquire into violation of human rights only in respect of subjects mentioned in the State List (List-II) and the Concurrent List (List-III) of the Seventh Schedule of the Constitution.

Composition : The State Human Rights Commission is a multi-member body consisting of a chairperson and two members. The chairperson should be a retired Chief Justice or a Judge of a High Court and members should be a serving or retired judge of a High Court or a District Judge.

Appointment : The chairperson and members are appointed by the Governor on the recommendations of a committee consisting of the chief minister as its head, the speaker of the Legislative Assembly, the state home minister and the leader of the opposition in the Legislative Assembly. In the case of a state having a Legislative Council, the chairman of the Council and the leader of the opposition in the Council would also be the members of the committee.

Term of office : The chairperson and members hold office for a term of three years or until they attain the age of 70 years, whichever is earlier.

Removal : They can be removed by the President.

Role and Function : It has all the powers of a civil court and its proceedings have a judicial character. The commission can recommend action but cannot punish or grant compensation.

The Commission submits its annual or special reports to the state government.

Government Bodies

The various types of Government bodies are -

Constitutional Bodies : They are specifically mentioned in the Constitution, meaning they have dedicated articles. Any change in the mechanism of these bodies would require a constitutional amendment. Some Constitutional bodies are as follows - Finance Commission, UPSC, Election Commission, CAG, National Commissions for SCs and STs, etc.

Statutory Bodies : They are created by an Act of Parliament. Some Statutory bodies are as follows - SEBI, NHRC, National Green Tribunal, Central Vigilance Commission, Competition Commission of India, National Commission for Women etc.

Regulatory Bodies : Regulatory bodies are public or government agencies responsible for exercising autonomous authority over some area of human activity in a regulatory or supervisory capacity. They are established by legislative acts. Some Regulatory Bodies are as follows - RBI, BCCI, TRAI, IRDAI, Central Board of Film Certification, National Housing Bank etc.

Executive Bodies : They are neither mentioned in the Constitution nor established by an act of Parliament. They are formed by executive resolution or action, which means that they are formed by the government's action only. Some Executive bodies are Niti Ayog, National Development Council, Central Bureau of Investigation etc.

Important Government Bodies

National Commission for Women : The National Commission for Women was set up as a statutory body in January 1992 under the National Commission for Women Act, 1990 to review the Constitutional and Legal safeguards for women, recommend remedial legislative measures, facilitate redressal of grievances and advise the Government on all policy matters affecting women.

National Commission for Scheduled Castes (NCSC): NCSC is a constitutional body that works to safeguard the interests of the scheduled castes (SC) in India. Article 338 of the constitution of India deals with this commission. Its structure consists of Chairperson,

Vice-chairperson and 3 other members. All of them are appointed by the President by warrant under his hand and seal.

It investigates and monitors all matters relating to safeguards provided for SC, to inquire into specific complaints and to participate and advise on the planning process of their socio-economic development etc.

National Commission for Scheduled Tribes (NCST): National Commission for Scheduled Tribes is a constitutional body set up on 19th February, 2004 by amending Article 338 and by inserting a new article 338A in the Constitution through the 89th Constitution Amendment Act, 2003.

Article 338A gives powers to the NCST to oversee the implementation of various safeguards provided to STs under the Constitution and to evaluate the working of such safeguards.

National Commission for Protection of Child Rights (NCPCR) : It is a statutory body that was set up in March 2007 under the Commission for Protection of Child Rights Act, 2005. It works under the administrative control of the Ministry of Women & Child Development. The Commission's Mandate is to ensure that all Laws, Policies, Programmes, and Administrative Mechanisms are in consonance with the Child Rights perspective as enshrined in the Constitution of India and also the UN Convention on the Rights of the Child.

Central Information Commission (CIC) : It was established by the Central Government in 2005, under the provisions of the Right to Information Act (2005). It is not a constitutional body. It consists of a Chief Information Commissioner and not more than ten Information Commissioners. They are appointed by the President on the recommendation of a committee consisting of the Prime Minister as Chairperson, the Leader of Opposition in the Lok Sabha and a Union Cabinet Minister nominated by the Prime Minister.

The jurisdiction of the Commission extends over all Central Public Authorities. The Chief Information Commissioner and an Information Commissioner shall hold office for such term as prescribed by the Central Government or until they attain the age

of 65 years, whichever is earlier. They are not eligible for reappointment.

State Information Commission : It is constituted by the State Government. It has one State Chief Information Commissioner (SCIC) and not more than 10 State Information Commissioners (SIC) to be appointed by the Governor on the recommendation of the Appointments Committee headed by the Chief Minister.

Inter-State Council : It is a constitutional body established under Article 263 of the Constitution of India on the recommendation of Sarkaria Commission. It facilitates coordination and cooperation between the central government and the state governments.

It composed of Prime Minister (Chairman), Chief Ministers of all States (Members), Chief Ministers of Union Territories having a Legislative Assembly and Administrators of UTs not having a Legislative Assembly and Governors of States under President's Rule (Members) and Six Ministers of Cabinet rank in the Union Council of Ministers to be nominated by the Prime Minister (Members).

Securities and Exchange Board of India : SEBI was initially formed in 1988 as a non-statutory body for the regulation of the securities market and later acquired statutory status on 30th January 1992. Headquartered in Mumbai, and has four regional offices located in Ahmedabad, Chennai, Delhi and Kolkata.

National Integration Council (NIC) : It was set up in June 1962 by Prime Minister Pt. Jawaharlal Nehru to address the problems of communalism and regionalism in India. It is chaired by the Prime Minister of India. It is an extra-constitutional body.

The members of the National Integration Council include union ministers, leaders of the opposition in the Lok Sabha and the Rajya Sabha, chief ministers of all states and Union Territories, leaders of national and regional political parties, chairpersons of national commissions, eminent journalists, and other public figures in India.

National Commission for Religious and Linguistic Minorities : It is also called Ranganath Misra Commission. It was constituted by the Government of India on 29 October 2004 to look into various

issues related to Linguistic and Religious minorities in India.

There shall be a Special Officer for linguistic minorities to be appointed by the President. It shall be the duty of the Special Officer to investigate all matters relating to the safeguards provided for linguistic minorities under this Constitution and report to the President upon those matters at such intervals as the president may direct, and the President shall cause all such reports to be laid before each House of Parliament, and sent to the Governments of the States concerned.

Parts of Indian Constitution

Originally, the Indian Constitution had 395 Articles, 22 Parts, and 8 Schedules.

Later Part VII (dealing with Part - B states) was repealed by the 7th Amendment Act (1956). On the other hand, both Part IV - A and Part XIV-A were added by the 42nd Amendment Act (1976), while Part IX-A was added by the 74th Amendment Act (1992), and Part IX-B was added by the 97th Amendment Act (2011). Presently, the Indian Constitution is made up of **448 Articles, 25 Parts, and 12 Schedules.**

The 25 Parts of the Indian Constitution :

Part I - The Union and its Territories (Article 1-4)

Part II - Citizenship (Article 5-11)

Part III - Fundamental Rights (Article 12-35)

Part IV - Directive Principles of State Policy (Article 36-51)

Part IV A - Fundamental Duties (Article 51A)

Part V - The Union (Article 52-151). It contains 5 chapters.

Chapter I - The Union Executive (Article 52-78)

Chapter II - Parliament (Article 79-122)

Chapter III - Legislative Powers Of The President (Article 123)

Chapter IV - The Union Judiciary (Article 124-147)

Chapter V - Comptroller And Auditor-General of India (Article 148-151)

Part VI - The States (Article 152-237). It contains 6 chapters.

Chapter I - General (Article 152)**Chapter II** - The Executive (Article 153-167)**Chapter III** - The State Legislature (Article 168-212)**Chapter IV** - Legislative Powers of Governor (Article 213)**Chapter V** - The High Courts (Article 214-232)**Chapter VI** - Subordinate Courts (Article 233-237)**Part VII** - The States in Part B of First Schedule (Article 238) **[Repealed]****Part VIII** - The Union Territories (Article 239-242).**Part IX** - The Panchayats (Article 243-243O)**Part IXA** - The Municipalities (Article 243P-243ZG)**Part IXB** - The Co-operative Societies (Article 243ZH-243ZT)**Part X** - The Scheduled and Tribal Areas (Article 244 -244A)**Part XI** - Relation between the Union and the States (Article 245-263). It contains 2 chapters.**Chapter I** - Legislative Relations (Article 245-255).**Chapter II** - Administrative Relations (Article 256-263).**Part XII** - Finance, Property, Contracts and Suits (Article 264-300A). It contains 4 chapters.**Chapter I** - Finance (Article 264-291)**Chapter II** - Borrowing (Article 292-293)**Chapter III** - Property, Contracts, Rights, Liabilities, Obligations and Suits (Article 294-300)**Chapter IV** - Right to Property (Article 300-A)**Part XIII** - Trade, Commerce and Intercourse within the territory of India (Article 301-307).**Part XIV** - Services Under the Union and the States (UPSC) (Article 308-323). It contains 2 chapters.**Chapter I** - Services (Article 308-314)**Chapter II** - Public Service Commissions (Article 315 to 323)**Part XIV A** - Tribunals (Article 323A-323B)**Part XV** - Elections (Article 324-329A)**Part XVI** - Special provisions relating to certain classes (Article 330-342)**Part XVII** - Official language (Article 343-351). It contains 4 chapters.**Chapter I** - Language of the Union

(Article 343-344)

Chapter II - Regional Languages (Article 345-347)**Chapter III** - Language of the Supreme Court, High Courts, etc. (Article 348-349)**Chapter IV** - Special Directives (Article 350-351)**Part XVIII** - Emergency Provisions (Article 352-360)**Part XIX** - Miscellaneous (Article 361-367)**Part XX** - Amendment of the Constitution (Article 368)**Part XXI** - Temporary, Transitional and Special Provisions (Article 369-392)**Part XXII** - Short Title, Commencement, Authoritative Text in Hindi and Repeals (Article 393-395).

Amendments

Amending the Constitution of India is the process of making changes to the nation's fundamental law or supreme law.

The procedure of amendment in the constitution is laid down in **Part XX** (Article 368) of the Constitution of India. The 1st Amendment of the Indian constitution was made in 1951. As of April 2024, the Constitution of India is amended 106 times.

Article 368 provides for two types of amendments : By a special majority of Parliament and by a special majority of the Parliament with the ratification by half of the total states.

Procedure for amending the Constitution

As per **Article 368** the procedure of the Constitutional Amendment is laid down as follows :

A bill for amendment can be introduced in either House of Parliament and not in the state legislature. It can be introduced by a minister or a private member and does not require the permission of the President to introduce.

The bill must be passed in each House by a special majority, which requires 2/3rd majority of members present and voting, supported by more than 50% of the total strength of the house.

Each House must pass the bill separately. In case of a disagreement between the two Houses, there is no provision for holding a joint sitting of the two Houses.

If the bill seeks to amend the provisions of the constitution, it must be ratified by the legislatures of half of the states by a simple majority.

After passage of the bill by both the houses, the bill is presented to the President, who must give assent. He can neither withhold his assent to the bill nor return the bill for reconsideration of the Parliament. After the President's assent, the **Bill** becomes an **Act**.

Types of Majorities

Simple Majority - This refers to the majority of more than 50% of the members present and voting. This is also known as functional majority or working majority. For example, amendments related to the official language can pass through Parliament with a simple majority without using Article 368.

Absolute majority - It refers to a majority of more than 50% of the total membership of the house. For example, as the total membership of Lok Sabha is 545, an absolute majority in Lok Sabha means - 50% of 545 plus 1, i.e. 273.

Effective Majority - Majority of 50% of seats of the house excluding vacant seats.

Special Majority - As per article 368, It requires a majority of 2/3rd members present and voting supported by more than 50% of the total strength of the house.

For example - To pass a constitution amendment bill in Lok Sabha, in addition to getting the support of 273 members, the bill should be favored by more than 2/3rd of the members present and voting.

Important Amendments

1st Amendment Act, 1951 - Article 15, 19, 85, 87, 174, 176, 341, 342, 372, and 376 were amended under this. The Ninth Schedule was added to protect certain laws, like land reforms, from judicial review. It allows states to take affirmative action for socially and economically backward groups, including Scheduled Castes and Scheduled Tribes, by limiting the application of fundamental rights.

2nd Amendment Act, 1952 - The scale of representation in the Lok Sabha was readjusted stating that 1 member can represent even more than 7.5 lakh people.

4th Amendment Act, 1955 - It made the scale of compensation given in lieu of compulsory acquisition of private property beyond the scrutiny of courts.

7th Amendment Act, 1956 - The State Reorganisation Act of 1956 was passed which established 14 states and 6 centrally administered territories in India. It also extended the jurisdiction of High Courts to Union territories and allowed for the appointment of the same person as Governor for two or more states.

10th Amendment Act, 1961 - It incorporated Dadra and Nagar Haveli as the seventh Union territory of India, by amending the First Schedule to the Constitution.

12th Amendment Act, 1962 - Goa, Daman, and Diu were incorporated into the Indian Union after being acquired from the Portuguese through police action in 1961. In 1987, Goa was conferred a statehood.

13th Amendment Act, 1962 - Formation of the State of Nagaland, with special protection under Article 371A.

14th Amendment Act, 1962 - The amendment added Pondicherry as India's ninth Union territory and established legislatures and councils of ministers for Himachal Pradesh, Manipur, Tripura, Goa, Daman and Diu, and Puducherry.

17th amendment Act, 1964 - It prohibited the acquisition of land under personal cultivation unless the market value of the land is paid as compensation. It also included 44 more Acts in the Ninth Schedule.

18th Amendment Act, 1966 - It includes Union Territories in Article 3 and hence permits reorganization of Union Territories. It created new states namely, Punjab and Haryana.

19th Amendment Act, 1966 - It abolished Election Tribunals and enabled trial of election petitions by regular High Courts.

22nd Amendment Act, 1969 - It facilitated the creation of a new autonomous State of Meghalaya within the State of Assam.

23rd Amendment Act, 1969 - The amendment ended Nagaland's reservation of seats for Scheduled Tribes in Lok Sabha and State Legislative Assembly. It also limited Governor nominations of Anglo-Indians in State Legislative Assemblies to one.

24th Amendment Act, 1971 - This amendment clarified Parliament's power to amend any part of the Constitution, including fundamental rights. It amended Article 368 to explicitly grant Parliament the authority to amend any provision. Additionally, it made the President's assent to Constitutional Amendment Bills compulsory.

26th Amendment Act, 1971 - It abolished the privy purses and privileges of the former rulers of princely states.

27th Amendment Act, 1971 - It reorganized Mizoram into a Union Territory with a legislature and council of ministers.

28th Amendment Act, 1972 - It deals with the service conditions of All-India service officers.

31st Amendment Act, 1973 - The Act increased the maximum Lok Sabha representation for states from 525 to 545 and reduced the maximum representation for Union Territories from 25 to 20.

35th Amendment Act, 1974 - Sikkim's status changed from a protectorate state to an 'Associate State' of India in 1974.

37th Amendment Act, 1975 - It refers to the formation of Arunachal Pradesh legislative assembly.

40th Amendment Act, 1976 - The amendment empowered Parliament to define India's territorial waters, continental shelf, exclusive economic zone (EEZ), and maritime zones, amending Article 297 of the Constitution.

41st Amendment Act, 1976 - It changed the age of retirement of the Chairman and members of the State Public Service Commissions from 60 to 62 years.

42nd Amendment Act, 1976 - The 42nd amendment, enacted by Indira Gandhi government, attempted to alter the basic structure of the Indian Constitution. It added 'Socialist', 'Secular' and 'Integrity' words to the Preamble, introduced Article 51A (10 Fundamental Duties), limited the power of judicial review, and added three new Directive Principles of State Policy. Due to its extensive changes, it's often called the '**Mini-Constitution**'.

43rd Amendment Act, 1977 - The Act, enacted by the Janata Government, aimed to rectify distortions introduced by the 42nd Amendment Act of 1976. It restored the jurisdiction of the Supreme

Court and High Courts for judicial review and writs, and removed Parliament's special powers to legislate on anti-national activities.

44th Amendment Act, 1978 - This Act reversed the provision of the 42nd Amendment Act, which allowed the government to amend the Constitution at will through Article 368. It restored checks on government power. Additionally, it reclassified the Right to Property from a fundamental right to a legal right under Article 300A.

47th Amendment Act, 1984 - It amended the Ninth Schedule to the Constitution, and added 14 legislations relating to land reforms.

52nd Amendment Act, 1985 - The 10th Schedule of the Indian Constitution popularly referred to as the 'Anti-Defection Law' was passed by Parliament in 1985 and reinforced in 2002.

53rd Amendment Act, 1986 - It gives special provision with respect to the State of Mizoram.

54th Amendment, Act 1986 - This amendment was made to increase the salary of the Chief Justice of India and other Judges and to determine future increases without the need for a constitutional amendment.

56th Amendment Act, 1987 - It confers Statehood on Goa and forms a new Union Territory of Daman and Diu.

58th Amendment Act, 1987 - It deals with the provision for an authoritative text of the Constitution in the Hindi language. It gave the same legal sanctity to the Hindi version of the Constitution.

61st Amendment Act, 1988 - Lowered the voting age of elections to the Lok Sabha and to the Legislative Assemblies of States from 21 years to 18 years. This was done by amending Article 326 of the Constitution, which concerns elections to the Lok Sabha and the Assemblies. The percentage of eligible voters who cast their vote in the election is termed as Voter turnout.

69th Amendment Act, 1991 - Union Territory of Delhi was given the special status of 'National Capital Territory of Delhi.' A 70-member legislative assembly and a 7-member council of ministers were established in Delhi.

70th Amendment Act, 1992 - The elected members of Pondicherry and National

Capital Territory of Delhi Legislative Assembly were included in the electoral college of the President of India.

71st Amendment Act, 1992 - It Includes Konkani, Manipuri and Nepali as official languages and amend schedule 8.

72nd Amendment Act, 1992 - It provides reservations to Scheduled Tribes in Tripura State Legislative Assembly.

73rd Amendment Act, 1992 - The Act, passed in 1992 and effective from 24 April 1993, empowered state governments to formalize and empower gram panchayats for self-governance. It added 'Part IX: The Panchayats' to the Constitution. Since 2010, 24th April has been observed as National Panchayati Raj Day by the central government.

74th Amendment Act, 1992 - P.V. Narasimha Rao's Government introduced the modified Municipalities Bill in the Lok Sabha in September 1991. It finally came into force on 1 June 1993 that gave the Urban local bodies distinct powers and superiorities.

76th Amendment Act, 1994 - The amendment included the Tamil Nadu Reservation Act of 1994, which mandates 69% reservation, in the Ninth Schedule to shield it from judicial review. This action was taken following a 1992 Supreme Court ruling that capped total reservation at 50%.

77th Amendment Act, 1995 - It provided for reservation in promotions in government jobs for Scheduled Castes and Scheduled Tribes. This amendment nullified the Supreme Court ruling with regard to reservation in promotions.

82nd Amendment Act, 2000 - By this amendment, states have been allowed to relax the minimum marks for candidates belonging to Scheduled Castes and Scheduled Tribes in cases of promotion to fill vacancies reserved for them in government jobs.

84th Amendment Act, 2001 - It extended the ban on the readjustment of seats in the Lok Sabha and the state legislative assemblies for another 25 years (i.e. up to 2026).

85th Amendment Act, 2002 - It was brought in for also giving consequential seniority to SCs and STs in matters of reservation in promotion.

86th Amendment Act, 2002 - The amendment added the 11th Fundamental Duty and inserted Article

21-A to provide free and compulsory education for children aged six to fourteen as a Fundamental Right.

87th Amendment Act, 2003 - Provided for the readjustment and rationalisation of territorial constituencies in the states on the basis of the population figures of 2001 census and not 1991 census as provided earlier by the 84th Amendment Act of 2001.

91st Amendment Act, 2003 - The size of ministerial councils at the Centre and in the states could not exceed 15% of the number of members of the Lok Sabha or state legislatures.

92nd Amendment Act, 2003 -The amendment added four languages (Bodo, Dogri, Mathili, and Santhali) to the Eighth Schedule, bringing the total number of constitutionally recognized languages to 22.

93rd Amendment Act, 2005 - It provided for the reservation of seats for Scheduled Castes and Scheduled Tribes in higher education institutions.

94th Amendment Act, 2006 - It is provided for a Minister of Tribal Welfare in newly created Jharkhand and Chhattisgarh States including Madhya Pradesh and Orissa.

95th Amendment Act, 2009 - Extended the reservation of seats for the SCs and STs and special representation for the Anglo-Indians in the Lok Sabha and the state legislative assemblies for a further period of ten years i.e., up to 2020 (Article 334).

96th Amendment Act, 2011 - The language of Orissa was changed from Oriya to Odia in the 8th schedule.

98th Amendment Act, 2012 - This amendment saw the introduction of Article 371-J into the Indian Constitution.

99th Amendment Act, 2014 - Replaced the collegium system of appointing judges to the Supreme Court and High Courts with a new body called the National Judicial Appointments Commission (NJAC). But the Supreme Court declared this Amendment Act as unconstitutional and void.

100th Amendment Act, 2015 - This Act facilitated the exchange of enclaves and territories between India and Bangladesh, operationalizing the Land Boundary Agreement and resolving border disputes.

101st Amendment Act, 2016 - It was enacted to facilitate the implementation of the Goods and Services Tax (GST) in India.

102nd Amendment Act, 2018 - This bill was introduced in Lok Sabha as the 123rd Amendment Bill. It gave the National Commission for Backward Classes (NCBC) constitutional validity, allowing the President to notify the list of Socially and Educationally Backward Classes (SEBC) for any state or union territory.

103rd Amendment Act 2019 - Introduced as the 124th Amendment Bill in Lok Sabha, it provided 10% reservation for Economically Weaker Sections (EWS) It modified Articles 15 and 16 to facilitate this reservation.

104th Amendment Act 2020- Introduced as 126 Amendment bill in the Parliament. It ended reservation of seats for Anglo-Indians in both the Lok Sabha and State Legislative Assemblies. It also extends the reservation of seats for scheduled castes, scheduled tribes in Lok Sabha and state legislative assemblies for next 10 years.

105th Amendment Act, 2021 - It restores the power of State Governments and Union Territories to identify the Socially and Educationally Backward Classes (SEBCs). The act Amended Articles 338B, 342A, and 366 of the constitution.

106th Amendment Act, 2023 - It reserves one-third of seats for women in Lok Sabha, State legislative assemblies, and the Legislative Assembly of the National Capital Territory of Delhi, including those reserved for SCs and STs. This reservation is effective after the publication of the census conducted following the Act's commencement and lasts for 15 years, with the possibility of extension through parliamentary action.

Schedules

Schedules are basically tables which contain additional details not mentioned in the articles. The Indian Constitution originally had **eight schedules**. Four more schedules were added by different amendments. Now the total number of Schedules in our constitution is twelve.

First Schedule - It contains the list of names of states and union territories and their extent and territorial jurisdictions.

Second Schedule - It contains provisions

related to allowances, privileges, and emoluments of the President, Governors of States, Speaker and the Deputy Speaker of the House of the People, the Chairman and the Deputy Chairman of the Council of States, the Speaker and the Deputy Speaker of the Legislative Assembly, the Chairman and the Deputy Chairman of the Legislative Council of a State, the Judges of the Supreme Court and of the High Courts, and the Comptroller and Auditor-General of India.

Third Schedule - It contains the forms of oath and affirmation for Union Ministers of India, the candidates for election to the Parliament, Members of Parliament (MPs), Supreme Court Judges, Comptroller and Auditor General, State Ministers, State Legislature Elections Candidates, State Legislature Members, High Court Judges.

Fourth Schedule - It contains provisions for the allocation of seats in the Rajya Sabha to the states and the union territories.

Fifth Schedule - It contains provisions relating to the administration and control of scheduled areas and scheduled tribes. It covers Andhra Pradesh, Chhattisgarh, Gujarat, Himachal Pradesh, Jharkhand, Madhya Pradesh, Maharashtra, Odisha, Rajasthan and Telangana.

Sixth Schedule - It contains provisions related to the Administration of Tribal Areas in the States of Assam, Meghalaya, Tripura and Mizoram as autonomous entities. 10 autonomous district councils in 4 states come under 6th schedule.

Seventh Schedule - It contains division of powers between the Union and the States. Presently, the Union List contains 98 subjects (originally 97), the State List contains 59 subjects (originally 66) and the Concurrent List contains 52 subjects (originally 47).

Union list - Defence of India, **Railways**, Atomic energy and mineral resources, Foreign affairs, Citizenship, naturalization and aliens, Highways, Foreign loans, Reserve Bank of India, Establishment of standard and weight measure, Central Bureau of Intelligence and Investigation, Extradition etc.

State list - Public order, Police, Publichealth and sanitation, Pilgrimages, Agriculture, Fisheries, Industries, Gas and gas-works, Sports, Entertainments and

amusements, Trade and commerce, Public debt of the State, Market and Fairs, Agricultural land, Electricity, State Public Service Commission etc.

Concurrent list - Criminal law, Preventive detention, Bankruptcy, Trust and Trustees, Civil procedure, Population control and family planning, Forests, Economic and social planning, Trade unions; Industrial and labor disputes, Education, Factories etc.

Eighth Schedule - It contains the list of official languages recognized by the Constitution. Originally, it had 14 languages but presently there are 22 languages. They are: Assamese, Bengali, Bodo, Dogri (Dongri), Gujarati, Hindi, Kannada, Kashmiri, Konkani, Mathili (Maithili), Malayalam, Manipuri, Marathi, Nepali, Odia, Punjabi, Sanskrit, Santhali, Sindhi, Tamil, Telugu and Urdu.

In **1967**, Sindhi was added by the 21st Constitutional Amendment Act. In 1992, Konkani, Manipuri and Nepali were added by the 71st Constitutional Amendment Act.

In **2003**, Bodo, Dogri, Maithili and Santhali were added by the 92nd Constitutional Amendment Act. In 2011, The spelling Oriya was replaced by Odia by the 96th Constitutional Amendment Act.

Ninth Schedule - It originally contained 13 laws. These laws pertain to land reforms, abolition of the zamindari system, and various other matters. Added by the 1st Amendment in 1951, it shielded included laws from judicial scrutiny for violating fundamental rights. However, in 2007, the Supreme Court ruled that laws added to the Ninth Schedule after April 24, 1973, are subject to judicial review.

Tenth Schedule - It contains provisions relating to the disqualification of members of Parliament and state legislatures on the grounds of defection. This schedule was added by the 52nd Amendment Act of 1985, also known as the Anti-defection law.

Eleventh schedule - It specifies the powers, authority and responsibilities of Panchayats. This schedule was added by the 73rd Amendment Act of 1992. It has 29 Items.

Twelfth Schedule - It specifies the powers, authority and responsibilities of

Municipalities. This schedule was added by the 74th Amendment Act of 1992. It has 18 Items.

Important Acts

Motor Vehicles Act, 1988

The Act, enforced on July 1, 1989, replaced the Motor Vehicles Act, 1939, which in turn replaced the initial enactment, the Motor Vehicles Act, 1914. It was later amended by The Motor Vehicles Act, 2019.

As per the Motor Vehicles Act, 1988, no individual under 18 years old shall receive a learner's license to operate a motorcycle without gears unless with the written consent of the person responsible for the applicant's care.

Section 43 of the Motor Vehicles Act, 1988, permits temporary vehicle registration, valid for up to one month and non-renewable.

Equal Remuneration Act, 1976

This Central Legislation ensures equal wages for work of similar nature for both male and female workers, prohibiting discrimination against female employees in transfers, training, promotions, etc.

Article 39 of the Indian Constitution addresses equal remuneration pay. The Equal Remuneration Act applies nationwide, covering all categories/types of employment.

Fugitive Economic Offenders (FEO) Act, 2018

This Act enables the prosecution of individuals who, while abroad, decline to return to India to face criminal charges. Under this Act, a person can be declared as an FEO if an arrest warrant has been issued against them for specified offenses involving a value exceeding Rs. 100 crore, and they have left the country and refuse to return for prosecution.

Mental Health Care Act 2017

This Act replaced the earlier Mental Health Act of 1987, enacted on May 22, 1987. The Mental Healthcare Act 2017, passed on April 7, 2017, became effective on May 29, 2018.

Section 20 of the Mental Healthcare Act 2017 guarantees every individual with mental illness the right to live with

dignity. It effectively decriminalized attempted suicide, previously punishable under Section 309 of the Indian Penal Code.

Right to Information Act 2005

It was introduced in 2005, receiving Presidential assent on June 15, 2005, and coming into force on October 12, 2005. It serves as a tool for individuals to access information that can improve their lives.

The Act mandates timely responses to citizen requests for Government information, aiming to combat corruption and enhance democracy. It extends throughout India, excluding Jammu and Kashmir, and applies to Public Authorities.

In 2019, an amendment to the Right to Information Act 2005 proposed that the Chief Information Commissioner and Information Commissioners be appointed for terms prescribed by the Central Government. The Lok Sabha passed the Right to Information (Amendment) Bill, 2019 on July 22, 2019.

Companies Act 2013

It regulates Indian Company Law, covering aspects such as company incorporation, corporate responsibilities, director duties, and company dissolution. According to this Act, any profit declared by a company but not paid or claimed within 30 days from the declaration date is transferred to the Unpaid Profit Account.

Khadi and village Industries Commission Act (1956)

The Act, passed in 1956, established the Khadi and Village Industries Commission, a statutory body operating under the Ministry of MSME. Its objective is to organize, promote, and plan the development and establishment of khadi industries in rural areas.

Juvenile Justice (Care and Protection of Children) Act, 2015

This Act, passed by the Indian Parliament, replaced the Juvenile Justice (Care and Protection of Children) Act 2000, allowing for juveniles aged 16-18 involved in heinous offenses to be tried as adults. It also aimed to establish a

universally accessible adoption law for India. Enforced on January 15, 2016. It was later replaced by the Juvenile Justice (Care and Protection of Children) Amendment Bill, 2021.

Prevention of Insults to National Honour Act, 1971

It is an act of the Parliament of India which prohibits the desecration of or insult to the country's national symbols, including the national flag, national emblem, national anthem, the constitution, and map of India including contempt of Indian constitution.

Rights of Persons with Disabilities Act, 2016

This Act, enacted in 2016, replaced The Person with Disabilities Act, 1995, which covered only seven disabilities. It ensures equal opportunities for individuals with disabilities amounting to 40% or more. The International Day of Persons with Disabilities is observed on December 3rd.

Right to Education Act (RTE), 2009

In 2009, the Right to Education Act was enacted, making free and compulsory education a fundamental right under Article 21-A. Originally, Article 45 and Article 39(f) of the Directive Principles of State Policy in Part IV of the Indian Constitution provided for state-funded and accessible education.

The Ramamurti Committee Report in 1990 was the first official document on the Right to Education. In a landmark judgment in 1993, the Supreme Court held in the Unnikrishnan JP vs State of Andhra Pradesh & Others case that education is a fundamental right under Article 21.

Contempt of Court Act 1971

Contempt of court refers to the offense of being defiant or disrespectful towards a court of law. This includes being impolite to legal authorities in the courtroom or willfully failing to follow a court order.

If found guilty of contempt of court, a judge can impose sanctions such as a penalty or imprisonment. Contempt of court can broadly fall into two categories: being disrespectful to legal

authorities in the courtroom and willfully failing to obey a court order.

The Supreme Court and the High Courts derive their power from the Constitution of India, while the procedure is outlined in the Contempt of Court Act, 1971.

Dowry Prohibition Act, 1961

It was enacted in **1961**. Its purpose was to prevent the giving and receiving of dowry. It is applicable to persons of all religions in India. Punishment for violation of law is not less than 5 years imprisonment and fine of not less than Rs 15000/- or the value of dowry whichever is higher.

Protection of Women from Domestic Violence Act, 2005

It was passed in 2005 while it was implemented in 2006. A woman who is a victim of domestic violence will have the right to the services of the police, shelter homes and medical establishments. She also has the right to simultaneously file a complaint under Section **498A** of the Indian Penal Code.

Representation of the People Act

It was passed in 1951. It provides for the actual conduct of elections in India. It deals with the election of the Parliament and State Legislatures, the qualifications and disqualifications for membership of Parliament and state legislatures. It was introduced in Parliament by the law minister Dr. B.R. Ambedkar. It was enacted under Article 327 of the Indian Constitution before the first general election in India.

Protection of Children from Sexual Offences (POCSO) Act, 2012

To address child sexual abuse cases, the Government introduced a special law, The Protection of Children from Sexual Offences (POCSO) Act, 2012, which came into effect on November 14, 2012.

This comprehensive law aims to protect children from sexual assault and pornography offenses. It ensures the child's interests are safeguarded at every stage of the judicial process by incorporating child-friendly mechanisms for reporting, recording evidence, investigation, and speedy trial of offenses through designated Special Courts.

National Food Security Act

In 2013, the Union Government of India passed the National Food Security Act to ensure food security for its citizens. According to this act, states are responsible for identifying eligible households, and the central government will provide necessary food grains accordingly. Eligible persons in priority households receive 5kg of food grain per person, while those under the Antyodaya Anna Yojana receive 35 kg per family. The scheme covers 75% of the rural population and 50% of the urban population.

Special Marriage Act, 1954

The Special Marriage Act, of 1954 replaced the **Old Act III, 1872**. It is an Indian law that provides a legal framework for the marriage of people belonging to different religions or castes.

The applicability of the act extends to the people of all faiths, including Hindus, Muslims, Sikhs, Christians, Jains, and Buddhists, across India. The minimum age to get married under the SMA is 21 years for males and 18 years for females.

Maintenance and Welfare of Parents and Senior Citizens (Amendment) Bill, 2019

The Maintenance and Welfare of Parents and Senior Citizens Bill, 2007, initially set a monthly maintenance amount of Rs. 10,000 payable by children to parents. However, the Maintenance and Welfare of Parents and Senior Citizens (Amendment) Bill, 2019, removed this upper limit on maintenance and broadened the definition of children to include step-children, adoptive children, children-in-law, and legal guardians of minor children.

Indian Forest Act, 1878

The Indian Forest Act of 1878 replaced its predecessor from 1865, expanding governmental control over forests. It introduced regulations on timber duties and aimed to eliminate local rights in reserved forests, reserving them solely for government use. Subsequently, the Indian Forest Act of 1927 superseded the 1878 Act, consolidating laws related to forests, transit of forest produce, and timber duties.

Indian Age of Consent Act

It was a legislation enacted in British India on 19 March 1891 that raised the age of consent for sexual intercourse from ten to twelve years for all girls in all jurisdictions. The Act was also known as **Act X** of 1891. It was an amendment to the Indian Penal Code and the Code of Criminal Procedure, Section 375, 1882.

Newsprint Control Order 2004

All domestic mills are registered under Schedule - I of Newsprint Control Order 2004 under the Department of policy for internal trade and industry, a wing of the ministry of commerce and industry.

First newsprint paper mill in India and Asia (1956, **NEPA** limited, Madhya Pradesh). NEPA - National Newsprint and Paper Mills. NEPA became a Central Government Company in 1959.

Surrogacy (Regulation) Bill 2016

The Bill permits altruistic surrogacy for certain Indian couples while prohibiting commercial surrogacy. It outlines eligibility conditions for surrogate mothers and intending couples and safeguards the rights of children born through surrogacy.

According to the Surrogacy (Regulation) Rules, 2022, a surrogate mother may undergo a maximum of three attempts at any surrogacy procedure. Additionally, the Surrogacy (Regulation) Amendment Rules 2023 mandate that andrologists involved in surrogacy procedures must possess a minimum of two years of experience in various surgical sperm retrieval procedures.

Hindu Succession Act, 1956

The Hindu Succession Act, 1956 is an Act of the Parliament of India enacted to amend and codify the law relating to intestate or unwilled succession, among Hindus, Buddhists, Jains, and Sikhs. It was amended in 2005. This act has been amended to provide equal rights to daughters in their ancestral property.

Muslim Women (Protection of Rights on Marriage) Act, 2019

It declares the instant divorce granted by pronouncement of talaq three times as void and illegal. It provides for imprisonment for a term up to 3 years and fine to the husband who practiced

instant Triple Talaq.

Unlawful Activities (Prevention) Act

It was passed in 1967. It provides for the more effective prevention of certain unlawful activities of individuals and associations dealing with terrorist activities, and was first put into effect in India in the year. It was amended several times, most recently in 2019, to include provisions related to terrorist financing, cyber-terrorism, individual designation, and seizure of property. It provides for the death penalty and life imprisonment as the highest punishments for terrorist acts. It allows for the detention of suspects without charge or trial for up to 180 days.

Data Protection Bill 2021

The Personal Data Protection Bill, introduced in the Lok Sabha by the Minister of Electronics and Information Technology, aims to safeguard personal data and establishes a Data Protection Authority. This proposed legislation covers the processing of digital personal data both within India and overseas, particularly focusing on activities involving offering goods or services to Indian residents or profiling individuals in India. It addresses the fundamental right to privacy.

Benami Transactions (Prohibition) Act 1988

It is an Act to prohibit benami transactions and the right to recover property held benami and for matters connected therewith or incidental.

Benami Transactions (Prohibition) Amendment Act 2016 amended the Original Act (Benami Transaction (Prohibition) Act 1988) and renamed it as Prohibition of Benami Property Transaction Act, 1988.

Taxation Law Amendment Act, 2021

It aims to withdraw tax demands made using a 2012 retrospective legislation to tax the indirect transfer of Indian assets. It amends the Income Tax Act of 1961.

Indian Arms Act of 1959

It aims to consolidate and amend the law relating to arms and ammunition. The main objective is to regulate and restrict

the circulation of arms and ammunition, which are illegal. It was enacted by the Indian government after independence, recognizing that certain law-abiding citizens must own and use firearms for sports, crop protection, and self-defense.

Landmark Judgements

Satbir Singh Vs State of Haryana

In the Satbir Singh vs State of Haryana case, 2021, the Supreme Court laid down guidelines for dowry death trials. The court in this case clarified the term '**soon before**' in section 304 B and also gave guidelines to be followed by the trial courts while recording the statement of the accused. Since the burden of proof is on the accused thus it creates more responsibility on courts to confirm that the principle of **Audi Alteram Partem** (Hear the other side) is followed.

Pranay Sethi Vs National Insurance Company Ltd.

In this case, the Supreme Court states that in the cases where the deceased is below the age of 40, and is self-employed, even then a 40% addition will be made to their income as future prospects.

Devilal Vs State of Madhya Pradesh

In this case, the court deliberated on the extent to which benefits under the Juvenile Justice Act could apply when the offender was between 16 and 18 years old at the time of the offence. The court ruled that even if the accused were found guilty, the case must be referred to the Jurisdictional Juvenile Justice Board.

Laxmi Bai Chandaragi Vs State of Karnataka

This case deals with the right to choose your life partner as a fundamental right under article 21 of the Indian Constitution. This case tells us that if there is consent between two adults to get married then there is no need for approval of the family members.

A.K. Gopalan Vs State of Madras (1950)

In 1950 the Supreme Court concluded that the provisions of the **Preventive**

Detention Act did not violate the Fundamental Rights entrenched in Articles 13, Article 19, Article 21, or Article 22, if the detention was carried out according to the law's procedures.

Shankari Prasad Vs Union of India (1951)

The constitutionality of the **First Amendment** was questioned in this case, which dealt with the amenability of fundamental rights. The Supreme Court argued that the authority to amend the constitution granted to Parliament under **Article 368** also extends to the Fundamental Rights protected in Part III of the Constitution.

Berubari Vs Union Case (1960)

This case challenged the Parliament's authority to transfer the territory of Berubai to Pakistan. The Supreme Court looked into Article 3 in depth and determined that the Parliament could not pass laws to carry out the Nehru-Noon agreement under this provision. As a result, the **9th Amendment Act (1960)** was enacted to make the agreement enforceable.

Judgements - Supreme Court concluded that the **Preamble** is not a part of the Indian Constitution. Therefore, it is not enforceable in a court of law.

Article 1(3) (c): Supreme Court, held that the article does not grant authority to India to acquire territories. It makes a provision for the absorption and integration of foreign territories that may be acquired by the Indian Union.

Golaknath Case (1967)

In this case, the Supreme Court held that Parliament could not amend Fundamental Rights, and this power would be only with a Constituent Assembly. The Court held that an amendment under Article 368 is "**law**" within the meaning of Article 13 of the Constitution and therefore, if an amendment "takes away or abridges" a Fundamental Right conferred by Part III, it is void.

Kesavananda Bharati Case (1973)

The core structure of the Constitution was defined by this decision. Although no component of the Constitution,

including Fundamental Rights, was outside Parliament's amending power, the Supreme Court decided that the **basic structure** of the Constitution could not be repealed even by constitutional amendment. The court held that democratic character of the polity was an essential component of the 'basic structure'.

Indira Nehru Gandhi Vs Raj Narain Case (1975)

The Supreme Court applied the theory of basic structure and struck down Clause(4) of Article 329-A, which was inserted by the **39th Amendment in 1975** on the grounds that it was beyond the Parliament's amending power as it destroyed the Constitution's basic features.

Maneka Gandhi case (1978)

A main issue in this case was whether the **right to go abroad** is a part of the Right to Personal Liberty under **Article 21**. The Supreme Court held that it is included in the Right to Personal Liberty. The Supreme Court also ruled that the mere existence of an enabling law was not enough to restrain personal liberty. Such a law must also be just, fair and reasonable.

Minerva Mills Vs Union of India (1980)

This incident reaffirms the concept of the Basic Structure once more. The 42nd Amendment Act of 1976 made amendments to the Constitution, two of which the court declared void for violating the fundamental structure. The Supreme Court's decision underscores that Parliament can amend Fundamental Rights to implement Directive Principles, provided such amendments do not harm or undermine the basic structure of the Constitution.

Waman Rao Case (1981)

The decision established a distinction between Acts included in the Ninth Schedule before the Kesavananda decision and those included after it. It determined that laws placed in the Ninth Schedule before the Kesavananda judgement are immune from scrutiny for violating Fundamental Rights, while laws enacted afterward can be challenged in court. This is also referred to as the Doctrine of Prospective Overruling.

Shah Bano Begum Case (1985)

The Supreme Court confirmed a **Muslim woman's right to alimony**, stating that the Code of Criminal Procedure, 1973, applies to all citizens, regardless of religion. This sparked a political debate, and the government at the time overturned the decision by passing the Muslim Women (Protection on Divorce Act), 1986, which stipulated that alimony be paid only during the **iddat period**.

Indra Sawhney Vs Union of India (1992)

The Supreme Court looked into the scope and application of Article 16(4), which provides for job reservation for backward classes. It confirmed the constitutional legitimacy of the OBC reservation of 27 percent, subject to specific limitations. (like creamy layer exclusion, no reservation in promotion, total reserved quota should not exceed 50 percent).

Sarla Mridugal Vs Union of India (1995)

It laid down the principles against the practice of solemnising second marriage by conversion to Islam, with first marriage not being dissolved. The verdict discusses the issue of bigamy, the conflict between the personal laws existing on matters of marriage, and invokes **Article 44** of the Indian Constitution. It is considered a landmark decision that highlighted the need for a **uniform civil code**.

Lily Thomas Vs Union Of India (2013)

The Supreme Court declared that any Member of Legislative Assembly, Member of Legislative Council or Member of Parliament convicted of a crime and sentenced to at least two years in jail will lose their seat in the House immediately.

Shayara Bano Vs Union of India and Others (2016)

The Supreme Court prohibited the antiquated practice of quick triple talaq which allowed Muslim males to break their marriages unilaterally by repeating the word talaq three times without providing for maintenance or alimony.

Justice K. S. Puttaswamy Vs. Union Of India (2017)

The Supreme Court of India proclaimed the right to privacy to be a Fundamental Right protected by the Indian Constitution. This decision safeguards individual rights against invasions of privacy. The Supreme Court of India held that the right to privacy is a part of the right to life under Article 21 of the Indian Constitution.

DK. Basu vs. The State of West Bengal (1997)

It has laid down some specific requirements and procedures that the police and other agencies have to follow for the arrest, detention and interrogation of any person. These guidelines have been named as **DK Basu guidelines**.

State of Kerala Vs Leesamma Joseph Case

The Supreme Court of India in its judgement in June 2021 affirmed that persons with disabilities have a right to reservation in promotion under Article 16(4) of the Constitution of India, 1950. The court held that allowing 3% reservation in promotions for persons with disabilities was within the scope of the Rights of Persons with Disabilities Act, 2016.

K.A. Najeeb Vs Union of India

The Supreme Court granted bail to an accused charged under the Unlawful Activities (Prevention) Act, 1967 ('UAPA'), India's primary anti-terror legislation. It deals with the violation of the right to life and personal liberty under Article 21 of the Indian Constitution.

Indian Young Lawyers Association & Ors vs. The State of Kerala & Ors

(Sabarimala Temple Entry case)

The Supreme Court invalidated provisions of the Kerala Hindu Places of Public Worship (Authorisation of Entry) Rules, 1965, permitting women of all ages to enter Sabarimala temple and worship the deity. A 4:1 majority of Supreme Court judges deemed the temple's practice of excluding women unconstitutional.

In this case, Justice Indu Malhotra penned a dissenting opinion that religious practices cannot be solely tested on the basis of Article 14. The right to equality under Article 14 does not override the fundamental right to religion under Article 25. The notions of rationality cannot be invoked in matters of religion.

Janhit Abhiyan Vs Union of India

The Supreme Court has upheld the validity of the 103rd Constitutional Amendment which provides 10% reservation for the Economically Weaker Sections (EWS) among forward castes in government jobs and colleges across India. The Constitutional Bench of the Supreme Court consists of 5 judges. This Act amended Articles 15 and 16 of the Constitution by inserting 15 (6) and 16 (6).

Indian Penal Code

The Indian Penal Code (IPC) is the official criminal code of India. It is a comprehensive code intended to cover all substantive aspects of criminal law. The code was drafted on the recommendations of the first law commission of India established in 1834 under the Charter Act of 1833 under the chairmanship of Thomas Babington Macaulay. It was enacted on 6 October 1860. It came into force in India during British rule on 1st January 1862.

The Code has been amended several times, latest is the Jammu and Kashmir Reorganisation Act, 2019. The Indian Penal Code of 1860, sub-divided into 23 chapters, comprises 511 sections.

The Indian Penal code is replaced by Bharatiya Nyaya (Second) Sanhita (BNS) 2023. Which is subdivided into 20 chapters, comprises 358 sections.

Important Sections of Indian Penal Code, 1860

(Corresponding BNS Sections).

Section 135 (Section 161) - Abetment of desertion of soldier, sailor or airman.

Section 143 [Section 187 (3)] - Punishment for unlawful assembly.

Section 147 [Section 189 (2) and (3)] - Punishment for rioting.

Section 159 (Section 192) - Affray.

Section 166A (Section 196) - Public servant disobeying direction under law.

Section 193 (Section 227) - Punishment for false evidence.

Section 223 (Section 259) - Escape from confinement or custody negligently suffered by a public servant.

Section 268 (Section 268) - Public nuisance.

Section 299 (Section 98) - Culpable homicide.

Section 300 (Section 99) - Murder.

Section 304-A (Section 104) - Causing death by negligence.

Section 304-B (Section 79) - Dowry death.

Section 306 (Section 106) - Abetment of suicide

Section 312 (Section 86) - Causing miscarriage.

Section 354 (Section 73) - Assault of criminal force on a woman with intent to outrage her modesty.

Section 354-A (Section 74) - Sexual harassment and punishment for sexual harassment.

Section 354-B (Section 75) - Assault or use of criminal force on a woman with intent to disrobe.

Section 354C (Section 76) - Voyeurism

Section 354D (Section 77) - Stalking.

Section 376 (Section 64) - Punishment for rape.

Section 377 (No Provision) - Punishment for voluntary carnal intercourse against the order of nature with any man, woman or animal shall be punished. The Supreme court in 'Navtej Singh Johar vs. Union Of India' case (2018) decriminalised the offences.

Code of Criminal Procedure

Criminal Procedure Code (CrPC) is the main legislation on procedure for administration of substantive criminal law in India. It was enacted in 1973 and came into force on 1 April 1974.

It lays the rules for the conduct of proceedings against any person who has committed an offense under any Criminal law, whether it is I.P.C or other Criminal law.

The 1973 Criminal Process Code is divided into 37 Chapters, 484 Sections, and two Schedules.

The Bharatiya Nagarik Suraksha Sanhita, 2023 (BNSS) seeks to replace the Criminal Procedure Code, 1973 (CrPC), which have 39 chapters, comprises 531 Sections.

Classification of Offenses

There are various types of Offences -
Bailable and Non-bailable offense,
Cognizable and non-cognizable offense,
Compoundable and non-compoundable offenses.

Bailable offenses : The accused can be released on bail without going through a court trial.

Non-bailable offenses : The accused must attend court proceedings and apply for bail in front of a judge.

Cognizable Offense : A cognizable offense is a serious offense in which the police have the authority to arrest the accused without a warrant and start the investigation without the need for the court's permission. Examples of cognizable offences include murder, rape, kidnapping, and theft.

Non-Cognizable Offense : A non-cognizable offense is a less serious offense in which the police do not have the authority to arrest the accused without a warrant, and the investigation can only be initiated with the court's permission. Examples include minor offenses like assault, defamation, and public nuisance.

Compoundable offenses : These are offenses that can be compromised, meaning the complainant can agree to withdraw the charges levied against the accused.

Non - compoundable offenses : These are more serious offenses in which the parties cannot compromise.

Important Articles

Article 152 - The definition of State does not include the state of Jammu and Kashmir.

Article 166 - Conduct of Business of the Government of a State.

Article 244(1) - It deals with provisions related to the administration and control

of Scheduled Areas and Scheduled Tribes in every State other than Assam, Meghalaya, Tripura, and Mizoram.

Article 244(2) - It deals with the administration and control of tribal areas of Meghalaya, Tripura, Assam, and Mizoram.

Article 245 - Parliament may make laws for the whole or any part of the territory of India, and the Legislature of a State may make laws for the whole or any part of the State.

Article 246 - States about the Subject matter of laws made by parliament and by the legislature of states.

Article 246A - Special provision with respect to goods and services tax.

Article 261 - Public acts, records and judicial proceedings.

Article 265 - Taxes not to be imposed save by authority of law.

Article 269A - GST on supplies in the course of inter-state trade or commerce shall be levied and collected by the Government of India.

Article 271 - Surcharge on certain duties and taxes for purposes of the Union.

Article 279A - Goods and Services Tax Council.

Article 290(A) - The expenses of any court or Commission, or the pension payable to or in respect of a person who has served before the commencement of this Constitution under the Crown in India or after such commencement in connection with the affairs of the Union or of a State, are charged on the Consolidated Fund of India or the Consolidated Fund of a State.

Article 300: It pertains to suits and proceedings by or against the Government of India or a state government.

Article 301 - Freedom of trade, commerce, and intercourse.

Article 302 - It talks about the power of the Parliament to impose limits on trade, commerce, and intercourse.

Article 307 - Deals with the appointment of authority for carrying out the purposes of Articles 301 to 304.

Article 326 - Elections to the House of the People and to the Legislative Assemblies of States to be on the basis of adult suffrage. Implicitly, guarantee

Right to vote. Right to vote in elections is a Constitutional Right.

Article 329 - Bar to interference by courts in electoral matters.

Article 330 - Reservation of seats for Scheduled Castes and Scheduled Tribes in the House of the People.

Article 336 - Special provision for Anglo-Indian community in certain services.

Article 337 - special provision concerning educational grants for the benefit of the Anglo Indian community.

Article 341 - Provides certain privileges and concessions to the members of Scheduled Castes.

Article 343 (1) - The official language of the Union shall be Hindi in Devanagari script

Article 343 (2) - The Constitution adopted in 1950 stipulated that English would be used for the Union's official business for a period of fifteen years.

Article 343 (3) - Empowered the Parliament to provide by law for continued use of English for official purposes even after **25 January 1965**.

Article 344 (1) - Constitution of an official language Commission by the President, which shall consist of a Chairman and such other members representing the different languages specified in the **Eighth Schedule** to make recommendations to the President for the progressive use of Hindi for official purposes of the Union.

Article 345 - Official language or languages of a State.

Article 347 - Special provisions provided to the language spoken in a section of the population.

Article 348(1) - English is the official language for the supreme court and high courts in the country.

Article 348 (2) - The Governor of a State with the previous consent of the President, authorizes the use of the Hindi language, or any other language used for any official purposes of the State, in proceedings in the High Court provided that judgments or orders passed by High Courts shall be in English.

Article 348 (3) - It states that the Legislature of a State has prescribed any language other than the English

language for use in Bills, acts or ordinances introduced in the state legislature.

Article 350 - States about the language to be used in representation for redress of grievances.

Article 350A - Facilities for instruction in mother-tongue at primary stage.

Article 350(B) - Special Officer for Linguistic Minorities

Article 361 - The President or the Governor of a State shall not be answerable to any court for the exercise and performance of the powers and duties of his office or for any act done or purporting to be done by him in the exercise and performance of those powers and duties.

Article 371 - special provision with respect to the States of Maharashtra and Gujarat.

Article 371A - Special provision with respect to the State of Nagaland.

Article 371B - Special provision with respect to the State of Assam.

Article 371C - Special provision with respect to the State of Manipur.

Article 371D - Special provisions with respect to the State of Andhra Pradesh or the State of Telangana.

Article 371E - Establishment of Central University in Andhra Pradesh.

Article 371F - Special provisions with respect to the State of Sikkim.

Article 371G - Special provision with respect to the State of Mizoram.

Article 371H - Special provision with respect to the State of Arunachal Pradesh.

Article 371 I - Special provision with respect to the State of Goa.

Article 371J - Special provisions with respect to the State of Karnataka.

Article 374 - Provides provisions as to Judges of the Federal Court and proceedings pending in the Federal Court.

Geography

Universe and Solar System

The universe is all of space and time and their contents, including all the planets, stars, galaxies, celestial bodies, the contents of the space between the galaxies, the subatomic particles, and all matter and all energy. Another word for the universe is cosmos.

Origin of the universe is explained by the **Big Bang theory**. According to this theory, the universe started from an unimaginably tiny "dot". This tiny structure was extremely hot and dense, and began expanding like a balloon after a supposed massive blast known popularly as the "big bang". This theory is given by **Georges Lemaitre**.

Steady state of Universe - It is a framework designed by Thomas Gold, Hermann Bondi, and Fred Hoyle devised the steady-state theory. The Steady-State Theory suggests that the density of the Universe does not change over time even if the Universe is continuously expanding. It suggests the universe is in a state of dynamic equilibrium.

The sun, the moon, the planets and all those objects shining in the night sky are called **celestial bodies**.

Supernova is a powerful and luminous explosion of a star. It happens when a star has reached the end of its life and explodes in a brilliant burst of light. The last supernova was directly observed in the Milky Way in 1604. It was also known as Kepler's Supernova.

Nebula is a giant cloud of dust and gas mostly made up of hydrogen and helium in space. The closest known nebula to Earth is called the **Helix Nebula**, which is located in the constellation Aquarius.

Orion Nebula is also known as Messier 42, is a diffuse nebula situated in the Milky Way.

Hydrogen is the most abundant element in the universe. It covers approximately 70% of the total mass of the universe.

Helium is the second most abundant element. It accounts for about 25 percent of the atoms in the universe. **Oxygen** is the third abundant element in the universe. Together these 3 elements compose 99% of the visible universe.

Galaxy

A galaxy is a collection of stars, stellar remnants, interstellar gas, dust, dark matter, bound together by gravity. Examples - Milky Way Galaxy, Andromeda Galaxy, Canis Major Dwarf Galaxy, Cygnus A.

The solar system is part of a galaxy known as the Milky Way Galaxy. The Milky Way is a large barred spiral galaxy.

The Canis Major Dwarf Galaxy was discovered in 2003, is the closest known galaxy to earth.

IC 1101 is a supergiant elliptical galaxy at the center of the Abell 2029 galaxy cluster and is one of the largest known galaxies.

The Roche limit, or Roche radius, is a critical distance from a celestial body where the tidal forces exerted by the larger body overpower the gravitational self-attraction of a smaller body.

A light-year - The distance that light travels in a vacuum in one Julian year (365.25 days) which is 9.4607×10^{15} m. Earth is about 8 light minutes from the Sun.

Star

It is huge celestial bodies made mostly of hydrogen and helium that produce light and heat from the churning nuclear forges inside their cores.

Proxima Centauri - The closest star to our planet other than the sun. It is about 4.2 light years away from the Earth.

The **luminosity of a star** is its energy that it radiates in all directions per second.

Star Shower - A transient rain of meteors, such as the Perseids, occurring at regular intervals and coming from a particular region in the sky. It is caused by the earth passing through a large number of meteoroids.

Habitable zone - The distance from a star at which liquid water could exist on orbiting planets' surfaces.

Kepler-22b - It is an exoplanet orbiting within the habitable zone of the Sun-like star Kepler-22. It is located about 600 light-years (180 parsecs) from Earth in the constellation of Cygnus.

Constellation

It is the collection of stars that looks like a particular shape in the sky and has been given a name. These stars are far away from Earth.

A group of stars that form a pattern smaller than a constellation is called Asterism.

Famous constellations among the 88 modern constellations -

Ursa Major, also known as the Big Dipper or the Great Bear, is a constellation easily spotted in the summer sky. It consists of seven bright stars that resemble a large ladle or a bear's body. It is the third largest constellation. Alioth is the brightest star of Ursa Major.

Ursa Minor - It is also known as the Little Bear with a long tail, is a constellation located in the far northern sky. It is also called the Little Dipper. The Indian name for the constellation Ursa Minor is **Laghu Saptharishi**.

Orion - It can be seen during winter in the late evening. It has seven or eight bright stars. Orion is also called the Hunter or Kalpurush.

Cassiopeia - A constellation of the northern sky easily recognized by a group of five bright stars forming a slightly irregular W, which is said to represent the Queen sitting on her throne. **Shedar** is the brightest star of Cassiopeia.

Sun

A naturally luminous body that has its own light source and emits its own light. It is a nearly perfect ball of hot plasma that radiates energy mainly as visible light.

The Sun is made of six layers. Inner layers are core, radiative zone and convection zone and the outer layers are photosphere, chromosphere and corona.

The **Sun's core** is the central region where nuclear reactions consume hydrogen to form helium.

Radiative zone - It is characterized by the method of energy transport - radiation.

Convection zone - The region of turbulent plasma between a star's core and its visible photosphere at the surface, through which energy is transferred by convection.

Corona - The outermost part of the Sun's atmosphere. The corona is hidden by the bright light of the Sun's surface.

Photosphere - It is the visible surface of the Sun, from which is emitted most of the Sun's light that reaches Earth directly.

Chromosphere is a thin layer of plasma that lies between the Sun's visible surface (the photosphere) and the corona (the Sun's upper atmosphere).

Transition region is a thin and very irregular layer of the Sun's atmosphere that separates the hot corona from the much cooler chromosphere.

Note - The surface temperature of the sun is approximately 6000°C.

Nuclear Fusion - It takes place in the Sun in which two hydrogen atoms fuse to produce Helium.

The Sun's magnetic field goes through a cycle, called the solar cycle. The average length of a solar cycle is 11 years. After two solar cycles, the Sun's magnetic field returns to its original state, completing what is known as a Hale cycle.

Most of the Sun's mass is Hydrogen and second most is Helium. The mass of the sun is about 3,33,000 times the mass of the Earth and the radius of the sun is 109 times the Earth's radius.

The gravitational attraction on the Sun is about 27.9 times that on the Earth. Density of the Sun is 1.41 g per cm³.

Aphelion is the point of the Earth's orbit that is farthest away from the Sun. It happens mainly on 4 July. The approximate distance between the Earth and the Sun is 152 million km during aphelion.

Perihelion is the point of the Earth's orbit that is shortest from the Sun. It happens on 3 January. The approximate distance between the Earth and the Sun is 147.5 million km during perihelion.

Solstice

Summer solstice - The longest day of the year as the Earth receives daylight for the longest duration. It occurs when the Sun is directly over the Tropic of Cancer. It occurs in the northern hemisphere on 21st June.

Winter solstice - The shortest day of the year as the Earth receives daylight for the

shortest duration. It occurs when the Sun is directly over the Tropic of Capricorn. It occurs in the southern hemisphere on 22 December.

Equinox - The situation when the equator receives the vertical rays of the Sun, and day and night are equal. The spring equinox occurs on 21st March. The autumn equinox occurs on 23 September.

Solar System

A solar system is a collection of various heavenly or celestial bodies that orbit the Sun and are bound because of the gravitational pull of the sun.

There are 8 known planets in our solar system and their order of distance from the Sun is as follows: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.

Planet

It is a large, rounded astronomical body that is neither a star nor its remnant.

The International Astronomical Union (IAU) defined in August 2006 that, in the Solar System, a planet is a celestial body that: is in orbit around the Sun, has sufficient mass to assume hydrostatic equilibrium (a nearly round shape), and it has "cleaned up the neighborhood" around its orbit.

The first four planets Mercury, Venus, Earth, and Mars are called Terrestrial Planets because they are composed of rocks and metals.

The last four planets Jupiter, Saturn, Uranus, and Neptune have massive gaseous atmospheres, so are often called gas giant planets or Jovian planets. They are composed primarily of hydrogen and helium, with traces of methane, ammonia, water, and other gasses in their atmospheres.

Asteroid belt - It is a region which divides the Terrestrial planet and Jovian planets which are in between Mars and Jupiter. Planets in order of size from smallest to largest are Mercury (smallest), Mars, Venus, Earth, Neptune, Uranus, Saturn and Jupiter (largest).

Venus and Uranus are the only two planets that rotate from east to west. Venus rotates slowest while Jupiter rotates fastest among all the planets.

Ascending order of planets on the basis of their density is : Saturn - 0.70 gm/cm³, Uranus - 1.17 gm/cm³, Jupiter - 1.33 gm/cm³, Neptune - 1.66 gm/cm³, Mars - 3.945 gm/cm³, Venus - 5.245 gm/cm³, Mercury - 5.44 gm/cm³, Earth - 5.517 gm/cm³.

Exoplanets (extrasolar planets) - The planets that orbit around other stars located outside our solar system.

Meso planets are planetary-mass objects with sizes smaller than Mercury but larger than Ceres (a dwarf planet).

Super-Earth is any rocky planet that is bigger than Earth and smaller than Neptune.

Planetarium - It is a theater built primarily for presenting educational and entertaining shows about astronomy and the night sky, or for training in celestial navigation.

Mercury

It is the closest planet to the Sun and also the smallest planet in our solar system. It does not have its own natural satellite. Rotation on its axis - 59 Earth days. Revolution - 88 days.

Venus

The second closest to the sun. It is the hottest planet because of It has a thick atmosphere full of the greenhouse gas carbon dioxide and clouds made of sulphuric acid that traps heat and becomes very hot.

It is also called Earth's twin sister because it is similar in size and density as of the Earth. It is also known as Morning and Evening Star. It does not have its own natural satellite like mercury.

Venus is also known by the name Lucifer (Light Bringer) due to its feature of being the brightest planet. It is the only planet whose period of rotation is longer than the period of revolution around the Sun. Rotation on its axis - 243 Earth days. Revolution - 225 days

Earth

The third planet from the Sun. It is called the blue planet as it is covered with over 71 percent water. Therefore from outer space it appears blue. The shape of earth is Geoid. It is the densest planet in the Solar System.

Earth is the only planet in our solar system with liquid water on the surface. It is a rocky planet with a solid and dynamic surface of mountains, canyons, plains and more.

The precise time taken by the earth rotates on its axis with respect to the Sun is 23 hr 56 min 4.09 sec. It is approximately 24 hours i.e one day.

Earth's rotation causes observable patterns like night and day. The Earth's axis of rotation is tilted by 23.5 degrees. This tilt causes the different seasons of the year.

The path of the Earth moving around the Sun is called an orbit. It revolves around the sun in 365 days, 5 hours, 59 minutes and 16 seconds. Earth revolves around the Sun, producing a year. Earth's revolution also causes a change of seasons.

Earth has its only natural satellite i.e. the Moon. Earth's orbital speed averages 30 km/s, which is fast enough to cover the planet's diameter in 7 minutes and the distance to the Moon in 4 hours.

The approximate average density of the Earth is 5.51 gm/cm³. The radius of the earth is 6370 km. Sunlight takes an average of 8 minutes and 20 seconds to travel from the Sun to Earth. Rotation on its axis - 24 hours. Revolution - 365 days.

Average distance from the Earth to the Sun is about 150 million km or 1 AU (Astronomical Unit).

Escape Velocity - The minimum velocity which a body must have in order to escape the gravitational attraction of a particular planet or other object. Escape velocity of the Earth is 11.2 km/s.

Mars

The 4th planet from the Sun. It is the second smallest planet in our solar system. It is called the Red planet due to the presence of iron oxide on its surface. Mars has two satellites that are Phobos and Deimos which were discovered by American astronomer Asaph Hall in 1877.

Mars is the site of Olympus Mons, the largest volcano and the highest known mountain (25 km approximately) in the Solar System. It is one of the largest canyons in the Solar System. It is about 3 times higher than Mount Everest.

Mars' atmosphere however is 95% carbon dioxide, 3% nitrogen, 1.6% argon, carbon monoxide, water, methane, and other gasses, along with a lot of dust. Its gravity is one third of earth.

One Martian day equals 24 hours 39 minutes which is approximately equals one Earth day.

Mars has four seasons just like Earth, but they last about twice as long. That's because it takes about two Earth years for Mars to go around the sun.

Rotation on its axis - 24.6 hours.
Revolution - 687 Earth days.

Jupiter

The 5th planet from the Sun. It is the largest planet in our solar system. It is the fastest spinning planet in our Solar System rotating on average once in just under 10 hours.

It has the shortest days of all the planets in the Solar System. Jupiter has a rotating oval of clouds, twice as wide as Earth, called the 'Great Red Spot', which has been observed for more than 300 years. It is a giant spinning storm, resembling a hurricane.
Rotation on its axis - 10 hours.
Revolution - 12 Earth years (4,333 Earth days).

It has around 95 moons. **Galileo Galilei** in early 1610 discovered that Jupiter has four moons. The four moons were Io, Europa, Ganymede, and Callisto.

Io - The moon of Jupiter. It is the most volcanically active world in the solar system, with hundreds of volcanoes.

Europa - It is one of the natural satellites of Jupiter. It is slightly smaller than Earth's Moon and barely one-quarter the diameter of Earth itself.

Ganymede - A satellite of Jupiter, is the largest and the most massive of the Solar System's moons. It is the ninth-largest object in the Solar System.

Callisto - The second largest moon of Jupiter. It is also known as Jupiter IV. It is the third largest moon in the solar system.

Carme - The moon of Jupiter. It was probably a D-type asteroid.

Saturn

It is the sixth planet from the Sun. It is the second-largest planet in our solar system. It is the only planet which has a ring system, with 7 rings made of chunks of ice and rock. It is also made mostly of hydrogen and helium.

Rotation on its axis - 10.7 hours.
Revolution - 29.4 Earth years (10,756 Earth days).

It is the only planet of our solar system whose density is less than water. It is the least dense planet of our solar system. Saturn has 146 moons.

Titan - The largest satellite of Saturn and the second largest among all the satellites which is composed mainly of water in the form of ice and rocky material. It is larger than Mercury.

Rhea - The second largest satellite of Saturn (largest without an atmosphere).

Atlas - The inner moon of Saturn.

Hyperion - The largest known nonspherical moon in the solar system. It is also known as Saturn VII.

Uranus

It is the seventh planet from the sun. It has the third largest diameter in our solar system. It is the coldest planet in the solar system and also called "ice giant". It is called the green planet due to the presence of methane in the atmosphere.

It is the first planet which was discovered with the aid of a telescope in 1781 by **William Herschel**.

Uranus has 28 known moons. Some important moons of Uranus are Ariel, Titania, Miranda, Oberon, etc. Most of which are named after characters that appear in, or are mentioned in, the works of William Shakespeare and Alexander Pope.

Rotation on its axis - 18 hours.
Revolution - 84 Earth years

Neptune

The eighth and farthest-known planet from the Sun. It is called 'the sleeping planet' because its axis is almost perpendicular to the sun. Its axis of rotation is tilted 28 degrees with respect to the plane of its orbit around the sun.

It is the windiest planet in the solar

system. It radiates more heat than it receives.

It has 16 known moons. Triton is the largest natural satellite of the planet Neptune. Namaka is the smaller, inner moon of the trans-Neptunian dwarf planet Haumea.

Rotation on its axis - 16 hours.
Revolution - 165 Earth years.

Dwarf Planet

According to the International Astronomical Union, a dwarf planet is a celestial body that - Orbits the sun, has enough mass to assume a nearly round shape, has not cleared the neighborhood around its orbit and is not a moon. They are smaller than the planets. The 5 dwarf planets in order of their decreasing size are Pluto, Eris, Makemake, Haumea and Ceres.

Pluto :- Discovered in 1930 by the astronomer Clyde Tombaugh. Since 2006, as per the International Astronomical Union's (IAU) planetary criteria, Pluto is not considered a planet as it had not cleared the area around its orbit, so it was reclassified as a dwarf planet.

The 5 moons of Pluto in order of their distance from Pluto are Charon, Styx, Nix, Kerberos, and Hydra.

Eris : The most massive and second-largest known dwarf planet in the Solar System.

Makemake : It is located in the Kuiper Belt, a donut-shaped region of icy bodies beyond the orbit of Neptune. It is the second-brightest object in the Kuiper Belt as seen from Earth (Pluto is the brightest).

Haumea : A dwarf planet located beyond Neptune's orbit.

Ceres : The smallest dwarf planet. It was the first dwarf planet to be discovered. It is the largest object in the asteroid belt. It's the only dwarf planet located in the inner solar system.

Satellite

A satellite is an astronomical body that orbits a planet, dwarf planet, or small Solar System body (or sometimes another natural satellite).

Natural satellites - It is natural objects

that orbit the other planets and our own Earth. The Moon is the natural satellite of the Earth.

Man made or artificial satellite :- A machine that is launched into space by man and orbits around a body in space in order to collect information. They are specially used for research, communications, weather information, and navigation.

Moon

It is the only natural satellite of our Earth. The Moon is moving around the Earth in an elliptical orbit that's why the distance between the object and the Earth changes throughout the orbit.

Moon is non-luminous as it does not emit its own light and glows by reflecting the light of the sun.

It is the fifth largest natural satellite in the solar system and the largest among planetary satellites relative to the size of the planet that it orbits. It is the second densest satellite after Jupiter's satellite named Io. Mass of the moon is $1/100$ times and the radius of the moon is 1.74×10^6 m or 1,737.4 km which is $1/4$ times that of the earth.

Therefore, the gravitational attraction on the moon is about one sixth of that on the Earth. Hence, the weight of an object on the moon is $1/6$ th of the weight on the Earth because the weight depends on the mass and gravity of the object.

The Moon travels around Earth once every 27.322 days in an elliptical orbit, but takes 29.5 days to change from New Moon to New Moon.

Neil Armstrong : The first person to land on this celestial body in 1969. The average distance between the moon and the Earth is 382,500 km.

Perigee - The point in the orbit of the moon or a satellite at which it is nearest to Earth. The perigee refers specifically to orbits around the Earth and is equivalent to the periapsis of a general orbit.

Apogee :- The point in the orbit of the moon or a satellite at which it is farthest from Earth. The apogee refers specifically to orbits around the Earth, and is equivalent to the apoapsis of a general orbit.

On Full Moon day the Sun and the Moon

are aligned on opposite sides of the Earth and there is a decrease in the amount of heat part of the moon that is visible from the earth.

On New Moon day the illuminated side of the Moon facing the Sun and the night side facing the Earth increases each day as the moon continues in its orbit around the Earth, away from the sun, increasingly more of its sunlit surface is visible.

The lunar phase from the new moon to the full moon is known as the **waxing phase**. The lunar phase from the full moon to the new moon is known as the **waning phase**.

There are eight phases of the moon which go in the order : New Moon, waxing crescent, first quarter (half moon), waxing gibbous, Full Moon, waning gibbous, third quarter (half moon), waning crescent. This cycle repeats once a month (every 29.5 days).

Half moon occurs twice in the moon cycle.

Blood Moon occurs on a total lunar eclipse. It happens when the Earth blocks all direct sunlight from illuminating the Moon's surface.

Solar Eclipse

A solar eclipse happens when the Moon passes between the Sun and the Earth, casting a shadow on Earth that either fully or partially blocks the Sun's light in some areas. It can only take place during a New Moon.

Types of solar eclipse:

Total eclipse - A total solar eclipse happens when the Moon passes between the Sun and Earth, completely blocking the face of the Sun.

Hybrid eclipse - It occurs when an eclipse shifts between annular and total as the Moon's shadow moves across the globe.

Partial eclipse - A partial solar eclipse happens when the Moon passes between the Sun and Earth but the Sun, Moon, and Earth are not perfectly lined up.

Annular eclipse - It happens when the Moon passes between the Sun and Earth, but when it is at or near its farthest point from Earth.

Baily's beads can be observed during the total solar eclipse. The Baily's beads

effect, or diamond ring effect, is a feature of total and annular solar eclipses. The effect is named after Francis Baily, who explained the phenomenon in 1836.

Lunar eclipse

Lunar Eclipse happens when the Earth comes between the Sun and the Full Moon and blocks the Sun's direct rays from lighting up the Moon. Lunar eclipse can only take place during a Full Moon.

Tides

Tides are caused by the combined effects of the gravitational force exerted on earth by the Sun, the Moon, and the rotation of the Earth.

Spring tides or High tides - It occurs twice a month, one during the full moon period and another during the new moon period. It happens when the Sun, Moon and Earth are in alignment.

Neap tides or Low tides - It occurs during the first and third quarter moon, when the moon appears "half full." It happens when the sun and moon are at right angles to each other.

Asteroids

The irregular objects found between the orbit of Mars and Jupiter and that orbit is known as the asteroid belt. **Lina** and **Klotho** are two major asteroids that are moving in the asteroid belt.

Comet

Comets are the mixture of ices and dust that revolve around the Sun in elongated orbits. They are the cosmic snowballs of frozen gases, rock and dust that orbit the Sun.

Halley : It is named after Edmond Halley, first astronomer who discovered the proper motion of stars and periodicity of comets. Halley is the only known short-period comet that is consistently visible to the naked eye from Earth. Halley's orbit period is, on average, 76 Earth years.

Halley was last seen in Earth's skies in 1986 and was met in space by an international fleet of spacecraft. It will return in 2061 on its regular 76-year journey around the Sun.

Meteors

Meteors are bits of rocks and ice ejected from comets as they move in their orbits about the sun.

Astronomers

Eratosthenes - Measured the Earth's circumference mathematically using the angle of the Sun rays at Syene and Alexandria in the third century BCE. Eratosthenes is called the father of geography.

Plato - The first person, as recorded in 'Timaeus' to have identified 'time' with the period of motion of the heavenly bodies. Plato is a great Greek philosopher and teacher of Aristotle.

Ptolemy - A 2nd century Greek mathematician, astronomer and geographer famous for his controversial geocentric theory of the universe, which would form the basis of our understanding of the motions of stars and planets for over a thousand years.

Simon Marius - A German astronomer who named the four largest moons of Jupiter which are Io, Europa, Ganymede and Callisto.

Tycho Brahe - A Danish astronomer who gave the theory of the solar system which is based on a stationary Earth around which the Moon and Sun revolve.

Johannes Kepler - A German mathematician and astronomer who discovered that the Earth and planets travel about the sun in elliptical orbits. He gave three fundamental laws of planetary motion.

First law says that planets move in elliptical orbits with the Sun as a focus.

Second law says that a planet covers the same area of space in the same amount of time no matter where it is in its orbit.

Third law says that the square of the period of any planet is proportional to the cube of the semimajor axis of its orbit.

Giovanni Cassini - Discovered the four moons of Saturn named as Iapetus, Rhea, Tethys, and Dione. He discovered the gap in the ring system of Saturn now known as the Cassini division in 1675.

Galileo Galilei - An Italian astronomer and scientist, the "father of modern science" was the first to report telescopic observations of the mountains on the moon, the moons of Jupiter, the phases of Venus, and the rings of Saturn.

Edwin Hubble - In 1929, he announced

that almost all galaxies appeared to be moving away from us. He found that the universe was expanding - with all of the galaxies moving away from each other. This phenomenon was observed as a redshift of a galaxy's spectrum. He gave the first evidence of the Big-Bang theory.

Charles Messier - A French astronomer. He published an astronomical catalog consisting of 110 nebulae and star clusters, which came to be known as the Messier objects.

Earth and Its Composition

The word geology means **Study of the Earth**. It is the primary Earth science that deals with the earth formation, its structure and composition.

Geological History of The Earth

The **Triassic period** is the first period of the Mesozoic Era that lasted from 245 to 208 million years ago. This period is of frogs and turtles.

The **Jurassic period** is the second period of the Mesozoic Era that lasted from 208 to 144 million years ago. It is called 'Age of Dinosaurs'.

The **Cretaceous** is a geological period that lasted from about 144 to 65 million years ago, Dinosaurs went extinct at the end of the Cretaceous period. It is the third and final period of the Mesozoic Era.

The **Permian period** is the last period of the Palaeozoic Era and it lasted from 245 to 286 millions years ago. In this period, reptiles dominated and replaced amphibians.

Geomorphology is the scientific discipline concerned with the description and classification of the Earth's topographic features.

The Earth is estimated to be roughly 4.54 billion years old. The earliest fossils of life on Earth date back to around 3.8 billion years ago which gives evidence that living organisms have been present on Earth for approximately 82 % of its history. The radius of the earth is 6371 km.

The Earth is made up of three different layers: the crust, the mantle and the core.

Crust

The outermost layer of the earth makes up 0.5 - 1.0 per cent of the earth's volume and less than 1 percent of Earth's mass.

It is the thinnest of all the layers. It is about 35 km on the continental masses and only 5 km on the ocean floors.

The outer covering of the crust is of sedimentary material and below that lie crystalline, igneous and metamorphic rocks which are acidic in nature. The lower layer of the crust consists of basaltic and ultra-basic rocks.

Almost 98% of Earth's crust is made up of eight elements which are -
Oxygen (O) - 46.6 %, Silicon (Si) - 27.7 %, Aluminium (Al) - 8.1%, Iron (Fe) - 5.0 %, Calcium (Ca) - 3.6 %, Sodium (Na) - 2.8 %, Potassium (K) - 2.6 %, Magnesium (Mg) - 1.5 % etc.

The earth's crust has only 0.02% carbon in the form of minerals (like carbonates, hydrogen- carbonates, coal and petroleum).

Minerals in Earth's Crust - The crust of the earth is made up of more than 2000 minerals, but out of these, only six are the most abundant and contribute the maximum. These six most abundant minerals are feldspar, quartz, pyroxenes, amphiboles, mica and olivine.

Feldspar - The most abundant mineral in the Earth's continental crust. It is a type of silicate mineral.

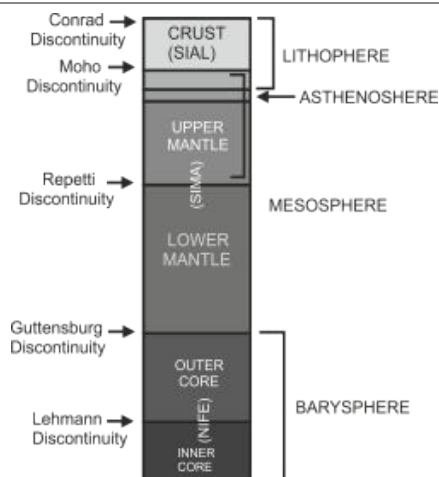
Oceanic crust - Oceanic crust is generally basaltic and is composed of rocks rich in silica and magnesium, it is referred to as SIMA. SIMA is found beneath the SIAL (Silicon- Aluminium) layer and it is heavier than SIAL.

Continental crust is rich in silica and aluminum, it is referred to as SIAL. The mean thickness of continental crust is 30 km.

Discontinuities Inside the Earth :

The discontinuities in the interior are boundaries between crust, mantle, and inner core. They also exist within crust, mantle, and core.

Discontinuities are formed due to variation in the composition of the earth's interior at different layers.



Conrad Discontinuity (named after the seismologist Victor Conrad) is the border between the upper crust (SIAL) and the lower crust (SIMA).

Moho Discontinuity - The Mohorovicic (Moho) Discontinuity forms the boundary between the crust and the Asthenosphere (upper reaches of the mantle).

Repetti Discontinuity - It is the transition zone between Outer mantle and Inner mantle.

Gutenberg Discontinuity - It separates the Core and the Mantle of the earth.

Lehmann Discontinuity - It is the Transition zone between outer and inner core.

Mantle

It forms about 84 percent of the Earth's volume and holds 67% of the earth's mass. It extends from Moho's discontinuity to a depth of 2,900 km.

The density of the upper mantle varies between 2.9 g/cm³ and 3.3 g/cm³. The lower mantle extends beyond the asthenosphere. It is in a solid state. The density ranges from 3.3 g/cm³ to 5.7 g/cm³ in the lower mantle.

Regarding its constituent elements, the mantle is made up of Oxygen (45%), Silicon (21%) and Magnesium (23%).

In the Mantle, temperatures range from approximately 200°C at the upper boundary with the crust to approximately 4,000 °C at the core-mantle boundary.

Core

The core forms the center of the earth. Its density is 13.0 g/cm³. Its temperature is about 5500° C to 6000° C. The core has two parts namely the inner core and the outer core.

Inner core extends from 5100 km below the earth's surface to the center of the earth. The inner core is generally believed to be composed primarily of iron (80%) and some nickel .

Earth's inner core rotates slightly faster relative to the rotation of the surface. The solid inner core is too hot to hold a permanent magnetic field. The purpose of the Earth's magnetic field is to shield the Earth from solar wind and cosmic radiation.

Outer Core - Surrounding the inner core, lies between 2900 km and 5100 km below the earth's surface, which is around 2200 km in thickness.

The outer core is composed of molten rocks, Iron mixed with Nickel and trace amounts of lighter elements. The Density of the outer core ranges from 9.9 g/cm³ to 12.2 g/cm³.

Dynamo theory suggests that convection in the outer core, combined with the Coriolis effect, gives rise to Earth's magnetic field. Outer core is in liquid form.

Types of Mechanical Layers of The Earth :

Lithosphere :- The rigid or rocky outer part of the earth with average thickness varying between 10-200 km. It is made of plates which fit together like a jigsaw puzzle. It includes the crust and the upper part of the mantle.

Asthenosphere :- The upper portion of the Mantle is called the Asthenosphere (astheno means weak). It lies just below the lithosphere extending up to 80-200 km. It is highly viscous, mechanically weak and ductile and its density is higher than that of the crust.

These properties of the asthenosphere aid in Plate tectonic movement. It is the main source of Magma that finds its way to the surface during volcanic eruptions. It is characterized by low seismic wave velocity and high attenuation.

Mesosphere :- The Mesosphere is beneath the Asthenosphere. It encompasses the Lower Mantle, where material still flows but at a much slower rate than the Asthenosphere.

Barysphere :- The innermost layer of Earth is called the Barysphere. It is divided into two parts: outer core and inner core.

Pyrosphere :- Pyrosphere is the middle zone of the interior of the earth with a thickness of 2780 km. It is located between the lithosphere and the barysphere. It is the zone of igneous activity and lava formation.

Distribution of Continents and Oceans

There are **3 theories** related to the distribution of Continents and Oceans around the world.

- 1) Continental drift theory
- 2) Sea floor spreading theory
- 3) Plate tectonic theory

Continental drift theory, proposed by Alfred Wegener in 1912, states that the continents were once joined together as a single supercontinent called Pangaea, and have since drifted apart over geological time.

Around 200 million years ago, Pangaea broke apart. The continents slowly drifted away on moving tectonic plates to their current positions. Evidence: matching coastlines, rock types, fossil distributions across continents. Plate tectonics later explained the driving mechanism.

Polar wandering (similar to Continental Drift Theory) - It is the relative movement of the earth's crust and upper mantle with respect to the rotational poles of the earth.

The **convection current theory** proposes that the movement of tectonic plates is driven by convection currents within the Earth's mantle. Hot mantle material rises, while cooler material sinks, creating circular convection currents. This convection is the driving force behind plate tectonics.

Seafloor Spreading Theory proposes that new oceanic crust is continuously formed at mid-ocean ridges as molten mantle material upwells, cools, and solidifies. As new crust is created, it pushes older crust aside, causing the seafloor to spread outward from the ridge.

At mid-ocean ridges, hot mantle material (Basaltic magma) rises, melts, and a new crust is formed.

This newly formed crust moves away from the ridge in opposite directions. Older oceanic crust gets recycled back into the mantle at deep ocean trenches (subduction zones).

This process of seafloor spreading, coupled with subduction, drives the continuous movement of tectonic plates.

Evidence for Seafloor Spreading - The rocks of the oceanic crust near the oceanic ridges are much younger than the rocks of the continental crust.

A **tectonic plate** is a massive, irregular slab of solid rock made of continental and oceanic lithosphere. **Tectonics** studies the forces (mantle convection) and processes (plate collisions, folding, faulting, volcanism) that shape Earth's crust and its evolution over time.

Plate Tectonics Theory : In 1967, McKenzie and Parker suggested the theory of plate tectonics. **Morgan** later outlined the theory in 1968.

According to the theory of plate tectonics, the earth's lithosphere is broken into distinct plates which are floating on a ductile layer called asthenosphere.

Plates move horizontally over the asthenosphere as rigid units. The lithosphere includes the crust and top mantle with its thickness range varying between 5 - 100 km in oceanic parts and about 200 km in the continental areas.

The oceanic plates contain mainly the **Simatic crust** and are relatively thinner, while the continental plates contain **Sialic material** and are relatively thicker.

Lithospheric plates (tectonic plates) vary from minor plates to major plates, continental plates (Arabian plate) to oceanic plates (Pacific plate), sometimes a combination of both continental and oceanic plates (Indo-Australian plate).

The movement of these crustal plates (due to convection currents in the mantle) causes the formation of various landforms and is the principal cause of all earth movements.

7 Major tectonic plates : Pacific plate (Largest), North American plate, Eurasia and the adjacent oceanic plate, Africa with the eastern Atlantic floor plate, Antarctica and the surrounding oceanic plate, India-Australia-New Zealand plate, South American plate (Smallest).

Minor Tectonic Plates

Cocos plate: Between Central America and the Pacific plate.

Nazca plate : Between South America and the Pacific plate.

Arabian plate : Mostly the Saudi Arabian landmass.

Philippine plate : Between the Asiatic and Pacific plate.

Caroline plate: Between the Philippine and Indian plate (North of New Guinea).

Fuji plate: North-east of Australia, Turkish plate, Aegean plate (Mediterranean region), Caribbean plate, Juan de Fuca plate (between Pacific and North American plates) and Iranian plate.

The **Somali plate** is the largest minor plate and Juan de Fuca is the smallest. Most of these minor plates were formed due to stress created by converging major plates.

Example: the Mediterranean Sea is divided into numerous minor plates due to the compressive force exerted by Eurasian and African plates.

Plate Boundaries

Divergent Boundaries : It occurs along spreading centres where plates are breaking apart and pushed upward from the mantle by magma to form a new crust. The sites where the plates move away from each other are called spreading sites.

Example - Mid-Atlantic Ridge separating American and Eurasian/ African plates. This single mid-ocean ridge system connects the world's oceans, making the ridge the longest mountain range in the world.

On land, giant troughs such as the Great Rift Valley in Africa form where plates are tugged apart. If the plates there continue to diverge, millions of years from now eastern Africa will split from the continent to form a new landmass.

There are many famous rift valleys like African rift valley, Baikal rift zone, West Antarctic rift system etc.

Convergent boundaries : It occurs where two tectonic plates move towards each other and collide.

There are three ways in which convergence can occur :

Oceanic-oceanic convergence occurs when two plates carrying oceanic crust move towards and collide with each other. The denser oceanic plate gets pushed downwards into the mantle (subduction). The subducting plate causes melting and volcanism in the overriding plate. This forms a volcanic arc of islands parallel to the trench. Example: Mariana Trench and Mariana

Island Arc in the western Pacific

Oceanic continental convergence - The denser oceanic plate is forced beneath the less dense continental plate at a convergent boundary. The subducting oceanic plate melts as it reenters the mantle. The magma rises and erupts. This creates a volcanic mountain range near the coast of the continent. Examples - The Andes Mountains (South America), Rocky mountains (North America).

Continental continental convergence - When two continental plates collide with each other, a collision occurs because both continental plates have similar densities, preventing subduction. This convergence leads to formation of Fold mountains. Examples - Himalayas, Alps.

Transform Boundaries : It occurs where two tectonic plates slide past each other horizontally. Transform faults are the planes of separation generally perpendicular to the mid oceanic ridges. The San Andreas Fault in California is an example of a transform boundary, where the Pacific Plate moves northward past the North American Plate.

Hotspots - Most of the volcanic activities are concentrated along or adjacent to plate boundaries, but there are some important exceptions in which this activity occurs within plates, called the Hotspots.

Hotspots exist over the Mantle Plumes. A mantle plume is an area under the crust of Earth, where magma is hotter than surrounding magma. Heat from this extra hot magma causes melting and thinning of the rocky crust, which leads to widespread volcanic activity on Earth's surface above the plume.

The hotspots are stationary. There are about 40 to 50 hot spots estimated to be around the world.

Major hotspots include : The Iceland hotspot, under the island of Iceland in the North Atlantic. The Reunion hotspot, under the island of Reunion in the Indian Ocean. The Afar hotspot, located under northeastern Ethiopia.

An elongated area of relatively low pressure extending from the center of a region of low pressure is called **Trough** while an elongated area of relatively high pressure extending from the center of a high-pressure region is called **Ridge**. U shaped lines indicate the presence of a ridge.

Mid - Oceanic Ridges: Underwater mountain ranges formed by tectonic plates moving apart. Magma rises from the mantle, creating a new crust. Examples: Mid-Atlantic Ridge, East Pacific Rise.

It is characterised by a central rift system at the crest, a fractionated plateau and flank zone all along its length. The rift system at the crest is the zone of intense volcanic activity.

The **Mid-Atlantic Ridge** is in effect an immensely long mountain chain extending for about 10,000 miles (16,000 km) in a curving path from the Arctic Ocean to near the southern tip of Africa. The ridge is equidistant between the continents on either side of it.

The **Lomonosov Ridge** is a trans-oceanic seafloor high that separates the Eurasia Basin from the Americas Basin. It extends for a distance of almost 1800 km across the Arctic Ocean from the Lincoln Shelf off Greenland and Canada to the East Siberian Shelf.

Aden Ridge is a part of an active oblique rift system located in the Gulf of Aden, between Somalia and the Arabian Peninsula to the north.

Geckel Ridge is a mid-oceanic ridge, a divergent tectonic plate boundary between the North American Plate and the Eurasian Plate. It is located in the Eurasian Basin of the Arctic Ocean, between Greenland.

Carlsberg Ridge is the northern section of the Central Indian Ridge, a divergent tectonic plate boundary between the African Plate and the Indo- Australian Plate, traversing the western regions of the Indian Ocean.

The **Arctic Ridge** has the slow rate (less than 2.5 cm/year) and the East Pacific Rise in the South Pacific (about 3,400 km west of Chile) has the fast rate (more than 15 cm/year).

Oceanic trough, an elongate depression in the seafloor that is characteristically shallower, shorter, narrower, and topographically gentler than oceanic trenches.

Maximal depths of oceanic troughs range between 2,300 m (7,500 feet) in the Papuan Trough and 7,440 m in the Banda Trough.

When one of the plates is an oceanic plate, it gets embedded in the softer

asthenosphere of the continental plate, and as a result, trenches or Oceanic Trough are formed at the zone of subduction.

Fault is a fracture or crack where two rock blocks slide past one another. If this displacement is rapid, it may cause an earthquake slowly due to sliding.

There are three kinds of faults : strike-slip (Transform), normal and thrust (reverse) faults.

The **Ring of Fire** is a string of volcanoes and sites of seismic activity, or earthquakes, around the edges of the Pacific Ocean. It is formed by tectonic plate movements, including subduction zones and volcanic arcs.

Volcano, Earthquake, Tsunami

Volcanoes

A volcano is an opening in the earth's crust through which lava, volcanic ash, and gases escape. They are built up by the liquid and solid rock erupting from the earth's interior.

Volcanism is the process of material escaping from Earth's interior to its surface.

Volcanoes are classified as active, dormant, or extinct.

Active Volcano has erupted recently or is expected to erupt soon.

Examples: Mauna Loa (Hawaii), Mount Vesuvius (Italy), Mount Nyiragongo (Congo), Taal (Philippines), Mount Merapi (Indonesia), Galeras (Colombia), Sakurajima (Japan), Santa Maria (Guatemala), Eyjafjallajokull (Iceland).

Dormant Volcano has not erupted in a long time but is expected to erupt again.

Examples - Mauna Kea (Hawaii), Mount Fuji (Japan), Mount Kilimanjaro (Africa), Mount Pelee of Martinique.

Extinct Volcano that has erupted in the past but is unlikely to erupt soon.

Example - Kyushu-Palau Ridge in the Philippine Sea, Huascarán in Peru, Mount Buninyong in Australia.

Types of Volcanoes on the basis of form developed at the surface.

Shield Volcanoes - Shaped like a bowl or shield, formed by the eruption of low-viscosity lava that can flow a great distance. Examples: Mauna Kea, Mauna Loa.

Cinder Cone Volcanoes - Circular or oval cones made up of small fragments of lava from a single vent that have been blown up.

Composite Volcanoes (Stratovolcanoes) - Steep-sided volcanoes composed of many layers of volcanic rocks, usually made from high-viscosity lava, ash and rock debris.

Calderas - The most explosive of the earth's volcanoes, formed when a volcano erupts and collapses.

Flood Basalt Provinces - Volcanoes that outpour highly fluid lava that flows for long distances. Example: Deccan Traps in India.

Mid-ocean ridge volcanoes occur in oceanic areas, forming the most extensive chain of underwater mountains, stretching over 65,000 km. The central portion of this ridge experiences frequent eruptions.

Largest Volcanoes in the world

Tamu Massif - Considered the largest volcano on Earth, located in the northwestern Pacific Ocean.

Mauna Loa - The largest active volcano in the world, located in Hawaii.

Mount Etna - The tallest active volcano in Italy.

Mount Vesuvius - The only active volcano in Europe.

Kilauea - An active shield volcano in the Hawaiian Islands.

Mount Merapi - Indonesia's most active volcano.

Volcanoes in India

Barren Island volcano - The only active volcano in India, located in the Andaman and Nicobar islands.

Narcondam - A small volcanic island located in the northern Andaman Sea.

Baratang islands in Andaman and Nicobar contain the only known mud volcanoes in India.

Dhosi Hill - An important Vedic period

site, an extinct volcano, in Aravalli mountain range in Haryana.

Dhinodhar Hill - A volcanic plug rising above the local sandstone in Gujarat.

Magma and Lava

Magma is molten rock beneath the Earth's surface, sourced from the asthenosphere. When it erupts through a volcano's vent, it becomes **lava**.

Batholiths are igneous rock formations formed when magma rises into the Earth's crust but doesn't erupt, creating underground structures.

Earthquake

An earthquake is the shaking of the Earth's surface due to sudden movement in the crust, releasing energy as seismic waves. Natural Earthquakes happen in the Lithosphere.

Endogenic forces, operating within the Earth's interior, create faults such as the San Andreas Fault, the world's longest, which in turn trigger seismic activity, resulting in earthquakes.

The location below the earth's surface where the earthquake starts is called the hypocenter, and the location directly above it on the surface of the earth is called the epicenter.

Earthquake Measurement

Seismograph - An instrument used to measure and register seismic waves, providing information about the epicenter, depth, and magnitude of an earthquake.

Richter scale - Measures earthquake magnitude from 1 to 10.

Mercalli Scale: Measures earthquake intensity on a scale of I to XII.

Causes of Earthquakes - Compressional or tensional stresses built up at the margins of the huge moving lithospheric plates, Sudden release of stress along faults, rock volume and density changes due to intense temperature and pressure, and human activities.

Primary Effects of Earthquakes - Ground Shaking, Differential ground settlement, Land and mudslides, Soil liquefaction, Ground lurching, Avalanches, displacement, Floods, Fires, Structural collapse, Tsunami.

Types of Earthquakes

Tectonic Earthquakes are generated due to sliding of rocks along a fault plane.

Volcanic Earthquakes occur due to volcanic activities and displacement caused by those.

Collapse Earthquakes caused by the collapse of underground mine roofs.

Explosion Earthquakes result from the collapse of underground mine roofs during intense mining activity, leading to minor tremors.

Reservoir Induced Earthquakes - Occur in the areas of large reservoirs.

Some of the great earthquakes in India - Cutch (Kutch) Earthquake (1819) - Gujarat, Assam Earthquake (1897), Bihar-Nepal Earthquake (1934), Koyna Earthquake (1967) - Maharashtra, Uttarkashi Earthquake (1991) - Uttarakhand, Killari Earthquake (1993) - Maharashtra, Bhuj Earthquake (2001) - Gujarat, Jammu Kashmir Earthquake (2005)

Seismic Waves (Earthquake Waves)

Seismic waves are the waves of energy caused by earthquakes or an explosion.

Types of Seismic Waves - Body waves and Surface waves.

Body waves are generated at the focus of an earthquake, traveling through the Earth's interior in all directions. There are two types of body waves -

P waves, or primary waves, are longitudinal seismic waves analogous to sound waves. They move faster, arrive first, and have high frequency. They can travel through all mediums, with velocities ranging from 1.5 to 8 km/s, faster in solids than in liquids and gases.

Secondary waves or S waves are transverse seismic waves similar to water ripples or light waves. They arrive after P waves and travel only through solids, being about 1.7 times slower.

Surface Waves are generated when body waves interact with surface rocks, moving along the Earth's surface.

There are two types of Surface waves - **L waves or Love waves** are confined to the

surface of the crust, producing entirely horizontal motion. They are much slower than body waves but are faster than Rayleigh.

Rayleigh waves roll along the ground like waves on water, causing both vertical and horizontal movement in the same direction as the wave. Rayleigh waves are more destructive than body waves.

Earthquake Zones in India

The Bureau of Indian Standards has divided the country into four seismic zones, Zone- II, III, IV and V.

Zone V (most active region) - Entire northeast India, parts of Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Rann of Kutch in Gujarat, part of North Bihar and Andaman & Nicobar Islands.

Zone IV - Parts of Jammu & Kashmir, Sikkim, parts of Gujarat and Bihar and small portions of Maharashtra near the west coast, Northern Parts of Uttar Pradesh, Bihar and West Bengal.

Zone III - Remaining parts of Uttar Pradesh, Gujarat, West Bengal, etc.

Zone II (least seismically active) - Covers remaining parts of the country

Tsunami

Tsunamis, caused by seismic events like earthquakes, undersea landslides, volcanic eruptions, or extraterrestrial impacts in the sea, are distinct from wind-driven sea waves. While wind primarily generates ordinary sea waves.

Tsunami in the Indian Ocean

Tsunamis are relatively rare in the Indian Ocean compared to the Pacific Ocean, as the seismic activity is less.

The 2004 Indian Ocean tsunami was caused by a massive 9.1 magnitude earthquake off the west coast of Sumatra, Indonesia, and it severely affected Tamil Nadu in India.

Longitudes and Latitudes

Latitude

The latitude of a place on the earth's surface is its distance north or south of the equator, measuring location north or south of the equator. Lines joining places with the same latitudes are called

parallels. The value of the equator is 0° and the latitude of the poles are 90° North and 90° South.

If parallels of latitude are drawn at an interval of one degree, there will be 89 parallels in the northern and the southern hemispheres each. The total number of parallels thus drawn, including the equator, will be 179. The distance between these lines is about 111 kilometers.

Some important parallels include the North and South Poles (at 90° N and 90°S respectively), which mark the farthest points from the equator. The Arctic and Antarctic Circles (at 66.5° North and South) define the boundaries of the Arctic and Antarctic regions.

Equator

Equator - An imaginary line located at 0° latitude. The equator is a circle of latitude that divides a spheroid, such as Earth, into the northern and southern hemispheres.

The equator passes through **13 countries and 3 continents** : In South America - Ecuador, Colombia, Brazil; in Africa - Sao Tome and Principe, Gabon, Republic of the Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia and in Asia - Maldives, Indonesia and Kiribati in Oceania.

The Congo River crosses the equator twice in Africa. Africa is the only continent through which the Tropic of Cancer, the Equator and the Tropic of Capricorn pass.

Tropic of Cancer

The Tropic of Cancer is the line of latitude (23 ½° N) in the Northern Hemisphere.

Tropic of Cancer passes through **16 countries**: Mexico, Bahamas, Mauritania, Mali, Algeria, Niger, Libya, Egypt, Saudi Arabia, United Arab Emirates, Oman, India, Bangladesh, Myanmar, China, Taiwan.

Tropic of Cancer passes through **8 states** in India : Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

It passes through Africa, Asia and the North America continent. The Tropic of Cancer divides India into almost two equal parts.

Mahi River is the only river in India that cuts the Tropic of Cancer twice, first in Madhya Pradesh from where it flows towards Rajasthan and enters Gujarat where it cuts for the second time.

Tropic of Capricorn

The Tropic of Capricorn is a parallel latitude on the Earth which is 23.5° latitude in the southern Hemisphere.

It passes through 10 countries : Argentina, Chile, Paraguay, Brazil, Madagascar, South Africa, Namibia, Botswana, Mozambique, Australia. The Limpopo River crosses the tropic of Capricorn twice in South Africa.

Zone

Torrid zone - 0° to 23 ½° north and south. The torrid zone is generally warm.

Temperate zone - 23 ½° to 66 ½° north and south.

Frigid zone - 66 ½° to 90° north and south. Frigid zones are areas that are very cold. It is located in the middle of the Arctic circle and North pole in the Northern Hemisphere and between the Antarctic circle and South pole in the Southern Hemisphere.

Longitude

Lines of longitude, also called meridians, are imaginary lines that divide the Earth. They run north to south from pole to pole, but they measure the distance east or west. The meridians of longitude are semicircles that converge at the poles.

The meridians intersect the equator at right angles. Unlike the parallels of latitude, they are all equal in length.

The time difference between each longitude (each degree) is 4 minutes. The distance between two longitudes is maximum at the equator (111.3km) and minimum at the poles (0 km).

The 90° East Meridian passes through state Meghalaya in India and it crosses Bhutan and Bangladesh.

Longitude and Time

The standard time is based on some longitude which usually passes through the center of the region or the country and all the clocks of the region regulate their timings with respect to this longitude or meridian.

A **prime meridian** is an arbitrary meridian (a line of longitude) in a geographic coordinate system at which longitude is defined to be 0°. It is used as the basis for the world's time zones. France has the most time zones with 12 (13 including its claim in Antarctica). Russia has 11 time zones.

Greenwich meridian, also called **prime meridian**, an imaginary line to indicate 0° longitude at London (England). On 1 November 1884, the Greenwich Meridian was adopted universally at the International Meridian Conference in Washington DC, USA.

Indian Standard Time is calculated on the basis of **82°30'E**, a longitude which is just west of the town of Mirzapur, near Prayagraj in the state of Uttar Pradesh.

The **standard meridian of India** passes through the following states- Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha and Andhra Pradesh.

Indian standard time = $82.5^\circ \times 4 = 330$ minutes or 5 hours 30 minutes. Indian Standard Time is **5:30 hours** ahead of Greenwich, London, UK.

International Date Line (IDL)

It is the Line that corresponds along the 180° meridian of longitude and is opposite to the Greenwich line. It passes through the Pacific Ocean.

It was accepted in the International Meridian Conference in 1884 in Washington, D.C. It is an imaginary and Zig-Zags line.

Isopleth

It is a line on a map connecting points having equal incidence of a specified meteorological feature.

Some Isopleth :

Isohaline - A line drawn on a map to indicate connecting points of equal salinity in the ocean.

Isochrones - The lines joining the places which reach at the same time on the map are called 'simultaneous lines'.

Isonomal - An imaginary line or a line on a chart connecting places that have the same anomalies especially of temperature or pressure.

Contours - Imaginary lines joining all the

points of equal elevation or altitude above mean sea level. They are also called "level lines".

Hachures - A short line used for shading and denoting surfaces in relief (as in map drawing) and drawn in the direction of slope.

Famous places and their latitudes and longitudes -

The **Ganga-Brahmaputra** basin lies between 10°N and 30°N latitudes.

Itanagar, the capital of Arunachal Pradesh, is located between 93° E longitude and 27° N latitude.

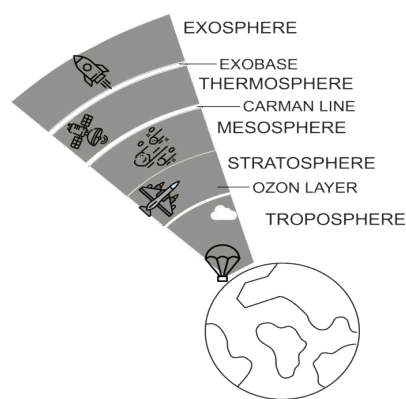
Mount Everest is within one degree of latitude from New Delhi. The latitude of Mount Everest is 27.9°N, 86.9°E while that of New Delhi is 28.61° N and 77.2°E.

Atmosphere

Atmosphere is a gaseous layer surrounding Earth that protects life by regulating temperatures and blocking harmful solar radiation. It consists of a mixture of gasses and aerosols, and its study, including weather phenomena, is called meteorology.

Earth's atmosphere is composed of Nitrogen (78.08%), Oxygen (20.95%), Argon (0.93%), Carbon dioxide (0.036%), Neon (0.002%), Helium (0.0005%), Hydrogen (0.00005%), etc. Noble gas **Radon** is not found in the atmosphere.

Layers of Atmosphere



Troposphere

The Troposphere is the lowest layer of Earth's atmosphere. It extends from ground level to 8 km at the poles and 18 km at the equator. Its average height is 13 km. It is the important layer for all biological activities.

It is also called "region of mixing"

because the entire **weather** is confined inside this layer of atmosphere. This layer is most liable for the greenhouse effect.

The troposphere absorbs the Earth's radiated heat, keeping the surface warm. Temperature decreases with height at the normal lapse rate of 6.5°C per km. The tropopause, about 1.5 km thick, separates the troposphere from the overlying stratosphere.

Stratosphere

The Stratosphere lies above the troposphere. It extends from ground level to about 50 km height. This region is known for its calm winds and crystal-clear skies, making it the perfect cruising altitude for airplanes.

It also consists of an **ozone layer** which acts as a shield for life on Earth; it absorbs ultraviolet rays of sunlight. In this way, it helps to reduce the temperature of Earth. The ozone layer extends from ground level to 15-30 Km height and 3 mm in thickness or about 300 Dobson Units.

The temperature increases with height in this layer due to the presence of ozone molecules. The upper layer of the stratosphere is known as stratopause.

Mesosphere

The Mesosphere is the third layer of Earth's atmosphere which lies above the stratosphere. It extends from ground level to 50-90 km height. In this layer, the temperature goes down with the height and makes it the coldest layer of the atmosphere. The temperature can reach the lowest of -90°C at a height of 80-90 km.

In this layer, the celestial bodies like meteorites burn up while entering into this layer of the atmosphere from outer space. The upper layer of the mesosphere is known as mesopause.

The carman line is a proposed conventional boundary between Earth's atmosphere and outer space set by the international record-keeping body FAI (Federation Aeronautique Internationale) at an altitude of 100 kilometers above mean sea level.

Thermosphere

The Thermosphere lies above the mesosphere. It extends from ground level to 80 - 640 km height. In this layer,

the temperature increases with the height due to the presence of ultraviolet and X-ray radiation from the sun. The temperature can reach the highest about 2000°C in the upper thermosphere. Weather events like northern lights, southern lights, and auroras happen in this layer.

This layer of the atmosphere is also called the Ionosphere due to the availability of electrically charged particles. This layer reflects back the radio waves to the Earth which are transmitted from the Earth. It is ionised by solar and cosmic radiation.

Exosphere

The Exosphere is the top most layer of Earth's atmosphere which lies above the Ionosphere. It extends from the ground level to 10,000 km in height. The air density is lesser in this layer and the temperature reaches up to 5568°C.

It is a very close layer to outer space. Lighter gaseous particles of hydrogen and helium move into outer space and make air density lesser. There is no distinct upper layer of the exosphere but the imaginary line of exopause separates the exosphere from outer space.

Ozone

Ozone is a toxic pale blue gas made up of three oxygen atoms, and it forms the ozone layer in the upper atmosphere.

Ozone layer depletion is the thinning of the ozone layer in the Earth's stratosphere, caused by Ozone Depleting Substances (ODSs) like chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), halons, and others.

An **Ozone hole** is a region in the stratosphere, predominantly over the poles, where ozone concentration becomes extremely low during certain months. The first ozone hole was discovered over Antarctica in 1970.

The **Montreal Protocol** is an international environmental agreement that seeks to regulate the production and consumption of ozone-depleting substances. It was signed in 1987 and came into force in 1989. India became a signatory to the Montreal Protocol in 1992.

Greenhouse effect

The greenhouse effect is a natural process where greenhouse gasses (GHGs) like carbon dioxide, methane, and tropospheric ozone absorb and re-radiate infrared radiation, warming the Earth's surface and atmosphere. While beneficial to an extent, excessive GHGs levels from human activities intensify this effect, contributing to global warming.

Global warming is a continuous increase in the average temperature of the planet.

The primary greenhouse gases in Earth's atmosphere are water vapour (H₂O), carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and ozone (O₃). Other GHGs are carbon monoxide, fluorinated gases, chlorofluorocarbons (CFCs), black carbon (soot), and brown carbon.

The Kyoto Protocol was an international agreement adopted in 1997 and enforced in 2005, where developed nations committed to reducing emissions of six major greenhouse gases: carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF₆), to mitigate climate change impacts.

Atmospheric pressure

The weight of a column of air contained in a unit area from the mean sea level to the top of the atmosphere is called the Atmospheric pressure. It is measured in force per unit area expressed in 'millibar' (mbar).

The aneroid barometer or mercury barometer is the instrument used to measure atmospheric pressure. The typical pressure at sea level is 1013.25 millibars (1.01×10^5 Pascal or 4.7 pounds per square inch or 760 mmHg). Barograph is used for continuous recording of atmospheric pressure.

Lightning

The bright flash of light which we see in the clouds is called lightning. It is an electric discharge in the atmosphere between oppositely charged clouds or between charged clouds and the earth. The study of lightning is called Fulminology.

Hydrosphere

A hydrosphere is the total amount of water on a planet. The hydrosphere includes water that is on the surface of the planet, underground, and in the air. A planet's hydrosphere can be Water, vapor, or ice.

On Earth, liquid water exists on the surface in the form of oceans, lakes and rivers. It also exists below ground — as groundwater, in wells and aquifers. Water vapor is most visible as clouds and fog.

The frozen part of Earth's hydrosphere is made of ice in the forms of glaciers, ice caps and icebergs. The frozen part of the hydrosphere has its own name, the cryosphere.

Water cycle - Water moves through the hydrosphere in a cycle. It collects in clouds, then falls as rain or snow. This water collects in rivers, lakes and oceans. It then evaporates into the atmosphere to start the cycle again.

The hydrological cycle is the circulation of water within the earth's hydrosphere in different forms i.e. the liquid, solid and the gaseous phases. It also refers to the continuous exchange of water between the oceans, atmosphere, land surface and subsurface and the organisms.

Water on the Earth's surface Reserve Percentage - Oceans - 97.25%, Ice Caps and Glaciers - 2.05%, Groundwater - 0.68%, Lakes - 0.01%, Soil Moisture - 0.005%, Atmosphere - 0.001%, Streams and Rivers - 0.0001%.

Oceans and Ocean Currents

The oceans constitute 97.3% of all water sources on the Earth. The percentage of freshwater sources is only 2.7%.

India receives nearly 4 per cent of the global precipitation and ranks 133 in the world in terms of water availability per person per annum.

Seawater constitutes a rich source of various commercially important chemical elements. Seawater is a rich source of Bromine.

Composition : An Ocean is a complex mixture of 96.5 percent water, 2.5 percent salts. Oceans and Seas contain salty water. They are not a source of sweet water.

Ocean Floor

The ocean floors can be divided into four major divisions: (i) the Continental Shelf; (ii) the Continental Slope; (iii) the Deep Sea Plain; (iv) the Oceanic Deep.

Continental shelf : It is the shallowest part of the ocean showing an average gradient of 1° or even less. Coral reefs are also common on continental shelf.

The **Siberian Shelf** is a coastal shelf in the Arctic Ocean and is the largest continental shelf of the Earth, a part of the continental shelf of Russia.

The continental shelf is the extended margin of each continent occupied by relatively shallow seas and gulfs.

Continental slopes : These are regions of steeply sloping seafloor that lie between continental shelves and the deep ocean basins. The gradient of the slope region varies between 2°-5°. The depth of the slope region varies between 200 and 3,000 m. The slope boundary indicates the end of the continents. Canyons and trenches are observed in this region.

The continental slope is a geological feature marking the divide between two ocean zones.

The deep-sea plains or Abyssal plains : These are gently sloping areas of the ocean basins. These are the flattest and smoothest regions of the world because of terrigenous and shallow water sediments that bury the irregular topography.

These are as irregular as the continental plains with submarine plateaus, hills, guyots, and seamounts. The floor of the abyssal plain is covered by sediments.

Oceanic deeps : It is a part of an ocean basin in which the depth greatly exceeds the average for the basin as a whole.

Seamount : It is a mountain with pointed summits, rising from the seafloor that does not reach the surface of the ocean. Seamounts are volcanic in origin. These can be 3,000 - 4,500 m tall. The Emperor seamount, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example.

Continental margins : It refers to the region of transition from the land to the deep seafloor, i.e. between continental and oceanic crust.

Oceanic ridges : These are the topographic expressions of linear or

curvilinear zones along which the Earth's oceanic crust is created or modified.

Ocean trenches : These are long, narrow depressions on the seafloor. Over the course of exploration, 57 deeps have been found, 32 of which are located in the Pacific Ocean, 19 in the Atlantic Ocean, and 6 in the Indian Ocean.

Thermocline : This boundary region, from where there is a rapid decrease of temperature, is called the thermocline. About 90 per cent of the total volume of water is found below the thermocline in the deep ocean.

Ocean acidification : It refers to a reduction in the pH of the ocean over an extended period of time, caused primarily by uptake of carbon dioxide (CO₂) from the atmosphere.

Ocean salinity

It is generally defined as the salt concentration in seawater. It increases by the loss of water to ice or evaporation and decreases by the input of fresh water. Saltiest ocean- Atlantic Ocean.

Factors affecting ocean salinity :- The salinity of water in the surface layer of oceans depends mainly on evaporation and precipitation.

Surface salinity is greatly influenced in coastal regions by the freshwater flow from rivers, and in polar regions by the processes of freezing and thawing of ice.

The lower salinity water rests above the higher salinity dense water. Salinity, generally, increases with depth and there is a distinct zone called the halocline, where salinity increases sharply.

Wind, ocean currents and Change in the temperature or density of water influences the salinity of water in an area.

Oceans of the world

According to size, the sequence of Oceans from largest to smallest is Pacific Ocean > Atlantic Ocean > Indian Ocean > Southern Ocean > the Arctic Ocean.

Pacific Ocean

It is the largest ocean on the Earth. It occupies one-third of the earth's surface and is larger than all the continents combined. It separates Asia and

Australia from North America and South America.

Ferdinand Magellan : Portuguese explorer Ferdinand Magellan gave it the name Pacific Ocean, which translates as "peaceful". Because he perceived the ocean to be quiet.

Mariana trench : It is the deepest oceanic trench in the world, which is located in the western Pacific Ocean. The Challenger Deep is the lowest point in the Mariana Trench, which is present in the Pacific ocean.

The Ring of Fire is a ring of volcanoes around the Pacific Ocean, which results from subduction of oceanic plates beneath lighter continental plates.

It is also referred to as the Circum-Pacific Belt, or Rim of the Pacific, is a path along the Pacific Ocean characterized by active volcanoes and frequent earthquakes. Its length is approximately 40,000 kilometres.

The Ring of Fire is home to more than 450 active and dormant volcanoes (75% of Earth's total volcanoes) forming a semicircle or horse shoe around the rim of the Pacific Ocean.

Location of Date line : The International Date Line is located in the Pacific Ocean. The Pacific Ocean and the Indian Ocean are connected by the Indonesian seaway (Torres strait and strait of Malacca).

Atlantic Ocean

It is the world's second-largest ocean. It covers about 20% of the earth's surface. Its name, Atlantis Thalassa, which means Sea of Atlas, comes from Greek mythology. The Atlantic Ocean extends into the Mediterranean Sea.

Location : It is located between Africa, Europe, the Arctic Ocean, the Americas, and the Southern Ocean. It is a body of water with an extended S shape. It separates North America from Europe.

Puerto Rico Trench: It is the deepest trench in the Atlantic Ocean (8,376 m). It is located on the boundary between the Caribbean Sea and the Atlantic Ocean.

Greenland is the largest island, found in the Atlantic Ocean. The Atlantic Ocean contains the Sargasso Sea, the only sea in the world that does not have a land boundary.

The **Gulf of Mexico** and **Labrador Sea** are part of the North Atlantic Ocean. The Northeast Canyons and Seamounts Marine National Monument is located in the Atlantic Ocean.

Importance of location : The Atlantic Ocean is an ideal location for natural harbours and ports because it has a long indented coastline, making it easier for ships to enter the port. Thus, the Atlantic Ocean is the busiest ocean in terms of trade and commerce.

Major seas in the Atlantic Ocean : Sargasso Sea, Caribbean Sea, Mediterranean Sea, Gulf of Mexico, Norwegian Sea - Greenland Sea, Labrador Sea, Irminger Sea, Baffin Bay, North Sea, Baltic Sea.

The Indian Ocean

The Indian Ocean is the third largest of the world's oceanic divisions. It lies to the south of India. It is smaller and less deep than the Atlantic Ocean.

Madagascar is the biggest island in the Indian Ocean (the fourth largest island in the world).

Location : It is situated between Asia, Australia, the Southern Ocean, and Africa. It is the world's hottest ocean and is enclosed by land in the northern hemisphere.

Trench : The deepest trench in the Indian Ocean is called the Sunda/Java Trench and is about 7450 metres deep.

Importance of location : The Indian Ocean's marginal seas are crucial for trade because they provide vital marine lanes that link the Middle East, East Asia, Africa, Europe, and the Americas.

An island, peninsula, or collection of islands that encloses a portion of an ocean is known as a marginal sea (group of islands). They are located close to land and are shallower than the ocean. The Indian Ocean separates Africa and Australia.

Area : The Indian ocean covers an area of about 28 million square miles (volume - 73 million km³) which is about 20% (19.8%) of the total area covered by the world's oceans.

The Indian Ocean has an average salinity of 35 parts per thousand. The Gulf of Oman and Red Sea are part of the Indian Ocean. The Bay of Bengal and Arabian Sea are the two northward extensions of

the Indian Ocean.

Antarctic Ocean

The southernmost parts of the world's oceans are collectively referred to as the Antarctic Ocean or Austral Ocean. It is also called the Southern Ocean. It is the second-smallest and fourth-largest ocean in the world, surrounding Antarctica south of 60°S latitude.

Characteristic : It is renowned for having severe storms. It stretches into the Southern Ocean and joins the southwest corner of the Atlantic (Scotia sea) with the southeast corner of the Pacific ocean.

Drake Passage : It is the body of water between Chile's Cape Horn in South America and Antarctica's South Shetland Islands.

The South Sandwich Trench is the deepest trench. Factorian Deep is the new deepest point in Antarctica's Southern Ocean.

Arctic Ocean

The Arctic Ocean is the smallest of all oceans covering an area of 1,55,58,000 km² and makes up only 4.3% of the global ocean.

Location : It is surrounded by Eurasia and North America.

Characteristics : It is mostly covered by sea ice throughout the year and almost completely in winter. Of the five great oceans, its salinity is the lowest on average.

The Greenland Sea, Labrador Sea, and Norwegian Sea connect the Arctic Ocean to the Pacific Ocean while the Bering Strait connects it to the Atlantic Ocean.

The deepest spot in the Arctic Ocean is the Molloy Deep, or "Molloy Hole," which is located in the Fram Strait (inside the Greenland Sea). The Arctic Ocean is centered approximately on the North Pole.

The Tethys Ocean, also called the Tethys Sea or the Neo-Tethys, was an ocean during much of the Mesozoic Era and early Cenozoic Era. It was present in place of the Himalayas before the tectonic plates shifted.

Deep Ocean Mission : It is a mission-mode project to support the Blue Economy Initiatives of the

Government of India. It is a high-level multi-ministerial, multi-disciplinary programme for a better understanding of the deep sea living and non-living resources of the Indian Ocean. The Ministry of Earth Sciences (MoES) will be the nodal Ministry implementing this multi-institutional Mission.

Ocean currents

Ocean currents are the continuous, predictable, directional movement of seawater driven by gravity, wind (Coriolis Effect), and water density. The primary forces that influence the currents are: (i) heating by solar energy; (ii) wind; (iii) gravity; (iv) coriolis force.

Types of ocean currents on the basis of depth :

Surface currents constitute about 10 percent of all the water in the ocean, these waters are the upper 400 m of the ocean.

Deep water currents make up the other 90 percent of the ocean water. These waters move around the ocean basins due to variations in the density and gravity.

Types of ocean currents on the basis of temperature :

Cold currents are usually found on the west coast of the continents in the low and middle latitudes (in both hemispheres) and on the east coast in the higher latitudes in the Northern Hemisphere. These bring cold water into warm water areas. The cold currents carry water from polar or higher latitudes to tropical or lower latitudes.

Example of Cold currents -

Labrador current - North Atlantic Ocean.
Canary current - North Atlantic Ocean.
Benguela current - South Atlantic Ocean.
Falkland current - South Atlantic Ocean.
Northeast monsoon current - North Indian ocean.
California current - Pacific Ocean.
West Australian current - South Indian Ocean.
Eastern Greenland current - Arctic sea and North Atlantic Ocean.

Warm currents bring warm water into cold water areas and are usually observed on the east coast of continents in the low and middle latitudes (in both hemispheres). In the northern hemisphere they are found on the west coasts of continents in high latitudes.

The warm ocean currents originate near the equator and move towards the poles.

Example of Warm Currents - North

Equatorial Current - Pacific Ocean & Atlantic Ocean.

Kuroshio Current - Pacific Ocean.

Alaskan current - North Pacific Ocean.

Equatorial counter current - Atlantic Ocean, Pacific Ocean, and the Indian Ocean.

Florida current - South Atlantic Ocean and Caribbean sea.

Gulf stream - North Atlantic sea.

Norwegian current - North sea.

Antilles current - North Atlantic ocean.

Mozambique current - Indian Ocean.

Agulhas current - Southwest Indian Ocean.

Brazilian current - South Atlantic Ocean

Southwest Monsoon current - Indian Ocean.

East Australian current - Southwest Pacific Ocean.

Effect of Ocean Current : The mixing of warm and cold currents help to replenish the oxygen and favour the growth of planktons, the primary food for fish population. The areas where the warm and cold currents meet provide the best fishing grounds in the world.

El Niño and La Niña: These are important phenomena associated with changes in ocean currents and sea surface temperatures in the Pacific Ocean. El Niño events involve warmer-than-average sea surface temperatures, while La Niña events involve cooler-than-average sea surface temperatures. These phenomena have significant impacts on weather patterns worldwide.

Sea

Bay of Bengal- It is the northeastern part of the Indian Ocean bounded by the west and northwest by India, on the north by Bangladesh, and on the east by Myanmar and the Andaman and Nicobar Islands of India.

Arabian Sea- It is the northwestern part of the Indian Ocean and forms part of the principal sea route between Europe and India. It is bounded to the west by the Horn of Africa and the Arabian Peninsula, to the north by Iran and Pakistan, to the east by India, and to the south by the remainder of the Indian Ocean.

Persian Gulf - It is sometimes called the Arabian Gulf, is a Mediterranean Sea in West Asia. It is connected to the Gulf of

Oman in the east by the Strait of Hormuz.

Red Sea - It is a seawater inlet of the Indian Ocean, lying between Africa and Asia. Its connection to the ocean is in the south, through the Bab el Mandeb strait and the Gulf of Aden.

The Adriatic Sea is a body of water separating the Italian Peninsula from the Balkan Peninsula.

Black Sea- It is a marginal sea of the Atlantic Ocean. It is a large inland sea situated at the southeastern extremity of Europe.

Caspian Sea- It is the world's largest inland body of water, often described as the world's largest lake or a full-fledged sea. It lies between Europe and Asia and borders Kazakhstan, Iran, Russia, Turkmenistan and Azerbaijan.

Mediterranean Sea- It lies between Southern Europe and North Africa and accounts for about 0.7% of the global ocean area.

Sargasso Sea- It is in the North Atlantic and is bounded by the Gulf Stream on the west, the North Atlantic Current on the north, the Canary Current on the east, and the North Equatorial Current on the south.

Amundsen Sea- It is an arm of the Southern Ocean off Marie Byrd Land in western Antarctica. It lies between Cape Flying Fish (the northwestern tip of Thurston Island) to the east and Cape Dart on Siple Island to the west.

Caribbean Sea- It is a body of water adjacent to the Atlantic Ocean, southeast of the Gulf of Mexico. It is bordered by Venezuela, Colombia and Panama to the south, Central American countries (Costa Rica, Nicaragua, Guatemala, Honduras and Belize) on the west; with the Greater Antilles (Cuba, Jamaica, the Dominican Republic and Puerto Rico) on the north and the Lesser Antilles on the east.

Sea of Azov- It is an inland sea situated off the southern shores of Ukraine and Russia. It forms a northern extension of the Black Sea, to which it is linked on the south by the Kerch Strait.

Andaman Sea- It is a marginal sea of the northeastern Indian Ocean bounded by the coastlines of Myanmar and Thailand.

South China Sea- It is an arm of western Pacific Ocean in Southeast Asia. It is south of China, east and south of

Vietnam, west of the Philippines and north of the island of Borneo.

Coral Sea- It is a marginal sea of the South Pacific off the northeast coast of Australia, and classified as an interim Australian bioregion.

Sea of Japan (East Sea)- It is the marginal sea between the Japanese archipelago, Sakhalin, the Korean Peninsula, and the mainland of the Russian Far East.

Yellow Sea - A marginal sea of the Western Pacific Ocean located between mainland China and the Korean Peninsula, and can be considered the northwestern part of the East China Sea.

Philippine Sea - Section of the western Pacific Ocean, lying east and north of the Philippines.

Ionian Sea - It is part of the Mediterranean Sea, lying between Albania (northeast), Greece (east), Sicily (southwest), and Italy (west and northwest).

Irish Sea- It is the arm of the North Atlantic Ocean that separates Ireland from Great Britain.

Ross Sea- It is the southern extension of the Southern Ocean, which, along with the vast Ross Ice Shelf at its head, makes a deep indentation in the circular continental outline of Antarctica.

East China Sea - A part of the Pacific Ocean and is a marginal sea in the east of China, Connects with the Sea of Japan through the Korea Strait and opens to the north into the Yellow Sea.

Tasman Sea - It is a section of the southwestern Pacific Ocean, between the southeastern coast of Australia and Tasmania on the west and New Zealand on the east.

Bohai Sea - It is a marginal sea on the east coast of Mainland China.

Celtic Sea - It is the area of the Atlantic Ocean off the southern coast of Ireland bounded to the north by Saint George's Channel.

White Sea - It is a southern inlet of the Barents Sea located on the northwest coast of Russia.

Weddell Sea - It is part of the Southern Ocean and contains the Weddell Gyre.

Baltic Sea - It is an arm of the Atlantic Ocean that is enclosed by Denmark, Estonia, Finland, Germany, Latvia, Lithuania, Poland, Russia, Sweden and the North and Central European Plain.

Bering Sea - It is a marginal sea of the Northern Pacific Ocean. It forms, along with the Bering Strait, the divide between the two largest landmasses on Earth (Eurasia and the Americas).

Aegean Sea - It is located between the Balkans and Anatolia.

Tyrrhenian Sea - It is part of the Mediterranean Sea off the western coast of Italy.

Arafura Sea - It lies west of the Pacific Ocean, overlying the continental shelf between Australia and Western New Guinea (also called Papua).

Java Sea - It is an extensive shallow sea on the Sunda Shelf, between the Indonesian islands of Borneo to the north, Java to the south, Sumatra to the west, and Sulawesi to the east.

Bismarck Sea - It lies in the southwestern Pacific Ocean within the nation of Papua New Guinea.

Tides

Tide is the periodical rise and fall of the sea level, once or twice a day, mainly due to the attraction of the sun and moon.

Movement of water caused by meteorological effects (winds and atmospheric pressure changes) are called surges. The moon's gravitational pull to a great extent and to a lesser extent the sun's gravitational pull, are the major causes for the occurrence of tides. The tides are most influenced by the moon.

Another factor is centrifugal force, which is the force that acts to counterbalance the gravity. Together, the gravitational pull and the centrifugal force are responsible for creating the two major tidal bulges on the earth.

The highest tides in the world occur in the Bay of Fundy (Canada). The highest tides in India are recorded at Okha, Gujarat. It takes 24 hours 50 minutes for the rotating Earth to bring the same meridian vertically below the Moon everyday. Hence, tides occur at regular intervals of 12 hours and 25 minutes.

Types of Tides

Tides Based on Frequency :-

Semi-diurnal tide - The most common tidal pattern, featuring two high tides and two low tides each day. The successive high or low tides are approximately of the same height.

Diurnal tide - There is only one high tide and one low tide during each day. The successive high and low tides are approximately of the same height.

Mixed tide - Tides having variations in height are known as mixed tides. These tides generally occur along the west coast of North America and on many islands of the Pacific Ocean.

Tides Based on the Sun, Moon and Earth Positions -

Spring tides - The position of both the sun and the moon in relation to the earth has direct bearing on tide height. When the sun, the moon and the earth are in a straight line, the height of the tide will be higher. These are called spring tides and they occur twice a month, one on full moon period and another during the new moon period.

Neap tides - Normally, there is a seven day interval between the spring tides and neap tides. At this time the sun and moon are at right angles to each other and the forces of the sun and moon tend to counteract one another. The Moon's attraction, though more than twice as strong as the sun's, is diminished by the counteracting force of the sun's gravitational pull.

Once in a month, when the moon's orbit is closest to the earth (perigee), unusually high and low tides occur. During this time the tidal range is greater than normal. Two weeks later, when the moon is farthest from earth (apogee), the moon's gravitational force is limited and the tidal ranges are less than their average heights.

When the earth is closest to the sun (perihelion), around 3rd January each year, tidal ranges are also much greater, with unusually high and unusually low tides. When the earth is farthest from the sun (aphelion), around 4th July each year, tidal ranges are much less than average.

The time between the high tide and low tide, when the water level is falling, is

called the ebb. The time between the low tide and high tide, when the tide is rising, is called the flow or flood.

Importance of Tides

Since tides are caused by the earth-moon-sun positions which are known accurately, the tides can be predicted well in advance. This helps the navigators and fishermen plan their activities. Tidal flows are of great importance in navigation. Tidal heights are very important, particularly for harbours near rivers and within estuaries having shallow 'bars' at the entrance, which prevent ships and boats from entering into the harbour.

Tides are also helpful in desilting the sediments and in removing polluted water from river estuaries. Tides are used to generate electrical power (in Canada, France, Russia, and China). India produces tidal energy in the Gulf of Khambhat, Kachchh region and the Sundarbans region.

Fact Related to Tides

Tidal bore - The tide crest looks like a vertical wall of water rushing upstream is known as a tidal bore.

Favorable conditions for tidal bore are that the river must be fairly shallow. The outlet into the sea should be narrow. The coast should have a high tidal range. In India, tidal bores are common in the Hooghly river. Most powerful tidal bores occur in Qiantang River in China.

Intertidal zone is the area where the ocean meets the land between high and low tides. These are Four physical divisions of the intertidal zone.

Spray zone is dampened by ocean spray and high waves and is submerged only during very high tides or severe storms.

The high intertidal zone is flooded during the peaks of daily high tides but remains dry for long stretches between high tides. It is inhabited by hardy sea life that can withstand pounding waves, such as barnacles, marine snails, mussels, limpets, shore crabs, and hermit crabs.

Middle intertidal zone - over which the tides ebb and flow twice a day, and which is inhabited by a greater variety of both plants and animals, including sea stars and anemones.

Low intertidal zone - The low intertidal

zone, which borders on the shallow subtidal zone, is only exposed to air at the lowest of low tides and is primarily marine in character. Life is more abundant there because of the protection provided by the water.

The tidal bulges on wide continental shelves have greater height. When tidal bulges hit the mid-oceanic islands they become low. The shape of bays and estuaries along a coastline can also magnify the intensity of tides.

River Drainage System

River drainage systems are like vast networks of streams and rivers that collect and drain water from a specific area. The Himalayas, for instance, are home to some of the world's largest drainage systems, including the Ganga, Indus, and Brahmaputra basins.

Drainage Pattern - It refers to the system of flow of surface water mainly through the forms of rivers and basins.

Types of Drainage Pattern

Dendritic Drainage Pattern - It is the most common form and resembles the branching pattern of tree roots. Examples - The rivers of the northern plains; Indus, Ganga and Brahmaputra.

Trellis drainage pattern - It develops in folded topography where hard and soft rocks exist parallel to each other. Examples - The rivers in the upper part of the Himalayan region; Indus, Ganga and Brahmaputra.

Rectangular Drainage Pattern - It is found in regions that have undergone faulting. Example - Streams found in the Vindhya mountain range; Chambal, Betwa and Ken.

Radial Drainage Pattern - It develops when streams flow in different directions from a central peak or dome like structure.

Centripetal drainage pattern - When rivers discharge their waters from all directions into a depression. Example - Loktak lake in Manipur.

Parallel pattern - Tributaries run parallel to each other in a uniformly sloping region with parallel, elongated landforms. Example - Godavari, Kaveri, Krishna, Tungabhadra.

River Landforms and Features

Estuary - It is a partially enclosed coastal body of freshwater with one or more rivers or streams flowing into it, and with a free connection to the open sea. Narmada, Periyar and Tapi are the major rivers that form estuaries.

Lagoon - A lagoon is a shallow body of water separated from a larger body of water by a narrow landform, such as reefs, barrier islands, barrier peninsulas, or isthmuses. Marovo Lagoon is the largest saltwater lagoon in the world.

Delta is the depositional feature of an almost triangular shape at the mouth of a river debouching either in the lake or the sea.

The Ganges-Brahmaputra Delta, also known as the Sundarbans Delta, is a massive triangle of land in Asia where the Ganges and Brahmaputra rivers meet the Bay of Bengal. It's the world's biggest river delta and a key habitat for Royal Bengal tigers. This swampy region is shared by Bangladesh and India, with Bangladesh holding the larger portion.

Godavari Krishna delta - It is spread across the Krishna river and Godavari river basins in Andhra Pradesh. It is the second largest delta in India.

Dam

A **Dam** is a hydraulic structure built across a river to create a reservoir on its upstream side for impounding water to store water, control flooding, and generate electricity.

Cofferdam is a temporary wall built underwater to create a dry workspace for construction. It pumps out water to allow building things like bridges or dams.

Detention dams are constructed to minimize the impact of flooding and to restrict the flow rate of a particular channel.

Rock dams are small dams made up of stone, bagged sand or gravel, or other long-lasting material through drainage.

Some terminology related to river

Snow-fed rivers - The rivers that started in high mountains from melting ice and snow are called "snow fed rivers". The Brahmaputra, Indus, Ganga and Yamuna

are the snow-fed rivers in India.

Exotic rivers are rivers that flow through dry environments where little or no freshwater exists. Example - The Nile river.

Episodic Rivers are rarely occurring rivers formed from run-off channels in very dry regions.

Rain-fed rivers are rivers that dry up in the summer, but it flows throughout the rainy season whereas a perennial river flows throughout the year in all the seasons.

India has many rain-fed rivers such as the Jhelum, Tapti, Narmada, Chenab, Sutlej, Beas, Mahanadi, Godavari, Krishna, Periyar and Kaveri.

Meanders are regular sinuous curves or bends formed by a river as it enters a plain and twists and turns. Over time, the meander loop gets cut off from the main river channel, forming an oxbow lake.

Waterfall is a point where water in a river or stream flows over a vertical drop or series of steep drops, including meltwaters dropping over icebergs or ice shelves. A cataract is a waterfall with an enormous volume of water, while a cascade waterfall has a series of rock levels it cascades over.

Rapids are areas of shallow, fast-flowing water in a stream.

Pond is a small, still, land-based body of water formed by pooling inside a depression, either naturally or artificially. A pond is smaller than a lake.

Gorge is a narrow valley with steep sides and a river running through it.

Flooding occurs when there is excessive stream discharge and water spills of the channel onto the adjacent food plain in all the directions.

Floodplains are an area of low-lying ground adjacent to a river, formed mainly by flooding of rivers and the deposition of sand sediments on the riverbanks.

A **levee** is a natural or artificial wall that blocks water from going where we don't want it to go. Levees are formed due to the deposition of sediments during flood periods when the water overtops the river banks and spreads over adjoining flood plains.

A **Sea cave** is a cave formed on a cliff by

the wave action of an ocean or lake.

A **sea arch** is a natural arch or bridge made of stone that has been created when water wears away the underside of a rock, leaving just the top behind.

Badland topography is a major feature of the Chambal valley and is characterized by an undulating floodplain, gullies, and ravines.

Gullies resemble large ditches or small valleys, but are meters to ten of meters in depth and width.

Ravine is a landform narrower than a canyon and is often the product of stream-cutting erosion.

Lift channel - A channel of the canal where water is forced to flow against the slope of land by upliftment.

A **Reservoir** is an artificial lake where water is stored.

Drainage System of India

There are broadly 2 types of Indian river systems based on origin, namely: Himalayan rivers Systems and Peninsular River Systems

Himalayan rivers basin includes the Ganga, the Indus and the Brahmaputra river basins. Most of the Himalayan rivers are perennial. These rivers also form V-shaped valleys, rapids and waterfalls in their mountainous course. These rivers perform intensive erosion activity in their upper courses and carry huge loads of silt and sand. In the middle and the lower courses, these rivers form meanders, oxbow lakes.

Peninsular River Systems – Godavari, Krishna, Cauvery, and Mahanadi River System. The peninsular rivers are characterized by fixed course, absence of meanders and nonperennial flow of water.

River treaty

Ganga Treaty (India-Bangladesh)- Agreement to share waters at the Farakka Barrage. India and Bangladesh have agreed to set up a Joint Technical committee for optimum utilization of Ganges water as being received by Bangladesh under the Ganga Water Sharing Treaty 1996. Bangladesh has the highest number of rivers (about 700) and is known as the 'land of rivers'.

Indus Water Treaty (India-Pakistan):

Distribution of waters in the Indus River basin. It was signed in Karachi on 19 September 1960 by then Indian Prime Minister Jawaharlal Nehru and then Pakistani President Ayub Khan.

River treaties with Nepal: Agreements on the Kosi, Gandak, and Mahakali rivers.

MoU on Brahmaputra river (India-China):

Agreement to share hydrological information signed in 2002.

Indus river

The Indus River, also known as Sindhu, is the westernmost of the Himalayan rivers in India. It originates from the Bokhar Chu glacier on Mount Kailash's northern slopes in Tibet, where it is called Singi Khamban or Lion's Mouth.

After flowing northwest into Ladakh, India, it enters Pakistan near Chilas.

The major tributaries - Sutlej, Beas, Ravi, Chenab, and Jhelum - join the Indus at Panjnad.

The Indus discharges into the Arabian Sea east of Karachi after a total length of 2,880 km, with 1,114 km flowing through India.

The mouth of the Indus river lies to the north of the Tropic of Cancer.

The Indus Basin falls in states of Jammu and Kashmir, Himachal Pradesh, Punjab, Haryana, Rajasthan.

According to the 1960 Indus Waters Treaty, the water of eastern rivers Sutlej, Ravi and Beas are used by India, and the water of western rivers Indus, Chenab and Jhelum are used by Pakistan.

Tributaries : Right bank - Shyok, Gilgit, Gumal and Kabul. **Left bank** :- Zaskar, Suru, Sohan, Shigar, Jhelum, Chenab (largest tributary), Ravi, Beas, Satluj.

Major rivers of Indus river system

Jhelum, also known as Vyath in Kashmir, rises from a spring at Verinag situated at the foot of the Pir Panjal Range.

It flows through Srinagar and the Wular lake before entering Pakistan. Jhelum ends in a confluence with Chenab near Jhang in Pakistan.

Left Bank Tributaries :- Ningal, Vishav,

Romshi, Doodhganga, Kishanganga, Rambiar, and Sukhnag. Right Bank Tributaries :- Pohru, Sandrin, Arapal, Sindh, Arapath, Bringi and Liddar.

Chenab river originates from the Lahaul valley in Himachal Pradesh formed by the confluence of two streams, the Chandra and the Bhaga, which join at Tandi near Keylong in Himachal Pradesh. In its upper reaches it is also known as Chandrabhaga. During ancient times, Chenab was known as Askini.

The river flows in a north-west direction parallel to Pir Panjal ranges and cuts the ranges near Kishtwar. It is the largest tributary of the Indus. Cities on its bank: Kishtwar, Jammu.

Ravi river is a 720 Km long perennial river originating in the Himalayas. It rises from the west of the Rohtang pass in the Kullu hills of Himachal Pradesh and flows through the Chamba valley.

Before entering Pakistan and joining the Chenab near Sarai Sidhu, it drains the area lying between the Pir Panjal and the Dhauladhar ranges. Lahore (Pakistan) is located on its bank. Parushni is the Rigvedic name of the river Ravi.

Beas river originates near the Rohtang Pass at an altitude of 4,000 meters above mean sea level. It flows through the Kullu valley and forms gorges at Kati and Largi in the Dhauladhar range.

It enters the Punjab plains where it meets Satluj near Harike. It is well known as the habitat of Dolphins. It is the only river of Panjnad which totally drains into Indian Territory. Vipasha is the ancient name of river Beas.

Sutlej or Shatadru river is an antecedent river originating in the 'Rakshas tal' near Mansarovar in Tibet where it is known as Langchen Khambab. It flows almost parallel to the Indus for about 400 km before entering India, and comes out of a gorge at Rupar. Ferozepur (Punjab) lies on the bank of Sutlej.

It passes through the Shipki La on the Himalayan ranges and enters the Punjab plains. There, it feeds the canal system of the Bhakra Nangal project.

Swaan is a tributary of the Sutlej, flows from the north to west also known as the "river of sorrow". Main Tributaries are Spiti, Beas, Chenab, Baspa.

Zaskar River is the first major tributary

of the Indus River, which flows entirely within Ladakh.

It originates northeast of the Great Himalayan range. Zaskar river, primarily known for its unique Chadar Trek or the Frozen River Trek, in Ladakh.

Suru is a tributary of the Indus river that flows largely through the Kargil district of Ladakh.

Shigar is a tributary to the Indus river formed from the melted water of the Baltoro Glacier and Biafo Glacier.

Doab

It is made up of two words – 'do' means two and 'ab' means water. It refers to the tract of land between two rivers.

Punjab is divided into five doabs - Rachna Doab (Chenab-Ravi), Chaj Doab (Chenab-Jhelum), Bist Doab (Sutlej-Beas), Bari Doab (Beas-Ravi), and Sindh Sagar Doab (Indus-Jhelum).

Major Dams on Indus River System

Diamer-Bhasha Dam : A concrete-filled gravity dam constructed on the Indus river between Kohistan district in Khyber Pakhtunkhwa and Diamer district in Gilgit Baltistan (POK).

Mangla Dam : A multipurpose dam situated on the Jhelum river in the Mirpur District of POK.

Baglihar Dam is a run-of-the-river power project on the Chenab river in the Ramban district of Jammu and Kashmir.

Bhakra Nangal Dam is a concrete gravity dam on the Sutlej River in Bhakra Village near Bilaspur, Himachal Pradesh. The dam has the third largest reservoir of India in terms of quantity of water named Gobind Sagar reservoir.

The Harike Barrage, situated at the confluence of the Beas and Sutlej rivers, channels water from these rivers into the Indira Gandhi Canal (Rajasthan canal). Harike Lake, the largest wetland in northern India, adjacent to the Harike Barrage.

Ranjit Sagar Dam (Thein Dam) is part of a hydroelectric project constructed on the Ravi River on the border of Jammu and Kashmir and Punjab.

Chamera Dam was built on the Ravi river.

It is located near Dalhousie, in the Chamba district of Himachal Pradesh. The reservoir of the dam is the Chamera Lake.

Ganga River

It is a trans-boundary river of Asia, which flows through India and Bangladesh. It is the longest river in India. It has a length of about 2525 km. The Ganga river system is the largest in India.

It rises in the Gangotri glacier near Gaumukh in the Uttarkashi district of Uttarakhand. Here, it is known as the Bhagirathi. Kedarnath is located on the banks of Bhagirathi. Bhilangna is a major tributary of Bhagirathi river.

The Ganga enters the plains at Haridwar. It flows through Uttarakhand, Uttar Pradesh, Bihar, Jharkhand and West Bengal.

In Bangladesh, it is called **Padma**. It is joined by the Jamuna, the lower stream of the Brahmaputra, and eventually the Meghna and joins in the Ganges Delta and empties into the Bay of Bengal.

Ambala (Haryana) is situated on the water divide between the Indus and Ganga river systems. Bihar is naturally divided by the Ganga river into two regions - the North Bihar Plains and South Bihar Plains.

Tributaries : Right tributaries - Yamuna, Son, Punpun, Damodar, Tamsa, Karamnasa, Falgu. Left tributaries - Mahananda, Ramganga, Gomti, Ghaghara, Gandak, Kali or Sharda, Kosi.

Ganga River Basin

It spreads across 11 States which are Uttarakhand, Himachal Pradesh, Haryana, Uttar Pradesh, Delhi, Rajasthan, Madhya Pradesh, Bihar, Chhattisgarh, Jharkhand, West Bengal.

Major rivers of Ganga river basin

Alaknanda - It has its source in the Satopanth glacier above Badrinath. This glacier is spread over an area of 21.17 sq km in Chamoli district of Uttarakhand.

Tributaries - Pindar and Dhauliganga are the tributaries of Alaknanda. Pindar originates from Pindari Glacier which is located in Bageshwar district in Uttarakhand.

The Garhwal region of Uttarakhand, India is home to "five holy confluences or Panch Prayags", where five rivers merge with the Alaknanda River. The five most important prayags on the Alaknanda River are:

Vishnuprayag - It is the place where Alaknanda meets Dhauli Ganga.

Nandprayag - It is the place where Alaknanda meets Nandakini.

Karnaprayag - It is the place where Alaknanda meets Pindar.

Rudraprayag - Alaknanda meets Mandakini or Kali Ganga (originates from Sumeru Glacier).

Devprayag - Alaknanda meets Bhagirathi at this prayag. Hereafter, it is known as the **Ganga**.

Rishiganga - It rises from the Uttari Nanda Devi Glacier on the Nanda Devi mountain in Uttarakhand's Chamoli district and flows through the Nanda Devi National Park.

Gandak - It is a northern Ganga tributary, rises in Nepal Himalayas between Dhaulagiri and Everest. Comprising Kaligandak and Trishul Ganga streams. Triveni canal was constructed on Gandak river for irrigation in Champaran, Bihar.

Ghaghara - It originates in the glaciers of Mapchachungo, in the northern slopes of the Himalayas in the Tibetan plateau. The river Sarda (MahaKali Ganga) joins Ghaghara in the plain before it finally meets the Ganga at Chhapra. Tila, Seti and Beri are the tributaries of Ghagra.

It is the largest tributary of the Ganges by volume and the second largest by length after Yamuna. It is called Karnali in Nepal which is known to be the longest river in Nepal.

Sharda - This river rises in the Milam glacier (Pithoragarh district, Uttarakhand) in the Nepal Himalayas where it is known as the Goriganga. Along the Indo-Nepal border, it is called Kali or Chauk, where it joins the Ghaghara.

Tributaries : Sarayu (Ayodhya is situated on its bank), Chameliya, Rmgun in Nepal and Kuthi, Dhauli, Gori, Ladhiya rivers in Uttarakhand.

Ramganga - It rises in the Garhwal hills near Gairsain (Uttarakhand), then crosses the Shivalik and enters into the plains of Uttar Pradesh near Najibabad. Its source is Namik Glacier. Finally, it joins the Ganga near Kannauj.

Mahananda - It rises in the Darjeeling hills (West Bengal) and joins the Ganga in West Bengal. Siliguri is situated on its bank.

Son - It is a large south bank tributary of the Ganga, originating from Amarkantak high lands in hills of Maikala range in Bilaspur district of Chhattisgarh. It reaches Maner, west of Patna, to join the Ganga. Rihand is one of its important tributaries.

Gomti - It originates from Gomati Taal (formerly known as Fulhaar Jheel) in Pilibhit, Uttar Pradesh. It meets the Ganga in Kaithi, Varanasi.

Kosi - It is a trans-boundary river which flows through Tibet, Nepal and India. This river originates from Nepal and meets the Ganga in Katihar, Bihar.

It carries the maximum amount of silt and sand after the Brahmaputra in India. It is also known as the **"Sorrow of Bihar"**. Its tributaries are Sun Kosi, Arun and Tamur. Purnia of Bihar is located on its bank.

Damodar - It originates in the Palamu hills of the Chota Nagpur plateau in the state of Jharkhand. It passes through two Indian states namely, Jharkhand and West Bengal. It is also known as the 'Sorrow of Bengal'. Panchet Dam is built on the Damodar river.

Tributaries - Left tributaries - Barakar, Konar, Jamunia, Nunia. Right tributaries - Sali River (West Bengal).

Karamnasa - It originates in Kaimur district of Bihar. It flows through Uttar Pradesh and Bihar. It is a tributary of Ganga River.

Varuna - It is a tributary of Ganga River originating in Phulpur in the Prayagraj district.

Meghna - This river originates in Kishoreganj district inside Bangladesh. It meets the Padma in Chandpur district, from where it is known as Lower Meghna. It empties into the Bay of Bengal. It is the widest river of Bangladesh.

Hooghly - It is a distributary or arm of the Ganges river about 260 km long. It is formed in Murshidabad (West Bengal), where Ganga splits into two parts at Farakka Barrage – the part flowing through Bangladesh is called the Padma.

It enters the plains of West Bengal in the northwest of Rajmahal hills.

Tributaries - Damodar, Rupnarayan, Haldi, Rasulpur, Shilabati, Kangsabati, Mayurakshi.

Mayurakshi - It is also called Mor river, is a major river in Jharkhand and West Bengal. It has its source on Trikuta Hills near Deogarh in Jharkhand. Brahmani, Dwaraka, Bakreshwar and Kopai are major tributaries.

Brahmani - It is an east flowing river. It originates near Nagri village in Ranchi district of Jharkhand. The basin is bounded in the North by Chhota nagpur plateau, in the West and South by the Mahanadi basin and in the East by the Bay of Bengal. Tributaries - Shankh, Tikra and Karo.

It has a length of 800 km. It covers the states of Jharkhand, Chhattisgarh and Orissa and has a Drainage area of 39,033 sq km. Rengali Dam in Odisha is constructed across the Brahmani river.

Yamuna - It originates from the Yamunotri Glacier in western Uttarakhand, which is located at Bandarpunch, which is the peak of lower Himalayas in Uttarakhand.

It is also the longest tributary of river Ganga. It is the second-largest tributary river of the Ganga river basin. The length of Yamuna river is around 1376 km.

It flows primarily in Uttarakhand and Uttar Pradesh states. It flows parallel to the Ganga and joins the Ganga at Prayagraj. This joining place is known as Sangam

Major Tributaries of Yamuna

Chambal - Chambal is one of the major right tributaries of the Yamuna river. It originates in the Vindhya Range just south of Mhow, near Indore, Madhya Pradesh.

It flows north into southeastern Rajasthan state. Turning northeast, it flows past Kota and along the Rajasthan–Madhya Pradesh border; shifting east-southeast, it forms a portion of the Uttar Pradesh–Madhya Pradesh border and flows through Uttar Pradesh to empty into the Yamuna after a 550-mile (900-km) course.

Tributaries : Banas, Kali Sindh, Kshipra,

and Parbati.

Shipra (Kshipra) - It rises in the Dhar district (Madhya Pradesh), then joins the Chambal river in Mandsaur district. Ujjain is situated on its bank.

Banas - It lies entirely within the state of Rajasthan. It is the tributary of the Chambal river. It is around 512 km in length. It dries up during summer.

Betwa - It is the right tributary of a river Yamuna in northern India, rising in the Vindhya Range (near Bhopal) just north of Hoshangabad, Madhya Pradesh. Orchha is the cultural tourism destination located on its banks.

It flows through Madhya Pradesh and Uttar Pradesh and empties into the Yamuna River just east of Hamirpur (UP) after a 610-km course. Jamni and Dhasan are major tributaries of Betwa.

Ken river - It is a right tributary of Yamuna, is one of the major rivers in the Bundelkhand region of central India and flows through the states of Madhya Pradesh and Uttar Pradesh. It merges with the Yamuna at district Banda in Uttar Pradesh.

Tributaries - Alona, Bearma, Sonar (largest tributary), Mirhasan, Shyamari, Banne, Kutri, Urmil, Kail and Chandrawal.

Sindh - It is a right tributary of the Yamuna River. It originates on the Malwa plateau in Vidisha district, then flows through Gwalior (Madhya Pradesh) to join the Yamuna river in Jalaun, Uttar Pradesh.

Hindon - It originates in Saharanpur between the Shivalik and Himalayan mountain ranges. It is a left tributary of Yamuna.

Tons - Tons is the left and longest tributary of the Yamuna. It flows through Garhwal, the western part of Uttarakhand. After that it joins the Yamuna below Kalsi near Dehradun, Uttarakhand. It is one of the most major perennial Indian Himalayan rivers.

Banganga - It is a tributary of Yamuna. This river originates from the hills of Aransar and Bairath (Virat Nagar) in Jaipur region of Rajasthan state and converges with Yamuna near Fatehabad in Agra, Uttar Pradesh.

Giri - It is an important right tributary of the Yamuna flows in Himachal Pradesh.

Its major tributaries are Ashani which meets Giri close to Chail and Jalal which joins it at Dachau. The Palar and Nait streams meet the river on its left bank.

Major Cities on the banks of Ganga river basin

Ganga - Haridwar, Rishikesh, Kannauj, Farrukhabad, Kanpur, Allahabad or Prayagraj, Phaphamau, Varanasi, Mirzapur, Patna, Munger and Hajipur.

Ramganga - Bijnor, Moradabad, Rampur, Bareilly, and Shahjahanpur.

Gomti - Lucknow, Lakhimpur Kheri, Jaunpur, Zafrabad, and Sultanpur.

Hooghly - Hugli, Kolkata, and Howrah.

Yamuna - Delhi, Agra, Mathura, Firozabad and Prayagraj.

Major Dams on Ganga river Basin

Tehri Dam - It is the tallest dam in India with a height of 260.5 m (855 ft). It is a multi-purpose dam on the Bhagirathi River in Uttarakhand.

Farakka Barrage - It is a barrage across the Ganga river located in Murshidabad district of West Bengal. This barrage bifurcates river Ganga to form its distributary called Bhagirathi-Hooghly.

Song Dam - The first gravity dam of Uttarakhand is about 148 m high. Song river is a tributary of Suswa river, which is the tributary of the Ganga.

Maithon Dam is located at Maithon, in Dhanbad district, Jharkhand. It was constructed on the Barakar River. It is 4,789 m long and 50 m high.

Tilaiya Dam - It was the first dam built across the Barakar river under the Damodar Valley Project.

Konar Dam - It was constructed on the Konar river.

Brahmaputra river

It is one of the largest rivers of the world, and has its origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake.

The Brahmaputra river flows through Tibet (China), India, and Bangladesh.

In Tibet it is known as the Tsangpo, which means 'the purifier'. The Rango

Tsangpo is the major right bank tributary of this river in Tibet.

It emerges as a turbulent and dynamic river after carving out a deep gorge by taking a U turn in the Central Himalayas near Namcha Barwa (7,755 m).

In India - It enters India through west of Sadiya town in Arunachal Pradesh as the Siang or Dihang river. Flowing southwest receives its tributaries Sikan or Dibang and Lohit thereafter known as Brahmaputra or Red river. After that it flows through Assam and West Bengal. It marks the easternmost boundary of the Himalayas.

In Bangladesh - It enters Bangladesh near Dhubri and flows southward. In Bangladesh, the Tista joins it on its right bank from where the river is known as the Jamuna. It finally merges with the river Ganga (known as Padma in Bangladesh) which falls in the Bay of Bengal.

Feature - It is the deepest and widest river in India also known as "Sorrow of Assam." It flows eastwards parallel to the Himalayas. The total length of Brahmaputra river is 2900 km. In India, it flows for 916 km which shows that most of its course lies outside India. It has the least storage capacity of usable water.

Right tributaries - Rango Tsangpo (Tibet), Subansiri (Tibet), Kameng (Arunachal Pradesh), Manas (Assam), Sankosh (Assam), Tista (Sikkim), Siang (Arunachal Pradesh), Tirap (Arunachal Pradesh), Noa Dihing (Arunachal Pradesh).

Left tributaries - Dibang or Sikan (Arunachal Pradesh), Lohit (Arunachal Pradesh), Burhi Dihing (Assam) and Dhansari (Assam).

Cities - Dibrugarh, Pasighat, Neamati and Guwahati are important urban centers. Kamakhya temple at Nilachal hills lies on the banks of Brahmaputra River. Majuli is the biggest inhabited riverine island in the world located on Brahmaputra river in Assam.

Major rivers in the Northeast India

Lohit - It rises in the Tibet region where it is called the Zayu river. It carries abundant amounts of lateritic soil, which gives it a reddish color.

It enters India flowing through Mishmi

hills where it joins the Siang river in Assam to form Brahmaputra.

Dhola Sadiya Bridge (Bhupen Hazarika setu) is the longest river bridge in India constructed over Lohit river in Tinsukia, Assam measuring 9.15 km.

Burhi Dihing - It originates at 2,375 m above sea level in the Eastern Himalayas (the Patkai Hills) in Arunachal Pradesh and flows through Tinsukia and Dibrugarh Districts in Assam to its confluence with the Brahmaputra at Dihingmukh.

Subansiri - It is the largest tributary of the Brahmaputra River enters India through Arunachal Pradesh and flows through the Miri hills. It flows through Bhutan and India. In India, it joins the river Brahmaputra.

Teesta - It originates from Zemu glacier in Sikkim's Kanchenjunga ranges. It's Sikkim's lifeline. Carving through Darjeeling hills, it divides them into Tiger Hill and Kalimpong ranges. Finally, it enters Bangladesh, joining the Brahmaputra.

Teesta river issue (India-Bangladesh): Dispute over sharing of Teesta river waters.

Barak - It is also known as the Surma River, originates from the Manipur hills, south of Mao in the Senapati district of Manipur state. It flows through the states of Manipur, Nagaland, Mizoram, and Assam. The city of Silchar is situated on the banks of this river.

Wang Chuu (Raidak River) - It is a tributary of Brahmaputra that flows through Bhutan, India and Bangladesh.

Muhuri (Little Feni) - It is a transnational river between India and Bangladesh. Rises in Lushai hills of Tripura. Thereafter it flows into Bangladesh where it merges with the Feni near the latter's mouth to the Bay of Bengal.

Torsa - It rises from the Chumbi Valley in Tibet, China, where it is known as Machu. Its course continues into Bhutan, India, and Bangladesh before joining Brahmaputra River into the Bay of Bengal.

Dibang River - This river, also known as Sikan by the Adi and Talo in Idu language, is an upstream tributary river of the Brahmaputra in the Indian state of Arunachal Pradesh. It originates and flows through the Mishmi Hills in the

Dibang Valley districts in Arunachal Pradesh.

Tributaries of Dibang river - Dri, Mathun, Talon, Eme, Ahi, Emra and Awa.

Rivers of Nagaland - Doyang (longest and biggest river in Nagaland), Dikhu and Dhansiri flow westward and join the Brahmaputra. Tizu river flows eastwards and joins Chindwin river in Burma, Myanmar. Other rivers - Milak, Dzu, Langlong, Zungki, Likimro, Lanye and Dzuza.

Major Bridges on Brahmaputra

Bogibeel Bridge - It is India's longest rail - cum - road bridge built between Dhemaji and Dibrugarh in Assam over Brahmaputra river measuring 4.94 km.

Kolia Bhomora Setu - It is a 3.015 km long concrete road bridge over the Brahmaputra river in Tezpur, Assam named after the Ahom General Kolia Bhomora Phukan. It connects Tezpur with Kaliabor.

Naranarayan Setu - It is a Double decker bridge having railway line on lower deck and road on upper deck constructed over Brahmaputra in Assam named after Nara Narayan Koch king. It has a length of 2.284 kms which connects Jogighopa with Pancharatna.

Other Bridges over Brahmaputra river are New saraighat Bridge, Saraighat Bridge.

Peninsular Rivers

Peninsula

It is an area of land surrounded by water on three sides. The largest peninsula in the world is the Arabian Peninsula. The Indian Peninsula is surrounded by the Arabian Sea, the Indian Ocean and the Bay of Bengal.

Peninsular Drainage System

Most of the major Peninsular rivers except Narmada and Tapi flow from west to east.

The other major river systems of the Peninsular drainage are Mahanadi, Godavari, Krishna and Kaveri.

Major River of Peninsular India

Narmada (Reva) river - It rises from the Amarkantak Plateau in Anuppur district of Madhya Pradesh. It is the fifth largest river in India and Peninsular longest west flowing river.

It flows through Madhya Pradesh, Maharashtra, Gujarat and drains through the gulf of Cambay in Arabian sea. It is also known as the "Life Line of Madhya Pradesh and Gujarat".

It is the largest flowing river in Madhya Pradesh with a length of 1312 km. It forms the traditional boundary between North India and South India.

Cities on its banks - Amarkantak, Maheshwaram, Narmadapuram (Hoshangabad), Jabalpur, Omkareshwar of Madhya Pradesh. Bharuch, Rajpipla, Sinor of Gujarat.

Tributaries : Left - Burner, Banjar, Sher, Shakkar, Dudhi, Tawa, Ganjal, Chhota Tawa, Kundi, Goi, and Karjan. Right - Hiran, Tendoni, Barna, Kolar, Man, Uri, Hatni, and Orsang.

Tapti river - It is the second largest west flowing river of Peninsular India with a length of 724 km. Rises from the Satpura range in the Gawilgarh hills of the Deccan plateau.

It flows through the Indian states of Maharashtra, Gujarat and Madhya Pradesh and drains into the Arabian sea through the gulf of Cambay (Khambhat). It flows nearly parallel to the Narmada river and both of them are separated by the Satpura range. Nearly 80% of the Tapi basin is in Maharashtra. It forms estuaries. Tapi originates from Multai in the Betul district of Madhya Pradesh. Surat is situated on the bank of Tapi river.

Tributaries : Left - Girna, Nesu, Amravati, Buray, Panjhara, Bori, Waghur, Purna, Mona, Sipna. Right - Vaki, Aner, Arunawati, Gomai

Luni river - It is also known as lavana river which means salty river. It is the largest west flowing river in Rajasthan and largest river in the Thar Desert of Northwest India.

It originates from Naga Pahar or Naga hills in the Pushkar valley of the Aravalli Range located in Ajmer Rajasthan and ends in the Marshy land of Rann of Kutch in Gujarat.

It forms the largest river system of Rajasthan. It is one of the major inland rivers because it is the only natural water source that drains inside a lake in the desert before reaching the Arabian Sea.

Cities situated on its banks - Ajmer, Barmer, Jalore, Jodhpur, Nagaur, Pali, and Sirohi districts of Rajasthan. Also Patan district of Gujarat.

Tributaries : Left - Sukri, Mithri, Bandi, Khari, Jawai, Guhiya, and Sagi. Right - Jojari River.

Mahi or Mahisagar river - It is a west flowing river originating from the northern slopes of Vindhya in Minda village, Dhar district of Madhya Pradesh. Tributaries are Som, Anas, Jakham, Panam.

Sabarmati river - It is one of the major west flowing rivers of India with a length of 371 km. It is a seasonal river whose flows are dominated by the monsoon.

It originates from Aravalli hills in Udaipur District of Rajasthan and covers the states of Rajasthan and Gujarat and finally drains into the Arabian sea through the Gulf of Cambay (Khambhat).

Cities on its banks - Gandhinagar and Ahmedabad.

Tributaries : Left - Wakal, Harnav, Hathmati, Vatrak. Right - Sei

Dhandhar river - It is one of the west flowing rivers in Gujarat with a length of 142 km. It is a monsoon river that has water only in the Monsoon season. It rises from Pavagadh Hills in Gujarat and runs through the districts of Vadodara and Bharuch and finally drains in the Arabian sea through the gulf of Khambhat.

Tributaries - Vishwamitri, Jambuo, Dev, and Surya.

Shetrunji river - It is one of the major rivers of Saurashtra. It rises at Chachai hills in Gir Forest of Junagadh district of Gujarat. It flows towards the east direction and empties into the gulf of Khambhat.

Arvari river - It originates from the Sakra Dam near Thanagazi in Alwar, Rajasthan. It is considered to be the smallest river of India having a total length of 45 km.

Godavari river (Dakshin Ganga or Vridha Ganga) - It is the largest river of peninsular India with a length of 1465

km. originates from Trimbakeshwar, Nashik, Maharashtra. At Dowlaiswaram, the river divides into two branches, the Gautami and Vasishta. The Gautami branch joins the Bay of Bengal flowing through the Yanam enclave of Union Territory of Puducherry. It is India's second longest river after the Ganga.

The Godavari basin extends over states of Maharashtra, Andhra Pradesh, Chhattisgarh and Odisha in addition to smaller parts in Madhya Pradesh, Karnataka and Union territory of Puducherry. About 50% of the Godavari Basin lies in Maharashtra.

The Godavari basin is the second largest basin after the Ganges basin and accounts for nearly 10% of the geographical area of India.

Cities - Nashik, Nagpur, Nanded of Maharashtra, Nizamabad of Telangana, Rajahmundry of Andhra Pradesh. The pilgrimage sites of Nasik, Triyambak and Bhadrachalam lie on the banks of Godavari river.

Tributaries : Right - Pravara, Manjira, Maner, Peddavagu and Mula. Left - Purna, Pranrita (largest tributary), Penganga, Wardha, Pench, Wainganga, Indravati (second largest tributary) and Sabari.

Pranrita river - It is the largest tributary of Godavari begins at the confluence of Penganga, Wardha and the Wainganga rivers. This junction lies on the border between the states of Maharashtra and Telangana. It flows along the border of Gadchiroli district in Maharashtra and Adilbad district in telangana.

Manjira river - This river is the longest tributary of Godavari river and Nizam Sagar is a reservoir built on the Manjira River.

Indravati river - It is a tributary of Godavari originating from the ghats of Dandakaranya range in the Kalahandi district of Odisha. It is known as the oxygen of the Bastar district of Chhattisgarh.

Krishna river - It is the second largest east flowing river of the peninsula with a length of 1400 km originating from the Western Ghats near Jor village of Satara district of Maharashtra, near Mahabaleshwar. Sahyadri mountain range is the source of Krishna river.

It flows eastwards through the states of

Maharashtra, Karnataka, Telangana and Andhra Pradesh, where it drains into the Bay of Bengal.

It is the third-longest river in India, after the Ganges and Godavari and the fourth-largest in terms of water inflows and river basin area after the Ganges, Indus and Godavari. Vijayawada and Amaravati are located on the bank of Krishna.

Pilgrimage centers - Mallikarjuna temple (Srisailem), Amareshwara Swamy temple (Amaravati), Datta Deva temple, Sangameshwara Shiva temple, Ramling temple and Durga Malleswara temple (Vijayawada).

Tributaries : Right - Ghatprabha, Malprabha, Tungabhadra, Venna, Koyna, Panchganga, Doodhganga. Left - Bhima, Musi river, Sina and Munneru.

Hyderabad is situated on Musi river. Yerla, Warana and Dindi are also the tributaries of the Krishna river system.

Bhima river - It is the longest tributary of Krishna River. It rises in the Bhimashankar heights of the Western Ghats and flows southeastward in Maharashtra to join the Krishna in Karnataka. **Sina** river is a large tributary of the Bhima river which starts near Ahmednagar city.

Kaveri river - It is also known as 'Ganga of south' is an east flowing river with a length of approximately 800 km.

It originates at Talakaveri on the Brahmagiri range in the Western Ghats, Coorg district of Karnataka and covers the states of Karnataka, Tamil Nadu and Kerala and drains into the Bay Of Bengal through Puducherry.

It is the third largest river of southern India after Godavari and Krishna also known as 'Pooni' in Tamil. The Kaveri Basin lies in the Indian states of Tamil Nadu, Karnataka and Kerala.

Cities on its banks are Tiruchirappalli and Erode of Tamil Nadu.

Disputes - Kaveri got associated with the water disputes tribunal of 1990 aimed at resolving the old issue of water sharing between Karnataka and Tamil Nadu.

Tributaries : Right - Lakshman Tirtha, Kabani (Kapila), Suvarnavati, Bhavani, Noyil and Amaravati. Left - Harangi, Hemavati, Shimsha and Arkavati.

Shimsha river - It originates from the

southern part of Devarayanadurga hill in Tumkur district, and flows for about 221 km before joining Kaveri.

Major rivers in Kerala - There are three east-flowing rivers in Kerala: the Kabini river flows into Karnataka, while the Bhavani and Pamba rivers flow into Tamil Nadu. These three major rivers in Kerala join the river Kaveri.

Kabini river is a tributary of the river Kaveri in southern India that originates in the Wayanad District of Kerala by the confluence of the Panamaram River and the Mananthavady River. It flows eastward to join the Kaveri River at Tirumakudalu Narasipura in Karnataka.

Mahanadi river - It is the largest river of Odisha and third largest river system of peninsular India with a length of 860 km. It is an east flowing river.

It originates from Sihawa Mountain near Pharsiya village in Raipur, Chhattisgarh and extends over the states of Chhattisgarh, Odisha Jharkhand, Maharashtra and Madhya Pradesh and drains into the Bay of Bengal.

Tributaries : Left - Seonath, Hasdeo, Mand. Right - Ong, Tel and Jonk.

Jonk river - It is a tributary of Mahanadi river that starts from the Sunabeda plateau in Odisha and flows through Chhattisgarh and Orissa. It joins Mahanadi at Shivrinarayan.

Subarnarekha river (streak of gold) - It is an East flowing river that originates from Chota Nagpur Plateau near Ranchi, Jharkhand forming a boundary between west Bengal and Odisha in its lower course.

Tributaries - Kharkai (Jharkhand), Roro, Kanchi, Harmu Nadi, Damra, Karru, Chinguru, Karakari, Gurma, Garra, Singaduba, Kodla, Dulunga and Khajjori.

Baitarani river - It is an east flowing river, originating from the Gonasika /Guptaganga (Cow Nose Shaped) hills in Kendujhar district of Odisha. It joins the Dhamra river and finally drains into the Bay of Bengal.

Tributaries - Deo, Salandi, Matai, Musal, Kusei, Ambajhara, Kukurkata, Gahira, Remal.

Kalinadi river - It is a west flowing river rises near Kushavali, a small village in Uttara Kannada district of Karnataka and drains into the Arabian sea. It supports

the livelihoods of thousands of people including fishermen on the coast of Karwar.

Mandovi river - It is also known as Mahadayi is a rain - fed river that is shared between Karnataka and Goa for their water needs. It joins with the Juari at a common creek at Cabo Aguada, forming the Mormugao harbor. Mandovi and Juari are the two primary rivers in the state of Goa.

Vaitarna river - It is a west flowing river that originates from the Trimbak hills in Nashik district of Maharashtra. It is one of the largest rivers in the Northern Konkan region. It supplies much of Mumbai's drinking water.

Periyar river - It is the largest river (west flowing) in Kerala also known as Lifeline of Kerala. It originates from Sivagiri hills of western ghats and flows through Periyar national park and Periyar lake and finally drains into the Arabian sea through Vembanad lake.

Tributaries - Muthirapuzha, Mullayar, Cheruthoni, Perinjankutti and Edamala rivers.

Rushikulya river - It is one of the major east flowing rivers in the state of Odisha. It originates from Daringbadi hills of the Eastern Ghats which is also known as 'Kashmir of Odisha' and finally drains into the Bay of Bengal.

Tributaries - the Baghua, Dhanei, Badanadi.

Penner river - It is also known as Uttara Pinakini is one of the major east flowing rivers of the peninsula with a length of 597 km that flows between Krishna and Kaveri. It rises from the Chenna Kesava hills of the Nandidurg ranges of Karnataka and drains into the bay of Bengal.

Tributaries - Jayamangali, Kunderu, Chiravati, Papagni.

Mithi river - It is a river on Salsette Island, Mumbai. It is a confluence of tail-water discharges of the Powai and Vihar lakes. The river is seasonal and rises during the monsoons.

Chandragiri river (Perumpuzha) - It is a west flowing river, originating from the northern slopes of the Greater TalaKaveri National Park in the Western Ghats of the Kodagu district, Karnataka and drains into the Arabian sea.

Tributaries - Kudumbur (largest) and Payaswini.

Netravati river - It is also known as Bantwal river that flows through Mangalore. The river has its origin from Gangamoola Kudremukh range of Karnataka and after it merges with the Kumaradhara river at Uppinangady before draining into the Arabian Sea. This river is the main source of water to Bantwal and Mangalore.

Vaigai river rises on the Eastern slopes of the Varusanadu hills near Kottaimalai in the Madurai district, Tamil Nadu. The Vaigai drains an area of 7,741 Sq.Km, which entirely lies in the state of Tamil Nadu.

Sharavathi - It is a west flowing river that originates and flows entirely within the state of Karnataka. It originates from the Shimoga district of Karnataka.

Ayiroor river - It originates from Vilangara in Navaikulam, flowing through Thiruvananthapuram district and emptying into the Edava- Nadayara Kayal.

Kallayi river - It originates in the Western Ghats of Kozhikode District of Kerala. It flows through Cherikkulathur, Kovur, Olavanna, Mankavu, Kallayi, Pallikkandi and Kothi and ultimately evacuates in the Arabian Sea at Kozhikode.

Ithikkara river - It is a 56 km long river in Kerala, India. It originates in the Kulathupuzha in the Western Ghats and empties into Paravur Lake.

Major Dams Project on Peninsular rivers

Sardar Sarovar Dam : The Sardar Sarovar Dam is a concrete gravity dam of length 1210 m built on the Narmada River in Navagam, Gujarat constructed to provide water and electricity to four Indian states: Gujarat, Madhya Pradesh, Maharashtra, and Rajasthan.

Tawa Dam : Dam on the Tawa River in Madhya Pradesh, forming the boundary of Satpura National Park.

Ukai Dam (Vallabh Sagar) : Second largest reservoir in Gujarat, built on the Tapi River.

Dharoi Dam : Gravity dam on the Sabarmati River in Gujarat, used for irrigation, power generation, and flood control.

Kaleshwaram Lift Irrigation Project : Largest multi-stage lift irrigation project in the world, on the Godavari River in Telangana.

Polavaram Project : Multi-purpose irrigation project on the Godavari River in Andhra Pradesh.

Kadam Dam : Reservoir on the Kadam River, a tributary of the Godavari River in Telangana.

Isapur Dam : Earthfill dam on the Penganga River in Maharashtra.

Nizam Sagar Dam : Reservoir on the Manjira River, a tributary of the Godavari River in Telangana.

Singur Dam : Major source of drinking water for Hyderabad, built on the Manjeera River in Telangana.

Nagarjuna Sagar Dam : Masonry dam across the Krishna River, between Andhra Pradesh and Telangana.

Lal Bahadur Shastri Dam (Almatti Dam) : Hydroelectric project on the Krishna River in Karnataka.

Srisailem Dam : Second-largest capacity hydroelectric station in India, on the Krishna River between Telangana and Andhra Pradesh.

Tungabhadra Dam : Multipurpose dam on the Tungabhadra River in Karnataka.

Somasila Dam : Dam on the Penna River, a tributary of the Krishna River in Andhra Pradesh.

Koyna Dam : One of the largest dams in Maharashtra, on the Koyna River.

Ujjani Dam (Bhima Dam) : Earthfill and masonry gravity dam on the Bhima River, a tributary of the Krishna River in Maharashtra.

Kallanai Dam (Grand Anicut) : One of the oldest dams in India, on the Kaveri River in Tamil Nadu.

Krishna Raja Sagara Dam : Dam on the Kaveri River in Karnataka.

Mettur Dam (Stanley Reservoir): Largest dam in Tamil Nadu, on the Kaveri River.

Hirakud Dam : Longest earthen dam in the world, on the Mahanadi River in Odisha. The Hirakud Captive Thermal Power plant is located in Odisha.

Tikrapara Dam : Dam on the Mahanadi River in Odisha.

Major Rivers of the World

Major Rivers of Asia

Yangtze - It is the longest river in China and the third-longest (6,300 km) river in the world. It sources the Tanggula Mountains and drains into the East China Sea. Shanghai is situated on this river.

Yellow or Huang river - It originates from the Bayan Har Mountains and drains into the Yellow Sea. It is the second-longest river in China. The color is caused by massive amounts of loose debris. The Grand Canal waterway (China) is the link of the Yellow river and Yangtze River. It is the world's longest man-made waterway.

Mekong - Its source on the Tibetan Plateau to the Mekong Delta. It is the longest river in Southeast Asia. It flows through six Asian nations - China, Vietnam, Laos, Myanmar, Thailand, and Cambodia. It drains into the South China Sea. It forms the border between Laos and Thailand.

Irrawaddy or Ayeyarwady - The Irrawaddy River flows in the country of Myanmar.

Salween (also known as the Nu and Thanlwin) - It is the longest undammed river in mainland Southeast Asia. It is spread throughout China, Myanmar, and Thailand.

Tigris - Originating from Lake Hazer in Turkey and emptying into the Persian Gulf (Indian Ocean).

Amur - It forms the border between northeastern China and Russia.

Mahaweli Ganga - It is the largest river basin of Sri Lanka.

Karnali is the longest river of Nepal.

The Manas River (Drangme Chhu) is the largest of Bhutan's four major river systems, with the other three being the Amo Chu (Torsa River), Wang Chu (Raidak), Puna Tshang Chu (Sankosh).

Major Rivers of Africa

Nile - The longest river in the world, length is roughly 6650 kilometers. It originates from Lake Victoria. It passes through Egypt, Uganda, Ethiopia, Kenya,

Tanzania, Rwanda, the Democratic Republic of the Congo, Eritrea, Burundi, Sudan, and South Sudan. It drains into the Mediterranean Sea. Tributaries - Blue and White Nile.

Congo or Zaire - It is the only major river which crosses the equator twice. It drains into the Atlantic Ocean.

Orange - It is the longest river in South Africa. It starts in the Drakensberg in neighboring Lesotho, where it is also known as the Senqu River. It drains into the Atlantic Ocean.

Limpopo - It drains to the Indian Ocean via Mozambique. It crosses the tropic of Capricorn twice.

Niger - It flows through West Africa, passing through countries Nigeria, Niger, Mali, and Guinea discharging through a massive delta, known as the Oil Rivers, into the Gulf of Guinea.

Zambezi - A major river in south-central Africa, It flows from western Zambia to the Indian Ocean in Mozambique. It passes through parts of Angola, Namibia, Botswana and Zimbabwe.

Major Rivers of Europe

Lena - The river originates in the Baikal range, flows in Siberia and drains into the Arctic Ocean.

Danube - It originates in Black Forest (Germany's) and drains into the Black Sea. It is Europe's second longest river. It flows through the ten countries: Germany, Austria, Slovakia, Hungary, Croatia, Serbia, Bulgaria, Romania, Moldova and Ukraine.

Dnieper - Its source is the Valdai Hills in central Russia and drain into the Black Sea.

Volga - Its source in the Valdai Hills to the Caspian Sea (the biggest inland body of water in the world). It is the longest river in Europe. It is the main river in western Russia.

Rhine - It originates from Alps mountain (Switzerland) and drains into the North Sea in the Netherlands. This transboundary river travels through six nations—Switzerland, Germany, Austria, Liechtenstein, Germany, and the Netherlands.

Thames - It originated in Cotswold Hills and drains into the North Sea. It is the

second-longest river in the United Kingdom and the longest in England.

Major Rivers of North America

Mississippi - The main river of the central United States and the second-longest river in North America after the Missouri River, flowing 2,320 km from its source at Lake Itasca to the Gulf of Mexico.

St. Lawrence - Flows from the Great Lakes to the Atlantic Ocean, forming the international border between Canada and the United States.

Mackenzie River - The longest river system in Canada, flowing 1,800 km from the Great Slave Lake to the Arctic Ocean.

Major Rivers of South America

Amazon - It originates from the Andes Mountains (Peru). In terms of total water flow volume, it is the largest (6575 Km) river in the world. It is the world's second-longest river.

It extensively drains into the Atlantic Ocean. Its maximum drainage system is in Brazil. The Amazon basin is the largest drainage basin in the world. It travels through Ecuador, Colombia, Venezuela, Bolivia, and Brazil.

Tributaries - Its important tributaries are Madeira, Ucayali, Purus, Rio Negro, Jurua, Japura, Tocantins, Araguaia. Madeira is the largest tributary of Amazon.

Parana - It is South America's second largest river (the Amazon being the largest). It passes through Argentina, Paraguay, Bolivia and Brazil and drains into the Atlantic Ocean.

Orinoco - The headwaters of the Orinoco River begin in Venezuela, near the border of Brazil. It passes through Colombia and Venezuela.

Major Rivers Of Australia/Oceania

Murray - It is the longest (2,530 km) river of Australia, flows through the southeast of the nation. It's also one of the world's driest major river basins. It originates from the Australian Alps mountain and drains into the Southern Ocean. Its major tributary is Darling river.

Fly River - It is situated in Papua New Guinea.

The **Waikato** River is the longest river in New Zealand.

Soils

The thin layer of grainy substance covering the Earth's surface is called soil. Pedology deals with the study of soils, including their formation, morphology, and classification. The Soil Survey of India was established in 1958 to carry out scientific surveys of soils.

Features of soil

Soils contain mineral particles, organic matter, water, and air. Their combinations determine soil properties like texture, structure, porosity, chemistry, and color.

Soils supply 13 essential nutrients to plants: 6 macronutrients (nitrogen, phosphorus, potassium, calcium, magnesium, sulfur) and 7 micronutrients (iron, manganese, boron, zinc, copper, molybdenum, chlorine).

Soil Profile

Soil layers are called horizons, and their sequence is the soil profile.

The main layers from top to bottom are: O-Horizon (organic matter), A-Horizon or Topsoil (humus-rich), E-Horizon (leached nutrients), B-Horizon or Subsoil (less organic matter), C-Horizon or Parent Rock/Bedrock, R-Horizon (compacted rocks).

Parent rock determines the soil's color, texture, chemical properties, mineral content, and permeability.

Soil Formation

Soils are formed through the interaction of five major factors: time, climate, parent material, topography and relief, and organisms.

Weathering is the breaking down of rocks. Rocks get broken down by agents including water, ice, wind, animals and growing plants.

Leaching is the removal of soluble materials, such as nutrients or contaminants, from one soil layer to a lower layer through the downward movement of water in the soil profile.

Eluviation involves the removal of nutrients and organic material from superficial horizons.

Deposition is the dropping of sediment in a new place. Some examples of deposition are the formation of an island or sand dunes.

Denudation is the wearing away of the terrestrial surface by processes including weathering and erosion.

Types of Soil

Alluvial Soil - The alluvial soil consists of various proportions of sand, silt and clay. These are formed due to the silt deposited by the Indo - Gangetic - Brahmaputra rivers. In the coastal area, the alluvial soil is formed due to the wave action. Alluvial Soils are widespread in the entire northern plains and the river valleys and cover about 43% of the total area of India. The color of the alluvial soils varies from the light grey to ash grey.

Deltas of Mahanadi, the Godavari and the Krishna are made from alluvial soil. It is mainly found in the plains of Gujarat, Punjab, Haryana, Bihar, Jharkhand, and Uttar Pradesh (maximum extent of alluvial soil).

Crops grown - Wheat, maize, sugarcane, pulses, oilseed. Rich in - Potash and Lime. Lacks in - Nitrogen and Phosphorous.

Types of alluvial soils -

Khadar - It is the new alluvium and occupies the flood plains of the rivers. It is enriched with fresh silt deposits every year. The texture of these soils is finer. Silt, mud, clay, and sand make up the Khadar land silt. Khadar soil is mainly found in eastern Uttar Pradesh, Bihar, and West Bengal.

Bhangar - It is the old alluvium, deposited away from the flood plains. The texture of these soils is rougher. It is made up of impure calcium carbonate nodules known as 'Kankars'. Example - The Ganga-Yamuna Doab and the Plains of Satluj.

Black soil - It covers about 15% of the total land area of the country. Black soil covers most of the Deccan Plateau which includes parts of Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh and some parts of Tamil Nadu. In the upper reaches of the Godavari and the Krishna, and the north western part of the Deccan Plateau, the black soil is

very deep. The black soil area of the Peninsular Plateau is known as Deccan Trap.

The lava plateaus are rich in black soil. These soils are also known as the 'Regur Soil' or the 'Black Cotton Soil'.

Gram can be grown on a wide range of soil. But well drained medium to black cotton soils are very suitable. It can also be grown in light alluvial soils which are rich in organic matter.

Crops grown - Cotton, sugarcane, jowar, tobacco, wheat, rice. Rich in - Lime, Iron, Magnesia and Alumina, Potash. Lacks in - Phosphorus, Nitrogen and organic matter.

Red or Yellow soil - It is also known as the "omnibus group". It covers about 18.5 % of the total land area of the country. It develops on crystalline igneous rocks in areas of low rainfall in the eastern and southern part of the Deccan Plateau. It is found in Orissa, Chattisgarh and southern parts of the middle Ganga plain, seen mainly in low rainfall areas.

Crops grown - Wheat, rice, sugarcane and pulses. Rich in - Iron and Potash. Lacks in - Nitrogen, Phosphorus and humus.

Laterite Soil - Laterite has been derived from the Latin word 'Later' which means brick. Laterite soil is found in areas of high temperature and high rainfall. These are the result of intense leaching due to tropical rains. With rain, Lime and silica is leached away from this soil. Humus content of the soil is removed fast by bacteria that thrives well in high temperatures. Thus, Humus content is low in this soil.

Laterite soils are widely cut as bricks for use in house construction. The laterite soils are commonly found in Karnataka, Kerala, Tamil Nadu, Madhya Pradesh and the hilly areas of Odisha and Assam. Laterite soil is reddish to yellow in color.

Crops grown - Cashew nuts, tea, coffee, rubber. Rich in - iron oxide, aluminum compounds and potash. Lacks in - Organic matter, Nitrogen, Phosphate and Calcium.

Red laterite soil found in Kerala, Tamil Nadu and Andhra Pradesh are well suited for tree crop cultivation like cashew nuts. The red color is due to the presence of iron in crystalline and metamorphic rocks. The soil appears yellow when it is in hydrated form.

Arid or Desert Soil - Cover 4.32 % of India, found in western Rajasthan. High salt content, lack moisture and humus. Suitable for drought-resistant and salt-tolerant crops. Arid soils exhibit arid topography that are developed in western Rajasthan.

Crops grown - Only drought resistant and salt tolerant crops such as barley, rape, cotton, millets maize and pulses. Rich in - Soluble salts, phosphate. Lacks in - Humus, Nitrogen.

Saline and Alkaline Soil - High salt content due to dry climate and poor drainage. Found in western Gujarat, eastern coastal deltas, Sunderbans, Punjab, and Haryana. These are also called Reh, Usara, Kallar, Rakar, Thur, and Chopan.

Crops grown - Unfit for agriculture. Rich in - Sodium, Potassium, Magnesium. Lacks in - Nitrogen and Calcium.

Other Soils

Loess soils are among the most fertile in the world because the abundance of silt particles ensures a good supply of plant available water, good soil aeration, extensive penetration by plant roots, and easy cultivation and seedbed production. It also consists of dust particles.

Podzols are soils with an ash grey subsurface horizon, bleached by organic acids, on top of a dark accumulation horizon with brown or black illuviated humus and reddish iron compound.

Pedalfer is a type of zonal soil deficient in lime but containing deposits of aluminium and iron, found in wet areas, especially those with high temperatures. The trees that lose their leaves each winter, need at least 65 cm of rain per year.

Pedocal is a type of zonal soil that is rich in lime and characteristic of relatively dry areas. The climates that form pedocals have less than 65 cm rainfall per year, compared to peddlers.

Sandy loam is a type of soil used for gardening. This soil type is normally made up of sand along with varying amounts of silt and clay. The best soils for cashew nuts are deep and well-drained sandy loams.

Clayey soil consists of very fine particles of clay and has a high capacity for water holding. It has little air spaces between

particles.

Peaty (Marshy) Soils - Formed from partially decomposed plant material under anaerobic conditions. Found in areas of heavy rainfall and high humidity, rich in humus and organic matter.

Forest Soils - Generally acidic, organic, and with limited chemical fertility. Found in regions of high rainfall.

Soil orders in India

As per Soils of India, National Bureau of Soil Survey and Land Use Planning reported that 'Inceptisols' type of soil (order) is most abundantly available. They form quickly through the alteration of the parent material.

Inceptisols are young soils, prevalent in semiarid to humid environments, which generally exhibit only moderate degrees of soil weathering and development. They are most widely distributed in India. The term "Inceptisols" is derived from Latin "inceptum," meaning "beginning."

Entisols are Immature soils that lack the vertical development of horizons. These soils are often associated with recently deposited sediments from wind, water, or ice erosion.

Aridisols are commonly light in color, and low in organic matter content.

Vertisols are soils with a high content of clay minerals that shrink and swell as they change water content.

Alfisols are moderately leached soils that have relatively high native fertility. These soils have mainly formed under forest and have a subsurface horizon in which clays have accumulated.

Ultisols are associated with humid temperate to tropical climates.

Mollisols are soils common to grassland environments.

Soil pH and Fertility

Soil pH measures the acidity or basicity (alkalinity) of soil. The right mix of minerals and organic matter makes the soil fertile. Factors affecting soil fertility include mineral composition, soil pH, soil texture, organic matter, manures, fertilizers, and bacteria like Rhizobium.

Soil Conservation

Soil conservation aims to protect soil from water runoff through methods like mulching, planting cover crops, and constructing contour ridges. Other methods include terrace farming, buffer strips, crop rotation, forest protection, afforestation, no-till farming, and rock dams.

Rock dams are a soil conservation method in which rocks are piled up to slow down the flow of water, preventing gullies and further soil loss.

Mulching is the process of covering the bare ground between plants with a layer of organic matter like straw, which helps to retain the soil's moisture.

Shelterbelts is the method of soil conservation in the coastal and dry regions where rows of trees are planted to check the wind movement to protect soil cover.

Contour barriers are contour strips or barricades created to slow down the water and soil particles flowing downwards.

Soil Erosion

Erosion is the movement of material from broken down rocks, including rocks, minerals, and plant and animal remains.

Soil Erosion is the main cause of Badlands, which are arid terrains with clay-rich soil that have been extensively eroded by wind and water.

Causes of soil erosion include heavy winds, rivers and streams, construction, logging and mining, agriculture, and overgrazing. Geological weathering also causes slow erosion of soil.

Sheet erosion is the least visible form of land erosion, involving the uniform removal of soil in thin layers by the forces of raindrops.

Overgrazing can lead to soil compaction, decreased water retention, increased salinity and loss of some nutrients, particularly nitrogen.

Landslides

Landslides are the movement of a mass of rock, debris or earth down a slope. Landslides are classified by their type of movement, including falls, topples, slides

(rotational and translational), and flows.

Landslides Mitigation Mechanisms include hazard mapping to identify areas prone to landslides, construction of retention walls to stop land slippage, increasing vegetation cover to arrest landslides, and ensuring surface drainage control works to restrict the movement of landslides.

Other important concepts of soils

Bearing capacity is the capacity of soil to support applied loads.

Field capacity is the amount of soil moisture held after excess water has drained away.

Grain size distribution can be determined through sieve analysis (for larger particles) and hydrometer analysis (for smaller particles).

Manures are decomposed organic matter derived from plants and animals. They are natural substances. Manures provide a lot of humus to the soil and they can be prepared in fields.

Humus is formed by the decomposition of plant and animal matter. It is rich in nutrients and retains soil moisture. It is considered decaying organic matter in soil.

Creep in geology means slow downslope movement of particles that occurs on every slope covered with loose, weathered material.

An **earthflow** is a downslope viscous flow of fine-grained materials that have been saturated with water and moves under the pull of gravity.

Infiltration occurs when surface water enters the soil.

Vegetation

Forests and Forest Cover

A forest is an area of land dominated by trees. In a forest, trees form the uppermost layer of vegetation and herb is the lowest layer of vegetation.

Forests protect soil, influence climate, water cycle, and air quality.

Forest Cover in India: According to the India State of Forest Report 2021, the total forest and tree cover in India constitutes 24.62% of the country's

geographical area, with the total forest cover being 21.71%. States with the largest forest cover include Madhya Pradesh, Arunachal Pradesh, Chhattisgarh, Odisha, and Maharashtra.

Deforestation and Afforestation

Deforestation, the large-scale removal of trees, can lead to various issues such as loss of biodiversity, decreased oxygen levels, habitat destruction, water cycle disturbances, increased pollution, and soil erosion.

Afforestation, the establishment of new forests, can help address these problems and provide benefits like a continuous supply of forest products, soil erosion prevention, flood control, climate change mitigation, and air pollution control.

Social Forestry

It refers to the management and protection of forests, afforestation on barren lands for environmental, social, and rural development.

It is classified into Urban Forestry, Rural Forestry, and Farm Forestry by the National Commission on Agriculture (1976).

Types of Forest in India

Tropical Deciduous Forests (dry and moist): Found in large parts of India, characterized by trees shedding leaves during the dry season. Teak, Sal, Shisham, etc are the main species.

Moist Tropical Forest in India include bamboos, epiphytes, aini, semul, gutel, mundane, hopea, benteak, kadam irul, rosewood, haldu, kanju, bijasal, kusum, bomsum, Indian chestnut, litsea, holloch, champa and mesua, etc.

The dry deciduous forests are found in areas having rainfall between 100 cm and 70 cm. These forests are found in the rainier parts of the Peninsular plateau and the plains of Bihar and Uttar Pradesh. Sal, acacia and bamboo are the main species.

Tropical Evergreen Forests: Found in the Western Ghats, Andaman and Nicobar Islands, and parts of Assam and Tamil Nadu. Well-stratified with trees up to 60m tall.

Moist Tropical semi-evergreen vegetation is found in the plateau of

Meghalaya, Sahyadri and the central and southern islands of the Nicobar Group.

Tropical Thorn Forests: Found in areas with rainfall less than 70cm, consisting of thorny trees and bushes like babool, ber, khejri, etc.

Thorny trees and bushes are found in the north western part of India. It includes semi-arid areas of south west Punjab, Haryana, Rajasthan, Gujarat, Madhya Pradesh and Uttar Pradesh. In these forests, plants remain leafless for most part of the year and give an expression of scrub vegetation.

The common animals are rats, mice, rabbits, fox, wolf, tiger, lion, wild ass, horses and camels.

Montane Forests: Found in mountainous regions, with vegetation changing from tropical to tundra with increasing altitude. Includes species like silver fir, deodar, spruce, etc.

Mangrove forests, also called mangrove swamps, Littoral forest, mangrove thickets or mangals, are productive wetlands that occur in coastal intertidal zones.

Mangrove forests grow mainly at tropical and subtropical latitudes because mangroves cannot withstand freezing temperatures. There are about 80 different species of mangroves.

In India, the mangrove forests spread over 6,740 sq. km which is 7 percent of the world's mangrove forests. Palm, coconut, keora, agar are the common trees found in these forests.

The Sundari trees (*Heritiera fomes*) are the dominant varieties of mangrove trees presently situated in the eastern state of West Bengal, and Bangladesh in the Ganga - Brahmaputra delta (Sundarban delta), Muthupet delta (Cauvery delta).

Coringa Wildlife Sanctuary is a part of Godavari estuary situated in Andhra Pradesh. It is the third-largest stretch of mangrove forests with 24 mangrove tree species and more than 120 bird species. It is home to the critically endangered white-backed vulture and the long-billed vulture.

Temperate Deciduous and Evergreen Forests: Found in mid-latitudes, with species like oak, ash, beech, and pine.

Biomes

A biome is a large naturally occurring community of flora and fauna occupying a major habitat. Biomes are defined by factors like plant structures, leaf types, plant and animal adaptations, and climate.

Major biomes of the world

Tundra Biome - Located in the Arctic regions, tundra biomes experience long, cold winters and cool summers.

Desert Biomes - Located in both cold and warm climates across the globe, deserts are defined by their dryness, and life in these areas are adapted to a lack of water and nutrients.

Grassland Biomes - The grassland biome, which is found on every continent except Antarctica, is characterized as being flat and grassy, with scant tree cover.

Coniferous Forest Biomes - These areas known as taiga or boreal forests - experience long, cold winters, short summers, and heavy precipitation. Located in the northern part of Asia, Europe. Within this biome, the primary vegetation types are conifers and evergreen trees.

Taiga or boreal forests span the higher latitudes between temperate forests and tundra, covering areas from 50 degrees north latitude to the Arctic Circle. These forests are abundant in northern Russia, Siberia, Canada, Alaska, Scandinavia, and the Himalayan region, featuring coniferous trees adapted to cool climates and steep slopes. Mosses and lichens characterize the nearby tundra biome.

Plants of the Taiga - The taiga is dominated by coniferous evergreen trees like spruce, oak, and fir, which form a dense canopy. These trees have needle-like leaves and produce seeds in cones, adapted to the harsh winters. Beneath the canopy, diverse plant life including ferns, sedges, mosses, and berries thrive in the moist environment, contributing to the rich biodiversity of the taiga.

Animals of the Taiga - During winter, taiga birds migrate south, while rodents inhabit the ground. The taiga hosts few large carnivores like bears and lynxes, and the world's largest cat, the Siberian

tiger, which preys on moose and wild boars.

Deciduous Forest Biomes - Located in Eastern North America, Western Europe, and northeastern Asia, this biome is marked by broad-leafed trees, such as maple and oak, that lose their leaves seasonally as the temperatures begin to drop.

Tropical Rainforest Biomes - These equatorial regions are warm and wet with diverse vegetation that forms a canopy.

Aquatic Biomes - There are numerous ways to classify aquatic biomes, and often freshwater and saltwater biomes are defined separately; factors used for classification include water depth, temperature, and salinity.

Grasslands

Grasslands are dry areas dominated by grasses and herbs, lying between deserts and forests.

Grasslands are subdivided into two types.

Tropical Grasslands - Tropical grasslands have dry and wet seasons that remain warm all the time. These regions are warm throughout the year, followed by dry and wet seasons. Tropical Grasslands are the ones which receive 50 cm to 130 cm of rainfall. The tropical grasslands contain quite short plants which makes it an excellent hunting ground. For instance, the African savanna is one of the tropical grasslands.

The tropical grassland is a home for elephants, giraffes, lions, cheetahs, zebras, and other spectacular species.

Tropical Grasslands are: East Africa-Savannah, Brazil- Campos, Venezuela-Llanos.

Temperate Grasslands - These are found in the midlatitudinal zones and in the interior part of the continents. Usually, grass here is short and nutritious. They have cold winters and warm summers with 25 cm and 75 cm. of annual rainfall. Shrublands are the best example of temperate grasslands.

Temperate grasslands are ideal for extensive wheat cultivation. For instance, some vegetation that grows here are, cacti, sagebrush, perennial

grasses, buffalo grass clovers, and wild indigos, etc.

Some Temperate Grasslands: Argentina - Pampas, USA - Prairies, South Africa - Veld, Asia - Steppe, Australia - Downs.

Other Grasslands of the world: Steppe - Europe and North Asia. Pustaz - Hungary. Canterbury - New Zealand. Taiga - Europe and Asia. Cerrados - South america.

Conservation and Management

In-situ conservation protects endangered species in their natural habitats, while ex-situ conservation involves methods like zoos, botanical gardens, and cryopreservation.

Joint Forest Management (JFM) involves local communities and forest agencies in the management of natural forests.

Other facts of Vegetation

Molai forest is a forest in Majuli district in the Brahmaputra River near Kokilamukh, Assam, India. House of Indian rhinoceros.

The **Kukrail Reserve Forest** (Lucknow) is famous for housing a nursery of endangered species of crocodiles, along with a deer park.

The **Amazon rainforest** is the world's largest tropical rainforest covering much of northwestern Brazil and extending into Colombia, Peru and other South American countries. Some other largest tropical rainforest are Congo, Australiasia, Sundaland, and Indo-Burma.

Gir Forest has approximately 600 Asiatic lions left in the of Western India, their last remaining natural habitat.

Bristlecone pines are the oldest living tree species on Earth. These strange trees are shaped by the wind, snow, and rain that have survived for thousands of years.

The Babul tree (*Vachellia nilotica*) is found in sandy or sterile regions. It is also known as gum Arabic tree.

Kelp are large brown seaweed that grow in shallow nutrient rich salt water, near coastal fronts around the world. They grow in dense groupings much like a forest on land. Underwater towers of kelp provide food and shelter for thousands of fish, invertebrates, and marine mammal species.

Cactus are mostly found in the dry region of India that are Gujarat and Thar desert of Rajasthan. The climatic condition of thorn forests is most ideal for the growth of Euphorbia.

Vertical distribution of different species occupying different levels is called stratification. For example, trees occupy top vertical strata or layer of a forest, shrubs the second and herbs and grasses occupy the bottom layers.

The mean annual average temperature in subtropical vegetation zone is 17°C to 24°C.

The wet temperate type of forests are found between a height of 1000 and 2000 meters.

Agriculture

The word Agriculture is derived from Latin words ager or agri meaning soil and culture meaning, cultivation.

Agriculture is a primary activity. It includes growing crops, fruits, vegetables, flowers and rearing of livestock. The land on which the crops are grown is known as Arable land.

The Planning Commission of India in 1989 divided India into 15 Agro-climatic regions - Western Himalayan Region, Eastern Himalayan Region, Lower Gangetic Plains Region, Middle Gangetic Plains Region, Upper Gangetic Plains Region, Trans - Gangetic Plains Region, Eastern plateau and Hill Region, Central Plateau and Hill, Western Plateau and Hill Region, Southern Plateau and Hill Region, East Coast Plain and Hill Region, West Coast Plains and hill Region, Gujarat Plain and hill Region, Western Dry region Plain and Island Region.

Farming Types/Methods

Primitive subsistence agriculture or shifting cultivation/slash and burn cultivation is practised in rainforest areas. It involves cutting down vegetation, burning the foliage, and using the ashes as nutrients for planting food crops.

It is practiced on small patches of land with primitive tools like hoe, dao, digging sticks, and family/community labor to meet the farmer's family needs.

Shifting cultivation method of cultivation is widely practised mostly by the tribal

people in the hilly area of North Eastern States like Assam, Nagaland, Arunachal Pradesh, etc.

The different names of slash and burn agriculture in different regions are:

'Jhumming or Jhum' in (north-eastern region) Assam, Mizoram, Meghalaya, Nagaland, 'Pamlou' in Manipur, 'Bewar or Dahiya' in Madhya Pradesh, 'Dipa' in Chhattisgarh and Andaman Nicobar islands, 'Podu or Penda' in Andhra Pradesh, 'Pama Dabi or Koman or Bringa' in Odisha, 'Kumari' in Western Ghats, 'Kuruwa' in Jharkhand, 'Valre or Waltre' in South eastern Rajasthan and 'Khil' in Himalayan belt.

It is also known as 'Milpa' in Mexico and Central America, 'Conuco' in Venezuela, 'Roca' in Brazil, 'Masole' in Central Africa, 'Ladang' in Indonesia, 'Ray' in Vietnam.

Intensive farming or Industrial agriculture involves large inputs of labour and money to maximise yields through heavy use of pesticides and chemical fertilizers.

The Intensive Agriculture District Programme was launched in 1960. Its major objectives were to achieve rapid and significant increase in agriculture production.

Sedentary cultivation refers to a method of agriculture where the same land is farmed every year.

Mixed farming involves the simultaneous cultivation of crops and the raising of livestock on the same farm.

It is found in highly developed parts of the world, such as North-western Europe, Eastern North America, parts of Eurasia, and temperate latitudes of Southern continents.

Commercial farming or Agribusiness is a farming method focused on growing high-demand crops and selling produce on the market.

Plantation is a type of commercial farming. In India, Tea, Coffee, Rubber, Sugarcane, Banana, etc. are important plantation crops.

Pastoral farming is aimed at producing livestock rather than growing crops. Examples include dairy farming, raising beef cattle, and raising sheep for wool.

Dairy farming's objective is production of safe, quality milk from healthy animals under generally acceptable conditions.

Extensive Farming is a system with large farms, moderately lower inputs, i.e. capital and labour.

Capitalistic farming is controlled and manipulated by individual entrepreneurs, prevalent in South America, North America, and Europe.

Organic farming or ecological farming uses organic fertilizers and emphasizes techniques like crop rotation and companion planting.

Sir Albert Howard gave the concept of organic farming in 1940. He is also known as the Father of Modern Organic Agriculture.

Sikkim was the first state declared as a fully organic state in 2016.

Wetland farming thrives in areas with abundant rainfall, growing crops like rice, jute, and sugarcane, and practicing aquaculture.

Terrace farming is the practice of carving flat regions out of hilly or mountainous terrain to produce crops, which helps in soil and water conservation.

Truck Farming is the farming where farmers specialize in vegetables only.

Intercropping is a multiple cropping practice involving growing two or more crops in proximity.

Contour ploughing is the process of farming by keeping the lines of contour on the sloppy regions to improve water distribution and avoid soil erosion.

Hydroponic farming is the technique of growing plants using a water-based nutrient solution rather than soil.

Crop rotation is the practice of growing different crops sequentially on the same land to improve soil health and optimise nutrients.

The 'Zabo' system is an indigenous farming system practiced largely in Phek district of Nagaland, also known as the "Dzüdü" or "Ruza" system.

Zabo combines forest, horticulture, agriculture, fishery, and animal husbandry with soil and water conservation measures on one hill.

Taungya Cultivation, started in Burma in 1856, involves growing forest crops alongside agricultural crops temporarily.

It was introduced in Bangladesh (Chittagong and Sylhet) in 1870.

Biofloc technology is a fish farming system that recycles waste nutrients as fish food.

Recirculatory Aquaculture System (RAS) is a technology where water is recycled and reused after mechanical and biological filtration.

Irrigation

Irrigation is the process of supplying water to the land at regular intervals using canals and other artificial methods to enhance agricultural growth.

The net irrigated area percentage in India has increased from 18% to 48% due to government interventions in irrigation technology.

Punjab has the highest total irrigated area in India at 98.7%, followed by Haryana, Uttar Pradesh, and Andhra Pradesh.

Irrigation Methods

Drip irrigation system delivers water to the crop using a narrow tube network of mainlines, sub-mains and lateral lines with emission points spaced along their lengths.

Advantages of the drip irrigation system include minimised fertilizer and nutrient loss, high water application efficiency, and prevention of soil erosion.

Sprinkler irrigation system allows application of water under high pressure with the help of a pump.

Bamboo drip irrigation system is a 200-year-old system of tapping stream and spring water by using bamboo pipes. Meghalaya uses the ancient bamboo drip irrigation method.

Water harvesting techniques like rainwater harvesting and traditional methods like Guls/Kuls, Johads, and Khadins are used for irrigation.

Guls or Kuls are used as channels. They are built in the hilly areas of Western Himalayas to serve the purpose of agriculture. 'Kulh' is the local method of canal irrigation found in Himachal Pradesh. Khadins are rainwater harvesting structures built in Gujarat.

Major Cropping Seasons in India

The three major cropping seasons in India are Kharif, Rabi, and Zaid.

Kharif crops are sown in the rainy season from June-July and harvested in September. It requires a temperature above 25 degrees Celsius with high humidity and annual rainfall above 100 cm.

Examples - Rice, Jowar, Pearl millet (Bajra), Maize, Cotton, Groundnut, Jute, Sugarcane, Turmeric, Soybean etc.

Kharif Pulses are Tur (Arhar), Blackgram (Urd), Greengram (Moong), Cowpea (Lobia).

Rabi crops are sown in winter from October-December and harvested in summer from April-June.

Rabi crops are benefitted from the temperate cyclones arising from the Mediterranean Sea that cause rainfall in Punjab, Haryana, Delhi and western Uttar Pradesh.

Examples - Wheat, barley, gram, mustard, and linseed, Lentil (Masoor), Pea (Matar), and Lathyrus (Khesari).

Zaid crops are summer season crops grown between Kharif and Rabi, from March-June.

Examples - Cucumber, bitter gourd, pumpkin, and watermelon.

Other Crops

Boro crops are a type of winter rice grown in the months of November-December and harvested in May-June. It is a rabi crop.

Pulses are all-season crops. They can be grown in any season.

India leads in both the production and consumption of pulses. Tur (arhar), urad, moong, masur, peas, and gram are the major pulses that are grown in India. Rajasthan, Madhya Pradesh, and Maharashtra are the top producers of pulses.

Sesamum is a Kharif crop in North India and Rabi crop in South India.

Fodder crops are specifically cultivated to nourish livestock. These include corn, maize, hay, and even the leftover pulp from sugar beet processing.

Crop Diseases

Crop disease is defined as a complication with the crop that reduces its yield or distorts its appearance. It can be caused by fungi, bacteria, viruses, mycoplasma, viroids, and nematodes.

Examples of crop diseases include Karnal bunt in wheat, blast disease in rice, late blight in potatoes and tomatoes, and Ikki disease in groundnut.

Pesticides

Pesticides are chemical compounds that are used to kill pests, including insects, rodents, fungi and unwanted plants.

Types of Pesticides : Insecticides - insects, Herbicides - plants, Rodenticides - rodents (rats), Bactericides - bacteria, Fungicides - fungi, and Larvicides - larvae.

Both synthetic and natural, are used to manage pests and diseases in crops.

Examples of synthetic pesticides include DDT, Heptachlor, Chlordane, Glyphosate, and Malathion, while natural pesticides include Rotenone and Pyrethrum.

In 1945, DDT was made available to farmers as an agricultural insecticide and played a role in the temporary elimination of malaria in Europe and North America.

In 1972, DDT was banned worldwide due to its harmful effects on human health and agriculture.

Fertiliser

It is any material of natural or synthetic origin that is applied to soil or to plant tissues to supply plant nutrients.

Types of fertilisers

Sole fertilisers - These are the fertilisers that supply only a single nutrient.

Complex fertilisers (Mixed fertilisers) - This type of fertiliser is a source of more than one macro-nutrient for the plant is known as mixed fertiliser.

The largest fertiliser producing states in India are Uttar Pradesh, Gujarat, Maharashtra, Rajasthan, Andhra Pradesh.

Commonly used mixed fertilisers - di-ammonium phosphate (18:46:0), nitro-phosphate (20:20:0), ammonium phosphate (28:0:0), ammonium

phosphate sulphate (16:20:0), calcium ammonium nitrate (8% Ca and 21–27% N), etc.

Biofertilizers are organisms that enrich the nutrient quality of the soil. The main sources of biofertilizers are bacteria, fungi and cyanobacteria. Rhizobium, Azospirillum, Azotobacter, Glomus, Anabaena, Nostoc, Oscillatoria are examples of biofertilizers.

Rhizobium, Azospirillum, Azotobacter are the main nitrogen fixing biofertilizers.

Various Forms of Agriculture

Pisciculture involves fish farming.

Sericulture is the rearing of silkworms.

Viticulture is the growing of grapes.

Floriculture deals with farming of different categories of flowers.

Olericulture refers to the study of vegetables and cultivation.

Silviculture is the practice of controlling the growth, composition, and quality of forests.

Apiculture is the scientific method of rearing honeybees.

Arboriculture is the cultivation, management, and study of individual trees, shrubs, vines, and other perennial woody plants.

Pomiculture is the cultivation of fruits and Pomology is the study of fruits and their cultivation. The father of pomology was Charles Downing.

Aviculture is the practice of keeping and breeding birds, especially of wild birds in captivity.

Horticulture is the science and art of growing fruits, vegetables, flowers, or ornamental plants, and plant propagation in horticulture involves sexual and asexual methods.

Grafting and budding are horticultural techniques used to join parts from two or more plants to form a single plant.

Land Reforms and Factors Affecting Agriculture

Land ceiling refers to fixing the specified limit of land that could be owned by an individual, with land above the limit being taken over by the government and allotted to landless cultivators and small farmers.

Factors that cause pressure on agricultural land include the right of inheritance, division of land, rising population, use of excessive chemical fertilizers, salinization, desertification, urbanization, waterlogging, climate change, droughts, and pollution.

Infrastructural factors of agriculture include irrigation, electricity, roads, credit, and marketing.

Institutional factors of agriculture include land reforms, crop insurance, establishment of Grameen banks, cooperative societies, and banks.

Green Revolution

The term 'Green Revolution' was coined by William Gaud in 1968 to describe the success achieved by the new agricultural technology in developing countries of Asia and Latin America.

Norman Borlaug was an American Agronomist known as the Father of the Green Revolution in the world and He got the Nobel prize in 1970 for his contribution in Agriculture.

In 1965, the Indian government launched the Green Revolution under geneticist M.S. Swaminathan, known as the Father of the Green Revolution in India.

The first phase (1967-1978) was restricted to wheat and rice, while the second phase (1980s) covered a wider range of crops.

The Green Revolution was based on the use of High Yielding Variety (HYV) seeds, double cropping in existing farmland, and the expansion of farming areas.

The Green Revolution led to a spurt in crop productivity, rise in acreage, expansion of irrigation facilities, increased use of chemical fertilisers and pesticides, and mechanisation of agricultural practices.

However, it also had negative impacts like decline in soil fertility, over-capitalization in agriculture, depletion of groundwater resources, and limited benefits to some farmers.

The National Saffron Mission was launched in 2010-11 for the cultivation of saffron in Jammu and Kashmir, and the government has decided to expand its cultivation to the northeast.

Major Crops in India

Wheat is the main food crop in the north and north-western parts of the country, requiring a temperature between 20°C to 25°C and 50 to 75 cm of annual rainfall. It is classified as fine grain.

There are two important wheat-growing zones in the country - the Ganga-Satluj plains in the north-west and black soil region of the Deccan.

Dilbagh Singh Athwal is known as Father of the Wheat Revolution in India.

India is the world's second-largest producer of wheat after China. The major wheat-producing states are Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Bihar and Rajasthan.

The state of Punjab is known as the bread basket of India.

Rice is the major cereal crop grown in about 44.6 million hectares in the country, requiring high temperature between 21°C to 37°C and high humidity with annual rainfall above 100 cm. Oryzenin is the principal protein of rice.

India is the world's second-largest producer and the largest exporter of rice.

In Assam, West Bengal and Odisha, three crops of paddy are grown in a year. Respectively, these are Aus, Aman and Boro.

Black rice which is mainly grown and consumed in Manipur where it is called Chakhao.

Some other famous varieties of rice are Birsa dhan-105, Karjat-2, Ratnagiri-3, Pant dhan-12, Jitendra, Dhanrasi, PNR-546, Amrut, Pusa etc.

The Kuttanad region of Alappuzha district of Kerala is known as the Rice bowl of the state.

Sugarcane is a tropical as well as a subtropical crop, requiring a temperature of 21°C to 27°C and an annual rainfall between 75cm and 100cm.

It is the main source of sugar, gur (jaggery), khandsari and molasses.

Sugarcane feedstock mainly consists of sugar in the form of disaccharide (sucrose), which is readily fermented into ethanol.

India is the second-largest producer of

sugarcane after Brazil, with Uttar Pradesh known as the 'Sugar bowl of India'.

Maize is a crop which is used both as food and fodder. It is a kharif crop which requires temperature between 21°C to 27°C and rainfall between 50 cm to 100 cm and lots of sunshine. It grows well in old alluvial soil.

The top maize producing states in India are Karnataka, Maharashtra, Madhya Pradesh, Tamil Nadu and Telangana. The United States of America is the largest producer of corn in the world.

Millets are a highly varied group of small-seeded grasses, widely grown in India as cereal crops or grains for fodder and human food. Kodon, Hraka, and Bauti are examples of millets.

India is the largest producer of millets in the world.

India's two varieties of millets namely Pearl Millet (Bajra) and Sorghum (Jowar) together contribute approx 19 per cent in world production.

The other varieties of millets are Finger Millet (Ragi/Mandua), Minor Millets i.e. Foxtail Millet (Kangani/Kakun), Proso Millet (Cheena), Kodo Millet (Kodo), Barnyard Millet (Sawa/Sanwa/Jhangora), Little Millet (Kutki) and two Pseudo Millets (Buck-wheat (Kuttu) and Amaranthus (Chaulai).

Jowar is the main food crop in semi-arid areas of central and southern India. It is sown in both kharif and rabi seasons in southern states. But it is a kharif crop in northern India where it is mostly grown as a fodder crop.

Jowar is a tropical crop and grows well at a temperature between 27° and 32°C but temperature below 16°C is harmful for the crop. It requires moderate rainfall between 30 and 65 cm.

Maharashtra produces more than half of the total jowar production of the country. Other leading producer states of jowar are Karnataka, Madhya Pradesh, Andhra Pradesh and Telangana.

Cotton requires a clear sky during the flowering stage, high temperature, light rainfall or irrigation, and 210 frost-free days.

It grows best in Black soil. India lost a large proportion of cotton growing area

to Pakistan during partition.

India grows both short staple (Indian) cotton as well as long staple (American) cotton, with major cotton-producing states being Gujarat, Maharashtra, Andhra Pradesh, Telangana, Haryana, and Karnataka.

Bt cotton is the only Genetically Modified (GM) crop that has been approved for commercial cultivation in India.

Coffee thrives in warm, wet climates with well-drained loamy soil. Ideal conditions include hill slopes, and temperatures ranging from 15°C to 28°C.

Rainfall between 150 to 250 cm is optimal, with dry weather necessary during berry ripening.

South India, particularly Karnataka and Kerala, is a major producer of coffee beans. It can be classed into four types: Arabica, Robusta, Liberica, and Excelsa.

The Arabica variety initially brought from Yemen is produced in the country. This variety is in great demand all over the world. The Cultivation of coffee was introduced on the Baba Budan hills in India.

The largest producer of coffee in the world is Brazil so it is called 'Coffee Bowl of the World'.

Tea plants grow well in tropical and subtropical climates. The Acidic Soil with around 4.5 - 5.5 pH is most suitable for Tea. Temperature above 32°C is disastrous for the crop if it is accompanied by low humidity.

India being the largest producer and consumer of black tea in the world. Assam, West Bengal, Tamil Nadu, and Kerala account for about 95 percent of total tea production in India.

Jute is a cash crop that grows well on fertile soils in the flood plains, with West Bengal being the largest producer.

It grows well on alluvial soil, high temperature, high rainfall. Humid Climate is required during the time of growth. The crop is grown in a temperature range of 20°C to 40°C but optimum temperature for its growth is 30°C to 34°C.

A rainfall of about 125 to 150 mm per month distributed well is required during the growth period.

India and Bangladesh are the leading producers of jute in the world. India is the

largest exporter of raw jute and jute goods.

West Bengal is the largest producer of jute and accounts for about three-fourth of the production in the country. Some other major Jute producing states are Bihar, Assam, Orissa, and Meghalaya.

Groundnut is a Kharif crop that requires 20-25 degrees Celsius temperature and moderate rainfall. Gujarat is the largest producer. India was the second largest producer of groundnut in the world in 2018.

Sunflower is an important oilseed grown in India. Its cultivation is concentrated in the regions of Karnataka, Andhra Pradesh, Telangana, and adjoining areas of Maharashtra.

Soybean is a Kharif crop, with Madhya Pradesh being known as 'Soya Pradesh'.

Rubber is an equatorial crop that requires a moist and humid climate with rainfall of more than 200 cm and temperature above 25°C.

Taramira oil or jamba oil is a seed oil which is pressed from the seeds of the Arugula. The plant is highly drought resistant. The oil is popular in West Asia, Pakistan and Northern India.

Natural fibres are fibres that are produced by geological processes, or from the bodies of plants or animals. They can be used as a component of composite materials

Common natural fibres sourced from the plant kingdom are Cotton, Flax, Hemp, Bamboo, Sisal, Jute. Their main component is cellulose. Jute and hemp are produced on a large scale in India.

The popular Animal fibres are Wool, Silk, Angora, and Mohair.

Viscose fibre is obtained from cellulose.

Viscose is made from tree wood pulp, like beech, pine, and eucalyptus.

Tamil Nadu is also known as the Yarn bowl of India.

Major Fruit Production Places in India

India is a producer of tropical as well as temperate fruits.

Mangoes - Maharashtra, Andhra Pradesh, Telangana, Uttar Pradesh and West

Bengal.

Oranges - Nagpur and Cherrapunjee (Meghalaya).

Litchi and Guava - Uttar Pradesh and Bihar.

Pineapples - Meghalaya,

Grapes - Andhra Pradesh, Telangana and Maharashtra.

Apples, Pears, Apricots and Walnuts - Jammu Kashmir and Himachal Pradesh.

Banana - It is the most produced fruit of India with an annual production of about 265 Lakh tonnes.

The major Banana producing states are Tamil Nadu, Maharashtra, Karnataka, Gujarat, Andhra Pradesh, Bihar, Assam and Madhya Pradesh.

Key varieties of Banana include Dwarf Cavendish, Robusta, Rasthali, Poovan, Nendran, Red Banana, Karpuravalli, Pachanadan, Virupakshi etc.

Agriculture Census

It is the main source of information on various agricultural parameters in India, such as number of operational holdings, land use, tenancy, and cropping patterns.

Conducted every 5 years by the Department of Agriculture and Farmers Welfare.

The first Agriculture Census was launched in 1970-71, and the latest one with reference year 2020-21 is the 11th Census.

Agricultural Boards, Institution in India

The Indian Council of Agricultural Research (ICAR) is the apex body for coordinating, guiding, and managing research and education in agriculture in the country.

The National Dairy Board (NDB) was responsible for enhancing the production of milk.

National Oilseeds and Vegetable Oils Development Board were responsible for the production enhancement of oilseeds in non-traditional areas.

National Mustard Research Center is located in Bharatpur, Rajasthan. It was established by ICAR on 20 October 1993.

National Horticulture Board (NHB) was set up by the Government of India in 1984.

Livestock Census

Livestock Census is conducted once in 5 years by the Animal Husbandry Statistics Division under the Ministry of Agriculture & Farmers Welfare. Latest census was done in the year 2019.

The main objective is planning and formulation of various schemes /programmes of Animal Husbandry & Veterinary Sector.

India is the **world's highest livestock owner**. It stands first in the buffalo population of the world, second in population of goats and fish, has the second largest poultry market in the world and stands fifth in the population of camels.

The total Livestock population is 536.76 million in the country. The total Livestock population in rural and urban areas is 514.11 million and 22.65 million respectively with percentage share of 95.78% for rural and 4.22% for urban area.

Miscellaneous Facts of Agriculture

The Water table is an underground boundary between the soil surface and the area where groundwater saturates.

Net sown area - The physical extent of land on which crops are sown and harvested.

Culturable Wasteland - Any land which is left fallow (uncultivated) for more than five years is included.

The Directorate of Rice Research (DRR) with the support of Indian Agricultural Research Institute (ICAR) developed the 'Rice Knowledge Management Portal (RKMP)' and launched it in 2010, with an aim of overall rice development in the country.

Food safety management system (FSMS) - It is a controlled process for managing food safety to ensure that all food that is produced is up to quality standards and safe to consume.

Madhuban Gajar, a biofortified carrot variety with high β -carotene and iron content developed in 1943 in Junagadh district, Gujarat by Shri Vallabhhai Vasrambhai Marvaniya.

Hari Krishan Jain was an Indian cytogeneticist and plant breeder, known for his contributions to the field of

genetic recombination and the control of interchromosome level.

Dr. Swapan Kumar Dutta is well known for his pioneering research on genetic engineering of Indica rice.

'Year 2023' is Declared as the International Year of Millets by the United Nations General Assembly.

IR8 is a high-yielding semi-dwarf rice variety developed by the International Rice Research Institute (IRRI). It is also called miracle rice, which helped save India from a massive famine in the 1960s. It was first harvested in 1967.

Rocks

Rock is a natural substance composed of solid crystals of different minerals that have been fused together into a solid lump. Rocks are the natural mass of mineral matter that makes up the earth's crust.

Structural geology deals with the form, classification, mechanism, and causes of rock structures' evolution.

Cliffs are areas of rock with a nearly vertical angle, formed by weathering, erosion, and the effect of gravity. Cliffs are usually composed of rock resistant to weathering and erosion.

Formation of rocks

Lithification - It refers to the process which converts unconsolidated sediments into sedimentary rocks. Example: Sandstone, limestone, shale, Halite etc.

Weathering describes the breaking down of rocks and minerals on the surface of the Earth. Water, ice, acids, salts, plants, animals, and changes in temperature are all agents of weathering.

Desertification is the degradation process by which a fertile land changes itself into a desert by losing its flora and fauna.

Abrasion is the physical process of rubbing, scouring, or scraping whereby particles of rock are eroded away by friction.

Main types of rocks - Igneous, Sedimentary, and Metamorphic.

Igneous Rocks

Igneous rock is formed through the cooling, solidification and crystallization of magma or lava. It may form with or without crystallisation, either below the surface as intrusive (plutonic) rocks or on the surface as extrusive (volcanic) rocks.

Types of igneous rocks include acid igneous rocks (high in silica, less dense, lighter in color), basic igneous rocks (higher in basic oxides, denser, darker in color), plutonic rocks (slow cooling, large mineral grains), and volcanic rocks (fast cooling, small mineral grains).

Examples - Batholith, Basalt, Basalt Porphyry, Dunite, Gabbro, Granite, Obsidian, Pegmatite, Mica and quartz.

Batholiths are large masses of intrusive igneous rock that form and cool deep in the Earth's crust.

Granite is a coarse-grained igneous rock commonly found throughout the continental crust, particularly in mountainous areas.

It is characterized by its light color and consists primarily of quartz, potassium feldspar, and sodium feldspar, making up more than 80% of the rock composition.

The formation of granite occurs from magma with a high silica and alkali metal oxide content, which slowly cools and solidifies underground. These rocks are classified based on the relative percentages of quartz, mica, alkali feldspar, and plagioclase.

Gabbro is a coarse-grained (phaneritic), dark-coloured, intrusive igneous rock that is widely used as a concrete aggregate, road base material, and crushed stone for railroad ballast. It is relatively low in silica and rich in iron, magnesium, and calcium.

Pumice is a light-colored, extremely porous igneous rock that forms during explosive volcanic eruptions. It floats in water.

Olivine is a yellowish green to green coloured gemstone found in mafic and ultramafic igneous rocks such as basalt, gabbro, dunite, diabase, and peridotite.

Sedimentary Rocks

Sedimentary rocks are formed by the deposition and consolidation of sediments obtained from the weathering

of pre-existing rocks.

Sedimentary rocks often contain fossils of plants, animals, and microorganisms.

The remains of the dead plant and animals trapped in the layers of rocks are called fossils. Almost all fossils are preserved in sedimentary rocks. The branch of biology that deals with the study of fossils is called paleontology.

Sedimentary rocks are classified into clastic (mechanically formed), chemical, and organic types.

Examples - Sandstone, limestone, lignite, Chalk, Clay, Shale.

Mechanically (Clastic) sedimentary rocks - Rocks are created from the remains of mechanical weathering. Example - Sandstone, shale, siltstone, and breccias.

Chemical sedimentary rocks - Chemical Sedimentary rocks are made up of dissolved substances that precipitate out of solutions. Example - Chert, some dolomites, flint, iron ore, limestones, and rock salt.

Organic Sedimentary Rocks - Plant and animal material accumulates to generate these types of rocks. Example - fossiliferous limestone, Shale, Lignite, chalk and coal.

The spread of Sedimentary Rocks in India : Alluvial deposits in the Indo-Gangetic plain and coastal plains is of sedimentary accumulation.

India has 26 major sedimentary basins, and the country is known for its coal, petroleum, and other mineral deposits in sedimentary formations. Mizoram is also known as the 'Molasses basin' which is made up of soft unconsolidated deposits.

Different varieties of **sandstone** are spread over Madhya Pradesh, eastern Rajasthan, parts of Himalayas, Andhra Pradesh, Bihar and Orissa.

Coal deposits occur in river basins of the Damodar, Mahanadi, the Godavari in the Gondwana sedimentary deposits.

The great Vindhyan highland in central India consists of sandstones, shales, limestones.

Metamorphic Rocks

Metamorphic rocks are formed by the transformation of existing rock types due to heat and/or pressure within the Earth's crust.

Metamorphic rocks are broadly classified into two major groups: Foliated rocks and non-foliated rocks.

Foliated Metamorphic Rocks - Granite, gneiss, and biotite schist.

Non-Foliated Metamorphic Rocks - Skarn, novaculite, marble, hornfels, and quartzite.

Metamorphic Rocks in India - The **gneisses** and **schists** are commonly found in the Himalayas, Assam, West Bengal, Bihar, Orissa, Madhya Pradesh and Rajasthan.

Graphite is found in Orissa and Andhra Pradesh.

Marble occurs near Alwar, Ajmer, Jaipur, Jodhpur in Rajasthan and parts of Narmada Valley in Madhya Pradesh

Metamorphism

It is the transformation of existing rock (the protolith) to rock with a different mineral composition or texture.

Various forms of metamorphism

Dynamic metamorphism :- Mechanical disruption and reorganization of the original minerals within rocks, without any appreciable chemical changes, is called dynamic metamorphism.

In contrast, thermal metamorphism involves the chemical alteration and recrystallization of rock materials due to the effects of heat.

There are two main types of thermal metamorphism: contact metamorphism and regional metamorphism.

Contact Metamorphism :- In contact metamorphism, rocks come into contact with hot, intruding magma or lava. Under these high-temperature conditions, the rock materials recrystallize. Quite often, new materials formed from the magma or lava are also added to the rocks.

Regional Metamorphism :- It occurs when rocks undergo recrystallization due to deformation caused by tectonic

shearing, combined with high temperature and/or pressure. This process can lead to the rearrangement of mineral grains or crystals in the rocks.

During metamorphism, the grains or minerals in some rocks may become arranged in layers or lines, a feature known as foliation or lineation.

Sometimes, minerals or materials of different compositions are arranged into alternating thin to thick layers, appearing as light and dark bands. This banded structure is called "banding," and rocks displaying it are referred to as "banded rocks."

Other Geological Features and Concepts

Horst is a ridge of the earth's crust that has been forced upward between two faults and so is higher than the surrounding land.

Karst landscapes are formed by the weathering of soluble rocks like limestone and dolomite, creating features like sinkholes and caves.

Lineations in structural geology are linear structural features within rocks.

Foliation in geology refers to repetitive layering in metamorphic rocks.

Banding means that the rock consists of alternating, thin layers of two different mineral compositions.

Flaking is a weathering process that operates on cavernous rock surfaces.

Mushroom rocks are a type of wind - eroded landform found in arid regions like the Thar Desert.

Mineral and Energy Resources

Energy

Energy is the ability to do work. Energy sources could be classified as Conventional sources of Energy and Non-conventional sources of Energy.

Conventional sources of Energy are also known as **non-renewable** sources of energy and are available in limited quantities. Its examples are coal, Fossil, fuels, Firewood, petroleum, natural gas, nuclear material etc.

Non-conventional sources are also known as **renewable** sources of energy. These are the resources that are continually renewed by natural processes. Its examples are solar energy, bioenergy, tidal energy, Geothermal energy and wind energy.

Mineral

Minerals are homogenous, and naturally occurring substances with a definable internal structure.

The study of the chemistry, crystal structure and physical properties of the mineral constituents of rocks is called Mineralogy.

Minerals are divided into 2 types: Metallic: Ferrous (Iron Ore, Manganese, Nickel, Chromite) and Non-Ferrous (Copper, Bauxite, Lead, Zinc, Gold) and Non-Metallic: Graphite, Mica, Monazite, Limestone, Dolomite, Gypsum, Silica, etc

The Mohs scale of mineral hardness is a qualitative ordinal scale, from 1 to 10, characterizing scratch resistance of minerals through the ability of harder material to scratch softer material. Diamond is the hardest known mineral, Mohs' 10. Talc is the softest.

Extraction of Mineral

Mining: The process of extracting metal ores buried deep underground.

Quarrying: The process of extracting minerals near the Earth's surface.

Drilling: The process of boring a hole through soil and rock to access oil and gas reservoirs.

Shaft Mining: Deep bores created to reach mineral deposits at great depths

Open - cast Mining : Surface mining technique of extracting rock or minerals from an open pit.

Major Mineral Resources in India

Mineral Exploration Corporation Limited (MECL) (1972) carries out mineral exploration activities and is the country's premier exploration agency. It is headquartered in Nagpur, Maharashtra.

Iron Ore - In India, communities like the Agarias specialized in the craft of iron

smelting. Following the directions of the Agarias, Dorabji Tata and Charles Weld trekked through the dense forest and found iron ores in the Rajhara Hills of Chattisgarh. Australia has the world's largest estimated reserves of iron ore.

Marmagao is the premier iron-ore exporting port situated in Goa.

95 per cent of total reserves of iron ore is located in the States of Odisha (largest producer), Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh and Tamil Nadu.

In India high grade iron ore deposits are found in Northern Odisha. Mayurbhanj and Keduja are important iron ore mines in Odisha. The Gua and Noamundi Iron ore mines are located in Jharkhand.

Haematite, magnetite, Iron pyrites, and siderite are the main ores of Iron. Iron Oxide is the primary component of naturally occurring mineral magnetite on Earth.

Haematite is the predominant iron ore variety, with the highest reserves found in Odisha, Jharkhand, Chhattisgarh, and Andhra Pradesh. Karnataka also has significant haematite reserves.

Bailadila is a hill range in Chhattisgarh famous for very high-grade hematites (Iron ore), which is used for steel making.

Magnetite is another important iron ore, with the highest reserves located in Karnataka, particularly the Kudremukh Iron Ore mine.

Philosopher's stone is an unknown substance, also called "the tincture" or "the powder,". It transforms baser metals like iron into precious ones, especially gold and silver.

Aluminium Ore - Bauxite, Cryolite, Kaolinite (a form of clay), Corundum are the ores of Aluminum. Bauxite is the primary raw material used in aluminum smelters and is found in abundance in India. Australia is the largest producer of Bauxite in the world.

Odisha is the largest producer of bauxite, with the Panchpatmali deposits in the Koraput district being the most important reserves. Kalahandi and Sambalpur are the leading producers in Odisha.

Jharkhand also has rich deposits of bauxite, particularly in the Lohardaga region.

Other states with significant bauxite reserves include Chhattisgarh, Madhya Pradesh, and Maharashtra.

Manganese Ore - The important manganese ores are pyrolusite, romanechite, manganite, and hausmannite and rhodochrosite.

Odisha has the highest reserves of manganese ore in India with the ore found in the Sundargarh, Koraput, and Ganjam districts.

According to the Indian Minerals Yearbook 2020, Madhya Pradesh has the highest production of manganese in India, with the ore found in the Balaghat, Chhindwara, and Jhabua districts.

Maharashtra also has manganese ore deposits, particularly in the Nagpur and Bhandara districts.

Copper Ore - Some ores of Copper are Copper pyrites, Malachite, Cuprite, Copper glance, Bornite, Azurite, Chalcocite, Chalcopyrite, Tetrahedrite, Digenite, Chrysocolla, Tennantite, Dioptase, Enargite etc.

Chile is the largest producer of copper in the world. Hindustan Copper Limited is vertically-integrated, primary copper producer in India.

Malanjkhand, open pit copper mine, in (Balaghat district) Madhya Pradesh is the largest copper mine in India. Khetri town in the Jhunjhunu district of Rajasthan is well-known for its Copper mines.

Zinc and Lead Ore - Some Zinc ore are Sphalerite, Zincite, Calamine etc. Galena is a lead sulfide mineral. It is the world's primary ore of lead. Rajasthan is the largest producer of zinc and lead in India while China is the largest producer of zinc and lead in the world.

Rampura Agucha is a zinc and lead mine located on a massive sulfide deposit in the Bhilwara district of Rajasthan.

Sindesar Khurd is a silver - zinc - lead underground mine located close to the Rajpura Dariba mine and Bamnia mine in Rajasthan.

Uranium - Kazakhstan is the largest uranium producing country in the world. Tummalapalle in Andhra Pradesh has the largest uranium reserves in India. The Jaduguda Mine in East Singhbhum district, Jharkhand, was the first uranium

mine in India, commencing operations in 1967.

Monazite - It is an important ore for thorium, lanthanum, and cerium, and Kerala has one of the richest monazite deposits in India.

Graphite - Arunachal Pradesh has the highest graphite deposits in India, accounting for 35% of the country's total graphite reserves.

Mica - India has the largest reserve of mica in the world. Andhra Pradesh is the largest producer of mica in India, with the Nellore district being famous for its mica (crude) production.

Koderma district in Jharkhand is renowned for its mica mining, especially for ruby mica. Mica (waste and scrap) is largely produced by the states of (Bhilwara) Rajasthan, Bihar and Jharkhand.

Natural Rubber - Kerala is the largest producer of natural rubber in India, while Thailand is the largest producer in the world.

Silver - The principal ores of silver are Argentine, horn silver and pyrargyrite. Rajasthan is the largest producer of silver in India, followed by Karnataka, Odisha, Jharkhand, and Andhra Pradesh.

Diamonds - Russia is the largest producer of diamonds in the world. Australia is famous for pink, purple and red diamonds. Madhya Pradesh, particularly the Panna region, is the only state in India that produces diamonds.

Gold - The United States has the highest gold reserves in the world, while China is the largest gold producer globally.

The largest resources of gold ore (primary) in India are located in Bihar, followed by Rajasthan, Karnataka, and other states. Karnataka is the largest producer of gold in India, with the Kolar Goldfield, Hutt Goldfield, and Ramagiri Goldfield being the primary gold-producing areas.

Tin - Tin ore is known as cassiterite. China, Malaysia, and Indonesia are among the world's leading tin producers, with the Asian continent producing more than half of the world's tin. Chhattisgarh is the only tin-producing state in India.

Silica - It is a widely available mineral resource in India. It is a major

component in cement, accounting for 17-25% of its composition.

Mineral-Rich Regions of India

Mysore Plateau - Large reserves of iron ore in the Baba Budan hills.

Krishna River Basin - Rich in mineral deposits such as oil, gas, coal, iron, limestone, dolomite, gold, granite, laterite, uranium, and diamond.

Godavari River Basin - Mineral-rich area with deposits of oil, gas, coal, and iron ore.

Chhota Nagpur Plateau - Nicknamed the "storehouse of minerals" in India, stretching across Jharkhand, Chhattisgarh, West Bengal, Odisha, and Bihar. Abundant mineral resources including mica, bauxite, copper, chromite, limestone, iron ore, and coal.

Amarkantak Plateau and Maikal Hills - Known for the presence of bauxite deposits.

Coal

Coal is an amorphous allotropic form of carbon. Coal is a sedimentary deposit composed predominantly of carbon that is readily combustible. The United States has the largest coal reserves in the world.

Significance of Coal in India - First coal mine was opened in 1774 at Raniganj along the western bank of the Damodar river in West Bengal. Coal is the largest contributor to the generation of electricity in India.

Types of Coal

Peat - The first stage of coal formation, with low carbon content (30-45%) and high moisture. Peat is a soft, dark brown coal that is formed from dead and decaying organic matter. It has a low heating capacity.

Lignite - A low-grade brown coal with 60-70% carbon content, used mainly for electricity generation. Tamil Nadu has the largest lignite reserves in India.

Bituminous Coal - The most widely available and used coal in India, with 60-85% carbon content. Jharia is the largest bituminous coal field in Jharkhand.

Anthracite Coal - The highest quality

hard coal, with 86-98% carbon content. Russia, China, Poland, and Ukraine have the largest estimated recoverable reserves of anthracite.

Major Coal Reserves in India

Jharkhand has the largest reserves of coal in India, followed by Odisha and Chhattisgarh. Jharia is the largest coal field in India, followed by Raniganj.

The most important Gondwana coal fields of India are located in the Damodar Valley, in the Jharkhand-Bengal coal belt. This region includes the major coal fields of Raniganj, Jharia, Bokaro, Giridih, and Karanpura.

Other important coal mining centers are Singrauli (Madhya Pradesh), Korba (Chhattisgarh), Talcher and Rampur (Odisha), Chanda-Wardha, Kamptee and Bander (Maharashtra), and Singareni (Telangana).

Coal occurs in rock sequences mainly of two geological ages, namely Gondwana and tertiary deposits.

Tertiary coals are found in Assam, Arunachal Pradesh, Meghalaya, and Nagaland.

Five major coal deposits have been discovered in **Bangladesh** in the Northwest region - Jamalganj, Barapukuria, Khalashpur, Phulbari and Dighipara.

Crude Oil

Crude oil is a naturally occurring, unrefined petroleum product composed of hydrocarbon deposits and other organic materials. products made from crude oil are Gasoline, jet fuel, Diesel, Asphalt etc.

Asphalt is a black or brown petroleum-like material that has a consistency varying from viscous liquid to glassy solid. It is obtained either as a residue from the distillation of petroleum or from natural deposits.

Major Oil Fields in India

Assam - India's first petroleum reserves were discovered in Assam, and the first oil refinery in Asia was set up in Digboi, Assam, in 1901.

Other important oil and gas fields in Assam include Naharkatiya, Moran,

Rudrasagar, Lakwa, Sonari, Amguri, Geleki, and Baghjan.

Bombay High Oil Fields - This is the largest petroleum-producing oilfield in India, discovered by ONGC (Oil and Natural Gas Corporation).

Gujarat - Major oil fields are located in Ankleshwar (largest oilfield in Gujarat), Kalol, Barkol, Katharina, Sanand, and Aliabet.

Rajasthan - Significant oil fields include Mangala, Guda, Saraswati, Kameshwari-I, Raag Oil, and Shakti.

Arunachal Pradesh - Oil and gas fields are found in Manabhaum, Kharsang, and Charai.

The basin and delta regions of the Godawari, Krishna, and Cauvery rivers hold great potential for oil and gas production, both onshore and offshore.

Refineries and Pipelines

The Naharkatia-Nunmati-Barauni Pipeline (1167 km) was the first pipeline constructed in India.

The Hazira-Vijaipur-Jagdishpur (HVJ or HBJ) Gas Pipeline connects Hazira in Maharashtra to Bijapur in Madhya Pradesh and Jagdishpur in Uttar Pradesh.

The Jamnagar-Loni LPG Pipeline (1410 km) is the longest LPG pipeline in India, connecting Jamnagar in Gujarat to Loni in Uttar Pradesh.

Other important pipelines include Salaya-Mathura Pipeline (SMPL) connects Gujarat to Uttar Pradesh; Paradip-Haldia-Barauni Pipeline (PHBPL) links Odisha to West Bengal and Bihar; and Mundra-Panipat Pipeline (MPPL) connects Gujarat to Haryana.

Assam has four major oil refineries: Digboi Refinery, Noonmati refinery in Guwahati, Dhaligoan refinery near Bongaigaon, and Numaligarh Refinery.

Note - India's first lithium refinery is planned to be set up in Gujarat, highlighting the strategic importance of mineral resources.

Climate

Climate refers to the long-term average weather patterns in a large area. It differs from weather, which consists of short-term (minutes to months) atmospheric changes.

Major controls of climate :- Latitude, Altitude, Pressure and wind system, Distance from the sea (continentality), Ocean currents, and Relief features.

Continentality refers to the climate characteristics influenced by a region's distance from the ocean, typically resulting in more extreme temperatures away from coastal areas.

Climatology or climate science is the scientific study of Earth's climate, typically defined as weather conditions averaged over a period of at least 30 years.

Branches of Climatology

Dendroclimatology is concerned with constructing records of past climates and climatic events by analysis of tree growth characteristics, especially growth rings.

Bioclimatology (biometeorology) is the study of the relationships between climate (weather) and living organisms.

Historical climatology is the study of historical changes in climate and their effect on civilization from the emergence of hominids to the present day.

Geo climatology is the study of climate and how it changes over time.

Though the climate of India is **monsoon** type, there are regional variations in climatic conditions within the country.

Regional variations in Temperature – Churu (Rajasthan) may record a temperature of 50°C or more on a June day while the temperature touches 19°C in Tawang (Arunachal Pradesh) on the same day. The temperature may drop down to -45°C in Drass (Ladakh) while Thiruvananthapuram or Chennai may record 20°C or 22°C on the same day.

Regional variations in Precipitation and its Amount – It snows in the Himalayan regions while the rest of the country gets rains. Cherrapunji and Mawsynram in the Khasi Hills (Meghalaya) receive rainfall over 1080 cm in a year while Jaisalmer (Rajasthan) rarely gets more than 9 cm

of rainfall during the same period.

Koeppen's climate classification, developed in 1884, is the most widely used climate classification which categorizes climates into major groups labeled A, B, C, D, E with an additional category H for highlands. These groups are further subdivided based on temperature and precipitation differences, with small letters indicating characteristics such as the presence of a No dry season (f), monsoon climate (m), winter dry season (w), or summer dry season (s).

Koeppen's climate classification

A - Tropical Humid Climates occur between the Tropic of Cancer and the Tropic of Capricorn, characterized by high temperatures, humidity, and significant annual rainfall due to the presence of the Inter Tropical Convergence Zone (ITCZ). They are divided into three types.

Tropical wet climate (Af) - No dry season. It is found near the equator. The major areas are the Amazon Basin in South America, western equatorial Africa and the islands of East Indies. Significant rainfall occurs throughout the year as afternoon thunder showers. Uniformly high temperatures with minimal annual temperature range.

Tropical monsoon climate (Am) - Monsoonal, short dry season. It is found over the Indian sub-continent, North Eastern part of South America and Northern Australia. Heavy rainfall occurs mostly in summer. Winter is dry.

Tropical wet and dry climate (Aw) - Winter dry season, located north and south of Af regions. Borders on dry climates to the west and Cf or Cw to the east. Found in peninsular India, parts of Brazil, Bolivia, Paraguay, Sudan, and Central Africa. Rainfall less than Af and Am, variable. Shorter wet season, longer dry season with severe drought. High temperatures year-round, with greatest diurnal temperature ranges in dry season. Features deciduous forests and tree-shredded grasslands. Also known as "savannah" climate, located just outside the ITCZ near the equator.

B - Dry Climate have very low rainfall unsuitable for plant growth, covering latitudes from 15° to 60° north and south of the equator. At latitudes 15° to 30°, they occur in subtropical high areas

where subsidence and temperature inversions inhibit rainfall.

Dry climates are divided into steppe or semi-arid climate (BS) and desert climate (BW). They are further subdivided as:

Subtropical steppe (BSh) - Low-latitude semi arid or dry.

Subtropical desert (BWh) - Low-latitude arid or dry.

Mid-latitude steppe (BSk) - Mid-latitude semi arid or dry.

Mid-latitude desert (BWk) - Mid-latitude arid or dry.

Subtropical Steppe (BSh) and Subtropical Desert (BWh) Climates:

Both have common precipitation and temperature characteristics. Subtropical steppe receives slightly more rainfall than desert, supporting sparse grasslands. Annual and diurnal temperature ranges are also high.

Coastal desert areas experience frequent fog due to cold currents. Very high maximum summer temperatures; Al Aziziyah, Libya recorded the highest shade temperature of 58°C in 1922.

C - Warm temperate (Mid-latitude) Climates, These climates extend from 30° - 50° of latitude mainly on the eastern and western margins of continents. These climates generally have warm summers with mild winters.

Humid subtropical (Cwa) - dry in winter and hot in summer. It occurs poleward of Tropic of Cancer and Capricorn, mainly in North Indian plains and South China interior plains. The climate is similar to Aw climate except that the temperature in winter is warm.

Humid subtropical (Cfa) - No dry season, warm summer. It lies on the eastern parts of the continent in subtropical latitudes. In this region the air masses are generally unstable and cause rainfall throughout the year.

They occur in the eastern USA, southern and eastern China, southern Japan, northeastern Argentina, coastal south Africa and eastern coast of Australia.

The annual averages of precipitation vary from 75-150 cm. Thunderstorms in summer and frontal precipitation in winter are common. Mean monthly temperature in summer is around 27°C, and in winter it varies from 5°-12° C. The daily range of temperature is small.

Mediterranean (Cs) - Dry hot summer. It occurs around the Mediterranean Sea and west coasts of continents in subtropical latitudes (30°-40°). Locations include Central California, Central Chile, southeastern and southwestern Australia.

These areas come under the influence of subtropical high in summer and westerly wind in winter. Hence, the climate is characterized by hot, dry summers and mild, rainy winters.

Monthly average temperature in summer is around 25° C and in winter below 10°C. The annual precipitation ranges between 35 - 90 cm.

Marine west coast climate (Cfb) - No dry season, warm and cool summer. It is located poleward from the Mediterranean climate on the west coast of the continents. The main areas are: Northwestern Europe, west coast of North America, north of California, southern Chile, southeastern Australia and New Zealand.

Due to marine influence, the temperature is moderate and in winter, it is warmer than for its latitude. The mean temperature in summer months ranges from 15°-20°C and in winter 4°-10°C. The annual and daily ranges of temperature are small. Precipitation occurs throughout the year. Precipitation varies greatly from 50-250 cm.

D - Cold Snow- forest Climates occur in the large continental area in the northern hemisphere between 40°-70° north latitudes in Europe, Asia and North America.

It is divided into two types:

Humid continental (Df) - No dry season, severe winter. It lies between latitude 30° south and 60° north of the equator with an average temperature of 65.5°F and 35°F during summer and winter, respectively. The weather condition in the humid continental climate varies because of its location in the mid-latitudes and polar fronts.

Subarctic (Dw) - Winter dry and very Severe. Cold climate with dry winter occurs mainly over Northeastern Asia. The development of a pronounced winter anticyclone and its weakening in summer sets in monsoon like reversal of wind in this region.

E - Polar Climates exist poleward beyond 70° latitude. Polar climates are those in

which the mean temperature of the warmest month is below 10°C.

This climatic group has been sub-divided into two types – the Tundra Climate is found under the cold climates in which the warmest month has a temperature above freezing point but is less than 10°C; and the Ice-caps climate in which the warmest month has temperature below 0°C.

Tundra (ET) - No true summer. Tundra regions typically get less than 25 centimeters (10 inches) of precipitation annually, which means these areas are also considered deserts. They have long, cold winters with high winds and average temperatures below freezing for six to ten months of the year.

Polar ice cap (EF) - Perennial Ice. Cold snow forest climates occur in the large continental area in the northern hemisphere between 40°- 70° north latitudes in Europe, Asia and North America. Even in summer, the temperature is below freezing point.

This area receives very little precipitation. The snow and ice get accumulated and the mounting pressure causes the deformation of the ice sheets and they break. They move as icebergs that float in the Arctic and Antarctic waters. Antarctica, 79°S, portrays this climate.

H - Highland Climates (Highland with snow cover) In the Highland Climate low temperature, low air pressure, large diurnal ranges of temperature and relatively large amount of precipitation at higher altitudes are common. This type of climate is found in the Alps, the Himalayas, the Tibetan Plateau, the Rockies and the Andes.

Based on temperature, the Earth is divided into three zones :-

Tropical zone or torrid zone is the region of the Earth between the Tropic of Cancer and the Tropic of Capricorn. It experiences the most direct sunlight throughout the year, resulting in consistently warm temperatures.

Temperate Zone is the climate zone between 35 and 50 north and south latitudes (between the subarctic and subtropical climates). These zones generally have wider temperature ranges throughout the year with moderate rainfall throughout the year, with sporadic

dry, mild to hot summer and cool to cold winters.

Polar zone or frigid zone is the region of the Earth near the poles, characterized by extremely cold temperatures and polar ice caps.

Climate of India

The climate of India is described as the 'monsoon' type which is characterized by the seasonal reversal of winds that bring rain to the Indian subcontinent.

There is a decrease in rainfall generally from east to west in the Northern Plains. These variations have given rise to variety in the lives of people – in terms of the food they eat, the clothes they wear and also the kind of houses they live in.

The peninsular region specifically, does not have a well-defined cold or winter season. There is hardly any noticeable seasonal change in the temperature pattern during winters.

In India generally, coastal areas experience less contrasts in temperature conditions. Seasonal contrasts are more in the interiors of the country.

For example, the average maximum temperature in January at Thiruvananthapuram is 31°C, and for June, it is 29.5°C. This is possible due to the moderating influence of the seas in the peninsular region.

Rajasthan has a higher annual range of temperature because it is in the interior and has a continental type of climate, very hot in summer and very cold in winter as it is far away from the oceanic effect.

The difference between the day and night temperatures is likely to be the highest in Thar Desert. Temperature can raise up to 50°C in day and about 15°C in night. While in Andaman and Nicobar Island we can hardly see any temperature difference.

In Northern India, the cold season is characterized by warm days and cold nights. The morning time is pleasant, while the result of the decrease in the temperatures leads to frost in some areas.

Typically coastal places will have a moderate climate throughout the year as the sea surface absorbs heat slowly and also loses the heat slowly (specific heat of water is more) which influences the adjacent land temperature.

Dras is the coldest place in India and often touted as second to the coldest place inhabited on Earth. Dras is a convenient staging point for treks to Amarnath, Suru valley and Sialkot. On an average during the freezing winters, the temperature can drop down to a bone-chilling - 45°C.

Seasons in India : Due to large latitudinal dimensions from north to south, India experiences six seasons :

Winter Season (Shishir or Shita Ritu) - January and February. As per the Hindu calendar, this season occurs in the months of Magha and Phalgun.

Summer Season - May and June. Hindu calendar month - Jyeshtha and Aashaadha.

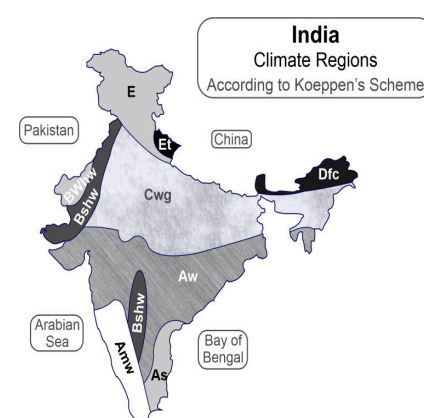
Rainy Season - July and August. Hindu calendar month - Shravana and Bhadrapada (Sawan and Bhado).

Spring Season - March and April. Hindu calendar month- Chaitra and Baisakh

Autumn - September, October and November. Hindu calendar, this season comes in Ashwin and Kartik months.

Pre-winter (Hemant Ritu) - November and December. Hindu calendar month - Agrahayana (Margashira) and Pausha (Agahan and Poos).

Koeppen's India Climate classification



Amw (Monsoon with Short Dry Winter Season): Found in western coastal regions of India and south of Goa, Average annual rainfall around 300 cm during southwest monsoon.

As (Monsoon Climate): Dry summers, rainy winters. Annual rainfall 75-100 cm during retreating monsoon. Geographically located in the Coromandel coast of Tamil Nadu.

Aw (Tropical Savanna Type): Rainfall during southwest monsoon, Average rainfall around 75 cm, Dry winters.

Bshw (Semi-Arid Steppe Type): Found in northwestern Gujarat, parts of western Rajasthan, and Punjab, average rainfall is 12-25 cm, Vegetation: Arid-steppe type, mainly fodder and grasses for cattle.

Bwhw (Hot Desert Type): Found in extreme western Rajasthan, Predominance of deserts with almost absent natural vegetation, average rainfall is less than 12 cm.

Cwg (Monsoon Type with Dry Winters): Found in most parts of Ganga Plain, Eastern Rajasthan, Northern Madhya Pradesh, and most of northeast India. Summer temperatures around 40°C, winter temperatures around 27°C. Rainfall mainly during summer with onset of southwest monsoon, Annual rainfall: 100-120 cm.

Dfc (Cold, Humid Winter with Short Summer): Found in Arunachal Pradesh, Sikkim, and parts of Assam. Winters are cold, humid, and longer. Annual rainfall: More than 200 cm. Average winter temperature: Around 10°C.

Et (Tundra Type): Found in mountainous regions of Uttarakhand, Temperature decreases with altitude, Average temperature: 0°C to 10°C.

E (Polar Type): Found in Uttarakhand, Himachal Pradesh, and Jammu & Kashmir. Precipitation mainly as snow due to extremely low temperatures. Warmest month temperatures range from 0°C to 10°C.

Tropical Monsoon Rainforest Climate

Heavy rainfall and high humidity characterized by heavy monsoon downpours. Summer temperatures are reduced by rain; foggy nights and mornings in winters.

"Tropical Monsoon Rainforest Climate, Assam (summer max. at 35–39°C and winter min. at 5–8 °C) and experiences heavy rainfall and high humidity.

Intertropical Convergence Zone (Doldrums) is a band of low pressure around the Earth that generally lies near the equator. It is the location where northeast winds in the Northern Hemisphere converge with the southeast winds from the Southern Hemisphere.

The duration of day and night is almost equal at the equatorial region.

The main reasons for the excessive cold in north India during the cold weather season are : being far away from the moderating influence of sea, some states (Punjab, Haryana, and Rajasthan) experience continental climate. The snowfall in the nearby Himalayan ranges creates a cold wave situation. Around February, the cold winds coming from the Caspian Sea and Turkmenistan bring a cold wave along with frost and fog over the northwestern parts of India.

El Nino and La Nina

El-Nino is a complex weather system that occurs every 3-7 years, bringing droughts, floods, and extremes worldwide. It increases sea-surface temperatures, weakens trade winds.

Warm equatorial current replaces cold Peruvian current, raising coastal temperatures by 10°C which distorts atmospheric circulation, reduces plankton, fish populations. It is named "Child Christ" for appearing around Christmas.

El nino is used in India for long-range monsoon forecasts. 1990-91 El Niño delayed southwest monsoon onset by 5-12 days. Winter El Niño brings warm conditions in the Indian subcontinent, summer leads to dryness and deficient monsoons.

La Nina Means 'little girl' in Spanish, also called El Viejo or 'cold event'. During La Nina, Eastern Pacific water temperature gets colder than normal. Due to this, Strong high pressure over eastern equatorial Pacific, low pressure in Western Pacific and off Asia. It Results in droughts in Ecuador and Peru, heavy floods in Australia, high temperatures in the Indian Ocean, Western Pacific, and off coast of Somalia.

La Nina years experience better than normal monsoons in India. La Nina events occur every **3 to 5 years** or so, but on occasion can occur over successive years.

El Nino and La Nina are opposite phases of the **El Nino-Southern Oscillation (ENSO) cycle**, which involves temperature changes in the eastern and central tropical Pacific Ocean, as well as alterations in wind patterns, sea level pressure, and tropical rainfall across the Pacific Basin.

El Nino: Warm phase of ENSO. La Nina:

Cold phase of ENSO. Both phases have significant impacts on global weather and climate. El Nino is often followed by La Niña, especially if El Nino is strong.

Mountains

A mountain is an elevated portion of the Earth's crust, generally with steep sides that show significant exposed bedrock. Generally, a steep hill with an elevation of more than 600 metres is classified as a mountain.

Mountaineers refer to altitudes above about **8000m** as the 'death zone'.

Major types of Mountains

Fold Mountains - Mountain ranges mainly consisting of uplifted folded sedimentary rocks are called Fold Mountains.

Old Fold Mountains - India's Aravali, The Appalachians in North America and the Ural Mountains in Russia are some examples.

Young Fold mountains - The Himalayas in Asia, the Alps in Europe, the Rockies in North America, and the Andes in South America are some examples. Young fold mountains in the northwest, north and northeast bind India.

Fold mountains are defined by complex rock formations known as folds.

Major Types of Folds

Anticline fold - Fold usually inclined upwards. Often the two foots are inclined in opposite directions from each other and in this the older strata are found in the middle.

Synclines fold - Folds are the reverse of anticlines in all details and the strata are downarched, that is, these become convex downwards.

Monocline - If the horizontal bed becomes inclined suddenly and becomes horizontal again after this inclined part.

Box fold - A rectangular, conjugate fold in which the inter-limb angles approximate to right angles.

Recumbent folds - Extreme types of overturned folds in which the axial plane acquires an almost horizontal attitude.

Isoclinal fold - When the two limbs of a fold are essentially parallel to each other and thus approximately parallel to the

axial plane.

Asymmetrical fold - Fold is one in which the axial plane is inclined.

Block mountains are formed due to the upward movement of the middle block between two normal faults.

Examples - The Sierra Nevada in North America, Black Forest Mountains in Germany, Vindhyas in India.

Volcanic Mountains - The mountains formed by the accumulation of volcanic materials are called Volcanic Mountains or 'Mountains of accumulation'.

Two types of volcanic mountains are shield volcanoes and Composite cones (Stratovolcanoes) volcanoes.

Composite cones is a cone-shaped volcano built from several layers of lava, pumice, ash, & tephra. Mount Fuji in Japan and Mount Shasta in California are composite volcanoes.

Shield volcanoes have a gentle slope cone because of the low viscosity of the emitted material. Mauna Kea and Mauna Loa are shield volcanoes.

Residual Mountains - It formed by the effects of weathering (as part of exogenic processes).

Examples - Hills like Nilgiri, Palkonda, Parasnath and Rajmahal, Aravalli, the Vindhya, the Satpura.

Dome mountains - These are formed when a large amount of magma floats up to the crust from beneath and pushes the rocks on the surface, it forms round swelling in the crust.

Examples - The Black Hills of South Dakota and the Adirondack Mountains of New York.

Major Mountain Ranges of The World

Rocky Mountains - The Rocky Mountains extend for a distance of about 4,800 km. They are a major mountain range and the largest mountain system in North America. The range's highest peak is Mount Elbert located in Colorado at 4,401 m above sea level.

Andes Mountains - The Andes is the longest (7000 km) mountain range in the world. Llamas are used for transportation. It is spread north to south across Venezuela, Colombia, Ecuador, Peru, Bolivia, Chile, and

Argentina. Mt. Aconcagua (6960 m) is the highest peak outside Asia. It lies in the Southern Andes. It is located on the Chilean border in west-central Argentina.

The Alaska Range is a 950 km long arc of mountains that stretches from the Alaska-Canada border all the way to the Alaska Peninsula in North America.

The highest peak in the Alaska Range is Denali (Mount McKinley), which stands at 6,190 meters and is the highest mountain in North America.

The Alps is 1200 km long, the highest and most extensive mountain range system that lies across eight Alpine countries (from west to east): Monaco, France, Switzerland, Italy, Liechtenstein, Germany, Austria and Slovenia. It is the natural boundary between France and Italy.

The highest peaks of the Western Alps and Eastern Alps are Mont Blanc (4,809 m), and Piz Bernina (4,049 m) respectively.

The Caucasus Mountains are located in the middle of the Eurasian plate between Europe and Asia. This mountain range lies in between Caspian Sea and Black Sea. Its length is about 1200 km.

Mount Elbrus (5642 m) is the highest peak of Caucasus Mountain in southwest Russia, and the highest point in Europe.

Ural Mountain (2500 km) stretches from north to south through western Russia, from the coast of the Atlantic Ocean to the Ural River and north western Kazakhstan. Ural mountains separate the continents of Europe and Asia.

Mount Narodnaya (1895 m) is the highest peak of the Urals in Russia. The highest mountain of the Polar Urals is Mount Payer (1472 m).

Arakan Mountains (Rakhine Yoma) - It is an extension of the Himalayas located in western Myanmar with a length of 950 km. This range is mainly found between the Arakan coast and the Irrawaddy River valley. Nat Ma Taung (Mount Victoria) is the highest peak.

Great Dividing Range - It is a series of plateaus and low mountain ranges in eastern Australia with a length of 3500 km. Its highest peak is Mount Kosciuszko (2228 m), also the highest peak in mainland Australia.

Hindukush Mountains - It is a vast mountain range that stretches between central Afghanistan and Northern Pakistan with a length of 800km. Its highest peak is Koyo Zom, 6,872 m.

Kunlun Mountains - They constitute one of the longest mountain chains in Asia, extending for more than 3,000 kilometers. The Kunlun Mountains stand parallel to the Tian Shan Mountains in the north and the Himalayan mountains in the south.

Transantarctic Mountain - It is an enormous mountain range in Antarctica that divides the continent into East Antarctica and West Antarctica. The mountain range stretches between the Ross Sea and the Weddell Sea. Its Highest Peak is Kirkpatrick (4,528 m).

Atlas Mountains - The Atlas Mountains Range stretches through Morocco, Algeria and Tunisia. It separates the Mediterranean and Atlantic coastlines from the Sahara Desert and is inhabited by Berber populations. Toubkal (4,167 m) is the highest peak of Atlas mountain in Morocco.

Major Hills Ranges in India Himalaya

Himalaya is a mountain range in Asia, separating the plains of the Indian subcontinent from the Tibetan Plateau. The Himalayan Mountain range was formed by the collision of Indian Plate and Eurasian Plate.

The Himalayas prevent the cold winds from Central Asia from entering the subcontinent.

Himalayas cover approximately 1,500 miles (2,400 km) and pass through the nations of India, Pakistan, Afghanistan, China, Bhutan and Nepal.

A beekeeper from New Zealand, Edmund Hillary, and the Nepalese Sherpa Tenzing Norgay became the first people to stand on the summit of the world's highest peak Mount Everest, known as 'Sagarmatha' is the Highest peak of the Himalaya.

Dhaulagiri is also known as the 'White Mountain'. It is the seventh highest mountain in the world.

Division of the Himalayas

The Himalayas consist of three parallel ranges, the northernmost range Greater Himalayas (Inner Himalayas or Himadri), the Lesser Himalayas (Middle Himalayas or Himachal), and the Shivalik hills (Outer Himalayas; Southernmost range).

Greater Himalayas (Inner Himalayas) -

The Great Himalayan range is also known as the central axial range. It is 2500 km from east to west. The mountains often rise to an average height of 6000 metres. The world's highest peaks are located in this range.

Nanda Devi (Highest peak of Uttarakhand) is located near the river Dhauliganga. Kamet peak lies in the Chamoli district of Uttarakhand.

Lesser Himalayas (Middle Himalayas) -

The range stretches between the Great Himalayas in the northeast and the Shivalik range in the southeast. Its expansions are in southeast Pakistan through Jammu and Kashmir, Himachal Pradesh, Western Uttar Pradesh, and Nepal.

The Pir Panjal, Dhauladhar, and Mahabharat ranges are part of Lesser Himalayas.

The most well-known hill towns in the Middle Himalayas are Shimla, Mussoorie, Almora, Ranikhet, Darjeeling and Nainital.

These mountains range from 3700 to 4500 m in height and have an average breadth of 50 km.

Valleys: The valleys in the Middle Himalayas, such as Kangra, Kashmir, and Kullu. Both Kangra and kullu valleys are in the Middle Himalayas and are in the state of Himachal Pradesh.

Pir Panjal Range is the largest range of the Lesser Himalayas that runs from east-southeast to west-northwest over the Indian states of Himachal Pradesh, Jammu and Kashmir, and Pakistan occupied Kashmir.

Shivalik hills (Outer Himalayas) - These ranges are the Himalayas' furthest-southern hills. Flat-bottomed valleys divide it from the Lesser Himalayas. The old name for this range was "Upgiri." It is the youngest and outermost range of the Himalayas.

The Shivaliks' width is 10-50 km, and its height rarely exceeds 1300 m. The

southern Shivalik range slopes in Punjab and Himachal Pradesh are unfrosted.

The expansion of Shivaliks from the Indus Gorge in the northwest to the Brahmaputra in Assam, the Outer Himalayas form a roughly continuous chain.

Longitudinal division of the Himalayas

Punjab/ Himachal Himalayas
The Kumaun Himalayas
The Central/ Nepal Himalayas
The Assam/ Eastern Himalayas

Punjab Himalayas - lying between the Indus and the Satluj rivers. It is also called Kashmir Himalaya. Karewa formations are found in Kashmir Himalayas.

All the major rivers of the Indus river system flow through Punjab Himalayas.

Mountains range (north to south) - Karakoram, Ladakh, Zaskar, Pir Panjal, Siwalik.

Kumaon Himalayas - The part of the Himalayas lying between Satluj and Kali rivers is known as Kumaon Himalayas (in Uttarakhand). The Lesser Himalaya in Kumaon Himalaya is represented by the Mussoorie and Nag Tibba ranges. It is also known as Garhwal Himalaya. Nanda Devi is part of the Garhwal Himalayas.

Peak: Peaks in the Kumaun Himalayas include Nandadevi, Kedarnath, Badrinath, Kamet.

Ranges - Dhauladhar, Mussoorie, Nag Tibba.

Nepal Himalayas - The Central (Middle) Himalayas is situated between the Tista River and the Kali River. It is also known as Central Himalaya.

Peak - Kanchenjunga, Makalu, Mt. Everest, Annapurna, Dhaulagiri, and Gosainthan.

Assam Himalayas - These Himalayan regions are divided on the basis of rivers and the mountain area which lies between the Tista river and Dihang (Brahmaputra) rivers is known as Assam Himalayas. It is known as the Eastern Himalayas.

Expansion: The Eastern Himalayas mostly cover areas of Bhutan and Arunachal Pradesh. There is a deficiency of glaciers and ice caps in this region as it is rather

close to the densely forested tropics. In the state of Arunachal Pradesh, the general direction of the Himalayas is from southwest to northeast.

Other Major Parts of Himalayas

Trans-Himalayan ranges - These ranges extend north of greater Himalaya and Zaskar range is parallel to it. North of Zaskar range lies Ladakh range. The Indus river flows between Zaskar and Ladakh range. The Nanga Parbat (8126 m) is an important range which is in the Zaskar Range.

Karakoram - The Karakoram Range, also known as Krishnagiri, is situated in the northernmost range of the Trans-Himalayan ranges. It is home to the second-highest peak of the world, K2 (also called Godwin Austen), which stands at 8,611 meters.

The Karakoram Range is known as the 'backbone of high Asia' and forms India's frontiers with Afghanistan and China, acting as a watershed between India and Turkestan.

The Purvanchal hills - The Purvanchal Range, or Eastern Mountains, is a sub-mountain range of the Himalayas in northeast India. It lies south of the Brahmaputra valley.

It comprises the Patkai hills, the Naga hills, the Manipur hills and the Mizo hills.

They extend in Arunachal Pradesh, Nagaland, Manipur, Mizoram, Tripura, and eastern Assam states.

Mishmi Hills are the highest range of Purvanchal hills situated in the North eastern part of Arunachal Pradesh.

Phawngpui is located in the Lushai Hills in the Purvanchal Range in the state of Mizoram.

Important Peaks of Himalaya

Major peaks of Himalaya in India:

Kanchenjunga - 8598 m
Nanga Parbat - 8126 m
Nanda Devi - 7817 m
Namcha Barwa - 7782 m
Kamet - 7756 m

Major peaks of Himalaya in Nepal

Mt. Everest - 8849 m
Makalu - 8481m
Dhaulagiri - 8172 m
Annapurna - 8078 m

Gurla Mandhata - 7728 m

Mount Kailash is located in the Himalayan range far west of Tibetan Plateau.

Saser Kangri is the easternmost subrange of the Karakoram range located within Ladakh.

Saltoro Kangri (Peak 36) is the highest peak of the Saltoro Mountains. It is located in the Siachen glacier at an altitude of 7742 m. K12 (7,428 m) is the second highest peak in the Saltoro Mountains.

The Kun Peak is a part of Nun Kun mountain in Ladakh.

Reo Purgyll belongs to the Western Himalayan Range.

Karoh Peak is part of Morni Hills in the Shivalik range of greater Himalayas. It is the highest point in the state of Haryana.

Hardeol is situated at the northern end of the Milam valley, in the Pithoragarh district of Uttarakhand.

Hathi Parbat, (also known as Elephant Peak) is a mountain in the Garhwal Himalayas in India.

Adi Kailash or **Jonglingkong** Peak is located in the Himalayan mountain range in the Pithoragarh district of Uttarakhand.

Kamet is the second highest mountain in the Garhwal region of Uttarakhand (Chamoli district).

Trishul is a group of three Himalayan mountain peaks located in Uttarakhand.

Kirat Chuli or **Tent Peak** and Kabru are mountains in the Himalayas. They lie on the border between Nepal and India.

Cho Oyu mountains are present in the Himalayas on the Nepalese–Tibetan (Chinese) border.

Annapurna (8078 m) is a mountain peak present in the Himalayas in north-central Nepal. Manaslu is located in the west-central part of Nepal.

Makalu is the fifth highest mountain in the world with an elevation of 8,481 m. It is located in the Mahalangur Himalayas in Nepal.

Mount Kanchenjunga (8,598 m) is the third highest mountain in the world and the highest mountain peak of the Himalayas in India. It lies between

Darjeeling and Sikkim Himalayas.

Mount Jopuno is situated in the western part of Sikkim. It lies under the Kanchenjunga National Park Sikkim.

Mount Siniolchu (6,888 m) is one of the tallest mountains of the Indian state of Sikkim.

Jomolhari or Chomolhari, sometimes known as "the bride of Kangchenjunga", is a mountain in the Himalayas, on the border of Bhutan and Tibet.

Kangto Peak (7090 m) and Gorichen Peak (6858 m) are among the higher peaks of Arunachal Pradesh.

Sandakphu is situated in the Darjeeling district in West Bengal. It lies in the Nepal-West Bengal border.

Toba Kakar Mountains are a southern branch of the Himalayas in the Balochistan region of Pakistan.

Region in Himalayas

Doons - The valleys lying between the lesser Himalayas and Shivaliks are called Doons.

Dehradun is the largest of all the doons of the shivaliks mountain range, Some other doons are Patli Doon, Kotli Doon, and Pinjore Doon.

Duars - These are the alluvial floodplains in eastern-northeastern India that lie south of the outer foothills of the Himalayas and north of the Brahmaputra River basin. It is divided by the Sankosh River into the Western and Eastern Duars.

Terai - It is a region of northern India and southern Nepal running parallel to the lower ranges of the Himalayas. Dudhwa national park is an example of the Terai region.

Bugyal or Meadow: It is a high altitude (between 3,300 metres and 4,000 metres) grassland which serves as a grazing field for livestock.

Some of the notable bugyals are - Auli near Joshimath, Garsi, Kwanri, Bedni, Panwali Kantha and Kush Kalyan, Dayara, Bagji Bugyal and Munsiyari. All are located in Uttarakhand.

Aravallis Range

They are aligned in a north-east to south-west direction. They run for about 800 km between Delhi and Gujarat.

It touches the states of Haryana, Delhi, Rajasthan and Gujarat. Mountains in the Aravalli range are divided into two main ranges – the Sambhar Sirohi Range and the Sambhar Khetri Range in Rajasthan.

Guru Shikhar (1722 m) is the highest peak in the Aravalli Range in Rajasthan.

Other peaks of Rajasthan: Ser Peak (Sirohi), Delwara (Sirohi), Jarga (Udaipur), Achalgarh (Sirohi), Kumbhalgarh (Rajsamand), Raghunathgarh (Sikar), Hrishikesh (Sirohi), Kamalnath (Udaipur), Khoh (Jaipur), Taragarh (Ajmer), Bhairach (Alwar), Babai (Jhunjhunu) and Baraith (Jaipur).

Eastern Ghats

The Eastern Ghats run about 1131 km from northern Odisha through Andhra Pradesh to Tamil Nadu in the south passing some parts of Karnataka.

Mahendragiri (1,501 m) is the highest peak in the Eastern Ghats, located in Odisha.

Jindhagada Peak, Deomali (1672m) and Solaikaradu peak are other peaks of Eastern ghats.

Hills in Eastern Ghats

The Nallamala Hills are part of the Eastern Ghats, forming the eastern boundary between Andhra Pradesh and Telangana. The highest peak is Bhairani Konda. These hills lie between the Pennar and Krishna rivers.

Garhjat Hills - It is a mountain range that stretches into Odisha from the Utkal Plains in the Chotanagpur region of Jharkhand and the Chhattisgarh Plains. Malayagiri is the highest peak of Garhjat Hills.

Shevaroy Hills - It is located in the south-east area in Tamil Nadu.

Javadi Hills - (Javadhu) is an extension of the Eastern Ghats, distributed in Tamil Nadu. Meltapattu is the highest peak of the Javadi hills. Bluish grey granite is found on these hills.

Seshachalam Hills are hilly ranges part of the Eastern Ghats in southern Andhra Pradesh.

Palkonda Hills are a range of hills that form a part of the Eastern Ghats in Andhra Pradesh.

Nagari Hills are located on the riverbank of Kusasthali, Andhra Pradesh.

Ananthagiri Hills is located in Telangana.

Western Ghats

It is also known as Sahyadri Hills and Sahya Parvatham in Kerala. The length of western ghats is around 1600 km. It covers the area of Tamil Nadu, Kerala, Karnataka, Goa, Maharashtra.

The central part of the hills is called Kanara and the southern part is called Malabar region or the Malabar Coast. It is a block mountain. The height of the Western Ghats progressively decreases from north to south.

Anamudi, (2,695 meters), known as the Everest of South India, lies at the border of Ernakulam and Idukki districts in Kerala, where the Cardamom, Anaimalai, and Palani hills meet. It's the highest peak in the Western Ghats and South India.

Hills in Western Ghats

Nilgiri Mountains form part of the Western Ghats in northwestern Tamil Nadu, Southern Karnataka, and eastern Kerala in India. The Western Ghats and Eastern Ghats meet at the Nilgiri Hills. **Doddabetta** is the highest mountain in the Nilgiri Mountains at 2,637 m.

The **Cardamom** Hills or Yela Mala are mountain ranges of southern India and part of the southern Western Ghats located in southeast Kerala and southwest Tamil Nadu.

Anaimalai Hills or Anamala are a range of mountains that form the southern border of the Western Ghats, through Tamil Nadu and Kerala. It is also known as the Elephant Hills. The **Palghat** gap separates Annamalai hills from Nilgiris.

The **Palani** Hills are a mountain range in the southern Indian states of Kerala and Tamil Nadu. **Vandaravu** peak is the highest peak in the Palani hills.

Mullayanagiri belongs to the Sahyadri Hills section of the Western Ghats in the state of Karnataka.

Satpura Range

It is a range of hills in central India. Its length is about 900 km. The seven mountain ranges or folds of Satpura - Maikal, Mahadeo Hills, Kalibhit, Asirgarh,

Bijargarh, Barwani and Arwani.

The range rises in eastern Gujarat running east through the border of Maharashtra and Madhya Pradesh and ends in Chhattisgarh.

It runs in an east-west direction south of the Vindhyas and in between the Narmada and the Tapi, roughly parallel to these rivers.

Mahadeo Hills are located in the northern plains of the Satpura range in Madhya Pradesh. Mount Dhoopgarh or Dhupgarh (1350 m) is the highest peak on Mahadeo Hills in Satpura Range, Madhya Pradesh. Amarkantak is another important peak of Satpura.

Vindhya Range

The Vindhyan range is bounded by the Satpura range on the south and the Aravallis on the northwest. At its eastern end, this range meets the hills of Chotanagpur plateau.

This range covers the area of Madhya Pradesh, Chhattisgarh, Gujarat, Southern parts of Uttar Pradesh, and Bihar.

These are non-tectonic mountains, they were formed not because of plate collision but because of the downward faulting of the Narmada Rift Valley (NRV) to their south. Geologically, they are younger than Aravallis and Satpura hills.

Other Famous Hills, Peaks and Range

Shatrunjaya Hills are located in Gujarat and one of the holiest of places for the 'Svetambara' Jains.

Other hills in Gujarat - Kalo Dungar (Black Hills, Highest point of Kutch), Pavagadh Hill, Wilson Hills, Saputara Hills, Girnar Mountain Range, Gabbar.

Karbi Anglong hills is located in Karbi Anglong plateau and is known as the oldest landform in Assam.

Nilachal Hills is situated in Guwahati, Assam. The famous Kamakhya temple (Goddess Durga) is situated at the top of Hill.

Bamuni Hills located in Tezpur of Sonitpur district in Assam.

The Borail Range is the highest Mountain Ranges in Assam. The Thumjang is the peak of the Barail range.

Abor Hills is in the region of Arunachal

Pradesh. The hills are bordered by the Mishmi Hills and Miri Hills and are drained by the Dibang River.

Dafla Hills is located on the border of Eastern Arunachal and Assam which is occupied by the Dafla tribe.

Patkai Bum are a series of mountains in the eastern part of India and Upper Burma region of Myanmar.

The Lushai Hills (or Mizo Hills) are a mountain range in Mizoram and Manipur. The range is part of the Patkai mountain range in Mizoram. Its highest point is Phawngpui, also known as 'Blue Mountain'.

Mount Tempu (Mount Iso) - This peak lies in the Naga Hills section of the Purvanchal Range in Manipur. It is the highest peak (2994 m) in Manipur.

Dongkyu range, is a mountain range in the lower Himalayas that forms the eastern border of Sikkim.

Gyemo Chen is the mountain that sits on the border between Sikkim and Bhutan.

Mount Tiya is a scenic hill located in Wokha, Nagaland. Mountains in Nagaland - Japfu Peak, Changkikong Range, Mount Saramati, Veda Peak, Ongpangkong Range.

Mount Saramati (3826 m) belongs to the Purvanchal range. It is the highest mountain peak in Nagaland.

Jamrai Tlang mountains are found in Tripura. They have the highest peak, Betling Sib (3,280 feet).

Saddle Peak is the highest point in the Andaman Nicobar Islands.

Other peaks in Andaman and Nicobar Islands are Saddle peak (North Andaman), Mount Diavolo (Middle Andaman), Mount Koyob (South Andaman) and Mount Thuiller (Great Nicobar).

Kalsubai (1646 m) peak is also known as the Everest of Maharashtra. It is the highest peak in Maharashtra. It is situated in the Sahyadri Hills.

Rajaji Range of the Shivalik Hills is located in the state of Uttar Pradesh.

Sonsogor (Sonsogodd or Darsingha) is the highest peak of Goa. It is a part of the Western Ghats.

Kodachadri is a mountain peak in the

Western Ghats that is located in the Shimoga district of Karnataka.

Kaimur Range is the eastern portion of the Vindhya Range, Madhya Pradesh.

Pachmarhi is a hill station in Narmadapuram, Madhya Pradesh.

Someshwar Fort, highest peak of Bihar, is situated in the Shivalik Hills.

Parasnath is situated in the Chota Nagpur plateau. This is the highest peak of Jharkhand.

Natarhat Mountains, also called 'Queen of Chotanagpur', located in Jharkhand.

Gaurlata is located in Dantewada district in Chhattisgarh. It is situated in the Bailadila Range in the Deccan Plateau.

Doli Gutta is located in the Deccan Plateau in the state of Telangana.

Perumal Malai Peak is a high peak at Kodaikanal in the Palani hills that are a part of the greater Western Ghat mountain range in Tamil Nadu.

Sri Pada mountain or Adam's Peak is an ancient pilgrimage site in Sri Lanka, which has attracted thousands of pilgrims from all faiths.

Pidurutalagala or Mount Pedro is an ultra prominent peak and the tallest mountain in Sri Lanka.

Tiger Hill (2,590 m) is a mountain located in Darjeeling, in the Indian State of West Bengal. It has a panoramic view of Mount Everest and Mount Kanchenjunga together.

Gangkhar Puensum (7570 m) is the highest mountain in Bhutan and the highest unclimbed mountain in the world.

Loi Leng is the highest mountain of the Shan Hills. Phangran, Phonyin Razi, Hikaru Bum, Madoi razi are situated in Myanmar.

Puncak Jaya (height 4884 m) is situated in West Papua, the western part of the island of New Guinea which belongs to Indonesia.

The **Chin Hills** are a range of mountains in Chin State of northwestern Burma (Myanmar).

Seven Summits

The Seven Summits are the highest mountains on each of the seven traditional continents. On 30 April 1985, Richard Bass became the first climber to reach the summit of all seven.

List of Seven Summit based on Continents

Mount Everest (8849 m) - Asia
Aconcagua (6960 m) - South America
Denali (6190 m) - North America
Mount Kilimanjaro (5,895 m) - Africa
Mount Elbrus (5642 m) - Europe
Mount Vinson (4892 m) - Antarctica
Mount Wilhelm (4509 m) - Oceania

Monsoon

The term "monsoon" is derived from the Arabic word "mawsim," meaning season.

Monsoons are seasonal winds that reverse direction with the change in seasons, causing heavy rainfall in summer and dry spells in winter. They blow from sea to land in summer and from land to sea in winter, forming a dual system of seasonal winds.

Monsoon in India : India experiences southwest monsoon winds in summer and northeast monsoons in winter. Southwest monsoons result from a low-pressure system over the Tibetan Plateau, while northeast monsoons are due to high-pressure cells over Siberian and Tibetan plateaus.

The intensity and position of the high-pressure area east of Madagascar impact the Indian Monsoon, with the Himalayas playing a crucial role in monsoonal rainfall.

Types of season : Meteorologists classify India into four seasons: Southwest Monsoon (Advancing Monsoon), Cold Weather, Hot Weather, and Retreating Southwest Monsoon.

Advancing monsoon (rainy season): The Southwest Monsoon, occurring from June to September, brings significant changes and the majority of rainfall to Indian weather conditions.

Factors influencing the onset of South - West monsoons : Intense low-pressure formation over the Tibetan Plateau, The permanent high-pressure cell in the South of the Indian Ocean, Subtropical jet

stream, African Easterly jet (Tropical easterly jet) and Inter - Tropical Convergence Zone (ITCZ).

Arrival in India : The southwest monsoon arrives in Kerala by early June, then swiftly moves to Mumbai and Kolkata. As the ITCZ shifts northward, it attracts southeasterly trade winds, which carry moisture from warm oceans into the Indian peninsula. This results in heavy rainfall, particularly in Kerala's Western Ghats and northeast India. By July, the ITCZ settles in the Indo-Gangetic Plain (20°-25° N latitude), forming the Monsoon Trough, and promotes the formation of a thermal low over Northern and North-West India.

Branches

The south - west monsoon winds arrive in two branches called the Bay of Bengal branch and the Arabian Sea branch.

Arabian Sea branch - These monsoon winds originate over the Arabian Sea. and first hit the Western Ghats of the coastal state of Kerala, thus making this area the first state in India to receive rain from the South - west Monsoon.

Bay of Bengal branch - The Arakan Hills along the coast of Myanmar deflect a big portion of this branch towards the Indian subcontinent. The monsoon, therefore, enters West Bengal and Bangladesh from south and southeast instead of from the south-westerly direction.

The Bay of Bengal Monsoon Branch and Arabian Sea Branch merge at the North Western Part of Ganga Plains.

Tamil Nadu gets less southwest monsoon rain due to its coast being parallel to the Bay of Bengal branch and in the rain shadow region of the Arabian Sea branch. The Western Ghats block monsoon winds, resulting in Coimbatore plateau receiving 50 cm and Nilgiri hills 50-100 cm of rain. Tamil Nadu receives rainfall from October - November during Monsoon Retreat or Northeast monsoon.

Monsoon burst is characterized by a sudden increase in rainfall at the onset of the monsoon, lasting for a few days. This is followed by pre-monsoon showers. The Easterly Jet Stream, known as the Somali Jet, sets in around 15°N latitude after the withdrawal of the western jet stream, and it triggers the monsoon burst in India.

Retreating/Post Monsoons (The Transition Season) : During October-November, as the sun moves southward, the monsoon trough weakens over the northern plains, replaced by a high-pressure system. Southwest monsoon winds gradually weaken and withdraw from the Northern Plains by the beginning of October. While crossing over the Bay of Bengal, these Northeast monsoon winds pick up moisture and cause torrential rainfall over the Tamil Nadu coast, southern Andhra Pradesh, southeast Karnataka and southeast Kerala. Here, October and November are the rainiest months of the year.

The Retreating south - west monsoon season is marked by clear skies and rise in temperature. Days are hot, while nights are cool and pleasant due to moist land. High temperature and humidity create oppressive daytime weather, known as 'October heat'. In the latter half of October, temperatures drop rapidly in northern India.

Due to the shift of low pressure area over the Indian ocean, cyclonic depressions occur over the Andaman sea. These cyclones generally cross the eastern coasts of India cause heavy and widespread rain. These tropical cyclones are often very destructive.

Break in the monsoon - When rain falls to occur for one or more weeks during the south west monsoon period.

Western disturbances - They are storms that originate in the Caspian or Mediterranean Sea, bringing non-monsoonal rainfall to northwest India. Labeled as extra-tropical storms, they're areas of low pressure that bring sudden showers, snow, and fog to northwest India, moving from west to east.

Some factors impacting India's climate: Himalayan mountains, Western Ghats, Indian Ocean.

Role of Himalayas in Monsoon rainfall

The Great Himalaya Range obstructs the passage of cold continental air from the north into India in winter and also forces the southwesterly monsoon(rain-bearing) winds to give up most of their moisture before crossing the range northward. The result is heavy precipitation (both rain and snow) on the Indian side but arid conditions in Tibet.

Leh experiences the lowest amount of rainfall during the monsoon season.

World's strongest monsoons - This region stretches from the South China Sea into the Indian Ocean and includes Asia and the northern end of Australia. From June until September, summer monsoon rains occur in South Asian countries such as Vietnam, Thailand, Cambodia, Bangladesh, Laos, India, and Pakistan.

Clouds

Clouds are formed due to the accumulation of tiny water droplets and ice crystals in the Earth's atmosphere.

Nephology is the science of clouds, which is undertaken in the cloud physics branch of meteorology.

Clouds are categorized into various types based on their altitude, stretch, density, and transparency.

According to their formation at different heights, clouds are grouped into four main categories.

High Clouds (16,000 - 43,000 ft): Cirrus, Cirrostratus, Cirrocumulus.

Middle clouds (7,000 - 23,000 ft) - Altostratus, Altocumulus.

Low clouds (surface - 7,000 ft) - Stratocumulus, Nimbostratus, Stratus.

Clouds with extensive vertical growth (surface - 43,000 ft) - Cumulus, Cumulonimbus.

Types of Clouds :-

Cirrus - Made of ice crystals, look like long, thin, fibrous, feathery streamers. Cirrus clouds don't produce precipitation. Height of these clouds ranges from 8000 - 12000 m.

Cirrostratus - High, very thin, uniform layer of ice crystals, often accompanied by a halo phenomenon and they also don't produce precipitation.

Cirrocumulus - Small rounded puffs shaped clouds in long rows, composed of ice crystals or supercooled water droplets.

Altostratus - Mid-level, gray or blue-gray clouds that usually cover the whole sky. Altostratus clouds can indicate an upcoming storm with continuous rain or snow.

Alto cumulus - Mid-level, grayish-white clouds with one part darker than the other, often form in groups. Alto cumulus clouds on a warm, humid morning can signal a thunderstorm by late afternoon.

Stratocumulus - Low, lumpy, and gray clouds that may line up in rows or spread out. Stratocumulus clouds only produce light rain or drizzle.

Nimbostratus - Dark gray clouds with ragged bases, sit low in the sky, associated with continuous rain or snow.

Stratus - Stratified or layered clouds that form due to mixing of air masses or loss of heat, look like a fog that doesn't reach the ground. Stratus clouds indicate chilly, overcast days.

Nimbus clouds formed at lower altitudes are usually black or dark grey in colour. Blocking the sunlight causes heavy rainfall and thunderstorms.

Cumulus clouds are puffy white or light gray clouds that look like floating cotton balls, cauliflower-shaped clouds, detached and dense clouds.

They form much closer to the ground and go vertically upwards. These can be associated with fair or stormy weather. These clouds grow upward, and they can develop into a giant **cumulonimbus**, which is a thunderstorm cloud. Height of these clouds ranges from 4000 m - 7000 m.

Rainfall

Rain is water in the form of droplets that have condensed from atmospheric water vapour. Rainfall is measured by Rain Gauge. Rain is liquid Precipitation.

Precipitation refers to the release of moisture that occurs after the condensation of water vapor. The main forms of precipitation include drizzling, rain, sleet, snow, ice pellets, graupel and hail.

Humidity is the amount of water vapor in the air.

Mahawat is the total amount of rainfall in the cold weather season, which is caused by the western cyclonic disturbances in the north and north-western part of India.

Types of Rainfall

Three main types – the conventional, orographic and the cyclonic.

Convectional Rain :- This type of rainfall occurs when warm air rises due to convection currents, expands, cools down, and subsequently undergoes condensation, resulting in precipitation.

This rainfall pattern is prevalent during the summer or in the warmer hours of the day. It is particularly common in equatorial regions and the inner regions of continents, especially in the northern hemisphere.

Heavy rainfall accompanied by thunder and lightning occurs, but it tends to be of shorter duration.

Orographic Rain (Relief Rain) :- It occurs due to the rise of air from highland. This type of precipitation occurs when warm, humid air strikes an orographic barrier (Mountains). Because of the initial momentum, the air is forced to rise.

The chief characteristic of this sort of rain is that the windward slopes receive greater rainfall. After giving rain on the windward side, when these winds reach the Leeward side, they descend, and their temperature rises. Then their capacity to take in moisture increases and hence, these leeward slopes remain rainless and dry.

Cyclonic (Frontal) Rain :- It occurs when warm, moist air meets cold air, causing rising, cooling, and condensation, resulting in prolonged and widespread precipitation associated with weather fronts and cyclones.

Rain - shadow area

The area situated on the leeward side, which gets less rainfall is known as the rain-shadow area.

Examples of rain shadow effect : Patagonian desert in Argentina, Eastern slopes of Western Ghats, The Deccan of the Western Ghats in India is drier because of the rain shadow effect.

Distribution of Rainfall in India

High Rainfall Zones: Regions with over 200 cm of rainfall include the west coast, Western Ghats, and the sub-Himalayan areas in the northeast, especially parts of the Khasi and Jaintia hills, which receive

more than 1,000 cm. The Brahmaputra valley and adjoining hills get slightly less, under 200 cm.

Medium Rainfall Zones: Areas receiving between 100-200 cm encompass southern Gujarat, east Tamil Nadu, northeastern Peninsula, and areas covering Odisha, Jharkhand, Bihar, eastern Madhya Pradesh, northern Ganga plain, and parts of the northeast.

Low Rainfall Zones: Areas receiving between 50-100 cm encompass Regions like western Uttar Pradesh, Delhi, Haryana, Punjab, Jammu and Kashmir, eastern Rajasthan, Gujarat, and the Deccan Plateau.

Areas of Rainfall in India

The average annual rainfall of India is 118 - 120 cm.

Mahabaleshwar, situated on the Western Ghats, receives more than 600 cm of rainfall.

Mawsynram town of Meghalaya receives the highest average annual rainfall in the world over 400 cm.

Jaisalmer is the spot that gets the least rainfall.

Dew Point

The temperature to which air must be cooled for water vapor in it to condense into dew or frost.

Winds

Wind is the movement of air, caused by the uneven heating of the Earth by the Sun and the Earth's own rotation. Anemometers are used for measuring the speed of the wind.

Denmark ranks First in the world to produce electricity from Wind. Therefore, Denmark is known as a country of winds.

Factors Controlling Wind Flow and Direction

Pressure Gradient Force - This force provides initial flow and direction to the wind. The flow is from high pressure to low pressure and it is at right angles to isobars. The velocity of wind is directly proportional to pressure gradient force.

Coriolis Force/Deflection Force - The force that deflects the direction of the

wind is called deflection force. They are deflected to the right in the northern hemisphere and the left in the southern hemisphere.

Friction force generated by surface: It is a restraining force on the flow of the wind. The amount of friction is an outcome of surface irregularities and the orientation of natural landforms. Friction is least above smooth oceanic surfaces.

Classification of Winds

Permanent Winds - The trade winds, westerlies and easterlies are the permanent winds.

Trade Winds : These are permanent winds flowing from east-to-west. It flows in the Earth's equatorial region (between 30°N and 30°S latitudes).

They flow as the north-eastern trade winds in the northern hemisphere and the south-eastern trade winds in the southern hemisphere. This deflection in their ideally expected north-south direction is because of Coriolis force (Ferrel law).

The trade winds from two hemispheres meet at the equator, and due to convergence they rise and cause heavy rainfall.

Westerlies - These winds blowing from the subtropical high-pressure belts (30°-35°) towards the subpolar low-pressure belts (60°-65°) in both hemispheres.

The westerlies are best developed between 40° and 65°S latitudes. These latitudes are often called Roaring Forties, Furious Fifties, and Shrieking Sixties - dreaded terms for sailors.

Easterlies - It is a prevailing wind blowing from the east. The trade winds in tropical regions and the prevailing winds in the polar regions are easterlies.

Secondary/Periodic Winds - The winds that change their direction with onsets of different seasons. Other examples of periodic winds include land and sea breeze, mountain and valley breeze, cyclones and anticyclones and air masses.

Seasonal winds - These winds change their direction in different seasons. Monsoons are the best example of large-scale modification of the planetary wind system.

Land Breeze - It is a wind that flows from the land towards the sea. It flows often at night.

Sea Breeze - It is a wind that blows towards land from the direction of a large water body. Sea breeze develops due to differences in air pressure created by the differing heat capacities of water and dry land.

Tertiary Winds or Local Winds - These blow only during a particular period of the day or year in a small area. Local differences in temperature and pressure produce local winds. Such winds are local and are confined to the lowest levels of the troposphere.

Examples of Indian Local winds - Norwester (Kolkata, Bengal, Assam), Kalbaisakhi (Bihar, Jharkhand, West Bengal and Odisha), Cherry Blossom (Karnataka), Loo (UP, Bihar, Rajasthan, Delhi, Punjab), Bordoisila or Bardoli Chheerha (Assam).

Mango Showers : These are Pre-monsoon showers which are common in the states of Karnataka and Kerala. These rains are usually seen at the end of the summer season. Locally, they are known as mango showers since they help in the early ripening of mangoes.

The Hot wind blowing in the northern plains in summers is known as Loo.

Northeast trade winds - They prevail over the country during the cold season. They blow from land to sea.

Kaal Baisakhi: It is a local rainfall and thunderstorm that occurs in Bihar, Assam, Jharkhand, West Bengal in India and in Bangladesh. It occurs in the summer season. It originated from Chota Nagpur Plateau.

Hot Winds

Chinook (Snow eaters) - The Rocky mountains (North America)
Foehn - The Alps (Europe)
Khamsin - Egypt.
Sirocco - Sahara to the Mediterranean Sea
Solano - Sahara to the Iberian Peninsula
Harmattan (Guinea doctor) - West Africa
Loo - North India (UP, Bihar) and Pakistan
Norwester - New Zealand
Santa Ana - South California
Karaburun (black storm) - Central Asia
Austru - Romania's east or southeast

Diablo - San Francisco Bay Area of Northern California
Brickfielder - Southern Australia
Leste - Madeira and Canary Islands
Ostria - Bulgarian coast
Shamal - Iraq and the Persian Gulf states
Sharki - Persian Gulf
Simoom - Sahara, Israel, Jordan, Syria, and the desert of Arabia
Kali Andhi - North India and Pakistan

Cold Winds

Bora - Blows from Hungary to North Italy
Mistral - The Alps and France
Pampero - Argentina
Punas - The western side of Andes Mountain
Blizzard - Tundra region
Purga - Russia
Levanter - Spain
Calima - Saharan Air Layer across the Canary Islands
Nor'easter - Northeast USA
Elephanta - Malabar coast.
Norte - Mexico.
Tramontana - West coast of Italy and Northern Corsica.
Vardar - Vardar valley in Greece.

Cyclones

A Cyclone is a large air mass that rotates around a strong center of low atmospheric pressure, anticlockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere (opposite to anticyclone).

The process of cyclone formation and intensification is called **Cyclogenesis**. High speed winds and air pressure difference is the reason for cyclones.

There are two types of cyclones:

Tropical cyclones - Tropical cyclones are violent storms that originate over oceans in tropical areas and move over to the coastal areas bringing about large scale destruction caused by violent winds, very heavy rainfall and storm surges.

They are known as Cyclones in the Indian Ocean, Hurricanes in the Atlantic, Typhoons in the Western Pacific and South China Sea, and Willy-willies in the Western Australia.

The conditions favorable for the formation and intensification of tropical storms are: (i) Large sea surface with temperature higher than 27° C; (ii) Presence of the Coriolis force; (iii) Small variations in the vertical wind speed; (iv)

A pre-existing weak low-pressure area or low-level-cyclonic circulation; (v) Upper divergence above the sea level system.

Extra Tropical Cyclones - The systems developing in the mid and high latitude, beyond the tropics are called the middle latitude or extra tropical cyclones.

Extra tropical cyclones form along the polar front. Initially, the front is stationary. In the northern hemisphere, warm air blows from the south and cold air from the north of the front. When the pressure drops along the front, the warm air moves northwards and the cold air move towards, south setting in motion an anticlockwise cyclonic circulation

The cyclonic circulation leads to a well developed extra tropical cyclone, with a warm front and a cold front.

Fronts

When two different air masses (having distinctly different properties) meet, the boundary zone between them is called a front.

There are four types of **fronts**:

Stationary front - The wind motion on both sides of the front is parallel to the front, it is called a stationary front.

Cold front - When the cold air moves towards the warm air mass, its contact zone is called the cold front.

Warm front - If the warm air mass moves towards the cold air mass, the contact zone is a warm front.

Occluded front - If an air mass is fully lifted above the land surface, it is called the occluded front.

Anticyclones

An anticyclone is the opposite of a cyclone; it has an outward-spiraling air circulation around a high pressure centre. An anticyclone's winds rotate clockwise in the Northern Hemisphere and anticlockwise in the southern Hemisphere around a center of high pressure.

Other Wind-Related Concepts

Doldrums - The place where trade winds of the two hemispheres meet is called the Intertropical Convergence Zone (ITCZ). The area around the ITCZ is called the doldrums. They approximately lies between 5°N - 5°S.

Horse Latitudes - The horse latitudes are a narrow zone of warm, dry climates between westerlies and the trade winds. The horse latitudes are located at about 30 degrees north and south of the equator.

Jet Stream - It is a type of geostrophic wind. It blows across the Asian continent at latitudes north of the Himalayas roughly parallel to the Tibetan highlands.

Blossom Shower - It helps the Coffee flowers to blossom in Kerala, Karnataka and nearby areas.

Nor Westers are dreaded evening thunderstorms in Bengal and Assam. Their notorious nature can be understood from the local nomenclature of 'Kalbaisakhi'. These showers are useful for tea, jute and rice cultivation. In Assam, these storms are known as "Bardoli Chheerha".

Principles of Atmospheric Pressure - Air pressure is highest at sea level and decreases with height. High pressure is associated with clear and sunny skies. The air always moves from high pressure areas to low pressure areas.

Islands

An Island is an area of land surrounded by water and an Archipelago is an area that contains a chain or group of islands scattered in lakes, rivers, or the ocean.

An Islet is a very small island. It has little or no vegetation. It may be made of rock, sand and hard coral.

Atoll is a group of islands found in the tropical oceans consisting of coral reefs and a central depression.

Examples of Atoll Caroline Islands are (the Coral Sea Islands, the Marshall Islands, the Tuamotu Islands, Kiribati, Tokelau and Tuvalu) and the Indian Ocean (the Chagos Archipelago, Lakshadweep, the atolls of the Maldives, and the Outer Islands of Seychelles).

Artificial island :- It is defined as an island that is made by human beings rather than naturally made ones. Willingdon Island is the largest artificial island in India and it forms a part of the city of Kochi in Kerala.

Major Islands of India

India has two major island groups, one in the Bay of Bengal and the other in the Arabian Sea.

Arabian Sea Islands

It includes Lakshadweep, an archipelago consisting of 36 islands, located between 8°N-12°N and 71°E-74°E. Entire group built from coral deposits, located off the coast of Kerala.

They were formerly known as Laccadive, Minicoy, and Amindive renamed as Lakshadweep in 1973. Minicoy is the largest island. Amini Island & Cannanore Island are the other islands.

Pitti (Pakshipitti) is an uninhabited coral islet in Lakshadweep. It has a bird sanctuary.

Bay of Bengal Islands

Comprising approximately 836 islands/islets, of which 31 are inhabited, located between 6°N-14°N and 92°E-94°E.

The entire group divided into the Andaman Islands in the north and the Nicobar Islands in the south. Other important islets : Ritchie's archipelago and Labyrinth island

Andaman Islands :- North Andaman Island, Little Andaman, Middle Andaman Island, South Andaman Island, Bluff Island.

Nicobar Islands - Great Nicobar, Car Nicobar. The Great Nicobar and Sumatra islands of Indonesia are connected by the Grand Channel.

The smaller Nicobars comprise some 22 main islands (10 inhabited). The Andamans and Nicobars are separated by the Ten Degree Channel.

Great Nicobar is the southernmost and largest of the Nicobar Islands of India.

Andaman and Nicobar Islands and Lakshadweep islands have recently been named as the new hope spots by the International Union for Conservation of Nature (IUCN) and Mission Blue (An organization involved in the study of oceans).

Renamed Islands: Ross Island- Netaji Subhash Chandra Bose Island; Havelock

Island- Swaraj Dweep; Neil Island- Shaheed Dweep.

Other famous Islands

Sriharikota Island of India is located in the Nellore district of Andhra Pradesh, between Pulicat Lake and the Bay of Bengal.

Dr Abdul Kalam Island :- It was formerly known as Wheeler Island and is located on the coast of Odisha. The Island was renamed to honour the late Indian president, Dr APJ Abdul Kalam in 2015.

Pamban Island (Rameswaram Island) of Tamil Nadu is located between peninsular India and Sri Lanka.

Diu Island is situated off the coast of Kathiawar.

Sagar Island is located in the Bay of Bengal's Ganga delta.

Phumdis/Floating Island is located in Manipur. It is well-known for its Eld's deer/Sangai.

Halliday Island is part of the Sundarbans region and is located in the state of West Bengal. It is situated on the Malta River.

Island of Divar lies in the Mandovi river in the Indian state of Goa.

Salsette Island is an island in the state of Maharashtra on India's west coast.

Peacock Island or **Umananda Island** is the smallest inhabited river island in Assam. It is also known as Bhasmachal.

Sadhu-Bet Island :- It is under construction and located between the Vindhya and Satpura ranges in the Narmada River.

Majuli is the largest river island in the world which is located on the Brahmaputra River in the state of Assam.

Mumbai is referred to as the **City of Seven Islands**. These seven islands are Isle of Bombay, Parel, Mazagaon, Mahim, Colaba, Worli, and Old Woman's Island.

Several smaller islands that lay to the east of the main seven islands are Butcher Island (Mumbai), Cross Island, Elephanta Island, Middle Ground, Oyster Rock.

Island in Andhra Pradesh :- Bhavani Island, Vijayawada (Bhivani), Diviseema, Sriharikota, Hope Island.

Island in Tamil Nadu :- The Island (Island Ground) Chennai, Srirangam Island, Quibble Island, Gulf of Mannar Marine National Park.

Some islands in Kerala :- Munroe Island, Vypin, Willington Island, Vallarpadam, **Mannar Island** lies off the north-western coast of Sri Lanka, is part of Mannar District, Sri Lanka.

Bluff Island - Port Shelter, Sai Kung District, Hong Kong.

The **Maldives Island** consists of about 1,190 coral islands grouped in a double chain of twenty-seven atolls situated in the Indian Ocean.

Ilha da Queimada Grande, also known as Snake Island, is an island off the coast of Brazil in the Atlantic Ocean.

Bluff Island - Prydz Bay, Antarctica.

Deserts

A desert is an area that receives less than 25 cm (10 inches) of precipitation per year.

Evaporation in deserts often greatly exceeds the annual rainfall, resulting in little available water for plants and other organisms.

Formation of Deserts - Deserts are formed by weathering processes as large variations in temperature between day and night put strains on the rocks which consequently break in pieces.

Major types of Deserts

Hot and Dry Deserts: Warm and dry year-round. Examples: Sahara Desert in Africa, Mojave Desert in the United States.

Semi-Arid Deserts: Comparatively cooler than hot and dry deserts, with long dry summers and some rainfall in the winter. Found in North America, Greenland, Europe, and Asia.

Coastal Deserts: Comparatively more humid than other desert types. Example: Atacama Desert in Chile, South America.

Cold Deserts: Dry and have extremely low temperatures. Examples: Antarctic, Ladakh, Gobi Desert.

Midlatitude deserts - These are found between 35° and 60° N and S, poleward of subtropical high-pressure areas. They

include the Tengger and Sonoran Deserts.

Desert in India

Thar Desert (India): Largest desert in India, located on the western margins of the Aravali Hills in Rajasthan and Gujarat.

Characteristics: Undulating sandy plain covered with sand dunes, low rainfall (<150 mm/year), arid climate, thorny bushes. Temperatures can range from 50°C during the day to 15°C at night.

Theri Desert - It is a small desert situated in the state of Tamil Nadu. It consists of red sand dunes and is confined to Thoothukudi district.

Largest deserts in the world, along with their locations:

Sahara desert - Largest hot desert in the world and the third-largest desert after Antarctica and the Arctic. Contains some saltwater lakes, with Lake Chad being the only freshwater lake.

It covers the parts - North Africa West Africa Middle Africa (Chad), East Africa (Eritrea)

Antarctica desert - Antarctica continent

Arctic - Northern America (Alaska, Canada, and Greenland), Northern Europe (Finland, Iceland, Norway, and Sweden), Eastern Europe (European Russia), North Asia (Siberia).

Great Victoria, Great Sandy, Tanami, Simpson, Gibson - Australia

Arabian Desert - Western Asia (Iraq, Jordan, Kuwait, Oman, Qatar, Saudi Arabia, United Arab Emirates, and Yemen)

Gobi Desert - East Asia (China and Mongolia)

Kalahari Desert - Southern Africa (Botswana, Namibia, and South Africa), Middle Africa (Angola)

Patagonian Desert - South America (Argentina and Chile)

Syrian Desert - Western Asia (Iraq, Jordan, and Syria)

Chihuahuan Desert - Central America (Mexico), Northern America (United States)

Karakum Desert - Turkmenistan

Sonoran Desert - Central America (Mexico), Northern America (United States)

Kyzylkum Desert - Central Asia (Kazakhstan, Turkmenistan, and Uzbekistan)
Taklamakan Desert - China

Thar Desert - Indian Subcontinent (India and Pakistan)

Dasht-e Margo and Registan Desert - Afghanistan

Atacama Desert - South America (Chile and Peru)

Great Basin, Colorado Plateau, Mojave Desert and Columbia Basin - United States

Namib Desert - Southern Africa (Namibia), Middle Africa (Angola)

Dasht-e Kavir and Dasht-e Loot - Iran

Features of Desert

Sand dunes - It is a land with undulating topography that is interspersed with barchans and longitudinal dunes. Shifting dunes are Locally called Dhrians in the Thar Desert.

Oasis - An Oasis is a fertile area of a desert or semi-desert environment that sustains plant life and provides habitat for animals.

Barchans - Made of desert sand, barchans are crescent-shaped sand dunes. Barchan constantly faces the wind is one of its distinguishing characteristics. It develops when there is a lot of sand in the desert.

Glaciers

A glacier is a huge mass of ice that moves slowly over land. They are often called "rivers of ice." It forms in areas where the amassing of snow goes beyond its ablation over many years. They are sensitive indicators of changing climate.

According to the topography and the location of the glacier, it can be categorized as Mountain Glacier (Alpine Glaciers) or Continental Glacier (Ice Sheets). The Continental Glacier moves outward in all directions whereas the Mountain Glacier moves from a higher to a lower altitude.

Over 68 percent of the freshwater on

Earth is found in icecaps and glaciers, and just over 30 percent is found in groundwater. Only about 0.3 percent of our freshwater is found in the surface water of lakes, rivers, and swamps.

Icebergs are large chunks of ice that break off from glaciers. Icebergs are made of frozen freshwater. Iceberg B-15 was the largest recorded iceberg by area in Antarctica.

According to the Space Applications Centre of the Indian Space Research Organization (ISRO), India has 16,627 glaciers.

Major Glaciers in India

Siachen Glacier - The longest glacier in India. It is the source for the Nubra River that waters the Nubra Valley near Leh in the Trans Himalayas. It is the Second - Longest glacier in the World's Non-Polar areas

Gangotri Glacier - It is Uttarakhand's largest glacier and second largest of India and the source of the Ganges (River Bhagirathi).

Bara Shigri Glacier - It feeds the Chandra River in Himachal Pradesh's Lahaul and Spiti valleys.

Baltoro Glacier - About 63 kilometer long glacier in the Karakoram range, west of the Siachen glacier. It gives rise to the Shigar River which is a tributary of the Indus.

Hispar Glacier - Located in Gilgit-Baltistan region. It is the third-largest glacier in the Karakoram area.

Zemu Glacier - The largest glacier in the Eastern Himalaya. It is the source of the Teesta River.

Pindari Glacier - A glacier found in the upper reaches of the Kumaon Himalayas, to the southeast of Nanda Devi and Nanda Kot. It is the source of the Pindar River.

Drang-Drung Glacier - This glacier is located in the Zaskar Range of the Himalayas, south of Kargil and east of Srinagar, the state capital of Jammu and Kashmir.

The Shafat Glacier - Also known as Parkachik Glacier. It is a 14 kilometer long glacier in Ladakh.

Machoi Glacier - It is the source of the

Sind and Dras rivers. It is situated in the Himalayan Range's northeastern region. It is a 9 kilometer long Himalayan glacier in Jammu and Kashmir, and Ladakh.

Milam Glacier is located in Kumaon Himalaya. It is a popular hiking site. It is the source of river Gori Ganga which is an important tributary of river Kali.

Satopanth Glacier - It is situated in Uttarakhand. The river Alaknanda, one of the primary headstreams of the Ganges, has its origin in the Satopanth glacier.

Other Glacier in India

Ladakh : Siachen Glacier, Parkachik Glacier, Chong Kumdan Glacier, Saltoro Glacier, Nubra Glacier.

Jammu and Kashmir :- Thajiwas glacier (located about 88 km from Srinagar, about 383 km from Jammu), Bhillan and Prui Glacier (Pir Panjal range), **Kolahoi** Glacier.

Uttarakhand: Gangotri Glacier, Yamunotri Glacier, Bandarpunch Glacier (source of Yamuna river), Alkapuri Glacier, Pindari Glacier, Kafni Glacier, Milam Glacier, Nanda Devi Glacier, Khatling Glacier, Doonagiri Glacier, Satopanth Glacier, Tipra Bamak, Kirti Glacier, Chorabari Glacier, Dokriani Glacier.

Himachal Pradesh : Bara Shigri Glacier, Chhota Shigri Glacier, Dudhon Glacier, Lady of Keylong Glacier, Chandranahan Glacier, Bhadal Glacier, Beas kund Glacier.

Arunachal Pradesh : Bichom Glacier, Kangto Glacier, and Mazgol Glacier.

Sikkim : Zemu Glacier, Talung Glacier, South Lhonak Glacier.

Famous glaciers of the world

Khumbu Glacier - Located in the Khumbu region of northeastern Nepal between Mount Everest and the Lhotse-Nuptse ridge. It is the world's highest glacier.

Lambert Fisher Glacier - The largest glacier of the world. It is in Antarctica.

Hubbard Glacier - It is a tidewater glacier in Alaska and, despite global warming, is expanding.

Luggye glacier - Bhutan.

Bering Glacier - Largest glacier in North America.

Fedchenko Glacier (Tajikistan) - Longest glacier outside of the Earth's polar regions.

Important terms related to Glaciers

Icebergs - Large chunks of ice that break off from glaciers. Icebergs are made of frozen freshwater. Iceberg B-15 was the largest recorded iceberg by area in Antarctica.

Glacial till is the sediment deposited by a glacier. It blankets glacier forefields, can be mounded to form moraines and other glacier landforms, and is ubiquitous in glacial environments.

Moraines - The material carried by the glacier such as rocks big and small, sand and silt gets deposited, which are called glacial Moraines.

Eskers - These are ridges made of Sands and gravels which are deposited by glacial melt water flowing through tunnels within and underneath glaciers.

Drumlins - Oval flat ridged landforms of glacial clay containing heaps of sand and gravel.

Demography

Demography is the statistical study of human populations. Demographers use census data, surveys, and statistical models to analyze the size, movement, and structure of populations.

Population geography is a branch of human geography that is focused on the scientific study of people, their spatial distributions, and density.

The Malthusian theory states that population grows in geometric progression while food production increases arithmetically, leading to the inability to produce enough food to sustain the population.

Neo Malthusianism is the advent of population control programmes to ensure resources for current and future populations.

Growth of population - It refers to the change in the number of inhabitants of a country/territory during a specific period of time.

There are three main processes of

change of population : birth rates, death rates and migration.

Birth rate is the number of live births per thousand persons in a year. It has been a major component of population growth in India. In countries like Japan, low birth rates make the population pyramid narrow at the base.

Death rate is the number of deaths per thousand persons in a year. The rapid decline in death rates has been the main cause of population growth in India.

Growth rate refers to the difference between birth rate and death rate. Nagaland recorded a negative population growth as per 2011 census.

Migration - It defines a migrant as any person who is moving or has moved across an international border or within a state away from his/her habitual place of residence.

Internal migration does not change the size of the population, but influences the distribution of population within the nation.

Factor of migration -

Push factors are forceful reasons that cause people to leave their residence. They include aspects such as : War, Poverty, Floods, Droughts, High crime, Lack of safety, Lack of services.

Pull factors entail the reasons individuals move to a certain location. This includes variables such as: Higher employment opportunities, More wealth, Safety, Education, Better services etc.

Immigration is the process of moving to a new country or region with the intention of staying and living there.

Brain drain is when highly qualified, trained and skilled people migrate from their home country to another country.

Migration in india

According to the 2011 Indian census, there are 5,71,721 Nepalis living in India.

Male migration in India includes education (2%), marriage (4%), and work /employment (24%).

According to the Census 2011, in India, the percentage of people (approximately) considered to be migrants (internal) is 37% (about 45 crore).

Uttar Pradesh has the highest share of out-migrants.

The number of inter-state migrants grew at 55% between the 1991 and 2001 Census. This came down to just 33% between the 2001 and 2011 Census.

Maharashtra at the top of the list with 23.8 lakh net migrants, followed by Delhi (17.6 lakh), Gujarat (6.8 lakh) and Haryana (6.7 lakh) as per Census 2011.

Phases of Indian population growth

1901-1921 - Stagnant or stationary phase with high birth and death rates.

The census year 1911- 1921 registered a negative growth rate of -0.31%.

1921 - The demographic divide or "Great Divide" with consistent population increase after this year.

1921-1951 - Steady population growth due to decline in mortality and high fertility.

1951-1981 - Period of population explosion due to rapid fall in mortality rate and high fertility rate.

1981-2011 - Period of high growth with definite signs of slowing down.

Factors of population distribution

Geographical factors - Landform, topography, climate, soil, availability of water, minerals, terrain.

Social factors - Housing, health, education, transport.

Cultural factors - Places with cultural and religious importance.

Economic factors - Educational institutions, employment opportunities, manufacturing, trade, commerce, and other facilities.

Population density

Population density is the average number of individuals in a population per unit of area or volume.

Density has increased from 117 persons/sq km in 1951 to 382 persons/sq km in 2011 (more than 3 times) in India.

Factors affecting density: Topography, climate, soil, water, minerals, social factors, industries/economic activities, cultural activities.

Different ways of measuring population density -

Arithmetic density: the total amount of persons per unit of land (calculated in sq. miles or km).

Urban density: The number of people living in a metropolitan region's entire area.

Environmental optimum: Density of population supported by environmental assets.

Physiological density: A population consists divided by the total amount of fertile farmland.

Physiological Density = Total Population/Total Cultivable Land.

Agricultural density: The ratio of total agrarian population to available farmland.

Residential density: The number of people who live in a certain geographic region or area of dwellings.

The livestock density index provides the number of livestock units (cattle, sheep, Goat, etc) per hectare of utilised agricultural area.

India's neighboring countries population densities - Maldives (1802 per Km²); Bangladesh (1260 per Km²); Mauritius (626 per Km²); Sri Lanka (341 per Km²); Pakistan (287 per Km²); Nepal (203 per Km²); China (153 per km²); Myanmar (83 per Km²); Bhutan (20 per Km²).

Demographic transition

It is a concept coined by Warren S. Thompson (1929) and later popularized by Frank W. Notestein (1945) to describe a historical process of change that explains trends in births, deaths, and population increase in today's industrialized civilizations.

The four stages of demographic transition:

Stage I (Pre-Transition): High birth and death rates, low population growth.

Stage II (Early Transition): Death rate begins to fall, birth rates remain high, rapid population growth.

Stage III (Late Transition): Low fertility and low mortality, deceleration in population growth.

Stage IV (Post-Transition): Population stability with low and nearly equal birth and death rates.

Population Categories

The population of a nation is generally grouped into three broad categories:

Child Population (usually younger than 15 years old) - They need to be given food, clothes, education, and medical care because they are economically unproductive.

As per 2011 Census, children (0-6 years) account for 13% of India's total population. 74.05% of the child population is in rural areas, 25.95% in urban areas.

Child sex ratio (0-6 years) is 914. Mizoram has the highest child sex ratio (971) followed by Meghalaya (970), while Haryana (830) and Punjab (846) have the lowest.

Literacy rate among children (7-18 years age group) is 88.3%. Kerala has the highest literacy rate (97.9%) and Arunachal Pradesh the lowest (78.8%) for this age group.

Working Population - The people belonging to the age group of 15 to 59 in a country, are referred to as the working population.

National workforce participation rate (as per the Census of India 2011) - female : 25.51%, males: 53.26%.

Aged/Senior Citizen (over 59 years): Even though they may be retired, they can still be economically active. They might be volunteering their time but are not open to being hired.

Population Terminology

Infant Mortality Rate (IMR) - It is defined as the 'number of deaths of children under the age of 1 year per 1000 live births for a given year.

Under 5 Mortality Rate (U5MR) - It refers to the probability of dying before the age of 5 years per 1,000 newborns.

Neonatal Mortality Rate (NMR): It is defined as the 'number of deaths during the first 28 completed days of life per 1,000 live births in a given year or period'.

The estimated national mortality rate of children under the age of five in the country as per Census of India 2011 is 55.

Empowered Action Group (EAG) states are those that lag behind in the demographic transition and have the highest infant mortality rates in the country.

According to the 2011 Census, 8 EAG states are Rajasthan, Uttar Pradesh, Uttarakhand, Bihar, Jharkhand, Madhya Pradesh, Chhattisgarh and Orissa.

Occupational structure - It is the distribution of the population according to different types of occupation.

Age Composition is the proportionate numbers of persons in successive age categories in a given population.

Gross reproduction rate (GRR) - It is the average number of daughters a woman would have if she survived all of her childbearing years, which is roughly to the age of 45.

Fertility rate - Average number of children born to women during their reproductive years.

Maternal mortality refers to deaths due to complications from pregnancy or childbirth.

According to UN inter-agency estimates the global maternal mortality ratio From 2000 to 2020 (MMR) declined by 34 percent - from 342 deaths to 223 deaths per 100,000 live births.

Economic carrying capacity - Number of individuals that can be supported without degrading the environment.

Closed population is a group that is not able to exchange members with different populations over time.

Conurbation is an urban area that forms when two or more urban areas grow together.

Demographic dividend is the economic growth potential from shifts in population age structure, with a larger working-age population.

According to the Economic Survey 2018-19, India's demographic dividend will peak around 2041, when the share of the working-age population is expected to hit 59%.

Lincoln Index provides a way to measure

population size of individual animal species. It is based on a capture / mark / recapture method.

National Population Commission

It was constituted in 2000 under the chairmanship of the Prime Minister.

Objectives: Monitor and guide implementation of National Population Policy, achieve synergy among different sectors, develop people-friendly population stabilization programs.

National Population Policy 2000 aims to achieve population stabilization by 2045 through voluntary, informed choice and target-free approach.

Human Settlement

Human settlement refers to clusters of dwellings where human beings reside, encompassing both rural and urban areas.

Rural settlements in India are broadly categorized into four types: clustered, semi-clustered, hamleted, and dispersed.

Clustered settlement - The settlement in which the houses are closely built up and compact.

Semi-clustered settlement - The settlement that is clustered in a restricted area of dispersed settlement. Examples - Gujarat plain and some parts of Rajasthan.

Hamleted settlement - The settlement that are fragmented into several units and physically separated from each other. Examples - Middle and lower Ganga plain, Chhattisgarh and lower valleys of the Himalayas.

Dispersed settlement - It appears in the form of isolated huts or hamlets of few huts in remote jungles, or on small hills with farms or pasture on the slopes. Examples - Parts of Meghalaya, Uttaranchal, Himachal Pradesh and Kerala.

Urban settlements in India are typically compact and larger in size. They can be classified based on their evolution into ancient towns, medieval towns, and modern towns.

The first urban settlement to reach a population of one million was London.

Miscellaneous

India – Size and Location

India's mainland extends from Kashmir in the north to Kanyakumari in the south, and Arunachal Pradesh in the east to Gujarat in the west. India's territorial limit further extends 12 nautical miles (about 21.9 km) from the coast.

India ranks seventh in the world in terms of size, after countries like Russia, Canada, the USA, China, Brazil, and Australia. India is also the second most populous country in the world after China, and Northern Indian plains are the most densely populated areas of the country.

Physiographic Divisions

Physiographically, India is divided into six main divisions: the Northern and North-eastern Mountains, the Northern Plain, the Peninsular Plateau, the Indian Desert, the Coastal Plains, and the Islands.

Geographical Coordinates

India is situated in the Northern and Eastern hemispheres, with the mainland extending between latitudes 8°4'N and 37°6'N, and longitudes 68°7'E and 97°25'E.

The land frontier of India is 15,200 km, and the coastline, including the mainland, Lakshadweep Islands, and the Andaman and Nicobar Islands, is 7,516.6 km long. India accounts for 2.4% of the total geographical area of the world.

India's Borders

India shares its land borders with Pakistan and Afghanistan in the northwest, China (Tibet), Nepal, and Bhutan in the north, and Myanmar and Bangladesh in the east. Arunachal Pradesh, in particular, shares a 1,129 km disputed border with China's Tibet Autonomous Region.

Coastline of India

The coastline of India is diverse, with the Western Coastal Plains including regions like Kathiawar, Konkan, Kannad, and Malabar, and the Eastern Coastal Plains comprising Coromandel, Northern Circars, and Utkal. Gujarat has the longest coastline in India, at about

1,214.7 km, while Goa has the smallest, at only 160 km.

Major Coastal Cities

Major coastal cities on the west coast include Surat, Porbandar, Vadodara, Karwar, Mangalore, Mumbai, Kochi, Udupi, and Panaji, while on the east coast, they include Gopalpur, Tuticorin, Chennai, Ennore, Visakhapatnam, Paradip, Haldia, and Kolkata.

Notable Bridges in India

India is home to several notable bridges, such as the Dumala Solapur Bridge in Pune, the Aman Setu connecting Kashmir Valley to PoK, the Maitri Setu between India and Bangladesh, the Jai Hind Bridge in Kolkata, the Digha-Sonpur Bridge in Bihar, the Mahatma Gandhi Setu over the Ganga in Bihar, and the Netravati Railway Bridge, the gateway to Mangalore.

Notable Species in India

India is home to several notable species, including the Susu (Ganges River Dolphin) found in the Indus, Ganga, and Brahmaputra rivers, and the Olive Ridley sea turtles on the Odisha coast.

Watershed Development in India

Watershed :- It's a land area that channels rainfall and snowmelt to creeks, streams, and rivers, and eventually to outflow points such as reservoirs, bays, and the ocean.

Watershed development initiatives, such as the Hariyali program, aim to harvest rainwater for irrigation, plantations, pasture development, fisheries, and drinking water, utilizing traditional water storage structures like Johad, Kund, and Tankas.

Examples of successful watershed development programs include Gendathur in Karnataka, the Jal Kranti Abhiyan, Neeru Merru in Andhra Pradesh, Arvary Pani Sansad in Rajasthan, and Ralegan Siddhi village in Maharashtra.

Important Geography Terminology

Gulf - A large inlet from the ocean into the landmass.

Isthmus - A narrow piece of land

connecting two larger areas.

Barchan (Barkhan) is crescent-shaped sand dune produced by the action of wind predominately from one direction.

Transhumance is a type of pastoralism or nomadism, a seasonal movement of livestock between fixed summer and winter pastures.

Human geography is the interaction and interplay between human activities and physical space.

Physical geography is the study of different landforms on the Earth's surface, the variety of life forms present in a location, and the natural activities that affect the Earth.

It encompasses all of the physical dimensions of Earth, including aerial (air), hydrological (water), pedological (soil), biogeographical (plants and animals), oceanography.

Stevenson screen is a unit designed for wet and dry bulb thermometers to protect temperature sensors from being affected by direct or reflected sunlight.

Physical geology is a branch of geology that focuses on the study of Earth's composition, structure, and the processes that shape it. The term physical geology was coined by William Hopkins.

An ice rink is a frozen body of water or an artificial sheet of ice created using hardened chemicals where people can ice skate or play winter sports.

Rideau Canal skateway is the World's Largest Natural Ice Rink with a length of 7.8 km in Canada.

Attrition is the process of rubbing aggregates with itself. Rounded shape aggregates are naturally occurring aggregates.

Example: Desert, Windblown and Seashore sands, River or Seashore Gravels.

The water table is the boundary between the unsaturated zone and the saturated zone underground. The saturated zone beneath the water table is called an aquifer.

A **hope spot** is an area of an ocean that needs special protection because of its wildlife and significant underwater habitats.

Guyot/Seamount :- It is an undersea mountain. Seamounts are formed by volcanic activity.

A **lagoon** is a shallow body of water separated from a larger body of water by a narrow landform, such as reefs, barrier islands, barrier peninsulas, or isthmuses.

Katabatic Wind - The Wind descending on the Leeward Side is heated adiabatically

The IUCN (International Union for Conservation of Nature) Red List is a critical indicator of the health of the world's biodiversity.

The Indian Pharmacopoeia (IP) has been recognised formally by the National Department of Regulation of Medicines and Health Products of the Ministry of Public Health of Islamic Republic of Afghanistan.

Green pages are used for those species that were formerly endangered but have now recovered to a point where they are no longer threatened.

The term '**Biodiversity hotspot**' was first coined by Norman Myers (1988).

Denudation involves the processes that cause the wearing away of the Earth's surface by moving water, by ice, by wind and by waves, leading to a reduction in elevation and in relief of landforms and of landscapes.

Weather radar (Doppler weather radar) is an instrument that sends pulses of electromagnetic energy into the atmosphere to find precipitation, determine its motion and intensity, and identify the precipitation type such as rain, snow or hail.

Cryosphere is the part of the Earth's climate system that contains snow, sea ice, lake and river ice, glaciers and ice caps.

Paleomagnetism :- Paleomagnetism is the study of the record of earth's magnetic field with the help of magnetic fields recorded in rocks, sediment, or archaeological materials.

Economics

Basics of Economics

Economics is a science that studies how to allocate scarce resources to satisfy consumer wants, producer profits, and social well-being.

Types of Economy

Microeconomics - It is the study of individuals, households and firms' behavior in decision making and allocation of resources. Example - Consumer equilibrium, individual income and savings. **Alfred Marshall** is known as the father of Microeconomics.

Macroeconomics - It is concerned with how the overall economy works. It studies broader aspects of the economy such as employment, gross domestic product, and inflation. **John Maynard Keynes** is known as the Father of Macroeconomics.

Sectors of Indian Economy

India's Economic sectors are classified into three categories on the basis of nature of activity, work condition and assets ownership.

On the basis of nature of activities

Primary Sector - This sector depends on the availability of natural resources to manufacture goods. Examples - Agriculture, Forestry, Dairy, fishing etc.

People engaged in primary activities are called red-collar workers.

According to the Periodic Labour Force Survey (PLFS), about 45.76% of the total workforce in India is engaged in agriculture and allied sectors during 2022-23.

Secondary Sector - Secondary Sector are industries where finished products are made from natural materials produced in the primary sector. It is also called the industrial sector. This sector jobs are called blue collar jobs. Examples - mining, manufacturing, and construction.

In 1950-51, the GDP of India contributed by the industrial sector was 13 percent. Which increased to 24.6 per cent in 1990-91.

Tertiary Sector - It is generally known as the service sector, and it is crucial for the development of the primary and secondary sectors. This sector jobs are called white collar jobs. Examples - Financial and business services, Hotels and restaurants, education, health etc.

Quaternary Sector - It includes activities like collection, production and dissemination of information. Examples - Financial planning, information generation, etc.

Quinary Sector - It focuses on high-level knowledge work. This includes creating, interpreting, and using new ideas and technologies, along with key decision-making.

On the basis of work condition

Organised Sector - The sector in which employment terms are fixed and regular.

Unorganised Sector - is characterised by small and scattered units which are largely outside the control of the government. The sector is marked by low incomes, unstable and irregular employment, and lack of protection either from legislation or trade unions.

On the basis of assets ownership

Public sector- The sector in which the government owns most of the assets and is the part of the economy concerned with providing various government services. The purpose of the public sector is social welfare.

Private sector- The sector in which ownership of assets and delivery of services are in the hands of private individuals or companies. The purpose of the private sector is to earn profit.

Basic Economic Activities

Production, consumption and investment (capital formation) are three basic economic activities of an economy.

Production - Land, labour, capital and entrepreneurship are called the factors of production. These factors are owned by the households of the country. The producers try to produce the maximum amount of goods and services by using various combinations of factors of production.

Land encompasses natural resources

vital for production, including minerals, water, wildlife, vegetation, and air. It forms the basis for economic activities like agriculture and mining.

Labour - It is the mental and physical efforts of humans (excluding entrepreneurship) used for production of goods and services.

Capital - Manufactured, synthetic goods used in the production of other goods, including machinery, equipment, tools, buildings, and vehicles.

Entrepreneurship - Human efforts which take on the risk of bringing labor, capital, and land together to produce goods.

Consumption - The consumption activity consists of the use of goods and services for satisfaction of human wants.

Investment (Capital formation) - It involves allocating capital today to increase its value in the future. This encompasses various components such as construction, machinery, and inventories.

Gross Investment is referred to as the total expenditure that is made for buying capital goods over a time period, without accounting for depreciation.

Gross Investment = Net Investment + Depreciation.

Net Investment is the total amount of money that a company spends on capital assets, minus the cost of the depreciation of those assets.

Economic variables

It is measurements that describe economic units, like the GDP, Inflation or Interest Rates. Variables used in economic analysis are classified as stock and flow. Both stock and flow variables may increase or decrease with time.

Stock variable - It is defined as any quantity measured at a particular point of time. e.g. number of machines in a plant, amount in the bank account on a specific date, amount of food grains in all available warehouses on the first day of its financial year in the country come under a stock variable.

Flow variables - It is defined as any quantity measured per unit at a particular period of time. Examples - production of rice, import of cloth, change in capital,

profit of a corporation, output produced by the firm, interest expense on loan, etc.

Capital and its types

Capital - The resource that can be in the form of cash or other kind which is used to further purchase raw materials and inputs.

Natural capital - The natural resources used in the production process. Examples of Natural capital are plants, animals, air, water, soils, minerals.

Human capital - The people's knowledge, skills, ability, expertise, education that help in the process. It is an intangible asset or quality not listed on a company's balance sheet.

Social capital - The group that helps in the maintenance of human capital like families.

Financial capital - The money that helps to purchase raw materials like debt and equity.

Market and its types

Market - A retail outlet, where the sellers and buyers can meet face-to-face, or in a virtual form like an online market, where there is an absence of direct physical contact between the buyers and sellers.

Monopoly Market - A single seller with no close substitutes. It is the exact opposite situation of perfect competition. The monopolist controls supply and price. Examples - Indian Railways, Microsoft.

Oligopoly Market - A market with only a few large sellers of homogeneous or differentiated products. There is high interdependence among sellers regarding pricing and output decisions. Examples include commercial air travel, auto industries, and cable television.

Perfect competition Market - A market with numerous buyers and sellers trading an identical product at a uniform price determined by the free interplay of supply and demand. No single firm can influence the market price. Examples are stock markets, grain markets, and raw gold markets.

Monopolistic Competition Market - A market situation that has elements of both perfect competition and monopoly. It is a mid-way situation between perfect competition and monopoly. For example,

hairdressers, restaurant businesses, hotels, and pubs.

Cournot Competition - It is an economic model used to describe an industry structure in which companies compete on the amount of output they will produce, which they decide on independently of each other and at the same time.

Branches / terminology of Economics

Depending upon the dominant view of a particular country, different forms of production patterns evolved and finally different economic systems came up to organize the economy.

Capitalism is an economic system where private businesses control production factors- capital goods, labor, resources, and entrepreneurship. It fosters market competition, aiming at public welfare, with minimal government intervention. Economic decisions stem from demand and supply, while factor rewards depend on productivity.

Socialist economy refers to government ownership and control over all economic activities in a country. The government decides what goods are produced and how they are distributed based on societal needs rather than profits. Economic decisions are made by central economic planners through a centralized planning system. It operates as a non-market economy without free market forces.

Mixed economy - An economy in which both the private sector (Capitalism) and the public sector (Socialism) are involved. A mixed economic system protects private property and allows a level of economic freedom in the use of capital, but also allows for governments to interfere in economic activities in order to achieve social aims.

Positive economics - The stream of economics that has an objective approach, relying on facts. It concentrates on the description, quantification, and clarification of economic developments, prospects, and allied matters. This subdivision of economics relies on objective data analysis and relevant facts and figures. An increase in money supply implies a price rise in an economy is related to positive economics.

Environmental economics - The study of the cost-effective allocation, use, and protection of the world's natural resources.

Development economics - A branch of economics that focuses on improving fiscal, economic, and social conditions in developing countries. Development economics considers factors such as health, education, working conditions, domestic and international policies, and market conditions with a focus on improving conditions in the world's poorest countries.

International economics - It is concerned with the effects upon economic activity from international differences in productive resources and consumer preferences and the international institutions that affect them. It seeks to explain the patterns and consequences of transactions and interactions between the inhabitants of different countries, including trade, investment and transaction.

Public economics or economics of the public sector - the study of government policy through the lens of economic efficiency and equity.

Gig economy - A temporary work system based on a short-term relationship between workers and companies. This is done to achieve advantage of cost, quality, and flexibility. For example, an Uber driver.

Parallel economy - The functioning of an illegal sector in the economy whose purposes run in opposition to the objectives of the official, sanctioned or legitimate sector.

The parallel economy has various aspects like political, commercial, legal, industrial, social and ethical aspects.

Circular economy - A system of resources utilization where reduction, reuse and recycling of elements prevail. In this, there are no trade activities with the rest of the world.

Important Principles and Laws of Economics

Monetary Principle - It states that you only record business transactions that can be expressed in terms of a currency. Thus, a company cannot record such non-quantifiable items as employee skill levels, the quality of customer service, or

the ingenuity of the engineering staff.

Gresham's Law - A monetary principle stating that "bad money drives out good". This means that if there are two types of money in circulation, one with a higher intrinsic value and one with a lower intrinsic value, people will tend to hoard the more valuable money and spend the less valuable money.

Heckscher-Ohlin model (H-O model) - The model says that countries export the products which use their relatively abundant and cheap factors of production (land, labor, and capital), and import the products which use the countries' relatively scarce factors.

The Solow-Swan model or exogenous growth model - The Solow-Swan model dives into long-run economic growth, analyzing how changes in population (growth rate), savings (rate), and technological progress impact an economy's output level over time.

Input-output model - This model is based on statistical information about the flow of goods and services among various industries.

An input-output model consists of three components -

1. A table showing the cost of inputs such as goods and services, labour and capital, consumed by each industry in the production process. This is called the input, or use, matrix.
2. A table showing which goods and services are produced by each industry. This is called the output, or make, matrix.
3. A table showing which goods and services are available for consumption by final users. This is called the final demand matrix.

Marshall's scissors analysis - The price and output of a goods are determined by supply and demand, which act like "blades of the scissors" in determining price.

Pareto efficiency or Pareto optimality - It is an economic state where resources cannot be reallocated to make one individual better off without making at least one individual worse off.

Mercantile - It is an economic policy that is designed to maximize the exports and minimize the imports for an economy.

Brain Drain - When highly qualified, trained and skilled people migrate from their home country to another country it is regarded as brain drain. Brain drain leads to reduced economic growth, limited innovative capacities and lack of skilled manpower.

Doughnut (economics Model) - The objective of the 'Doughnut Model' of development is to envision a world in which people and the planet can thrive in balance. The Doughnut model of economics was proposed by Kate Raworth.

Major Economists

Adam Smith is known as the father of economics. He is the first person who gave a well defined shape to economics as a new discipline. He is the author of 'Wealth of Nations'. It was published in 1776.

Ernst F. Schumacher, an economist, authored the book "Small is Beautiful," published in 1973 by Harper. The central premise of the book is that economics should prioritize serving people rather than the other way around.

Karl Marx - Karl Marx used labor theory of value (LTV) to explain relative differences in market prices.

Karl Marx popularised the following phrase - "From each according to his ability, to each according to his needs". He is famous for his theories about **capitalism and communism**. Famous work-"The Communist Manifesto" and "Das Kapital".

Amartya Sen took the subject of economics away from approaches that focus exclusively on income, growth, and utility, with an increased emphasis on individual entitlements, capabilities, freedoms, and rights.

Lionel Robbins - "Economics is a science that studies human behaviour as a relationship between end and scarce means that have alternative uses".

Paul A. Samuelson - "Economics is the study of how people and society choose, with or without the use of money, to employ scarce productive resources which could have alternative uses, to produce various commodities over time and distribute them for consumption now and in the future among various persons and groups of society".

Alfred Marshall defines economics as the study of how individuals earn and spend income, focusing on wealth accumulation and well-being.

James Heckman developed theory and methods used in the analysis of individual or household behavior.

Milton Friedman is known as the most influential advocate of free-market capitalism and monetarism. He gave the theory of permanent income hypothesis (PIH), a model to explain the formation of consumption patterns.

As per **Mankiw's** Principles of Economics, the standard of living of a country depends on the country's ability to produce goods and services. Gregory Mankiw in his Principles of Economics outlines ten Principles of Economics.

David Ricardo, an economist, is famous for his theories on wages, profit, comparative advantage, and rents. He suggested that during high deficits, people save more, and he equated taxation with borrowing as forms of government spending.

Esther Duflo (Nobel Prize 2019) is a French economist who is renowned for committing her work towards finding solutions to end poverty and improve the welfare of the poor.

Simon Kuznets, a Russian-American economist, won the 1971 Nobel Prize in Economics for his research on economic growth. He's renowned for developing GDP as the primary indicator of a nation's welfare.

Gottfried Haberier gave the theory of the opportunity cost.

British economist John Maynard Keynes (1883-1946), who is regarded as the founder of modern macroeconomics. His most famous work, The General Theory of Employment, Interest and Money, was published in 1936.

Kenneth Arrow was an American economist who won the Nobel Prize in Economics in 1972 for his work in general equilibrium analysis and welfare economics.

Raj Krishna coined the terminology 'Hindu Rate of Growth'. **The Hindu rate of growth** is a term used to describe India's low rate of economic growth for a prolonged period (1950 - 80).

Money

Money is a medium of exchange for various goods and services. Before money there was a system called barter system, where direct exchange of goods took place without using any medium of exchange. An example of a barter system is exchanging rice for wheat.

Problems with the barter system-

Inability to make deferred payments, lack of common measure value, difficulty in storage of goods, Difficulty in exchanging indivisible goods, lack of double coincidence of wants, difficulty in storing wealth and difficult to form a measure of value which is accepted by all.

Money serves as a widely accepted medium of exchange, facilitating transactions and expressing values. The characteristics of money are durability, portability, divisibility, uniformity, limited supply, and acceptability.

According to Prof. Walker, "Money is what money does". It emphasizes on different functions of Money.

The functions of money are divided into Primary, Secondary, and Contingent. Primary functions - Medium of exchange, Unit of value. Secondary function - Store and transfer of value, Standard of deferred payments. Contingent Function - Credit Creation, Facilitates distribution of income.

Money illusion states that individuals usually tend to view their income and wealth in nominal terms (face value), as opposed to real terms (purchasing power). Money illusion was first coined by American economist **Irving Fisher** in his book "Stabilizing the Dollar."

Demerits of money - Money has created many social and moral disadvantages. It has led to instability in the value system in modern society and results in corruption and bribery. The craze to earn more and more money has disrupted the basic fabric of family life.

Types of money

Fiat money, comprising legal tender notes and coins, is accepted by law as a medium of exchange despite lacking intrinsic value. It's issued without the backing of gold or silver.

Credit money - It is money whose intrinsic value (as a commodity) is much

lower than its face value, like credit cards, demand deposits, etc.

Commodity money - It is money whose value comes from a commodity of which it is made. Examples are gold, silver, cotton, oil and many more.

Metallic money - It refers to coins made of various metals such as gold, silver, bronze, nickel, and so on.

Reserve Money - It is currency in circulation plus deposits of commercial banks with the central bank. It includes notes, rupee and coins in circulation.

Fiduciary money is money that is accepted as a medium of exchange due to the trust that exists between the payer and the payee. Example - Cheque.

Narrow money is the most liquid and includes currency with public and demand deposits in the banks and post offices.

Broad Money refers to the money supply whose liquidity extends from currency to time deposits with the banking system.

Digital currency - It is a form of currency that is available only in digital or electronic form. It is also called digital money, electronic money, electronic currency, or cybercash. It does not have physical attributes and is available only in digital form. This type of currency is issued by the Reserve **Bank of India**.

Hot money is the flow of funds from one country to another in order to earn a short-term profit on interest rate differences.

Vault cash refers to the cash reserves held by the bank itself, over and above the mandated Cash Reserve Ratio (CRR) amount. It is NOT part of the M1 money supply, instead it is included in the monetary base.

Hard Currency is expected to remain relatively stable and is the strongest currency of the world, which has a high level of liquidity.

Soft currency is the currency that is easily available in any economy in its forex market.

Helicopter Money involves the central bank of the country printing currency notes and distributing it to the people free of cost. The idea is to promote demand in the economy during

recession.

Cheap currency means a period of comparatively lower/softer interest rates regime.

Dear currency means a period of comparatively higher/costlier interest rates regime.

Legal Tender Money is a coin or a banknote that is legally tenderable for discharge of debt or obligation.

Money Supply

The total stock of money in circulation among the public at a particular point of time is called **money supply**. Money Supply is a flow concept.

RBI publishes figures for four alternative measures of money supply M1, M2, M3 and M4.

These measures represent the decreasing order of liquidity, with M1 being the most liquid and easiest for transactions, while M4 is the least liquid.

M1 and M2 are known as **narrow money**. M3 and M4 are known as **broad money**.

M1 includes currency, demand deposits, and other liquid deposits, including savings deposits.

M2 = M1 + Savings deposits with Post Office savings banks.

M3 = M1 + Net time deposits of commercial banks

M4 = M3 + Total deposits with Post Office savings organisations (excluding National Savings Certificates)

Money Multiplier

It is the ratio of money supply to the reserve money. Money multiplier is a term in monetary economics that is a phenomenon of creating money in the form of credit creation, which is based on the fractional reserve banking system.

The **money multiplier** formula uses the reserve ratio.

$$\text{Money Multiplier Effect} = \frac{1}{\text{Reserve Ratio}}$$

The **expenditure multiplier** shows what impact a change in autonomous spending will have on total spending and aggregate demand in the economy,

$$\text{Expenditure Multiplier} = \frac{1}{1 - c}$$

where 'c' represents the marginal propensity to consume (MPC).

Circulation of money

The circulation of money refers to all the currency issued by a country's monetary authority, minus any cash that has been withdrawn from circulation.

Velocity of circulation of money - The number of times a unit of money changes hands during the unit period.

Income velocity of money is the number of times one unit of currency is spent over a given period of time. The income velocity of the hoarded money (idle) by the public is equal to zero.

Fisher's Quantity Theory of Money

In his book "The Purchasing Power of Money" in 1911, Irving Fisher introduced the "Equation of Exchange," which states that the total quantity of money (MV) in a country during a given period equals the total value of all goods and services bought and sold (PT).

The equation is represented as $MV = PT$, where M is the money supply or quantity of money, V is the velocity of money, P is the price level, and T is the volume of transactions.

Currency Notes

There are a total **17 languages** mentioned on Indian Banknotes. There are **15 languages** written on the Indian currency language panel. These languages are Assamese, Bengali, Gujarati, Kannada, Kashmiri, Konkani, Malayalam, Marathi, Nepali, Oriya, Punjabi, Sanskrit, Tamil, Telugu, and Urdu. Apart from these languages **Hindi and English** languages are also written on the Indian currency.

Clause on the Indian bank notes- "I promise to pay the bearer the sum of rupees" is the clause on the Indian notes.

Dimension, colour and Motif of Indian Currencies

Currency (Rs.)	Dimension (in mm)	Colour	Motif
10	63 × 123	Chocolate Brown	Sun temple
20	63 × 129	Greenish yellow	Ellora caves

50	66 × 135	Fluorescent blue	Hampi
100	66 × 142	Lavender	Rani ki Vav
200	66 × 146	Bright yellow	Sanchi Stupa
500	66 × 150	Stone grey	Red fort
2000	66 × 166	Magenta	Managalyaan

Printing press and Mints - There are currently four printing presses in India where currency notes are printed - Dewas, Salboni, Nasik and Mysore. Further, there are four mints where coins are minted at Kolkata, Hyderabad, Mumbai, Noida. Security Paper Mill was established in 1968 at Hoshangabad, Madhya Pradesh to make papers for bank notes.

The **Ministry of Finance issues coins** (under the coinage act 1906) and the one rupee note (bears the signatures of the Finance Secretary), while other notes are printed by the Reserve bank of India (which bears the signatures of the RBI governor).

Note - The Indian rupee symbol (₹) was adopted on 15 July 2010. This symbol was conceptualized and designed by Uday Kumar.

Keynes's Theory of Demand for Money - According to Keynesian theory of demand for money, people hold money in the form of cash with three motives.

Transaction motive is the motive of holding money to carry out transactions, which is further of two types- Income motive and business motive.

Precautionary motive is holding money for various contingencies like Unemployment, sickness, and accidents.

Speculative motive arises when investing the money in some asset or bond is considered riskier than simply holding the money.

Demonetisation

Demonetisation is the act of canceling the legal tender status of a currency unit in circulation, replacing it with a new currency. It aims for outcomes like a corruption-free economy, removing black money, controlling inflation, and stopping funding of illegal activities.

In India, on 8 November 2016, under Section 26(2) of the RBI Act, Prime Minister Narendra Modi declared ₹500 and ₹1000 notes invalid. Previously, ₹1,000 and ₹10,000 notes were

demonetized in January 1946, reintroduced ₹1,000, ₹5,000, and ₹10,000 notes in 1954, and again demonetized in January 1978. India's first republic banknotes were issued in ₹2, ₹5, ₹10, and ₹100 denominations in 1950.

Cost, Production, Consumption and Market

Cost

Cost refers to the expenditure incurred by a producer on the factor as well as non-factor inputs for a given amount of output of a commodity.

Types of costs- Long run and short run cost

In the short run, some factors of production cannot be varied and remain fixed and In the long run, all inputs are variable, so total cost behaves as the total variable cost

Other types of costs

Economic cost is the total expenditure a firm faces when using economic resources to produce goods and services.

Explicit costs refer to the money a firm spends on input costs. Some examples of explicit costs include salaries, rent payments, raw materials, etc.

Implicit costs refer to the costs that do not require an explicit outflow of money.

Fixed costs and **variable costs** are also important types of economic costs. They play an important role when a firm decides how to allocate its resources so that it can maximize its profit.

Fixed costs, or total fixed costs (TFC), are business expenses that remain constant regardless of changes in production or sales volume. These costs include items like rent, interest payments, and insurance premiums, and they do not vary with output levels.

A variable cost (TVC) is a corporate expense that changes in proportion to how much a company produces or sells.

Adding the fixed and the variable costs, we get the **total cost (TC)** of a firm

$$TC = TVC + TFC$$

Note - Total Variable cost = total quantity of output × variable cost per unit.

If output is equal to zero then the variable cost is equal to zero.

Marginal costs refer to the increase in cost as a result of increasing production by one unit.

Average total cost (ATC) is found by dividing a firm's total cost (TC) by the quantity of total output (Q) produced, expressed as $AC = TC/Q$.

Replacement cost is the expense incurred in replacing an asset, such as plant machinery or equipment.

Imputed costs are hypothetical expenses used solely for decision-making purposes and do not require any actual cash outlay.

Sunk cost is the expenditure that a company has already made and can't recover. Sunk cost is ignored when making future economic decisions.

Opportunity costs represent the potential benefits an individual, investor, or business misses out on when choosing one alternative over another.

Choosing to see a movie means missing out on the time and money that could have been used for other activities, representing the opportunity cost.

Real cost refers to the actual expenses involved in creating a product or service, including the value of tangible resources like raw materials and labour.

Production

Production is the provision of a product or service to satisfy consumer's wants and needs. The process involves adding value to a product or raw material.

This relationship between the variable input and output, keeping all other inputs constant, is often referred to as **Total Product (TP)** of the variable input.

Average product is defined as the output per unit of variable input.

Marginal product of an input is defined as the change in output per unit of change in the input when all other inputs are held constant.

The production possibility frontier gives the combinations of two goods that can be produced when the resources of the economy are fully utilised. It illustrates the production possibilities of the economy. Example - the curve gives the

maximum amount of cotton that can be produced in the economy for any given amount of cotton and vice-versa.

The production function gives the maximum quantity of output (Q) that can be produced by using different combinations of these two factors of productions - Labour (L) and Capital (C).
 $Q = f(L, C)$

The Production Possibilities Curve addresses three central economic problems- full resource utilisation, economic efficiency, and economic growth. It illustrates combinations of goods producible in the economy with existing technology and resources.

Consumption

Consumption is defined as the use of goods and services by an individual. It is a component in the calculation of the Gross Domestic Product (GDP).

The key determinants of consumption include income, savings, expectations, changes in fiscal policy, debt levels, and the availability of goods and services.

Domestic consumption is defined as the products and services that are bought and used in the country that makes or offers them.

Foreign consumption is defined as the products and services used by non-residents.

Consumption Function refers to an economic formula that represents the functional relationship between total consumption and gross national income (GNI). The consumption function is represented as-
 $C = A + MY$, where- C = consumer spending, A = autonomous consumption, M = marginal propensity to consume, Y = real disposable income.

The consumption Function has the same slope as the Aggregate Demand Function. It is also parallel to the Aggregate Demand Function.

Market

A market is a place where buyers and sellers can meet to facilitate the exchange or transaction of goods and services. Markets can be physical like a retail outlet, or virtual like an e-retailer.

A financial market is a market for the creation and exchange of financial

assets. Financial markets exist wherever a financial transaction occurs. Financial markets are classified into Capital and Money Market.

Capital markets are financial markets that bring buyers and sellers together to trade stocks, bonds, currencies, and other financial assets.

Money market is a market for short term funds which deals in monetary assets whose period of maturity is upto one year. These assets are close substitutes for money.

It is a market where low risk, unsecured and short term debt instruments that are highly liquid are issued and actively traded everyday.

The instruments of the money market are Treasury Bills, Commercial Paper, Call Money, Certificates of Deposit, and Commercial Bill.

Domestic market is a nation's internal market representing the mechanisms for issuing and trading securities of entities domiciled within that nation.

The market in which national currencies are traded for one another is known as the **foreign exchange market**.

The major participants in the foreign exchange market are commercial banks, foreign exchange brokers and other authorised dealers and monetary authorities.

Product market is the market where final goods or services are sold to the customer. Example - Pharmaceutical sector, smart-phones.

A factor market is a market where businesses purchase the items needed to produce goods or services. Example - Entrepreneurial talent, capital, land, and natural resources.

The equity market is a place where stocks and shares of companies are traded. The equities that are traded in an equity market are either over the counter or at stock exchanges. Often called the stock market or share market, an equity market allows sellers and buyers to deal in equity or shares in the same platform.

A commodity market is a marketplace where raw materials or primary items can be bought, sold, and traded. Commodities are typically classified as hard commodities or soft commodities.

Hard commodities include natural resources such as gold, rubber, and oil. On the other hand, soft commodities are agricultural goods or livestock like maize, wheat, coffee, sugar, soybeans, and pork.

Marketable Surplus is known as the difference between farmer's output over and above his on-farm consumption. It is an indication of the Commercialisation of Agriculture.

The marketed surplus is more than the marketable surplus when the farmer retains a smaller quantity of the crop than his actual requirements.

A consumer surplus happens when the price that consumers pay for a product or service is less than the price they're willing to pay.
 Consumer's surplus = Potential price – Actual price.

Hoarding - Purchasing and storing a large amount of product belonging to a particular market, creating scarcity of that product, and ultimately driving the price of that product up.

Auxiliaries to trade are essential services and functions that assist the business or facilitate smooth exchange of goods and services which include transport, banking, warehousing, insurance, advertising etc.

Arbitrage is termed as buying an asset in one market and simultaneously selling an identical asset in another market at a higher price, profiting from the temporary difference in prices.

Mortgage refers to the process of offering something as a guarantee or collateral against a loan. One may come across the term when looking for secured loans.

E - Commerce is a business model that lets firms and individuals buy and sell things over the Internet. Its **Advantages** are increased productivity and perfect competition, global market, cuts the product distribution chain, managing business easy and simple.

Types of e - commerce include B2C (Business to Consumer), B2B (Business to business), C2C (Consumer to Consumer), P2P (Peer to Peer), D2D (Distributor -to- Distributor) and M - Commerce (Mobile Commerce). Examples of E-Commerce are Amazon, Flipkart, eBay, Fiverr, Upwork, Olx, Quikr, etc.

JioMart is an Indian e-commerce company, headquartered in Navi Mumbai, Maharashtra, India, that started as a joint venture between Reliance Retail and Jio Platforms.

Game Theory

It is a field of economics that analyses the behaviour of consumers and firms as if they were players in a simple game in which their outcomes are affected by their own choices and the choices of other players.

Zero-sum game- A zero-sum game is a situation in which one person's gain is equivalent to another's loss, so that the net change in wealth or benefit is zero.

Non zero-sum game- A non zero sum game is a situation where there is a net benefit or net loss to the system based on the game's outcome.

Public goods game- A public goods game investigates behavioural economics.

Sunk cost game- The sunk cost game can be defined as the financial cost which is already invested and now it cannot be incurred or money you cannot get back.

Goods

Goods are tangible items sold to customers, while services are tasks performed for the benefit of the recipients. Examples of goods are automobiles, appliances, and clothing.

There are two types of goods, Private goods and Public goods.

A Private good, a product or service produced by a privately owned business and purchased to increase the utility, or satisfaction, of the buyer. Clothes, cosmetics, footwear, cars, electronic products and food are examples.

A Public good refers to a commodity or service that is made available to all members of a society. Typically, these services are administered by governments and paid for collectively through taxation. There is a possibility of free ridership in its consumption.

Examples of public goods include law enforcement, national defence, Roads and Police.

A public good has two key characteristics - it is nonexcludable and nonrivalrous.

Nonexcludable means that it is costly or impossible for one user to exclude others from using a goods.

Nonrivalrous means that when one person uses a good, it does not prevent others from using it.

Spillover benefits are free benefits that third parties or society receive from the actions of others.

Other types of goods

Intermediate goods are incomplete goods which go as input for further finishing. The raw materials that a firm buys from another firm which are completely used up in the process of production are also called intermediate goods.

Examples of intermediate goods are steel, crude oil, sugar and wheat.

Inferior goods are a type of goods whose demand decreases with an increase in the consumer's income or expansion of the economy. These include "store-brand" grocery products, instant noodles, and certain canned or frozen foods.

Final goods are directly consumed goods and do not undergo further production stages or transformations.

Final goods are further divided into consumer goods and capital goods. Capital goods, such as machinery and vehicles, are used in production. Consumer goods, like food and clothing, satisfy human wants directly.

Consumer goods are directly used for the satisfaction of human wants and not used later to produce another consumer good. Example - Food, Beverages, Clothing, Vehicles.

Consumer goods are categorised as durable or non-durable. Durable goods, like refrigerators and televisions, last a long time and are relatively expensive. Non-durable goods, such as milk and vegetables, have a shorter shelf life and lower price.

Demand and Supply

Demand - The quantity of a commodity that a consumer is willing to buy and is

able to afford, given prices of goods and income of the consumer, is called demand for that commodity.

In the case of normal goods, income and demand are directly related, which means an increase in income will cause demand to rise and a decrease in income causes demand to fall.

Human Wants - In economics, human wants encompass all desires, aspirations, and motives of individuals. Examples include necessities such as food, shelter, and clothing.

Major Types of Demand

Aggregate demand measures the total demand for all goods and services in an economy. To gauge aggregate demand, one assumes constant prices for final goods and a steady interest rate in the short run. Aggregate supply is presumed to be perfectly elastic. Reductions in taxes typically have a smaller multiplier effect on aggregate demand compared to increases in government spending.

Precautionary demand for money arises mainly due to the uncertainty of future receipts and expenses.

The amount of money required for current transactions of companies and individuals is known as transaction demand for money.

Speculative demand arises from the perception that money is optimally part of a portfolio of assets being held as investments. The speculative demand for money is relatively interest inelastic.

Generic demand refers to the demand that arises for Generic brand products lacking widely recognized names or logos due to minimal advertising.

Joint demand occurs when demand arises simultaneously for two or more goods to satisfy a particular consumer want, such as the demand for milk, coffee beans, and sugar.

Derived demand is when the demand for a commodity depends on the demand for other goods, like the demand for workers producing bags depending on the demand for bags.

Direct demand refers to the demand for commodities that directly satisfy consumer needs, such as books, stationery, and clothes.

Competitive demand arises between close substitute commodities, where an increase in the demand for one decreases the demand for the other, like an increase in tea demand decreasing coffee demand.

Alternative demand can be satisfied using different alternatives, as seen in various clothing options like jeans, shirts, and trousers.

Elasticity of Demand

The elasticity of demand at any point on a straight line demand curve is given by the ratio of the lower segment and the upper segment of the demand curve at that point.

Elasticity of Demand (eD) =

$$\frac{\text{percentage change in demand of the goods}}{\text{percentage change in the price of the goods}}$$

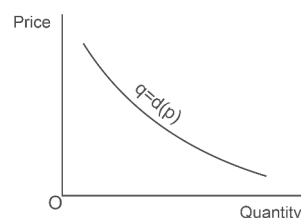
Price elasticity of demand is a measure of the change in the demand for a product in relation to a change in its price. If the price elasticity of demand is less than one, then the demand for the goods is said to be inelastic.

Income elasticity of demand is the change in quantity demanded of a good or service in relation to the change in real income of a consumer. There is a direct relationship between income and demand.

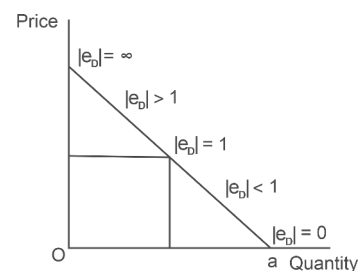
Demand Curve

It shows the quantity of goods demanded by consumers at each price, always downward sloping.

The **Law of Demand** states that when the price of a commodity rises, then its demand falls and vice versa keeping all other things constant. So, the demand and price are inversely proportional to each other.



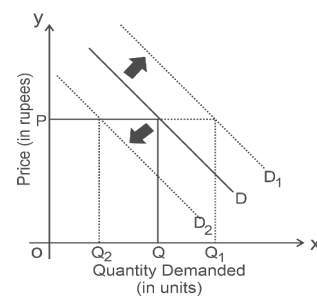
The demand curve is called elastic at a point where price elasticity is greater than unity, inelastic at a point where the price elasticity is less than unity and unitary elastic when price elasticity is equal to 1, or a price change is equal to the change in supply quantity.



Shifts in Demand Curve

Rightward shift - Quantity demanded increases at each price level.

Leftward shift - Quantity demanded decreases at each price level.

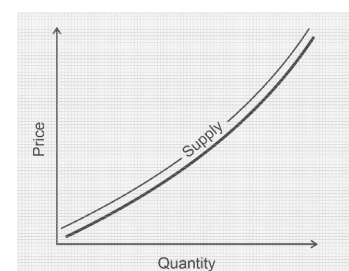


Supply

A firm's 'supply' is the quantity that it chooses to sell at a given price, given technology, and given the prices of factors of production.

The law of supply states that the price of a commodity is directly related to its supply. If the price of the product increases, its supply will increase. Similarly, the lower the price of a commodity, the lower its supply.

The **supply curve** shows the levels of output the firm chooses to produce at different market prices.



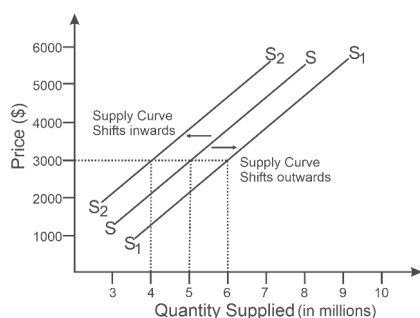
The **price elasticity of supply** of a good measures the responsiveness of quantity supplied to changes in the price of the good.

price elasticity of supply =

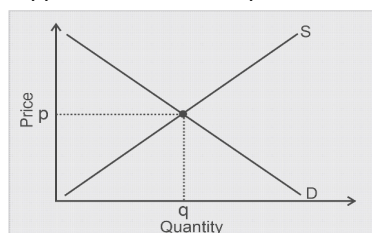
$$\frac{\text{percentage change in quantity supplied}}{\text{percentage change in price}}$$

Shifts in supply curve - An increase in the change in supply shifts the supply

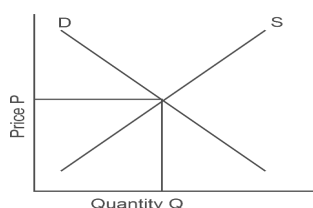
curve to the right, while a decrease in the change in supply shifts the supply curve left.



Market Equilibrium - It occurs when quantity demanded equals quantity supplied at the market price.



Change in Equilibrium Price :- An decrease in demand and an increase in supply will decrease the equilibrium price and vice versa but the effect on equilibrium quantity cannot be determined.



Profit

Normal profit refers to the profit level sufficient to cover both explicit costs and opportunity costs of a firm.

Super-normal profit is the profit earned by a firm beyond the normal profit level.

The break-even point of a firm occurs at the point on the supply curve where the firm earns normal profit.

Utility

It is a measure of satisfaction derived from consuming goods or services.

Total Utility is the overall satisfaction obtained from consuming a fixed quantity of a commodity. It's calculated by summing the marginal utilities of each unit consumed. For instance, if total utility increases from 12 to 18, the marginal utility would be 6.

Marginal Utility represents the additional satisfaction gained from consuming one more unit of a commodity, diminishing as consumption increases.

National Income

National income is the monetary value of all goods and services produced in a country over a specific period. It's a key measure of economic growth, reflecting an economy's ability to satisfy human needs. However, traditional measures often overlook the value of housewives' unpaid work, such as caregiving and household tasks, which can underestimate the true economic contribution of households.

National income at constant prices is a true measure of the purchasing power of the people. Therefore, it is also known as the **real national income**. The subject of the Study of Macroeconomics is based on the principle of National Income.

After independence in August 1949, the government of India established the National Income Committee to compute the national income of the economy. P.C. Mahalanobis was the chairperson of the Committee.

Now, the Central Statistics Office under the Ministry of Statistics and Programme Implementation (MOSPI) is responsible for measuring National Income.

Per capita income (PCI) is the total income of the country divided by its total population. The per capita income is also called **average income**.

$$PCI = \frac{\text{Total income of a country}}{\text{Total population of a country}}$$

Concepts of National Income

Gross Domestic Product (GDP) - It is the market value of all final goods and services produced within a financial year by factors of production located within a country irrespective of ownership. It is released by the National Statistical Office (NSO) with the Base year of 2011-2012.

Two types of GDP -

The Nominal GDP refers to the GDP evaluated at current market prices.

Real GDP is an inflation-adjusted measure that reflects the value of all goods and services produced by an economy in a given year.

Calculation of GDP

There are three methods to calculate GDP in India.

Expenditure Method (Income disposable Method) is commonly used for calculating GDP for the manufacturing sector. It measures the final expenditure on GDP.

$$GDP = C + I + G + (X - M)$$

Here, C denotes consumption expenditure, I denotes investment, G is for government expenditure, and (X - M) signifies net exports.

The Value Added Method, also known as the Production or Output Method, calculates GDP at Market Price by multiplying final output by market price. It deducts the value of intermediate goods to prevent double counting. This method is commonly used for calculating GDP in the Primary sector and for national income calculations in India.

If there are N firms in the economy, each assigned with a serial number from 1 to N, then as per the product method, GDP can be written as $GVA_1 + GVA_2 + \dots + GVA_N$.

$$GDP = \text{Real GDP (GDP at constant prices)} - \text{Taxes} + \text{Subsidies}$$

The Income Method calculates GDP for the services sector by aggregating payments made by firms to households, households and firms to the government, and banks to households/firms. It measures income, including wages, rents, interest, and profits received by all factors of production involved in producing final goods.

GDP calculated based on this approach is known as Gross Domestic Income (GDI).

$$GDI = \text{GDP at factor cost} + \text{Taxes} - \text{Subsidies}$$

Some Macroeconomic Identities

The base year is used as a standard year to make comparisons by calculating GDP in real terms. The base year has been shifted from 2004-05 to 2011-12 in India.

The factor cost method assesses the performance of eight different industries. In 2015, the Central Statistics Office (CSO) discontinued GDP at factor cost and embraced the worldwide practice of GDP at market price and the Gross Value Addition (GVA) measure.

$$GDP \text{ at market price} = GDP \text{ at factor cost}$$

+ Indirect Taxes – Subsidies.

Gross Value Added (GVA) - It is defined as the value of output less than the value of intermediate consumption. The GVA at basic prices will include production taxes and exclude production subsidies available on the commodity. The Central Statistical Office adopted the GVA concept in India in 2015.

NVA (Net Value Added) method of measuring national income is also known as net output method, production method and industry of origin method.

SNA (System of National Accounts) is the most accepted and globally followed national income accounting format prepared by the UN and ratified by the IMF, World Bank, OECD and EC.

Operating Surplus is a concept used in national accounts statistics which refers to the measure of surplus accruing from production of the output before deducting property income from it like land, rent, and interest. It is a component of value added and gross domestic product.

Gross National Product (GNP) - It is defined as the total market value of all final goods and services, produced within a financial year by factors of production owned by a country's citizens regardless of where the output is being produced.

GNP = GDP + Factor income earned by the domestic Factors of production employed in the rest of the world - Factor Income earned by the factor of production of the rest of the world employed in the domestic economy.

If the factor income from abroad is greater than the factor income paid abroad, then GNP would be greater than GDP.

Per Capita GNP - It is the total value of all the goods and services produced by a country in a year, including income from foreign investments, divided by the number of people living there. It is the most generally used indicator of overall economic performance.

Net Domestic Product (NDP) and Net National Product (NNP)

NDP = GDP - Depreciation

NNP = GNP - Depreciation

The amount by which an asset's value falls during the production period is called **Depreciation**.

Net Domestic Product at factor cost (NDP at FC) is the income earned by the factors in the form of wages, profits, rent, interest etc.

NDP at FC = GDP (Gross Domestic product) at FC - Depreciation.

NNP (factor cost) = NNP (market price) - Net Indirect Taxes, where Net Indirect taxes = Indirect taxes - Subsidies.

Gross national income (GNI) is defined as gross domestic product, plus net receipts from abroad of compensation of employees, property income and net taxes less subsidies on production.

Gross National Happiness (GNH) is a measure of economic and moral progress that the King of Bhutan introduced in the **1970s** as an alternative to gross domestic product. It is a qualitative measure.

Gross State Domestic Product (GSDP) is the sum total of value added by different economic sectors (Agriculture, Industry & Services) produced within the boundaries of the state calculated without duplication during a year.
GSDP = Gross Value Added + Taxes – Subsidies.

Green Gross Domestic Product (Green GDP) is an economic growth indicator that incorporates environmental factors alongside the traditional GDP of a country. China introduced this concept by publishing its first Green GDP data for the year 2004 in 2006.

Calculating Gross national income (GNI) using GDP & GNP -

GNI (from GDP) = GDP + (Income earned from abroad – Income remitted by foreigners living in the country back to their home countries).

GNI (from GNP) = GNP + (Income spent by foreigners within the country – Foreign income not remitted by citizens)

Gross National Disposable Income (DPI)

Disposable income is defined as the income remaining after payment of tax and fines etc. If tax payment is high, disposable income will be lower and vice versa. If tax payment and fines are zero then disposable income and total income will be the same.

DPI = PI – Personal Taxes

National Disposable Income = Net National Product at market prices + Other current transfers.

Domestic income (NDP_{FC}) is the net money value of all the final goods and services produced within the domestic territory of a country during a period of one year.

Current income refers to cash flows that are anticipated in the immediate to short-term.

Private income is referred to as the total of all the factors incomes and transfer earnings received by the private sector from all sources.

Private Income = NDP at FC - Income from Property and Entrepreneurship accruing to Government Administrative Departments - Savings of Non Departmental Enterprise + Net Factor Income from Abroad (NFIA) + National Debt Interest + Current Transfers from Government + Net Current Transfers from Rest of the World.

Personal income refers to an individual's total earnings from wages, investment enterprises, and other ventures.

Personal Income = National Income - Corporate Income Taxes - Undistributed Corporate Profits - Social Security Contribution + Transfer Payments.

Personal Disposable Income (PDI) equals the amount of money available to the household for saving and spending after the deduction of all forms of taxes.
PDI = Personal Income (PI) – Personal tax payments – Non-tax payments.

Factor income and Non -Factor income

Factor income refers to the income earned by factors of production, such as land, labour, capital, and entrepreneurship, in return for their productive services. Examples include wages, rent, interest, and profit.

Non-Factor Income, on the other hand, comprises money receipts that do not require any sacrifice on the part of the recipients, such as gifts and donations. These incomes, also known as transfer incomes, are not included in national income calculations.

The Circular Flow of Income model depicts the continuous circulation of money, goods, and services in an

economy. It involves households providing factors of production to businesses in exchange for compensation. In a simplified two-sector economy, consumption equals factor income, facilitating the estimation of aggregate income through income, product, and expenditure methods.

Inventory

Inventory refers to the stock of unsold finished goods, semi-finished goods, or raw materials carried by a firm from one year to the next. It can increase or decrease based on changes in sales and production.

Unplanned accumulation of inventory- In case of an **unexpected fall in sales**, the firm will have unsold stock of goods which it had not anticipated.

Unplanned decumulation of inventory- where there is an unexpected rise in the sales.

Planned accumulation of inventories based on the anticipated fall in sales for a particular period or year.

Planned decumulation of inventories based on the **anticipated rise in the sales** for a particular period or year.

GDP and Welfare

GDP and welfare are intertwined, as GDP growth can provide more resources for social welfare. However, GDP alone doesn't capture all aspects of welfare. Environmental degradation and unequal income distribution are often overlooked by GDP growth. So, while GDP growth is crucial, it doesn't always guarantee overall welfare improvement.

Net Economic Welfare (NEW) is a concept proposed by Paul Samuelson to provide a more comprehensive measure of a nation's well-being than Gross National Product (GNP).

$$NEW = GNP + \text{Value of Housewives Services} + \text{Value of leisure} - \text{Expenditure on defence} - \text{Cost of Environment Degradation}.$$

Equitable Distribution of Income (EDI) aims to ensure fairness in wealth accumulation by providing equal opportunities for all members of the economy. Governments redistribute tax revenue to support low-income earners without reducing the incentive to work and accumulate wealth.

Inflation

Inflation is defined as a rise in the general level of prices of goods and services in an economy over a period of time. When the price of goods rises, each unit of money buys fewer goods. Thus, Inflation also reflects the erosion in the purchasing power of money.

Types of Inflation

Demand-Pull Inflation - This type of inflation occurs when The total demand for goods and services in an economy exceeds the supply of the same when supply is less, the price of these goods and services rises.

Causes of Demand-Pull Inflation - An increase in money supply, government expenditure, black money, population pressure, and rising income.

Cost-Push Inflation - This type of inflation arises from increased production costs, such as higher wages or raw material costs, resulting in higher prices for finished goods and services.

Causes of Cost-Push Inflation - Rising input costs, indirect taxes, seasonal fluctuations in agriculture, administered price increases, infrastructural bottlenecks, and higher import prices.

Rate-Based Inflation is a category within the broader classification of inflation based on its rate of increase. This subheading would include the following sub-categories

Creeping Inflation - It occurs when the inflation rate is in the range of 1% to 5%. Such a type of inflation erodes the purchasing power of money but is referred to as manageable and inevitable in a growing economy.

Trotting Inflation - It occurs when the inflation rate is in the range of 5% to 10%.

Galloping Inflation - This is very high inflation running in the range of double or triple digits (i.e. 20% or 200% a year). It is also known as hopping, jumping, running, or runaway inflation.

Hyperinflation - Refers to a situation of exceptionally rapid and uncontrolled price increases.

Bottleneck Inflation - This inflation takes place when the supply falls drastically

and the demand remains at the same level.

Core Inflation - Core Inflation calculates the price of all the goods and services in the economy excluding the energy and food items. Such a measure doesn't include the volatile items, which may distort the true picture of inflation in the economy.

Headline Inflation- A measure of the total inflation within the economy, including commodities such as food and energy prices(oil and gas), which tend to be much more volatile and prone to inflationary spikes.

Economic Concepts Related to Inflation

Inflationary Gap - Excess of total government spending above the national income (i.e. fiscal deficit). It is done to increase the production level which ultimately increases the price because of the extra-creation of money. It results in over-employment.

Deflationary Gap The shortfall in total spending of the government (i.e. fiscal surplus) over the national income. It is a situation of producing more than the demand. It is also known as the **output gap**.

Phillips curve - It is a graphic curve according to which there is an inverse relationship between inflation and unemployment. Higher inflation is associated with lower unemployment and vice versa.

Reflation - This is a situation often deliberately brought by the government to reduce unemployment and increase demand by going for higher levels of economic growth. Government goes for higher public expenditure, tax cuts, interest rate cuts, etc.

Stagflation - It refers to a situation in an economy when inflation and unemployment both are at high levels. It is a combination of high inflation and low growth.

Deflation - When there is a decrease in the prices of goods and services over a period of time. It is the opposite of inflation. Deflation occurs when the inflation rate falls below 0%.

Disinflation - It is defined as a slowdown in the inflation rate.

Base Effect - It refers to the impact of the rise in the price level (i.e. last year's inflation) in the previous year over the corresponding rise in the price levels in the current year (i.e. current inflation).

Inflation Targeting - It means that the central bank of a country targets to contain inflation at a certain level or between a certain range and uses its monetary policy to achieve that target within a particular time frame. The 'healthy range of inflation' is to be 2 - 6%.

Effects of Inflation

During inflation, **lenders suffer** and **borrowers benefit** as real purchasing power declines. The lenders who have lent money at a fixed interest rate suffer as the value of the money they receive back decreases. Whereas During Deflation (when inflation falls), the opposite happens. During inflation investment in the country is boosted.

Inflation **erodes the purchasing power of money**, meaning that with the same amount of money, individuals can buy fewer goods and services so consumption falls.

The **currency** of the economy **depreciates**. Inflation leads to increase in imports and decrease in exports as the prices of the products rise in the domestic market making it less attractive for export.

Inflation can **affect the real value of taxes**. When prices rise, individuals may move into higher tax brackets, leading to increased tax revenue for the government.

Holding money might result in loss. People put the maximum in banks in savings accounts as savings rates increase. But this happens in the short run. In the longer run, it depletes the value.

Indices to Measure Inflation

Wholesale Price Index (WPI) - It represents the price of various goods at a wholesale stage. The base year of WPI has been changed to **2011-12** and the number of items has been increased to 697. The index is used by RBI to decide **repo rate and measure inflation**. It is published by the Office of Economic Adviser, Ministry of Commerce and Industry.

$$WPI = \frac{\text{Current Price}}{\text{Base Period Price}} \times 100$$

Limitations of WPI - WPI does not take into account services such as Health, IT, Education, transport. It does not account for the products of the unorganized sector in India, which constitutes about 35% of the manufactured output of the Indian economy.

Consumer Price Index (CPI) - A Price Index computed using a bundle that is meant to represent the 'market basket' purchased by a typical consumer. CPI is calculated by taking into account price changes for each item in the basket of goods and averaging them (monthly basis).

In India, RBI uses CPI (combined) released by the Central Statistical Office, with the base year as 2012.

Analysis of WPI and CPI - WPI is purely a commodity index and does not take into account services for measuring inflation, whereas CPI includes both commodities and services, though the number of commodities covered by it is much smaller as compared to WPI.

Index of Industrial Production (IIP) - Index of Industrial Production is a composite indicator that measures the short-term changes in the volume of production of a basket of industrial products during a given period with respect to that in a chosen base period.

National Statistical Office (NSO), Ministry of Statistics and Programme Implementation compiles and publishes it monthly.

Broad sectors are Mining, Manufacturing, and Electricity. Use-based sectors are Basic Goods, Capital Goods, and Intermediate Goods. The base year for the Index of Industrial Production is **2011-2012**.

The eight Core Industries in decreasing order of their weightage- Refinery Products> Electricity> Steel> Coal> Crude Oil> Natural Gas> Cement> Fertilizers.

Budget and Economic Survey

Budget is an annual financial statement mentioned in Article 112 (or Article 202 for state budgets), outlines estimated revenues and expenditures for the

upcoming financial year (1 April to 31 March). It shows projected expenditure and receipts for the period.

Historical Background

Budget word comes from a French word 'bougette' which means a leather bag or suitcase.

It was first introduced in India on April 7, 1860, when Scottish economist and politician James Wilson from the East India Company presented it to the British Crown.

Independent India's first budget was presented on November 26, 1947 by the then Finance Minister RK Shanmukham Chetty.

The Indian Union Budget is prepared by the Department of Economic Affairs under the Ministry of Finance.

So far, **Morarji Desai** has presented the maximum number of budgets (10 times), followed by P Chidambaram (9 times) and Pranab Mukherjee and Yashwant Sinha (8 times).

Mr. Arun Jaitley, the Minister of Finance in the BJP led NDA Government under the Prime Ministership of Mr. Narendra Modi, changed the date of the Annual Budget from 2017 to the first working day of February.

2021-22 year's budget was the first budget to be presented as a paperless budget and It was presented by Finance Minister Nirmala Sitharaman. In 2021, Uttar Pradesh became the first Indian state to present a paperless budget.

The Government of India approved the merger of the Railway Budget with the General Budget on September 21, 2016 with effect from the 2017-18 budget. A separate budget for Railways was started in 1924. It was done on the recommendations of the Acworth Committee.

In **2005-06** the first gender Budgeting Cell was introduced in parliament. In December 2004, the Department of Economic Affairs, Ministry of Finance, issued instructions to all Ministries and Departments to establish a '**Gender Budgeting Cell**' by 1st January 2005.

Along with the Budget, the Finance Minister also places the Macro-Economic Framework Statement in Parliament as per the Fiscal

Responsibility and Budget Management Act, 2003. It assesses the country's GDP growth, external balance and fiscal balance.

Objectives of Government Budgets - Reallocation of Resources, Reducing inequalities in income and wealth, Economic Stability, Management of Public Enterprises, Economic Growth, Reducing regional disparities, Balanced regional growth.

Functions of Budget

Allocation Function of Government Budget - Government provides certain goods and services which cannot be provided by the market mechanism (by exchange between individual consumers and producers). These are referred to as public goods. Examples of such goods are national defence, roads, government administration etc.

Redistribution of income - To close the income gap between rich and poor, several budgetary schemes are launched from the government's end. Fiscal instruments like subsidies, taxations, etc. are effectively used to achieve this goal.

Stabilisation function - It is a strategy implemented by the government of a nation to ensure that the economy is steady, it helps to reduce price fluctuations in an economy through the implementation of certain measures and monitoring the economic cycle.

Interim Budget and Vote on Account

Interim Budget - During an election year, the incumbent government cannot present a full budget owing to which the finance ministers present an interim Budget that covers the government's expenses and revenues for a short period. This is till a new government is elected and takes charge.

Vote on Account - According to Article 116 of the Indian constitution, vote on account is a grant in advance for the Central government to meet short-term expenditure, generally lasting for a few months till the new financial year starts.

Vote on Account and Interim Budget are different as a vote-on-account contains only the expenditure of the government whereas the interim budget deals with both receipts and expenditure.

Types of Budget

Gender Budgeting - It is defined as a gender-based assessment of Budgets, incorporating a gender perspective at all levels of the budgetary process and restructuring revenues and expenditures in order to promote gender equality". India introduced Gender-Responsive Budgeting (GRB) in 2005-06 to design public spending in a way that ensures that the benefits accrue as much to the country's women as they do to the men.

Zero based budgeting (ZBB) - A method of budgeting in which all expenses are evaluated each time a Budget is made and expenses must be justified for each new period. It was introduced by Peter Pyhrr in 1970's. It was first introduced in the US. In India, the ZBB was introduced in the Department of Science and Technology in 1983, and adopted by the Indian Government in 1986. Maharashtra is the first state to implement Zero-based budgeting.

Line-Item Budget - In this, expenditures for the budgeted period are listed according to objects of expenditure, called "line items".

Performance Budget - It uses performance information for the allocation of funds and management of a program.

Outcome Budget - It focuses on specific outcomes. The first outcome budget in India was passed by the Parliament on August 25, 2005.

Components of Budget

A Government Budget consists of Two parts - Revenue Budget and Capital Budget.

Revenue Budget consist of - Revenue Receipt and Revenue Expenditure.

Revenue receipts - It consists of the money earned by the government through direct or indirect tax (such as excise duty, income tax, GST) and non-tax sources (such as dividend income, profits, interest receipts).

Tax revenue is a compulsory payment that is made to the government by the people or the companies without having any direct benefit in return. They are either from direct taxes or indirect taxes.

The 5 major sources of tax revenue are Income tax, Goods and Services Tax

(GST), corporate tax, Customs duties, and Union Excise duties.

Non-Tax Revenue is the recurring income that is earned from sources other than taxes by the government. Profit and Dividends accrued from public sector enterprises are the parts of non-tax revenue. Examples are income from interest, dividend, fees, fines, etc

Revenue Expenditure is expenditure incurred for purposes other than the creation of physical or financial assets of the central government. It relates to those expenses incurred for the normal functioning of the government departments and various services, interest payments on debt incurred by the government, and grants given to state governments and other parties (even though some of the grants may be meant for creation of assets).

Capital Budget consist of - Capital Receipt and Capital Expenditure.

Capital Receipt - This is income received from the sale of non-current assets of the business, such as the proceeds received from selling a motor vehicle.

Capital expenditure refers to government spending that creates physical or financial assets or reduces liabilities. It includes expenses like land acquisition, building construction, and investments in shares. Examples include school construction and military equipment purchases.

GFCF - Gross fixed capital formation refers to the net increase in physical assets (investment minus disposals). It does not account for the consumption (depreciation) of fixed capital. GFCF is an indicator for gauging the fixed capital formation.

Plan and Non Plan Expenditure

Plan Expenditure - The expenditure incurred on the programmes that are detailed in the five-year plan of a country. For example expenditure on electricity generation, irrigation and rural developments, construction of roads, bridges, canals, etc.

Non-plan Expenditure - All expenditures other than plan expenditure were known as non-plan expenditure. It is of 2 types.

Non-plan revenue expenditure- Includes expenses like interest payments, subsidies (especially on food and fertilisers), wages and salaries for

government employees, defence spending, grants to state and union territory governments, pensions, police, tax collection, and social services.

Non-plan capital expenditure - It includes defence, loans to public enterprises, loans to States, Union Territories and foreign governments.

The Indian Government declared the removal of the classification of plan expenditure and non-plan expenditure in the fiscal year 2017-18 on the recommendations of C Rangarajan Committee 2011.

Deficit

A deficit can occur when a government, corporation, or person spends more than earned in a given period, which is usually a year.

Types of Deficit in India

Budget deficit occurs when total expenditure exceeds total receipts.

Fiscal deficit is defined as the excess of total expenditures over the total receipts, excluding the borrowings in a year. In other words, this can be defined as the amount that the government needs to borrow in order to meet all expenses.

$\text{Fiscal deficit} = \text{Total expenditures} - \text{Total receipts excluding borrowings}$

If both the primary deficit and interest payments are doubled then the fiscal deficit increases by 100 %.

Zero fiscal deficit is a fiscal deficit that is estimated as the difference between total expenditure and total receipts. The Gross fiscal deficit is also estimated as the sum total of borrowing from RBI. The fiscal deficit in the economy will be zero if there is no provision for borrowings in the budget.

Revenue deficit - When the government's revenue expenditure exceeds its revenue receipts.

$\text{Revenue deficit} = \text{Total revenue expenditure} - \text{Total revenue receipts}$

Effective Revenue Deficit is the difference between revenue deficit and grants for the creation of capital assets. Every year the Central Government gives grants to State Government and Union Territories and with the help of these grants both create capital assets

however these capitals are not added to the capital expenditure of the central government.

The primary deficit is calculated by subtracting the interest payments on previous borrowings from the fiscal deficit of the current year.

$\text{Gross primary deficit} = \text{Gross fiscal deficit} - \text{Net interest liabilities}$

Zero primary deficit means the government resort to borrowing only to clear the existing backlog of interest payments. It is a sign of fiscal discipline or fiscal responsibility on the part of the government. A high primary deficit, on the other hand, reflects the fiscal irresponsibility of the government.

A budget is said to be a **deficit budget** if the estimated government expenditure exceeds the expected government revenue in a particular financial year.
 $\text{Budget deficit} = \text{Budget expenditure} - \text{Budget receipt}$

A budget is said to be a **surplus budget** if the expected government revenues exceed the estimated government expenditure in a particular financial year.

A budget is said to be a **balanced budget** if the estimated government expenditure is equal to expected government receipts in a particular financial year.

Economic Survey

Economic Survey is the flagship annual document of the Ministry of finance. It gives a detailed account of the various sectors of the economy and overall economic scenario of the country in the past years and provides an outline for the year ahead.

In 1951 the Economic Survey of India was presented as a part of the Union Budget for the first time in India. It was presented by RK Shanmukham Chetty (First Finance Minister of India). Up to 1964, it was presented along with the Union Budget. From 1964 onwards, it had been delinked from the Budget.

The **Department of Economic Affairs**, Ministry of Finance presents the Economic Survey of India in the Parliament every year, just before the Union Budget. It is prepared under the guidance of the Chief Economic Adviser attached to the Ministry of Finance.

The **Economic Survey 2017-18** is printed in pink colour to underline the importance of gender issues to the economy. The survey points out that the North-Eastern states (a model for the rest of the country) consistently out-perform others.

The **Economic Survey 2020** coined a new term called 'Thalinomics' and the government pitched the term as "economics for the common man".

Basically the survey attempts to calculate the cost that an average worker incurs based on his actual plate of food in India. And on calculating that cost, the survey concludes that workers were able to save due to moderation in prices of items that form part of a regular Indian thali.

Other Terms of Budget

Government borrowing is a loan taken by the government and falls under capital receipts in the Budget document. Usually, the Government borrows through issue of government securities called G-secs and Treasury Bills.

Disinvestment - An act of selling or liquidating of assets. The process of dilution of a government's stake in a PSU (Public Sector Undertaking) is disinvestment. It allows the transferring of the government's enormous public debt of PSU to the private sector. Disinvestments, in most cases, are primarily motivated by the optimisation of resources to deliver maximum returns.

Income statement of an organisation or business entity is the financial statement which contains financial information about the three important components, which are revenues, profit or loss and expenses incurred during the accounting period.

A **balance sheet** is known as a statement of financial position as it shows the position of assets, liabilities and equity at the end of an accounting period.

Cash flow statement reveals the movement of cash in an organisation. It comprises cash inflows and outflows. Cash flow can be classified into three activities - operating activities, investing activities and financing activities.

A **budget line** shows the possible combinations of two commodities within a given income and price range, all

offering the same level of satisfaction. It must ensure that the cost of each combination doesn't exceed the consumer's income. Also known as Price Line or Budget Constraint Line, it depicts the constraints consumers face when making purchasing decisions.

Budget set is the collection of all bundles that the consumer can buy with her income at the prevailing market prices.

Taxation System of India

The origin of the word 'Tax' is from '**Taxation**' which means an estimate. Tax is a payment compulsorily collected from individuals or firms by the government.

Major types of Taxes

1. Direct tax - It can be defined as a tax that is paid directly by an individual or organization to the imposing entity. It is levied on the income or profits of an individual or the company. The liability of payment and the burden of the tax falls on the same person. It is Progressive in nature. For Higher Income, higher tax and for Lower Income, less or no tax. The Direct Tax Code Bill introduced in the Parliament of India in 2010.

Direct Tax example - Income tax, Corporate tax, Minimum alternate tax and wealth tax.

Income tax - It is levied on the income of individuals, Hindu undivided families, unregistered firms, and other associations of people. In India, the nature of income tax is progressive. For taxation purposes income from all sources is added and taxed as per the income tax slabs of the individual.

The Taxation Laws (Amendment) Bill, 2021 was introduced in Lok Sabha by the Minister of Finance, Nirmala Sitharaman, on August 5, 2021 to amend the Income Tax Act, 1961 (IT Act) and the Finance Act, 2012.

The 2012 Act had amended the IT Act to impose tax liability on the income earned from the sale of shares of a foreign company on a retrospective basis. The Bill proposes to nullify this retrospective basis for taxation.

The Income Tax Act, 1961 is an act to levy, administer, collect, and recover income tax in India. The act is effective from 1 April 1962. It consists of 298 sections and 14 schedules. The act helps

determine a taxpayer's taxable income, tax liability, appeals, penalties, and prosecution.

ITNS (Income tax new series) Challan - It is a form used for paying tax to the government. Tax paid through a challan is first deposited with the authorised banks, and subsequently with the Income Tax department. Payment through a challan can be either made online via net banking or offline by submitting the challan at the designated bank branches.

ITNS 281 Challan form is issued when the taxpayer deposits Tax Deducted at Source (TDS) and Tax collection at source (TCS).

ITNS 282 Challan form is filled to deposit gift tax, estate duty, expenditure tax, and other miscellaneous tax.

ITNS 283 Challan is used for payment of Banking Cash Transaction Tax and Fringe Benefit Tax.

ITNS 284 Challan is used for payment of Income Tax on Companies (Corporation Tax) and Income Tax on other than Companies.

Income Tax Slab - It is defined as the Individual taxpayers will need to pay the income tax based on the slab system which they fall under. Depending on the income of the individual, he or she may fall under a different tax slab. Therefore, individuals with a higher income will need to pay more tax. The slab system was introduced to maintain a fair tax system in the country.

Income Tax Slabs and rates for the **FY 2024 - 2025**- Rs0.0 – Rs 3.0 lakh (Nil), Rs.3 lakh – Rs.6 lakh (5%), Rs.6 lakh - Rs.9 lakh (10%), Rs.9 lakh - Rs.12 lakh (15%), 12 lakh - Rs.15 lakh (20%). Above Rs.15 lakh (30%).

Corporation tax - In India, companies registered under the Companies Act, 1956, are subject to corporation tax, a levy on their net profits. This applies to both public and private companies. The tax is calculated by subtracting allowable expenses like operating costs and depreciation from the company's total revenue, resulting in a taxable net profit. Corporation tax serves a dual purpose - generating government revenue and promoting wealth redistribution by placing a higher tax burden on more profitable companies, thereby helping to reduce income inequality.

Corporate tax includes the following (MAT and FBT) taxes -

Minimum Alternative Tax (MAT) - It is imposed by the Government on Zero Tax companies. The accounts of these companies are made according to the Companies Act.

Fringe benefit tax (FBT) - It is a type of tax, which is levied on the employees for the other benefits that they receive from their employers as salary. This consists of things such as tickets, reimbursements, superannuation contribution, company vehicles etc. The tax was introduced in the financial year 2005-06.

Originally, the Finance Act of 2009 abolished the fringe benefit tax in India, which was previously a levy imposed on business owners or employers. However, this abolishment only came into effect from the financial year 2010-11 onwards.

Dividend Distribution Tax (DDT) - It is the tax imposed by the Government on domestic companies which pay dividends to their investors. A dividend is a return given by a company to its shareholders out of the profits earned by the company in a particular year. In Budget 2020, the Finance Minister abolished the Dividend Distribution Tax (DDT).

Securities Transaction Tax (STT) - It is a levy imposed on the buying and selling of securities traded on Indian stock exchanges. These securities, which can be bought and sold, are financial instruments like shares (stocks), corporate and government bonds, debentures, and equity-oriented mutual funds. Introduced in the 2004 Union Budget, STT came into effect on October 1, 2004. Notably, both the buyer and seller pay this tax, typically at a rate of 0.1% of the transaction value.

Advance Tax - Mechanism requires people to pay their income tax and Corporation tax in advance installments on a quarterly basis if their annual tax liability is Rs. 10,000.

Wealth tax - It is imposed on the value of the Asset that a person possesses which can take several forms, such as property tax, inheritance or gift tax and capital gains tax. It is, therefore, decided to abolish the levy of wealth tax under the Wealth-tax Act, 1957 with effect from the 1st April, 2016.

Estate duty - It is a type of tax which is calculated at the time of death of an individual whose property is going to be acquired by his/her successors. It is also called inheritance tax.

It is a liability on the people who will inherit their parents property, they have to pay estate duty as a tax. Estate duty was introduced in India in the Estate duty act 1953 but was abolished by the Rajiv Gandhi Government in 1985.

Gift tax - It is a provision introduced by the Parliament of India in 1958. It was introduced to impose tax on giving and receiving gifts under certain circumstances which is specified under the act. These gifts can be in any form including cash, jewellery, property, shares, vehicle, etc. Though gift tax is applicable on gifts whose value exceeds Rs.50,000, the gift is exempted from tax if it was given by a relative.

Capital gain tax - It is levied on the profits made on investments. It covers real estate, gold, stocks, mutual funds, and various other financial and non-financial assets. Under the Income Tax Act, gains from sale of capital assets both movable and immovable are subject to 'capital gains tax'.

Types of Capital gains tax -

STCG (Short-term capital Gain) - An asset held for a period of 36 months or less is a short-term capital Gain.

LTCG (Long-term capital Gain) - An asset held for more than 36 months is a long-term capital Gain. Assets held over 24 months (land, property) or 1 year (shares) qualify for long-term capital gains tax rate (usually 10%).

2. Indirect tax - An Indirect Tax is a tax collected by an intermediary (such as a retail store) from the person who bears the ultimate economic burden of the tax (such as the customer).

Example - Excise duty, Customs duty, Sales tax, VAT, and GST (goods and services tax).

Excise duty - It is a form of tax imposed on goods for their production, licensing and sale. Excise duty was levied on manufactured goods and levied at the time of removal of goods.

At the central level, excise duty earlier used to be levied as Central Excise Duty, Additional Excise Duty, etc.

The GST introduced in July 2017 subsumed many types of excise duty. Now excise duty applies only on **petroleum and liquor**.

Customs Duty - It is a tax collected on imports and exports by the customs authorities of a country. It is usually based on the value of goods that are imported. Depending on the context, import duty may also be referred to as tariff, import tax, customs duty, and import tariff. Moreover, it helps reduce fraudulent activities and the circulation of black money. Customs duty in India falls under the Customs Act 1962 and Customs Tariff Act of 1975.

India's tariff system is based on the **Harmonised System of Nomenclature (HSN)** of the Customs Co-operation Council.

Customs duties are divided into different taxes, such as -

Basic Customs Duty - It is levied on imported items that are part of **Section 12 of the Customs Act, 1962**. The tax rate is levied as per First Schedule to Customs Tariff Act, 1975.

Additional Customs Duty - It is levied on goods that are stated under Section 3 of the **Customs Tariff Act, 1975**. The tax rate is more or less similar to the Central Excise Duty charged on goods produced within India.

Protective Duty - This is levied for the purpose of protecting indigenous businesses and domestic products against overseas imports. The rate is decided by the Tariff Commissioner.

Paper tax - Taxes, such as gift tax Estate duty, and wealth tax in India which carry their significance only on paper and do not have significance in terms of revenue yield are referred to as paper taxes.

Cess is an additional tax on tax in basic terms. This tax is imposed on all taxpayers. Cess must only be applied to the purpose for which it is levied and is paid to the Consolidated Fund of India. For example, the Indian government collects an education cess and uses it solely for education.

Health and education cess - It is tax collected by the government of India with an aim to improve the healthcare and educational facilities in the deprived areas of rural and semi-urban India.

Health and education cess was introduced by late Finance Minister Arun

Jaitley in Union Budget 2018 replacing the earlier secondary and higher education cess of 3 per cent. Health and Education Cess is levied at the rate of 4% on the amount of income-tax plus surcharge.

Anti-dumping Duty - Anti - dumping duty is a tariff imposed on imports manufactured in foreign countries that are priced below the fair market value of similar goods in the domestic market.

Safeguard Duty - This is levied if the customs authorities feel that the exports of a particular good can damage the economy of the country.

VAT - It is a common form of indirect tax levied on services and goods. It is paid to the government by the producers at every stage in the supply chain. Haryana was the first state to introduce VAT in 2003. VAT was introduced into the Indian taxation system from 1 April 2005.

VAT = Output Tax - Input Tax

Service tax - It is a tax levied by the government on service providers on certain service transactions, but is actually borne by the customers. It is categorised under Indirect Tax and came into existence under the Finance Act, 1994.

Stamp Duty and Registration taxes are a major tax share in states in India. Stamp duty is a tax imposed on the sale of property or property ownership by the state government. It is payable under Section 3 of the Indian Stamp Act, 1899 and Registration fee is the cost levied by the government to record the property details. However, the stamp duty and registration charges depend on the ownership type and the transaction value of the property.

Commodity transaction Tax (CTT) - It is introduced to tax commodity trading in India where both parties—buyer & seller of contract—will be taxed depending on the amount of the contract size.

Securities Transaction Tax (STT) - A tax payable in India on the value of securities (excluding commodities and currency) transacted through a recognized stock exchange levied on the purchase and sale of equities in the stock market.

Pigovian tax - It is a tax on any market activity that generates negative impacts on the environment or society. Examples

include tobacco taxes, sugar taxes, and carbon taxes.

A carbon tax is a tax levied on the carbon content of goods and services.

Tobin tax - It is a duty proposed on spot currency trades to penalize short-term currency trading in order to stabilize markets and disincentive speculation, but not investment flows.

Goods and Services Tax (GST)

The Goods and Services Tax (GST), which has replaced the Central and State indirect taxes such as VAT, excise duty and service tax, was implemented from 1st July 2017.

According to the 101st Constitutional amendment act 2016, Goods and Services Tax (**One Nation- One Market- One Tax**) is an indirect tax used in India on the supply of goods and services. The Goods and Service Tax was first introduced in France in 1954.

Four tax slabs under GST viz. 5%, 12%, 18%, and 28%. However, petroleum products, alcoholic drinks, and electricity are not taxed under GST and instead are taxed separately by the individual state governments. Tobacco is the product on which both GST and central excise duty are applicable in India.

Assam is the first state of India to pass the GST Bill. Puducherry became the First Union Territory to pass the goods and services tax bill in 2017.

The **3 types of GST** are CGST (central level), IGST (inter-state level), and SGST (state level).

Central level taxes merged under GST are Central Excise Duty and Additional Excise Duty, Service Tax, Additional Customs Duty commonly known as Countervailing Duty, Special Additional Duty of Customs, and Central Surcharges and Cess.

State taxes subsumed under GST are Value Added Tax (VAT), Entertainment Tax, Central Sales Tax, Octroi and Entry tax, Purchase Tax, Central Sales Tax, Luxury tax, and Taxes on lottery, betting, and gambling.

GST Council - It is the governing body of GST. It has 33 members, out of which 2 belong to the union and 31 members are from states and Union territories (with legislature). As per Article 279A (4), the Council will make recommendations to

the Union and the States on important issues related to GST, like the goods and services that may be subjected or exempted from GST, model GST Laws, principles that govern Place of Supply, threshold limits, GST rates including the floor rates with bands, special rates for raising additional resources during natural calamities/disasters, special provisions for certain States, etc. **The Union Finance Minister acts as the chairperson of the GST council.**

Goods and Services Tax Identification Number (GSTIN) - Under GST a service tax registration number by the Centre is given to each and every service Board of Indirect Taxes and Customs (CBIC) provider. A GSTIN number is a 15-digit code that identifies a GST registered business in India.

CBIC - Central Board of Indirect Taxes and Customs (erstwhile Central Board of Excise and Customs). It deals with the tasks of formulation of policy concerning levy and collection of customs, central excise duties, Central Goods & Services Tax (CGST) and Integrated GST (IGST).

Other types of Tax

Proportional Income Tax- It is an income tax system that levies the same percentage tax to everyone regardless of income.

Professional tax is a tax that is levied by a state government on all individuals who earn a living through any medium.

Progressive tax - It is a tax system that collects a greater share of income from those with high incomes than from those with lower incomes.

Proportional tax - It is an income tax system that helps to reduce the autonomous expenditure multiplier because taxes reduce the MPC (Marginal Propensity to Consume). It acts as an automatic stabiliser in the economy.

Regressive tax - It is the one in which the tax rate decreases as the amount subject to taxation increases; and the tax rate progresses from high to low.

Reserve Bank of India

The Reserve Bank of India (RBI) is India's central bank and regulatory body responsible for regulation of the Indian banking system. It is under the ownership of the Ministry of Finance.

Historical Background

RBI was established on **1 April 1935**. It was set up on the basis of the recommendations of the 'Hilton Young Commission' (1926) and RBI Act 1934.

The Central Office of the Reserve Bank was initially established in **Calcutta** but was permanently moved to **Mumbai** in 1937. RBI was nationalised with effect from 1st January, 1949. Insignia of the Reserve Bank of India consists of a tiger and a palm tree.

The Reserve Bank of India also acted as the central bank for Burma till April 1947.

As per the RBI Act 1934, the objective of RBI are as follows -

To run the nation's currency and credit System.

Maintaining monetary reserves to achieve monetary stability in India.

To govern the issue of bank notes.

To maintain financial stability or credit by engaging in effective activities and keeping itself free from any political impact.

To perform central banking functions by acting as Banker's bank, Banker to government, and note-issuing authority.

To promote economic growth and support planned advancement of the economy of the country.

Functions of Reserve Bank of India

The Monetary Authority steers the economy by implementing policies for price stability and balanced growth.

The RBI acts as the regulator and administrator of the financial system, overseeing crucial aspects like license issuance and asset liquidity

Custodian of Foreign Exchange Reserves - The Reserve Bank sells and buys foreign currencies to keep foreign exchange rates stable.

Issuer of Currency - RBI is in charge of issuing and exchanging of currency and coins. In case of notes, the Reserve Bank follows the Minimum Reserve System.

Regulator and Supervisor of Payment and Settlement Systems- Introduces and upgrades safe and efficient modes of

payment systems in the country to meet the requirements of the public at large.

Banker Function - The RBI manages public debt on behalf of the Central and the State Governments.

RBI's Developmental role- RBI organises promotional functions to support national objectives, encouraging rural and agricultural development. It directs commercial banks to lend to small-scale industries.

Central Board of RBI

The Central Board of Directors of the RBI (Reserve Bank of India) is appointed /nominated for a period of **four years**. As of its Preamble the Deputy Governor of RBI can't be more than 4.

List of governor of RBI - Sir Osborne Smith (Overall 1st), Sir James Taylor, Sir C D Deshmukh (1st after independence and 1st Indian), Sir Benegal Rama Rau, K G Ambegaonkar, H V R Iyengar, P C Bhattacharya, L K Jha, B N Adarkar, S Jagannathan, N C Sen Gupta, K R Puri, M Narasimham, Dr. I G Patel, Dr. Manmohan Singh, A Ghosh, R N Malhotra, S Venkitaramanan, Dr. C Rangarajan, Dr. Bimal Jalan, Dr. Y V Reddy, Dr. D. Subbarao, Dr. Raghuram Rajan, Dr. Urjit R. Patel, **Shri Shaktikanta Das (25th)**.

Important Sections related to RBI Act, 1934

Section 20 - Obligation of the Bank to transact Government business.

Section 22 of Reserve Bank of India Act 1934 grants power to RBI to issue the currency.

Section 24 - Denominations of notes.

Section 25 - The design, form and material of bank notes.

Section 27 - Re-issue of notes.

Section 32 - Penalty (Omitted by the Reserve Bank of India (Amendment) Act, 1974).

Monetary policy committee

The Monetary Policy Committee (MPC) is a committee in India responsible for fixing the benchmark interest rate in India. The meetings of the Monetary Policy Committee are held at least four times a year (specifically, at least once a quarter).

The committee comprises six members, three officials of the Reserve Bank of India (RBI) and three external members nominated by the government of India. The MPC determines the policy repo rate required to achieve the inflation target.

Some Initiatives of RBI

The Lead Bank Scheme, introduced in 1969 by RBI based on recommendations from the Gadgil Study Group and Banker's Committee, aims to provide banking and credit in rural areas via a 'service area approach,' assigning one bank per area.

Indian Financial Technology & Allied Services (IFTAS), a wholly-owned subsidiary of RBI, was established in February 2015 by Institute for Development and Research in Banking Technology (IDRBT), funded by RBI, following the Rangarajan Committee's (2009) recommendation. It provides essential IT services to RBI, banks, and financial institutions.

Jammu and Kashmir, State Government transactions are carried out by RBI in terms of the agreement entered into with the State Governments in terms of section 21A of the Act only till 2019. As of now, such agreements exist between RBI and all the State Governments except Sikkim.

RBI liberalised the licensing policy for Urban Co-operative Banks in 1993.

The '**Clean Note Policy**' was announced by the Reserve Bank of India in the year 1999. The objective of the Reserve Bank's Clean Note Policy is to give the citizens good quality currency notes and coins while the soiled notes are withdrawn out of circulation. In 2002, under the clean note policy, the RBI advised banks to stop stapling currency notes.

The Reserve Bank of India unveiled Payments Vision 2025 with intent to increase e-payments. The vision document has five key themes- Integrity, Inclusion, Innovation, Institutionalisation and Internationalisation.

Banks

Bank is a financial institution that accepts deposits from the public and creates credit. Lending activities can be

performed either directly or indirectly through capital markets.

Three Presidency banks were set up under charters from the British East India Company- Bank of Calcutta (1806), Bank of Bombay (1840) and Bank of Madras (1843). These banks worked as quasi central banks in India for many years.

Schedule and Non-scheduled banks

Schedule banks are those that are listed in the Second Schedule of the RBI Act, 1934 and thus meet the following requirements- a bank must have a paid-up capital and reserve of at least Rs. 5 lakh and a bank must satisfy the Reserve Bank of India (RBI) that its affairs are not conducted in a manner that is detrimental to the interest of its deposits.

Non-scheduled banks are those that are not listed in the second schedule of the RBI Act, 1934. The term "scheduled banks" refers to both "scheduled commercial banks" and "scheduled cooperative banks."

Commercial bank - A commercial bank is a financial institution that provides services like loans, certificates of deposits, saving bank accounts, bank overdrafts, etc. to its customers. These institutions make money by lending loans to individuals and earning interest on loans. Credit creation is the function performed by Commercial Banks.

Various types of loans given by a commercial bank are business loans, car loans, house loans, personal loans, and education loans. Agency function and credit creation are also notable works of commercial banks. Commercial banks help mobilize savings through a banking network. The 1949 Banking Regulation Act governs commercial banks in India.

There are three main types of commercial banks- public sector banks, private banks, and foreign banks.

Public sector banks (PSBs) - Those banks in which the government owns more than 50% of the stock. The government regulates the financial guidelines for these banks.

Imperial bank of india - In 1921, the three presidency banks were amalgamated to form Imperial Bank of India. In 1955, this Imperial Bank of India was nationalized

and renamed as **State Bank of India (SBI)**. Thus, SBI is the oldest and largest public sector Bank of India among the banks that exist today.

Banking Merger in India

Finance Minister Nirmala Sitharaman announced the merger of 10 Public Sector Banks into four on August 30, 2019. The union cabinet approved the merger on March 4, 2020, effective from April 1, 2020. As of February 2024, there are 12 Public Sector Banks.

List of Merger banks -

Punjab National Bank - Oriental Bank of Commerce + United Bank of India

Canara Bank - Syndicate Bank

Indian Bank - Allahabad Bank

Bank Of Baroda - Dena Bank + Vijaya Bank

Union Bank of India - Andhra Bank + Corporation Bank

State Bank of India (SBI) - State Bank of Bikaner and Jaipur (SBBJ) + State Bank of Hyderabad (SBH) + State Bank of Mysore (SBM) + State Bank of Patiala (SBP) + State Bank of Travancore (SBT) + Bharatiya Mahila Bank

Important Banks of India

State Bank of India (SBI) - The State Bank of India Act, 1955, established the State Bank of India by converting the then-existing Imperial Bank of India into the State Bank of India. SBI headquarter in Mumbai. The tagline of the bank is "The Banker to Every Indian", "The Nation Banks on us", "A Bank of Common Man", "With You all the Way", "Pure Banking Nothing Else".

Yono is the official app of SBI. SBI Life Insurance is a joint venture life insurance company between State Bank of India (SBI) and the largest state-owned banking French financial institution BNP Paribas Cardif.

Union Bank of India (UBI) - Union Bank of India was established on 11th November 1919 with its headquarters in Mumbai. It was promoted by Seth Sitaram Poddar. The tagline of the bank is "Good People to Bank with".

Uco Bank (UB) - UCO Bank, formerly United Commercial Bank, was incorporated on January 6, 1943, with the vision of Mr. G.D. Birla established a truly

Indian Bank, which materialized after the 'Quit India' movement. Headquartered in Kolkata, it was nationalized in 1969. The bank's tagline is "Honors your Trust". It became the first bank to get the Reserve Bank of India's approval for rupee trade.

Punjab National Bank - Punjab National Bank (PNB) is an Indian multinational bank headquartered in New Delhi, founded in 1894. Lala Lajpat Rai, a prominent Punjab freedom fighter, opened the first account at PNB's initial office in Lahore. PNB was India's first national and swadeshi bank with the tagline "You can bank upon".

Punjab and Sind Bank - Punjab and Sind Bank, established on June 24, 1908, in Amritsar, Punjab, was founded by Bhai Vir Singh, Sir Sunder Singh Majitha, and Sardar Tarlochan Singh. Headquartered in New Delhi, the bank's tagline is "Where Service is a Way of Life".

Indian Overseas Bank - Indian Overseas Bank (IOB), founded in February 1937, specializes in foreign exchange business and overseas banking. Headquartered in Chennai, Tamil Nadu, its tagline is "Good People to Grow With".

Indian Bank - Indian Bank, established in 1907 and headquartered in Chennai, is a public sector bank founded by S. Rm. M. Ramaswami Chettiar. Its tagline is "Your Own Bank, Banking That's Twice As Good".

Allahabad Bank - Established in 1865 known as India's oldest joint stock bank merged into the Indian bank.

Central Bank of India - Central Bank of India, established in 1911 and known as India's First Swadeshi bank, was founded by Sir Sorabji Pochkhanawala, with Sir Pherozeshah Mehta as its first chairman. This bank is headquartered in Mumbai. its tagline is "Central to You Since 1911, Build a better life around us".

Canara Bank - Canara Bank, founded in July 1906 by Shri Ammembal Subba Rao Pai in Mangalore, Karnataka, is headquartered in Bengaluru. Its tagline is "Together We Can".

Bank of Maharashtra - The bank was founded by V.G.Kale and D. K. Sathe in Pune. The bank was registered on 16 September 1935. The bank is headquartered in Pune. The tagline of the bank is "Ek Parivaar, Ek Bank" (One Family One Bank) .

Bank of India - Bank of India, founded in 1906 by Mumbai businessmen, was private until nationalized in July 1969 along with 13 other banks. It was the first Indian bank to open a branch outside India, in London in 1946. Based in Mumbai, its tagline is "Relationship beyond Banking".

Bank of Baroda - Bank of Baroda, an Indian state-owned international banking and financial services company, headquartered in Vadodara (formerly Baroda) in Gujarat, was established by the Maharaja of Baroda, Sir Sayajirao Gaekwad - III, on 20 July 1908. It was nationalized on 19 July 1969, by the Government of India. Its tagline is "India's International Bank".

Private Sector Banks in India - Private sector banks are those in which private individuals or private corporations own a significant portion of the bank's equity.

AXIS Bank - Founded on December 3, 1993, as UTI Bank, it became Axis Bank in July 2007, headquartered in Mumbai. Its tagline is "Badhti Ka Naam Zindagi". Axis Bank introduced the Iris Scan Authentication feature for Aadhaar-based transactions, boasting a contactless technology with a 98.2 percent authentication success rate according to the International Centre for Biometric Research.

HDFC Bank - HDFC Bank, one of India's largest private banks by total sales, received early approval from the Reserve Bank of India (RBI) to establish in the private sector. Established in 1994, headquartered in Mumbai, its tagline is "We understand your world."

ICICI Bank - Established - 1994, Headquarter - Mumbai, Tagline - Khayaal Aapka.

IndusInd Bank - Established in 2004, Headquarter - Mumbai, Tagline - We make money simple

Yes Bank - Established - 1993, Headquarter - Mumbai, Tagline - Experience our expertise

Bandhan Bank - Bandhan bank is a small micro lender as Bandhan Financial Services, In April 2014, Bandhan Bank commenced operations on August 23, 2015. Headquarter - Kolkata, Tagline - Aapka Bhalai, Sabki Bhalai.

Federal Bank - Established - 1931, Headquarter - Kochi, Kerala, Tagline - Your perfect banking partner.

IDFC First Bank - Established in 2015 - Headquarter - Mumbai, Tagline - Always You First.

Karnataka Bank - Established in 1924 - Headquarter - Mangaluru, Tagline - Your family bank across India.

Kotak Mahindra Bank - Established in 2003 - Headquarter - Mumbai, Tagline - Let's make money simple.

Jammu & Kashmir Bank - Established in 1938 - Headquarter - Srinagar, Tagline - Serving to Empower.

Nainital Bank - Established in 1938 - Headquarter - Nainital, Uttarakhand, Tagline - Banking with personal touch.

RBL Bank - Established in 1943 - Headquarter - Mumbai, Tagline - Apno ka Bank.

Karur Vysya Bank - Established in 1916 - Headquarter - Karur, Tamil Nadu, Tagline - Smart way to bank.

CSB Bank - Established in 1920 - Headquarter - Thrissur, Kerala, Tagline - Support all the way.

City Union Bank - Established in 1904 - Headquarter - Kumbakonam, Tamil Nadu, Tagline - Trust and Excellence since 1904.

DCB Bank - Established in 1930 - Headquarter - Mumbai, Tagline - We Value You.

Dhanlaxmi Bank - Established in 1927 - Headquarter - Thrissur, Kerala, Tagline - Tann. Mann. Dhan.

IDBI Bank - Established in 1964 - Headquarter - Mumbai, Tagline - Banking For all. Aao Sochein bada; Bank Aisa Dost Jaaisa.

South Indian Bank - Established in 1929 - Headquarter - Thrissur, Kerala, Tagline - Experience next generation banking.

Tamilnadu Mercantile Bank - Established in 1921 - Headquarter - Thoothukudi, Tamil Nadu Tagline - Be a step ahead in life.

Small Finance Banks in India - They are the financial institutions which provide financial services to the unserved and unbanked region of the country.

List of (As of March 2024) small Finance banks, Commenced year, and

Headquarters.

ESAF Small Finance Bank (2017) - Thrissur.

Equitas Small Finance Bank (2016) - Chennai.

Capital Small Finance Bank (2016) - Jalandhar

AU Small Finance Bank (2017) - Jaipur.

Suryoday Small Finance Bank (2017) - Navi Mumbai

Ujjivan Small Finance Bank - Bangalore.

Utkarsh Small Finance Bank (2017) - Varanasi

North East Small Finance Bank (2017) - Guwahati.

Jana Small Finance Bank (2018) - Bangalore

Shivalik Small Finance Bank (2021) - Noida.

Unity Small Finance Bank (2021) - Mumbai

Payment Bank - Payment banks are banks that do not take on any credit risk. As of April 2024 Active Payment bank and their Headquarters.

Airtel Payments Bank - New Delhi

India Post Payments Bank - Delhi

Fino Payments Bank - Mumbai

Jio Payments Bank - Navi Mumbai

NSDL Payments Bank - Mumbai

Paytm Payment Bank - Noida

RRBs - The Regional Rural Banks (RRBs) were established based on recommendations of the 'Narasimham Committee' on Rural Credit. The Regional Rural Banks Amendment Act (2014) amended The RRBs Act 1976, stipulating that 50% of the capital issued by an RRB shall be held by the central government, 15% by the concerned state government, and 35% by the sponsor bank. This bill permits RRBs to raise capital from sources other than the central and state governments, as well as sponsor banks.

The first Regional Rural Bank, "Prathama Grameen Bank," was established on October 2, 1975, in Moradabad, Uttar Pradesh.

Banking Committees

Narasimham Committee- I (1991) - The Committee was set up by Finance minister Manmohan Singh, to analyse India's banking sector and recommend reforms, under the chairmanship of Maidavolu Narasimham. He was the 13th governor of the Reserve Bank of India

(RBI) from 2 May 1977 to 30 November 1977.

Narasimham Committee- II (1998) - This committee was set up by P Chidambaram as the finance minister, to review the progress of the implementation of reforms and to suggest a design for further strengthening of the Banking sector.

Rangarajan Committee (1988) - The Reserve Bank of India set up a Committee on computerization in banks headed by Dr. C. Rangarajan. Banks initially started using information technology with standalone PCs and then moved to Local Area Network (LAN) connectivity.

Saraf Committee (1994) - It was formed to find the latest issue with the use of modern technology in the Indian banking sector. The concept of online banking in India was introduced by this committee.

The **Shere Committee (1995)** had recommended a set of EFT Regulations by the Reserve Bank under the Reserve Bank of India Act, 1934 and amendment to the Bankers' Books Evidence Act, 1881 as short term measures and promotion of a few Acts like the Electronic Funds Transfer Act, the Computer Misuse and Data Protection Act etc. as long term.

Pannir Selvam Committee (1998) was appointed to provide a detailed explanation on the non performing assets. Non Performing Assets is a loan according to the Reserve bank of India that is overdue from more than 90 Days.

Nachiket Mor committee (2013) - It was headed by Nachiket Mor. It was formed by former Reserve Bank of India Governor Raghuram Rajan. It is also known as the Committee on Comprehensive Financial Services for Small Businesses and Low Income Households. It recommended Differentiated Banks in India - Payment and Small Finance Banks.

Bimal Jalan Committee (2013) - The committee was constituted by RBI under the chairmanship of Bimal Jalan in 2013. RBI had constituted the committee to examine the criteria, business plans and corporate governance practices of applicants applying for new bank Licences.

The Urjit Patel Committee (2014) was largely responsible for modifying and enhancing the RBI's Monetary Policy

Framework. Raghuram Rajan, the former governor of the Reserve Bank of India (RBI), formed it.

Chandrababu Naidu Committee (2016) - The committee was formed to promote digital payment systems and financial inclusion in India. The committee was headed by Chandrababu Naidu. The union government established a 13-member Committee of Chief Ministers. The Committee submitted its interim report to the Hon'ble Prime Minister

Sunil Mehta Committee (2018) - It was set up to look into the faster resolution of stressed assets. It has recommended the creation of an asset management company for the resolution of stressed loans worth more than Rs.500 crore. The committee had also laid out a plan to resolve SME loans within 90 days.

Bimal Jalan Committee (2018) - Reserve Bank of India (RBI) had constituted a six member "Expert Committee to Review the Extant Economic Capital Framework of the RBI" under the Chairmanship of Dr. Bimal Jalan.

Nandan Nilekani Committee (2019) - The Committee on Deepening of Digital Payments was set up by the Reserve Bank of India under the Chairmanship of Nandan Nilekani.

Karve Committee- Also known as Rural and Small Industries Committee, it was established in 1955. It was established to explore the potential of using small scale industries to promote rural development. It was presided over by Dattatreya Gopal Karve.

Fiscal Policy and Monetary Policy

Fiscal policy - It is the policy under which the government uses the instrument of taxation, public spending and public borrowing to achieve various objectives of economic policy. Fiscal policy is based on the theories of the British economist John Maynard Keynes.

Fiscal year is a year that represents the financial reporting period for the government. The fiscal year period in India is from 1st April of a year to 31st March of the next upcoming year.

Objectives of Fiscal Policy

Reducing Inequality - Government reduces income and wealth disparities by taxing the affluent more and providing subsidies to the less privileged through Direct Benefit Transfer (DBT) schemes.

Creating Employment Opportunities - The government boosts job prospects by establishing public sector enterprises, offering subsidies, tax incentives, and promoting small-scale industries, encouraging employment-oriented ventures.

Promoting Economic Growth - Government initiatives include setting up basic and heavy industries, developing infrastructure such as transportation networks, education, healthcare, and utilities, all of which contribute to economic expansion.

Ensuring Price Stability - To maintain stable prices of essential goods and services, the government regulates their supply through various mechanisms.

Ensuring Effective Administration - Government expenditure on essential services like law enforcement, defense, judiciary, and legislative bodies ensures efficient governance and administration.

Addressing Balance of Payments Deficit - Measures to rectify deficits in the balance of payments involve regulating imports and exports, although import taxes are now less favored due to their potential to hinder international trade flow.

Types of Fiscal Policy

Neutral Fiscal Policy - This policy implies a balanced budget where government spending is equal to tax revenue. It means that government spending is fully funded by tax revenue and overall the budget outcome has a neutral effect on the level of economic activity.

Contractionary Fiscal policy - This policy involves raising taxes or cutting government spending, so that (Government spending < Tax revenue) cuts upon the aggregate demand (thus, economic growth) and reduces the inflationary pressures in the economy.

Expansionary Fiscal Policy - It aims to stimulate the economy by boosting GDP growth or during recessions when national income growth is inadequate. It involves tax cuts and/or increased

government spending to spur economic growth and reduce unemployment rates. However, this policy can lead to budget deficits and is not sustainable, so caution must be exercised when implementing it.

Escape Clause - Provided under the FRBM Act 2003, it allows the government to exceed fiscal deficit target by 0.5%, such as national security threats, calamities, or economic downturns.

Fiscal Instruments - Governments influence the economy using spending and revenue tools. Spending tools involve government expenditure, while revenue tools encompass tax collection.

Monetary Policy

Monetary Policy refers to the measures pertaining to policy undertaken by the Central Bank (RBI) to influence the availability; determine the size and rate of growth of the money supply in the economy. In other words, monetary policy can be defined as a process of managing a nation's money supply to contain/control inflation, achieving higher growth rates, and achieving full employment. The general target of the monetary policy is to ensure price stability in the economy.

In most modern economies, the central bank deals with monetary policy while a government is responsible for fiscal policy.

Types of Monetary Policy

Contractionary monetary policy - It decreases the total money supply in the market when the market is under inflationary pressure. This is also known as Tight Monetary Policy or Hawkish monetary policy.

Expansionary Monetary Policy - It increases the total money supply in the economy which creates more economic activities and results in a sharp rise in GDP, which accelerates the growth of the economy. It is also known as Easy monetary policy or Dovish monetary policy.

Monetary policy committee

There are six members in the RBI's Monetary Policy Committee (MPC) of which 3 are from RBI and 3 are appointed by the Central government. The Members of the Monetary Policy Committee appointed by the Central Government

shall hold office for a period of four years. The Monetary policy committee is required to meet at least four times in a year. Monetary policy committee is chaired by Governor of the Reserve bank of India

Tools of Monetary Policy

Following are the major instruments that RBI uses for conducting its monetary policy, they are of two broad categories - Quantitative and Qualitative Tools .

1. Quantitative Tools - As the name suggests, these are related to the volume of the money supply. They are indirect in nature and influence the quantity of money supply in the economy. Following are the types of Quantitative tools used by RBI -

Cash Reserve Ratio (CRR) - The CRR is the percentage of a bank's total deposits that it must maintain as reserves with the RBI in the form of cash. Banks are not allowed to use this fund for commercial purposes, and the RBI does not pay interest on the CRR balances held by banks. Higher the Cash Reserve Ratio (CRR), lower will be the liquidity in the system. RBI determines the CRR.

One of the main objectives of CRR is to remove excess cash from the economy. The range of this ratio will be from 0-15% which is decided by RBI. CRR is governed by the provisions of Section 42 of the Reserve Bank of India Act, 1934. The current CRR in India is 4.50% (Feb 2024). This means that for every ₹100 deposited by customers, banks have to maintain ₹4.50 with the RBI.

Statutory Liquidity Ratio (SLR) - It is the minimum percentage of deposits that a commercial bank has to maintain in the form of liquid cash, gold or other securities. It is basically the reserve requirement that banks are expected to keep before offering credit to customers. Lower the Statutory Liquidity Ratio (SLR), higher will be the liquidity in the system.

Liquidity adjustment facility (LAF) - It is a tool used in monetary policy, primarily by the Reserve Bank of India (RBI) that allows banks to borrow money through repurchase agreements (repos) or to make loans to the RBI through reverse repo agreements. This arrangement is effective in managing liquidity pressures and assuring basic stability in the financial markets.

The RBI introduced the LAF as a result of the Narasimham Committee on Banking Sector Reforms (1998).

Repo rate - It refers to the rate at which commercial banks borrow money by selling their securities to the Reserve bank of India (RBI). It is one of the main tools of RBI to keep inflation under control. When RBI decides to decrease the repo rate, money supply will increase.

If **inflation** in an economy is rising steadily then the central bank might increase the repo rate. An increased repo rate acts as a disincentive for banks to borrow from the RBI, which results in reduced money supply in the economy and eventually helps control inflation.

Reverse Repo Rate (RRR) - It is defined as the rate at which the Reserve Bank of India (RBI) absorbs liquidity from banks against the collateral of eligible government securities under the LAF. It is an important monetary policy tool employed by the RBI to maintain liquidity and check inflation in the economy.

The Reverse Repo Rate helps the RBI to borrow money from commercial banks within its own territory when it needs. In return, the RBI offers attractive interest rates to them. The banks also voluntarily park excess funds with the central bank as it provides them with an opportunity to earn higher interest on surplus money. The Reverse Repo Rate is decided by the Monetary Policy Committee (MPC), headed by the RBI Governor. The decision is taken in the bi-monthly meeting of the Committee.

Open market operations (OMO) - A monetary policy tool used by central banks to regulate the money supply in the economy by buying and selling securities or treasuries on the open market. There are two types of OMOs - permanent OMO and temporary OMO.

Permanent Open Market Operations (POMO) - These involve the central bank of any country selling and buying securities or treasuries on the open market in order to change the money supply. It is a means of influencing the economy.

Temporary Open Market Operations (TOMO) - These are used to add or subtract reserves from or into the banking system on a short-term basis. Repurchase agreements, also known as Repos or reverse repurchase

agreements, or RRP, are used for short-term open market transactions .

Bank rate - A bank rate is the interest rate at which a nation's central bank lends money to domestic banks, often in the form of very short-term loans. Managing the bank rate is a method by which central banks control economic activity. Such loans are given out either by direct lending or by rediscounting (buying back) the bills of commercial banks and treasury bills. It is aligned to the Marginal standing facility rate. It is increased by the RBI to discourage credit in the economy. The bank rate is also known as the **discount rate** .

Marginal standing facility (MSF) - It is a window for banks to borrow from the Reserve Bank of India in an emergency situation when interbank liquidity dries up completely. Banks borrow from the central bank by pledging government securities at a rate higher than the repo rate under liquidity adjustment facility (LAF). The MSF rate is pegged 100 basis points or a percentage point above the repo rate. Under MSF, banks can borrow funds up to one percent of their net demand and time liabilities (NDTL).

Market Stabilization Scheme (MSS) - This scheme is issued with the objective of providing the RBI with a stock of securities with which it can intervene in the market for managing liquidity. These securities are issued not to meet the government's expenditure. It was launched in April 2004 to strengthen the RBI's ability to conduct exchange rate and monetary management. The bills/bonds issued under MSS have all the attributes of the existing treasury bills and dated securities. These securities will be issued by way of auctions to be conducted by the RBI. The timing of issuance, amount and tenure of such securities will be decided by the RBI.

Bank reserves are a commercial bank's cash holdings physically held by the bank, and deposits held in the bank's account with the central bank.

2. Qualitative Tools - These instruments do not directly impact the money supply but instead attempt to influence the direction of credit and behavior of financial institutions.

Credit Rationing - Under credit rationing, RBI fixes a ceiling (maximum limit) on loans and advances of various

categories, which the commercial banks cannot exceed. This controls the amount of credit for certain sectors and ensures that all sectors get adequate credit. This is required for inclusive growth of all sectors of the economy.

Margin requirements - The RBI can set minimum margin requirements for specific types of loans, such as loans for buying cars or houses.

Regulating Consumer Credit - Consumer credit refers to loans taken by the public for purchase of goods and services. RBI regulates this by either fixing a minimum time frame for repayment or increasing down payment required for availing loan.

Moral Suasion - RBI uses persuasion and request, giving suggestions and advice to commercial banks to undertake certain actions in the economic interests of the country. The advice is morally binding, but not mandatory for the banks.

Monetary policy stance to combat Inflation -

Hawkish stance - To combat inflation, the central bank is willing to hike interest rates to curb the money supply and thus reduce the demand because tight monetary policy decreases the money supply.

Dovish stance - to support economic growth, the central bank is willing to lower interest rates or keep them low, because loose monetary policy increases the money supply.

Accommodative stance - The Central bank is prepared to expand the money supply by cutting interest rates to boost economic growth.

Neutral stance - It suggests that the central bank can either cut rate or increase rate. This stance is typically adopted when the policy priority is equal to both inflation and growth. During neutral policy, the central bank doesn't commit to hike rates or cuts. The interest rate can move to either side depending on incoming data.

Financial Institutions

A financial institution is an establishment that completes and facilitates monetary transactions, such as loans, mortgages, and deposits.

Major Financial Institutions of World

World Bank Group

The World Bank Group, a major source of funding and development knowledge for poorer nations, aims to fight poverty and promote sustainable growth. Founded alongside the IMF in 1944, it has 189 members.

The World Bank group consists of 5 different Organizations which are -

International Bank for Reconstruction and Development (IBRD) - Established in 1944 and headquartered in Washington, D.C, this arm lends to middle-income developing countries.

International Development Association (IDA) - Founded in 1960, IDA provides grants and low-interest loans to the world's poorest countries.

The World Bank is the collective name for the International Bank for Reconstruction and Development (IBRD) and International Development Association (IDA).

International Finance Corporation (IFC) - Established in 1956, IFC promotes private sector growth in developing countries.

International Centre for Settlement of Investment Disputes (ICSID) - Founded in 1966, this institution facilitates legal dispute resolution between investors and governments.

Multilateral Investment Guarantee Agency (MIGA) - Established in 1988, MIGA offers political risk insurance for foreign investments in developing countries.

IMF (International Monetary Fund)

The International Monetary Fund (IMF) is an international organization that promotes global economic growth and financial stability, encourages international trade, and reduces poverty. The IMF was originally created in 1945 as part of the **Bretton Woods Agreement**, which attempted to encourage international financial cooperation by introducing a system of convertible currencies at fixed exchange rates. The

dollar was redeemable for gold at \$35 per ounce at the time. Quotas of member countries are a key determinant of the voting power in IMF decisions.

Quotas and basic votes comprise one vote per 100,000 Special Drawing Rights (SDR). The Special Drawing Right (SDR) is an interest-bearing international reserve asset created by the IMF in 1969 to supplement other reserve assets of member countries. The SDR is based on a basket of international currencies comprising the U.S. dollar, Japanese yen, euro, pound sterling and Chinese Renminbi.

Major Financial Institution In India

EXIM Bank - Established in 1982 under the Export-Import Bank of India Act 1981. EXIM Bank was previously a branch of IDBI before becoming an independent body wholly owned by the Government of India. It facilitates and promotes foreign trade financing and is regulated by the RBI. Headquarters - Mumbai.

National Housing Bank (NHB) - It was established on July 9, 1988, under the National Housing Bank Act, 1987, envisioned by the National Housing Policy, 1988, as the apex institution for housing. Initially, the Reserve Bank of India provided the entire paid-up capital, but following a 2019 amendment to the NHB Act, shares were transferred from the RBI to the Government of India, resulting in the government contributing the entire capital of NHB.

Small Industries Development Bank of India (SIDBI) - It was set up on 2nd April 1990 under a special Act of the Parliament, acts as the Principal Financial Institution for Promotion, Financing and Development of the Micro, Small and Medium Enterprise (MSME) sector as well as for co-ordination of functions of institutions engaged in similar activities. Its headquarter is in Lucknow, Uttar Pradesh.

National Bank for Agriculture and Rural Development (NABARD) - It was established on 12 July, 1982. It was established on the recommendation of B. Sivaraman committee to implement the National Bank for Agriculture and Rural Development Act 1981.

It provides developmental credit in rural areas. It is a specialized bank for Rural

and agricultural development in India. Its headquarter is in Mumbai. M. Rama Krishannaya was the first chairman of NABARD.

NABARD launched the project E-Shakti for the digitisation of all members of self groups in India. The project was launched for the first time in Jharkhand and Maharashtra.

NABARD with MYRADA (Mysore Resettlement and Development Agency) launched an Action Research Programme in 1987 based on which a pilot project for linking 500 SHGS with banks was launched.

LEDPs (Livelihood and Enterprise Development Programmes) was launched by NABARD in 2015 to promote the livelihood of the people in both farm and off-farm activities.

The Industrial Development Bank of India (IDBI) was established under the Industrial Development Bank of India Act, 1964. It was the principal financial institution for providing credit and other facilities for developing industries and assisting development institutions. Until 1976, IDBI was a subsidiary bank of the Reserve Bank of India (RBI). In 1976, it was separated from the RBI and the ownership was transferred to the Government of India. The head office of IDBI is located in Mumbai.

In 2004, IDBI was converted into a commercial bank when the RBI incorporated it as a 'scheduled bank' under the RBI Act, 1934.

The Industrial Finance Corporation of India (IFCI) was initially established in 1948. It was converted into a public company on 1 July 1993 and is now known as Industrial Finance Corporation of India Ltd. The main aim is to provide assistance to the industrial sector to meet their medium and long-term financial needs.

The Industrial Credit and Investment Corporation of India (ICICI) - It is a government institution established on 5 January 1955. And in 2002, ICICI merged with the ICICI bank and no longer served as a developmental financial institution.

National Bank for Financing Infrastructure and Development (NaBFID) - It is a development financial institution for infrastructure financing. It was announced in the Budget 2021 and

was established under the NaBFID Act, 2021. NaBFID shall be regulated and supervised as an All India Financial Institution (AIFI) by the Reserve Bank under Sections 45L and 45N of the Reserve Bank of India Act, 1934.

Microfinance Institutions (MFIs) provide small loans to underserved borrowers, with the RBI issuing guidelines for their operation in India. These institutions cater to low-income populations, offering services like microloans, microsavings, and microinsurance. In India, microloans, defined by the RBI as collateral-free, target households with annual incomes up to Rs. 3 lakh, under a regulatory framework introduced in 2022.

The interest rates offered by MFIs are typically lower than those of traditional banks, benefitting the "missing middle" – low-income individuals or groups lacking collateral for formal credit. Microfinance borrowers in rural areas have an upper limit of Rs. 1.25 lakh annual household income, while in urban/semi-urban areas, it's Rs. 2 lakh.

Established as Rural Cooperatives post-independence, these institutions play a vital role in financial inclusion. SEWA Bank, founded in Ahmedabad, became the first MFI in India in 1974, initially formed as a trade union in 1972.

Some of the Other NGOs involved in microfinance-related activities are MYRADA in Bangalore, ADITHI in Patna.

The objectives of MFIs include supporting lower-income segments, fostering rural entrepreneurship, eradicating poverty, promoting financial inclusion, enhancing rural skills, and improving living standards.

The RBI's Board of Directors formed the Malegam Committee to address microfinance sector concerns. Established on October 15, 2010, under Mr. Y.H. Malegam's chairmanship, it submitted its report in December 2011. The report offers recommendations on regulating the sector, interest rate pricing, enhancing transparency, and mitigating issues related to multiple lending and over-borrowing.

Non-Banking Financial Company (NBFC) is a company registered under the Companies Act, 1956 engaged in the business of loans and advances, acquisition of shares/ stocks/ bonds/ debentures/ securities issued by

Government or local authority or other marketable securities.

It does not include any institution whose principal business is that of agriculture activity, industrial activity, purchase or sale of any goods (other than securities) or providing any services and sale/ purchase/ construction of immovable property.

In December 2011, **Reserve Bank of India** introduced a new category of Non-banking financial companies (NBFCs) called Non-banking Financial Companies - Microfinance Institutions (NBFC - MFIs) with specific regulations for the micro - credit sector, focusing on Customer protection. Capital adequacy ratio for a NBFC-MFI is **15%** of the aggregate risk weighted assets.

As per guidelines issued by the Reserve Bank of India in June 2021 each Non-banking financial companies-Microfinance institution (NBFC-MFI) is required to have less than 85% of its Net Assets in the nature of Qualifying Assets. The maximum limit on microfinance loans for NBFCs other than NBFC-MFIs is 25% of the total assets.

The Qualifying Assets are those assets which have a substantial period of time to be ready for its intended use or sale.

Sa-Dhan (Association of Community Development Finance Institutions) is designated as the self-regulatory organization for Microfinance institutions in India.

It was founded in 1999 as the Association of Community Development Finance Institutions by SEWA Bank, BASIX, Dhan Foundation and others.

Self-Help Group (SHG) - These are voluntary associations of economically disadvantaged individuals, typically from similar backgrounds, joining forces to address common issues through self-help and mutual assistance. These groups encourage small savings among members, which are deposited in a bank under the SHGs name. Typically, SHGs consist of fewer than twenty members, with a female member often designated as a banking agent, known as a Bank Sakhi, a concept pioneered by Ilaben Bhat in 1970.

The primary objective of SHGs is to facilitate collateral-free loans, providing financial support without requiring collateral. The SHG-Bank linkage program aims to integrate informal

savings and credit groups into mainstream banking systems, spearheaded by NABARD with support from RBI. In April 1996, RBI designated SHG-BLP as a priority sector lending activity.

Joint Liability Group (JLG) - It typically comprises 4 to 10 individuals seeking loans with mutual guarantee, often for agricultural purposes or related activities. Borrowers, including farmers, rural workers, and tenants, form these informal groups. In JLGs, each member shares equal responsibility for loan repayment without requiring financial administration. Social pressure within the group ensures a high repayment rate, reaching 98% in lending to groups of poor women.

Microcredit - It is delivered through Scheduled commercial banks (SCBs) (including small finance banks (SFBs) and regional rural banks (RRBs)), Cooperative banks, Non-banking financial companies (NBFCs), Microfinance institutions (MFIs) registered.

Self - Regulatory Organisation (SRO) - In 2014, the Reserve Bank of India (RBI) acknowledged the Microfinance Institutions Network (MFIN) as a Self-Regulatory Organization (SRO) for NBFC-MFIs. This designation made MFIN the inaugural industry association recognized as an SRO by the RBI.

The National Payments Corporation of India (NPCI) manages retail payment and settlement systems in India, established by the RBI and the Indian Banks' Association under the Payment and Settlement Systems Act, 2007. It's a "Not for Profit" entity incorporated under the Companies Act, 1956 (now Section 8 of the Companies Act, 2013), focusing on robust payment infrastructure. NPCI's regulatory board, based in Mumbai, includes nominees from the RBI and ten core promoter banks.

External Sectors

The external sector refers to the part of the economy that deals with interactions between a country's domestic economy and the rest of the world. It encompasses international trade, investment flows, foreign exchange markets, and international financial transactions.

Balance of Payment (BOP) - It is a comprehensive record of a country's economic transactions with the world over a specified period, typically a year. It assesses whether a country has a trade surplus or deficit. Improving the BOP often involves devaluation when export and import demand elasticities exceed unity. It categorizes transactions into the Current account, Capital account, and Errors and Omissions, and tracks changes in Foreign Exchange Reserves.

Current Account - It records a country's global transactions excluding financial dealings, involving economic values between resident and non-resident entities. It encompasses Trade in Goods (exports, imports, and balance of trade) and Net Invisibles (services, income, and transfers), including elements like Remittances and Current Account Surplus/Deficit.

Capital account - It records the value of long-term transactions like foreign loans, FDIs, FIIs, etc. and these transactions are two-way and multiple transactions.

Foreign Direct Investment (FDI) - FDI involves investments in businesses located in another country. Main investing countries in India include Mauritius, Singapore, USA, Netherlands, Japan, UK, Germany, and UAE. Investments are focused on services, software/hardware, telecommunications, trade, automobiles, construction, and chemicals.

Department for Promotion of Industry and Internal Trade (DPIIT) oversees policies to promote and regulate foreign investment in various sectors in India.

In India, there are two FDI routes -

Automatic Route - Foreign investors can invest in sectors without prior approval from Indian authorities, only needing to inform the Reserve Bank of India (RBI) within a set timeframe.

Government Route - This route requires prior approval from the Indian government or relevant ministries, typically for sectors involving national security, strategic interests, or specific regulatory concerns.

100% FDI is allowed in India in infrastructure - development projects, industrial park, Airport, manufacturing, mining and various other sectors by automatic route.

Advantages and Disadvantages of FDI - FDI can drive economic growth in both recipient and investing countries. Developing nations often rely on FDI to fund infrastructure and generate employment. However, FDI entails navigating regulations and oversight from multiple governments, increasing political risk.

Foreign Portfolio Investment (FPI) - It comprises securities and financial assets held by foreign investors without providing direct ownership. It's relatively liquid but influenced by market volatility. Examples include stocks, bonds, mutual funds, ETFs, American Depositary Receipts (ADRs), and Global Depositary Receipts (GDRs).

Balance of Payment Crisis - It occurs when a country's international reserves are insufficient to offset a sustained deficit in both the current and capital accounts. This deficit is calculated as the sum of the current account deficit and the capital account deficit.

Trade Barriers - These are implemented by governments to safeguard domestic economies, and aim to restrict imported goods and services. They can take various forms, including subsidies, standardization, tariffs, quotas, embargoes, export licenses, and other regulatory measures.

Embargo - It is a trade restriction implemented by a government, group of countries, or international organization as an economic sanction. It can involve prohibiting all trade or applying restrictions to specific categories, such as arms imports.

Tariffs Barriers - Tariffs are taxes imposed by governments on imported and exported goods, increasing their prices and making them less competitive, thus discouraging imports.

Non-Tariff Barriers - These are the administrative measures, other than tariffs, imposed by a domestic Government to discriminate against foreign goods in favor of domestic goods.

Subsidies like Government subsidies, including production and export subsidies, are provided to exporters to enhance the competitiveness of their goods in the global market.

Quotas are a restriction on the amount of goods that is allowed to be imported into

a country. This is usually enforced by issuing import licenses.

Foreign Exchange Market

The market where different currencies can be bought and sold. The supply of a foreign currency (dollars) is determined by exports, FDI/FII, remittances, and NRI deposits. The demand for foreign currency (dollars) comes from imports, outward tourism, banks, government, etc.

Exchange Rate - The exchange rate is the price of one nation's currency in terms of another nation's currency.

Depreciation implies a fall in the value of the exchange rate of the rupee with respect to other currencies like the dollar. It encourages exports and makes imports more costly.

Appreciation implies a rise in the exchange rate of the rupee with respect to other currencies like the dollar. It increases imports and makes exports costlier.

Devaluation occurs when a government (such as RBI in India) reduces the exchange rate of its domestic currency against foreign currencies.

Devaluation of the Indian Rupee has taken place 3 times since 1947.

The rupee was first devalued by 30.5% in 1949.

The rupee was devalued by 36.5% approx 37% in 1966 on weak macro-economic indicators.

In July of 1991 the Indian government devalued the rupee by between 18 and 19 percent. It was carried out in two steps - on July 1 and July 3.

Revaluation is when the exchange rate of a domestic currency is increased by its government (RBI in the case of India) against any foreign currency.

Foreign exchange (Forex) reserves - These reserves, held by central banks like the Reserve Bank of India, consist of foreign currency assets used to back their currency liabilities and influence monetary policy. They include banknotes, deposits, and government securities in reserve currencies, with some reserves possibly in gold and special drawing rights.

Special Drawing Right (SDR) - It is an international reserve asset established

by the International Monetary Fund (IMF) to bolster the official reserves of its member countries. The value of SDR is determined by a basket of major international currencies, including the US dollar, Euro, British pound sterling, Japanese yen, and Chinese renminbi (added in 2016). SDRs are allocated to IMF member countries based on their quotas, enhancing liquidity and supplementing their reserves.

The SDR serves as the unit of account of the IMF and some other international organizations. The currency value of the SDR is determined by summing the values in US dollars, based on market exchange rates, of a SDR basket of currencies.

Exchange rate Regime in India -

Liberalised Exchange Rate Management System (LERMS)- It was introduced by Manmohan Singh in the 1992 budget as an initial step towards transitioning to a market-determined exchange rate system. It allowed exporters to retain 60% of their earnings at market rates while surrendering the rest at the official rate.

India shifted from a fixed to a dual exchange rate system, where exchange rates are primarily determined by the market but RBI can intervene during periods of volatility. Fixed exchange rates are set by central banks, while floating rates are determined by market demand and supply.

Foreign Exchange Regulation Act, 1973 The Act aims to regulate certain dealings in foreign exchange, impose restrictions on certain kinds of payments and to monitor the transactions impinging the foreign exchange and the import and export of currency. It came into force as a replacement of Foreign Exchange Regulation Act, 1947.

The companies IBM and Coca Cola shut down their operations for not being able to comply with the Foreign Exchange Regulation Act that mandated foreign investors cannot own over 40% in Indian enterprises in 1977.

The Foreign Exchange Management Act, 1999 (FEMA) is legislation enacted by the Parliament of India to consolidate and update laws pertaining to foreign exchange. It aims to facilitate external trade, payments, and orderly

development of the foreign exchange market.

FEMA replaced the Foreign Exchange Regulation Act (FERA) and reclassified offenses related to foreign exchange as civil offenses. Applicable across India, FEMA established a new foreign exchange management framework aligned with the principles of the World Trade Organisation (WTO).

Foreign Draft - It is a banking instrument used in international trade and finance as an alternative to foreign currency. It's essentially a bank draft drawn on a financial institution in a non-home country. Foreign drafts can be obtained from commercial banks, typically for a fee depending on the institution and account type. Essentially, they're bills of exchange drawn in one country but payable in another.

National Investment and Manufacturing Zones (NIMZ) - The NIMZs are envisaged as integrated industrial townships with state-of-the-art infrastructure, land use on the basis of zoning, clean and energy-efficient technology, necessary social infrastructure, skill development facilities, etc. to promote world-class manufacturing activities.

It contains Special economic zones, Industrial parks and warehousing zones, Export oriented units etc.

Special economic zone (SEZ) - It is a specially marked territory or enclave within the national borders of a country that has more liberal economic laws than the rest of the country. The economic regulations of special economic zones (SEZs) tend to be conducive to—and attract—foreign direct investment (FDI). SEZ Developers or Co-Developers and Units enjoy tax benefits as prescribed in the SEZs Act, 2005.

There were 7 Central Government Special Economic Zones (SEZs) and 12 State/Private Sector SEZs prior to the enactment of the SEZs Act, 2005.

Infrastructure investments - It involves real assets such as bridges, roads, highways, sewage systems, and energy facilities. They serve as the backbone of a nation's economic growth and contribute to enhancing citizens' quality of life. In India, it's projected that by 2030, transport infrastructure will attract over 60% of investment.

Property of income - It is the concept of comparing the income of any two particular groups like countries, states and societies.

Property of income is calculated based on per capita income, purchasing power parity, gdp, infant mortality rate, distribution of income etc. are the other properties of income which are important in comparing two or more societies.

Purchasing Power Parity (PPP) - It is the rate of currency conversion that equalizes the purchasing power of different currencies by eliminating the differences in price levels between countries. In their simplest form, PPPs are simply price relatives that show the ratio of the prices in national currencies of the same goods or services in different countries.

PPPs are also calculated for product groups and for each of the various levels of aggregation up to and including GDP. The Purchasing Power Parity (PPP) theory is used to make long-run predictions about exchange rates in a flexible exchange rate system.

Remittance refers to money sent by foreign migrant workers to relatives in their home countries. It serves as a significant income source for people in low-income and developing nations, often surpassing direct investment and international development assistance received by a country.

Policies of Government

India's economic policy went from centrally planned to market-oriented. After independence, they focused on government control and limited imports. This strategy had limited success. A 1991 crisis led to a shift towards globalization. Since then, India has reduced trade barriers and integrated with the world economy, fueling growth particularly in the service sector.

Industrial policy

Industrial policy involves government actions shaping industry ownership, structure, and performance through subsidies, finance, or regulation. It encompasses procedures, policies, incentives, tariff and labor policies, and attitudes toward foreign capital.

Its objectives include sustaining

productivity growth, enhancing employment, optimizing human resource utilization, achieving international competitiveness, and positioning India as a significant global player.

Industrial Policies in India since Independence

Industrial Policy 1948 - Industrial policy resolution of 1948 was first officially documented in India on 6 April, 1948. The Industries (Development and Regulation) Act was passed in 1951 to implement the Industrial Policy Resolution, 1948. It outlined India's adoption of a Mixed Economic Model.

It classified industries into four broad areas-

Strategic Industries (Public Sector) - It included three industries in which the Central Government had a monopoly. These included Arms and ammunition, Atomic energy and Rail transport.

Basic / Key Industries (Public-cum-Private Sector) - 6 industries viz. coal, iron & steel, aircraft manufacturing, ship-building, manufacture of telephone, telegraph & wireless apparatus, and mineral oil.

Important Industries (Controlled Private Sector)- It included 18 industries including heavy chemicals, sugar, cotton textile & woollen industry, cement, paper, salt, machine tools, fertiliser, rubber, air and sea transport, motor, tractor, electricity etc.

Other Industries (Private and Cooperative Sector) - All other industries which were not included in the above mentioned three categories were left open for the private sector.

Industrial Policy 1956 - Industrial Policy Resolution of 1956 (IPR 1956, Economic Constitution of India or "The Bible of State Capitalism") is a resolution adopted by the Indian parliament in April 1956. It led to the establishment of a socialistic pattern in society.

According to the policy 1956, the industries were classified as-

The **first category (Schedule A)** consisted of 17 industries which could be established and developed exclusively as public sector enterprises.

The **second category (Schedule B)** consist of 12 industries that could be set up as both public sector and private sector enterprises, with the private sector being given a secondary role.

The **third category (Schedule C)** consisted of all industries other than those specified in the above categories. To meet new challenges, from time to time, it was modified through statements in 1977, 1980 and 1991.

Industrial Policy 1977 - In December 1977, the Janata Government introduced its New Industrial Policy in Parliament. The primary focus was on promoting cottage and small industries dispersed in rural areas and small towns. The small sector was categorized into three groups—cottage and household sector, tiny sector, and small-scale industries.

The 1977 policy designated specific areas for large-scale industrial sectors, including Basic industries, Capital goods industries, High technology industries, and Other industries outside the reserved items for the small-scale sector. Additionally, the government encouraged worker participation in management from shop floor to board level.

Industrial Policy 1980 - The policy sought to promote the concept of economic federation, to raise the efficiency of the public sector and to reverse the trend of industrial production of the past three years and reaffirmed its faith in the Monopolies and Restrictive Trade Practices (MRTP) Act and the Foreign Exchange Regulation Act (FERA). It increased the investment limit for a tiny industry/unit to ₹2 lakh.

New Industrial Policy (NIP) 1991 - Economic reforms, or the NIP, were initiated on July 24, 1991, aiming to liberalize the economy and accelerate economic growth. These reforms sought to increase private sector participation in India's economic development.

Key changes included upgrading technology, abolishing industrial licensing, lifting restrictions on the private sector, encouraging foreign investment, and promoting foreign trade.

The core features of the reforms are Liberalization, Privatization, and Globalization, known as LPG. Former Prime Minister PV Narasimha Rao, often called the father of Indian economic reforms, along with Finance Minister Dr.

Manmohan Singh, introduced the LPG policy, which revitalized the Indian economy.

The broad range of reforms under the economic reform policy 1991 (LPG model) included-

Liberalization - This involved making economic policies less restrictive and reducing tariffs or eliminating non-tariff barriers. It also included abolishing the industrial licensing system, reducing import tariffs, and promoting a more open market.

Privatization - This entailed transferring ownership of government-owned property or businesses to private entities and deregulating markets. While it aimed at fostering competition and efficiency, it faced criticism for potentially limiting access to essential services for marginalized populations.

Globalization - This encompassed expanding economic activities beyond national borders, correcting exchange rates, liberalizing foreign direct investment and trade policies, and removing mandatory convertibility clauses.

Rise in share of industry sector in the GDP is a good indicator to identify the effect of policies on the industrial sector. Industrial Indicator is a statistical combination of multiple factors.

Features of New Industrial Policy

De - licensing - It involved abolishing industrial licensing for most projects, except for a short list of industries related to security, strategic, and environmental concerns. Currently, only four industries require an industrial license- Electronic aerospace and defense equipment, Specified hazardous chemicals, Industrial explosives, and Cigars and cigarettes of tobacco and manufactured tobacco substitutes.

The Monopolies and Restrictive Trade Practices (MRTP) Act of 1969 prevents the concentration of economic power in a few hands and regulates monopolies and restrictive trade practices. It was replaced by the Competition Act, 2002, following the Raghavan committee's recommendations.

Disinvestment in Public Sector Units - (DIPAM) is a Department under the

Ministry of Finance. It deals with all matters relating to management of Central Government investments in equity including disinvestment of equity in Central Public Sector Undertakings. It is an act of Privatization.

Disinvestment Commission, Ministry of Industry (Department of Public Enterprises) established in 1996. This commission was set up for three years.

The first Chairman of Disinvestment Commission was **GV Ramakrishna**.

The Ramkrishna commission, formed in August 1996, advised on the gradual disinvestment of Indian PSUs until its discontinuation in May 2004.

The Rangarajan Committee of 1993 recommended disinvesting up to 49% of PSEs equity in industries reserved for the public sector and over 74% in other industries. However, the government did not act on the Committee's recommendations.

National Manufacturing Policy - It was notified by the Department of Industrial Policy and Promotion (DIPP) on November 4, 2011, aims to increase the growth of the manufacturing sector to 12-14% over the medium term. It also seeks to enhance the global competitiveness of Indian manufacturing, boost domestic value addition, and enhance technological depth in manufacturing.

NIIF - National Investment and Infrastructure Fund, India's first sovereign wealth fund, was announced in 2015-16. It is an investor-owned fund manager, anchored by the Government of India (GoI) in collaboration with leading global and domestic institutional investors. The funds are registered as Alternative Investment Fund (AIF) with the Securities and Exchange Board of India (SEBI). Types of NIIF Funds- Master Fund, Fund of Funds, Strategic Fund.

The National Investment Fund (NIF) was established by the Government of India on November 3, 2005, to channelize proceeds from the disinvestment of Central Public Sector Enterprises. 75% of its annual income was designated for schemes promoting health, education, and employment.

List of Maharatna, Navratna, Miniratna PSUs

The Indian government categorizes Central Public Sector Enterprises (CPSEs) into three tiers- Maharatna, Navratna and Miniratna. This classification is based on CPSE's annual turnover, net worth, and net profit.

Maharatna CPSEs can invest between INR 1,000 crore and INR 5,000 crore and allocate up to 15% of their net worth to a project without prior permission from the Government of India. The Maharatna Scheme for CPSEs was introduced on May 19, 2010.

Maharatna PSUs - Bharat Heavy Electricals Limited (BHEL), Coal India Limited (CIL), Indian Oil Corporation limited (IOCL), Gas Authority Of India Ltd (GAIL), National Thermal Power Corporation Limited (NTPC), Oil and Natural Gas Corporation Limited (ONGC), Steel Authority of India Limited (SAIL), Bharat Petroleum Corporation Limited (BPCL), Power Grid Corporation of India Limited (PGCIL), Hindustan Petroleum Corporation Limited (HPCL), Power Finance Corporation Limited (PFCL), Rural Electrification Corporation Limited (RECL), Oil India Limited (OIL).

Navratna companies possess the autonomy to invest up to INR 1000 crores without official permission from the central government. Navratna Scheme was launched in 1997.

Guidelines for upgradation from Navratna to Maharatna - Moreover, a Navratna can upgrade into a Maharatna after it has gained an annual profit of INR 5000 crores in the last 3 years with a net worth of INR 15,000 crores. The turnover of INR 25,000 crores in the past 3 years is also considered during the respective status upgradation. It should be listed on the Indian Stock exchange.

The Top navratna PSUs - Bharat Electronics Limited (BEL), Container Corporation of India Limited (CCIL), Engineers India Limited (EIL), Hindustan Aeronautics Limited (HAL), Mahanagar Telephone Nigam Limited (MTNL), National Aluminium Company Limited (NALCO), National Buildings Construction Corporation Limited (NBCC), National Mineral Development Corporation (NMDC), Neyveli Lignite Corporation Limited (NLCL), Rashtriya

Ispat Nigam Limited (VIZAG STEEL), Shipping Corporation of India Limited (SCIL).

'Miniratna' Scheme was launched in October 1997.

Category I CPSEs - These CPSEs have made consistent profits in the last three years. The respective CPSE must have made a net profit of INR (Rs.) 30 crores or more in one of the three previous years.

Category II CPSEs - Central Public Sector Enterprises should have made profits in three consecutive years and have a positive net worth. The respective Central Public Sector Enterprises (CPSEs) must not have defaulted in the repayment of loans in any form. Another important aspect is that the CPSE must not depend on Government guarantees or budgetary support.

Guidelines of Government of India to upgrade Miniratna to the Navratna PSU - The company must have the status of Miniratna Category - The Memorandum of Understanding (MoU) scores must fall into 'very good' or 'excellent' category in three out of five years. The PSU company must obtain a score of 60 or above.

The Top Miniratna PSUs are Airports Authority of India, Antrix Corporation Limited, Balmer Lawrie & Co. Limited, Bharat Coking Coal Limited, Bharat Dynamics Limited, Bharat Earth Movers Limited (BEML), Bharat Sanchar Nigam Limited, Braithwaite & Company Limited, Bridge & Roof Company (India) Limited, Central Warehousing Corporation, Central Coalfields Limited, Central Mine Planning & Design Institute Limited.

Stocks, Debentures and Foreign Trade

Stock - It is also known as equity, is a security that represents the ownership of a fraction of the issuing corporation. Units of stock are called "**shares**" which entitles the owner to a proportion of the corporation's assets and profit equal to how much stock they own. Stocks are bought and sold predominantly on stock exchanges and are the foundation of many individual investors' portfolios.

Types of Stocks

Common stocks are securities that represent individuals' ownership in a said

corporation and their claim on the venture's accrued profits.

Preferred stock is a special type of stock that pays a set schedule of dividends and does not come with voting rights. Preferred stock combines aspects of both common stock and bonds in one security, including regular income and ownership in the company.

Market capitalization (M-cap) - It is the total value of a company's shares of stock. Market cap allows investors to evaluate a company based on how valuable the public perceives it to be. Investing across market caps can help create a diversified portfolio.

Large-cap or big cap refers to a company with a market capitalization value of more than \$10 billion.

Mid-cap stocks are shares of companies with total market capitalization in the range of about \$2 billion to \$10 billion.

Small-cap stocks refers to a company with a market capitalization value of less than \$2 billion.

Blue chip stocks are renowned for their stellar reputation and solid financial standing. They are typically large, well-established companies with consistent earnings and a history of paying dividends. Example - HUL, ITC, etc.

Red Chip stocks are enterprises that are incorporated outside of the Mainland and are controlled by Mainland Government entities.

White chip stocks are companies that are considered to be of poor quality. These stocks often have less value and produce lower yields.

Green chip stocks are the stocks of those companies that are listed on the exchanges and operate in environment friendly sectors.

Stock market - It is a platform where you can invest in various financial instruments, including shares, bonds, futures and derivatives. The Securities and Exchange Board of India (SEBI) is the regulatory authority established under the SEBI Act 1992 and is the principal regulator for Stock Exchanges in India.

There are two major stock exchanges in India

Bombay Stock Exchange (BSE) - It is one of the largest and oldest securities markets in India, it is also the first stock exchange in Asia and was established in 1875 by cotton merchant Premchand Roychand as the Native Share and Stock Brokers' Association based in Mumbai, India. The BSE lists close to 6,000 companies.

It became the first stock exchange in India to launch commodity derivatives contracts in gold and silver. BSE created history on June 9, 2000 by launching the first Exchange-traded Index Derivative Contract in India i.e. futures on the capital market benchmark index - the BSE Sensex.

Indices of BSE SENSEX- S&P BSE SmallCap, S&P BSE MidCap, S&P BSE LargeCap, BSE 500.

National Stock Exchange (NSE) - It is India's largest financial market. Incorporated in 1992, the NSE has developed into a sophisticated, electronic market, which ranked fourth in the world by equity trading volume. Trading commenced in 1994 with the launch of the wholesale debt market and a cash market segment shortly thereafter. Indices of NSE - NIFTY 50, NIFTY Next 50, NIFTY 500

SENSEX - Sensex, otherwise known as the S&P BSE Sensex index, is the benchmark index of India's BSE. The Sensex comprises 30 of the largest and most actively-traded stocks on the BSE, providing a gauge of India's economy. The index's composition is reviewed in June and December each year. Created in 1986, the Sensex is the oldest stock index in India. Analysts and investors use it to observe the cycles of India's economy and the development and decline of particular industries.

NIFTY - Nifty is derived from the term National Stock Exchange Fifty and it comprises 50 companies that are traded on NSE. It is the benchmark index of NSE and was introduced in 1996.

Demat account - It was first introduced in 1996 by NSE. It helps investors hold shares and securities in an electronic format. It is a type of account for citizens in India so that they can trade in stocks which are listed in the stock market. This

kind of account is also called a dematerialised account.

It also helps to keep proper track of all the investments an individual makes in shares, exchange-traded funds, bonds, and mutual funds in one place. In India, depositories such as NSDL and CDSL provide Demat account services. There are two types of Demat accounts—Repatriable Demat account and Non-repatriable Demat account.

Initial public offering (IPO) - When a private company sells shares to the public, transitioning from private to public ownership. This process, also called "going public," raises capital for debt repayment, growth initiatives, and increasing public visibility. It also allows insiders to diversify holdings or create liquidity by selling private shares.

A Follow-on Public Offer (FPO) is when a publicly-traded company sells new shares to raise capital or reduce debt. There are two types: dilutive (adding new shares) and non-dilutive (selling existing private shares publicly). At-the-market (ATM) offerings allow companies to sell secondary shares based on market prices.

Mutual fund - It is a fund that pools money from investors to invest in securities like stocks, bonds, and short-term debt, forming its portfolio. Investors buy shares in mutual funds. SEBI and RBI regulate and supervise mutual funds to safeguard investor interests.

The Mutual Fund industry in India started in 1963 with the formation of Unit Trust of India (UTI) by an Act of Parliament and functioned under the Regulatory and administrative control of RBI.

Advantages of mutual funds - Advanced portfolio management, dividend reinvestment, risk reduction.

An index fund is a type of mutual fund or exchange-traded fund (ETF) that mirrors market indexes like BSE Sensex, Nifty50, or S&P 500. It offers broad exposure, low costs, and stable tracking of its benchmark regardless of market conditions.

Provident fund (PF) - employer-employee investment fund for long-term retirement savings, crucial for employees in various sectors. It involves deducting a portion of an employee's salary, with matching

contributions from the employer, all deposited into a savings account.

Debt Instruments

A debt instrument is a fixed-income asset that allows the lender to earn a fixed interest on it besides getting the principal back while the issuer can use it to raise funds at a cost.

Bonds, debentures, leases, certificates, bills of exchange and promissory notes are examples of debt instruments.

Bonds

Bonds are issued generally by the government, central bank or large companies and are backed by a security. On maturity of the bond, the principal amount is paid back to the bondholder. Bonds essentially work the way loans do.

Carbon Bonds - Also known as 'Green Bonds,' finance environmentally friendly projects and record reductions in carbon footprints. They offer a straightforward way to offset emissions without requiring investors to alter their activities. However, they are not included in Forex reserves.

Masala bonds are Rupee-denominated bonds issued abroad, named by the International Finance Corporation. The first one, backed by the World Bank's IFC, raised 1,000 crore in November 2014 for Indian infrastructure. Residents of the issuing country, members of the Financial Action Task Force, can subscribe to these bonds. Kerala was the first Indian state to issue Masala bonds.

Debentures

Debentures are debt financial instruments issued by private companies, but any collaterals or physical assets do not back them up.

Difference between Bonds and Debentures -

Bonds are generally secured by the collateral or physical assets of the issuing company, while debentures are unsecured and are not backed by any collateral.

Bonds are long term investments and their tenure is generally higher than debentures which are generally short to medium term.

International trade

It refers to the exchange of goods & services between two or more nations or within boundaries. India has been one of the major trading countries from the time of independence.

During the colonial period, Britain held a monopoly over India's imports & exports. International trade may be categorized into two types -

Bilateral trade - Bilateral trade is done by two countries with each other. They enter into agreement to trade specified commodities amongst them.

Multilateral trade - A country can trade with a number of many other countries.

World Trade Organisation - It is an intergovernmental organization that regulates and facilitates international trade. It began operations on January 1, 1995, succeeding the General Agreement on Tariffs and Trade (GATT) established in 1948. Headquartered in Geneva, Switzerland, the WTO oversees global trade affairs. India, an original contracting party to GATT, has been a WTO member since January 1995.

Finance Commission

The Finance Commission is a constitutional body set up under Article 280 to promote fiscal federalism in India. Its core responsibilities include evaluating the finances of the Union and State Governments, recommending tax sharing between them, and laying down principles for tax distribution among States.

The first Finance Commission was constituted in 1951 under KC Neogy's chairmanship. The President appoints the chairman and 4 other members, including a High Court judge. It is reconstituted every five years.

Notably, the 14th Finance Commission recommended increasing the States' share in the Center's divisible tax pool from 32% to 42%.

15th Finance Commission

The 15th Finance Commission was constituted in November 2017 by the President of India. It covers the period of 2021-22 to 2025-26 financial year and is headed by NK Singh, with members

Shaktikanta Das, Anoop Singh, Ramesh Chand, and Ashok Lahiri.

The Commission has recommended that states be given 41% share of the divisible tax pool of the Center from 2021-22 to 2025-26. This is a 1 percentage point reduction from the previous 42% share, factoring in the conversion of Jammu and Kashmir into two Union Territories.

The criteria used by the Commission to determine each state's share in central taxes, and the weight assigned is as follows - Income distance - 45%, Population (2011) - 15%, Area - 15%, Forest and Ecology - 10%, Demographic performance - 12.5%, Tax Effort - 2.5%.

Income distance is the distance between the income of a state from the state which has the highest income.

Demographic performance As per its Terms of Reference, the Commission used the population data of 2011 for its recommendations.

Demographic performance criteria are used to reward states for their efforts to control their population.

Tax and fiscal efforts This criterion has been used to reward states with higher tax collection efficiency.

The State Finance Commission (SFC) is a constitutional body set up under Article 243-I by the Governor of a state every five years. It was created by the 73rd and 74th Constitutional Amendments to rationalize fiscal relations between the state and its local bodies.

The SFC has functions similar to the Central Finance Commission, including allocating the state's resources in terms of taxes, duties, and levies to the Panchayati Raj Institutions at all three levels. It reviews the financial position of panchayats in the state and recommends principles for the distribution of tax proceeds among them to the Governor.

Taxes, duties, levies, toll fee collected by the state between the state and its Panchayati Raj Institutions at all three levels – village level, block level and district level.

16th Finance Commission

The 16th Finance Commission was constituted on 31.12.2023 with Arvind Panagariya, former NITI Aayog Vice-Chairman, as its Chairman. The

Commission has been asked to submit its recommendations by October 31, 2025, covering an award period of 5 years starting from April 1, 2026.

Its members are Ajay Narayan Jha, Annie George Mathew, Niranjan Rajadhyaksha, and Soumya Kanti Ghosh.

Niti Aayog / Planning Commission and Five - Year Plans

National Planning Committee (1938)

Planning was first initiated in India in 1938 by Congress President Netaji Subhash Chandra Bose, later on Jawaharlal Nehru was made head of the National Planning Committee. It was the first attempt on the part of the people of India to examine the fundamental economic problems and draw up coordinate plan for upliftment of the people.

Bombay Plan was presented in 1944 by Eight Industrialists of Bombay (Mr. JRD Tata, GD Birla, Purshottamdas Thakurdas, Lala Shriram, Kasturbhai Lalbhai, John Mathai, AD Shroff, Ardesir Dalal). Dalal, & John Mathai working together prepared "A Brief Memorandum Outlining a Plan of Economic Development for India". This is known as "Bombay Plan".

This plan envisaged doubling the per capita income in 15 years and tripling the national income during this period.

Gandhian Plan (1944) This plan was drafted by Sriman Narayan Agrawal, principal of Wardha Commercial College. It emphasized the economic decentralization with primacy to rural development by developing the cottage industries.

People's Plan (1945), a ten year plan was prepared by the M.N. Roy. It was different from the Bombay plan in methodology and priorities. Its chief emphasis was on agricultural and consumer goods industries through collectivization and setting up of state owned industrialization.

Sarvodaya Plan (1950) was drafted by Jaiprakash Narayan. This plan itself was inspired by Gandhian Plan and Sarvodaya Idea of Vinoba Bhave. This plan emphasized on agriculture and small & cottage industries.

Planning Commission

The Industrial policy 1948 recommended setting up of a planning commission. Planning Commission was set up on 15 March 1950 and the plan started from 1 April 1951 with the launching of the First Five Year Plan (1951-56).

The concept of the 5-year plan was borrowed from the USSR into the Indian constitution. Joseph Stalin implemented the first five year plan (FYP) in the Soviet Union (USSR) in the late 1920s.

The goals of the five-year plans in India were growth, modernisation, self-reliance and equity, high GDP growth, equal benefits to all the people, self-reliance.

The five-year plans in India laid down not only specific objectives to be attained in the five years of a plan, but also what was to be achieved over a period of 20 years. This long-term plan is called a perspective plan.

Before being replaced by NITI Aayog, Planning Commission was responsible for collecting data about unemployment using **Lakdawala formula**.

First Five-Year Plan (1951-1956)

Mainly focused on the development of the primary sector and self-sufficiency in food along with controlling inflation. First five years plan's president was Jawaharlal Nehru and vice-president Gulzarilal Nanda.

The motto of the first five years plan was '**Development of agriculture**' and the aim was to solve different problems that formed due to the partition of the nation and second world war.

This plan was drafted by the economist K.N Raj and based on the Harrod-Domar economic model to focus on agriculture and irrigation to boost farm output.

This plan was a success, with a 3.6% growth rate (more than its target of 2.1%).

Many irrigation projects were initiated during this period, including the Bhakra, Hirakud and Damodar Valley project. At the end of the plan period in 1956, five Indian Institutes of Technology (IITs) were started as major technical institutions.

Contracts were signed to start five steel plants, which came into existence in the middle of the Second Five-Year Plan. The plan was quasi-successful for the government.

Second Five-Year Plan (1956 -1961)

Introduced under Jawaharlal Nehru's Prime Ministership, this plan aimed at public sector development and rapid industrialization.

It was based on the Mahalanobis model by Prof. Prasanta Chandra Mahalanobis, emphasizing heavy industry for long-term growth. The actual growth rate slightly fell short of the targeted rate.

Hydroelectric power projects and three steel plants at Bhilai, Durgapur, and Rourkela were established with the help of the Soviet Union, Britain (U.K) and Germany respectively. Coal production was increased. More railway lines were added in the north east.

The Tata Institute of Fundamental Research and Atomic Energy Commission of India were established as research institutes.

During the Second Five Year plan On 22 March 1957, India adopted the National Currency on Decimal basis system.

Third Five - Year Plan (1961-1966)

It was introduced under the Prime Ministership of Jawaharlal Nehru. The plan is also known as the '**Gadgil Yojna**,' after D.R. Gadgil, the Deputy Chairman of the Planning Commission.

The primary goal of the Third Plan was to establish India as a **self-reliant and self-generating economy**. The Third Plan included agricultural development as one of its objectives to achieve balanced, regional development.

Unfortunately, this period had many misfortunes like Indo - China war in 1961-62, Indo-Pak war in 1965-66 which ruined this plan. The statement 'Planning is a continuous movement towards desired goals' given by Pandit Jawahar Lal Nehru for this plan.

Plan Holiday

Due to miserable failure of the Third Plan

the government was forced to declare "plan holidays" (**from 1966 to 1967, 1967-68, and 1968-69**), three Annual Plans were devised.

While the Fourth Plan was designed in 1966, it was abandoned under the pressure of drought, Indo-Pakistan conflict, currency devaluation, and inflationary recession on the economy. The agricultural crisis and severe food shortages necessitated a focus on agriculture during the Annual Plans.

Fourth five - year Plan (1969 - 1974)

During this Plan, Indira Gandhi was the Prime Minister of India. The fourth plan aimed at growth with stability and greater self-reliance, with a focus on agriculture's growth rate were the objectives.

During this period, the Indo-Pakistani War of 1971 happened and Bangladesh became an independent Nation.

Indira Gandhi government nationalised **14 major Indian banks** (Allahabad Bank, Bank of Baroda, Bank of India, Bank of Maharashtra, Central Bank of India, Canara Bank, Dena Bank, Indian Bank, Indian Overseas Bank, Punjab National Bank, Syndicate Bank, UCO Bank, Union Bank and United Bank of India).

Fifth five - year plan (1974 - 1978)

It was introduced under the Prime Ministership of Indira Gandhi. This plan focussed on **Garibi Hatao**, employment, justice, agricultural production and defense. In other words we can say The fifth five-year plan was launched with twin objectives of poverty eradication and attainment of self-reliance. The plan was launched and prepared by DP Dhar.

The Electricity Supply Act was amended in 1975. A **Twenty-point program** was launched in 1975, the Minimum Needs Programme (MNP) and the Indian National Highway System was introduced. This plan was overall successful, with a growth rate of 4.8% vs the aim of 4.4%. This plan was terminated in 1978 by the newly elected Prime Minister Moraji Desai(Janta party) government .

Rolling Plan (1978 - 1980)

It was the sixth five-year plan introduced

by the Janata Government in 1978-83, after suspending the fifth five-year plan.

According to the Rolling Plan, the plan's performance will be evaluated each year, and a new plan will be developed the following year based on this evaluation. The Rolling Plan is made up of three different types of plans that were proposed.

The rolling plan concept was coined by Gunnar Myrdal. This plan was rejected again by the Indian National Congress government when it came to power in 1980 and a new sixth plan was made (1980- 1985).

Sixth five - year plan (1980-1985)

It was introduced under the Prime Ministership of Indira Gandhi. The basic objective of this plan was economic liberalization by eradicating poverty and achieving technological self-reliance.

The Rural Landless Employment Guarantee Programme (RLEGP) was launched on **15 August 1983** by Prime Minister of India Indira Gandhi during this plan.

During this period To prevent overpopulation, family planning was introduced and also On the recommendation of the Shivaraman Committee, the National Bank for Agriculture and Rural Development (NABARD) was established.

It was based on investment Yojna, infrastructural changing, and trend to the growth model. Its growth target was 5.2% but it achieved 5.7% growth.

Seventh Five - Year Plan (1985 -1990)

Started during Rajiv Gandhi's tenure, with Dr. Manmohan Singh as Deputy Chairman of the planning commission, the Seventh Plan aimed for speedy development, modernization, self-reliance and social justice.

It focused on increasing food grain production and continued programs like NREP and RLEGP while introducing Jawahar Rojgar Yojana. Despite targeting 5.0% growth, it achieved a successful growth rate of 6.1%.

Annual Plans (1990 - 92)

The Eight Five Year Plan was not

introduced in 1990 and the following years 1990-91 and 1991-92 were treated as Annual Plans. This was largely because of the economic instability. India faced a crisis of foreign exchange reserves during this time.

Eighth Five - Year Plan (1992 - 1997)

Introduced during P.V. Narasimha Rao's tenure, this plan prioritized human resource development and industrial modernization. India's accession to the World Trade Organization in 1995 was a key aspect.

With an annual growth rate of 6.8%, surpassing the target of 5.6%, it also introduced indicative planning and the New Economic Policy. Pranab Mukherjee served as the Deputy Chairman of the Planning Commission of India during this period.

Ninth Five Year Plan (1997 - 2002)

It was introduced under the Prime Ministership of Atal Bihari Vajpayee. The Ninth plan was started with the objective of **"Growth with Social Justice and Equality"**. It also assigned importance to agriculture growth. Regulation of the debt programs was emphasized to improve the government's financial position. It was launched in the 50th year of independence of India. Its growth target was 6.5% but it achieved a 5.4% growth

Tenth Five Year Plan (2002 - 07)

Introduced during Atal Bihari Vajpayee's tenure, this plan aimed to double India's Per Capita Income in 10 years, reduce the poverty ratio to 15% by 2012, and narrow gender gaps in education and wages by 2007. While targeting 8.0% growth, it achieved 7.6%.

Eleventh Five-Year Plan (2007 - 2012)

Introduced during Manmohan Singh's tenure, the Eleventh Plan aimed for rapid and inclusive growth amidst the global financial crisis. Led by C. Rangarajan, it prioritized higher education enrollment and emphasized inclusive growth.

Twelfth Five - Year Plan (2012 - 2017)

Introduced during Manmohan Singh's tenure, India's last Five Year Plan focused on "Faster, More Inclusive and Sustainable Growth". It prioritized infrastructure development and universal electricity access in villages. Montek Singh Ahluwalia served as the last Deputy Chairman of the Planning Commission of India during this period.

NITI Aayog

The National Institution for Transforming India, also called NITI Aayog, was formed via a resolution of the Union Cabinet on January 1, 2015 and replaced the Planning Commission of India instituted in 1950. It is a policy **'Think Tank'** of the Government of India.

Neither a constitutional body nor a statutory body. In other words, it is a non - constitutional or extra - constitutional body (not created by the Constitution) and a non- statutory body (not created by an Act of the Parliament). It comes under the Ministry of Planning. The NITI Aayog's creation has **two hubs**

Team India Hub - It leads to the participation of Indian states with the central government. The **Knowledge and Innovation Hub** - It builds the institution's think tank capabilities.

Composition of NITI Aayog

Prime Minister of India is the Chairperson of NITI Aayog. The Prime Minister of India serves as the ex-officio chairperson of the NITI (National Institution for Transforming India) Aayog. Narendra Modi is the First chairman and Sindhushree Khullar was First CEO of NITI Aayog.

Vice-Chairperson - To be appointed by the Prime-Minister.

Governing Council - Chief Ministers of all states and Lt. Governors of Union Territories.

Regional Council - To address specific regional issues, Comprising Chief minister and Lt. Governors Chaired by the Prime Minister or his nominee.

Ad-hoc Membership - Two members in ex-officio capacity from leading Research institutions on a rotational manner.

Ex-Officio membership - Maximum four from the Union council of ministers to be nominated by the Prime Minister.

Chief Executive Officer (CEO)- Appointed by the Prime-minister for a fixed tenure, in the rank of Secretary to Government of India.

Special Invitees Member - Experts nominated by the Prime Minister, experts with domain knowledge.

There are three documents published by NITI Aayog are as follows-

Fifteen-Year Vision, that encompasses overall goals and objectives of the country for next 15 years.

Seven Year Strategy which lays the roadmap of development for next seven years dividing those goals and objectives into two parts.

Third and Final is a **"Three Year Action Agenda"** which states the tasks and targets to be accomplished in next three years' time frame, further dividing the strategy into two parts.

NITI Aayog has launched indigenously developed India's first Digital Asset Management and Storage platform on 23 December 2020, called **'DigiBox India'**.

Unemployment

Unemployment is a situation where a person is able and willing to work at the prevailing wage rate but cannot find a job. It impacts the economy and society.

The most common measure of unemployment is the unemployment rate, calculated as-

Unemployment rate = $\left(\frac{\text{Number of unemployed workers}}{\text{Total labor force}} \right) \times 100$.

The unemployment rate is a key indicator of an economy's health.

Survey and committee related to unemployment

The Periodic Labour Force Survey (PLFS) provides estimates of employment and unemployment indicators like Labour Force Participation Rates, Worker Population Ratio, Unemployment Rate. A person is considered unemployed if they don't work for at least 1 hour on any day during the week.

The Employment-Unemployment Survey (EUS) is a comprehensive household survey on labour force statistics in India, first conducted in 1955. It became a quinquennial (every 5 years) survey from the 27th round in 1972-73, based on the M.L. Dantwala Committee Report.

Bhagwati Committee (1973) is related to Unemployment and Public Welfare. Upon recommendations of the **Dantwala Committee**, 3 approaches are used for measurement of Unemployment in India, such as Usual Status Approach, Weekly Status Approach, and Daily Status Approach.

CMIE (Centre for Monitoring Indian Economy) is a leading business information company established in 1976. It produces economic and business databases, develops analytical tools, analyzes data to decipher economic trends.

The National Sample Survey

Organisation (NSSO) conducts the survey on employment and unemployment in India, the largest independent organization for socio-economic surveys since 1950.

There are three main measures/estimates of unemployment developed by NSSO-

Usual Status Unemployment - Estimates the number of people who remained unemployed for a major part of the year, considering principal and subsidiary activity status.

Weekly Status Unemployment - Measure unemployment with respect to one week, classifying those who did not have any gainful work for even an hour in the preceding week and were seeking/available for work.

Current Daily Status Unemployment - Considers a person's activity status for each day of the preceding week. If someone did not find work on a day or some days, they are regarded as unemployed. Considered the most comprehensive measure.

Main causes of unemployment include slow economic growth, rising labor force, lack of infrastructure and housing, rising population, seasonality of agriculture, joint families, sluggish industrialization, low savings and investment, and labor immobility.

Unemployment is broadly categorized as voluntary or involuntary. Voluntary unemployment occurs when able workers are unwilling to take available jobs. Involuntary unemployment arises when willing and able workers cannot find work at the prevailing wage rate due to market conditions.

Types of unemployment

Disguised Unemployment refers to underemployment or hidden unemployment, wherein more people are employed than actually needed, primarily seen in agriculture and the unorganized sectors in rural India. It doesn't affect economic output.

Seasonal Unemployment refers to the situation when people are unemployed during some months of the year. Eg. Agriculture Sector .

Structural Unemployment is involuntary unemployment due to a mismatch between available jobs and workers' skills (skills gap).

Cyclical unemployment occurs due to changes in economic activity over business cycles, like layoffs during recessions. It's mostly found in capitalist economies.

Technological unemployment arises from labor-saving technological changes and automation that minimize the human role.

Educated Unemployment refers to educated youth lacking employability skills or being unable to find jobs matching their qualifications.

Frictional unemployment is the time lag between the jobs when an individual is searching for a new job or is switching between the jobs. It is a short term and voluntary unemployment which arises due to breakdown of machines, strikes, shortage of raw materials, power failure etc.

Partial Unemployment Workers hired full time who have their weekly hours temporarily reduced by no more than 60 percent.

Chronic unemployment-If unemployment continues to be a long term feature.

Casual Unemployment is when the worker is employed on a day-to-day basis for a contractual job and has to leave it once the contract terminates.

Natural Unemployment refers to the lowest unemployment level a healthy economy can sustain causing inflation. It exists when the labor market is in perfect equilibrium.

Real wage unemployment occurs when wages are set above the equilibrium level causing the supply of labour to be greater than demand. It is also called classical unemployment.

Marginal workers- Those who had not worked for the major part of the year (i.e. those who had worked for less than 6 months or less than 183 days in a year) were termed as In India, 8.7% are marginal workers.

Poverty

According to the World Bank, Poverty is pronounced deprivation in well-being, and comprises many dimensions. It includes low incomes and the inability to acquire the basic goods and services necessary for survival with dignity. Poverty also encompasses low levels of health and education, poor access to clean water and sanitation, inadequate physical security, lack of voice, and insufficient capacity and opportunity to better one's life. In 2011 as much as 21.9% of India's population was below the poverty line.

Types of Poverty

Absolute Poverty - A condition where household income is below a necessary level to maintain basic living standards like food, shelter, safe drinking water, education, healthcare, etc.

Relative poverty - Relative poverty is when households earn substantially less than the average household income, often around 50% less, leaving them unable to afford anything beyond basic necessities.

Situational poverty is a type of poverty that occurs as a result of an adverse event such as an environmental disaster, job loss, or a severe health problem

Generational Poverty is passed down from one generation to the next to individuals and families. This is more challenging because there is no way out because everyone is caught up in the root of the problem and lacks access to the necessary resources.

Urban poverty, prevalent in megacities, showcases dire living conditions, inadequate incomes, and essential utility shortages. Causes include Rural-Urban migration, Skill deficits, Indebtedness, Inflation, etc.

Rural poverty is often a product of poor infrastructure that hinders development and mobility. Rural areas tend to lack sufficient roads that would increase access to agricultural inputs and markets. Without roads, the rural poor are cut off from technological development and emerging markets in more urban areas.

Measurement of Poverty in India

A common method used to estimate poverty in India is based on the income or consumption levels and if the income or consumption falls below a given minimum level, then the household is said to be Below the Poverty Line (BPL).

Monthly per capita expenditure can be referred to in the estimation of poverty. The proportion of people below the poverty line is not the same for all social groups and economic categories in India.

Poverty Line Basket - The basket of goods and services necessary to satisfy basic human needs is the Poverty Line Basket (PLB).

Poverty Ratio - The proportion of the population below the poverty line is called the poverty ratio or headcount ratio (HCR).

The poverty threshold, poverty limit, poverty line or breadline is the minimum level of income deemed adequate in a particular country. The poverty line is the minimum amount of money a person needs to fulfill the basic necessities of life, like shelter and food.

In India, the poverty line is typically determined every five years through sample surveys, specifically the Consumption Expenditure Survey (CES) carried out by the National Statistical Office (NSO).

Need for Poverty Estimations

Impact of Welfare Schemes - Poverty estimates are not just important for academic purposes but are also crucial to track the impact and success of various government policies, especially

social welfare schemes that are intended to eliminate poverty.

Poverty Elimination Plan - The Poverty estimates in the form of poverty line are used to formulate poor centric poverty elimination plans.

Pre-Independence Poverty Estimation - Dadabhai Naoroji through his book, **"Poverty and Unbritish Rule in India"** made the earliest estimation of the poverty line (₹16 to ₹35 per capita per year).

The poverty line proposed by him was based on the cost of a subsistence or minimum basic diet (rice or flour, dal, mutton, vegetables, ghee, vegetable oil, and salt).

The National Planning Committee's (1938) poverty line (ranging from ₹15 to ₹20 per capita per month) was also based on a minimum standard of living perspective in which nutritional requirements were implicit.

In 1938, the National Planning Committee was set up by Subhash Chandra Bose under the chairmanship of Jawaharlal Nehru for the purpose of drawing up an economic plan with the fundamental aim to ensure an adequate standard of living for the masses.

The Bombay Plan (1944) proponents had suggested a poverty line of ₹75 per capita per year. The Bombay Plan was a set of proposals of a small group of influential business leaders in Bombay for the development of the post-independence economy of India.

Post-Independence Poverty Estimation - Planning Commission Expert Group (1962), a working group constituted by the Planning Commission formulated the separate poverty lines for rural and urban areas (₹20 and ₹25 per capita per year respectively).

VM Dandekar and N Rath (1971), made the first systematic assessment of poverty in India, based on National Sample Survey (NSS) data.

Unlike previous scholars who had considered subsistence living or basic minimum needs criteria as the measure of poverty line, VM Dandekar and N Rath were of the view that poverty line must be derived from the expenditure that was adequate to provide 2250 calories per day in both rural areas.

Expenditure based Poverty line estimation, generated a debate on minimum calorie consumption norms.

Alagh Committee (1979) - Task force constituted by the Planning Commission under the chairmanship of **Y K Alagh**, constructed a poverty line for rural and urban areas on the basis of nutritional requirements and related consumption expenditure.

Poverty estimates for subsequent years were to be calculated by adjusting the price level for inflation.

The minimum consumption expenditure (per capita per month) set as a benchmark of the poverty line for rural and urban India in 1979 was Rs. 49.09 and Rs. 56.64 respectively.

Lakdawala Committee (1993) - Task Force chaired by DT Lakdawala, based on the assumption that the basket of goods and services used to calculate Consumer Price Index-Industrial Workers (CPI-IW) and Consumer Price Index-Agricultural Laborers (CPI-AL) reflect the consumption patterns of the poor.

Suggestions made by this committee -

Calculate consumption expenditure based on calorie consumption.

Construct state-specific poverty lines, updated using CPI-IW for urban areas and CPI-AL for rural areas.

Discontinue scaling of poverty estimates based on National Accounts statistics.

Tendulkar Committee (2005) - Expert group constituted by the Planning Commission and, chaired by Suresh Tendulkar, was constituted to review methodology for poverty estimation and to address the following shortcomings of the previous methods- Obsolete Consumption Pattern, Inflation Adjustment, Health and Education Expenditure. It submitted the report in December 2009

Recommendations - Shift from Calorie Consumption based Poverty Estimation, Uniform Poverty line Basket, Private Expenditure, Price Adjustment Procedure, Mixed Reference Period.

Tendulkar committee computed poverty lines for 2004-05 at a level that was equivalent, in Purchasing Power Parity (PPP) terms, to Rs 33 per day.

For **2011-12**, for rural areas the national poverty line using the Tendulkar

methodology is estimated at Rs. 816 per capita per month and Rs. 1,000 per capita per month in urban areas.

Rangarajan Committee - The committee was set up in the backdrop of national outrage over the Planning Commission's suggested poverty line of ₹22 a day for rural areas.

Objectives - To review international poverty estimation methods and indicate whether based on these, a particular method for empirical poverty estimation can be developed in India.

Methodology Used - The Rangarajan committee estimation is based on an independent large survey of households by Center for Monitoring Indian Economy (CMIE). It has also used different methodology wherein a household is considered poor if it is unable to save.

As per the report, the poverty line is estimated as Monthly Per Capita Expenditure of Rs 1407 in urban areas and Rs 972 in rural areas.

Multidimensional Poverty Index

The Global MPI is a poverty measure that reflects the multiple deprivations that poor people face in the areas of education, health, and living standards. The Global MPI Report is jointly published by the Oxford Poverty and Human Development Initiative (OPHI) and the United Nations Development Programme (UNDP).

National Multidimensional Poverty Index
The National MPI Project is aimed at deconstructing the Global MPI and creating a globally aligned as well as customized Indian MPI, which is aimed to create a more diverse action plan to improve India's position in the global MPI rankings. Its report is published by NITI Aayog.

It is based on 3 dimensions. Health (Nutrition, Child and Adult Mortality, Antenatal Care) Standard of living (Cooking Fuel, Drinking Water, Sanitation, Electricity, Housing, Assets, Bank Account) Education (Year of schooling, School attendance)

Types of Poor

Chronic poor - People who are leading constant lives of poverty and who are normally poor but may have a small

amount of money with them (for example, casual workers) are classified collectively as the chronic poor.

Churning poor - The churning poor are the people who go in and out of poverty (for example, small farmers and seasonal workers).

Transient poor - The poor who are well off most of the time but may be subject to bad luck or difficult times at times. They are known as the transient poor.

Agriculture and Food Processing

Agriculture is a primary activity, which produces most of the food that we consume. Besides food grains, it also produces raw material for various industries.

India is an agriculturally important country. Around Two-thirds of its population is engaged in agricultural activities.

Green Revolution

The term "green revolution" describes the dramatic rise in food grain output, particularly for wheat and rice. It began in the middle of the 20th century. It was largely attributed to the introduction of new, high-yielding variety (HYV) seeds into developing nations.

In the **first phase** of the green revolution (approximately mid 1960s upto mid 1970s), the use of HYV seeds was primarily restricted to the more affluent states such as Punjab, Andhra Pradesh, and Tamil Nadu. The first phase of the Green Revolution was restricted largely to wheat and rice growing regions.

In the **second phase** of the green revolution (mid-1970s to mid-1980s), the HYV technology spread to a larger number of states and benefited a wider variety of crops.

Indira Gandhi released a special stamp entitled 'Wheat Revolution' in July 1968. The Wheat Revolution (Green Revolution) was initiated in the 1960s {under the Third Five Year Plan (1961-66)} by introducing high-yielding varieties of rice and wheat. Father of the Green Revolution in India - M S Swaminathan.

In India, between 1950 and 1990, the proportion of GDP contributed by agriculture declined significantly (67.5%

in 1950 to 64.9% by 1990). As of 2024, the agriculture sector contributes roughly 15% of the country's total GDP. The service sector contributes most to the Indian Economy, i.e around 53 % of total India's GVA (gross value added). The Industry sector contributes around 28%.

ICAR (Indian Council of Agricultural Research) has its headquarters in New Delhi. It is formerly known as the Imperial Council of Agricultural Research. It was founded on July 16, 1929. It is an autonomous organization under the Department of Agricultural Research and Education (DARE), Ministry of Agriculture and Farmers Welfare.

MSP (Minimum Support Price) is a form of market intervention by the Government of India to insure agricultural producers against a sharp fall in agricultural prices. It is the guaranteed amount paid to farmers when the government buys their produce.

The MSP (Minimum Support Price) is announced by the Department of Agriculture and Cooperation, Government of India at the beginning of the sowing season for certain crops based on the recommendations of the Commission for Agricultural Costs and Prices (CACP).

Government announces minimum support prices (MSPs) for 22 mandated crops and fair and remunerative price (FRP) for sugarcane. The mandated crops are 14 crops of the kharif season, 6 rabi crops and two other commercial crops.

In addition, the MSPs of toria and de-husked coconut are fixed on the basis of the MSPs of rapeseed/mustard and copra, respectively.

The list of crops are as follows - **Cereals (7)** - paddy, wheat, barley, jowar, bajra, maize and ragi, **Pulses (5)** - gram, arhar/tur, moong, urad and lentil, **Oilseeds (8)** - groundnut, rapeseed/mustard, toria, soyabean, sunflower seed, sesamum, sunflower seed and niger seed, **Other crops** - Raw cotton, Raw jute, Copra, De-husked coconut, **Sugarcane (Fair and remunerative price)**, Flue-cured Virginia (FCV) tobacco

Food Security

The Indian Food Security System designed by the Government of India has 2 components. One is the Buffer Stock

and the other is the Public Distribution System.

Buffer Stock - The stock of foodgrains, namely rice and wheat, procured by the Government through the Food Corporation of India (FCI) is known as the Buffer Stock. The purchased food grains are stored in granaries. Buffer stock will help in resolving the food shortage issues faced during natural calamities like droughts, floods, earthquakes etc. Buffer stock is used to meet the food requirements of poorer strata of society, by selling the food grains at lower price when compared to market price. This is known as Issue Price.

FCI was set up in 1965 under The Food Safety Act 1964 with a primary objective of buying, storing, moving and selling food grains and food stuffs.

PDS - The Public Distribution System (PDS) is an Indian food security system that was established by the Government of India under the Ministry of Consumer Affairs, Food and Public Distribution to distribute food and non-food items to India's poor at subsidized rates.

Major commodities distributed include staple food grains, such as wheat, rice, sugar and essential fuels like kerosene, through a network of fair price shops (also known as ration shops) established in several states across the country. Food Corporation of India, a government-owned corporation, procures and maintains the PDS.

The introduction of rationing in India dates back to the 1940s Bengal famine. This rationing system was revived in the wake of acute food shortage during the early 1960s, before the Green Revolution.

It involves two types, **RPDS** and **TPDS**. In 1992, PDS became RPDS (Revamped PDS) focusing the poor families, especially in the far-flung, hilly, remote and inaccessible areas. In 1997 RPDS became TPDS (Targeted PDS) which established Fair Price Shops for the distribution of food grains at subsidized rates.

The **National Food Security Act 2013** legally entitled upto 75% of the rural population and 50% of the urban population to receive subsidized food grains under Targeted Public Distribution System. Wheat costs Rs. 2 per kg, rice costs Rs. 3 per kg and coarse grains cost

Rs. 1 per kg. It was notified by the Union government on September 10, 2013, ensures food and nutritional security at affordable prices, enabling a life with dignity.

The four features of food security are availability, accessibility, utilization, and stability.

NFWP - The **National Food for Work Programme**, 2004 was launched by the Ministry of Rural Development, central government on 14 November 2004 in 150 of the most backward districts of India to generate supplementary wage employment.

Food Safety and Standards Authority of India - It is an autonomous statutory body established under the Food Safety and Standards Act, 2006. It comes under the Administrative control of the Ministry of Health & Family Welfare. Headquarter - New Delhi. It holds the responsibility for setting the maximum residue limits (MRLs) of pesticides in food materials.

List of Agricultural Revolutions in India

Revolution	Products/Aim	Father of the Revolution
Green Revolution	Food Grains sufficiency	M.S. Swaminathan
White Revolution	Milk Production	Verghese Kurien
Blue Revolution	Fish Production	Dr Arun Krishnan
Yellow Revolution	Oilseed Production	Sam Pitroda
Golden Revolution	production of honey and horticulture	Nirpakh Tutej
Pink Revolution	Prawn or onion production	Durgesh Patel
Red Revolution	Meat or Tomato Production	Vishal Tiwari
Silver Revolution	Egg Production / Poultry Production	Indira Gandhi (Mother of the Revolution)
Grey Revolution	Increase Fertilizers production	-
Protein Revolution	Higher Production (Technology-driven 2nd Green revolution)	-
Silver fiber revolution	Cotton	-
Evergreen Revolution	Integration of ecological principles in technology development	M.S. Swaminathan

Purple Revolution (Lavender Revolution) was initiated in 2016 through the **Aroma Mission** of the Council of Scientific and Industrial Research (CSIR). Main product

is Lavender oil which sells for at least Rs. 10,000 per litre. In India, the lavender (scientific name - *Lavandula*) crop is cultivated in low rainfall regions and on the slopes of hills of Himachal Pradesh(HP) and Uttar Pradesh(UP).

Committees related to Agriculture

The **Rangarajan Panel** was set up by the Government of India for total decontrol of the sugar industry in November 2012. In April 2013, the government announced partial decontrol of sugar.

Swaminathan Aayog or The National Commission on Farmers - The National Commission on Farmers is an Indian commission constituted on 18 November 2004 under the chairmanship of Professor **M.S. Swaminathan** to address the nationwide calamity of farmers suicides in India. The Terms of Reference reflected the priorities listed in the **Common Minimum Programme**. Swaminathan, submitted five reports through the period December 2004 - October 2006. Following from the first four, the final report focused on causes of farmer distresses and the rise in farmer suicides, and recommends addressing them through a holistic national policy for farmers.

The Kumarappan committee (1949) was appointed to look into the problem of land in West Bengal and Kerala. Land Reforms consisted of four elements- The Abolition of the Intermediaries, Tenancy Reforms, Fixing Ceilings on Landholdings and Consolidation of Landholdings.

Other Important agriculture related Information

e-Nam - It is an important initiative of GoI to use **pan-India electronic trading portal** for bidding and network the existing APMC mandis to create a unified national market for agricultural commodities.

Operation Greens was launched in 2018-19 to promote Farmer Producers Organizations, agri-logistics, processing facilities, and professional management for Tomato, Onion, and Potato (TOP) crops.

To help in the structural and infrastructure part of the scheme, **agriculture market committee (APMCs)**

promoted markets will be connected to the e-NAM platform. The government will also help in the development of 22,000 agricultural markets. It was Announced in the Budget speech of 2018 - 19.

National Agricultural Cooperative Marketing Federation of India Ltd. (NAFED) was established on 2nd October 1958. Nafed was set up with the objective to promote Cooperative marketing of Agricultural Produce to benefit the farmers. It is registered under the Multi-State Co-operative Societies Act.

AGMARK is a certification mark for agricultural produce, assuring that they conform to a grade standard notified by Directorate of Marketing & Inspection (DMI), Department of Agriculture, Cooperation and Farmers Welfare, Ministry of Agriculture & Farmers Welfare under Agricultural Produce (Grading Marking).

The FPO mark is a certification mark mandatory on all processed fruit products sold in India such as packaged fruit beverages, fruit-jams, squashes, pickles, dehydrated fruit products, and fruit extracts, following the Food Safety and Standards Act of 2006.

Amul (The Kaira District Co-operative Milk Producers' Union) is an Indian dairy cooperative society, based at Anand in the Indian state of Gujarat. Formed in 1946, it is a cooperative brand managed by a cooperative body, the Gujarat Co-operative Milk Marketing Federation Ltd. (GCMMF). Tribhuvandas Kishibhai Patel became the founding chairman of the organization.

Zero-budget natural farming - (ZBNF) It is a method of cultivating crops without relying on chemical fertilizers, pesticides, or external inputs. The word Zero Budget refers to the zero cost of production of all crops. It originated in Karnataka, championed by the Karnataka Rajya Raitha Sangha, and popularized by agriculturist Subhash Palekar in the mid-1990s.

Economic Acts

MGNREGA 2005 (Mahatma Gandhi National Rural Employment Guarantee Act) - In 1991, the Act was proposed by the then Prime Minister of India P.V. Narasimha Rao.

The Mahatma Gandhi Rural Employment Act (MGNREGA), formerly known as the National Rural Employment Act, is a significant Indian labor law aiming to ensure the 'right to work' for people. Enacted in September 2005, it began rolling out in Anantapur district of Andhra Pradesh on 2 February 2006 and was implemented in all districts of India by 1 April 2008.

MGNREGA aims to improve rural livelihoods by offering 100 days of wage employment annually to households with willing adult members for unskilled manual labor within 5 km of their homes, ensuring payment of minimum wages.

Consumer Protection Act, 2019 - This bill replaces the Consumer Protection Act, 1986, enforcing consumer rights and facilitating complaint redressal for product defects and service deficiencies. Consumer Disputes Redressal Commissions will operate at District, State, and National levels. Appeals follow a hierarchical process, with the Supreme Court handling National Commission appeals.

Additionally, a Central Consumer Protection Authority is established to safeguard consumer rights collectively. It possesses powers to issue safety notices, mandate refunds, recall products, and counter misleading advertisements. Moreover, consumers experiencing harm due to product defects or service shortcomings can pursue product liability claims against relevant parties.

Competition Act, 2002 - The Competition (Amendment) Act, 2023 ("the Act") was recently passed in India. The Act makes a number of changes to the Competition Act, 2002. The Competition Act was enacted to encourage and preserve market competition, protect consumer interests, and guarantee trade freedom for other Indian market participants. It replaced the erstwhile Monopolies and Restrictive Trade Practices Act, 1969.

Industries Development and Regulation Act of India, 1951 - This Act effective from May 8, 1952, empowers the Central Government to regulate specified industries' production and development. It mandates the registration of existing undertakings and establishes the Central Advisory and Development Councils.

Section 12 of this Act grants the Central

Government authority to revoke or modify licenses for establishing new undertakings or manufacturing new articles if they fail to materialize within the stipulated time frame without valid reasons.

Companies (Second Amendment) Act, 2002 - This Act provided for the constitution of the National Company Law Tribunal, and led to the repeal of Sick Industrial Companies Act (SICA) 1985.

Sick Industrial Companies Act (SICA) 1985 - This act defined a sick industrial unit as one that had existed for at least five years and had incurred accumulated losses equal to or exceeding its entire net worth at the end of any financial year. This was based on the recommendations furnished by Tiwari Committee Report (1981), which dealt with widespread industrial sickness in India. Industrial Reconstruction Bank of India was established primarily for the rehabilitation of Sick units.

Regional Rural Banks (Amendment) Act 2014 - The Act amended the Regional Rural Banks Act, 1976, which governs the establishment, regulation, and dissolution of Regional Rural Banks (RRBs). This amendment increased the authorized capital of each RRB from Rs 5 crore to Rs 2000 crore, with Rs 200 crore fully paid-up shares of Rs 10 each. Additionally, it stipulated that the authorized capital of any RRB could not fall below Rs 1 crore, with shares always fully paid up at Rs 10 each. The tenure of directors was extended from 2 years to 3 years, and no director could serve for more than six years in total.

National Bank for Financing Infrastructure and Development Act, 2021 (NBFID) - Under this Act, the National Bank for Financing Infrastructure and Development has been established as an infrastructure focused development financial institution (DFI) to support the development of long-term non-recourse infrastructure financing in India. It was announced in the Budget 2021 and the headquarters of this bank are located in Mumbai.

Insurance (Amendment) Act, 2021 - This act amends the Insurance Act, 1938 and increases the limit on foreign investment in an Indian insurance company from 49% to 74%, and removes restrictions on ownership and control.

Fugitive Economic Offenders Act, 2018 -

This act aims to confiscate assets of economic offenders who flee the country to evade criminal prosecution or refuse to return for trial. It deters such offenders from avoiding Indian courts and upholds the rule of law.

Prevention of Money - laundering Act, 2002 (PMLA) -

This act was enacted in 2002 and it came into force in 2005, to curb money laundering (process of converting black money into white) and to provide for seizure of property derived from money-laundering. Money laundering is defined as the illegal process of converting money generated through criminal activities, such as drug trafficking or terrorist funding, to appear to have come from a legitimate source.

Benami Property Transactions Act, 1988 -

This Act prohibits benami transactions and allows for the recovery of properties held benami. A benami transaction involves the illegal transfer of property from one person to another. It refers to a property purchased by someone not in their name, lacking legal proof of the transaction's validity.

Black Money (Undisclosed Foreign Income and Assets) and Imposition of Tax Act of 2015

targets undisclosed foreign income and assets. It provides an opportunity for holders of such income to disclose it and pay taxes within a compliance window. Failure to do so subjects them to severe penalties and prosecution measures outlined in the law.

Agricultural and Processed Food Products Export Development Authority Act, 1985 (APEDA) -

It is an Act to provide for the levy and collection, by way of a cess, of a duty of customs on the exports of certain agricultural and processed food products for the development and promotion of their export and for matters connected therewith.

Fiscal Responsibility and Budget Management Act (FRBM) -

This Act was enacted in August 2003, with rules notified in July 2004, coming into effect on July 5, 2004. It aims to ensure inter-generational equity and long-term macroeconomic stability. The Act sets limits on the Central government's debt and deficits, restricting the fiscal deficit to 3% of GDP.

Additionally, the 12th Finance Commission's recommendations tied debt relief to States with their enactment of similar laws. States have since implemented their own Financial Responsibility Legislation, capping annual budget deficits at 3% of Gross State Domestic Product (GSDP). Karnataka was the first state to enact this legislation. The FRBM Act also mandates enhanced transparency in fiscal operations and medium-term fiscal policy framework.

Insolvency and Bankruptcy Code (Second Amendment) Act, 2020 -

This act amends the Insolvency and Bankruptcy Code, 2016. The Code provides a time-bound process for resolving insolvency in companies and among individuals. Insolvency is a situation where individuals or companies are unable to repay their outstanding debt.

As per the Insolvency and Bankruptcy Code, 2016 (the Code), the procedure involved in the Corporate Insolvency Resolution Procedure (CIRP) should be completed within 180 days or within the extended period of 90 days and mandatorily be completed within 330 days including any extension and the time taken in legal proceedings.

Factories Act, 1948 - The Factory (Amendment) Act of 1948 played an effective and more important role in improving India. A five-year plan was developed during the Interim Congress Regime to improve certain labor conditions in India, and it also referred to the Factory Act of 1934, the Great Britain Factory Act, and the most recent ILO (International Labour Organization) Convention in matters of safety, health and welfare, working hours, industrial hygiene, medical examination of young people, and submissions of factory building plans.

Section 48 of factory act provides that In every factory wherein more than 30 women workers are ordinarily employed there shall be provided and maintained a suitable room or rooms for the use of children under the age of six years of such women.

First Factories Act, 1881 - It was adopted in 1881 during Lord Ripon. It aimed to improve the working conditions of labour, to regulate health, safety, welfare, and annual leave and enact special provision in respect of young

persons, women and children who work in the factories. It also prohibited the employment of children under the age of 7 years.

Central Board of Revenue Act, 1963 is

An Act to provide for the constitution of separate Boards of Revenue for Direct Taxes and for Excise and Customs and to amend certain enactments for the purpose of conferring powers and imposing duties on the said Boards.

Real Estate (Regulation and**Development) Act, 2016 -**

It is an Act of the Parliament of India which seeks to protect home-buyers as well as help boost investments in the real estate industry. It was passed by the Parliament in 2016 that came into effect fully from 1st May, 2017.

State Financial Corporation Act, 1951 -

The Union Parliament passed the State Financial Corporation Act on September 28, 1951 in order to support the entrepreneurial activities of small to medium enterprises in India.

The act involves significant members of the board of financial corporations of different states in India. Where it promotes small and medium industries of the states. Besides, help in ensuring balanced regional development, higher investment, more employment generation and broad ownership of various industries.

Coinage Act, 2011 -

This is An Act to consolidate the laws relating to coinage and the Mints, the protection of coinage and to provide for the prohibition of melting or destruction of coins and prohibit the making or the possession thereof for issue and for matters connected therewith or incidental thereto. Coins can be issued up to the denomination of Rs.1000 as per the Coinage Act, 2011 in India.

Industrial Dispute Act, 1947 -

The act conducts labour regulations for all workers working on the Indian mainland. This Act came into effect on April 1, 1947. It regulates Indian labour law that concerns trade unions as well as individuals working in any industry in India.

Banking Regulation Act, 1949 -

This act was originally enacted as Banking Companies Act, 1949 on 16 March 1949. From 1st March, 1966, it was renamed as Banking Regulation Act, 1949. The act was initially enacted to regulate the

banking companies. In 1965, it was amended and made applicable to cooperative banks also. The act initially extended to entire nation except Jammu and Kashmir. It consists of **56 sections**.

In 1956, the word 'except the State of Jammu and Kashmir' was deleted from the Banking Regulation Act, 1949.

Important Sections of Banking Regulation Act, 1949 -

Section 8 (Prohibition of trading) - Notwithstanding in section 6 or in any contract, no banking company shall directly or indirectly deal in the buying or selling or bartering of goods, except in connection with the realization of security given to or held by it, or engage in any trade, or buy, sell or barter goods for others otherwise than in connection with bills of exchange received for collection or negotiation or with such of its business as is referred to in clause (i) of sub-section (1) of section 6.

Section 16 (Prohibition of common directors) - No banking company incorporated in India shall have as a director in its Board of directors any person who is a director of any other banking company.

Section 25 (Assets in India) - The assets in India of every banking company at the close of business on the last Friday of every quarter or, if that Friday is a public holiday under the Negotiable Instruments Act, 1881 (26 of 1881), at the close of the business on the preceding working day, shall not be less than seventy-five percent of its demand and time liabilities in India. Every banking company shall, within one month from the end of every quarter, submit to the Reserve Bank a return in the prescribed form and manner of the assets and liabilities.

Economic Terminologies

Banking Terminology

Commercial bank - These banks offer financial services like depositing money for safekeeping, providing interest on deposits, managing savings accounts, and extending loans to individuals and businesses.

NOSTRO Account - An account maintained by a domestic bank in a foreign country's currency for carrying out transactions there.

VOSTRO Account - An account that domestic banks hold for foreign banks in the former's domestic currency.

Recurring Deposit is the savings through regular monthly deposits of a fixed sum over a period.

Savings bank deposit - It is held at a financial institution that provides principal security and a modest interest rate.

Reinvestment deposit - It is a reinvestment of the interest earned on your deposit.

Fixed deposit - It is an investment instrument that allows you to invest a lump-sum amount for a chosen tenure to earn interest at a fixed rate.

Debit Card is a payment card linked to the cardholder's bank account.

Credit Card is a payment card issued by a financial institution that allows the cardholder to borrow funds to make purchases.

Overdraft is the short term financing facility. It is a type of financial product through which money can be withdrawn from your savings or current account, even if the account balance is zero.

Collateral - The term refers to an asset that a lender accepts as security for a loan.

Cheque - A paper instructing the bank to pay a specific amount from a person's account to another person in whose name it has been issued is known as a cheque.

Bank Statement is a list of all transactions for a bank account over a set period, usually monthly.

Small Savings - It helps individuals achieve their financial goals over a particular period. Examples of small savings are Kisan Vikas Patras, National Savings Certificates, Post Office Deposits.

MMID (Mobile Money Identifier) - It is a 7-digit code that is issued by a bank to its customer in order to avail the mobile banking services. Each mobile banking account has a unique MMID allotted to the user by the bank itself.

Skimming is a type of fraud where a small electronic device is used to steal debit/credit card information during an

authorised banking transaction.

The Lender of last resort (LoR) - It is an institution, usually a country's central bank, that offers loans to banks or other eligible institutions that are experiencing financial difficulty or considered highly risky or near collapse.

Basel III - These are the regulatory norms for setting common standards for banks across different countries.

Lending rate - It is the bank rate that usually meets the short - and medium-term financing needs of the private sector.

Prime Lending rate (PLR) - It is also known as prime lending rate. It is the interest rate at which banks lend to its most secure and most credit worthy customers.

Sub-prime lending Rate - It usually refers to the practice of giving loans to those who do not qualify for regular loans at market interest rates because of their poor credit history.

Rate of Interest - The amount charged by a lender to a borrower for the use of assets, or earned by a bank from a deposit account.

Annual Percentage Rate (APR) - The interest rate earned from savings accounts or certificates of deposit (CDs) at a bank or credit union. It helps consumers compare different financial products by providing a standardised measure of the cost of borrowing.

Base Rate - The minimum lending rate set by the Reserve Bank of India, below which banks cannot lend to customers except in certain cases.

Morbidity Rate - The frequency of specific diseases within a defined community over a specified period.

Amortized Rate (AR) - Calculated from an amortization schedule, it represents the decreasing percentage of each interest payment and increasing percentage of principal payment over time. This gradual shift occurs as loan balances decrease with each payment.

Variable Rate (VR) - Also known as adjustable or floating rate, it fluctuates based on an underlying benchmark interest rate or index. Borrowers benefit from lower interest payments if the benchmark rate declines, but payments

can increase if the benchmark rate rises.

Finance related Terminology

Interest is the additional amount earned on a loan or deposit. When the general interest rate reaches a very low level, most people will expect the interest rate to rise in the future.

Soft Loan - A loan with comparatively more liberal terms and conditions as compared to other loans available in the market.

Instant loans - It can get you cash quickly— sometimes within the same day even if you don't have strong credit.

Corporate finance refers to activities and transactions related to raising capital to create, develop and acquire a business.

Private equity typically refers to investment funds, generally organised as limited partnerships, that buy and restructure companies.

Investment Banking - A type of banking that conducts underwriting of large, complex financial transactions such as mergers or initial public offerings (IPOs).

Lease - It is a contract, or part of a contract, that conveys the right to use an asset (the underlying asset) for a period of time in exchange for consideration.

Credit terms are the payment terms mentioned on the invoice at the time of buying goods.

Working capital - In financial terms, working capital is the Current Assets and the current Liabilities.

Balance sheet - It is a financial statement that reports a company's assets, liabilities, and shareholder equity at a moment in time.

Assets = Liabilities + Equity.

Loan-to-value (LTV) ratio is a number that lenders use to determine how much risk they're taking on with a secured loan.

Doubtful asset - An asset that has remained in the substandard category for a period of 12 months is classified as a doubtful asset by a bank.

Bankruptcy is a legal declaration of a person who is unable to pay off debts. The Insolvency and Bankruptcy Code, 2016 (IBC) is the bankruptcy law of India.

Z-score - The Z-score formula for predicting bankruptcy was published in 1968 by **Edward Altman**. Altman's Z-Score model is a numerical measurement that is used to predict the chances of a business going bankrupt in the next two years.

Bailout - An act of giving financial assistance to a failing business or economy to save it from collapse.

Debt financing - It is a method for companies to raise capital by borrowing money from a lender with the obligation to repay the loan amount plus interest in the future.

Hedging is a strategy used to reduce or eliminate the risk of price fluctuations associated with an asset by taking offsetting positions. It serves as a stabilising force for investments.

Scalability refers to the ability of financial institutions to control the higher demands from the market.

Self-reliance is the social and economic ability of an individual, a household or a country to meet economic needs.

Money related Terminology

Fiat money is a government-issued currency that is not backed by a commodity such as gold. The value of representative full-bodied money is much higher than its value as a commodity.

Fiduciary money is the money which is accepted as a medium of exchange because of the trust between the payer and the payee.

Monetary base - The total liability of the monetary authority of the country is called the monetary base or high powered money.

Credit money refers to a future monetary claim against an individual who has used the credit facility to buy goods and services.

Indian anna - Anna was a currency unit formerly used in British India equal to 16th Part of a Rupee. It was subdivided into four Paisa or twelve pies (thus there were 192 pies in a rupee).

Treasury Bills (T-Bills) are money market instruments. These are short-term debt instruments issued by the Government of India via auctions conducted by the

Reserve Bank of India. These are presently issued in three tenors, namely, 91 day, 182 day and 364 day. These are zero coupon securities and pay no interest. Instead, they are issued at a discount and redeemed at the face value at maturity.

Other Terms

Backwardation is a situation when the futures price of a commodity is lower than the spot price today. The commodity's spot price can be high due to the sudden rise in demand for the commodity.

Merchant discount rate is the rate charged to a merchant for payment processing services on debit and credit card transactions.

Angel tax is essentially the tax that unlisted companies (startups) are liable to pay on the capital they raise through issue of shares. It was introduced in the budget of 2012.

Trade liberalisation - It has helped India improve its comparative advantage in industries with medium-to-high technology content.

Escheat - It is the income of the Government, which arises out of the property left by the people without a legal heir.

Giffen Good is a product that people consume more of as the price rises and vice versa—violating the basic law of demand in microeconomics. It is a type of good which demand increases as its price rises.

Transfer Payments are payments made by the government to individuals without corresponding production of goods or services. They are used to redistribute income within society.

Blue Economy - The blue economy is the "sustainable use of ocean resources for economic growth, improved livelihoods, and jobs while preserving the health of the ocean ecosystem."

GDP Deflator - It is the ratio between GDP at Current Prices and GDP at Constant Prices. Value of '1' implies that there is no change in the price level. It is a better measure of price behaviour because it covers all goods and services produced in the country.

Mortgage - When property, land or any other commodity is used as collateral to borrow money or to take a loan from a lender, it is known as Mortgage.

A type of loan that allows borrowers to make low payments in the initial phases but repayment of the balance amount in lump sum at maturity is known as **Balloon Mortgage**.

Invoice financing is a form of short term borrowing which is extended by the bank or a lender to its customers based on unpaid invoices. Invoice financing is often carried out to meet short-term liquidity needs of the company.

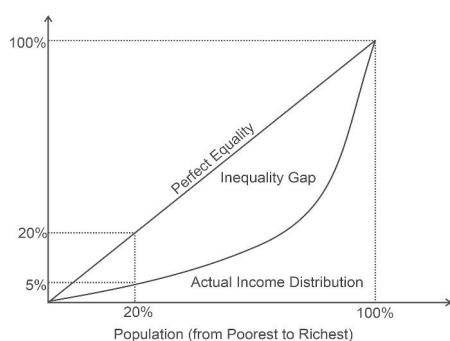
Liquidity Trap - It is a situation of very low rate of interest in the economy where every economic agent expects the interest rate to rise in future and consequently bond prices to fall, causing capital loss to render economic policy ineffective.

Paradox of Thrift - The paradox of thrift, or paradox of savings, is an economic theory that proposes that personal savings are a net drag on the economy during a recession. This theory relies on the assumption that prices are not clear or that producers fail to adjust to changing conditions, contrary to the expectations of classical microeconomics. The paradox of thrift was popularised by British economist John Maynard Keynes.

Economic Curves

The Lorenz Curve (the actual distribution of income curve), a graphical distribution of wealth, was developed by Max Lorenz in 1906. It shows the proportion of income earned by any given percentage of the population.

The line at the 45° angle shows perfectly equal **income distribution**, while the other line shows the actual distribution of income. The further away from the diagonal, the more unequal the size of the distribution of income.



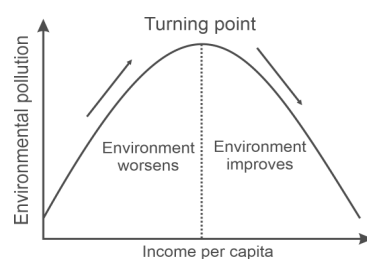
Gini Coefficient - It is derived from the **Lorenz Curve**, and can be used as an indicator of economic development in a country. The Gini Coefficient measures the degree of income equality in a population. The Gini Coefficient can vary from 0 (perfect equality) to 1 (perfect inequality). A Gini Coefficient of zero means that everyone has the same income, while a Higher Coefficient of 1 represents a single individual receiving all the income.

The distribution of Income in an economy is represented by the **Lorenz Curve** and the degree of income inequality is measured through the **Gini Coefficient**.

Kuznets curve - Kuznets Curve is used to demonstrate the hypothesis that economic growth initially leads to greater inequality, followed later by the reduction of inequality.

This curve is meant to illustrate economist Simon Kuznets' hypothesis about the behavior and relationship of these two variables (economic growth and Income inequality) as an economy develops from a primarily rural agricultural society to an industrialized urban economy.

The environmental Kuznets curve depicts the relationship between environmental degradation and per capita income.



Demand curve - It is a graphical representation of a change in product demand brought out by a change in price. A product's price is inversely related to demand, provided other factors remain constant.

Any increase or decrease in demand due to a fall or rise in price is depicted by a downward or upward movement.

Supply curve - It is a graphical representation of the relationship between the number of products that manufacturers or producers are willing to sell or supply and the price of those items at any given time. While the price of the products is indicated on the X-axis, the quantity is indicated on the Y-axis

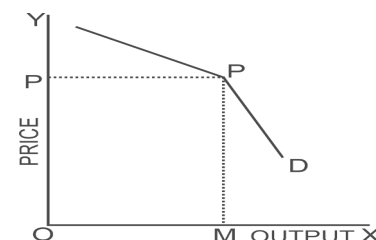
when the other conditions affecting the elements remain constant.

Sweezy's Kinked Demand Curve Model -

The kinked demand curve of oligopoly was developed by Paul M. Sweezy in 1939.

Instead of laying emphasis on price-output determination, the model explains the behavior of oligopolistic organizations.

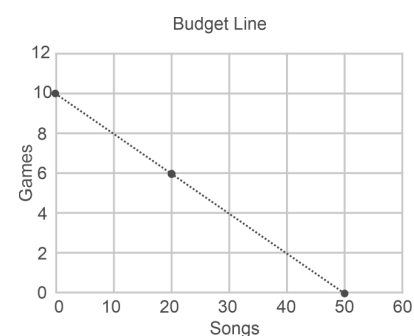
The model advocates that the behavior of oligopolistic organizations remain stable when the price and output are determined.



The Budget Line - It is also called as Budget Constraint shows all the combinations of two commodities that a consumer can afford at given market prices and within the particular income level.

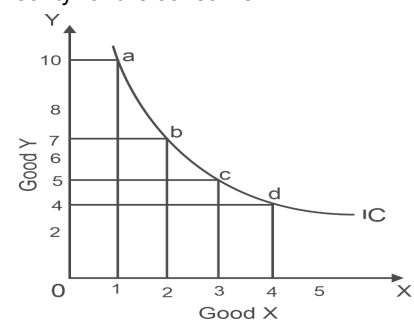
The equation of a budget line is given by- $M = P_X \cdot Q_X + P_Y \cdot Q_Y$.

Where, M = Money income of the consumer; P_X = Price of Good-X; P_Y = Price of Good-Y; Q_X = Quantity of Good-X; Q_Y = Quantity of Good-Y.

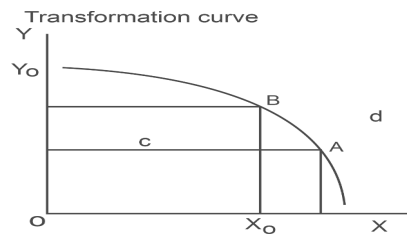


Indifference curve (IC) -

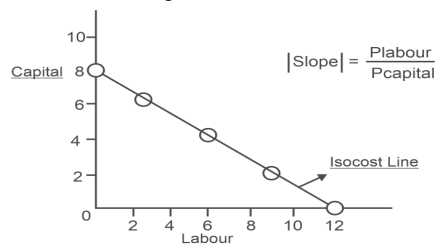
It is a graphical representation of different combinations or consumption bundles of two goods or commodities, providing equal levels of satisfaction and utility for the consumer.



Transformation curve - It is defined as the maximum amount of commodity X obtainable for any given amount of commodity Y, and vice versa. It is also known as **production - possibility curve**. The other way to represent transformation curves is using an **Edgeworth box**.

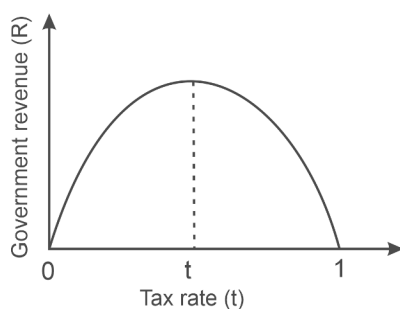


Isocost line - It is a graph showing various possible combinations of inputs (labor and capital) that can be purchased for an estimated total cost. For the two production inputs labor and capital, with fixed unit costs of the inputs, the isocost curve is a straight line.

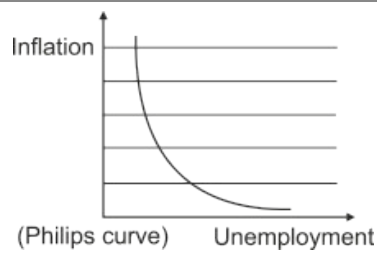


Laffer Curve - It is a graphic representation of the relationship between rates of taxation and the resulting levels of government revenue. The theory tries to arrive at an optimal tax rate beyond which tax revenues for an economy tend to fall.

Ronald Reagan, the 40th President of the US, signed the largest tax cut in the history of their country in 1981. This policy was centered around the economic theory of the Laffer Curve.



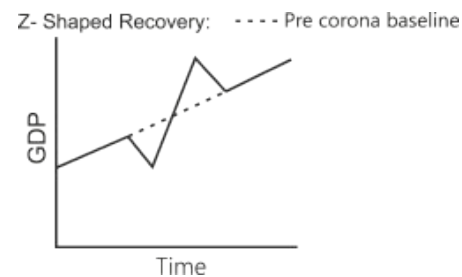
Phillips Curve - It was pioneered by A. W. Phillips, describes an inverse relationship between inflation and unemployment. It suggests that when inflation is high, unemployment tends to be low, and conversely, when inflation is low, unemployment tends to be high.



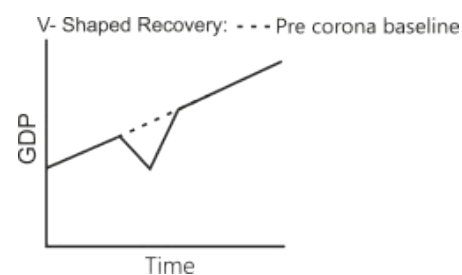
Shapes of Economic Recovery

During Covid 19 pandemic, economic activities went to a halt. The recovery of the Indian economy was termed by many as K shaped recovery. Economic recovery can take many forms, which is depicted using alphabetic notations.

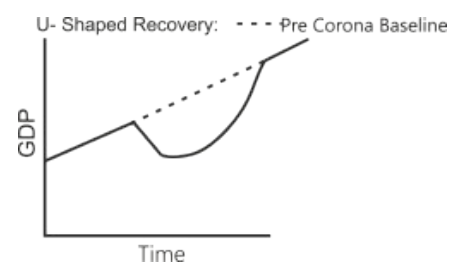
Z - Shaped Recovery - It is the most-optimistic scenario in which the economy quickly rises after an economic crash.



V-shaped Recovery - In a V-shaped recession, the economy suffers a sharp but brief period of economic decline with a clearly defined Trough, followed by a strong recovery.

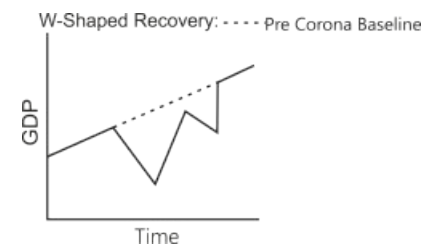


U-shaped recovery - It is a scenario in which the economy, after falling, struggles around a low growth rate for some time, before rising gradually to usual levels.

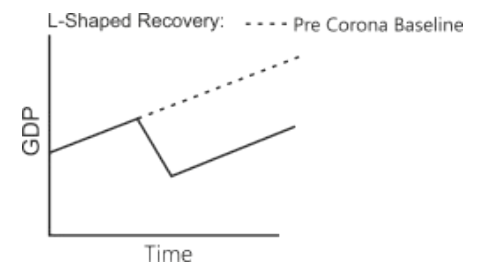


W-shaped recovery - A W-shaped recovery is a dangerous creature. In this, growth falls and rises, but falls again

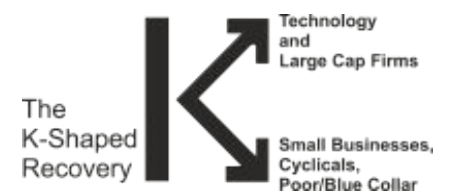
before recovering, thus forming a W-like chart.



L-shaped recovery - In this, the economy fails to regain the level of GDP even after years go by.



K-shaped recovery - It occurs when, following a recession, different parts of the economy recover at different rates, times, or magnitudes.



Banking Full Forms

BHIM - Bharat Interface for Money is an initiative to enable fast, secure, reliable cashless payments through your mobile phone.

UPI - Unified Payments Interface - Unified Payments Interface is an instant real-time payment system developed by National Payments Corporation of India.

MICR - Magnetic Ink Character Recognition is a character recognition technology used mainly by the banking industry to streamline the processing and clearance of cheques and other documents.

IFSC - Indian Financial System Code - IFSC is a unique eleven-digit number which is a combination of alphabets and numerals. It is used to transfer funds online for NEFT, IMPS and RTGS transactions.

NEFT - National Electronic Funds Transfer is an electronic funds transfer system maintained by the Reserve Bank

of India. Started in November 2005, the setup was established and maintained by Institute for Development and Research in Banking Technology.

SIDBI - Small Industries Development Bank - SIDBI is the Primary Financial Institution for promoting, developing, and financing MSME (Micro, Small, and Medium Enterprise) sector. Founded- 2 April 1990, Headquarters- Lucknow, Sivasubramanian Ramanan, (Chairman & MD).

RTGS - Real Time Gross Settlement - It is a money transfer system that allows the transfer of funds from one bank account to another in real-time and on a gross settlement basis.

CTS - Cheque Truncation System - Cheque Truncation System or Image-based Clearing System, in India, is a project of the Reserve Bank of India, commenced in 2010, for faster clearing of cheques.

IMPS - Immediate Payment Service - Immediate Payment Service is an instant payment inter-bank electronic funds transfer system in India. IMPS offers an inter-bank electronic fund transfer service through mobile phones. The service is available 24x7 throughout the year including bank holidays. NEFT was also made available 24x7 from December 2019.

NBFC - Non Banking Financing Companies - A non-banking financial institution or non-bank financial company is a financial institution that does not have a full banking license or is not supervised by a national or international banking regulatory agency.

PCA - Prompt Corrective Action - It is a system that the RBI imposes on banks showing signs of financial stress.

SWIFT - Society for Worldwide Interbank Financial Telecommunication - It is a member-owned cooperative that offers secure and safe money transfers for its representatives. Its headquarter is in Belgium and has an extensive network of distributors that includes numerous countries, including Australia, India, Japan, Germany, Korea, Italy, France, South Africa, Sweden, Spain, the US, the UK, UAE and the Russian Federation.

SAT - The Securities Appellate Tribunal (SAT) was established in 1992 in India. Securities Appellate Tribunal was mainly

established to hear an appeal against the order passed by the SEBI (Securities and Exchange Board of India) or by an adjudicating officer under the SEBI Act. It is a three-member tribunal and is currently headed by Justice D.R. Dhanuka (2022).

KCC - Kisan Credit Card- Launched in 1998; Model Scheme - National Bank for Agriculture and Rural Development on the recommendations of the R. V. Gupta Committee; Implemented by Commercial Banks, RRBs, Small Finance Banks, and Cooperatives.

UTR - Unique transaction reference is a 22 character code used to identify a transaction in RTGS system.

PPF - Public Provident Fund (PPF) is a savings-cum-tax-saving instrument in India, introduced by the National Savings Institute of the Ministry of Finance in 1968. The main objective of the scheme is to mobilize small savings by offering an investment with reasonable returns combined with income tax benefits.

BFSI - Banking, Financial, Services, and Insurance (BFSI) is the industry's umbrella term for companies that provide a range of such financial products or services. This includes universal banks that provide a range of financial services or companies that operate in one or more of these financial sectors. BFSI comprises commercial banks, insurance companies, non-banking financial companies, cooperatives, pensions funds, mutual funds and other smaller financial entities.

Other Full Forms

CPCL - Chennai Petroleum Corporation Limited is a subsidiary of Indian Oil Corporation which is under the ownership of the Ministry of Petroleum and Natural Gas of the Government of India. Headquarters- Chennai, Founded- 18 November 1965,

MOOC - Massive Open Online Courses (MOOCs) are free online courses available for anyone to enroll. MOOCs provide an affordable and flexible way to learn new skills, advance your career and deliver quality educational experiences at Scale.

Industries

Industries are sectors of the economy focused on producing goods or services. They include manufacturing, agriculture, construction, technology, finance, and more. Industries play a key role in economic growth and job creation.

Classifications of Industries

On the basis of Ownership

Private-Sector Industries- These are industries which are owned and run by an individual or group of individuals. Examples- Reliance Industries Limited, Tata Consultancy Services, Infosys Technologies Ltd.

Public-Sector Industries - These are industries which are owned and run by the government. Examples- Steel Authority of India Limited and Hindustan Aeronautics Limited.

Joint-Sector Industries - These are industries which are owned and run by the state as well as individuals. Examples- Oil India Ltd, Maruti Udyog Limited.

MSME

Micro, Small and Medium Enterprises (MSME) contribute significantly to the development process and acts as a vital link in the industrialisation in terms of production, employment and exports for economic prosperity by widening entrepreneurial base and use of local raw materials and indigenous skills. The Micro, Small and Medium Enterprises Development (MSMED) Act was passed in 2006. The National Board for Micro, Small, and Medium Enterprises meets once every 6 months in a year.

Micro Enterprises - A micro-enterprise is defined as an enterprise where the investment in Plant and Machinery or Equipment does not exceed one crore rupees and turnover does not exceed five crore rupees.

Small Enterprises - A small enterprise is defined as an enterprise where the investment in Plant and Machinery or Equipment does not exceed ten crore rupees and turnover does not exceed fifty crore rupees.

Medium Enterprises - A medium enterprise is defined as an enterprise where the investment in Plant and Machinery or Equipment does not exceed fifty crore rupees and turnover

does not exceed two hundred and fifty crore rupees.

Industrial regions - These areas have a high concentration of industries that have developed due to favorable geo-economic conditions.

The major industrial regions of India-

The Mumbai-Pune Industrial Region, The Kolkata-Hugli Industrial Region, The Ahmedabad -Vadodara Industrial Region, The Madurai-Coimbatore-Bangalore Industrial Region, The Chota Nagpur Industrial Region, The Agra- Delhi- Kalka-Saharanpur Industrial Region, Vishakhapatnam -Guntur Region, Kollam-Thiruvananthapuram Region,

Industrial corridor projects -

Following 11 industrial corridors are being developed by Government of India in a phased manner, which will be implemented through National Industrial Corridor Development and Implementation Trust (NICDIT).

1. Delhi Mumbai Industrial Corridor (DMIC)
2. Chennai Bengaluru Industrial Corridor (CBIC)
3. Amritsar Kolkata Industrial Corridor (AKIC)
4. East Coast Industrial Corridor (ECIC) with Vizag Chennai Industrial Corridor (VCIC)
5. Bengaluru Mumbai Industrial Corridor (BMIC)
6. Extension of CBIC to Kochi via Coimbatore
7. Hyderabad Nagpur Industrial Corridor (HNIC)
8. Hyderabad Warangal Industrial Corridor (HWIC)
9. Hyderabad Bengaluru Industrial Corridor (HBIC)
10. Odisha Economic Corridor (OEC)
11. Delhi Nagpur Industrial Corridor (DNIC)

Core Industries - There are Eight Core Industries which includes Natural Gas, Coal, Refinery Products, Crude Oil, Cement, Electricity, Steel, and Fertilizers. The Eight Core Industries comprise 40.27 percent of the weight of items included in the Index of Industrial Production (IIP).

IIP - The Index of Industrial Production (IIP) is an index which shows the growth rates in different industry groups of the economy in a stipulated period of time. The IIP index is computed and published

by the Central Statistical Organisation (CSO) on a monthly basis.

Distribution of Major Industries

Cotton Textile Industry

The **1st modern mill** was established at **Fort Gloster** near Calcutta 1818, but it was a commercial failure. Cowasjee Nanabhoy Davar started the first successful cotton textile mill in Mumbai in 1854. Mumbai is called the 'Cottonopolis of India'.

Maharashtra is the largest producer of cotton textile in the country followed by Gujarat and Telangana. Warangal, Secunderabad and Hyderabad are centers of cotton-textile industry in Telangana. Narma is the American cotton variety known in the north western part of India. Cotton Industry is an example of an agro-based industry. Silk and cotton goods from India dominated the international textile market before the age of mechanized industries.

Jute Industry

National Jute Policy 2005 was formulated with the objective of increasing productivity, improving quality, ensuring good prices for the jute farmers, and enhancing the yield per hectare. The policy is undertaken by the Ministry of Textiles.

The leading world's jute-producing countries are India, Bangladesh, China, and Thailand. It is known as **golden fiber**. It is cultivated mainly in Eastern India because of the rich alluvial soil of the Ganga-Brahmaputra delta. Jute industry is an example of Agro based Industry.

The **Acland Mill** was the first jute mill established in India in 1855 by British entrepreneur **George Acland** and Bengali financier Babu Bysumber Sen in Rishra, West Bengal. **West Bengal** is the largest jute producing state in India. Bihar, Assam, Andhra Pradesh, Odisha are major jute producing states.

Iron and Steel Industry

The iron and steel industry is one of the most crucial sectors in India. In January 2019, India surpassed Japan to become

the world's second-largest producer of crude steel.

Different Types of Carbon Steel-

Low Carbon, also known as "mild steel," comprises up to 0.3% carbon.

Medium carbon comprises 0.31% to 0.60% carbon.

High Carbon, also known as "carbon tool steel," is the strongest carbon steel comprising 0.60% to 1.5% carbon.

Most of the iron and steel plants are located in a region that spreads over four states West Bengal, Jharkhand, Odisha and Chhattisgarh.

In the **second five-year plan** three steel plants were established under Hindustan Steel Limited (HSL)- Rourkela (Odisha) with the help of Germany in 1959. Bhilai (Chhattisgarh) with the help of Russia; It was established in 1959. Durgapur Steel Plant was set up in 1956 in West Bengal, with the help of the UK.

Tata Steel Limited - It is formerly known as Tata Iron and Steel Company (TISCO) is in Jamshedpur, Jharkhand founded on 25 August 1907.

SAIL - Five Integrated Steel Plants are at Bhilai (Chhattisgarh), Bokaro (Jharkhand), Rourkela (Odisha), Durgapur (West Bengal) & Burnpur (West Bengal) and Special Steel Plant at Salem (Tamil Nadu)

Three more steel plants were started during the fourth Five Year Plan - Salem Iron and Steel Plant in Tamil Nadu, Vizag Iron and Steel Plant in Andhra Pradesh, Vijayanagar Iron and Steel Plant in Karnataka.

Three plants at **Hirapur**, **Kulti**, and **Burnpur** in West Bengal were set up in 1864, 1908, and 1937 respectively. The Indian Iron and Steel Company (IISCO) is the result of the merger of these plants in 1918.

In 1964, the Bokaro steel plant (Jharkhand) was established with the help of **Russia**.

Bhadravati, Karnataka, is primarily famous for its iron and steel plant which was started in 1923.

Aluminum is the second most important metallurgical industry in India after the iron and steel industry.

Companies in India associated with Aluminium - Hindustan Aluminium Company (Mumbai, Maharashtra), Vedanta Limited (Jharsuguda, Odisha), India Foils Limited (Kolkata, West Bengal), Sacheta Metals (Mumbai, Maharashtra), Jindal Aluminium (Bengaluru, Karnataka), National Aluminium Company Limited (NALCO) (Bhubaneswar, Odisha), Bharat Aluminium Company Limited (BALCO) (New Delhi, India), Madras Aluminium Company Limited (MALCO) (Mettur, Tamil Nadu), Indian Aluminium Company Limited (INDAL) (Mumbai, Maharashtra), Century Extrusions Limited (Kolkata, West Bengal).

Facts related to Industry in India

Madras Atomic Power Station (MAPS) - India's first indigenous nuclear power station located near Chennai.

Hindustan Aeronautics Limited (HAL) - Established in 1940, headquartered in Bangalore, plays a vital role in India's aerospace industry.

Hindustan Copper Limited (HCL) - A PSU under the Ministry of Mines, headquartered in Kolkata, incorporated in 1967.

ONGC Videsh - Wholly owned subsidiary of ONGC, under the Ministry of Petroleum & Natural Gas, engaged in overseas oil exploration.

National Fertilizers Limited (NFL) - Mini Ratna PSU, headquartered in New Delhi, operates gas-based ammonia-urea plants across India.

Integral Coach Factory (ICF) - Situated in Chennai, it manufactures railway coaches and is a key contributor to Indian Railways.

Muppandal Wind Farm - India's largest operational wind farm, located in Tamil Nadu's Kanyakumari district.

Asia's First Solar-Powered Textile Mill - Located in Maharashtra's Parbhani district, operated by Jai Bhawani women's cooperative, spread across 30 acres.

Pharmaceutical Industry (SPI) Schemes - Implemented by the Ministry of Chemicals and Fertilizers to support technology upgradation and

infrastructure in the pharmaceutical sector.

HMT Limited - A PSU under Ministry of Heavy Industries, headquartered in Bangalore, with manufacturing units across India, founded in 1953.

Mizoram's Sports Industry Status - Mizoram becomes the first Indian state to grant sports industry status, known for traditional textile variety Pawnsui and Puans.

Paithani Saree Weaving Industry - Maharashtra's handwoven sarees made from fine mulberry silk.

Kauna Craft - Flourishes in Manipur's Imphal Valley, with other centers in Godna, Butta, and Miri Jims.

Cement Production in Beawar - Rajasthan's Beawar is the largest producer of cement in northern India.

Asia's First Cross-Country Pipeline - Constructed by Oil India Limited for transporting crude oil from Assam to Bihar.

Hindustan Shipyard Limited (HSL) - Established in 1941, taken over by the Government in 1952, and now a central PSU, located in Visakhapatnam.

Physics

Units and Measurements

The base units for length, mass and time in CGS, FPS, and MKS systems are In the CGS system they are centimetre, gram and second respectively.

In the FPS system they are foot, pound and second respectively.

In the MKS system they are metre, kilogram and second respectively.

SI (International System) unit

SI unit is an international system of measurements that are used universally in technical and scientific research to avoid the confusion with the units.

7 SI Fundamental units are Length (l) - Meter (m), Mass (M) - Kilogram (Kg), Time (T) - Seconds (s), Electric current (i) - Ampere (A), Thermodynamic temperature (Θ) - kelvin (K), Amount of substance (n) - Mole (mol), Luminous intensity (J) - Candela (cd).

Physical Quantity and its Dimensional Formulas

◆ Area (A)= Length × Breadth , [M⁰L²T⁰]
SI Unit - m²

◆ Volume (V)= Length × Breadth × Height
[M⁰L³T⁰], SI Unit - m³

◆ Density (d) = $\frac{\text{Mass}}{\text{Volume}}$, [M¹L⁻³T⁰],
SI Unit - kg m⁻³

◆ Speed (s) = $\frac{\text{Distance}}{\text{Time}}$, [M⁰L¹T⁻¹],
SI Unit - ms⁻¹

◆ Velocity (v)= $\frac{\text{Displacement}}{\text{Time}}$, [M⁰L¹T⁻¹],
SI Unit - ms⁻¹

◆ Acceleration (a) = $\frac{\text{Change in velocity}}{\text{Time}}$
[M⁰L¹T⁻²], SI Unit - ms⁻²

◆ Acceleration due to gravity (g) = $\frac{\text{Change in velocity}(\text{gravity})}{\text{Time}}$,
[M⁰L¹T⁻²], SI Unit - ms⁻²

◆ Specific gravity = $\frac{\text{Density of body}}{\text{density of water at } 4^{\circ}\text{C}}$
No dimensions [M⁰L⁰T⁰], No units

◆ Linear momentum (p)= Mass x Velocity, [M¹L¹T⁻¹], SI Unit - kg ms⁻¹

◆ Force (F) = Mass × Acceleration,
[M¹L¹T⁻²], SI Unit - Newton

◆ Work (W) = Force × Distance, [M¹L²T⁻²],
SI Unit - J (Joule)

◆ Energy (E) = Capacity to do work.
[M¹L²T⁻²], SI Unit - J (Joule)

◆ Impulse (I) = Force × Time, [M¹L¹T⁻¹],
SI Unit - Newton-second (N-s).

The dimensional formulas of momentum and impulse are the same.

◆ Pressure (P) = $\frac{\text{Force}}{\text{Area}}$, [M¹L⁻¹T⁻²],
SI Unit - Nm⁻² or Pascal (Pa)

◆ Power (P) = $\frac{\text{Work}}{\text{Time}}$, [M¹L²T⁻³],
SI Unit - watt (W)

◆ Moment of inertia (I) = Mass × (distance)² ;
[M¹L²T⁰], SI Unit - kgm²

◆ Moment of force, moment of couple = Force × distance , [M¹L²T⁻²], SI Unit - Nm

◆ Surface tension (T) = $\frac{\text{Force}}{\text{Length}}$, [M¹L⁰T⁻²],
SI Unit - Nm⁻¹, CGS Unit - Dyne/cm

◆ Surface energy (E) = $\frac{\text{Energy}}{\text{unit area}}$, [M¹L⁰T⁻²],
SI Unit - Nm⁻¹

◆ Force constant (x) = $\frac{\text{Force}}{\text{Displacement}}$
[M¹L⁰T⁻²], SI Unit - Nm⁻¹

◆ Coefficient of viscosity (η) = $\frac{\text{Force}}{\text{area} \times \text{velocity gradient}}$, [M¹L⁻¹T⁻¹],
SI Unit - Nsm⁻²

◆ Thrust (F) = Force, [M¹L¹T⁻²], SI Unit - N

◆ Tension (T) = Force, [M¹L¹T⁻²], SI Unit - N

◆ Stress= $\frac{\text{Force}}{\text{Area}}$, [M¹L⁻¹T⁻²], SI Unit- Nm⁻²

◆ Strain = $\frac{\text{Change in dimension}}{\text{Original dimension}}$,
No dimensions [M⁰L⁰T⁰], SI Unit - No unit

◆ Modulus of Elasticity (E) = $\frac{\text{Stress}}{\text{strain}}$,
[M¹L⁻¹T⁻²], SI Unit - Nm⁻²

◆ Radius of gyration (k) = Distance,
[M⁰L¹T⁰], SI Unit - m

◆ Angle (θ), Angular displacement = $\frac{\text{Arc length}}{\text{Radius}}$, dimensionless [M⁰L⁰T⁰],
SI Unit - rad

◆ Trigonometric ratio (sin θ, cos θ, tan θ, etc) = $\frac{\text{Length}}{\text{length}}$, No dimensions
[M⁰L⁰T⁰], No unit.

◆ Angular velocity (ω)= $\frac{\text{Angle}}{\text{Time}}$, [M⁰L⁰T⁻¹],
SI Unit - rad/s

◆ Angular acceleration (α) = $\frac{\text{Angular velocity}}{\text{Time}}$, [M⁰L⁰T⁻²],
SI Unit - rad/ s²

◆ Angular momentum (J)=Moment of

inertia × Angular velocity , [M¹L²T⁻¹],
SI Unit - kgm²s⁻¹

◆ Planck's constant (h) = $\frac{\text{Energy}}{\text{Frequency}}$,
[M¹L²T⁻¹], SI Unit - Js

◆ Buoyant force = Force, [M¹L¹T⁻²],
SI Unit - N

◆ Pressure gradient= $\frac{\text{Pressure}}{\text{Distance}}$,
[M¹L⁻²T⁻²], SI Unit - Nm⁻³

◆ Torque = Moment of inertia × Angular acceleration , [M¹L²T⁻²], SI Unit - Nm

◆ Velocity gradient ($\frac{dv}{dx}$) = Velocity / Distance, [M⁰L⁰T⁻¹], SI Unit - s⁻¹

◆ Rate flow = $\frac{\text{Volume}}{\text{Time}}$, [M⁰L³T⁻¹],
SI Unit - m³s⁻¹

◆ Wavelength = Length of a wavelength,
[M⁰L¹T⁰], SI Unit - m

◆ Angular frequency (ω) = 2π × frequency , [M⁰L⁰T⁻¹], SI unit - rad s⁻¹

◆ Pressure energy= Pressure × Volume ,
[M¹L²T⁻²], SI Unit - J

◆ Temperature = [M⁰L⁰T⁰K¹],
SI Unit - K (Kelvin)

◆ Heat (Q) = [M¹L²T⁻²], SI Unit - J (Joule)

◆ Latent heat (L)= Heat / Mass , [M⁰L²T⁻²],
SI Unit - Jkg⁻¹

◆ Specific heat (S)= $\frac{\text{Heat}}{\text{Mass} \times \text{temperature}}$,
[M⁰L²T⁻²K⁻¹], SI Unit - Jkg⁻¹K⁻¹

◆ Thermal expansion coefficient or thermal expansivity= $\frac{\text{change in dimension}}{\text{original dimension} \times \text{temperature}}$,
[M⁰L⁰T⁰K⁻¹], SI Unit - K⁻¹

◆ Thermal conductivity = $\frac{\text{Heat energy} \times \text{thickness}}{\text{Area} \times \text{Temperature} \times \text{Time}}$,
[M¹L¹T⁻³K⁻¹], SI Unit - Wm⁻¹K⁻¹

◆ Centripetal acceleration = $\frac{(\text{Velocity or speed})^2}{\text{radius}}$, [M⁰L¹T⁻²]

◆ Stefan constant (σ) = $\frac{\text{Energy/Area} \times \text{Time}}{(\text{temperature})^4}$,
[M¹L⁰T⁻³K⁻⁴], SI Unit - Wm⁻²K⁻⁴

◆ Power of lens = $\frac{1}{\text{focal length}}$,
[M⁰L⁻¹T⁰], SI Unit - D (diopetre)

◆ Magnification = $\frac{\text{Image distance}}{\text{Object distance}}$,
No Dimensions , No unit

◆ Fluid flow rate = $\frac{(\pi/8)(\text{pressure}) \times (\text{radius})^4}{\text{viscosity coefficient} \times \text{length}}$,
[M⁰L³T⁻¹], SI Unit - m³s⁻¹

◆ Capacitive reactance (X_c) =

$\frac{1}{2\pi \times (\text{Angular frequency} \times \text{capacitance})}$
 (Angular frequency $\times$ capacitance),
 $[M^1L^2T^{-3}A^{-2}]$, SI Unit - ohms (Ω)

♦ Inductive reactance (X_L) = Angular frequency $\times$ inductance, $[M^1L^2T^{-3}A^{-2}]$, SI Unit - ohms (Ω)

♦ Wien constant = Wavelength $\times$ temperature, $[M^0L^1T^0K^1]$, SI Unit - mK

♦ Gas constant (R) = $\frac{\text{Pressure} \times \text{Volume}}{\text{number of moles} \times \text{temperature}}$,
 $[M^1L^2T^{-2}K^{-1}]$, SI Unit - $JK^{-1}mol^{-1}$

♦ Boltzmann constant (K) = $\frac{\text{Energy}}{\text{temperature}}$,
 $[M^1L^2T^{-2}K^{-1}]$, SI Unit - JK^{-1}

♦ Charge (q) = Current $\times$ time,
 $[M^0L^0T^1A^1]$, SI Unit - Coulomb (C)

♦ Current density = $\frac{\text{Current}}{\text{area}}$, $[M^0L^{-2}T^0A^1]$,
 SI Unit $\rightarrow Am^{-2}$

♦ Electric potential (V), voltage, electromotive force = $\frac{\text{Work}}{\text{Charge}}$,
 $[M^1L^2T^{-3}A^{-1}]$, SI Unit - Weber per second which is equivalent to volt.

♦ Resistance (R) = $\frac{\text{Potential difference}}{\text{Current}}$,
 $[M^1L^2T^{-3}A^{-2}]$, SI Unit $\rightarrow$ ohms (Ω)

♦ Capacitance = $\frac{\text{Charge}}{\text{potential difference}}$,
 $[M^{-1}L^{-2}T^4A^2]$, SI Unit $\rightarrow$ F (Farad)

♦ Electrical resistivity or (electrical conductivity) $^{-1}$ = $\frac{\text{Resistance} \times \text{Area}}{\text{length}}$,
 $[M^1L^3T^{-3}A^{-2}]$, SI Unit - Ωm (resistivity)

♦ Electric field (E) = $\frac{\text{Force}}{\text{Charge}}$, $[M^1L^1T^{-3}A^{-1}]$, SI Unit - NC^{-1}

♦ Electric flux = Electric field $\times$ area,
 $[M^1L^3T^{-3}A^{-1}]$, SI Unit - Nm^2C^{-1}

♦ Electric dipole moment = $\frac{\text{Torque}}{\text{electric field}}$,
 $[M^0L^1T^1A^1]$, SI Unit - C m

♦ Electric field strength or electric intensity = $\frac{\text{Potential difference}}{\text{distance}}$,
 $[M^1L^1T^{-3}A^{-1}]$, SI Unit - NC^{-1}

♦ Magnetic field (B), magnetic flux density, magnetic induction = $\frac{\text{Force}}{\text{current} \times \text{length}}$, $[M^1L^0T^{-2}A^{-1}]$,
 SI Unit - T (Tesla)

♦ Magnetic flux (Φ) = Magnetic field $\times$ area, $[M^1L^2T^{-2}A^{-1}]$, SI Unit - Wb (Weber)

♦ Inductance = $\frac{\text{Magnetic flux}}{\text{current}}$, $[M^1L^2T^{-2}A^{-2}]$, SI Unit - H (Henry)

♦ Magnetic dipole moment = $\frac{\text{Torque}}{\text{field}}$
 Or current $\times$ area, $[M^0L^2T^0A^1]$, SI Unit - Am^2

♦ Magnetic field strength (H), magnetic intensity or magnetic moment density - Magnetic moment / volume, $[M^0L^{-1}T^0A^1]$, SI Unit - Am^{-1} .

The unit of magnetic intensity in the CGS system is Oersted.

♦ Hubble constant = $\frac{\text{Recession speed}}{\text{distance}}$,
 $[M^0L^0T^{-1}]$, SI Unit - s^{-1}

♦ Intensity of wave = (Energy / time) / area, $[M^1L^0T^{-3}]$, SI Unit - Wm^{-2}

♦ Radiation pressure = Intensity of wave / speed of light, $[M^1L^{-1}T^{-2}]$

♦ Energy density = $\frac{\text{Energy}}{\text{volume}}$, $[M^1L^{-1}T^{-2}]$,
 SI Unit - Jm^{-3}

♦ Critical velocity = $\frac{\text{Reynold's number} \times \text{coefficient of viscosity}}{\text{Mass density} \times \text{radius}}$
 $[M^0L^1T^{-1}]$, SI Unit - ms^{-1}

♦ Escape velocity = (acceleration due to gravity $\times$ planet's radius) $^{1/2}$, $[M^0L^1T^{-1}]$, SI Unit - ms^{-1}

♦ Kinetic energy = $\frac{1}{2} \text{mass} \times (\text{velocity})^2$
 $[M^1L^2T^{-2}]$, SI Unit - J

♦ Potential energy - Mass $\times$ acceleration due to gravity $\times$ height, $[M^1L^2T^{-2}]$, SI Unit - J

♦ Rotational kinetic energy = $\frac{1}{2} \times$ moment of inertia $\times$ (angular velocity) 2 ,
 $[M^1L^2T^{-2}]$, SI Unit - J

♦ Efficiency = $\frac{\text{output work or energy}}{\text{input work or energy}}$, No dimensions $[M^0L^0T^0]$, No unit

♦ Angular impulse = Torque $\times$ time,
 $[M^1L^2T^{-1}]$, SI Unit - Js (Joule second)

♦ Permeability constant (of free space) = $\frac{2\pi \times \text{force} \times \text{distance}}{\text{current} \times \text{current} \times \text{length}}$, $[M^1L^1T^{-2}A^{-2}]$,
 SI Unit - NA^{-2}

♦ Refractive index = $\frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$
 It is a dimensionless and unitless quantity.

♦ Faraday constant (F) = Avogadro constant $\times$ elementary charge,
 $[M^0L^0T^1A^1mol^{-1}]$, SI Unit - $Cmol^{-1}$

♦ Wave number = $\frac{2\pi}{\text{wavelength}}$, $[M^0L^{-1}T^0]$

♦ Radiant flux, Radiant power = Energy emitted / time, $[M^1L^2T^{-3}]$, SI Unit - W (Watt)

♦ Luminosity of radiant flux or radiant intensity =

$\frac{\text{Radiant power or radiant flux of source}}{\text{solid angle}}$,

$[M^1L^2T^{-3}]$, SI Unit - Wsr^{-1} (Watt/steradian)

♦ Luminous power or luminous flux of source = $\frac{\text{Luminous energy emitted}}{\text{time}}$,
 $[M^1L^2T^{-3}]$, SI Unit - lm (lumen)

♦ Luminous intensity or illuminating power of source - Luminous flux / Solid angle, $[M^1L^2T^{-3}]$, SI Unit - cd (candela)

♦ Intensity of illumination or luminance (L_v) = $\frac{\text{Luminous intensity}}{(\text{distance})^2}$, $[M^1L^0T^{-3}]$,
 SI Unit - cdm^{-2} .

"nit" is a non SI unit of Luminance used to measure a multitude of light intensity.

♦ Luminous efficiency = (Total luminous flux / Total radiant flux), No dimensions, No unit

♦ Illuminance = $\frac{\text{Luminous flux incident}}{\text{Area}}$,
 $[M^1L^0T^{-3}]$, SI Unit - lx (lux)

♦ Mass defect = (Sum of masses of nucleons) - (mass of the nucleus),
 $[M^1L^0T^0]$

♦ Binding energy of nucleus = mass defect $\times$ (speed of light in vacuum) 2 ,
 $[M^1L^2T^{-2}]$

♦ Decay constant = 0.693 / half-life,
 $[M^0L^0T^{-1}]$

♦ Resonant frequency = $\frac{1}{2\pi \times \sqrt{(\text{inductance} \times \text{capacitance})}}$,
 $[M^0L^0T^{-1}A^0]$

♦ Quality factor or Q-factor of coil = $\frac{2\pi \times \text{Resonant frequency} \times \text{inductance}}{\text{Resistance}}$,

It is a dimensionless and unitless quantity.

Derived SI units :-

Force - Newton (N) or (Kg. m /s 2), and CGS - Dyne (g.cm /s 2)

Electrical conductance - Siemens (Ω).
 1 calorie of heat energy is approximately equal to **4.2 Joules**. CGS unit of work - Erg (dyne-cm) or Dyne-centimetre.

Electron-volt is a unit of energy. It is commonly used in atomic and nuclear physics. One electron volt is equal to 1.602×10^{-12} erg, or 1.602×10^{-19} joule.

Potential difference unit is Joules per coulomb.
 Coulomb per second is equivalent to ampere.

Plane angle is the physical quantity for the expression arc/radius. It is measured

in radians (2π radians in a circle) or degrees (360 degrees to a circle).

Solid Angle is measured in steradian.

Radioactivity - Becquerel (Bq)

A physical quantity not having dimensions called a **dimensionless quantity**. A dimensionless quantity may possess a unit or it may be unitless.

Example - Angular displacement (θ) coefficient of friction, Strain, Mass fraction, Molar fraction, Relative humidity, Mach number, refractive index, Specific gravity. Angle of Friction are dimensionless quantities.

The physical quantities which have only magnitude are known as **scalar quantities**. Example - mass, length, time, speed, temperature, electric current, volume, density, energy.

Vector Quantities are those which have both magnitude and direction. Examples of Vector Quantities – Force, Electric field, Angular Momentum, Magnetic Moment, Linear Momentum, Average Velocity, Torque, Impulse, Acceleration and Gravitational Intensity.

Chain and statute mile was a unit of length.

A rod, a perch or a pole is a unit of length in the imperial and US systems.

Stevenson screen is a unit designed for wet and dry bulb thermometers to protect temperature sensors from being affected by direct or reflected sunlight.

Light year is the distance light travels in 1 yr (3.15×10^7 s) with a speed of 3.00×10^8 m/s. Larger stellar and extragalactic distances are measured in Light year. One light-year is about 6 trillion miles (9 trillion km) (9.46×10^{12} km).

Astronomical unit (AU) is a unit of length. It is effectively equal to the average distance between Earth and the Sun. 1 Astronomical Unit = 1.496×10^{11} m (92,955,807.3 miles)

The parsec is a unit of length used to measure vast distances between astronomical objects outside the Solar System. **1 parsec** = $3.085677581 \times 10^{16}$ metres. One parsec is approximately equal to 3.26 light-years.

The unit of calorific value of fuel is kilojoule per kilogram i.e. kJ/kg.

A fathom is a unit of length, it is used especially for measuring the depth of water. 1 fathom equal to 6 feet or 1.8288 m.

Angstrom (Å), unit of length, equal to 10^{-10} metre, or 0.1 nanometre. It is used in measuring wavelengths of light.

The **knot** is a unit of speed.

1 Knot equal to 1.852 km/h.

The **pound** is a unit of mass. 1 pound is equal to 16 ounces. One pound is approximately equal to 0.45359237 kilograms (kg).

The SI unit of Pressure is Pascal (Pa), which is N/m^2 (Newton per square meter).

CGS unit of pressure - Dyne / cm^2 .

Torr is a unit of pressure.

1 torr is equal to 133.32 Pa.

The standard atmospheric pressure is 101325 Pa.

Horsepower is a unit of measurement of power. 1 Horsepower is equal to 746 watt.

Kilowatt hour (KWh) commercial unit of electric energy. 1 kilowatt hour = 3.6×10^6 joules. 1kw = 3600 KJ/h.

Watt-hour is a unit of energy. It is often used to measure the amount of electrical energy consumed.

Eight original SI prefixes were officially adopted: deca, hecto, kilo, myria, deci, centi, milli, and myrio.

SI Prefixes -

(10^{24} , Yotta, Y), (10^{21} , Zetta, Z), (10^{18} , Exa, E), (10^{15} , Peta, P), (10^{12} , Tera, T), (10^9 , Giga, G), (10^6 , Mega, M), (10^3 , kilo, k), (10^2 , Hecto, h), (10^1 , deca, da), (10^{-1} , deci, d), (10^{-2} , centi, c), (10^{-3} , milli, ml), (10^{-6} , micro, μ), (10^{-9} , nano, n), (10^{-12} , pico, p), (10^{-15} , femto, f), (10^{-18} , atto, a), (10^{-21} , zepto, z), (10^{-24} , yocto, y).

The **Dobson Unit** is the most common unit for measuring ozone concentration.

Candela is the SI base unit for photometry.

'**mho**' is the unit of electrical conductivity of a substance.

Resistance and conductance are inverse of each other that is why $\text{mho} \propto \frac{1}{\text{ohm}}$.

Ohmmeter is used for measuring electrical resistance.

The **Svedberg** unit is a measurement of the sedimentation rate of a particle when it is centrifuged. It is actually a measure of time and is defined as exactly 10^{-13} seconds.

Dioptre is the unit of measurement of power of a lens. Diopter is equal to the reciprocal of focal length of the lens.

Ophthalmology is a branch of medicine which deals with diagnosis of eye disorders.

Tonometer is used to measure the

pressure inside the eyes of a person.

The **Hubble Constant** is the unit of measurement used to describe the expansion of the universe.

The cosmos has been getting bigger since the Big Bang kick-started the growth about 13.82 billion years ago.

A nautical mile is equal to 1852 metres.

Nautical mile is used in air and water navigation as a unit of measurement.

Important Scientific Instruments and their Usage :

Accumulator - It is used to store electrical energy.

Wattmeter - It is an instrument for measuring the electric active power in watts of any given circuit.

Ammeter - It is used to measure either direct or alternating electric current.

Galvanometer - It measures the electric current of low magnitude in the circuit.

Potentiometer - It is used for comparing the electromotive force of cells.

Voltmeter - It measures the electric potential difference between two points.

Commutator - An instrument to change or remove the direction of an electric current. In dynamo it is used to convert alternating current into direct current.

Electroscope - It detects the presence of an electric charge.

Transistor - A small device which may be used to amplify currents and perform other functions usually performed by a thermionic valve.

Anemometer - It measures the force and velocity of the wind.

Wind vane - It is an instrument that measures the direction of wind.

Barograph - It is a barometer that records the barometric pressure over time in graphical form.

Turbidity Meter - It is used to measure the turbidity. Turbidity is the measure of relative clarity of a liquid.

Binocular - It is used to view distant objects.

Altimeter - It measures altitudes and is used in aircraft.

Calorimetry - It is the process of measuring the amount of heat released or absorbed during a chemical reaction.

Carburettor - It is used by an internal combustion engine to control and mix air and fuel entering the engine.

Cinematography - It is an instrument

used in cinema making to throw on screen an enlarged image of a photograph.

Quartz Clock - A highly accurate clock used in astronomical observations and other precision work.

Colorimeter - It is an instrument that compares the amount of light getting through a solution with the amount that can get through a sample of pure solvent.

Cyclotron - A charged particle accelerator which can accelerate charged particles to high energies.

Dynamo - It converts mechanical energy into electrical energy.

Dynamometer - It measures force, torque, and power of engine and motor.

Endoscope - It is a thin tube like instrument with an attached camera used to examine the internal parts of the body.

Eudiometer - A glass tube for measuring volume changes in chemical reactions between gases.

Fathometer - It measures the depth of the ocean.

Hydrometer - It measures the specific gravity of liquids.

Hydrophone - It is used to detect acoustic signals underwater.

Hygrometer - It measures relative humidity in the air. The concentration of Moisture present in the air is known as humidity.

Kymograph - It graphically records physiological movements (Blood pressure and heartbeat).

Lactometer - It determines the purity of milk.

Manometer - It measures the pressure of gases.

Mariner's compass - It is an instrument used by the sailors to determine the direction.

Microphone - It converts the sound waves into electrical vibrations.

Phonograph - An instrument for producing sound.

Intensity of sound (noise)- It is measured in Decibels.

Machmeter- It is used to measure the speed of an aircraft in terms of the speed of sound.

Speedometer - It is used to measure the speed of the vehicle.

Tachometer - An instrument used in measuring speeds of aeroplanes and motor boats.

Odometer - It is used to measure distance travelled by vehicle.

Cathetometer - It is a device used for accurate measurement of small vertical distances.

Photometer - It is used to measure the intensity of light.

Radar - It uses radio waves to determine the distance (ranging), angle, or velocity of objects. It can be used to detect aircraft, ships, spacecraft, guided missiles, motor vehicles.

Radiometer - It measures the emission of radiant energy.

Rain Gauge - An apparatus for recording rainfall per unit area at a particular place.

Udometer - It is used to measure the amount of liquid precipitation over a set period of time. It is also called Rain Gauge.

Pirani gauge - It measures pressure in the vacuum systems.

Rectifier - An instrument used for the conversion of Alternating Current into Direct Current.

Saccharimeter - It measures the amount of sugar in the solution.

Salinometer - It determines the salinity of solution.

Seismograph - It measures the intensity of earthquake shocks.

Sextant - This is used by navigators to find the latitude of a place by measuring the elevation above the horizon of the sun or another star. It measures the angle between two visible objects.

Spectrometer - It is an instrument for measuring the energy distribution of a particular type of radiation.

Spectroscope - An instrument used for spectrum analysis.

Spherometer- It is an instrument used for the precise measurement of the radius of curvature of a sphere or a curved surface.

Sphygmomanometer - It measures blood pressure.

Stereoscope - It is used to view two dimensional pictures.

Stethoscope - An instrument which is used by the doctors to hear and analyse heart and lung sounds.

Stroboscope - It is used to view rapidly moving objects.

Teleprinter - This instrument receives and sends typed messages from one place to another.

Telescope - It views distant objects in space.

Theodolite - It measures horizontal and vertical angles.

Thermostat - It regulates the temperature at a particular point.

Venturi meter - It measures flow of liquid.

Vernier - An adjustable scale for measuring small subdivisions of scale.

Viscometer - It measures the viscosity of liquids.

Planimeter - It is also known as a platometer, is a measuring instrument used to determine the area (irregular) of an arbitrary two-dimensional shape.

Cartometer- It consists of a small wheel and a calibrated dial used to measure distances on a map by following the desired route.

Segmometer- It is a modified tape measure with tips used for body measurements (anthropometry) and lengths of bones.

Piezoelectric meter - It is a device that uses the piezoelectric effect to measure changes in pressure, acceleration, temperature, strain, or force by converting them to an electrical charge.

Pyranometer - It is an instrument that is used by meteorologists to measure the solar radiation from the region of the hemisphere of incidence on a flat surface.

Thermopile - It is an instrument that measures small quantities of radiant heat.

Bolometer - It is a device to measure infra - red radiation. Bolometer is used for measuring radiant heat by means of a material having a temperature-dependent electrical resistance.

Chronometer- It is an extraordinarily accurate timepiece, with an original focus on the needs of maritime navigation.

Hypsometer - It is an instrument for measuring height or elevation.

Pyrheliometer - It is an instrument designed specifically to measure the direct beam solar irradiance.

Sling Psychrometer- It is used to measure both the dry bulb and wet bulb temperatures at times.

Pyrometer - It measures a very high temperature of the surfaces.

Reaumur Scale- It is used to measure Temperature.

Electrocardiogram (ECG)- It is a simple test that can be used to check your heart's rhythm and electrical activity.

Polygraph (commonly referred to as a lie detector)- It is an instrument that measures and records several physiological responses such as blood.

Chandrasekhar limit- It is the maximum mass of a stable white dwarf star.

Vernier caliper - It is an instrument used to measure the diameter of round and cylindrical objects by positioning the jaws of the caliper on either side of the circumference. It was introduced in 1631 by Pierre Vernier.

In a vernier caliper, a retainer is used to block a movable part to allow the easy transferring of a measurement.

Least Count of Vernier Caliper is 0.1 mm.

Least count =

$$\frac{\text{Value of one main scale division}}{\text{Total number of division on vernier scale}}$$

Speed and Laws of Motion

Motion is a change in position of an object with respect to time.

Distance - The total length traveled by an object from one point to another in a given time interval is called distance. It is a scalar quantity. SI unit is meter.

Distance-time graphs are used to find the speed of objects. The slope of the curve in the distance-time graph represents the instantaneous speed.

Displacement is defined as the change in position of an object or the shortest length between initial and final position. It doesn't depend on the path. It has both direction and magnitude. It is a vector quantity. The SI unit is meter.

Displacement is the area under the curve of a velocity-time graph included between itself and two time co-ordinates.

Speed is defined as the distance traveled per unit time. It is a scalar quantity and its SI unit is metre per second (m/s).

If an object covers a distance 'd' in time 't', then its speed, $s = \frac{d}{t}$

When the distance traveled by an object is directly proportional to the time, it is

said to travel with constant speed.

Average speed is the total distance traveled by the object in a particular time interval.

If an object is moving with an actual speed of 'x' in uniform motion its average speed will also be 'x'.

Velocity (v) is defined as the speed at which something moves in a particular direction. It is displacement per unit time. The SI unit of velocity is metre per second (m/s).

The velocity of an object can be uniform or variable. It can be changed by changing the object's speed, direction of motion or both.

Average velocity is defined as the change in position or displacement (Δx) divided by the time intervals (Δt), when an object is moving along a straight line.

In case the velocity of the object is changing at a uniform rate, then

Average Velocity

$$= \frac{\text{initial velocity} + \text{final velocity}}{2}$$

Acceleration (a) is the rate of change of velocity. In non-uniform motion, velocity varies with time. It has different values at different instants and at different points of the path. When an object moves in a circular path at a constant speed, it is still accelerating, because the direction of its velocity is changing.

The SI unit of acceleration is the meter per second square or ms^{-2} .

Acceleration is defined as the change in the velocity of an object per unit time.

$$\text{Acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$$

Newton's First Law of motion : It is also called the law of Inertia.

Newton's first law states that every object will remain at rest or in uniform motion in a straight line unless compelled to change its state by the action of unbalanced external forces.

Example - Passengers sitting in a stopped vehicle tend to lean backwards when a stopped vehicle starts moving, shaking the tree causes its fruits to break and fall down etc.

Newton's Second Law of motion :

The rate of change of momentum of any body is proportional to the force applied and its (momentum change) direction is the same as that of the force. Newton's second law gives the magnitude of force.

$$F \propto \frac{\Delta p}{\Delta t}$$

Which can be further derived in terms of Force, mass and acceleration.

Force = mass x acceleration ($F = m \times a$) speed.

Example - An aircraft's motion resulting from aerodynamic forces, aircraft weight, and thrust.

Newton's Third Law of motion : It is also called Action and Reaction law.

The third law states that for every action (force) in nature there is an equal and opposite reaction acting on different bodies.

Example - Movement of horse drawn carriage. Rocket propulsion technology. The motion of lift from an airfoil. The motion of a spinning ball, the air is deflected to one side, and the ball reacts by moving in the opposite direction.

In the Jet Airplanes, the gas is released backwards in the form of a jet by the explosion. The gas propels the airplane forward by applying a reaction force, which makes the airplane fly forward with a high velocity.

Uniform Motion - It is defined as the motion of an object in which the object travels in a straight line, and its velocity remains constant along that line as it covers equal distances in equal intervals of time, irrespective of the duration of the time.

If a body is involved in rectilinear motion and the motion is consistent, then the acceleration of the body must be zero.

Non Uniform Motion - The motion of an object in which the object travels with varied speed and it does not cover the same distance in equal time intervals, irrespective of the time interval duration.

If a body is involved in rectilinear motion, and if the motion is not consistent, then the acceleration of the body must be non-zero.

Example - If a car covers 10 meters in the first 2 seconds and 15 meters in the next 2 seconds.

Uniform Circular Motion - When a body moves with a constant speed in a circular path. Its motion is said to be the uniform circular motion. Velocity is not constant due to continuous change in the direction of motion.

Example - Revolution of the earth around the sun. Second, minute and hour hands of watch.

Non Uniform Circular Motion - It is a

circular motion when the speed of the body doesn't remain constant. The body has both normal as well as tangential acceleration. Its velocity would change every second.

Example - Motion of a body on a vertical circle.

Rectilinear motion is the motion in which objects move along the straight path. So, a car moving in the straight path, lifting of an object up and down, the free fall of a heavy object are examples of a Rectilinear motion.

The motion of an object moving in a curved path is called **curvilinear motion**. A stone thrown up in the air at an angle is an example of Curvilinear Motion.

Projectile Motion or Parabolic motion is the motion of an object traveling along a path in the shape of a parabola, such as the path of a golf ball and the throwing of a stone.

Rotational motion can be defined as the motion of an object around a circular path, in a fixed orbit.

Example - The motion of the wheel, gears, motors, is rotational motion. The motion of the blades of the helicopter is also rotatory motion.

Three equations of motion

Velocity time equation -

First Equation of motion : $v = u + at$.

Position time equation -

Second Equation of motion :

$$s = ut + \frac{1}{2}at^2.$$

Position velocity equation -

Third Equation of motion : $v^2 = u^2 + 2as$.

u = initial velocity, v = final velocity,
 a = acceleration, s = displacement,
 t = time.

When an object is traveling in a straight line with equal increase in velocity at equal intervals of time, then the object is said to be in uniform acceleration.

The motion of a freely falling body is an example of uniform accelerated motion.

The motion of the body moving along a circular path is an accelerated motion.

Retardation is when a moving object is losing speed continuously with time. Retardation is negative acceleration.

Apparent weight of a person inside a lift:

(i) When lift accelerates upwards, the apparent weight of the person inside

increases.

(ii) When lift accelerates downwards, the apparent weight of the person inside decreases.

(iii) When lift is at rest or moving with uniform velocity upward or downward then apparent weight is equal to actual weight.

(iv) When a person in the lift falls freely, he feels weightlessness.

The weight of the body at the centre of the earth is zero.

$$W = mg = 0$$

(g at the centre of the earth is zero).

Oscillatory Motion

A motion that repeats itself at equal intervals of time is called periodic or oscillatory motion. The to-and-fro motion of an object relative to its mean position is referred to as oscillatory motion.

Examples of Oscillatory Motion: Oscillation of a simple pendulum, vibrating strings of musical instruments, movement of spring, freely hanging bob, alternating Current.

Simple Pendulum - The simple pendulum comprises a small metallic ball, called bob, of mass ' m ' suspended by a thin string secured to a platform at its upper end of length (l).

Amplitude of a simple pendulum is defined as the distance travelled by the pendulum from the equilibrium position to one side.

Length of a simple pendulum is defined as the distance between the point of suspension to the centre of the bob and is denoted by " l ".

Note: If the temperature of a system changes then the time period of the simple pendulum changes due to a change in length of the pendulum.

Examples: In summer, the pendulum clocks takes more time to complete one oscillation because length of string increases (due to thermal expansion) as Time period $\propto \sqrt{\text{length}}$.

Time period of a pendulum increases at the Moon. Gravity on moon is one sixth of the gravity on the earth and time period is inversely proportional to g i.e.

$$\text{Time period} \propto \frac{1}{\sqrt{g}}.$$

Simple harmonic motion - A simple form of oscillatory motion is called Simple Harmonic Motion (SHM).

According to Hooke's law, "the restoring force is directly proportional to the displacement from the equilibrium position".

$F_s = -kx$, the negative sign shows that the force is acting against the displacement (x).

Inertia is the inherent property of a body that makes it oppose any force that would cause a change in its motion.

According to the law of inertia, an object will continue to be in the state of rest or in a state of motion unless an external force acts on it.

Types of Inertia :-

Inertia of Rest - When the resistance is offered by the body to continue in the state of rest unless an external force acts on it.

Example - Dust particles remain at rest until the carpet is shaken.

Inertia of direction - When the resistance is offered by the body to continue the motion in the same direction unless an external force acts on it.

Example - When a moving bus takes a sharp turn at a circular path, we fall in the opposite direction.

Inertia of Motion - A body in uniform motion does not get accelerated or retarded on its own. It also can not come to rest on its own.

Example - When a car comes to stop suddenly the passenger sitting inside the car leans forward.

Other Examples of inertia - Tendency of moving back when the stationary bus starts to move, jerk experienced when the lift operates suddenly, moving forward when a sudden break is applied.

Moment of inertia is the phenomenon by which the body in motion opposes the change in its rotational motion.

The magnitude of MOI is directly proportional to the square of the distance between the mass and the axis of rotation.

It is also called the rotational inertia of the body. It is a scalar quantity.

Inertia is directly proportional to mass. so Inertia will increase if mass increases.

Moment of Inertia, $I = m \times r^2$. where, m = sum of the product of the mass, r = distance from the axis of the rotation.

The moment of inertia depends on the

following factors - The density of the material, shape and size of the body, axis of rotation (distribution of mass relative to the axis).

Moment of inertia of a thin rod of mass 'M' and length 'L' about the axis perpendicular to the rod and passing

through its centre is $\frac{ML^2}{12}$.

Relativity is a theorem formulated by Albert Einstein, which states that space and time are relative, and all motion must be relative to a frame of reference.

Albert Einstein's Theory of Relativity encompasses two theories:

Special Theory of Relativity - It deals with the structure of space-time (time interval). The laws of physics are the same for all, irrespective of the observer's velocity.

Special relativity implies consequences of mass-energy equivalence, relativity of simultaneity, length contraction, and a universal speed limit.

General Theory of Relativity - It states that being at rest in the gravitational field and accelerating are identical physically. For example, an observer can see the ball fall the same way on the rocket and on Earth. This is due to the rocket's acceleration, which equals 9.8 m/s^2 .

Force and Pressure

Force is an external agent capable of changing a body's state of rest or motion. It has a magnitude and a direction.

The Force can be measured using a spring balance.

The SI unit of force is Newton (N).

$1 \text{ kg} = 9.81 \text{ N}$.

If two forces act on an object from opposite directions, then the magnitude of the net force acting on it is the difference of the magnitudes of the two forces and which is equal to, mass x acceleration.

Types of Forces :

Frictional Force - The force of friction always opposes the applied forces. Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. It depends on the smoothness of the surfaces. When a surface is rough, frictional force increases.

Friction force, $f = \mu N$.

μ = coefficient of friction, N = normal force.

An object slips less on rough surfaces than a smooth surface.

Example - Applying brakes to stop a moving vehicle.

Type of Friction :-

Static friction is defined as the frictional force that acts between the surfaces when they are at rest with respect to each other. Example - Table lamp resting on the table.

Rolling friction or **Rolling drag** is the force resisting the motion when a body rolls on a surface. Example - Wheels of the moving vehicles.

Sliding friction is a frictional force that appears when two bodies come into sliding contact. Example - Sliding of the block across the floor.

Note: Static friction > Sliding friction > Rolling friction

The substances which reduce friction are called **Lubricants**.

Fluid friction (viscosity) is defined as the friction that exists between the layers of the fluid when they are moving relative to each other. Examples - The flow of ink in pens, break oil and engine oil.

Air resistance force - It is a force that is caused due to air when an object moves through it.

Muscular Force - When force is caused or carried out by the action of muscles is called muscular force. It is a contact force.

Other types of contact forces are: Frictional force, Tension force, Normal Force, Air Resistance Force, Applied Force, Spring Force.

Gravitational Force - Earth exerts a pull on all objects or bodies towards itself. This pull is known as gravitational force. As a result of gravitational force, every object in this universe, irrespective of its size and shape, exerts some force on every other object.

Electrostatic Force - It is a force applied by a charged body on another charged or uncharged body called electrostatic force. In other words, the electrostatic force is an attractive or repulsive force

acting between two electrically charged objects.

Gravitational force, magnetic force, electrostatic force and Nuclear force are non-contact forces.

Buoyant Force - It is an upthrust force exerted by fluid opposing the weight of the object that is fully or partially immersed in that fluid.

Buoyancy is generally caused by the differences in pressure acting on opposite sides of an object immersed in a static fluid.

Tension Force - Tension force acts when a rope or string or any similar object is pulled by forces acting from the opposite direction. It is directed over the length of the wire or the string.

Drag Force - It is the resistant force caused by the motion of a body through a fluid. It also acts in a fluid like water or air. Drag is an example of mechanical force.

Spring Force - It is a type of restoring force. This is because the force exerted by the spring is always in the opposite direction to the displacement.

Nuclear Force - It is the force acting between two neutrons, between two protons or between a neutron and a proton. A nuclear force is always an attractive force.

Shear force is a force acting in a direction that's parallel to (over the top of) a surface or cross section of a body.

Shearing Stress is defined as a type of stress that acts coplanar with a cross section of material. The SI unit is pascal.

A longitudinal strain is defined as the ratio of change in the length due to applied force to the original length of an object. It is caused due to longitudinal stress.

Tensile stress or compressive stress can also be termed as Longitudinal stress.

Centripetal force - When an object moves in a circular path, a force acting towards the center of the object is called centripetal force. It is responsible for the motion of the Moon around the Earth.

Exogenic forces - The forces which derive their strength from the earth's exterior or originate within the earth's atmosphere are called exogenic forces or external forces.

Example - Weathering and Erosion.

London forces are a temporary attractive force, and the interaction energy is inversely proportional to the sixth power of the distance between two interacting particles. These forces are also known as dispersion forces.

These are the weakest intermolecular attractive forces. Examples - gasoline and petroleum jelly, which are mostly made of alkanes of varying lengths.

Conservative force - It is a force with the property that the total work done in a moving particle between two points is independent of the path taken.

Properties of Conservative Forces :
Work done by it is completely reversible. A conservative force is one in which work done by or against it depends only on the starting and ending points of a motion. The stored energy is reversible as work.

Torsion balance is used to measure small forces. It is based on the principle that a wire or thread resists twisting with a force that is proportional to the stress.

Pressure - It is defined as the force (**thrust**) exerted on a surface divided by the area.

$$\text{Pressure (P)} = \frac{\text{Force}}{\text{Area}}$$

The SI unit is Newtons per meter square (N/m^2), and this unit is given a new name, Pascals (Pa) or Bar or Torr, $1 \text{ Pa} = 1 \text{ N/m}^2$, $1 \text{ Bar} = 0.986923 \text{ atm}$, $1 \text{ atm} = 1.013 \text{ bar}$.

The air pressure always moves from high pressure areas to low pressure areas.

Thrust - It is the force applied to a surface in a direction that is normal or perpendicular to the surface. Thrust acting per unit area is known as pressure i.e.

$$\text{Pressure} = \frac{\text{Thrust}}{\text{Area}}$$

Momentum (p) is defined as the quantity of motion of the body. Momentum of a body is the product of its mass (m) and velocity (v), and is denoted by p.

$$p = m v$$

The SI unit of momentum is a kilogram meter per second (kg m/s). Momentum is a vector quantity. The Jet engine works on the phenomenon of Conservation of Linear Momentum.

Conservation of linear momentum :

According to the conservation of linear momentum, If the net external force acting on a system of bodies is zero, then the momentum of the system remains constant.

The Jet engine works on the principle of Conservation of linear momentum.

Impulse is a large amount of force you apply for a very short amount of time to cause a change in linear momentum. The SI unit of impulse is Newton-seconds (Ns) or kg m/s .

Work and Energy

Work is said to be done when a force is applied on an object and the object moves due to the force in the direction of the force.

Work Done = Force $\times$ Displacement.

The SI unit of work is Joule (J). It is a scalar quantity.

When the Force of 2N moves an object of 2 gram, 2 meters in the direction of force then Work is done 4 joules.

A **conservative force** is a force with the property that the total work done in moving a particle between two points is independent of the path taken.

Gravitational force and elastic spring forces are two examples of conservative forces.

Work Done by conservative forces is completely reversible.

Non-Conservative force is a force with the property that the total work done in moving a particle between two points is dependent on the path taken. Examples of non conservative forces include frictional force, viscous force, etc.

Energy and **Torque** both of these physical quantities have the same dimensions.

Energy is the stored ability to do work . According to the law of conservation of energy, energy can neither be created nor be destroyed. We can only transfer energy from one form to another .

Torque is defined as the vector product of linear instantaneous force and the distance of that force with the axis around which it is rotating. Different points on the same body may experience different torque. It is also called moment of force.

$$\tau = r \times F$$

SI Unit of Torque is Newton meter.

The unit of work and energy is also Newton-meter or Joule.

The measure of the force that can cause an object to rotate about an axis is called Torque.

Type of energy :-

Radiant energy is the physical energy resulting from electromagnetic radiation, usually observed as it radiates from a source into the surrounding environment. Its sources include the entire electromagnetic radiation spectrum, including gamma rays, x-rays, radio frequencies, microwaves, light and heat. Example - Solar Thermal plant.

Nuclear energy - It is a form of energy released from the nucleus, the core of atoms, made up of protons and neutrons. This source of energy can be produced in two ways: fission (when nuclei of atoms split into several parts) or fusion (when nuclei fuse together).

Geothermal energy is the energy derived from underground heat of the earth. Geothermal power plants use steam to produce electricity.

Solar energy is created by nuclear fusion that takes place in the sun. It is necessary for life on Earth, and can be harvested for human uses such as electricity. Solar cell panels are used to convert this energy into electricity.

Mechanical energy is the energy that is possessed by an object due to its motion or due to its position. It is the sum of potential energy and kinetic energy.

The Energy possessed by a body due to its motion is called **kinetic Energy (K.E)**. It is a property of a moving object that depends not only on its motion but also on its mass. Example - Wind Turbine, falling Coconuts, speeding cars, rolling stones and flying aircraft.

For an ideal gas at a temperature of 25°C , the average kinetic energy per molecule in SI units is $6.17 \times 10^{-21} \text{ J}$.

$$K.E = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2KE}{m}}$$

Potential Energy is stored energy that depends upon the relative position of various parts of a system. It is denoted by (U) = mgh.

Example - Spring has more potential energy when stretched or compressed, Hydropower plant.

Biological Energy is a form of renewable energy that is derived from recently living organic materials known as biomass, which can be used to produce transportation fuels, heat, electricity, and products.

Light Energy is a kind of kinetic energy with the ability to make types of light visible to human eyes.

Chemical Energy is the energy which is stored in the bonds of chemical compounds (molecules and atoms). It is released in the chemical reaction and mostly produces heat as a by-product. Example - Diesel Generator.

Magnetic Energy is the movement of the electrons in the different particles. It is the movement that generates the current that produces the behaviour of the electron like that of a small magnet. The earth also possesses a magnetic field generating magnetic energy on the earth.

Electrical Energy is the energy generated by the movement of electrons from one point to another point.

The following types of energies can be harnessed from the sea -

Ocean wave energy- Kinetic energy in the waves is used to generate electricity.

Ocean thermal energy - The temperature difference at different depths of the ocean is converted into mechanical and then into electrical energy.

Tidal energy is power produced by the surge of Ocean waters during the rise and fall of tides. Tidal energy is a renewable source of energy.

Sound Energy is the movement of energy through a substance and is caused by vibrations. Solids, liquids and gasses transmit sound as waves.

Thermal Energy arises due to the motion of atoms or molecules in the body, and it is measured as average kinetic energy of molecules.

Wind Energy is the most mature and developed renewable energy. It generates electricity through wind, by using the kinetic energy produced by the effect of air currents.

Hydel Energy (Hydroelectric energy) - It is a form of renewable energy that uses the power of moving water to generate electricity.

Example - water flowing over a waterfall.

Conversion of energy - Scientific Principle behind the generation of Hydroelectric power.

In a hydropower plant, the water stored in the dam has gravitational potential energy.

When the water flows from higher altitude, the Potential energy is converted into Kinetic energy.

When the flowing water rotates the **turbine**, the Kinetic energy is converted into Mechanical energy.

The **generator** converts Mechanical energy into Electrical energy.

When a stone is thrown upwards, kinetic energy would be converted into gravitational potential energy.

Example of Electrical Energy to Kinetic Energy conversion - electric motors (Fan, Mixer, elevators).

In the process of **photosynthesis**, Solar Energy from the sun is converted into Chemical Energy.

Electrochemical cell or Battery is a device which converts a Chemical energy into Electrical energy through a redox reaction.

In **Photovoltaic** cells, the Light Energy is directly converted to Electrical Energy. Applications in which photovoltaic cells are used : radio, space probes and artificial satellites.

Photovoltaic cells generate clean and green energy as no harmful gasses such as CO₂ and NO₂ are released.

Relation between Energy (E), Capacitance (C) and Potential difference (V) -

The capacity of a capacitor to store the electric charge is called Capacitance.

The capacitance of a conductor is the ratio of charge (Q) to it by a rise in its Potential (V)

$$C = \frac{Q}{V}$$

The unit of capacitance is Farad.

Energy stored in the capacitor is given by

$$E = \frac{1}{2} CV^2$$

If we simplify then $2E = CV^2$.

The English physicist James Prescott Joule outlined the basis of the principle of conservation.

The **law of conservation of energy** states that the total energy of an isolated system remains constant with time.

Wind speed above the surface of the liquid is called **Wind gradient**. It is the rate of increase of wind strength with unit increase in height above ground level.

Power is the amount of energy transferred or converted per unit time or The rate of doing work is called Power.

$$P = \frac{W}{t}$$

W= Work Done, T = Time taken.

Sound

Sound is defined as vibrations that travel through the air or another medium as an audible mechanical wave. Sound waves are longitudinal waves and are progressive in nature. SI unit of sound is decibel (dB).

Characteristics of Sound Wave :

Sound is produced by vibrating objects. The matter or substance through which sound is transmitted is called a medium. It can be solid, liquid or gas. It does not travel in vacuum.

Loudness is a measure of the response of the ear to the sound.

The loudness of sound is directly proportional to the square of the amplitude of vibration. Unit of loudness is a bell. A practical unit of loudness is the decibel (dB) which is $\frac{1}{10}$ th of bel.

Another unit of loudness is the phon.

In human beings, the vibration of vocal cords produce sound.

Unpleasant sounds are called Noise. A sound of a single frequency is called a tone. The sound which is produced due to a mixture of several frequencies is called a note.

The amount of sound energy passing each second through the unit area is called the intensity of sound.

Solids are packed together tighter than liquids and gases, hence sound travels fastest in solids. Liquids are more dense than gases, but less than solids, so sound travels second fastest in liquids.

In gases, sound travels the slowest because they are the least dense, the molecules in gases are very far apart, compared to solids and liquids.

Speed of sound in a particular medium depends on the temperature and pressure of that medium. In any medium, if the temperature increases, the speed of sound also increases.

The properties of sound are amplitude and frequency because amplitude determines the loudness of sound and frequency determines its pitch. The loudness and pitch affect the quality of sound.

The number of oscillations made by the wave in one second is called the frequency of the wave. It is measured in hertz (Hz).

The maximum displacement of a wave on either side of its mean position is called **Amplitude**.

The speed of sound in a medium can be determined by the equation:

$$v = (K/p)^{1/2}$$

Where

v = speed of sound,

K = compressibility and p = density.

Speed of sound in air at 0°C = 331 m/s, at 20° = 343 m/s.

Speed of sound in water (at normal temperature and pressure) = 1498 m/s and at 0°C = 1402 m/s.

Speed of sound in solid = 5000 m/s.

The speed of sound in air at 0°C is about 331 meters per second, or a kilometer in 2.9 s or a mile in 4.7 s.

1 kilohertz = 1000 oscillation per second

$$10 \text{ oscillation per second} = \frac{10}{1000}$$

= 0.01 Kilohertz

The **audible range** of sound for an average adult human being is 20 Hz - 20000 Hz. More than 20000 Hz sound is considered damaging.

Sound with a frequency below 20 Hz is known as an **infrasonic sound**. Sound with a frequency above 20,000 Hz is known as an **ultrasonic sound**.

The bats emit ultrasonic waves of frequency 20 to 100 Kilohertz and the returning echo of the sound waves give information about prey.

Supersonic speed is a rate of travel of an object that exceeds the speed of sound.

$$\text{Mach number (m)} = \frac{\text{Object speed}}{\text{Speed of Sound}}$$

If $m < 1$ it is subsonic

$m = 1$ transonic

$m > 1$ supersonic

$m > 5$ hypersonic

The study of the production and propagation of sound waves called **Acoustics**.

Speed (in m/s) of sound in different media at 25°C : **Solids** - Aluminium (6420), Nickel (6040), Steel (5960), Iron (5950), Brass (4700), Glass (3980); **Liquids** - Sea water (1531), Distilled water (1498), Ethanol (1207), Methanol (1103); **Gases** - Hydrogen (1284), Helium (965), Air (346), Oxygen (316), Sulfur dioxide (213).

Echo : When sound is reflected it causes an echo. To hear a distinct echo the time interval between the original sound and the reflected one must be at least 0.1s. Echoes may be heard more than once due to successive or multiple reflections.

To hear echo, the minimum distance between source of sound and reflecting body is 17 m.

Reverberation - A sound created in a big hall will persist by repeated reflection from the walls until it is reduced to a value where it is no longer audible. The repeated reflection that results in this persistence of sound is called reverberation.

To reduce reverberation, the roof and walls of the auditorium are generally covered with sound-absorbent materials like compressed fibreboard, rough plaster or draperies.

The term **SONAR** stands for Sound Navigation And Ranging. It is a device used for measuring distance using ultrasonic waves. Sound waves travel farther in the water than do radar and light waves.

As light travels faster than sound, hence light is seen before sound is heard during lightning. Light travels at a speed of 3×10^8 km/s (186,000 mi/s) from a stationary source. Light from Moon to Earth: 1.3 seconds, Sun to Earth (1 AU) : 8.3 minutes.

Gravitation

Gravitational force - Newton's Law of Universal Gravitation states that the force of attraction between any two bodies is directly proportional to the product of their masses and is inversely proportional to the square of the distance between them.

$$\text{Gravitational force (F)} = G \frac{m_1 m_2}{r^2}$$

where, m_1 is the mass of the first body, m_2 is the mass of the second body, r is the distance between the centers of two bodies.

Universal gravitational constant,
 $G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ or $\text{N m}^2 / \text{Kg}^2$.
 It is given by the English scientist Henry Cavendish.

The acceleration due to gravity on Earth or the value of g on Earth is 9.8 m/s^2 .

Nature of Gravitational force :-

Gravitation is a non-contact force. It acts because of mass. It is a long-range force. It is the weakest fundamental force in nature. It is always directed towards the center of mass.

Gravitational force doesn't depend upon the medium between the objects. It is always attractive in nature. Therefore the gravitational potential energy, which is the negative of work done, is always negative.

Importance of gravitational force -

Earth pulls everything towards itself due to gravity. It binds us to the Earth.

The tides in the ocean are caused by the gravitational pull of the moon and the sun.

Gravitational force of earth keeps earth's atmosphere in place.

The Moon revolves around the Earth because of gravity.

Planets move around the sun because of gravity.

Once a satellite is launched into orbit, the motion of a satellite is because of gravity.

It is the Gravitational force that is holding all the gasses in the Sun.

Gravity is a force that attracts a body towards the center of the earth or any other physical body having mass. Gravitational force exerted by the earth on the moon provides the necessary centripetal force required for the moon to orbit the earth.

Centrifugal force is the apparent outward force on a mass when it is rotated.

Since Earth rotates around a fixed axis, the direction of centrifugal force is always outward away from the axis, opposite to the direction of gravity at the equator. At Earth's poles it is zero.

Numerical Example - Let, the mass of the two bodies be m_1 and m_2 respectively and the distance between their centers be r .

Then, the gravitational force between them is given by:

$$F = G \frac{m_1 m_2}{r^2}$$

Now, when the distance between them is made three times, the gravitational force will be:

$$F^* = G \frac{m_1 m_2}{(3r)^2} \Rightarrow F^* = G \frac{m_1 m_2}{9r^2} \Rightarrow F^* = \frac{F}{9}$$

(Force becomes one - nine of original force).

Galileo Galilei was the first to conclude that in a vacuum all objects fall with the same acceleration g and reach the ground at the same time. It means that objects move with a constant speed when no force acts on them.

Three laws of planetary motion

Kepler's First law (The Law of Orbits) - According to Kepler's first law, "All the planets revolve around the Sun in elliptical orbits, and the Sun is at one of the focus of that ellipse".

Kepler's Second Law (The Law of Equal Areas) - "A line segment joining a planet and the Sun sweeps out equal areas in equal intervals of time".

It can also be stated as "The areal velocity of a planet revolving around the sun in elliptical orbit remains constant, which implies the angular momentum of a planet remains constant".

Kepler's Third Law (The Law of Periods) - According to Kepler's law of periods, "The square of the time period of revolution of a planet around the sun in an elliptical orbit is directly proportional to the cube of its semi-major axis".

$$T^2 \propto a^3$$

Escape velocity is the minimum velocity required by a body to be projected to overcome the gravitational pull of the earth.

Escape velocity at the earth's surface is 11.2 km/s.

$$\text{Escape velocity } (V) = \sqrt{\frac{2GM}{R}}$$

where, 'G' is the gravitational constant, 'M' is the mass of the planet, 'R' is the radius from the center of gravity.

The orbital velocity of a satellite is defined as the minimum velocity required to place or maintain a satellite in a given

orbit.

The orbital velocity is expressed as $V_o = \sqrt{GM/R}$.

Relation Between Escape Velocity And Orbital Velocity : $V_{\text{escape}} = \sqrt{2} V_{\text{orbital}}$.

The escape velocity is approximately 1.414 times the orbital velocity. This relationship holds true for a circular orbit around a celestial body.

Weightlessness: It is a situation, in which the effective weight of the body becomes zero. Weightlessness is achieved -
(i) during freely falling under gravity
(ii) inside a spacecraft or satellite
(iii) at the centre of the earth

Weight of a Body at the Moon: The mass and radius of the moon are less than that of the earth, so the force of the gravity at the moon is also less than that of the earth. The weight of an object on the moon will be about one - sixth of what it is on earth.

The **Coriolis force** is an inertial or fictional force that acts on objects that are in motion within a frame of reference that rotates with respect to an inertial frame.

Coriolis force is responsible for the rotation of the earth around its axis. Foucault's pendulum is an easy experiment demonstrating the Earth's rotation.

The **twin paradox** is a thought experiment in special relativity involving identical twins, one of whom makes a journey into space in a high-speed rocket and returns home to find that the twin who remained on Earth has aged more.

LIGO stands for "Laser Interferometer Gravitational - wave Observatory". It was used to detect gravitational waves for the very first time and to develop gravitational - wave observations as an astronomical tool.

Electric Current

A **switch** is a simple device used to either break an electrical circuit or complete it.

The purpose of a switch in a series circuit is to make it easy to open or close the electrical circuit, turning the flow of electricity on or off.

Ohm's law states that The potential difference, V , across the ends of a given metallic wire in an electric circuit is

directly proportional to the current flowing through it, provided its temperature remains the same.

This law was discovered by Georg Simon Ohm in 1827. $V \propto I$, $V/I = \text{constant}$, $V/I = R$ or $V = IR$, Where, V - voltage, I - Electric current, R - Electrical Resistance and its SI unit = ohm(Ω)

So, if the resistance becomes doubled then-current becomes half because the current is inversely proportional to resistance.

Resistance of wire depends on the length of the wire, temperature, cross-sectional area of the wire and resistivity of the material.

It is given by the relation ($R = \rho \frac{l}{A}$).

The Electrical resistivity (ρ) of a material is defined as the resistance offered to current flow by a conductor of unit length having unit area of cross section. ρ depends on the material of the conductor.

Among conductors, Silver has least electrical resistivity and Nichrome has high electrical resistivity.

Manganese has the highest electrical resistivity at 20°C, as compared to the Iron, Mercury and Chromium.

The heat generated in a resistive wire is determined by three factors: The square of the current (I^2) flowing through the wire. The resistance (R) of the wire material. The time duration (t) of current flow.

These are related by the equation:

$H = I^2 R t$, where H is the heat produced.

This equation is called the **Joule's equation** of electrical heating.

The coil of wire in an electric heater and Electric iron is called a heating element. Nichrome wire is used as a heating element because it is very stable, even at high temperatures.

A current - carrying wire composed of a composite material that melts at a moderate temperature is called a fuse wire.

Fuse wire has a low melting point and high conductivity. A lead and tin alloy with a low melting point and high conductivity is used to make fuse wire.

A fuse protects an electric circuit from overloading. An electrical fuse is a safety device that operates to provide

protection against the overflow of current in an electrical circuit.

A fuse rating is the current need to blow (break) the fuse. The fuse rating is usually defined in Ampere. The current rating of the fuse mainly depends on its operating temperature.

Electrons move for the flow of current in a conducting metallic wire only if there is a difference of electric potential. An electron is a negatively charged subatomic particle.

1 Ampere of current carries 6.25×10^{18} electrons.

Electric charge is a scalar quantity. Electric charge is the physical property of matter that causes charged particles to experience a force when placed in an electromagnetic field. The SI unit of electric charge is the coulomb (C).

If a charge Q flows through the cross-section of a conductor in time t , then the current $I = \frac{Q}{t}$.

In a circuit, electrons flow from negative to positive.

One ampere is constituted by the flow of one coulomb of charge per second. Ampere (A) is the SI unit of electrical current.

An **electrical circuit** is a network consisting of a closed loop, giving a return path for the current. The current in the bulb will stop flowing if the circuit is broken.

In Series connection, The total resistance is equal to the sum of the individual resistances. Equivalent resistance $R_n = R_1 + R_2 + R_3 + R_4 + \dots + R_n$.

In a parallel circuit, the voltage is the same across all resistors, but the current is divided among them.

The total resistance (R_{total}) in a parallel combination is the sum of the individual resistances $(\frac{1}{R_1}, \frac{1}{R_2}, \frac{1}{R_3}, \dots, \frac{1}{R_n})$:

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

Electric circuits use two forms of electric current, Alternating Current (AC) and direct current (DC).

A **rectifier** is a device that converts an oscillating two-directional alternating current (AC) into a single - directional

direct current (DC).

PVC (Polyvinyl chloride) is an electric insulator. Electric wires are generally coated with Polyvinyl chloride.

The **electric power** is given by $P = VI$, where V is the potential difference, I is the electric current and P is the electric power.

SI Unit of electric power is watt and its larger unit is kilowatt. kilowatt hour is the commercial unit of energy.

Watt is the unit of electrical power equal to one ampere under the pressure of one volt.

The relation between power (P), current (I) and resistance (R) is $P = I^2R$.

Potential difference is defined as the amount of work (W) done by an external agent in moving a unit charge (Q) from one point to another point in the circuit.

It is given by: $V = \frac{W}{Q}$.

Kirchhoff's first law (Junction Rule) is based on the conservation of charge because the sum of current entering the junction is equal to sum of current leaving the junction.

$\Sigma I = 0$ (Junction Rule)

Kirchhoff's second law (mesh rule) (Loop Rule) states that the algebraic sum of potential drops in a closed electrical circuit is zero. It follows the Principle of Conservation of Energy. $\Sigma V = 0$ (Loop Rule)

Coulomb's Law - According to this law, the force between two charged bodies is directly proportional to the product of the magnitude of the two charges and inversely proportional to the square of the distance between them. It acts along the line joining the two charges considered to be point charges. This Law is valid for only point charges.

$$F = k \frac{q_1 q_2}{r^2}$$

Force between the two electric charges reduces to a quarter of its former value when the distance between them is doubled.

Gauss law states that the total flux linked with a closed surface is $1/\epsilon_0$ times the charge enclosed by the closed surface. Electric flux through a closed surface S

$$= \frac{q}{\epsilon_0}$$

Faraday's first law of electromagnetic

induction states that The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

EMF (Electromotive Force (ϵ)) is the voltage or electrical potential difference produced by a cell or battery when it is not connected to an external circuit. EMF is measured in volts (V).

Michael Faraday introduced the concept of electric field for the first time.

An electric field is created by a charged body in the space that surrounds it and results in a force exerted on any other charges placed within the field.

An **electric dipole** is a pair of equal and opposite point charges q and $-q$, separated by a distance $2a$. By convention, the direction from $-q$ to q is said to be the direction of the dipole. The mid-point of locations of $-q$ and q is called the centre of the dipole. The total charge in an electric dipole will be zero.

An **ammeter** is an instrument used to measure the electric current in a circuit.

Voltmeter is a device used to measure the electric potential difference between two points or nodes in a circuit.

A **Galvanometer** is a sensitive device used to detect very low current. Its working is based on the principle that a coil placed in a uniform magnetic field experiences a torque when an electric current is set up in it.

Rheostat is a variable resistor with two terminals, used to control or limit the flow of current in an electrical circuit by manually adjusting its resistance value.

Wheatstone bridge is an arrangement of four resistors used for accurate measurement of resistance. The Wheatstone Bridge Circuit comprises two known resistors, one unknown resistor and one variable resistor connected in the form of a bridge. This bridge is very reliable as it gives accurate measurement. The Wheatstone bridge works on the principle of null deflection, i.e. the ratio of their resistances is equal, and no current flows through the circuit.

Capacitance - It is the ability of a circuit to collect and store energy in the form of an electrical charge. The Wien bridge is used to measure the value of Capacitance.

A **capacitor** is a device that stores electrical energy in an electric field.

Capacitors are used in digital circuits so the information stored in large computer memories is not lost during a momentary electric power failure.

The electric field in the region between the conductors is proportional to the charge Q .

$$C = \frac{Q}{V}$$

The constant C is called the capacitance of the capacitor. C is independent of Q or V . The capacitance C depends only on the geometrical configuration (shape, size, separation) of the system of two conductors. The SI unit is 1 farad.

Resistor is a passive electrical component with two terminals that are used for either limiting or regulating the flow of electric current in electrical circuits.

A **diode** is a two-terminal semiconductor device. It allows the unidirectional flow of current and offers high resistance in the opposite direction.

Inductance - It is the tendency of an electrical conductor to oppose a change in the electric current flowing through it.

A **transistor** is a semiconductor device used to amplify or switch electrical signals and power. It is also called Bipolar Junction Transistor (BJT). There are mainly two types of transistors: n-p-n transistor and p-n-p transistor. Three segments of a transistor are emitter, base, collector.

A **thermistor** is a resistance thermometer, or a resistor whose resistance is dependent on temperature.

Conductors are the materials, through which an electric charge is readily transferred. They conduct electricity because they have free electrons which move easily inside them. Example: Aluminum, copper or any metal.

Materials that don't have free charge electrons and provide high resistance to electricity are called poor conductors of electricity or Insulators, such as Hard Rubber, wood and Plastic, Glass, Diamond etc.

Semiconductors are the materials which have a conductivity between conductors and non-conductors or insulators, such as Germanium and silicon.

Semiconductors can be pure elements, such as silicon, germanium, gallium etc.

The resistivity of a semiconductor decreases with the increase of temperature.

When a small amount of a suitable impurity is added to the pure semiconductor. The conductivity of the semiconductor is increased manifold.

The deliberate addition of a desirable impurity is called doping and the impurity atoms are called dopants. Such a material is also called a doped semiconductor.

There are two types of dopants used in doping the tetravalent silicon (Si) or Germanium(Ge):

Pentavalent (valency 5); like Arsenic (As), Antimony (Sb) and Trivalent (valency 3); like Indium (In), Boron (B).

The dopant behaves as an electron donor or electron acceptor.

Metals are good conductors due to the presence of free electrons and when we increase the temperature the metal ions start vibrating which leads to the increase in resistance so the conductivity is decreased.

Silver, Copper and Gold are good conductors of electricity.

Mercury is a heavy, silvery-white metal. Compared to other metals, it is a poor conductor of heat, but a fair conductor of electricity. It is the only metal which is in liquid state at room temperature.

Non - metals are bad conductors of electricity because they do not have free electrons except Graphite. At room temperature, bromine is the only non-metal that exists as a liquid.

A set of equations involving electric and magnetic fields, and their sources, the charge and current densities are known as **Maxwell's equations**.

Maxwell first used the equations to propose that light is an electromagnetic phenomenon.

In 1785, **Charles Augustin Coulomb** used the calibrated torsion balance to measure the force between electric charges.

Albert Einstein was awarded a Nobel Prize in 1921 for his services to Theoretical Physics, and especially for his discovery of the Law of the Photoelectric Effect.

Robert Andrews Millikan was an American physicist honored with the

Nobel Prize for Physics in 1923 for the measurement of the elementary electric charge and for his work on the photoelectric effect.

The photoelectric effect is a phenomenon where electrons are emitted from the metal surface when light of sufficient frequency is incident upon it.

An **Inverter** is an electronic device that converts low voltage direct current into high voltage alternating current; it is also a battery that converts chemical energy into electrical energy.

Electrical conductivity, a measure of a material's ability to conduct electric current. 'mho' is the unit of electrical conductivity of a substance.

Transformer is a device used to change the A.C. voltage without changing the frequency between the circuits.

A transformer consists of two sets of coils, insulated from each other. One of the coils called the primary coil has N_p turns. The other coil is called the secondary coil, it has N_s turns. Often the primary coil is the input coil and the secondary coil is the output coil of the transformer. There are two types of transformer:

Step up transformer, converts low voltage AC into high voltage AC.

Step down transformer, converts high voltage AC into low voltage AC.

A transformer is an inductive electrical device for changing the voltage of alternating current

Transformer works on Faraday's law of mutual induction.

The full form of LED is Light Emitting Diode.

A **cell** is a device that generates electricity by using chemical energy.

In the symbol for the electric cell, the thicker, shorter line represents the negative terminal. The longer, thinner vertical line represents the positive terminal.

Types of cells are Primary Cell, Secondary cell, Reserve cell, and Fuel cell.

Primary cells are non-rechargeable and must be discarded after use, while secondary cells (like lead-acid batteries) can be recharged and reused multiple times by reversing the current flow.

Common AA batteries have a nominal voltage of 1.5V and are widely used in household gadgets like remotes, toys, and wireless appliances. These are typically alkaline or zinc-carbon primary cells.

Silver oxide cells, Mercury cells, and Zinc air cells are used in watches.

Domestic electric circuits - Normally in houses, we receive AC electric power of 220 V with a frequency of 50 Hz. The neutral wire has black insulation. Livewire has a voltage of 220 V and is covered with red insulation. Earth wire has a voltage of 0 V (same as Earth) and is covered with green insulation.

Magnetic Field

Magnetic Field is the region around a magnetic material or a moving electric charge within which the force of magnetism acts. Magnetic field is a vector quantity that has both direction and magnitude.

The radius of curvature of a charged particle's passage in a homogenous magnetic field is proportional to momentum of particles.

Magnetic Field Lines - Magnetic field lines are imaginary lines, which depict the strength and direction of the magnetic field.

Properties of magnetic fields lines -

They emerge from the north pole and merge at the south pole. They never cross each other. The density of the field lines indicates the strength of the field, meaning when field lines are crowded it shows a strong magnetic field. They always make closed loops. A tangent line drawn at any point to the magnetic field line gives the direction of the resultant magnetic field at that point.

Magnetic lines of forces inside a bar magnet emerge from the south pole to the north pole of the magnet.

Magnetic field due to a current-carrying conductor - French scientist André Marie Ampère suggested that a magnet also exerts an equal and opposite force on a current carrying conductor.

Magnetic field due to a straight current carrying conductor - When current is passed through a straight current-carrying conductor, a magnetic field is

produced around it. The magnetic field produced by a current carrying straight wire depends inversely on the distance from it.

Right-Hand Thumb rule- If the thumb of the right hand points along the direction of current, then the remaining curled fingers of the same hand gives the direction of the magnetic field due to the current. It is also known as Maxwell's Corkscrew rule.

Magnetic field at the center of a circular current carrying coil - Magnetic field produced will be more concentrated in the center of the loop than outside the loop.

Magnetic field due to current in a Solenoid - The direction of the flow of the magnetic field in a solenoid is from the north pole to the south pole and Current flowing through the coil produces a magnetic field that has north and south poles like a bar magnet.

Magnetic flux is defined as the number of magnetic field lines passing through a given closed surface.

It provides the measurement of the total magnetic field that passes through a given surface area. The net magnetic flux through any closed surface is zero. It is a Scalar Quantity. The SI unit of magnetic flux is Weber (Wb).

Magnetic flux (ϕ_B) = $B \cdot A \cos\theta$.

ϕ_B = Magnetic flux, B = Magnetic field force, A = Area of the surface through which flux passes.

The **Neutral point** in the magnetic field of a horizontally positioned bar magnet is the point where the magnetic field is zero.

Magnetic flux density is the amount of magnetic flux in an area taken perpendicular to the magnetic flux's direction. It is denoted by the symbol B and its unit is tesla.

The magnetic field strength around a current-carrying wire increases with higher current and decreases with greater distance from the wire. When the distance between the wire and a point decreases, the magnetic field strength at that point increases.

Force on a Current Carrying Conductor in a Magnetic Field

Ampere's experiment - When an electric

conductor is placed in a magnetic field, it experiences a force. This force is directly proportional to the current and is also perpendicular to its length and magnetic field.

Fleming's left-hand rule - If the thumb, forefinger and middle finger of the left hand stretched such that they are mutually perpendicular to each other, then the thumb points towards the direction of the magnetic force, the forefinger points towards the direction of the magnetic field and the middle finger points towards the direction of current.

Electric motor- It converts electrical energy into mechanical energy.

Maxwell Effect - It is a phenomenon of electromagnetic induction in which an electric charge, near a solenoid in which current changes slowly, feels an electromotive force (e.m.f.) even if the magnetic field is practically static inside and null outside.

Lenz's law - "The direction of the induced current is always such that it opposes the cause by which it is produced." Lenz's law is based on the law of conservation of energy. This law was derived by the German physicist Heinrich Friedrich Lenz in 1834.

Induce e.m.f = Rate of change of flux

$$e.m.f = -N \frac{d\phi}{dt}$$

e.m.f is the induced voltage (also known as electromotive force).

$d\phi$ = change in magnetic flux, dt = change in time.

In other words Lenz law states that the current induced by a change in flow opposes the effect producing the change.

The induced current is highest when the direction of motion of the coil is at right angles to the magnetic field.

Electromagnets - A type of magnet that is created by passing an electric current through a coil of wire. They work on the principles of electromagnetism and they create electromagnetic fields. They are temporary magnets with a strong magnetic field.

Electromagnetic Induction - It is a current produced because of voltage production (electromotive force) due to a changing magnetic field. Michael Faraday discovered electromagnetic induction.

Applications of Electromagnetic Induction - AC generators, magnetic flow meter, electrical transformers work on the principle of electromagnetic induction.

A Solenoid is a type of electromagnet formed by a helical coil of wire whose length is substantially greater than its diameter, which generates a controlled magnetic field.

The magnetic field inside a solenoid is proportional to the current, number of turns and inversely proportional to length. The magnetic field inside the solenoid is uniform throughout.

The Toroid - It is a doughnut-shaped hollow circular ring with several turns of insulated wire twisted so close together that there is no room between them. It is used as an inductor in electronic circuits, especially at low frequencies where comparatively large inductance is necessary.

Electric generator - The device that converts mechanical energy into electrical energy. It operates on the principle of electromagnetic induction.

Electric generator and electric motors work on the principle of Fleming's right and left-hand rule respectively.

Biot-Savart's Law - The magnetic field is directly proportional to the length of the element and the sine of the angle formed by the line joining the element to that point P. The magnetic field is inversely proportional to the square of the distance r from the point P. This relation is called Biot-Savart's law.

Magnetic Force - It is an attraction or repulsion that arises between electrically charged particles because of their motion.

Lorentz force - It is defined as the force exerted on a charged particle moving through an electric field and a magnetic field. Hendrik Lorentz derived the modern formula of the Lorentz force in 1895.

Moving Coil Galvanometer - A galvanometer is an electrical device used to measure the small value of currents. It works on the principle that a current-carrying coil experiences torque when placed in a magnetic field.

Galvanometer as Voltmeter - connecting a shunt resistance in series connection with a galvanometer, it can be converted

into a voltmeter. A voltmeter is always connected in a parallel connection with the circuit.

Galvanometer as Ammeter - connecting a shunt resistance in parallel connection with a galvanometer, it can be converted into an ammeter. An ammeter is always connected in a series connection with the circuit.

Cyclotron - It is a machine that boosts charged particles' energy levels. Created in 1934 by E.O. Lawrence and M.S. Livingston, it explores nuclear structures. By using electric and magnetic fields together, it accelerates these particles.

Magnetic Length (L_m) - The Magnetic length is the distance between the two poles of the magnet. **Geometric length (L_g)** - It is the actual length of the magnet.

The relationship between geometric length (L_g) and magnetic length (L_m) is

$$L_m = \frac{5}{6} L_g.$$

The Law of Orsted, in electromagnetism, is the physical law stating that a magnetic field is created by an electric current.

Retentivity - The capacity of a magnet to hold its magnetism even after being exposed to demagnetizing forces.

The ability of a magnetic field to magnetize a material medium is called its **magnetic intensity (H)**.

Magnetic properties of materials :

Ferromagnetic Materials are those which strongly magnetize in the direction of an applied magnetic field. Its examples are Iron (Fe), Nickel (Ni), Cobalt (Co).

Permanent magnets are made from special alloys (ferromagnetic materials) such as iron, nickel and cobalt, several alloys of rare-earth metals and minerals such as lodestone. Steel is highly effective for making permanent magnets.

Diamagnetic Materials are those which are easily magnetized in the opposite direction when placed in a magnetic field. Examples: Copper (Cu), Lead (Pb), Silicon (Si), Nitrogen (N) etc.

Paramagnetic Materials are those which weakly magnetize in the direction of an applied magnetic field. Examples : Aluminium (Al), Sodium (Na), Calcium (Ca) etc.

Heat and Thermodynamics

Thermodynamics is the branch of physics that deals with the concepts of heat and temperature and the inter-conversion of heat and other forms of energy. Thermodynamics is a macroscopic science.

Heat is the form of energy transferred between two or more systems and its surroundings by virtue of temperature difference. It is transferred from a warmer substance to a colder one. The SI unit of heat energy transferred is expressed in joule (J).

The **Zeroth law of thermodynamics** states that 'When two systems are in thermal equilibrium with a third system, then those two systems will also be in thermal equilibrium with each other'.

The term Zeroth law of thermodynamics was coined by Ralph H Fowler in 1931.

William Thomson is known as the Lord Kelvin in the honour of finding the correct value of Absolute zero is - 273.15 degrees Celsius or - 459.67 degrees Fahrenheit.

Around 1850 Rudolf Clausius and William Thomson (Kelvin) stated both the First Law and the Second Law of Thermodynamics.

The **First law of thermodynamics** is also known as the Law of Conservation of Energy. It states that energy can neither be created nor be destroyed, energy can only be transferred or changed from one form to another.

The equation of the first law of thermodynamics:

$$\Delta Q = \Delta U + \Delta W,$$

where,

ΔU - change in internal energy of the system, ΔQ = Heat supplied to the system by the surroundings, ΔW = Work done by the system on the surroundings.

The **Law of Conservation of Energy** states that energy can neither be created nor be destroyed; it can only be transformed from one form to another form.

The equation represents conservation of energy

$$E = mc^2$$

where, E is energy, m is mass, c is speed of light

The Second Law of Thermodynamics gives a fundamental limitation to the efficiency of a heat engine and the coefficient of performance of a refrigerator. The Second Law was originally formulated in terms of the fact that heat does not spontaneously flow from a colder body to a hotter.

There are two statements on the second law of thermodynamics:

Kelvin - Planck statement - it is impossible to devise a heat engine that can absorb energy in the form of heat as well as produce work of an equivalent amount i.e. no device bears 100% thermal efficiency.

Clausius statement - It is impossible to construct a system which will operate in a cycle, transfers heat from the low-temperature reservoir (or object) to the high-temperature reservoir (or object) without any external effect or work interaction with the surrounding.

Metals are good conductors of heat. Example: Steel rod, Silver, Iron, Copper and Aluminium etc.

Woolen thread, water, and wooden plank all are insulators (poor conductors of heat).

Modes of Heat transfer

Conduction - The process of transmission of energy from one particle of the medium to another with the particles being in direct contact with each other.

Example: When frying vegetables in a pan heat transfer takes place from flame to the pan and then to the vegetables.

The rate of conduction can be calculated by the following equation:

$$Q = \frac{[K.A.(T_{hot} - T_{cold})]}{L}$$

where, Q is the transfer of heat per unit time, K is the thermal conductivity of the body, A is the area of the cross section, T_{hot} is the temperature of the hot region, T_{cold} is the temperature of the cold region, L is the length of the body.

Convection - The movement of fluid molecules from higher temperature regions to lower temperature regions. Example: Heating of milk in a pan. The equation to calculate the rate of convection is as follows:

$$Q = h_c.A.(T_s - T_f)$$

where, Q is the heat transferred per unit time, h_c is the coefficient of convective heat transfer, A is the area of heat transfer of the body, T_s is the surface temperature, T_f is the fluid temperature.

Radiation - It is a form of heat transfer that is moved from one place to another by electromagnetic radiation waves or rays no medium is required for heat transfer in this method. Example: In a microwave, the substances are heated directly without any heating medium.

The equation to calculate the rate of radiation is as follows:

$$P = e.\sigma.A.(T^4 - T_s^4)$$

Where, P is the net power of radiation, A is the surface area of the body, T is the radiator temperature, T_s is the surrounding temperature, e is emissivity and σ is Stefan's constant ($\sigma = 5.67 \times 10^{-8} \text{ Wm}^2\text{K}^{-4}$)

Temperature is a physical quantity that expresses the degree of hotness or coldness of a body.

The **dew point** is the temperature to which air must be cooled to become saturated with water vapor, assuming constant air pressure and water content.

Air temperature is a measure of how hot or cold the air is. It is the most commonly measured weather parameter. Temperature is usually expressed in degrees Fahrenheit or Celsius.

Relative humidity (RH) is a measure of the percentage of moisture present in the atmosphere as compared to its full capacity at a given temperature is known as the relative humidity.

Surface Air temperature - The temperature of the air near the surface of the earth.

Sea Surface temperature - The temperature of water close to the ocean's surface.

A clinical thermometer (medical thermometer) reads the temperature from 35°C to 42°C.

Sir Thomas Clifford Allbutt was a British physician and he invented a small clinical thermometer.

Gabriel Daniel Fahrenheit invented a quicker mercury thermometer in 1714, and the Fahrenheit temperature scale.

There are some thermometers that are

used on a daily basis: Clinical thermometer / medical thermometer, Laboratory thermometer, Mercury thermometer, Pyrometer thermometer.

Steam point is the temperature at which the maximum vapor pressure of water is equal to one atmosphere. It has the value of 100°C on the Celsius scale and 212°F on the Fahrenheit scale.

Triple point is defined as the condition of the temperature under which the gaseous, liquid and solid phase of a substance can exist in equilibrium.

The **Celsius Scale** is also called the centigrade scale based on 0° for the freezing point of water and 100°C for the boiling point of water. Invented in 1742 by the Swedish astronomer Anders Celsius.

Thermal expansion is the tendency of matter to change its shape, area, volume, and density in response to a change in temperature.

The Eiffel tower gets taller in summer because of the thermal expansion. Thermal expansion is a phenomenon due to which the metal expands when heated and contracts when cooled.

There are three types of thermal expansion:

Linear expansion - The expansion in length due to heat is called linear expansion.

$$\frac{\Delta L}{L} = \alpha . \Delta T$$

Where, L = original length, α = coefficient linear expansion, Δt = temperature difference, ΔL = change in length

Area expansion - The expansion in an area due to heat is called area expansion.

$$\frac{\Delta A}{A} = \alpha_A . \Delta T$$

where,

A = original area, ΔA = change in the area, α_A = coefficient of area expansion, ΔT = temperature difference

Volume expansion - The expansion in volume due to heat is called volume expansion.

$$\frac{\Delta V}{V} = \alpha_v . \Delta T$$

where,

V = original volume, ΔV = expanded volume, α_v = coefficient of volume expansion, ΔT = temperature difference.

The sun heats the earth through radiation. The Sun is the star at the center of the Solar System. The Sun radiates this energy mainly as light, ultraviolet, and infrared radiation, and is the most important source of energy for life on Earth. The radiation is a form of energy that does not require a medium to travel.

Condensation - The process of transformation of a substance in a gaseous state into a liquid state due to change in pressure or temperature of the substance is called condensation. It depends upon the amount of cooling and the relative humidity of the air.

Passive solar heating is the concept of using sunshine to keep houses warm in freezing winters. It refers to the use of the sun's energy for the heating and cooling of living spaces by exposure to the sun.

The latent solar heating required to convert a solid into a liquid or vapor, or a liquid into a vapor, without change of temperature.

Relation between Kelvin, Celsius and Fahrenheit scale

$$\frac{F - 32}{180} = \frac{C}{100} = \frac{K - 273}{100}$$

Humidity is the concentration of water vapour present in the air. Humidity is inversely proportional to the rate of evaporation, decreasing humidity increases the rate of evaporation.

Fluid Mechanics

Fluid - It is a liquid or any substance that continuously flows or deforms under applied shear stress. Example - gasses (oxygen, hydrogen), liquids (water, petrol, sulphuric acid).

Fluid statics is the part of fluid mechanics that deals with fluids when there is no relative motion between the fluid particles.

Example - Drink using a straw.

Fluid dynamics is the study of the movement of fluids, including their interactions as two fluids come into contact with each other.

Example - Water flowing through a pipeline, or across a surface.

Newtonian fluids are fluids having a constant viscosity and a zero shear rate at zero shear stress. These fluids obey

Newton's law of constant viscosity.

Example - water and air.

Non-Newtonian fluids are the fluids that have a variable viscosity and a variable relationship with shear stress. These fluids do not obey Newton's law of constant viscosity.

Example - salt solutions, blood, paint, toothpaste, and starch solutions.

Properties of fluid:

Viscosity (μ) - It is the property of fluid (liquid or gas) which offers resistance to the movement of one layer of fluid over another layer of fluid.

Newton's viscosity formula is as follows:

$$\text{Force (F)} = \mu A \frac{u}{y} \text{ here,}$$

μ = viscosity of the fluid, A = area of each plate, $\frac{u}{y}$ = rate of shear deformation.

The SI unit is the poiseuille (Pl), with other units being N s m⁻² or Pa s. Its dimensions are [ML⁻¹T⁻¹].

Types of viscosity -

There are 2 types of Viscosity:

Dynamic viscosity is the resistance to movement of one layer of a fluid over another. The SI unit is newton-second per square meter (N.s/m²) or pascal second.

Kinematic viscosity is expressed as the ratio of fluid dynamic viscosity to its density. The SI unit is square meter per second or m²/s. The Gaussian unit of Kinematic viscosity is Stokes (St).

Fluidity - It is the ability of a substance to flow. It is reciprocal of viscosity.

The SI unit of fluidity is (Pa s)⁻¹ or 1/Pa s.

Turbidity - It is the measure of relative clarity of a liquid. It is measured by a nephelometer or turbidimeter.

NTU is the Nephelometric Unit of turbidity.

Specific Weight (γ) - It is defined as weight per unit volume. The SI unit for specific weight is (N/m³).

Specific Gravity (ρ) or Relative density is the ratio of the density (mass of a unit volume) of a substance to the density of a given reference material.

Relative density of some substances:

Silver - 10.8, Gold - 19.3, Mercury - 13.6, Copper - 8.4, Iron - 7.8, Lead - 11.3, Water - 1, Honey - 1.42, Milk - 1.03.

Density (ρ) - It is defined as the ratio of the mass to the volume. The SI unit of density is measured in kilograms per cubic meter (kg/m³).

Adhesion: The attraction of two separate molecules, such as the water molecules, consisting of hydrogen and oxygen, and straws made of plastic.

Cohesion: The attraction of two identical molecules, such as in between molecules of water.

Surface tension - It is the tendency of liquid surfaces to shrink into the minimum surface area possible. These inward forces on the surface molecules of the liquid droplet tend to cause the surface to volume ratio to be as small as possible.

Examples of surface tension: water droplets of rain always take a round shape, insects walking on water and Needles floating (or paper clip) on water.

Capillary action occurs when the adhesion between liquid and solid is greater than the cohesion of the fluid. It is also known as capillary effect or motion. The liquid flows through narrow spaces without external forces, such as gravity. The liquid's movement is aided by intermolecular forces present in between the liquid and solid surface(s).

Examples of Capillary action: In pen, ink rises due to capillarity. In plants, the water is transferred to the leaves from the roots by capillarity. Cotton clothes absorb water by capillarity. The rise of any oil in the wick. Water moving up in straw or glass tubes against gravity.

Fluid at rest - When the fluid is at rest the velocity gradient is zero, therefore no shear stress/force will be acting and no external force is acting on it, the only force present is due to gravity and pressure of the fluid.

Pascal's Law - According to this law any force applied to a confined fluid is transmitted uniformly in all directions throughout the fluid regardless of the shape. Hydraulic lift and hydraulic brakes are based on Pascal's law.

When a glass bottle with water is kept in a freezer, the water inside the bottle expands due to lowering of the temperature, which can lead to sudden bursting of the water bottle.

Water contracts on heating from 0°C to 4°C and expands from 4°C to 100°C, it has the minimum volume and maximum

density of 4°C i.e. anomalous behavior of water.

Archimedes' principle - The buoyant force acting on an object in a liquid medium will be equal to the weight of the liquid displaced by that object. In other words, the buoyant force acting on an object partially or fully immersed in a liquid medium is equal to the weight of the liquid displaced by the object.

Applications of Archimedes' Principle: Ships, Submarines, Floating (Ice Cube), Hydrometer, Lactometers, Swimming, Hot Air Balloon.

Buoyant Force is an upward force exerted on an object that is partially or completely immersed in water. It helps swimmers float in water.

Buoyant force is given by : $F_b = \rho \times g \times V$. Here, F_b = buoyant force. ρ = density of the fluid. V = submerged volume, g = acceleration due to gravity.

Bernoulli's theorem - It states that as the speed of a moving fluid (liquid or gas) increases, the pressure within the fluid decreases.

It is applicable to steady, incompressible, irrotational flow along a streamline when the effect of viscosity is neglected. Mathematically,

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant.}$$

where, P is pressure, ρ is the density, v is the velocity, g is the acceleration due to gravity and h is the height.

This equation is based on the conservation of energy.

Application - spray guns, venturi meters, nozzles.

The Swiss Physicist Daniel Bernoulli developed this relationship in 1738.

Poiseuille's Equation - It is a physical law that gives the pressure drop in an incompressible and Newtonian fluid in laminar flow flowing through a long cylindrical pipe of constant cross section.

Fick's law states that the rate of diffusion is proportional to both the surface area and concentration difference and is inversely proportional to the thickness of the membrane'.

Fluid mixing can be described as the incorporation of one immiscible liquid into another immiscible liquid.

Waves

Wave is a form of disturbance which carries energy that travels through a material medium due to the repeated motion of the particles of the medium about their mean positions without any actual transportation of matter.

Wave Speed = Frequency $\times$ Wavelength

$$v = f \times \lambda$$

$$v = \frac{\lambda}{T} \quad \{\therefore f = \frac{1}{T}\}$$

Properties of waves :

Amplitude - The maximum displacement of an oscillating particle from mean position is amplitude. The amplitude of a wave is measured in meters.

It denotes the energy contained in the wave. The wave that has more amplitude is more powerful and energetic.

Wavelength - The minimum distance between two points in a wave having the same phase at a particular instant of time.

Wavelength is calculated as the distance between two successive troughs or crests of the wave.

$$\text{Wavelength } (\lambda) = \frac{\text{velocity } (v)}{\text{frequency } (f)}$$

Wave frequency - Number of waves passing by a specific point per second. Frequency of a wave is the reciprocal of the time period. Its SI unit is hertz (Hz).

$$\text{frequency } (f) = \frac{1}{\text{Time Period } (T)}$$

Time Period of Wave - It is the total time that a wave takes to complete a single cycle. It is measured in seconds.

$$\text{Time Period } (T) = \frac{1}{\text{frequency } (f)}$$

Wave velocity - It is the distance traveled by waves per unit time.

Wavelength divided by the time period is equal to wave velocity.

Wave velocity (v) = frequency (f) $\times$ wavelength (λ).

Wave height - It is the overall vertical change in height between the crest and the trough.

Wave speed - It is the distance a wave travels in a given amount of time.

Wave crest and trough - A crest is a point on a surface wave where the displacement of the medium is at a maximum. A trough is the opposite of a

crest, so the minimum or lowest point in a cycle.

Based on necessity of medium

Mechanical Waves - These waves propagate through a material medium (solid, liquid or gas) at a wave speed that depends on the elastic and inertial properties of that medium. Newton's laws of motion govern these waves. These waves cannot pass through vacuum.

Examples - waves on a string, water waves, sound waves, and seismic waves.

L waves are earthquake waves that are surface waves.

Non-mechanical Waves do not need any medium to propagate and can easily travel through vacuum. Example - Electromagnetic waves, Gravitational Waves.

Based on direction of particle motion waves are divided into 2 types

Transverse waves - The wave in which particles do not oscillate along the line of wave propagation but oscillate up and down (perpendicular) about their mean position as the wave travels. Example - Light, radio waves, microwaves, ocean waves, wiggling strings and polarized waves.

Longitudinal waves propagate in a direction parallel to the direct displacement or disturbances. Here, the wave particles vibrate to and fro in the direction of the propagation of the wave.

There is a zero - degree angle formed between the wave particles and the energy particles. It can also be put as the displacement of the medium occurs in the direction of the movement of the wave. Examples - sound waves, pressure waves, seismic P waves.

Doppler Effect - When there is relative motion between a sound source and a listener, the frequency of the sound heard by the listener is lower or higher than the original frequency. This is called the Doppler effect. Christian Johann Doppler gave the Doppler effect in 1842.

One of the examples is the pitch of the siren on an ambulance or fire engine. You may have noticed that as the high-speed siren passes you, the pitch of the siren suddenly drops in pitch. First, the siren is coming towards you when the pitch is high.

Uses of Doppler Effect - Sirens, Radar, Astronomy, Medical Imaging.

Water waves are an example of a combination of both longitudinal and transverse motions.

Electromagnetic waves - These are the waves that require no material medium for their production and propagation. They can pass through a vacuum and any other material medium.

All electromagnetic waves have the same speed in a vacuum (c) whose value is: $c = 299,792,458 (3 \times 10^8) \text{ m/s}$.

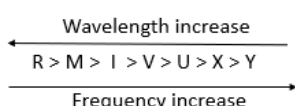
A changing magnetic field will induce a changing electric field and vice-versa, whereas the two are linked. These changing fields form electromagnetic waves.

Examples - gamma rays, X-rays, ultraviolet light, Visible light, infrared waves, microwaves radio waves.

Properties of Electromagnetic Waves - travel with the speed of light, can be polarized, show a transverse nature, carry momentum.

Electromagnetic Spectrum :

The electromagnetic spectrum is generally divided into seven regions, in order of decreasing wavelength and increasing energy and frequency.



Where,

R - Radio waves, M - Microwave waves, I - Infrared waves, V - Visible spectrum, U - Ultraviolet rays, X - X-Rays, Y - Gamma rays

Types of Electromagnetic Wave

Radio Waves have the longest wavelengths in the electromagnetic spectrum. Its frequency range is approximately from 500 kHz to 1000 MHz and wavelengths approximately from 1 mm to 100 km.

Examples - Cellular Networks, RADAR, TV, Radio Astronomy, Satellite Communication etc.

In the 1890s Guglielmo Marconi used radio waves to transmit signals over distances of several kilometers.

Microwave is a form of electromagnetic radiation with wavelengths ranging from

about 0.1 meter to 1 mm corresponding to frequencies between 300 MHz and 300 GHz respectively.

Microwaves are generally used in RADAR systems due to the fact that they have longer wavelengths and low frequencies. So they can be focussed along a straight line without much deviation.

Microwaves fall in the range of the EM spectrum between radio and infrared (IR). It is used in modern technology such as wireless networks, spacecraft communication, medical diathermy, and cancer treatment.

Klystron - the thermionic tube used to oscillate and amplify microwave frequency signals.

Infrared Rays - 'Heat Waves' is another name for Infrared rays. Water molecules present in most materials readily absorb these rays. After absorption, their thermal motion increases which increases their heat and that of their surroundings.

Infrared waves are electromagnetic radiation with wavelengths longer than those of visible light.

Many physical therapy treatments use Infrared lamps. These rays also play an important role in maintaining the earth's average temperature through the greenhouse effect.

Ultraviolet Rays - It is a form of electromagnetic radiation with wavelengths shorter than that of visible light, but longer than X-rays.

It covers wavelengths ranging from about $4 \times 10^{-7} \text{ m}$ (400 nm) down to $6 \times 10^{-10} \text{ m}$ (0.6 nm).

UV waves are invisible to the human eye, some insects, such as bumblebees, can see.

Ordinary Glass absorbs Ultraviolet (UV) rays. Many water purifiers use UV lamps to kill germs in water. They are used to destroy the bacteria and for sterilizing surgical instruments. They help to find the structure of atoms.

X - Rays are electromagnetic radiation similar to visible light. X - rays have higher energy and can pass through most objects including our body.

X - rays are used as a diagnostic tool in medicine and as a treatment for certain forms of cancer. It covers wavelengths from about 10^{-8} m to 10^{-13} m . X - rays can damage or destroy living tissues.

Gamma Rays - The wavelengths of these waves range from about 10^{-10} m to 10^{-14} m . Gamma rays are the shortest wavelengths and the highest frequencies of electromagnetic induction. It has no charge, they are considered neutral. It has a strong penetrating ability. It is the strongest form of radiation. An important application of Gamma rays is their extensive use in medicine to destroy cancer cells.

Matter waves - These waves are related to the constituents of matter - protons, electrons, and neutrons. They are more conceptual than mechanical or electromagnetic waves.

These waves are also referred to as De Broglie waves. They find application in electron microscopes.

Oscillations motion - When any object moves over a point repetitively then this type of motion of the object is the Oscillatory Motion. The oscillating movement takes place around an equilibrium point or a mean value.

The to and fro or 'back and forth' motion of an object is termed as vibration. Example - The motion of the pendulum, Alternating Current, Tuning Fork, Freely hanging bob.

Magnetic Resonance Imaging uses a combination of a large magnet and radio waves to view the organs and organs structures inside our bodies.

It is often used for disease detection, diagnosis, and treatment monitoring.

Computed tomography (CT) is an imaging procedure that uses special x-ray equipment to create detailed pictures, or scans, of areas inside the body.

Fluoroscopy is an imaging technique that uses X-rays to obtain real - time moving images of the interior of an object.

A **Positron Emission Tomography** (PET) scan is an imaging test that can help reveal the metabolic or biochemical function of your tissues and organs.

The technique is based on the detection of radioactivity emitted after a small amount of a radioactive tracer is injected into a peripheral vein.

Fresnel's equations - It describes the reflection and transmission of electromagnetic waves at an interface between different optical media.

Huygens' principle states that every

point on a wavefront may be considered as a source of secondary waves.

Schrodinger wave equation - It is a mathematical expression describing the energy and position of the electron in space and time, taking into account the matter wave nature of the electron inside an atom.

Light and Optics

Light - It is a type of electromagnetic radiation that allows the human eye to see or makes objects visible. Light always moves in a straight line. Light travels at a faster rate than sound.

Optics is the branch of physical science that deals with the properties and phenomena of both visible and invisible light and with vision.

Photonics is the physical science of light waves.

Incident ray: The light ray which strikes the surface.

Reflected ray: The light ray which returns from the surface after reflection.

Normal: An imaginary line perpendicular to the surface passing from the point of contact.

Angle of incidence: Angle between the incident ray and the normal.

Reflected Angle: Angle between the reflected ray and the normal.

Basic Properties of Light :

Light has the dual nature of a wave and a particle (known as photon). It can act as a particle at times, that explains why light travels in straight lines.

The Photoelectric effect demonstrates the particle nature of light.

It can function like a wave at times, that explains how light bends (or diffracts) around an object.

The speed of light is the same for all electromagnetic waves.

Light radiation travels at a speed of approximately 29,97,92,458 (m/s) or 3×10^8 m/s in a vacuum.

The wavelengths and frequencies of electromagnetic waves distinguish them from one another {wavelength (λ) is inversely related to frequency (f)}.

$$\lambda \propto \frac{1}{f}$$

Relative Speed of light :

Solid < Liquid < Gas < Vacuum.

Example - Air > Water > Glass > Diamond.

In air (3×10^8 m/s), in water (2.26×10^8 m/s), in glass (2×10^8 m/s), in diamond (1.25×10^8 m/s).

Sunlight takes about 8 minutes 20 seconds to reach the earth.

Optics Ray

Reflection of light - The phenomenon of bouncing back of light in the same medium on striking the surface of any object is called reflection.

Types of Reflection -

Specular or regular reflection - It is defined as light reflected from a smooth surface at a definite angle like a reflection on a bathroom mirror, the lake and the glare on a pair of glass.

Diffuse reflection - It is the reflection of light from a surface such that a ray incident on the surface is scattered at many angles rather than at just one angle.

Multiple reflections - When we take two mirrors, a single source of light can be reflected multiple times like in a barber's shop.

The size of the irregularities on the surface in relation to the wavelength of the light incident on the surface determines whether rays are reflected in a regular or irregular pattern.

Laws of Reflection -

First Law - According to the first law of reflection, when a ray of light strikes a mirror and gets reflected back then the angle of incidence is equal to the angle of reflection.

$I = R$, Where, I = Angle of incidence, R = Angle of reflection.

Second Law - According to the second law of reflection the incident ray, reflected ray, and the normal lie on the same plane on the surface of reflection.

Application of Light Waves: It is mainly used in telescopic equipment used for research purposes. Obstacle Detection by Cleaning Robot, Atmospheric observations are made using laser beams, early detection of cancer and dementia, multifunction sensors for smartphones, exploring the interior of objects without destruction, product manufacturing and eyesight correction.

Scattering of light - The process by which small particles present in the atmosphere cause scattering of light resulting in optical phenomena such as the blue color of the sky.

Types of Scattering :

Elastic Scattering - If the energy of the scattered light is conserved, then such scattering is called elastic scattering. Example - Rayleigh Scattering and Mie Scattering.

Inelastic Scattering - If the energy of the scattered light is not conserved, then such scattering is called inelastic scattering. Example - Raman Scattering and Brillouin Scattering.

Factors affecting scattering of light - Size of particle, Wavelength of light.

The relationship between scattering and wavelength is given by

scattering $\propto \frac{1}{\lambda^4}$, where, λ is the wavelength of light.

And, wavelength of light is given by the following equation

$$\text{Wavelength } (\lambda) = \frac{\text{wave velocity } (v)}{\text{frequency } (f)}$$

Application of Scattering of Light :

The sky appears blue as the light gets scattered by the particles present in the atmosphere. During sunset and sunrise, the sky appears red due to the scattering of light. Light is scattered least at Noon because the sun is overhead, and the distance between the Sun and the Earth becomes least.

The sequence of colours in light is arranged from shortest wavelength to longest and is called the visible spectrum of light. The sequence is: Violet, Indigo, Blue, Green, Yellow, Orange and Red (VIBGYOR). The light wave which has less wavelength or more frequency will scatter more.

Raman Effect, discovered in 1928, deals with the scattering of light by molecules of a medium when they are excited to vibrational energy levels. Raman was awarded the Nobel prize in Physics in 1930 for his discovery of Raman scattering (scattering of light).

Rayleigh Effect is a phenomenon of scattering of light or other electromagnetic radiation by particles much smaller than the wavelength of the radiation.

Refraction of light - When the light travels from one medium to another medium other than the vacuum, it deflects from its path. This deflection of light is called refraction of light.

Laws of Refraction :

First law - The first law of refraction states that the incident ray, the refracted ray and the normal to the interface of two transparent media at the point of incidence, all lie in the same plane.

Second law - The second law states that the ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant. This law of refraction is also termed as **Snell's law**.

$$\frac{\sin i}{\sin r} = \text{constant.}$$

This constant is the refractive index (n).

When a ray of light passes through a rarer medium to a denser medium, the refracted ray comes closer to the normal while the ray of light passes through a denser medium to a rarer medium, the refracted ray moves away from the normal.

Example - When a pencil is placed in a glass filled with water, the pencil appears crooked. A coin is placed in a tub full of water, then the coin appears slightly above the bottom of the tub.

Application of Refraction of Light :

In nature, the formation of rainbows and mirages are because of the refraction of light. Concave and convex glasses are used to correct the refractive errors of human eyes.

Peepholes in the doors, magnifying glasses, binoculars, cameras, lenses in the projectors use the principle of refraction of light.

The Sun is visible to us two minutes before the sunrise and two minutes after the sunset because of the bending of the light due to atmospheric refraction.

Light coming from the star undergoes continuous refraction due to change in density of air. This results in the twinkling of stars.

Refractive index is a property of a material that describes how the speed of light is affected when it travels through that material. It is denoted by the symbol μ .

$$\mu = \frac{\text{speed of light in vacuum } (c)}{\text{speed of light in the material } (v)}$$

Canada balsam has its refractive index

closest to that of crown glass. The refractive index (n) of crown glass is about 1.52.

Refractive index of different mediums - Ethyl Alcohol (1.36), Ice (1.31), Water (1.333), Air (1.0003), Diamond (2.417), Vacuum (1) and Glass (1.50).

Total Internal Reflection (TIR) - Phenomenon in which waves arriving at the interface from one medium to another are not refracted into the second medium, but completely reflected back into the first medium.

Optical fibers work on the principle of total internal reflection. When a light ray strikes at the internal surface of an optical fiber cable such that the incidence angle is greater than critical angle, then the incident light ray reflects in the same medium and this phenomenon repeats. In this way a light signal travels from one end of the cable to another end.

Mirage is also an application of TIR.

Plane Mirror

The most common mirrors that are flat in shape are called 'plane mirrors'.

Plane mirrors are built by putting a thin layer of silver nitrate or aluminium behind a flat piece of glass. In a plane mirror, when a light ray strikes, the incidence angle equals the reflection angle.

Properties and Characteristics of a Plane Mirror :

Image is formed behind the mirror.

The image is virtual and erect.

The image is of the same size as that of the object, i.e., magnification = 1.

The distance of the object from the mirror (object distance, o) is the same as that of the image from the mirror (image distance, i).

Hence, the plane mirror equation is: $i = o$

Plane mirrors show a lateral inversion of light.

Multiple Reflections of Light

If the object is placed **symmetrically** to the system, then the number of images

$$\text{formed by this system, } (n) = \frac{360^\circ}{\theta} - 1$$

If the object is placed **asymmetrically** to the system, then the number of images

$$\text{formed by this system, } (n) = \frac{360^\circ}{\theta}$$

Angle **Number of images**

$$30^\circ \rightarrow 11$$

45°	→	7
60°	→	5
120°	→	2 (symmetrically)
120°	→	3 (asymmetrically)
180°	→	1
0°	→	infinite

Lateral inversion is the phenomenon of left side appearing right side and right side appearing left side on reflection in a plane mirror.

A real image can be obtained on a screen, whereas a virtual image cannot.

Uses and applications of Plane Mirror - At home in daily life, especially in the bathroom and bedroom and as a wall mirror.

In homemade periscope and kaleidoscope used by children. In SLR camera and overhead projector. As optical levers in scientific instruments.

Spherical Mirror :

The spherical mirror forms the virtual or real image because spherical mirrors have a consistent curve and constant radius of the curvature. On the basis of the surface painting, spherical mirrors are of two types, concave and convex mirrors.

Terms used in Spherical Mirror -

Pole - It is the midpoint of a mirror.

Centre of Curvature - It is the centre of the sphere of which the mirror forms a part.

Radius of Curvature - It is the distance between the pole and the centre of the curvature. It is twice the focal length of the mirror.

Principal Axis - It is an imaginary line passing through the pole and the centre of curvature of the spherical mirror.

Focus - It is the point on the principal axis on which all the light rays after reflection meet actually or appear to meet.

Concave Mirror - When the inner surface of a spherical mirror is reflected and the outer surface is painted, it is called a concave mirror. Concave mirrors reflect and focus incoming light rays (parallel) at a point, called the focus point.

Concave mirrors are also called converging mirrors since light rays converge after reflection at concave mirrors. Uses - Vehicle headlights, searchlights, torches, solar furnaces, microscopes, telescopes, shaving mirrors, dental mirror and makeup

mirrors.

Image formation by a concave mirror for different positions of the object :

Position of the object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus F	Highly diminished point-sized	Real and inverted
Beyond C	Between F and C	Diminished	Real and inverted
At C	At C	Same size	Real and inverted
Between C and F	Beyond C	Enlarged	Real and inverted
At F	At infinity	Highly enlarged	Real and inverted
Between P and F	Behind the mirror	Enlarged	Virtual and erect

Convex Mirror - When the outer surface of a spherical mirror is reflected and the inner surface is painted, it is called a convex mirror.

Convex mirrors are also known as diverging mirrors since they reflect light outwards. They do not converge light to one prime focus point.

Uses - Rear-view mirrors in vehicles, Security purposes in buildings, side mirrors of cars and ATMs.

Nature, position and relative size of the image formed by a convex mirror :

Position of the object	Position of the image	Size of the image	Nature of the image
At infinity	At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect
Between infinity and the pole P of the mirror	Between P and F, behind the mirror	Diminished	Virtual and erect

Sign Conventions :

Distances measured along the direction of the incident ray are taken as positive. Distances measured opposite the direction of the incident ray are taken as negative. Distances measured above the principal axis are positive. Distances measured below the principal axis are negative.

Mirror formula - The mirror formula is the relationship between the focal length of the mirror, the object's distance u from the pole of the mirror, and the image's distance v from the pole.

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

where, u = Object distance, v = Image distance and f = Focal length.

Introduction lens : A lens is an optical element made of transparent material like glass and bounded by the two curved surfaces which can refract the light rays such that either they converge or diverge.

Two types of lenses -

Convex lenses - The lens which has two convex surfaces and converges the light rays is known as convex lens. Convex lens also known as a converging lens or Magnifying lens.

The convex lens is thin at the corner of the lens and thick at the middle of the lens. This lens can produce real and virtual both types of images.

Concave lenses - A lens which has concave surfaces and diverge rays of light is known as concave lens, concave lens also known as a diverging lens. A concave lens is the lens which is thin at the center and thick at the corner. Concave lenses always form virtual images.

Related Terms -

Center of curvature - The center of the sphere of which the surface of a lens is a part, is called center of curvature. As the lens has two surfaces it has two centers of curvatures.

Radius of curvature - Distance between center of curvature and surface of lens is called radius of curvature.

Optical Center - The point in the lens from which a ray passes through without deviation. Optical center situated at the center of the lens.

Aperture - refers to the opening of the diaphragm of a lens that spatially limits the propagation of light.

Principal Axis - An imaginary axis which passes through the centers of curvature of the lens.

Principle Focus - It is different for both the lenses. A lens has two focuses.

Focal length is the distance of the principal focus from the pole.

For convex lenses - A point on the principle axis, the rays coming from become parallel to the principal axis after the reflection is called first principle axis. And a point on the principle axis on which a parallel ray of light will converge after the reflection is called second principal focus.

For concave lens - A point on the principle axis at which the incident ray appears to meet will become parallel

after the refraction is called first principal focus. And a point on the principle focuses about which a parallel ray of beam appears to diverge after the refraction is called second principal focus.

Lens formula for convex and concave lenses -

The lens formula is the relation between the focal length of lens, position of object and position of image. The lens Formula is $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

SI unit of power of lens = Diopter.

where, f = focal length, v = distance of image from the lens, u = distance of object from the lens.

Magnification formula = Height of Image/ Height of Object. $m = h'/h$
another formula for magnification in lenses $\text{Magnification} = v/u$
where v is image distance, u is object distance.

Nature, position and relative size of the image formed by a convex lens for various positions of the object :

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_2	Highly diminished, point-size	Real and inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real and inverted
At $2F_1$	At $2F_2$	Same size	Real and inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real and inverted
At focus F_1	At infinity	Infinitely large or highly enlarged	Real and inverted
Between focus F_1 and optical centre O	On the same side of the lens as the object	Enlarged	Virtual and erect

Nature, position and relative size of the image formed by a concave lens for various positions of the object :

Position of the object	Position of the image	Relative size of the image	Nature of the image
At infinity	At focus F_1	Highly diminished, point-sized	Virtual and erect
Between infinity and optical centre O of the lens	Between focus F_1 and optical centre O	Diminished	Virtual and erect

The Human Eye

It is a natural optical instrument which is used to see objects by human beings.

It is like a camera which has a lens and screen system. In humans, specialized sense organs are capable of receiving visual images, which are then carried to the brain.

Eyes lens form a real and inverted image on the retina.

Optic nerves - The retina is connected to the brain through the optic nerve. The sensation of image formation on the retina is transferred to the brain by the optic nerve.

Retina - The lens system of an eye forms an image on a light-sensitive screen known as the retina. It contains two photoreceptor cells namely cone and rod cells.

Cones - Cones are the nerve cells that are more sensitive to bright light. They help in detailed central and colour vision.

Rods - Rods are the optic nerve cells that are more sensitive to dim lights. They help in peripheral vision.

The minimum distance of distinct vision for a young adult person with no defect in eyes is 25 cm.

We cannot see objects in dim light when coming from a bright place because the iris contracts the pupil to allow less light to enter the eye.

Defects of vision and their corrections

Myopia (Short sightedness)- It is a kind of defect in the human eye due to which a person can see near objects clearly but he cannot see distant objects clearly.

It happens due to Excessive curvature of cornea and Elongation of the eyeball.

Correction- Myopia or short-sightedness can be corrected by wearing spectacles containing concave lenses.

Hypermetropia (Long sightedness) - It is a kind of defect in the human eye due to which a person can see distant objects properly but cannot see the nearby objects clearly.

It happens due to Decrease in power of eye lens i.e., increase in focal length of eye lens and Shortening of eyeball.

Correction - Hypermetropia or long - sightedness can be corrected by wearing spectacles containing Convex lenses.

Presbyopia - It is caused by a hardening of the lens of the eye and weakening of ciliary muscle, which occurs with ageing.

Correction - It can be corrected by wearing spectacles having bifocal lens and progressive lens.

Astigmatism - It is a kind of defect in the human eye due to which a person cannot see (focus) simultaneously horizontal and vertical lines both.

Astigmatism can be corrected by using cylindrical lens which have different curvature in horizontal as well as in vertical direction.

Shadow

A shadow is a dark area where light from a light source is blocked by an opaque object.

Opacity is defined as the quality of not letting light pass through.

Substances can be categorised into transparent, translucent, and opaque, based on the transmission of light through them.

Opaque - These substances do not allow the transmission of light through them. The causes behind opacity are absorption, scattering and reflection. Example - Wood, metal, A wooden bat and frying pan.

Transparent - The property of transmitting light without appreciable scattering so that bodies lying beyond are seen clearly.

Example - Water, Sunroof, Air, Sunglasses, Window Glass, Lense, Prism.

Luminous Objects are which emit light on their own. Example - Sun emits its own light, stars, torch, candle.

Non-luminous objects are which can not emit light on their own. Example - The moon, earth and other planets, paper, tree.

The Moon is visible only due to reflected light.

Translucent - Objects through which we can only see partially and the objects on the other side are not clearly visible, are known as translucent. Example: Oiled paper, tissue paper, colored sunglasses.

Optical instruments

Periscope is used for the navigation of the ships while remaining under water or submerged. It works on the principle of reflection of light. Periscope makes use of two plane mirrors at each end positioned parallel to each other at a 45° angle.

Microscope - It is an instrument that makes an enlarged image of a small object, thus revealing details too small to be seen by the unaided eye. The most familiar kind of microscope is the optical microscope, which uses visible light focused through lenses.

Uses - Analyzing Tissues, Studying Forensic Data, Examining the Structure of an Atom.

The **gnomon** is the part of a sundial that casts the shadow, and it is probably also the world's oldest astronomical instrument. It is used in solar clocks.

Dispersion

When white light passes through a glass prism, it separates into its spectrum of seven colors and this process is known as Dispersion.

Rainbow - It is an optical phenomenon that can occur under certain meteorological conditions. It is caused by reflection, refraction and dispersion of light in water droplets resulting in an effect similar to a spectrum of light appearing in the sky. It takes the form of a multicoloured circular arc.

Rainbows caused by sunlight always appear in the section of sky directly opposite the Sun.

Sun rays are white in color and a mixture of seven colors. It is usually called VIBGYOR.

Violet light has the shortest wavelength, which means it has the highest frequency and energy.

Red has the longest wavelength, the shortest frequency, and the lowest energy.

Violet is the most dispersed when white light passes through a prism.

Gold and copper happen to absorb blue and violet light, leaving yellow light. So gold and copper are yellowish in color.

The white colour of light is formed when red, blue, and green colours of light meet

in equal proportion. The **primary colors** of light are red, blue and green.

A **secondary colour** is a color made by mixing two primary colours in a given colour space.

Example - Red + Green = Yellow, Green + Blue = Cyan, Blue + Red = Magenta.

Different colours of light bend through different angles with respect to the incident ray, as they pass through a prism. White light is composed of seven different colours. All these colours, deviate by a different angle. Thus, we see these colours when light passes through a prism.

Tyndall effect - The Tyndall effect is the scattering of light as a light beam passes through a colloid. The individual suspension particles scatter and reflect light, making the beam visible. The Tyndall effect was first described by 19th-century physicist John Tyndall.

Examples - Visible rays of sunlight, Scattering of car light in fog, light shined through milk, blue coloured Iris, smoke from motorcycles, opalescent glass, blue colour of sky.

Light-emitting diode

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it.

Determines the colour - The two primary materials used in LEDs are aluminium, gallium, indium, phosphide alloys and indium gallium nitride alloys. Aluminium alloys are used to obtain red, orange and yellow light, and indium alloys are used to get green, blue and white light. Slight changes in the composition of these alloys change the colour of the emitted light.

Applications - TV Backlighting, Smartphone Backlighting, LED displays, Automotive Lighting, Dimming of lights

Apart from the red and green wavelengths of sunlight, the wavelength of yellow in sunlight is absorbed by water molecules in the ocean. When sunlight hits the ocean, some of the light is reflected back directly but most of it penetrates the ocean surface and interacts with the water molecules.

Alpenglow - It is the phenomenon by which mountain tops acquire a rosy or orange hue around sunrise and sunset. It is either the indirect sunlight reflection

off of clouds after sunset or before sunrise, or to direct sunlight that occurs near sunset or sunrise.

Stimulated Brillouin scattering - It is a nonlinear scattering effect involving acoustic phonons.

Circle of confusion - In optics, a circle of confusion (CoC) is an optical spot caused by a cone of light rays from a lens not coming to a perfect focus when imaging a point source. It is also known as disk of confusion, circle of indistinctness, blur circle, or blur spot.

Barrel Distortion - When straight lines are curved inwards in the shape of a barrel, this type of aberration is called "barrel distortion".

Zeeman Effect is a phenomena in physics and astronomy which involves the splitting of a spectral line into two or more components of slightly different frequency when the light source is placed in a magnetic field.

The lumen (lm) is the unit of luminous flux, a measure of the total quantity of visible light emitted by a source per unit of time, in the International System of Units (SI).

Fiber optics - It is the science of transmitting data, voice, and images by the passage of light through thin, transparent fibers.

Polarisation of light - The process of transforming unpolarized light waves to polarized light waves is called the polarisation of light. Polarisation of light is a property shown by transverse waves.

Polarised light has electric fields oscillating in one direction, whereas unpolarised light has electric fields oscillating in all directions.

Polarised light is coherent in nature whereas unpolarised light is incoherent in nature.

Interference of light is the phenomenon in which two monochromatic waves superpose to form the resultant wave.

Monochromatic light contains light with a single wavelength. Example - Sodium lamp is a source of monochromatic light. White light is not monochromatic but polychromatic instead.

A Non-Monochromatic light is used in an experiment on the photoelectric effect.

Nuclear Physics

Nuclear physics is the study of the nucleus of atoms and the forces that govern its behavior. It deals with nuclear interactions and reactions like: Nuclear fusion, Nuclear fission, Radioactive decay etc.

Nuclear fusion is a reaction in which two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particles.

Nuclear fusion characteristics

Nuclear fusion produces much more energy than nuclear fission.

Electricity production through this process is commercially less viable than nuclear fission.

Nuclear Fusion reactions power the Sun and other stars.

The device named '**Tokamak**' is associated with Atomic fuel. Tokamak is a device used in nuclear-fusion research for magnetic confinement of plasma. The fusion reaction in the Tokamak will be powered with deuterium and tritium, two isotopes of hydrogen.

The **hydrogen bomb** (or **H-bomb**) is also known as the thermonuclear bomb. These bombs have greater destructive power compared to atomic bombs. These hydrogen bombs are the result of nuclear fusion reactions.

'Magnetic confinement plasma' approach is the scientific principle of Fusion reactor.

It is an approach to generate thermonuclear fusion reactor power that uses magnetic fields to confine fusion fuel in the form of a plasma.

Nuclear Fission

When a heavy nucleus splits into two lighter nuclei, such a reaction is called nuclear fission.

Uncontrolled nuclear fission is used in nuclear bombs.

Controlled nuclear fission used to generate heat in nuclear reactors.

Nuclear fission was discovered in December 1938 by chemists Otto Hahn and Fritz Strassmann and physicists Lise Meitner and Otto Robert Frisch.

Deuterium oxide (Heavy water, $^2\text{H}_2\text{O}$ or D_2O) is a form of water that contains only deuterium (^2H or D, also known as heavy

hydrogen) rather than the common hydrogen-1 isotope (^1H or H , also called protium) that makes up most of the hydrogen in normal water. Pure heavy water is not a radioactive substance.

Heavy Water is an excellent moderator due to its high moderating ratio and low absorption cross section for neutrons.

Heavy Water is used as a primary coolant to transport heat generated by the fission reaction to secondary coolant, light water.

Graphite is used as a moderator because they reflect the fast moving neutrons. Nuclear power reactors do not produce direct carbon dioxide emissions.

Tritiated water is a radioactive form of water in which the usual protium atoms are replaced with tritium. In its pure form it may be called tritium oxide (T_2O or $^3\text{H}_2\text{O}$) or super-heavy water. Pure T_2O is corrosive due to self-radiolysis.

Nuclear fusion and fission release energy through Einstein's equation, $E=mc^2$, E = Energy, m = mass, c = Speed of light. It's called the theory of relativity and this phenomenon illustrates the principle of mass-energy equivalence, where mass can be converted into energy and vice versa, as described by Einstein's equation.

A **particle accelerator** is a machine that uses electromagnetic fields to propel charged particles to very high speeds and energies, and to contain them in well-defined beams.

A **betatron** is a cyclic particle accelerator. that uses the electric field induced by a varying magnetic field to accelerate electrons (beta particles) to high speeds in a circular orbit.

The **Giant Metrewave Radio Telescope (GMRT)**, located near Narayangaon, Pune in India, is a low-frequency radio telescope that helps investigate various radio astrophysical problems ranging from nearby solar systems to the edge of the observable universe.

Liquid sodium is used as a coolant in nuclear reactors to absorb the energy released from the reaction and transfer it to the water in steam generators to produce steam which is then used to turn the turbines to generate electricity. Other coolants in nuclear reactors - Heavy Water, Carbon dioxide.

Boron, Cadmium, and Gadolinium are the best neutron-absorbing substances by the capture process.

A **control rod** is a device that is used to absorb neutrons so that the nuclear chain reaction taking place within the reactor core can be slowed down or stopped completely by inserting the rods further, or accelerated by removing them slightly.

A **nuclear reactor shield** is fitted around the reactor to absorb any radiation from leaking into its immediate environment. Nuclear reactor shield is made of Lead or special types of plastics, shielding is necessary for both the transportation and storage of radioactive neutron sources.

Alpha particles consist of two Protons and two Neutrons bound together into a particle exactly like a helium nucleus. They are generally produced in the process of alpha decay. Alpha particle has a charge $+2$ and can be represented as He^{2+} or ^4_2He .

William Thomson invented the current balance, also known as the Kelvin balance or Ampere balance (SiC), for the precise specification of the ampere.

Marie Curie won the 1911 Nobel Prize in Chemistry for her discovery of the elements polonium and radium, using techniques she invented for isolating radioactive isotopes.

Solar power is the conversion of energy from sunlight into electricity, either directly using photovoltaics (PV) or indirectly using concentrated solar power.

The process by which Earth makes heat is called **radioactive decay**. It involves the disintegration of natural radioactive elements inside Earth - like Uranium. Uranium is a special kind of element because when it decays, heat is produced. It is the heat that keeps Earth from cooling off completely. Radioactive energy powers the Earth's internal heat engine.

Radioactivity is the phenomenon in which nuclei of a given species transform by giving out α or β or γ rays; α -rays are helium nuclei; β -rays are electrons. γ -rays are electromagnetic radiation of wavelengths shorter than X-rays.

There are two units used for measurement of radioactivity: becquerels (Bq, international unit) and curies (Ci, U.S. unit).

Transducer is a device that converts energy from one form to another. Usually a transducer converts a signal in one form of energy to a signal in another.

Piezoelectric transducers are used in seismographs to measure vibrations in rockets. Also used by automotive industries to measure detonations in engines and ultrasonic imaging in medical applications.

Temperature transducer is used to measure the temperature of the air such that to control the temperature of several control systems like air-conditioning, heating, ventilation, and so on.

Ultrasonic transducers are key components in sensors for distance, flow and level measurement as well as in power, biomedical and other applications of ultrasound.

The main advantage of an ultrasonic temperature transducer is that it can measure rapid temperature fluctuation.

Pressure sensors for monitoring pressure & levels in plant/factory automation.

Discoveries

Zacharias Jansen made the first compound microscope.

X - Ray was discovered by **Wilhelm Roentgen** in 1895. He received the first Nobel Prize in Physics in 1901.

Hideki Yukawa, received the nobel prize in 1949, for the discovery of the theory of nuclear forces.

Electromagnetic Telegraph was invented by Carl Fredrich Gauss and Wilhelm Eduard Weber in 1833.

Guglielmo Marconi was awarded the Nobel prize in Physics in 1909, along with Karl Ferdinand Braun for the development of Practical wireless telegraphy.

Joseph Henry conducted early experiments on electromagnetic induction and inductance. Faraday and Henry collaborated in 1831 to further explore and understand this phenomenon.

Isaac Newton was the first to use a glass prism to split sunlight into its spectrum. His book "Philosophiae Naturalis Principia Mathematica" explains Newton's laws of motion and universal gravitation. He also discovered that prisms split light and invented the Reflecting Telescope, using mirrors instead of lenses.

In 1820, **Hans Christian Oersted** linked electricity and magnetism by showing an electric current could deflect a compass needle.

Andre - Marie Ampere named the science of electrodynamics, now known as electromagnetism. Ampere was named after Andre - Marie Ampere.

Devid Gross said that "Physics is always a gamble." He was awarded the 2004 Nobel Prize in Physics for their discovery of asymptotic freedom.

Stephen Hawking is best known for his discovery that black holes emit radiation which can be detected by special instrumentation.

The father of black hole theory is **John Michell**.

Alfred Nobel was the inventor of dynamite in 1866. Dynamite is an explosive made of nitroglycerin, sorbents and stabilizers.

Louis-Victor de Broglie discovered the wave nature of electrons. He was awarded the Nobel Prize in Physics for this discovery in 1929.

John Bardeen discovered the theory of superconductivity.

William Bradford Shockley, John Bardeen and Walter Houser Brattain jointly received the nobel prize in 1956 for the discovery of the Transistor effect.

Victor Francis Hess, discovered cosmic radiation.

In 1928, **Paul Dirac** established the most general theory of quantum mechanics and discovered the relativistic equation for the electron.

Meghnad Shah made important contributions to the theory of thermal ionization and its application to stellar spectra.

Christiaan Huygens built the first pendulum clock in 1656.

Percy Spencer discovered the Microwave.

Henri Becquerel was awarded half of the Nobel Prize for the discovery of radioactivity Physics in 1903, the other half being given to Pierre and Marie Curie for their study of the Becquerel radiation.

Carl Anderson received the Nobel Prize in Physics in 1936 for the discovery of a positively charged particle called the 'positron' in 1932.

J J Thomson discovered the electron in 1897. He won the Nobel prize in Physics in 1906 for his work on the conduction of electricity in gasses.

James Chadwick discovered the neutron in 1932. In 1935, he won the Nobel Prize in Physics for his discovery of the neutron.

The photoelectric effect was discovered by **Heinrich Rudolf Hertz** in 1887. In this effect it is observed that shining ultraviolet light on the electrodes caused a change in voltage between them.

James Watt established the unit of horsepower that is equal to one horse doing 33,000 foot pounds of work in one minute.

Doppler weather radar is an instrument that sends pulses of electromagnetic energy into the atmosphere to detect rainfall, determine its speed and intensity, and identify precipitation types such as snow, rain or hail.

Benjamin Franklin Inventions - The Franklin stove, The lightning rod, and bifocals.

Luigi Galvani is known for Bioelectricity (animal electricity).

Otto von Guericke invented the world's first air pump in 1650.

John Cockcroft and Ernest Walton developed a device, an accelerator, to generate more penetrating radiation.

Enrico Fermi won the 1938 Nobel Prize in Physics for his work on induced radioactivity by neutron bombardment and for the discovery of transuranium elements. He started the world's first man - made nuclear reactor in 1942.

Auguste Bravais demonstrated that there are fourteen space lattices, or regularly repeating arrangements of points in

space, that differ in symmetry and geometry.

Geiger and Nuttall found an empirical relationship between the half life of alpha decay and the energy of the emitted alpha particles in 1911.

Lord Rayleigh is famous for the discovery and isolation of Argon.

John Tyndall showed that gases including carbon dioxide and water vapour can absorb heat.

Lewis Waterman invented the capillary feed fountain pen.

Ian Donald was an obstetrician who developed ultrasound diagnostics.

David Hughes invented the Carbon Microphone.

Rudolf Diesel invented an efficient, compression ignition, internal combustion engine in the 1890s.

Max Planck won the Nobel prize in Physics in 1918 for the discovery of 'Energy Quanta'.

Werner Karl Heisenberg received the Nobel Prize in Physics 1932 for the creation of quantum mechanics.

Superconductivity was discovered by **Heike Kamerlingh Onnes** in 1911. He was also awarded the Nobel Prize for Physics in 1913.

John Ambrose Fleming invented the first thermionic valve or vacuum tube.

Eli Whitney invented the cotton gin in 1794.

Stephanie Louise Kwolek was an American chemist who is known for inventing Kevlar. She invented the stronger - than - steel fiber(Kevlar) in 1965.

George Eastman invented flexible roll film and in 1888 introduced the Kodak camera shown to use this film.

Steve Wozniak designed Apple's first products, the Apple I and II computers and he helped design the Macintosh.

Henry Moseley invented the atomic battery in 1913.

Alessandro Volta invented the electric battery or power battery in 1800. He also invented the voltaic pile and Hydrogen lamp.

Louis Pasteur discovered the principles of vaccination, microbial fermentation and pasteurization.

Niels Bohr won the 1922 Nobel Prize in Physics for investigating the structure of atoms and the radiation emanating from them.

Homi Jehangir Bhabha is known as the 'Father of Indian Nuclear Programme'

Vikram Sarabhai is considered as the Father of the Indian space program.

In 1851, **Leon Foucault** discovered the phenomenon of eddy currents, which flow in closed loops within conductors in planes perpendicular to the magnetic field.

APJ Abdul Kalam is known as 'Missile Man of India' for his work in the development of ballistic missile technology.

The Wright brothers (**Wilbur** and **Orville Wright**) completed the world's first successful controlled powered flight in 1903.

The Mercury Barometer was discovered by **Evangelista Torricelli** in 1643.

Richard Salter discovered Spring Balance.

Hooke's law, the law of elasticity discovered by **Robert Hooke** (1660). Example - A ball and spring model of chemical bonds obey Hooke's law.

The first reliable measurement of 'G' was made by **Henry Cavendish** in 1798. He discovered hydrogen in 1766.

Michael Faraday discovered electromagnetic induction, diamagnetism and electrolysis. He discovered benzene in illuminating gas in 1825.

Jean Baptiste Joseph Fourier is known for discovering that any wave can be represented as an infinite number of weighted sinusoids, i.e., the sum of sine and cosine waves.

Blaise Pascal founded Pascal's law (Pascal's principle or Principle of transmission of Fluid Pressure) in 1653.

Eugene Bourdon invented the pressure-measuring device (the Bourdon tube) in 1849.

The first commercial electric bulb was first invented by **Thomas Alva Edison** in the year 1879. His other inventions included the phonograph, the carbon-button transmitter for the telephone speaker and microphone, the first commercial electric light and power system, an experimental electric railroad, and key elements of motion-picture equipment.

The first induction motor was invented by **Nikola Tesla** in 1887.

Frank Shuman was known for his work on solar engines.

John Logie Baird invented the first live working television system on 26 January 1926.

In 1851, **George Gabriel Stokes**, founded the science of hydrodynamics using his law of viscosity.

Thomas Savery invented the steam engine.

Edwin Howard Armstrong developed FM (frequency modulation) radio and the superheterodyne receiver system.

Hendrik Antoon Lorentz who shared the 1902 Nobel Prize in Physics with **Pieter Zeeman** for the discovery and theoretical explanation of the Zeeman effect.

Max Von Laue received the Nobel Prize for Physics in 1914, for his discovery of the diffraction of X-rays on crystals.

In 1859 **Robert Wilhelm Bunsen** and **Gustav Robert Kirchhoff** invented the flame spectroscope.

Satyendra nath Bose in 1924 to derive the blackbody radiation law with respect to agas of light quanta (photon). S.N. Bose collaborated with Albert Einstein in developing a theory regarding the gaslike qualities of electromagnetic radiation.

In 1924, Albert Einstein predicted a new state of matter, the Bose - Einstein Condensate (BEC), based on a quantum formulation by the Indian physicist Satyendra nath bose.

In 1912, **Casimir Funk** proposed the 'vitamine hypothesis'.

Pierre and Marie Curie discovered Radium in 1898.

Auguste Laurent discovered and synthesized a large number of aromatic

organic compounds such as anthracene, phenanthrene, stilbene, benzil, phthalic acid, phthalic anhydride, and pyrene.

Mr. Dicas of Liverpool invented the lactometer.

Ultraviolet light was discovered by **Johann Wilhelm Ritter** in 1801.

William Crookes discovered the element thallium.

Albany magnet, world most powerful electromagnet, that can lift 750 pounds of metal at a time created by **Joseph Henry** in 1830.

Luis Alvarez invented a radio distance and direction indicator. He was awarded the Nobel Prize in Physics in 1968 for his discovery of resonance states in particle physics using the hydrogen bubble chamber.

Chemistry

Structure of Atom

John Dalton is considered as the Father of Modern Atomic Theory. In 1808, John Dalton published a paper titled "A New System of Chemical Philosophy" describing principles such as the uniqueness of atoms of the same element, relative atomic mass and the laws of chemical combination.

Dalton's Atomic Theory :

Atoms are the smallest bits of elements, and can't be broken apart. They combine in simple ratios to make compounds. All matter is made of these tiny, unbreakable blocks. Atoms of the same element are identical in size, mass, and properties, but atoms of different elements vary. This idea follows the laws of mass conservation and constant composition.

Drawbacks of Dalton's Atomic Theory

- This theory did not define subatomic (such as electrons, protons and neutrons) particles. It does not define the isotopes and isobars. Elements need not combine in simple, whole-number ratios to form compounds. Allotropes were not defined.

Thomson's Atomic model : According to J. J. Thomson Atomic model, the atom is made of a positively charged sphere and the electrons are embedded in it. Negative and positive charges have the same result. Therefore atoms are electrically neutral.

J.J. Thomson's atomic model, derived from his cathode ray tube experiments, revealed that atoms harbor minute, negatively charged subatomic particles known as electrons. Consequently, an atom comprises a positively charged sphere in which electrons are embedded.

Drawbacks of Thomson's Atomic Model :

Thomson's atomic model could not explain how the positive charged body holds on the electrons situated inside the atom. It also fails to explain the stability that an atom possesses. The concept of the nucleus was not present in this theory.

Millikan's oil drop experiment was performed to measure the charge of an electron. Electrically charged oil droplets entered the electric field and were balanced between two plates by altering

the field. When the charged drops fell at a constant rate, the gravitational and electric forces on it were equal.

Rutherford Model : According to his model, the mass of the atom and the positive particles lie at the center of the atom called the nucleus. He also defined that the negatively charged particles revolved around the nucleus in circular paths called orbits.

Alpha Rays was discovered by Ernest Rutherford in 1899. Alpha Rays is used to refer to helium atoms that have lost two electrons. Most alpha particles are in the range 4–8 MeV.

Drawbacks of Rutherford's Atomic Model : This model could not explain the motion of electrons in circular orbitals. A charged particle would release energy during acceleration and eventually, it will fall into the nucleus.

Henry Moseley observed regularities in the characteristic X-ray spectra of the elements. Based on the X-ray experiment Moseley proposed that the physical and chemical properties of the elements are periodic functions of their atomic numbers.

Bohr's Model of Atom (1913) : Niels Bohr proposed a theory for the hydrogen atom based on the quantum theory that some physical quantities take only discrete values.

According to Niels Bohr's Atomic model, an electron will have more energy if it is located away from the nucleus whereas electrons will have less energy if it is located near the nucleus.

Postulates of Bohr Atomic Model

Electrons revolve around the nucleus in a fixed circular path termed orbits or shells. The orbits are termed as stationary orbits.

The electrons will not radiate energy as long as they continue to revolve around the nucleus in the fixed orbital shells.

The different energy levels are denoted by integers such as $n = 1$ or $n = 2$ or $n = 3$ and so on. These are called **quantum numbers**. The range of quantum numbers may vary and begin from the lowest energy level (nucleus side $n = 1$) to the highest energy level.

The different energy levels or orbits are represented in two ways such as 1, 2, 3, 4... or K, L, M, N.... shells. The lowest energy level of the electron is called the ground state.

The change in energy occurs when the electrons jump from one energy level to another.

Electronic distribution of various orbits or energy levels can be calculated by the formula $2n^2$. Here, 'n' denotes the number of orbits.

The number of electrons in the K shell (1st orbit) can be calculated by $2n^2 = 2 \times 1^2 = 2$. Thus, maximum number of electrons in 1st orbit = 2.

The number of electrons in the L shell (2nd orbit) = $2 \times 2^2 = 8$. Thus, maximum number of electrons in 2nd orbit = 8.

Similarly, number of electrons in M shell = 18, number of electrons in N shell = 32 and so on.

Example - Number of electrons in a Boron atom is 5. On applying the Bohr's model a Boron atom has 5 electrons so it gets the number of electrons in K shell = 2, L shell = 3. The valency of Boron is 3.

Schrodinger's wave equation : It is a mathematical expression describing the energy and position of the electron in space and time, taking into account the matter wave nature of the electron inside an atom.

A **Molecule** is a group of two or more atoms that are chemically bonded together. Atoms of the same element or of different elements can join together to form molecules.

The **atomic radius** of a chemical element is a measure of the size of its atom, usually the mean or typical distance from the center of the nucleus to the outermost isolated electron.

Calculations by Rutherford showed that the volume occupied by the nucleus is negligibly small as compared to the total volume of the atom. The radius of the atom is about 10^{-10} m, while that of the nucleus is 10^{-15} m. If a cricket ball represents a nucleus, then the radius of the atom would be about 5 km.

Atomic mass of the atom of an element is defined as the average mass of the atom, as compared to $1/12$ th the mass of one carbon-12 atom. Its unit is the atomic Mass unit.

The mass of any isotope of any element is expressed in relation to the Carbon-12 standard. The mass of an electron is approximately 1/2000 times that of a proton.

Atomic Mass of **Hydrogen** is 1.00784 u. Hydrogen was discovered by Henry Cavendish in 1766.

Hydrogen atoms have only one proton, one electron and no neutron. Since hydrogen is devoid of neutrons the mass number of hydrogen is equal to its atomic number.

All elements have atoms with neutrons except Hydrogen atoms. Hydrogen compounds are commonly called hydrides.

As **water** (H_2O) contains 2 atoms of Hydrogen and one atom of Oxygen, the mass of one atom of Hydrogen is 1u and of oxygen is 16u. In water, the mass ratio of hydrogen to oxygen is always 1 : 8. The ratio by mass of carbon and oxygen in carbon dioxide is 3 : 8.

One mole of anything with 6.022×10^{23} particles is known as Avogadro's number or Avogadro's constant. The correct representation of the number of the atoms is : $\frac{\text{given mass}}{\text{molar mass}} \times \text{Avogadro number}$.

An ion is an atom or group of atoms with a positive or negative electrical charge. A negatively charged ion is called an anion, while a positively charged ion is called a cation.

In sodium chloride ($NaCl$) crystals, ions present are Na^+ and Cl^- . They arrange in a 1:1 ratio with a coordination number of 6, meaning six anions surround each sodium ion in the crystal lattice.

Cathode rays are streams of electrons seen in discharge tubes, known for their negative charge, straight path, and ability to ionize gas in the tube. Cathode ray tubes contain electron beams focused on phosphor-coated screens, emitting light when struck. In atoms, protons and neutrons reside in the nucleus, while negatively charged electrons occupy the extranuclear region.

A positive charge on an object indicates an excess of protons, while a negative charge indicates an excess of electrons.

John Thomson discovered the electron. Mass of an electron is 9.1×10^{-31} kg.

The charge of an electron is -1.6×10^{-19} coulomb.

Ernest Rutherford in 1917 discovered the proton. Mass of a proton is 1.67262×10^{-27} kg. The charge of a proton is $+1.6 \times 10^{-19}$ coulomb (C).

James Chadwick discovered the neutron. Mass of a neutron is 1.675×10^{-27} kg.

The relative charge of a proton and neutron is +1 and 0 respectively.

The electronic configuration of an element describes how electrons are distributed in its atomic orbitals.

The maximum number of electrons that can be accommodated in a shell is based on the principal quantum number (n) and is represented by the formula $2n^2$, where 'n' signifies the number of given shells from the nucleus.

Each shell can contain only a fixed number of electrons. The first shell can hold up to two electrons, The second shell can hold up to eight (2 + 6) electrons, the third shell can hold up to 18 (2 + 6 + 10) electrons and so on.

Shell levels and their electron capacity :
Energy Level Index 1 (Shell - K, Number of electrons - 2), Index 2 (Shell - L, electrons - 8), Index 3 (Shell - M, Number of electrons - 18) Index 4 (Shell - N ; electrons - 32), Index 5 (Shell - O, electrons - 50), and Index 6 (Shell - P, electrons - 72).

Frederick Soddy discovered the concept of isotopes in 1913. **Isotopes** are elements that all have the same atomic number, but different mass number.

There are two main types of isotopes: stable and unstable (radioactive). Stable isotopes have stable nuclei whereas unstable isotopes have unstable nuclei.

Some important Isotopes :-

Hydrogen has three naturally occurring isotopes, which are indicated as Protium (${}_1H^1$), Deuterium (${}_1H^2$) and Tritium (${}_1H^3$). The most stable radioisotope of hydrogen is tritium, with a half-life of 12.32 years.

Protium is the most abundant common isotope of Hydrogen. Deuterium contains one proton and one neutron in its nucleus. The nucleus of deuterium is called a deuteron.

Deuterium reacts with oxygen to form a

compound called deuterium oxide (D_2O). It's also called heavy water.

Carbon has two stable isotopes Carbon-12 and Carbon-13. Carbon-14 is radioactive or unstable isotopes of the carbon.

Uranium is composed of three major isotopes, uranium-238, uranium-235, and uranium-234.

Titanium is composed of five stable isotopes; (${}^{46}Ti$), (${}^{47}Ti$), (${}^{48}Ti$), (${}^{49}Ti$) and (${}^{50}Ti$). Atomic mass of Titanium (Ti) is 47.86.

Scandium has 13 isotopes whose half-lives are known, with mass numbers 40 to 52. Naturally occurring scandium consists of its one stable isotope, ${}^{45}Sc$.

Gold (Au_{79}) has a stable isotope Au^{197} . Among the radioisotopes, Au^{195} is the most stable one whose half life is 186 days.

Nitrogen has two stable isotopes, ${}^{14}N$ and ${}^{15}N$. ${}^{14}N$ is the more abundant of the two, comprising 99.63% of the nitrogen found in nature.

The element oxygen (O) is found in three naturally occurring stable isotopes ${}^{18}O$, ${}^{17}O$, and ${}^{16}O$.

Chlorine is the element of group 17 and has two isotopes of masses 35 and 37 amu with the average abundance of 75.77% and 24.23% respectively.

Atomic number is equal to the number of protons in the nucleus of an atom. A Helium nucleus has two protons and two neutrons. Beryllium atoms have four protons in an atom. The number of protons in a Lithium nucleus is three.

Mass number = Sum of number of neutrons and protons.

Isobars are the atoms of different chemical elements having the same atomic mass but different atomic number (or number of protons). Examples - ${}_{18}Ar^{40}$ & ${}_{20}Ca^{40}$ represent isobars.

Calcium and **argon** are isobars as they have same mass number i.e. 40 but different atomic numbers.

Isotones are atoms that have the same neutron number but different proton number. For example - ${}_{16}S^{36}$, ${}_{17}Cl^{37}$, potassium-39.

Atomicity is the total number of atoms present in a molecule. For example - each molecule of oxygen (O_2) is composed of two oxygen atoms. Therefore, the atomicity of oxygen is 2. The atomicity of argon is 1. The atomicity of all noble gases (He, Ne, Ar, Kr, Xe, Rn) is 1.

Monatomic elements are elements that are stable as single atoms. Example - Ne, Ar (noble gases).

Diatomic :- The elements that have two atoms in each molecule are called diatomic. There are seven diatomic elements. Example - Hydrogen (H_2), Nitrogen (N_2), Oxygen (O_2), Fluorine (F_2), Chlorine (Cl_2), Iodine (I_2), and Bromine (Br_2).

Polyatomic :- The elements that have more than two atoms per molecule are called polyatomic. Example - S_8 and P_4 . Polyatomic molecules are electrically neutral groups of three or more atoms held together by covalent bonds. Selenium, Sulphur and Phosphorus are only three polyatomic elements (all non metals) in the periodic table.

Tetra-atomic elements - They are made up of four atoms. Example - N_4 , CN_2O , BFN_2 , C_2O_2 .

Phosphorus is represented as (P_4). The number of atoms present in a molecule of phosphorus (P) is 4. So the atomicity of phosphorus is 4 as it is tetra-atomic or aliphatic.

The **Higgs boson** is often called the God particle, after the title of Nobel physicist Leon Lederman's.

The Higgs boson is linked to the Higgs field, which provides mass to particles like electrons and quarks. Mass dictates an object's resistance to changes in speed or position when acted upon by force.

Crystal field theory explains how ligands (atoms around a metal) affect the energy of the metal's d-orbitals, influencing the complex's geometry and color.

Ligand field theory describes the electronic structure of coordination or complex compounds.

Valence bond theory describes chemical bonding.

Molecular orbital theory states that each atom tends to combine together and

form molecular orbitals.

Polar molecules are molecules in which one end of the molecule is slightly positive, while the other end is slightly negative. Example - H_2O , HCl .

Nonpolar molecules are molecules that do not have any electrical charges or partial charges. Nonpolar molecules are made of atoms. Example - H_2 , O_2 .

Atoms and non-polar molecules are electrically symmetrical and have no dipole moment because their electronic charge cloud is symmetrically distributed.

Bond dipole moment is a measure of the polarity of a chemical bond between two atoms in a molecule. It involves the concept of electric dipole moment, which is a measure of the separation of negative and positive charges in a system. Example - Water molecule.

Intermolecular forces (IMF) are forces of attraction between neighboring particles (atoms, molecules or ions). For Example - (H_2O).

Some common types of intermolecular forces :-

Dispersion Forces, also called London forces - operate for a short distance and it is the weakest force. This kind of force arises due to the movement of electrons thus creating temporary positive and negative charged regions.

Dipole-dipole forces are attractive forces between the positive end of one polar molecule and the negative end of another polar molecule. There is a dipole-dipole force between chlorine and hydrogen between two different polar ends of molecules.

Dipole-induced dipole force is a type of attractive force that operates between the polar molecules having permanent dipole and the molecules lacking permanent dipole.

A **hydrogen bond** is an Intermolecular force (IMF) that forms a special type of dipole-dipole attraction when a hydrogen atom bonded to a strongly electronegative atom exists in the vicinity of another electronegative atom with a lone pair of electrons.

Polar molecule : A molecule containing polar bonds where the sum of all the bond's dipole moments is not zero.

Examples of Polar Molecules : Water (H_2O), Ethanol (C_2H_5OH), Ammonia (NH_3), Sulphur dioxide (SO_2), Hydrogen sulphide (H_2S).

Van der Waals Force : It includes attraction and repulsion between atoms, molecules, and surfaces, as well as other intermolecular forces.

The order of strength in the decreasing order - ion-dipole > hydrogen bonds > dipole-dipole > Van Der Waals Forces.

Dash-wedge formula : A method of representing the three-dimensional structure of a molecule in which simple lines represent bonds in the plane of the image, wedge-shaped lines represent bonds towards the viewer, and dashed lines represent bonds away from the viewer.

A bond line structural formula is another way of structural representation of organic compounds. Here, every bond is represented as a line in a zigzag manner. If not specified, every terminal is assumed to be a methyl ($-CH_3$) group.

Lewis Structure is a very simplified representation of the valence shell electrons in a molecule. It is used to show how the electrons are arranged around individual atoms in a molecule. Example ; methane is CH_4 and propanol is $CH_3(CH_2)_2OH$.

Electrons can change from a metallic atom of one type to another. In 1950 **Henry Taube** showed that this does not happen through a direct transition, instead a molecule serves as a bridge between the metallic atoms.

Acid, Base and Salt

Acid

The word Acid is taken from the Latin word acidus or acere which means sour. Acids are the substances that give off hydrogen ions (H^+) in water, produce salts when combined with metals and have a pH value from 0 to less than 7.

Acids are separated into **two categories** based on where they are found:

Natural Acids: These come from organic sources like fruits and animal byproducts, Citric, tartaric, and lactic acid.

Mineral Acids: Acids made from minerals are known as mineral acids. Examples of

this are Hydrochloric acid (HCl), Sulphuric Acid (H_2SO_4), and nitric acid (HNO_3).

The name of some Acids and products in which they found are as follows

Oxalic Acid ($\text{C}_2\text{H}_2\text{O}_4$) is found in spinach, rhubarb, rice bran, miso, tomato, raspberry, dates.

Erucic Acid is found in mustard oil.

Carbonic Acid is found in soda water.

Lauric Acid is found in Coconut milk.

Butyric Acid is found in Butter.

Lactic Acid ($\text{C}_3\text{H}_6\text{O}_3$) is found in milk products and beer. It is responsible for muscle fatigue in humans.

Citric Acid ($\text{C}_6\text{H}_8\text{O}_7$) is found in orange, tangerine, lemon, grapefruit, berries, and kiwifruit.

Tartaric Acid ($\text{C}_4\text{H}_6\text{O}_6$) naturally occurs in plants like grapes, apricots, apples, bananas, avocados and tamarinds. It is added in foods giving a sour taste and serving as an antioxidant.

Malic acid ($\text{C}_4\text{H}_6\text{O}_5$) is found in apples and apricots.

Ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$) is found in tomatoes, red and black currant, and broccoli.

Formic acid (HCOOH) in bee, nettle, and ant stings can be neutralized with moist baking soda (NaHCO_3).

Folic Acid ($\text{C}_{19}\text{H}_{19}\text{N}_7\text{O}_6$) is also called Pteroylglutamic acid, Folacin, Vitamin B_9 . It is used by the body to manufacture DNA.

Acetic acid (CH_3COOH) is in vinegar, used for cooking, cleaning, and more. It might help with weight loss and health. In 1845, chemist Hermann Kolbe made it from inorganic materials.

Amino acids are organic compounds that contain both amino and carboxylic acid functional groups. Examples of Amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

Methyl Alcohol (CH_3OH) is also called Methanol or Wood spirit.

Muriatic acid, also known as hydrochloric acid (HCl), serves various purposes like

synthesizing organic compounds, purifying table salt and sugar, tanning leather, adjusting the pH of medicines, and electroplating.

Human body stomach contains hydrochloric acid (HCl). It helps to digest food. Hydrochloric acid destroys the bacteria that enter the stomach.

Perchloric acid (HClO_4) is also called Hyperchloric acid.

Sulphuric Acid (H_2SO_4), also called oil of vitriol or hydrogen sulfate.

Nitric Acid (HNO_3) is a strong acid and also known as aqua fortis.

Benzoic Acid is used as antimicrobial preservatives in food and beverages, as it presents its strong antibacterial activity at pH 2.5 - 4. Other preservatives are Acetic acid and Natamycin.

Benzoic Acid is produced commercially by partial oxidation of toluene ($\text{C}_6\text{H}_5\text{CH}_3$) with Oxygen.

Propionic acid ($\text{C}_3\text{H}_6\text{O}_2$) is used in herbicides, fine chemical intermediates, rubber chemicals, emulsions.

Tartronic acid ($\text{C}_3\text{H}_4\text{O}_5$) is used as a reactant in the catalytic oxidation with air to form mesoxalic acid.

Acrylic acid ($\text{C}_3\text{H}_4\text{O}_2$) is used in production of acrylic esters and resins, which are used primarily in coatings and adhesive.

Base

A substance is called base if its aqueous solution tastes bitter, soapy to touch, releases a hydroxide ion (OH^-) into water, and neutralizes acids.

Base has a pH value greater than 7 to 14.

All bases do not dissolve in water. An Alkali is a corrosive base that dissolves in water.

Examples - Calcium Hydroxide or slaked lime $\text{Ca}(\text{OH})_2$ is used in production of paper, lime water, and root canal fillings, cement industry.

Sodium Hydroxide or Caustic soda (NaOH) is used in production of soap and explosives. Concentrated Sodium hydroxide (NaOH) is a highly basic compound and its pH is nearest to 14.

Ammonium Hydroxide (NH_4OH) is used

in the production of window cleaner and hair dye.

Milk of Magnesia ($\text{Mg}(\text{OH})_2$) has a pH of 10.5. It's used as an antacid to balance stomach acid and as a fire retardant.

pH

pH is a measure of the concentration of Hydrogen ions, the acidity or alkalinity of a solution.

It was first described by Dr. Soren Sorensen in 1909. The p stands for the German word "Potenz", meaning power or concentration, and the H for the Hydrogen ion (H^+).

The pH range goes from 0 - 14, with 7 being neutral. pH of less than 7 indicates acidity, whereas a pH of greater than 7 indicates a base.

The normal pH of human blood is 7.4. The pH of milk is about 6.7 to 6.9, Lemon Juice (2 to 3), Shampoo (7 to 10). At room temperature (25°C) the pH value of neutral water is 7.

Concentrated Acids have pH value nearest to zero. Some examples are concentrated Hydrochloric Acid (HCl), concentrated Nitric Acid (HNO_3).

Concentrated Bases have pH value nearest to 14. Its examples are concentrated Sodium Hydroxide (NaOH), concentrated Potassium Hydroxide (KOH).

Tooth enamel is made up of Calcium Phosphate $\text{Ca}_3(\text{PO}_4)_2$. Tooth decay starts when the pH of the mouth is lower than 5.5.

Living organisms can survive only in a narrow range of pH change. The normal range of pH in human beings is from 7.35 to 7.45. When the body pH falls below 6.9, it can lead to coma.

Acidic nature of soil is shown by high concentration of hydrogen. Slightly acidic soils are more suitable for plant growth. Maximum nutrients are available to crops when the pH ranges from 6.5 to 7.

When the pH of rainwater comes down below 5.6 then it is termed as acid rain. Acid rain is a byproduct of sulphur dioxide (SO_2) and nitrogen dioxide (NO_2) emitted into air. Acid Rain is harmful for human beings, animals, plants, aquatic ecosystems.

Litmus paper is a type of pH paper made

by treating the paper with natural dyes from lichens. Litmus paper is either red or blue. Base turns Red litmus to blue, while acid turns blue litmus to red.

Acid-Base indicators

Litmus solution is a purple dye, which is extracted from lichen, a plant belonging to the division Thallophyta, and is commonly used as an indicator. When the litmus solution is neither acidic nor basic, its color is purple.

There are many natural materials like red cabbage leaves, turmeric, coloured petals of some flowers such as Hydrangea, Petunia and Geranium, which indicate the presence of acid or base in a solution. These are called acid-base indicators or sometimes simply indicators.

There are some substances whose odour changes in acidic or basic media. These are called olfactory indicators. Examples of olfactory indicators are vanilla, onion and clove.

China rose is a natural Acid - Base indicator which changes the acidic solution to magenta (dark pink) and basic to green.

Baking soda solution is basic in nature, it gives green colour with china rose.

Starch in potatoes reacts with iodine, turning to a blue colour.

Anthocyanins are colored water-soluble pigments belonging to the phenolic group. Anthocyanins are responsible for the colors, red, purple, and blue and are in fruits and vegetables.

Red Cabbage contains anthocyanins that turn pink/reddish in the presence of acid, turn purple when neutral and turn blue or green when mixed with alkaline substances.

When an Acidic solution is mixed with a Basic solution, both the solutions neutralize the effect of each other, this type of reaction is called a **Neutralization reaction**. The products of this reaction are salt and water.

An example of neutralization is formation of Sodium chloride (NaCl) and Water (H₂O) when Sodium Hydroxide (NaOH) reacts with Hydrochloric acid (HCl).

When electricity is passed through an aqueous solution of sodium chloride

(called brine), it decomposes to form sodium oxide.

Dilution is the process of reducing the concentration of a solute in solution, usually simply by mixing with more solvent.

When Acid or Base is mixed with water it leads to decrease in the concentration of ions (H₃O⁺/OH⁻) per unit volume, during this process the acid or the base is said to be diluted.

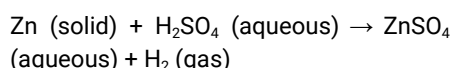
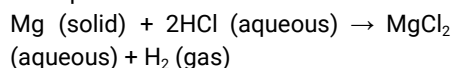
The process of dissolving an acid or a base in water is an **exothermic reaction**.

Both acidic and basic solutions in water conduct electricity. Acids, when dissolved in water release the Hydrogen ions (H⁺), and bases when dissolved in water release the Hydroxide ions (OH⁻). These ions are charged species and so act as charge carriers.

Hydronium ions (H₃O⁺) are formed by the combination of H₂O and H⁺.

A **Dilute Acid** is an acid that has been weakened, typically by adding water to it. When dilute acid reacts with a metal, hydrogen gas is evolved and a corresponding salt is formed.

Examples -



When Carbon dioxide (CO₂) is passed through lime water (Ca(OH)₂), it turns lime water milky due to the formation of a white precipitate of Calcium carbonate (CaCO₃).

Some materials which contain Calcium carbonate are chalk, limestone, and marble.

Aqua regia, (Latin for 'royal water') is a freshly prepared mixture of concentrated Hydrochloric acid (HCl) and concentrated nitric acid (HNO₃) in the ratio of 3:1. It can dissolve Gold (Au), even though neither of these acids can do so alone.

Generally all metals combine with oxygen to form metal oxides. Metal oxides are basic in nature since they react with dilute acids to form water and salt.

Some metal oxides, such as Aluminum oxide (Al₂O₃), Zinc oxide (ZnO) show both acidic as well as basic behavior. Such

metal oxides are known as Amphoteric oxides.

The metals which do not react with any other elements to form compounds are called Noble metals.

Salts of a strong acid and weak base are acidic with pH value less than 7.

Salt

It is a **neutral substance** whose aqueous solution does not affect litmus.

Salinity is the term used to define the total content of dissolved salts in seawater. It is calculated as the amount of salt (in gm) dissolved in 1,000 gm (1 kg) of seawater. It is usually expressed as parts per thousand.

The two most common elements in seawater, after Oxygen and Hydrogen, are Sodium and chlorine.

The important salts found in seawater are Sodium Chloride (NaCl), Magnesium Chloride (MgCl₂), Magnesium Sulphate (MgSO₄), Potassium Chloride (KCl).

Types of Chemical Reactions

Chemical reaction is a process that involves rearrangement of the molecular or ionic structure of a substance, as distinct from a change in physical form or a nuclear reaction.

During a chemical reaction, the substances that react are known as **reactants** whereas the substances that are formed during a chemical reaction are known as **products**.

A chemical equation is the symbolic representation of a chemical reaction in the form of symbols and chemical formulas.

The formation of bubbles, or rather a gas, is another indicator of a chemical reaction taking place. When bubbles form, a temperature change could also be taking place.

Burning of wood is a chemical change as new substances which cannot be changed back (e.g. carbon dioxide) are formed. For example, if wood is burned in a fireplace, there is no wood anymore but ash.

Other examples include burning of a

candle, rusting of iron, and baking a cake.

Boiling of milk is an example of chemical change because during boiling of milk, water vapors release.

Crushing a can and folding a paper, both are examples of a physical change.

Common types of chemical reactions

Combination reaction - Two or more compounds combine to form one compound. Reaction: $A + B \rightarrow AB$, where A and B are reactants and AB is product. Example: Burning of coal is a combination reaction.

Decomposition reaction - It is the opposite of a combination reaction, here a complex molecule breaks down to make simpler ones. Reaction: $AB \rightarrow A + B$.

For instance, when food digests in our bodies or when an egg turns rotten, it's due to this type of reaction, where the original substance breaks down into simpler components.

Precipitation reaction - Two solutions of soluble salts are mixed resulting in an insoluble solid (precipitate) forming. Reaction: $A + \text{Soluble salt B} \rightarrow \text{Precipitate} + \text{soluble salt C}$.

Neutralization reaction - An acid and a base react with each other. Generally, the product of this reaction is salt and water. Reaction: $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$.

Combustion reaction - Oxygen combines with a compound to form carbon dioxide and water. These reactions are exothermic, meaning they give off heat. Reaction: $A + O_2 \rightarrow H_2O + CO_2$.

Displacement reaction - One element takes place with another element in the compound. Reaction: $A + BC \rightarrow AC + B$.

Types of Combustion

Rapid combustion: It is a process in which a large amount of heat and light is released in a very short span of time. The substances which undergo rapid combustion have lower ignition temperatures. Example - When we bring a matchstick near the gas stove it rapidly catches fire.

Explosive Combustion happens when the reaction occurs very rapidly. The reaction

occurs when something ignites to produce heat, light and sound energy. The simple way to describe it is to call it an explosion. Some classic examples are firecrackers or blowing up dynamite.

Spontaneous combustion is the phenomenon in which a hydrocarbon substance unexpectedly bursts into flame without apparent cause.

Slow combustion (smoldering) is the slow, low-temperature, flameless form of combustion, sustained by the heat evolved when oxygen directly attacks the surface of a condensed-phase fuel.

Facts

Corrosion slowly coats the surfaces of metallic objects with oxides or other salts of the metal. The rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion.

In corrosion, a metal is oxidized by loss of electrons to oxygen and formation of oxides.

Haber-Bosch process (Haber process) is industrial procedures to be adopted for the production of ammonia. In this process the atmospheric nitrogen (N_2) is converted to ammonia (NH_3) by reacting it with hydrogen (H_2).

Producer gas is the substance that is produced by burning biomass with an air deficit and a regulated amount of humidity. $\text{Producer gas} = CO + N_2 + H_2$.

Potassium chlorate ($KClO_3$) decomposes in the presence of catalyst manganese dioxide to create potassium chloride and oxygen gas.

Chemical balance is a delicate instrument which is used to weigh small amounts of substances accurately up to the fourth place of decimal.

The **Maillard reaction** is also known as non-enzymatic browning. It is an extremely complex process and is the reaction between reducing sugars and proteins by the impact of heat.

Caramelization is a process of browning of sugar used extensively in cooking for the resulting sweet nutty flavor and brown color.

Dextrinization is a special type of acid hydrolysis that occurs by heating acidified starch with reduced moisture or

heating of the aqueous starch slurry with or without pH change.

Diffusion is the mixing of molecules due to their random motion, whether in a liquid or a gas. A drop of food coloring spreading out in water is an example of diffusion. The gases from the atmosphere reach the aquatic animals and plants through Diffusion.

Effusion is the process in which a gas escapes from a container through a hole of diameter considerably smaller than the mean free path of the molecules.

Vapor pressure is defined as the pressure exerted by a vapor in thermodynamic equilibrium with its condensed phases (solid or liquid) at a given temperature in a closed system.

Osmosis is when solvent molecules move from low to high concentration through a semipermeable membrane. Reverse osmosis pushes water through a semipermeable membrane to remove contaminants.

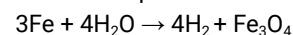
Water gas ($CO + H_2$) is an equimolar mixture of carbon monoxide and hydrogen. It is prepared by passing steam over red hot coke.

An **exothermic reaction** is a reaction in which energy is released in the form of light or heat. Thus, in an exothermic reaction, energy is transferred into the surroundings. Respiration is an example of it.

Endothermic reactions are the chemical reactions in which the reactants absorb heat energy from the surroundings to form products. These reactions lower the temperature of their surrounding area, thereby creating a cooling effect. Examples include melting ice, melting salts, and photosynthesis.

Galvanization or galvanizing is the process of applying a protective zinc coating to steel or iron, to prevent rusting. The most common method is hot-dip galvanizing, in which the parts are submerged in a bath of hot, molten zinc.

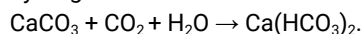
Process of formation of hydrogen by passing steam over red hot iron is presented by the following balanced chemical equation:



Hydrogenation reaction is a chemical reaction between hydrogen and other compounds in presence of catalyst

barium.

Calcium carbonate reacts with carbon dioxide and water to produce Calcium hydrogen carbonate.



Pasteurisation is a process of food preservation in which milk is boiled at 72°C for 15 to 30 seconds to eliminate pathogens and extend shelf life.

Coordination isomerism exists in compounds containing two or more complexes, each of which contains a different set of ligands that can in principle be replaced with a ligand of the other complex. Example - $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is a coordination isomer with $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$.

Ionization isomerism involves the exchange of ions inside and outside the coordination sphere. Examples of ionization isomerism are $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$.

Solvate (Hydrate) isomerism are the isomers which have the same composition but differ with respect to the number of solvent ligand molecules as well as the counterion in the crystal lattice. For example $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ is violet colored and its solvent $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2\text{H}_2\text{O}$ is gray-green.

Linkage isomers are the isomers which have the same molecular formula but differ in the linkage of ligand atom to the central metal atom. Example - $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$.

Chemical Formulas and Common Names

The chemical name of **bleaching powder** is Calcium hypochlorite or Calcium Oxychloride. Its chemical formula is $\text{Ca}(\text{OCl})_2$. Chlorine gas is used for the manufacturing of bleaching powder.

Ethanoic acid, also called glacial acetic acid, freezes in cold climates and is a carboxylic acid. It's formed from ethanol oxidation and is found in vinegar. Vinegar contains a 5% to 8% solution of acetic acid (CH_3COOH). Ascorbic acid is known as Vitamin C.

Detergent Powder's Chemical name is Sodium dodecyl benzene sulphonate and Chemical formula is $\text{C}_{18}\text{H}_{29}\text{NaO}_3\text{S}$.

Baking Soda (sodium hydrogen carbonate) - Chemical name - Sodium

bicarbonate, Chemical Formula - NaHCO_3 .

Milk of magnesia $\text{Mg}(\text{OH})_2$ also known as Magnesium dihydroxide or Magnesium hydroxide.

Plaster of Paris is a fine white powder of Calcium sulphate hemihydrate. The chemical formula of Plaster of Paris is $\text{CaSO}_4 \cdot \frac{1}{2} \cdot \text{H}_2\text{O}$.

Chemical formula of **Gypsum** is Calcium sulphate dihydrate ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). Gypsum is the product of the reaction when anhydrite undergoes hydration.

Slaked lime is also known as Calcium Hydroxide. The Chemical formula of Slaked lime is $\text{Ca}(\text{OH})_2$.

Asbestos is the common name for Calcium Magnesium silicate. The Chemical formula of Asbestos is $\text{Mg}_3(\text{Si}_2\text{O}_5)(\text{OH})_4$.

Dry ice is solid carbon dioxide. It undergoes sublimation and changes directly into gas.

Zinc oxide (ZnO) is commonly known as philosopher's wool. It is an inorganic compound, which is white in colour and has a powdery texture. Zinc oxide is insoluble in water.

Calcium Nitrate or Norgessalpeter (Norwegian saltpeter) is the inorganic compound with the formula $\text{Ca}(\text{NO}_3)_2$.

Calcium phosphate $\text{Ca}_3(\text{PO}_4)_2$ is a calcium salt of phosphoric acid.

Calcium chloride is a salt with the chemical formula CaCl_2 .

Calcium carbonate (CaCO_3) is abundant in rocks as calcite and aragonite and makes up eggshells, gastropod shells, and shellfish skeletons. Limestone, chalk, and marble are different forms of calcium carbonate.

Calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) is the primary mineral in human bones and teeth. Pearls consist mainly of calcium carbonate (CaCO_3), along with 2-4 % water and 10 % conchiolin (an organic binding agent).

Sodium nitrate (NaNO_3) is the chemical compound. This alkali metal nitrate salt is also known as Chile saltpeter.

Sodium hydroxide is sometimes called caustic soda or lye. It is a common

ingredient in cleaners and soaps.

Chloroform (CHCl_3 , trichloromethane) is a non-flammable, clear, colourless liquid that is denser than water and has a pleasant ether-like odour.

It was first prepared in 1831. It was first used in 1847 by James Young Simpson as an anesthesia.

Potassium Hydroxide (KOH) is an inorganic compound, commonly referred to as Caustic Potash. It is used in the thickening of food, in chip fabrication for semiconductors and as an electrolyte in some alkaline batteries.

The scientific name of **chromite** is Iron chromium oxide. Chromite is used as a refractory in the production of steel, copper, glass, and cement.

Silicon dioxide, also known as silica, is an oxide of silicon with the chemical formula SiO_2 , commonly found in nature as quartz. In many parts of the world, silica is the major constituent of sand.

Methane (CH_4) is also known as Marsh Gas because it is found in marshy land where anaerobic bacterial decomposition of vegetable matter produces methane gas.

Calamine is the common name of a natural admixture mineral of zinc carbonate (ZnCO_3 or smithsonite) and hydrous zinc silicate ($\text{Zn}_4\text{Si}_2\text{O}_7(\text{OH})_2 \cdot \text{H}_2\text{O}$ or hemimorphite).

Washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$), also called sodium carbonate, serves as a potent cleaning agent for laundry and effectively removes permanent water hardness.

Salt Petre is a common name for Potassium Nitrate (KNO_3). Common name of Potassium chloride is Sylvite. Common name of Sodium Chloride (NaCl) is common salt or table salt. Nitrous oxide is referred to as laughing gas (N_2O).

Sulfuric acid (H_2SO_4), also known as oil of vitriol, is a clear, odorless liquid made from sulfur, oxygen, and hydrogen. It dissolves easily in water and forms through very hot reactions.

Epsom salt is also known as magnesium sulfate (MgSO_4). It gets its name from the town of Epsom in Surrey, England, where it was originally discovered.

Quicklime is also known as Calcium oxide (CaO) is a widely used chemical compound. Boiling point of CaO is very high. It is a white, caustic, alkaline crystalline solid at room temperature. It is widely used in purification of sugar, in production of cement and bleaching powder.

3 : 1 mixture of NaOH (Sodium Hydroxide) and CaO (Calcium Oxide) is called soda lime. CaO makes NaOH less reactive.

CaO (Quicklime) + H₂O (Water) → Ca(OH)₂ (Calcium Hydroxide/Slaked lime).

Bauxite method - Industrial preparation - Mixing bauxite in concentrated H₂SO₄ and heating it, filter the solution and add appropriate amount of potassium sulphate to it. On cooling, crystals of alum are obtained.

The Chemical formula of **Borax** is Na₂[B₄O₅(OH)₄]·8H₂O. It is also known as sodium borate, sodium tetraborate, or disodium tetraborate. It can be used in the testing of cleaning agent, insecticide, water softening agent and diabetes mellitus.

Potash Alum or potassium aluminum sulfate has a molecular formula of K₂(SO₄)·Al₂(SO₄)₃·24H₂O or KAl(SO₄)₂·12H₂O. It can be used as Fire retardant, for tanning, Iron and Steel dissolving, Gourmet food, used for dyeing, pigmentation of the lake, blocking chemicals.

Na₂SO₄Al₂(SO₄)₃·24H₂O - (Soda alum/ Sodium aluminum sulfate).

Butylated Hydroxyanisole is a chemical compound that is used as a yeast defoaming agent, commonly found in butter, meat, cereals, snack foods, dehydrated potatoes and beer.

Acetone with chemical formula (CH₃)₂CO or C₃H₆O has 1 oxygen atom and 3 carbon atoms.

Latin Name of Chemical Elements - Iron (Fe) - Ferrum, Copper (Cu) - Cuprum, Zinc (Zn) - Zincum, Antimony (Sb) - Stibium, Gold (Au) - Aurum, Lead (Pb) - Plumbum, Mercury (Hg) - Hydrargyrum, Potassium (K) - Kalium, Silver (Ag) - Argentum, Sodium (Na) - Natrium, Tin (Sn) - Stannum, Tungsten (W) - Wolfram.

Chemical Formulas

Ethyl alcohol or Ethanol - C₂H₅OH or

CH₃CH₂OH, Ammonium Chloride - NH₄Cl, Phosphoric Acid - H₃PO₄, Sodium Peroxide - Na₂O₂, Hydrogen Sulphide - H₂S, Aluminium Oxide - Al₂O₃, Barium Peroxide - BaO₂, Barium sulfate - BaSO₄, Barium Oxide - BaO, Cyanogen - C₂N₂, Acetonitrile - CH₃CN, Sodium Aluminate - NaAlO₂, Magnesium Hydroxide - Mg(OH)₂, Ammonium Hydroxide - NH₄OH, Topaz - Al₂SiO₄(F,OH)₂, Opal - SiO₂·nH₂O, Agate - SiO₂, Benzene - C₆H₆, Methanol - CH₃OH, Magnesium chloride - MgCl₂, Ammonia - NH₃.

Polyatomic positive ions often have common names ending with the suffix -onium.

Negative ions that have a single atom are named by adding suffix -ide.

Halocarbonyl is the prefix of acyl halides compounds.

In alchemy, aqua fortis is the classical name of **Nitric acid** that is used chiefly in the manufacture of explosives and fertilizers, and in organic synthesis.

The preferred IUPAC name of Allyl bromide is 3-Bromopropene.

Resorcinol is the name of the 1,3-isomer (or meta-isomer) of benzenediol with the chemical formula C₆H₄(OH)₂.

Oxolane is a synonym of **Tetrahydrofuran** having the molecular formula (CH₂)₃CH₂O.

2-Propen-1-amine is the IUPAC name of **allylamine** which is used to make pharmaceuticals and other chemicals.

U is the symbol of the element whose IUPAC name is **Unnilennium**.

Acrolein is the name of the chemical compound having the molecular formula C₃H₄O which is mostly used for the production of acrylic acid and acrylate ester.

Carbaldehyde suffix is used when the -CHO group is attached to a carbon atom of a ring or ring system, or to a heteroatom.

Chemical Properties

Chemical property is a characteristic of a particular substance that can be observed in a chemical reaction.

Examples of chemical properties include flammability, toxicity, acidity, reactivity

(many types), and heat of combustion.

Chemists categorize chemical changes into three main classes: inorganic chemical changes, organic chemical changes and biochemical changes.

Chemical compound is a material formed by chemically bonding between two or more chemical elements. The type of bond keeping elements in a compound together may vary: covalent bonds and ionic bonds are two common types.

Ionic bond, also called electrovalent bond, is a type of linkage formed from the electrostatic attraction between oppositely charged ions in a chemical compound.

The ionic bond, when a positively charged ion forms a bond with a negatively charged ion and one atom passes electrons to another.

Covalent bond is formed by equal sharing of electrons from both the participating atoms.

Examples - CHCl₃ (Chloroform), N₂ (Nitrogen gas), and CH₄ (Methane Gas).

Important characteristics of a compound :-

Every molecule of a particular compound always has a fixed composition.

The properties of the resulting substance differ from the properties of its constituent elements.

Energy is either absorbed or evolved when a compound is formed.

Salt (Sodium Chloride, NaCl) is a chemical compound consisting of an ionic assembly of positively charged cations and negatively charged anions.

Properties :- Salt is a non-volatile solute which is responsible for elevation in the boiling point as well as depression in the freezing point. Due to this, the freezing point of water decreases and boiling point increases when common salt is mixed with water.

Sodium salts are the most soluble because sodium readily combines with water forming covalent bonds.

Potassium salt is used to regulate hypokalemia as a primary condition or secondary to other medical conditions. Potassium salts produce a characteristic

purple or violet colour in a flame.

Cobalt oxide gives blue colour to glass. A related material is cobalt (II, III) oxide, a black solid with the formula Co_3O_4 .

Copper sulphate (CuSO_4) solution is Blue in colour. When copper sulphate is heated it loses 5 molecules of water of crystallisation and becomes anhydrous copper sulphate which is white in colour. It is also called Blue vitriol.

Ferrous sulphate - It is also known as Green Vitriol.

Lithium treatment remains the "gold standard" of treatment for preventing recurrences in bipolar disorder (It is a type of mental illness, in which there is a rapid change in the mood of the person).

Cyanides are chemical compounds which are fast acting poisons with a chemical formula CN^- .

Ammonia molecule can bind another proton (H^+) and become an ammonium ion, NH_4^+ .

Beryllium (Be, 4) has high polarising power and forms covalent compounds. It is a bivalent and highly toxic element.

Use : Beryllium is used in alloys with copper or nickel to make gyroscopes, springs, electrical contacts, spot-welding electrodes and non-sparking tools.

Rubidium (Rb, 37) is a soft alkali metal that ignites in air and reacts violently with water.

In 2018, an experiment on the International Space Station cooled Rubidium atoms to near absolute zero, forming a Bose-Einstein Condensate in space.

Use: It is used in glasses, used in fireworks to give purple-red colour, fibre optics telecommunication systems and in night vision devices.

Francium (Fr; 87) is a highly radioactive alkali metal. It is the second rarest naturally occurring element.

Sodium is a highly electropositive element that readily gives up an electron in order to obtain a stable electronic configuration. It yields a yellow colour. Its compound Sodium carbonate is used in the manufacture of detergents, soaps, and paper.

Silver articles turn black after exposure

to air due to the reaction of sulphur present in air. Silver reacts with sulphur to form dark grey to black silver sulphide.

Use: Silver facilitates the transformation of carbon dioxide into carbon monoxide (CO), which is used to synthesise other useful chemicals.

When magnesium ribbon is burnt in air that is the combustion reaction of the magnesium metal. On burning in presence of oxygen magnesium gives white colour magnesium oxide.

Flames colour - An orange flame means the gas is not burning correctly. A blue flame means complete combustion of the gas. With complete combustion, LPG (Propane) burns with a blue flame.

Barium nitrate is used in signal flares and fireworks that burn in bright green colour.

Zinc (419.5 °C) has a higher melting point than Tin. The melting points of Tin, Lead and Silver are 232° C, 327.5 °C and 961.8 °C respectively.

At lower pressure or higher altitudes, the boiling point is lower. At sea level, pure water boils at 212 °F (100°C).

The Neon gas is used in Neon Lamp. A neon lamp (also neon glow lamp) is a miniature gas discharge lamp. When sufficient voltage is applied and sufficient current is supplied between the electrodes, the lamp produces an orange glow discharge.

The fluorescent lamp is perhaps the best known gas-discharge glow lamp. The fluorescent lamp consists of a glass tube filled with a mixture of argon and mercury vapour.

Hydrogen gas at low pressure with an electrode at each end inside a slim tube of the hydrogen discharge tube. If a high voltage (5000 volts) is applied in the tube, the tube lights up with a bright pink glow.

Phosphine gases are naturally formed via the anaerobic decay of phosphorus - containing organic matter.

Phosgene - Phosgene (COCl_2) is a colourless gas with a suffocating odour. Exposure to phosgene may cause irritation to the eyes, dry burning throat, vomiting, cough, foamy sputum, breathing difficulty and chest pain.

Phosphorus pentafluoride can be

prepared by direct combination of phosphorus and fluorine.

Carotenoids are pigments that give yellow, orange, and red fruits and vegetables their colour. Our body is able to convert some carotenoids into the vitamin (A).

Nitrogen is a odourless, colourless, tasteless and chemically unreactive gas which was invented by Daniel Rutherford in 1772 and was independently isolated at the same time by CW Scheele and H Cavendish.

Potato chips manufacturers usually flush bags of chips with gas such as nitrogen primarily to prevent the chips from getting oxidised.

At normal temperature and atmospheric pressure, **carbon dioxide** has a density of 1.87 kg/m³, which is 1.5 times heavier than air and exists as a liquid below the critical temperature of 31°C.

Calomel (Hg_2Cl_2) is the popular name for mercury (I) chloride which is used as a laboratory reagent and as a depolarizer in dry batteries.

Boiling point of **Chloroform** is 334 K and that of **Methane** is 111 K.

Melting point of Acetic acid is 290 K and that of Ethanol is 156 K.

Alkynes ($\text{C}_n\text{H}_{2n-2}$) have higher boiling points than alkanes and alkenes due to their stronger London dispersion forces resulting from their linear shape and increased electron density from the triple bond.

Oleic acid makes up 55-80% of olive oil, making it a good choice for most cooking methods.

Oxidation and Reduction

Redox reaction :- A reaction in which oxidation and reduction occur simultaneously.

Rusting of iron is a redox reaction. It is an electrochemical process. Here oxygen is an oxidizing agent as it gains electrons. Iron is a reducing agent as it loses electrons. Rust needs three components to occur - Iron, oxygen and moisture.

Oxidation is a process which involves the addition of oxygen or any electronegative element or the removal of hydrogen or any electropositive element. Oxidation

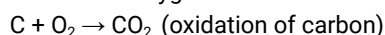
reaction releases energy.

Antoine Lavoisier first used the term oxidation to describe a substance's reaction with oxygen.

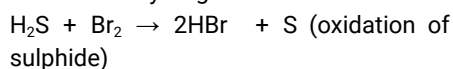
Oxidising agents - Ozone, Bleach, peroxide, O_2 , S, Cl_2 , Br_2 and H_2O_2 .

Oxidation reactions involve -

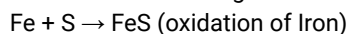
Addition of oxygen



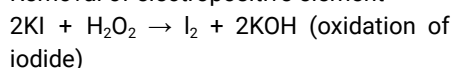
Removal of hydrogen



Addition of electronegative element



Removal of electropositive element



Oxidation Number - Charge found on the atom of a molecule or ion. It can never be higher than the number of valence electrons found in an element.

Effects of oxidation reactions in everyday life -

Corrosion- It is the slow process in which the surfaces of metallic objects are coated with oxides, hydroxide, carbonate or sulphide of the metal.

It causes the destruction of metal by chemical or electrochemical reactions with the environment.

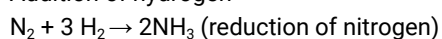
Examples - Development of green coating on copper, tarnishing of silver, iron rusting and wood-rotting are good examples of gradual oxidation or slow oxidation.

Reduction is a process which involves the addition of hydrogen or any electropositive element or the removal of oxygen or any electronegative element. This reaction stores energy.

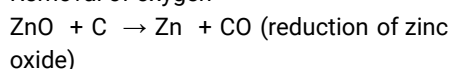
Reducing agents - Metal, H_2 , $HgCl_2$ and C.

Reduction reactions involve -

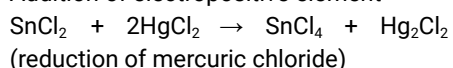
Addition of hydrogen



Removal of oxygen



Addition of electropositive element



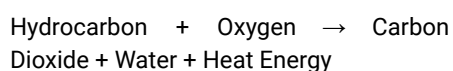
Removal of electronegative element
 $2FeCl_3 + H_2 \rightarrow 2FeCl_2 + 2HCl$ (reduction of ferric chloride)

Combustion - It is a chemical process or a reaction between fuel (hydrocarbon) and oxygen. When fuel and oxygen react it releases the heat and light energy. Heat and light energy then result in the flame.

Oxygen is the necessary constituent for any combustion reaction.

Examples - Burning of wood or coal to heat your home, burning of petrol or diesel to run your car, for the production of energy in thermal power plants and fireworks.

The general exothermic reaction of combustion can be expressed as



Uses - Car motors, rocket engines and many other machineries.

Carbon Monoxide (CO) is a colorless and odorless gas, which is produced by the combustion of materials that contain carbon.

In any **electrochemical cell** (electrolytic or galvanic) the electrode at which reduction occurs is called the cathode.

In any electrochemical cell the anode is the electrode at which oxidation occurs.

States of Matter

Matter is anything that takes up space and can be weighed. Matter has volume and mass. There are five states of matter.

Solids: They have a fixed shape and a fixed volume, they cannot be easily compressed, they have high density, force of attraction between the particles is very strong, the space between the particles of solids is smaller compared to liquids and gasses.

Liquids : They have a fixed volume but no fixed shape, they can be compressed. Large pressure is required to compress them, they have lesser densities than solids, intermolecular forces of attraction are weaker than solids, they have considerable space between the particles.

Gasses : They have neither a fixed shape nor a fixed volume, they can be

compressed easily, they have the least density among the three, intermolecular forces of attraction are weakest, and the space between the gas particles is large.

Plasma is called the fourth state of matter along with solid, liquid and gas. It is made up of ionised gas, which consists of electrons, positive ions and neutral particles.

Bose-Einstein condensates - It is the fifth state of matter. It is made after cooling a gas of very low density at a very low temperature.

Process for changing state of matter

Melting or fusion is a physical process that results in the phase transition of a substance from a solid to a liquid. The point on which this process occurs is called Melting point.

The melting point of Ice at sea level at normal atmospheric pressure and at 45-degree latitude is 273.16° K (0°C).

At sea level, pure water boils at 212 °F (100°C). At the lower atmospheric pressure on the top of Mount Everest, pure water boils at about 154 °F (68°C).

Evaporation is the process by which a liquid turns into a gas. It is the process by which the common salt is obtained from sea water. When the sea water is heated or evaporated, the solid salt is left behind.

Freezing is a process in which the element turns into solid from liquid state. The point where this process happens is called Freezing Point. Water freezes at 0°C.

Sublimation is a physico-chemical process in which a substance changes directly from its solid state to a gas. The state of matter does not change to an intermediate liquid state during this whole process.

The vaporization of camphor directly from the solid state is an example of sublimation. Due to the process of sublimation the dry ice (Solid CO_2) directly converted to gas from solid skipping the middle phase of converting into liquid.

The **Mpemba effect** states that hot water may freeze faster than cold water, under similar conditions.

The **dew point** is the temperature to

which air must be cooled to become saturated with water vapor, assuming constant air pressure and water content.

Triple point :- It is defined as the condition of temperature and pressure under which the gaseous, liquid and solid of the substance can exist in equilibrium.

The sublimation, vapourisation and fusion curves for a pure substance meet at a triple point. The triple point of water is 273.16 K, 0.01° C, or 32.018° F.

Distillation is the process of separating the components or substances from a liquid mixture by using vapourisation and condensation.

The process of **distillation** begins with heating a liquid to boiling point. The liquid evaporates, forming a vapor. The vapor is then cooled, usually by passing it through pipes or tubes at a lower temperature.

Decantation is the process of separating liquids from solids and other immiscible (non-mixing) liquid substances.

For Example - Oil floats on water. Oil can be drained off of water after letting an oil and water mixture settle for a while.

Crystallization is the process by which solid forms, where the atoms or molecules are highly organized into a structure known as a crystal.

Condensation - It is the change of the state of matter from the gas phase into the liquid phase. It is the reverse of vaporization. Condensation is crucial to the water cycle because it is responsible for the formation of clouds.

Precipitation is the process behind the condensation of water vapour in the atmosphere that falls down under the influence of gravity, for example drizzle, rain.

Vaporization :- The substance changes from a liquid to gas. The latent heat of vapourization of water at 100 degree celsius is 540 cal/g.

Deposition is the transition of a substance directly from the gas to the solid state on cooling, without passing through the liquid state. An example of deposition is frost formation.

On the basis of the nature of order present in the arrangement of their constituent particles, Solids are

classified into Amorphous and Crystalline.

Crystalline solids have well defined edges and faces whereas amorphous solids have irregular or curved surfaces. A crystalline solid changes abruptly from solid to liquid when heated.

Humidity is the concentration of water vapour present in the air. Humidity is inversely proportional to the rate of evaporation decreasing humidity increases rate of evaporation.

The **Ignition temperature** can be defined as the minimum temperature at which the substance catches fire for spontaneous combustion. It is also known as the Ignition point.

Ambient temperature is the air temperature of any object or environment where equipment is stored. It is also known as room temperature.

Ideal Gas Laws and Molar Mass

The gas laws were developed towards the end of the 18th century by numerous scientists (after whom, the individual laws are named).

Boyle's Law (1662) - gives relation between the pressure and the volume of a gas. Basically, the volume of a gas is inversely proportional to the pressure of a gas at a constant temperature.

We can express the same in an equational form $P \propto 1/V$

Charles's Law (1780) - It is also known as the law of volumes. When the pressure on a sample of a dry gas is held constant, the Kelvin temperature and the volume will be in direct proportion.

This relationship of direct proportion can be written as: $V \propto T$ or $V/T = k$ or $V = kT$ where, V is the volume of the gas, T is the temperature of the gas (measured in kelvins), and k is a non-zero constant.

Gay-Lussac's Law (1808) - It is a gas law which states that the pressure exerted by a gas (of a given mass and kept at a constant volume) varies directly with the absolute temperature of the gas.

The mathematical expression of Gay-Lussac's law can be written as follows: $P \propto T$; $P/T = k$ where P is the pressure exerted by the

gas, T is the absolute temperature of the gas, k is a constant.

Avogadro's Law (1811) - It is a gas law which states that the total number of atoms/molecules of a gas (i.e. the amount of gaseous substance) is directly proportional to the volume occupied by the gas at constant temperature and pressure.

Avogadro's law is closely related to the ideal gas equation since it links temperature, pressure, volume and amount of substance for a given gas. Avogadro constant - $6.02214076 \times 10^{23} \text{ mol}^{-1}$.

At constant pressure and temperature, Avogadro's law can be expressed via the following formula: $V \propto n$ or $V/n = k$ where V is the volume of the gas, n denotes the amount of gaseous substance (often expressed in moles), k is a constant.

Ideal gas Law - The ideal gas law, also known as the general gas equation, is an equation of the state of a hypothetical ideal gas. It is a combined form of Boyle's, Charles's and Avogadro's law.

The empirical form of ideal gas law is given by :

$$PV = nRT$$

$$PV = NkT$$

where, P is the pressure, V is the volume, n is the amount of substance, k is Boltzmann constant, R is the ideal gas constant, T is the absolute temperature of the gas.

One mole of an ideal gas occupies a volume of 22.4 liter at 273 K and 1 atm pressure.

$$\text{Boltzmann constant } k = 1.38 \times 10^{-23} \text{ J/K.} \\ \text{J/K} = \text{m}^2 \cdot \text{kg} / (\text{s}^2 \cdot \text{K}^{-1})$$

Other important Laws -

Raoult's law states that for a solution of volatile liquids, the partial vapor pressure of each component of the solution is directly proportional to its mole fraction present in solution.

$$P_{\text{solution}} = X_{\text{solvent}} P_{\text{solvent}}^0$$

where,

P_{solution} = vapor pressure of the solution,

X_{solvent} = mole fraction of the solvent,

P_{solvent}^0 = vapor pressure of the pure solvent.

Dalton's law says the total pressure of a mixture of gasses is equal to the sum of

partial pressure of individual gasses.

Mathematically, it can be expressed as $P_{\text{total}} = P_1 + P_2 + P_3$ ———, where P_{total} is the total pressure and $P_1 + P_2 + P_3$ ——— are the partial pressure of the individual gases.

Henry's law is one of the gas laws that states that the amount of dissolved gas in a liquid is proportional to its partial pressure above the liquid.

Mathematically, it can be expressed as $C = kP$, where C is the concentration of the dissolved gas, k is Henry's constant (which depends on the specific gas and solvent), and P is the partial pressure of the gas.

So in order to increase the solubility of CO_2 in soft drinks, the bottle is sealed under high pressure.

Molar mass (M) of a chemical compound is defined as the ratio between the mass and the amount of substance (measured in moles) of any sample of said compound.

Molar mass of some elements -

Hydrogen (1.008), Helium (4.0026), Lithium(6.94), Beryllium (9.0122), Boron (10.81), Carbon (12.011), Nitrogen (14.007), Oxygen (15.999), Fluorine (18.998), Neon (20.180), Sodium (22.990), Magnesium (24.305), Aluminum (26.982), Silicon (28.085), Phosphorus (30.974), Sulfur (32.06), Chlorine (35.45), Argon (39.948).

The Molar mass of nitric acid is 63u.

Molar mass of **Sulphurous acid** is 82.0791 g/mol.

Carbon monoxide contains approximately 1.33g of oxygen for each 1.00 g of carbon. Every sample of carbon monoxide contains 16.0 g of oxygen for every 12.0 g of carbon.

Carbon Monoxide is used in the purification of nickel. It is used as a reducing agent. It is also used in water gas shift reactions to produce hydrogen.

The ratio by mass of nitrogen and hydrogen in ammonia is 14 : 3.

Temperature Conversion -

Fahrenheit to Centigrade (Celsius) - $^{\circ}\text{C} = (F - 32) \times 5/9$

Celsius to Fahrenheit - $F = (^{\circ}\text{C} \times 9/5) + 32$

Kelvin to celsius - $^{\circ}\text{C} = (K - 273)$

Kelvin to Fahrenheit - $F = (K - 273) \times 9/5 + 32$

Celsius to Kelvin - $K = (^{\circ}\text{C} + 273)$

In the 1860s, Thomas Andrews undertook a thorough investigation of the gas laws expressing the relationships of pressure, temperature and volume in carbon dioxide.

Robert Boyle described in 1662 that when a gas was pumped into an enclosed space, it would shrink to fit in that space, but the pressure exerted by the gas on the container would increase.

Helium is used as a cooling medium for the Large Hadron Collider (LHC) and the superconducting magnets in MRI scanners and NMR spectrometers.

The lowest temperature at which a substance catches fire is called its **ignition temperature**.

Solutions

Solution is a homogeneous mixture of two or more substances. It consists of a solute and a solvent. The solute dissolved in the solvent. For example, in a saline solution, salt is the solute dissolved in water as the solvent.

In a **homogeneous solution**, the solute gets dissolved completely in the solvent. For example a cup of coffee, perfume, cough syrup, a solution of salt or sugar in water, etc.

Heterogeneous solutions are solutions with non-uniform composition and properties throughout the solution. A solution of oil and water, water and chalk powder and solution of water and sand, etc.

Types of Solutions based on Physical State

Solid solutions

In these solutions, the solvent is a solid or a substance present in its solid state and solute can be solid, liquid or gas.

Solid in solid solutions - The physical state of both solute and solvent is solid. Example - Copper dissolved in gold.

Liquid in solid - The physical state of solute is liquid and that of solvent is solid. Example - Amalgam of mercury

with sodium.

Gas in solid - The physical state of solute is gas and that of solvent is solid. Example - Solution of hydrogen in palladium.

Liquid solutions

In these solutions, the solvent is a liquid and solute can be solid, liquid or gas.

Solid in liquid - The physical state of solute is solid and that of solvent is liquid. Example - Glucose dissolved in water.

Liquid in liquid - The physical state of both solute and solvent is liquid. Example - Milk in water.

Gas in liquid - The physical state of solute is gas and that of solvent is liquid. Example - Oxygen dissolved in water.

Gaseous solutions

In these solutions, the solvent is a gas and solute can be solid, liquid or gas. Solid in gas - The physical state of solute is solid and that of solvent is gas. Example - Camphor in nitrogen gas.

Liquid in gas - The physical state of solute is liquid and that of solvent is gas. Example - Chloroform mixed with nitrogen gas.

Gas in gas - The physical state of both solute and solvent is gas. Example - Mixture of various gases.

In a **vinegar solution** for 20 - 24 hours, the eggshell will absorb the acid and break down, or dissolve. The calcium carbonate will become carbon dioxide gas, which will go into the air.

Colloid is a mixture in which one substance of microscopically dispersed insoluble particles is suspended throughout another substance.

The types of colloids include sol, emulsion, foam, and aerosol.

Sol is a colloidal suspension with solid particles in a liquid. Examples are blood, paint, mud, etc.

Emulsion is between two liquids. Examples are Milk, cream, mayonnaise, etc.

Foam is formed when many gas particles are trapped in a liquid or solid. Examples are Whipped cream, soap Lather, sponge,

etc.

Aerosol contains small particles of liquid or solid dispersed in a gas. Examples are Cloud, mist, fog.

Different factors

Suspension is defined as a heterogeneous mixture in which the solid particles are spread throughout the liquid without dissolving in it.

The particles of a suspension do not pass through a filter paper, so they can be separated by filtration. It scatters a beam of light passing through it because of its large particle size.

Winnowing - It is the process of separating heavier and lighter components of a mixture by wind or by blowing air.

Filtration is a process used to separate solids from liquids or gases using a filter medium that allows the fluid to pass through but not the solid.

Decantation is a process for the separation of mixtures of immiscible liquids or of a liquid and a solid mixture such as a suspension.

Distillation is a process involving the conversion of a liquid into vapour that is subsequently condensed back to liquid.

Sublimation is the transition of a substance directly from the solid phase to the gas phase without passing through the intermediate liquid phase. Naphthalene balls disappear after some days, solid CO_2 , burning of camphor are some examples of sublimation.

In **Evaporation**, water changes from a liquid to a gas or vapour.

In **Boiling**, a liquid turns into a vapour when it is heated and reaches its boiling point.

Transpiration is a biological process by which the aerial parts of the plants lose water in the form of water vapour.

Chromatography is a physical method of separation in which the constituents to be separated are distributed between two phases, one of which is stationary while the other moves in a definite direction.

Column chromatography is a technique that is used to separate a single

chemical compound from a mixture dissolved in a fluid. Column chromatography can be used to separate the impurities in milk.

Autoclaves provide a physical method for disinfection and sterilization.

Steam distillation is a separation process that consists of distilling water together with other volatile and non-volatile components. Substances that are steam volatile can be separated by steam distillation.

Ion Exchange and Distillation techniques can be used for reducing the total dissolved solids (TDS) in water. The TDS (Total Dissolved Solids) level between 50-150 is considered as the most suitable and acceptable.

Sedimentation is the process of allowing particles in suspension in water to settle out of the suspension under the effect of gravity.

The particles that settle out from the suspension become sediment, and in water treatment is known as **sludge**.

Solubility of gases in liquids increases with decrease in temperature and it decreases with the increase in temperature.

Adding heat to the solution provides thermal energy that overcomes the attractive forces between the gas and the solvent molecules, thereby decreasing the solubility of the gas and when we decrease the temperature, solubility increases.

Calcium and magnesium combination is present in higher amounts in hard water. When water percolates through limestone or gypsum which is high in calcium and magnesium content then Hard water is formed.

Abatement is reducing the degree or intensity of or eliminating pollution.

Absorption is a physical or chemical phenomenon or a process in which atoms, molecules or ions enter some bulk phase – liquid or solid material.

Aeration is the process by which air is circulated through, mixed with or dissolved in a liquid.

Chalk powder (CaCO_3) does not dissolve in water. It is a soft, white, porous

sedimentary rock, a form of limestone composed of the mineral calcite.

Seawater is a solution of pure water and dissolved ionic substances. Water is a very good solvent.

Periodic Table

Periodic Table of Elements

		s Block																								p Block									
Group Period	1																						18												
	1	2																					13	14	15	16	17	18							
	1	2																					3	4	5	6	7	8							
	1	2																					5	6	7	8	9	10							
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	151	152																					125	126	127	128	129	130							
	169	170																					133	134	135	136	137	138							
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	201	202																					141	142	143	144	145	146							
	217	218																					145	146	147	148	149	150							
	231	232																					149	150	151	152	153	154							
	247	248																					153	154	155	156	157	158							
	261	262																					157	158	159	160	161	162							
	277	278																					161	162	163	164	165	166							
	293	294																					165	166	167	168	169	170							
	309	310																					169	170	171	172	173	174							
	325	326																					173	174	175	176	177	178							
	341	342																					177	178	179	180	181	182							
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	469	470																					209	210	211	212	213	214							
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	501	502																					217	218	219	220	221	222							
	517	518																					221	222	223	224	225	226							
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	549	550																					229	230	231	232	233	234							
	565	566																					233	234	235	236	237	238							
	581	582																					237	238	239	240	241	242							
	597	598																					241	242	243	244	245	246							
	613	614																					245	246	247	248	249	250							
	629	630																					249	250	251	252	253	254							
	645	646																					253	254	255	256	257	258							
	661	662																					257	258	259	260	261	262							
	677	678																					261	262	263	264	265	266							
	693	694																					265	266	267	268	269	270							
	709	710																					269	270	271	272	273	274							
	725	726																					273	274	275	276	277	278							
	741	742																					277	278	279	280	281	282							
	757	758																					281	282	283	284	285	286							
	773	774																					285	286	287	288	289	290							
	789	790																					289	290	291	292	293	294							
	805	806																					293	294	295	296	297	298							
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	853	854																					305	306	307	308	309	310							
	869	870																					309	310	311	312	313	314							
	885	886																					313	314	315	316	317	318							
	901	902																					317	318	319	320	321	322							
	917	918																					321	322	323	324	325	326							
	933	934																					325	326	327	328	329	330							
	949	950																					329	330	331	332	333	334							
	965	966																					333	334	335	336	337	338							
	981	982																					337	338	339	340	341	342							
	997	998																					341	342	343	344	345	346							
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	1285	1286																					413	414	415	416	417	418							
	1301	1302																					417	418	419	420	421	422							
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	1861	1862																					557	558	559	560	561	562							
	1877	1878																					561	562	563	564	565	566							
	1893	1894																					565	566	567	568	569	570							
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	1957	1958																					581	582	583	584	585	586							
	1973	1974																					585	586	587	588	589	590							
	1989	1990																					589	590	591	592	593	594							
	2005	2006																					593	594	595	596	597	598							
	2021	2022																					597	598	599	600	601	602							
	2037	2038																					601	602	603	604	605	606							
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	2069	2070																					609	610	611	612	613	614							
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	2133	2134																					625	626	627	628	629	630							
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The **Law of Triads** was the first attempt to classify elements based on their material and chemical properties. It was proposed by Johann Dobereiner, a German chemist, in 1829.

Dobereiner made the first observation on platinum (Pt) as a catalyst and discovered similar triads of elements which led to the development of the Periodic Table of elements.

Dobereiner triads are defined as sets of three chemically same elements, the atomic weight of one, which is nearly equal to the mean of the atomic weights of the other two elements.

Such triads, including the Lithium (Li) - Sodium (Na) - Potassium (K), Calcium (Ca) - Strontium (Sr) - Barium (Ba), and Chlorine (Cl) - Bromine (Br) - Iodine (I).

The triad method worked only for a few elements so it was dismissed.

Dobereiner is also known for his discovery of Furfural ($C_5H_4O_2$), and for the invention of lighter.

In 1865 the English chemist J.A.R. Newlands given the **Law of Octaves**. It states that when chemical elements are arranged according to increasing atomic masses, those with similar physical and chemical properties occur after each interval of seven elements.

That is, the elements presented by Newlands start with the lowest molecular mass (hydrogen) and end with thorium which is the 56th element.

He compared this to the Octaves found in music. Therefore, he called it the Law of Octaves. Its examples are Lithium and Sodium, Beryllium and Magnesium.

This law failed because it was applicable only upto Calcium, as after Calcium every eighth element did not possess properties similar to that of the first.

The **Periodic law** was developed independently by Dmitri Mendeleev and Lothar Meyer in 1869. They both arranged the elements by their atomic mass and proposed that certain properties periodically reoccur.

2019 was designated by UNESCO as the International Year of the Periodic Table (IYPT) to mark the 150th anniversary of the Mendeleev Periodic Table.

German chemist, Lothar Meyer was the first to show a graphical representation of the periodicity of an atomic volume plotted against atomic weight.

The First Periodic Table to become generally accepted was that of the Russian chemist Dmitri Mendeleev. He formulated the periodic table as a dependence of chemical properties on atomic mass.

The number of elements known at the time of Russian chemist Dmitri Ivanovich Mendeleev was 63.

Mendeleev divided the table into Eight columns (vertical) and Seven rows (horizontal).

Mendeleev boldly predicted the existence of some elements that had not been discovered at that time. Mendeleev named them by prefixing a Sanskrit numeral, Eka.

Gallium was discovered later and replaced Eka aluminium. Similarly Scandium replaced Eka Boron and Germanium replaced Eka Silicon. The atomic mass predicted by Mendeleev: Mass of Eka aluminium was 68, mass of Eka Boron was 44, mass of Eka silicon was 72, mass of Eka magnesium was 100.

Mendeleev's periodic table failed because he was unable to tell the position of a Hydrogen atom in the periodic table.

In the 1920s, Charles Janet created the Left-Step Periodic Table (LSPT) of the chemical elements, basing the positions of the elements on patterns in the arrangement of electrons around the atomic nucleus.

Henry Moseley carried out a systematic series of experiments which showed that the frequencies of the X-rays emitted from an elemental target under bombardment by cathode rays were characteristic of that element and could be used to identify the charge on its atomic nucleus.

This led to a reorganization of the periodic table, with chemical elements now arranged on the basis of atomic

number (Z) rather than atomic weight (A).

The **Modern Periodic Table** is the present form of the periodic table which was given by Henry Mosley in 1913.

It consists of 18 vertical columns and 7 horizontal rows. These columns are known as groups and rows are known as periods.

The elements are arranged from left to right and top to bottom in order of increasing atomic number.

In the Modern Periodic Table, a zig-zag line separates metals from non-metals. The borderline elements Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium and Polonium are intermediate in properties and are called Metalloids or Semimetals.

Among all the elements in the periodic table, there are only 11 gases present under normal atmospheric conditions. These gases are Hydrogen, Nitrogen, Oxygen, Fluorine, Chlorine, Helium, Neon, Argon, Krypton, Xenon and Radon.

There are 14 elements in the Modern Periodic table which are represented by a single letter. Out of which 10 elements are in the first 30 elements of the modern periodic table and the remaining four are Yttrium (Y), Iodine (I), Tungsten (W) and Uranium (U).

Metals are generally found on the left side and nonmetals are on the right side in the Modern periodic table.

The total number of elements present in the Modern periodic table is 118. The number of non-metals is 22. The number of metalloids is 7 and the number of metals is 93.

Most nonmetals are gases. These include elements such as Hydrogen, Helium, Nitrogen, Oxygen, Fluorine, Neon, Chlorine, Argon, Krypton, Xenon and Radon.

The non-metals Carbon, Phosphorus, Sulfur, Selenium, and Iodine are solids. The element Astatine (At) is the radioactive element.

Atomic radius or atomic size is the distance from the nucleus of an atom to the outermost orbital of its electron.

In general, the atomic radius decreases as we move from left to right in a period and it increases when we go down a group.

The atomic radius of Lithium is greater

than beryllium, nitrogen and carbon.

Atomic radius of Helium is the smallest and the atomic radius of Francium is the largest.

Francium is the 87th element in the periodic table and has no biological role because it has half life of only 22 minutes.

Nuclear charge experienced by valence electrons decreases down the group because the outermost electrons are farther away from the nucleus. Therefore, these electrons can be lost easily.

Metallic character decreases across a period moving from left to right and increases down the group.

Electronegativity is a chemical property that describes the tendency of an atom or a functional group to attract electrons toward itself.

The electronegativity of an atom is affected by both its atomic number and the distance that its valence electrons reside from the charged nuclei.

Electronegativity increases from bottom to top in groups, and increases from left to right across periods.

Fluorine is the most electronegative element, while Francium is one of the least electronegative.

In the periodic table, highly electronegative halogens and the highly electropositive alkali metals are separated by the Noble gases.

The **ionisation energy** of an atom is the amount of energy that is required to remove an electron from a mole of atoms in the gas phase.

The ionisation energy decreases from top to bottom in groups, and increases from left to right across a period. Thus, helium has the largest ionization energy, while Caesium has the lowest.

Atomic Number is the number of an element that is determined by the number of protons in it. The atomic number of Neon is 10 and the atomic number of sulphur is 16. The atomic number of Oganesson (Og) is 118.

Atomic mass of the atom of an element is defined as the average mass of the atom, as compared to 1/12 th the mass of one carbon-12 atom. Its unit is amu (Atomic Mass unit).

Alkali metals

Lithium (Li), Sodium (Na), Potassium (K),

Rubidium (Rb), Caesium (Cs), Francium (Fr). They have one electron in their outermost shell.

When Alkali metals react with water they form metal hydroxide and hydrogen gas.

Lithium was discovered by Swedish chemist Johan August Arfwedson in 1817. Lithium is soft, silvery-white, lightest solid metal. Its atomic number is 3. It is used in production of batteries of vehicles, laptops, mobile phones.

A diagonal relationship is found between the pair of Lithium (Li) and Magnesium (Mg).

Sodium ($^{23}\text{Na}_{11}$) is a soft, silvery-white, highly reactive metal.

Sodium is the soft metal in group 1 of the periodic table which tarnishes within a few seconds of exploration to air.

Pure sodium was first isolated by Sir Humphry Davy in 1807 through the electrolysis of Caustic soda (NaOH). It is the main content of Common salt (NaCl).

Alkaline earth metals

Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium (Ba), Radium (Ra). They have two electrons in their outermost shells.

Example - The Beryllium Atom has two electrons in its outermost shell.

Beryllium is a silvery-white metal that is used in an alloy with copper or nickel to make gyroscopes, springs, electrical contacts, spot-welding electrodes, and non-sparking equipment.

Magnesium (Mg) is a gray-white lightweight metal. Its atomic number is 12. It is an element of the IIA group and third period.

Strontium (Sr) is a soft silver-white yellowish metallic element that is highly chemically reactive. Its atomic number is 38 and belongs to the IIA group and fifth period.

Radium (Ra) is the heaviest known alkaline earth metal and is the only radioactive member of its group. Its atomic number is 88 and atomic mass is 226.

Noble gases

Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), Radon (Rn).

Xenon (Xe) is the last element of the fifth period.

Element (Neon) with the electronic configuration $[\text{He}] 2s^2 2p^6$ will be put in the 18th group.

The Noble gases or Inert gases, are the elements that belong to group 18 or zero group of the Modern periodic table. Van der Waals forces and London dispersion forces exist between them.

These are non-reactive, monoatomic elements with extremely low boiling points. Generally, due to completely filled shells they do not chemically react.

Radon is a naturally occurring radioactive noble Gas. The atomic mass of Radon is 222 amu making it the heaviest known gas. It is 220 times heavier than the lightest gas, Hydrogen.

Helium (He) is a chemical element with atomic number 2. It is a colorless, odorless, tasteless, non-toxic, inert gas and the first in the Noble gas group in the periodic table. Its boiling point is the lowest among all the elements.

Noble gases are used in lamps, lasers, deep sea diving tanks etc.

Transition metals

Scandium (Sc), Titanium (Ti), Vanadium (V), Chromium (Cr), Manganese (Mn), Iron (Fe), Cobalt (Co), Nickel (Ni), Copper (Cu), Zinc (Zn), Zirconium (Zr), Niobium (Nb), Molybdenum (Mo), Technetium (Tc), Ruthenium (Ru), Rhodium (Rh), Palladium (Pd), Silver (Ag), Cadmium (Cd), Hafnium (Hf), Tantalum (Ta), Tungsten (W), Rhenium (Re), Osmium (Os), Iridium (Ir), Platinum (Pt), Gold (Au), Mercury (Hg), Rutherfordium (Rf), Dubnium (Db), Seaborgium (Sg), Bohrium (Bh), Hassium (Hs).

Bohrium (Bh_{107}) is named after physicist Niels Bohr, one of the founders of quantum theory.

Chromium (Cr) is a steely-grey, lustrous, hard, and brittle transition metal. Its atomic number is 24 and atomic mass is 52.

Technetium, a radioactive element with atomic number 43 in the modern periodic table, is unstable, and the half-lives of all its isotopes are relatively short, from 4.2 million years to 5.0 s.

Seaborgium (Sg) with the atomic number 106, is a synthetic chemical element created by bombardment of californium-249 with oxygen-18. Element 106 was provisionally named seaborgium by American researchers in

honour of Nobel Laureate Glenn T Seaborg.

Copper (Cu) is an element which is a soft, ductile metal with very high thermal and electrical conductivity. Its atomic number is 29 and atomic mass is 63.5.

Zirconium (Zr) is a very strong, malleable, ductile, lustrous silver-gray metal. Its atomic number is 40 and atomic mass is 91.2. It is used in the production of abrasives, bullet proof jackets.

Molybdenum (Mo) is a silvery-white metal and highly resistant to corrosion. Its atomic number is 42 and atomic mass is 95.9. It is mostly used in the manufacturing of various types of steels.

Silver (Ag) is a soft, white, lustrous transition metal. Its atomic number is 47 and atomic mass is 107.9. It has the highest thermal conductivity of any element and the highest light reflectance.

Mercury (Hg) is an element which is also known as quicksilver and was formerly named hydrargyrum taken from the Greek words hydor (water) and argyros (silver). Its atomic number is 80 and atomic mass is 200.6 u. It is the only metal which is liquid at room temperature.

The Densest naturally occurring element is Osmium (atomic number - 76) with density of 22.59 g/cm^3 and the density of Iridium (atomic number - 77) is 22.56 g/cm^3 .

Rutherfordium (Rf, Atomic number - 104) is named in honor of New Zealand Chemist Ernest Rutherford. It is a radioactive transuranium element. It is created by bombarding Californium (Cf) - 249 with Carbon (C) - 12.

Post transition metals

Aluminium (Al), Gallium (Ga), Indium (In), Tin (Sn), Thallium (Tl), Lead (Pb), Bismuth (Bi), Polonium (Po), Astatine (At).

Gallium (Ga) is a soft, silvery-white metal. Its atomic number is 31 and atomic mass is 69.7. It becomes liquid when heated slightly. It has important uses in doping of semiconductors, blu-ray technology.

Gallium (Ga) has a high boiling point, making it ideal for recording temperatures that would vaporise a thermometer.

Lead (Pb) is a bluish-white lustrous metal. Its atomic number is 82 and

atomic mass is 207.2. It is used in the production of the batteries.

Metalloids

Boron (B), Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te).

Boron (B) is the only non-metal of the group 13 of the Modern Periodic Table of the elements. Its atomic number is 5 and atomic mass is 10.8. It is used in production of Eye drops, rocket fuel.

Germanium (Ge) is a lustrous, hard-brittle, grayish-white, metalloid element. Its atomic number is 32 and atomic mass is 72.6. It is used in the semiconductor industry.

Arsenic (As) is a metalloid that occurs in many minerals, usually in combination with Sulfur and metals, but also found as a pure elemental crystal. Its atomic number is 33 and atomic mass is 74.9. It is highly toxic and used as a wood preservative.

Reactive non-metals

Hydrogen (H), Carbon (C), Nitrogen (N), Oxygen (O), Fluorine (F), Phosphorus (P), Sulfur (S), Chlorine (Cl), Selenium (Se), Bromine (Br), Iodine (I).

Hydrogen (H) is a colourless, odorless, tasteless, flammable gaseous Element. It is the lightest gas in the universe. Its atomic number is 1. Hydrogen resembles the properties of group 1 and group 17 elements.

It is an element of the first group and first period of the modern periodic table. It is not an Inert gas. It has three known Isotopes. The mass numbers of Isotopes are 1, 2, and 3.

The least dense naturally occurring element is hydrogen with density of 0.089 g/cm^3 .

Carbon (C) is a nonmetallic and tetravalent element. Its atomic number is 6 and atomic mass is 12. Diamond, graphite or graphene and fullerene (C-60) are the allotropes of carbon.

In diamond, each carbon atom is bonded to four other carbon atoms forming a rigid three-dimensional structure.

In graphite, each carbon atom is bonded to three other carbon atoms in the same plane giving a hexagonal array.

In fullerene (C-60), carbon atoms are arranged in the shape of a football.

Daniel Rutherford discovered Nitrogen in 1772. The atomic number of Nitrogen is 7 and atomic mass is 14.

Chlorine (Cl) is a yellow-green gas at room temperature. Its atomic number is 17. It is an extremely reactive element and a strong oxidising agent.

Lanthanides

Lanthanides are also known as rare earth elements.

Lanthanum (La), Cerium (Ce), Praseodymium (Pr), Neodymium (Nd), Promethium (Pm), Samarium (Sm), Europium (Eu), Gadolinium (Gd), Terbium (Tb), Dysprosium (Dy), Holmium (Ho), Erbium (Er), Thulium (Tm), Ytterbium (Yb), Lutetium (Lu).

Cerium (Ce) is the most abundant of the rare earth elements.

Praseodymium is a soft, malleable and silvery-yellow metal with atomic number 59 in the lanthanide series.

Actinides

The 15 elements that are part of the Actinides series are : Actinium (Ac), Thorium (Th), Protactinium (Pa), Uranium (U), Neptunium (Np), Plutonium (Pu), Americium (Am), Curium (Cm), Berkelium (Bk), Californium (Cf), Einsteinium (Es), Fermium (Fm), Mendelevium (Md), Nobelium (No), and Lawrencium (Lr).

The atomic number of Actinium is 89. Actinium is a d-block element that is represented by the Greek symbol 'alpha' and the metallic 'rays'.

Curium (Cm) is a transuranic radioactive element. Its atomic number is 96 and atomic mass is 247.

Uranium is the heaviest naturally occurring element of the Periodic Table with an atomic weight of 238.

Einsteinium (Es) is a synthetic element of the Actinide Series. Its atomic number is 99 and atomic mass is 254. It is a soft silver radioactive element.

Mendelevium (Md) is a highly radioactive, synthetic element in the Actinide series. Its atomic number is 101 and Atomic mass is 256.

Nobelium (No) is a Radioactive, synthetic chemical element. Its atomic number is 102. It is named in honor of Alfred Nobel, the inventor of Dynamite.

The Elements of Group 9 are Cobalt (Co),

Rhodium (Rh), Iridium (Ir) and Meitnerium (Mt).

The Boron are the chemical elements in group 13 of the Modern periodic table, comprising Boron (B), Aluminium (Al), Gallium (Ga), Indium (In), Thallium (Tl), and Nihonium (Nh).

Nihonium is an element of group 13 which has the atomic number 113 and its electronic configuration is $[Rn] 5f^{14}6d^{10}7s^27p^1$.

Chalcogens are the elements found in group 16 of the Modern periodic table. The names of these elements are Oxygen (O), Sulfur (S), Selenium (Se), Tellurium (Te), Polonium (Po), and Livermorium (Lv). There are six electrons in the outermost shell of a group 16 element.

The Halogens are non-Metallic elements found in group 17 (second from the right) of the Modern periodic table. The halogen elements are Fluorine (F), Chlorine (Cl), Bromine (Br), Iodine (I), Astatine (At), and Tennessine (Ts).

At room temperature, Fluorine (F) and Chlorine (Cl) are gases and Bromine (Br) is liquid. Iodine (I) and Astatine (At) are solids.

Halogens are very reactive, the reactivity decreases from Fluorine to Astatine. Halogens have seven electrons in their outermost shells. Glycerine is not a halogen.

The order of electron affinity of halogens is $I > Br > F > Cl$.

s-block elements

The elements in which the last electron enters the s - orbital of their respective outermost shells are called s - block elements.

The First period starts with the filling of the lowest level (1s) and therefore it has two elements Hydrogen (1s¹) and Helium (1s²) when the first shell is completed.

The second period starts with lithium and the third electron enters the 2s orbital.

p-block elements

The elements in which the last electron enters any one of the three p orbitals of their respective outermost shells are called p block elements. All elements from group 3 to group 12 are the p block

elements. The non-metals and metalloids exist only in the p-block of the periodic table.

d-block elements

The elements in which the last electron enters any one of the five d orbitals of their respective penultimate shells are called d- block elements.

f-block elements

The Elements in which the last electron enters any one of the 7 f orbitals of their respective ante - penultimate shell are called f block elements.

Valency is the combining power of an element. Elements in the same group of the periodic table have the same valency.

Valency of some elements : Aluminium (+3), Boron (+3), fluorine (-1).

The valencies of aluminium and chlorine in the compound $AlCl_3$ are 3 and 1 respectively.

The valency of argentic (containing silver) is +2.

Metals, Non-metals and Alloys

The word 'metal' derives from the Greek word 'metallon,' which means to mine, excavate or extract from the ground.

Metal - Those substances which are hard, shiny, malleable, ductile, acoustic and good conductors of heat and electricity are called metals.

Physical properties of metals :- High melting points, good conductors of electricity, good conductors of heat (lead exception), High density, high tensile strength and pure metals have a shining surface.

Malleability :- The property of metals by which they can be beaten into thin sheets is called malleability.

Lustrousness - The quality or state of having or giving off light.

Ductility - The physical property of the metal which means if we pull the metal it's going to stretch rather than break. Most ductile metal of the earth is Gold and Platinum.

Sonority (of a material) - The property of producing a ringing sound when struck. All metals are highly sonorous.

Due to very low electronegativity, electron bonding in metals is highly delocalized. When hit, the electron cloud moves easily and the dissipation of energy is less.

Tensile refers to a type of external physical force applied to a material in a pulling direction away from its surface.

Chemical properties of metals : Metals react with air and corrode. Generally metals are in a solid state at room temperature.

Metals + Water $\Rightarrow$ Metals oxide + Hydrogen

Metals + Air $\Rightarrow$ Metals Oxide

Metals + Acids $\Rightarrow$ Salt + Hydrogen

Reactivity Series

The Activity Series is a list of elements in decreasing order of their reactivity. Since metals replace other metals, while nonmetals replace other nonmetals, they each have a separate reactivity.

Metals in decreasing order in Reactivity series are Potassium (K - Most reactive metal), Sodium (Na), Calcium (Ca), Magnesium (Mg), Aluminium (Al), Zinc (Zn), Iron (Fe), Lead (Pb), Hydrogen (H), Copper (Cu), Mercury (Hg), Silver (Ag) and Gold (Au).

Due to high reactivity, Potassium and Sodium are kept in Kerosene. Sodium can catch fire if left in the open.

Potassium and sodium are stored in kerosene because they are highly reactive. They are less dense than water. When these metals react with cold water, they release hydrogen gas and form metal hydroxides. Sodium reacts more vigorously than potassium.

Calcium is a s-block element which is derived from the latin word Calx.

Potassium is the 7th most common metal in the Earth's crust. Its atomic number is 19, atomic mass 39.098 g/mol. The word "potassium" comes from medieval Latin "Kalium."

Aluminum (Al) is a lightweight metal. It is soft and malleable, has good conductor heat and electricity.

Magnesium (Mg) is a metal that has a tendency to lose electrons and is electropositive in nature. It has atomic number 12. It is a shiny gray metal having a low density.

Gold (Au) is derived from the Anglo

Saxon word Aurum. Philosopher's stone converts all baser metals like iron into gold.

Arsenic is a naturally found Metalloid on the Earth's layer. The chemical symbol of Arsenic is As.

Sodium is a silver like white metal. It has a strong metallic glow. It is so soft at room temperatures that it can be cut with a knife. It is also a good conductor of electricity. Its compounds tarnish rapidly on exposure to the air, though less rapidly than potassium.

Lithium is a soft, silvery metal with atomic number 3. Its nucleus has 3 protons and 4 neutrons. It's the lightest metal, very reactive, not found freely in nature. It's the only alkali metal reacting with nitrogen. Despite being a metal, it's soft enough to cut with a knife and light enough to float on water.

Mercury (Hg, Atomic Number - 80), turns from liquid to gas at 357°C. It's called quicksilver and was once a hydrargyrum. It's the only metal liquid at room temp. Used in dental fillings, paints, soaps, batteries, and fluorescent lights.

Hardest metal of earth is Chromium (Atomic number 24). Chromium ion released from the tannery industry causes acute water pollution.

Softest metal - Caesium (Atomic number 55).

Most dense metal - Osmium (Atomic number 76). Its density is 22.59 g/cm³.

Copper (Cu) reacts with moist carbon dioxide, losing its shiny brown surface and turning green from basic copper carbonate. Its name comes from Cyprus, depicted on maps resembling the metal's color.

Astatine is the rarest element in the earth's crust occurring only as the decay product of various heavier elements.

Pyrite is considered the most common of the sulfide minerals. Pyrite's metallic luster and pale brass-yellow colour give it a superficial resemblance to gold, hence the well-known nickname of fool's gold.

Drinking water contaminated with **arsenic** causes skin cancer and several internal cancers such as lung, bladder and kidney cancer, as well as cardiovascular disease and others.

Iron (Fe, atomic number 26) is the most abundant metal on Earth, part of the first

transition series and group 8. It has variable valency. Melting point: 1538°C, boiling point: 2861°C.

Silver - The single most conductive metal. It conducts heat and electricity efficiently because of its unique crystal structure.

Tungsten, also known as wolfram (W), has atomic number 74 and the highest melting point (3422°C). It's used for electric bulb filaments.

Important Melting points of metals - Aluminum 660°C (1220°F), Brass 930°C (1710°F), Aluminum Bronze 1027-1038°C (1881-1900°F), Chromium 1860°C (3380°F), Copper 1084°C (1983°F), Gold 1063°C (1945°F).

Copper and its alloys (brasses, bronzes, cupronickel, copper-nickel-zinc, and others) are used as natural antibacterial materials.

Rare Earth Elements or Rare Earth Metals are a set of 17 chemical elements in the periodic table. These are Cerium (Ce), Dysprosium (Dy), Erbium (Er), Europium (Eu), Gadolinium (Gd), Holmium (Ho), Lanthanum (La), Lutetium (Lu), Neodymium (Nd), Praseodymium (Pr), Promethium (Pm), Samarium (Sm), Scandium (Sc), Terbium (Tb), Thulium (Tm), Ytterbium (Yb), and Yttrium (Y). Their chemical properties are similar.

Metallic bonding is a type of chemical bonding that arises from the electrostatic attractive force between conduction electrons and positively charged metal ions.

Metals are the only substances that use metallic bonds among their atoms.

Positive (+) charge ions - Cation

Negative (-) Charged ions - Anion

The **ammonium cation** is a positively charged polyatomic ion with the chemical formula [NH₄]⁺. It is formed by the protonation of ammonia.

Ionic solids are formed by the three-dimensional arrangements of cations and anions bound by strong electrostatic forces. They have very high melting point and are poor conductors in the solid state. Example of ionic solid is table salt.

Bismuth forms metallic bonds. It is used in fire extinguishers, electric fuses and fire detectors.

Metalloids can be defined as chemical

elements whose physical and chemical properties fall in between the metal and non-metal elements. Boron (B, 5), Germanium (Ge, 32), Silicon (Si, 14), Antimony (Sb, 51), Arsenic (As, 33), Tellurium (Te, 52) and Polonium (Po, 84) are the seven most widely recognized metalloids.

Mica is a group of silicate minerals whose outstanding physical characteristic is that individual mica crystals can easily be split into extremely thin elastic plates.

Properties of Mica - Translucent, highly fireproof, highly tough, chemically inert, platy, reflective, refractive, resilient.

Silicon dioxide (SiO₂) has a high melting point. It has melting and boiling points as 1713° C and 2950° C, respectively. The density is about 2.648 g/cm³.

Gallium (Ga) melts on your palm due to its low melting point (29.77°C). It's used in LED semiconductor tech along with metals like Arsenic, Indium, and rare-earth elements (Cerium, Europium, Gadolinium, etc.).

Non metals :- Non metals are the substances that do not conduct electricity and heat. They can be gas, liquid or solids. They are dull in colour.

Hydrogen, carbon, nitrogen, oxygen, phosphorus, arsenic and selenium are the non-metallic elements in the periodic table.

Carbon is a non-metal found in various forms. It makes up about 18.5% of the human body.

Nitrogen (N) is a nonmetallic element of Group 15 of the periodic table. It is a colourless, odourless tasteless gas.

Fluorine exists in gas form. It is critical for the production of nuclear material for nuclear power plants and for the insulation of electric towers.

Bromine (Br) is a liquid non-metal at room temperature, part of group 17 with atomic number 35. It's used in agriculture, dyes, insecticides, pharmaceuticals, and chemicals.

Physical properties of Non metals - Poor conductor of electricity, Non ductile, Brittle Solids, Non sonorous.

Chemical properties of Non metals - Number of electrons in outer shell is 4-8. They easily gain electrons. They form acidic oxides when they react with

oxygen. They are highly electronegative material and a great oxidizing agent. It also doesn't react with water. Non-metals do not react with acid.

Alloy

It is defined as the metallic substance composed of two or more elements, as either a compound or a solution.

Type of Alloys :-

Steel - Steel is an alloy of carbon and iron. Steel is highly effective for making a permanent magnet.

Stainless steel is a mixture of Iron with Chromium and Nickel.

Bronze is an alloy of tin and copper. It is commonly used in heavy tools and gears, coins, and medals.

Brass - It is an alloy of copper and zinc. In this, the quantity of zinc is kept at 45%. Metals such as Sn, Mn, Al, Fe and Pb are also present.

Bell metal - It is an alloy made of 78% copper (Cu) and 22% tin (Sn). It's used for making bells and similar instruments. It can be considered a form of bronze that contains a relatively large amount of tin in its composition.

Rose Metal - It is a fusible alloy with a low melting point. Rose's metal consists of 50% bismuth, Lead and tin. Its melting point is between 94 and 98 °C (201 and 208 °F).

German silver is an alloy of copper, zinc and nickel, sometimes also containing lead and tin. German silver is widely used because of its hardness, toughness, and resistance to corrosion.

Solder is a fusible metal alloy used to create a permanent bond between metal workpieces. It is made up of 60% tin and 40% lead.

Rolled Gold - Cu + Al. It is used in making cheap ornaments.

Gun Metal - Cu + Sn + Zn + Pb. It is used in making guns, barrels, gears and bearings.

Duralumin - Al + Cu + Mg + Mn. It is used for making utensils and aircraft frames.

Dutch metal - Cu + Zn. It is used in making artificial ornaments.

Electrum is the combination of gold and silver.

Amalgam is an alloy of mercury with one or more metals. Sodium along with liquid mercury is known as sodium amalgam. Amalgam of mercury with tin, silver and zinc is widely used in dental filling.

Quartz is a tough mineral made of silicon dioxide (SiO₂), forming a continuous framework of SiO₄ tetrahedra. It's Earth's second most abundant mineral after feldspar and the most weathering-resistant.

Nichrome is generally used to make the coils used in water heaters because of its relatively high resistance and high melting point. It consists of 80% nickel, 20% chromium.

Topaz (Al₂SiO₄(F,OH)₂) is a silicate of fluorine and aluminum. It is used as a gemstone in jewellery and other adornments.

Constantan is a copper-nickel alloy also known as Eureka, Advance, and Ferry. It usually consists of 55% copper and 45% nickel. Its main feature is the low thermal variation of its resistivity, which is constant over a wide range of temperatures.

Talonite is a combination of Cobalt and Chromium. It is used for manufacturing knife blades.

Potash alum, also called "fitkari," is a metal sulfate with potassium, aluminum, and sulfate ions in a 1:1:2 ratio. Its formula is K₂SO₄.Al₂(SO₄)₃.24H₂O. It's used as a flame retardant, mordant, and astringent.

Rhombic sulphur is crystalline in nature and has an octahedral shape. It is of yellow colour with a melting point of 385.8K and specific gravity of 2.06. Rhombic sulphur cannot be dissolved in water but can be dissolved in benzene, ether, alcohol.

Galinstan is used as a mercury replacement in thermometers due to its nontoxic properties.

Ozone (O₃) is a triatomic allotrope of oxygen. Ozone is a powerful oxidant (far more so than dioxygen) and has many industrial and consumer applications related to oxidation.

Sapphire and ruby are varieties of natural aluminium oxide. Aluminium forms one stable oxide, known by its mineral name corundum (Al₂O₃). Sapphire and ruby are impure corundum contaminated with trace amounts of other metals.

Ores and Minerals

Minerals - Minerals are natural substances with specific chemical formulas, solid and inorganic, with a crystal structure. Rocks are made of minerals or non-minerals and lack a specific chemical composition. Examples include Dolomite, Bauxite, and Magnetite.

Ores - Ores are rocks containing valuable minerals, especially metals, that can be economically extracted. Not all minerals are ores, but all ores are minerals. Examples: Cinnabar (HgS) is a mercury ore, and sphalerite (ZnS) is a zinc ore.

Minerals are divided into two types namely metallic and non-metallic.

Metallic minerals are shiny and contain metals in their chemical composition. They're mined for metal extraction. Examples: Manganese, iron ore, and bauxite.

Metallic minerals are divided into ferrous and non-ferrous metallic minerals.

Ferrous minerals are those minerals that contain iron such as iron ore, manganese, nickel, and chromite.

Non-ferrous minerals are ones that do not contain iron. Example - copper, bauxite, lead, zinc and gold.

Bauxite ore mostly exists in deeply weathered rocks. Volcanic rocks contain bauxite deposits in some regions.

Manganese ore is a silvery brittle or grey-white metallic ore that occurs in many forms.

Non-metallic minerals don't have a metallic shine and lack extractable metals. Examples include limestone, gypsum, and mica.

Physical Characteristics of Minerals - Crystal form, luster, colour, harness and specific gravity.

Streak is the colour of the ground powder of any mineral. It may be of the same color as the mineral or may differ. Malachite is green and gives a green streak, fluorite is purple or green but gives a white streak.

Minerals on the basis of colour :-

Idiochromatic minerals are "self colored" due to their composition. The color is a constant and predictable component of

the mineral. Examples are blue Azurite, red Cinnabar, and green Malachite.

Allochromatic minerals that are colorless in their pure form are known as Allochromatic Minerals. These minerals show colors due to the presence of small quantities of "Coloring" agents. Examples - Quartz, Diamond, Beryl, Corundum etc.

Pseudochromatic minerals are "false colored" due to tricks in light diffraction. In these cases, color is variable but a unique property of the mineral. Examples are the colors produced by precious opal and the shiller reflections of labradorite.

Important metals and their ores :-

The most common metal ores are oxides and sulphides.

Sodium (Na) - Chile saltpetre, Trona Borax, Common salt.

Potassium (K) - Sylvine, Carnallite.

Strontium (Sr) - Strontianite, Silestone.

Silver (Ag) - Ruby silver, Horn silver, Argentite.

Tin (Sn) - Cassiterite.

Antimony (Sb) - Stibnite.

Nickel (Ni) - Milarite.

Barium (Ba) - Barytes.

A **mineral's hardness** is a measure of its relative resistance to scratching. It is measured by scratching the mineral against another substance of known hardness on the Mohs Hardness Scale.

Mineral Hardness on Mohs Scale - Talc (1), Gypsum (2), Calcite (3), Fluorite (4), Apatite (5), Orthoclase (6), Quartz (7), Topaz (8), Corundum (9), Diamond (10).

Iron ore is the most common element in the world and makes up close to 5% of the Earth's crust. This makes it one of the most abundant rock elements.

Primary constituent is iron oxide - Hematite - Fe_2O_3 , Magnetite - Fe_3O_4 , Siderite - FeCO_3 , Iron pyrites - FeS_2 . Limonite, one of the major iron minerals, hydrated ferric oxide ($\text{FeO}(\text{OH}) \cdot n\text{H}_2\text{O}$).

Galena is basically a Sulfide mineral. It is the natural mineral form of Lead Sulfide (PbS). It has the lowest electrostatic potential in sulfide minerals. It is also called lead glance.

Carnotite is a radioactive, bright-yellow,

soft, and earthy vanadium mineral that is an important source of uranium. The primary uranium ore mineral is Uraninite (UO_2), which was previously known as pitchblende.

Magnesium is found in minerals such as magnesite, dolomite ($\text{MgCO}_3 \cdot \text{CaCO}_3$), brucite, serpentine, Epsom salt, Kieserite, Carnallite, Pyrolusite, Magnate.

The main minerals of **platinum** ores are pyrrhotite, chal-copyrite, pentlandite, and cubanite.

Gold is usually found alone or alloyed with mercury or silver, but can also be found in ores such as calaverite, sylvanite, nagayagite, petzite or krennerite.

Calaverite or Gold telluride is an uncommon telluride of gold. Its chemical formula is AuTe_2 .

Mercury is either found as a native metal (rare) or in cinnabar, metacinnabar, corderoite, livingstonite and other minerals. Cinnabar is a cochineal-red, toxic ore of mercury, composed of mercury sulfide (HgS).

Greenockite is a rare cadmium (Cd) bearing metal sulfide mineral consisting of cadmium sulfide in crystalline form.

Corundum is a crystalline form of aluminum oxide (Al_2O_3) with traces of iron, titanium and chromium. It is a rock-forming mineral.

Smelite (CoAsS_2) is an ore of Cobalt.

Copper is found in many minerals which occur in deposits large enough to mine. These include: Azurite, Cuprite, Copper glance, Copper pyrites, malachite, chalcocite, acanthite, chalcopyrite and bornite.

Azurite is a soft, deep-blue copper mineral produced by weathering of copper ore deposits.

Pyrolusite (MnO_2) is an important ore of manganese. It is a black, amorphous appearing mineral, often with a granular, fibrous, or columnar structure, sometimes forming reniform (kidney-shaped) crusts.

Calcium ores include Dolomite, Calcite, Gypsum, Fluorspar, and Asbestos. Dolomite is a mineral made of calcium magnesium carbonate ($\text{CaMg}(\text{CO}_3)_2$), and it's also a sedimentary rock mostly composed of this mineral.

Kieserite is a magnesium sulphate

mineral. It is used in the production of Epsom salt and as a fertilizer in agriculture.

Granite is the most widely known igneous rock. It is an intrusive rock with visible grains of feldspar, quartz, mica, and amphibole minerals.

Sphalerite is also called zinc blende, zinc sulfide (ZnS). It is the chief ore mineral of zinc. Other ores - Zincite, Calamine.

Metallurgy

Metallurgy can be defined as the extraction of metal in its pure form from their ore.

Mineralogy is the study of the chemistry, crystal structure and physical properties of the mineral constituents of rocks.

The study of rocks is called Petrology (in Greek, *petra* means rock, *logos* means science). Petrogenesis is the study of the origin of rocks.

Minerals closest to the surface of the earth can only be extracted by the process of **Quarrying**.

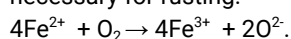
Opencast mining, also known as open-pit mining, is a surface mining technique that extracts minerals from an open pit in the ground.

Shaft mining or shaft sinking is the action of excavating a mine shaft from the top down.

Corrosion is a natural process that converts a refined metal into a more chemically stable oxide.

Corrosion of copper - Copper forms a green deposit on its surface when exposed to air, which is usually copper carbonate (CuCO_3) and Copper hydroxide ($\text{Cu}(\text{OH})_2$).

Rusting of iron refers to the formation of rust, a mixture of iron oxides, on the surface of iron objects or structures. This corrosion is a redox reaction between oxygen and iron in a water-rich environment. Presence of oxygen (O_2) is necessary for rusting.



Pure iron does not rust in dry and carbon dioxide free air.

Anodising is the process of forming a thick oxide layer of aluminum. This aluminum oxide coat makes it resistant to further corrosion. It is useful in preventing the metal from corrosion, in architectural finishing.

Electroplating - It is the process of applying a metal coating on another piece of metal through an electro-deposition process. Uses - kitchenware, mobile phones, Jewellery, Coins, building structures, machinery parts.

Electrolysis - It is the breakdown of ionic compounds into elements using an electric current in a liquid. It involves oxidation at the anode and reduction at the cathode. For instance, in molten sodium chloride, sodium ions migrate to the cathode, where they gain electrons to form sodium atoms.

Kohlrausch Law of independent migration of ions states that limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte.

Crystallisation - It is the process by which a solid forms, where the atoms or molecules are highly organized into a structure. It can also be used to obtain pure alum crystals from an impure alum.

Annealing is a heat treatment process that changes the physical and the chemical properties of a material to increase ductility and reduce the hardness to make it more workable.

Welding is a fabrication process that joins materials, usually metals or thermoplastics, by using high heat to melt the parts together and allowing them to cool, causing fusion.

Smelting is a process of applying heat to ore in order to extract a base metal.

Roasting is the process of heating ore in the presence of air, whereas calcination is the process of heating ore in the absence of air.

Cinnabar (HgS) is mercury ore found near volcanic areas and hot springs. When heated in air, it transforms into mercury oxide (HgO) and releases sulfur dioxide (SO_2). The chemical reaction is $2\text{HgS} + 3\text{O}_2 \rightarrow 2\text{HgO} + 2\text{SO}_2$

Magnetite is a mineral whose primary component is an iron oxide that contains equal amounts of iron (II) and iron (III). Magnetite is in the hexoctahedral crystal. It is a rock mineral.

Aluminum oxide (Al_2O_3) is a white, inert material used in ceramics, sourced from ores like bauxite and kaolinite, which are

abundant in the Earth's crust. Major aluminum ores include bauxite ($\text{Al}_2\text{H}_2\text{O}_4$), corundum, feldspar, cryolite, alunite, and Kaolinite ($\text{Al}_2(\text{OH})_4\text{Si}_2\text{O}_5$), often found in clay deposits.

Kaolinite is the product as a result of the process of hydrolysis of orthoclase.

Thermodynamics

The branch which deals with the movement of energy from one form to the other and the relation between heat and temperature with energy and work done is called **thermodynamics**.

The Universe = The System + The Surrounding.

System refers to the portion of the universe which is under observation.

Three types of systems in thermodynamics are open, closed, and isolated.

Open System :- In a system, when there is exchange of energy and matter taking place with the surroundings, then it is called an open system. Example - The presence of reactants in an open beaker is an example of an open system.

Closed System :- Across the boundary of the closed system, the transfer of energy takes place but the transfer of mass doesn't take place. Example - When we boil water with a closed lid, the heat can exchange but matter cannot.

Isolated System :- An isolated system cannot exchange energy and mass with its surroundings. The universe is considered an isolated system. For example - The presence of reactants in a thermos flask, or substance in an insulated closed vessel.

Insulation means creating a barrier between hot and cold objects that reduces heat transfer by either reflecting thermal radiation or decreasing thermal conduction and convection from one object to the other. Insulation is used in building and in manufacturing processes to prevent heat loss or heat gain.

The state of the system :- The state of a thermodynamic system means its macroscopic or bulk properties which can be described by state variables: Pressure (P), volume (V), temperature (T) and amount of substance (n).

Isothermal process :- When the operation is carried out at constant

temperature, the process is said to be isothermal. For isothermal processes, $dT = 0$, where dT is the change in temperature.

Adiabatic process :- It is a process in which no transfer of heat between the system and surroundings takes place.

Isobaric process :- When the process is carried out at constant pressure, it is said to be isobaric. For an Isobaric process, $dP = 0$, where dP is the change in pressure.

Isochoric process :- A process when carried out at constant volume, it is known as isochoric in nature.

Cyclic process :- If a system undergoes a series of changes and finally returns to its initial state, it is said to be a cyclic process.

Reversible Process :- When in a process, a change is brought in such a way that the process could, at any moment, be reversed by an infinitesimal change, the change is called reversible.

Enthalpy is the measurement of energy in a thermodynamic system. The quantity of enthalpy equals the total heat content (H) of a system, equivalent (E) to the system's internal energy plus the product of volume (V) and pressure (P).
 $H = E + PV$.

Hess's law of constant heat summation :- If the same chemical change is completed in one or more steps, the heat produced or absorbed in the complete change is the same irrespective of the method by which the change is effected.

The amount of heat that is required for a phase change of a unit mass of a substance without any change in temperature is known as latent heat.

The latent heat of fusion for ice is 80 calories per gram.

Latent heat of vaporization of water or latent heat of steam is 540 cal/g or 2256 kJ/Kg.

Specific heat efficiency is measured by the amount of heat energy required to raise one gram of one degree Celsius of a product.

Water has a specific heat capacity of 4182 J/kg°C.

Specific heat of some substances : Iron/Steel - (450 Jkg⁻¹K⁻¹), Aluminium - (900.0 Jkg⁻¹K⁻¹), Carbon - (506.5 Jkg⁻¹K⁻¹),

Tungsten - (134.4 Jkg⁻¹K⁻¹).

Organic Chemistry

The compounds in solid, liquid or gaseous state which contain carbon in its molecules are known as **organic compounds**. The most common elements present in it are carbon, hydrogen, oxygen and nitrogen.

Condensed structural formulas - A system for writing organic compounds in a single line. Example : methane is CH₄ and propanol is CH₃(CH₂)₂OH.

IUPAC (International Union of Pure and Applied Chemistry) is an internationally recognized body that has systematically named all chemical organic substances. IUPAC publishes many books which contain its complete list of definitions. They are divided into 7 - "**colour books**".

Blue Book - Nomenclature of organic chemistry commonly referred to chemists.

Green Book - Quantities, Units and Symbols in physical chemistry.

Red Book - Nomenclature of inorganic chemistry, by chemists commonly.

White Book - Pertaining to biochemical research.

Silver Book - Titled compendium of terminology and nomenclature of properties in clinical laboratory sciences.

Gold Book - It contains internationally accepted definitions for terms in chemistry.

Purple Book - It contains the nomenclature of polymer.

The Word **root** of IUPAC name indicates the number of carbon atoms in the longest possible continuous carbon chain also known as **parent chain**.

Carbon is found in nature both in free state and combined state. Forms of diamond, graphite and fluorine are kept under the crystalline form of carbon while charcoal under amorphous form.

3 isotopes of Carbon : ¹²C (6 neutrons and 6 protons), ¹³C (6 protons and 7 neutrons), and ¹⁴C (6 protons and 8 neutrons).

Allotropy is the property of certain elements to exist in two or more different forms, called allotropes, while remaining in the same physical state.

In allotropes, atoms of the same element are bonded to each other in different ways.

There are several allotropes of carbon like Diamond, Graphite, Lonsdaleite, C-60 Fullerenes (Buckminsterfullerene or buckyball), C-540, C-70, Amorphous carbon, single-walled carbon nanotube, or buckytube. Methane is not an allotrope of carbon. Chaoite is related to Carbon.

Catenation is the ability of an atom to form bonds with similar atoms of the same element. This is demonstrated by both carbon and silicon.

It occurs most readily with carbon, which forms **covalent bonds** with other carbon atoms to form longer chains and structures.

Diamond :- Four atoms of carbon are found in it. It is the solid form of carbon. It is the hardest natural substance. It is obtained naturally in the form of **kimberlite**. Diamonds shine due to total internal reflection of light (TIR). The Crystal system of Diamond is Cubic.

There are a total of **7 crystal structures** - triclinic, monoclinic, orthorhombic, tetragonal, trigonal, hexagonal and cubic.

Graphite - It consists of rings of six carbon atoms. It crystallizes in the hexagonal lattice. It is insoluble in water and organic solvents.

The melting point of graphite is slightly higher than the melting point of diamond because in its C-C bonds have a partial **double-bond** character and hence are stronger and more difficult to break.

The densities of **graphite** and **diamond** at 298K are 2.25 g cm⁻³ and 3.51 g cm⁻³ respectively.

Fullerenes - Contains 60 carbon atoms linked together in the shape of a football.

Uses - They are used as insulators and superconductors and in the treatment of cancer and AIDS.

Aliphatic Hydrocarbon - It is an organic compound containing hydrogen and carbon atoms that are usually linked together in chains via single, double or triple bonds.

Eicosane (C₂₀H₄₂) is an unsaturated aliphatic hydrocarbon composed of 20 carbon atoms, used to make candles and paraffin wax with solar energy storage capacity.

It has two types: **Saturated** and **Unsaturated** Hydrocarbons.

Saturated hydrocarbons have single bonds between two carbon atoms. They burn with a Blue colour non-sooty flame. They are fairly unreactive and are inert in the presence of most reagents.

Alkanes are saturated open chain hydrocarbons containing carbon - carbon single bonds. General formula is C_nH_{2n+2} . Molecular Formula of Alkane - Methane (CH_4), Ethane (C_2H_6), Propane (C_3H_8), Butane (C_4H_{10}), Pentane (C_5H_{12}), Hexane (C_6H_{14}).

Dodecane is a straight-chain alkane containing 12 carbon atoms that are used as a major component in kerosene and some jet fuel research and also in the rubber and paper processing industry.

Unsaturated hydrocarbons have double or triple bonds between carbon atoms. Unsaturated carbon compounds burn with a yellow sooty flame.

Unsaturated hydrocarbons add hydrogen in the presence of catalysts such as **palladium** or **nickel** to give saturated hydrocarbons. It is of two types - **Alkene** and **Alkyne**.

Alkenes - In this double bond is found between the carbon atoms. Its general formula is C_nH_{2n} . Examples - Ethene or Ethylene (C_2H_4), Butene (C_4H_8), Octene (C_8H_{16}).

Alkyne - It has a triple bond between two carbon atoms. Its general formula is C_nH_{2n-2} . The triple bond is known as the '**acetylenic bond**'. Alkynes have very high boiling points. Examples - Ethyne /Acetylene - C_2H_2 , Pentyne (C_5H_8).

Methane (CH_4) is also called **Marsh gas**. It is colourless, odorless and tasteless. It is lighter than water due to which it is insoluble in water, but soluble in organic compounds. It is used in domestic fuel. It is the main component of gobar gas or fossil fuel. The boiling point of methane is 111k.

The major component of bio-gas and **Compressed Natural Gas (CNG)** is Methane. Bio-CNG contains about 92-98% of methane and only 2-8% carbon dioxide.

In 1856, Berthelot synthesised the **Methane** compound by the reaction of a mixture of sulfated hydrogen with carbon disulfide on red-hot copper.

Propane (C_3H_8) - It is also known by the name of **Dimethyl Methane**. It is a colourless, odourless gas. It is a gas at standard temperature and pressure, but is compressible to a transportable liquid.

Butane (C_4H_{10}) is an alkane with four carbon atoms. used as fuel for cigarette lighters, portable stoves, propellant in aerosols, etc. It is also found in Liquefied Petroleum Gas (LPG). **Methylpropane** is an isomer of n-butane.

Isopentane, also called 2-methylbutane, is a branched-chain saturated hydrocarbon with the chemical formula C_5H_{12} . 2, 3-Dimethylbutane is an isomer of hexane. 1, 3-Dimethylbenzene also known as m-xylene.

CCl_2F_2 is the chemical formula of dichlorodifluoromethane.

Nitromethane (CH_3NO_2) is used to make industrial antimicrobials and pharmaceuticals and is also used as a soil fumigant and as a fuel in race car engines. Some other compounds - CH_4NO_2 (Oxidoamino methanol), CH_3NO_3 (Methyl Nitrate).

Ethylene (Ethene) is the simplest alkaline (hydrocarbon) having the formula C_2H_4 or $H_2C=CH_2$. It is a colorless flammable gas with a mild (sweet and musky) odor when pure.

Ethylene is synonymous with olefiant gas which is used to make anesthetics, refrigerators and other chemicals. It is mostly used in the manufacturing of polythene and to artificially ripen fruits.

Ethynyl has the molecular formula C_2H which is derived from the hydride of acetylene.

Acetylene (C_2H_2) was discovered by Edmund Davy (in 1836). It is the first stable member of the alkyne series. It is also called ethyne. It is a colourless gas with a faint, ethereal odour. Used in the manufacturing of synthetic rubber, and as an additive to preserve food.

Succinic Acid (butanedioic acid) is a dicarboxylic acid that occurs naturally in plant and animal tissues. The chemical is also known as "**Spirit of Amber**".

Butanoic acid (Butyric acid): It is present in animal fats and oils, milk, butter. It is an anaerobic fermentation product.

Butyric acid ($C_4H_8O_2$) includes a saturated short-chain fatty acid with 4-carbon molecules, commonly found in

esterified form in animal fats and plant oils.

Acetic Acid (CH_3COOH) - It's Purely known as glacial acetic acid as it freezes like ice.

Other uses - Used in ketchup, mustard, mayonnaise, salad dressings, meat, chicken, soups, and cheeses, Production of Vinyl Acetate Monomer, Production of Acetic Anhydride, Production of Ester, As a Solvent.

Trichloromethane better known as Chloroform ($CHCl_3$) is a colorless liquid that quickly evaporates into gas. It is used as a solvent, to make other chemicals, as a fumigant Carbon Tetrachloride (CCl_4). The boiling point of Chloroform is 334 K.

Other uses - In making **fluorocarbon-22**, a refrigerant during surgery. Today the primary use of chloroform is as a solvent in industry and in the production of the refrigerant freon.

Chlorophenol (C_6H_5ClO) is a chlorinated aromatic compound and has been extensively used as a fungicide.

2- Chloroethanol (C_2H_5ClO) is primarily used for synthesis of ethylene oxide and a number of synthetic reactions for preparation of dyes, pharmaceuticals, biocides and plasticizers.

An **ester** is a chemical compound derived from an acid (organic or inorganic) in which at least one -OH hydroxyl group is replaced by an -O-alkyl group. The term ester was introduced by German chemist Leopold Gmelin. **Examples** - Ethyl acetate, propyl methanoate.

Toluene (Toluol) - It is a colorless, flammable liquid that is about 15% lighter than water. It has a sweet, pungent benzene-like odor. It is used for the production of **Trinitrotoluene (TNT)**.

Paradichlorobenzene ($C_6H_4Cl_2$) is used as a fumigant insecticide to control cloth moths.

Ethylbenzene ($C_6H_5CH_2CH_3$) is to manufacture styrene, a compound used to make plastics.

Benzoyl chloride (C_7H_5ClO) : It is used for peroxide production. It is used primarily as a solvent, degreasing agent, and chemical intermediate.

Ethanol (CH_3CH_2OH) is used in the

manufacture of drugs, plastics, lacquers, polishes, plasticizers, and cosmetics. The boiling point of Ethanol is 156K.

Heating ethanol with excess concentrated sulphuric acid produces ethene.

$\text{CO}_2 + \text{H}_2\text{O} + \text{heat and light}$ are the products formed when $\text{CH}_3\text{CH}_2\text{OH}$ reacts with O_2 .

Benzophenone is a naturally occurring organic compound with the formula $(\text{C}_6\text{H}_5)_2\text{CO}$.

Monosaccharides include glucose (dextrose), fructose, galactose, xylose and ribose. Monosaccharides are the simplest form of carbohydrates with only one simple sugar.

Oligosaccharides include sucrose, lactose and maltose. Starch, cellulose, and glycogen are some examples of polysaccharides.

Lactitol is derived through the hydrogenation of lactose.

Liquefied Petroleum Gas (LPG) is a blend of light hydrocarbon compounds. It mainly consists of **butane** (C_4H_{10}) or **propane** (C_3H_8) or a mixture of both. It is used for cooking in homes.

Ethyl Mercaptan is used as an additive to odorless gases like butane, propane, and petroleum to give them a warning odor.

It is an organic sulfur compound. It has a distinct odor which is used to detect the leakage of different types of Odorless LPGs.

Aldehyde ($\text{R}-\text{CH}=\text{O}$) - It is an organic compound containing a functional group $-\text{CHO}$. Example - formaldehyde (CH_2O) Acetaldehyde ($\text{C}_2\text{H}_4\text{O}$). **Propanal** is an example of an aldehyde functional group.

Benzaldehyde has a pleasant almond aroma that is commonly used to impart almond flavour to chocolate and baked goods.

Carbaldehyde suffix is used when the $-\text{CHO}$ group is attached to a carbon atom of a ring or ring system, or to a heteroatom.

Carboxylic acid ($\text{R}-\text{COOH}$) - An organic compound containing a carboxyl functional group. Example - acetic acid (CH_3COOH).

Ketone ($\text{RC}=\text{OR}$) - It is a class of organic compounds characterized by the presence of a carbonyl group in which

the carbon atom is covalently bonded to an oxygen atom.

They are most widely used as solvents, especially in industries manufacturing explosives, lacquers, paints and textiles. Methyl ethyl ketone is also known as 2-butanone.

Alcohols are organic compounds which contain a hydroxyl ($-\text{OH}$) functional group covalently bonded to a carbon atom. Example - CH_3OH (Methanol, Methyl alcohol).

Alcohol is used within medicine as an antiseptic, disinfectant, antidote, high-efficiency fuel, antifungal, and antibacterial properties. Alcohol is inflammable in nature.

Potassium bitartrate (potassium acid salt) is obtained as a by-product of alcohol manufacture during the fermentation process.

The compound in which the hydroxy group, $-\text{OH}$, is attached to a saturated carbon atom which has two other carbon atoms attached to it is called secondary alcohol. Butan-2-ol is a secondary alcohol.

Ethers - Ethers are a class of organic compounds that mostly contain an ether group wherein the oxygen atom is bonded to two alkyl or aryl groups.

(-OR- Alky groups, -OAr Aryl groups) - They have the general formula $\text{R}-\text{O}-\text{R}'$, where R and R' represent the alkyl or aryl groups. Example - $\text{CH}_3\text{OC}_2\text{H}_5$ (Ethylmethyl ether).

Oxygen-containing functional groups are alcohols, ethers, aldehydes, ketones and carboxylic acids as well as a variety of derivatives of **carboxylic acids** such as the amides, esters and acid halides.

Benzene (C_6H_6): Organic compound with a planar ring of six carbon atoms and one hydrogen atom per carbon. It is a regular hexagon, with bond angles -120° (sp^2).

Esters (alkoxy group $-\text{O}-\text{R}$) - These are sweet-smelling substances and are used in making perfumes. They are produced when carboxylic acids are heated with alcohol in the presence of an acid catalyst.

Ethyl acetate is used in decaffeinating tea, glues, nail polish removers and coffee. It is colourless liquid and has a

strong fruity fragrance.

Cyanides (CN^-) is the only compound that contains carbon but isn't organic.

The few carbon - containing compounds classified as organic - carbon monoxide (CO), sodium bicarbonate (NaHCO_3), Carbon dioxide(CO_2), and carbon tetrachloride(CCl_4).

Ethanoic acid is a weak electrolyte. Acetic acid (**CH_3COOH**), also called ethanoic acid, is the most important of the carboxylic acids. The boiling point of acetic acid is 290K.

Uses - Making dyes, pigments, paint, and coating additives. Acetic acid is the main component of vinegar.

Proteins are large biomolecules and macromolecules that comprise one or more long chains of amino acid residues.

Anisole (methoxybenzene, $\text{CH}_3\text{OC}_6\text{H}_5$) and **Acetophenone** ($\text{C}_6\text{H}_5\text{C}(\text{O})\text{CH}_3$) are used for fragrance in soaps and perfumes and as a flavoring agent in foods.

Aniline ($\text{C}_6\text{H}_5\text{NH}_2$) used to make dyes, drugs, explosives, plastics, and photographic and rubber chemicals.

Aniline reacts with nitrous acid at 273-278 K to prepare benzenediazonium chloride.

Aniline is a primary arylamine in which an amino functional group is substituted for one of the benzene hydrogens.

Formic acid (methanoic acid (HCOOH)) - It is a simple carboxylic acid found in stings of other insects and Ant sting.

Oxalic acid is found in leafy greens, vegetables, fruits, cocoa, nuts and seeds.

Lithium aluminum hydride (LiAlH_4) is a strong, unselective reducing agent for polar double bonds. It can reduce aldehydes, ketones, esters, carboxylic acid chlorides, carboxylic acids and even carboxylate salts to alcohols.

Other strong reducing agents are Sodium Borohydride (NaBH_4), Titanium hydride (TiH_2), Calcium hydride (CaH_2), Bismuth hydride (BiH_3).

Cumene (C_9H_{12} isopropylbenzene) is a constituent of crude oil and refined fuels. Uses - Manufactures other chemicals (phenol, acetone), thinner in paints, a component of high-octane motor fuels, in manufacturing of rubber, iron, steel, and pulp and paper.

Furan is used in the production of resins and lacquers, agrochemicals, and pharmaceuticals.

Styrene is used to make latex, synthetic rubber, and polystyrene resins.

Markovnikov's-rule:- The rule states that the negative part of the addendum (adding molecule) gets attached to that carbon atom which possesses a lesser number of hydrogen atoms.

Bredt's rule states that a double bond cannot be placed at the bridgehead of a bridged ring system unless the rings are large enough.

Baldwin's-rule states a series of guidelines outlining the relative favorabilities of ring closure reactions in alicyclic compounds.

Cornforth's-rule :- He is renowned for solving the details of the chemistry of the complex biosynthetic pathway in which acetic acid, containing two carbon atoms, is converted in nature into the steroid cholesterol, which contains 27 carbon atoms and has a structure containing four rings.

Lactic acid ($C_3H_6O_3$) is a natural preservative often found in foods like yogurt, baked goods, and pickled vegetables.

Citric acid ($C_6H_8O_7$) is found naturally in citrus fruits, especially lemons.

Nitric Acid (HNO_3) is a strong acid. It is also known as the spirit of niter and aqua fortis.

Vinegar is an aqueous solution of acetic acid (CH_3COOH) that typically contains 5-8% acetic acid by volume. Natural sources of acetic acid include apples, grapes, pineapple, strawberries, and oranges.

A **homologous series** is a collection of compounds with the same general formula that differ only in the carbon chain length. Examples - C_4H_8 (Butene) and C_3H_6 (Propene), and C_5H_{10} , belong to the alkene homologous series.

Compounds in which an - OH group is attached directly to an aromatic ring are called **Phenols** and can be abbreviated ArOH in chemical equations.

It differs from alcohols in that they are slightly acidic in water. They react with aqueous sodium hydroxide (NaOH) to form salt and water.

Aliphatic compounds can be cyclic (closed chains) or acyclic (open chain) compounds. Examples are isooctane, acetylene, propene, propane.

Benzenoid compounds that contain a benzene ring in their structure. Examples are Mesitylene and cumene.

Resorcinol is a 1, 3- isomer (or meta-isomer) of benzenediol with the formula $C_6H_4(OH)_2$ or Benzene- 1, 3 diol.

Uses - As an antiseptic and disinfectant in topical pharmaceutical products in the treatment of skin disorders and infections such as acne, seborrheic dermatitis, eczema, etc.

Picric acid (2,4,6-trinitrophenol) is a pale yellow, odourless crystalline solid that has been used as a military explosive, as a yellow dye, and as an antiseptic.

Glycerol ($C_3H_8O_3$) is a chemical compound, generally nontoxic, sweet-tasting viscous liquid. It possesses the property of humectant - a substance that attracts water. It can draw in moisture, preventing drying of the skin.

Glycerol is used in moisturising creams and soaps because of its hygroscopic properties.

Benzene hexachloride ($C_6H_6Cl_6$) is a commonly used insecticide. It has toxic effects like ataxia, central nervous system dysfunction etc.

Diene is a covalent compound that contains two double bonds. Diene is based on the relative position of the double bond in the overall molecule.

A **cumulated diene** has the double bonds sharing a common atom. The result is more specifically called an allene.

A **conjugated diene** is a compound with two conjugated double bonds that are spaced apart by just one bond.

Two double bonds that are spaced apart by two or more double bonds make up an **isolated diene**.

Terpenes are a group of organic compounds with the general formula $(C_5H_8)_n$ where, $n > 1$.

In a **combustion reaction**, carbon compounds are oxidized in the presence of oxygen. It is generally an oxidation reaction, not all oxidation reactions are combustion reactions. Oxidation is also carried out by using oxidizing agents (Oxidants).

Alexander Zaitsev's rule is an empirical rule used to predict the major products of elimination reactions. It states that in an elimination reaction the major product is the more stable alkene with the more highly substituted double bond.

Traube's rule relates the surface tension of capillary active organic compounds to the number of the hydrocarbon CH_2 groups present in their molecules.

Victor Grignard was conferred the Nobel Prize for Chemistry in 1912 for his discovery of "Grignard reagent".

Grignard reagent is an organomagnesium compound which can be described by the chemical formula "R-Mg-X" where R refers to an alkyl or aryl group and X refers to a halogen.

$$H_2O + C \quad H_3CH_2MgBr \rightarrow CH_3CH_3 + MgBr(OH).$$

Aromatic compounds contain a benzene ring or benzene-like properties (but not necessarily a strong aroma). Examples - Naphthalene and anthracene.

Anthracene ($C_{14}H_{10}$) is a white to yellowish solid with a weak aromatic odour purified by the process of sublimation.

Tertiary butyl hydroquinone is a synthetic aromatic organic compound which inhibits the chemical breakdown of food that occurs in the presence of oxygen, commonly used in foods such as crackers, microwave popcorn, butter and chicken nuggets.

Antiaromatic compounds - The molecules that are cyclic, planar and completely conjugated but consist of $(4n + 2)\pi$ electrons. Examples - Pentalene, Biphenylene.

Non Aromatic compound - A cyclic compound which doesn't necessitate a continuous form of overlapping ring of p orbitals need not be considered as aromatic or even anti aromatic and hence these are termed as non aromatic. Examples - alkanes, alkenes etc.

Graphene is the rings hexagonal in shape. It resembles a single layer of graphite. It is also considered as a basic unit for graphite and fullerene.

Tribromomethane is a synonym for Bromoform which is a brominated organic solvent. Bromoform is a colorless liquid with a sweet odor. Uses - fire extinguishers, fumigants, and solvents.

In **Phthalaldehyde** $\{C_6H_4(CHO)_2\}$, there are two formyl groups attached to adjacent carbon centres on a benzene ring and form fluorescent conjugation products with primary amines.

Fluorine was arbitrarily assigned a value of 4.0 in the 'Halogens' section in 1922 by Linus Pauling, an American scientist. Dry ice, ammonium chloride, naphthalene balls and camphor, all these compounds are examples of **Sublimable substances**. Pectin, which is responsible for the firmness and softness of fruits, is mainly composed of Galacturonic acid $(C_6H_{10}O_7)$.

The strength of a hydrogen bond is determined by interaction between the lone-pair electrons of the electronegative atom of one molecule and the hydrogen atom of another molecule.

Chemical Tests

Chemical test - A chemical test is a qualitative or quantitative procedure designed to identify, quantify, or characterize a chemical compound or chemical group.

Lassaigne's test (sodium fusion test) helps to detect nitrogen, sulphur, and halogens in organic compounds. It is used where the covalent compounds are converted into ionic compounds by fusion with metallic **sodium**.

Flame test is a qualitative analysis used by the chemist to identify the metal and metalloid ion in the sample.

D-dimer test is a blood test that can be used to help rule out the presence of a serious blood clot.

When a magnesium ribbon is burnt in oxygen, magnesium oxide (MgO) is formed accompanied by the emission of light. $2Mg + O_2 \rightarrow 2MgO$

Magnesium oxide (MgO) is a white hygroscopic solid mineral that is physically and chemically stable at high temperatures. It has white flame when it burns.

Zinc is used in the **Veri-Green process**. It is stated that the addition of the metal ions result in vegetables that are greener than the conventionally processed.

In **Kolbe's reaction**, Ortho - hydroxybenzoic acid is formed as the primary product by heating sodium phenoxide at $125^\circ C$ in the presence of carbon dioxide gas under a pressure of

about 100 atm.

Carbon dioxide gas is used by chemical leaveners, such as baking powder and baking soda, to lighten and aerate baked goods.

Baking powder is heated or mixed in water, carbon dioxide gas is produced. When sodium bicarbonate (**baking soda**) is moistened and heated, it releases carbon dioxide gas.

Bromine test - If bromine water is added to a solution of phenol in water, the bromine water is decolorized and a white precipitate is formed which smells of antiseptic.

Lucas Test is a chemical test that differentiates and categorizes primary, secondary and tertiary alcohols using a solution of anhydrous zinc chloride in concentrated hydrochloric acid.

Iodoform test (triiodomethane) is used to check the presence of carbonyl compounds in a molecule. The reaction of iodine, a base and a methyl ketone gives a yellow precipitate along with an "antiseptic" smell. Iodine test is used for the detection of starch.

Ferric chloride test is used to determine the presence of phenols in a given sample or compound.

Phenols form a complex with ferric ions. This complex has an intense colour, which may vary from blue, green or even red.

Benedict's test is a chemical test that can be used to check for the presence of reducing sugars in a solution.

Tollens test - Formation of a layer of silver metal on the inner surface of the test tube which shines like a mirror, indicates the presence of an aldehyde.

Schiff test: Take 3-4 drops of the liquid compound or dissolve a few crystals of organic compound in alcohol and add 2-3 drops of the Schiff's reagent. Appearance of pink colour indicates the presence of an aldehyde.

Hinsberg Test: Distinguishes between primary, secondary, and tertiary amines by shaking with Hinsberg reagent in the presence of aqueous alkali.

Carbon Dioxide (CO_2) is a colourless, odourless gas. Uses - In the food industry, oil industry, and chemical industry.

Hydrogen Sulfide (H_2S) is a colourless gas with the smell of rotten eggs. Uses - In the water purifying process.

Sulphur Dioxide (SO_2) is a Colourless gas with a pungent smell. Uses - In food preservatives, refrigerants, and beach products. It is very toxic and can cause death.

Radioactivity

Radioactivity is defined as the emission of particles and electromagnetic rays from the nucleus of an unstable atom.

Radioactivity provides the base phenomena on which the nuclear energy and bomb are prepared. Important minerals used for the generation of nuclear energy are uranium and thorium.

A. H. Becquerel discovered radioactivity in 1896. Becquerel (Bq) is the SI unit of radioactivity.

Marie Curie coined the term radioactivity for the spontaneous emission of ionizing, penetrating rays by certain atoms.

Radioactive Elements - Uranium, Polonium, Astatine, Randon, Francium, Radium.

Uranium 235 is one of the most popular isotopes of uranium that is used to prepare nuclear energy in power plants.

The rays of radiation are classified as Alpha, Beta and Gamma rays.

In 1899, **Ernest Rutherford** demonstrated that there were at least two distinct types of radiation: **alpha radiation** and **beta radiation**.

Alpha particles consist of two protons and two neutrons. Alpha particles are double charged ions of **helium**.

Beta particles are high energy, fast electrons (-) or positrons (+) released from the nucleus by some radioactive elements during a kind of radioactive decay.

"Positron" is a positively charged particle, discovered by Carl Anderson in 1932 whose mass appears to be equal to the mass of an electron.

If a proton is converted to a neutron, it is known as β^+ decay. Similarly, if a neutron is converted to a proton, it is known as β^- decay.

Electromagnetic spectrum in order from

highest to lowest energy - gamma rays, X-rays, ultraviolet radiation, visible light, infrared radiation, and radio waves.

Gamma rays were first observed in 1900 by the French chemist Paul Villard. Gamma rays are a form of electromagnetic radiation like x-rays, except much higher in energy. Gamma rays have the highest energy and shortest wavelength.

Radio waves have the lowest energy and longest wavelength.

The function of moderator and coolant is performed by **heavy water** in the Canada Deuterium Uranium (CANDU) reactor. CANDU is a natural uranium-fuelled reactor.

There are three isotopes of hydrogen namely, protium (P), deuterium (D) and tritium (T). Tritium is radioactive in nature which emits low-energy beta particles. It has one proton and two neutrons in its nuclei.

Hans Geiger and **J. M. Nuttall** developed the Geiger-Nuttall rule in 1910. It relates the decay constant of a radioactive isotope with the energy of the alpha particles emitted. It states that short-lived isotopes emit more energetic alpha particles than long-lived ones.

^{14}C is a radioactive isotope which has a half life of 5770 years, which is commonly used to estimate the age of organic materials such as paper and parchment.

Fuels

Fuel is any substance that can provide heat and produce energy when it is burned. This energy that releases is generally in the form of chemical energy or heat energy. Example - fossil fuels, biogas, nuclear energy.

Non-renewable source of energy is one which is exhaustible. Example - Fossil Fuels.

Fossil Fuels are the dead and decayed remains of plants and animals subjected to decades of pressure and temperature under the earth's crust.

Primarily fossil fuels are hydrocarbons. Types of fossil fuels are coal, oil and natural gas.

Plant and animal deposits are mostly found in sedimentary rocks and these

deposits lead to the formation of fossil fuels like petroleum.

Natural Gas is a fossil fuel. It occurs naturally and approximately consists of 95% hydrocarbon methane, the other 5% consists of nitrogen, carbon dioxide, helium or hydrogen sulfide.

Compressed Natural Gas (CNG) is composed primarily of methane (CH_4). Methane is called Marsh gas.

LPG is a **Liquefied Petroleum Gas**. It is most commonly used for cooking, in heating appliances and in vehicles. The components of LPG are Butane and Propane.

Fractional Distillation is used in oil refineries to separate the crude oil into petrol, diesel, kerosene oil and other petroleum products.

Coal is a black hard rock, flammable in nature. It belongs to the category of solid fossil fuels because of its ignition property.

It consists of 65-95% carbon. The other constituents include hydrogen, sulphur, oxygen and nitrogen. Coal that is used commercially is bituminous coal. This coal is graded based on plasticity, moisture content, volatility and ash content.

Anthracite is the highest rank of coal. It is a hard, brittle and black lustrous coal. It is often referred to as hard coal, containing a high percentage of fixed carbon and a low percentage of volatile matter. It contains 80 to 95 percent carbon.

Bituminous coal or black coal is a relatively soft coal containing a tar-like substance. It is soft coal. It contains 40 to 80 percent carbon.

Lignite is formed from compressed peat, and is often referred to as brown coal. Lignite is a low ranking and highly volatile coal which is used mainly in power stations. It contains 40 to 55 percent carbon.

Peat is the middle stage in the process of coal formation. It has more impurities. The parts of the wood are remarkably present in it. It contains less than 40 to 55 percent carbon.

Coke is the purest form of carbon as 98% of coke is carbon. Coke is a fuel with few impurities and a high carbon content.

Coal Tar is a thick dark liquid which is a by-product of the production of coke and coal gas from coal.

Coal gas is a gaseous mixture, mainly consisting of hydrogen, methane, and carbon monoxide formed by the destructive distillation of bituminous coal and used as a fuel.

When Coal or charcoal is burnt in an unventilated room without sufficient oxygen supply, carbon monoxide gas will be produced and it will lead to poisoning living things.

Biogas, a clean-burning renewable fuel, is made by decomposing food scraps and animal waste without oxygen (anaerobic digestion). Biogas converts the harmful methane gas produced during decomposition, into less harmful carbon dioxide gas.

The components in biogas are as follows: (**Methane 50-75%**), (Carbon dioxide 25-50%), (Nitrogen 0-10%), (Hydrogen 0-1%). Hence, methane is the major component in biogas.

The energy produced by the combustion of one kg of fuel is known as its **calorific value**.

Calorific value of some fuels - (Cow Dung, 8000 kJ/kg), (Wood, 22000 kJ/kg), (Coal, 33000 kJ/kg), (Biogas, 40000 kJ/kg), (Diesel, 45000 kJ/kg), (Kerosene, 45000 kJ/kg), (Petrol, 45000 kJ/kg), (Methane, 50000 kJ/kg), (LPG, 55000 kJ/kg), (Hydrogen, 150000 kJ/kg).

Wobbe Index indicates the heating characteristics of fuel gases and is used to compare the combustion energy output.

Polymers

Polymers are high molecular mass substances consisting of large numbers of repeating structural units. They are also called macromolecules. Examples - Polythene, bakelite, rubber, nylon 66.

Semi-synthetic polymers are those that are derived from nature itself but are made to undergo chemical processes to enhance their quality. Some semi-synthetic polymers examples are Rayon, vulcanized rubber and gun cotton.

Rayon (artificial silk) is a synthetic fiber manufactured from the cellulose of wood pulp or cotton. Used in making various articles of clothing such as blankets,

sheets and curtains.

Nylons (polyamides): It is the collective name for synthetic polymers based on certain aliphatic compounds. Used in dental brushes, clothing, socks. Like other thermoplastics, it melts at high temperatures.

Acrylic is a transparent plastic material with outstanding strength, stiffness, and optical clarity.

Polyester is a synthetic or man-made fiber material. Its technical name is polyethylene terephthalate (PET). It is made by mixing ethylene glycol and terephthalic acid. It contains the ester functional group in every repeat unit of their main chain.

Carbon Fiber (graphite fiber): It is a very strong material and lightweight.

Natural rubber becomes soft and sticky at high temperatures and brittle at low temperatures.

Vulcanization is a process of heating a mixture of raw rubber with sulphur and an appropriate additive at a temperature range between 373K and 415K to improve physical properties of rubber.

Natural fibres - Fibres obtained from natural sources like plants and animals. Examples - Cotton, Silk, Wool, Linen, Jute and Hemp.

Artificial fibres - Fibres which are manufactured in laboratories. Examples - Rayon, Nylon, Polyester, Acrylic and Aramid.

Plastics are a wide range of synthetic or semi-synthetic materials that use polymers as a main ingredient. Plastics can be divided into two major categories: thermoplastics and thermosetting plastics.

Thermoplastics are a class of polymers that can be deformed easily and melted by the application of heat, and can be processed either in the heat-softened state or in the liquid state. Thermoplastic substances can be recycled or reused. Examples - Acrylic, polyester, polypropylene, polystyrene, nylon and Teflon.

A thermosetting polymer is a permanent setting polymer as it hardens and sets during the moulding process and cannot be softened again. Such a material is also known as a thermoset or

thermosetting plastics. The examples are **bakelite** and melamine-formaldehyde polymers.

Thermo resistant plastic is generally any plastic that has a continuous-use temperature of above 150°C (302°F) or a short-term exposure resistance of 250°C (482°F) or more.

Bioplastics are plastic materials produced from renewable biomass sources, such as vegetable fats and oils, corn starch, straw, woodchips, sawdust, recycled food waste.

Polytetrafluoroethylene (PTFE) is a synthetic fluoropolymer of tetrafluoroethylene (**Teflon**).

Teflon is a brand name for a type of plastic that is very smooth and does not have chemical reactions with other substances.

It is used in industry and as a surface for cooking pans so that food does not stick to the pan. Polytetrafluoroethylene (PTFE) was discovered in 1938 by **Roy Plunkett**.

Discoveries

Ernest Rutherford was awarded the Nobel Prize in Chemistry **1908** "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances". In 1911, Ernest Rutherford published the results of the gold-foil experiment.

Positron was discovered by Carl Anderson in 1932.

Thomas Alva Edison invented the electric bulb.

Issac Newton gave three laws of motion and universal law of gravity.

Nikola Tesla is best known for his contributions to the design of the modern alternating current (AC) electricity supply system.

Joseph Proust is best known for his enunciation of the law of definite proportions or law of constant proportions in 1797. He summarised his experiment in 1806.

Robert Boyle discovered that the volume of a gas decreases with increasing pressure and vice versa—the famous Boyle's law in 1662.

Berzelius discovered three chemical elements: cerium (1803), selenium (1817), thorium (1828) and devised the modern method by which one or two letters are used to symbolize the elements.

Antonie Lavoisier is most noted for his discovery of 'the role of oxygen plays in combustion'. He recognized oxygen (1778) and hydrogen (1783). He is known as "the father of modern chemistry".

Neil Bohr and Max Planck gave quantum theory and they were awarded the Nobel Prize for their research on the structure of atoms in 1922.

Amedeo Avogadro is famous for his research on gas volume, pressure, and temperature which he discovered in 1811.

Friedrich Wohler showed that heating silver cyanide (an inorganic compound) with ammonium chloride (another inorganic compound) produced urea without the aid of any living organism or part of a living organism, in 1828.

Robert Millikan discovered the elementary charge of an electron using the oil-drop experiment.

George Brandt discovered Cobalt in 1735.

Wilhelm Ostwald, founder of Physical chemistry, explained the role of catalyst in a reaction.

Roger Boscovich gave the first coherent description of atomic theory in 1758.

Fredrick Sanger is the Scientist to win two Nobel prizes in Chemistry. He won his first Nobel prize in 1958 and second in 1980.

Johann Wolfgang Dobereiner tried to sort the elements with the same properties in groups of three elements (triads) in 1817.

Sir William Crookes discovered the element Thallium (1861) and invented the radiometer.

Enrico Fermi is called the "architect of the nuclear age" and the "architect of the atomic bomb".

Germanium was discovered by Clemens Winkler in 1886.

Beryllium was discovered by Louis

Nicolas Vauquelin in 1798.

Alessandro Volta was the first person to isolate methane and discovered methane mixed with air could be exploded using an electric spark. This is the basis of the internal combustion engine.

Auguste Laurent is known for discoveries of anthracene (1832), phthalic acid (1836) and carbolic acid. He synthesized a large number of aromatic organic compounds.

Hennig Brandt is known for the discovery of phosphorus in 1669.

Langmuir reforms the Lewis concept by abandoning the idea of a stable cubical arrangement of the octet and introducing the term 'covalent bond' in 1919.

Paul Dirac was known to have introduced the concept of Antiparticle in 1930.

Victor Franz Hess discovered cosmic rays in 1912.

Alfred Werner was the first to understand the molecular structures of inorganic substances in 1893.

Sir William Ramsay discovered helium in 1895.

William Ramsay with **Lord Rayleigh** showed in 1894 that air also contained an element unknown until then, which was named Argon.

George de Hevesy with **Dirk Coster** discovered the element hafnium (Hf) in 1923.

Newlands stated the law of octaves in 1865. He was awarded the Davy medal by Royal Society, London, in 1887 for the discovery of the periodic law of chemical elements.

Lothar Meyer developed a first periodic classification of the chemical element in 1862.

Mendeleev was considered the father of the Periodic table.

AEB de Chancourtois prepared Vis Tellurique (an early form of the periodic table, 1862). He was the first scientist to see the periodicity of elements when they were arranged in order of their atomic weights.

Oxygen was discovered by Joseph Priestley (1774).

Lev Davidovich Landau received the

1962 Nobel Prize in Physics for his development of a mathematical theory of superfluidity that accounts for the properties of liquid helium.

Edwin Mattison McMillan and Glenn Theodore Seaborg discovered transuranium elements. The Nobel Prize in Chemistry 1951 was awarded to them jointly.

Plutonium is an actinide, discovered by **Glenn T Seaborg** in 1940 that is used as a heat source for sensitive electrical components in satellites.

Petrus (Peter) Josephus Wilhelmus Debye won the Nobel Prize in Chemistry 1936 for studying the molecular structure through the investigation on dipole moments and the diffraction of X-rays and electrons in gasses.

Sir William Ramsay and **Morris W. Travers** discovered Neon in 1898.

In 1929, **Ernest Lawrence** and **MS Livingston** invented the cyclotron.

Deuterium, a stable isotope of hydrogen, was discovered in 1932 by Harold C. Urey.

Tritium, an unstable isotope of hydrogen, was discovered in 1934 by Ernest Rutherford, M.L. Oliphant, and Paul Harteck.

Prafulla Chandra Ray known as "Father of Indian Chemistry". He discovered the stable compound Mercurous Nitrite in 1895.

Marie and Pierree Curie first discovered Polonium named after Marie homeland Poland in July 1898. Then they discovered Radium. Marie was the first woman to receive the nobel prize in 1903.

In 1820 **Michael Faraday** produced the first known compounds of carbon and chlorine. In 1825 he isolated and described benzene.

In 1800, **Alessandro Volta** invented the electric battery (known as the voltaic pile).

Emil Fischer in 1891 determined the structures of glucose and seven other D-aldohexoses. Emil Fischer discovered purines.

Lithium was discovered by **Johan Arfvedson** in 1817.

Marcellin Berthelot French chemist was one of the first scientists to synthesize organic compounds such as formic acid, methane and acetylene from their elements.

Adolph Wilhelm Hermann Kolbe was a major contributor to the birth of modern organic chemistry. He published a method for the synthesis of acetic acid, a natural product containing two carbon atoms.

Antonie Lavoisier published the book 'Methode de nomenclature (System of Chemical Nomenclature)' along with Claude Berthollet, and Louis Bernard Guyton that determined the modern method of naming substances.

Nobel Prize-winning chemist **Herbert Brown**, who shared the 1979 Nobel prize in chemistry 1979 with Georg Wittig "for their development of the use of boron and phosphorus-containing compounds, respectively, into important reagents in organic synthesis". He discovered that diborane reacts with aldehyde and ketone to produce dialkoxyboranes, which are hydrolysed by water to produce alcohols.

Erich Huckel proposed Huckel's Rule in 1931.

Alfred Nobel was a swedish chemist who discovered dynamite. It also instituted the nobel awards which are awarded to individuals who have contributed to the development of humankind.

Fluorine was arbitrarily assigned a value of 4.0 in the 'halogens' section in 1922 by **Linus Pauling**, an American scientist.

A **Pauling scale** was introduced in 1932 that determined the electronegativity of fluorine to be 4.0.

Carl Wieman with **Eric Cornell** in 1995 experimentally produced the first Bose Einstein condensate in a rarefied gas of rubidium atoms at extremely low temperatures.

Albert Einstein predicted a new state of matter, the Bose-Einstein condensate (BEC), based on a quantum formulation by the Indian physicist Satyendra Nath Bose in 1924.

Geiger and Nutall found an empirical relationship between the half-life of alpha decay and the energy of the emitted alpha particles in 1911.

Thomas Andrews, in 1860, undertook a thorough investigation of the gas laws expressing the relationship of pressure, temperature and volume in carbon dioxide.

In 1937, **Edward Davies Hughes along with Sir Christopher Ingold** proposed nucleophilic substitution of alkyl halides and related compounds.

In 1950, **Otto Schmidt**, a Russian astronomer revised the 'nebular hypothesis' proposing that the sun is surrounded by a solar nebula consisting mainly of hydrogen and helium as well as dust.

The atomic number Bohrium is 107 which is named after a physicist **Neil Bohr**.

Sidgwick and Powell first introduced the idea of a relationship between molecular geometry and the number of valence electrons in the 1940s.

The equation $C = KP_{\text{gas}}$ was formulated in 1803 by **William Henry**.

Dennis Gabor is known for developing the principle of holography in 1947 to improve the resolution of the electron microscope.

In 1784, **Henry Cavendish** discovered the composition of water through his experiment with hydrogen and oxygen.

Philipp Lenard conducted an experiment in the late 19th century and observed that a glass tube filled with a low-pressure gas emitted radiation when a voltage was applied between two metal plates.

In 1957, **Nyholm and Gillespie** redefined a model that predicted the shape of individual molecules based on the extent of electron-pair electrostatic repulsion.

In 1932, **Carl Anderson** discovered the positron, a particle described as 'carrying a positive charge but having a mass of the same order of magnitude as that normally possessed by a free negative electron'.

In 1848, **Louis Pasteur** discovered molecular chirality and observed that crystals of some compounds exist as mirror images which laid the foundation for modern stereochemistry.

Erich Huckel proposed that an aromatic compound must have an odd number of pairs of electrons, which can mathematically be written as $4n + 2$ ($n =$

0, 1, 2, 3 etc.) in 1931.

HC Brown discovered that diborane reacts with aldehydes and ketones to produce dialkoxyboranes, which are hydrolysed by water to produce alcohols.

Vladimir Prelog received the Nobel Prize 'for his research in the stereochemistry of organic molecules and reactions' in 1975.

Chemistry in Everyday Life

The sticks of safety matches consist of a mixture of **Antimony trisulphide** and **Potassium chlorate**. Sweden is associated with the invention of safety matches.

Antimony trisulfide (Sb_2S_3) is found in nature as the crystalline mineral stibnite and the amorphous red mineral metastibnite. It is used in manufacturing of safety matches, military ammunition, and explosives.

Potassium sulphides are important intermediates in many pyrotechnic effects and some glitter formulations.

Nitrate esters are used in making explosives. Some nitrate ester explosives are pentaerythritol tetranitrate (PETN), erythritol tetranitrate (ETN), and mannitol hexanitrate (MHN).

Rancidity is a condition in which fat and oils present in food are oxidised and gives it an unpleasant flavour and odour. When unsaturated fats are exposed to air it converts into **hydroperoxide**. Oxygen gas causes rancidity in food products.

Lactic acid buildup causes muscle fatigue and soreness. Some common foods that naturally contain lactic acid are curd and sour Milk, pickled vegetables, kefir, yoghurt, cheese, miso, kimchi, and sauerkraut.

Acetic acid produced by Humans naturally in small amounts. It plays an important role in the **metabolism** of fats and carbohydrates in the body.

Nitric acid (HNO_3) is used in the purification of gold and silver since it dissolves the impurities like various metal oxides present on the surface of gold. Gold is insoluble in HNO_3 .

Hydrochloric acid aids digestion by supplying H^+ which activates pepsinogen.

Lemon is used as a natural food preservative.

Malic acid is most abundantly found in apples, other sources of malic acid are grapes, watermelons, cherries, carrots and broccoli. The dominant element found in apples is Potassium (K). Malic acid is the primary acid present in **ripe bananas**.

Latex is composed of water and rubber material. It is most commonly used to make items such as gloves, swim caps, chewing gum, and mattresses.

Terylene fibre is used, as polyester knit, as a fashion garments fabric. It is mainly used in making plastic bottles, sheets, ropes, nets and sails.

Polystyrene is widely used to protect consumer products. CD and DVD cases, food packaging, meat poultry trays and egg cartons.

Pentane is used to create a foam known as polystyrene.

Ethylene gas, acetylene gas liberated from calcium carbide, and ethephon are used for artificial ripening of fruits.

Chloretone acts as a hypnotic and nervous sedative.

Thinner is a volatile solvent used to make paint or other solutions less viscous. Thinner used in plastic paint is water.

Diclofenac, a veterinary drug, is largely responsible for nearly wiping out vultures from India.

Tylenol known as acetaminophen is used to relieve mild to moderate pain in the body.

Eszopiclone (Lunesta) is used in the treatment of insomnia.

Adderall can help increase your ability to pay attention, stay focused on an activity, and control behaviour problems.

Oxidising bleaching agents that do not contain chlorine are usually based on peroxides such as hydrogen peroxide, sodium percarbonate, and sodium perborate.

Hydrogen peroxide is used as a bleaching agent in some foods such as wheat flour, edible oil, and egg white.

Lindane is the gamma-isomer of

Benzene hexachloride. It is used in some creams and shampoos for the treatment of lice infection.

Freon is a generic descriptor for a number of halocarbon products. They are stable, nonflammable, low toxicity gases or liquids which have generally been used as refrigerants and as aerosol sprays.

These include the chlorofluorocarbons (CFCs) that cause ozone depletion and chlorodifluoromethane (HCFCs).

Bromomethane is used to kill a variety of pests including rats, insects, and fungi. It is also used to make other chemicals or as a solvent to get oil out of nuts, seeds, and wool.

The **six elements** of life are Carbon, Hydrogen, Nitrogen, Oxygen, Phosphorus, and Sulphur. These six elements are the most abundant element in cells, thus they are called macronutrients.

Methane emissions from livestock (including cattle) are the largest sources of agricultural emissions worldwide. A single cow can make up to 99 kg of methane gas per year.

The substances used in **black and white** photography are Silver bromide and silver chloride.

Plaster of Paris should be stored in a moisture - proof container because it absorbs moisture (or water) to form a hard substance called gypsum which makes Plaster of Paris useless.

Kerosene, also called paraffin oil, is a flammable hydrocarbon liquid commonly used as a fuel. It is obtained from petroleum.

Antiseptics are the chemical drugs which either kill or prevent the growth of microorganisms.

Boric acid is used as antiseptic.

Phosphoric acid is an ingredient used for electrolyte replenishment and total parenteral nutrition (TPN) therapy, and for the relief of upset stomach associated with nausea.

Coconut oil and **palm kernel oil** have lauric acid which is antimicrobial, antifungal and antiviral.

Seawater constitutes a rich source of **bromine**, magnesium.

Hydrogen can be used in fuel cells to

generate electricity, or power and heat. Nowadays hydrogen is most commonly used in petroleum refining and **fertiliser** production.

Helium is often used to fill decorative party balloons, weather balloons and airships as it is lighter than air.

Carbon dioxide is used for **making soft drinks** because soft drink manufacturers add bubbles by dissolving carbon dioxide into the water at high pressures.

Carbon monoxide (CO) may be used in low oxygen packaging of fresh meat to maintain a satisfactory colour.

Nitrogen gas is used in electric bulbs. A bulb is filled with an Inert gas or Nitrogen.

Argon is an inert gas used in light bulbs. It slows down the evaporation of the **Tungsten filament** compared to operating it in a vacuum.

Nitrogen Gas molecules are larger and slower moving than those of compressed air. So it is used to inflate the car tyres.

Chips packets are filled with **nitrogen** because it is an inert gas, which prevents the oxidation of oils present in chips.

Nitrogen trihydride is the systematic name of **Ammonia** (NH₃). It is used in many industrial processes as a fertiliser (max usage), refrigerant gas, for purification of water supplies, and in the manufacture of plastics, pesticides, dyes.

Urea (CH₄N₂O), a key ingredient in fertilizers, animal feed, and even plastics, is made by combining ammonia and carbon dioxide. It's also known as carbamide.

Sodium is used as a heat exchanger in some nuclear reactors, and as a reagent in the chemical industry.

Salt (NaCl), **sodium nitrite** (NaNO₂), butylated hydroxy anisole (C₁₁H₁₆O₂), butylated hydroxytoluene (C₁₅H₂₄O), tert-butylhydroquinone (C₁₀H₁₄O₂), **vinegar** (CH₃COOH), citric acid (C₆H₈O₇), **spices**, and calcium propionate (C₆H₁₀CaO₄) are all chemicals that preserve foods.

Sodium benzoate is added to the preserving tomato sauce and pickle so that it can be stored for a longer period of time. It is a preservative used mainly in foods that have an acidic pH.

Kitchen salt is used as a natural preservative for pickles that triggers **osmosis**.

Formic acid is in use as a preservative and antibacterial agent, it is also useful in the manufacturing of leather and rubber. It is useful as a miticide by beekeepers.

Calcium metabisulfite is used as natural preservatives for shelf life extension of fruit juices.

Some food items that can be used as natural food preservatives are Garlic, lemon, sugar and vinegar.

Sulphites are used as a preservative to slow browning and discolouration in foods and beverages during preparation, storage and distribution.

Potassium meta bisulphite is used as a food preservative in some sauces and beverages, retaining their natural color and protecting against bacteria.

Some common **food preservatives** are sugar, garlic, sodium metabisulphite, potassium nitrate, Erythorbic acid, benzoic acid, Sodium Benzoate.

Sodium palmate can be used in bar soaps and cleansers as a gentler alternative to some harsher surfactants.

Sodium Hydroxide is also used to remove the skins of tomatoes, potatoes and other fruits and vegetables for canning and as an ingredient in food preservatives that helps prevent mold and bacteria from growing in food.

Sodium thiosulphate (Na₂S₂O₃·xH₂O) is used in photography, it is used as a fixing agent to dissolve the silver salts from the negatives.

Sodium sulphate (Na₂SO₄) - Glauber's salt is a decahydrate sodium sulfate.

Sodium Peroxide (Na₂O₂) is a yellowish-white powder. It is used as a bleaching agent, disinfectant, laboratory chemical, and in the production of other chemicals.

Sodium carbonate is the dominant chemical present in detergent powder. It is used to remove permanent hardness of water. Sodium carbonate (Na₂CO₃) is used in the manufacture of sodium compounds such as borax (Na₂H₂₀B₄O₁₇).

Sodium chlorite (NaClO₂) is a chemical compound used in the manufacturing of paper and as a disinfectant.

Sodium ferrocyanide can also be made

from the reaction among calcium cyanide, iron sulphate and sodium carbonate. It is a common **anti-caking agent**. It is used as a stabiliser for the coating on welding rods.

Disodium phosphate is made by combining phosphoric acid, derived from phosphate rock, with soda ash. It can be used to regulate the acidity of food, stabilise it, thicken it and also maintain moisture level.

Magnesium bisulphite ($\text{H}_2\text{MgO}_6\text{S}_2$) is used in the enzymatic production of fructose.

Magnesium silicate is widely used to remove free fatty acids and other polar compounds from used cooking oils to extend their life.

Magnesium Hydroxide ($\text{Mg}(\text{OH})_2$) is used to treat constipation and acid indigestion.

Magnesium Oxide (MgO) can be used as an antacid to relieve heartburn, sour stomach, or acid indigestion.

Magnesium chloride (MgCl_2) is a mineral supplement used to increase your intake of magnesium.

Aluminium carbide can be used as an abrasive in high-speed cutting tools.

Aluminium phosphide is a highly toxic inorganic compound, primarily used as a pesticide and as fumigant for cereals and as an igniting agent in flares.

Aluminium chloride is used widely in manufacturing rubber, lubricants, wood preservatives, and paints. It is used in pesticides and pharmaceuticals.

Potassium helps regulate fluid balance, muscle contractions and nerve signals.

Potassium sulphate is generally used as fertiliser.

Potassium chloride: White crystalline solid, soluble in water. Used as fertilizer, in food processing, to harden glass, in plastics production, and as a saline solution for IV injection. It is a mineral supplement used to treat or prevent low levels of potassium in the blood.

Sulphur is found in some plants including fruits and vegetables. The characteristic garlicky odour of garlic is due to Sulphur. Uses of sulfur include car batteries, fertilizers, oil refining, water processing, and mineral extraction.

Sulphur is used in the vulcanisation of black rubber, as a fungicide and in black

gunpowder. Sulphur compounds in petroleum can cause corrosion to parts of internal combustion engines and refineries.

Fluorine is used in rocket fuels. Fluorine is a very reactive element which makes it good for very explosive reactions. It is found as fluoride in toothpaste and drinking water.

Calcium chloride (CaCl_2) is used as a firming agent in canned vegetables, to firm soybean curds into tofu, and to make a caviar substitute from vegetable or fruit juices. It is used in waste water treatment applications to remove undesired impurities.

Calcium hydroxide is an odorless white powder. It also has medical and dental uses. For example, root canal fillings often contain calcium hydroxide.

Whitewashed walls appear to be shiny after 3-4 days due to the formation of **calcium carbonate** (CaCO_3).

Calcium carbonate (also known as chalk), mined as calcite, is the most commonly used filler for PVC.

Calcium nitrate is used in fertilisers, explosives and pyrotechnics. It is used to make fireworks and incandescent lamp chimneys.

Calcium nitrate ion leads to formation of iron hydroxide, whose protective layer reduces corrosion of the concrete reinforcement.

Calcium silicate (Ca_2SiO_4), also known as calcium silicon oxide, is a product derived from limestone and diatomaceous earth, and is used as an antacid and as an anticaking agent in the preparation of food.

Limestone is usually added to acid soils to raise soil pH.

Copper sulphate is used as a drying agent in the anhydrous form, as an additive for fertilisers and foods, and several industrial applications such as textiles, leather, wood, batteries, ink, petroleum, paint, and metal.

Heated calcium sulphate forms two types of compounds on hydration. The compound with half a water of crystallisation is also known as **bassanite**.

Zinc supports a number of functions in the human body. It plays a major role in the creation of DNA, growth of cells,

building proteins, healing damaged tissue, and helping to build a healthy immune system.

Cadmium is a natural element found in tiny amounts in air, water, soil, and food. All soils and rocks, including coal and mineral fertilisers, contain some cadmium.

Barium is a soft, silvery metal that rapidly tarnishes in air and reacts with water. It produces green colour in fireworks and can also help stabilise other volatile elements.

Sodium produces yellow colour. Copper produces blue colour and Zinc produces bluish-white colour.

Bromine: Liquid at room temp, brownish-red with bleach-like odor, dissolves in water. Used in swimming pools as chlorine alternative, agriculture, sanitation, and fire retardants.

Bromopropane is a colourless liquid. It is currently used as a solvent in the adhesives, dry cleaning, vapour degreasing, and electronic and metal cleaning industries.

Silver is used for jewellery and silver tableware, where appearance is important. It is also used to make mirrors, as it is the best reflector of visible light known.

Iron is a mineral that the body needs for growth and development to make haemoglobin.

Iron carbide (Fe_3C), commonly known as cementite, is used in welding to create a hard, wear-resistant coating on metals.

Iodine is a mineral found in some foods. The body needs iodine to make thyroid hormones. Iodine deficiency is the most common cause of Goiters (irregular growth of the thyroid gland).

Iodide salts are used in pharmaceuticals and disinfectants, printing inks and dyes, catalysts, animal feed supplements and photographic chemicals.

Solar cell, also called photovoltaic cell, a device that directly converts the energy of light into electrical energy through the photovoltaic effect.

Silicon is used to make solar cells and silver is used for interconnection of the cells in the panel. Silicon dioxide is found mostly in Sand.

Neon is used in electrically controlled advertisement signboards so as to

enable the readability of its contents during the night.

Xenon gas is used as anesthesia. Xenon exerts its anesthetic properties through the noncompetitive inhibition of N-methyl-D-aspartate receptors.

Hardness in water is generally caused by compounds of Calcium (Ca^{2+}) and Magnesium (Mg^{2+}). Hardness of water is expressed in terms of ppm.

Temporary hardness is caused by the dissolved bicarbonates of calcium and magnesium. **Permanent hardness** is caused by the dissolved sulphates and chlorides of calcium and magnesium.

Public water treatment cleans harmful substances using a multi-step process: coagulation-flocculation, sedimentation (settling), filtration, and disinfection, to make water safe for consumption.

Central Ground Water Board guidelines state that the pH of tap water should be between **6.5 and 8.5**.

Water disinfection means the removal, deactivation or killing of pathogenic microorganisms.

Chlorination is the process of adding chlorine to drinking water to kill parasites, bacteria, and viruses.

Water pollutants include contamination due to domestic wastes, food processing waste, insecticides and herbicides, volatile organic compounds (VOCs), pollutants from livestock operations, heavy metals, and chemical waste.

Pure water contains 88.89% of oxygen by mass and 11.11% of hydrogen. This is explained by the law of Constant Composition.

Potable water is stored water that is suitable for human consumption.

Distilled water is water that has been boiled into vapor and condensed back into liquid in a separate container.

Dehydration is a condition caused by the loss of too much fluid from the body. It happens when the body loses more fluids out than taking in, and the body does not have enough fluids to work properly.

Cooking vessels are made from metals like copper, aluminium as they conduct heat well, and are chemically unreactive, so that they do not alter the flavour of the

food.

Perishable foods are those likely to spoil, decay or become unsafe to consume if not kept refrigerated at **40 °F** or below, or frozen at 0°F or below.

Eugenol, also called clove oil, is an aromatic oil extracted from cloves that is used widely as a flavouring for foods and teas.

Casein and **whey protein** are the major proteins of milk. Casein constitutes approximately 80% (29.5 g/L) of the total protein in bovine milk, and whey protein accounts for about 20% (6.3 g/L).

Curcumin, is a naturally orange/yellow colour extracted from the turmeric (*Curcuma longa*) root, which is used as a spice, in food colouring and as a traditional herbal medicine.

Ginger is used to treat nausea and vomiting from motion sickness, pregnancy, and cancer chemotherapy.

Black pepper contains a chemical called piperine. It seems to reduce pain, improve breathing, and reduce inflammation.

Allicin is one of the main active sulphur-containing compounds present in garlic and onion. Allicin is an oily, slightly yellow liquid that gives garlic its unique odour.

Maltose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is a disaccharide consisting of two units of glucose.

Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is a simple sugar monosaccharide having two isoforms, alpha and beta.

Starch ($\text{C}_6\text{H}_{10}\text{O}_5$)_n is a polysaccharide produced by most green plants for energy storage.

Lactose (glucose + galactose) is generally called as milk sugar as it is found in milk.

Glucose is generally called blood sugar as it is found in fruit (including berries), vegetables and honey.

Fructose is a generally called fruit sugar found primarily in fruits (such as apples, dates, figs, pears and prunes).

Maltose is a disaccharide that's made up of 2 glucose units joined together. It is found naturally in a range of different foods (like cooked sweet potato, pears and honey).

Honey is composed of water and types of sugar that is a combination of glucose and fructose.

Dextrose (a form of glucose) and levulose (also known as fructose) rotate the plane of linearly polarised light in clockwise and counterclockwise directions.

Glucose and **sucrose** are natural sweeteners.

Aspartame ($\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5$) is an artificial non-saccharide sweetener 200 times sweeter than sucrose and is commonly used as a sugar substitute in foods and beverages. It is used in ingredients in beverages, dairy products, nutrition bars, desserts, chewing gum, sauces and syrups.

Sucralose ($\text{C}_{12}\text{H}_{19}\text{Cl}_3\text{O}_8$) is a no-calorie sweetener that can be used to lower one's intake of added sugars while still providing satisfaction from enjoying the taste of something sweet.

Neotame ($\text{C}_{20}\text{H}_{30}\text{N}_2\text{O}_5$) - 8000 times sweeter than sucrose.

The active ingredient used in cough syrups is usually **dextromethorphan** (DM). Other cough suppressants include camphor, eucalyptus oil, and menthol.

Beta carotene is a red-orange pigment found in plants and fruits, and acts as a powerful antioxidant that can benefit our brain, skin, lungs, and eye health.

The human body converts beta carotene into vitamin A.

Lutein and **zeaxanthin** work as supportive antioxidants in our skin.

Quinoline Yellow is generally artificial colouring agents used in dietary supplements, sauces, soups and broths, bakery, dairy fats and oils, seafood, condiments, breath fresheners, desserts, convenience foods and beverages.

Acid fuchsin is an acidic magenta dye.

Rock salt is the common name for halite.

Orange G is a valuable acid dye that finds application in many staining processes.

Oil Red O is a fat-soluble dye that stains neutral triglycerides and lipids.

Gelatinisation process occurs when starch granules are heated in a liquid, causing them to swell and burst, which results in the liquid thickening.

Dextrinization is the process involving

the browning of starch foods when subjected to dry heat.

Maillard reaction is an organic chemical reaction in which reducing sugars react with amino acids to form a complex mixture of compounds.

Epsom salt (magnesium sulphate): It is composed of magnesium, sulphur and oxygen and is commonly used in detox baths to relax muscles and relieve pain in the shoulders, neck, back and scalp.

Potash is the group of minerals and chemicals that contain potassium (K), mostly used as fertiliser.

Phosphine (PH_3) is used in the semiconductor industry to introduce phosphorus into silicon crystals.

Saw dust (wood dust) is used to adulterate coriander powder. It is a waste product of woodworking operations such as sawing, sanding, milling, planing, and routing.

Blast furnaces, pig iron, and shaping metal are a part of the process of the manufacture of steel.

Ginning is the process of removing the seeds and debris from cotton.

Alitame ($\text{C}_{14}\text{H}_{25}\text{N}_3\text{O}_4\text{S}$) is currently marketed in some countries under the brand name Aclame. It is 2000 times sweeter than sucrose, 12 times sweeter than aspartame ($\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5$), and six times sweeter than saccharin ($\text{C}_7\text{H}_5\text{NO}_3\text{S}$) and Sucralose ($\text{C}_{12}\text{H}_{19}\text{Cl}_3\text{O}_8$).

Water fluoridation prevents tooth decay safely and effectively by making use of water with consistent contact with low levels of fluoride. Fluoride helps to rebuild and strengthen the tooth's surface, or enamel.

Salt iodisation has been the preferred strategy to control and eliminate iodine deficiency disorders.

The gradual process of coal formation from dead vegetation is known as **carbonisation**.

Acidic drain cleaners made of sulphuric or hydrochloric acid are powerful enough to clear heavy-duty hair, food, grease, soap scum or paper-based clogs in 15 minutes or less.

Acrylic is a transparent plastic material with outstanding strength, stiffness, and optical clarity. It is used in lenses, acrylic nails, paint, security barriers, medical devices, LCD screens, and furniture.

Ethanol is used in cough syrups and tonics. It is naturally produced by the fermentation of sugars by yeasts (scientific name *Saccharomyces cerevisiae*) or via petrochemical processes such as ethylene hydration.

Dipropylene Glycol is used to absorb extra water and maintain moisture in certain medicines, cosmetics, or food products.

Butter is about 80% fat, and the rest is mostly water.

Convection is a process in which hot, less dense materials rise upward and are replaced by colder, more dense materials.

Toothpastes generally contain the components such as water, abrasives, including aluminium hydroxide, calcium hydrogen phosphates, limestone, silica and hydroxyapatite.

Fluoride (usually 1450 ppm) mainly in the form of sodium fluoride.

Ascorbic acid, also known as vitamin C, is found in citrus fruits, blackcurrants, red peppers, and leafy green vegetables. It functions as an antioxidant.

Maleic acid is an industrial raw material for the production of glyoxylic acid by ozonolysis.

Formic acid or Methanoic acid (HCOOH) is injected into the skin of a person during an ant's bite. Its effect can be neutralized by rubbing a mild base like baking soda solution on the bite area of the skin.

The Maillard reaction takes place during the cooking of meat with contributions from other chemical reactions, including the breakdown of the tetrapyrrole rings of the muscle protein myoglobin. Maillard reactions also occur in dried fruit.

The Finkelstein reaction is a Substitution Nucleophilic Bimolecular reaction that involves the exchange of halogen atoms.

Swartz reaction is used to get alkyl fluorides from alkyl chlorides or alkyl bromides.

Gattermann reaction is a chemical reaction in which aromatic compounds are formulated by a mixture of hydrogen cyanide and hydrogen chloride in the presence of a Lewis acid catalyst such as AlCl_3 .

Gelatin is used as the fining agent in wine. Fining is a practice in which a

material that aids clarification is added to the wine. Some of the most commonly used fining agents for wine are isinglass, egg white, casein, bentonite, carbon.

Baking soda is a **leavening agent** used in baked goods like cakes, muffins, and cookies.

When we cut an onion, the synthase enzyme converts the **amino acid sulfoxides** of the onion into **sulfenic acid**.

The acids found in grapes are tartaric, malic and citric acids.

Black gram proteins are rich in essential amino acids.

Hydrogen sulphide gas has an odor that is similar to the odor of rotten eggs. Some common names for the gas include sewer gas, stink damp, swamp gas and manure gas.

A traditional Battery contains sulphuric acid. Lead and lead dioxide are the active materials on the battery's plates, which react with sulfuric acid in the electrolyte to form lead sulphate.

Manganese dioxide is used to remove green colour from glass that results from the presence of iron.

Bromate, Chlorite, and Chlorate are examples of disinfection byproducts formed at traditional drinking water treatment plants.

Titanium dioxide (TiO_2), is derived from the mineral Ilmenite. It is used as a white pigment in products such as toothpaste.

When an iron nail is dipped into a blue coloured copper sulphate solution, Fe being more reactive than copper, it displaces copper from Copper sulphate and forms Ferrous Sulphate which is greenish blue, and the copper displaces deposits around nails giving it colour of reddish brown.

The word **elektron** in Greek means **amber**, the yellow fossilised resin of evergreen trees.

Capsaicin is the inflammatory compound in pepper spray which causes burning, pain and tears when it comes into contact with a person's eyes.

Gingerols are phenolic compounds in ginger. It has a role as an antineoplastic agent and a plant metabolite.

Antacid is a substance that neutralises stomach acidity and is used to relieve heartburn, indigestion, or an upset stomach.

Agonist - An agonist is a chemical that activates a receptor to produce a biological response. Receptors are cellular proteins whose activation causes the cell to modify. Examples: oxycodone, morphine, heroin, fentanyl etc.

Analgesics are medications that relieve different types of pain.

Antiseptic is an antimicrobial substance that reduces the possibility of infection, sepsis, and putrefaction.

Antibiotics are chemical substances that inhibit the growth of or destroy microorganisms.

Antidepressants are a class of drugs used to treat major depressive disorder, anxiety disorders, chronic pain, and addiction. Examples are citalopram, escitalopram, fluoxetine, etc.

Central Nervous System (CNS)

Depressants - A type of drug that slows down brain activity, which causes the muscles to relax and calms and soothes a person. Examples are benzodiazepines, barbiturates, etc.

Cloud seeding (artificial rain) is the process of spreading silver iodide aerosols, or dry ice (frozen carbon dioxide), potassium iodine, and liquid propane, into the upper part of clouds to try to stimulate the precipitation process and form rain.

The sour taste of curd, lemon juice, orange juice, and vinegar is because of the presence of acids.

Conversion of milk into curd is a type of chemical change.

Arginine is an amino acid that helps the body build protein.

Leucine amino acids can be used by skeletal muscle to give energy during exercise.

Phenylalanine is used for depression, Parkinson's disease, chronic pain, osteoarthritis, alcohol withdrawal symptoms, and a skin disease called vitiligo.

Monosodium glutamate (MSG) is commonly added to Chinese food,

canned vegetables, soups and processed meats to enhance the flavour.

Gloxazone is a drug tested against anaplasmosis and cowdriosis, both tick-borne diseases. It was found to cause toxicity in the kidneys of rats.

Gobar gas is a clean, unpolluted and cheap source of energy in rural areas. It consists of 55-70% methane which is inflammable.

The main greenhouse gases whose concentrations are rising are carbon dioxide, methane, nitrous oxide, hydrochlorofluorocarbons (HCFCs), hydrofluorocarbons (HFCs) and ozone in the lower atmosphere.

Sulphur hexafluoride (SF₆) is a synthetic fluorinated compound with an extremely stable molecular structure known to be the most potent greenhouse gas ever found.

Texanol is an ideal choice for retarder solvents used in coil coatings and high-bake enamels. Texanol balances performance characteristics well for ink applications that require good opening.

Butanol is used in plastics, polymers, lubricants, brake fluids, and synthetic rubber.

Isopropyl rubbing alcohol is also known as spirit. It can be used as a disinfectant and cleaning agent for a house.

Natural pesticides are pesticides that come from plant or mineral derivatives. Examples - Rotenone, pyrethrum, neem, spinosad, abamectin, bacillus thuringiensis, garlic, cinnamon, pepper.

Phosphoric Acid is responsible for the sharp flavour in soft drinks.

The atmosphere of **Venus** is made up mainly of carbon dioxide, and thick white and yellowish clouds of **sulfuric acid**.

Photoreceptors of a printer or a photocopy machine are made of amorphous silicon.

Iodine solution is the most commonly used method to detect starch adulteration in milk.

To check the presence of adulterated sugar solution in honey, a cotton wick dipped in the sample is ignited with a match stick. Adulteration is confirmed if it burns with a crackling sound.

In medicine, **electrode** is a device such as a small metal plate or needle that carries electricity from an instrument to a patient for treatment or surgery. Nitrous oxide gas is used as an anaesthetic in dental surgery.

Saponification value is used to determine the molecular weight of fatty acids.

Pepperoni, ethyl acetate, butyraldehyde and nitrate are some common adulterants used in ice cream.

The IUPAC name of allylamine is 2-Propen-1-amine, which is used to make pharmaceutical and other chemicals.

Chemical Constituents of spices :
Coriander - Linalool, borneol, terpineol, cumene. Saffron - Crocins, safranal, salicylic, vanillic acid. Mustard seeds - Allyl isothiocyanate. Garlic - Allicin. Ginger - Gingerol. Cumin - Cumin aldehyde, Curcuminoids - Turmeric, 1, 8-cineole - Cardamom, Eugenol - Cloves, Piperine - Black pepper.

Bisphenol A is present in hard plastic such as drinking water bottles and many household items.

Pesticides residue is considered a chemical hazard in food.

Esters have a sweet smell like fruits.

Kaolinite is the product as a result of the process of hydrolysis of orthoclase.

Benzaldehydes have a pleasant almond aroma that is commonly used to impart almond flavour to chocolate and baked goods.

Halite (sedimentary mineral) known to everyone as 'salt', usually forms in arid climates where sea water evaporates.

The colour of the outer zone of complete combustion of a candle flame is blue.

The pH of normal milk is 6.7 - 6.9.

Biology

Cell : Basic Unit of life

Robert Hooke discovered cells in cork in 1665. Schleiden and Schwann proposed the cell theory in 1838-39, stating that cells are the basic units of life in all organisms. The study of cells is called cytology.

Cells consist of three main parts: the cell membrane, nucleus, and cytoplasm. The entire content of a living cell is known as protoplasm.

Unicellular organisms are single-celled, like bacteria and protozoa. Multicellular organisms, such as humans and trees, are made of many cells.

Types of cell: Prokaryotic and Eukaryotic

Prokaryotic Cells

Prokaryotic cells lack a membrane-bound nucleus and organelles. Examples : Bacteria and Archaea {cyanobacteria (blue-green algae), Escherichia coli, mycoplasma, Streptococcus, Nostoc, Anabaena}. Length of the bacterial cell varies from 3 to 5 µm.

Prokaryotic cells: Small size, replicate entire genome at once, have single loop DNA with plasmids, composed of nucleic acid, store hereditary information.

Prokaryotes have a complex cell wall, plasma membrane, and cytoplasm. Growth rate is faster.

Eukaryotic Cells

Eukaryotic cells have a nucleus with a nuclear membrane. Examples - animals, plants, fungi, protozoans, and algae.

Eukaryotic cells: Large size (10-100 micrometers), regulated DNA replication with selective origins, multiple chromosomes, and the ability to store hereditary information.

Cell wall is simple (present in plants and fungi). Ribosomes are large. Growth rate is slower. Organelles are present. Plasma membrane and cytoplasm are present.

Parts of the cell

Cell membrane: Discovered by Robert Hooke in 1665. Present in all cells, separating the interior from the external environment.

The cell membrane transports nutrients into the cell and removes toxic substances. It offers structural support and is selectively permeable to certain molecules. Comprising 40-60% proteins, cell membranes play crucial roles in cellular function.

The primary function of **Aquaporins protein** is to transport water across the cell membrane in response to the osmotic gradient created by active solute transport.

Cell walls: Present in plant cells, bacteria, algae, fungi, and some archaea. Rigid, semi-permeable structure adjacent to the cell membrane. Composed of cellulose (in plants) or chitin (in fungi, not in plants).

Cell wall is composed mainly of carbohydrates like cellulose, hemicellulose and lignin in the ratio 4 : 3 : 3 and some fatty substances like wax. Cellulose is responsible for most of the mechanical strength of the cell wall.

Cytoskeleton: Found in the cytoplasm, comprises microfilaments, microtubules, and intermediate filaments. Gives cell shape, anchors organelles, and facilitates cell movement.

Nucleus: Discovered by Robert Brown in 1831. Spherical, central organelle, easily visible with microscope staining.

Nucleus is separated from the cytoplasm by a double layered membrane called the nuclear membrane. This membrane is also porous and allows the movement of materials between the cytoplasm and the inside of the nucleus [diffusion].

DNA in the nucleus forms chromosomes, tightly coiled around histone proteins for structure support.

Cytoplasm - It is the jelly-like substance present between the cell membrane and the nucleus. It contains organelles like mitochondria, Golgi bodies, ribosomes, each with specific functions.

Cytoplasmic membrane (plasma membrane) is a selective barrier controlling what enters and leaves the cell via diffusion, osmosis, and various transport mechanisms (active/passive), endocytosis, and exocytosis.

Cytosol is known as the matrix of the cytoplasm. It surrounds the cell organelles in eukaryotes. In prokaryotes, all the metabolic reactions occur here.

Protoplasm: Living substance of the cell.

Bacterial cell nucleus lacks organization and a nuclear membrane. Each cell has a membrane to separate its contents from the external environment.

The entire content of a living cell is known as protoplasm [cytoplasm + nucleus].

Nucleoid : In some organisms like bacteria, the nuclear region of the cell may be poorly defined due to the absence of a nuclear membrane. Such an undefined nuclear region containing only nucleic acids is called a nucleoid.

Vacuole: Empty structure in the cytoplasm is called vacuole. Varies in size; large in plant cells (e.g., onion), smaller in animal cells (e.g., cheek). Stores waste in animals, maintains water balance in plants.

Vacuoles are storage sacs for solid or liquid contents. In plant cells vacuoles are full of cell sap and provide turgidity (swollen and hard) and rigidity to the cell.

Vacuoles in plant cells store essential substances like amino acids, sugars, organic acids, and proteins.

In single-celled organisms like amoeba, food vacuoles contain consumed food items. Specialized vacuoles in unicellular organisms aid in expelling excess water and waste from the cell.

Endoplasmic Reticulum (ER) - It is a large network of membrane-bound tubes and sheets. It looks like long tubules or round or long bags (vesicles). Largest organelle, similar to plasma membrane.

Key functions: protein synthesis, transport, folding, lipid/steroid synthesis, carb metabolism, calcium storage. Helps build cell membrane (membrane biogenesis), produces enzymes and hormones. Acts as channels for material transport between cytoplasmic regions and nucleus.

The ER has distinct domains: tubules, sheets, and the nuclear envelope. It lacks ribosomes and is involved in fat and steroid hormone production and metabolism.

Rough Endoplasmic Reticulum (RER) : Ribosomes RER looks rough under a microscope because it has particles called ribosomes attached to its surface. contains millions of membrane bound ribosomes that are involved in the production, folding, quality control and dispatch of certain proteins.

Smooth Endoplasmic Reticulum (SER): It produces fats, lipids, phospholipids (like

those in cell membranes), and steroids crucial for cell function.

The Endoplasmic Reticulum acts as a cytoplasmic framework, supporting various biochemical activities within cell.

In the liver cells of the group of animals called vertebrates, SER plays a crucial role in detoxifying many poisons and drugs.

Golgi Apparatus or Golgi Complex - The Golgi apparatus, resembling a stack of membrane-bound vesicles known as cisterns, acts as the cell's post office. Located in the cytoplasm, it processes proteins and lipids for utilization within and outside the cell.

Varied Number of cisternae present in Golgi Apparatus. Cisternae is a membrane-bound cell organelle consisting of a series of flattened membranous sacs that look like stacked pouches.

Principal Functions - Golgi apparatus transport, modify, and package proteins and lipids into vesicles for delivery to target destinations. The golgi apparatus is also involved in the formation of lysosomes.

The Golgi apparatus has concentric cisternae arranged near the nucleus, featuring a convex cis (forming) face and a concave trans (maturing) face.

The Golgi apparatus (discovered by Camillo Golgi in 1898) is located close to the nucleus and can be very large in secretory cells, where it fills almost the complete cytoplasm.

Lysosomes: Discovered by Christian de Duve, act as cell's waste disposal, breaking down material and organelles with hydrolytic enzymes. Known as "suicidal bags" due to potential burst and self-digestion during damage. Enzymes produced by Rough Endoplasmic Reticulum.

Mitochondria: The cell's powerhouse, release energy in the form of ATP molecules, known as the cell's energy currency. They have two membranes.

Mitochondria was coined by **Carl Benda** in 1898. Mitochondria was discovered by **Albert von Kolliker**.

Mitochondria feature folds called cristae in their inner membrane, creating a large surface area for ATP generation. They possess their own DNA and ribosomes, allowing them to synthesize some proteins independently.

Mitochondria cell organelle appears as rounded or sausage shaped structures under electron microscopy measuring about 0.5 - 1.0 μm in diameter and 2 - 8 μm in length. It converts nutrients into energy. It can make copies of itself. It is found in yeast, algae and Fungi.

Plastids - Plastids, found exclusively in plant cells, are small colored bodies in the cytoplasm. Based on the type of pigments plastids can be classified into chloroplasts, chromoplasts and leucoplasts. Similar to mitochondria, plastids have a double-membraned external structure.

Leucoplasts, a type of plastid, serve as storage organelles for starch, oils, and protein granules. They lack pigments, making them colorless. The internal structure of plastids includes multiple membrane layers within a stroma matrix.

Three types - Amyloplasts: Store and synthesize starch. Proteinoplasts: Store proteins, common in seeds. Elaioplasts: Store fats and oils needed by the plant.

Chloroplasts, like mitochondria, conduct photosynthesis and produce energy. They share traits such as endosymbiotic origin, independent genetic systems, and division. Hugo von Mohl discovered chloroplasts.

Chromoplasts are carotenoid - accumulating plastids conferring color to many flowers and fruits as well as to some tubers and roots.

Ribosomes are organelles made of RNA and protein, translating genetic code into amino acid chains. George E. Palade discovered Ribosomes.

Polysomes: These are clusters of ribosomes strung along an mRNA molecule, resembling beads on a thread. They efficiently produce multiple copies of a polypeptide by translating the mRNA's instructions simultaneously.

Centrosome: Cellular structure aiding cell division, controlling motility, adhesion, and polarity during interphase, and organizing spindle poles in mitosis.

Centrioles aid in cell division in animal cells. They typically have a 9 + 0 structure of triple microtubules, arranged in pairs and orthogonal after duplication.

Peroxisomes are organelles managing various oxidative reactions vital for metabolism, detoxification of reactive oxygen species, and signaling. They host

pathways like fatty acid β -oxidation, crucial for processes like embryogenesis, seedling growth, and stomatal opening.

Cell Cycle

The cell cycle encompasses events occurring as a cell grows and divides. Interphase, where the cell spends most of its time, involves growth, chromosome replication, and preparation for division. Following interphase, the cell undergoes mitosis and completes division.

Interphase is the metabolic phase of the cell where it obtains nutrients, grows, replicates DNA for mitosis, and performs normal cell functions.

Phases of Cell cycle - Interphase {G₁ phase (Gap 1), S phase (Synthesis), and G₂ phase (Gap 2)} and Mitosis (M) phase {Prophase, Metaphase, Anaphase, and Telophase}.

S phase is where DNA is replicated, sandwiched between G₁ and G₂ phases in the cell cycle. It's tightly controlled due to the crucial need for accurate genome duplication for successful cell division.

During the G₂ phase of the cell cycle, metabolic changes assemble the cytoplasmic material required for mitosis and cytokinesis.

Mitosis: Part of cell cycle where replicated chromosomes divide into two new nuclei. Results in genetically identical cells with maintained chromosome count. Also called equational division.

Prophase initiates mitosis, where duplicated genetic material is divided into two daughter cells. Chromatin, the DNA-protein complex within the nucleus, condenses during this phase.

Prometaphase, the second phase of mitosis, involves the breakdown of the nuclear envelope, the barrier enclosing the nucleus, facilitating the separation of duplicated genetic material into two daughter cells.

Metaphase is the third phase of mitosis, the process that separates duplicated genetic material carried in the nucleus of a parent cell into two identical daughter cells.

Anaphase is the fourth phase of mitosis, the process that separates the

duplicated genetic material carried in the nucleus of a parent cell into two identical daughter cells.

Telophase: Final phase of cell cycle. Chromosomes move to opposite poles, unwind into DNA strands. Spindle fibers disappear, nuclear membrane reappears.

Cytokinesis is the physical process of cell division, which divides the cytoplasm of a parental cell into two daughter cells.

There are mainly **two types** of cell division, such as Mitosis and Meiosis.

Mitosis is the process by which cells make copies of themselves. It occurs in many cells of the body such as skin and muscles. This creates two new cells that are genetically identical to each other and the original parent cell.

Meiosis produces sperm and egg cells, not identical daughter cells like mitosis. Through two divisions, it generates four haploid cells with half the original genetic material, reducing chromosome number for sperm and egg formation.

During cell division, the formation of a new cell wall is followed by the accumulation of **Phragmoplasts**.

Leptotene: First substage of Prophase I, where replicated chromosomes begin condensing into long strands within the nucleus.

Zygotene: Second phase of Prophase I, where homologous chromosomes pair or synapse.

Pachytene: Third stage of Prophase I in meiosis, where chromosome pairs separate into sister chromatids with gene breakage and crossing over.

Diplotene: Fourth stage of Prophase I in meiosis, where chromatid pairs of tetrads start separating, and chiasmata become visible.

Diakinesis: Final stage of Prophase I, involving the disappearance of the nuclear envelope around the cell.

Synaptonemal complexes are meiosis-specific nuclear structures that mediate the tight association between homologous chromosomes.

Synapsis is the pairing of two homologous chromosomes that occur during meiosis.

Tissues

Tissues (French word 'tissu' means to weave) are groups of cells that have a similar structure and act together to perform a specific function.

Characteristics of Permanent Tissue:

They have cells lacking division ability, well-developed and shaped, with thick cell walls, large nuclei, and dense cytoplasm.

Permanent tissues are of three main types: Parenchyma, Collenchyma, and Sclerenchyma.

Parenchyma: Simple permanent tissue in plants, forms a major part of ground tissues. Non-vascular, composed of simple, living, undifferentiated cells. Modified for various functions.

Collenchyma is plant tissue aiding support and protection, part of ground tissue (parenchyma), crucial for plant structure.

Sclerenchyma tissue is made of tough, thick-walled dead cells with lignin and cellulose, giving plants structural support.

There are 4 basic types of tissue: Connective tissue, epithelial tissue, muscle tissue and nervous tissue.

Reticular tissue, a loose connective tissue with prominent reticular fibers, provides support for bone marrow, liver, and lymphoid organs like the spleen, lymph nodes, and tonsils.

Epithelium is a tissue covering internal and external body surfaces, lining cavities and organs, and is predominant in glands.

Adipose tissue, or body fat, is a connective tissue found under the skin (subcutaneous fat), around organs (visceral fat), and in bone marrow cavities.

Connective tissue types: cartilage, bone, collagen, reticular, elastic fibers, blood, hematopoietic/lymphatic, adipose tissue, bone marrow, lymphoid tissue.

Epithelial tissue is a protective tissue in the animal body. Types: simple squamous, simple cuboidal, simple columnar, and pseudostratified.

Meristem is plant tissue with undifferentiated cells, located in growth zones enabling plant growth.

Important Facts

Membrane proteins are essential for cell function.

Glycolipids are a type of complex lipids comprising carbohydrates, fatty acids, sphingolipids or a glycerol group.

Phagocytosis is a process wherein a cell binds to the item it wants to engulf on the cell surface and draws the item inward while engulfing around it.

Pseudopodia are temporary cell projections used for movement and feeding in organisms like amoebas and white blood cells.

Columnar epithelial cells are elongated and column-shaped and have a height of at least four times their width.

Safranin is used as a counterstain in some staining protocols, colouring cell nuclei red.

Pectin is found in the middle lamella, primary cell walls, and secondary walls of plants. It accumulates during initial cell expansion stages.

Choanocytes are responsible for a number of important metabolic processes in sponges, such as digestion, reproduction, and circulation.

Somatic cells are the cells in the body other than sperm and egg cells (which are called germ cells).

Gametes are the scientific name for **sex cells** (egg or sperm).

Glyoxysomes are plant peroxisomes with enzymes for the glyoxylate cycle. They convert lipids into sugar during the initial growth of oilseeds.

Tubulin is a protein that polymerizes into long chains of filament that form microtubules, hollow fibers that act as the skeletal system for living cells.

Elastin is an extracellular matrix (ECM) protein responsible for the extensibility and elastic recoil of many vertebrate tissues, such as large arteries, heart valves, pulmonary tissues, skin, and certain ligaments and cartilages.

Kupffer cells are found in the liver of mammals.

Flow cytometry is a cell analysis technique used to rapidly analyze single cells or particles as they flow through

single or multiple lasers while suspended in a buffered salt-based solution.

J.E. Purkinje in 1839 coined the term 'protoplasm' for the fluid substance of the cell. **Protoplasm** is the living substance of the cell surrounding the cell membrane, nucleus, and cytoplasm.

Inheritance

Inheritance is the passing on of **genetic** traits from parents to their offspring.

DNA and RNA are the two types of nucleic acids in living systems. DNA serves as genetic material in most organisms, while RNA acts as genetic material in some viruses and mainly functions as a messenger.

Mendel's Laws of Inheritance

Gregor Mendel, the Father of Genetics, conducted groundbreaking pea plant experiments from 1856 to 1863, laying the foundation for modern genetics.

Conclusions - Genes are inherited in pairs known as alleles, maintaining distinct traits across generations without blending.

Contrasting Traits Studied by Mendel in Pea - Stem height (Tall / dwarf), Flower color (Violet / white), Flower position (Axial / terminal), Pod shape (Inflated / Constricted), Pod color (Green / Yellow), Seed shape (Round / Wrinkled), Seed color (Yellow / Green).

Mendel proposed three laws

Law of Dominance states that when purebred parents with contrasting traits are crossed, only one trait appears in the next generation, with the dominant allele expressing over the recessive. In heterozygous conditions, the dominant allele's traits are expressed. F1 generation shows dominant traits, while recessive traits emerge in the F2 generation.

Law of Segregation - It states that during the production of gametes, two copies of each hereditary factor segregate so that offspring acquire one factor from each parent.

The Law of Independent Assortment states that traits segregate independently during gamete formation, giving different traits an equal chance to occur together.

Genetics

Chromosomes :

Chromosomes are threadlike structures in the cell nucleus containing genes and DNA. They become visible as rod-shaped structures during cell division. Chromosomes, composed of DNA and proteins, carry genetic information from parents to the next generation. DNA molecules within chromosomes provide the essential instructions for building and organizing cells.

Humans have **23 pairs** of chromosomes. Chromosomes numbered from 1 to 22 are common for both sexes and called **Autosomes**. The 23rd pair, the **sex chromosomes**, differ between males and females.

This unique structure of the chromosome keeps DNA tightly packed with **histone** proteins to fit inside the cell and it also helps in giving more stability to it.

Some animals have a lot of chromosomes, but many of them have blank DNA. The blank DNA is called "**Junk DNA**".

Chromosomes can be classified on the basis of the position of the centromere. These types are Metacentric, Sub-metacentric, Telocentric, and Acrocentric.

DNA - DNA is a long polymer of deoxyribonucleotides. The length of DNA is usually defined as the number of nucleotides (a pair of nucleotides referred to as base pairs) present in it.

It is an acidic substance in the nucleus identified by **Friedrich Meischer**. Its double helical structure was given by **Watson and Crick**.

DNA is comprised of nucleotides, which consist of a phosphate molecule and a five-carbon sugar called deoxyribose. Nucleotides serve as the basic building blocks of DNA and RNA.

DNA is mainly stored in the nucleus (where it is called nuclear DNA), but a small amount of DNA is also stored in the **mitochondria** (where it is called mitochondrial DNA or mtDNA).

Nucleosides are the structural subunit of nucleic acids such as DNA and RNA.

A **nucleotide** is composed of one of the **4 bases**, one sugar molecule, and one phosphate molecule.

A **sugar-phosphate chain** acts as a backbone and bases are on the inside. There are two types of nitrogenous bases - **Purines** (Adenine and Guanine), and **Pyrimidines** (Cytosine, Uracil and Thymine).

Cytosine is common for both DNA and RNA and Thymine is present in DNA. Uracil is present in RNA at the place of Thymine.

The ratio between Adenine and Thymine and Guanine and Cytosine are constant and equal.

Purines and pyrimidines are vital for energy, DNA, RNA, protein production, enzyme regulation, and cell signaling.

Adenine : A chemical compound that is used to make one of the building blocks of DNA and RNA. It is also a part of many substances in the body that give energy to cells.

Cytidine is a pyrimidine that, besides being incorporated into nucleic acids, can serve as a substrate for the salvage pathway of pyrimidine nucleotide synthesis.

Thymine helps stabilize nucleic acid structures.

Guanosine is a purine nucleoside which has neuroprotective properties. It is released in the brain under physiological conditions and even more during pathological events.

Uracil helps in the synthesis of enzymes that are necessary for cell functioning. It also acts as a coenzyme and allosteric regulator for reactions in animals and plants.

Cytosine is a key nucleobase in DNA and RNA, pairing with guanine.

Difference between Nuclear DNA and mitochondrial DNA : Nuclear DNA, found in the nucleus of eukaryotic cells, typically has two copies per cell, whereas mitochondrial DNA, located in the mitochondria, contains 100 to 1,000 copies per cell.

Plasmids are small, circular DNA molecules separate from chromosomal DNA, often found in bacteria and sometimes in multicellular organisms. They replicate independently and are

valuable in cloning short DNA segments and replicating proteins.

Functions of DNA

Transcription: The process of copying genetic information from one DNA strand into RNA by using one strand of DNA.

DNA Replication: Copies genetic information for cell division and inheritance, creating exact duplicates. It's rapid in prokaryotes, taking minutes, and slower in eukaryotes, taking hours.

DNA Packaging: DNA is wrapped around histone proteins to form nucleosomes, compacting into chromatin in eukaryotes.

DNA Fingerprinting : The unique DNA sequence of each individual enables DNA fingerprinting, a technique developed in 1984 by British geneticist Alec Jeffreys, to identify individuals based on their DNA.

Mutations : It is defined as change in the genetic material. A point mutation is a change of a single base pair in DNA. The term "mutation" was coined by Hugo de Vries.

Types of DNA

B-form DNA: Most common DNA form, with two right-hand helical strands coiled around the same axis.

Z-form DNA: Zigzag structure formed by alternating purines and pyrimidines, potentially involved in gene regulation and transcription initiation.

A-form DNA: Right-handed double helix like B-form, but thicker with shorter base pair distance. Can form when protein binding removes solvent from DNA.

The **central dogma** was proposed by Crick. The central dogma states that the DNA is converted into RNA and the RNA is converted into proteins.

In retroviruses, the flow of information is opposite, i.e., RNA to DNA to mRNA to Protein. The central dogma of molecular biology: DNA → RNA → protein.

Genes

Genes, made of DNA, are the functional units of heredity, determining traits. Abnormal or mutated genes also influence traits. Genes come in pairs, with each parent passing one copy to

their child, shaping inherited characteristics like hair and eye color.

Functions of Genes - Genes control the functions of DNA and RNA. Genes consist of a particular set of instructions or specific functions. For example, the globin gene was instructed to produce hemoglobin.

Sex Determination (Males and Females)

Male - 46 chromosomes (44 autosomes + an X and Y chromosome) in his body cells. 23 pairs of chromosomes (half of total or 22 autosomes + an X or Y chromosomes) in his sperm cells.

Females- 46 chromosomes (44 autosomes + 2 copies of X chromosome) in her body cells. 23 pairs of chromosomes (half of total or 22 autosomes + an X chromosome) in her egg cells.

RNA - RNA, a ribonucleic acid, aids protein synthesis and cell production from DNA. It has a single strand with ribose sugar, unlike DNA's double strands with deoxyribose. RNA serves enzymatic functions in body reactions. Its bases include adenine, guanine, cytosine, and uracil, replacing thymine.

RNA resembles a **hairpin structure** and like the nucleotides in DNA, nucleotides are formed in this ribonucleic material (RNA).

Functions of RNA - RNA facilitates the translation of DNA into proteins by acting as an adapter molecule in protein synthesis. It serves as a messenger between DNA and ribosomes, promoting the selection of the correct amino acids needed to build new proteins in the body.

Types of RNA

tRNA (Transfer RNA): Selects the correct amino acids for protein synthesis.

rRNA (Ribosomal RNA): Forms the structure of ribosomes, located in the cell cytoplasm.

mRNA (Messenger RNA): Carries genetic information from DNA to ribosomes, instructing protein synthesis.

Some Facts in Inheritance

Genome refers to the complete set of DNA (genetic material) in an organism.

The Human Genome Project (HGP), launched in 1990 and completed in 2003, aimed to map, sequence, and identify all human genes and their functions. It stands as the world's largest collaborative biological endeavor, often referred to as a Mega Project.

Polymorphism, as related to genomics, refers to the presence of two or more variant forms of a specific DNA sequence that can occur among different individuals or populations.

Haploid refers to the presence of a single set of chromosomes in an organism's cells.

Diploid is a term that refers to the presence of two complete sets of chromosomes in an organism's cells, with each parent contributing a chromosome to each pair.

Recombination - Exchange of genetic material between chromosomes.

Hybrid DNA is formed from complementary single DNA strands from two different parental molecules.

Dihybrid cross is a breeding experiment between two organisms which are identical hybrids for two traits.

A monohybrid cross is a genetic cross that takes into account the effect of alleles at a single locus (single gene).

Nutrition in Plants

Nutrition in Plants - The process of obtaining food and utilizing it to grow, stay healthy and repair any damaged body part is known as **nutrition**. Plants require nutrients for their growth which can be supplied through soil, air and water.

Plants must obtain the following mineral nutrients from their growing medium :

Macronutrients : nitrogen (N) , magnesium (Mg), phosphorus (P), potassium (K), calcium (Ca), sulfur (S), carbon (C), oxygen (O), hydrogen (H).

Nitrogen is an essential element used in the synthesis of proteins in plants because plants are not capable of producing nitrogen, so, plants obtain nitrogen with the help of nitrogen-fixing bacteria like Rhizobium.

Magnesium (Mg) is required in chlorophyll synthesis, production,

transportation, and utilization of photoassimilates, enzyme activation, and protein synthesis.

Phosphorus plays a major role in photosynthesis and respiration. It helps in energy storage and transfers them as ADP, ATP, DPN, and TPN.

Micronutrients (or trace minerals): iron (Fe), boron (B), chlorine (Cl), manganese (Mn), zinc (Zn), copper (Cu), molybdenum (Mo), nickel (Ni).

There are two modes of nutrition -

Autotrophic - Plants exhibit autotrophic nutrition and are called primary producers. Plants synthesize their food by using light, carbon dioxide and water.

Heterotrophic - Both animals and human beings are called heterotrophs, as they depend on plants for their food.

Autotrophic Nutrition in Plants

Plants are able to produce their own food through a process called photosynthesis.

Jan Ingenhousz is known as the 'father of photosynthesis'.

Photosynthesis provides food to all living beings. Food production is primarily carried out in leaves. Water and minerals from the soil are absorbed by the root and transported to the leaves through vessels.

The green plant in the terrestrial ecosystem captures about **1 percent** of the sunlight that falls on their leaves and converts it into food energy.

A plant converts the atmospheric CO₂ into simple sugar by using the energy of the sun. Carbon dioxide (CO₂) enters through the stomata. Oxygen (O₂) is also a product of photosynthesis. The exchange of gases takes place due to the opening and closing of stomata. Carotenoid helps to trap light energy that is essential for photosynthesis.

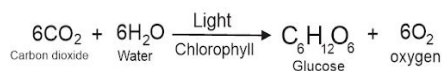
Conditions necessary for Photosynthesis : Sunlight, Water, Carbon dioxide, Chlorophyll.

Site for photosynthesis : Photosynthetic pigment chlorophyll occurs in green coloured cell organelles called chloroplasts. Mesophyll contains many chloroplasts.

Calvin Cycle: It is the term used for the reactions of photosynthesis that use the

energy stored by the light-dependent reactions to form glucose and other carbohydrate Molecules.

Photosynthesis Equation -



In photosynthesis, the role of ATP (adenosine triphosphate) and NADPH (nicotinamide adenine diphosphate) is to carry the energy of the Calvin cycle that occurs in the absence of light. They are also called light-independent reactions.

ATP and NADPH are energy storage and electron carrier/donor molecules. Both ATP and NADPH are used in the next stage of photosynthesis. The chlorophyll molecule regains the lost electron from a water molecule through a process called photolysis, which releases dioxygen (O₂) molecules.

Steps in Photosynthesis -

The major steps involved in the process of photosynthesis are:

1. Absorption of sunlight by chlorophyll
2. Absorption of water, and obtaining carbon dioxide for conversion.
3. Conversion of Chemical energy into light energy and splitting of water into hydrogen and oxygen by light energy.
4. Reduction of carbon dioxide by hydrogen to form carbohydrates like glucose by utilizing the chemical energy (obtained by transformation of light energy)

Heterotrophic Nutrition in Plants

Plants do not contain chlorophyll and depend on other plants for their food through the heterotrophic mode of nutrition. These types of nutrition in plants are referred to as Heterotrophic nutrition in plants.

Heterotrophic Plants - Parasitic, Insectivorous, Saprophytic, Symbiotic. They exhibit various types of nutrition :

Parasitic Nutrition - Heterotrophic plants depend on other plants and animals for nutrition. Such plants are known as parasitic plants. Examples - Cuscuta, Cassytha, Viscum, Loranthus, Corpse flower, Thurber's stemsucker, Dodder, Dwarf mistletoe, Australian Christmas tree.

Insectivorous Nutrition - Plants have special structural features that help them

to trap insects and are commonly known as carnivorous or heterotrophic plants.

These plants digest the insects by secreting digestive juices and absorb the nutrients from them. Example: Pitcher plant, Venus flytrap, Cobra Lily, Butterwort, Monkey Cup, Australian Sundew and Big Floating Bladderwort.

Saprophytic Nutrition - The saprophytic plants derive nutrition from dead and decaying plants and animals. They dissolve the dead and decaying matter by secreting digestive juices and absorb the nutrients. Example - mushrooms, molds, yeast, Corallorhiza orchids, mycorrhizal fungi etc.

Symbiotic Nutrition - When two different plants belonging to two different categories show a close association, they are termed as symbiotic. In this, both the plants are benefitted from each other. Example - the association of fungi and trees, the blue green algae Anabaena lives in the leaves of Azolla (a freshwater fern).

Nutrition through Root

The main function of the root system is to transport water and minerals to the plant. Food is transported by phloem tissue in the vascular system of plants.

Xylem and phloem are complex permanent tissues found in vascular bundles in plants.

Xylem is the water transporting tissue in plants. The stele is the central part of the root or stem containing the tissues derived from the procambium.

The Types of Xylem are -

Tracheids serve for support and for upward conduction of water and dissolved minerals in all vascular plants and are the only such elements in conifers and ferns.

Xylem vessels - Their primary function is to transport water and nutrients from the roots to the leaves.

Xylem parenchyma - Xylem parenchyma cells can store water, which is essential for the proper functioning of a plant.

Xylem fibres - They are the thick lignified fibers that are made up of the dead sclerenchyma cells.

Phloem is the vascular tissue responsible for the transport of sugars from source tissues (example

photosynthetic leaf cells) to sink tissues. It transports food from the leaves to the.

The Types of Phloem are :

Sieve tubes - The sieve tube is the conducting element of the phloem.

Sieve elements are specialized cells that are important for the function of phloem. They are the major conducting cells in phloem.

Companion cells are large elongated cells of phloem with large nucleus.

Phloem fibres (dead cells) - The fibres of sclerenchyma associated with phloem are called phloem fibres.

Phloem parenchyma - The main function of phloem parenchyma is to store food and other substances like resins, latex and mucilage. They also help in transport of food.

Transpiration

It is a process that involves loss of water vapor through the stomata of plants. The loss of water vapor from the plant cools the plant down when the temperature is very high, and water from the stem and roots moves upwards or is 'pulled' into the leaves.

The Types of transpiration are -

Stomatal Transpiration - It is the evaporation of water from the stomata of the plants.

Lenticular Transpiration - Lenticular transpiration is the evaporation of water from a plant's lenticels.

Cuticular Transpiration - It is the evaporation of water from the cuticle of the plants. The cuticle is a waxy covering on the surface of the leaves of the plants.

Chromoplasts

Chromoplasts are **carotenoid** - accumulating plastids conferring color to many flowers and fruits as well as to some tubers and roots. Carotenoids are pigments present in plants, algae, and photosynthetic bacteria. Beta-carotene, found in carrots, converts to vitamin A in the body.

Based on the carotenoid bearing components, they are classified into four types:

Membranous Chromoplasts - Carotene

pigments are present in about 20 concentric membranes. Examples: Citrus sinensis- petals, daffodils, tulip

Globular Chromoplasts - It consists of plastoglobuli, which bear carotenoid pigments. Often they are concentrated in the peripheral stroma. Examples: Ranunculus repens- petals, Capsicum- yellow fruits, Citrus fruit- perianth, etc.

Tubular Chromoplasts: Carotenoids are present in lipoprotein tubules. Examples: Red fruits of capsicum, hypanthium and rose.

Crystalline Chromoplasts: Pure carotene is embedded as crystals. Examples: Carrot roots- β -carotene, Tomato fruits- lycopene

One more chromoplasts is **Fibrillar chromoplasts**.

Terms related to Plants

Leucoplast- It is a colourless organelle found in plant cells, used for the storage of starch or oil.

Leucoplasts are of 3 types- amyloplasts, aleuroplasts and elaioplasts.

Amyloplast is a colorless plant plastid that forms and stores starch. example- potato.

Aleuroplasts store proteins, example - aleurone cells of maize grain.

Elaioplasts store oils and fats, e.g., oil Seeds. It specializes in oil synthesis and storage and is mainly found in the layer of cells in pollen grains that develop pollen grains.

Stomata or stoma (plural stomata or stomas), any of the microscopic openings or pores in the epidermis of leaves and young stems. Stomata are generally more numerous on the underside of leaves. A stomata opens and closes in response to the internal pressure of two sausage-shaped guard cells that surround it. It amounts to 80-90% of total transpiration.

Pneumatophores are specialised roots in hydrophytes. These are aquatic plants that are especially suited for living in aquatic environments. Examples - Lotus (Nelumbo), Vallisneria, Potamogeton, Salvinia, hydrilla, water hyacinths etc.

Glycolysis is the process in which glucose is broken down to produce energy.

Tropism is a phenomenon indicating growth or turning movement of an organism, usually a plant, in response to an environmental stimulus.

Cork cells protect the tree from bacterial or fungal infection, it prevents water loss through the bark and also from external injury.

Velamen - A plant tissue that helps in absorption of water.

Guttation process - In plants excess water is released from plants through leaves.

Pyrenoid is a protein body in the chloroplasts of algae and hornworts that are involved in carbon fixation and starch formation and storage of Chlorophyceae.

Anthocyanins are colored water-soluble pigments belonging to the phenolic group. Anthocyanins are responsible for the colors in fruits and vegetables. Berries, currant grapes, and some tropical fruits have high anthocyanins content.

Hornwort is a flowerless, spore producing plant that is usually a thin, horn-like, or needle-like capsule with spores that develops from a flat, green leaf.

Palisade is a layer of closely packed cells found under the upper epidermis and cuticle which is the outermost layer of a leaf.

Guard cells are another type of plant single-cell models to study early signal transduction and stress tolerance mechanisms in plants. Guard cells control influx and efflux of CO_2 and water from leaves, respectively. Guard cells are located in leaf epidermis.

Mesophyll represents the middle layer of cells located between the epidermal layers of a leaf. The spongy mesophyll's function is to allow for the interchange of gasses (CO_2) that are needed for photosynthesis.

Chlorophyll is a green pigment present in leaves which helps the leaves capture energy from sunlight to prepare their food.

Chlorophyll molecules contain a central magnesium atom encircled by a nitrogen-containing porphyrin ring, with a phytol chain attached. While porphyrins assist in electron transport by enhancing the redox properties of iron ions, their

role in photosynthesis is distinctive and essential.

Chloroplasts are found as green dots in the palisade and spongy cells of mesophyll.

Nutrition in Animals

Nutrition involves acquiring food for energy, growth, and body repair. Animals utilize nutrients such as carbohydrates, lipids, nucleic acids, proteins, minerals, and vitamins.

Animals are divided into **three groups** on the basis of their food habits :

Herbivores (animals which eat only plants like grasses, leaves, fruits etc. Examples - Goat, Cow, Sheep, Horse, Ox, Elephant etc).

Carnivores are animals that eat meat. Examples include lions, tigers, frogs, vultures, and lizards.

Omnivores are animals that eat both plants and animals. Examples include humans, dogs, crows, bears, and ants.

There are **five** steps in the process of Nutrition in animals :

1.Ingestion: Food enters the body through the mouth.

2.Digestion: Food is broken down into nutrients and waste.

3.Absorption: Nutrients are absorbed into the bloodstream.

4.Assimilation: Nutrients are taken into the cells for use.

5.Egestion: Undigested food is expelled from the body through the anus.

Amino acids

Amino acids are tiny organic compounds that are the basic building blocks of proteins. Proteins, in turn, are crucial for our body's structure and functions, made up of chains of these amino acids.

There are twenty-one amino acids in total, with selenocysteine being the 21st proteinogenic one. Selenocysteine is found in selenoproteins, where it replaces sulfur in cysteine. Most amino acids have nonpolar side chains, making up the largest group.

Amino acids can be placed in **three**

different groups -

Essential amino acids

1. Histidine: Supports tissue growth, repair, and blood cell production, also used in histamine synthesis.

2. Isoleucine: Vital for overall physiological functions, including growth, immunity, and metabolism.

3. Leucine: Regulates protein synthesis, tissue repair, and metabolic processes.

4. Lysine: Aids in calcium absorption and collagen formation, crucial for bone strength and connective tissues.

5. Methionine: Acts as an antioxidant, protecting against radiation damage and detoxifying harmful substances.

6. Phenylalanine: Essential for protein, catecholamine, and melanin synthesis, contributing to brain chemical production.

7. Threonine: Essential for healthy skin, teeth, and structural proteins like collagen and elastin.

8. Tryptophan: Used to produce melatonin and serotonin, regulating sleep, mood, and appetite.

9. Valine: Supports muscle growth and tissue repair.

Nonessential amino acids

Alanine: Used in protein synthesis and as an energy source for muscles and the nervous system. It also supports immune function and assists in breaking down tryptophan and vitamin B-6.

Asparagine: Assists in the breakdown of ammonia, protein modification, and the production of neurotransmitters in the brain.

Aspartic acid: Essential for cellular function and hormone production and release.

Glutamic acid: Converts into glutamate, a neurotransmitter that aids in communication between nerve cells in the brain.

Conditional amino acids

Arginine - It plays an important role in cell division, wound healing, removing ammonia from the body, immune function, and the release of hormones.

Cysteine is vital for protein synthesis and metabolism. It's a key component of beta-keratin, found in nails, skin, and hair, and is essential for collagen production.

Beta-carotene acts as an antioxidant, It protects the body from harmful free radicals.

Glutamine - Glutamine helps in the synthesis of proteins in the human body.

Tyrosine produces key brain chemicals that help in nerve cell communication.

Glycine - It acts as neurotransmitter in the central nervous system and it has many roles such as antioxidant, anti-inflammatory, cryoprotective, and immunomodulatory in peripheral and nervous tissues.

Proline is crucial for protein synthesis, wound healing, antioxidative reactions, and immune responses. Actin, found in all eukaryotic cells, is a highly abundant intracellular protein.

Serine is essential for plant metabolism, development, and cell signaling. It's involved in biosynthesizing amino acids, nucleotides, phospholipids, and sphingolipids.

Pyrrolysine, sometimes considered the "22nd amino acid", is not used by the human body.

Three important building blocks of the body are, Amino acids, lipids, and nucleotides. Every human cell contains proteins, which are made up of chains of amino acids. Protein in the diet aids cell repair and growth.

The Nutrition Care Process (NCP)

The NCP is a structured way to give top-quality nutrition care. It has four steps: Nutrition Assessment, Diagnosis, Intervention, and Monitoring/Evaluation.

Minerals - The elements present in food that are required by our body to develop and function properly.

Macro minerals, required by the body in large quantities, include calcium, chloride, sodium, potassium, magnesium, phosphorus, and sulfur. They are vital for metabolism and proper body function.

Micro minerals (trace minerals) are needed in small amounts by the body.

Examples include iron, iodine, copper, manganese, fluoride, zinc, cobalt, and selenium. Excessive intake can lead to mineral toxicity, causing health issues like nausea, diarrhea, and discoloration.

Biomacromolecules in food need to be broken down into simple substances like glucose and amino acids in the digestive system before the body can utilize them. Seaweed, eggs, milk, and milk products are rich sources of iodine, essential for the production of thyroid hormones, which regulate metabolism. Adequate iodine intake is crucial for proper bone and brain development, particularly during pregnancy and infancy.

Milk contains two main proteins: casein (80%) and whey (20%). Glutelin is found in wheat, while myosin is present in muscles.

Milk and milk products have a good balance of protein, fat and carbohydrate and are a very important source of essential nutrients, including calcium, riboflavin, phosphorus, vitamins A and B12, potassium, magnesium, zinc, iodine.

Eggs: High-quality protein, heart-healthy unsaturated fats, rich in vitamin B6, B12, and vitamin D.

Red meat: Rich in iron, zinc, and B vitamins, especially vitamin B12.

Myosins: Motor proteins crucial for muscle contraction and various cellular movements in eukaryotes. ATP-dependent, driving action-based motility.

Rennin (Chymosin): Proteolytic enzyme in digestion of milk, secreted as inactive Prorenin, activated by Hydrochloric acid. Common in young children.

Troponin : Protein in heart muscles, typically absent in blood. Released into the bloodstream when heart muscle damage occurs.

Glucose: Primary cellular fuel; excess stored as glycogen in liver and skeletal muscle.

Lactose is a sugar that is naturally found in milk and milk products, like cheese or ice cream.

Cellulose: Abundant organic compound ($C_6H_{10}O_5$)_n, complex carbohydrate with oxygen, carbon, and hydrogen. It is tasteless and odorless. Discovered by

French chemist Anselme Payen in 1838.

Hair: Composed mainly of keratin, a fibrous and helical protein making up 95% of hair. Keratin is a type of scleroprotein, with alpha-keratin (α -keratin) found in vertebrates.

Tubulin, abundant in the cell cytoplasm, serves as a crucial structural component in various tissues like hair, nails, and skin among vertebrates. These proteins form microtubules, narrow hollow tubes within cells, which play essential roles in cell division and movement.

Dystrophin: Rod-shaped cytoplasmic protein crucial for linking muscle fiber cytoskeleton to extracellular matrix through cell membrane, forming the dystrophin-associated protein complex (DAPC) or costamere.

Calorific value is the energy released during complete combustion, measured in units like kJ/kg or kcal/g. Fats have the highest energy density, providing 9 calories per gram, while protein and carbohydrates offer 4 calories per gram each.

Parenteral nutrition (PN), or total parenteral nutrition (TPN): Delivering specialized nutrition intravenously to correct or prevent malnutrition.

Plasma is a straw-colored, viscous fluid it has about 90-92% water and 6-8% proteins. The major proteins of blood plasma are Albumin, globulin, and fibrinogen.

Insulin: Pancreatic hormone facilitating glucose entry into cells for energy. Zinc crucial for insulin synthesis, storage, and secretion, maintaining its structure.

Leptin: Hormone mainly from fat cells, key in long-term energy balance regulation. Named from Greek "leptos," meaning thin.

Carbohydrates

Starch and **sugar** are forms of carbohydrates in our food. Carbohydrates are the main source of energy for the body.

Carbohydrates are mainly classified into **three groups** -

Monosaccharides -

Glucose is a sugar with the molecular formula $C_6H_{12}O_6$.

Fructose ($C_6H_{12}O_6$), or fruit sugar, is a ketonic simple sugar found in many plants, often bonded with glucose to form sucrose. Honey contains water and two sugars: fructose and glucose.

Galactose ($C_6H_{12}O_6$), is a monosaccharide sugar that is about as sweet as glucose, and about 65% as sweet as sucrose.

Disaccharides -

Maltose ($C_{12}H_{22}O_{11}$) is a sugar made out of two glucose molecules bound together.

Lactose ($C_{12}H_{22}O_{11}$): Sugar naturally occurring in milk and dairy, composed of one molecule of glucose and galactose each.

Sucrose ($C_{12}H_{22}O_{11}$) is a molecule composed of two monosaccharides, namely glucose and fructose.

Polysaccharides -

Starch ($C_6H_{10}O_5$)_n : Tasteless white powder insoluble in cold water, alcohol, etc. Polysaccharide composed of glucose monomers linked by 1,4 bonds.

Fiber is a type of carbohydrate that the body can't digest.

Glycogen is the stored form of glucose that's made up of many connected glucose molecules.

Omega-3 sources: Fish, seafood (salmon, mackerel, tuna, etc.), nuts, seeds (flaxseed, chia, walnuts), plant oils (flaxseed, soybean, canola). Adult body contains about 25 g of magnesium.

Magnesium: Vital for bone health, blood pressure regulation, heart rhythm, muscle and nerve function, blood sugar levels, protein, bone, and DNA synthesis.

Fat : Body's densest energy source, yielding over twice the energy of carbohydrate or protein. During exercise, stored fat in adipose tissue breaks down into fatty acids for fuel.

Autotrophic nutrition: Organisms produce food from inorganic materials like water, carbon dioxide, and minerals, using sunlight. Bacteria are autotrophs, converting light, chemicals, or inorganic substances into usable energy.

Fungi are **eukaryotic species** that derive nutrition through the heterotrophic mode of nutrition.

Protozoa are unicellular, **heterotrophic**, eukaryotic organisms comprising four organization types: amebae, flagellates, ciliates, and parasitic sporozoans.

Sources of nutrients/minerals

Soybeans are a rich source of plant-based protein, containing approximately 36 grams of protein per 100 grams.

Apples are extremely rich in important antioxidants, flavonoids, and dietary fiber. Wheat is rich in carbohydrates.

Groundnuts are rich in the vitamins niacin, folate, pantothenic acid, thiamin, riboflavin, choline, Vitamin B6 and Vitamin E.

Nuts, vegetable oil and **fish** are good sources of Healthy fats (like Omega-3). Monounsaturated fats and polyunsaturated fats are known as the “**good fats**” because they are good for heart, cholesterol, and overall health.

Carbohydrates sources - Fresh fruits, vegetables, corn, potatoes, milk and milk products.

Protein sources - Animal-based foods (meat, poultry, fish, eggs, and dairy foods) & plant-based foods (fruits, vegetables, grains, nuts, and seeds).

Minerals - Cereals, bread, meat, fish, milk, dairy, nuts, fruit (especially dried fruit) and vegetables.

Chlorella (space algae): High protein content (45%) and essential nutrients. Dried form: 20% fat, 20% carbohydrate, 5% fiber, 10% minerals/vitamins. Used in space for CO₂ consumption and oxygen production.

Micronutrients: Essential for growth, required in smaller amounts (e.g., Iron, Iodine, Zinc, vitamins).

Macronutrients: Needed in large amounts for body functions (e.g., fat, carbohydrates, protein).

The five major minerals in the human body are **calcium, phosphorus, potassium, sodium, and magnesium**.

Calcium: Abundant mineral (1.5-2% body weight), vital for bone and teeth health, muscle contraction regulation, and blood clotting.

Sodium is required to conduct nerve impulses, contract and relax muscles,

and maintain the proper balance of water and minerals.

Potassium: Electrolyte aiding nerve and muscle function, maintaining regular heartbeat, facilitating nutrient and waste transport in cells.

Phosphorus: Crucial for bone and teeth formation.

Fat emulsification: Increases fat surface area in the small intestine by forming small clusters. Bile salts, partially lipid-soluble and water-soluble, emulsify fat particles, increasing surface area. Lipase enzyme digests fat into glycerol and fatty acids.

Adolescence (between the ages of 10 and 19 years) is the period of transition between childhood and adulthood.

Childhood (age range of 2 - 9).

Adulthood: Early Adulthood Stage (age range of 22-34), Early Middle Age (age range of 35-44), Late Middle Age (age range of 45-64), Late Adulthood (age range of 65 and above). The nutrition requirements of adolescents are higher than adults.

Recommended Dietary Allowance (RDA) = Requirements + Margin of Safety. The RDAs are the amounts of essential nutrients that are judged by the Food and Nutrition Board to be adequate to meet the known nutrient needs of practically all healthy persons.

Metabolism: Body's cellular chemical reactions converting food into energy. Proteins can serve as fuel. Nucleus controls cell metabolism.

The most common proteins used are **pepsin, trypsin, and chymotrypsin**.

Colostrum is the first form of **breastmilk** that is released by the mammary glands after giving birth.

Sebum is an oily, waxy substance produced by your body's **sebaceous** glands.

Synovia is a lubricating fluid resembling the white of an egg, secreted by certain membranes, as those of the joints.

Cerumen (Earwax), is a brown, orange, red, yellowish or gray waxy substance secreted in the ear canal of humans and other mammals.

Oral rehydration therapy (ORT): Fluid replacement to prevent/treat

dehydration, especially from diarrhea. Involves water with sugar and salts (sodium, potassium), can be given orally or via nasogastric tube. Reduces diarrhea-related death risk by up to 93%.

Some facts related to Animal Nutrition

Cholesterol: Waxy, fat-like substance in all body cells. Needed for hormone, vitamin D, and digestive substance production.

Gluten: Protein in wheat, rye, barley. Found in various products like supplements, lip balms, medicines.

Glycemic index (GI) measures how a carbohydrate-containing food raises blood sugar.

HDL (High-density lipoproteins): "Good" cholesterol. Transports cholesterol from body to liver for removal.

LDL (low-density lipoproteins). It is also known as “bad” cholesterol.

Monounsaturated fat is a type of fat found in avocados, canola oil, nuts, olives and olive oil, and seeds.

Polyunsaturated fat: Liquid at room temperature. Includes omega-6 and omega-3 fatty acids. Omega-6 found in vegetable oils like corn, safflower, soybean oil.

Triglycerides are a type of fat found in our blood. Too much of this type of fat may raise the risk of coronary artery heart disease, especially in women.

Food and its Varieties

Sucrose: It is a natural sweetener found in plants and commercially produced from sugar cane and sugar beets.

Yeast: A unicellular fungus, converts sugar into alcohol through fermentation, crucial in making beverages like wine and beer. *Saccharomyces cerevisiae* is commonly used for this purpose, producing ethanol and carbon dioxide, which inhibit other microorganisms.

Kahwa chai: It is a traditional Kashmiri green tea blend infused with spices, nuts, and saffron, often brewed in a brass kettle called a samovar.

Sake Rice (Japanese wine): An Alcoholic beverage made by fermenting rice.

Soya Foods are extremely beneficial for weight loss, reducing cholesterol, and supporting bone and heart health. It can be divided into Unfermented Foods and Fermented Foods.

Unfermented Foods - Tofu, soyamilk, edamame, soya nuts and sprouts.

Fermented Foods - Miso, tempeh, natto and soya sauce.

Caffeine : It is a methylxanthine alkaloid. It is derived from the beans of the coffee plant arabica. It is used for stimulation of the brain and nervous system.

Maillard reaction: It is a chemical reaction between reducing sugar and amino acid in the presence of heat. It was first discovered by **Louis Camille Maillard**.

Hydrogen peroxide : It is used as a bleaching agent in some foods such as wheat flour, edible oil, egg etc. It is used as an oxidizer and antiseptic.

Streptococcus thermophilus is a gram-positive bacterium found in fermented milk products, often used in yogurt production.

Calcium silicate is a white free-flowing powder used as an anticaking agent in food. It is derived from limestone and diatomaceous earth.

Pickling preserves food using anaerobic fermentation in brine or vinegar immersion. Its solutions are typically highly acidic, with a pH of 4.6.

Roughage refers to the indigestible part of plant foods like whole grains, nuts, seeds, and fruits and vegetables like carrots and broccoli.

Starches: It found in breads, cereals and Pasta are made up of long chains of glucose.

Varieties of Cheese: Mozzarella, cheddar, camembert, Asiago, Bocconcini, Brie, Burrata, Colby, Cotija, Emmental, Feta, Gouda, Halloumi, Havarti, Jarlsberg, Limburger, Mascarpone etc.

Butylated hydroxyanisole is a chemical compound used as a defoaming agent in foods like butter, meat, cereals, chewing gum, baked goods, snack foods, dehydrated potatoes, and beer.

Monosodium glutamate (MSG) is a flavor enhancer commonly added to Chinese food, canned vegetables, soups, and processed meats.

Sodium benzoate: It is a sodium salt of benzoic acid and well soluble in water. It is used to preserve tomato sauces. Due to its antifungal and antibacterial properties, it is tasteless, and odorless.

Certification marks issued for different products in India

AGMARK - It is a certification mark used for agricultural products in India.

The ISI mark is a certification indicating that industrial products in India conform to Indian Standards, issued by the Bureau of Indian Standards (BIS).

The BIS Hallmark certifies the purity of gold jewelry and coins in India.

Food Process Order (FPO) mark is mandatory for all processed fruit products sold in India, indicating they were produced in a hygienic environment and are safe for consumption.

The green and brown dot system indicates the vegetarian (green) or non-vegetarian (brown) nature of food products.

India Organic Certification label is given to farm products processed in India.

Ecomark certifies ecologically safe products by Bureau of Indian Standards.

Vitamins

Vitamin is a nutrient that the body needs in small amounts to function and stay healthy. Some vitamins are made in the human body from food products. Vitamins are also known as the "Accessory dietary factor".

Plants can synthesize almost all vitamins but humans can not produce sufficient amount of vitamins on their own. Vitamins and some minerals are organic compounds essential for metabolic reactions.

Vitamins are of two types : Water soluble (Vitamin B and C) and Fat Soluble (Vitamin A, D, E and K).

Fat-soluble vitamins, stored in the body for long-term, pose a higher risk of toxicity than water-soluble vitamins if overconsumed.

Vitamin A : It is important for normal vision, the immune system, reproduction,

and growth and development. Scientific name : Retinol.

Sources : Green leafy vegetables, ripe yellow fruits, guava, milk, liver, nuts, tomatoes, oranges, carrots, broccoli, watermelon and papaya.

Deficiency disease: Hyperkeratosis, night blindness (nyctalopia), Xerophthalmia, and keratomalacia.

Keratomalacia is an eye disorder characterized by drying and clouding of the cornea.

Glaucoma is a group of eye diseases that can lead to vision loss and blindness by damaging the optic nerve.

Carr-Price reagent, a chloroform solution of antimony trichloride, has been widely used in the determination of vitamin A.

Vitamin B - There are 8 types of vitamin B: Thiamin (B₁), Riboflavin (B₂), Niacin (B₃), Pantothenic acid (B₅), Pyridoxine (B₆), Biotin (B₇), Folate or 'folic acid' when included in supplements (B₉), Cyanocobalamin (B₁₂).

Vitamin B1 - It is essential for glucose metabolism and nerve, muscle, and heart function. It was the first vitamin B to be discovered in 1897.

Sources: Wholemeal cereal grains, seeds (especially sesame seeds), legumes, wheatgerm, nuts, yeast.

Diseases/Deficiency : Beriberi, Wernicke-Korsakoff syndrome. It affects Transketolase activity. Due to its deficiency, the concentration of pyruvic acid and lactic acid in blood is increased.

Vitamin B2 - It helps in improving vision and skin health. Niacin or nicotinic acid is a monocarboxylic acid derivative of pyridine. It is synthesized in the body from Tryptophan.

Sources: milk, yoghurt, cottage cheese, wholegrain breads and cereals, egg white, leafy green vegetables, meat, yeast.

Symptoms of deficiency: skin disorders, angular stomatitis (sores at the corners of the mouth), swelling of the mouth and throat, Cheilosis (swollen, cracked lips), hair loss, red, itchy eyes, and Magenta tongue.

Disease : Ariboflavinosis.

Ariboflavinosis : A disease characterized

by sores on the mouth.

Angular cheilitis is a common inflammatory skin condition. It affects one or both corners of your mouth and causes cracked sores.

Vitamin B3 - It converts carbohydrates, fat, and alcohol into energy, maintains skin health, and supports the nervous and digestive systems. Unlike other B-group vitamins, niacin is very heat stable and little is lost in cooking.

Sources - meats, fish, poultry, milk, eggs, wholegrain breads and cereals, nuts, mushrooms, all protein-containing foods.

Symptoms: brown discoloration on skin exposed to sunlight, patches of skin with a rough appearance, a bright red tongue.

Disease - Pellagra (It is marked by the 3 Ds – dementia, diarrhoea and dermatitis.)

Vitamin B5 - It is required to metabolise carbohydrates, proteins, fats, and alcohol as well as produce red blood cells and steroid hormones. It is a constituent of coenzymes involved in Acetylation.

Sources: liver, Meats, milk, Kidneys, eggs, yeast, peanuts. Legumes.

Symptoms - numbness in hands and feet, burning foot syndrome, headache.

Disease: Paresthesia.

Vitamin B6 - It is essential for red blood cell formation, and hormone activity. It forms part of the coenzyme for amino acid synthesis and glycogen synthesis.

Sources - Cereal, grains, legumes, green and leafy vegetables, fish and shellfish, meat and poultry, nuts.

Deficiency - Anaemia, scaling on the lips, cracks at corners of the mouth, swollen tongue.

Vitamin B7 - It is needed for energy metabolism, amino acid metabolism and glycogen synthesis. It is a coenzyme of the enzyme Carboxylase. High biotin intake can contribute to raised blood cholesterol levels.

Sources: cauliflower, egg yolks, peanuts, chicken, yeast, mushrooms.

Symptoms: thinning of the hair, a scaly rash around eyes, nose, and mouth, brittle nails.

Disease: Dermatitis, enteritis.

Vitamin B9 - It helps in fetal nervous system development, DNA synthesis, and cell growth. It is crucial for women of child-bearing age to prevent birth defects.

Symptoms: heart palpitations, sores on the tongue or in the mouth.

Diseases : Megaloblastic anaemia.

Megaloblastic anemia, a type of macrocytic anemia, results from inhibited DNA synthesis during red blood cell production.

Vitamin B12- Essential for myelin production around nerve cells, mental ability, red blood cell formation, and energy production from fatty acids and amino acids. It reduces homocysteine levels.

Sources: meat, milk, cheese, eggs etc.

Symptoms of Deficiency - weight loss, constipation, numbness and tingling in the hands and feet, memory problems.

Deficiency of vitamin B12 can be diagnosed by the Schilling test.

Disease: Megaloblastic anemia.

Vitamin C - It is an antioxidant that aids in healing infections, scavenges free radicals, forms collagen, and aids in water purification. It also promotes cell cohesion and influences iron absorption.

Scientific name: Ascorbic acid or ascorbate.

Sources: Fresh fruits, black currant, broccoli, goat milk and chestnuts, citrus fruits and juices such as lemons, oranges and sweet limes/mousambis, amla, tomatoes.

Deficiency disease: Scurvy, anorexia, poor wound healing, and hyperkeratosis, swelling of gums.

Scurvy : It is a sign of severe malnutrition.

Anorexia - It is an eating disorder characterised by an abnormally low body weight, an intense fear of gaining weight and a distorted perception of weight.

Hyperkeratosis: It refers to the increased thickness of the stratum corneum, the outer layer of the skin.

Vitamin D - It aids in calcium and phosphorus absorption for bone health.

It may also help reduce cancer cell growth, control infections, and decrease inflammation. It is synthesized in the body with sunlight exposure.

Types of Vitamin D :-

Vitamin D1 : Mixture of molecular compounds of ergocalciferol with lumisterol.

Vitamin D2: Ergocalciferol. It is an inactivated vitamin D analog, treats hypoparathyroidism, rickets, and familial hypophosphatemia. It is synthesized by plants in UVB light. It is used to treat hypoparathyroidism. It is produced by invertebrates, some plants, and fungi.

Vitamin D3: Cholecalciferol. It is synthesised in the skin by the reaction of 7 -dehydrocholesterol with UVB radiation, present in sunlight with an UV index of three or more. Kidneys play a major role in turning vitamin D into its active form.

Vitamin D4: 22-Dihydroergocalciferol.

Vitamin D5: Sitocalciferol.

Scientific name of Vitamin D: Calciferol.

Sources: Fish, beef, cod liver oil, egg yolk, liver, chicken breast and cereals.

Deficiency disease: Rickets, Osteomalacia, Cystic fibrosis, Crohn's disease and Celiac disease.

Rickets is the softening and weakening of bones in children.

Osteomalacia is the softening of the bones.

Cystic fibrosis is an inherited disorder that causes severe damage to the lungs, digestive system and other organs in the body.

Crohn's disease is a chronic disease that causes inflammation and irritation in your digestive tract.

Celiac disease is a chronic digestive and immune disorder that damages the small intestine.

Vitamin E - It maintains healthy skin and eyes, and the immune system.

Scientific name: Tocopherol.

Sources: Potatoes, pumpkin, guava, mango, milk, nuts and seeds.

Deficiency disease: Heart problems, Haemolysis and sterility, Nerve damage, muscle damage, sterility, vision loss, and

Creatinuria.

Vitamin K - It is vital for blood clotting, bone health, and regulating calcium levels. It is essential for producing healthy bone tissue through proteins like osteocalcin.

Vitamin K1 (phytonadione), Vitamin K2 (menaquinone) are types of Vitamin K.

Scientific name: Phytonadione.

Sources: Tomatoes, broccoli, mangoes, grapes, chestnuts, cashew nuts, apples.

Deficiency disease: Haemorrhage, Poor bone development, excessive bleeding from wounds, osteoporosis and cardiovascular diseases.

Long lasting vitamins

Vitamin B4 (adenine) is vital for protein synthesis and accompanying chemical processes. It is a key component of DNA and RNA.

Vitamin B8 (inositol): It is also called myo-inositol, D-chiro-inositol, or hexaphosphate (IP6). It plays a critical function in the body's cellular growth. It is not a vitamin. Inositol is a type of sugar that helps the body to process insulin.

Vitamin B10 (para amino benzoic acid – PABA): It is an organic compound found in certain foods and produced by the chemical industry.

Vitamin B11 (salicylic acid) : It often works in tandem with B12 to contribute to the formation of DNA and RNA.

Vitamin K : It is involved in posttranslational modification of the blood clotting factors by acting as a cofactor for the Carboxylase enzyme.

Deficiency and Diseases

Disease can be defined as a state of illness or sickness that produces specific signs or symptoms.

Diseases can be broadly grouped into infectious and non-infectious.

Infectious diseases

Diseases which are easily transmitted from one person to another, are called infectious diseases.

A wide range of organisms belonging to

bacteria, viruses, fungi, protozoans, helminths, etc., could cause diseases in humans. Such disease causing organisms are called pathogens

Diseases caused by various Bacteria

Diphtheria is a highly contagious, infectious disease caused by a bacterium called *Corynebacterium diphtheriae*. People with diphtheria have serious breathing and swallowing problems.

Cholera is an acute diarrheal illness caused by infection of the intestine with *Vibrio cholerae* bacteria. People can get sick when they swallow food or water contaminated with cholera bacteria.

Whooping cough (pertussis) is a very contagious respiratory illness caused by a type of bacteria called *Bordetella pertussis*. The disease is only found in humans. The treatment for whooping cough is usually antibiotics like azithromycin, erythromycin and clarithromycin.

Tetanus - It is caused by bacteria called *Clostridium tetani*. It makes muscles tighten up, which can make it hard to open the mouth or swallow. This condition is often called "lockjaw" because of how it affects the jaw muscles.

Plague is a bacterial disease caused by *Yersinia pestis*, often transmitted by flea bites from infected rodents. Antibiotics like streptomycin and doxycycline are used for treatment.

Anthrax, caused by *Bacillus anthracis* bacteria, is a rare yet severe illness primarily affecting livestock and wild animals. It typically enters the body through skin wounds or ingestion of contaminated meat, or inhalation of spores.

Leprosy, also known as Hansen's disease, is a chronic infectious disease caused by the bacterium *Mycobacterium leprae*. This bacteria damages nerves, skin, eyes, and the respiratory tract. Treatment typically involves a combination of antibiotics, including dapsone with rifampicin, and clofazimine, known as multidrug therapy.

Botulism is a severe illness caused by a toxin from the bacterium *Clostridium botulinum*, leading to muscle weakness

and paralysis.

Syphilis is a sexually transmitted infection caused by the bacterium *Treponema pallidum* subspecies *pallidum*. Syphilis spreads from person-to-person by direct contact with a syphilitic sore, known as a chancre.

Tuberculosis (TB) is caused by *Mycobacterium tuberculosis* and mainly affects the lungs. It spreads through droplet infection. TB has three stages: exposure, latent, and active disease. Diagnosis can be done through skin or blood tests. Bacille Calmette-Guerin (BCG) vaccine helps prevent TB.

Typhoid fever is a bacterial infection that can spread throughout the body, affecting many organs. It's caused by a bacterium called *Salmonella typhi*, which is related to the bacteria that cause salmonella food poisoning. Antibiotic therapy is the only effective treatment for typhoid fever.

Gonorrhea is an infection caused by a sexually transmitted bacterium that infects both males and females. Gonorrhea most often affects the urethra, rectum or throat. In females, gonorrhea can also infect the cervix. Gonorrhea is most commonly spread during vaginal, oral or anal sex. Affected organ - reproductive tract.

Diseases caused by various Fungus

Onychomycosis, also known as tinea unguium, is a fungal infection of the nail.

Jock itch is an infection that's caused by a fungus. It's a form of ringworm that likes to live in warm, moist places on its body.

Diseases caused by various Protozoa

Diarrhea is loose, watery stool occurring frequently, leading to fluid loss and depletion of essential salts and minerals like sodium, chloride, and potassium. The primary cause of acute diarrhea is typically intestinal infections such as gastroenteritis or food poisoning.

Malaria is caused by infection with parasites in the genus *Plasmodium*. It is transmitted to people through the bites of infected female *Anopheles* mosquitoes. It is preventable and curable.

In humans, malaria is caused by six *Plasmodium* species: *P. falciparum*, *P. malariae*, *P. ovale curtisi*, *P. ovale wallikeri*, *P. vivax* and *P. knowlesi*. Among those infected, *P. falciparum* is the most common species identified (~75%) followed by *P. vivax* (~20%).

Kala Azar is also known as Visceral leishmaniasis (VL). Protozoa causes the disease 'kala-azar'. It is the most severe form of leishmaniasis and is highly fatal. It is transmitted by the bite of an infected female phlebotomine sandfly.

Diseases caused by various Virus

Ebola, or Ebola virus disease (EVD) and Ebola hemorrhagic fever (EHF), is a viral illness affecting humans and primates, caused by viruses in the Ebolavirus and Filoviridae family. It leads to severe inflammation and tissue damage. Diagnosis involves testing blood samples for viral RNA, antibodies, or the virus itself.

Zika Virus Disease is caused by a virus transmitted primarily by *Aedes* mosquitoes (AM), mainly *Aedes aegypti*. This is the same mosquito that transmits dengue, chikungunya, and yellow fever. It was first identified in Uganda in 1947 in monkeys. Zika virus is named after the Zika Forest in Uganda.

Jaundice (Icterus) is a condition in which the skin, whites of the eyes and mucous membranes turn yellow because of a high level of bilirubin, a yellow-orange bile pigment. Jaundice has many causes, including hepatitis, gallstones and tumors.

Bilirubin is formed from the breakdown of red blood cells.

Herpes results from infection with the herpes simplex virus (HSV). It causes sores or blisters in or around the mouth or genitals. There is no cure for herpes, but treatment can help manage symptoms. Genital herpes is a common sexually transmitted infection (STI).

Polio caused by the poliovirus, can quickly paralyze people by attacking the nervous system. It spreads person-to-person, affecting the spinal cord and causing paralysis. India, along with the WHO South-East Asia Region, received polio-free certification from the World Health Organization on March 27, 2014.

The common cold is an infection of the nose, sinuses, throat and windpipe. Many different respiratory viruses can cause the common cold, but rhinoviruses are the most common.

Smallpox is an acute contagious disease caused by the variola virus, a member of the orthopoxvirus family.

Rabies is a fatal but preventable viral disease. It can spread to people and pets if they are bitten or scratched by rabid animals like dogs, bats, monkeys, raccoons, skunks, and foxes. Risks from monkey bites include serious wound infections, herpes B virus, and rabies.

Hepatitis - Inflammation of the liver (the irritation or swelling of the liver cells from any cause). Caused by a group of viruses, "hepatotropic" viruses, including A, B, C, D, and E.

Hepatitis B is a potentially life-threatening liver infection caused by the hepatitis B virus (HBV). It is a major global health problem. It can cause chronic infection and puts people at high risk of death from cirrhosis and liver cancer.

Rubella, also called German measles, is a contagious disease caused by a virus. Most people who get rubella usually have a mild illness, with symptoms that can include a low-grade fever, sore throat, and a rash that starts on the face and spreads to the rest of the body.

Acquired immunodeficiency syndrome (AIDS) is a chronic, potentially life-threatening condition caused by the human immunodeficiency virus (HIV). AIDS is the late stage of HIV infection that occurs when the body's immune system is badly damaged because of the virus. HIV infection causes an acquired immunodeficiency syndrome, principally because of depletion of CD4 lymphocytes.

Symptoms of AIDS include - Rapid weight loss, recurring fever or profuse night sweats, extreme and unexplained tiredness, prolonged swelling of the lymph glands in the armpits, groin, or neck.

There is no cure for HIV, but it can be controlled with HIV treatment. There are three types of HIV tests: antibody tests, antigen/antibody tests, and nucleic acid tests (NAT).

Japanese encephalitis (JE) is an infectious disease of the central nervous system caused by the Japanese encephalitis virus (JEV), a zoonotic mosquito-borne flavivirus.

Mumps is a disease which is caused by a viral infection that affects our salivary glands. It can be prevented easily by a vaccine.

Swine flu (H1N1) is a type of viral infection. It's called swine flu because it resembles a respiratory infection that pigs can get. The H1N1 flu is a type of influenza A virus. The symptoms of H1N1 are similar to those of regular flu.

Nipah virus infection is a zoonotic illness that is transmitted to people from animals, and can also be transmitted through contaminated food or directly from person-to-person. It causes a range of illnesses from asymptomatic (subclinical) infection to acute respiratory illness and fatal encephalitis.

SARS (Severe Acute Respiratory Syndrome) - It is a viral respiratory disease caused by a SARS-associated coronavirus. It was first identified at the end of February 2003. SARS is an airborne virus and can spread through small droplets of saliva in a similar way to the cold and influenza.

Immunity

Immunity refers to the ability of an organism to resist or defend against harmful pathogens, such as bacteria, viruses, fungi, and parasites, as well as against foreign substances or toxins.

Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity.

Innate immunity is a non-specific type of defense that is present at the time of birth. This is accomplished by providing different types of barriers to the entry of the foreign agents into our body.

The body's defense mechanisms against pathogens include physical barriers like skin and mucous membranes, physiological barriers such as stomach acid and tears, cellular barriers involving white blood cells, and cytokine barriers like interferons secreted by virus-infected cells.

Acquired Immunity or Adaptive Immunity - It is the immunity that our body acquires or gains over time. Unlike the

innate immunity, this is not present by birth.

Non-infectious diseases

Non-infectious diseases, also known as non-communicable diseases (NCDs), are medical conditions usually developed due to a combination of genetic, environmental, and lifestyle factors.

Chronic obstructive pulmonary disease (COPD) is a chronic inflammatory lung disease that causes obstructed airflow from the lungs. Symptoms include breathing difficulty, cough, mucus (sputum) production and wheezing.

Diabetes (Diabetes mellitus) is caused by Insulin Deficiency. It is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces.

There are three main types of diabetes: type 1, type 2, and gestational diabetes (diabetes while pregnant). Diabetes damages the nerves and causes problems with sensation. Diabetes is not a water or hygiene-related disease.

Hyperglycaemia, also called raised blood glucose or raised blood sugar, is a common effect of uncontrolled diabetes.

Hypoglycaemia is an abnormally low level of glucose in the blood.

Alzheimer's disease is a brain disorder that slowly destroys memory and thinking skills and, eventually, the ability to carry out the simplest tasks. In most people with the disease those with the late-onset type symptoms first appear in their mid-60s. Alzheimer's disease is the most common cause of dementia.

Arthritis is a condition that affects the joints. Osteoarthritis is the most common type of arthritis.

Hemophilia is an inherited bleeding disorder in which the blood does not clot properly. This can lead to spontaneous bleeding as well as bleeding following injuries or surgery. It is an example of X-linked recessive disorder.

The major types of this condition are hemophilia A (also known as classic hemophilia or factor VIII deficiency) and hemophilia B (also known as Christmas disease or factor IX deficiency).

A **stroke**, sometimes called a brain attack, occurs when something blocks blood supply to part of the brain or when a blood vessel in the brain bursts. In either case, parts of the brain become damaged or die. A stroke can cause lasting brain damage, long-term disability, or even death.

Heart disease is a general term that includes many types of heart problems. It's also called cardiovascular disease. Coronary artery disease (also called coronary heart disease) is the most common type of heart disease. Other heart problems:

Angina is a type of chest pain caused by reduced blood flow to the heart.

Heart attack occurs when the flow of blood to the heart is severely reduced or blocked.

Heart failure - When the heart can't pump enough blood to meet the body's needs.

Arrhythmia - A problem with the rate or rhythm of the heartbeat.

Kwashiorkor is a type of malnutrition characterized by severe protein deficiency. It causes fluid retention and a swollen, distended abdomen. It mostly affects children. Children who develop kwashiorkor may not grow or develop properly and may remain stunted for the rest of their lives.

Cancer

Cancer is one of the most dreaded diseases of human beings and is a major cause of death all over the globe.

Some common types of cancer

Polycythemia causes bone marrow to produce too many red blood cells. Too many red blood cells can make blood thick and sluggish and increase risk of blood clots and complications such as heart attack and stroke.

Leukaemia is a cancer of the Blood. Leukaemia affects white blood cells and can be classified by the type of white cell affected (myeloid or lymphatic) and by the way the disease progresses (acute or chronic). Acute and chronic do not refer to how serious the disease is but to how rapidly it progresses. Acute - short time; Chronic - long duration.

Malignant tumors are cancerous and can spread cancer cells throughout one's body through the blood or lymphatic system, a process known as metastasis.

Cervical cancer originates in cervix cells, the lower end of the uterus. Long-term infection with certain HPV strains is its primary cause, transmitted during sexual contact. Screening tests and HPV vaccination aid in cervical cancer prevention.

Colon cancer (also known as colorectal cancer) is a type of cancer that starts in the colon (large intestine) or rectum.

Ovarian cancer occurs when abnormal cells in the ovaries or fallopian tubes grow and multiply out of control.

Thyroid cancer is a rare type of cancer that affects the thyroid gland, a small gland at the base of the neck that produces hormones.

Blood cancer initiates in the bone marrow where blood forms. Symptoms include fever, fatigue, weight loss, and anemia. Treatments include chemotherapy, radiation, stem cell transplant, surgery, and immunotherapy, with CAR T-cell therapy for specific blood cancers.

Skin cancer occurs when skin cells grow uncontrollably, with three main types: basal cell carcinoma, squamous cell carcinoma, and melanoma. Basal and squamous cell carcinomas are often called "non-melanoma skin cancer" and are mainly caused by overexposure to sunlight's UV rays, damaging skin DNA. Radon (Rn) is a noble gas used in cancer treatment.

Lymphoma is a broad term for cancer that begins in cells of the lymph system.

Genetic disorders

Genetic disorders occur when a mutation affects the genes or when we have the wrong amount of genetic material.

Some Common Genetic disorders -

Down syndrome (Trisomy 21) - It is a genetic condition where a person is born with an extra copy of chromosome 21. This means that they have a total of 47 chromosomes instead of 46. It causes learning difficulties and physical health development. It is an inherited disease.

Fragile X syndrome - It is the leading

cause of inherited intellectual disabilities like **autism**. There are behavioral, physical, intellectual and mental health symptoms.

Klinefelter syndrome - It is a common genetic condition where a male is born with an extra X chromosome.

Trisomy X (Triple-X syndrome) - It is a genetic condition where a female is born with an extra X chromosome.

Turner syndrome - It is a genetic disorder affecting girls and women. The cause of Turner syndrome is a completely or partially missing X chromosome. Turner syndrome symptoms include short stature and lack of breast development and periods.

Edwards syndrome (Trisomy 18) - Also called Edwards syndrome, is a chromosomal condition associated with abnormalities in many parts of the body.

Patau syndrome (Trisomy 13) - It is also called Patau syndrome. It is a chromosomal condition associated with severe intellectual disability and physical abnormalities in many parts of the body.

Other Types of Disease and Disorder

Air borne diseases - The common cold, Influenza, Diphtheria, Pneumonia, Chickenpox, Mumps, Measles, Whooping cough (pertussis), Tuberculosis (TB).

Water borne diseases - Diarrhea, Cholera, Typhoid, Amoebiasis, Meningitis, Dysentery, E. Coli, Guinea worm disease, Salmonella, gastroenteritis, campylobacteriosis, giardiasis, worm infections, Hepatitis A Etc.

Lymphatic filariasis, considered globally as a Neglected Tropical Disease (NTD), is a parasitic disease caused by microscopic, thread-like parasitic worms known as filarial worms.

Itai-itai disease was the name given to the mass cadmium poisoning of Toyama Prefecture, Japan, starting around 1912. It is the most severe form of chronic Cadmium poisoning caused by prolonged oral Cadmium ingestion.

Gingivitis - The gums become swollen and red and may bleed. It is considered the main cause of tooth loss for adults worldwide. In its more serious form,

called periodontitis. The gums can pull away from the tooth, bone can be lost, and the teeth may loosen or fall out.

Convulsion is sudden, violent, irregular movement of the body, caused by involuntary contraction of muscles and associated especially with brain disorders such as epilepsy.

Osteoporosis is a condition of fragile bone with an increased susceptibility. There could be many reasons for osteoporosis, but the most predominant reason is lack of enough calcium in the bones.

Myxovirus is responsible for causing influenza.

Meningitis is an acute or chronic inflammation of the protective membranes covering the brain and spinal cord, collectively called the **meninges**. The most common symptoms are fever, headache, and neck stiffness. It is usually caused by a viral or bacterial infection. In most cases, there is no specific treatment for viral meningitis.

Dysentery is an intestinal infection causing diarrhea with blood or mucus, accompanied by stomach cramps, vomiting, and fever. It's caused by bacterial or parasitic infections. Treatment includes fluid intake and oral rehydration solutions to prevent dehydration.

Guinea worm disease (GWD), also called **Dracunculiasis** is a neglected tropical disease (NTD), is caused by the parasite *Dracunculus medinensis*. A person becomes infected by drinking water containing water fleas infected with guinea worm larvae.

Hemosiderosis disease is caused by excessive iron deposition within the reticuloendothelial cells of the liver, spleen and bone marrow.

Marasmus is a severe form of malnutrition specifically, protein-energy undernutrition. It results from an overall lack of calories. Marasmus is a deficiency of all macronutrients: carbohydrates, fats, and protein.

Hypoxia is a condition in which the body or a region of the body is deprived of adequate oxygen supply at the tissue level. Hypoxia may be classified as either generalized, affecting the whole body, or local, affecting a region of the body.

Dementia is a term used to describe a group of symptoms affecting memory, thinking and social abilities severely. It is characterized by impairment of at least two brain functions, such as memory loss and judgment. There is currently no cure for dementia. But there are medicines and other treatments that can help with dementia symptoms.

Minamata disease is a neurological disease caused by severe mercury poisoning. Symptoms include ataxia, numbness in the hands and feet, general muscle weakness, loss of peripheral vision, and damage to hearing and speech.

A congenital form of the disease can also affect fetuses in the womb and may cause cerebral palsy. Mercury poisoning factor is responsible for the disease called '**Mad hatter**' disease.

Gallstones are hard, pebble-like pieces of material, usually made of cholesterol or bilirubin, that develop in gallbladder.

A benign tumor is a mass of cells (tumor) that lacks the ability to either invade neighbouring tissue or get spread throughout the body.

Scleroderma is a long-lasting disease that affects skin, connective tissue and internal organs. It happens when the immune system causes the body to make too much of the protein collagen.

Dissociative disorders are mental disorders that involve experiencing a disconnection and lack of continuity between thoughts, memories, surroundings, actions and identity.

Somatoform disorders are a group of psychiatric disorders in which patients present with a myriad of clinically significant but unexplained physical symptoms.

Mood disorder (affective disorder) is any of a group of conditions of mental and behavioral disorder, where a disturbance in the person's mood is the main underlying feature.

Gaucher disease is a rare, inherited metabolic disorder caused due to the deficiency of the enzyme glucocerebrosidase.

Hunter syndrome is a genetically associated lysosomal storage disorder due to the deficiency of the iduronate 2-sulfatase enzyme (IDS).

Phenylketonuria (PKU) is an inherited

error of metabolism caused by a deficiency in the enzyme phenylalanine hydroxylase.

Tay Sachs disease (TSD) is a progressive, lethal neurodegenerative disorder caused by a deficiency of enzyme hexosaminidase A.

Short bowel syndrome is a condition in which the body cannot absorb enough nutrients from foods because part of the small intestine is missing or damaged.

Cirrhosis is scarring (fibrosis) of the liver caused by long-term liver damage. Cirrhosis is not transmitted from one person to another.

Obsessive-compulsive disorder (OCD) is a common, chronic, and long-lasting disorder in which a person has uncontrollable, reoccurring thoughts ("obsessions") and/or behaviors ("compulsions") that he or she feels the urge to repeat over and over.

Pellagra is a disease caused by severe vitamin B3 (niacin) deficiency, characterized by the 3D's: diarrhea, dermatitis, and dementia. It can also develop due to gastrointestinal diseases, weight loss surgery, anorexia, excessive alcohol use, and carcinoid syndrome.

Treatment and Prevention

Antibiotics are used to treat or prevent some types of bacterial infection. They kill bacteria or prevent them from reproducing and spreading. Antibiotics aren't effective against viral infections.

An allergy is a reaction by the immune system to something that does not bother most other people. People who have allergies often are sensitive to more than one thing. Substances that often cause reactions are: Pollen, Dust mites.

Oral rehydration solutions (ORS) are used to treat dehydration caused by diarrhea. Most cases of diarrhea are due to bacteria, viruses, or parasites. Digestive system disorders can also cause chronic diarrhea.

The **measles-mumps-rubella (MMR)** vaccine is safe and highly effective in preventing rubella. The vaccine provides lifelong protection against rubella.

Diagnostic Tools and Tests

A C-reactive protein (CRP) test measures the level of C-reactive protein

— a protein made by the liver — in blood. The liver releases CRP into the bloodstream in response to inflammation.

Vivax antigen test detects malaria antigens in the blood sample.

An electroencephalogram (EEG) is a test that measures electrical activity in the brain using small, metal discs (electrodes) attached to the scalp. Brain cells communicate via electrical impulses. EEG can help diagnose a number of conditions including epilepsy, sleep disorders and brain tumors.

A Pap smear (Pap test) is an exam a doctor uses to test for cervical cancer in women.

Widal test is one method that is used to help make a presumptive diagnosis of enteric fever, also known as typhoid fever.

ELISA (enzyme-linked immunosorbent assay) Test is used to detect HIV infection.

Plant Diseases

Plant Diseases caused by viruses and viroids : Tobacco mosaic, cucumber mosaic, barley yellow dwarf, tomato spotted wilt, prunus necrotic ring spot, potato spindle tuber.

Plant Diseases caused by bacteria : Granville wilt, fire blight, soft rot, crown gall, aster yellows, citrus canker.

Plant Diseases caused by fungi : Chestnut blight, Dutch elm disease, black stem rust of wheat, coffee rust, white-pine blister rust, corn smut, apple scab, nectria canker, brown rot.

Other Common Plant Disease are Black Spot, Other Leaf Spots, Powdery Mildew, Downy Mildew, Canker.

Phobias

A phobia is an uncontrollable, irrational, and lasting Fear of a certain object, situation or activity.

Acrophobia - Fear of heights

Aerophobia - Fear of flying

Aichmophobia - Fear of needles or pointed objects

Ailurophobia - Fear of cats

Alektorophobia - Fear of chickens

Algophobia - Fear of pain

Anginophobia - Fear of angina or choking

Anthophobia - Fear of flowers

Arachnophobia - Fear of spiders

Ataxophobia - Fear of disorder or untidiness

Bacteriophobia - Fear of bacteria

Barophobia - Fear of gravity

Batrachophobia - Fear of amphibians

Botanophobia - Fear of plants

Cacophobia - Fear of ugliness

Chromophobia - Fear of colors

Cibophobia - Fear of food

Claustrophobia - Fear of confined spaces

Cynophobia - Fear of dogs

Dendrophobia - Fear of trees

Dentophobia - Fear of dentists

Entomophobia - Fear of insects

Equinophobia - Fear of horses

Heliophobia - Fear of the sun

Hemophobia - Fear of blood

Herpetophobia - Fear of reptiles

Hydrophobia - Fear of water

Hypochondria - Fear of illness

Latrophobia - Fear of doctors

Melanophobia - Fear of the color black

Microphobia - Fear of small things

Mysophobia - Fear of dirt and germs

Necrophobia - Fear of death or dead things

Noctiphobia - Fear of the night

Nosocomophobia - Fear of hospitals

Nyctophobia - Fear of the dark

Ombrophobia - Fear of rain

Ophidiophobia - Fear of snakes

Ornithophobia - Fear of birds

Pathophobia - Fear of disease

Pedophobia - Fear of children

Phobophobia - Fear of phobias

Photophobia - Fear of the light

Pteromerhanophobia - Fear of flying

Pyrophobia - Fear of fire

Selenophobia - Fear of the moon

Somniphobia - Fear of sleep

Tachophobia - Fear of speed

Thalassophobia - Fear of deep water

Tonitrophobia - Fear of thunder

Trypanophobia - Fear of needles or

Injections.

Verminophobia - Fear of germs

Zoophobia - Fear of animals

Taxonomy

Taxonomy is a science that deals with naming, describing and classification of all living organisms including plants. Carolus Linnaeus is considered as the Father of Taxonomy. A P de Candolle in 1813, coined the term Taxonomy.

Linnaeus classified the universe into three broad kingdoms namely Animals, Plants, and Minerals. He classified nature into a hierarchy. The hierarchy is Kingdom, Phylum or division, Class, Order, Family, Genus, and Species.

Biological Classification

In Linnaeus' time, the two Kingdom system of classification with Plantae and Animalia kingdoms was developed. The system did not distinguish between the eukaryotes and prokaryotes, unicellular and multicellular organisms and photosynthetic (green algae) and non-photosynthetic (fungi) Organism.

Biologists, such as Ernst Haeckel (1894), Robert Whittaker (1969) and Carl Woese (1977) have tried to classify all living organisms into broad categories called kingdoms.

R.H. Whittaker (1969) proposed a Five Kingdom Classification. The kingdoms defined by him were named Monera, Protista, Fungi, Plantae and Animalia.

Kingdom Monera

Bacteria, sole members of the Kingdom Monera, are abundant microorganisms. They are prokaryotes and unicellular. Blue- green algae or cyanobacteria have cell walls, but Mycoplasma lacks a cell wall.

Monera organisms can be autotrophic, synthesizing their own food, or heterotrophic, obtaining it from the environment. Many also live as parasites on other organisms.

Bacteria are classified into four shapes: cocci (spherical), bacilli (rod-shaped), vibrios (comma-shaped), and spirilla (spiral).

The filament is the longest portion of the bacterial flagellum which extends from

the cell surface to the outside.

Archaeobacteria thrive in extreme environments like salty areas(halophiles), hot springs (thermoacidophiles), and marshes (methanogens).

Archaeobacteria have unique cell wall structures. Methanogens, found in the guts of ruminant animals like cows and buffaloes, produce methane from their dung, contributing to biogas production.

Eubacteria (true bacteria) are prokaryotic microorganisms consisting of a single cell, lacking a nucleus and containing DNA. Examples - E. coli, Lactobacilli, and Azospirillum.

Photosynthetic bacteria - The cyanobacteria have chlorophyll similar to green plants and are photosynthetic autotrophs.

The cyanobacteria are unicellular, colonial, freshwater, marine or terrestrial algae. Some of these organisms can fix atmospheric nitrogen in specialized cells called Heterocysts. Examples - Nostoc and Anabaena.

Chemosynthetic bacteria oxidize inorganic substances like nitrates and ammonia for ATP production, crucial for recycling nutrients like nitrogen, phosphorus, iron, and sulfur.

Heterotrophic bacteria are abundant and serve as key decomposers. They aid in curd formation, antibiotic production, and nitrogen fixation in legume roots.

Some are pathogens causing damage to human beings, crops, farm animals and pets. Cholera, typhoid, tetanus, and citrus canker are the diseases that are caused by different bacteria.

Reproduction - Bacteria reproduce mainly by fission. Under unfavorable conditions, they produce spores. They also reproduce by adopting a primitive type of DNA transferred from one bacterium to the other.

Mycoplasma - Organisms that completely lack a cell wall. These are the smallest living cells and can survive without oxygen.

Protista Kingdom

Protists, A single-celled eukaryotes, have diverse nutrition modes, including autotrophic and heterotrophic. Mostly aquatic, some have flagella or cilia for

movement. They reproduce via asexual or sexual methods involving cell fusion and zygote formation.

Chrysophytes, including diatoms and golden algae, are mainly photosynthetic. Diatoms, important ocean producers, are microscopic and passive plankton.

Dinoflagellates (fire algae), mainly marine and photosynthetic, often possess two flagella. Examples include Noctiluca and Gonyaulax.

Euglenoids are mostly freshwater organisms in stagnant water, lacking a cell wall but having a flexible protein-rich layer called a Pellicle. They're photosynthetic in sunlight but act as heterotrophs in darkness, preying on smaller organisms.

Slime molds are non-photosynthetic saprophytes that form plasmodium aggregates in favorable conditions, spreading widely. Examples include Stemonitis, Physarum, Dictyostellida, Lycogala, and Tubifera. In unfavorable conditions, plasmodium differentiates into fruiting bodies with spores at their tips, possessing true walls.

Protozoans - All protozoans are heterotrophs and live as parasites. There are four major groups of protozoans.

Amoeboid protozoans inhabit freshwater, seawater, or moist soil, capturing prey using pseudopodia. Silica adorns marine species. Some, like Entamoeba, are parasitic.

Flagellated protozoans- The members of this group are either free-living or parasitic. They have flagella. The parasites cause sleeping sickness. Example: Trypanosoma.

Ciliated protozoans - These are aquatic organisms with thousands of cilia for active movement. They have a gullet opening to the cell surface, where coordinated cilia movement directs food-laden water into the gullet. Example: Paramecium

Sporozoans, in the phylum Sporozoa, are single-celled parasites. The notable Plasmodium, a malarial parasite, is classified in the phylum Apicomplexa.

Kingdom Fungi

Fungi are heterotrophic eukaryotes, mostly saprophytes, absorbing organic matter from dead substrates. Parasitic fungi rely on living organisms. They can also live symbiotically with algae (lichens) or plant roots (mycorrhiza).

Fungi reproduce vegetatively through fragmentation, fission, and budding.

Fungi have chitin cell walls and prefer warm, humid conditions. They're mostly filamentous, forming mycelium from hyphae, but yeast is unicellular. Button Mushrooms and toadstools are also fungi.

Phycomycetes reproduce asexually via zoospores (motile) or aplanospores (non-motile). Examples - Mucor and Rhizopus (bread mold) and Albugo (parasitic fungus on mustard).

Ascomycetes (sac fungi), include multicellular forms like Penicillium and unicellular forms like yeast (Saccharomyces).

Basidiomycetes, like mushrooms and bracket fungi, thrive in diverse habitats including soil, logs, and living plants as parasites causing diseases like rusts and smuts. They lack sex organs and typically reproduce vegetatively through fragmentation.

Deuteromycetes, known as imperfect fungi, exhibit only the asexual or vegetative phases.

Plantae Kingdom

These are autotrophs and use chlorophyll for photosynthesis. Some, like insectivorous plants (e.g., Bladderwort, Venus flytrap), or parasites like Cuscuta, are partially heterotrophic. The plant life cycle alternates between diploid sporophytic and haploid gametophyte phases.

The lengths of the haploid and diploid phases are either free-living or dependent on others. It varies among different groups in plants. This phenomenon is called alternation of generation.

Animalia Kingdom

These include multicellular eukaryotes without cell walls and are heterotrophs, often relying on plants for food. They digest internally, storing energy as glycogen or fat, and come in varied shapes and sizes. Most can move and reproduce sexually through copulation, followed by embryological development.

Viruses, Viroids, Prions and Lichens

Whittaker's five kingdom classification does not include lichens or acellular like viruses, viroids, and prions.

Viruses - It is found dead in the environment. They are obligate parasites, taking over host cell machinery to replicate, often killing the host.

It contains genetic material, either RNA or DNA. Plant viruses typically have single-stranded RNA, while those infecting animals may have single-stranded or double-stranded RNA, or double-stranded DNA. Bacterial viruses or bacteriophages are usually double stranded DNA viruses.

The capsid, composed of capsomeres, protects the nucleic acid. Capsomeres form helical or polyhedral geometric structures.

Viroids are discovered by T.O. Diener. They are infectious agents smaller than viruses. They consist of free RNA without a protein coat. Viroid RNA is low in molecular weight. Viroids, like the one causing potato spindle tuber disease, are known pathogens.

Prions are unique proteins capable of inducing disease by causing normally healthy brain proteins to fold abnormally. Unlike bacteria and viruses, prions lack genetic material and consist solely of proteins. Notable diseases caused by prions include Bovine Spongiform Encephalopathy (BSE), commonly known as mad cow disease in cattle, and its human variant, Creutzfeldt-Jakob disease (CJD).

Lichens - Lichens are symbiotic associations between algae and fungi. Lichens are very good pollution indicators - they do not grow in polluted areas.

The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. Algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

Amoeba - It is a single-celled organism capable of changing shape. It reproduces asexually through binary fission. Examples - Entamoeba Histolytica, Amoeba Proteus, Chaos carolinense etc.

Amoeba acquires food through endocytosis. Pseudopodia are called the false feet as they are actually extensions of the cytoplasm or the thick liquid that is inside organisms like an Amoeba.

Zooplankton are tiny aquatic organisms like crustaceans and rotifers that feed on phytoplankton and serve as vital food for fish and larger creatures.

Plant Kingdom

Plant Kingdom - They are eukaryotic, multicellular and autotrophic organisms.

Paleobotany - It is the branch of botany dealing with the recovery and identification of plant remains from geological contexts. **Birbal Sahni** is the father of Indian Paleobotany.

Classification of Plant Kingdom

Plant kingdom is divided into Algae, Bryophytes, Pteridophytes, Gymnosperms, and Angiosperms.

Thallophyta- It comprises non-motile organisms without a well-differentiated body, forming a polyphyletic group based on shared characteristics but lacking a common ancestor. They represent the lowest plants in the plant kingdom. Examples include Ulothrix, Spirogyra, Cladophora, Fucus, Volvox, Ulva, Chara, and Lichen.

The division of Thallophyta is classified into two subdivisions: Algae and Fungi.

Fungi

Fungi are eukaryotic microorganisms encompassing yeasts, molds, and mushrooms, reproducing via spores. Lacking chlorophyll, they cannot perform photosynthesis and store food as glycogen. Biosynthesis of chitin occurs in fungi. Morchella is an example of a fungus.

Use of Fungi - Fungi, derived from Penicillium for penicillin, play vital roles: medicinal (antibiotics), pest control, decomposition (nutrient recycling), and food production (mushrooms, fermentation).

Algae

Algae are chlorophyll-bearing, thalloid, autotrophic organisms primarily found in aquatic environments. They are simpler than aquatic plants, lacking a true root, leaf, and stem system. Algae inhabit oceans, lakes, rivers, ponds, and moist soil.

They reproduce by vegetative (fragmentation), asexual and sexual

methods(fusion of two gametes). FE Fritsch explained the structure and reproductive system of algae in 1935.

Use of Algae - Algae, essential in ecological systems, fix half of Earth's carbon dioxide through photosynthesis, regulating the carbon cycle and oxygen levels. They serve as nutritious food, organic fertilizers, and aid in sewage treatment, contributing to pollution control and environmental health.

Types of Algae :-

Chlorella (Chlorella pyrenoidosa) is a type of algae that grows in freshwater. It's used for nutrition and as medicine as it is rich in proteins and B-complex vitamins. It is used as food supplements by space travelers.

Chlorophyceae - They are commonly called green algae. Their body may be unicellular, colonial or filamentous. They are usually grass green due to the dominance of pigments chlorophyll a and b. Examples are Chlamydomonas, Volvox, Ulothrix, Spirogyra, Chara, Cladophora, Ulva, etc.

Phaeophyceae - The members of phaeophyceae or brown algae are found primarily in marine habitats. They show great variation in size and form. Asexual reproduction occurs through Zoospores. Examples are Ectocarpus, Dictyota, Laminaria, Sargassum and Fucus.

Laminaria algae is also called Kelp found in the deep sea.

Rhodophyceae - The members of rhodophyceae are commonly called red algae because of the predominance of the red pigment, r-phycoerythrin in their body.

The water holding compounds produced in large quantities by marine brown and red algae are called **Hydrocolloids**. The stored food material is in the form of floridean starch. Examples are Polysiphonia, Porphyra, Gracilaria and Gelidium.

The **Cyanophyceae** (Cyanobacteria or blue green algae) are a group of photosynthetic microorganisms.

Note - Blue-Green Algae are not algae, they are **Cyanobacteria** (prokaryotes).

Bryophytes

They are found both on land and in water so they are amphibians like liver warts,

horn warts, moss etc. These plants can live in soil but are dependent on water for sexual reproduction. Bryophyta are the amphibians of the **plant kingdom**. Examples are Funaria, Martia, Rechesia, etc

Reproduction - They reproduce asexually by fragmentation and sexually by spores. **Antheridium** is a structure that produces and holds sperm cells in bryophytes and ferns.

Uses of Bryophytes - They are known for their excellent water absorption, serve as effective flood preventives and aid in soil erosion control. Moss plants, with stem and leaf-like structures, find utility as peat energy and antiseptics. Some bryophytes, like Sphagnum, double as packaging materials.

The bryophyta division consists of three groups of non-vascular land plants, namely – Mosses, liverworts and hornworts.

Mosses - Life cycle of a moss consists of two stages: The first stage is the Protonema Stage, Which develops directly from a spore. The second stage is the Leafy Stage, which develops from the secondary protonema as a lateral bud. Examples are Funaria (lives in many environments and it has small, flattened leaves, root-like rhizoids, and peristome), Polytrichum and Sphagnum.

Liverworts - The liverworts grow usually in moist, shady habitats. The plant body of a liverwort is **thalloid**, e.g., Marchantia.

Marchantia form umbrella shaped structures that raise gametangia above the main gametophyte body and sporophytes develop below these structures.

Hornworts - They typically grow in moist soil or on damp rocks. Hornworts reproduce via spores produced in sporophyte structures at the base of the plant. Example: Anthoceros punctatus (Dotted Hornwort).

Pteridophytes

Pteridophytes are vascular plants that reproduce via spores, lacking flowers or seeds. Despite this, they possess roots, leaves, and vascular tissues like xylem and phloem. Notable examples include Ferns, Fern Allies, Clubmosses, Spike Mosses, Quillworts, Funaria (Moss), Horsetails, and Lycophytes, representing an early stage in terrestrial plant

evolution. They reproduce sexually via spores.

Benefits - Use in medicine, ornamentation, food, phytoremediation, and agriculture.

Types of pteridophytes - Psilopsida (Psilotum), Lycopsidea (Selaginella, Lycopodium), Sphenopsida (Equisetum) and Pteropsida (Dryopteris, Pteris, Adiantum).

Equisetum is the only living representative of Sphenopsida that has an underground creeping and perennial rhizome that gives off aerial as well as underground branches.

The Transport of male gametes in bryophytes and pteridophytes occurs through **water**.

Gymnosperms

Gymnosperms are plants characterized by seeds that lack a protective covering and ovaries. They often manifest as medium to tall trees or shrubs, with tap roots being common. Typically found in colder regions experiencing snowfall, some exceptions like cycads thrive in dry and tropical areas.

Their leaves are needle-like, a feature adapted to reduce surface area. Unlike angiosperms, they produce seeds but no fruits. Notable examples include Conifers, Gnetophytes, Pine, Ginkgo, and Pinus Trees.

Reproduction - It can occur both sexually and asexually, albeit at a slower pace compared to many other plant species. The process from pollination to seed maturation may take up to three years.

Benefits - Gymnosperms also offer nutritional benefits, with species like ginko, pinus, and cycas serving as food sources.

Types of Gymnosperms - Cycadophyta, Ginkgophyta, Gnetophyta, Coniferophyta, among others.

Angiosperms

These are seed-producing flowering plants whose seeds are enclosed within an ovary. Their ovules are developed in specialized structures called flowers. The **seeds** are enclosed in fruits. Female sex organ is Pistil or Carpel. Their vascular tissues are extremely well developed. It is the largest and most

diverse group in the plant kingdom. Examples are Rose, Tomato, Corn, Rubiaceae, Euphorbiaceae, Malvaceae etc.

Reproduction - Angiosperms reproduce sexually via the fusion of male and female gametes during pollination and fertilization. They also employ asexual reproduction through vegetative structures like stems, roots, and leaves.

Benefits - They provide food, fodder, fuel, medicines and other products.

Type of Angiosperm

Monocot - They are having an embryo with a single cotyledon. Examples are Grasses, Cereals, Corn, Wheat etc.

Dicot - Dicot seeds are seeds with two embryonic leaves and cotyledons. Examples are bitter gourd seeds, Castor seeds, Mango seeds, Neem seeds, Night Jasmine seeds, Papaya seeds, and Tamarind seeds, etc.

Types of Plants

Creepers - The plant with weak stems that grow along the ground, around another plant, or up a wall by means of extending stems or branches. Examples are Bottle gourd, Watermelon, Pumpkin, Strawberry, Bougainvillea, Bignonia, Cucumber, etc.

Climbers - These are plants with weak, long, and thin green stems that require support like walls or trees to grow or carry weight. They have flexible, soft tissues, enabling rapid growth. Examples include grapes, sweet gourds, bitter gourds, and bottle gourds.

Shrubs - These are characterized by woody stems, equal-sized branches, and stems close to the soil, resulting in a bushy appearance and maximum height of about 3 meters. Examples include lemon, tulsi, rose, jasmine, bougainvillea, china rose, pomegranate, and henna.

Herbs - They are those plants which have green and tender stems. They are usually short and may not have many branches. They have a very short life span of one or two seasons. Examples are Tomato, Mustard, Radish, Sunflower, Carrot, Ginger, Turnip, Wheat, Paddy (Rice), Cabbage, Mint etc.

Trees - They are big and tall plants. They have very thick, woody and hard stems

called the trunk. Examples are Banyan, Mango, Neem, Cashew, Teak and Oak.

Plant types based on adaptations

Hydrophytes - Plants that are adapted to live in aquatic environments are called hydrophytes. They have floating leaves with long, fine and dissected petioles to prevent flooding of water.

Mesophytes - They are terrestrial plants which are neither adapted to particularly dry nor particularly wet environments. Examples are corn, rose, clover, squash, etc.

Xerophytes (dry + plant) are plants which are able to live in very dry places. When water is abundant, their rate of transpiration is about the same as other plants. Examples are pineapple, acacia, prickly pear and alfalfa.

Halophyte - A plant adapted to growing in **saline conditions**. Examples are Mangrove trees, thrift (Armeria), sea lavender (Limonium), and rice grass (Spartina).

Heliophytes - These are the plants that are adapted to live under plenty of sunlight.

Sciophytes - The Plants which are grown under shades are known as Sciophytes.

Plant Morphology Root system in plants

The root is the part of a plant that anchors it to the ground and absorbs water and nutrients from the soil. It also provides support, prevents soil erosion, and offers food and shelter for various organisms. The root system consists of different types of roots, each serving specific functions.

Types of Root Systems:

Taproots - Found in dicotyledonous plants, taproots grow vertically downward and serve as the primary root system. Examples include carrots, radishes, and turnips.

Modification of tap roots -

For Storage of Food - Tap roots of some plants become fleshy due to storage of reserve food.

Aerial Roots - Grow on bark of other

plants, Banyan and Peepal

For Nitrogen Fixation - Nodular root members of the family have swellings called root nodules on their main roots and their branches.

These nodules store nitrogen fixing bacteria called **Rhizobium** which converts atmospheric nitrogen into nitrates and supply them to the plants. These nodules are red due to the existence of a pigment well-known as leg hemoglobin.

Respiration Process - The roots emerge out of the ground and grow upwards to get oxygen for respiration. The root tips of these plants have minute pores called lenticels through which they respire. For example **Rhizophora**.

Fibrous roots - Bushy and moderately branching, fibrous roots grow from the stem in monocotyledonous plants like rice and wheat.

Adventitious root - These roots grow from various parts of the plant other than the radicle. They can sprout from stems, branches, or even leaves. Examples include money plants and bamboo.

Breathing root - These roots grow above the soil surface to access oxygen for respiration. They are common in plants like Rhizophora.

Napiform roots - The type of root that becomes swollen and spherical at the upper end and tapered like a thread at their lower end examples turnip (Brassica rapa), sugarbeet.

Buttress root - These roots grow from the trunk to support the tree, as observed in jackfruit trees.

Modification of Adventitious Roots -

Storage Roots : Some roots become fleshy due to the storage of reserve food.

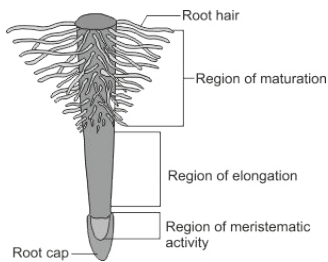
Photosynthetic or Assimilatory Roots - These roots develop chlorophyll when exposed to sun and can prepare food for plants by photosynthesis. Example is Tinospora (orchid).

Reproductive Roots - Adventitious roots of some plants help in vegetative propagation by developing buds that give rise to new shoots. Example is sweet potato.

Sucking Roots - These are microscopic, developed by the roots to absorb nutrients from the host. These are also

known as parasitic roots.

Regions of the Root



Root cap - The root is covered at the apex by a thimble-like structure called the root cap.

Region of meristematic activity- A few millimeters above the root cap is the region of meristematic activity. The cells of this region are very small, thin-walled and with dense protoplasm.

Region of elongation - The cells that are formed increase in length and thereby increase the length of the root. The area where these cells are formed is called the zone of elongation.

Region of maturation - In this region, cells mature into the various types of primary tissues.

Root hairs - Root hairs are long tubular-shaped outgrowths from root epidermal cells (provide a protective barrier against mechanical injury, water loss, and infection). The root hairs absorb water and minerals from the soil.

Periblem - It is the middle layer in the root apical meristem (regions of unspecialised cells in plants that are capable of cell division) that gives rise to the cortex.

Calyptragen - It is a layer of rapidly dividing cells at the tip of a plant root, from which the root cap is formed.

Dermatogen - It is the outer primary meristem of a plant or plant part that, according to the histogen theory, gives rise to epidermis.

Plerome - It is the central core of the primary meristem of a plant or plant part that according to the histogen theory gives rise to the stele. The term meristem was coined by C. Nageli 1858.

Stem

Stem is that portion of plant which grows upwards. It may be **herbaceous** (composed of vascular bundles (xylem

and phloem) arranged in a circle around a central core of spongy tissue made up of parenchyma cells, called the pith) or **woody**.

Functions of Stem - Stems support plants and transport fluids between root and shoot via xylem and phloem. They also store nutrients, produce new cells, and aid in food assimilation.

Types of Stems -

Underground Stem - The stem that grows inside the soil is known as the underground stem. Such types of stems store food for contingency periods. Example- Potatoes.

Subaerial Stem - The stem, which partially remains inside the soil and partial above (i.e. in the air), is known as subaerial stem. Example-Cynodon.

Aerial Stem - The stem, which entirely remains in the air (i.e. out-side of soil or water), is known as aerial stem. Examples - passiflora, grapes, etc.

Modification of Stems -

In some plant stems perform some specific task (other than their regular task), for which they change their shapes and sizes. Some examples are:

Underground Modified Stem - Bulb (Garlic, Onion, etc.), Corm (Saffron, Crocus, etc), Stem tuber (Potato), Rhizome (Ginger).

Subaerial Modified Stem - Stolon (Jasmin, Straberi, etc), Sucker (Roses, Musa, etc.), Runner (Mereilia, Cynodon, etc.)

Aerial Modified Stem - Stem thorn (Lemon, Citrus), Stem tendril (Grape) , Phylloclade (Cactus, Opuntia).

Leaf

Leaves are flat lateral outgrowths in plants, vital for photosynthesis and transpiration. They vary in shape, size, and color, with a typical structure comprising a leaf base, petiole, and lamina. Stipules, small leaf-like structures, are often found at the base. Venation, the pattern of veins in the blade, facilitates nutrient transport.

Leaves, vital for plants, are lateral outgrowths vital for photosynthesis and transpiration. They vary in shape, size, and color, with a structure including a

base, petiole, and lamina. Stipules at the base and venation aid nutrient transport.

Parts of a Leaf

Leaf Base - This is the part where a leaf attaches to the stem. Leaf base has two small leaf-like structures called **stipules**.

Petiole - It is a long, thin stalk connecting the leaf blade to the stem. It contains tiny tubes that connect the leaf veins to the stem, facilitating water transport to the leaf and food transport away from the leaf to other parts of the plant.

Lamina - Also known as leaf blade. It is the green, flat surface of the leaves. It consists of a small branched vein and veinlets.

Functions of Leaves -

Photosynthesis - This is the most important function of a leaf. They contain chloroplasts which have the pigment chlorophyll (a pigment that gives plants their green color) that is responsible for helping in photosynthesis. Mesophyll cells (help in photosynthesis) also found in leaves.

Helps the plant breathe - The leaf epidermis contains guard cells that regulate stomata, small pores on the leaf undersurface. Stomata control water and gas exchange and pores get open when water flows into the guard cells.

Storage of food - In some plants, the leaves are modified to store food. These plants generally have succulent leaves as seen in xerophytic plants (any plant adapted to life in a dry or physiologically dry habitat).

The Fruit

Fruit is the sweet, fleshy product of a plant containing seeds and edible as food. It consists of the pericarp, the outer covering layers, and the seeds within.

The **Pericarp** of a fruit can be further divided into-

Exocarp - It is the outermost layer which is formed from the outer layer or the epidermis.

Mesocarp - It is the second or the middle layer which is often juicy and varies in thickness in different fruits.

Endocarp - It is the innermost layer and also is different in different kinds of fruits.

Types of Fruit -

True Fruits - When the fruit forms in the ovary (of the flower) through fertilization is known as true fruit. **Example-** Mango, Banana, Tomato, Kiwi fruit, Blueberry, Plum etc.

False Fruits - The fruits formed by some other means (other than ovary), such as calyx, thalamus, corolla, etc. known as false fruits. Example- Pear, Apple, Strawberry, Pineapple, Mulberry, etc.

Simple Fruits - These fruits are formed from a single pistil only. They are divided into Fleshy (have a fleshy and juicy pericarp) and Dry fruits (do not have juicy or thick pericarps) based on the nature of their pericarp and its layers. Examples are Apple, banana, cherry pear, plum, tomato, etc.

Fruits and their Edible Parts - Apple (Thalamus), Banana (Mesocarp), Coconut (Endosperm), Coriander (Thalamus), Chinese date (Epicarp & Mesocarp), Custard apple (Pericarp), Guava (Pericarp), Grape (Pericarp), Ground nut (Seed leaves), Jackfruit (Sepals), Lemon (Juicy pore), Litchi (Pulpy aerie), Mango (Mesocarp), Mulberry (Bract, sepals), Orange (Juicy hair), Pear (Thalamus), Papaya (Pericarp), Tomato (Pericarp), Wood apple (Mesocarp).

Seed

Seed is a ripened ovule that contains an embryonic plant with sufficient reserve food for the development of an embryo, the ovules after the fertilization process develops into seeds.

Types of Seed -

Monocotyledonous seeds - Any group of flowering plants having an embryo with a single cotyledon and usually leaves with parallel veins and flower parts in groups of three compare dicotyledon. Examples: Orchids, Alisma, corn, rice, Grasses, etc.

Structure of Monocotyledonous seeds - Seed Coat - In cereals such as maize, the seed coat is membranous and generally fused with the fruit wall (pericarp), known as Hull. The integuments are the outer layer of the ovule and develop into a seed coat as the ovule matures following fertilization.

Endosperm - The endosperm stores

food. In general, monocot seeds are endospermic but some exceptions are there like orchids which are non-endospermic.

Aleurone layer - The outer covering of the endosperm detaches the embryo by a proteinous layer called the aleurone layer.

Embryo - The embryo is the small fleshy body seen after removing the seed coat, it is located in a groove at one end of the endosperm.

Dicotyledonous seeds - Any member of the flowering plants, or angiosperms, that has a pair of leaves, or cotyledons, in the embryo of the seed. Examples: Gram, pea, bean, mustard, mango, Rose, etc.

Structure of Dicotyledonous seeds -

Testa - It is the outer, thick, protective layer. It protects the seed from pathogens like bacteria, fungi, insects, etc.

Tegman - It is an inner, thin, membranous covering that protects the inner part of the seed.

Hilum - It is a scar on the seed coat through which the developing seed is attached to the fruit.

Micropyle - This is a small pore present above the hilum. It is the entry for oxygen, water, etc., into the seed.

Raphe - It is a ridge-like structure present around the median groove. It represents the portion of the stalk which remains fused with the test.

Embryo - It has an embryonal axis and two cotyledons.

Important, Harmful and Valuable Plants

Parthenium hysterophorus (carrot grass or congress grass) - It is native to the American tropics. It triggers allergic respiratory problems, asthma, bronchitis, contact dermatitis, and mutagenicity in cattle and humans.

Water hyacinth is a major freshwater weed in most of the frost-free regions of the world and is generally regarded as the most troublesome aquatic plant. But it can be used to feed animals, or processed to improve the soil, as compost or biochar.

Pitcher plant - A plant with a deep

pitcher-shaped pouch that contains fluid into which insects are attracted and drowned. Nutrients are then absorbed from their bodies by the plant. They are mainly found in Meghalaya (Khasia & Jaintia hills).

Turmeric - Turmeric is a rhizome (root) that comes from the plant of the ginger family native to India and is used as a spice in cooking. It has a yellow pigment known as Curcumin.

Ficus Religiosa (Pipal) - Used for treating many diseases in traditional medicines like Ayurveda, Homeopathy, Unani, and Siddha.

The Venus flytrap - They are a perennial carnivorous plant of the sundew family that attracts prey and usually traps insects and then breaks them down with digestive enzymes. It is native to North Carolina and South Carolina.

Opuntia (cactus) - It shows adaptation in the desert area through leaves that are reduced to spines that reduce water loss.

Mimosa (Touch me not) - It helps in the treatment of many disorders like piles, insomnia, diarrhea, etc.

Hibiscus (Orhul) - It can help boost the immune system.

Pea (Pisum sativum) - It is an herbaceous annual plant in the family Fabaceae.

Fern - A fern is a member of a group of vascular plants that reproduce via spores and have neither seeds nor flowers.

Oleo-gum-resin (Asafoetida) can be obtained from the exudates of the roots of the Iranian endemic medicinal plant, Ferula asafoetida.

Sugarcane (Saccharum officinarum) - It is perennial grass of the family Poaceae, primarily cultivated for its juice from which sugar is processed.

Cryptogamae - A cryptogam (scientific name Cryptogamae) is a plant (in the wide sense of the word) that reproduces by spores, without flowers or seeds. Example - Pteridophyta.

Crocus Sativus (Saffron) - It is widely cultivated in Iran and other countries such as India and Greece and has great economic value.

Cocos nucifera (Coconut tree) - The husk

of coconut is made of sclerenchymatous tissue (provide strength to plants). It is eaten as food or used as medicine.

Monoecious plant - Monoecious plants have both male and female reproductive parts in the same plant. Examples are Maize, Cucumber, Fig, Melon, etc.

Calotropis - It is a poisonous plant that causes vision loss. It is a noxious weed that has a chemical defense mechanism to protect it from predators.

Animal Kingdom

Carl Linnaeus introduced a complete biological system of naming and classification of animals.

Linnaeus divided the animal kingdom into several classes: Mammalia, Aves, Amphibia, Pisces, Insecta and Vermis.

Animal kingdom is classified on the basis of arrangement of cells, body symmetry, nature of coelom, patterns of digestive, circulatory or reproductive systems etc.

Classification of Animal Kingdom

Phylum Porifera - Members of this phylum are commonly known as sponges. They are generally marine and mostly asymmetrical animals. They have a cellular level of organization.

Sponges have a water transport or canal system. Water enters through minute pores (ostia) in the body wall into a central cavity, spongocoel, from where it goes out through the osculum.

Sponges are multicellular animals with no regular tissues, featuring numerous pores on their bodies and a central cavity called the sponge cavity.

Phylum Cnidaria, or Coelenterata: It comprises mostly marine, radially symmetrical animals that can be sessile or free-swimming. They're named for cnidoblasts or cnidocytes, specialized cells containing stinging capsules called nematocytes. These are used for defense and capturing prey. Cnidarians are diploblastic and exhibit two body forms: polyp (sessile, cylindrical, like Hydra) and medusa (umbrella-shaped, like jellyfish).

Some live in colonies (corals), and some have solitary lives (hydra). Examples

include Physalia (Portuguese man-of-war), Adamsia (Sea anemone), Pennatula (Sea-pen), Gorgonia (Sea-fan), Meandrina (Brain coral), and Hydra.

Those cnidarians which exist in both forms exhibit alternation of generation (Metagenesis), i.e., polyps produce medusae asexually and medusae form the polyps sexually (e.g., Obelia).

Phylum Ctenophora - Ctenophora are commonly known as sea walnuts or comb jellies. They are exclusively marine, radially symmetrical, diploblastic. They have biradial symmetry, an oral-aboral axis of symmetry, and three layers (two cell layers and one thick cellular mesoglea). Some true organs are present.

Examples - Ctenoplane, Hormiphora, Pleurobrachia (sea-gooseberry), Cestum (Venus's girdle), etc.

Phylum Platyhelminthes (flatworms) have a flattened body, making them bilaterally symmetrical, triploblastic, and acoelomate. They lack a coelom, skeleton, and anus.

Flatworms include endoparasites found in animals, such as tapeworms (e.g., Taenia), liver fluke (Fasciola), and Planaria, a freshwater flatworm. They possess specialized cells called flame cells for osmoregulation and excretion, with some members like Planaria exhibiting high regeneration capacity.

Phylum Nematoda, or roundworms, have a circular cross-section, distinguishing them by their cylindrical shape. They possess a pseudocoelom, with a distinct alimentary canal featuring both mouth and anus.

These bilaterally symmetrical, triploblastic organisms exhibit sexual dimorphism and organ-system level organization. Examples include Ascaris, Wuchereria, and Ancylostoma.

Ascaris has a well defined muscular Pharynx.

Phylum Annelida - They are aquatic (marine and freshwater) or terrestrial; free-living, and sometimes parasitic. They are triploblastic, metamerically segmented and coelomate animals. Their body surface is distinctly marked out into segments or metamerites. Aquatic annelids like Nereis possess **lateral appendages**, parapodia, which

help in swimming.

Phylum Arthropoda - This is the largest phylum of Animalia which includes insects. Over two-thirds of all named species on earth are arthropods. Class Insecta is the largest class of Phylum Arthropoda.

Arthropod means jointed legs. Animals which have jointed appendages belong to this phylum. The body consists of the head, thorax and abdomen. Excretion takes place through malpighian tubules. Examples are Apis (Honey bee), Bombyx (Silkworm), Laccifer (Lac insect), Vectors - Anopheles, Culex and Aedes (Mosquitoes), Gregarious pest - Locusta, Living fossil - Limulus (King crab), Scorpion, Millipedes, Centipede.

Class Insecta examples: Cockroach, mosquito, silverfish, Crayfish.

Phylum Mollusca - The second largest animal phylum, comprises terrestrial or aquatic organisms with an organ-system level of organization. They are bilaterally symmetrical, triploblastic, and coelomate, possessing a calcareous shell covering an unsegmented body with a distinct head, muscular foot, and visceral hump.

Molluscs feature a radula, a file-like rasping organ, in the mouth for feeding. They are typically dioecious and oviparous, often with indirect development. Examples include Pila, Pinctada, Sepia, Loligo, Octopus, Aplysia, Dentalium, and Chaetopleura.

Phylum Echinodermata - The term Echinodermata is derived from the Greek words, echinos meaning hedgehog and derma meaning skin. Thus, echinoderms are spiny-skinned animals. The adult echinoderms are radially symmetrical but larvae are bilaterally symmetrical. It has a peculiar water driven tube system that they use for moving around.

They have a water **vascular system** which helps in locomotion, capture and transport of food and respiration. An excretory system is absent. Sexes are separate. Reproduction is sexual. Examples are Asterias (Star fish), Echinus (Sea urchin), Antedon (Sea lily), Cucumaria (Sea cucumber) and Ophiura (Brittle star).

Phylum Hemichordata - Hemichordates have a rudimentary structure in the collar region called stomochord, a structure

similar to notochord. This phylum consists of a small group of worm-like marine animals with organ-system level of organization. They are bilaterally symmetrical, triploblastic and coelomate animals. The body is cylindrical and is composed of an anterior proboscis, a collar and a long trunk. Earlier they were part of Chordata. Examples are Balanoglossus and Saccoglossus.

Phylum Chordata - Animals belonging to phylum Chordata are fundamentally characterized by the presence of a notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits. They are bilaterally symmetrical, triploblastic, coelomate with organ-system level of organization.

Phylum Chordata is divided into three subphyla: Urochordata, Cephalochordata and Vertebrata.

Urochordata and Cephalochordata - They are referred to as protochordates and are exclusively marine. Examples are Urochordata Ascidia, Salpa, Doliolum; Cephalochordata - Branchiostoma (Amphioxus or Lancelet).

Notochord is an embryonic midline structure common to all members of the phylum Chordata that serves as a source of midline signals to surrounding tissues and as a major skeletal element of the developing embryo.

Vertebrata - All vertebrates are chordates but all chordates are not vertebrates. The notochord is replaced by a cartilaginous or bony vertebral column in the adult. Vertebrates are bilaterally symmetrical, triploblastic, coelomic and segmented, with complex differentiation of body tissues and organs. Smooth endoplasmic reticulum plays a crucial role in detoxifying the liver from many drugs.

Class Cyclostomata - All living members of the class Cyclostomata are ectoparasites on some fishes. They have an elongated body bearing 6-15 pairs of gill slits for respiration. Examples are Petromyzon (Lamprey) and Myxine (Hagfish).

Class Chondrichthyes - They are marine animals with streamlined bodies and cartilaginous endoskeletons. They have ventrally located mouths, persistent notochords, and separate gill slits without opercula.

Their tough skin is covered in placoid scales, and some species have electric organs or poison stings. Chondrichthyes are cold-blooded animals. Examples include Scoliodon, Pristis, Carcharodon, and Trygon.

Class Osteichthyes: It includes both marine and freshwater fishes with bony endoskeletons. They have streamlined bodies, mostly terminal mouths, and four pairs of gills covered by opercula. Their skin is covered in cycloid or ctenoid scales, and they possess an air bladder for buoyancy regulation. Examples include Exocoetus, Hippocampus, Labeo, Catla, Clarias, Betta, and Pterophyllum.

Class Amphibia: It includes animals capable of living in both aquatic and terrestrial environments. They typically have two pairs of limbs and a body divided into head and trunk. Their ear is represented by a tympanum, and they possess a cloaca, where the alimentary canal, urinary, and reproductive tracts open.

Amphibians have a three-chambered heart and are cold-blooded, with external fertilization. Examples include Bufo, Rana, Hyla, Salamandra, and Ichthyophis.

Class Reptilia: It comprises mostly terrestrial animals with dry, cornified skin covered in epidermal scales or scutes. Their hearts are typically three-chambered, although crocodiles have four chambers. Reptiles are poikilothermic.

Snakes and lizards shed their scales as skin casts. Examples include Chelone, Testudo, Chameleon, Calotes, Crocodilus, Alligator, Hemidactylus, and Poisonous snakes such as Naja, Bangarus, and Vipera.

Class Aves - The characteristic features of Aves (birds) are the presence of feathers and most of them can fly except flightless birds (e.g., Ostrich). The forelimbs are modified into wings. The hind limbs generally have scales and are modified for walking, swimming or clasping the tree branches. The heart is completely four chambered. They are warm-blooded (homoiothermous) animals, i.e., they are able to maintain a constant body temperature. Respiration is by lungs.

Examples are Corvus (Crow), Columba (Pigeon), Psittacula (Parrot), Struthio (Ostrich), Pavo (Peacock), Aptenodytes

(Penguin), Neophron (Vulture), Parrot, etc.

Class Mammalia: It is diverse, inhabiting various habitats from polar ice caps to deserts and forests. They exhibit adaptations for walking, running, climbing, burrowing, swimming, or flying with two pairs of limbs.

Mammals are characterized by milk-producing mammary glands for nourishing young, hair-covered skin, external ears, and a four-chambered heart. They are warm-blooded and mostly viviparous, with direct development. Examples include Ornithorhynchus, Macropus, Pteropus, Camelus, Macaca, Rattus, Canis, Felis, Elephas, Equus, Delphinus, Balaenoptera, Panthera tigris, and Panthera leo.

Some facts related to animal kingdom and animals

Benthos, also known as benthon, is the community of organisms that live at the bottom of a sea, river, lake, or stream. Types of benthos - Sea stars, sea squirts, sea slugs, corals, crabs, whelks (sea snails). The Benthos name was coined by **Haeckel** in 1891. Most benthos feed on food as it floats by scavenging for food on the ocean floor.

Types of Benthos

Hyperbenthos - These are the organisms that have the ability to swim and live near the bottom but are not attached to it.

Epibenthos - Epibenthos spend their lives attached to the floor, on rocks, or on shells and they include sponges.

Camel - Llama hybrids (or the Cama) is a genetic hybrid between an old world camelid (Dromedary or Bactrian camel) and a new world camel (Llama or alpaca).

Nicobar Pigeon comes under the category of endemic species.

Cilia are small hair-like structures found on the surface of mammalian cells, involved in locomotion and mechanoreception. Organisms with cilia are called ciliates, with Ciliophora being one such lineage recognized as monophyletic. Cilia are present in all ciliates during some stage of their life cycle, arranged in lines called kineties.

Viviparous animal - Animals giving birth to offspring are known as viviparous

animals. Example - Dogs, Cats, Cows and Humans etc.

Oviparous animals - These animals are egg-laying creatures. – Birds, frogs, all fishes. Except for some snake species, all other reptiles are oviparous.

Nocturnal animals - They are animals that are active during the night and sleep during the day. Examples are bats, skunks, aardvarks, and owls.

Olive ridley turtle - The olive ridley sea turtle (*Lepidochelys olivacea*), also known commonly as the Pacific ridley sea turtle. The term Arribada is associated with it.

Honey bee - The honey bee has five eyes, two large compound eyes and three smaller ocelli eyes in the center of its head.

White ant- Termite is another name for 'white ant' or 'wood ant'. A termite nest is known as a termitary or termitarium.

Frog - Metamorphosis in Frog is controlled by Thyroxine which is secreted by Thyroid gland. It can breathe through both skin and lungs.

Ruminant animals, belonging to the suborder Ruminantia within the order Artiodactyla, encompass a diverse group including cows, pronghorns, giraffes, okapis, deer, chevrotains, cattle, antelopes, sheep, and goats.

In ruminants like cattle, bacteria present in **rumen** helps in digestion of cellulose.

Burrowing animals - The most well-known borrowers are mammals, especially the mole, gopher, groundhog (woodchuck), and rabbit. Bears are most likely the largest burrowing animals. Some amphibians (some species of frogs), and a number of reptiles, (including assorted snakes) are also among them.

Bird and mammal hearts have four chambers (two atria and two ventricles). A frog, which is an amphibian, has a heart with three chambers (one ventricle and two atria), and fish hearts have two chambers (one atrium and one ventricle).

Cockroaches have a 13-chambered heart that is long, thick, and muscular. A single heart is represented by each chamber. It can be found in the hemocoel's pericardial sinus.

The interior structure of a leech is divided into 32 different segments, each of which has its own brain.

Facts about camels: Camels, adapted to desert life, conserve water by sweating less and store fat in their hump for energy and water release during scarcity. Found in deserts, they include dromedary camels (Arabian), Bactrian camels (Asian), and wild Bactrian camels (critically endangered).

Silkworm Life stages

Egg - An egg is the first stage of the life cycle of the silkworm.

Silkworm - A hairy silkworm arises after the eggs crack. In this stage of silkworms, the growth happens. They feed on mulberry leaves and consume a large amount of these leaves.

Cocoon - In this stage, silkworms spin a protective cocoon around itself. It is the size of a small cotton ball and is made of a single thread of silk.

Pupa - The pupa stage is a motionless stage. In this stage, people kill the pupa by plunging the cocoon into boiling water and unwind the silk thread.

Moth - In this stage, the pupa changes into an adult moth. The female moth lays eggs after mating and thus the life cycle of silkworm begins again.

Muga silkworm (*Antheraea assamensis* Helfer) is endemic to Assam and adjoining areas in North-Eastern India, and naturally produces golden silk.

Microorganisms

An organism that can be seen only through a microscope is termed as microorganism. Microorganisms include bacteria, protozoa, algae, and fungi. Although viruses are not considered living organisms, they are sometimes classified as microorganisms.

Living things are classified into five kingdoms: Animalia, Plantae, Fungi, Protista, and Monera.

Bacteria are ubiquitous, mostly free-living organisms often consisting of one biological cell. They constitute a large domain of prokaryotic microorganisms. Typically a few micrometers in length, bacteria were among the first life forms to appear on Earth, and are present in most of its

habitats.

Pleuro Pneumonia Like Organisms (PPLO) are a group of bacteria that belong to the class mollicutes.

Bacteria flagella have 3 components.

Filament, hook and basal body are three parts of bacterial flagellum.

Basal Body: It is attached to the cell membrane and cytoplasmic membrane.

Hook : It is a broader area present at the base of the filament, which connects filament to the motor protein in the base.

Filament: Thin hair-like structure arising from the hook.

Azotobacter are Gram-negative, free-living, aerobic bacteria found in soil. They fix nitrogen and form cysts in harsh conditions. They are oval or spherical in shape.

Gram-positive bacteria are classified by the color they turn after a chemical called Gram stain is applied to them. Gram-positive bacteria stain blue when this stain is applied to them. (Other bacteria stain red. They are called gram-negative).

Gram-positive and gram-negative bacteria stain differently because their cell walls are different. They also cause different types of infections, and different types of antibiotics are effective against them.

Sulfur-reducing bacteria are microorganisms able to reduce elemental sulfur to hydrogen sulfide. These microbes use inorganic sulfur compounds as electron acceptors to sustain several activities such as respiration, conserving energy and growth, in absence of oxygen.

Lactobacillus, (genus *Lactobacillus*), any group of rod-shaped, gram-positive, non-spore-forming bacteria of the family Lactobacillaceae. *Lactobacillus* are characterized by their ability to produce lactic acid as a by-product of glucose metabolism.

Endospores are highly retractile and thick-walled structures formed inside the bacterial cells. They protect and allow the bacteria to exist in "**suspended animation**". They also form lower plants like bryophytes, lichens, algae, and Fungi.

Bacillus cereus is a Gram-positive,

rod-shaped, anaerobic, motile, beta-hemolytic, spore-forming bacterium commonly found in soil, food, and marine sponges. It can cause two types of gastrointestinal illness: emetic (vomiting) syndrome and diarrhoeal syndrome.

Coliform bacteria are organisms that are present in the environment and in the feces of all warm-blooded animals and humans. Coliform bacteria will not likely cause illness. However, their presence in drinking water indicates that disease-causing organisms could be in the water system.

Rhizobia are nitrogen-fixing bacteria which invade the root hairs of leguminous plants and induce, in a specific manner, the formation of root nodules in which they fix nitrogen.

Protozoa are microscopic unicellular eukaryotes that have a relatively complex internal structure and carry out complex metabolic activities.

Paramecium or Paramoecium is a genus of unicellular ciliated protozoa.

Sargassum, a type of brown algae, is commonly known as gulfweed or sea holly. It's found in the Sargasso Sea in the Atlantic. Rich in nutrients like vitamins, carotenoids, fibers, proteins, and minerals, it's both edible and nutritious.

Spirogyra, a multicellular organism belonging to the green algae group, forms chains of cells. Its cell walls consist of cellulose and pectin layers. Within its cells are chloroplasts, vacuoles, nucleus, and cytoplasm. Commonly found near the surface of streams and ponds, it forms floating masses.

Chlamydomonas is a genus of green algae consisting of about 150 species all unicellular flagellates, found in stagnant water and on damp soil, in freshwater, seawater, and even in snow as "snow algae". It is a haploid unicellular eukaryote cells has a diameter of 5-10 μ m, and about half of their volume is occupied by cup shaped chloroplasts.

Eudorina is a paraphyletic genus in the volvocine green algae clade. Eudorina colonies consist of 16, 32 or 64 individual cells grouped together. It is an algae that undergoes anisogamous fusion of gametes.

Fungi, belonging to the eukaryotic

kingdom Fungi, encompass microorganisms like yeasts, molds, and mushrooms. They're omnipresent, possessing cell walls and classified as heterotrophs.

Characteristics of fungi: Non-vascular, non-motile, and heterotrophic. Can be unicellular or filamentous, reproducing via spores. Lack chlorophyll, hence unable to perform photosynthesis. Store food as starch, produce chitin. Have small nuclei.

A lichen is a composite organism that arises from algae or cyanobacteria living among filaments of multiple **fungi** species in a mutualistic relationship.

Yeasts are eukaryotic, single-celled microorganisms classified as members of the fungus kingdom. It is a popular kitchen ingredient called '**sugar eating fungus**'.

The yeast species *Saccharomyces cerevisiae* converts carbohydrates to carbon dioxide and alcohols through the process of fermentation.

Deuteromycetes, commonly called molds, are "second-class" fungi that have no known sexual state in their life cycle. They reproduce only by producing spores via mitosis, this asexual state is called the anamorph state.

Phycomycetes organisms are found throughout the world in soil, animal manure, and on fruits.

Basidiomycetes are commonly known as club fungi.

Ascomycetes are spore shooters, they are called sac fungi because their sexual spores, called ascospores, are produced in a sac or ascus.

Penicillium is a genus of saprophytic (feeding on dead and decaying materials) fungi.

Puccinia is a genus of fungi. All species in this genus are obligate plant pathogens and are known as rusts.

A **virus** is a submicroscopic infectious agent that replicates only inside the living cells of an organism. Viruses infect all life forms, from animals and plants to microorganisms, including bacteria and archaea. Virus considered to be at the boundary of living or non living. Human viruses can vary in size but are generally in the range of **20–200 nm** in diameter.

Melatonin encodes proteins in cells to prevent virus entry. It is synthesized in the lungs and acts as a barrier against Covid-19 infection.

The **contractile vacuole (CV)** complex is an osmoregulatory organelle of free-living amoebae and protozoa. It controls the intracellular water balance by accumulating and expelling excess water out of the cell, allowing cells to survive under hypotonic stress as in pond water.

Food vacuole is a membrane-enclosed sac, which has a digestive function. It is present in unicellular protozoans such as amoeba, plasmodium, etc. They work as an intracellular stomach, digesting the ingested food.

The **endoplasm** of an amoeba contains the food vacuoles, a granular nucleus, and a clear contractile vacuole. The endoplasm is used to digest food, carry the genetic material for its reproduction, and to make itself move.

Sarcodina - A taxonomic group within Phylum Sarcomastigophora. It includes the amoebas, heliozoan, rediozoa and foraminifera, which are characterized by their ability to move by cytoplasmic flow or by pseudopods.

Heliozoa, commonly known as sun-animalcules, are microbial eukaryotes with stiff arms radiating from their spherical bodies, which are responsible for their common name.

Lobosa is a taxonomic group of amoebae in the phylum Amoebozoa. Most loboseans possess broad, bluntly rounded pseudopods, although one genus in the group, the recently discovered *Sapocribrum*, has slender and threadlike pseudopodia.

Zooplankton are a type of heterotrophic plankton that range from microscopic organisms to large species, such as jellyfish, larval crustaceans, krill, copepods, protozoans, etc. They are free-swimming and bottom-dwelling in forms.

Phytoplankton are mostly microscopic, single-celled photosynthetic organisms that live suspended in water, such as algae and diatoms. Fungi (eukaryotic organisms), jelly fishes (Phylum Cnidaria Cnidarians).

Redworms do not have teeth. A redworm can eat food equal to its own weight, in a day. They have a structure called gizzard,

which helps them in grinding their food.

Some Microorganisms and diseases caused by them - *Campylobacter jejuni* (Diarrhea), *Salmonella typhi* (Typhoid high fever), *Staphylococcus aureus* (Infected eczema, psoriasis or any other pus draining lesion), *Escherichia coli* (Diarrhea), *Listeria monocytogenes* (Septicemia and meningitis), *Shigella* (Shigellosis), *Paratyphi* (Paratyphoid fever), *Staphylococcus aureus* (illness, skin infections), *Clostridium botulinum* (Flaccid paralysis of muscles), *Bacillus cereus* (Emetic toxin), *Yersinia enterocolitica* (Yersiniosis), *Vibrio parahaemolyticus* (Cholera), *Norovirus* (Dehydration, malnutrition and even death).

Digestive System

Digestive systems - The digestive system processes food and liquids, breaking them down into nutrients for energy, growth, and tissue repair. It comprises the alimentary canal and digestive glands.

The digestion consists of the following process : Ingestion → Digestion → Absorption → Assimilation → Egestion

The Digestive system consists of the mouth and continues through the esophagus, pancreas, stomach, small intestine, large intestine, liver, gallbladder, and ends with the anus.

Different parts of Digestive system

Alimentary Canal -

The alimentary canal, a continuous passage, includes the buccal cavity, esophagus, stomach, small intestine, large intestine, rectum, and anus. It spans 6 to 9 meters in total length.

Mouth (Buccal Cavity or Oral Cavity) -

Digestion commences in the buccal cavity, where food is broken down by teeth into smaller particles. Saliva, produced by salivary glands, moistens the food, aiding in the initial stages of carbohydrate digestion. The tongue then pushes the food into the pharynx, initiating the swallowing process.

A digestive enzyme in saliva called amylase (**optimum pH 6.8**) starts to break down some of the carbohydrates (starches and sugars) in the food. The saliva secreted into the oral cavity

contains electrolytes (Na⁺, K⁺, Cl⁻, HCO₃⁻) and enzymes.

Salivary Amylase and **Lysozyme** (present in saliva acts as an antibacterial agent that prevents infections). The normal daily production of saliva varies between 0.5 and 1.5 liters.

The tongue, a muscular organ located in the buccal cavity, assists in chewing, mixing saliva with food, and swallowing. It houses taste buds within papillae, allowing for the detection of different flavors through gustatory receptors.

Food Pipe/Oesophagus

The pharynx, a shared passage for food and air, connects the oral cavity to the oesophagus and trachea. The epiglottis, a cartilaginous flap, prevents food from entering the glottis during swallowing. Digestion does not occur in the oesophagus.

Mucus in saliva helps in lubricating and adhering the masticated food particles into a bolus. The bolus is then conveyed into the pharynx and then into the esophagus by swallowing. The bolus further passes down through the oesophagus by successive waves of muscular contractions called **peristalsis**.

Stomach - The stomach, positioned on the left side of the abdominal cavity below the diaphragm, serves as a muscular reservoir for food, allowing time for digestion. Its inner lining secretes mucus, hydrochloric acid (with a pH of 1.5-2.0), and digestive enzymes. The acidic environment aids in bacterial sterilization and activates pepsin for protein digestion. Approximately 2 liters of gastric juice are produced daily, initiating the breakdown of proteins in the stomach.

Pelvic cavity is a bowl-like structure that sits below the abdominal cavity.

Small Intestine - Complete digestion of food takes place in the small intestine. Small intestine is distinguishable into three regions, a 'C' shaped duodenum, a long coiled middle portion jejunum and a highly coiled ileum. Ileum opens into the large intestine.

The small intestine is highly coiled and is about 7 meters long. The digested food passes into the blood vessels in the wall of the intestine. This process is called **absorption**.

The inner walls of the small intestine have thousands of finger-like outgrowths, called **villi** (singular villus). The villi increases the surface area for absorption of the digested food. Villi are supplied with a network of capillaries and a large lymph vessel called the **lacteal**.

Complete digestion of fats also takes place in the small intestine. The absorbed substances are transported via the blood vessels to different organs of the body. Then they are used to build complex substances such as the proteins required by the body. This is called **assimilation**.

In the cells, glucose breaks down with the help of oxygen into carbon dioxide and water, and energy is released. The food that remains undigested and unabsorbed then enters into the large intestine.

Large Intestine - The large intestine is wider and shorter than the small intestine. It is about 1.5 metre in length. Its function is to absorb water and some salts from the undigested food material.

The remaining waste passes into the rectum and remains there as semi-solid faeces. The faecal matter is removed through the anus from time-to-time. This is called **egestion**.

The large intestine has four parts: cecum, colon, rectum and anal canal.

Cecum - A pouch that forms the first part of the large intestine. It connects the small intestine to the colon, which is part of the large intestine.

Colon - The longest part of the large intestine (a tube-like organ connected to the small intestine at one end and the anus at the other). The colon removes water and some nutrients and electrolytes from partially digested food.

Rectum - A chamber that begins at the end of the large intestine, immediately following the sigmoid colon, and ends at the anus.

Anal canal - The part that connects the rectum to the anus, located below the level of the pelvic diaphragm.

The appendix, a muscular structure connected to the colon, is a vestigial organ derived from the cecum. Although its function is not fully understood, it may assist in digesting roughage, aiding the passage of food and waste through the

gut.

Probiotics can be defined as live microorganisms that are intended to have health benefits when consumed or applied to the body.

Digestive Glands

The **glands** associated with the digestion include salivary glands, the liver and the pancreas.

Salivary glands - Saliva is mainly produced by three pairs of salivary glands, the parotids (cheek), the sub-maxillary (lower jaw) and the sub-linguals (below the tongue). These glands secrete salivary juice into the buccal cavity. The saliva breaks down the **starch** into sugars.

Liver - It is a reddish brown gland situated in the upper part of the abdomen on the right side. It is the **largest gland** in the body and has the regenerative property.

It secretes bile juice (pH = 7 to 8.6 (alkaline)) that is stored in a sac called the gallbladder. The bile plays an important role in the digestion of fats. It has two lobes.

The **hepatic lobules** are the structural and functional units of the liver containing hepatic cells. Liver converts glucose into glycogen and amino acids into proteins.

Liver converts some quantity of protein into glucose during deficiency of carbohydrates. If the necessity of glucose arises the liver converts reserve glycogen into glucose. Thus, it regulates the quantity of glucose in the blood.

Pancreas

The pancreas, situated just below the stomach, is a gland with both exocrine and endocrine functions. Its alkaline pancreatic juice, with a pH of 8.3 to 8.6, acts on carbohydrates and proteins, breaking them down into simpler forms. This gland, located between the limbs of the 'U' shaped duodenum, releases enzymes from its exocrine portion and hormones such as insulin and glucagon from its endocrine portion. Together, these substances facilitate digestion and regulate blood sugar levels.

Enzymes

Mouth - Salivary **Amylase** or **ptyalin** (Breakdown starch into maltose).

Stomach - **Pepsin** (Breakdown proteins

into dipeptides), **Renin** (Breakdown caseinogen to casein found in milk).

Pancreas - **Lipase** (Breakdown emulsified fats into fatty acids and glycerol), **Trypsin** (Breakdown Protein and peptone into polypeptides and amino acids), **Amylase** (Breakdown starch into maltose).

Small Intestine - **Maltase** (Breakdown Maltose into glucose), **Sucrase** (Breakdown Sucrose into glucose and fructose), **Peptidase** (Breakdown Dipeptides into amino acid), **Lactase** (Breakdown Lactose into glucose and galactose), **Erepsin** (Breakdown protein and peptone into amino acids).

Roughage - The **indigestible** portion of our diet essential to the health of the digestive system. Roughage includes all plant products like grains, nuts, legumes, fruits, etc. They are indigestible but act as a source of food for the beneficial bacteria present in our gut.

Fiber is a type of carbohydrate that the body can't digest. Fiber cannot be broken down by the gastrointestinal (GI) tract.

Glycogen is the storage form of carbohydrates in mammals.

Starch or **amylum** is a polymeric carbohydrate consisting of numerous glucose units joined by glycosidic bonds. It is mainly found in the pulp of seeds, fruits, tubers, roots and stems of plants, especially in corn, potatoes, wheat and rice.

Respiratory System

Respiration is the metabolic process through which living cells obtain energy (ATP) by using oxygen and releasing carbon dioxide from the oxidation of organic substances. This exothermic process releases large amounts of energy.

Components of the Respiratory System

Airways: It delivers air to lungs. Airways are a complicated system that include -

Mouth and nose - Openings that pull air from outside the body into the respiratory system.

Sinuses - Hollow areas between the bones in the head that help regulate the temperature and humidity of the air

inhale.

Pharynx (throat) - A tube that delivers air from mouth and nose to the trachea (windpipe).

Trachea - Passage connecting throat and lungs.

Bronchial tubes - Tubes at the bottom of the windpipe that connect into each lung.

Lungs - Two organs that remove oxygen from the air and pass it into blood. From the lungs, the bloodstream delivers oxygen to all organs and other tissues. Regular traditional breathing exercises can increase the capacity of lungs to take in more oxygen.

Muscles and bones help in moving the air we inhale into and out of lungs. Some of the bones and muscles in the respiratory system includes -

Diaphragm is the primary muscle used in respiration, which is the process of breathing. This dome-shaped muscle is located just below the lungs and heart. It contracts and expands continually as we breathe in and out.

Ribs - Bones that surround and protect lungs and heart. Rib bones are flat bones. When we breathe out, our blood carries carbon dioxide and other waste out of the body. During inhalation, ribs move up and outwards and diaphragm moves down.

Other components that work with the lungs and blood vessels include:

Alveoli - Tiny air sacs in the lungs where the exchange of oxygen and carbon dioxide takes place.

Bronchioles - Small branches of the bronchial tubes that lead to the alveoli.

Capillaries - Blood vessels in the alveoli walls that move oxygen and carbon dioxide.

Lung lobes - Sections of the lungs - three lobes in the right lung and two in the left lung.

Pleura - Thin sacs that surround each lung lobe and separate lungs from the chest wall.

Some other components of respiratory system include :

Cilia - The respiratory system's mucous membrane secretes mucus, which traps particles like pollen or smoke. Cilia, tiny

hairlike structures lining the mucous membrane, move trapped particles out of the nose in a wave-like motion, filtering dust and irritants from airways.

Epiglottis - Tissue flap at the entrance to the trachea that closes when you swallow to keep food and liquids out of the airway (trachea).

Larynx (voice box) - Larynx is part of the respiratory system. It's a hollow tube that lets air pass from the throat (pharynx) to the trachea on the way to the lungs. Larynx also contains vocal cords and is essential to human speech, so it's often called the voice box.

When the larynx grows larger during puberty, it sticks out at the front of the throat. It is called 'Adam's apple (Kanthamani)'.

Thyroid Cartilage - The tough, flexible tissue that forms the front part of the larynx (voice box). It supports and protects the vocal cords and helps create the sound of a person's voice. The thyroid cartilage also forms Adam's apple.

Types of Respiration

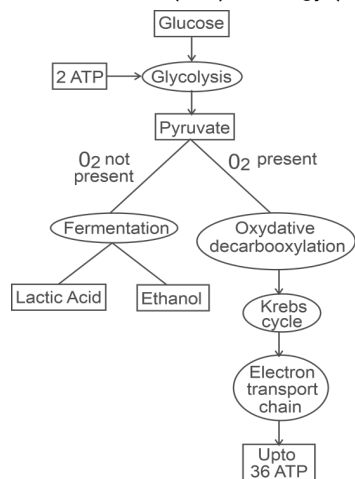
There are two types of respiration.

Aerobic respiration - It is a type of cellular respiration that takes place in the presence of oxygen to produce energy. It is a continuous process that takes place within the cells of animals and plants.

Glucose ($C_6H_{12}O_6$) + Oxygen ($6O_2$) → Carbon dioxide ($6CO_2$) + Water ($6H_2O$) + Energy (ATP).

Anaerobic respiration - It is a type of cellular respiration that takes place in the absence of oxygen to produce energy.

Glucose ($C_6H_{12}O_6$) → Alcohol ($2C_2H_5OH$) + Carbon dioxide ($2CO_2$) + Energy (ATP).



The three stages of aerobic respiration are -

Glycolysis initiates the breakdown of glucose into pyruvic acid, yielding 2 ATP and forming pyruvates (3 carbon molecules). It progresses into aerobic respiration in the presence of oxygen or anaerobic respiration in its absence.

The Krebs cycle, also known as the Citric acid cycle, is a series of biochemical reactions occurring in the mitochondria. In this cycle, acetyl-CoA is oxidized, and ADP is converted to ATP.

Oxidative phosphorylation is a cellular process that utilizes oxygen reduction to create high-energy phosphate bonds, forming adenosine triphosphate (ATP).

In anaerobic respiration this process of breaking of pyruvates is called fermentation and it leads to the release of either ethanol + energy (Yeast or Bacteria) or lactic acid + energy (During heavy workouts when oxygen supply is low) depending upon the organism in which the process happens.

Respiration in animals

Simple diffusion - Occurs via the cell membrane in simple unicellular creatures like Amoeba.

Skin - Earthworms and other soil animals use their skin to absorb oxygen from the air and expel carbon dioxide.

Gills - An organ used for breathing in many water-dwelling animals, including most fish and many arthropods.

Lung - These are the respiratory organs of land creatures such as humans, birds, lizards, dogs, and frogs. Frogs can breathe through both their lungs and their skin.

Respiratory failure arises when the lungs cannot adequately oxygenate the blood. Oxygen is inhaled into the lungs, and carbon dioxide, a waste gas, is exhaled.

Parts of respiration in insects

Spiracles - This is the respiratory organ of insects such as grasshoppers, cockroaches, houseflies, and mosquitos as tiny pores on their bodies and air tubes called tracheae.

Trachea - Insects take in Oxygen and expel Carbon Dioxide using a series of

internal air tubes, the tracheae. It connects the spiracles to the lungs. It is made of **chitin**.

Tracheoles - Thin-structured branch of the trachea. Its function is nearly identical to that of vertebrates' blood capillaries.

Air bag - The trachea, a form of breathing mechanism also present in birds, is attached to this pouch, which is shaped like a balloon.

Respiration in Plants

Plants use photosynthesis to create glucose and oxygen from carbon dioxide and water, while respiration utilizes these products to release energy. The two processes are interdependent, with the outputs of one becoming the inputs of the other. The glucose and oxygen produced during photosynthesis are used in aerobic respiration to release energy.

The roots, stems, and leaves of a plant all respire and have specialized structures for this purpose. Leaves have tiny pores called stomata on their surfaces, which open and close depending on the plant's need for gas exchange with the atmosphere. Oxygen from the air enters into a leaf through the stomata and reaches all the cells by the process of diffusion.

Lenticels are openings on the surfaces of stems and roots through which gas exchange takes place with the atmosphere.

Nervous System

Nervous System is the body's command center. Originating from the brain, it controls movements, thoughts and automatic responses.

It also controls other body systems and processes, such as digestion, breathing and sexual development (puberty). Diseases, accidents, toxins and the natural aging process can damage the nervous system.

The nervous system has two main parts :

Central Nervous System is made up of the brain and spinal cord.

Peripheral nervous system is made up of nerves that branch off from the spinal cord and extend to all parts of the body.

Neurons, or nerve cells, are the building blocks of the brain and nervous system. They receive sensory input, send motor commands to muscles, and relay electrical signals. Neurons consist of dendrites and axons and typically have long, branched structures.

Each neuron in the brain has one long cable that snakes away from the main part of the cell. This cable is thinner than a human hair and is called an **axon**. It is where electrical impulses from the neuron travel away to be received by other neurons.

Dendrites are projections of a neuron (nerve cell) that receive signals (information) from other neurons. The transfer of information from one neuron to another is achieved through chemical signals and electric impulses, that is, electrochemical signals.

Myelin sheaths are made of myelin, and myelin is produced by different types of neuroglia : oligodendrocytes and Schwann cells, where oligodendrocytes myelinate axons in the central nervous system, and Schwann cells myelinate axons in the peripheral nervous system.

An **axon** typically develops side branches called axon collaterals, so that one neuron can send information to several others. These collaterals, split into smaller extensions called terminal branches. Each of these has a synaptic terminal on the tip.

Synapses refer to the points of contact between neurons where information is passed from one neuron to the next. Synapses most often form between axons and dendrites and consist of a presynaptic neuron, synaptic cleft, and a postsynaptic neuron.

Nerve ending is one of the fine branches at the end of a neuron, by which it makes contact either with another neuron or with a muscle or gland cell.

The sarcolemma, the muscle cell's plasma membrane, is enveloped by the basement membrane and endomysial connective tissue. Excitable like neuronal cell membranes, it shares numerous properties with them.

Microglial cells, specialized macrophages in the central nervous system (CNS), play a vital role in removing damaged neurons and infections, thereby maintaining CNS

health.

Human Brain

The human brain is well protected by the skull. Inside the skull, the brain is covered by cranial meninges consisting of an outer layer called dura mater, a very thin middle layer called arachnoid and an inner layer (which is in contact with the brain tissue) called pia mater. The brain can be divided into three major parts: forebrain, midbrain, and hindbrain

Forebrain - It consists of cerebrum, thalamus and hypothalamus. These are the largest part of the brain containing 2/3rd part of the brain.

Cerebrum is a major part of our brain, and it's split into two halves: left and right. These halves are joined by a bunch of nerve fibers called the corpus callosum.

The outer layer of the cerebrum is called the cerebral cortex, and it looks wrinkly because it's folded. This part is grayish and has a lot of nerve cells, so it's called gray matter. It's where a lot of the thinking happens. Inside, there are nerve fibers covered in a white substance called myelin, which makes it look white.

The cerebral cortex does lots of things, like controlling movement and sensing stuff. There are also areas that help with things like remembering stuff and understanding language.

The cerebrum wraps around a structure called thalamus, which is a major coordinating center for sensory and motor signaling. Another very important part of the brain called hypothalamus lies at the base of the thalamus.

The hypothalamus contains a number of centers which control body temperature, urge for eating and drinking. It also contains several groups of neurosecretory cells, which secrete hormones called hypothalamic hormones.

Midbrain - (or mesencephalon) It connects forebrain to hindbrain. It is located between the thalamus and pons. The midbrain consists of the center for visual and auditory reflexes. It also helps in maintaining posture and muscle tone.

The midbrain is a complex structure housing neuron clusters, neural pathways, and colliculi, enabling

functions like hearing, movement, and environmental response. It includes the substantia nigra, rich in dopamine neurons and implicated in Parkinson's disease, as part of the basal ganglia , crucial for movement and coordination.

Hindbrain - The hindbrain consists of three main parts. They are the pons, cerebellum and medulla. The cerebellum and pons develop from the metencephalon and the medulla develops from the myelencephalon.

Pons - The pons is the connection between the midbrain and the medulla. It is the origin for four of the 12 cranial nerves. It enables a range of activities such as tear production, chewing, blinking, focusing vision, balance, hearing and facial expression.

Medulla - At the bottom of the brainstem, the medulla is where the brain meets the spinal cord. Functions of the medulla regulate many bodily activities, including heart rhythm, breathing, blood flow, and oxygen and carbon dioxide levels. The medulla produces reflexive activities such as sneezing, vomiting, coughing and swallowing.

Cerebellum - The cerebellum, often referred to as the "little brain," is located at the back of the head, below the temporal and occipital lobes and above the brainstem. It coordinates voluntary muscle movements, maintains posture, and balances the body, crucial for walking in a straight line.

Brain Coverings

Meninges- Three layers of protective covering called meninges surround the brain and the spinal cord (Dura mater, Arachnoid mater, pia mater).

The outermost layer, the dura mater, is thick and tough. The **arachnoid mater** is a thin, weblike layer of connective tissue that does not contain nerves or blood vessels. The **pia mater** is a thin membrane that hugs the surface of the brain and follows its contours.

Lobes of the Brain and What They Control

Each brain hemisphere (parts of the cerebrum) has four sections, called lobes: frontal, parietal, temporal and occipital. Each lobe controls specific functions.

Frontal lobe

The frontal lobe, situated at the front of the head, is the brain's largest lobe, influencing personality traits, decision-making, and movement. Smell recognition and speech ability, governed by Broca's area, are also located here.

Parietal lobe (middle part of the brain)

The parietal lobe aids in object identification, spatial awareness, and interpreting pain and touch sensations. Additionally, it houses Wernicke's area, crucial for understanding spoken language.

Occipital lobe - It is the back part of the brain that is involved with vision.

Temporal lobes - These are involved in short-term memory, speech, musical rhythm and some degree of smell recognition.

Deeper Structures within the Brain

Hypothalamus - Part of the forebrain and situated above the pituitary gland, the hypothalamus governs body temperature, sleep patterns, hunger, thirst, and contributes to memory and emotion.

Amygdala manages the processing of information between prefrontal-temporal association cortices and the hypothalamus.

Hippocampus - A curved seahorse-shaped organ on the underside of each temporal lobe, the hippocampus is part of a larger structure called the hippocampal formation. It supports memory, learning, navigation and perception of space.

The Ventricles and Cerebrospinal Fluid

The brain has four cavities known as ventricles, filled with cerebrospinal fluid. This fluid serves various functions: absorbing physical shocks, distributing nutrients to nervous tissue, removing waste, and maintaining a stable chemical environment.

Cranium is the part of the skull that encloses the brain. Cranium consists of **8 bones** - 1 frontal bone, 2 parietal bones, 2 temporal bones, 1 occipital bone, 1

sphenoid bone, 1 ethmoid bone.

The **spinal cord** extends from the bottom of the medulla and through a large opening in the bottom of the skull. Supported by the vertebrae, the spinal cord carries messages from the brain to the rest of the body.

Spinal nerve in vertebrates, any one of many paired peripheral nerves that arise from the spinal cord. In humans there are 31 pairs: 8 cervical (neck), 12 thoracic (chest), 5 lumbar (lower back), 5 sacral (hips), and 1 coccygeal. Each pair connects the spinal cord with a specific region of the body.

Terminology in Nervous system

Action potential is an electrical impulse transmitted across a plasma membrane or nerve fiber.

Afferent neurons (also called sensory neurons or input neurons) carry impulses from the peripheral receptors (skin and sensory organs) to the central nervous system (spinal cord and the brain).

Anterior horn is a horn-like projection of gray matter into the white matter of the spinal cord.

The **brachial plexus** is a network of nerves and blood vessels that animate the upper limbs.

The **dorsal horn** is the horn shaped projection of gray matter in the posterior region of the spinal cord.

Efferent neurons (also called motor neurons or output neurons) carry and transmit impulses from the central nervous system (brain and spinal cord) to muscles and glands.

The **glial cells** (neuroglia) are specialized cells of the nervous tissue. These cells are the support cells of the nervous system.

Schwann cells are cells that form the myelin sheath around the peripheral nerve fibers. Schwann cells are also called neurilemma cells, are the main glial cells in the peripheral nervous system and play an essential role in the survival and functions of neurons.

The **Wernicke center** is a section of the posterior temporal lobe where the sensory speech center is located.

Dyskinesia is an involuntary movement

of a part or whole of the body that you cannot control.

Dysphagia is a condition where patients have difficulty swallowing food.

An **aneurysm** is an abnormal swelling or bulge in the wall of a blood vessel, such as an artery.

Dysphonia is difficulty speaking due to voice impairment or altered voice production.

Epilepsy is a neurological disorder characterized by paroxysmal transient disturbances of brain function, resulting in convulsions and loss of consciousness.

Guillain-Barre syndrome is an autoimmune disease which attacks the peripheral nerve myelin.

Hemiparesis means weakness on one side of the body. Symptoms include loss of balance and lack of coordination.

Myasthenia Gravis is an autoimmune disorder affecting the neuromuscular junction, which causes muscle weakness.

The mammalian brain depends on glucose as its main source of energy. In the adult brain, neurons have the highest energy demand, requiring continuous delivery of glucose from blood.

Circulatory system

The **circulatory system** (cardiovascular system) is responsible for the transport of substances (nutrients, oxygen, carbon dioxide, and cellular waste material) throughout the body.

William Harvey was the first person to discover the circulation of blood in our body.

There are two types of circulation systems -

Pulmonary Circulation - It controls the movement of blood between the heart and the lungs only. **4.4 percent** of carbon dioxide in the air exhaled by human lungs.

Systemic Circulation - This controls the movement of blood between the heart and the rest of the body.

Human Heart

The human heart is made up of special muscles called cardiac muscles. The heart is responsible for pumping blood throughout the body. It receives carbon dioxide-rich blood from the rest of the body and sends it to the lungs and receives oxygenated blood from the lungs and sends it to the rest of the body.

Deoxygenated blood enters the right atrium through the aorta and then into the right ventricle from where it is pumped to the lungs.

Oxygenated blood enters the Left Auricle from lungs and then it is pushed into the Left Ventricle. It is then pushed into the main artery called Aorta. The blood then reaches different parts of the body through smaller arteries.

The heart pumps around 7,600 litres of blood in a day. An adult heart beats about 60 to 80 times per minute, and newborn babies heart beats 70 to 190 beats per minute. On an average the heart beats about 100,000 times a day, and around 3 billion beats in a lifetime.

The **pericardium** is a fibroserous sac that contains the heart and the roots of the great vessels.

The **conduction system** of the heart is a system within the heart that is controlled by the autonomic nervous system and that delivers **electrical impulses** and motivates the rhythmic contractions of the heart.

Heart wall - The heart wall consists of three layers: the outer epicardium, which covers the heart and attaches it to a protective sac; the middle myocardium, a strong muscle tissue responsible for pumping action; and the inner endocardium, which lines the interior structures of the heart.

Heart chambers

Right atrium - Two large veins deliver oxygen-poor blood to the right atrium. The **superior vena cava** carries blood from the upper body. The **inferior vena cava** brings blood from the lower body. Then the right atrium pumps the blood to the right ventricle.

Right ventricle - The lower right chamber pumps the deoxygenated blood to the lungs through the pulmonary artery.

Left atrium - The pulmonary veins carry the oxygenated blood from lungs to the left atrium and it pumps the blood to the left ventricle.

Left ventricle - The left ventricle is slightly larger than the right. It pumps oxygen-rich blood to the rest of the body.

Heart Valves are the tissues located in the cardiac chambers between the veins. It allows the blood to flow in a single direction (unidirectional). It also prevents the blood from flowing backwards.

Types of valves based on their function -

The **tricuspid valve** regulates blood flow between the right atrium and right ventricle.

The **pulmonary valve** controls blood flow from the right ventricle into the pulmonary arteries (the only arteries that carry deoxygenated blood to lungs).

The **mitral valve** allows oxygenated blood from the lungs to pass from the left atrium to the left ventricle.

The SA (sinoatrial) node generates an electrical signal that causes the upper heart chambers (atria) to contract. The signal then travels from the AV (atrioventricular) node to the lower heart chambers (ventricles), causing them to contract or pump. The SA node is considered the pacemaker of the heart. AV (atrioventricular) nodes collect signals from SA nodes. The AV node serves as a gate that slows the electrical current before the signal is permitted to pass down through to the ventricles.

Cardiac Pacemaker - An electronic device that is implanted in the body to control abnormal heart rhythms. Provides electrical stimulation to slow heart rate.

The **aortic valve** opens the way for oxygen-rich blood to pass from the left ventricle into the aorta. It is the largest artery of the body.

Septum - The heart is separated by a thin wall called the septum. It is of two types:

The septum which is present between the right and left atrium is known as the **atrial septum**.

The septum between the right and left ventricles is known as the **ventricular septum**.

Atrial natriuretic factor (ANF) - It is known as the hormone of the heart. It is made up of protein hormone which is secreted by the cardiac in counter to the high blood pressure through the heart muscle cells.

Blood Vessels

A tube through which the blood circulates in the body. **Blood vessels** include a network of arteries, arterioles, capillaries, venules, and veins.

Veins supply deoxygenated (carbon dioxide-rich) blood to the heart via inferior and superior **vena cava**, and it eventually drains into the right atrium.

The **Pulmonary vein** is the only vein to carry oxygenated blood. It carries blood from the lungs to the left auricle of the heart.

Venules are the smallest veins and receive blood from capillaries.

Superficial veins are located in the fatty layer under skin. They carry blood from surrounding tissues to the deep veins.

Deep veins are found in muscles or along bones. They have the important job of receiving blood from the superficial veins and pumping it to the heart.

Systemic veins transport blood from the body tissue to the right atrium of the heart. This blood has a reduced oxygen content because the oxygen has been used for metabolic activities in the tissue cells.

Varicose veins are a common condition caused by weak or damaged vein walls and valves. Veins have one-way valves inside them that open and close to keep blood flowing toward the heart. Weak or damaged valves or walls in the veins can cause blood to pool and even flow backwards.

Capillaries are the smallest and most numerous blood vessels that form the connection between the vessels that take blood away from the heart (arteries) and the vessels that bring blood to the heart (arteries).

Arteries are muscular-walled tubes mainly involved in supplying oxygenated blood away from the heart to all other parts of the body.

The aorta is the largest of the arteries

and it branches into many smaller arteries throughout the body. After this tissue is formed, it divides into capillaries.

The **Pulmonary Artery** is the only artery to carry deoxygenated blood. Right pulmonary artery is longer than the left. It carries blood from the right ventricle of the heart to the lung.

The **coronary arteries** are major blood vessels in the body, supplying blood to the heart. They make it possible for the heart to beat and pump blood throughout the body.

Arterioles is a very small blood vessel that branches off from the artery and carries blood away from the heart to the tissues and organs.

Blood

It is a fluid in humans and other animals that carries essential substances such as nutrients and oxygen to cells and carries metabolic waste products away from those same cells. It regulates body temperature. It is a connective tissue.

Human Blood constitutes 7% of the total weight of the human's body. Its pH value is nearly 7.4, so it is considered base in nature. There is an average of 5-6 litres of blood in the human body. Blood cells are produced in Bone Marrow. The study of blood is called Hematology.

Main Components of Blood

Plasma - It is a slightly sticky, yellowish coloured liquid that contains the dissolved proteins, mineral ions, glucose, hormones, carbon dioxide, including the blood cells. Around 90% of the plasma is made up of water and around 6 % of it constitutes the proteins.

Albumin is the main protein in the plasma, which regulates osmotic blood pressure. 55% of the blood fluid is made up of plasma.

Globulin plays an important role in liver functioning, blood clotting, and fighting from infection. It also contains fibrinogen, which aids in blood clotting.

Coagulation is also known as blood clotting. It is the process by which blood changes from a liquid and becomes thicker, like a gel. The cessation of blood loss from a damaged tissue caused hemostasis.

Anticoagulants are commonly known as blood thinners.

The other three types of cells make up about 45% of blood cells.

Red Blood cells (RBC) or Erythrocytes - It is a type of blood cell found in the blood that is made in the bone marrow. The lifespan of a red blood cell (RBC) is 120 days. The diameter of red blood cells is about 7.5 µm.

Spleen is called the graveyard of RBC. It is red in color due to the presence of hemoglobin, which is an iron-containing complex protein. Iron is responsible for the production of hemoglobin.

The first observation of red blood cells in humans and other animals was done by **Anton Van Leeuwenhoek**.

Glycated hemoglobin is the form of hemoglobin that is covalently linked to glucose. Hemoglobin is linked together with glucose in the blood through the process of glycation. The levels of glycated hemoglobin are influenced by many factors such as Anaemia, Iron deficiency, Pregnancy etc.

A healthy individual has 12 - 16 grams of hemoglobin per 100 ml of blood. It varies from (12-16) in women to (13.5-18) in Men.

White Blood Cells (WBC) or Leucocytes - It has nuclei. It is a part of the immune system. It helps the body to fight from infection and defend against other foreign materials. It is also known as '**soldiers of the human body**'.

Due to lack of hemoglobin in the blood these cells are colorless. The life span of WBC is about 12-20 days.

Type of WBC

Granulocytes - It is the type of white blood cells. It has granules which secrete enzymes are used by it to kill pathogens. Neutrophils, eosinophils, and basophils are the types of granulocytes which have large nuclei.

Monocytes - A type of WBC which resides in blood and tissues to find and destroy germs and eliminate infected cells. Monocytes help treat injury and prevent infection.

Lymphocytes - It is a type of immune cell that is created in the bone marrow and is found in the blood and in lymph

tissue.

Platelets or Thrombocytes - These are blood cell fragments produced in the bone marrow. They play a crucial role in controlling bleeding and blood clotting. Platelets have a lifespan of 8-9 days and contain dense granules, alpha granules, and lambda granules as their components.

A normal platelet count in adults ranges from 150,000 to 450,000 platelets per microliter of blood.

Blood Groups

The Groupings of Blood for transfusion are ABO and Rh .

The **Rh blood group system** is one of the prominent blood grouping systems. About two-thirds of the population normally contains the third antigen called Rh factor or Rh antigen on the surface of their red blood cells.

This antigen was discovered by **Landsteiner and Weiner** in the red blood cells of **rhesus monkeys** and human beings. People who have Rh antigen in their RBCs have positive blood groups or are termed as **Rh-positive**. People who do not have this antigen in their RBCs have negative blood groups or are termed as **Rh-negative**.

The **ABO blood group** system consists of 4 types of blood group – A, B, AB, and O and is mainly based on the antigens and antibodies on red blood cells and in the plasma. Blood group is determined by the genes inherited from parents.

Blood group A: A person having blood group A has the A antigen on the surface of the red blood cell (RBC) and shows the B antibodies in plasma.

Blood group B : A person having blood group B has B antigen on the surface of the red blood cell and shows A antibodies in plasma.

Blood group AB : A person having blood group AB has both antigen A and B on the surface of the red blood cell and has no antibodies in the plasma.

Blood group O : A person having blood group O lacks antigens on the surface of the red blood cells and shows the presence of both A and B antibodies in the plasma.

Eight possible blood group types are

formed after combining the ABO and the Rh groups: A+, A-, AB+, AB-, B+, B-, O+, and O-.

A **type of human's blood group** A, B, AB or O are based on the blood types of their mother and father.

Parent 1	Parent 2	Baby's possible blood type
A	A	A, O
A	B	A, B, AB, O
A	AB	A, B, AB
A	O	A, O
B	B	B, O
B	AB	A, B, AB
B	O	B, O
AB	AB	A, B, AB
AB	O	A, B
O	O	O

AB positive (AB+) and O negative (O-) are considered **universal blood** types. People who are AB+ are **universal recipients**. O negative (O-) individuals are **universal donors**.

Bombay blood group - The Bombay blood group is a rare blood group, phenotypes of this group lacking H antigen on the red cell membrane and have anti-H in the serum. It fails to express any A, B or H antigen on their red cells or other tissues.

The rare Bombay blood group was first discovered in Mumbai (Bombay) in 1952 by **Dr Y M Bhende**.

Blood pressure - The pressure exerted by circulating blood on the walls of blood vessels is called blood pressure. **120/80 mm of Hg** is considered as the normal systolic/ diastolic blood pressure of a human being.

Systolic blood pressure measures the pressure in arteries when the heart beats.

Diastolic blood pressure measures the pressure in arteries when the heart rests between beats.

The **Spleen** is a bean-shaped organ that stores erythrocytes, lymphocytes, and phagocytes. It ensures smooth flow of blood by protecting it from blood-borne pathogens in its cells.

The **Lymph nodes** are small solid structures located at different points along the lymphatic system. It also filters substances that travel through the lymphatic fluid.

The gas that reduces the capacity of the

blood to carry oxygen is **Carbon Monoxide**. It gets deposited in the haemoglobin of Red Blood Cells (RBCs) and prevents proper oxygen carrying capacity.

Thrombocytopenia - The disease caused due to lack of platelets in the blood is called Thrombocytopenia.

Haemophilia is also called bleeder's disease. In this condition, the blood clot is not able to form in the body of the patient.

Arrhythmia - It is the condition of the irregular rhythm of the heartbeat.

Rapid arrhythmias - In this condition the heart beats greater than 100 beats per minute. It is called tachycardias.

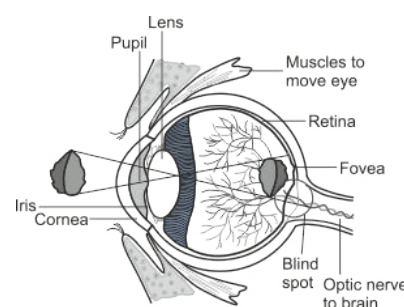
Slow arrhythmias - In this condition the heart beats slower than 60 beats per minute. It is called bradycardias.

Sensory Organs

The sensory organs detect all types of changes in the environment and send appropriate signals to the central nervous system. There are five sensory organs - Eye, Ear, Nose, Tongue, and Skin.

Eye

Eye is a visual sensory organ of our body.



The various parts of the eye are -

Eyeball - It is a spheroidal structure containing sense receptors for vision.

Blind Spot - It is the area where the optic nerve exits the eye through the optic disk. It lacks photoreceptor cells. Vision is not possible in this region.

Retina - It is the inner layer. It converts images into electrical impulses. Images on the retina are inverted and real. Owls possess a retina with numerous cones and fewer rods.

Cornea - It is the anterior portion of this

layer. Light enters the eye through the cornea.

Fovea - It is a spot in the retina where images are focused. It has a high density of cones but has no rods.

Choroid - It is a thin layer of tissue between Sclera and the Retina. Its function is to provide nutrition and to prevent reflection of light.

Sclera - It is the outer covering, a protective tough white layer called the sclera (white part of the eye). It helps to maintain the eyeball's shape, and protects it from injury.

Pupil - It is situated in the iris. It regulates the amount of light reaching the retina. Its black appearance stems from light either being absorbed by eye tissues directly or after diffuse reflections within the eye.

Iris - It is a dark muscular tissue behind the cornea. It determines eye color and regulates pupil size. Its muscles are examples of striated muscles.

Lens - It is located behind the pupil. It is a convex, transparent, and flexible structure made of jelly-like material. Controlled by ciliary muscles, it alters its shape to focus light onto the retina.

Optical Nerve - These nerves take the image to the brain in the form of electrical signals from Retina.

Photoreceptor cells - They are of two types - Cones and Rods.

Cones - They are more sensitive to bright light. They are responsible for color vision as well as eye color sensitivity.

Rods - They are more sensitive to dim lights. They help in peripheral (the ability to see things where you are not directly looking) vision.

Aqueous Humour - It is a fluid similar to blood plasma that fills the space between the cornea and the lens of the eye.

Ciliary Muscle - It holds the eye lens in its position and brings about changes in the shape (curvature) of the eye lens. The lens and ciliary muscles are attached to each other via suspensory ligaments.

Accommodation Power - The ability of the eye to change the focal length of the

eye lens with the help of ciliary muscles to get the clear view of nearby objects (about 25 cm) and far distant objects (at infinity).

Rhodopsin - It is also called visual purple, found in the retina of the eyes. It is a type of protein. It enables vision in low light conditions.

Superior rectus - It is an eye muscle, controlling the eye as it moves up. It is one of the four muscles that control eye movement. The other three are medial rectus, lateral rectus, and inferior rectus.

Stereopsis - It is also known as stereoscopic depth perception. It is the ability of both eyes to see the same object as one image and to create a perception of depth.

Ophthalmology - It is a branch of medicine which deals with diagnosis of eye disorders.

Defects of vision and their corrections

Myopia (Short sightedness) - It is an eye defect where near objects are clear but distant ones appear blurry, caused by excessive corneal curvature or elongation of the eyeball. It can be corrected with concave lenses in spectacles.

Hypermetropia (Long sightedness) - A common eye defect, results in clear vision of distant objects but blurry vision of nearby ones. It occurs due to the shortening of the eyeball. It can be corrected with convex lenses.

Presbyopia - Age-related lens hardening and ciliary muscle weakening lead to this condition, correctable with bifocal spectacles.

Astigmatism - It causes difficulty in focusing on both horizontal and vertical lines simultaneously. It is corrected with cylindrical lenses.

Cataract - Membrane growth over the eye lens causes haziness or opacity, resulting in vision loss. In old aged people, crystalline lenses become milky and cloudy.

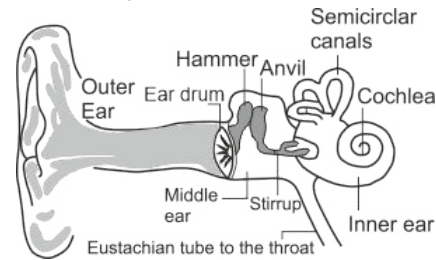
Amblyopia (lazy eye) - It is a type of poor vision that usually happens in just 1 eye but less commonly in both eyes.

Trachoma - It is a disease of the eye caused by infection with the bacterium *Chlamydia trachomatis*. Blindness from

trachoma is irreversible.

Ear

The main function of the ear is to maintain our sense of balance and to detect the various sounds. It converts the pressure variations into electronic signals that travel through the brain with the auditory nerve.



The ear can be divided into three major sections called the outer ear, the middle ear and the inner ear.

Inner Ear - It contains fluid and the cochlea, responsible for hearing. Situated in the temporal bone, it translates vibrations into electric signals for the auditory nerve. It consists of two parts, the bony and the membranous labyrinths.

Semicircular Canals - It senses head rotations, arising either from self-induced movements or from angular accelerations of the head imparted by external forces.

The Middle Ear - It is a small air filled space on the inside of the eardrum. It has three bones - Hammer, the Anvil, and the Stirrup. The central part of the eardrum links to the hammer, which transmits vibrations to the anvil and stirrup. These bones amplify sound waves, increasing their volume.

The middle ear contains three ossicles called malleus, incus and stapes. The stapes is the smallest bone in the human body.

Tegmen tympani - It is present on the roof of the middle ear. It separates the middle ear from the meninges and temporal lobe of the brain.

Eardrum - It behaves like a stretched rubber sheet. It is a part of the ear that separates the middle ear from the outer ear.

Eustachian tube - It regulates middle ear pressure to match that outside the body. It is a canal linking the middle ear to the nasopharynx, comprising the upper throat and back of the nasal cavity.

The Outer Ear - It consists of ear pinna and external auditory canal.

Pinna - This is the part of the ear which is outside of our head and that we can see. It helps us to gather sound vibrations.

Ear Wax - It is the protective yellow waxy substance secreted in the passage of the outer ear. Also called Cerumen.

External auditory canal - It leads inwards and extends up to the tympanic membrane (the eardrum).

Otoscopes - It is used to screen for illness during regular check-ups and also to investigate ear symptoms. It gives a view of the ear canal and tympanic membrane or eardrum.

Cochlear nerve - The sensory nerve that transfers auditory information from the cochlea (auditory area of the inner ear) to the brain.

Ear blockage during cold - During a cold, bacteria and viruses can enter our body and reach the middle ear, potentially blocking the Eustachian tube. This tube regulates air movement in the middle ear, preventing pressure buildup.

Ear Problems

Ear infection (otitis media) - It results from bacterial or viral entrapment. It is more prevalent in children than adults.

Eustachian tube dysfunction - The tubes get clogged. Symptoms include tinnitus, muffled hearing, sensation of fullness and possible ear pain.

Otitis externa - Swimmer's ear is an ear canal infection caused by bacteria or fungi. Getting water in the ear can cause this condition also when the ear canal is damaged with cotton swabs.

Ruptured eardrum - A hole in the tympanic membrane is called a ruptured eardrum.

Otosclerosis - It occurs when abnormal bone remodels in the middle ear.

Vestibular neuritis - It occurs when the vestibular nerve in the inner ear becomes inflamed.

Tests related to Ears

Pure-tone testing - This simple hearing test involves wearing headphones and raising the hand when you hear a "beep".

Auditory brainstem response (ABR) - During this test, electrodes are placed on the head. These electrodes record brain activity.

Otoacoustic emissions (OAEs) - This test determines how well cochlea works. When sound reaches the inner ear, the tiny hairs inside the cochlea vibrate.

Why do our ear have pain during flight?

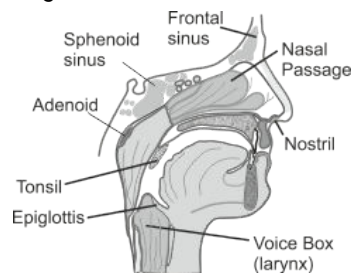
When an airplane climbs or descends, the air pressure changes rapidly. The eustachian tube often can not react fast enough, which causes the symptoms of airplane ear. It usually heals with time.

Nose

The prominent structure between the eyes that serves as the entrance to the respiratory tract and contains the olfactory organ.

Function of Nose- The nose helps in the sense of Smell. It also helps in the sense of taste. It works in coordination with the taste buds to identify the actual flavor of the food. Nose helps lungs to create pressure which is used to generate speech.

Nose helps in purification of inhaled air with the help of Hair or cilia. Nose hair also moisturizes and warms air, mirroring lung conditions.



The various parts of Nose are -

Bone - It supports the bridge of the nose.

Cartilage - The upper cartilage provides support to the sides of the nose. The lower cartilage adds width and height to the nose. It provides shape to the nostrils and nose tip.

Nasal Cavity - It is the hollow space through which the air flows.

Septum - The septum divides the inside of the nose into two chambers. It is a thin wall made of bones and cartilage.

Sinuses - The bone around the nose contains hollow, air-filled chambers known as sinuses. The mucus flows into

the nasal cavity from the sinuses.

Mucous Membrane - Mucous membranes line body cavities and cover internal organ surfaces. Examples include lips, mouth, nasal passages, middle ear, and the eustachian tube.

Turbinate - Each side of the nose contains curved turbinates, and the bony ridges are lined with mucous membranes.

Hair and cilia - Hair and cilia are tiny hair-like structures inside the nose that traps dirt and particles.

Adenoid - Traps germs coming through the mouth and nose.

External acoustic meatus - It is a bony canal for the ear canal.

External nostrils - Two chambers divided by the septum.

Problems related to Nose

Sinusitis is an inflammation and infection in the sinuses.

Deviated Septum - A deviated septum occurs when nasal septum and the thin wall that separates the right and left nasal passages is displaced to one side.

Nasal Polyps - Nasal polyps are soft, non-cancerous growths on the lining of the nose or sinus caused by inflammation often due to allergies.

Tongue

The tongue, a muscular organ covered with moist, pink mucosa. It facilitates licking, tasting, breathing, swallowing, and speaking. Its rough texture is due to papillae, housing thousands of taste buds comprised of nerve-like cells connected to brain-bound nerves.

Functions of Tongue- The tongue aids in mixing food with saliva during chewing, crucial for digestion. It assists in swallowing food into the esophagus and discerning taste. It secretes mucous and serous fluid which keeps the mouth moist.

Structure of Tongue

The human tongue is about 3.3 inches in men and 3.1 inches in women. It is located in the oral cavity. Tongue is divided into the anterior and posterior part.

The tongue's anterior portion, called the, attaches to the oral cavity floor. Its posterior segment, the pharyngeal or postsulcal part, forms the base of the oropharynx's ventral wall.

Anterior part (oral or presulcal part) - It includes the apex and body of the organ and it ends at the Sulcus terminalis.

Posterior part (pharyngeal or postsulcal part) - It is part of the organ that is made up of the base of the organ. It is located behind the palatoglossal folds.

Taste Buds is an end organ mediating the sensation of taste and lying chiefly in the epithelium of the tongue. Most of the taste buds on the sides of the tongue detect sour taste. Most of the taste buds at the back of the tongue detect bitter taste.

In front of the tongue it detects sweet and salty tastes. The maximum effect of sweet taste is felt at the tip of the tongue whereas the maximum effect of salty taste is felt just behind the tip of the tongue.

Tongue Diseases

Oral Thrush - It is an oral yeast infection that mainly affects newborns and younger children.

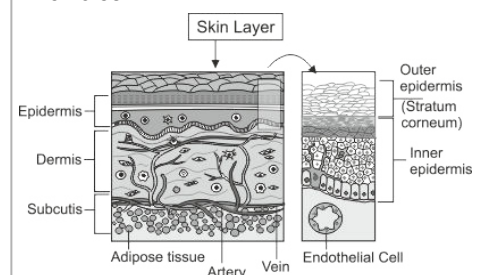
Kawasaki Disease - It is a rare and serious systemic inflammatory condition that affects children.

Oral leukoplakia - It is a potentially malignant disorder affecting the oral mucosa. It appears as thick, white patches on the inside surfaces of the mouth.

Macroglossia - Macroglossia is an abnormal enlargement of the tongue.

Skin

Skin, the body's largest organ, comprises a fleshy surface with hair, nerves, glands, and nails. It acts as a barrier between the internal and external environments. It is known as the cutaneous membrane. It anchors hair strands through hair follicles.



Structure of Skin

Epidermis - The outermost layer of the skin consists of keratinocytes (cells containing keratin), a protein that strengthens and waterproofs the skin.

The Types of Epidermal Cells are -

Melanocytes – These cells are located in the basal cell layer and produce a pigment called melanin. Melanocytes, responsible for producing melanin. More melanin, more darker is the skin.

Keratinocytes – They are produced in the basal layer and approach towards the outermost layer.

Langerhans - These are specialized cells that protects the body against foreign particles.

Dermis - It has connective tissue, blood vessels, oil and sweat glands, nerves, hair follicles. It is made up of a thin upper layer called the papillary dermis, and a thick lower layer called the reticular dermis.

It synthesizes Vitamin D to absorb calcium on exposure to sunlight. It acts as a lubricant and protects the skin by acting against the microbes.

Hypodermis - It is the bottom layer of skin. Also known as subcutaneous tissue, the hypodermis insulates and protects the body, stores energy (fat), helps to regulate body temperature, and connects the skin to muscles and bones.

Further, Skin is subdivided into Stratum corneum, Spinous cell layer, Basal cell layer.

Stratum corneum - The outermost layer of the epidermis is the stratum corneum. These cells contain abundant keratin and provide rigidity to the skin.

Spinous cell layer - found in the superficial layers of the skin.

Basal layer - It is beneath all other layers of the epidermis, where new cells are produced continuously.

Functions of Skin - It can store fats and water in its tissues. These provide extra insulation to our body. It regulate temperature through sweating.

Arrector pili - Small muscles attached to hair follicles in mammals. Contraction of these muscles causes the hairs to stand

on end, known colloquially as goose bumps.

Excretory System

Excretory system is a vital biological system that removes excess and waste products from the body to maintain homeostasis.

In humans, the excretory system consists of a pair of kidneys, one pair of ureters, skin, a urinary bladder and a urethra.

Kidneys - The kidneys are a pair of bean-shaped organs on either side of our spine, below our ribs and behind our belly. The left kidney is usually a little larger than the right.

Each kidney has nearly one million complex tubular structures called **nephrons**, which are the structural and functional units. Each nephron has two parts – the glomerulus and the renal tubule.

The kidneys remove **nitrogenous wastes** such as urea, as well as salts and excess water from the blood and excrete them in the form of urine.

Bowman's capsule - The Bowman capsule is the part of the nephron in the human kidney. Bowman's capsule carries out the function of ultra-filtration. It is also called glomerular capsule, renal corpuscular capsule, or capsular glomeruli.

Glomerulus is the main filtering unit of the kidney. It is formed by a network of small blood vessels (capillaries) enclosed within a sac called the Bowman's capsule. The space inside the capsule that surrounds the glomeruli is known as the Bowman's space.

Creatinine is a waste product that comes from the normal wear and tear on muscles of the body. Normally, kidneys filter creatinine from the blood and send it out of the body in the urine.

Treatment of kidney - There are two treatment options for kidney failure: Dialysis (hemodialysis or peritoneal dialysis) and Kidney transplantation.

Dialysis is a vital treatment for kidney failure, removing waste products and excess fluid from the blood when the kidneys malfunction. Without it, the

accumulation of waste in the blood can harm other organs.

There are two ways to get dialysis

Hemodialysis and **Peritoneal dialysis** :

Hemodialysis is a procedure where a machine removes blood from the body, filters it through a dialyzer (artificial kidney) and returns the cleaned blood to the body.

Peritoneal dialysis is a procedure where tiny blood vessels inside the abdominal lining (peritoneum) filter blood with the help of the aid of a dialysis solution.

Kidney transplantation is the ultimate method in the correction of acute renal failures (kidney failure). A functioning kidney is used in transplantation from a donor, preferably a close relative, to minimise its chances of rejection by the immune system of the host.

Function of the Kidney tubules -

Proximal convoluted tubule - It is a part of the kidney leading from the Bowman's capsule to the loop of Henle. It is lined by a simple cuboidal brush border epithelium, which increases the surface area for reabsorption.

Distal Convoluted Tubule : It is the shortest segment of the nephron segment. It plays a critical role in reabsorption, potassium secretion, and calcium and magnesium handling.

Henle's loop is a long U-shaped portion in the kidney to reabsorb water and important nutrients in the filtrate. It aims to reduce the volume of water and solutes within the urine without any change in its concentration.

Collecting Duct : This long duct extends from the cortex of the kidney to the inner parts of the medulla. Large amounts of water is reabsorbed from this part to produce a concentrated urine. It also plays a role in the maintenance of pH and ionic balance of blood by the selective secretion of H⁺ and K⁺ ions.

Role of other organs in excretion -

The ureters are thin bilateral tubes (3 to 4 mm) linking the kidneys to the urinary bladder, facilitating the transport of urine from the renal pelvis to the bladder.

The urinary bladder, situated in the lower abdomen, is a hollow, balloon-shaped organ responsible for storing urine. Its muscular wall expands to accommodate urine and contracts to expel it from the

body. Positioned in the pelvic cavity, its size and shape adjust according to urine volume and surrounding organ pressure. During urination, urine exits the bladder through the urethra.

Skin plays a role in excretion through the production of sweat by sweat glands. The sweat and sebaceous glands in the skin can eliminate certain substances through their secretions.

Sebaceous eliminates certain substances like sterols, hydrocarbons and waxes through sebum. This secretion provides a protective oily covering for the skin.

The urethra is a slender, fibromuscular tube responsible for expelling urine from the body, draining it from the bladder. In males, it serves the dual function of transporting urine and semen, while in females, it solely conducts urine from the bladder to the exterior.

Urine formation - Urine formation involves three main processes, i.e. filtration, reabsorption and secretion, that takes place in different parts of the nephron.

The liver produces several chemicals (enzymes) that change ammonia into urea, that the body can remove by the urine.

Glomerular filtration is the renal process in which fluid in the blood is filtered across the capillaries of the glomerulus. On an average, 1100-1200 ml of blood is filtered by the kidneys per minute which constitute roughly 1/5th of the blood pumped out by each ventricle of the heart in a minute.

The parietal epithelium of **Bowman's capsule** is the outer layer and is composed of simple squamous epithelial cells called "parietal cells." The parietal layer is not directly involved with filtration from the capillaries.

Glomerular filtration rate (GFR) - It is a blood test that checks how well kidneys are working. A GFR test estimates how much blood passes through glomerulus filters each minute.

Reabsorption - The nephron removes water and solutes from the tubular fluid (pre-urine) and returns them to the circulating blood. It is called reabsorption.

Secretion - Secretion is sending or eliminating the waste material from the body.

Urochrome, also known as **urobilin**, is a tetrapyrrole dicarboxylic acid that causes the yellow colour in urine. It is formed by the breakdown of bile pigments in the liver, preceded by the breakdown of haemoglobin.

Important facts and terms used for excretory system -

Albumin is a protein made by the liver. It enters the bloodstream and helps fluid from leaking out of blood vessels into other tissues.

Bilirubin is a yellowish substance made during the process of breaking down old red blood cells by the body. Bilirubin is a fluid that helps in digesting food and is found in the liver.

Glomerulonephritis - It damages the tiny filters inside the kidneys (the glomeruli) and causes Inflammation in the kidney.

Renal calculi - Stone or insoluble mass of crystallised salts (oxalates, etc.) formed within the kidney.

Uremia is a condition of Malfunctioning of kidneys which can lead to accumulation of urea in blood.

Renin is an enzyme made by special cells in kidneys. An enzyme that helps control the blood pressure and maintain healthy levels of sodium and potassium in the body.

Ultrafiltration occurs when fluid passes across a semipermeable membrane due to a driving pressure.

Columns of Bertini - They are also known as renal columns. Its main function is to make the cortex anchored in a better manner to the medullary region.

Renal cortex is also known as the kidney cortex. The cortex surrounds the inside of the organ, which is called the **medulla**. The renal cortex is covered by the **renal capsule**.

Mature renal medulla - It is the inner part of the kidney, consists of the medullary collecting ducts, loops of Henle, vasa recta and the interstitium.

Renal calyx are chambers of the kidney through which urine passes. The minor calyx surrounds the apex of the renal pyramids.

Efferent arterioles are blood vessels that are part of the urinary system of many animals.

Afferent arterioles are a group of blood vessels that supply the nephrons in many excretory systems.

Reptile kidneys regulate internal balance by excreting waste, controlling salt and water levels, and producing hormones and vitamins, but they cannot produce hypertonic urine.

Skeleton System

The skeletal system encompasses all the bones in the human body, providing structural support and framework. It includes bones, cartilage, ligaments, and tendons, which connect bones to each other and to muscles.

At the time of birth, infants have more bones (~300 bones), which due to fusion in some of the bones, form larger bones and 206 bones remain in a human adult.

Functions of bones

The bones give shape to the body, protect internal organs such as the brain, heart, and lungs, provide support and anchor muscles, store minerals like calcium and phosphate, and act as the birthplace for red blood cells (RBCs).

Fluids - Fluids provide lubrication, reduce friction between bones and make the joint movements smooth. The fluids present in body are -

Synovial Fluid - It is found between joints. It facilitates smooth movement and aids in various body movements, including locomotion. It acts as a lubricant, reducing friction between bones in joints like the ball and socket, pivot, gliding, hinge, and saddle joints.

Cerebrospinal Fluid (CSF) - Cerebrospinal fluid (CSF) is a clear, colorless fluid that circulates around the brain and spinal cord, playing vital roles in protecting these structures from injury, providing nutrients, and removing waste products. It is produced by specialized cells known as choroid plexus, located in the lateral, third, and fourth ventricles of the brain.

Amniotic fluid, contained within the amniotic sac of a pregnant amniote, cushions the fetus and facilitates nutrient exchange between mother and child. Normal levels at 36-38 weeks

gestation range between 800-1000, originating from maternal plasma and passing through fetal membranes via osmotic and hydrostatic forces.

Interstitial Fluid: It fills spaces between cells. It is composed of water, amino acids, sugars, fatty acids, coenzymes, hormones, salts, etc. It facilitates material delivery, intercellular communication, and waste removal.

Shoulder Girdle - It is also referred to as the pectoral girdle. It is a set of bones found in the appendicular skeleton connecting the arm on either side. The pectoral girdle in humans comprises the clavicle (collarbone) and the scapula (shoulder blade). The girdles support the appendages to the axial skeleton.

Joints in human body

Based on the functional classification, three types of joints are present in the human body -

1. Synarthroses (immovable)
2. Amphiarthrosis (slightly movable)
3. Diarthrosis (freely movable)

Synarthroses (immovable): They are defined as two or more bones in close contact that have no movement. Example - Bones of the skull.

Amphiarthrosis (slightly movable), also known as cartilaginous joints, these joints are defined as two or more bones held so tightly together that only limited movement can take place. Example - vertebrae of the spine.

Diarthroses (freely movable): It is also called synovial joints. These joints have synovial fluid for smooth movement. Examples: knee, shoulder joints.

Synovial joints are further classified by the type of movements they permit. There are six such classifications - hinge (elbow), saddle (carpometacarpal joint), planar (acromioclavicular joint), pivot (atlantoaxial joint), Condylod (metacarpophalangeal joint), ball and socket (hip joint).

Hinge joints permit movement primarily in one direction, resembling the action of a door hinge. They allow bones to move back and forth with limited motion along other planes. Examples of hinge joints in the body include those found in the fingers, toes, elbows, knees, and ankles.

Saddle joints are synovial joints where one bone resembles a saddle, with the

other bone resting on it like a rider on a horse. They offer stability while allowing more flexibility than hinge or gliding joints. In humans, saddle joints are found only in two locations: the carpal bone of the thumb and the tarsal bone of the foot.

Gliding joints, formed between flat or nearly flat articular surfaces, enable bones to move in multiple directions along the plane of the joint. Commonly found in the wrists and ankles, these joints facilitate smooth gliding motions between bones.

Pivotal joints (rotary or trochoid joints) are freely movable joints that only allow rotation on a single axis. It is a type of synovial joint in which the ends of two bones are connected together.

Three pivot joints in the human body: neck, wrist, elbow. These joints allow the bending of the lower arm against the upper arm and movement of head. The neck is joined by the head by a Pivotal joint.

Ball and socket joints feature a partially spherical end fitting into a socket, allowing versatile movement and rotation, including flexion, extension, adduction, and rotation. Examples include the shoulder and hip joints, providing extensive mobility.

Bones in Human Body

The patella is the largest sesamoid bone in the body, located in front of the knee joint within the quadriceps tendon. It attaches to both the quadriceps tendon and the patellar ligament.

Babies are born without the **kneecap or patella**. The kneecap starts out as cartilage and starts significantly hardening into bone between the ages of 2 and 6 years old.

Femur Bone - The femur is the thigh's sole bone, being the strongest and longest bone in the human body. It links the pelvic bone to the lower leg bones and forms a ball-and-socket joint with the hip.

The tibia, or shinbone, is one of the lower leg bones connecting the knee to the ankle.

Fibula bone - The fibula is called the calf bone and connects to the tibia above and below. It runs laterally to the tibia and is

smaller than the two. The bottom part of the fibula extends well past the tibia and forms the lateral part of the ankle.

Tarsal bones - These bones are found in the leg which are organized into three rows: proximal, intermediate and distal. The tarsal bones consist of seven short bones located at the proximal region of the foot in the ankle area - calcaneus, talus, cuboid, navicular, and the medial, middle and lateral cuneiforms.

Metatarsals connect the phalanges to the tarsals. It helps the foot withstand the weight of the body whilst standing and in motion.

Wrist Bone - Wrist is made up of eight small bones (carpal bones) plus two long bones in the forearm, the radius and the ulna. The bones comprising the wrist include the distal ends of the radius and ulna, 8 carpal bones, and the proximal portions of the 5 metacarpal bones. The trapezoid bone is the smallest bone in the distal row of carpal bones that give structure to the palm of the hand.

Shoulder Bone - The shoulder is made up of three bones, scapula (shoulder blade), clavicle (collarbone), and humerus (upper arm bone). Two joints in the shoulder allow it to move, the acromioclavicular joint, where the highest point of the scapula (acromion) meets the clavicle, and the glenohumeral joint.

Four joints are present in the shoulder: the sternoclavicular, acromioclavicular, and scapulothoracic joints, and glenohumeral joints.

Collar Bone - The collarbone (clavicle) connects our arms to the body. It runs horizontally between the top of the breastbone (sternum) and shoulder blades (scapula). Sometimes called the "beauty bone" in women. It is the only long bone in the body that lies horizontally.

Stapes Bone - There are three bones in the middle ear - Malleus, Stapes and Incus. The Stapes is the **smallest** of the three bones and it looks like a bell. The size of the staples is 3mm × 2.5 mm. It is responsible for transmitting sound waves from the air outside to the fluid-filled labyrinth (cochlea).

Forelimb Bones - Forelimb / upper limb bones include the humerus, radius, ulna, carpals, and metacarpals. There are 30 bones in each upper limb.

The rib cage is a structure of ribs connecting the spine and sternum in the chest, protecting vital organs like the heart and lungs.

Coxal Bone - The hip bone (os coxae, innominate bone, pelvic bone, or coxal bone) is a large irregular bone, constricted in the center and expanded above and below. In some vertebrates (including humans before puberty), it is composed of three parts: the ilium, ischium, and the pubis.

Coccyx Bone (tailbone) : It is the terminal part of the vertebral column. The coccyx serves as an attachment site for tendons, ligaments, and muscles. It also functions as an insertion point of some of the muscles of the pelvic floor. Its function is to support and stabilize a person while sitting.

Pelvis Bone : The pelvis or pelvic girdle is a basin-shaped complex bone placed right between the abdomen and pelvic region. The pair of hip bones (coxal bone) join the lower limbs and the spine.

Skull Bone - The skull or the cranium is made up of 22 bones. Except for the mandible, all other bones of the skull are joined together. The bones of the skull can be divided into 2 main segments - Cranial bones (8), Facial skeletal bones (14).

Spinal Cord Bone - The spine has 33 stacked vertebrae (small bones) that form the spinal canal. The spinal canal is a tunnel that houses the spinal cord and nerves, protecting them from injury.

Chest Bone - In a healthy adult human, there are 26 chest bones, but some individuals have a congenital condition where they possess an extra rib (called cervical ribs).

Human Hand Bones - The human hand is made up of 27 bones. Bones in the hand can be divided into three categories

Carpal bones (proximal) - 8 bones that make up the wrist. The 2 rows of carpal bones are attached to the 2 bones of the hand - the ulna bone and the radius bone.

Metacarpals - 5 bones that make up the middle part of the hand.

Phalanges (distal) - 14 bones found in the fingers of each hand and also in the toes of each foot. Each finger has 3 phalanges (distal, middle and proximal); The thumb has only 2 phalanges.

Human Arm Bones - Arm is made up of 3 bones: the upper arm bone (humerus) and two forearm bones (the ulna and the radius).

Pelvis Bone - There are 3 bones of the pelvis: the hip bone, sacrum and coccyx.

Legs Bone - There are a total of 60 bones in the legs.

Feet Bone - Each foot is made up of 26 bones, 30 joints and more than 100 muscles.

The passage surrounded by the bony pelvis, termed the pelvic cavity, is a particular body division below the abdomen and essentially contains and supports the urinary bladder, intestines, and other internal reproductive organs.

The skeletal tissue is a connective tissue that possesses a solid matrix and the living cells in the fluid-filled spaces called **lacunae**.

There are two types of skeletal tissues, namely - **Bones** and **Cartilages**.

Bones are strong and nonflexible tissues. The cells are embedded in a hard matrix consisting of a collagenous protein complex called **ossein** and minerals. The bones provide shape and skeletal support to the body.

Cartilage is a type of skeletal tissue. It consists of a ground substance called matrix that contains cartilage cells. The primary function of cartilage is to connect the bones of a body together. Cartilages are found in the joints between bones like elbows, knees, ankles, rib cage, discs of the spine, ear, nose, throat, and bronchioles.

Tissues in Skeletal System

Tendon - Tendons join muscles to bones. Tendon tissue is also known as sinew. It is a band of fibrous connective tissues. Tendon is made up of **collagen**.

Ligament - Ligaments connect bone to bone. They are made up of bundles of elastic fibres and few collagen fibres.

Sprain is caused due to excessive pulling of ligaments.

Muscles in the Human Body

Masseter is the strongest muscle of the body which is located in the jaw. The muscle masseter is a facial muscle that plays a major role in the chewing of solid

foods.

The muscle has a parallelogram-like shape, binding to the mandible (lower jaw bone) and the cheekbone. The elevation of the mandible happens when the jaws are closed.

Soleus - This is one of the leg muscles that have an extension from the knee's back to the heel. This muscle is important for standing and walking.

Deltoid Muscles - This muscle is located over shoulder joints or glenohumeral joints and are large, triangular-shaped muscles. It is also known as the 'common shoulder muscle'.

The deltoid muscle is forming the round shape of the human shoulder. Deltoids appear to be made up of three sets of fibres : Anterior, Posterior, Intermediate.

The function of Deltoid - When all its fibers contract simultaneously, the deltoid is the prime mover of arm abduction along the frontal plane.

Gluteus maximus - It is the largest muscle in the body, is located at the back of the hip and is responsible for hip rotation and extension. It's one of three gluteal muscles, along with the Gluteus minimus and Gluteus medius. The Sartorius, running from the upper thigh to the inside of the knee, is the longest muscle, while the Latissimus dorsi, in the lower and middle back, is the widest

Teeth Skeletal System

Teeth are considered as a part of the skeleton system even though they are not bone. Teeth are the strongest substance in the body being made up of enamel and dentin. There are 32 teeth in an adult, and 28 in children. There are 20 milk teeth in Children.

4 Types of Teeth

Incisors (8) - It is used for cutting the food.

Canines (4) - It is used for Grip and tear food.

Premolars (8) - It is used for chewing the food.

Molars (12) - It is used for chewing, crushing and grinding food.

The 3rd Molar tooth is called **Wisdom Teeth**. Ivory Tusk of elephants is the second Incisor tooth of the upper jaw. Lion large teeth are Canines.

The layers of a tooth include -

Dentin is the largest dental tissue. Dentin is a composite substance of organic and inorganic material found in the tooth's interior. Dentin is pale yellow and becomes darker with age. When dentin is damaged or exposed, a variety of problems can occur, such as tooth sensitivity.

Enamel is the hardest substance in the human body and contains the highest percentage of minerals (at 96%), with water and organic material composing the rest. The primary mineral is hydroxyapatite, which is crystalline calcium phosphate.

Dental pulp forms the tooth's inner core, containing the blood vessels and nerves.

Cementum is a hard connective tissue that covers the roots of the tooth.

Endocrine/Exocrine Glands

Glands are important organs located in the body. They produce and secrete substances, called Hormone, that perform certain specific functions.

Hormones are chemicals that coordinate different functions in the body by carrying messages through the blood to organs, skin, muscles and other tissues. All glands in human body are divided into two systems:

1. Endocrine system
2. Exocrine system

Endocrine System

The human endocrine system consists of ductless glands that release hormones directly into the bloodstream, regulating essential functions like growth, metabolism, and reproduction.

The nervous system regulates the function of endocrine glands, with hormones serving as messengers between the two systems and organs. Key coordination centers for this interaction are the hypothalamus and the pituitary gland.

Hypothalamus - It is the part of the forebrain. It contains many groups of hormone-secreting cells called nuclei. These hormones regulate the synthesis and secretion of pituitary hormones.

The hormones produced by hypothalamus are of two types, the releasing hormones (which stimulate secretion of pituitary hormones) and the inhibiting hormones (which inhibit secretions of pituitary hormones). It connects both endocrine and neurological systems.

Pituitary gland - It is located at the bottom of the brain. It is about the size of one pea. It is known as the "Master gland" because it controls several other hormone glands in the body.

Several hormones are produced by the human pituitary gland, such as:

Growth hormone (Somatotropin hormone-STH) - It controls the development of bones and muscles.

Antidiuretic hormone - It helps control body water balance through its effect on the kidneys.

Oxytocin - It triggers the contractions of the uterus that happen during labour pain and childbirth.

Luteinizing hormone - It manages oestrogen in women and testosterone in men.

Endorphins are molecules that work on the nerve system and relieve sensations of pain, and are secreted by the pituitary gland and hypothalamus.

The pituitary gland expels hormones that stimulate the reproductive system to produce sex hormones. The pituitary gland effectively regulates ovulation as well as the menstruation cycle in women.

It secretes hormones from both the front part (anterior) and the back part (posterior) of the gland.

Pineal gland - The pineal gland, conarium, or epiphysis cerebri, is a small endocrine gland in the brain of most vertebrates. The pineal gland is located on the dorsal side of the forebrain. It is the smallest gland in the human body.

The pineal gland secretes melatonin, a hormone crucial for regulating the body's rhythms. Melatonin helps maintain the sleep-wake cycle, body temperature, and influences metabolism, pigmentation, the menstrual cycle, and the body's defense mechanisms.

Thyroid gland - Thyroid gland is located at the larynx just below in the front of the

lower neck. It has a similar shape to a butterfly.

The thyroid gland produces two main hormones: thyroxine (T4) and triiodothyronine (T3), both of which contain iodine.

Thyroid hormone regulates carbohydrate, fat, and protein metabolism, and iodine deficiency can lead to its deficiency and result in goitre.

Hashimoto's disease is related to the thyroid gland.

Cretinism - The condition that causes severe mental retardation due to the deficiency of iodine and thyroid hormones, especially during early pregnancy.

Para-Thyroids - There are four tiny glands that work together called the parathyroid glands. The parathyroid gland secretes a hormone called parathormone. It helps to regulate calcium and phosphate levels in the blood with the help of calcitonin.

Parathyroid hormones - It secretes when there is deficiency of calcium in blood. Insufficient level of parathyroid hormone and low amount of calcium in the blood causes **Hypocalcemia** conditions.

Calcitonin hormones - It controls the quantity of calcium in blood.

Thymus Gland - The thymus gland, located in the lower neck and upper chest, secretes thymus hormone, aiding in immune system development. It produces T-lymphocytes, vital in combating infections and fostering a child's immune system. Following puberty, the thymus gradually diminishes in size.

Adrenal Gland - Adrenal glands (triangular shaped) are located at the top of each kidney. This gland is also called glands of emergency. This gland has two major parts - Adrenal medulla and Adrenal Cortex. In absence of Adrenal cortex a person can die within one or two weeks due to Addison disease. Adrenal glands produce Adrenaline hormone (also called Epinephrine).

The inner part of the adrenal gland (adrenal medulla) produces Adrenaline hormone which is also known as 'fight or flight' hormone. This hormone is

secreted in large amounts when the person is excited or frightened.

Cortisol (Corticosteroids) Hormone : The adrenal cortex, the outer portion of the adrenal gland, regulates the body's salt and water balance. As the primary stress hormone, it elevates blood sugar levels and boosts the brain's glucose utilisation.

Pancreas - It is a part of the digestive system and produces insulin and other important enzymes and hormones that help break down foods. It acts as both endocrine and exocrine gland.

This gland is present just below the stomach and secretes a hormone called insulin. The function of insulin is to lower the sugar level of the blood.

Reproductive Glands (Gonads) - The gonads are the main source of sex hormones.

Testes : Male gonads are known as testes. They secrete hormones called androgens, the most important of which is testosterone. The **testosterone** controls the development of male sex organs and male features such as deeper voice, moustache, beard, making of sperm etc.

Ovaries : Female gonads known as the ovaries. They make eggs and secrete the female hormones estrogen (oestrogen) and progesterone.

Estrogen helps in controlling the development of female sex organs and female features such as feminine voice, soft skin, body fat around the hips and thighs, breast growth during puberty.

The progesterone hormone controls the uterus changes in the menstrual cycle. It helps in the maintenance of pregnancy.

Proboscis gland : It helps in excretion of nitrogenous waste in hemichordates.

Exocrine System

Exocrine glands release (secrete) substances through openings (ducts) onto your body surfaces. Exocrine glands secrete sweat, tears, saliva, milk and digestive juices.

Salivary glands - Salivary glands, found throughout the mouth, produce saliva that aids in chewing, swallowing, and digestion, while also helping to kill germs with its antibodies.

Parotid glands - It is located in front of and just below the ears. It produces saliva (spit) to keep your mouth lubricated, and to aid in chewing and digestion.

Sublingual glands - It is located just under the tongue. The paired sublingual glands are major salivary glands in the mouth.

Submandibular glands - It is located below the jaw. Its main function is to contribute to the production of saliva.

Sweat glands - It is also known as sudoriferous glands. Skin contains two types of sweat glands. They are eccrine and apocrine.

Eccrine glands - It is found in almost all parts of the skin.

Apocrine glands - It is found in hair-bearing areas, such as the skin, armpits, and groyne. It secretes a milky fluid, usually as during stress.

Sebaceous Glands - These glands, present in the skin excluding the palms and soles, produce an oily substance called sebum, which lubricates the skin, retains moisture, and regulates body temperature by collaborating with sweat glands.

Mammary glands - These are a type of sweat gland, responsible for the production of breastmilk. While males possess underdeveloped mammary glands, they typically don't produce substantial amounts of hormones within the breasts. However, the oestrogen generated during puberty in females prompts the growth of this tissue.

Lacrimal gland - The lacrimal gland is a bilobed, teardrop-shaped gland that secretes tears. Onions produce the chemical irritant known as syn-Propanethial-S-oxide. It stimulates the eyes' lachrymal glands so they release tears.

Brunner glands - The Brunner glands are located in the first part of the small intestine. It produces mucus that protects the duodenum from stomach acid.

Ceruminous glands - Ceruminous glands are glands in ears. They help to produce ear wax (cerumen).

Enzymes and Hormones

Hormones are non-nutrient chemicals which act as intracellular messengers and are produced in trace amounts.

Hormones in endocrine system

The endocrine glands and hormone producing diffused tissues/cells located in different parts of the body constitute the endocrine system. Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females) are the organised endocrine bodies in the body.

Hypothalamus controls the body temperature, regulates emotions, hunger, thirst, sleep, moods and allows the production of hormones.

Thyrotropin-releasing hormone (TRH), gonadotropin-releasing hormone (GnRH), growth hormone-releasing hormone (GHRH), corticotropin-releasing hormone (CRH), somatostatin, and dopamine are released from the hypothalamus into the blood and travel to the anterior pituitary.

Gonadotropin-releasing hormone (GNRH) is produced and secreted by specialised nerve cells in the hypothalamus of the brain. It is released into tiny blood vessels that carry this hormone from the brain to the pituitary gland, where it stimulates the production of two more hormones – follicle stimulating hormone (FSH) and luteinising hormone (LH).

Corticotropin-releasing hormone (CRH) is the main element that drives the body's response to stress. It is also present in diseases that cause inflammation. Too much or too little corticotropin-releasing hormone can have a range of negative effects.

Growth hormone-releasing hormone (GHRH) is a hypothalamic hormone that is essential for normal expansion of the somatotrope lineage during pituitary development.

Thyrotropin-releasing hormone (TRH) is produced by the hypothalamus. It plays an important role in the regulation of thyroid gland activity.

Somatostatin inhibits secretion of growth hormone, thyroid stimulating hormone, cholecystokinin, and insulin.

Dopamine is a type of neurotransmitter and hormone. It regulates movement, memory, pleasure, motivation.

Pineal glands (thalamus) secretes melatonin, which plays a very important role in the regulation of 24-hour (diurnal) rhythms of our body.

The parathyroid glands secrete a peptide hormone called parathyroid hormone (PTH). The secretion of PTH is regulated by the circulating levels of calcium ions.

The thymus gland is a lobular structure located between lungs behind the sternum on the ventral side of aorta. It secretes the peptide hormones called thymosins.

Thymosins play a major role in the differentiation of T-lymphocytes, which provide cell-mediated immunity.

Thymopoietin is a polypeptide hormone of the thymus consisting of 49 amino acids. It fuels the production of T-cells and tells the pituitary gland to release hormones.

Thymulin (formerly called FTS) is a well defined nonapeptide hormone produced by thymic epithelial cells. It is used for various diseases associated with immune dysfunction, viral and bacterial infections, regeneration normalization, immunodepression, and the depression of hematogenesis after chemical and radiotherapy.

Thyroid gland plays an important role in the regulation of the basal metabolic rate. It supports the process of red blood cell formation. It controls the metabolism of carbohydrates, proteins and fats. It also secretes a protein hormone called thyrocalcitonin (TCT) which regulates the blood calcium levels.

Thyroid gland is composed of follicles and stromal tissues. Each thyroid follicle is composed of follicular cells, enclosing a cavity. These follicular cells synthesise two hormones, tetraiodothyronine or thyroxine (T₄) and triiodothyronine (T₃).

Adrenal gland produces the hormones that control the sex drive, cortisol and stress hormone. It makes the hormones cortisol, aldosterone, adrenaline, and noradrenaline.

Adrenal medulla secretes two hormones called adrenaline or epinephrine and noradrenaline or norepinephrine. Adrenaline and noradrenaline are rapidly secreted in response to stress of any kind and are called emergency hormones or hormones of Fight or Flight.

Aldosterone, a type of steroid hormone, is produced by the adrenal glands and serves to moderate the levels of salt and water in the human body, which in turn affects blood pressure.

Low aldosterone levels are found in a rare condition called **Addison's disease**.

Cortisol is a steroid hormone that is produced by 2 adrenal glands, which sit on top of each kidney. When you are stressed, increased cortisol is released into the bloodstream.

Pituitary gland is located in a bony cavity called sella tursica. It is divided anatomically into an adenohypophysis and a neurohypophysis.

The pars distalis region of pituitary, called anterior pituitary, produces growth hormone, prolactin, thyroid stimulating hormone, adrenocorticotrophic hormone (ACTH), luteinizing hormone (LH) and follicle stimulating hormone (FSH).

Prolactin regulates the growth of the mammary glands and formation of milk in them.

Neurohypophysis (posterior pituitary) stores and releases two hormones called oxytocin and vasopressin.

Oxytocin acts on the smooth muscles of our body and stimulates their contraction. In females, it stimulates a vigorous contraction of the uterus at the time of child birth, and milk ejection from the mammary gland.

Vasopressin or antidiuretic hormone (ADH) acts mainly at the kidney and stimulates resorption of water and electrolytes by the distal tubules and thereby reduces loss of water through urine (diuresis).

SIADH is a disorder related to Vasopressin Hormone.

Luteinizing hormone (LH) is a gonadotropin hormone secreted by the pituitary and is essential for puberty and normal sexual function. Its deficiency results in sexual dysfunction and infertility.

Follicle-stimulating hormone (FSH) is a hormone that plays a significant role in sexual development and reproduction by affecting the function of the ovaries and testes. It works alongside luteinizing hormone (LH).

Growth hormone (GH), also called

somatotropin or human growth hormone, is a peptide hormone secreted by the anterior lobe of the pituitary gland.

Over-secretion of GH stimulates abnormal growth of the body leading to gigantism and low secretion of GH results in stunted growth resulting in pituitary dwarfism.

Acromegaly is a disorder related to Growth Hormone.

Thyroid stimulating hormone (TSH) stimulates the synthesis and secretion of thyroid hormones from the thyroid gland.

Adrenocorticotrophic hormone (ACTH) stimulates the synthesis and secretion of steroid hormones called glucocorticoids from the adrenal cortex.

Pancreas gland: It is a composite gland which acts as both exocrine and endocrine gland. The endocrine pancreas consists of 'Islets of Langerhans'.

The two main types of cells in the Islet of Langerhans are called α -cells and β -cells. The α -cells secrete a hormone called glucagon, while β -cells secrete insulin.

Insulin helps the body to lower blood glucose levels and 'store' the sugar away in fat, muscle, liver and other body tissues. Insulin is a peptide hormone secreted by the β cells of the pancreatic islets of Langerhans.

Islets of Langerhans, also called islands of Langerhans, irregularly shaped patches of endocrine tissue located within the pancreas of most vertebrates. They are named for the German physician Paul Langerhans, who first described them in 1869.

The hormones secreted by the islets of Langerhans are α -cells-Secrete glucagon, β -cells-Secrete insulin, δ -cells-Secrete somatostatin.

Glucagon, a peptide hormone, plays an important role in maintaining the normal blood glucose levels. It acts mainly on the liver cells (hepatocytes) and stimulates glycogenolysis resulting in an increased blood sugar (hyperglycemia).

Testes glands - In men, the testes secrete the male sex hormone to testosterone. It also produces sperm.

Testosterone is the male sex hormone that is made in the testicles. Testosterone hormone levels are important to normal male sexual development and functions.

During puberty (in the teenage years), testosterone helps boys develop male features like body and facial hair, deeper voice, and muscle strength.

Ovaries: In the female reproductive system, the ovaries release estrogen, progesterone, and other female sex hormones.

Estrogen is a category of sex hormone responsible for the development and regulation of the female reproductive system and secondary sex characteristics.

Progesterone supports pregnancy. Progesterone also acts on the mammary glands and stimulates the formation of alveoli (sac-like structures which store milk) and milk secretion.

The steroid hormone is a steroid that acts as a hormone. Steroids work by decreasing inflammation and reducing the activity of the immune system.

There are five major classes of steroid hormones: testosterone (androgen), estradiol (estrogen), progesterone (progestin), cortisol/corticosterone (glucocorticoid), and aldosterone (mineralocorticoids).

Glucocorticoids are steroid hormones. It stimulates gluconeogenesis, proteolysis, erythropoiesis, lipolysis, cardio-vascular system, blood pressure, and glomerular filtration rate and inhibit inflammatory reactions by suppressing the immune response

Other important Hormones

Auxin Hormone is a plant hormone responsible for plant growth and development, orchestrating cell division, elongation and differentiation, embryonic development, root and stem tropisms, apical dominance, and transition to flowering.

Ethylene as a Plant Hormone that regulates and mediates complex cycles in plants, regarding their growth and development, and their survival throughout their life cycle.

Cytokinin is a class of plant growth hormones that are active in promoting cell division, growth, differentiation, and other physiological processes. Examples are kinetin, zeatin, and benzyl adenine.

Gibberellin Hormone regulates various developmental processes, including germination, dormancy, flowering, flower

development, and leaf and fruit senescence.

Atrial natriuretic factor (ANF) or Atrial Natriuretic Peptide (ANP) is a 28 amino acid polypeptide hormone secreted mainly by the heart atria in response to atrial stretch. ANF acts on the kidney to increase sodium excretion and GFR, to antagonise renal vasoconstriction, and to inhibit renin secretion.

Cholecystokinin (CCK) is a hormone secreted by the I-cells of the upper small intestine in response to fat, protein, and some non nutrients, for example, camostat and a peptide/neurotransmitter secreted by neurons of the central and peripheral nervous systems.

Gastric inhibitory polypeptide (GIP) is a 42-amino-acid gastrointestinal hormone. It is a member of the incretin family mediating increases of insulin secretion prior to the postabsorptive nutrient-induced increase in blood glucose.

GIP is secreted primarily from the stomach and the duodenum K cells in response to ingestion.

Renin is a hormone made by the kidneys. It controls the production of aldosterone.

Erythropoietin (EPO) is a glycoprotein hormone, naturally produced by the peritubular cells of the kidney, that stimulates red blood cell production. Renal cortex peritubular cells produce most EPO in the human body. PO2 directly regulates EPO production.

Calcitonin is a hormone that helps to control the level of calcium in blood. Calcitonin is made in the thyroid gland by cells called "C cells".

Higher-than-normal levels of calcitonin can occur in people with kidney disease, smokers, and higher body weight.

Enzymes

Enzymes catalyse biochemical reactions. It breaks down complex molecules like carbohydrates, protein, fats, etc. into simple molecules.

Polyphenol oxidase, a copper-containing metalloprotein enzyme, responsible for the oxidation reaction that occurs in some foods, mostly fruits and vegetables, that turns the food brown. For example when an apple is cut, it gets discoloured due to the presence of the polyphenol oxidase enzyme.

Pyruvate kinase is an enzyme that catalyses the conversion of phosphoenolpyruvate and ADP to pyruvate and ATP in glycolysis and plays a role in regulating cell metabolism.

The carbonic anhydrases form a family of enzymes that catalyse the interconversion between carbon dioxide and water and the dissociated ions of carbonic acid. The active site of most carbonic anhydrases contains a zinc ion.

Nitrate reductase (NR) is the key enzyme for nitrogen assimilation in plant cells and also works as an important enzymatic source of nitric oxide (NO), which then regulates plant growth and resistance to biotic and abiotic stresses.

Lipids are fatty compounds. They're part of cell membranes and help control what goes in and out of cells. They help with moving and storing energy, absorbing vitamins and making hormones.

Lipoproteins are particles made of protein and fats (lipids). They carry cholesterol through the bloodstream to cells. The two main groups of lipoproteins are called HDL (high-density lipoprotein) or "good" cholesterol and LDL (low-density lipoprotein) or "bad" cholesterol.

Papain is an enzyme found in the white fluid (latex) that occurs in raw papaya fruit. It is a protease, meaning it breaks down proteins.

Papain contains substances that might help fight infection and heal wounds. Papain is used as a popular ingredient in meat tenderizer.

Glucose oxidase (GOx) is an oxidoreductase enzyme with many important roles in biological processes. It is considered an "ideal enzyme" and is often called an oxidase "Ferrari" because of its fast mechanism of action, high stability and specificity.

Amylase is a digestive enzyme predominantly secreted by the pancreas and salivary glands and found in other tissues at very small levels. Amylase is considered one of the first enzymes in history to be scientifically investigated. Amylase breaks down starches (complex carbohydrates) into sugars, which our body can absorb more easily.

Lipase is an enzyme the body uses to break down fats. Lipase is produced in the pancreas, mouth, and stomach.

Lipase breaks down dietary fats into smaller molecules known as glycerol and fatty acids.

Steapsin, the enzyme that converts fats into fatty acids and glycerol, is secreted by Pancreas.

Lysozyme, an enzyme present in bodily secretions like tears, mucus, sweat, saliva, and milk, as well as in high concentrations in avian egg whites, plays a crucial role in the innate immune system. By hydrolyzing the polysaccharides of bacterial cell membranes, lysozyme acts as an antibacterial agent, contributing to the body's defence against infections.

Trypsin is an enzyme that breaks down proteins, continuing the process of digestion that began in the stomach. It is produced by the pancreas in an inactive form called trypsinogen.

Bromelain is a protein-digesting enzyme mixture derived from the stem, fruit, and juice of the pineapple plant. It is used to remove dead skin from burns, and orally, to reduce inflammation and swelling – particularly of the nasal passages. It is also used as a digestive aid for osteoarthritis.

Maltase is an enzyme that catalyses the disaccharide maltose hydrolysis to the simple sugar glucose. This enzyme is present in bacteria, yeast, and plants, and is generated by cells of the mucous membrane lining the intestinal wall in humans and other vertebrates. Maltase is present in the Saliva and breaks the sugar maltose into glucose.

Proteases are enzymes that break the peptide bonds of proteins; they are divided into acid, neutral, and alkaline proteases.

The human saliva contains an enzyme called salivary protease that breaks down starch, a complex molecule, into sugar.

Saliva is a digestive juice secreted by the salivary glands and is present in the mouth.

Peptidases are enzymes capable of cleaving, and thereby often inactivating, small peptides. They are widely distributed on the surface of many different cell types, with the catalytic site exposed only at the external surface.

Antimicrobial enzymes are widespread in nature, they play a critical role in

defending living organisms from bacterial attack.

The major antimicrobial proteins of milk are lysozyme, lactoferrin (Lf), lactoperoxidase (LP), and immunoglobulins. Lactoferrin, is a glycoprotein, which consists of small fractions of milk proteins.

Invertase is an enzyme that is widely distributed among plants and microorganisms and that catalyzes the hydrolysis of the disaccharide sucrose into glucose and fructose.

Lactase is an enzyme that helps the body to digest lactose. Lactose is found in dairy foods like milk and cheese. The lactase supplement prevents bloating, diarrhoea and gas caused by lactose intolerance.

Fungal enzymes can degrade detritus into simpler inorganic substances.

Alliinase enzyme is released during peeling, cutting or crushing onion tissue.

Some other enzymes/proteins and their role in DNA replication:

Helicase is an enzyme/protein that opens up the DNA double helix.

RNA primase - enzyme/protein that helps in RNA primer synthesis.

DNA polymerase - helps in nucleotide polymerisation.

DNA ligase - joins the 3' end of the new DNA fragment to the 5' end of the previous one.

Other Biomolecules

Serotonin is a chemical that carries messages between nerve cells in the brain and throughout the body. Serotonin plays a key role in such body functions as mood, sleep, digestion, nausea, wound healing, bone health, blood clotting and sexual desire.

Pepsin is an **endopeptidase** that breaks down proteins into smaller peptides. It is produced in the gastric chief cells of the stomach lining and is one of the main digestive enzymes in the digestive system.

Collagen is a protein molecule made up of amino acids. It provides structural support to the extracellular space of connective tissues. Due to its rigidity and resistance to stretching, it is the perfect

matrix for skin, tendons, bones, and ligaments.

Luciferin is an organic substance, present in luminescent organisms such as fireflies, that produces light when oxidised by the action of the enzyme luciferase.

Meconium is a newborn's first poop. This sticky, thick, dark green poop is made up of cells, protein, fats, and intestinal secretions, like bile.

Emulsifiers, in foods, any of numerous chemical additives that encourage the suspension of one liquid in another, as in the mixture of oil and water in margarine, shortening, ice cream, and salad dressing.

Mannose is a sugar monomer of the aldohexose series of carbohydrates. It is a C-2 epimer of glucose. Mannose is important in human metabolism, especially in the glycosylation of certain proteins.

Bile secreted by the hepatic cells passes through the hepatic ducts and is stored and concentrated in a thin muscular sac called the gall bladder.

Serrapeptase (serratiopeptidase) is produced naturally within the digestive tract of silkworms. Silkworms use this enzyme to break down their cocoons at birth.

Reproduction in Plants

Reproduction is the biological process by which organisms produce new individuals of the same species.

Plant reproductive system - Plants are living organisms. It means they need to reproduce in order to pass on their genes to future generations. Plants can create offspring through either sexual or asexual reproduction.

Type of Reproduction in Plant -

Asexual Reproduction - Asexual reproduction involves a single parent and the offspring produced are identical to the parent. It excludes gamete fusion and zygote formation, occurring instead through asexual spores, which can be either flagellate or non-flagellate.

Different types of asexual reproduction -

Vegetative reproduction creates new plants from somatic cells, buds, or plant

organs. A detached part like a root, stem, leaf, or bud grows into a new independent plant. Examples - Begonia, Bryophyllum, etc.

Fragmentation - In filamentous algae, an accidental breaking of the filament into many fragments, each fragment having at least one cell, may give rise to a new filament of the algae by cell division e.g. **Spirogyra**.

Spore formation is a method of asexual reproduction which is found in non-flowering plants such as fungi (Rhizopus) and bacteria. In this method of reproduction, the parent plant produces hundreds of tiny spores which can grow into new plants. For example bread mold, mushroom, hydra etc.

Budding - Budding is the mode of asexual reproduction wherein a new plant is developed from an outgrowth known as the bud. A bud is generally formed due to cell division at one particular site.

Hydra generally reproduces asexually. They do this through budding. A bud begins to form on the tubular body of an adult Hydra. The bud develops a mouth and tentacles. The bud detaches from its parent and the new Hydra is fully developed.

Apomixis in flowering plants is defined as the asexual formation of a seed from the maternal tissues of the ovule, avoiding the processes of meiosis and fertilization, leading to embryo development. It is a unique mechanism of asexual reproduction in certain plants (e.g. **dandelions**).

Asexual reproduction in plants can be done artificially by Cutting, Grafting (the act of joining two plants together), Layering and Micropropagation.

Advantages of Asexual reproduction - It produces two offspring which are the exact copies of the parents. A large number of individuals (plants) can be produced in a very small time.

Sexual reproduction - It involves fusion of male and female reproductive cells (gametes) which are haploid and are produced by male and female reproductive organs. This fusion is known as fertilization.

Features of Sexual Reproduction - Two parents are involved (both male and a female) and Gamete formation, fertilization take place. Offspring are

different from parents, genetically and physically.

Gametogamy is Sexual reproduction involves the formation of male and female reproductive cells which fuse to form a zygote, it is also called **syngamy**.

Ovule : Seed precursor in flower's ovary, consisting of nucellus protected by integuments. Small bead-like structures that develop into seeds when fertilized.

Pollen : Essential for plant sexual reproduction, containing male gametes. palynology is the scientific study of living and fossilised pollen grains.

The yellow particles present in the middle of a flower is called **pollen**, which is present on the anthers pollen grains, and the stick that holds it, is called a **stamen**. Flowers reproduce when bees or other pollinators carry pollen between flowers.

Zygote is the progenitor stem cell that gives rise to all the embryonic and post embryonic tissues and organs. Unlike animals, both maternal and paternal gene products participate in the initial development of zygotes in plants.

After fertilization, zygotes in dicots undergo a period of growth and elongation before the first asymmetric zygote division.

Directional growth movement in Plant-

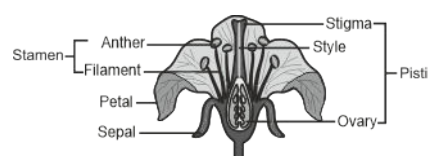
Thigmonasty is a form of directional growth movement by a plant or a fungus in response to touch or vibration.

Geotropism (gravitropism) is directional growth of plants in response to gravity.

Phototropism is when plants bend toward light sources, optimizing their photosynthesis.

Chemotropism- Plants and some bacteria grow towards helpful chemicals or away from harmful ones.

Flower plant Reproduction



The flowers are the reproductive parts of a plant.

The **stamens** are the male reproductive

part and the **pistil** is the female reproductive part.

The flowers which contain either only the pistil or only the stamens are called **unisexual flowers**.

The flowers which contain both stamens and pistils are called **bisexual flowers**.

Flowering plants reproduce sexually through a process of pollination. A plant has a male part called **androecium** and a female part called a **gynoecium**.

Androecium consists of stamens. Each stamen represents the male reproductive organ and it is made up of a stalk or a filament and an anther.

The anther is present at the tip of the filament and each one is bilobed. Every lobe has two chambers which are called pollen sacs. The production of pollen grains takes place in pollen sacs.

Anther - This is a four-lobed sac-like structure responsible for pollen formation.

Filaments - These thread-like structures are attached to the anther, and they keep the anther in place.

Gynoecium - The gynoecium consists of a single pistil (monocarpellary) or has more than one pistil (multicarpellary). When there are more than one, the pistils may be fused together (syncarpous) or may be free (apocarpous).

Pistil is the innermost part of the flower. The pistil has three parts which are stigma, style, and ovary.

Style - It is a tubular structure that connects the ovary and the stigma. It is responsible for the transportation of pollen from the stigma to the ovary and holding the stigma in place.

The ovary is the enlarged basal part on which presents an extended long tube called the **style**. Ovary and stigma are connected through style.

Stigma is present on the tip of the style and it receives the incoming pollen grains. Some of the most recognizable examples of these woody shrubs and trees include pines, spruces, firs, and ginkgoes.

Monocarpellary - The gynoecium consists of a single pistil. For example, peas and beans.

Multicarpellary - Gynoecium comprises

more than one pistil.

Syncarpous - It is the gynoecium with combined pistil. For eg., Tomato, cucumber.

Apocarpous - It is the gynoecium with free pistil. For example Lotus Vinca.

Fertilization takes place in the ovary of the female flower. When a carrying agent like wind/water/animals carry pollen grains from male anthers to the female flower part i.e. stigma, a pollen tube forms and the pollen grain travels down the style and finally reaches the ovary where it fertilizes an ovule.

Parts of a flower

A typical flower bears four whorls born on a thalamus or stalk.

These whorls from outside are Calyx - consisting of **sepals**, Corolla - consisting of **petals**, Androecium - consisting of **stamens** and Gynoecium or pistil - consisting of **carpels**.

Sepals (collectively called the calyx) are modified leaves that encase the developing flower. They are sterile floral parts and may be either green or leaflike or composed of petal-like tissue.

Petals are modified leaves that surround the reproductive parts of flowers. They are often brightly colored or unusually shaped to attract pollinators. All of the petals of a flower are collectively known as the corolla.

Stamen is the male part of the flower. The pistil is a plant's female part. It is generally shaped like a bowling pin and is located in the flower's center. It consists of a stigma, style and ovary.

Carpels is the female reproductive organ of a flower, consisting of an ovary, a stigma, and usually a style. It may occur singly or as one of a group.

Petiole, a slender stalk, connects the leaf blade to the stem, allowing the leaf to twist and position itself to capture sunlight for efficient photosynthesis.

Pollination is transfer of pollen grains from the anther to the stigma of a flower. Pollination is of two types:

Self-pollination is transfer of Pollen grains to stigma of the same or from another flower borne on the same plant as in the pea family examples pea and

gram.

Cross-pollination is transfer of Pollen grains from a flower to stigma of another flower borne on another plant of the same species, for example in palm and maize.

Characteristics in Flowers which favor Cross Pollination -

Wind Pollination (Anemophily) describes the process of the transfer of pollen from one individual plant to another, whereby the pollen is carried by air currents.

Entomophily is a form of pollination whereby pollen is distributed by insects, particularly bees, Lepidoptera (e.g. butterflies and moths), flies and beetles.

Entomophilous species frequently evolve mechanisms to make themselves more appealing to insects, for example brightly colored or scented flowers, nectar, and appealing shapes and patterns.

Propagation in plant

Plant propagation is the process of creating new plants. There are two types of propagation: sexual and asexual.

Sexual propagation is the union of the pollen and egg, drawing from the genes of two parents to create a new, third individual. Sexual propagation involves the floral parts of a plant.

Asexual propagation: Creating new plants from parent parts, genetically identical. Uses stems, roots, or leaves. Methods: cuttings, layering, division, budding/grafting.

Artificial Propagation - When we use the vegetative parts for propagating crops or ornamental plants it is termed as artificial vegetative propagation. Artificially propagated plants vegetatively by using the specialized parts by cutting, grafting and layering.

Vegetative propagation creates new plants from roots, stems, or leaves of a single parent plant without gamete formation or fusion.

Grafting - Asexual propagation method joining plant parts from different plants to grow as one. Used for plants with poor rooting, to utilize superior roots, or maintain clonal production.

Layering - Asexual propagation method where a stem on the parent plant forms roots while in contact with a rooting

medium. Once rooted, the stem is cut to form a new plant.

Important term and fact used to reproduction of plant -

Parthenocarpy - Fruit development without fertilization, resulting in seedless, sterile fruit. Examples include pineapple, banana, cucumber, grape, orange, grapefruit, persimmon, and breadfruit.

Regeneration is a type of asexual reproduction where specialized cells generate many new cells.

Hybridogenesis is an unusual form of reproduction that is found in hybrids between different species.

Hydrophytes - Plants living in water, adapt to surroundings. Fully submerged (e.g., Hydrilla, Valisneria) or mostly underwater (e.g., trapa, lotus). Includes water lilies, sedges, crowfoots.

Endophily and exophily refer to the degree of preference mosquitoes exhibit for indoor or outdoor resting places throughout various stages of their life cycle.

Dithecous is a typical angiosperm anther that is bilobed with each lobe having two theca.

Microsporangia, or pollen sacs, are borne on the lower surfaces of the microsporophylls.

Tapetum is the innermost cell layer in the anther. It nourishes the developing pollen grains.

Sporogenous tissue - When the anther is young, a group of compactly arranged homogenous cells, it occupies the center of each microsporangium.

Pollen grain - Pollen grains are microscopic structures, which bear androecium - a male reproductive organ of a flower.

Exine - The hard outer layer in pollen grain is made up of sporopollenin which is one of the most resistant organic materials known.

Intine - The inner wall of the pollen grain is called the intine. It is a thin and continuous layer made up of cellulose and pectin.

The vegetative cell is bigger and has abundant food reserves and a large irregularly shaped nucleus.

The generative cell is small and floats in the cytoplasm of the vegetative cell. It is spindle shaped with dense cytoplasm and a nucleus.

Pericarp - It is the part of a fruit formed from the wall of the ripened ovary. Examples - peach, cherry, and apple.

Endocarp is the innermost layer of the pericarp which surrounds a seed in a fruit. Examples- Coconut, mango, Walnut, Peach, Apricot, Olive, Date, Pistachio, Blackberry, and Cherry.

Mesocarp is the middle layer of the pericarp of a fruit, between the endocarp and the exocarp. **Examples** - drupes and tomatoes.

Radicle is defined as the embryonic root of the plant, which develops into the future root of the plant.

Cotyledon is an embryonic leaf in seed-bearing plants, one or more of which are the first leaves to appear from a germinating seed.

Primary endosperm cell (PEC) is formed from the central cell after the double fertilization and later it develops into endosperm.

Primary endosperm nucleus (PEN) - The nucleus formed by the fusion of two polar nuclei in the embryo sac of a seed plant prior to fertilization.

Syngamy is the fusion of gametes to initiate the development of a new individual organism.

Polyembryony is the formation of more than one embryo from a single fertilized ovum or in a single seed.

Scutellum - Embryos of monocotyledons possess only one cotyledon. In the grass family the cotyledon is called scutellum that is situated towards one side (lateral) of the embryonal axis.

The coleorhiza, or root sheath, is a protective tissue layer surrounding the primary root (radicle) in monocot seeds, and it is also present in adventitious roots for protection.

Coleoptile - The portion of the embryonal axis above the level of attachment of scutellum is the epicotyl. Epicotyl has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure.

Non-albuminous or ex-albuminous seeds are those which do not have

endosperm and they store their food in cotyledons and kernels.

Perisperm - In some seeds such as black pepper and beet, remnants of nucellus are also persistent. This residual, persistent nucellus is called perisperm.

Achene is a one-seeded, dry, indehiscent fruit with seed attached to the pericarp at one point only. Such as the unit fruits of sunflowers.

The endosperm is a tissue produced inside the seeds of most of the flowering plants following double fertilization.

Globular embryo - The embryo proper is initially in the globular stage, which is a spherical mass of cells seen growing inside the oval-shaped ovule.

Post fertilization is the process of development of endosperm and embryo followed by the formation of fruit after fertilization.

Synergid cells are two specialized cells that lie adjacent to the egg cell in the female gametophyte of angiosperms and play an essential role in pollen tube guidance and function.

Suspensor is a terminally differentiated embryonic region that connects the embryo to surrounding tissues during early seed development.

Bagging - Covering emasculated flowers with a suitable-sized bag, typically made of butter paper, to prevent stigma contamination with unwanted pollen.

Emasculation is a biological term given for the procedure of removing anthers from the flower bud before the anther dehiscence using a pair of forceps is necessary.

Artificial hybridisation is one of the major approaches of crop improvement programmes.

Filiform apparatus guides the entry of the pollen tube towards the egg. They guide the entry of the pollen tube into the embryo sac.

Chasmogamous flowers which are similar to flowers of other species with exposed anthers and stigma.

Cleistogamous flowers which do not open at all. In such flowers, the anthers and stigma lie close to each other.

Reproduction in Animals

Reproduction is defined as a biological process in which an organism gives rise to young ones (offspring) similar to itself.

The main reproductive hormones estrogen, testosterone, and progesterone are instrumental in sexuality and fertility.

They are responsible for pregnancy, puberty, menstruation, menopause, sex drive, sperm production etc. These hormones are produced in the ovaries (in females) and testes (in males).

Brood parasitism is a reproductive strategy in which a parasite lays its eggs in the nest of another species of the same species or a different species and the host parasite raises the young.

Modes of Reproduction - Depending on the number of parents involved, there are different modes of reproduction. In animals, there are two types of reproduction:

1. **Asexual Reproduction.**
2. **Sexual Reproduction.**

Asexual Reproduction in Animals

Asexual Reproduction is a mode of reproduction in which the new offspring are produced from a single parent. The different kind of asexual reproduction are:

Budding : It is the formation of an outgrowth (bud) from an organism. The bud is capable of developing into a new individual. The "**offspring**" is a clone of the parent. Rather than sex cells, somatic cells are involved. They divide mitotically and give rise to new sets of cells with the same genetic composition.

Examples - Hydra, Bacteria, yeast, corals, flatworms, Jellyfish and sea anemones.

Fragmentation : It refers to the process by which a living organism splits into parts that later grow identical to the original organism.

Examples - liverworts, mosses, cyanobacteria, fungi, many plants, and also in animals including, sponges, some annelid worms and sea stars.

Binary Fission : It is a form of asexual reproduction in which an organism divides themselves into two, each part carrying one copy of genetic material. Example - Amoeba, Paramecium etc.

Vegetative Propagation : It is an asexual method of plant reproduction that occurs in its leaves, roots and stem. Examples - the rhizome of ginger, the tuber of the potato, guava tree, etc.

Sporogenesis : It is the process of reproduction via spores. It comprises three processes: meiosis, cytokinesis and centrifugal deposition of wall layers. Examples - Conidial fungi *Aspergillus* and *Penicillium*.

Parthenogenesis : It is the reproduction in which the embryo grows and develops without fertilization from unfertilised eggs. **Examples** - lizards, and birds.

Apomixis : It is a form of asexual reproduction that occurs via seeds, in which embryos develop without fertilisation. **Examples** - Blackberries, meadow grasses, mat grass, hawkweeds, hawthorns, shadbush, Sorbus, brambles etc.

Cloning : It is the production of an exact copy of a cell, any other living part, or a complete organism. It is a man-made method to produce organisms without natural reproduction. For instance, Dolly the Sheep was cloned by Ian Wilmut and team in Scotland.

The three different types of cloning are:

Gene cloning - Which creates copies of genes or segments of DNA.

Reproductive cloning - Which creates copies of whole animals.

Therapeutic cloning - Which creates embryonic stem cells.

Sexual reproduction

Sexual reproduction occurs when male and female gametes fuse. Animals' reproductive parts produce gametes that fuse to form a zygote, which grows into a new individual.

Zygote is a cell made when two gametes fuse during fertilization. It has DNA from both gametes. In multicellular organisms, the zygote is the earliest developmental stage.

Female Reproductive system

The female reproductive system consists of a pair of ovaries along with a pair of oviducts, uterus, cervix, vagina and the external genitalia located in the pelvic region.

Ovaries are two oval-shaped organs that lie to the upper right and left of the uterus. The ovary is the primary female reproductive organ in which the ova or eggs are produced.

Uterus is shaped like an upside-down pear, with a thick lining and muscular walls. The uterus muscles are able to expand and contract to accommodate a growing fetus.

Vagina is a closed muscular canal that extends from the outside of the female genital area (vulva) to the neck of the uterus (cervix).

Fallopian tubes are the tubes connecting the ovaries to the uterus. There are two fallopian tubes, each attached to a side of the uterus. Fertilisation usually takes place in a fallopian tube.

Ovum is a mature female reproductive cell which can divide to give rise to an embryo usually only after fertilisation by a male cell. The human egg, or ovum, is one of the largest cells in the human body.

Some other parts in the female reproductive system

The female external genitalia consists of the accessory reproductive organ structures that are called the vulva. The female external genitalia includes the clitoris, labia majora, labia minora, and the Bartholin's glands.

Clitoris is located at the point where the labia majora meet near the pubic bone.

Cervix is the lower, narrow end of the uterus that forms a canal between the uterus and vagina.

Oviduct or fallopian tube is used to collect an ovum from the ovaries and transfer it to the uterus. The eggs can be fertilised by sperm to become zygotes in the oviduct.

Hymen is a thin piece of mucosal tissue that surrounds or partially covers the external vaginal opening.

Bartholin's glands produce a mucoid secretion that aids in vaginal and vulvar lubrication.

The outer folds are called the labia majora. The inner folds are called the labia minora. These skin folds protect the opening of the urethra and the vagina.

Umbilical cord: Bundle of blood vessels connecting fetus to placenta. Enclosed in amnion sheath. Umbilical arteries carry deoxygenated blood from fetus to placenta.

There are two arteries in an umbilical cord. It normally contains two arteries (the umbilical arteries) and one vein (the umbilical vein).

Menstrual Cycle : The reproductive cycle in the female primates is called the menstrual cycle. The first menstruation begins at puberty and is called menarche. Menstruation is repeated at an average interval of about 28/29 days, and the cycle of events starts from one menstruation till the next one.

Male Reproductive System

The male reproductive system is located in the pelvis region. It includes a pair of testes along with accessory ducts, glands and the external genitalia.

Penis is the male organ for sexual intercourse. It has three parts: The root, The body or shaft, The glans.

Sperm is the male reproductive cell or gamete. The term "gamete" implies that the cell is half of a whole. Sperm may be tiny cells, but they are 50% responsible for reproduction.

Scrotum is the loose pouch-like sac of skin that hangs behind the penis. It holds the testicles (also called testes), as well as many nerves and blood vessels.

Epididymis is a long, coiled tube that rests on the backside of each testicle. It carries and stores sperm cells that are created in the testes.

Urethra is the tube that carries urine from the bladder to the outside of the body. In males, it has the additional function of expelling (ejaculating) semen.

Vas deferens is a long, muscular tube that travels from the epididymis into the pelvic cavity, to just behind the bladder.

Ejaculatory ducts are formed by the fusion of the vas deferens and seminal vesicles. It empties into the urethra.

Seminal vesicles are sac-like pouches that attach to the vas deferens near the base of the bladder. The seminal vesicles make a sugar-rich fluid (fructose) that provides sperm with a source of energy and helps with the sperms' ability to

move (motility).

Prostate gland is a walnut-sized structure that is located below the urinary bladder in front of the rectum. The prostate gland contributes additional fluid to the ejaculate.

Bulbourethral glands, or **Cowper's glands**, are pea-sized structures located on the sides of the urethra, just below the prostate gland.

Functions of Hormones

The hormones that controlled the Human reproductive system are -

Progesterone hormones are chemical messengers that tell the body how to work. They aid in menstruation and help to maintain the early stages of pregnancy.

Estrogen hormone is a sex hormone that is necessary for maintaining sexual and reproductive health. Estrogen levels naturally fluctuate during the menstrual cycle and decline during menopause.

Testosterone hormone is the primary male hormone responsible for regulating sex differentiation, producing male sex characteristics, spermatogenesis, and fertility. Testosterone's effects are first seen in the fetus. During the first 6 weeks of development, the reproductive tissues of males and females are identical.

Relaxin: It is a reproductive hormone produced by ovaries and the placenta. It loosens muscles, joints and ligaments during pregnancy to help in body stretch.

Development of Embryo.

Fertilisation is the fusion of a sperm with an egg. It results in the formation of zygote which begins to develop into an embryo. The types of fertilisation are Internal (takes place inside the female body) and External (takes place outside the female body).

Embryo is the stage when cells produced by the division of the zygote begin to form groups that develop into different tissues and organs of the body.

Embryo Development - When the sperm successfully fertilises the egg, an embryo forms and begins to grow. When the embryo is 2 to 3 days old it is at the 'cleavage stage'; at 5 to 6 days old, it is called a '**blastocyst**'.

The stage of embryo in which all the body parts can be identified is called **Fetus**.

Oviparous and Viviparous animals -

Oviparous animals produce eggs, and the young hatch after being expelled from the body. Examples - Frogs, snakes, lizards, hens, duck, fishes, shark, penguins, butterflies, and octopus.

Viviparous animals give birth directly to their kind. Examples - Humans, bears, giraffes and cattle.

Metamorphosis is the process of transformation of the eggs into an adult through drastic changes. All butterflies have "complete metamorphosis." To grow into an adult they go through 4 stages: egg, larva, pupa and adult.

In Vitro Fertilisation and Embryo Transfer (IVF-ET) is a two-part assisted reproductive technology. In IVF, a human egg is fertilised with sperm in a laboratory then implanted into a uterus. If the fertilised egg (embryo) successfully implants in the uterus, this will result in pregnancy.

The IVF Cycle consists of Egg Retrieval (A minor surgical procedure in which a doctor obtains eggs from the follicles of the ovaries), Fertilisation, Embryo Transfer (embryos are placed into uterus)

Important term in Reproduction system in Animal

Donor Insemination: It is a process in which the sperm is placed directly inside the vagina at the cervix or inside the uterus (called Intrauterine Insemination or IUI). Donor insemination was previously called Artificial Insemination.

Totipotency is the ability of a single cell to produce a fertile adult individual.

Pluripotency describes the ability of a cell to develop into the three primary germ cell layers of the early embryo and therefore into all cells of the adult body, but not extra-embryonic tissues such as the placenta.

Mutation is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied or as the result of environmental factors such as UV light and cigarette smoke. Mutations can occur during DNA replication if errors are made and not corrected in time.

Seminiferous tubules are located within the testes, and are the specific location of meiosis, and the subsequent creation of male gametes, namely spermatozoa.

Sertoli cells are the epithelial supporting cells of the seminiferous tubules. They are derived from the epithelial sex cords of the developing gonads.

Leydig cells, which are found in interstitial cells, are responsible for the production of male sex hormones.

Spermatogonial stem cells (SSCs) are the germ stem cells of the seminiferous epithelium in the testis. Through the process of spermatogenesis, they produce sperm.

Prepuce is a movable sheath of skin that covers the head of the penis.

Mons pubis is a cushion of fatty tissue covered by skin and pubic hair.

Semen is the thick, white fluid released during ejaculation that contains sperm.

Oogenesis is the process by which the female gametes, or ova, are created. The female gamete is called an ovum.

Antrum, the lowermost part of the stomach, is funnel-shaped, with its wide end joining the lower part of the body and its narrow end connecting with the pyloric canal.

Graafian follicle is a fluid-filled, spherical vesicle in the ovary of a mammal, containing an oocyte attached to its wall, and responsible for oestrogen production in the ovary.

An **oocyte** or ovocyte is a female gametocyte or germ cell involved in reproduction.

Zona pellucida glycoproteins are responsible for species-restricted binding of sperm to unfertilized eggs, inducing sperm to undergo acrosomal exocytosis, and preventing sperm from binding to fertilised eggs.

Ovulation is a phase of the female menstrual cycle that involves the release of an egg (ovum) from one of the ovaries.

Corpus luteum is a vital yet temporary organ that plays a crucial role in fertility during the luteal phase.

Cleavage is a period after fertilisation,

when a 1-cell embryo starts developing into a multicellular organism.

Blastomere is a type of cell produced by cell division (cleavage) of the zygote after fertilisation; blastomeres are an essential part of blastula formation, and blastocyst formation in mammals.

Trophoblasts are cells that form the outer layer of a blastocyst, which provides nutrients to the embryo, and then develop into a large part of the placenta.

Implantation is when a fertilised egg cell attaches to the wall of the uterus in early pregnancy.

Placenta is an organ that develops in the uterus during pregnancy. This structure provides oxygen and nutrients to a growing baby.

Parturition (childbirth) is the process of delivery of the foetus.

Endoderm is found in both vertebrate and invertebrate embryos, and is responsible for the formation of the gut and associated organs.

Ectoderm is the outermost layer of cells or tissue of an embryo in early development, or the parts derived from this, which include the epidermis, nerve tissue, and nephridia.

Ampulla : The ampulla of vas deferens, also called the ampulla of ductus deferens, is an enlargement of the vas deferens at the fundus of the bladder which acts as a reservoir for sperm.

Gamete is a reproductive cell of an animal or plant. In animals, female gametes are called ova or egg cells, and male gametes are called sperm. Ova and sperm are haploid cells, with each cell carrying only one copy of each chromosome.

Follicle is the functional unit of the ovary, which is composed of three types of cells: oocytes, granulosa cells, and theca cells.

Granulosa and theca cells are a site of action for the gonadotropins and a site for production of steroid hormones.

Vestibule encompasses the vaginal orifice, the external urethral orifice, and in some females, the hymen.

Vaccines

Anthrax Vaccine - The anthrax vaccine was discovered by Louis Pasteur in 1881. Louis Pasteur was an eminent French chemist who also coined the term 'vaccine'. Anthrax is an infection caused by the bacterium *Bacillus anthracis*.

Bacille Calmette Guerin (BCG TB vaccine)- It serves as a vaccine against the disease and promotes the development of tuberculosis (TB) resistance. It is made from a weakened form of a bacterium called *Mycobacterium bovis* (*Bacillus Calmette-Guérin*), which is similar to the bacteria that cause tuberculosis (TB). Leon Charles Albert Calmette and Camille Guerin created the vaccine.

Cholera Vaccine - Jaime Ferran was the original developer of the preparation of the cholera vaccine. In 1885, a vaccine composed of live attenuated bacteria was first inoculated in humans during that year's epidemic. Cholera is an acute diarrheal illness caused by infection of the intestine with *Vibrio cholerae* bacteria.

DTP (Diphtheria, Tetanus, Pertussis) - Diphtheria infects the throat and tonsils, making it hard for children to breathe and swallow. Tetanus causes very painful muscle contractions. It can cause children's neck and jaw muscles to lock (lockjaw), making it hard for them to open their mouth, swallow (breastfeed) or breathe. Pertussis (whooping cough) causes coughing spells that can last for weeks.

Hep B - This vaccine is given to prevent Hepatitis B, a serious liver infection that can remain silent for decades, especially when contracted in infancy.

Hib - This vaccine is given to prevent *Haemophilus influenza* type b (Hib), it is a bacterium that causes pneumonia, meningitis and other severe infections almost exclusively in children under 5 years old.

HPV Vaccine - HPV (Human papillomavirus) usually has no symptoms, but some strains can cause cervical cancer - the fourth most common cancer in women.

MMR (Measles, Mumps, Rubella)- Measles is a highly contagious disease with symptoms that include fever, runny nose, white spots in the back of the

mouth and a rash. Mumps can cause headache, malaise, fever, and swollen salivary glands. Rubella infection in children and adults is usually mild, but in pregnant women it can cause miscarriage, stillbirth, infant death or birth defects.

Pneumococcal - Pneumococcal diseases range from serious diseases such as meningitis and pneumonia to milder but more common infections like sinusitis and ear infections.

Polio Vaccine - The first effective polio vaccine was developed by Jonas Salk in 1952. Polio, or poliomyelitis, is a disabling and life-threatening disease caused by the poliovirus.

Rotavirus Vaccine - Rotaviruses cause severe diarrhoea and vomiting, which can lead to dehydration, electrolyte imbalance and shock in young children.

Smallpox Vaccine - The first vaccine was introduced by a British physician, Edward Jenner, in 1796. He used the cowpox virus (*vaccinia*) to confer protection against smallpox, in humans. Smallpox is caused by the variola virus.

There are vaccines to prevent hepatitis A and hepatitis B; however, there is no vaccine for hepatitis C.

Scientific studies

Agrostology - Scientific study of the grasses.

Agrology - The branch of agriculture that deals with the origin, structure, analysis, and classification of soils especially in their relation to crop production.

Agronomy - The branch of agriculture that deals with field crop production and soil management.

Allometry - study of the relationship of body size to shape, anatomy, physiology and finally behaviour.

Anthology - Study of flowers and flowering plants.

Anatomy - study of the structure inside the body as well as outside the body.

Aphidology - study of Aphids (Plant Lice)

Archaeobiology - Study of the biology of ancient times through archaeological materials.

Apiculture - The technique of scientific rearing of honey bees in a specially designed wooden box. It is the study of bees.

Anthropology - Study of the origin and development of human societies and cultures.

Agriology - Study of customs of primitive man.

Arthology - Study of joints.

Aerobiology - The study of flying organisms in relation with other flying objects.

Agraphia - Loss of the ability to write.

Aphasia - Loss of the ability to speak and understand.

Bryology - The study of ancient plants, like mosses, that grow in moist, humid environments.

Botany (phytology) - The branch of Biology that deals with the study of plants.

Biocybernetics - Application of Cybernetics to Biological Sciences.

Bacteriology - Study of bacteria.

Biophysics - Study of physical aspects of living systems.

Biochemistry - Study of chemicals constituting the body and chemical reactions.

Biotechnology - Utilisation of living organisms in Industrial Processes

Biogeography - Study of the geographic distributions of living organisms.

Chronobiology - Study of time-dependent phenomena in living organisms.

Ctetology - A branch of biology that deals with the origin and development of acquired characters.

Craniology - The study of shape and size of the skulls of different human races.

Cryobiology - The branch of biology which deals with the properties of organisms and tissues at low temperatures.

Chiropody (Podiatry) - It is a medical specialty focused on diagnosing and treating foot diseases and disorders.

Carpology - The study of structure of seeds and fruit.

Cytology - Study of the detailed structure of cells.

Cynology - Study of canines or domestic dogs.

Cardiology - Study of the heart and its functioning.

Criminalistics - Scientific tests or techniques used in connection with the detection of crime, forensics.

Dendrology - Study of shrubs and trees.

Eidonomy - Study of the external appearance of an organism.

Endocrinology - Study of ductless glands and their hormones.

Ethology - Study of behaviour of animals

Ethnobotany - scientific study of the traditional knowledge and customs of a people concerning plants and their medical, religious, and other uses.

Etiology - Study of the causative agent of disease.

Entomology - The study of insects and their relationship to humans, the environment, and other organisms.

Eugenics - Study refers to the use of procedures like selective breeding and forced sterilisation is an attempt to improve the genetic purity of the human race.

Ethnology - The study of the characteristics of different peoples and the differences and relationships between them.

Euthenics - Science deals with the improvement of the future of humanity by changing the environment.

Euphenics - Treatment of defective heredity through genetic engineering.

Enzymology - Study of enzymes.

Embryology - The study of development of embryo and its growth and repair.

Exobiology - Study of the possibility of life in space.

Esthesiology - Scientific study of sensations and sense organs.

Floriculture - Cultivation of flower yielding plants.

Forensic Biology - Application of Biology to law enforcement.

Gerontology - Study of the social, psychological, cognitive and biological aspects of aging.

Gynecology - Study of the female reproductive organ.

Gastroenterology - Study of alimentary canal or stomach, intestine and their disease.

Glycobiology - Study of the structure, biosynthesis and biology of glycans (carbohydrates).

Histology - The branch of biology which studies the microscopic anatomy of biological tissues.

Haematology - Study of blood and disease caused by it.

Hepatology - Study of liver.

Helminthology - Study of parasitic worms.

Hydroponics - Study of growing plants without soil in water which contain nutrients.

Herpetology - Study of reptiles and amphibians.

Hydrobiology - Science of life and life processes in water.

Immunology - Study of natural or acquired resistance to specific diseases.

Ichthyology - Study of fish and its culture.

Karyology - Study of the nucleus.

Katology - Study of Human Beauty.

Kinesiology - Study of muscle movements.

Koniology - A science that deals with atmospheric dust and its effects on plant and animal life.

Limnology - Study of inland aquatic ecosystems.

Lichenology - The study of the structure, physiology, and ecology of lichens.

Lipidology - Study of lipids (fatty compounds that perform a variety of functions in your body).

Mastology - Study of breasts.

Morphology - Study of size, shape, and structure of animals, plants and microorganisms.

Myology - The study of the structure, arrangement, and action of muscles.

Mycology - Study of fungi.

Microbiology - The study of the biology of microscopic organisms - viruses, bacteria, algae, fungi, slime molds, and protozoa.

Mammalogy - Study of mammals.

Malacology - Study of molluscs and their shells.

Medicine - Study of treating disease by drug.

Mammography - Branch of science which examines the breast for the early detection of cancer and other breast diseases.

Mechanobiology - Study of interface of Biology and Engineering.

Monoecious - Plants which have both male and female flowers.

Nephrology - The branch of medicine which deals with the kidneys.

Neonatology - Study of newborn up to the age of 2 months.

Neurology - Study of neurons and nerve rings.

Nosology - Classification of diseases.

Osteology - The study of the structure and function of the skeleton and bony structures.

Odontology - Study of teeth.

Organology - Study of different organs.

Obstetrics - It is the field of study concentrated on pregnancy, childbirth, and the postpartum period.

Oneirolgy - The study of dreams.

Ontogeny - Study of various life-cycles.

Ornithology - Study of birds.

Ophthalmology - Study of eyes.

Ophiology - Study of snakes.

Olericulture - The branch of horticulture which deals with the production, storage, processing and marketing of vegetables.

Orthopaedic - Relating to the branch of medicine dealing with the correction of deformities of bones or muscles.

Oncology - Study of tumours.

Oecologia (coined by Ernst Haeckel) - Study of relationships of organisms with their environment.

Physiology - Study of process and function associated with life.

Pathology - Study of diseases, effects, causable agents and transmission of pathogens.

Phylogeny - Study of evolutionary relationships among different groups of organisms.

Palaeontology - Study of prehistoric species, mostly ones that are extinct.

Palynology - The study of plant pollen, spores, and certain microscopic plankton organisms in both living and fossil forms.

Parasitology - Study of parasites (Organisms which depend on other living organisms for their food and shelter).

Pulmonology - Study of lungs

Primatology - Study of primates.

Protozoology - Study of protozoa.

Psychiatry - Treatment of mental disease.

Pisciculture - The culturing of fish (fish farming).

Pedology - Study of different types of soils.

Phenomics - Study of the physical and biochemical traits of organisms.

Pediatrics - Branch of medicine dealing with children.

Phenology - It refers to the timings of cyclical or seasonal biological events, such as migrations, egg laying, flowering, and hibernation.

Pomology - Study of fruits.

Phycology - Study of algae.

Psychobiology - Study of behavioural aspects of animals.

Phytogeography - Study of plant distribution on the earth.

Pharmacology - Scientific study of drugs.

Photobiology - Study of interaction of light and living organisms.

Protistology - Scientific study of unicellular eukaryotes and their relatives - single cells as living organisms.

Presbycusis - Loss of the ability to hear.

Psychology - Study of human mind and behaviour.

Paleontology is the study of the history of life on Earth as based on fossils.

Paleobotany is the scientific study of ancient plants, using plant fossils found in sedimentary rocks.

Radiology - Study of effects of radiation on the living being.

Rhinology - Study of nose and olfactory organs.

Scotobiology - Study of Biological effects of darkness.

Sonography - Study of ultrasound imaging.

Sociobiology - Study of the biological bases of sociology.

Syndesmology - Study of bone joints and ligaments.

Saurology - Study of lizards.

Serology - Study of blood serum.

Sphygmology - Study of pulse and arterial pressure.

Synantherology - A branch of botany that deals with the study of the plant family Asteraceae (also called Compositae).

Silviculture - Study of cultivation of timber yielding plants.

Sericulture - Study of the production of silk and the rearing of silkworms.

Splanchnology - Study of visceral organs.

Tectology - Study of structural organisation of the body.

Trophology - Study of nutrition science.

Traumatology - Study of wounds.

Trichology - study of the hair and scalp.

Toxicology - Study of poisons and methods of treatment.

Taxonomy - Study of classification, nomenclature and identification of the organism.

Taxonomic - The study is related to a group of individual organisms with fundamental similarities as a species is called taxonomic.

Teratology - The scientific study of congenital abnormalities and abnormal formations.

Therapeutics - Treatment of disease.

Urology - Study of urine and its diseases.

Virology - Study of viruses.

Venereology - Study and treatment of venereal diseases.

Veterinary Science - Science of health care and treatment of domestic animals.

Xylotomy - Study of anatomy of wood/xylem.

Zoology - The scientific study of animals.

Zoogeny - The doctrine of the formation of living beings.

Zoogeography - It is the study of the distribution of animals in different geographic regions.

Zymology - Study of the fermentation process.

Zoonosology - study of animal disease.

Zoopathology - A branch of pathology dealing with the diseases of the lower animals.

Scientific names and Botanical names

Animals

Ant - Formicidae

Cow - Bos taurus

Wild yak - Bos mutus

Horse - Equus caballus

Wolf - Canis lupus

Hen - Gallus gallus domesticus

African wild cat - Felis lybica

Goat - Capra aegagrus hircus

Gaur - Bos gaurus

Bear - Ursidae

Blackbuck - Antelope cervicapra
 Cheetah - Acinonyx jubatus
 Cobra - Naja
 King Cobra - Ophiophagus hannah

Dog - Canis lupus familiaris

Buffalo - Bubalus bubalis

Cat - Felis catus

Deer - Cervidae

Indian Elephant - Elephas maximus indicus

African elephants - Loxodonta

Fish - Vertebrata

Red fox - Vulpes vulpes

Leopard or panther - Panthera pardus

Monkey - Cercopithecidae

Lion - Panthera leo

Pig - Sus

Rabbit - Oryctolagus cuniculus

Rat - Rattus

Tiger - Panthera tigris

Grasshopper - Caelifera

Honey Bee - Apis

Snake - Serpentes

Blind Snake - Typhlops

The lion tailed macaque or Beard Ape (Baboon) - Macaca silenus

Humpback whale - Megaptera novaeangliae

Chimpanzee - Pan troglodytes

Zebra - Equus quagga

Giraffe - Giraffa camelopardalis

Birds

Peacock - Pavo cristatus

Alexandrine parakeet or Alexandrine

Parrot - Psittaciformes

Koel - Eudynamis scolopaceus

Sparrow - Passer Domesticus or Passeridae

Pigeon - Columba livia

Lark - Alaudidae

Owl - Strigiformes

Crow - Corvus

Myna - Acridotheres tristis

Bulbul - Pycnonotidae

Indian Vulture - Gyps indicus

Indian Cuckoo - Cuculus micropterus

Dodo - Raphidae columbiformes

Duck - Anatidae anseriformes

Eagle - Aquila accipitridae

Hummingbird - Trochilidae apodiformes

Ostrich - Struthio camelus

Penguin - Spheniscidae
 Swan - Cygnus coscoroba

Fruits

Apple - Malus domestica

Papaya - Carica papaya

Coconut - Cocos nucifera (Botanical name)

False Ashoka - Monoon longifolium, Polyalthia longifolia

Orange - Citrus sinensis

Watermelon - Citrullus lanatus

Banana - Musa paradisiacum

Chiku - Achras sapota (Manilkara zapota)

Guava - Psidium guajava

Lemon - Citrus Limonium

Mulberry - Morus

Pear - Pyrus

Pineapple - Ananus sativus

Pomegranate - Punica granatum

Strawberry - Fragaria ananassa

Sweet Potato - Ipomoea batatas

Mango - Mangifera indica (Scientific name) is the national fruit of India, Pakistan and the Philippines. It is the national tree of Bangladesh.

Vegetables

Tomato - Solanum Lycopersicum

Potato - Solanum tuberosum

Onion - Allium cepa

Garlic - Allium sativum

Pumpkin - Cucurbita pepo

Brinjal - Solanum melongena

Radish - Raphanus sativus

Carrot - Daucus carota

Capsicum - Capsicum annuum

Cucumber - Cucumis sativus

Ginger - Zingiber officinale

Turmeric - Curcuma longa

Clove - Syzygium aromaticum

Black Pepper - Piper nigrum

Coriander - Coriandrum sativum

Pea - Pisum sativum

Broccoli - Brassica oleracea italica group

Broccoli rabe - Brassica ruvo

Calabash - Lagenaria siceraria

Cauliflower - Brassica oleracea

Chives - Allium schoenoprasum

Cluster bean - Cyamopsis tetragonoloba

Luffa - Luffa aegyptiaca

Okra or Lady Finger - Abelmoschus esculentus

Rutabaga - Brassica napus

Soybean - Glycine max

Spinach - Spinacia oleracea

Trees

Margosa tree or Neem tree - Azadirachta indica (Botanical name)

Pine tree - Pinus roxburghii

Jackfruit - Artocarpus integra

Teak - Tectona grandis Linn

Silver Oak - Grevillea robusta

Sausage tree - Kigelia africana

Flowers

Daffodil - Narcissus (Botanical name)

Rose - Rosa

Lily - Lilium

Tulip - Tulipa

Orchid - Orchidaceae

Carnation - Dianthus

Hyacinth - Hyacinthus

Peruvian Lily - Alstroemeria

Sunflower - Helianthus

Barberson - Daisy gerbera

Plants

Bamboo - Bambusoideae

Banyan - Ficus benghalensis

Black Gram - Vigna mungo

Black Pepper - Piper nigrum

Clove - Syzygium aromaticum

Cotton - Gossypium herbaceum

Curry plant - Murraya koenigii

Drumstick - Moringa oleifera

Green Gram - Vigna radiata

Henna - Lawsonia inermis

Horse Gram - Macrotyloma uniflorum

Jowar - Sorghum vulgare

Maize - Zea mays

Mint - Mentha arvensis

Money Plant - Epipremnum aureum

Red Gram - Cajanus cajan

Sandalwood - Santalum album

Tobacco - Nicotina tabacum

Tulsi - Ocimum sanctum

Wheat - Triticum aestivum

Carrot grass - Parthenium Hysterophorus

Lemon grass - Cymbopogon

Lawn Grass - Zoysia

Horsetails - Equisetum
 Brown algae - Phaeophyceae
 Green algae - Chlorophyta
 Red algae - Rhodophyta
 Comb jellies - Ctenophora
 Flatworms - Platyhelminthes
 Bumblebee - Bombus
 Tree frog - Hylidae

Medical/Scientific Terms

Intravenous - Indicates medication or fluid that's delivered by vein.

Sutures - Stitches, which are used to join tissues together as they heal.

Radio therapy - It is a cancer treatment that uses high doses of radiation to kill cancer cells and shrink tumors.

Chronic - Something that persists or continues for at least six months.

Laparotomy - Any surgery involving an incision in the abdominal wall.

Thoracotomy - Surgery on the chest cavity.

Intubation - Medical insertion of a tube into the body, for example, into the throat to assist with breathing.

Hysterectomy - Surgical procedure to remove the uterus.

Appendectomy - Surgical procedure to remove the appendix.

Detoxification - The removal of toxins from the body, such as drugs or alcohol.

Neonatal period - The time from a baby's birth to 4 weeks of age.

Magnetic resonance imaging (MRI) - A type of scan that uses strong magnetic fields and radio waves to produce detailed images of the inside of the body.

Venipuncture - The drawing of blood from a vein.

Thrombosis - A blood clot within a blood vessel that affects normal blood flow.

Cesarean birth - An operation in which a baby is born through a cut made in the wall of the abdomen and the uterus.

Chemotherapy - It is a drug treatment that uses powerful chemicals to kill fast-growing cancer cells in our body.

Surgery - A procedure to remove or repair a part of the body or to find out whether disease is present. It is also called an operation.

Gastric Bypass and other weight loss surgeries are collectively known as

Bariatric surgery. It involves making changes to our digestive system to help lose weight.

Alcoholic fermentation - Alcoholic beverages like beer, wine, brandy, whisky, etc are made using fermentation of grains, grapes, rice, etc by yeast.

Pasteurization (heat-treatment process) destroys pathogenic microorganisms in certain foods and beverages.

Melanin - It is a skin pigment. It occurs in both humans and animals.

Microtomy is a method for the preparation of thin sections for materials such as bones, minerals, and teeth.

Physiotherapy - Treatment of body defects through massage and exercise.

Sentinel surveillance - It is a type of surveillance specifically aimed at identifying missing cases of disease.

Agonist - It is a type of drug that mimics the natural messenger by switching on the receptor.

Antagonists are the drugs that bind to the receptor site and inhibit its natural function.

Medical Instruments

Glucometer is generally used as a monitor by a Diabetic patient.

Defibrillator - A medical device that uses electric shocks to restore normal heartbeat.

Microscope - It is mainly used to magnify small objects.

Dialyser - It is used to remove toxic materials from the blood that are generally removed by the kidneys.

Endoscope - It is used to look inside the gastrointestinal tract.

Nebulizer - It is used to produce aerosols of drugs to be administered by respiratory route.

Otoscope - It is used to look into the external ear cavity.

Stethoscope - It is used to hear sounds from movements within the body like heart beats, intestinal movement, breath sounds, etc.

Electrocardiograph machine - It is used to record the electrical activity of the heart over a period of time.

Ophthalmoscope - It is used to look at the retina.

Laxmeter - It is used for measuring knee joint laxity.

Sphygmometer - It measures Blood Pressure by balancing air pressure against the pressure of blood in the artery.

Holter Monitor - A device that measures the electrical activity in the patient's heart for a period of 24 to 48 hours.

Medical Specialists

Dermatologist - A doctor who specializes in conditions that affect the skin, hair, and nails.

Pulmonologist - A physician who specializes in the respiratory system.

Hematologists - Doctors or pediatricians who have extra training in disorders related to your blood, bone marrow, and lymphatic system.

Anesthesiologist - They specialize in anesthesia care, pain management, and critical care medicine.

Ophthalmologist - A specialist in the branch of medicine concerned with the study and treatment of disorders and diseases of the eye.

Gynecologist - A doctor who specializes in female reproductive health.

Andrologist - The physicians who specialize in treating men's reproductive-related issues.

Dentist - A dental surgeon, is a healthcare professional who specializes in dentistry.

Orthopedic surgeon - A doctor who specializes in the musculoskeletal system (the bones, joints, ligaments, tendons, and muscles).

Neurologist - A doctor who specializes in neurology. It treats disorders that affect the brain, spinal cord, and nerves.

Cardiologist - A physician who is expert in the care of heart and blood vessels.

Pediatrician - A doctor who focuses on the health of infants, children, adolescents and young adults.

Psychiatrist - A doctor who specializes in mental health, including substance use disorders.

Endocrinologists - A doctor who has specifically trained in the tools and techniques needed to diagnose and treat endocrine disorders.

Oncologist - A doctor who treats cancer and provides medical care for a person diagnosed with cancer.

Miscellaneous

Tanner-whitehouse, greulich-pyle, FELS are an age verification test method that makes an assessment based on hand and wrist radiographs.

Brain fingerprinting is also known as Brain Electrical Oscillation Signature Profiling (BEOSP).

EEG - The electroencephalogram (EEG) is a medical test used to measure the electrical activity of the brain.

Scientists and their famous works

Alfred D Hershey was awarded the 1969 Nobel Prize in Physiology or Medicine jointly with Max Delbruck and Salvador E. Luria for their discoveries concerning the replication mechanism and the genetic structure of viruses.

Alexander Fleming discovered Penicillins, a group of antibiotics derived from *Penicillium* fungi in 1928. Penicillin V, a type of penicillin antibiotic, treats various bacterial infections like ear, throat, and skin infections, as well as preventing rheumatic fever.

Penicillins are effective against throat infections, meningitis, syphilis, and other bacterial infections. Fleming, Chain, and Florey shared the 1945 Nobel Prize in Physiology or Medicine for their pioneering work on penicillin.

Allan Octavian Hume was an Ornithologist too and is fondly remembered as the father of Indian ornithology.

Albert Sabin developed the Oral Polio Vaccine (OPV) first used in 1961.

Angelina (Lina) Fanny Hesse was the first to propose the use of agar as a culture media for growing and isolating bacteria.

Antonie van Leeuwenhoek, the father of microbiology, used homemade single-lens microscopes to make groundbreaking observations of bacteria and protozoa in 1676. He also discovered blood cells and nematodes in blood vessels, laying the foundation for our understanding of microscopic life forms.

Aristotle is recognized as both the Father of Biology and the Father of Zoology.

August Wilhelm Eichler's innovative system classified plants based on evolution, a major leap in plant taxonomy.

Barbara McClintock was awarded the Nobel Prize in Medicine in 1983 for her discovery of genetic transposition.

Barry Marshall and Robin Warren were jointly awarded the Nobel Prize in Physiology or Medicine in 2005 for their discovery of the bacterium *helicobacter pylori* and its role in gastritis and peptic ulcer disease.

Blumberg, in 1976 he won the Nobel Prize in Medicine for his discovery of the hepatitis B virus. He and his colleagues discovered the virus in 1967, developed the blood test that is used to detect the virus, and invented the first hepatitis B vaccine in 1969.

Camillo Golgi, In 1873, he published the first photograph of nerve tissue stained with the black reaction technique, describing a whole nerve cell.

Camillo Golgi and Santiago Ramon y Cajal shared the Nobel Prize in 1906 for their work on the histology of the nerve cell, but both held diametrically opposed views about the Neuron Doctrine.

Carolus Linnaeus developed the system of scientific naming of organisms in 1751.

Casimir Funk originally coined the term "vitamin", and he is considered as the father of vitamin therapy.

Cesare Lombroso gave the biological theory of criminology.

Charak is considered the father of ancient Indian science of medicine. He was the Raj Vaidya (royal doctor) in the court of Kanishka. His *Charak Samhita* is a remarkable book on medicine. He was the first to talk about digestion, metabolism and immunity.

Charles Robert Darwin discovered the Theory of Evolution, proposing the concept of natural selection in the 19th century.

Christiaan Barnard performed the first human heart transplant on December 3, 1967, in Cape Town, South Africa.

Dr. Edwards and Steptoe - In 1978, they announced the first successful live birth from IVF in England.

Edward Jenner pioneered the concept of vaccination and developed the smallpox vaccine in 1796.

Ernst Boris Chain isolated and purified penicillin (which had been discovered in 1928 by Sir Alexander Fleming) and performed the first clinical trials of the antibiotic.

Ernst Mayr published his famous book "Systematics and the Origin of Species" in 1942, which heavily contributed to population genetics and the evolutionary synthesis theory. He is also regarded as the Darwin of the 20th century.

Ettore Marchiafava is known for his study of Malaria.

Frederick Banting and Charles West - Discovered Insulin in 1921.

Friedrich Miescher was first who isolated Deoxyribonucleic Acid (DNA), analysed and reorganised as a unique macro-molecule in 1869.

George E Palade demonstrated the role of endoplasmic reticulum in protein processing and sorting in the 1960s.

Gregor Mendel, known as the "father of modern genetics", chose to study variation in plants. He worked with seven characteristics of pea plants: plant height, pod shape and color, seed shape and color, and flower position and color.

Har Gobind Khorana was one of the first scientists to demonstrate the role of nucleotides in protein synthesis and helped crack the genetic code. He also helped to develop custom-designed pieces of artificial genes and methods that anticipated the invention of the polymerase chain reaction (PCR) process, a biochemical technology used to amplify a single or a few copies of a piece of DNA.

In 1968 Khorana was awarded the Nobel Prize for Physiology or Medicine with Marshall W. Nirenberg and Robert W. Holley for "breaking the genetic code" and describing how it operates in protein synthesis.

Henry Chandler Cowles conducted pioneering studies on plant life in the Indiana Dunes in 1896.

Herbert Spencer initiated a philosophy of "Social Darwinism". He was the one who coined this term "survival of the fittest" seven years before Darwin published his theory of natural history.

H. J. Muller (Hermann Joseph Muller) was an American geneticist. He was awarded the Nobel Prize for the discovery of the production of mutations by means of X-ray irradiation, in 1946.

Ian Wilmut and Keith Campbell cloned Dolly the Sheep, the first mammal, in 1996.

Jagadish Chandra Bose discovered the Crescograph. It is a mechanical device to measure the growth or blossom of plants.

Joe Hin Tjio published research in 1956 defining the exact number of human chromosomes as $2n = 46$.

Joseph Banks was an English botanist and patron of the natural sciences. He was the first person to introduce the Western world to acacia, mimosa, eucalyptus and Banksia, a genus named after him.

Jonas Salk developed the Inactivated Polio Vaccine (IPV) in 1952.

Joseph Dalton Hooker was an English botanist. He was noted for his botanical travels and studies and for his encouragement of Charles Darwin and of Darwin's theories.

J.B.S. Haldane suggested that life must have developed from simple inorganic molecules. He was one of the founders of Neo-Darwinism.

Linus Pauling and Robert Corey

proposed the alpha helix and beta sheet secondary structures of proteins in 1951.

Louise Brown, born in 1978 in England, was the first baby conceived through in vitro fertilization (IVF). Kanupriya Agarwal (Durga), also born in 1978 in India, was the first Indian baby conceived through IVF.

Luc Montagnier first identified HIV/AIDS in 1983.

Marie Curie, a pioneering scientist, discovered Polonium and Radium alongside her husband Pierre Curie. Their work, along with Henri Becquerel, earned them the 1903 Nobel Prize in Physics. Marie's solo efforts in chemistry led to her winning the Nobel Prize in Chemistry in 1911.

Martinus Willem Beijerinck pioneered the recognition of viruses as distinct reproducing entities and discovered new soil bacteria, while also describing biological nitrogen fixation.

Matthias Jacob Schleiden and Theodor Schwann, co-founded the cell theory. In 1838, Schleiden defined the cell as the basic unit of plant structure, and a year later Schwann defined the cell as the basic unit of animal structure.

Mildred C. Rebstock was the first to synthesise active chloromycetin. Parke-Davis and Company publicly announced the synthesis of chloramphenicol on 26 March 1949.

Oliver Smithies, Mario R. Capecchi, Sir Martin J. Evans was jointly awarded The Nobel Prize in 2007 for their discoveries of principles for introducing specific Gene Modifications in Mice by the use of embryonic stem cells.

Patanjali introduced the first Yoga Sutra and is also known as the father of Yoga. He gave 195 yoga sutras which are considered as the backbone of Yoga followed today.

Paul Erlich is credited with coining the term "Chemotherapy".

Paul Lauterbur pioneered magnetic resonance imaging (MRI) in the early 1970s, allowing for multi-dimensional imaging of organs and soft tissues. His work, along with that of Peter Mansfield, earned them the Nobel Prize in Physiology or Medicine in 2003.

Rachel Louise Carson catalyzed the modern environmental movement with her book "Silent Spring" published in 1962.

Richard Axel was awarded the 2004 Nobel Prize in Medicine or Physiology for his pioneering work on the olfactory system.

Richard Willstätter won the Nobel Prize in 1915 for elucidating the structure of chlorophyll in plants, identifying magnesium as its central element. He also researched anthocyanins, the blue and red pigments found in flowers.

Robert Brown was a botanist. He collected, studied and classified thousands of plant flora from the Flinders expedition to Australia in 1801 – 1805. He described Brownian motion, the movement of small particles in solution, which is named after him and he described and named the plant cell nuclei (nucleus).

Rudolf Virchow's greatest accomplishment was his observation that a whole organism does not get sick, only certain cells or groups of cells get

sick. He stated that all diseases involve changes in normal cells.

He published an essay in 1855 titled 'Cellular Pathology' in which he formulated his famous aphorism '**Omnis cellula-e cellula**' which means that "every cell arises from another cell".

Samuel Hahnemann founded Homeopathy in 1796.

Selman Waksman is considered 'Father of Antibiotics'.

Sushruta, an ancient Indian physician, is hailed as the Father of Surgery and the Father of Indian Medicine and plastic surgery. He pioneered surgical techniques and authored the Sushruta Samhita, one of the earliest texts on plastic surgery.

Watson & Crick revealed the helical shape of the DNA molecule in 1953.

Walther Flemming is credited with founding cytogenetics through his systematic study of chromosome behavior.

Wilhelm Conrad Roentgen discovered X-ray in 1895.

William Harvey was the first person to describe the circulation of blood in the body. He showed that arteries and veins form a complete circuit. The circuit starts at the heart and leads back to the heart.

Environment and Ecology

Ecology and Ecosystem

Ecology studies how organisms interact with each other and their environment, from single creatures to whole ecosystems on Earth.

Ernst Haeckel coined "Oecologie" (ecology) in 1869, defining the study of organisms and their environment. Eugene Odum (father of modern ecology). Charles Elton (father of animal ecology). Ramdeo Mishra (Father of ecology in India).

Synecology is the study of the structure, development, and distribution of ecological communities.

Gote Turesson- Swedish botanist who coined "ecotype" and "agamospecies" and advanced our understanding of plant adaptation and genetics.

Raymond Lindeman - In the end of 1942, Raymond Lindeman submitted "The trophic- dynamic aspect of ecology" (the final chapter from his Ph. D. thesis) to the journal Ecology.

The Ecological component can be classified into two parts **Biotic and Abiotic**.

Biotic components are living factors of an ecosystem. Examples : Bacteria, animals, birds, fungi, plants, etc.

Abiotic components are non-living chemical and physical factors of an ecosystem. These components could be acquired from the atmosphere, lithosphere and hydrosphere. Examples : Sunlight, soil, air, moisture minerals etc.

The **Biosphere** is the narrow contact zone between the Lithosphere, hydrosphere and Atmosphere where we find land, water and air together, which contains all forms of life.

Ecological succession is a long-term process in which one biological community is replaced by another. The 5 stages of Ecological Succession include the following - Nudation, Invasion, Competition and Co-action, Reaction, and Climax or Stabilization.

Stratification in the field of ecology refers to the vertical layering of various

species; the arrangement of vegetation in layers. It classifies the layers (sing. stratum, pl. strata) of vegetation largely according to the different heights to which their plants grow.

The **green plants** in a terrestrial ecosystem capture **about 1%** of the energy of sunlight that falls on their leaves and converts it into food energy.

The sealable or open glass container containing soil and plants that can be opened for maintenance and to access the plants inside is called **terrarium**.

A **biome** is a collection of different ecosystems which share similar climatic conditions.

Biomass, in ecology, is the mass of living biological organisms in a given area or ecosystem at a given time, measured in metric tons of organic carbon per hectare (t – C / ha).

Biomass productivity is determined by dividing the biomass per unit area by the age of the forest ecosystem.

Flora is all the plant life present in a particular region or time, generally the naturally occurring (indigenous) native plants.

Arthur Tansley : The term 'Ecosystem' was first used in 1935 by him.

An ecosystem is where plants, animals, and landscapes cooperate to support life. Every element depends on others, directly or indirectly. Ecology is broader, with ecosystems being a part of it. Sunlight fuels ecosystems, which can be natural (like forests) or man-made (like gardens).

Eutrophication is the process in which a water body becomes overly enriched with nutrients, leading to plentiful growth of simple plant life. The excessive growth (or bloom) of algae and plankton in a water body are indicators of this process.

Planktonic - All those animals or plants, which keep flowing in water by water waves or water current.

Net primary productivity - It is the total amount of biomass accessible for heterotrophic use.

Gross primary productivity - It is the rate at which organic matter is produced through photosynthesis.

Decomposition - The breakdown of

complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition.

Leaching is the process by which soluble constituents present in the soil are dissolved in water and carried through the soil to the lower strata with percolation water.

Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called **catabolism**.

Humification means the formation of humus. Humus is a black amorphous substance produced by the decomposition of dead and decaying organic matter by microorganisms.

Mineralisation refers to the conversion of organic matter to mineral nutrients, which are readily absorbed by plants, whereas **immobilization** refers to mineral uptake by soil microbes and conversion to organic matter, which is unavailable to plants.

Mautam (Northeast India): Devastating famines every 48 yrs due to "Rat Flood" caused by mass bamboo flowering.

The IUCN Red List Categories and Criteria are intended to be an easily and widely understood system for classifying species at high risk of global extinction. It was founded in 1964.

It divides species into nine categories : Not Evaluated, Data Deficient, Least Concern, Near Threatened, Vulnerable, Endangered, Critically Endangered, Extinct in the Wild and Extinct.

Some examples of critically endangered species : Pygmy Hog (*Porcula salvania*), Kondana Rat (*Millardia kondana*), Great Indian Bustard (*Ardeotis nigriceps*), Siberian Crane (*Grus leucogeranus*), Indian Vulture (*Gyps indicus*), Gharial (*Gavialis gangeticus*), Leatherback Turtle (*Dermochelys coriacea*), Tiger toad (*Xanthophryne tigerinus*), Ganges Shark (*Glyphis gangeticus*), Fire corals (*Millepora boschmai*) etc.

The movement of nutrient elements through various components of an ecosystem is called **nutrient cycling**. Another name of nutrient cycling is biogeochemical cycles (bio: living organism, geo: rocks, air, water).

Nutrient cycles are of two types : (a) gaseous and (b) sedimentary. The

reservoir for gaseous type of nutrient cycle (e.g., nitrogen, carbon cycle) exists in the atmosphere and for the sedimentary cycle (e.g., sulphur and phosphorus cycle), the reservoir is located in Earth's crust.

Climax Community - An ecological community in which populations of plants or animals remain stable and exist in balance with each other and their environment. Examples are tundra, grassland, desert, and the deciduous, coniferous, and tropical rainforests.

Sere - The entire sequence of communities that successively change in a given area.

Hydrarch succession takes place in wet areas and the successional series progresses from hydric to the mesic conditions.

Xerarch succession takes place in dry areas and the series progresses from xeric to mesic conditions.

Pioneer species are hardy species that are the first to colonize barren environments or previously biodiverse steady-state ecosystems that have been disrupted, such as by wildfire. Common examples include bacteria, lichen, mosses, and fungi.

Standing state - The amount of nutrients, such as carbon, nitrogen, phosphorus, calcium, etc., present in the soil at any given time.

Adaptation

Adaption is the process an organism goes through in order to adapt or change based on its environment. An organism will adapt in a way that increases the organism's survivability. A camel has adapted physically to have a hump that stores excess fat.

There are three types of adaptations:-

Morphological adaptation is a structural change which gives an organism a greater chance of survival in its habitat. Giraffes develop their long neck is a prime example of this.

Behavioral adaptations are changes in behavior that certain organisms or species use to survive in a new environment. Changes in bird cells are a prime example of this.

Physiological adaptation is an internal

change that animals and plants go through to ensure their survival in their environment. Plants, such as nightshade, have developed properties that are poisonous and deadly to other organisms if they ingest the plant.

Henry Chandler Cowles studied ecological succession in the 1890s in the Indiana Dune of the USA.

Land Reclamation is the process of reclaiming or creating new land from oceans, riverbeds or lake beds. This method is used for conservation of land rescore.

Environment Acts/Conventions/Protocols

Wild Birds and Animals Protection Act -

This law made it illegal to possess and sell wild birds which were either killed or captured. A second new law was enacted in 1912 called the Wild Birds and Animals Protection Act. This act was amended in 1935.

The Wild Life Protection Act, 1972 was enacted for the protection of plants and animal species. It has six schedules that give varying degrees of protection.

Schedule I and part II of Schedule II provide absolute protection - offenses under these are prescribed the highest penalties. Species listed in Schedule III and Schedule IV are also protected, but the penalties are much lower. Schedule V includes the animals which may be hunted. The specified endemic plants in Schedule VI are prohibited from cultivation and planting.

The act has been amended in 1982, 1986, 1991, 1993, 2002, 2006 and 2013.

Wild Life (Protection) Amendment Act, 2006 - Provides for creating National Tiger Conservation Authority and Other Endangered Species Crime Control Bureau (Wildlife Crime Control Bureau).

Endemic species are those species of plants and animals which are found exclusively in a particular area.

Diclofenac has been held largely responsible for nearly wiping out the population of vultures from India.

The Water (Prevention and Control of Pollution) Act, 1974 aims to provide prevention and control of water pollution. Maintaining or restoring of wholesomeness and purity of water in

the various sources of water. It vests regulatory authority in Centre Pollution Control Boards (CPCB) and State Pollution Control Board (SPCB).

Forest Conservation Act, 1980 is targeted to limit deforestation, conserve biodiversity and save wildlife. Under this act, central permission is necessary to practice sustainable agro-forestry in forest areas. Violation or lack of permit was treated as a criminal offence. Forest (Conservation) Rules, 2022 is conferred by Section 4 of the Forest (Conservation) Act, 1980.

The **Air** (prevention and control of pollution) act, 1981 - The act targets to control and prevent air pollution in India and its main objectives are - prevention, control, and abatement of air pollution, To provide for the establishment of the boards at the central and state levels to implement the act. Central and State Pollution Control Boards (CPCB & SPCBs) were given the responsibility.

Environment Protection Act (EPA), 1986

- After the 1984 Bhopal Tragedy, the Government of India enacted the Environment Protection Act of 1986, which was passed in March 1986. The purpose of the Act is to implement the decisions of the UN Conference on the Human Environment.

National Forest Policy 1988 focuses primarily on forest preservation, conservation, and development. The key themes of the National Forest Policy 1988 were the preservation and restoration of ecological equilibrium, as well as the protection of forests as national heritages containing numerous types of flora and animals.

Biodiversity Act 2002 was enacted to conserve biological diversity in India. The act was passed on February 5, 2003. Biodiversity Act 2002 is the first attempt in India to legislate the conservation of biodiversity. To meet the requirements of the United Nations Convention on Biological Diversity (CBD), the Biodiversity Act 2002 was enacted.

The Disaster Management Act of 2005 (DMA 2005) is an act passed by the government of India for the 'efficient management of disasters and other matters connected to it.

The Forest Rights Act was passed in 2006. It was intended to address the 'historical injustice' done to 'traditional forest dwellers' of India. It was enacted

to protect the right to life and livelihood of scheduled tribes and other traditional forest dwellers.

Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 (FRA) – The act recognizes and vests the forest rights and occupation in forest land for Forest Dwelling Scheduled Tribes (FDST) and Other Traditional Forest Dwellers (OTFD) who have been residing in such forests for generations. This act comes under the aegis of the Ministry of Tribal Affairs.

National Green Tribunal (NGT) was established in 2010 under the NGT Act of 2010 as a specialized body for handling any environmental disputes that involve multi-disciplinary issues. It was formed by replacing the National Environment Appellate Authority.

Indian Forest (Amendment) Ordinance, 2017 : Bamboo has ceased to be a tree anymore. For 90 years Bamboo (which, taxonomically, is a grass) was registered as a tree in the Forest Act so that there will be no requirement of felling/transit permit for its economic use.

National Mission for **Green India** 2014 is one of the eight missions that come under the National Action Plan on Climate Change (NAPCC). The primary aim is to protect, restore and enhance India's diminishing forest cover.

Its objectives are growth in forest or tree cover to 5 million hectares (mha) and increase the quality of forest cover in another 5 million hectares of forest or non-forest lands.

The National Mission on Sustainable Habitat is one of the eight climate missions of the National Action Plan on Climate Change (NAPCC) mitigation strategy. Its main aim is to achieve a balance between the economic and social development of human habitats together with the protection of the environment. It was launched in the year 2010 and is governed by the Ministry of Urban Development.

Conventions/Protocols

Minamata Convention- Global treaty to fight mercury pollution (protecting health & environment) adopted in 2013 (Kumamoto, Japan).

Stockholm Convention is a convention on Persistent Organic Pollutants (POPs).

It was adopted in 2001 in Geneva, Switzerland. It came into force in 2004.

CITES (The Convention on International Trade in Endangered Species): International agreement protecting endangered wildlife trade since 1975. 184 countries (including the EU) work together to regulate trade in species at risk.

The 1984 Geneva or **EMEP Protocol** on Long-term Financing of the Cooperative Programme for Monitoring and Evaluation of the Long-range Transmission of Air Pollutants in Europe (EMEP).

The Protocol on Environmental Protection to the **Antarctic Treaty** was signed in Madrid on October 4, 1991 and entered into force in 1998. It designates Antarctica as a "natural reserve, devoted to peace and science". The Madrid Protocol was signed by India which came into force on 14th January, 1998.

Convention on Biological Diversity (CBD) is a convention for the conservation of biological diversity. It was adopted in 1992. It came into force in 1993.

Bonn Convention is a convention on the Conservation of Migratory Species of Wild Animals. It was adopted in 1979. It came into force in 1983.

The Vienna Convention is a convention for the Protection of Ozone Layers. It was adopted in 1985. It came into force in 1988.

Montreal Protocol is an international environment protocol on substances that deplete the Ozone Layer. It was adopted in 1987. It came into force in 1989.

The **Kyoto Protocol** is an international protocol to reduce greenhouse gas emissions. It was adopted in Kyoto, Japan in 1997 at CoP3 of UNFCCC. India ratified the protocol in 2002. It came into force in 2005.

United Nations Framework Convention on Climate Change (**UNFCCC**) is an international environmental treaty governing actions to combat climate change through adaptation and mitigation efforts directed at control of emission of Green House Gases (GHGs) that cause global warming. It was adopted in 1992. It came into force in 1994.

Rio Summit is a United Nations

Conference on Environment and Development. It was held in 1992 at Rio de Janeiro, Brazil.

UNCCD is a United Nations Convention to Combat Desertification. It was adopted in 1994. It came into force in 1996.

Basel Convention is a convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal. It was adopted in 1989. It came into force in 1992.

Rotterdam Convention is an international environmental convention on Prior Informed Consent (PIC) Procedure for Certain Hazardous Chemicals and Pesticides in International Trade. It was adopted in 1998. It came into force in 2004.

Cartagena Protocol is an international environmental protocol on Biosafety to the Convention on Biological Diversity. It was adopted in 2000. It came into force in 2003.

UN-REDD is a United Nations Programme on Reducing Emissions from Deforestation and Forest Degradation. It was created in 2008.

Nagoya Protocol is an international environment protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits Arising from their Utilization (ABS) to the Convention on Biological Diversity (CBD). It was adopted in 2010. It came into force in 2014.

The Kigali Agreement is an amendment to the Montreal Protocol. It was adopted in 2016. It came into force in 2019.

Agenda 21, originating from the 1992 Rio Earth Summit, is a non-binding action plan by the United Nations to foster global sustainable development, encompassing economic, social, and environmental concerns. It's been complemented by the Sustainable Development Goals since 2015 under Agenda 2030.

Few important environmental movements in India include, Reni Village in Garhwal saw the **Chipko movement** in 1970, led by Sunderlal Bahuguna. The **Save Silent Valley campaign** in 1973 aimed to protect Silent Valley in Kerala. In the early 1980s, the **Jungle Bachao Andolan** in Singhbhum District,

Jharkhand, and the **Appiko Movement** in Uttara Kannada, Karnataka, advocated for forest conservation.

The Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity.

SDGs	Goals
SDG 1	No poverty
SDG 2	Zero hunger
SDG 3	Good health and well-being
SDG 4	Quality education
SDG 5	Gender equality
SDG 6	Clean water and sanitation
SDG 7	Affordable and clean energy
SDG 8	Decent work and economic growth
SDG 9	Industry, innovation and infrastructure
SDG 10	Reduced inequalities
SDG 11	Sustainable cities and communities
SDG 12	Responsible consumption and production
SDG 13	Climate action
SDG 14	Life below water
SDG 15	Life on land
SDG 16	Peace, justice, and strong institutions
SDG 17	Partnerships for the goals

Food Chain and Food Web

Food Chain: The flow of nutrients and energy from one organism to another at different trophic levels is known as the food chain. Each step in the food chain is called **trophic level**. Example : Grasses - Grasshopper - Frog - Snake - Hawk.

Major parts of Food Chain :

Sun : This is the initial source of energy, which provides energy for everything on the planet.

Producers (Autotrophs) : They are first in trophic levels (maximum energy). These are the plants or other organisms that produce their own nutrients through photosynthesis.

Producers of the food chain include Typha bulrushes, Sagittaria, Hydrilla, Rupia, Chara, occurring in concentric layers from the periphery to the deep regions. Examples - Phytoplankton, grass, trees, Green plants.

Consumers : These are all organisms

that are dependent on plants or other organisms for food. This is the largest part of a food web, as it contains almost all living organisms. In a food chain, consumers are categorized into three categories.

Primary Consumer (Herbivores): They are the cattle, sheep, rabbits, deer, insects (grasshopper) and snails which feed on the green plants in the terrestrial ecosystems and the zooplankton, fishes and animals which feed on phytoplankton in the aquatic ecosystems. They are second in trophic levels.

Secondary Consumer (Carnivores): They are the organisms who eat primary consumers. They occupy the third trophic level of the energy pyramid. The secondary consumers are, sometimes, also known as carnivores and they usually consume meat. Examples - Birds, fish, wolf, etc.

Tertiary consumer (Top carnivores): They are Fourth in trophic levels. It consume the Secondary consumers. Examples - Man, Lion, tigers, pumas, jaguars.

Decomposers are the organisms that get energy from dead or waste organic material. This is the last stage in a food chain, and they convert organic waste materials into inorganic materials like nutrients - rich soil or land. Examples - Bacteria and fungi.

There are two types of food chains :

Grazing food chain :

In this energy flow, photosynthesis fuels the lowest trophic level, beginning with green plants producing food for herbivores, which then sustain carnivores. This food chain relies on energy transfer from autotrophs to herbivores, forming the basis for most ecosystems.

Detritus food chain :

This food chain involves dead organic matter at its primary trophic level, where microorganisms decompose decaying substances. It encompasses various organisms such as algae, bacteria, fungi, protozoa, mites, insects, and worms.

In an ecosystem the two chains are interconnected and make a **y - shaped** food chain. These two types of food chains are : Producers - Herbivores - Carnivores, Producers - Detritus, Feeders

- Carnivores.

Ecological pyramids, or Eltonian pyramids, were developed by Charles Elton in 1927. They visually represent trophic levels within ecosystems and come in three types: Pyramid of Number, Pyramid of Biomass, and Pyramid of Energy.

Pyramid of numbers represents the number of organisms at each trophic level.

Pyramid of biomass indicates a decrease in biomass in each trophic level from base to apex.

The biomass is maximum at the base where producers are found while the biomass at the apex or top is minimum where tertiary consumers are found.

A gradual decrease in the biomass from producers to tertiary consumers in the grazing food chain pyramid represents the ascending trophic level. In most of the terrestrial ecosystems the pyramid of biomass is upright. But in case of aquatic ecosystems the pyramid of biomass may be inverted.

Pyramid of energy represents the total amount of energy at each trophic level. Energy is expressed in terms of rate such as kcal/unit area /unit time or cal/unit area/unit time.

10 Percent Energy Law

The transfer of the energy in the food chain is limited and hence, the number of trophic levels in the food chain is limited.

There is only **10 percent** of the transfer of energy from each lower trophic level to the next or higher trophic level. This law, known as the 10 percent energy law, was proposed by **Raymond Lindeman**.

Other Terms Related to Food Chain

Bioaccumulation is the process by which toxic substances enter the food chain by building up in individual organisms.

Biomagnification, or biological magnification, is the process whereby certain chemical substances or toxins, like heavy metals and mercury, accumulate at higher trophic levels in a food chain. As these substances build up, they pose increasing risks to organisms higher up in the chain.

Food Chain in a forest: In this food chain, the grass is the producer organism that uses sunlight energy to prepare food like

carbohydrates by the process of photosynthesis. This grass is then consumed by a herbivore like deer. And the deer is consumed by a carnivore like a lion.

Food chain in grassland: In this food chain, grass is the producer. The insects (herbivores) are the primary consumer, the frog (small carnivore) and snakes are the secondary consumers whereas the bird (top carnivore or large carnivore) is the tertiary consumer.

Grass - Insects - Frog - Snake - Birds,
Grass - Mouse - Snake - Hawk.

Food chain in a pond: The food chain operates in the aquatic ecosystem like a pond, lake, or sea. In this aquatic food chain, algae is the producer. The scorpion is the primary consumer, small fish is the secondary consumer. Swan is the tertiary consumer.

Detritivores are heterotrophs that feed on detritus, aiding in decomposition and nutrient cycling within ecosystems. Predators consume detritivores, maintaining ecological balance. Examples include millipedes, springtails, woodlice, dung flies, and slugs.

Human beings are known as the producers of resources because they can produce food through agricultural activities. They are also known as the consumers of resources because they are dependent on plants and other animals for their food.

Primary producers including bacteria, phytoplankton, and algae form the lowest trophic level, the base of the aquatic food web.

Food Web

Trophic levels in an ecosystem are not linear rather they are interconnected and make a food web. Hence, Food Web is a web of interconnected food chains existing in an ecosystem.

One animal may be a member of several different food chains. Food webs are more realistic models of energy flow through an ecosystem. The flow of energy in an ecosystem is always linear or one way. The quantity of energy flowing through the successive trophic levels decreases.

Biomass energy

Biomass energy is derived from living or once-living organisms, commonly plants like corn and soy. This energy can be harnessed for heat or electricity generation. Examples of biomass materials include corn stover, wheat straw, oat straw, barley straw, sorghum stubble, and rice straw.

The process of getting energy from organic matter greatly depends on the type of biomass used. The main types are as follows :

Natural - This type of biomass is produced without human intervention.

Residual - It consists of the biological remains from farming and various types of industries. A few examples are sawdust and wood pellets from the forestry industry, cereal straw, olive pits, almond shells, or corn stover from farming.

Resulting from Energy Crops - It is the industrial growth of certain types of plants with the specific purpose of using them to obtain energy.

The International Energy Agency (IEA) estimates bioenergy accounts for roughly 1/10th of the world's total energy supply.

Conversion of biomass

Thermal Conversion of biomass - There are three main thermal processes - combustion, gasification, and pyrolysis - to convert the biomass into various energy products. Combustion is well established and widely practiced with many examples of dedicated plant and co-firing applications.

Chemical conversion - A range of chemical processes may be used to convert biomass into other forms, such as to produce a fuel that is more practical to store, transport and use, or to exploit some property of the process itself.

Biochemical processes have developed in nature to break down the molecules of which biomass is composed, and many of these can be harnessed. In most cases, microorganisms are used to perform the conversion. The processes are called anaerobic digestion, fermentation, and composting.

Wood in all of its forms including logs, sawdust, wood chips, wood pellets and tree bark can be used to produce energy. These are the main processes used to obtain energy from wood.

Agricultural Products - This category includes plant-based materials, both biological residue and energy crops. These are burned, and the heat is used to heat up a boiler.

Solid Waste - Garbage can be put to good use by burning it to produce energy. The system is essentially the same as the one described for wood combustion.

Biogas - It is produced by microscopic organisms, such as bacteria and fungi as they feed on dead plants and animals, as a waste product. The gasses generated through this process, particularly methane gas can then be used to produce energy because they are flammable.

Landfills - Landfills are deposit sites for waste material that is buried in the ground. This method has been used to dispose of waste for many years, and it is still quite common today.

Alcohol Fuel - Methanol, ethanol, propanol and butanol, the first four aliphatic alcohols, can be used as alternative fuels. Out of these four types, ethanol is by far the most accessible, because it can be produced in large quantities from fermenting plant-based material.

Biodiesel - It is yet another central alternative fuel option. It is produced from alcohol and vegetable oil or animal oil. In fact, recycled restaurant oil is one of the most important sources of biodiesel.

Pollution

Pollution - It is an undesirable change in physical, chemical or biological change in our surroundings. Agents that bring such changes are known as Pollutants.

Pollutants are substances or materials which impact the environment and its creatures thereby hampering the quality of life. **Examples** are Ozone (O₃), Nitrogen dioxide (NO₂), Radioactive substances, chemicals, biological wastes, flammable wastes, and explosives.

8 pollutants are included to measure the **Air Quality Index** in India. These eight pollutants are particulate matter (PM_{10}), ($PM_{2.5}$), Ozone (O_3), Sulphur dioxide (SO_2), nitrogen dioxide (NO_2), carbon monoxide (CO), lead (Pb) and ammonia (NH_3). These act as major parameters in deriving the AQI of an area.

Classification of Pollutants

On the basis of their existence in nature, pollutants are of two types – quantitative and qualitative.

Quantitative Pollutants - These are the pollutants that normally occur in nature. Example - Carbon dioxide.

Qualitative Pollutants - Those pollutants which are generated due to human activity. Examples - Insecticide, fungicide, herbicide etc.

There are two types on the basis of their forms in nature : Primary and Secondary.

Primary pollutants - Pollutants that enter directly from the sources into the atmosphere are called primary pollutants. It can be Natural or Anthropogenic. Example – **Particulate matter** like aerosols, soot, dust, hydrocarbons, sulphur dioxide, nitrogen oxide, carbon monoxide, carbon dioxide, fly ash.

Secondary pollutants - Pollutants that are formed due to interactions between primary air pollutants and other atmospheric components are called secondary pollutants. These interactions can be physical or chemical. Examples : Acid rain, photochemical smog, peroxyacetyl nitrate, ozone.

From an ecosystem point of view : Biodegradable and Nonbiodegradable.

Bio degradable - The substances that are degraded quickly by natural means in nature. Ex - Domestic Sewage

Non Biodegradable - The substance that does not degrade quickly by natural process or takes time to degrade. Ex - DDT, Aluminum Cans.

Types of Pollutions

Air Pollution, Noise Pollution, Water Pollution, Soil Pollution, Radioactive Pollution.

Air pollution: Contamination of air with harmful substances like gases &

particles that can harm people and the environment.

Pollutants of Air Pollution

Greenhouse gasses such as carbon dioxide and methane occur naturally in the atmosphere. They absorb sunlight reflected from Earth and prevent it from escaping into space. Greenhouse gasses are Water vapor, Carbon dioxide, Methane, Ozone, Nitrous oxide, Chlorofluorocarbons, etc.

Greenhouse effect - A process that occurs when gasses in Earth's atmosphere trap the Sun's heat.

Global warming - Earth's gradual warming due to human activities like burning fossil fuels, trapping heat and causing climate change.

Smog formation in general depends on both primary and secondary pollutants.

Ozone is called a "secondary" pollutant because it is produced by chemical reactions between two primary pollutants, namely, oxides of nitrogen and volatile organic compounds.

Volatile organic compounds (VOCs) are carbon-containing chemicals released by petrol and diesel vehicles. They impact air quality and human health.

Acid Rain - When rainwater's pH drops below 5.6 due to interactions with atmospheric carbon dioxide, it's termed acid rain. Major components include sulfur dioxide, nitrogen oxides, formic acid, carbon dioxide, and carbon monoxide.

Noise Pollution - Unwanted or excessive sound that can have deleterious effects on human health and environmental quality. Noise is measured in decibels (dB), as per WHO (World Health Organization) has prescribed optimum noise level as 45 dB by day to 35 dB at night.

Water pollution is the contamination of natural water bodies like lakes, rivers, and oceans due to the influx of pollutants. Any changes in water's chemical, physical, or biological properties that harm living organisms and the environment constitute water pollution. Ultraviolet (UV) treatment effectively neutralizes bacteria in water.

Soil pollution refers to the presence of toxic chemicals, or contaminants, in soil at concentrations high enough to endanger human health and the ecosystem. Even if naturally occurring contaminants in soil don't reach harmful levels, soil pollution occurs when their concentrations exceed natural levels.

Radioactive pollution results from the release of radioactive substances into the environment during various human activities, including nuclear explosions, weapon testing, mining of radioactive ores, and accidents at nuclear power plants. This pollution poses risks to both living organisms and ecosystems.

Terms related to Pollution

Smog Towers - Smog towers refer to large-scale structures designed to act as air purifiers. There are often several layers of air filters present in it, which clean the polluted air. The **world's largest** smog tower is located in China.

Anti-smog Gun is a kind of machine which sprays fine drops of nebulized water into the air. It is attached to a water tank and through a high-pressure propeller, tiny droplets of 50 to 100 microns are thrown into the air. Due to this, dust and pollution particles start getting absorbed.

Fly ash is a substance produced by burning many things (such as coal) which is made up of fine particles. These lighter particles rise up with the exhaled gasses. Fly ash contains substantial amounts of micron-sized earth elements such as oxides of silica, aluminum and calcium.

The National Air Quality Index (AQI) in India was launched on 17 October 2014 in New Delhi by the Environment Minister Shri Prakash Javadekar under the Swachh Bharat Abhiyan. Air Quality Index 8 pollutants - PM_{10} , $PM_{2.5}$, NO_2 , SO_2 , CO, O_3 , NH_3 and Pb).

Bharat Stage Norms (BS norms) - Bharat Stage Emission Standards have been instituted by the Central Pollution Control Board (CPCB), instituted within the Ministry of Environment Forests and Climate Change.

Vermicomposting- It is an environment friendly and useful method of conversion of rural and urban organic waste into good quality manure.

Computer

A computer is an electronic device that processes data according to instructions that are provided by computer programs.

Types of Computers

Analogue Computers - They are designed to process analogue data. Analogue data is continuous data that changes continuously and cannot have discrete values. Examples - Speedometer, Mercury Thermometer, Slide Rules, Differential Analysers, Castle Clock,

Digital computers - It processes data quickly using binary inputs (0s and 1s) and stored programs to perform calculations and logic operations, yielding output. Examples - Laptops, Desktops, Smartphones.

Hybrid Computers - Hybrid computers have features of both analogue and digital computer. It is fast like an analogue computer and has memory and accuracy like digital computers. Examples - Electrocardiogram machine, Echocardiogram machine, Airlines sector, Radar systems.

Supercomputers - They are the largest and fastest computers, engineered to handle vast amounts of data. Capable of processing trillions of instructions per second, they excel in tackling complex computations and data-intensive tasks. Examples - Fugaku (Japan), Summit (U.S.), Sierra, U.S., Sunway TaihuLight (China), Selene (U.S.), Tianhe-2A (China), Frontera (US). India's Supercomputers-AIRAWAT, PARAM Series, Pratyush.

Generations of Computers

There are five computer generations known till date.

First Generations of Computers (1940-1956) - Vacuum tubes were widely used in computers from 1940 through 1956. For Example ENIAC (Electronic Numerical Integrator And Computer), EDVAC (Electronic Discrete Variable Automatic Computer), EDSAC (Electronic Delay Storage Automatic Calculator), UNIVAC I (Universal Automatic Computer I), IBM 701, IBM 650 etc.

Second Generations of Computers (1956 - 1963) -The second generation of computers saw the use of transistors instead of vacuum tubes. Examples

include IBM 1620, IBM 7094, CDC 1604, CDC 3600, UNIVAC 1108, Honeywell 400 etc.

Third Generations of Computer (1964-1971) - The third generation of computers introduced the use of IC (integrated circuits) in computers. Examples include IBM 370, PDP-11, Honeywell - 6000, PDP8, IBM 360, ICL 2900 and DEC series.

Fourth Generations of Computers (1971-2010) - The fourth generation computers were made using very large scale integration technology. The computers made in the fourth generation were based upon the microprocessor, which was the extended version of the third generation of computers. Examples include DEC 10, STAR 1000, IBM 4341 and PUP 11.

Fifth Generations of Computers (2010-present) - This generation is based on parallel processing hardware and AI (Artificial Intelligence) software. Examples include IBM laptops, notebooks, Chromebook, Ultrabook Param 1000, Intel P4 and PCs of Pentium.

Input Device and Output Device

An **input device** is any hardware component or peripheral that allows users to provide data or commands to a computer. Examples - Keyboard, Mouse, JoyStick, Light pen, Track Ball, Scanner, Graphic Tablet, Microphone, Magnetic Ink Card Reader(MICR), Optical Character Reader(OCR), Barcode Reader Optical Mark Reader(OMR).

An **output device** is any peripheral device that receives data from a computer and presents it in a human-readable form. These devices convert electronic information into a format that humans can understand, such as text, images, or sound. Examples- Monitor, Printers, Headphone, Computer Speaker, Projector, GPS (Global Positioning System), Sound card.

Memory

Computer memory refers to the internal storage space within a computer system where data, programs, and instructions are stored for processing and retrieval. It is a crucial component of any computing device and comes in various forms,

including RAM (Random Access Memory) and ROM (Read-Only Memory).

Memory is primarily of three types :-

Cache Memory - Cache memory is a very high speed semiconductor memory which can speed up the CPU. It acts as a buffer between the CPU and the main memory. It consumes less access time as compared to main memory.

Primary Memory/Main Memory - Primary memory holds only those data and instructions on which the computer is currently working. It has a limited capacity and data is lost when power is switched off.

RAM (Random Access Memory) is a part of the computer's Main Memory which is directly accessible by the CPU. RAM is used to Read and Write data into it which is accessed by the CPU randomly.

The SRAM (Static Random Access Memory) memories consist of circuits capable of retaining the stored information as long as the power is applied. That means this type of memory requires constant power. Uses a flip-flop circuit to store data

DRAM (Dynamic Random Access Memory) stores the binary information in the form of electric charges applied to capacitors.

Read-Only Memory (ROM), is a type of non-volatile memory that stores data permanently on a computer or electronic device.

Types of ROM :

Mask ROM - The most basic type, permanently programmed during manufacturing.

PROM (Programmable Read-Only Memory) - Can be programmed once using a special high-voltage pulse.

EPROM (Erasable Programmable Read-Only Memory) - Can be erased with ultraviolet light and then reprogrammed.

EEPROM (Electrically Erasable Programmable Read-Only Memory) - Can be erased and reprogrammed electronically, offering more flexibility.

Flash memory - A more advanced type of EEPROM commonly used in storage devices like USB drives and solid-state drives (SSDs). It offers faster erase and rewrite capabilities.

Secondary Memory - It is a type of computer memory that is used to store data and programs that can be accessed or retrieved even after the computer is turned off.

Some examples of secondary memory include hard disk drives (HDDs), solid-state drives (SSDs), optical discs (such as CDs and DVDs), and flash memory (such as USB drives and memory cards).

Computer Memory Measurement Units

1 bit - Binary Digit (0/1)

1 nibble - 4 bits

1 byte - 8 bits

1 kilobyte (KB) - 1024 byte - 2^{10} bytes

1 Megabyte (MB) - 1024 KB - 2^{20} bytes

1 Gigabyte (GB) - 1024 MB - 2^{30} bytes

1 Terabyte (TB) - 1024 GB - 2^{40} bytes

1 Petabyte (PB) - 1024 TB - 2^{50} bytes

1 Exabyte (EB) - 1024 PB - 2^{60} bytes

1 Zettabyte (ZB) - 1024 EB - 2^{70} bytes

1 Yottabyte (YB) - 1024 ZB - 2^{80} bytes

1 Brontobyte - 1024 YB - 2^{90} bytes

1 Geopbyte - 1024 Brontobyte - 2^{100} bytes

Central Processing Unit - It is the primary component of a computer that acts as its "control center." It is called the brain of a computer.

There are three components of a Central Processing Unit (CPU) that are listed as follows -

Arithmetic and Logical Unit (ALU) - It is the brain of a Central Processing Unit (CPU) that performs arithmetic and logical operations.

Control Unit (CU) - It is circuitry within a computer's processor that directs operations.

Memory or storage unit - It holds all of the necessary data and instructions for processing.

Softwares - It is a collection of instructions, data, or computer programs that are used to run machines and carry out particular activities.

Types of Softwares :-

System Softwares - It is software that directly operates the computer hardware

and provides the basic functionality to the users as well as to the other software to operate smoothly. Examples - Operating System, Device Drivers, Firmware, BIOS and UEFI, Programming Language Translator, Utilities.

Application Softwares - Software that performs special functions or provides functions that are much more than the basic operation of the computer is known as application software. Examples - A collection of Microsoft software including MS Office, PowerPoint, Word, Excel, and Outlook. Common Internet browsers like Google Chrome, Safari, Firefox, etc. Graphics and designing software such as Canva, Adobe Photoshop, CorelDraw, and AutoCAD.

Hardware - It refers to the computer's tangible components or delivery systems that store and run the written instructions provided by the software. Examples - Motherboard, Central processing unit (CPU), Graphics processing unit (GPU), Power supply unit (PSU), Internal disk drive, Optical disk drive, Memory or RAM (think USB sticks).

Microsoft Office

Microsoft Word Terminologies

Alignment - It is how text flows in relation to the rest of the page (or column, table cell, text box, etc.). There are four main alignments- left, right, center, and justified.

Bullets - Any small character that appears before a listed item.

Case - It refers to whether letters are capitalized or not.

Clip Art - Pictures and drawings that can be inserted into a document.

Clipboard - A temporary storage place in memory.

Drag and Drop - A quick method to copy or move text a short distance.

Formatting Toolbar - Contains buttons for changing character and paragraph formatting, such as alignment and type styles.

Standard Toolbar - Located below the Menu bar, contains buttons you can use to perform common tasks, such as printing and opening documents.

Template - A designed and formatted document on which new documents are based.

Thesaurus - A feature that provides the user with synonyms for selected words.

Word Wrap - A feature that automatically advances text to the next line without pressing the Enter key.

Some important shortcuts in MS Word

Ctrl + x - Cut, Ctrl + c - Copy, Ctrl + v - Paste, Ctrl + z - Undo, Ctrl + y - Redo, Ctrl + B - Bold, Ctrl + I - Italics, Ctrl + U - Underline, Ctrl + L - Align left, Ctrl + R - Align Right, Ctrl + E - Align centered, Ctrl + J - Justify, Ctrl + > - Increase font size, Ctrl + < - Decrease font size.

Microsoft Word Excel Terminologies

3-D reference - A reference to a range that spans two or more worksheets in a workbook.

Active cell - The selected cell in which data is entered when you begin typing. Only one cell is active at a time.

Address - The path to an object, document, file, page, or other destination.

Argument - The values that a function uses to perform operations or calculations.

Array - Used to build single formulas that produce multiple results or that operate on a group of arguments that are arranged in rows and columns.

AutoSum - It is a built-in function in Microsoft Excel that automatically calculates and displays the sum of a selected range of cells.

Cell - A box formed by the intersection of a row and column in a worksheet or a table, in which you enter information.

Cell reference - The set of coordinates that a cell occupies on a worksheet. For example, the reference of the cell that appears at the intersection of column A and row 3 is A3.

Field (PivotTable) - In a PivotTable or PivotChart, a category of data that's derived from a field in the source data. PivotTables have row, column, page, and data fields. PivotCharts have series,

category, page, and data fields.

Fill handle - The small black square in the lower-right corner of the selection.

Formula - A sequence of values, cell references, names, functions, or operators in a cell that together produce a new value. A formula always begins with an equal sign (=).

Index - A database component that speeds up searching for data.

Iteration - Repeated calculation of a worksheet until a specific numeric condition is met.

Worksheet - A worksheet is a single spreadsheet in an Excel workbook where users can enter, manipulate, and analyze data. Each workbook can contain multiple worksheets.

Some important functions :-

ABS (number) - Returns the absolute value of a number, a number without its sign.

BAHTTEXT (number) - Converts a number to text (baht).

CHAR (number) - Returns the character specified by the code number from the character set for your computer.

CONCAT (text1,...) - Concatenates a list or range of text strings.

COUNTA (value1, [value 2],...) - Counts the number of cells in a range that are not empty.

COUNTBLANK (range) - Counts the number of empty cells in a specified range of cells.

COUNTIF(range, criteria) - Counts the number of cells within a range that meets the given conditions.

INDEX (...) - Returns a value or reference of the cell at the intersection of a particular row and column, in a given range.

INT (number) - Rounds a number down to the nearest integer.

LEN (text) - Returns the number of characters in a text string.

MATCH - Returns the relative position of an item in an array that matches a specified value in a specified order.

MEDIAN (number1, [number2], ..) - Returns the median, or the number in the middle of the set of given numbers.

NOW () - Returns the current date and time formatted as a date and time.

RAND () - Returns a random number greater than or equal to 0 and less than 1, evenly distributed (changes or recalculation).

SUMIF (range, criteria, [sum_range])- Adds the cells specified by a given condition or criteria.

TRANSPOSE (array) - Converts a vertical range of cells to a horizontal range, or vice versa.

TRIM (text) - Removes all spaces from a text string except for single spaces between words.

XIRR (values, dates, [guess]) - Returns the internal rate of return for a schedule of cash flow.

YIELD - Returns the yield on a security that pays periodic interest.

Some Important shortcuts in MS Excel

Shift + F10 - To open the context menu.

F2 - To edit selected cell.

F9 - To calculate the workbook.

Ctrl + Enter - To fill the selected range with the current entry.

Alt + Enter - To start a line in the same cell.

Shift + F3 - To insert function.

Ctrl + X or Shift + Delete - To cut.

Ctrl + C or Ctrl + Insert - To copy.

Ctrl + V or Shift + Insert - To paste.

Ctrl + P or Ctrl + Alt + Shift + F2 - To print.

Ctrl + F or Shift + F5 - To find.

Ctrl + Alt + O - To zoom your document to 100%.

Ctrl + Alt + 9 - To hide rows.

Ctrl + Alt + 0 - To hide columns.

MS Powerpoint Terminologies

Animation - It refers to visual effects that are added to individual items (such as titles, bulleted text and graphics) on a slide rather than to the slide itself.

Handouts - Handouts are basically thumbnails of slides printed together on a sheet of paper.

Layout (Slide Layout) - It typically refers to the predefined arrangement of content elements on a slide in presentation software such as Microsoft PowerPoint or Google Slides.

Master (Slide Master)- It allows users to define and customize the overall layout, formatting, and design elements that will be consistent across all slides in a presentation.

Placeholder - A placeholder on the slide is a holding area where you will add your content.

Slide - A slide is an individual page (or screen) in a slideshow.

Slide Show - A slide show is a series of slides that are saved together in a presentation and are displayed to an audience in sequence.

Slide Sorter View - The Slide Sorter view in PowerPoint allows you to view thumbnail versions of all slides contained in a presentation on one screen.

Template - A template is a file that you begin with when creating a new presentation.

Theme (Design Theme)- A design theme is a predefined set of formatting options that control the look and feel of your presentation.

Transition - Transitions in a slide show refer to the animation that is applied to make one slide forward to the next.

Some important shortcut keys

Ctrl + M - Insert a new slide.

Page down - Go to the next slide.

Page up - Go to the previous slide.

Ctrl + Minus sign (-) - Zoom out.

Ctrl + Plus sign (+) - Zoom in.

Ctrl + Alt + O - Zoom to fit.

Ctrl + Shift + D - Make a copy of the selected slide.

Ctrl + O - Open a presentation.

Ctrl + D - Close a presentation.

Alt + F - Open the File menu.

Alt + H - Open the Home tab.

Alt + N - Open the Insert tab.

Alt + G - Open the Design tab.

Alt + K - Open the Transitions tab.

Alt + A - Open the Animations tab.

Alt + S - Open the Slide Show tab.

Alt + R - Open the Review tab.

Alt + W - Open the View tab.

Electronic Mail

Electronic Mail (e-mail) is one of the most widely used services on the Internet. This service allows an Internet user to send a message in a formatted manner (mail) to another Internet user in any part of the world..

Structure of Email

Header - An email header is a code snippet that consists of essential details to authenticate an email message. The email header consists of all the information about the sender, the recipient, and more :

The sender, The recipient, The date, The subject, The server responsible for the transmission.

Image (optional). The email header includes the From, To, Cc, Bcc, Date, and Subject fields.

From - The email address, and, optionally, the name of the author(s). Some email clients are changeable through account settings.

To - Is the main recipient of the email.
Subject - A brief summary of the topic of the message.

Cc (Carbon copy) - The CC field allows you to send a copy of the email to any recipient of your choice.

Subject - A brief summary of the subject of the message.

Bcc (Blind carbon copy)- For security and privacy reasons, it is best to use the Blind Carbon Copy (BCC) feature when sending an email message to a large number of people.

Archived-At - A direct link to the archived form of an individual email message.

Body - The email body includes the message content and any attachments.

Parts of Email Address

1. Username - The username is the first part of an email address.

2. @ Symbol - The @ symbol, which means "at," separates the username from the rest of the email address. It means that the username belongs to the domain on the other side of the @ symbol.

3. Mail Server - The name of the person's mail server is on the right side of the @ symbol. It indicates the organization that hosts the server where their emails go.

4. Top-Level Domain - The mail server and top-level domain (TLD) combine to make a user's domain name. A top-level domain (TLD) is the part of a domain that comes after the dot. Top-level domains-
.com – company, .gov – government institution, .edu – educational institution, .org – organization, .net – network, .mil – military

Uniform Resource Locator

A URL (Uniform Resource Locator) is a unique identifier used to locate a resource on the Internet. It is the mechanism used by browsers to retrieve any published resource on the web. For Example

<https://www.sscgipinnacle.com/>

Components of URL

<https://www.sscgipinnacle.com-443/blog/article/search?docid=890&tll=yn#dayten>

Here:- Scheme (https- //), Sub Domain (www.), Domain Name (sscgipinnacle), Top- Level Domain (.com), Port Number (443), Path (blog/article/search), Query String Separator (?), Query String (docid=890&tll=yn), Fragment (dayten).

Network Protocols

Network protocols are the essential building blocks of network communication. They are sets of rules that define how data is formatted, packaged, addressed, transmitted and received by devices on a network.

TCP (Transmission Control Protocol) - This is a fundamental protocol that operates at the transport layer of the OSI model. TCP establishes a connection between two devices and guarantees reliable, in-order delivery of data packets.

IP (Internet Protocol)- This protocol works at the network layer and is responsible for addressing and routing data packets across networks.

UDP (User Datagram Protocol)- Unlike TCP, UDP is a connectionless protocol that prioritizes speed over reliability. It transmits data packets without establishing a connection or error checking, making it faster but less reliable than TCP.

HTTP (Hypertext Transfer Protocol) - This is the foundation of web communication. HTTP defines how web browsers and servers communicate to request and deliver web content.

FTP (File Transfer Protocol) - This protocol allows for the transfer of files between computers on a network.

DNS (Domain Name System)- DNS acts like a phonebook for the internet.

SNMP (Simple Network Management Protocol)- This protocol is used for network management and monitoring.

SSL/TLS (Secure Sockets Layer /Transport Layer Security) - These protocols provide secure communication by encrypting data transmission between devices.

SSH (Secure Shell) - This protocol allows for secure remote access and management of computer systems.

POP3/IMAP (Post Office Protocol & Internet Message Access Protocol) - These protocols are used for retrieving emails from a mail server.

BGP (Border Gateway Protocol)- This protocol is responsible for routing data packets between different autonomous systems (AS) on the internet.

SMTP (Simple Mail Transfer Protocol) - This protocol is used for sending emails.

DHCP (Dynamic Host Configuration Protocol) - This protocol helps automate IP address assignment on a network.

Data Transmission in Computer Network

Data transmission is sending and receiving digital or analog data between devices. This can be achieved through different mediums, such as cables, optical fibers, or wireless signals.

The way in which data is transmitted from one device to another device is known as transmission mode.

The Transmission mode is divided into three categories-

Simplex Mode - In Simplex mode, the communication is unidirectional, i.e., the data flow in one direction. Examples - Radio Broadcasts, Television, Security Cameras, Traffic Sensors, Keyboard and Mouse.

Half-duplex mode - In a Half-duplex channel, direction can be reversed, i.e., the station can transmit and receive the data as well. Examples - Walkie-Talkies, Satellite Communication Systems.

Full-duplex mode - In Full duplex mode, the communication is bi-directional, i.e., the data flow in both the directions. Examples - Landline Telephones, Mobile Phones (Modern), Video Conferencing, Online Gaming.

OSI (Open Systems Interconnection) Model

Layer 1 - Physical Layer - Deals with the physical transmission of data bits across a network medium like cables or wireless signals. Examples - Cables (coaxial, fiber optic), connectors (RJ-45), network cards.

Layer 2 - Data Link Layer - Focuses on error-free data transmission between directly connected network devices. Examples - Ethernet, Wi-Fi (IEEE 802.11).

Layer 3 - Network Layer - Responsible for routing data packets across networks. Examples- IP protocol, routing protocols (BGP).

Layer 4 - Transport Layer - Provides reliable and ordered data delivery between applications on different computers. Examples - TCP (Transmission Control Protocol) for reliable data transfer, UDP (User Datagram Protocol) for faster but connectionless communication.

Layer 5 - Session Layer - Establishes, manages, and terminates sessions between applications communicating across a network. Examples - Session initiation protocols used for establishing connections.

Layer 6 - Presentation Layer - Deals with data presentation and formatting for applications. Examples - Encryption protocols (SSL/TLS), image/video compression formats.

Layer 7 - Application Layer - Provides network services directly to user applications. Examples - HTTP protocol for web browsing, FTP protocol for file transfer.

The OSI model serves as a valuable tool for understanding network communication.

A **firewall** is a security system that monitors incoming and outgoing network traffic on a computer or network. It acts as a barrier between your internal network and the potentially dangerous external world, like the internet.

Antivirus software is a program designed to protect your computer system from malicious software (malware) such as viruses, worms, Trojan horses, spyware, and ransomware. Examples - Avast, AVG, Bitdefender Total Security, Kaspersky, McAfee, Malwarebytes, Norton, Quick Heal.

A **programming language** is a system of communication used to give computers instructions. Examples - Python, C, C++, Java, C# (C Sharp), JavaScript, HTML (HyperText Markup Language) & CSS (Cascading Style Sheets), Ruby, Swift, Kotlin, R.

Web browsers are software applications that allow you to access and interact with websites on the internet. The most popular browsers are Mozilla Firefox, Microsoft Internet Explorer, Google Chrome, Opera browser, Apple Safari, and Tor.

Network Topology

A network topology refers to the physical or logical layout of how devices in a network are interconnected.

Types of Network Topology :

Point to Point Topology - This is the simplest type of network, connecting two devices directly with a cable.

Mesh Topology - Devices are interconnected with multiple pathways, providing redundancy and fault tolerance.

Star Topology - A central hub or switch acts as a central connection point for all devices.

Bus Topology - All devices are connected to a single central cable.

Ring Topology - Ring Topology- Devices are connected in a closed loop, where data travels in one direction around the ring.

Tree Topology - Combines characteristics of star and bus topologies.

Hybrid Topology - Combines two or more of the basic topologies to create a network that meets specific needs.

Abbreviations

ADSL - Asymmetric Digital Subscriber Line

ALGOL - Algorithmic Language

ARPANET - Advanced Research Projects Agency Network

ASCII - American Standard Code for Information Interchange

BASIC - Beginner's All-purpose symbolic instruction code

BIOS - Basic Input Output System

COBOL - Common Business Oriented Language

CMOS - Complementary Metal Oxide Semiconductor

DML - Data Manipulation Language

DCL - Data Control Language

EBCDIC - Extended Binary Coded Decimal Interchange Code

ENIAC - Electronic Numerical Integrator and Calculator

EDVAC - Electronic Discrete Variable Automatic Computer

FORTRAN - Formula Translation

FLOPS - Floating Point Operations Per Second

GML - General Markup Language

HTML - HyperText Markup Language

IMAP - Internet Message Access Protocol

JDBC - Java Database Connectivity

MIME - Multipurpose Internet Mail Extensions

MODEM - Modulator And Demodulator

NTFS - New Technology File System

PLA - Programmable Logic Array

RDP - Remote Desktop Protocol

RIP - Routing Information Protocol

SOAP - Simple Object Access Protocol

SGML - Standard Generalized Markup Language

UTP - Unshielded Twisted Pair Cable

VLSI - Very Large-Scale Integration

WAP - Wireless Application Protocol

WML - Wireless Markup Language

XML - Extensible Markup Language

XGA - Extended Graphics Array